CHEMICAL THERMODYNAMICS

VOLUME 15

SELECTED ANCILLARY COMPOUNDS OF INTEREST TO RADIOACTIVE WASTE MANAGEMENT

OECD
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TDB
THERMOCHEMICAL DATA BASE PROJECT

NEA
NUCLEAR ENERGY AGENCY

CHEMICAL THERMODYNAMICS OF SELECTED ANCILLARY COMPOUNDS OF INTEREST TO RADIOACTIVE WASTE MANAGEMENT

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Preface

This volume is the fifteenth of the series of expert reviews of the chemical thermodynamics of key chemical elements in nuclear technology and waste management. Tables in Chapters III and IV list the selected thermodynamic values derived largely by the authors of this NEA TDB Project.

Ancillary derives from the Latin word *ancilla*, in this case providing support that supplements data on elements of prime interest to nuclear waste isolation. In other words, such data are of secondary importance, but require the same level of accuracy and authenticity as for the primary data.

This volume deals with the thermodynamic data of many inorganic compounds and some solid solutions that are likely to be involved in the reactions occurring if stored radioactive waste material is accidentally released into the environment. The work described therein builds essentially on the material covered in Chapter VI of Volume 1 of this series where brief mention was made of the thermodynamics of selected auxiliary compounds and provisional values were derived that were used in the analyses of the data for the main radionuclides of primary interest. The fact that this work also was limited to data available 30 years ago provided another incentive for the current study.

In the original scope of work proposed by Malcolm Rand and Jean Fuger, values of the existing TDB auxiliary data were to be compared with those in the NBS CODATA, SUPCRT and PSI databases. The final selected systems for study are as follows:

- **Phosphates:** It had been suggested that there were problems with the data for the enthalpy of formation and entropy of $H_3PO_4(aq)$ as well as with the interaction coefficients in aqueous phosphate species. That may have led to inconsistencies in the equilibrium constants recommended for aqueous phosphate species and solids in the NEA TDB database. The ion interaction coefficients for the corresponding arsenate were also to be investigated to establish similarities with the values for the corresponding phosphates.
- **Carbonates:** The ion interaction coefficients and dissociation constants of $CO_2(aq)$ and HCO_3^- were also re-examined in order to treat the data for magnesium and calcium aqueous species available to high temperatures and ionic strengths.
- **$SiO_2(s)$ and $SiO_2(aq)$ species:** In this review we considered only quartz, cristobalite, tridymite and vitreous silica in detail, with a short note on the amorphous phase. Note that in the preliminary NEA review, a choice was made

between two divergent datasets for the solubility of quartz. However, shortly thereafter an extensive study established unequivocally that the other dataset was indeed correct. In contrast to the accepted policy of the NEA, solubility data for amorphous silica have been treated because of the inherent stability of this phase, even at moderate temperatures. These results provide excellent, reproducible Si(IV) hydrolysis constants, whereas quartz is far too inert at ambient temperatures to allow the attainment of equilibrium.

- **Aqueous borate species**: Most of the thermodynamic data of borate hydrolysis obtained in the twentieth century are considered to be reliable today and have been included in the analysis of the hydrolysis of boric acid. In regard to the polyborates, only very recent experiments using Raman spectroscopy and conductance measurements have provided excellent data for the simpler polyborates, even to relatively high temperatures. The latter arrived too late to allow us time to re-treat them.

- **Aluminium species**: Since the last decade of the past century, studies of the solubility of gibbsite improved dramatically our understanding of the behaviour of gibbsite, and its polymorph, bayerite, over a wide range of pH and temperature, thus affecting the hydrolysis data from Al^{3+}. Boehmite is the dominant solid phase above *ca.* 350 K, but its passivity at ambient temperature prevented us from accepting the large amount of data available.

- **Magnesium and calcium:** The aqueous ions and their hydrolysed products have been well studied as have the chloride and sulfate species, the data for all of which are presented in this review. Interactions with carbonate in solution is extremely important in nature and have also been treated in detail in this review, mainly invoking the solid phases, brucite, magnesite, calcite, portlandite, anhydrite, gypsum, aragonite, and dolomite. The existence of solid phosphato complexes has been acknowledged.

- **Solid sodium and potassium carbonates and hydrocarbonates:** The data for sodium and potassium carbonates and hydrocarbonates were also assessed. The enthalpies of formation were based principally on the dissolution measurements by Berg and Vanderzee [1984BER/VAN]. It may be noted that no experimental data were found on the heat capacity and entropy of $KHCO_3$(cr) and no such data are selected for this compound.

- **Solid sodium borate:** The examination of the existing data for borax leads to the surprising conclusion that no value can be selected for the Gibbs energy of formation of this rather common compound.

- **Magnesium and calcium containing minerals**: The properties of the magnesium mineral chrysotile and of the structurally related antigorite have been examined in detail and the limitations of our knowledge of sepiolite and

related species have been outlined. Among calcium-containing minerals, we have examined two hydrated borates, colemanite and inyoite and the better known hydroxylapatite and fluorapatite.

The contributions of the authors may be summarised as follows: Malcolm Rand dealt with the data for the heat capacities and entropies of all the solid species, while Jean Fuger assessed their enthalpies of formation. Donald Palmer was responsible for the assessment of aqueous species, while Tamás Gajda dealt with the aqueous carbonate species, noting that all authors have approved of the selections and text. The initial editing was done by the chairmen with very efficient technical editing by Maria-Eleni Ragoussi (former TDB Project Co-ordinator), Davide Costa (former Junior Scientist) and Jesus S. Martinez (who contributed first as Junior Scientist and later on as TDB Project Co-ordinator). The entire manuscript has undergone a peer review by an independent international group of scientists, according to the procedures in the TDB-6 Guidelines, available from the NEA.

The review team held four in-person meetings in Paris. The first meeting in 2011 counted with the participation of Dmitri Chevdov (who left the team at the beginning of the project); the second meeting in 2012 counted with the participation of Cherry Tweed and Kastriot Spahiu (both were then members of the TDB IV Executive Group) and Robert Lemire; Cherry Tweed joined as well for the third meeting in 2013; the fourth and last in-person meeting took place in 2014 and counted with Rebecca Beard as a representative of the TDB V Executive Group. Three teleconferences were held after 2014. The COVID-19 pandemic has prevented any face-to-face interactions in recent years.

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The entire manuscript has undergone a peer-review by an independent, international group of reviewers, according to the procedures described in the TDB Project Guidelines, available from the NEA. The peer-reviewers have approved the modifications made by the authors in response to their comments. The peer-review comment records may be obtained upon request from the NEA.

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Their valuable contributions to this review are gratefully acknowledged.

Editor’s note

This is Volume 15 of a series of expert reviews of the chemical thermodynamics of key chemical elements in nuclear technology and waste management. This volume is devoted to the *Chemical Thermodynamics of Selected Ancillary Compounds of Interest to Radioactive Waste Management*. The tables contained in Chapters III and IV list the currently selected thermodynamic values within the NEA TDB Project. The database system developed at the NEA Data Bank assures consistency among all the selected and auxiliary data sets.

The recommended thermodynamic data are the result of a critical assessment of published information. The values in the auxiliary data set (see Tables IV-1 and IV-2) have been adopted from CODATA key values or have been critically reviewed in this or earlier volumes of the series.

How to contact the NEA TDB Project

Information on the NEA and the TDB Project is available at www.oecd-nea.org/dbtdb/. Selected data, as well as many documents in electronic format, can also be found on the NEA website.

To contact the TDB Project co-ordinator and the authors of the review reports, send comments on the TDB reviews, or to request further information, please send e-mails to tdb@oecd-nea.org or write to:

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Part 1

Introductory material

Chapter I

Introduction

I.1 Background

The term ‘Ancillary Data’ covers data on mineral and related species that, besides the traditional Auxiliary Data used in previous volumes, are required for the evaluation of potential chemical interactions in waste storage facilities.

In the field of radioactive waste management, the hazardous material consists to a large extent of actinides, fission products and activation products from nuclear reactors, in addition to lesser amounts from other sources such as waste from medicine, industry and research facilities. The scientific literature on thermodynamic data, mainly on equilibrium constants and redox potentials in aqueous solution, has been contradictory in a number of cases, especially in actinide chemistry. A critical and comprehensive review of the available literature is necessary to establish a reliable thermochemical database that fulfils the requirements for proper modelling of the behaviour of actinides and fission products in the environment.

Between 1963 and 1967, the International Atomic Energy Agency (IAEA) in Vienna published three technical reports with thermochemical assessments of the nuclear fuel systems: U-C and Pu-C [1963IAE], UO_2 and related phases [1965IAE], and Pu-O and U-Pu-O [1967IAE]. These reports were followed by nine special issues of Atomic Energy Review with compilations of physicochemical properties of compounds and alloys of elements important in reactor technology: Pu, Nb, Ta, Be, Th, Zr, Mo, Hf and Ti. In 1976, the IAEA also started publication of the series “The Chemical Thermodynamics of Actinide Elements and Compounds”, oriented towards nuclear engineers and scientists. This international effort resulted in the publication of ten volumes, each concerning the thermodynamic properties of a given type of compounds for the entire actinide series. These reviews cover the literature up to about 1984, although the last volume published in this series, Part 12: The Actinide Aqueous Inorganic Complexes [1992FUG/KHO] did not appear until 1992. Unfortunately, four of the reviews planned for inclusion in the IAEA series, reviews of importance for

radioactive waste management, were never published (for example, Part 10: The Actinide Oxides and Part 14: Aqueous Organic Complexes).

The Radioactive Waste Management Committee (RWMC) of the OECD Nuclear Energy Agency (NEA) recognised the need for an internationally acknowledged, high quality thermochemical database for application in the safety assessment of radioactive waste disposal, and undertook the development of the NEA Thermochemical Data Base (TDB) project [1988WAN], [2003MOM/WAN], [2015RAG/BRA], [2019RAG/COS]. The RWMC assigned high priority to the critical review of relevant chemical thermodynamic data of compounds and complexes containing any of the actinides uranium, neptunium, plutonium or americium, or the fission product technetium. The first four books in this series on the chemical thermodynamics of uranium [1992GRE/FUG], americium [1995SIL/BID], technetium [1999RAR/RAN] and neptunium and plutonium [2001LEM/FUG] originated from this initiative.

In 1998, Phase II of the TDB Project (TDB-II) was started to provide for further needs of radioactive waste management programs by updating the existing database and applying the TDB review methodology to other elements (nickel, selenium, zirconium) and to simple organic compounds and complexes. In TDB-II the overall objectives were set by a Management Board, including the representatives of 17 organisations from the field of radioactive waste management. These participating organisations, together with the NEA, provided financial support for TDB-II. The TDB-II Management Board was assisted in technical matters by a group of experts in chemical thermodynamics (the Executive Group). The NEA acted in this phase as Project Co-ordinator ensuring the implementation of the Project Guidelines and liaising with the Review Teams. Five publications resulted from TDB-II: the update on the chemical thermodynamics of uranium, neptunium, plutonium, americium and technetium [2003GUI/FAN], the books on chemical thermodynamics of nickel [2005GAM/BUG], selenium [2005OLI/NOL] and zirconium [2005BRO/CUR] and the volume devoted to compounds and complexes of the above mentioned metals with selected organic ligands [2005HUM/AND].

In 2003 TDB-III, the third phase of the Project began, and it was organised in a way similar to that used for TDB-II. An expert group was constituted to prepare a state-of-the-art report on the chemical thermodynamics of solid solutions of interest in radioactive waste management. That report [2007BRU/BOS] was published as volume 10 of the Chemical Thermodynamics Series. Also, three Review Teams were established to focus on the elements thorium, tin and iron. The volumes devoted to inorganic species and compounds of thorium [2008RAN/FUG] and tin [2012GAM/GAJ] have been published as volumes 11 and 12 of the Chemical Thermodynamics Series. However, it was recognised early in Phase III that assessment of the chemical thermodynamics of inorganic species and complexes of iron was too large an endeavour for a single book. Therefore, in 2008 the NEA RWMC and the

Executive Group endorsed deferral of the assessment of data for some iron species to Phase IV of the Project. The first of two volumes on iron species and compounds, was the final deliverable from TDB-III, and was published as volume 13a in the Chemical Thermodynamics Series.

The fourth phase (2008-2013), included in its program of work three reviews concerning iron (part 2), molybdenum and ancillary data, as well as the initiation of two state-of-the-art reports on cement minerals and high ionic strength solutions. The fifth phase of the TDB project entered into force in 2014. The work programme of TDB-V consisted in the completion of the reviews and state-of-the-art reports initiated in the fourth phase, and the preparation of an update of the phase II actinide volumes, including technetium. The present volume, the fifteenth in the Series, is the third published within TDB-6 after the recent release of the *Chemical Thermodynamics of Iron, Part 2* (volume 13b) and that of the *2nd Update on the chemical thermodynamics of uranium, neptunium, plutonium, americium and technetium* (volume 14).

The earlier volumes of the series are listed in the frontispiece of this volume; it may be noted that they and the companion volume on the modelling of aqueous solutions [1997ALL/BAN], are available without charge on the NEA website, www.oecd-nea.org/tdb-publications.

Simultaneously with the NEA TDB project, other reviews on the physical and chemical properties of actinides have appeared, including the book by Cordfunke *et al*. [1990COR/KON], the series edited by Freeman *et al*. [1984FRE/LAN], [1985FRE/LAN], [1985FRE/KEL], [1986FRE/KEL], [1987FRE/LAN], and [1991FRE/KEL], Part 12 of the IAEA review series by Fuger *et al*. [1992FUG/KHO] mentioned above and the two comprehensive books by Katz *et al*. [1986KAT/SEA] and Morss *et al*. [2006MOR/EDE] (particularly Chapter 19 [2006KON/MOR]). In 2010, a sixth volume [2010MOR/EDE] completed the series edited by Morss *et al*. All the compilations mentioned, although much less detailed than the NEA reviews (particularly with regard to documentation and discussion of published data), are important sources of information, and complement the thermodynamic data contained in the TDB volumes (though only with due regard to consistency).

I.2 Focus of the Review

The first and most important step in the modelling of chemical reactions is to decide whether they are controlled by chemical thermodynamics or kinetics, or possibly by a combination of the two. This also applies to the modelling of more complex chemical systems and processes, such as waste repositories of various kinds, the processes describing transport of toxic materials in ground and surface water systems, global geochemical cycles, *etc*.

As the data in the earlier TDB reviews have been accumulated, it became clear that equivalent thermodynamic assessments of the important auxiliary data of many of the solid and aqueous species, which either interact with the nuclides or control the prevailing aqueous chemistry, were required. Within this range of species there are in some cases new experimental data and inconsistences between the results of various experimental studies that have been identified and may have created doubts about the choices made in the previous TDB volumes.

This review considers the aqueous species formed in the presence of the following anions: OH^-, Cl^-, CO_3^{2-}, SO_4^{2-}, PO_4^{3-}, and $B(OH)_4^-$ as well as the individual hydrolysis behaviour of the latter four. The corresponding cations of interest are: H^+, Na^+, K^+, Mg^{2+}, Ca^{2+} and Al^{3+}, with again the latter three involving the need for examining additional hydrolysis data. Emphasis has been on the information at and near ambient conditions, in preference to that at higher temperatures and pressures (studies involving elevated temperatures were limited arbitrarily to 473.15 K). The aqueous ion interaction parameters for the SIT model for the relevant solutions have also been re-assessed, in some cases as a function of temperature as well as ionic strength. In addition, the stability of $Si(OH)_4(aq)$ previously assessed from solubility measurements of α-quartz, was in question and this analysis also necessitated determining the acid dissociation constants of this species.

The following generalised solid compounds were identified for an in-depth analysis of their thermodynamic properties: selected compounds of the oxides, hydroxides, carbonates, chlorides, sulfates and phosphates of Al, Mg, Ca and the carbonates of sodium and potassium. For many of these solids, we have made considerable use of the earlier assessments by the Russian IVTAN teams (*e.g.* [1981GLU/MED], [1989GUR/VEY]), especially for their references to earlier Russian studies. It may be noted that the original Russian volumes edited by Glushko and Medvedev are no longer easily available outside Russia, but that all of these volumes were reprinted in 1999 with a few helpful English translations in the eight-volume set noted in reference [1999YUN]. Moreover, for both the aqueous and solid species of calcium, we have made considerable use of the CODATA assessment of these [1987GAR/PAR].

In addition, the existing thermodynamic data for several minerals of interest for the storage of nuclear waste were also examined, though many of these data are much less reliable than those for the simpler solids noted above and no reliable data could be selected for some minerals.

I.3 Review procedure and results

The objective of the present review is to present an assessment of the sources of published thermodynamic data in order to decide on the most reliable values and their

uncertainties that can be recommended for modelling purposes. Experimental measurements published in the scientific literature are the main source for the selection of recommended data. Previous reviews are not neglected, since they form a valuable source of critical information on the quality of primary publications. When necessary, experimental source data are re-evaluated by using chemical models which are either found to be more realistic than those used by the original author, or are consistent with subsequent experimental information. For example, data on solubilities might need to be re-interpreted to take into account the crystal structure and particle size of the phases actually investigated.

Re-evaluation of literature values might be also necessary to correct for known systematic errors (for example, if complex formation was neglected in the original publication) or to make extrapolations to standard state conditions ($I = 0$) by using the specific ion interaction (SIT) equations (*cf.* Appendix B). For convenience, these SIT equations are referred to in some places in the text as "the SIT". To ensure that consistent procedures are used for the evaluation of primary data, a number of guidelines have been developed. They have been updated and improved since 1987, and their most recent versions are available at the NEA [1999WAN], [1999WAN/OST], [2013GRE/MOM], [2000OST/WAN], [2000WAN/OST]. Some of the procedures also are outlined in this volume, *cf.* Chapter II, Appendix B, and Appendix C. Parts of these sections were recently revised in the thorium [2008RAN/FUG] and tin [2012GAM/GAJ] reviews. Further minor changes were made in this review, including corrections to a few entries in the Tables of Appendix B.

Once the critical review process for each volume in the NEA TDB project is completed, the resulting manuscript is reviewed independently by qualified experts nominated by the NEA. The independent peer review is performed according to the procedures outlined in the TDB-6 guideline [1999WAN]. The purpose of the additional peer review is to obtain an independent view of the judgements and assessments made by the primary reviewers, to verify assumptions, results and conclusions, and to check whether the relevant literature has been exhaustively considered. The independent peer review is performed by scientists having technical expertise in the subject matter to be reviewed.

The thermodynamic data selected in the present review (see Chapter III) refer to the reference temperature of 298.15 K and to standard conditions, *cf.* Section II.3. For the modelling of real systems it is, in general, necessary to re-calculate the standard thermodynamic data to non-standard state conditions. For aqueous species a procedure for the calculation of the activity factors is required. As noted earlier, this review uses an approximate specific ion interaction treatment (SIT) for the extrapolation of experimental data to the standard state in the data evaluation process, and in some cases this requires the re-evaluation of original experimental values (solubilities, cell-potential data, *etc*.). For maximum consistency, the selected data presented in this review must always be used in conjunction with this method as described in Appendix B. The

thermodynamic data selected in this review are provided with uncertainties representing the 95% confidence level. As discussed in Appendix C, there is no unique way to assign uncertainties, and the assignments made in this review are to a large extent based on the subjective choices of the reviewers, supported by their scientific and technical experience in the corresponding area.

The quality of thermodynamic models cannot be better than the quality of the data on which they are based. The quality aspect includes both the numerical values of the thermodynamic data used in the model and the "completeness" of the chemical model used, *e.g.* the inclusion of all the relevant dissolved chemical species and solid phases. For the user it is important to note that the selected data set presented in this review (Chapter III) is certainly *not* "complete" with respect to all the conceivable systems and conditions; there are gaps in the information. The gaps are pointed out in the main text in Chapter V, and this information may be used as a basis for the assignment of future research priorities. Addition of thermodynamic values from other sources for modelling purposes without proper consideration of compatibility with the values in Chapters III and IV may lead to incorrect results.

Chapter II

Standards, conventions and contents of the tables

This chapter outlines and lists the symbols, terminology and nomenclature, the units and conversion factors, the order of formulae, the standard conditions, and the fundamental physical constants used in this volume. They are derived from international standards and have been specially adjusted for the TDB publications.

II.1 Symbols, terminology and nomenclature

II.1.1 Abbreviations

Abbreviations are mainly used in tables where space is limited. Abbreviations for methods of measurement are listed in Table II-1.

Table II-1: Abbreviations for experimental methods.

aix	Anion exchange
AES	Atomic Emission Spectroscopy
BET	Brunauer-Emmett-Teller specific surface area measurement
cal	Calorimetry
chr	Chromatography
cix	Cation exchange
col	Colorimetry
con	Conductivity
cou	Coulometry
cry	Cryoscopy
dis	Distribution between two phases
DSC	Differential Scanning Calorimetry
DTA	Differential Thermal Analysis
EDS	Energy Dispersive Spectroscopy
em	Electromigration
E_{cell}	Potential difference of an electrochemical cell

(Continued on next page)

Table II-1 (continued)

EPMA	Electron Probe Micro Analysis
ESR	Electron Spin Resonance Spectroscopy
EXAFS	Extended X-ray Absorption Fine Structure
FTIR	Fourier Transform Infra Red
IDMS	Isotope Dilution Mass–Spectroscopy
IR	Infrared
gl	Glass electrode
ise–x	Ion selective electrode with ion X stated
isop	Isopiestic
Ix	Ion exchange
Kin	Rate of reaction
LIBD	Laser Induced Breakdown Detection
MAS NMR	Magic Angle Spinning Nuclear Magnetic Resonance
MVD	Molar Volume Determination
NMR	Nuclear Magnetic Resonance
PAS	Photo Acoustic Spectroscopy
pol	Polarography
pot	Potentiometry
prx	Proton relaxation
qh	Quinhydrone electrode
red	E_{cell} with redox electrode
rin	Refractometry (refractive index measurements)
RAM/DFT	Raman spectroscopy and Density functional theory
SEM	Scanning Electron Microscopy
sp	Spectrophotometry
sol	Solubility
TC	Transient Conductivity
TEM	Transmission Electron Microscopy
TGA	Thermo Gravimetric Analysis
TLS	Thermal Lensing Spectrophotometry
Tj	Temperature jump
TRLFS	Time Resolved Laser Fluorescence Spectroscopy
UV	Ultraviolet
vlt	Voltammetry
XANES	X-ray Absorption Near Edge Structure
XRD	X-ray Diffraction
?	Method unknown to the reviewers

Other abbreviations may also be used in tables, such as SHE for the standard hydrogen electrode or SCE for the saturated calomel electrode. The abbreviation NHE has been widely used for the “normal hydrogen electrode”, which is by definition identical to the

SHE. It should nevertheless be noted that NHE customarily refers to a standard state pressure of 1 atm, whereas SHE always refers to a standard state pressure of 0.1 MPa (1 bar) in this review.

II.1.2 Symbols and terminology

The symbols for physical and chemical quantities used in the TDB review follow the recommendations of the International Union of Pure and Applied Chemistry, IUPAC [1979WHI], [1993MIL/CVI], [2007COH/CVI]. They are summarised in Table II-2.

Table II-2: Symbols and terminology.

Symbols and terminology	
length	l
height	h
radius	r
diameter	d
volume	V
mass	m
relative atomic mass	A_r
molar mass	M
density (mass divided by volume)	ρ
molar surface area	$\mathcal{A}$
time	t
frequency	ν
wavelength	λ
internal transmittance (transmittance of the medium itself, disregarding boundary or container influence)	T_i
internal transmission density, (decadic absorbance): $\log_{10}(1/T_i)$	A
molar (decadic) absorption coefficient: $A / c_B l$	ε
relaxation time	τ
Avogadro constant	N_A
relative molecular mass of a substance[(a)]	M_r
thermodynamic temperature, absolute temperature	T
Celsius temperature	t
(molar) gas constant	R
Boltzmann constant	k_B
Faraday constant	F
(molar) entropy	S_m
(molar) heat capacity at constant pressure	$C_{p,m}$
(molar) enthalpy	H_m
(molar) Gibbs energy	G_m

(Continued on next page)

Table II-2 (continued)

Symbols and terminology	
chemical potential of substance B	μ_B
pressure	p
partial pressure of substance B: $x_B p$	p_B
fugacity of substance B	f_B
fugacity coefficient: f_B/p_B	γ_{f_B}
amount of substance[(b)]	n
mole fraction of substance B:	x_B
molarity or concentration of a solute substance B (amount of B divided by the volume of the solution)[(c)]	c_B, [B]
molality of a solute substance B (amount of B divided by the mass of the solvent)[(d)]	m_B
factor for the conversion of molarity to molality of a solution: m_B / c_B	ξ
mean ionic molality[(e)], $m_\pm^{(\nu_+ + \nu_-)} = m_+^{\nu_+} m_-^{\nu_-}$	$m_\pm$
activity of substance B	a_B
activity coefficient, molality basis: a_B / m_B	γ_B
activity coefficient, concentration basis: a_B / c_B	y_B
mean ionic activity[(e)], $a_\pm^{(\nu_+ + \nu_-)} = a_B = a_+^{\nu_+} a_-^{\nu_-}$	$a_\pm$
mean ionic activity coefficient[(e)], $\gamma_\pm^{(\nu_+ + \nu_-)} = \gamma_+^{\nu_+} \gamma_-^{\nu_-}$	$y_\pm$
osmotic coefficient, molality basis	ϕ
ionic strength: $I_m = \frac{1}{2}\sum_i m_i z_i^2$ or $I_c = \frac{1}{2}\sum_i c_i z_i^2$	I
SIT ion interaction coefficient between substance B_1 and substance B_2	$\varepsilon(B_1, B_2)$
stoichiometric coefficient of substance B (negative for reactants, positive for products)	ν_B
general equation for a chemical reaction	$0 = \sum_B \nu_B B$
equilibrium constant[(f)]	K
charge number of an ion B (positive for cations, negative for anions)	z_B
charge number of a cell reaction	n
potential difference of an electrochemical cell	E
$\mathrm{pH} = -\log_{10}[a_{H^+} / (\mathrm{mol \cdot kg^{-1}})]$	pH
molar conductivity	Λ
molar conductivity at $I = 0$	Λ_0
superscript for standard state[(g)]	$^\circ$

(a) Ratio of the average mass per formula unit of a substance to $\frac{1}{12}$ of the mass of an atom of nuclide ^{12}C.

(b) *cf.* sections 1.2 and 3.6 of the IUPAC manual [1979WHI].

(c) This quantity is called "amount–of–substance concentration" in the IUPAC manual [1979WHI]. A solution with a concentration equal to 0.1 $\mathrm{mol \cdot dm^{-3}}$ is called a 0.1 molar solution or a 0.1 M solution.

(d) A solution having a molality equal to 0.1 moles of solute per kg solvent is called a 0.1 molal solution or a 0.1 m solution.

(e) For an electrolyte $N_{\nu_+} X_{\nu_-}$ which dissociates into $\nu_\pm$ $(= \nu_+ + \nu_-)$ ions, in an aqueous solution with molality m, the individual cationic molality and activity coefficient are $m_+ (= \nu_+ m)$ and $\gamma_+ (= a_+ / m_+)$. A similar definition is used for the anionic symbols. Electrical neutrality requires that $\nu_+ z_+ = \nu_- z_-$.

(f) Special notations for equilibrium constants are outlined in Section II.1.7. In some cases, K_c is used to indicate a concentration constant in molar units, and K_m a constant in molal units.

(g) See Section II.3.1.

II.1.3 Chemical formulae and nomenclature

This review follows the recommendations made by IUPAC [1971JEN], [1977FER], [1990LEI] on the nomenclature of inorganic compounds and complexes, except for the following items:

- The formulae of co-ordination compounds and complexes are not enclosed in square brackets [1971JEN] (Rule 7.21). Exceptions are made in cases where square brackets are required to distinguish between co-ordinated and unco-ordinated ligands.
- The prefixes "oxy-" and "hydroxy-" are retained if used in a general way, *e.g.*, "gaseous uranium oxyfluorides". For specific formula names, however, the IUPAC recommended citation [1971JEN] (Rule 6.42) is used, *e.g.*, "uranium(IV) difluoride oxide" for UF_2O(cr).

An IUPAC rule that is often not followed by many authors [1971JEN] (Rules 2.163 and 7.21) is recalled here: the order of arranging ligands in co-ordination compounds and complexes is the following: central atom first, followed by ionic ligands and then by the neutral ligands. If there is more than one ionic or neutral ligand, the alphabetical order of the symbols of the ligands determines the sequence of the ligands. For example, $(UO_2)_2CO_3(OH)_3^-$ is standard, $(UO_2)_2(OH)_3CO_3^-$ is not recommended.

It is convenient to have a shorthand method of referring to an ionic medium where the anion concentration is kept constant, but the cation concentrations vary, since such solutions are often used to minimise changes in interionic attractions. For this we enclose the major cation in parentheses. For example, a solution described as "3.00 M (Na)ClO_4" could be a mixture of 0.01 M $Th(ClO_4)_4$ and 2.96 M $NaClO_4$ with a total perchlorate concentration of 3.00 M.

Abbreviations of names for organic ligands appear sometimes in formulae. Following the recommendations by IUPAC, lower case letters are used, and if necessary, the ligand abbreviation is enclosed within parentheses. Hydrogen atoms that can be replaced by the metal atom are shown in the abbreviation with an upper case "H", for example: H_3edta^-, Am(Hedta)(s) (where edta stands for ethylenediaminetetraacetate).

II.1.4 Phase designators

Chemical formulae may refer to different chemical species and are often required to be specified more clearly in order to avoid ambiguities. For example, UF_4 occurs as a gas, a solid, and an aqueous complex. The distinction between the different phases is made by phase designators that immediately follow the chemical formula and appear in parentheses. The only formulae that are not provided with a phase designator are aqueous ions. They are the only charged species in this review since charged gases are not considered. The use of the phase designators is described below.

- The designator (l) is used for pure liquid substances, *e.g.*, $H_2O(l)$.
- The designator (aq) is used for undissociated, uncharged aqueous species, *e.g.*, $U(OH)_4(aq)$, $CO_2(aq)$. Since ionic gases are not considered in this review, all ions may be assumed to be aqueous and are not designed with (aq). If a chemical reaction refers to a medium other than H_2O (*e.g.*, D_2O, 90% ethanol/10% H_2O), then (aq) is replaced by a more explicit designator, *e.g.*, "(in D_2O)" or "(sln)". In the case of (sln), the composition of the solution is described in the text.
- The designator (sln) is used for substances in solution without specifying the actual equilibrium composition of the substance in the solution. Note the difference in the designation of H_2O in Eqs. (II.2) and (II.3). $H_2O(l)$ in Reaction (II.2) indicates that H_2O is present as a pure liquid, *i.e.*, no solutes are present, whereas Reaction (II.3) involves an HCl solution, in which the thermodynamic properties of $H_2O(sln)$ may not be the same as those of the pure liquid $H_2O(l)$. In dilute solutions, however, this difference in the thermodynamic properties of H_2O can be neglected, and $H_2O(sln)$ may be regarded as pure $H_2O(l)$.

  Example:

$$UO_2Cl_2(cr) + 2\,HBr(sln) \rightleftharpoons UO_2Br_2(cr) + 2\,HCl(sln) \quad \text{(II.1)}$$

$$UO_2Cl_2{\cdot}3H_2O(cr) \rightleftharpoons UO_2Cl_2{\cdot}H_2O(cr) + 2\,H_2O(l) \quad \text{(II.2)}$$

$$UO_3(\gamma) + 2\,HCl(sln) \rightleftharpoons UO_2Cl_2(cr) + H_2O(sln) \quad \text{(II.3)}$$

- The designators (cr), (am), (vit), and (s) are used for solid substances. (cr) is used when it is known that the compound is crystalline, (am) when it is known that it is amorphous, and (vit) for glassy substances. Otherwise, (s) is used.
- In some cases, more than one crystalline form of the same chemical composition may exist. In such a case, the different forms are distinguished by separate designators that describe the forms more precisely. If the crystal has a mineral name, the designator (cr) is replaced by the first four characters of the mineral name in parentheses. If there is no mineral name, the designator (cr) is replaced by a Greek letter preceding the formula and indicating the structural phase, *e.g.*, α-UF_5, β-UF_5.

Phase designators are also used in conjunction with thermodynamic symbols to define the state of aggregation of a compound to which a thermodynamic quantity refers. The notation is in this case the same as outlined above. In an extended notation (*cf.* [1982LAF]) the reference temperature is usually given in addition to the state of aggregation of the composition of a mixture.

Example:

$\Delta_f G_m^o(Na^+, 298.15\ K)$	standard molar Gibbs energy of formation of aqueous Na^+ at 298.15 K
$S_m^o(UO_2(SO_4)\cdot 2.5H_2O, cr, 298.15\ K)$	standard molar entropy of $UO_2(SO_4)\cdot 2.5H_2O$(cr) at 298.15 K
$C_{p,m}^o(UO_3, \alpha, 298.15\ K)$	standard molar heat capacity of α-UO_3 at 298.15 K
$\Delta_f H_m(HF, sln, HF\cdot 7.8H_2O)$	enthalpy of formation of HF diluted 1:7.8 with water.

II.1.5 Processes

Chemical processes are denoted by the operator Δ, written before the symbol for a property, as recommended by IUPAC [1982LAF]. An exception to this rule is the equilibrium constant, *cf.* Section II.1.7. The nature of the process is denoted by annotation of the Δ, *e.g.*, the Gibbs energy of formation, $\Delta_f G_m$, the enthalpy of sublimation, $\Delta_{sub} H_m$, *etc*. The abbreviations of chemical processes are summarised in Table II-3.

Table II-3: Abbreviations used as subscripts of Δ to denote the type of chemical process.

Subscript of Δ	Chemical process
at	separation of a substance into its constituent gaseous atoms (atomisation)
dehyd	elimination of water of hydration (dehydration)
dil	dilution of a solution
f	formation of a compound from its constituent elements
fus	melting (fusion) of a solid
hyd	addition of water of hydration to an unhydrated compound
mix	mixing of fluids
r	chemical reaction (general)
sol	process of dissolution
sub	sublimation (evaporation) of a solid
tr	transfer from one solution or liquid phase to another
trs	transition of one solid phase to another
vap	vaporisation (evaporation) of a liquid

The most frequently used symbols for processes are $\Delta_f G$ and $\Delta_f H$, the Gibbs energy and the enthalpy of formation of a compound or complex from the elements in their reference states (*cf.* Table II-6).

II.1.6 Nomenclature in log terms

It is of course clear that the arguments used in log functions must be dimensionless, and this practice has been used in writing pressures, usually as $\log_{10} p$/bar. However, since the units of the equilibrium constants, K_m and K_c, are frequently quite cumbersome, they have been omitted in the log terms of these ($\log_{10} K_m$ and $\log_{10} K_c$) to facilitate reading of the text.

II.1.7 Equilibrium constants

IUPAC has not explicitly defined the symbols and terminology for equilibrium constants of reactions in aqueous solution. The NEA TDB Project has therefore adopted the conventions that have been used in the work *Stability Constants of Metal Ion Complexes* by Sillén and Martell [1964SIL/MAR], [1971SIL/MAR]. An outline is given in the paragraphs below. Note that, for some simple reactions, there may be different correct ways to index an equilibrium constant. It may sometimes be preferable to indicate the number of the reaction to which the data refer, especially in cases where several ligands are discussed that might be confused. For example, for the equilibrium:

$$m\ \mathrm{M} + q\ \mathrm{L} \rightleftharpoons \mathrm{M}_m\mathrm{L}_q \qquad \text{(II.4)}$$

both $\beta_{q,m}$ and β (II.4) would be appropriate, and $\beta_{q,m}$ (II.4) is accepted, too. Note that, in general, K is used for the consecutive or stepwise formation constant, and β is used for the cumulative or overall formation constant. In the following outline, charges are only given for actual chemical species, but are omitted for species containing general symbols (M, L).

The expressions in sub-Sections II.1.7.1.1 through II.1.7.1.4 are expressed in terms of (amount substance) concentrations (*i.e.*, [X] ≡ c_X and the constants are K_c; *cf.* footnote *f* to Table II-2), but parallel nomenclature has been used in the rest of the review for equilibria expressed using molalities (K_m) or using activities in thermodynamic equilibrium constants (K°). The units here for

$$K_c = \prod_{\mathrm{B}} c^{\nu_\mathrm{B}}$$

are $(\mathrm{mol\cdot dm^{-3}})^{\Sigma_\mathrm{B}\nu_\mathrm{B}}$.

II.1.7.1 Protonation of a ligand

$$\mathrm{H}^+ + \mathrm{H}_{r-1}\mathrm{L} \rightleftharpoons \mathrm{H}_r\mathrm{L} \qquad K_{1,r} = \frac{[\mathrm{H}_r\mathrm{L}]}{[\mathrm{H}^+][\mathrm{H}_{r-1}\mathrm{L}]} \qquad \text{(II.5)}$$

$$r\ \mathrm{H}^+ + \mathrm{L} \rightleftharpoons \mathrm{H}_r\mathrm{L} \qquad \beta_{1,r} = \frac{[\mathrm{H}_r\mathrm{L}]}{[\mathrm{H}^+]^r[\mathrm{L}]} \qquad \text{(II.6)}$$

This notation has been proposed and used by Sillén and Martell [1964SIL/MAR], but it has been simplified later by the same authors [1971SIL/MAR] from $K_{1,r}$ to K_r.

For the addition of a ligand, the notation shown in Eq. (II.7) is used.

$$HL_{q-1} + L \rightleftharpoons HL_q \qquad K_q = \frac{[HL_q]}{[HL_{q-1}][L]} \tag{II.7}$$

Eq. (II.8) refers to the overall formation constant of the species H_rL_q.

$$r\,H^+ + q\,L \rightleftharpoons H_rL_q \qquad \beta_{q,r} = \frac{[H_rL_q]}{[H^+]^r[L]^q} \tag{II.8}$$

In Eqs. (II.5), (II.6) and (II.8), the second subscript *r* can be omitted if *r* = 1, as shown in Eq. (II.7).

Example:

$$H^+ + PO_4^{3-} \rightleftharpoons HPO_4^{2-} \qquad \beta_{1,1} = \beta_1 = \frac{[HPO_4^{2-}]}{[H^+][PO_4^{3-}]}$$

$$2\,H^+ + PO_4^{3-} \rightleftharpoons H_2PO_4^- \qquad \beta_{1,2} = \frac{[H_2PO_4^-]}{[H^+]^2[PO_4^{3-}]}$$

II.1.7.2 Formation of metal complexes

$$ML_{q-1} + L \rightleftharpoons ML_q \qquad K_q = \frac{[ML_q]}{[ML_{q-1}][L]} \tag{II.9}$$

$$M + q\,L \rightleftharpoons ML_q \qquad \beta_q = \frac{[ML_q]}{[M][L]^q} \tag{II.10}$$

For the addition of a metal ion, *i.e.*, the formation of polynuclear complexes, the following notation is used, analogous to Eq. (II.5):

$$M + M_{m-1}L \rightleftharpoons M_mL \qquad K_{1,m} = \frac{[M_mL]}{[M][M_{m-1}L]} \tag{II.11}$$

Eq. (II.12) refers to the overall formation constant of a complex M_mL_q.

$$m\,M + q\,L \rightleftharpoons M_mL_q \qquad \beta_{q,m} = \frac{[M_mL_q]}{[M]^m[L]^q} \tag{II.12}$$

The second index can be omitted if it is equal to 1, *i.e.*, $\beta_{q,m}$ becomes β_q if *m* = 1. The formation constants of mixed ligand complexes are not indexed. In this case, it is necessary to list the chemical reactions considered and to refer the constants to the corresponding reaction numbers.

It has sometimes been customary to use negative values for the indices of the protons to indicate complexation with hydroxide ions, OH^-. This practice is not adopted in this review. If OH^- occurs as a reactant in the notation of the equilibrium, it is treated like a normal ligand L, but in general formulae the index variable *n* is used instead of *q*. If H_2O occurs as a reactant to form hydroxide complexes, H_2O is considered as a protonated ligand, HL, so that the reaction is treated as described below in Eqs. (II.13) to (II.15) using *n* as the index variable. For convenience, no general form is used for the stepwise constants for the formation of the complex $M_mL_qH_r$. In many experiments, the formation constants of metal ion complexes are determined by adding a ligand in its protonated form to a metal ion solution. The complex formation reactions thus involve a deprotonation reaction of the ligand. If this is the case, the equilibrium constant is supplied with an asterisk, as shown in Eqs. (II.13) and (II.14) for mononuclear and in Eq. (II.15) for polynuclear complexes.

$$ML_{q-1} + HL \rightleftharpoons ML_q + H^+ \qquad {}^*K_q = \frac{[ML_q][H^+]}{[ML_{q-1}][HL]} \qquad \text{(II.13)}$$

$$M + q\,HL \rightleftharpoons ML_q + qH^+ \qquad {}^*\beta_q = \frac{[ML_q][H^+]^q}{[M][HL]^q} \qquad \text{(II.14)}$$

$$m\,M + q\,HL \rightleftharpoons M_mL_q + qH^+ \qquad {}^*\beta_{q,m} = \frac{[M_mL_q][H^+]^q}{[M]^m[HL]^q} \qquad \text{(II.15)}$$

Example:

$$UO_2^{2+} + HF(aq) \rightleftharpoons UO_2F^+ + H^+ \qquad {}^*K_1 = {}^*\beta_1 = \frac{[UO_2F^+][H^+]}{[UO_2^{2+}][HF(aq)]}$$

$$3\,UO_2^{2+} + 5\,H_2O(l) \rightleftharpoons (UO_2)_3(OH)_5^+ + 5\,H^+ \qquad {}^*\beta_{5,3} = \frac{[(UO_2)_3(OH)_5^+][H^+]^5}{[UO_2^{2+}]^3}$$

Note that an asterisk is only assigned to the formation constant if the protonated ligand that is added is deprotonated during the reaction. If a protonated ligand is added and co-ordinated as such to the metal ion, the asterisk is to be omitted, as shown in Eq. (II.16).

$$M + q\,H_rL \rightleftharpoons M(H_rL)_q \qquad \beta_q = \frac{[M(H_rL)_q]}{[M][H_rL]^q} \qquad \text{(II.16)}$$

Example:

$$UO_2^{2+} + 3\,H_2PO_4^- \rightleftharpoons UO_2(H_2PO_4)_3^- \qquad \beta_3 = \frac{\left[UO_2(H_2PO_4)_3^-\right]}{\left[UO_2^{2+}\right]\left[H_2PO_4^-\right]^3}$$

II.1.7.3 Solubility constants

Conventionally, equilibrium constants involving a solid compound are denoted as "solubility constants" rather than as formation constants of the solid. An index "s" to the equilibrium constant indicates that the constant refers to a solubility process, as shown in Eqs. (II.17) to (II.19).

$$M_aL_b(s) \rightleftharpoons a\,M + b\,L \qquad K_{s,0} = [M]^a[L]^b \tag{II.17}$$

$K_{s,0}$ is the conventional solubility product, and the subscript "0" indicates that the equilibrium reaction involves only uncomplexed aqueous species. If the solubility constant includes the formation of aqueous complexes, a notation analogous to that of Eq. (II.12) is used:

$$\frac{m}{a}M_aL_b(s) \rightleftharpoons M_mL_q + \left(\frac{mb}{a} - q\right)L \qquad K_{s,q,m} = \left[M_mL_q\right]\left[L\right]^{\left(\frac{mb}{a}-q\right)} \tag{II.18}$$

Example:

$$UO_2F_2(cr) \rightleftharpoons UO_2F^+ + F^- \qquad K_{s,1,1} = K_{s,1} = \left[UO_2F^+\right]\left[F^-\right] .$$

Similarly, an asterisk is added to the solubility constant if it simultaneously involves a protonation equilibrium:

$$\frac{m}{a}M_aL_b(s) + \left(\frac{mb}{a} - q\right)H^+ \rightleftharpoons M_mL_q + \left(\frac{mb}{a} - q\right)HL$$

$$^*K_{s,q,m} = \frac{\left[M_mL_q\right]\left[HL\right]^{\left(\frac{mb}{a}-q\right)}}{\left[H^+\right]^{\left(\frac{mb}{a}-q\right)}} \tag{II.19}$$

Example:

$$U(HPO_4)_2{\cdot}4H_2O(cr) + H^+ \rightleftharpoons UHPO_4^{2+} + H_2PO_4^- + 4\,H_2O(l)$$

$$^*K_{s,1,1} = {}^*K_{s,1} = \frac{\left[UHPO_4^{2+}\right]\left[H_2PO_4^-\right]}{\left[H^+\right]}.$$

II.1.7.4 Equilibria involving the addition of a gaseous ligand

A special notation is used for constants describing equilibria that involve the addition of a gaseous ligand, as outlined in Eq. (II.20).

$$\mathrm{ML}_{q-1} + \mathrm{L(g)} \rightleftharpoons \mathrm{ML}_q \qquad K_{p,q} = \frac{[\mathrm{ML}_q]}{[\mathrm{ML}_{q-1}]p_\mathrm{L}} \tag{II.20}$$

The subscript "p" can be combined with any other notations given above.

Example:

$$\mathrm{CO_2(g)} \rightleftharpoons \mathrm{CO_2(aq)} \qquad K_p = \frac{[\mathrm{CO_2(aq)}]}{p_{\mathrm{CO_2}}}$$

$$3\ \mathrm{UO_2^{2+}} + 6\ \mathrm{CO_2(g)} + 6\ \mathrm{H_2O(l)} \rightleftharpoons (\mathrm{UO_2})_3(\mathrm{CO_3})_6^{6-} + 12\ \mathrm{H^+}$$

$${}^*\beta_{p,6,3} = \frac{[(\mathrm{UO_2})_3(\mathrm{CO_3})_6^{6-}][\mathrm{H^+}]^{12}}{[\mathrm{UO_2^{2+}}]^3\ p_{\mathrm{CO_2}}^6}$$

$$\mathrm{UO_2CO_3(cr)} + \mathrm{CO_2(g)} + \mathrm{H_2O(l)} \rightleftharpoons \mathrm{UO_2(CO_3)_2^{2-}} + 2\ \mathrm{H^+}$$

$${}^*K_{p,\mathrm{s},2} = \frac{[\mathrm{UO_2(CO_3)_2^{2-}}][\mathrm{H^+}]^2}{p_{\mathrm{CO_2}}}$$

In cases where the subscripts become complicated, it is recommended that K or β be used with or without subscripts, but always followed by the equation number of the equilibrium to which it refers.

II.1.7.5 Redox equilibria

Redox reactions are usually quantified in terms of their electrode (half cell) potential, E, which is identical to the potential difference of a galvanic cell in which the electrode on the left is the standard hydrogen electrode, SHE[1], in accordance with the "1953 Stockholm Convention" [1993MIL/CVI]. Therefore, electrode potentials are given as reduction potentials relative to the standard hydrogen electrode, which acts as an electron donor. In the standard hydrogen electrode, $H_2(g)$ is at unit fugacity (an ideal gas at unit pressure, 0.1 MPa), and H^+ is at unit activity. The sign of the electrode potential, E, is that of the observed sign of its polarity when coupled with the standard hydrogen electrode. The standard electrode potential, E°, *i.e.*, the potential of a standard galvanic cell relative to the standard hydrogen electrode (all components in their standard state, *cf.* Section II.3.1, and with no liquid junction potential) is related to the standard Gibbs energy change $\Delta_r G_m^\circ$ and the standard (or thermodynamic) equilibrium constant K° as outlined in Eq. (II.21):

[1] The definitions of SHE and NHE are given in Section II.1.1.

$$E^{\circ} = -\frac{1}{nF}\Delta_r G_m^{\circ} = \frac{RT}{nF}\ln K^{\circ} \quad \text{(II.21)}$$

and the potential, E, is related to E° by:

$$E = E^{\circ} - (RT/nF)\sum \nu_i \ln a_i. \quad \text{(II.22)}$$

For example, for the hypothetical galvanic cell:

$$\text{Pt} \;\Big|\; \text{H}_2(\text{g}, p = 1\ \text{bar}) \;\Big|\; \text{HCl}(\text{aq}, a_{\text{H}^+} = 1,\ f_{\text{H}_2} = 1) \;\Big\vdots\; \begin{matrix}\text{Fe(ClO}_4)_2\,(\text{aq}, a_{\text{Fe}^{2+}} = 1)\\ \text{Fe(ClO}_4)_3\,(\text{aq}, a_{\text{Fe}^{3+}} = 1)\end{matrix} \;\Big|\; \text{Pt} \quad \text{(II.23)}$$

where $\Big\vdots$ denotes a liquid junction and $\Big|$ a phase boundary, the reaction is:

$$\text{Fe}^{3+} + \tfrac{1}{2}\,\text{H}_2(\text{g}) \rightleftharpoons \text{Fe}^{2+} + \text{H}^+ \quad \text{(II.24)}$$

Formally Reaction (II.24) can be represented by two half cell reactions, each involving an equal number of electrons, (designated "e^-"), as shown in the following equations:

$$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+} \quad \text{(II.25)}$$

$$\tfrac{1}{2}\,\text{H}_2(\text{g}) \rightleftharpoons \text{H}^+ + \text{e}^- \quad \text{(II.26)}$$

The terminology is useful, although it must be emphasised that "e^-" here does not represent the hydrated electron.

Equilibrium (II.26) and Nernst law can be used to introduce a_{e^-}:

$$E = E^{\circ}(\text{II.26}) + \frac{RT}{F}\ln\left(\sqrt{f_{\text{H}_2}} / (a_{\text{H}^+} a_{e^-})\right) \quad \text{(II.27)}$$

According to the SHE convention $E^{\circ}(\text{II.26}) = 0$, $f_{\text{H}_2} = 1$, $a_{\text{H}^+} = 1$, hence

$$E = -\frac{RT}{F}\ln a_{e^-} \quad \text{(II.28)}$$

This equation is used to calculate a numerical value of a_{e^-} from potential-difference measurements *vs.* the SHE; hence, as for the value of E (V *vs.* the SHE), the numerical value of a_{e^-} depends on the SHE convention. Equilibrium constants may be written for these half cell reactions in the following way:

$$K^{\circ}(\text{II.25}) = \frac{a_{\text{Fe}^{2+}}}{a_{\text{Fe}^{3+}} \cdot a_{\text{e}^-}} \quad \text{(II.29)}$$

$$K^{\circ}(\text{II.26}) = \frac{a_{\text{H}^+} \cdot a_{\text{e}^-}}{\sqrt{f_{\text{H}_2}}} = 1 \quad \text{(by definition)} \quad \text{(II.30)}$$

In addition, $\Delta_r G_m^o$ (II.26) = 0, $\Delta_r H_m^o$ (II.26) = 0, $\Delta_r S_m^o$ (II.26) = 0 by definition, at all temperatures, and therefore $\Delta_r G_m^o$ (II.25) = $\Delta_r G_m^o$ (II.24). From $\Delta_r G_m^o$ (II.26) and the values given at 298.15 K in selected auxiliary data for $H_2(g)$ and H^+, the corresponding values for e^- can be calculated to be used in thermodynamic cycles involving half cell reactions. The following equations describe the change in the redox potential of Reaction (II.24), if p_{H_2} and a_{H^+} are equal to unity *(cf.* Eq. (II.22)):

$$E(\text{II.24}) = E^o(\text{II.24}) - RT \ln\left(\frac{a_{Fe^{2+}}}{a_{Fe^{3+}}}\right) \qquad \text{(II.31)}$$

For the standard hydrogen electrode $a_{e^-} = 1$ (by the convention expressed in Eq. (II.30)), while rearrangement of Eq. (II.29) for the half cell containing the iron perchlorates in cell (II.23) gives:

$$-\log_{10} a_{e^-} = \log_{10} K^o(\text{II.25}) - \log_{10}\left(\frac{a_{Fe^{2+}}}{a_{Fe^{3+}}}\right)$$

and from Eq. (II.27):

$$-\log_{10} a_{e^-} = \log_{10} K^o(\text{II.24}) - \log_{10}\left(\frac{a_{Fe^{2+}}}{a_{Fe^{3+}}}\right) \qquad \text{(II.32)}$$

and

$$-\log_{10} a_{e^-} = \frac{F}{RT\ln(10)} E(\text{II.24}) \qquad \text{(II.33)}$$

which is a specific case of the general equation (II.28).

The splitting of redox reactions into two half cell reactions by introducing the symbol "e^-", which according to Eq. (II.27) is related to the standard electrode potential, is arbitrary, but useful (this e^- notation does not in any way refer to solvated electrons). When calculating the equilibrium composition of a chemical system, both "e^-", and H^+ can be chosen as components and they can be treated numerically in a similar way: equilibrium constants, mass balance, *etc.* may be defined for both. However, while H^+ represents the hydrated proton in aqueous solution, the above equations use only the activity of "e^-", and never the concentration of "e^-". Concentration to activity conversions (or activity coefficients) are never needed for the electron (*cf.* Appendix B, Example B.3).

In the literature on geochemical modelling of natural waters, it is customary to represent the "electron activity" of an aqueous solution with the symbol "pe" or "pε" ($= -\log_{10} a_{e^-}$) by analogy with pH ($= -\log_{10} a_{H^+}$), and the redox potential of an aqueous solution relative to the standard hydrogen electrode is usually denoted by either "Eh" or "E_H" (see for example [1981STU/MOR], [1982DRE], [1984HOS], [1986NOR/MUN]).

In this review, the symbol $E^{o\prime}$ is used to denote the so-called "formal potential" [1974PAR]. The formal (or "conditional") potential can be regarded as a

standard potential for a particular medium in which the activity coefficients are independent (or approximately so) of the reactant concentrations [1985BAR/PAR] (the definition of $E^{o\prime}$ parallels that of "concentration quotients" for equilibria). Therefore, from

$$E = E^{o\prime} - \frac{RT}{nF} \sum \nu_i \ln c_i \tag{II.34}$$

$E^{o\prime}$ is the potential E for a cell when the ratio of the *concentrations* (not the activities) on the right-hand side and the left-hand side of the cell reaction is equal to unity, and

$$E_c^{o\prime} = E_c^{o} - \frac{RT}{nF} \sum \nu_i \ln \gamma_{c,i} \tag{II.35a}$$

$$E_c^{o\prime} = E_c^{o} - \frac{RT}{nF} \sum \nu_i \ln(\xi \cdot \gamma_{m,i} \cdot \rho^*_{H_2O}) \tag{II.35b}$$

$$\gamma_{c,i} = \left(\frac{m_i / m^o}{c_i / c^o} \right) (\rho^*_{H_2O} / \rho^o) \gamma_{m,i} \tag{II.35c}$$[1]

where the $\gamma_{c,i}$ and $\gamma_{m,i}$ are the activity coefficients on amount concentration and molality basis, respectively [2008GAM/LOR], related by Eq. (II.35c), ρ^*_{H2O} is the density of pure water, and ξ is (m_i / c_i), the ratio of molality to molarity (*cf.* Section II.2). The medium must be specified. Equation (II.35c) has been derived most elegantly by [1959ROB/STO].

II.1.8 pH

Because of the importance that potentiometric methods have in the determination of equilibrium constants in aqueous solutions, a short discussion on the definition of "pH" and a simplified description of the experimental techniques used to measure pH will be given here. For a comprehensive account, see [2002BUC/RON].

The acidity of aqueous solutions is often expressed in a logarithmic scale of the hydrogen ion activity. The definition of pH as:

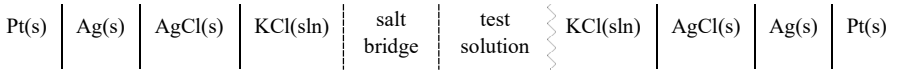
$$\mathrm{pH} = -\log_{10} a_{H^+} = -\log_{10}(m_{H^+} \gamma_{H^+})$$

can only be strictly used in the range of the limiting Debye-Hückel equation (that is, in extremely dilute solutions). In practice the use of pH values requires extra assumptions as to the values for single ion activities. In this review values of pH are used to describe qualitatively the ranges of acidity of experimental studies, and the assumptions described in Appendix B are used to calculate single ion activity coefficients.

The determination of pH is often performed by potential-difference measurements of galvanic cells involving liquid junctions [1969ROS], [1973BAT]. A

[1] ρ^*_{H2O} is a variable depending on T, m^o = 1 mol·kg^{-1}, c^o = 1 mol·dm^{-3}.

common setup is a cell made up of a reference half cell (*e.g.*, Ag(s)|AgCl(s) in a solution of constant chloride concentration), a salt bridge, the test solution, and a glass electrode (which encloses a solution of constant acidity and an internal reference half cell):



(II.36)

where ⟨ stands for a glass membrane (permeable to hydrogen ions).

The potential difference of such a cell is given by:

$$E = E^* - \frac{RT}{nF} \ln a_{H^+} + E_j$$

where E^* is a constant, and E_j is the liquid junction potential. The purpose of the salt bridge is to minimise the junction potential in junction "*b*", while keeping constant the junction potential for junction "*a*". Two methods are most often used to reduce and control the value of E_j. An electrolyte solution of *high* concentration (the "salt bridge") is a requirement of both methods. In the first method, the salt bridge is a saturated (or nearly saturated) solution of potassium chloride. A problem with a bridge of high potassium concentration is that potassium perchlorate might precipitate[1] inside the liquid junction when the test solution contains a high concentration of perchlorate ions.

In the other method the salt bridge contains the same *high* concentration of the same inert electrolyte as the test solution (for example, 3 M $NaClO_4$). However, if the concentration of the background electrolyte in the salt bridge and test solutions is reduced, the values of E_j are dramatically increased. For example, if both the bridge and the test solution have $[ClO_4^-]$ = 0.1 M as background electrolyte, the dependence of the liquid junction at *"b*" on acidity is $E_j \approx -440\,[H^+]$ mV·dm^3·mol^{-1} at 25 °C [1969ROS] (p.110), which corresponds to an error of ≥ 0.07 in pH at a pH value of 2.

Because of the problems in eliminating the liquid junction potentials and in defining individual ionic activity coefficients, an "operational" definition of pH is given by IUPAC [1993MIL/CVI]. This definition involves the measurement of pH differences between the test solution and standard solutions of known pH and similar ionic strength (in this way similar values of γ_{H^+} and E_j cancel each other when potential-difference values are subtracted).

Another method of determining the molal H^+ concentration, $\log_{10} m_{H^+}$, in chloride solutions up to high ionic strength was proposed by Knauss *et al.* [1990KNA/WOL]. The activity of HCl (a_{HCl}) can be measured with a liquid junction free cell consisting of a H^+ sensitive glass electrode and a chloride sensitive electrode from the relation:

[1] $KClO_4$(cr) has a solubility of ≈ 0.15 M in pure water at 25 °C

$$\log_{10} a_{HCl} = ½ (\log_{10} m_{H^+} + \log_{10} m_{Cl^-} + \log_{10} \gamma_{H^+} + \log_{10} \gamma_{Cl^-}) \qquad \text{(II.37)}$$

The value of $\log_{10} m_{H^+}$ in the test solution can then be derived from the measured value of $\log_{10} a_{HCl}$, the given chloride concentration $\log_{10} m_{Cl^-}$ and the activity coefficients $\log_{10} \gamma_{H^+}$ and $\log_{10} \gamma_{Cl^-}$ which can be calculated either with the SIT or, for higher chloride concentrations, with the Pitzer equations as proposed by [1990KNA/WOL].

The measurement and use of pH in equilibrium analytical investigations creates many problems that have not always been taken into account by the investigators, as discussed in many reviews in Appendix A. In order to deduce the stoichiometry and equilibrium constants of complex formation reactions and other equilibria, it is necessary to vary the concentrations of reactants and products over fairly large concentration ranges under conditions where the activity coefficients of the species are either known, or constant. Only in this manner is it possible to use the mass balance equations for the various components together with the measurement of one or more free concentrations to obtain the desired information [1961ROS/ROS], [1990BEC/NAG], [1997ALL/BAN], pp. 326–327. For equilibria involving hydrogen ions, it is necessary to use concentration units, rather than hydrogen ion activity. For experiments in an ionic medium, where the concentration of an "inert" electrolyte is much larger than the concentration of reactants and products we can ensure that, as a first approximation, their trace activity coefficients remain constant even for moderate variations of the corresponding total concentrations. Under these conditions of fixed ionic strength the free proton concentration may be measured directly, thereby defining it in terms of $-\log_{10}[H^+]$ (also often referred to as pH_c, and correspondingly pH_m for $\log_{10} m_{H^+}$) rather than on the activity scale as pH, and the value of $-\log_{10}[H^+]$ and pH will differ by a constant term, *i.e.*, $\log_{10} \gamma_{H^+}$. Equilibrium constants deduced from measurements in such ionic media are therefore *conditional* constants, because they refer to the given medium, not to the standard state. In order to compare the magnitude of equilibrium constants obtained in different ionic media it is necessary to have a method for estimating activity coefficients of ionic species in mixed electrolyte systems to a *common* standard state. Such procedures are discussed in Appendix B.

Note that the precision of the measurement of $-\log_{10}[H^+]$ and pH is virtually the same, in very good experiments, ± 0.001. However, the accuracy is generally considerably poorer, depending in the case of glass electrodes largely on the response of the electrode (linearity, age, pH range, *etc.*), and to a lesser extent on the calibration method employed, although the stoichiometric $-\log_{10}[H^+]$ calibration standards can be prepared far more accurately than the commercial pH standards.

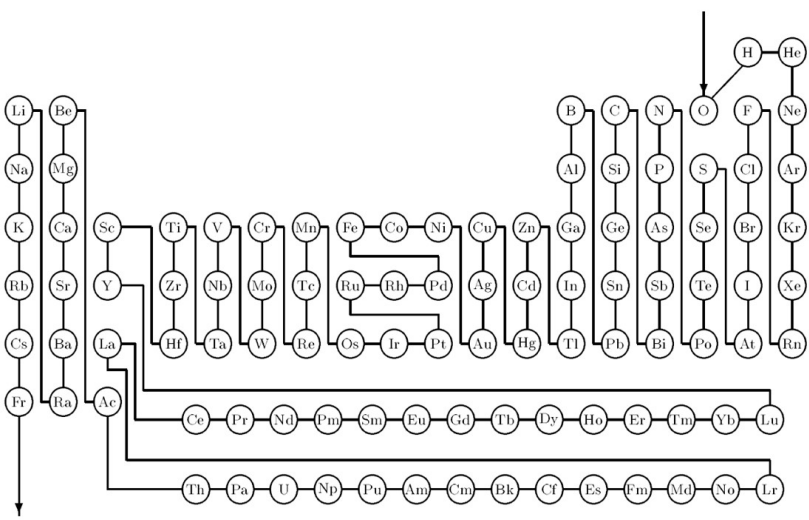
II.1.9 Order of formulae

To be consistent with CODATA, the data tables are given in "*Standard Order of Arrangement*" [1982WAG/EVA]. This scheme is presented in Figure II-1 below, and

shows the sequence of the ranks of the elements in this convention. The order follows the ranks of the elements.

For example, for uranium, this means that, after elemental uranium and its monoatomic ions (*e.g.*, U^{4+}), the uranium compounds and complexes with oxygen would be listed, then those with hydrogen, then those with oxygen and hydrogen, and so on, with decreasing rank of the element and combinations of the elements. Within a class, increasing coefficients of the higher rank elements go before increasing coefficients of the lower rank elements. For example, in the U–O–F class of compounds and complexes, a typical sequence would be $UOF_2(cr)$, $UOF_4(cr)$, $UOF_4(g)$, $UO_2F(aq)$, UO_2F^+, $UO_2F_2(aq)$, $UO_2F_2(cr)$, $UO_2F_2(g)$, $UO_2F_3^-$, $UO_2F_4^{2-}$, $U_2O_3F_6(cr)$, *etc*. [1992GRE/FUG]. Formulae with identical stoichiometry are in alphabetical order of their designators.

Figure II-1: Standard order of arrangement of the elements and compounds based on the periodic classification of the elements (from [1982WAG/EVA]).



II.1.10 Reference codes

The references cited in the review are ordered chronologically and alphabetically by the first two authors within each year, as described by CODATA [1987GAR/PAR]. A reference code is made up of the four digits of the year of appearance, followed by the first three letters of the surnames of the first two authors, separated by a slash.

If there are multiple reference codes, a “2” will be added to the second one, a “3” to the third one, and so forth. Reference codes are always enclosed in square brackets.

II.2 Units and conversion factors

Thermodynamic data are given according to the *Système International d'unités* (SI units). The unit of energy is the joule. Some basic conversion factors, also for non-thermodynamic units, are given in Table II-4.

Table II-4: Unit conversion factors.

To convert from (non–SI unit symbol)	to (SI unit symbol)	multiply by
ångström (Å)	metre (m)	1×10^{-10} (exactly)
standard atmosphere (atm)	pascal (Pa)	1.01325×10^{5} (exactly)
bar (bar)	pascal (Pa)	1×10^{5} (exactly)
thermochemical calorie (cal)	joule (J)	4.184 (exactly)
entropy unit e.u. $\triangleq \text{cal}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$	$\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$	4.184 (exactly)

Since a large part of the NEA TDB Project deals with the thermodynamics of aqueous solutions, the units describing the amount of dissolved substance are used very frequently. For convenience, this review uses "M" as an abbreviation of "$\text{mol}\cdot\text{dm}^{-3}$" for molarity, c, and, in Appendices B and C, "m" as an abbreviation of "$\text{mol}\cdot\text{kg}^{-1}$" for molality, m. It is often necessary to convert data from molarity to molality and *vice versa*. This conversion is used for the correction and extrapolation of equilibrium data to zero ionic strength by the specific ion interaction treatment, which works on molality basis (*cf.* Appendix B). This conversion is made in the following way.

Molality[1], m_B, is defined as the amount of substance, n_B dissolved in the mass of pure solvent, m_A, which in most cases is pure water

$$m_B = n_B / m_A .$$

Molarity (concentration), c_B, is defined as the amount of substance, n_B, dissolved in the volume, V, of solution

$$c_B = n_B / V.$$

Density of solution, ρ, is given by mass of solution divided by its volume,

$$\rho = (n_B M_B + m_A)/V$$

where M is the molar mass of the solute.

[1] In this derivation the symbol m is used with two different meanings: m_B denotes the *molality* of solute B and m_A denotes the *mass* of solvent A.

From this it follows that:

$$m_{\mathrm{B}} = \frac{c_{\mathrm{B}}}{\rho - c_{\mathrm{B}} M_{\mathrm{B}}} .$$

When the ionic strength is kept high and constant by an inert electrolyte, I, the ratio $m_{\mathrm{B}}/c_{\mathrm{B}}$ can be approximated by

$$\frac{m_{\mathrm{B}}}{c_{\mathrm{B}}} = \frac{1}{\rho - c_{\mathrm{I}} M_{\mathrm{I}}}$$

where c_{I} is the concentration of the inert electrolyte and M_{I} its molar mass.

Baes and Mesmer [1976BAE/MES], (*p*.439) give a table with conversion factors (from molarity to molality) for nine electrolytes at various ionic strengths. Conversion factors at 298.15 K for twenty one electrolytes, calculated using the density equations reported by Söhnel and Novotný [1985SOH/NOV], are reported in Table II-5.

Example:

$$\begin{aligned} 1.00\ \mathrm{M\ NaClO_4} &\triangleq 1.05\ \mathrm{m\ NaClO_4} \\ 1.00\ \mathrm{M\ NaCl} &\triangleq 1.02\ \mathrm{m\ NaCl} \\ 4.00\ \mathrm{M\ NaClO_4} &\triangleq 4.95\ \mathrm{m\ NaClO_4} \\ 6.00\ \mathrm{M\ NaNO_3} &\triangleq 7.55\ \mathrm{m\ NaNO_3} \end{aligned}$$

It should be noted that equilibrium constants need also to be converted if the concentration scale is changed from molarity to molality or *vice versa*. For a general equilibrium reaction, $0 = \sum_{\mathrm{B}} \nu_{\mathrm{B}} \mathrm{B}$, the equilibrium constants can be expressed either in molarity or molality units, K_c or K_m, respectively:

$$\log_{10} K_c = \sum_{\mathrm{B}} \nu_{\mathrm{B}} \log_{10} c_{\mathrm{B}}$$

$$\log_{10} K_m = \sum_{\mathrm{B}} \nu_{\mathrm{B}} \log_{10} m_{\mathrm{B}}$$

With $(m_{\mathrm{B}}/c_{\mathrm{B}}) = \xi$, or $(\log_{10} m_{\mathrm{B}} - \log_{10} c_{\mathrm{B}}) = \log_{10} \xi$, the relationship between K_c and K_m becomes very simple, as shown in Eq. (II.38).

$$\log_{10} K_m = \log_{10} K_c + \sum_{\mathrm{B}} \nu_{\mathrm{B}} \log_{10} \xi \qquad \text{(II.38)}$$

$\sum_{\mathrm{B}} \nu_{\mathrm{B}}$ is the sum of the stoichiometric coefficients of the solutes and the values of ξ are the factors for the conversion of molarity to molality as tabulated in Table II-5 for several electrolyte media at 298.15 K. The differences between the values in Table II-5 and the values listed in the uranium NEA TDB review [1992GRE/FUG] (*p*.23) are found at the highest concentrations, and are no larger than ± 0.003 $\mathrm{dm^3 \cdot kg^{-1}}$, reflecting the accuracy expected in this type of conversion. The uncertainty introduced by the use of Eq. (II.38) in the values of $\log_{10} K_m$ will be no larger than $\pm 0.001 \sum_{\mathrm{B}} \nu_{\mathrm{B}}$.

Table II-5: Factors ξ for the conversion of molarity, c_B, to molality, m_B, of a substance B, in various media at 298.15 K (calculated from densities in [1985SOH/NOV]).

	$\xi = m_B / c_B$ (dm^3 of solution per kg of H_2O)							
c (M)	$HClO_4$	$NaClO_4$	$LiClO_4$	NH_4ClO_4	$Ba(ClO_4)_2$	HCl	NaCl	LiCl
0.10	1.0077	1.0075	1.0074	1.0091	1.0108	1.0048	1.0046	1.0049
0.25	1.0147	1.0145	1.0141	1.0186	1.0231	1.0076	1.0072	1.0078
0.50	1.0266	1.0265	1.0256	1.0351	1.0450	1.0123	1.0118	1.0127
0.75	1.0386	1.0388	1.0374	1.0523	1.0685	1.0172	1.0165	1.0177
1.00	1.0508	1.0515	1.0496	1.0703	1.0936	1.0222	1.0215	1.0228
1.50	1.0759	1.0780	1.0750	1.1086	1.1491	1.0324	1.0319	1.0333
2.00	1.1019	1.1062	1.1019		1.2125	1.0430	1.0429	1.0441
3.00	1.1571	1.1678	1.1605		1.3689	1.0654	1.0668	1.0666
4.00	1.2171	1.2374	1.2264			1.0893	1.0930	1.0904
5.00	1.2826	1.3167				1.1147	1.1218	1.1156
6.00	1.3547	1.4077				1.1418		1.1423
c (M)	KCl	NH_4Cl	$MgCl_2$	$CaCl_2$	NaBr	HNO_3	$NaNO_3$	$LiNO_3$
0.10	1.0057	1.0066	1.0049	1.0044	1.0054	1.0056	1.0058	1.0059
0.25	1.0099	1.0123	1.0080	1.0069	1.0090	1.0097	1.0102	1.0103
0.50	1.0172	1.0219	1.0135	1.0119	1.0154	1.0169	1.0177	1.0178
0.75	1.0248	1.0318	1.0195	1.0176	1.0220	1.0242	1.0256	1.0256
1.00	1.0326	1.0420	1.0258	1.0239	1.0287	1.0319	1.0338	1.0335
1.50	1.0489	1.0632	1.0393	1.0382	1.0428	1.0478	1.0510	1.0497
2.00	1.0662	1.0855	1.0540	1.0546	1.0576	1.0647	1.0692	1.0667
3.00	1.1037	1.1339	1.0867	1.0934	1.0893	1.1012	1.1090	1.1028
4.00	1.1453	1.1877	1.1241	1.1406	1.1240	1.1417	1.1534	1.1420
5.00		1.2477		1.1974	1.1619	1.1865	1.2030	1.1846
6.00					1.2033	1.2361	1.2585	1.2309
c (M)	NH_4NO_3	H_2SO_4	Na_2SO_4	$(NH_4)_2SO_4$	H_3PO_4	Na_2CO_3	K_2CO_3	NaSCN
0.10	1.0077	1.0064	1.0044	1.0082	1.0074	1.0027	1.0042	1.0069
0.25	1.0151	1.0116	1.0071	1.0166	1.0143	1.0030	1.0068	1.0130
0.50	1.0276	1.0209	1.0127	1.0319	1.0261	1.0043	1.0121	1.0234
0.75	1.0405	1.0305	1.0194	1.0486	1.0383	1.0065	1.0185	1.0342
1.00	1.0539	1.0406	1.0268	1.0665	1.0509	1.0094	1.0259	1.0453
1.50	1.0818	1.0619	1.0441	1.1062	1.0773	1.0170	1.0430	1.0686
2.00	1.1116	1.0848		1.1514	1.1055	1.0268	1.0632	1.0934
3.00	1.1769	1.1355		1.2610	1.1675		1.1130	1.1474
4.00	1.2512	1.1935		1.4037	1.2383		1.1764	1.2083
5.00	1.3365	1.2600			1.3194		1.2560	1.2773
6.00	1.4351	1.3365			1.4131			1.3557

II.3 Standard and reference conditions

II.3.1 Standard state

A precise definition of the term "standard state" has been given by IUPAC [1982LAF]. The fact that only changes in thermodynamic parameters, but not their absolute values, can be determined experimentally, makes it important to have a well–defined standard state that forms a base line to which the effect of variations can be referred. The IUPAC [1993MIL/CVI] definition of the standard state has been adopted in the NEA TDB Project. The standard state pressure, p° = 0.1 MPa (1 bar), has therefore also been adopted, *cf.* Section II.3.2. The application of the standard state principle to pure substances and mixtures is summarised below. It should be noted that the standard state is always linked to a reference temperature, *cf.* Section II.3.3.

- The standard state for a gaseous substance, whether pure or in a gaseous mixture, is the pure substance at the standard state pressure and in a (hypothetical) state in which it exhibits ideal gas behaviour.
- The standard state for a pure liquid substance is (ordinarily) the pure liquid at the standard state pressure.
- The standard state for a pure solid substance is (ordinarily) the pure solid at the standard state pressure.
- The standard state for a solute in a liquid or solid solution is referenced to the ideal dilute behaviour of the solute. It is the (hypothetical) state of solute B at the standard molality m°, standard pressure p° and exhibiting infinitely dilute solution behaviour.

It should be emphasised that the use of superscript, $^\circ$, *e.g.*, in $\Delta_f H_m^\circ$, implies that the compound in question is in the standard state and that the elements are in their reference states. The reference states of the elements at the reference temperature (*cf.* Section II.3.3) are listed in Table II-6.

Table II-6: Reference states for some elements at the reference temperature of 298.15 K and standard pressure of 0.1 MPa [1982WAG/EVA], [1989COX/WAG], [1991DIN], [2005GAM/BUG], [2005OLI/NOL].

O_2	gaseous	Zn	crystalline, hexagonal
H_2	gaseous	Cd	crystalline, hexagonal
He	gaseous	Hg	liquid
Ne	gaseous	Cu	crystalline, cubic
Ar	gaseous	Ag	crystalline, cubic
Kr	gaseous	Ni	crystalline, fcc
Xe	gaseous	Fe	crystalline, cubic, bcc
F_2	gaseous	Tc	crystalline, hexagonal
Cl_2	gaseous	V	crystalline, cubic
Br_2	liquid	Ti	crystalline, hexagonal
I_2	crystalline, orthorhombic	Am	crystalline, dhcp
S	crystalline, orthorhombic	Pu	crystalline, monoclinic
Se	crystalline, trigonal	Np	crystalline, orthorhombic
Te	crystalline, hexagonal	U	crystalline, orthorhombic
N_2	gaseous	Th	crystalline, cubic
P	crystalline, cubic (“white”)	Be	crystalline, hexagonal
As	crystalline, rhombohedral (“grey”)	Mg	crystalline, hexagonal
Sb	crystalline, rhombohedral	Ca	crystalline, cubic, fcc
Bi	crystalline, rhombohedral	Sr	crystalline, cubic, fcc
C	crystalline, hexagonal (graphite)	Ba	crystalline, cubic
Si	crystalline, cubic	Li	crystalline, cubic
Ge	crystalline, cubic	Na	crystalline, cubic
Sn	crystalline, tetragonal (“white”)	K	crystalline, cubic
Pb	crystalline, cubic	Rb	crystalline, cubic
B	crystalline, rhombohedral	Cs	crystalline, cubic
Al	crystalline, cubic		

II.3.2 Standard state pressure

The standard state pressure chosen for all selected data is 0.1 MPa (1 bar) as recommended by IUPAC [1982LAF].

However, the majority of the thermodynamic data published in the scientific literature, and used for the evaluations in this review, refers to the old standard state pressure of 1 “standard atmosphere” (= 0.101325 MPa). The difference between the thermodynamic data for the two standard state pressures is not large and lies in most cases within the uncertainty limits. It is nevertheless essential to make the corrections for the change in the standard state pressure in order to avoid inconsistencies and

propagation of errors. In practice the parameters affected by the change between these two standard state pressures are the Gibbs energy and entropy changes of all processes that involve gaseous species. Consequently, changes occur also in the Gibbs energies of formation of species that consist of elements whose reference state is gaseous (H, O, F, Cl, N, and the noble gases). No other thermodynamic quantities are affected significantly. A large part of the following discussion has been taken from the NBS tables of chemical thermodynamic properties [1982WAG/EVA], see also Freeman [1984FRE].

The following expressions define the effect of pressure on the properties of all substances:

$$\left(\frac{\partial H}{\partial p}\right)_T = V - T\left(\frac{\partial V}{\partial T}\right)_p = V(1-\alpha T) \tag{II.39}$$

$$\left(\frac{\partial C_p}{\partial p}\right)_T = -T\left(\frac{\partial^2 V}{\partial T^2}\right)_p \tag{II.40}$$

$$\left(\frac{\partial S}{\partial p}\right)_T = -V\alpha = -\left(\frac{\partial V}{\partial T}\right)_p \tag{II.41}$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V \tag{II.42}$$

$$\text{where } \alpha \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \tag{II.43}$$

For ideal gases, $V = RT/p$ and $\alpha = R/pV = 1/T$. The conversion equations listed below (Eqs. (II.44) to (II.51)) apply to the small pressure change from 1 atm to 1 bar (0.1 MPa). The quantities that refer to the old standard state pressure of 1 atm are assigned the superscript $^{(\mathrm{atm})}$, and those that refer to the new standard state pressure of 1 bar are assigned the superscript $^{(\mathrm{bar})}$.

For all substances the changes in the enthalpy of formation and heat capacity are much smaller than the experimental accuracy and can be disregarded. This is exactly true for ideal gases.

$$\Delta_f H^{(\mathrm{bar})}(T) - \Delta_f H^{(\mathrm{atm})}(T) = 0 \tag{II.44}$$

$$C_p^{(\mathrm{bar})}(T) - C_p^{(\mathrm{atm})}(T) = 0 \tag{II.45}$$

For gaseous substances, the entropy difference is:

$$S^{(\mathrm{bar})}(T) - S^{(\mathrm{atm})}(T) = R \ln\left(\frac{p^{(\mathrm{atm})}}{p^{(\mathrm{bar})}}\right) = R \ln 1.01325$$
$$= 0.1094 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} \tag{II.46}$$

This is exactly true for ideal gases, as follows from Eq. (II.41) with $\alpha = R/pV$. The entropy change of a reaction or process is thus dependent on the number of moles

of gases involved:

$$\Delta_r S^{(\text{bar})} - \Delta_r S^{(\text{atm})} = \delta R \ln\left(\frac{p^{(\text{atm})}}{p^{(\text{bar})}}\right)$$
$$= 0.1094\ \delta\ \text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} \qquad \text{(II.47)}$$

where δ is the net increase in moles of gas in the process.

Similarly, the change in the Gibbs energy of a process between the two standard state pressures is:

$$\Delta_r G^{(\text{bar})} - \Delta_r G^{(\text{atm})} = -\delta RT \ln\left(\frac{p^{(\text{atm})}}{p^{(\text{bar})}}\right)$$
$$= -0.03263\ \delta\ \text{kJ}\cdot\text{mol}^{-1} \text{ at } 298.15\ \text{K}. \qquad \text{(II.48)}$$

Eq. (II.48) applies also to $\Delta_f G^{(\text{bar})} - \Delta_f G^{(\text{atm})}$, since the Gibbs energy of formation describes the formation process of a compound or complex from the reference states of the elements involved:

$$\Delta_f G^{(\text{bar})} - \Delta_f G^{(\text{atm})} = -0.03263\ \delta\ \text{kJ}\cdot\text{mol}^{-1} \text{ at } 298.15\ \text{K}. \qquad \text{(II.49).}$$

The changes in the equilibrium constants and cell potentials with the change in the standard state pressure follows from the expression for Gibbs energy changes, Eq. (II.48),

$$\log_{10} K^{(\text{bar})} - \log_{10} K^{(\text{atm})} = -\frac{\Delta_r G^{(\text{bar})} - \Delta_r G^{(\text{atm})}}{RT \ln 10}$$
$$= \delta\frac{\ln\left(\frac{p^{(\text{atm})}}{p^{(\text{bar})}}\right)}{\ln 10} = \delta\log_{10}\left(\frac{p^{(\text{atm})}}{p^{(\text{bar})}}\right)$$
$$= 0.005717\ \delta \qquad \text{(II.50)}$$

$$E^{(\text{bar})} - E^{(\text{atm})} = -\frac{\Delta_r G^{(\text{bar})} - \Delta_r G^{(\text{atm})}}{nF}$$
$$= \delta\frac{RT \ln\left(\frac{p^{(\text{atm})}}{p^{(\text{bar})}}\right)}{nF}$$
$$= \delta\frac{0.0003382}{n}\ \text{V} \quad \text{at } 298.15\ \text{K} \qquad \text{(II.51)}$$

It should be noted that the standard potential of the hydrogen electrode is equal to 0.00 V exactly, by definition.

$$\text{H}^+ + \text{e}^- \rightleftharpoons \tfrac{1}{2}\text{H}_2(\text{g}) \qquad E^{\circ} \stackrel{\text{def}}{=} 0.00\text{V} \qquad \text{(II.52).}$$

This definition will not be changed, although a gaseous substance, $H_2(g)$, is involved in the process. The change in the potential with pressure for an electrode potential conventionally written as:

$$Ag^+ + e^- \rightleftharpoons Ag(cr)$$

should thus be calculated from the balanced reaction that includes the hydrogen electrode,

$$Ag^+ + \tfrac{1}{2}H_2\,(g) \rightleftharpoons Ag\,(cr) + H^+ .$$

Here $\delta = -0.5$. Hence, the contribution to δ from an electron in a half cell reaction is the same as the contribution of a gas molecule with the stoichiometric coefficient of 0.5. This leads to the same value of δ as the combination with the hydrogen half cell.

Example:

$Fe(cr) + 2\,H^+ \rightleftharpoons Fe^{2+} + H_2(g)$	$\delta = 1$	$E^{(bar)} - E^{(atm)} = 0.00017$ V
$CO_2(g) \rightleftharpoons CO_2(aq)$	$\delta = -1$	$\log_{10} K^{(bar)} - \log_{10} K^{(atm)} = -0.0057$
$NH_3(g) + \tfrac{5}{4}O_2(g) \rightleftharpoons NO(g) + \tfrac{3}{2}H_2O(g)$	$\delta = 0.25$	$\Delta_r G^{(bar)} - \Delta_r G^{(atm)} = -0.008$ kJ·mol^{-1}
$\tfrac{1}{2}Cl_2(g) + 2\,O_2(g) + e^- \rightleftharpoons ClO_4^-$	$\delta = -3$	$\Delta_f G^{(bar)} - \Delta_f G^{(atm)} = 0.098$ kJ·mol^{-1}

II.3.3 Reference temperature

The definitions of standard states given in Section II.3 make no reference to fixed temperature. Hence, it is theoretically possible to have an infinite number of standard states of a substance as the temperature varies. It is, however, convenient to complete the definition of the standard state in a particular context by choosing a reference temperature. As recommended by IUPAC [1982LAF], the reference temperature chosen in the NEA TDB Project is T = 298.15 K or t = 25.00 °C. Where necessary for the discussion, values of experimentally measured temperatures are reported after conversion to the IPTS–68 [1969COM]. The relation between the absolute temperature T (K, kelvin) and the Celsius temperature t (°C) is defined by t / °C = T /K – 273.15.

II.4 Fundamental physical constants

To ensure the consistency with other NEA TDB Reviews, the fundamental physical constants are taken from a publication by CODATA [1986COD]. Those relevant to this review are listed in Table II-7. Updated values of the fundamental constants can be obtained from CODATA, notably through its Internet site. In most cases, re-calculation of the NEA TDB database entries with the updated values of the fundamental constants will not introduce significant (with respect to their quoted uncertainties) excursions from the current NEA TDB selections.

Table II-7: Fundamental physical constants. These values have been taken from CODATA [1986COD]. The digits in parentheses are the one-standard-deviation uncertainty in the last digits of the given value.

Quantity	Symbol	Value	Units
speed of light in vacuum	c_0	299 792 458	$m \cdot s^{-1}$
permeability of vacuum	μ_0	$4\pi \times 10^{-7} = 12.566\ 370\ 614\ldots$	$10^{-7}\ N \cdot A^{-2}$
permittivity of vacuum	ϵ_0	$1/\mu_0\ c^2 = 8.854\ 187\ 817\ldots$	$10^{-12}\ C^2 \cdot J^{-1} \cdot m^{-1}$
Planck constant	h	6.626 0755(40)	$10^{-34}\ J \cdot s$
elementary charge	e	1.602 177 33(49)	$10^{-19}\ C$
Avogadro constant	N_A	6.022 1367(36)	$10^{23}\ mol^{-1}$
Faraday constant	F	96 485.309(29)	$C \cdot mol^{-1}$
molar gas constant	R	8.314 510(70)	$J \cdot K^{-1} \cdot mol^{-1}$
Boltzmann constant, R/N_A	k_B	1.380 658(12)	$10^{-23}\ J \cdot K^{-1}$
Non–SI units used with SI:			
electronvolt, (e/C) J	eV	1.602 177 33(49)	10^{-19} J
atomic mass unit, $1\,u = m_u = \frac{1}{12} m\,(^{12}C)$	u	1.660 5402(10)	10^{-27} kg

II.5 Uncertainty estimates

One of the principal objectives of the NEA TDB development effort is to provide an idea of the uncertainties associated with the data selected in the reviews. In general the uncertainties should define the range within which the corresponding data can be reproduced with a probability of 95%. In many cases, a full statistical treatment is limited or impossible due to the availability of only one or a few data points. Appendix C describes in detail the procedures used for the assignment and treatment of uncertainties, as well as the propagation of errors and the standard rules for rounding.

II.6 The NEA TDB system

A database system has been developed at the NEA Data Bank that allows the storage of thermodynamic parameters for individual species as well as for reactions [2019MAR/SAN]. The structure of the database system allows for internal consistency checks and produces alerts when the implemented mathematical constraints are not met. The literature sources of the data are also stored in the database.

The following thermodynamic parameters, valid at the reference temperature of 298.15 K and at the standard pressure of 1 bar, are stored in the database:

$\Delta_f G^o_m$ the standard molar Gibbs energy of formation from the elements in their reference state ($kJ \cdot mol^{-1}$)

$\Delta_f H^o_m$ the standard molar enthalpy of formation from the elements in their reference state ($kJ \cdot mol^{-1}$)

S^o_m the standard molar entropy ($J \cdot K^{-1} \cdot mol^{-1}$)

$C^o_{p,m}$ the standard molar heat capacity ($J \cdot K^{-1} \cdot mol^{-1}$) at constant pressure.

For aqueous neutral species and ions, the values of $\Delta_f G^o_m$, $\Delta_f H^o_m$, S^o_m and $C^o_{p,m}$ correspond to the standard partial molar quantities, and for individual aqueous ions they are relative quantities, defined with respect to the aqueous hydrogen ion, according to the convention [1989COX/WAG] that $\Delta_f H^o_m(H^+, T) = 0$ and that $S^o_m(H^+, T) = 0$. Furthermore, for an *ionised solute* B containing any number of different cations and anions:

$$\Delta_f H^o_m(B_\pm, aq) = \sum_+ \nu_+ \, \Delta_f H^o_m(\text{cation, aq}) + \sum_- \nu_- \, \Delta_f H^o_m(\text{anion, aq})$$

$$S^o_m(B_\pm, aq) = \sum_+ \nu_+ \, S^o_m(\text{cation, aq}) + \sum_- \nu_- \, S^o_m(\text{anion, aq})$$

As the thermodynamic parameters vary as a function of temperature, provision is made for including the compilation of the coefficients of empirical temperature functions for these data, as well as the temperature ranges over which they are valid. In many cases the thermodynamic data measured or calculated at several temperatures were published for a particular species, rather than the deduced temperature functions. In these cases, a linear regression method is used in this review to obtain the most significant coefficients of the following empirical function for a thermodynamic parameter, *X*:

$$\begin{aligned} X(T) \;=\; & a_X + b_X T + c_X T^2 + d_X T^{-1} + e_X T^{-2} + f_X \ln T + g_X T \ln T \\ & + h_X \sqrt{T} + \frac{i_X}{\sqrt{T}} + j_X T^3 + k_X T^{-3}. \end{aligned} \tag{II.53}$$

Most temperature variations can be described with three or four parameters. In the present series, only $C_{p,m}(T)$, *i.e.*, the thermal functions of the heat capacities of individual species are considered and stored in the database. They refer to the relation:

$$C^o_{p,m}(T) = a + bT + cT^2 + dT^{-1} + eT^{-2} + fT^3 \tag{II.54}$$

(where the subindices for the coefficients have been dropped) and are listed in the selected value tables.

The pressure dependence of thermodynamic data has not been the subject of critical analysis in the present compilation. The reader interested in higher temperatures and pressures, or the pressure dependency of thermodynamic functions for geochemical applications, is referred to the specialised literature in this area, *e.g.*, [1982HAM], [1984MAR/MES], [1988SHO/HEL], [1988TAN/HEL], [1989SHO/HEL], [1989SHO/HEL2], [1990MON], [1991AND/CAS].

Selected standard thermodynamic data referring to chemical reactions are also compiled in the database. A chemical reaction "r", involving reactants and products "B", can be abbreviated as:

$$0 = \sum_{B} \nu_B \, B \qquad \text{(II.55)}$$

where the stoichiometric coefficients ν_B are positive for products, and negative for reactants. The reaction parameters considered in the NEA TDB system include:

$\log_{10} K_r^o$	the equilibrium constant of the reaction, logarithmic
$\Delta_r G_m^o$	the molar Gibbs energy of reaction (kJ·mol^{-1})
$\Delta_r H_m^o$	the molar enthalpy of reaction (kJ·mol^{-1})
$\Delta_r S_m^o$	the molar entropy of reaction (J·K^{-1}·mol^{-1})
$\Delta_r C_{p,m}^o$	the molar heat capacity of reaction (J·K^{-1}·mol^{-1}) at constant pressure.

The temperature functions of these data, if available, are stored according to Eq. (II.53).

The equilibrium constant, K_r^o, is related to $\Delta_r G_m^o$ according to the following relation:

$$\log_{10} K_r^o = -\frac{\Delta_r G_m^o}{RT \ln(10)}$$

and can be calculated from the individual values of $\Delta_f G_m^o(B)$ (for example, those given in selected values tables), according to:

$$\log_{10} K_r^o = -\frac{1}{RT \ln(10)} \sum_{B} \nu_B \, \Delta_f G_m^o(B) \qquad \text{(II.56)}$$

II.7 Presentation of the selected data

The selected data are presented in Chapters III and IV. Unless otherwise indicated, they refer to standard conditions (*cf.* Section II.3) and 298.15 K (25 °C) and are provided with an uncertainty which should correspond to the 95% confidence level (see Appendix C).

Chapter III contains a table of selected thermodynamic data for individual compounds and complexes (Table III-1), a table of selected reaction data (Table III-2) and a table containing the heat capacities of individual species (Table III-3) that have been used in the evaluations. The selection of all these data is discussed in Chapters V to XI.

Chapter IV contains tables of thermodynamic data for individual species (Table IV-1) and reaction data (Table IV-2), for auxiliary compounds and complexes.

Most of these values are the CODATA Key Values [1989COX/WAG]. The selection of the remaining auxiliary data is discussed in [1992GRE/FUG], [1999RAR/RAN], [2001LEM/FUG], [2005GAM/BUG], [2005OLI/NOL] and [2008RAN/FUG], [2020LEM/PAL] and in this review.

All the selected data presented in Table III-1, Table III-2, Table III-3, Table IV-1 and Table IV-2 are internally consistent. Therefore, when using the selected data, the auxiliary data of Chapter IV must be used together with the data in Chapter III to ensure internal consistency of the data set.

It is important to note that Table III-2 and Table IV-2 include only those species for which the primary selected data are reaction data. The formation data derived therefrom and listed in Table III-1 are obtained using auxiliary data, and their uncertainties are propagated accordingly. In order to maintain the uncertainties originally assigned to the selected data in this review, the user is advised to make direct use of the reaction data presented in Table III-2 and Table IV-2, rather than taking the derived values in Table III-1 and Table IV-1 to calculate the reaction data. The latter approach would imply a twofold propagation of the uncertainties and result in reaction data whose uncertainties would be considerably larger than those originally assigned.

The thermodynamic data in the selected set refer to a temperature of 298.15 K (25.00 °C), but they can be re-calculated to other temperatures if the corresponding data (enthalpies, entropies, heat capacities) are available [1997PUI/RAR]. For example, the temperature dependence of the standard reaction Gibbs energy as a function of the standard reaction entropy at the reference temperature (T_0 = 298.15 K), and of the heat capacity function is:

$$\Delta_r G_m^o(T) = \Delta_r H_m^o(T_0) + \int_{T_0}^{T} \Delta_r C_{p,m}^o(T)\,\mathrm{d}T - T\left(\Delta_r S_m^o(T_0) + \int_{T_0}^{T} \frac{\Delta_r C_{p,m}^o(T)}{T}\mathrm{d}T\right),$$

and the temperature dependence of the standard equilibrium constant as a function of the standard reaction enthalpy and heat capacity is:

$$\log_{10} K^o(T) = \log_{10} K^o(T_0) - \frac{\Delta_r H_m^o(T_0)}{R\ln(10)}\left(\frac{1}{T} - \frac{1}{T_0}\right) - \frac{1}{RT\ln(10)}\int_{T_0}^{T} \Delta_r C_{p,m}^o(T)\,\mathrm{d}T + \frac{1}{R\ln(10)}\int_{T_0}^{T} \frac{\Delta_r C_{p,m}^o(T)}{T}\mathrm{d}T,$$

where R is the gas constant (*cf.* Table II-7).

In the case of aqueous species, for which enthalpies of reaction are selected or can be calculated from the selected enthalpies of formation, but for which there are no selected heat capacities, it is in most cases possible to re-calculate equilibrium constants to temperatures up to 100 to 150 °C, with an additional uncertainty of perhaps about ± 1 to ± 2 in $\log_{10} K$, due to neglect of the heat capacity contributions to the temperature

correction. For isocoulombic/isoelectric reactions the uncertainty may be smaller; for a detailed discussion see [1997PUI/RAR]. However, it is important to observe that "new" aqueous species, *i.e.*, species not present in significant amounts at 298.15 K and therefore not detected, may be significant at higher temperatures, see for example the work by Ciavatta *et al.* [1987CIA/IUL]. Additional high–temperature experiments may therefore be needed in order to ascertain that proper chemical models are used in the modelling of hydrothermal systems. For many species, experimental thermodynamic data are not available to allow a selection of parameters describing the temperature dependence of equilibrium constants and Gibbs energies of formation. The user may find information on various procedures to estimate the temperature dependence of these thermodynamic parameters in [1997PUI/RAR]. The thermodynamic data in the selected set refer to infinite dilution for soluble species. Extrapolation of an equilibrium constant K, usually measured at high ionic strength, to K° at $I = 0$ using activity coefficients γ, is explained in Appendix B. The corresponding Gibbs energy of dilution is:

$$\Delta_{\text{dil}} G_{\text{m}} = \Delta_{\text{r}} G_{\text{m}}^{\circ} - \Delta_{\text{r}} G_{\text{m}} \tag{II.57}$$

$$= - RT \, \Delta_{\text{r}} \ln \gamma_{\pm} \tag{II.58}$$

Similarly $\Delta_{\text{dil}} S_{\text{m}}$ can be calculated from $\ln \gamma_{\pm}$ and its variations with T, while:

$$\Delta_{\text{dil}} H_{\text{m}} = RT^2 \frac{\partial}{\partial T} (\Delta_{\text{r}} \ln \gamma_{\pm})_p \tag{II.59}$$

depends only on the variation of γ with T, which is neglected in this review, when no data on the temperature dependence of γs are available. In this case the Gibbs energy of dilution $\Delta_{\text{dil}} G_{\text{m}}$ is entirely assigned to the entropy difference. This entropy of reaction is calculated using $\Delta_{\text{r}} G_{\text{m}}^{\circ} = \Delta_{\text{r}} H_{\text{m}}^{\circ} - T \Delta_{\text{r}} S_{\text{m}}^{\circ}$, the value of $\Delta_{\text{dil}} G_{\text{m}}$ and with the approximation that $\Delta_{\text{dil}} H_{\text{m}} = 0$.

Part 2

Tables of selected data

Chapter III

Selected ancillary data

This chapter presents the chemical thermodynamic data set for ancillary species that has been selected in this review. Table III-1 contains the recommended thermodynamic data of the ancillary compounds and species, Table III-2 the recommended thermodynamic data of chemical equilibrium reactions by which these compounds and complexes are formed, and Table III-3 the temperature coefficients from the available heat capacity data for species listed in Table III-1.

The species and reactions in the tables appear in standard order of arrangement. Table III-2 contains information only on those reactions for which primary data selections are made in Chapter V of this review. These selected reaction data are used, together with data for key ancillary species and auxiliary data selected in this review, to derive the corresponding formation data in Table III-1. The uncertainties associated with values for key species and the auxiliary data are in some cases substantial, leading to comparatively large uncertainties in the formation quantities derived in this manner.

The values of $\Delta_r G_m^o$ for many reactions are known more accurately than would be calculated directly from the uncertainties of the $\Delta_f G_m^o$ values in Table III-1 and auxiliary data. The inclusion of a table for reaction data (Table III-2) in this report allows the use of equilibrium constants with total uncertainties that are based directly on the experimental accuracies. This is the main reason for including both Table III-1 and Table III-2.

The selected thermal functions of the heat capacities, listed in Table III-3 refer to the relation

$$C_{p,m}^o(T) = a + bT + cT^2 + eT^{-2}$$

A detailed discussion of the selection procedure is presented in Chapter V. It may be noted that these chapters contain data on more species or compounds than are present in the tables of Chapter III. The main reasons for this situation are the lack of information for a proper extrapolation of the primary data to standard conditions in some systems and lack of solid primary data in others.

A warning: the addition of any aqueous species and their data to this internally consistent database can result in a modified data set, which is no longer rigorous and can lead to erroneous results. The situation is similar when gases or solids are added.

Table III-1: Selected thermodynamic data for ancillary compounds and complexes. All ionic species listed in this table are aqueous species. Unless noted otherwise, all data refer to the reference temperature of 298.15 K and to the standard state, *i.e.*, a pressure of 0.1 MPa and, for aqueous species, infinite dilution ($I = 0$). The uncertainties listed below each value represent total uncertainties and correspond in principle to the statistically defined 95% confidence interval. New compounds and data that have been changed from previous selections of TDB auxiliary data are denoted with an asterisk (*). Values in bold typeface are CODATA Key Values and are taken directly from [1989COX/WAG] without further evaluation. Values obtained from internal calculation, *cf.* footnotes (a) and (b), are rounded at the third digit after the decimal point and may therefore not be exactly identical to those given in Chapter V. Systematically, all the values are presented with three digits after the decimal point, regardless of the significance of these digits. The data presented in this table are available on computer media from the OECD Nuclear Energy Agency [2019MAR/SAN].

Compound	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
PO_4^{3-} [1992GRE/FUG]	– 1025.491 ± 1.576	– 1284.400 ± 4.085	– 220.970 ± 12.846	
HPO_4^{2-} **[1992GRE/FUG]**	**– 1095.985**[(a)] **± 1.567**	**– 1299.000** **± 1.500**	**– 33.500** **± 1.500**	
$H_2PO_4^-$ **[1992GRE/FUG]**	**– 1137.152**[(a)] **± 1.567**	**– 1302.600** **± 1.500**	**92.500** **± 1.500**	
H_3PO_4(aq)	– 1149.438[(b)] * ± 1.567	– 1294.756[(b)] * ± 1.509	160.016[(b)] * ± 1.599	
SiO_2(α-quar) **[1992GRE/FUG]**	**– 856.287**[(a)] **± 1.002**	**– 910.700** **± 1.000**	**41.460** **± 0.200**	44.500[(c)] * ± 0.200
SiO_2(cris)*	– 853.854[(a)] * ± 1.101	– 907.700 * ± 1.100	43.363 * ± 0.150	44.885[(c)] * ± 0.300
SiO_2(trid S-I)*	– 853.095[(a)] * ± 2.511	– 906.900 * ± 2.500	43.500 * ± 0.800	44.642[(c)] * ± 0.500
SiO_2(vit) *	– 849.188[(a)] * ± 1.237	– 901.600 * ± 1.200	48.170 * ± 1.000	44.061[(c)] * ± 0.800
$SiO_2(OH)_2^{2-}$	– 1177.088[(b)] * ± 1.140	– 1386.070[(b)] * ± 15.051	– 10.453[(b)] * ± 50.402	
$SiO(OH)_3^-$	– 1253.236[(b)] * ± 1.034	– 1431.753[(b)] * ± 1.352	91.727[(b)] * ± 3.153	
$Si(OH)_4$(aq)	– 1309.201[(b)] * ± 1.017	– 1461.070[(b)] * ± 1.239	181.105 [(b)] * ± 2.501	195.000 * ± 3.000
$Si_2O_3(OH)_4^{2-}$	– 2272.500 * ± 3.000			

(Continued on next page)

Table III-1 (continued)

Compound	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$Si_2O_2(OH)_5^-$	– 2335.400 * ± 3.000			
$Si_3O_6(OH)_3^{3-}$ (cyclic)	– 3053.100 * ± 4.000			
$Si_3O_5(OH)_5^{3-}$ (linear)	– 3296.100 * ± 4.000			
$Si_4O_8(OH)_4^{4-}$ (cyclic)	– 4080.700 * ± 6.000			
$Si_4O_7(OH)_5^{3-}$	– 4142.200 * ± 5.000			
$B(OH)_3$(aq) [1992GRE/FUG]	**– 969.268**[a] **± 0.820**	**– 1072.800 ± 0.800**	**162.400 ± 0.600**	
$B(OH)_4^-$ *	– 1153.755[b] * ± 1.075	– 1345.283[b] * ± 1.058	100.514[b] * ± 3.067	
Al^{3+}	– 488.824[a] * ± 1.614	**– 538.400 ± 1.500**	– 334.000 * ± 2.000	
Al_2O_3(cr) [1992GRE/FUG]	**– 1582.257**[a] **± 1.302**	**– 1675.700 ± 1.300**	**50.920 ± 0.100**	79.033[c] ± 0.100**
$AlOH^{2+}$*	– 697.880[b] * ± 1.663	– 767.670[b] * ± 2.106	– 168.541[b] * ± 5.512	
$Al(OH)_2^+$ *	– 902.991[b] * ± 1.736	– 1005.946[b] * ± 5.801	– 46.521[b] * ± 19.022	
$Al(OH)_3$(gibb)*	– 1155.775[a] * ± 1.503	– 1294.000 * ± 1.500	68.440 * ± 0.300	91.700[c] * ± 0.300
$Al(OH)_3$(baye)*		– 1289.000 * ± 2.500		
$Al(OH)_3$(aq)*	– 1106.682[b] * ± 2.438	– 1242.511[b] * ± 6.659	76.476[b] * ± 22.695	
$Al(OH)_4^-$ *	– 1306.404[b] * ± 1.521	– 1499.018[b] * ± 1.884	119.275[b] * ± 3.911	
Mg^{2+}	– 455.703[a] * ± 0.786	**– 467.000 ± 0.600**	– 135.900 * ± 1.700	
MgO(cr) [1992GRE/FUG]	**– 569.312**[a] **± 0.305**	**– 601.600 ± 0.300**	**26.950 ± 0.150**	**37.237**[c] **± 0.200**
$MgOH^+$*	– 626.683[b] * ± 1.031	– 685.934[b] * ± 3.426	– 63.481[b] * ± 11.657	

(Continued on next page)

Table III-1 (continued)

Compound	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$Mg(OH)_2$(cr)*	– 833.418[(a)] * ± 0.324	– 924.450 * ± 0.300	63.180 * ± 0.400	76.900[(c)] * ± 0.300
$MgCl_2$(cr)*	– 594.768[(a)] * ± 0.740	– 644.300 * ± 0.700	89.620 * ± 0.800	71.384[(c)] * ± 0.200
α-$MgSO_4$*		– 1288.600 * ± 0.600		
β-$MgSO_4$*	– 1170.481[(a)] * ± 0.671	– 1284.800 * ± 0.600	91.600 * ± 1.000	96.400 * ± 0.800
$MgSO_4$(aq)*	– 1212.962[(b)] * ± 0.895	– 1370.579 [(b)] * ± 0.976	– 53.620[(b)] * ± 2.830	
$MgCO_3$(cr)*	– 1028.409[(a)] * ± 1.313	– 1112.200 * ± 1.300	65.100 * ± 0.600	76.108[(c)] ± 0.760
$MgCO_3$(aq)*	– 1000.533[(b)] * ± 1.304	– 1128.126[(b)] * ± 0.810	– 81.811[(b)] * ± 4.123	
$Mg_3Si_2O_5(OH)_4$(cr)*	– 4036.371[(a)] * ± 3.431	– 4364.000 * ± 3.400	221.300 * ± 1.500	273.700[(c)] * ± 1.500
CaO(cr) [1992GRE/FUG]	**– 603.296[(a)] ± 0.916**	**– 634.920 ± 0.900**	**38.100 ± 0.400**	**42.049[(c)] ± 0.400**
$CaOH^+$*	– 717.676[(b)] * ± 3.115	– 765.470[(b)] * ± 4.123	– 16.134[(b)] * ± 16.665	
$Ca(OH)_2$(cr) *	– 897.497[(a)] * ± 1.014	− 985.160 * ± 1.000	83.400 * ± 0.400	87.488[(c)] * ± 0.100
$CaCl_2$(cr)*	– 748.788[(a)] * ± 1.561	– 795.390 * ± 1.000	108.368 * ± 4.000	72.857[(c)] * ± 0.400
$CaSO_4$(cr)*	– 1322.242[(a)] * ± 1.307	– 1434.510 * ± 1.300	107.400 * ± 0.200	101.200[(c)] * ± 0.200
$CaSO_4 \cdot 2H_2O$(cr)*	– 1797.402[(a)] * ± 1.309	– 2023.000 * ± 1.300	193.800 * ± 0.300	187.200 * ± 0.200
$CaSO_4$(aq)*	– 1310.213[(b)] * ± 1.159	– 1445.461[(b)] * ± 1.124	30.328[(b)] * ± 1.751	
β-$Ca_3(PO_4)_2$*	– 3887.597[(a)] * ± 12.016	− 4123.600 * ± 12.000	236.000 * ± 1.600	228.481[(c)] * ± 1.600
$Ca_5(PO_4)_3OH$(cr)*	− 6340.686[(a)] * ± 15.025	− 6740.400 * ± 15.000	389.400 * ± 2.000	381.150[(c)] * ± 3.000
$Ca_5(PO_4)_3F$(cr)*	− 6494.786[(a)] * ± 20.024	− 6875.800 * ± 20.000	385.600 * ± 2.500	373.000[(c)] * ± 3.500

(Continued on next page)

Table III-1 (continued)

Compound	$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}	$C_{p,m}^o$ /J·K^{-1}·mol^{-1}
$Ca(H_2PO_4)_2 \cdot H_2O$(cr)*	– 3071.840[(a)] * ± 2.306	− 3423.405 * ± 2.295	259.840 * ± 0.420	258.840 * ± 0.250
$CaCO_3$(calc)*	– 1128.983[(a)] * ± 1.208	– 1207.500 * ± 1.200	91.710 * ± 0.210	83.472[(c)] * ± 0.420
$CaCO_3$(arag)*	– 1128.074[(a)] * ± 1.307	– 1207.700 * ± 1.300	87.990 * ± 0.210	82.320[(c)] * ± 0.200
$CaCO_3$(aq)*	– 1098.795[(b)] * ± 1.257	– 1202.340[(b)] * ± 1.820	7.766[(b)] * ± 5.565	
$CaMg(CO_3)_2$(cr)*	– 2163.402[(a)] * ± 2.019	– 2326.200 * ± 2.000	155.170 * ± 0.800	157.510[(c)] * ± 0.800
$Ca_2B_3O_4(OH)_3 \cdot H_2O$(cr)*	– 3167.438[(a)] * ± 20.005	– 3470.000 * ± 20.000	191.800 * ± 1.500	217.100 * ± 1.000
$CaB_3O_3(OH)_5 \cdot 4H_2O$(cr)*	– 4116.064[(a)] * ± 12.093	– 4648.000 * ± 12.000	355.500 * ± 5.000	361.000 * ± 5.000
Na_2CO_3(cr)*	– 1045.450[(a)] * ± 0.345	– 1129.250 * ± 0.300	135.000 * ± 0.400	112.300[(c)] * ± 0.500
$NaHCO_3$(cr)*	– 851.175[(a)] * ± 0.365	– 949.000 * ± 0.200	102.000 * ± 1.000	87.960[(c)] * ± 0.500
$Na_2B_4O_7 \cdot 10H_2O$(cr)*		− 6289.600 * ± 3.400		608.000 * ± 6.000
K_2CO_3(cr)*	– 1065.833[(a)] * ± 2.060	– 1151.500 * ± 2.000	155.500 * ± 1.600	114.430[(c)] * ± 1.200
$KHCO_3$(cr)*		– 960.700 * ± 2.000		

(a) Value calculated internally using $\Delta_f G_m^o = \Delta_f H_m^o - T \sum S_{m,i}^o$.
(b) Value calculated internally from reaction data see (Table III-2).
(c) Temperature coefficients of this function are listed in Table III-3.
(**) Only the uncertainty has changed.

Table III-2: Selected thermodynamic data for reactions involving ancillary compounds and complexes. All ionic species listed in this table are aqueous species. Unless noted otherwise, all data refer to the reference temperature of 298.15 K and to the standard state, *i.e.*, a pressure of 0.1 MPa and, for aqueous species, infinite dilution I = 0). The uncertainties listed below each value represent total uncertainties and correspond in principle to the statistically defined 95% confidence interval. New reactions and data that have been changed from previously selected TDB Auxiliary Data are denoted with an asterisk. Values in bold typeface are CODATA Key Values and are taken directly from [1989COX/WAG] without further evaluation. Values obtained from internal calculation, *cf.* footnote a), are rounded at the third digit after the decimal point and may therefore not be exactly identical to those given in Chapter V. Systematically, all the values are presented with three digits after the decimal point, regardless of the significance of these digits. The data presented in this table are available on computer media from the OECD Nuclear Energy Agency [2019MAR/SAN].

Species	Reaction $\log_{10} K^{\circ}$	$\Delta_r G_m^{\circ}$ /kJ·mol^{-1}	$\Delta_r H_m^{\circ}$ /kJ·mol^{-1}	$\Delta_r S_m^{\circ}$ /J·K^{-1}·mol^{-1}
$H_3PO_4(aq)$ [(d)]	$H_2PO_4^- + H^+ \rightleftharpoons H_3PO_4(aq)$			
	2.152 *	– 12.286 *	7.844 *	67.516[(a)]
	± 0.005	± 0.029	± 0.163	± 0.555*
$SiO_2(OH)_2^{2-}$	$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+$			
	– 23.145 *	132.113 *	75.000 *	– 191.558[(a)]
	± 0.090	± 0.514	± 15.000	± 50.340*
$SiO(OH)_3^-$	$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+$			
	– 9.804 *	55.965 *	29.317 *	– 89.378[(a)]
	± 0.033	± 0.188	± 0.541	± 1.921*
$Si(OH)_4(aq)$	$SiO_2(\alpha\text{-quar}) + 2H_2O(l) \rightleftharpoons Si(OH)_4(aq)$			
	– 3.743 *	21.366 *	21.290 *	– 0.255[(a)]
	± 0.027	± 0.154	± 0.727	± 2.492*
$B(OH)_4^-$ [(e)] *	$B(OH)_3(aq) + OH^- \rightleftharpoons B(OH)_4^-$			
	4.777 *	– 27.267 *	– 42.468 *	– 50.986*
	± 0.121	± 0.692	± 0.691	± 3.001
Al^{3+}*	$Al(OH)_3(gibb) + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l)$			
	7.793 *	– 44.483 *	– 101.890 *	– 192.544[(a)]
	± 0.098	± 0.558	± 2.120	± 7.353*
$AlOH^{2+}$*	$Al^{3+} + H_2O(l) \rightleftharpoons AlOH^{2+} + H^+$			
	– 4.920 *	28.084 *	56.560 *	95.509[(a)]
	± 0.070	± 0.400	± 1.478	± 5.136*
$Al(OH)_2^+$ *	$Al(OH)_3(gibb) + H^+ \rightleftharpoons Al(OH)_2^+ + H_2O(l)$			
	– 2.741 *	15.644 *	2.224 *	– 45.011[(a)]
	± 0.152	± 0.868	± 5.604	± 19.020*

(Continued on next page)

Table III-2 (continued)

Species	Reaction	$\log_{10} K^{o}$	$\Delta_r G_m^o$ /kJ·mol^{-1}	$\Delta_r H_m^o$ /kJ·mol^{-1}	$\Delta_r S_m^o$ /J·K^{-1}·mol^{-1}
$Al(OH)_3(aq)$*	$Al(OH)_3(gibb) \rightleftharpoons Al(OH)_3(aq)$				
		– 8.601 *	49.093 *	51.489 *	8.036[a]
		± 0.336	± 1.919	± 6.488	± 22.693*
$Al(OH)_4^-$ *	$Al(OH)_3(gibb) + H_2O(l) \rightleftharpoons Al(OH)_4^- + H^+$				
		– 15.156 *	86.511 *	80.812 *	– 19.115[a]
		± 0.040	± 0.228	± 1.140	± 3.900*
$MgOH^+$*	$Mg^{2+} + H_2O(l) \rightleftharpoons MgOH^+ + H^+$				
		– 11.591 *	66.160 *	66.896 *	2.469[a]
		± 0.117	± 0.666	± 3.373	± 11.532*
$MgSO_4(aq)$*	$Mg^{2+} + SO_4^{2-} \rightleftharpoons MgSO_4(aq)$				
		2.322 *	– 13.255 *	5.761 *	63.780[a]
		± 0.016	± 0.090	± 0.658	± 2.227*
$MgCO_3(aq)$*	$Mg^{2+} + CO_3^{2-} \rightleftharpoons MgCO_3(aq)$				
		2.966 *	– 16.930 *	14.104 *	104.089[a]
		± 0.169	± 0.965	± 0.484	± 3.621*
$CaOH^+$*	$Ca^{2+} + H_2O(l) \rightleftharpoons CaOH^+ + H^+$				
		– 12.661 *	72.269*	63.360 *	– 29.884[a]
		± 0.514	± 2.932	± 4.000	± 16.635*
$CaSO_4(aq)$*	$Ca^{2+} + SO_4^{2-} \rightleftharpoons CaSO_4(aq)$				
		2.348 *	– 13.403 *	6.879 *	68.028[a]
		± 0.045	± 0.258	± 0.321	± 1.381*
$CaCO_3(aq)$*	$Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3(aq)$				
		3.166 *	– 18.089 *	15.890 *	113.966[a]
		± 0.100	± 0.570	± 1.500	± 5.382*

(a) Value calculated internally using $\Delta_r G_m^o = \Delta_r H_m^o - T\,\Delta_r S_m^o$.
(b) Value calculated from a selected standard potential.
(c) Value of $\log_{10} K^{o}$ calculated internally from $\Delta_r G_m^o$.
(d) $\Delta_r C_{p,m}^o = -(114.300 \pm 4.600)$ J·K^{-1}·mol^{-1}.
(e) $\Delta_r C_{p,m}^o = -(24.000 \pm 3.000)$ J·K^{-1}·mol^{-1}.

Table III-3: Selected temperature coefficients for heat capacities marked with [c] in Table III-1, according to equation $C_{p,\mathrm{m}}(T) = a + bT + cT^2 + eT^{-2}$. The functions are valid between the temperatures T_{min} and T_{max} (in K). The notation E ± *nn* indicates the power of 10. All compounds and values included in this table have been selected in this review. $C_{p,\mathrm{m}}$ units are J·K^{-1}·mol^{-1}. $\Delta_{\mathrm{trs}}H$ is the phase-transition enthalpy in kJ·mol^{-1}.

Compound	*a*	*b*	*c*	*e*	T_{min}	T_{max}	$\Delta_{\mathrm{trs}}H$	
SiO_2(α-quar)	4.653598E+01	3.32117E–02	3.88114E–06	– 1.091883E+06	298.15	847.3	0.742	
SiO_2(β-quar)	7.489141E+01	– 1.037879E–02	6.375226E–06	– 1.819987E+06	847.3	1650		
SiO_2(α-cris)	3.3350E+01	5.9066E–02	0	– 5.4004E+05	298.15	540	1.280	
SiO_2(β-cris)	6.6930E+01	4.8429E–03	0	– 2.5419E+06	540	2000		
SiO_2(trid S-I)	8.748693E+01	– 4.427262E–02	3.020355E–05	– 2.873920E+06	298.15	850		
SiO_2(vit)	4.88375E+01	3.4958E–02	– 1.20830E–05	– 1.255648E+06	298.15	1480		
	8.1373E+01				1480	2000		
Al_2O_3(cr)	5.413155E+01	1.807542E–01	– 1.28383E–04	– 1.56257E+06	298.15	501.2		
	1.139367E+02	1.902946E–02	– 4.186664E–06	– 4.061287E+06	501.2	1162		
	1.17909E+02	1.11923E–02	– 4.688480E–07	– 3.906959E+06	1162	2250		
$Al(OH)_3$(gibb)	7.739936E+01	9.87237E–02	5.99677E–05	– 1.819167E+06	298.15	500		
MgO(cr)	4.6374E+01	7.0518E–03	– 1.57771E–06	– 9.8661E+05	298.15	1000		
$Mg(OH)_2$(cr)	9.33928E+01	3.0121E–02	0	– 2.2659E+06	298.15	600		
$MgCl_2$(cr)	7.5655E+01	9.975E–03	0	– 6.440E+05	298.15	987		
$MgCO_3$(cr)	7.3335E+01	6.399E–02	0	– 1.4495E+06	298.15	800		
CaO(cr)	5.7801E+01	– 1.08436E–02	5.34523E–06	– 1.155008E+06	298.15	1500		
$Ca(OH)_2$(cr)	8.9264E+01	3.3112E–02	0	– 1.0355E+06	298.15	700		
$CaCl_2$(cr)	8.79697E+01	– 2.98445E–02	2.83591E–05	– 7.76527E+05	298.15	1045		
$CaSO_4$(cr)	9.89745E+01	7.67322E–02	1.514915E–05	– 1.71613E+06	298.15	1000		
β-$Ca_3(PO_4)_2$	2.08755E+02	1.59058E–01	0	– 2.46211E+06	298.15	1383		
$CaCO_3$(calc)	7.878075E+01	5.68682E–02	0	– 1.09019E+06	298.15	1200		
$CaCO_3$(arag)	7.94282E+01	4.9320E–02	0	– 1.05010E+06	298.15	600		
$CaMg(CO_3)_2$(cr)	8.279486E+01	3.47996E–01	– 2.03776E–04	– 9.71209E+05	298.15	650		
α-Na_2CO_3	6.4118E+01	1.61604E–01	0	0	298.15	628	0.800	
β-Na_2CO_3	6.4118E+01	1.61604E–01	0	0	628	754	2.1	
γ-Na_2CO_3	3.6836E+01	1.36800E–01	0	0	754	1129	28.0	
$NaHCO_3$(cr)	4.5898E+01	1.4108E–01	0	0	298.15	360		
α-K_2CO_3	6.7798E+01	1.56403E–01	0	0	298.15	693	2.1	
β-K_2CO_3	1.08922E+02	6.8852E–02			693	1173	30.0	
$Mg_3Si_2O_5(OH)_4$(cr)	3.17000E+02	1.323E–01	0	– 7.3555E+06	298.15	800		
$Ca_5(PO_4)_3OH$(cr)	4.10119E+02	1.82626E–01	0	– 7.41505E+06	298.15	480	0.25	
$Ca_5(PO_4)_3OH$(cr)	4.490467E+02	1.24878E–01	– 1.04128E–05	– 8.75879E+06	480	1200		
$Ca_5(PO_4)_3F$(cr)	4.597936E+02	7.5546E–02	0	– 9.72048E+06	298.15	1200		

Chapter IV

Selected auxiliary data

This chapter presents the chemical thermodynamic data for auxiliary compounds and complexes which are used within the NEA TDB Project. Most of these auxiliary species are used in the evaluation of the recommended ancillary data in Tables III-1, III-2 and III-3. It is therefore essential to always use these auxiliary data in conjunction with the selected data for ancillary. The use of other auxiliary data can lead to inconsistencies and erroneous results.

The values in the tables of this chapter for auxiliary compounds and complexes are either CODATA Key Values, taken from [1989COX/WAG], or were evaluated within the NEA TDB Project, as described in the corresponding chapters of the uranium review [1992GRE/FUG], the technetium review [1999RAR/RAN], the neptunium and plutonium review [2001LEM/FUG], the Update review [2003GUI/FAN], the nickel review [2005GAM/BUG], the selenium review [2005OLI/NOL], the thorium review [2008RAN/FUG], the tin review [2012GAM/GAJ], and the iron reviews [2013LEM/BER] and [2020LEM/PAL].

Table IV-1 contains the selected thermodynamic data of the auxiliary species and Table IV-2 the selected thermodynamic data of chemical reactions involving auxiliary species. The reason for listing both reaction data and entropies, enthalpies and Gibbs energies of formation is, as described in Chapter III, that uncertainties in reaction data are often smaller than the derived S_m^o, $\Delta_f H_m^o$ and $\Delta_f G_m^o$, due to uncertainty accumulation during the calculations.

All data in Table IV-1 and Table IV-2 refer to a temperature of 298.15 K, the standard state pressure of 0.1 MPa and, for aqueous species and reactions, to the infinite dilution standard state ($I = 0$).

The uncertainties listed below each reaction value in Table IV-2 are total uncertainties, and correspond mainly to the statistically defined 95% confidence interval. The uncertainties listed below each selected value in Table IV-1 have the following significance:

- for CODATA values from [1989COX/WAG], the ± terms have the meaning: "it is probable, but not at all certain, that the true values of the thermodynamic quantities differ from the recommended values given in this

report by no more than twice the ± terms attached to the recommended values".

- for values from [1992GRE/FUG], [1999RAR/RAN], [2003GUI/FAN], [2005GAM/BUG], [2005OLI/NOL], [2008RAN/FUG] and [2012GAM/GAJ], the ± terms are derived from total uncertainties in the corresponding equilibrium constant of reaction (*cf.* Table IV-2), and from the ± terms listed for the necessary CODATA key values.

CODATA [1989COX/WAG] values are available for CO_2(g), HCO_3^-, CO_3^{2-}, $H_2PO_4^-$ and HPO_4^{2-}. From the values given for $\Delta_f H_m^o$ and S_m^o the values of $\Delta_f G_m^o$ and, consequently, all the relevant equilibrium constants and enthalpy changes can be calculated. The propagation of errors during this procedure, however, leads to uncertainties in the resulting equilibrium constants that are significantly higher than those obtained from experimental determination of the constants. Therefore, reaction data for CO_2(g), HCO_3^-, CO_3^{2-}, which were absent form the corresponding Table IV-2 in [1992GRE/FUG], are included in this volume to provide the user of selected data with the data needed to obtain the lowest possible uncertainties on reaction properties.

Note that the values in Table IV-1 and Table IV-2 may contain more digits than those listed in either [1989COX/WAG] or in the chapters devoted to data selection in [1992GRE/FUG], [1995SIL/BID], [2001LEM/FUG], [2003GUI/FAN], [2005GAM/BUG], [2005OLI/NOL], [2008RAN/FUG], [2012GAM/GAJ] and [2013LEM/BER] and [2020LEM/PAL] because the data in the present chapter are retrieved directly from the computerised database and rounded to three digits after the decimal point throughout.

Table IV-1: Selected thermodynamic data for auxiliary compounds and complexes adopted in the NEA TDB Project, including the CODATA Key Values [1989COX/WAG]. All ionic species listed in this table are aqueous species. Unless noted otherwise, all data refer to 298.15 K and a pressure of 0.1 MPa and, for aqueous species, a standard state of infinite dilution ($I = 0$). The uncertainties listed below each value represent total uncertainties and correspond in principle to the statistically defined 95% confidence interval. New compounds and data that have been changed from previous selections of TDB auxiliary data are denoted with an asterisk (*). Values in bold typeface are CODATA Key Values and are taken directly from [1989COX/WAG] without further evaluation. Values obtained from internal calculation, *cf*. footnotes (a) and (b), are rounded at the third digit after the decimal point and may therefore not be exactly identical to those given in the chapters devoted to data selection in [1992GRE/FUG], [1995SIL/BID], [2001LEM/FUG], [2003GUI/FAN], [2005GAM/BUG], [2005OLI/NOL], [2008RAN/FUG] and [2012GAM/GAJ]. Systematically, all the values are presented with three digits after the decimal point, regardless of the significance of these digits. The reference listed for each entry in this table indicates the NEA TDB Review where the corresponding data have been adopted as NEA TDB Auxiliary data. The data presented in this table are available on computer media from the OECD Nuclear Energy Agency [2019MAR/SAN].

Compound and review where adopted	$\Delta_f G^\circ_m$ /kJ·mol^{-1}	$\Delta_f H^\circ_m$ /kJ·mol^{-1}	S°_m /J·K^{-1}·mol^{-1}	$C^\circ_{p,m}$ /J·K^{-1}·mol^{-1}
O(g) [1992GRE/FUG]	**231.743** [(a)] **± 0.100**	**249.180** **± 0.100**	**161.059** **± 0.003**	**21.912** **± 0.001**
O_2(g) [1992GRE/FUG]	**0.000**	**0.000**	**205.152** **± 0.005**	**29.378** **± 0.003**
H(g) [1992GRE/FUG]	**203.276** [(a)] **± 0.006**	**217.998** **± 0.006**	**114.717** **± 0.002**	**20.786** **± 0.001**
H^+ [1992GRE/FUG]	**0.000**	**0.000**	**0.000**	**0.000**
H_2(g) [1992GRE/FUG]	**0.000**	**0.000**	**130.680** **± 0.003**	**28.836** **± 0.002**
D_2(g) [2008RAN/FUG]	0.000 ± 0.000	0.000 ± 0.000	144.958 ± 0.005	29.195 ± 0.005
T_2(g) [2008RAN/FUG]	0.000 ± 0.000	0.000 ± 0.000	153.327 ± 0.005	29.199 ± 0.005
OH^- [1992GRE/FUG]	**– 157.220** [(a)] **± 0.072**	**– 230.015** **± 0.040**	**– 10.900** **± 0.200**	
H_2O(g) [1992GRE/FUG]	**– 228.582** [(a)] **± 0.040**	**– 241.826** **± 0.040**	**188.835** **± 0.010**	**33.609** **± 0.030**

(Continued on next page)

Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
H_2O(l) [1992GRE/FUG]	**– 237.140** [(a)] **± 0.041**	**– 285.830 ± 0.040**	**69.950 ± 0.030**	**75.351 ± 0.080**
H_2O_2(aq) [1992GRE/FUG]		– 191.170 ± 0.100		
He(g) [1992GRE/FUG]	**0.000**	**0.000**	**126.153 ± 0.002**	**20.786 ± 0.001**
Ne(g) [1992GRE/FUG]	**0.000**	**0.000**	**146.328 ± 0.003**	**20.786 ± 0.001**
Ar(g) [1992GRE/FUG]	**0.000**	**0.000**	**154.846 ± 0.003**	**20.786 ± 0.001**
Kr(g) [1992GRE/FUG]	**0.000**	**0.000**	**164.085 ± 0.003**	**20.786 ± 0.001**
Xe(g) [1992GRE/FUG]	**0.000**	**0.000**	**169.685 ± 0.003**	**20.786 ± 0.001**
F(g) [1992GRE/FUG]	**62.280** [(a)] **± 0.300**	**79.380 ± 0.300**	**158.751 ± 0.004**	**22.746 ± 0.002**
F^- [1992GRE/FUG]	**– 281.523** [(a)] **± 0.692**	**– 335.350 ± 0.650**	**– 13.800 ± 0.800**	
F_2(g) [1992GRE/FUG]	**0.000**	**0.000**	**202.791 ± 0.005**	**31.304 ± 0.002**
HF(aq) [1992GRE/FUG]	– 299.675 ± 0.702	– 323.150 ± 0.716	88.000 [(a)] ± 3.362	
HF(g) [1992GRE/FUG]	**– 275.400** [(a)] **± 0.700**	**– 273.300 ± 0.700**	**173.779 ± 0.003**	**29.137 ± 0.002**
HF_2^- [1992GRE/FUG]	– 583.709 ± 1.200	– 655.500 ± 2.221	92.683 [(a)] ± 8.469	
Cl(g) [1992GRE/FUG]	**105.305** [(a)] **± 0.008**	**121.301 ± 0.008**	**165.190 ± 0.004**	**21.838 ± 0.001**
Cl^- [1992GRE/FUG]	**– 131.217** [(a)] **± 0.117**	**– 167.080 ± 0.100**	**56.600 ± 0.200**	
Cl_2(g) [1992GRE/FUG]	**0.000**	**0.000**	**223.081 ± 0.010**	**33.949 ± 0.002**
ClO^- [2005OLI/NOL]	– 37.669 [(b)] ± 0.962	– 108.300 [(b)] ± 1.500	42.560 [(b)] ± 9.238	
ClO_2^- [1992GRE/FUG]	10.250 ± 4.044			

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
ClO_3^- [1992GRE/FUG]	– 7.903 [(a)] ± 1.342	– 104.000 ± 1.000	162.300 ± 3.000	
ClO_4^- [1992GRE/FUG]	**– 7.890 [(a)]** **± 0.600**	**– 128.100** **± 0.400**	**184.000** **± 1.500**	
HCl(g) [1992GRE/FUG]	**– 95.298 [(a)]** **± 0.100**	**– 92.310** **± 0.100**	**186.902** **± 0.005**	**29.136** **± 0.002**
HClO(aq) [2005OLI/NOL]	– 80.023 [(a)] ± 0.613	– 125.700 ± 1.500	– 126.255 ± 5.434	
$HClO_2$(aq) [1992GRE/FUG]	– 0.938 ± 4.043			
Br(g) [1992GRE/FUG]	**82.379 [(a)]** **± 0.128**	**111.870** **± 0.120**	**175.018** **± 0.004**	**20.786** **± 0.001**
Br^- [1992GRE/FUG]	**– 103.850 [(a)]** **± 0.167**	**– 121.410** **± 0.150**	**82.550** **± 0.200**	
Br_2(aq) [1992GRE/FUG]	4.900 ± 1.000			
Br_2(g) [1992GRE/FUG]	**3.105 [(a)]** **± 0.142**	**30.910** **± 0.110**	**245.468** **± 0.005**	**36.057** **± 0.002**
Br_2(l) [1992GRE/FUG]	**0.000**	**0.000**	**152.210** **± 0.300**	
BrO^- [1992GRE/FUG]	– 32.095 ± 1.537			
BrO_3^- [1992GRE/FUG]	19.070 [(a)] ± 0.634	– 66.700 ± 0.500	161.500 ± 1.300	
HBr(g) [1992GRE/FUG]	**– 53.361 [(a)]** **± 0.166**	**– 36.290** **± 0.160**	**198.700** **± 0.004**	**29.141** **± 0.003**
HBrO(aq) [1992GRE/FUG]	– 81.356 [(b)] ± 1.527			
I(g) [1992GRE/FUG]	**70.172 [(a)]** **± 0.060**	**106.760** **± 0.040**	**180.787** **± 0.004**	**20.786** **± 0.001**
I^- [1992GRE/FUG]	**– 51.724 [(a)]** **± 0.112**	**– 56.780** **± 0.050**	**106.450** **± 0.300**	
I_2(cr) [1992GRE/FUG]	**0.000**	**0.000**	**116.140** **± 0.300**	
I_2(g) [1992GRE/FUG]	**19.323 [(a)]** **± 0.120**	**62.420** **± 0.080**	**260.687** **± 0.005**	**36.888** **± 0.002**

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
IO_3^- [1992GRE/FUG]	– 126.338 [(a)] ± 0.779	– 219.700 ± 0.500	118.000 ± 2.000	
HI(g) [1992GRE/FUG]	**1.700** [(a)] **± 0.110**	**26.500 ± 0.100**	**206.590 ± 0.004**	**29.157 ± 0.003**
HIO_3(aq) [1992GRE/FUG]	– 130.836 ± 0.797			
S(cr)(orthorhombic) [1992GRE/FUG]	**0.000**	**0.000**	**32.054 ± 0.050**	**22.750 ± 0.050**
S(g) [1992GRE/FUG]	**236.689** [(a)] **± 0.151**	**277.170 ± 0.150**	**167.829 ± 0.006**	**23.674 ± 0.001**
S^{2-} [1992GRE/FUG]	120.695 ± 11.610			
S_2(g) [1992GRE/FUG]	**79.686** [(a)] **± 0.301**	**128.600 ± 0.300**	**228.167 ± 0.010**	**32.505 ± 0.010**
SO_2(g) [1992GRE/FUG]	**– 300.095** [(a)] **± 0.201**	**– 296.810 ± 0.200**	**248.223 ± 0.050**	**39.842 ± 0.020**
SO_3^{2-} [1992GRE/FUG]	– 487.472 ± 4.020			
$S_2O_3^{2-}$ [1992GRE/FUG]	– 519.291 ± 11.345			
SO_4^{2-} [1992GRE/FUG]	**– 744.004** [(a)] **± 0.418**	**– 909.340 ± 0.400**	**18.500 ± 0.400**	
HS^- [1992GRE/FUG]	**12.243** [(a)] **± 2.115**	**– 16.300 ± 1.500**	**67.000 ± 5.000**	
H_2S(aq) [1992GRE/FUG]	**– 27.648** [(a)] **± 2.115**	**– 38.600 ± 1.500**	**126.000 ± 5.000**	
H_2S(g) [1992GRE/FUG]	**– 33.443** [(a)] **± 0.500**	**– 20.600 ± 0.500**	**205.810 ± 0.050**	**34.248 ± 0.010**
HSO_3^- [1992GRE/FUG]	– 528.684 ± 4.046			
$HS_2O_3^-$ [1992GRE/FUG]	– 528.366 ± 11.377			
H_2SO_3(aq) [1992GRE/FUG]	– 539.187 ± 4.072			
HSO_4^- [1992GRE/FUG]	**– 755.315** [(a)] **± 1.342**	**– 886.900 ± 1.000**	**131.700 ± 3.000**	

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
Se(cr) (trigonal) [2005OLI/NOL]	0.000	0.000	42.090 ± 0.330	25.090 [(d)] ± 0.300
Se(monoclinic) [2005OLI/NOL]	1.281 [(a)] ± 0.184	2.140 ± 0.100	44.970 ± 0.400	25.090 [(d)] ± 0.800
Se(l) [2005OLI/NOL]				[(d)]
Se(g) [2005OLI/NOL]	195.927 [(a)] ± 1.524	236.070 [(b)] ± 1.521	176.729 ± 0.006	20.819 [(d)] ± 0.001
Se^{2-} [2005OLI/NOL]	128.600 [(b)] ± 3.000			
Se_2(g) [2005OLI/NOL]	92.442 [(a)] ± 3.009	141.100 ± 3.000	247.380 ± 0.400	41.710 [(d)] ± 0.200
Se_2^{2-} [2005OLI/NOL]	112.670 [(b)] ± 6.294			
Se_3(g) [2005OLI/NOL]	123.549 [(a)] ± 10.439	178.000 ± 10.000	308.900 ± 10.000	
Se_3^{2-} [2005OLI/NOL]	100.590 [(b)] ± 9.198			
Se_4(g) [2005OLI/NOL]	111.647 [(a)] ± 13.406	163.000 ± 12.000	340.000 ± 20.000	
Se_4^{2-} [2005OLI/NOL]	97.580 [(b)] ± 12.149			
Se_5(g) [2005OLI/NOL]	88.393 [(a)] ± 5.256	144.400 ± 4.300	398.300 ± 10.000	
Se_6(g) [2005OLI/NOL]	83.638 [(a)] ± 4.866	136.100 ± 3.800	428.500 ± 10.000	
Se_7(g) [2005OLI/NOL]	92.845 [(a)] ± 4.891	150.200 ± 4.500	487.000 ± 6.000	
Se_8(g) [2005OLI/NOL]	100.307 [(a)] ± 3.580	156.800 ± 3.000	526.200 ± 6.000	
SeO(g) [2005OLI/NOL]	30.855 [(a)] ± 6.218	57.400 ± 6.210	233.700 ± 1.000	32.520 [(d)] ± 0.250
SeO_2(cr) [2005OLI/NOL]	– 171.797 [(a)] ± 0.620	– 225.390 ± 0.600	67.490 ± 0.400	58.230 [(d)] ± 0.180
SeO_2(g) [2005OLI/NOL]	– 115.166 [(a)] ± 2.600	– 110.590 [(b)] ± 2.571	262.590 ± 1.250	43.360 [(d)] ± 0.150

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
SeO_3(cr) [2005OLI/NOL]	– 86.154 [(a)] ± 2.222	– 163.100 ± 2.200	91.740 ± 1.000	77.240 [(d)] ± 0.790
SeO_3^{2-} [2005OLI/NOL]	– 362.392 [(b)] ± 1.756	– 507.160 ± 1.130	– 5.055 [(a)] ± 7.011	
SeO_4^{2-} [2005OLI/NOL]	– 439.485 [(b)] ± 1.431	– 603.500 [(b)] ± 3.500	32.965 [(b)] ± 12.687	
Se_2O_5(cr) [2005OLI/NOL]		– 414.590 [(b)] ± 2.577		
HSe^- [2005OLI/NOL]	43.471 [(b)] ± 2.024			
H_2Se(g) [2005OLI/NOL]	15.217 [(a)] ± 2.003	29.000 ± 2.000	219.000 ± 0.100	34.700 [(d)] ± 0.100
H_2Se(aq) [2005OLI/NOL]	21.495 [(b)] ± 2.003	14.300 [(b)] ± 2.022	148.637 [(b)] ± 1.029	
$HSeO_3^-$ [2005OLI/NOL]	–410.112 [(b)] ± 1.166	–512.330 ± 1.010	137.656 [(a)] ± 5.184	
$HSeO_4^-$ [2005OLI/NOL]	–449.474 [(b)] ± 1.312	–582.700 [(d)] ± 4.700	136.232 [(a)] ± 16.370	
H_2SeO_3(cr) [2005OLI/NOL]		–524.720 [(b)] ± 0.650		
H_2SeO_3(aq) [2005OLI/NOL]	–425.181 [(b)] ± 0.849	–505.320 ± 0.650	211.710 [(a)] ± 3.601	
H_2SeO_4(cr) [2005OLI/NOL]		–530.500 ± 1.880		
Te(cr) [1992GRE/FUG]	0.000	0.000	49.221 ± 0.050	25.550 ± 0.100
TeO_2(cr) [2003GUI/FAN]	– 265.996 [(a)] ± 2.500	– 321.000 ± 2.500	69.890 ± 0.150	60.670 ± 0.150
N(g) [1992GRE/FUG]	**455.537 [(a)]** **± 0.400**	**472.680** **± 0.400**	**153.301** **± 0.003**	**20.786** **± 0.001**
N_2(g) [1992GRE/FUG]	**0.000**	**0.000**	**191.609** **± 0.004**	**29.124** **± 0.001**
N_3^- [1992GRE/FUG]	348.200 ± 2.000	275.140 ± 1.000	107.710 [(a)] ± 7.500	
NO_3^- [1992GRE/FUG]	**– 110.794 [(a)]** **± 0.417**	**– 206.850** **± 0.400**	**146.700** **± 0.400**	

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
HN_3(aq) [1992GRE/FUG]	321.372 ± 2.051	260.140 ± 10.050	147.381 ± 34.403	
NH_3(aq) [1992GRE/FUG]	– 26.673 ± 0.305	– 81.170 ± 0.326	109.040 ± 0.913	
NH_3(g) [1992GRE/FUG]	**– 16.407** [(a)] **± 0.350**	**– 45.940** **± 0.350**	**192.770** **± 0.050**	**35.630** **± 0.005**
NH_4^+ [1992GRE/FUG]	**– 79.398** [(a)] **± 0.278**	**– 133.260** **± 0.250**	**111.170** **± 0.400**	
P(am)(red) [1992GRE/FUG]		– 7.500 ± 2.000		
P(cr)(white, cubic) [1992GRE/FUG]	**0.000**	**0.000**	**41.090** **± 0.250**	**23.824** **± 0.200**
P(g) [1992GRE/FUG]	**280.093** [(a)] **± 1.003**	**316.500** **± 1.000**	**163.199** **± 0.003**	**20.786** **± 0.001**
P_2(g) [1992GRE/FUG]	**103.469** [(a)] **± 2.006**	**144.000** **± 2.000**	**218.123** **± 0.004**	**32.032** **± 0.002**
P_4(g) [1992GRE/FUG]	**24.419** [(a)] **± 0.448**	**58.900** **± 0.300**	**280.010** **± 0.500**	**67.081** **± 1.500**
PO_4^{3-} [1992GRE/FUG]	– 1025.491 ± 1.576	– 1284.400 ± 4.085	– 220.970 ± 12.846	
$P_2O_7^{4-}$ [1992GRE/FUG]	– 1935.503 ± 4.563			
HPO_4^{2-} [1992GRE/FUG]	**– 1095.985** [(a)] **± 1.567**	**– 1299.000** **± 1.500**	**– 33.500** **± 1.500**	
$H_2PO_4^-$ [1992GRE/FUG]	**– 1137.152** [(a)] **± 1.567**	**– 1302.600** **± 1.500**	**92.500** **± 1.500**	
H_3PO_4(aq) This review	– 1149.438[(b)] * ± 1.567	– 1294.756[(b)] * ± 1.509	160.016[(b)] * ± 1.599	
$H_3PO_4 \cdot 100H_2O$* [1989COX/WAG]		**– 1296.200** [(e)] * **± 1.000**		
$HP_2O_7^{3-}$ [1992GRE/FUG]	– 1989.158 ± 4.482			
$H_2P_2O_7^{2-}$ [1992GRE/FUG]	– 2027.117 ± 4.445			
$H_3P_2O_7^-$ [1992GRE/FUG]	– 2039.960 ± 4.362			

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$H_4P_2O_7$(aq) [1992GRE/FUG]	– 2045.668 ± 3.299	– 2280.210 ± 3.383	274.919 ± 6.954	
As(cr) [1992GRE/FUG]	0.000	0.000	35.100 ± 0.600	24.640 ± 0.500
AsO_2^- [1992GRE/FUG]	– 350.022 [a] ± 4.008	– 429.030 ± 4.000	40.600 ± 0.600	
AsO_4^{3-} [1992GRE/FUG]	– 648.360 [a] ± 4.008	– 888.140 ± 4.000	– 162.800 ± 0.600	
As_2O_5(cr) [1992GRE/FUG]	– 782.449 [a] ± 8.016	– 924.870 ± 8.000	105.400 ± 1.200	116.520 ± 0.800
As_4O_6(cubic) [1992GRE/FUG]	– 1152.445 [a] ± 16.032	– 1313.940 ± 16.000	214.200 ± 2.400	191.290 ± 0.800
As_4O_6(monoclinic) [1992GRE/FUG]	– 1154.009 [a] ± 16.041	– 1309.600 ± 16.000	234.000 ± 3.000	
As_4O_6(g) [2005GAM/BUG]	– 1092.716 [a] ± 16.116	– 1196.250 ± 16.000	408.600 ± 6.000	
$HAsO_2$(aq) [1992GRE/FUG]	– 402.925 [a] ± 4.008	– 456.500 ± 4.000	125.900 ± 0.600	
$H_2AsO_3^-$ [1992GRE/FUG]	– 587.078 [a] ± 4.008	– 714.790 ± 4.000	110.500 ± 0.600	
H_3AsO_3(aq) [1992GRE/FUG]	– 639.681 [a] ± 4.015	– 742.200 ± 4.000	195.000 ± 1.000	
$HAsO_4^{2-}$ [1992GRE/FUG]	– 714.592 [a] ± 4.008	– 906.340 ± 4.000	– 1.700 ± 0.600	
$H_2AsO_4^-$ [1992GRE/FUG]	– 753.203 [a] ± 4.015	– 909.560 ± 4.000	117.000 ± 1.000	
H_3AsO_4(aq) [1992GRE/FUG]	– 766.119 [a] ± 4.015	– 902.500 ± 4.000	184.000 ± 1.000	
$(As_2O_5)_3{\cdot}5H_2O$(cr) [1992GRE/FUG]		– 4248.400 ± 24.000		
Sb(cr) [1992GRE/FUG]	0.000	0.000	45.520 ± 0.210	25.260 ± 0.200
Bi(cr) [2001LEM/FUG]	0.000	0.000	56.740 ± 0.420	25.410 ± 0.200
C(cr) [1992GRE/FUG]	**0.000**	**0.000**	**5.740** **± 0.100**	**8.517** **± 0.080**

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
C(g) [1992GRE/FUG]	**671.254 [(a)]** **± 0.451**	**716.680** **± 0.450**	**158.100** **± 0.003**	**20.839** **± 0.001**
CO(g) [1992GRE/FUG]	**– 137.168 [(a)]** **± 0.173**	**– 110.530** **± 0.170**	**197.660** **± 0.004**	**29.141** **± 0.002**
CO_2(aq) [1992GRE/FUG]	**– 385.970 [(a)]** **± 0.270**	**– 413.260** **± 0.200**	**119.360** **± 0.600**	
CO_2(g) [1992GRE/FUG]	**– 394.373 [(a)]** **± 0.133**	**– 393.510** **± 0.130**	**213.785** **± 0.010**	**37.135** **± 0.002**
CO_3^{2-} [1992GRE/FUG]	**– 527.900 [(a)]** **± 0.390**	**– 675.230** **± 0.250**	**– 50.000** **± 1.000**	
HCO_3^- [1992GRE/FUG]	**– 586.845 [(a)]** **± 0.251**	**– 689.930** **± 0.200**	**98.400** **± 0.500**	
CN^- [2005OLI/NOL]	166.939 [(b)] ± 2.519	147.350 [(b)] ± 3.541	101.182 [(b)] ± 8.475	
HCN(aq) [2005OLI/NOL]	114.368 [(b)] ± 2.517	103.750 [(b)] ± 3.536	131.271 [(b)] ± 8.440	
HCN(g) [2005OLI/NOL]	119.517 [(a)] ± 2.500	129.900 ± 2.500	201.710 ± 0.100	
SCN^- [1992GRE/FUG]	92.700 ± 4.000	76.400 ± 4.000	144.268 [(a)] ± 18.974	
Si(cr) [1992GRE/FUG]	**0.000**	**0.000**	**18.810** **± 0.080**	**19.789** **± 0.030**
Si(g) [1992GRE/FUG]	**405.525 [(a)]** **± 8.000**	**450.000** **± 8.000**	**167.981** **± 0.004**	**22.251** **± 0.001**
SiO_2(α– quar) [1992GRE/FUG]	**– 856.287 [(a)]** **± 1.002**	**– 910.700** **± 1.000**	**41.460** **± 0.200**	**44.500 [(c)]** **± 0.200**
SiO_2(cris)* This review	– 853.854 [(a)] * ± 1.101	– 907.700 * ± 1.100	43.363 * ± 0.150	44.885 [(c)] * ± 0.300
SiO_2(trid S-I)* This review	– 853.095 [(a)] * ± 2.511	– 906.900 * ± 2.500	43.500 * ± 0.800	44.642 [(c)] * ± 0.500
SiO_2(vit)* This review	– 849.188 [(a)] * ± 1.237	– 901.600 * ± 1.200	48.170 * ± 1.000	44.061 [(c)] * ± 0.800
$SiO_2(OH)_2^{2-}$ This review	– 1177.088 [(b)] * ± 1.140	– 1386.070 [(b)] * ± 15.051	– 10.453 [(b)] * ± 50.402	
$SiO(OH)_3^-$ This review	– 1253.236 [(b)] * ± 1.034	– 1431.753 [(b)] * ± 1.352	91.727 [(b)] * ± 3.153	

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$Si(OH)_4$(aq) This review	– 1309.201[b] * ± 1.017	– 1461.070[b] * ± 1.239	181.105 [b] * ± 2.501	195.000 * ± 3.000
$Si_2O_3(OH)_4^{2-}$ This review	– 2272.500 * ± 3.000			
$Si_2O_2(OH)_5^-$ This review	– 2335.400 * ± 3.000			
$Si_3O_6(OH)_3^{3-}$(cyclic) This review	– 3053.100 * ± 4.000			
$Si_3O_5(OH)_5^{3-}$(linear) This review	– 3296.100 * ± 4.000			
$Si_4O_8(OH)_4^{4-}$(cyclic) This review	– 4080.700 * ± 6.000			
$Si_4O_7(OH)_5^{3-}$ This review	– 4142.200 * ± 5.000			
SiF_4(g) [1992GRE/FUG]	**– 1572.773** [a] **± 0.814**	**– 1615.000** **± 0.800**	**282.760** **± 0.500**	**73.622** **± 0.500**
Ge(cr) [1992GRE/FUG]	**0.000**	**0.000**	**31.090** **± 0.150**	**23.222** **± 0.100**
Ge(g) [1992GRE/FUG]	**331.209** [a] **± 3.000**	**372.000** **± 3.000**	**167.904** **± 0.005**	**30.733** **± 0.001**
GeO_2(tetragonal) [1992GRE/FUG]	**– 521.404** [a] **± 1.002**	**– 580.000** **± 1.000**	**39.710** **± 0.150**	**50.166** **± 0.300**
GeF_4(g) [1992GRE/FUG]	**– 1150.018** [a] **± 0.584**	**– 1190.200** **± 0.500**	**301.900** **± 1.000**	**81.602** **± 1.000**
Sn(cr) [2012GAM/GAJ]	**0.000**	**0.000**	**51.180** **± 0.080**	**27.110** **± 0.080**
Sn(g) [1992GRE/FUG]	**266.223** [a] **± 1.500**	**301.200** **± 1.500**	**168.492** **± 0.004**	**21.259** **± 0.001**
Sn^{2+} [2012GAM/GAJ]	**– 27.390** [a] **± 0.300**	**– 9.617** **± 1.236**	**– 19.889** **± 4.266**	
Sn^{4+} [2012GAM/GAJ]	**– 46.711** [b] **± 3.871**			
SnO(tetragonal) [2012GAM/GAJ]	**– 255.446** [a] **± 0.763**	**– 284.240** **± 0.760**	**57.180** **± 0.220**	**47.760** **± 0.080**
SnO_2(cassiterite, tetragonal) [2012GAM/GAJ]	**– 516.640** [a] **± 0.206**	**– 577.630** **± 0.200**	**51.770** **± 0.140**	**55.260** **± 0.090**

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
Pb(cr) [1992GRE/FUG]	**0.000**	**0.000**	**64.800** **± 0.300**	**26.650** **± 0.100**
Pb(g) [1992GRE/FUG]	**162.232** [(a)] **± 0.805**	**195.200** **± 0.800**	**175.375** **± 0.005**	**20.786** **± 0.001**
Pb^{2+} [1992GRE/FUG]	**– 24.238** [(a)] **± 0.399**	**0.920** **± 0.250**	**18.500** **± 1.000**	
$PbSO_4$(cr) [1992GRE/FUG]	**– 813.036** [(a)] **± 0.447**	**– 919.970** **± 0.400**	**148.500** **± 0.600**	
B(cr) [1992GRE/FUG]	**0.000**	**0.000**	**5.900** **± 0.080**	**11.087** **± 0.100**
B(g) [1992GRE/FUG]	**521.012** [(a)] **± 5.000**	**565.000** **± 5.000**	**153.436** **± 0.015**	**20.796** **± 0.005**
B_2O_3(cr) [1992GRE/FUG]	**– 1194.324** [(a)] **± 1.404**	**– 1273.500** **± 1.400**	**53.970** **± 0.300**	**62.761** **± 0.300**
$B(OH)_3$(cr) [1992GRE/FUG]	**– 969.667** [(a)] **± 0.820**	**– 1094.800** **± 0.800**	**89.950** **± 0.600**	**86.060** **± 0.400**
$B(OH)_3$(aq) [1992GRE/FUG]	**– 969.268** [(a)] **± 0.820**	**– 1072.800** **± 0.800**	**162.400** **± 0.600**	
$B(OH)_4^-$ * This review	– 1153.755[(b)] * ± 1.075	– 1345.283[(b)] * ± 1.058	100.514[(b)] * ± 3.067	
BF_3(g) [1992GRE/FUG]	**– 1119.403** [(a)] **± 0.803**	**– 1136.000** **± 0.800**	**254.420** **± 0.200**	**50.463** **± 0.100**
Al(cr) [1992GRE/FUG]	**0.000**	**0.000**	**28.300** **± 0.100**	**24.200** **± 0.070**
Al(g) [1992GRE/FUG]	**289.376** [(a)] **± 4.000**	**330.000** **± 4.000**	**164.554** **± 0.004**	**21.391** **± 0.001**
Al^{3+} This review	– 488.824 [(a)] * ± 1.614	**– 538.400** **± 1.500**	– 334.000 * ± 2.000	
Al_2O_3(cr) [1992GRE/FUG]	**– 1582.257** [(a)] **± 1.302**	**– 1675.700** **± 1.300**	**50.920** **± 0.100**	79.033[(c)]** ± 0.100
$AlOH^{2+}$* This review	– 697.880[(b)] * ± 1.663	– 767.670[(b)] * ± 2.106	– 168.541[(b)] * ± 5.512	
$Al(OH)_2^+$ * This review	– 902.991 [(b)] * ± 1.736	– 1005.946 [(b)] * ± 5.801	– 46.521[(b)] * ± 19.022	
$Al(OH)_3$(gibb)* This review	– 1155.775 [(a)] * ± 1.503	– 1294.000 * ± 1.500	68.440 * ± 0.300	91.700[(c)] * ± 0.300

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$Al(OH)_3$(baye)* This review		– 1289.000 * ± 2.500		
$Al(OH)_3$(aq)* This review	– 1106.682[(b)] * ± 2.438	– 1242.511[(b)] * ± 6.659	76.476[(b)] * ± 22.695	
$Al(OH)_4^-$ * This review	– 1306.404[(b)] * ± 1.521	– 1499.018[(b)] * ± 1.884	119.275[(b)] * ± 3.911	
AlF_3(cr) [1992GRE/FUG]	**– 1431.096 [(a)] ± 1.309**	**– 1510.400 ± 1.300**	**66.500 ± 0.500**	**75.122 ± 0.400**
Tl^+ [1999RAR/RAN]	– 32.400 ± 0.300			
Zn(cr) [1992GRE/FUG]	**0.000**	**0.000**	**41.630 ± 0.150**	**25.390 ± 0.040**
Zn(g) [1992GRE/FUG]	**94.813 [(a)] ± 0.402**	**130.400 ± 0.400**	**160.990 ± 0.004**	**20.786 ± 0.001**
Zn^{2+} [1992GRE/FUG]	**– 147.203 [(a)] ± 0.254**	**– 153.390 ± 0.200**	**– 109.800 ± 0.500**	
ZnO(cr) [1992GRE/FUG]	**– 320.479 [(a)] ± 0.299**	**– 350.460 ± 0.270**	**43.650 ± 0.400**	
Cd(cr) [1992GRE/FUG]	**0.000**	**0.000**	**51.800 ± 0.150**	**26.020 ± 0.040**
Cd(g) [1992GRE/FUG]	**77.230 [(a)] ± 0.205**	**111.800 ± 0.200**	**167.749 ± 0.004**	**20.786 ± 0.001**
Cd^{2+} [1992GRE/FUG]	**– 77.733 [(a)] ± 0.750**	**– 75.920 ± 0.600**	**– 72.800 ± 1.500**	
CdO(cr) [1992GRE/FUG]	**– 228.661 [(a)] ± 0.602**	**– 258.350 ± 0.400**	**54.800 ± 1.500**	
$CdSO_4$ ⅔ H_2O(cr) [1992GRE/FUG]	**– 1464.959 [(a)] ± 0.810**	**– 1729.300 ± 0.800**	**229.650 ± 0.400**	
Hg(g) [1992GRE/FUG]	**31.842 [(a)] ± 0.054**	**61.380 ± 0.040**	**174.971 ± 0.005**	**20.786 ± 0.001**
Hg(l) [1992GRE/FUG]	**0.000**	**0.000**	**75.900 ± 0.120**	
Hg^{2+} [1992GRE/FUG]	**164.667 [(a)] ± 0.313**	**170.210 ± 0.200**	**– 36.190 ± 0.800**	
Hg_2^{2+} [1992GRE/FUG]	**153.567 [(a)] ± 0.559**	**166.870 ± 0.500**	**65.740 ± 0.800**	

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
HgO(montroydite, red) [1992GRE/FUG]	**– 58.523** [(a)] **± 0.154**	**– 90.790 ± 0.120**	**70.250 ± 0.300**	
Hg_2Cl_2(cr) [1992GRE/FUG]	**– 210.725** [(a)] **± 0.471**	**– 265.370 ± 0.400**	**191.600 ± 0.800**	
Hg_2SO_4(cr) [1992GRE/FUG]	**– 625.780** [(a)] **± 0.411**	**– 743.090 ± 0.400**	**200.700 ± 0.200**	
Cu(cr) [1992GRE/FUG]	**0.000**	**0.000**	**33.150 ± 0.080**	**24.440 ± 0.050**
Cu(g) [1992GRE/FUG]	**297.672** [(a)] **± 1.200**	**337.400 ± 1.200**	**166.398 ± 0.004**	**20.786 ± 0.001**
Cu^{2+} [1992GRE/FUG]	**65.040** [(a)] **± 1.557**	**64.900 ± 1.000**	**– 98.000 ± 4.000**	
CuCl(g) [2003GUI/FAN]		76.800 ± 10.000		
$CuSO_4$(cr) [1992GRE/FUG]	**– 662.185** [(a)] **± 1.206**	**– 771.400 ± 1.200**	**109.200 ± 0.400**	
Ag(cr) [1992GRE/FUG]	**0.000**	**0.000**	**42.550 ± 0.200**	**25.350 ± 0.100**
Ag(g) [1992GRE/FUG]	**246.007** [(a)] **± 0.802**	**284.900 ± 0.800**	**172.997 ± 0.004**	**20.786 ± 0.001**
Ag^+ [1992GRE/FUG]	**77.096** [(a)] **± 0.156**	**105.790 ± 0.080**	**73.450 ± 0.400**	
AgCl(cr) [1992GRE/FUG]	**– 109.765** [(a)] **± 0.098**	**– 127.010 ± 0.050**	**96.250 ± 0.200**	
Ni(cr) [2005GAM/BUG]	0.000 ± 0.000	0.000 ± 0.000	29.870 ± 0.200	26.070 ± 0.100
Ni^{2+} [2005GAM/BUG]	– 45.773 ± 0.771	– 55.012 [(a)] ± 0.878	– 131.800 ± 1.400	– 46.100 ± 7.500
NiF_2(cr) [2005GAM/BUG]	–609.852 [(a)] ± 8.001	–657.300 ± 8.000	73.520 ± 0.400	63.210 ± 2.000
α-Fe [2020LEM/PAL]	0.000	0.000	27.085 ± 0.160	25.084 ± 0.500
Fe^{2+} [2020LEM/PAL]	– 90.719 [(b)] ± 0.641	– 90.295 ± 0.522	– 102.171 [(a)] ± 2.778	– 23.000 ± 10.000
Fe^{3+} [2020LEM/PAL]	– 16.226 [(b)] ± 0.650	– 50.056 [(b)] ± 0.973	– 282.404 [(b)] ± 3.927	– 108.000 ± 20.000

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$Fe_{0.932}O$(cr) [2020LEM/PAL]	– 244.593 [(a)] ± 1.906	– 265.800 ± 1.900	56.690 ± 0.490	48.970 ± 0.290
α-Fe_2O_3(cr) [2020LEM/PAL]	– 744.448 [(a)] ± 2.632	– 826.290 ± 2.630	87.400 ± 0.160	103.930 ± 0.170
Ti(cr) [1992GRE/FUG]	**0.000**	**0.000**	**30.720 ± 0.100**	**25.060 ± 0.080**
Ti(g) [1992GRE/FUG]	**428.403 [(a)] ± 3.000**	**473.000 ± 3.000**	**180.298 ± 0.010**	**24.430 ± 0.030**
TiO_2(rutile) [1992GRE/FUG]	**– 888.767 [(a)] ± 0.806**	**– 944.000 ± 0.800**	**50.620 ± 0.300**	**55.080 ± 0.300**
$TiCl_4$(g) [1992GRE/FUG]	**– 726.324 [(a)] ± 3.229**	**– 763.200 ± 3.000**	**353.200 ± 4.000**	**95.408 ± 1.000**
Am^{3+} [1995SIL/BID]	– 598.698 [(a)] ± 4.755	– 616.700 ± 1.500	– 201.000 ± 15.000	
Pu^{3+} [2001LEM/FUG]	– 578.984 ± 2.688	– 591.790 ± 1.964	– 184.510 [(b)] ± 6.154	
Np^{4+} [2001LEM/FUG]	– 491.774 [(a)] ± 5.586	– 556.022 ± 4.185	– 426.390 [(b)] ± 12.386	
NpO_2^+ [2001LEM/FUG]	– 907.765 [(a)] ± 5.628	– 978.181 ± 4.629	– 45.904 ± 10.706	– 4.000 ± 25.000
U(cr) [1992GRE/FUG]	0.000	0.000	50.200 ± 0.200	27.660 ± 0.050
U^{4+} [2003GUI/FAN]	– 529.860 [(b)] ± 1.765	– 591.200 ± 3.300	– 416.895 [(a)] ± 12.553	– 220.000 ± 50.000
UO_2^{2+} [1992GRE/FUG]	**– 952.551 [(a)] ± 1.747**	**– 1019.000 ± 1.500**	**– 98.200 ± 3.000**	**42.400 ± 3.000**
Be(cr) [1992GRE/FUG]	**0.000**	**0.000**	**9.500 ± 0.080**	**16.443 ± 0.060**
Be(g) [1992GRE/FUG]	**286.202 [(a)] ± 5.000**	**324.000 ± 5.000**	**136.275 ± 0.003**	**20.786 ± 0.001**
BeO(bromellite) [1992GRE/FUG]	**– 580.090 [(a)] ± 2.500**	**– 609.400 ± 2.500**	**13.770 ± 0.040**	**25.565 ± 0.100**
Mg(cr) [1992GRE/FUG]	**0.000**	**0.000**	**32.670 ± 0.100**	**24.869 ± 0.020**
Mg(g) [1992GRE/FUG]	**112.521 [(a)] ± 0.801**	**147.100 ± 0.800**	**148.648 ± 0.003**	**20.786 ± 0.001**

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
Mg^{2+} This review	– 455.703[a] * ± 0.786	**– 467.000** **± 0.600**	– 135.900 * ± 1.700	
MgO(cr) [1992GRE/FUG]	**– 569.312** [a] **± 0.305**	**– 601.600** **± 0.300**	**26.950** **± 0.150**	**37.237**[c] **± 0.200**
$MgOH^+$* This review	– 626.683 [b] * ± 1.031	– 685.934 [b] * ± 3.426	– 63.481[b] * ± 11.657	
$Mg(OH)_2$(cr)* This review	– 833.418[a] * ± 0.324	– 924.450 * ± 0.300	63.180 * ± 0.400	76.900[c] * ± 0.300
MgF_2(cr) [1992GRE/FUG]	**– 1071.051** [a] **± 1.210**	**– 1124.200** **± 1.200**	**57.200** **± 0.500**	**61.512** **± 0.300**
$MgCl_2$(cr)* This review	– 594.768[a] * ± 0.740	– 644.300 * ± 0.700	89.620 * ± 0.800	71.384[c] * ± 0.200
α-$MgSO_4$* This review		– 1288.600 * ± 0.600		
β-$MgSO_4$* This review	– 1170.481[a] * ± 0.671	– 1284.800 * ± 0.600	91.600 * ± 1.000	96.400* ± 0.800
$MgSO_4$(aq)* This review	– 1212.962[b] * ± 0.895	– 1370.579[b] * ± 0.976	– 53.620 [b] * ± 2.830	
$MgCO_3$(cr)* This review	– 1028.409[a] * ± 1.313	– 1112.200 * ± 1.300	65.100 * ± 0.600	76.108[c] ± 0.760
$MgCO_3$(aq)* This review	– 1000.533[b] * ± 1.304	– 1128.126[b] * ± 0.810	– 81.811[b] * ± 4.123	
$Mg_3Si_2O_5(OH)_4$(cr)* This review	– 4036.371[a] * ± 3.431	– 4364.000 * ± 3.400	221.300 * ± 1.500	273.700[c]* ± 1.500
Ca(cr) [1992GRE/FUG]	**0.000**	**0.000**	**41.590** **± 0.400**	**25.929** **± 0.300**
Ca(g) [1992GRE/FUG]	**144.021** [a] **± 0.809**	**177.800** **± 0.800**	**154.887** **± 0.004**	**20.786** **± 0.001**
Ca^{2+} [1992GRE/FUG]	**– 552.806** [a] **± 1.050**	**– 543.000** **± 1.000**	**– 56.200** **± 1.000**	
CaO(cr) [1992GRE/FUG]	**– 603.296** [a] **± 0.916**	**– 634.920** **± 0.900**	**38.100** **± 0.400**	**42.049**[c] **± 0.400**
$CaOH^+$* This review	– 717.676 [b] * ± 3.115	– 765.470 [b] * ± 4.123	– 16.134[b] * ± 16.665	
$Ca(OH)_2$(cr)* This review	– 897.497 [a] * ± 1.014	− 985.160 * ± 1.000	83.400 * ± 0.400	87.488[c] * ± 0.100

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
CaF(g) [2003GUI/FAN]	– 302.118 ± 5.104	– 276.404 ± 5.100	229.244 ± 0.500	33.671 ± 0.500
CaF_2(cr)* [1987GAR/PAR]		– 1228.004 * ± 2.000		
CaCl(g) [2003GUI/FAN]	– 129.787 ± 5.001	– 103.400 ± 5.000	241.634 ± 0.300	35.687 ± 0.010
$CaCl_2$(cr)* This review	– 748.788 [(a)] * ± 1.561	– 795.390 * ± 1.000	108.368 * ± 4.000	72.857[(c)] * ± 0.400
$CaCl_2 \cdot 2H_2O$(cr)* [1987GAR/PAR]		– 1404.000 * ± 1.800		
$CaSO_4$(cr)* This review	– 1322.242[(a)] * ± 1.307	– 1434.510 * ± 1.300	107.400 * ± 0.200	101.200[(c)] * ± 0.200
$CaSO_4 \cdot 2H_2O$(cr)* This review	– 1797.402[(a)] * ± 1.309	– 2023.000 * ± 1.300	193.800 * ± 0.300	187.200 * ± 0.200
$CaSO_4$(aq)* This review	– 1310.213 [(b)] * ± 1.159	– 1445.461[(b)] * ± 1.124	30.328[(b)] * ± 1.751	
$Ca(NO_3)_2 \cdot 4H_2O$(cr)* [1987GAR/PAR]	– 1711.890 * ± 1.400	– 2132.030 * ± 1.300	372.521 * ± 3.400	
$Ca(H_2PO_4)_2 \cdot H_2O$(cr)* This review	– 3071.840[(a)] * ± 2.306	− 3423.405 * ± 2.295	259.840 * ± 0.420	258.840 * ± 0.250
β-$Ca_3(PO_4)_2$* This review	– 3887.597[(a)] * ± 12.016	− 4123.600 * ± 12.000	236.000 * ± 1.600	228.481[(c)] * ± 1.600
$Ca_5(PO_4)_3OH$(cr)* This review	− 6340.686[(a)] * ± 15.025	− 6740.400 * ± 15.000	389.400 * ± 2.000	381.150[(c)] * ± 3.000
$Ca_5(PO_4)_3F$(cr)* This review	− 6494.786[(a)] * ± 20.024	− 6875.800 * ± 20.000	385.600 * ± 2.500	373.000[(c)] * ± 3.500
$CaCO_3$(calc)* This review	– 1128.983[(a)] * ± 1.208	– 1207.500 * ± 1.200	91.710 * ± 0.210	83.472[(c)] * ± 0.420
$CaCO_3$(arag)* This review	– 1128.074[(a)] * ± 1.307	– 1207.700 * ± 1.300	87.990 * ± 0.210	82.320[(c)] * ± 0.200
$CaCO_3$(aq)* This review	– 1098.795[(b)] * ± 1.257	– 1202.340[(b)] * ± 1.820	7.766[(b)] * ± 5.565	
$CaMg(CO_3)_2$(cr)* This review	– 2163.402[(a)] * ± 2.019	– 2326.200 * ± 2.000	155.170 * ± 0.800	157.510[(c)] * ± 0.800

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
$CaB_3O_4(OH)_3 \cdot H_2O$(cr)*	– 3167.438 [(a)] *	– 3470.000 *	191.800 *	217.100 *
This review	± 20.005	± 20.000	± 1.500	± 1.000
$CaB_3O_3(OH)_5 \cdot 4H_2O$(cr)*	– 4116.064[(a)] *	– 4648.000 *	355.500 *	361.000 *
This review	± 12.093	± 12.000	± 5.000	± 5.000
Sr(cr)	0.000	0.000	55.700	
[1992GRE/FUG]			± 0.210	
Sr^{2+}	– 563.864 [(a)]	– 550.900	– 31.500	
[1992GRE/FUG]	± 0.781	± 0.500	± 2.000	
SrO(cr)	– 559.400 [(a)]	– 590.600	53.630	45.150
[2020GRE/GAO]	± 0.903	± 0.900	± 0.100	± 0.100
$SrCl_2$(cr)	– 784.974 [(a)]	– 833.850	114.850	
[1992GRE/FUG]	± 0.714	± 0.700	± 0.420	
$Sr(NO_3)_2$(cr)	– 783.146 [(a)]	– 982.360	194.600	
[1992GRE/FUG]	± 1.018	± 0.800	± 2.100	
Ba(cr)	0.000	0.000	62.420	
[1992GRE/FUG]			± 0.840	
Ba(g)	152.852	185.000	170.245	20.786
[2003GUI/FAN]	± 5.006	± 5.000	± 0.010	± 0.001
Ba^{2+}	– 557.656 [(a)]	– 534.800	8.400	
[1992GRE/FUG]	± 2.582	± 2.500	± 2.000	
BaO(cr)	– 519.780 [(a)]	– 548.100	70.010	47.060
[2020GRE/GAO]	± 2.513	± 2.500	± 0.100	± 0.100
BaF(g)	– 349.569	– 324.992	246.219	34.747
[2003GUI/FAN]	± 6.705	± 6.700	± 0.210	± 0.300
$BaCl_2$(cr)	– 806.953 [(a)]	– 855.200	123.680	
[1992GRE/FUG]	± 2.514	± 2.500	± 0.250	
Li(cr)	**0.000**	**0.000**	**29.120**	**24.860**
[1992GRE/FUG]			**± 0.200**	**± 0.200**
Li(g)	**126.604 [(a)]**	**159.300**	**138.782**	**20.786**
[1992GRE/FUG]	**± 1.002**	**± 1.000**	**± 0.010**	**± 0.001**
Li^+	**– 292.918 [(a)]**	**– 278.470**	**12.240**	
[1992GRE/FUG]	**± 0.109**	**± 0.080**	**± 0.150**	
Na(cr)	**0.000**	**0.000**	**51.300**	**28.230**
[1992GRE/FUG]			**± 0.200**	**± 0.200**
Na(g)	**76.964 [(a)]**	**107.500**	**153.718**	**20.786**
[1992GRE/FUG]	**± 0.703**	**± 0.700**	**± 0.003**	**± 0.001**

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
Na^+ [1992GRE/FUG]	**– 261.953** [(a)] **± 0.096**	**– 240.340** **± 0.060**	**58.450** **± 0.150**	
NaF(cr) [2001LEM/FUG]	– 546.327 [(a)] ± 0.704	– 576.600 ± 0.700	51.160 ± 0.150	
NaCl(cr) [2001LEM/FUG]	– 384.221 ± 0.147	– 411.260 ± 0.120	72.150 ± 0.200	50.500
$NaNO_3$(cr) [2003GUI/FAN]		– 467.580 ± 0.410		
Na_2CO_3(cr)* This review	– 1045.450 [(a)] * ± 0.345	– 1129.250 * ± 0.300	135.000 * ± 0.400	112.300[(c)] * ± 0.500
$NaHCO_3$(cr)* This review	– 851.175 [(a)] * ± 0.365	– 949.000 * ± 0.200	102.000 * ± 1.000	87.960[(c)] * ± 0.500
$Na_2B_4O_7{\cdot}10H_2O$(cr)* This review		− 6289.600 * ± 3.400		608.000 * ± 6.000
K(cr) [1992GRE/FUG]	**0.000**	**0.000**	**64.680** **± 0.200**	**29.600** **± 0.100**
K(g) [1992GRE/FUG]	**60.479** [(a)] **± 0.802**	**89.000** **± 0.800**	**160.341** **± 0.003**	**20.786** **± 0.001**
K^+ [1992GRE/FUG]	**– 282.510** [(a)] **± 0.116**	**– 252.140** **± 0.080**	**101.200** **± 0.200**	
KCl(cr) [2005GAM/BUG]		– 436.461 ± 0.129		
KBr(cr) [2005GAM/BUG]		– 393.330 ± 0.188		
KI(cr) [2005GAM/BUG]		– 329.150 ± 0.137		
K_2CO_3(cr)* This review	– 1065.833 [(a)] * ± 2.060	– 1151.500 * ± 2.000	155.500 * ± 1.600	114.430[(c)] * ± 1.200
$KHCO_3$(cr)* This review		– 960.700 * ± 2.000		
Rb(cr) [1992GRE/FUG]	**0.000**	**0.000**	**76.780** **± 0.300**	**31.060** **± 0.100**
Rb(g) [1992GRE/FUG]	**53.078** [(a)] **± 0.805**	**80.900** **± 0.800**	**170.094** **± 0.003**	**20.786** **± 0.001**
Rb^+ [1992GRE/FUG]	**– 284.009** [(a)] **± 0.153**	**– 251.120** **± 0.100**	**121.750** **± 0.250**	

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Table IV-1 (continued)

Compound and review where adopted	$\Delta_f G^o_m$ /kJ·mol^{-1}	$\Delta_f H^o_m$ /kJ·mol^{-1}	S^o_m /J·K^{-1}·mol^{-1}	$C^o_{p,m}$ /J·K^{-1}·mol^{-1}
Cs(cr) [1992GRE/FUG]	**0.000**	**0.000**	**85.230** **± 0.400**	**32.210** **± 0.200**
Cs(g) [1992GRE/FUG]	**49.556** (a) **± 1.007**	**76.500** **± 1.000**	**175.601** **± 0.003**	**20.786** **± 0.001**
Cs^+ [1992GRE/FUG]	**– 291.456** (a) **± 0.535**	**– 258.000** **± 0.500**	**132.100** **± 0.500**	
CsCl(cr) [2001LEM/FUG]	– 413.807 (a) ± 0.208	– 442.310 ± 0.160	101.170 ± 0.200	52.470
CsBr(cr) [2001LEM/FUG]	– 391.171 ± 0.305	– 405.600 ± 0.250	112.940 ± 0.400	52.930

(a) Value calculated internally using $\Delta_f G^o_m = \Delta_f H^o_m - T\sum_i S^o_{m,i}$.
(b) Value calculated internally from reaction data (see Table IV-2).
(c) Temperature coefficients of this function are listed in Table III-3 of this review.
(d) Temperature coefficients of this function are listed in Table III-3 of [2005OLI/NOL].
(e) See note 51 (p. 37) of [1989COX/WAG].
(**) Only the uncertainty has changed.

Table IV-2: Selected thermodynamic data for reactions involving auxiliary compounds and complexes used in the evaluation of thermodynamic data for the NEA TDB Project data. All ionic species listed in this table are aqueous species. The selection of these data is described in Chapter VI of [1992GRE/FUG] and [2005OLI/NOL]. Unless noted otherwise, all data refer to 298.15 K and a pressure of 0.1 MPa and, for aqueous species, a standard state of infinite dilution (I = 0). The uncertainties listed below each value represent total uncertainties and correspond in principle to the statistically defined 95% confidence interval. Systematically, all the values are presented with three digits after the decimal point, regardless of the significance of these digits. The reference listed for each entry in this table indicates the NEA TDB Review where the corresponding data have been adopted as NEA TDB Auxiliary data. New reactions and data that have been changed from previously selected TDB Auxiliary Data are denoted with an asterisk (*). The data presented in this table are available on computer media from the OECD Nuclear Energy Agency [2019MAR/SAN].

Species and review where adopted	Reaction			
	$\log_{10} K^\circ$	$\Delta_r G_m^\circ$ /kJ·mol^{-1}	$\Delta_r H_m^\circ$ /kJ·mol^{-1}	$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}
HF(aq)	$F^- + H^+ \rightleftharpoons HF(aq)$			
[1992GRE/FUG]	3.180 ± 0.020	– 18.152 ± 0.114	12.200 ± 0.300	101.800 [a] ± 1.077
HF_2^-	$F^- + HF(aq) \rightleftharpoons HF_2^-$			
[1992GRE/FUG]	0.440 ± 0.120	– 2.511 ± 0.685	3.000 ± 2.000	18.486 [a] ± 7.090
ClO^-	$HClO(aq) \rightleftharpoons ClO^- + H^+$			
[2005OLI/NOL]	– 7.420 ± 0.130	42.354 ± 0.742	17.400 ± 2.100	– 83.695 [a] ± 7.470
ClO_2^-	$HClO_2(aq) \rightleftharpoons ClO_2^- + H^+$			
[1992GRE/FUG]	– 1.960 ± 0.020	11.188 ± 0.114		
HClO(aq)	$Cl_2(g) + H_2O(l) \rightleftharpoons Cl^- + H^+ + HClO(aq)$			
[1992GRE/FUG]	– 4.537 ± 0.105	25.900 ± 0.600		
$HClO_2$(aq)	$H_2O(l) + HClO(aq) \rightleftharpoons 2H^+ + HClO_2(aq) + 2\,e^-$			
[1992GRE/FUG]	– 55.400 [b] ± 0.700	316.230 ± 3.996		
BrO^-	$HBrO(aq) \rightleftharpoons BrO^- + H^+$			
[1992GRE/FUG]	– 8.630 ± 0.030	49.260 ± 0.171	30.000 ± 3.000	– 64.600 [a] ± 10.078

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Table IV-2 (continued)

Species and review where adopted	Reaction			
	$\log_{10} K^{o}$	$\Delta_r G_m^o$ /kJ·mol^{-1}	$\Delta_r H_m^o$ /kJ·mol^{-1}	$\Delta_r S_m^o$ /J·K^{-1}·mol^{-1}
HBrO(aq)	$Br_2(aq) + H_2O(l) \rightleftharpoons Br^- + H^+ + HBrO(aq)$			
[1992GRE/FUG]	– 8.240 ± 0.200	47.034 ± 1.142		
HIO_3(aq)	$H^+ + IO_3^- \rightleftharpoons HIO_3(aq)$			
[1992GRE/FUG]	0.788 ± 0.029	– 4.498 ± 0.166		
S^{2-}	$HS^- \rightleftharpoons H^+ + S^{2-}$			
[1992GRE/FUG]	– 19.000 ± 2.000	108.450 ± 11.416		
SO_3^{2-}	$H_2O(l) + SO_4^{2-} + 2\,e^- \rightleftharpoons 2OH^- + SO_3^{2-}$			
[1992GRE/FUG]	– 31.400 [b] ± 0.700	179.230 ± 3.996		
$S_2O_3^{2-}$	$3H_2O(l) + 2SO_3^{2-} + 4\,e^- \rightleftharpoons 6OH^- + S_2O_3^{2-}$			
[1992GRE/FUG]	– 39.200 [b] ± 1.400	223.760 ± 7.991		
H_2S(aq)	$H_2S(aq) \rightleftharpoons H^+ + HS^-$			
[1992GRE/FUG]	– 6.990 ± 0.170	39.899 ± 0.970		
HSO_3^-	$H^+ + SO_3^{2-} \rightleftharpoons HSO_3^-$			
[1992GRE/FUG]	7.220 ± 0.080	– 41.212 ± 0.457	66.000 ± 30.000	359.590 [a] ± 100.630
$HS_2O_3^-$	$H^+ + S_2O_3^{2-} \rightleftharpoons HS_2O_3^-$			
[1992GRE/FUG]	1.590 ± 0.150	– 9.076 ± 0.856		
H_2SO_3(aq)	$H^+ + HSO_3^- \rightleftharpoons H_2SO_3(aq)$			
[1992GRE/FUG]	1.840 ± 0.080	– 10.503 ± 0.457	16.000 ± 5.000	88.891 [a] ± 16.840
HSO_4^-	$H^+ + SO_4^{2-} \rightleftharpoons HSO_4^-$			
[1992GRE/FUG]	1.980 ± 0.050	– 11.302 ± 0.285		
Se(g)	$\frac{1}{2}\,Se_2(g) \rightleftharpoons Se(g)$			
[2005OLI/NOL]			–165.520 ± 0.250	

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Table IV-2 (continued)

Species and review where adopted	Reaction			
	$\log_{10} K^\circ$	$\Delta_r G_m^\circ$ /kJ·mol^{-1}	$\Delta_r H_m^\circ$ /kJ·mol^{-1}	$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}
Se^{2-}	$Se(cr) + 2\,e^- \rightleftharpoons Se^{2-}$			
[2005OLI/NOL]	–22.530 ± 0.526	128.600 ± 3.000		
Se_2^{2-}	$2Se^{2-} \rightleftharpoons Se_2^{2-} + 2\,e^-$			
[2005OLI/NOL]	25.320 ± 0.333	–144.530 ± 1.900		
Se_3^{2-}	$3Se^{2-} \rightleftharpoons Se_3^{2-} + 4\,e^-$			
[2005OLI/NOL]	49.966 ± 0.333	–285.210 ± 1.900		
Se_4^{2-}	$4Se^{2-} \rightleftharpoons Se_4^{2-} + 6\,e^-$			
[2005OLI/NOL]	73.023 ± 0.333	–416.820 ± 1.900		
$SeO_2(g)$	$SeO_2(cr) \rightleftharpoons SeO_2(g)$			
[2005OLI/NOL]			114.800 ± 2.500	
SeO_3^{2-}	$HSeO_3^- \rightleftharpoons H^+ + SeO_3^{2-}$			
[2005OLI/NOL]	–8.360 ± 0.230	47.719 ± 1.313		
SeO_4^{2-}	$HSeO_4^- \rightleftharpoons H^+ + SeO_4^{2-}$			
[2005OLI/NOL]	–1.750 ± 0.100	–9.989 ± 0.571	–20.800 ± 3.200	–103.267 [(a)] ± 10.902
$Se_2O_5(cr)$	$SeO_2(cr) + SeO_3(cr) \rightleftharpoons Se_2O_5(cr)$			
[2005OLI/NOL]			–26.100 ± 1.200	
HSe^-	$H_2Se(aq) \rightleftharpoons H^+ + HSe^-$			
[2005OLI/NOL]	–3.850 ± 0.050	21.976 ± 0.285		
$H_2Se(aq)$	$H_2Se(g) \rightleftharpoons H_2Se(aq)$			
[2005OLI/NOL]	–1.100 ± 0.010	6.279 ± 0.057	–14.700 ± 0.300	–70.363 [(a)] ± 1.024
$HSeO_3^-$	$H_2SeO_3(aq) \rightleftharpoons H^+ + HSeO_3^-$			
[2005OLI/NOL]	–2.640 ± 0.140	15.069 ± 0.799		

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Table IV-2 (continued)

Species and review where adopted	Reaction			
	$\log_{10} K^\circ$	$\Delta_r G_m^\circ$ /kJ·mol^{-1}	$\Delta_r H_m^\circ$ /kJ·mol^{-1}	$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}
$HSeO_4^-$	$H_2O(l) + H_2SeO_3(aq) \rightleftharpoons 3H^+ + HSeO_4^- + 2\ e^-$			
[2005OLI/NOL]	–37.289 [(c)] ± 0.175	212.847 ± 1.000		
$H_2SeO_3(cr)$	$H_2O(l) + SeO_2(cr) \rightleftharpoons H_2SeO_3(cr)$			
[2005OLI/NOL]			–13.500 ± 0.180	
$H_2SeO_3(aq)$	$3H_2O(l) + Se(cr) \rightleftharpoons 2H_2(g) + H_2SeO_3(aq)$			
[2005OLI/NOL]	–50.147 ± 0.147	286.240 ± 0.840		
$HN_3(aq)$	$H^+ + N_3^- \rightleftharpoons HN_3(aq)$			
[1992GRE/FUG]	4.700 ± 0.080	– 26.828 ± 0.457	– 15.000 ± 10.000	39.671 [(a)] ± 33.575
$NH_3(aq)$	$NH_4^+ \rightleftharpoons H^+ + NH_3(aq)$			
[1992GRE/FUG]	– 9.237 ± 0.022	52.725 ± 0.126	52.090 ± 0.210	– 2.130 [(a)] ± 0.821
$HNO_2(aq)$	$H^+ + NO_2^- \rightleftharpoons HNO_2(aq)$			
[1992GRE/FUG]	3.210 ± 0.160	– 18.323 ± 0.913	– 11.400 ± 3.000	23.219 [(a)] ± 10.518
PO_4^{3-}	$HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+$			
[1992GRE/FUG]	– 12.350 ± 0.030	70.494 ± 0.171	14.600 ± 3.800	– 187.469[(a)] ± 12.758
$P_2O_7^{4-}$	$HP_2O_7^{3-} \rightleftharpoons H^+ + P_2O_7^{4-}$			
[1992GRE/FUG]	– 9.400 ± 0.150	53.656 ± 0.856		
$H_2PO_4^-$	$H^+ + HPO_4^{2-} \rightleftharpoons H_2PO_4^-$			
[1992GRE/FUG]	7.212 ± 0.013	– 41.166 ± 0.074	– 3.600 ± 1.000	126.000 [(a)] ± 3.363
$H_3PO_4(aq)$[(d)]	$H^+ + H_2PO_4^- \rightleftharpoons H_3PO_4(aq)$			
This review	2.152 * ± 0.005	– 12.286 * ± 0.029	7.844 * ± 0.163	67.516[(a)] * ± 0.555
$HP_2O_7^{3-}$	$H_2P_2O_7^{2-} \rightleftharpoons H^+ + HP_2O_7^{3-}$			
[1992GRE/FUG]	– 6.650 ± 0.100	37.958 ± 0.571		

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Table IV-2 (continued)

Species and review where adopted	Reaction			
	$\log_{10} K^\circ$	$\Delta_r G_m^\circ$ /kJ·mol^{-1}	$\Delta_r H_m^\circ$ /kJ·mol^{-1}	$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}
$H_2P_2O_7^{2-}$	$H_3P_2O_7^- \rightleftharpoons H^+ + H_2P_2O_7^{2-}$			
[1992GRE/FUG]	– 2.250 ± 0.150	12.843 ± 0.856		
$H_3P_2O_7^-$	$H_4P_2O_7(aq) \rightleftharpoons H^+ + H_3P_2O_7^-$			
[1992GRE/FUG]	– 1.000 ± 0.500	5.708 ± 2.854		
$H_4P_2O_7(aq)$	$2H_3PO_4(aq) \rightleftharpoons H_2O(l) + H_4P_2O_7(aq)$			
[1992GRE/FUG]	– 2.790 ± 0.170	15.925 ± 0.970	22.200 ± 1.000	21.045 [(a)] ± 4.673
$CO_2(aq)$	$H^+ + HCO_3^- \rightleftharpoons CO_2(aq) + H_2O(l)$			
[1992GRE/FUG]	6.354 ± 0.020	– 36.269 ± 0.114		
$CO_2(g)$	$CO_2(aq) \rightleftharpoons CO_2(g)$			
[1992GRE/FUG]	1.472 ± 0.020	– 8.402 ± 0.114		
HCO_3^-	$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$			
[1992GRE/FUG]	10.329 ± 0.020	– 58.958 ± 0.114		
CN^-	$HCN(aq) \rightleftharpoons CN^- + H^+$			
[2005OLI/NOL]	– 9.210 ± 0.020	52.571 ± 0.114	43.600 ± 0.200	– 30.089 [(a)] ± 0.772
HCN(aq)	$HCN(g) \rightleftharpoons HCN(aq)$			
[2005OLI/NOL]	0.902 ± 0.050	– 5.149 ± 0.285	– 26.150 ± 2.500	– 70.439 [(a)] ± 8.440
$SiO(OH)_3^-$	$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+$			
This review	– 9.804 * ± 0.033	55.965 * ± 0.188	29.317 * ± 0.541	– 89.378[(a)] * ± 1.921
$SiO_2(OH)_2^{2-}$	$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+$			
This review	– 23.145 * ± 0.090	132.113 * ± 0.514	75.000 * ± 15.000	– 191.558[(a)] * ± 50.340
$Si(OH)_4(aq)$	$2H_2O(l) + SiO_2(\alpha\text{-quar}) \rightleftharpoons Si(OH)_4(aq)$			
This review	– 3.743 * ± 0.027	21.366 * ± 0.154	21.290 * ± 0.727	– 0.255[(a)] * ± 2.492
$Si_2O_3(OH)_4^{2-}$	$2Si(OH)_4(aq) \rightleftharpoons 2H^+ + H_2O(l) + Si_2O_3(OH)_4^{2-}$			
[1992GRE/FUG]	– 19.000 ± 0.300	108.450 ± 1.712		

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Table IV-2 (continued)

Species and review where adopted	Reaction $\log_{10} K^\circ$	$\Delta_r G_m^\circ$ /kJ·mol^{-1}	$\Delta_r H_m^\circ$ /kJ·mol^{-1}	$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}
$Si_2O_2(OH)_5^-$	$2Si(OH)_4(aq) \rightleftharpoons H^+ + H_2O(l) + Si_2O_2(OH)_5^-$			
[1992GRE/FUG]	– 8.100 ± 0.300	46.235 ± 1.712		
$Si_3O_6(OH)_3^{3-}$	$3Si(OH)_4(aq) \rightleftharpoons 3H^+ + 3H_2O(l) + Si_3O_6(OH)_3^{3-}$			
[1992GRE/FUG]	– 28.600 ± 0.300	163.250 ± 1.712		
$Si_3O_5(OH)_5^{3-}$	$3Si(OH)_4(aq) \rightleftharpoons 3H^+ + 2H_2O(l) + Si_3O_5(OH)_5^{3-}$			
[1992GRE/FUG]	– 27.500 ± 0.300	156.970 ± 1.712		
$Si_4O_8(OH)_4^{4-}$	$4Si(OH)_4(aq) \rightleftharpoons 4H^+ + 4H_2O(l) + Si_4O_8(OH)_4^{4-}$			
[1992GRE/FUG]	– 36.300 ± 0.500	207.200 ± 2.854		
$Si_4O_7(OH)_5^{3-}$	$4Si(OH)_4(aq) \rightleftharpoons 3H^+ + 4H_2O(l) + Si_4O_7(OH)_5^{3-}$			
[1992GRE/FUG]	– 25.500 ± 0.300	145.560 ± 1.712		
$B(OH)_4^-$ [(c)] *	$B(OH)_3(aq) + OH^- \rightleftharpoons B(OH)_4^-$			
This review	4.777 * ± 0.121	– 27.267 * ± 0.692	– 42.468 * ± 0.691	– 50.986* ± 3.001
Al^{3+}*	$Al(OH)_3(gibb) + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l)$			
	7.793 * ± 0.098	– 44.483 * ± 0.558	– 101.890 * ± 2.120	– 192.544[(a)] * ± 7.353
$AlOH^{2+}$*	$Al^{3+} + H_2O(l) \rightleftharpoons AlOH^{2+} + H^+$			
This review	– 4.920 * ± 0.070	28.084 * ± 0.400	56.560 * ± 1.478	95.509[(a)] * ± 5.136
$Al(OH)_2^+$ *	$Al(OH)_3(gibb) + H^+ \rightleftharpoons Al(OH)_2^+ + H_2O(l)$			
This review	– 2.741 * ± 0.152	15.644 * ± 0.868	2.224 * ± 5.604	– 45.011[(a)] * ± 19.020
$Al(OH)_3(aq)$*	$Al(OH)_3(gibb) + H^+ \rightleftharpoons Al(OH)_3(aq)$			
This review	– 8.601 * ± 0.336	49.093 * ± 1.919	51.489 * ± 6.488	8.036[(a)] * ± 22.693
$Al(OH)_4^-$ *	$Al(OH)_3(gibb) + H_2O(l) \rightleftharpoons Al(OH)_4^- + H^+$			
This review	– 15.156 * ± 0.040	86.511 * ± 0.228	80.812 * ± 1.140	– 19.115[(a)] * ± 3.900
$MgOH^+$*	$Mg^{2+} + H_2O(l) \rightleftharpoons MgOH^+ + H^+$			
This review	– 11.591 * ± 0.117	66.160 * ± 0.666	66.896 * ± 3.373	2.469[(a)] * ± 11.532

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Table IV-2 (continued)

Species and review where adopted	Reaction $\log_{10} K^{\circ}$	$\Delta_r G_m^{\circ}$ /kJ·mol^{-1}	$\Delta_r H_m^{\circ}$ /kJ·mol^{-1}	$\Delta_r S_m^{\circ}$ /J·K^{-1}·mol^{-1}
$MgSO_4(aq)$*	$Mg^{2+} + SO_4^{2-} \rightleftharpoons MgSO_4(aq)$			
This review	2.322 * ± 0.016	– 13.255 * ± 0.090	5.761 * ± 0.658	63.780[a] * ± 2.227
$MgCO_3(aq)$*	$Mg^{2+} + CO_3^{2-} \rightleftharpoons MgCO_3(aq)$			
This review	2.966 * ± 0.169	– 16.930 * ± 0.965	14.104 * ± 0.484	104.089[a] * ± 3.621
$CaOH^+$*	$Ca^{2+} + H_2O(l) \rightleftharpoons CaOH^+ + H^+$			
This review	– 12.661 * ± 0.514	72.269* ± 2.932	63.360 * ± 4.000	– 29.884[a] * ± 16.635
$CaSO_4(aq)$*	$Ca^{2+} + SO_4^{2-} \rightleftharpoons CaSO_4(aq)$			
This review	2.348 * ± 0.045	– 13.403 * ± 0.258	6.879 * ± 0.321	68.028[a] * ± 1.381
$CaCO_3(aq)$*	$Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3(aq)$			
This review	3.166 * ± 0.100	– 18.089 * ± 0.570	15.890 * ± 1.500	113.966[a] * ± 5.382

(a) Value calculated internally using $\Delta_r G_m^{\circ} = \Delta_r H_m^{\circ} - T\,\Delta_r S_m^{\circ}$.
(b) Value calculated from a selected standard potential.
(c) Value of $\log_{10} K^{\circ}$ calculated internally from $\Delta_r G_m^{\circ}$.
(d) $\Delta_r C_{p,m}^{\circ} = -(114.300 \pm 4.600)$ J·K^{-1}·mol^{-1}.
(e) $\Delta_r C_{p,m}^{\circ} = -(24.000 \pm 3.000)$ J·K^{-1}·mol^{-1}.

Part 3

Discussion of data selection

Chapter V

Detailed assessments

This chapter contains the detailed assessments of the experimental data for the following species:

- Aqueous phosphoric acid, $H_3PO_4(aq)$ (Chapter V.1)
- Aqueous carbonic acid $H_2CO_3(aq)$ (Chapter V.2)
- SiO_2(quartz), SiO_2(cristobalite) and $SiO_2(aq)$ (Chapter V.3)
- Aqueous borate species (Chapter V.4)
- Aluminium species, including $Al_2O_3(cr)$, $Al(OH)_3(cr)$ and aqueous species (Chapter V.5)
- Mg and Ca species (Chapter V.6)
- Na and K species (Chapter V.7)
- Selected Mg and Ca minerals (Chapter V.8)

The critical review of the most relevant literature references used for the realisation of this work is addressed in the entries of Appendix A.

Chapter V.1

Aqueous phosphate solutions

It is recognised that problems exist with the data for the enthalpy of formation and entropy of $H_3PO_4(aq)$ as well as with the interaction coefficients in aqueous phosphate solutions. Rard and Wolery [2007RAR/WOL] noted that [1982WAG/EVA] and [1992GRE/FUG] contain discrepant data for $H_2PO_4^-$ and HPO_4^{2-} that may have led to inconsistencies in the equilibrium constants recommended for aqueous phosphate complexes and solids in the NEA TDB database. It is now clear that the enthalpies of formation of many, perhaps most, of the phosphate species quoted in the NBS Tables [1982WAG/EVA] are too positive by *ca.* 6-7 $kJ \cdot mol^{-1}$, most probably due to a dependence on an erroneous enthalpy of solution of $P_2O_5(cr)$. It was recognised that the CODATA tables [1989COX/WAG], contain values for $\Delta_f H_m^o(HPO_4^{2-}, 298.15\,K)$ and $\Delta_f H_m^o(H_2PO_4^-, 298.15\,K)$, respectively, which are 6.86 and 6.31 $kJ \cdot mol^{-1}$ more negative than the NBS values. Grenthe *et al.* [1992GRE/FUG] accepted the CODATA values and also assessed a value for $\Delta_f H_m^o(H_3PO_4, aq, 298.15\,K)$, which was 5.78 $kJ \cdot mol^{-1}$ more negative than the NBS value.

V.1.1 Data for H_3PO_4(aq)

As discussed by Rard and Wolery [2007RAR/WOL], all the CODATA phosphate values are based on $\Delta_f H_m^o(H_3PO_4 \cdot 100H_2O, sln, 298.15\,K) = -(1296.2 \pm 1.0)\ kJ \cdot mol^{-1}$, derived principally from the enthalpy of formation of $PCl_5(cr)$ determined by Schumm *et al.* [1974SCH/PRO] and its enthalpy of dissolution [1968BIR/SKI], supported by the enthalpy of combustion and dissolution of white phosphorus into water [1970HEA/LEW]. In the preliminary stages of this review, this problem was investigated by combining this CODATA value for $\Delta_f H_m^o(H_3PO_4 \cdot 100H_2O, sln, 298.15\,K)$ with the interaction parameters in $H_3PO_4(sln)$ given by Pitzer and Silvester [1976PIT/SIL], based on the Pitzer ion interaction model, to back-calculate the values $\Delta_f H_m^o(H_3PO_4, aq, 298.15\,K) = -1295.14\ kJ \cdot mol^{-1}$ and $S_m^o(H_3PO_4, aq, 298.15\,K) = 158.506\ J \cdot K^{-1} \cdot mol^{-1}$, respectively, *ca.* 1.0 $kJ \cdot mol^{-1}$ and 3.4 $J \cdot K^{-1} \cdot mol^{-1}$ different from the current TDB values (see Section V.1.3 for further discussion).

Therefore, there was a need to repeat these calculations, if possible, using the SIT model with the appropriate ε values for H_3PO_4(sln). However, attempts to use SIT in conjunction with the osmotic coefficients for H_3PO_4(sln) [1946ELM/MAS], [1975PLA], while constraining our optimal value of $\log_{10} K_1^\circ$ (V.1) to – (2.153 ± 0.003) at 298.15 K and invoking a $\Delta\varepsilon$ value with two SIT interaction coefficient terms in a series of iterations with respect to ionic strength, failed to produce reasonable ε_1 and ε_2 values. One reason for the failure of the simple SIT model is that the Pitzer and Silvester model [1976PIT/SIL] took into account the formation of a H_3PO_4-$H_2PO_4^-$ dimer which was also identified in earlier studies. Wood and Platford [1975WOO/PLA] interpreted their isopiestic measurements of binary mixtures of NaH_2PO_4 and $NaClO_4$ at 298.15 K as giving strong evidence for the formation of a $(H_2PO_4)_2^{2-}$ dimer with a dimerisation constant K° = (0.25 ± 0.10).

V.1.2 Interaction coefficients in alkali phosphate solutions

The SIT ion interaction coefficients of alkali metal cations with $H_2PO_4^-$, HPO_4^{2-} and PO_4^{3-} are known to be dependent on ionic strength [1992GRE/FUG] but their values needed to be verified.

In existing NEA thermodynamic volumes, the ion interaction coefficients for phosphate anions with alkali metal cations are quoted as single epsilon values in Table B-5 and as two-term ionic-strength dependent values in Table B-6. It is apparent from Table V-1 to Table V-6 below that the currently recommended NEA TDB values can be attributed to [1980CIA] and [1988CIA] ([1988CIA] is a personal communication from Ciavatta and gives more significant figures for the two-term epsilon values). Ciavatta [1980CIA] cited [1973PIT/MAY] as the source of data for $H_2PO_4^-$ and HPO_4^{2-}. However, in that paper Pitzer and Mayorga simply listed Pitzer ion interaction coefficients for a large range of acids, bases and salts, and cited Robinson and Stokes [1965ROB/STO] as their source of "experimental" data, which in turn seemed to have come from Scatchard and Breckenridge [1954SCA/BRE] for the phosphate and arsenate salts treated in this review. The tabulated osmotic and activity coefficients in the landmark text of Robinson and Stokes are in fact graphically smoothed results. In this review we have examined most of the data available to confirm the origin and establish the validity of ion interaction coefficients derived by Ciavatta.

V.1.2.1 NaH_2PO_4(aq)

The available ion interaction coefficient data for $\varepsilon(Na^+, H_2PO_4^-)$ are summarised in Table V-1 and were re-fitted by this review to ionic strengths as high as 6 mol·kg^{-1} using the SIT.

Table V-1: Values of the $\varepsilon(Na^+, H_2PO_4^-)$ ion interaction coefficients at 298.15 K based on osmotic coefficients taken from the literature and those calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(Na^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(Na^+, H_2PO_4^-)$ /kg·mol^{-1}	$\varepsilon_2(Na^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.08 ± 0.04)		– (0.109 ± 0.001)	(0.095 ± 0.003)		NEA currently selected
– (0.11 ± 0.09)	2.0 [(a)]				[1980CIA]
– (0.08 ± 0.04)	2.0 [(a)]	– (0.109 ± 0.001)	(0.095 ± 0.003)	3.0 [(a)]	[1988CIA], NEA
– (0.08 ± 0.04)	2.0 [(a)]	– (0.109 ± 0.001)	(0.089 ± 0.004)	2.0	[1954SCA/BRE] [1965ROB/STO]
– (0.082 ± 0.008)	3.0	– (0.110 ± 0.001)	(0.096 ± 0.004)	3.0	[1965ROB/STO]
		– (0.112 ± 0.002)	(0.105 ± 0.002)	6.0	[1965ROB/STO]
– (0.063 ± 0.005)	2.0	**– (0.113 ± 0.001)**	**(0.107 ± 0.002)**	5.0	[1945STO2] [1954SCA/BRE] [1965ROB/STO] [1973CHI/DOW] [1976PLA] [2013SCH/MUN][(b)]

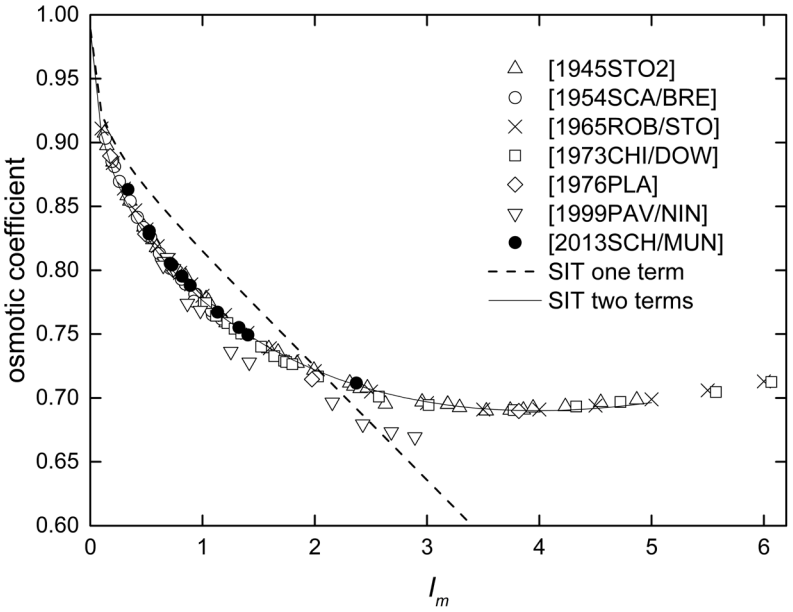
(a) The limits of these previous regressions are not stated but are assumed to be the values cited above.
(b) A mean solubility of (7.095 ± 0.098) mol·kg^{-1} was derived from nine investigations of the anhydrous phase [2013SCH/MUN].

It was assumed that dissociation of the $H_2PO_4^-$ anion could be ignored in the fitting process. Also, $H_3PO_4(aq)$ interactions with Na^+ or Cl^- were not taken into account.

The inadequacy of the single epsilon fits is highlighted in Figure V-1 even over a limited range of ionic strength and is likely the cause for the discrepancy which would exist between the sodium and potassium interaction coefficients if this approach were adopted. On the other hand, if the ionic strength dependence is recognised, the various values of ε_1 and ε_2 can be readily reconciled within the different ranges of I_m fitted in each case. A reasonable compromise of limiting the applicability of the two-term SIT treatment to $I_m = 5$ is suggested by this review.

A regression of the osmotic coefficients from the sources given in Figure V-1 with the exception of those from [1999PAV/NIN] that were discarded, utilised the two-epsilon fit ($\varepsilon_1 + \varepsilon_2 \log_{10}(I_m)$) to give the following values of $\varepsilon_1(Na^+, H_2PO_4^-) = -(0.113 \pm 0.001)$ kg·mol^{-1} and $\varepsilon_2(Na^+, H_2PO_4^-) = (0.107 \pm 0.002)$ kg·mol^{-1}. These values are very similar to those recommended currently by the NEA TDB, *viz.*, $\varepsilon_1(Na^+, H_2PO_4^-) = -(0.109 \pm 0.001)$ kg·mol^{-1} and $\varepsilon_2(Na^+, H_2PO_4^-) = (0.095 \pm 0.003)$ kg·mol^{-1}, based presumably on two rather than six independent studies.

Figure V-1: Plot of the osmotic coefficients of NaH_2PO_4(sln) *versus* I_m at 298.15 K based on SIT treatments (see Table V-1) after rejecting data in [1999PAV/NIN].



V.1.2.2 KH_2PO_4

Table V-2 and Figure V-2 show that the situation is simpler for the analogous potassium salt because the available data in [1965ROB/STO] are limited by solubility to *ca*. 1.8 mol·kg^{-1}, although measurements extend to supersaturated solutions (*ca*. 2.2 mol·kg^{-1}). There is excellent agreement between all three analyses when the full molality range is treated, but again the inadequacy of an ionic-strength independent $\varepsilon(K^+, H_2PO_4^-)$ interaction coefficient is obvious. In addition, it is apparent that the choices made by Ciavatta for these coefficients were based on a fit of all the available data.

Table V-2: Values of the $\varepsilon(K^+, H_2PO_4^-)$ ion interaction coefficients at 298.15 K based on osmotic coefficients taken from the literature and those calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(K^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(K^+, H_2PO_4^-)$ /kg·mol^{-1}	$\varepsilon_2(K^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.14 ± 0.04)	1.8 [(a)]	– (0.1473 ± 0.0008)	(0.121 ± 0.004)	1.8 [(a)]	NEA currently selected

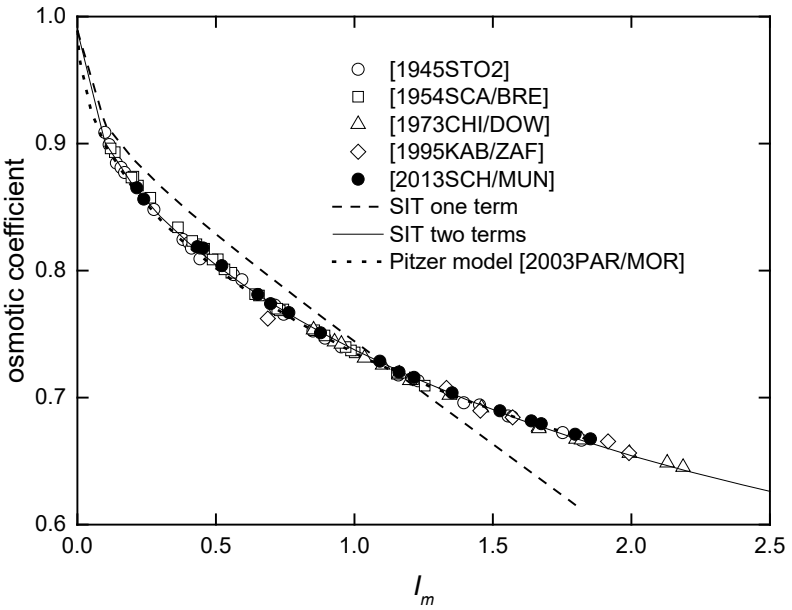
(Continued on next page)

Table V-2 (continued)

$\varepsilon(K^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(K^+, H_2PO_4^-)$ /kg·mol^{-1}	$\varepsilon_2(K^+, H_2PO_4^-)$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.14 ± 0.04)		– 0.15	0.12	1.8 [(a)]	[1980CIA]
– (0.14 ± 0.04)		– (0.1473 ± 0.0008)	(0.121 ± 0.004)		[1988CIA]
– (0.136 ± 0.011)	1.8	– (0.1475 ± 0.0005)	(0.123 ± 0.003)	1.8	[1954SCA/BRE] [1965ROB/STO]
– (0.187 ± 0.015)	0.6				[1954SCA/BRE] [1965ROB/STO]
		– (0.148 ± 0.001)	(0.123 ± 0.003)	2.0	[1945STO2]
– (0.141 ± 0.005)	1.8	**– (0.147 ± 0.001)**	**(0.122 ± 0.003)**	2.2	[1945STO2] [(b)] [1954SCA/BRE] [1973CHI/DOW] [1974CHI/DOW] [1995KAB/ZAF] [2013SCH/MUN] [(c)]

(a) The limits of these previous regressions are not stated but are assumed here to be I_m = 1.8.
(b) A solubility limit of 1.83 mol·kg^{-1} was proposed.
(c) A mean solubility of (1.864 ± 0.038) mol·kg^{-1} was derived from sixteen investigations.

Figure V-2: Osmotic coefficients of KH_2PO_4 (sln) *vs.* I_m at 298.15 K based on the SIT.



A regression of the osmotic coefficients from the sources given in Figure V-2 utilising the two-epsilon fit ($\varepsilon_1 + \varepsilon_2 \log_{10}(I_m)$) gave the following values of $\varepsilon_1(K^+, H_2PO_4^-) = -(0.147 \pm 0.001)$ kg·mol^{-1} and $\varepsilon_2(K^+, H_2PO_4^-) = (0.112 \pm 0.003)$ kg·mol^{-1}, in excellent agreement with those recommended currently by the NEA TDB. It is even more significant that the Pitzer ion interaction model gave a virtually identical fit.

V.1.2.3 Na_2HPO_4

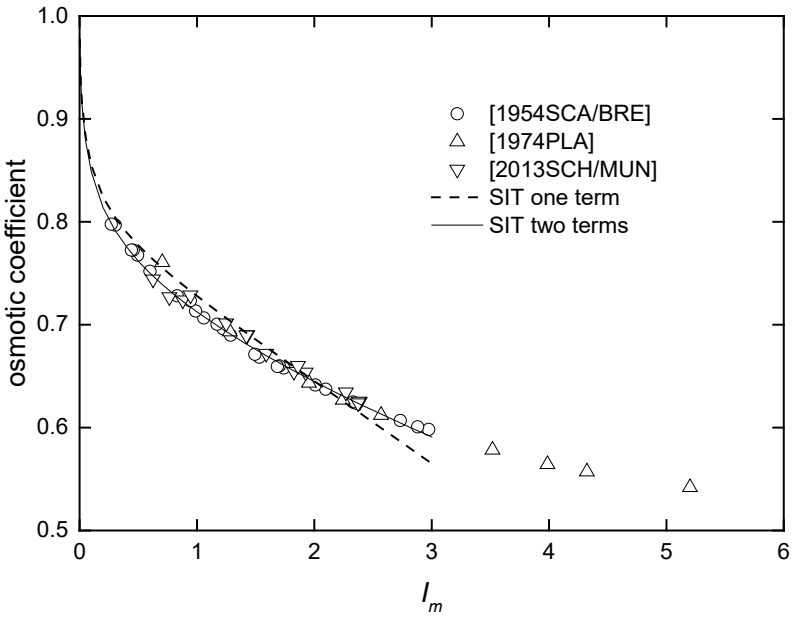
From Table V-3 it is apparent that the ion interaction terms for Na^+ with HPO_4^{2-} recommended in [1992GRE/FUG] stemmed from [1988CIA], which may have been derived from the smoothed data tabulated in [1965ROB/STO]. Re-analysis using the SIT treatment of the isopiestic data in [1954SCA/BRE], [1976PLA] and [2013SCH/MUN] led to the coefficients in Table V-3 that varied only slightly from the current values accepted by the NEA TDB as illustrated in Figure V-3.

Table V-3: Values of the $\varepsilon(Na^+, HPO_4^{2-})$ ion interaction coefficients at 298.15 K based on osmotic coefficients taken from the literature and calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(Na^+, HPO_4^{2-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(Na^+, HPO_4^{2-})$ /kg·mol^{-1}	$\varepsilon_2(Na^+, HPO_4^{2-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.15 ± 0.06)		– (0.19 ± 0.01)	(0.11 ± 0.03)		NEA currently selected
– (0.14 ± 0.04)		– 0.19	0.06		[1980CIA]
– (0.15 ± 0.06)		– (0.19 ± 0.01)	(0.11 ± 0.03)		[1988CIA]
– (0.168 ± 0.008)	3	**– (0.196 ± 0.004)**	**(0.095 ± 0.010)**	3	[1954SCA/BRE] [1965ROB/STO] [1974PLA] (a) [2013SCH/MUN](a,b)

(a) These data actually extend to much higher concentrations.
(b) A mean solubility of (0.833 ± 0.013) mol·kg^{-1} was derived from eleven investigations of the anhydrous phase.

Figure V-3: Plot of the osmotic coefficients of $Na_2HPO_4(sln)$ *versus* I_m at 298.15 K based on the SIT.



A regression of the osmotic coefficients from the sources given in Figure V-3 utilising the two-epsilon fit ($\varepsilon_1 + \varepsilon_2 \log_{10}(I_m)$) limited to $I_m \leq 3$, which embraced all the data from [1954SCA/BRE], gave the following values of $\varepsilon_1(Na^+, HPO_4^{2-}) = -(0.196 \pm 0.004)$ kg·mol^{-1} and $\varepsilon_2(Na^+, HPO_4^{2-}) = (0.095 \pm 0.010)$ kg·mol^{-1}, in good agreement with those recommended currently by the NEA TDB.

V.1.2.4 K_2HPO_4

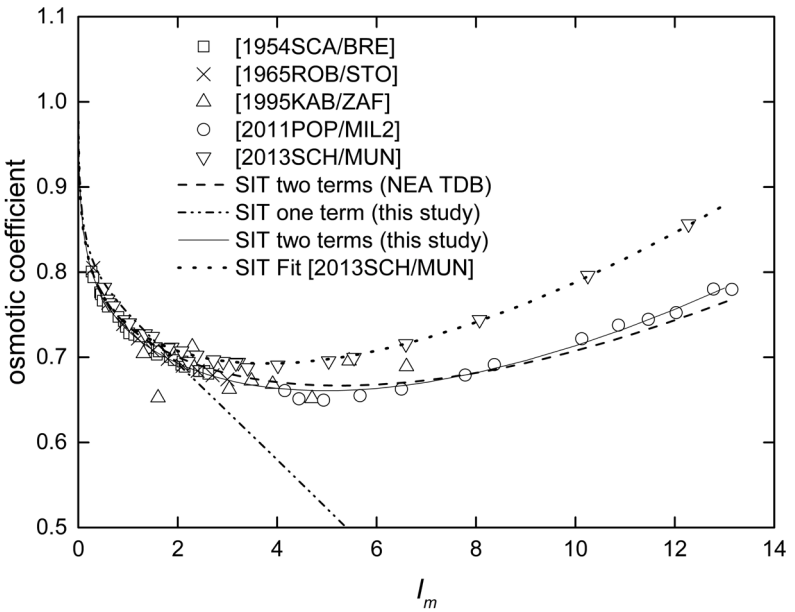
From the results summarised in Table V-4 it is apparent that the $\varepsilon(K^+, HPO_4^{2-})$ ion interaction coefficients at 298.15 K accepted by the NEA TDB were taken from [1988CIA]. The SIT treatment was applied to the osmotic coefficients in [1954SCA/BRE] and those in [2011POP/MIL2] truncated at 4.4 mol·kg^{-1}. The results presented in [1995KAB/ZAF] were considered to be too scattered to be included in this analysis. An SIT treatment of more recent osmotic coefficient data [2013SCH/MUN] led to the systematic trend in Figure V-4 when compared to those in [2011POP/MIL2]. Popović *et al.* [2014POP/MIL] subsequently suggested that this trend was consistent with the presence of $KH_2PO_4(s)$ in the solid phase used by Scharge *et al.* [2013SCH/MUN] and consequently these results were not included in the final SIT treatment.

Table V-4: Values of the $\varepsilon(K^+, HPO_4^{2-})$ ion interaction coefficients at 298.15 K cited in the literature and those calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(K^+, HPO_4^{2-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(K^+, HPO_4^{2-})$ /kg·mol^{-1}	$\varepsilon_2(K^+, HPO_4^{2-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.10 ± 0.06)		– (0.152 ± 0.007)	(0.123 ± 0.016)		NEA currently selected
		– 0.15	0.09		[1980CIA]
– (0.10 ± 0.06)		– (0.152 ± 0.007)	(0.123 ± 0.016)		[1988CIA]
– (0.122 ± 0.008)	2.62	**– (0.159 ± 0.002)**	**(0.129 ± 0.008)**	2.62	[1954SCA/BRE] [1965ROB/STO]
		– (0.163 ± 0.003)	(0.135 ± 0.003)	13.15	[1954SCA/BRE] [1965ROB/STO] [2011POP/MIL2]
		– (0.149 ± 0.003)	(0.135 ± 0.003)	12.28	[2013SCH/MUN][(a)]

(a) A mean solubility limit of (9.587 ± 0.632) mol·kg^{-1} was derived from six investigations of the trihydrate phase.

Figure V-4: Plot of the osmotic coefficients of K_2HPO_4(sln) *versus* I_m at 298.15 K based on the SIT.



V.1.2.5 Na_3PO_4 and K_3PO_4

The general paucity of results led to the decision to combine the results for these salts in one section. The results of applying the SIT treatment are given in Table V-5 and Table V-6, respectively, and are shown in Figure V-5. From Table V-5 it is apparent that the experimental data of Scatchard and Breckenridge [1954SCA/BRE] formed the sole basis for the ion interaction coefficients $\varepsilon(Na^+, PO_4^{3-})$ and $\varepsilon(K^+, PO_4^{3-})$ derived by Ciavatta [1980CIA] and adopted by the NEA. Scharge *et al.* [2013SCH/MUN] re-calculated the osmotic coefficients reported in [1974REZ/VIT] at concentrations from 0.820 to 6.196 mol·kg^{-1}. The thermodynamics of formation of the ion pair KPO_4^{2-} were studied calorimetrically at 298.15 K [1994KOC/DUS], but were not considered explicitly in the present evaluation of osmotic coefficient data.

Table V-5: Values of the $\varepsilon(Na^+, PO_4^{3-})$ ion interaction coefficients at 298.15 K cited in the literature and those re-calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(Na^+, PO_4^{3-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(Na^+, PO_4^{3-})$ /kg·mol^{-1}	$\varepsilon_2(Na^+, PO_4^{3-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.25 ± 0.03)		– (0.29 ± 0.02)	(0.10 ± 0.01)		NEA currently selected
		– 0.29	0.10		[1980CIA]
– (0.25 ± 0.06)		– (0.29 ± 0.01)	(0.10 ± 0.01)		[1988CIA]
– (0.23 ± 0.02)	4.2	**– (0.32 ± 0.02)**	**(0.16 ± 0.04)**	4.2	[1954SCA/BRE] (a)

(a) A mean solubility limit of (0.809 ± 0.084) mol·kg^{-1} was derived by Scharge *et al.* [2013SCH/MUN] from six investigations of the anhydrous phase.

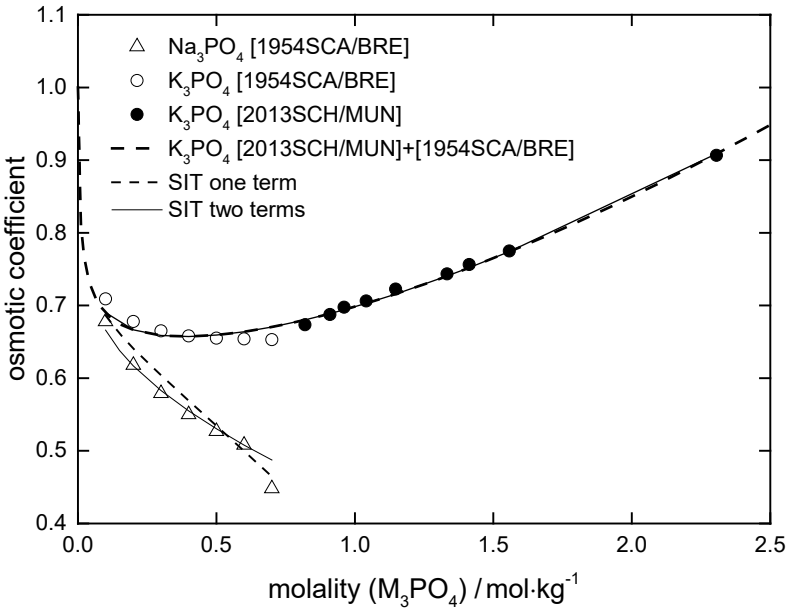
Table V-6: Values of the $\varepsilon(K^+, PO_4^{3-})$ ion interaction coefficients at 298.15 K cited in the literature and those calculated by this review (uncertainties are 2σ), where the final values result from a fit to the common I_m max value of the datasets cited. The values selected by this review are shown in bold.

$\varepsilon(K^+, PO_4^{3-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(K^+, PO_4^{3-})$ /kg·mol^{-1}	$\varepsilon_2(K^+, PO_4^{3-})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	Reference
– (0.09 ± 0.02)					NEA currently selected
– 0.09					[1980CIA]
– 0.09					[1988CIA]
– (0.09 ± 0.01)	4.2	**– (0.13 ± 0.02)**	**(0.08 ± 0.03)**	4.2	[1954SCA/BRE]
		– (0.183 ± 0.012)	(0.176 ± 0.013)	13.8 (a)	[1954SCA/BRE]
					[1974REZ/VIT]
					[2013SCH/MUN] (b)

(a) The tabulated data in [2013SCH/MUN] were truncated arbitrarily at I_m = 13.8.

(b) A mean solubility limit of (4.885 ± 0.069) mol·kg^{-1} was derived from six investigations of the heptahydrate phase.

Figure V-5: Plot of the osmotic coefficients of $Na_3PO_4(sln)$ and $K_3PO_4(sln)$ *versus* I_m at 298.15 K based on the SIT.



Due to the unknown reliability of the osmotic coefficients presented in [2013SCH/MUN] and derived from the experimental data in [1974REZ/VIT], only the ion interaction coefficients from [1954SCA/BRE] were retained by this review.

V.1.2.6 Comparison of the alkali metal phosphate and arsenates

The ion interaction coefficients for the sodium and potassium phosphates are summarised for convenience in Table V-7. Coefficients for the corresponding arsenate salts are not currently listed by the NEA TDB and therefore it was considered reasonable to apply the SIT treatment to the osmotic coefficient data in [1954SCA/BRE] for comparison with the phosphate data. It is clear that the ion interaction coefficients for these salts are ionic strength dependent. The differences between the coefficients for the potassium and sodium salts of the P(V) and As(V) oxyanions should be at least similar. The last column in Table V-7 lists such differences calculated at a randomly chosen ionic strength of 1 mol·kg^{-1} (*i.e.*, where $\varepsilon_2(M^+, H_nPO_4^{(n-3)})\ \log_{10} I_m = 0$). The trend in these differences is the same and the absolute values are indeed "similar".

Table V-7: Summary of $\varepsilon(M^+, H_nPO_4^{(n-3)})$ and $\varepsilon(M^+, H_nAsO_4^{(n-3)})$ ion interaction coefficients at 298.15 K re-calculated showing the maximum ionic strength of the data treated with the SIT (uncertainties are 2σ), and the current NEA recommended values are shown in italics. The new values selected by this review are shown in bold. $\varepsilon(M^+, H_nAsO_4^{(n-3)})$ values are included as they provide a good analogue to the phosphate with the same charge and cation.

n	M	$\varepsilon(M^+, H_nPO_4^{(n-3)})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon_1(M^+, H_nPO_4^{(n-3)})$ /kg·mol^{-1}	$\varepsilon_2(M^+, H_nPO_4^{(n-3)})$ /kg·mol^{-1}	I_m max /mol·kg^{-1}	$\varepsilon(K^+)-\varepsilon(Na^+)$ /kg·mol^{-1} [(a)]
2	Na	**– (0.063 ± 0.005)**	2.0	**– (0.113 ± 0.001)**	**(0.107 ± 0.002)**	5.0	– (0.034 ± 0.001)
		– (0.08 ± 0.04)		*– (0.109 ± 0.001)*	*(0.095 ± 0.003)*		
2	K	**– (0.141 ± 0.005)**	1.8	**– (0.147 ± 0.001)**	**(0.122 ± 0.003)**	2.2	
		– (0.14 ± 0.04)		*– (0.1473 ± 0.0008)*	*(0.121 ± 0.004)*		
1	Na	**– (0.168 ± 0.008)**	3	**– (0.196 ± 0.004)**	**(0.095 ± 0.010)**	3	(0.040 ± 0.008)
		– (0.15 ± 0.06)		*– (0.19 ± 0.01)*	*(0.11 ± 0.03)*		
1	K	**– (0.122 ± 0.008)**	2.62	**– (0.159 ± 0.002)**	**(0.129 ± 0.008)**	2.62	
		– (0.1 ± 0.06)		*– (0.152 ± 0.007)*	*(0.123 ± 0.016)*		
0	Na [(a)]	**– (0.23 ± 0.02)**	4.2	**– (0.32 ± 0.02)**	**(0.16 ± 0.04)**	4.2	(0.19 ± 0.03)
		– (0.25 ± 0.03)		*– (0.29 ± 0.02)*	*(0.10 ± 0.01)*		
0	K [(a)]	**– (0.09 ± 0.01)**	4.2	**– (0.13 ± 0.02)**	**(0.08 ± 0.03)**	4.2	
		– (0.09 ± 0.02)					
		$\varepsilon(M^+, H_nAsO_4^{(n-3)})$ /kg·mol^{-1}		$\varepsilon_1(M^+, H_nAsO_4^{(n-3)})$ /kg·mol^{-1}	$\varepsilon_2(M^+, H_nAsO_4^{(n-3)})$ /kg·mol^{-1}		$\varepsilon(K^+)-\varepsilon(Na^+)$ /kg·mol^{-1} [(a)]
2	Na	**– (0.08 ± 0.00)**	1.3	**– (0.08 ± 0.00)**	**(0.02 ± 0.00)**	1.3	
2	K	**– (0.13 ± 0.02)**	1.3	**– (0.13 ± 0.02)**	**(0.06 ± 0.01)**	1.3	– (0.04 ± 0.02)
1	Na	**– (0.11 ± 0.00)**	3	**– (0.12 ± 0.01)**	**(0.03 ± 0.02)**	3	
1	K	**– (0.04 ± 0.00)**	3	**– (0.04 ± 0.01)**	**– (0.01 ± 0.02)**	3	(0.08 ± 0.01)
0	Na	**– (0.17 ± 0.02)**	4.2	**– (0.25 ± 0.02)**	**(0.15 ± 0.04)**	4.2	
0	K	**– (0.01 ± 0.01)**	4.2	**(0.01 ± 0.02)**	**– (0.04 ± 0.03)**	4.2	(0.26 ± 0.04)

(a) These values were calculated at I_m = 1 chosen for the sake of simplicity.

V.1.3 First dissociation constant of phosphoric acid

$$H_3PO_4(aq) \rightleftharpoons H^+ + H_2PO_4^- \quad (V.1)$$

Grenthe *et al*. [1992GRE/FUG] derived the first dissociation constant for $H_3PO_4(aq)$ from data reported by Perrin [1982PER] at 298.15 K and $I = 0$, and by Mesmer and Baes [1974MES/BAE] for the corresponding hydrolysis equilibrium. Grenthe *et al*. proposed a conservative uncertainty to reflect the largest deviation from their selected value, *viz*., $\log_{10}K_1^\circ = -(2.140 \pm 0.030)$. The results of SIT analyses of the dissociation constants provided as a function of ionic strength in the literature are given in Table V-8.

Table V-8: Values of $\log_{10}K_1^\circ$ and the corresponding $\Delta\varepsilon$ obtained at 298.15 K from SIT fits (uncertainties are 2σ).

Ionic medium	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$-\log_{10}K_1^\circ$ SIT	I_m max /mol·kg^{-1}	$-\log_{10}K_1^\circ$ reported	Reference
KCl	$-(0.052 \pm 0.014)$ [(a)]	(0.304 ± 0.102) [(a)]	2.154 [(a)]	1.0	(2.153 ± 0.004)	[1974MES/BAE]
NaCl	(0.024 ± 0.020)	(0.013 ± 0.031)	2.153	6.0	2.146 [(b)]	[1989HER/FER]
KCl	$-(0.049 \pm 0.080)$ [(b)]	(0.130 ± 0.250) [(b)]	2.148	0.36	(2.147 ± 0.004)	[1991DAN/DER]
NaCl	(0.001 ± 0.025) [(b)]	$-(0.089 \pm 0.075)$ [(b)]	2.141	0.36	(2.147 ± 0.004)	
KCl	$-(0.030 \pm 0.010)$	(0.121 ± 0.004)			2.140	NEA currently selected
NaCl	(0.010 ± 0.001)	(0.095 ± 0.003)			2.140	

(a) Extrapolated from data obtained in the range 313.15 to 473.15 K assuming that the ion interaction parameters are independent of temperature.
(b) The authors took this value from [1976PIT/SIL].

It was shown in Section V.1.2.1 and Section V.1.2.2 that the values of $\varepsilon_1(Na^+, H_2PO_4^-)$, $\varepsilon_2(Na^+, H_2PO_4^-)$, $\varepsilon_1(K^+, H_2PO_4^-)$ and $\varepsilon_2(K^+, H_2PO_4^-)$ recommended currently by the NEA are validated and are accepted by this review. Moreover, SIT treatments based on these values at 298.15 K effectively reproduce the values of $\log_{10}K_1$ for the three studies cited in Table V-1 within the combined experimental uncertainties.

Treatments of the thermodynamics of the acid dissociation of $H_3PO_4(aq)$, Reaction (V.1), by Goldberg *et al.* [2002GOL/KIS] and Rard and Wolery [2007RAR/WOL] were based largely on the landmark publication by Bates [1951BAT]. Bates used his Harned cell to study this equilibrium from 273.15 to 333.15 K at 5 K intervals to ionic strengths of < 0.4 mol·kg^{-1} with 0.005 mol·kg^{-1} KCl present and varying ratios of $KH_2PO_4(sln)$ and $HCl(sln)$. These results and those available in the literature are summarised in Table V-9.

Table V-9: Summary of available values of $\log_{10}K_1^\circ$ and corresponding thermodynamic quantities for Reaction (V.1) with uncertainties estimated in this review at 2σ.

T /K	$-\log_{10}K_1^\circ$ SIT	$-\log_{10}K_1^\circ$ reported	$-\Delta_r H_{m,1}^\circ$ /kJ·mol^{-1} retained	Method	$-\log_{10}K_1^\circ$ retained	$-\Delta_r H_{m,1}^\circ$ /kJ·mol^{-1} retained	Reference
291.15		2.120		pot	(2.12 ± 0.05)		[1929BJE/UNM]
298.15		2.161			(2.16 ± 0.05)		
310.15		2.232			(2.23 ± 0.05)		
298.15			(8.16 ± 0.34) (7.87 ± 0.34) [a]	cal		(7.87 ± 0.34)	[1937PIT]
273.15		(2.056 ± 0.010)		pot	(2.06 ± 0.05)		[1951BAT]
278.15		(2.073 ± 0.005)			(2.07 ± 0.04)		
283.15		(2.088 ± 0.005)			(2.09 ± 0.03)		
288.15		(2.107 ± 0.005)			(2.11 ± 0.03)		
293.15		(2.127 ± 0.005)			(2.13 ± 0.03)		
298.15		(2.148 ± 0.005)	7.650		(2.15 ± 0.03)		
303.15		(2.171 ± 0.005)			(2.17 ± 0.03)		
308.15		(2.196 ± 0.005)			(2.20 ± 0.03)		
313.15		(2.224 ± 0.005)			(2.22 ± 0.03)		
318.15		(2.251 ± 0.005)			(2.25 ± 0.03)		
323.15		(2.277 ± 0.005)			(2.28 ± 0.03)		
328.15		(2.308 ± 0.005)			(2.31 ± 0.04)		
333.15		(2.338 ± 0.005)			(2.34 ± 0.05)		
298.15	(2.153 ± 0.002)	(2.153 ± 0.004)	(8.48 ± 0.35)	est	(2.15 ± 0.04)		[1974MES/BAE]
323.15	(2.277 ± 0.020)	(2.289 ± 0.004)		pot	(2.28 ± 0.03)		
373.15	(2.567 ± 0.020)	(2.617 ± 0.006)			(2.57 ± 0.03)		
423.15	(2.897 ± 0.020)	(2.961 ± 0.016)			(2.90 ± 0.04)		
473.15	(3.256 ± 0.020)	(3.348 ± 0.024)			(3.26 ± 0.05)		
298.15		2.146		est			[1976PIT/SIL]
298			(8.6 ± 0.7) [b]	cal			[1981VAC/SAB]
298.15		(2.148 ± 0.002)	(7.95 ± 0.42)	est	(2.15 ± 0.03)	(7.95 ± 0.42)	[1982MAR/SMI]

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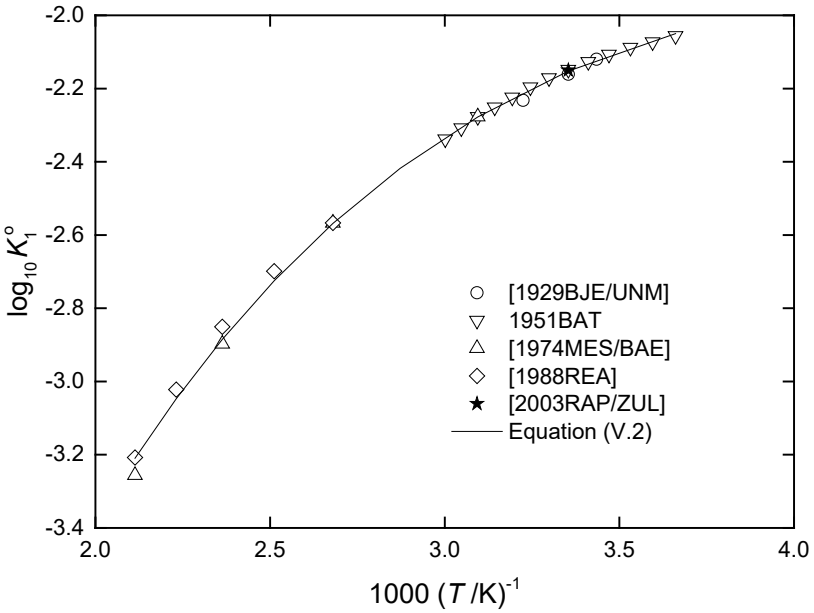
The accepted values for $\log_{10}K_1^\circ$ and $\Delta_r H_m^\circ$ listed in Table V-9 were subjected to a weighted regression whereby the uncertainties were modified by the reviewers to reflect the real experimental values rather than the statistical ones. The regression resulted in Equation (V.2) and these results are illustrated in Figure V-6 with the derived thermodynamic quantities at 298.15 K shown below.

Figure V-6: Plot of the temperature dependence of $\log_{10}K_1^\circ$ (V.1) .



$$\log_{10}K_1^\circ(\text{V.1}) = (36.46171 \pm 0.83359) - (1370.401 \pm 39.245)(T/\text{K})^{-1} - (5.97053 \pm 0.12314)\ln(T/\text{K}) \quad \text{(V.2)}$$

Equation (V.2) resulted in the following thermodynamic quantities at 298.15 K, which are our selected values for Reaction (V.1):

$$\log_{10}K_1^\circ((\text{V.1}), 298.15\ \text{K}) = \mathbf{-\ (2.152 \pm 0.005)},$$

$$\Delta_r G_m^\circ((\text{V.1}), 298.15\ \text{K}) = \mathbf{(12.286 \pm 0.029)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^\circ((\text{V.1}), 298.15\ \text{K}) = \mathbf{-\ (7.844 \pm 0.163)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^\circ((\text{V.1}), 298.15\ \text{K}) = \mathbf{-\ (67.516 \pm 0.555)}\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C_{p,m}^\circ((\text{V.1}), 298.15\ \text{K}) = -\ (114.300 \pm 4.600)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

which compare with those in [1992GRE/FUG] at 298.15 K: $\log_{10}K_1^\circ = -(2.140 \pm 0.030)$; $\Delta_r G_m^\circ(\text{V.1}) = (12.215 \pm 0.171)$ kJ·mol^{-1}; $\Delta_r H_m^\circ(\text{V.1}) = -(8.480 \pm 0.600)$ kJ·mol^{-1}; $\Delta_r S_m^\circ(\text{V.1}) = (69.412 \pm 2.093)$ J·K^{-1}·mol^{-1}.

Larsen *et al*. [1982LAR/ZEE] reported – 128 J·K^{-1}·mol^{-1} for $\Delta_r C_{p,m}^\circ(\text{V.1})$ at 298.15 K based on their careful measurements of the phosphate system with a Picker flow calorimeter, *cf*. – (190 ± 29) J·K^{-1}·mol^{-1} reported by Mesmer and Baes [1974MES/BAE].

The following thermodynamic quantities at 298.15 K result from using the corresponding CODATA values from [1989COX/WAG] for the data for $H_2PO_4^-$:

$$\Delta_f G_m^\circ(H_3PO_4, aq) = \Delta_f G_m^\circ(H_2PO_4^-) - \Delta_r G_m^\circ(\text{V.1}) =$$
$$-(1137.152 \pm 1.567) - (12.286 \pm 0.029) = \mathbf{-(1149.438 \pm 1.567)}\ \text{kJ·mol}^{-1},$$

$$\Delta_f H_m^\circ(H_3PO_4, aq) = \Delta_f H_m^\circ(H_2PO_4^-) + \Delta_r H_m^\circ(\text{V.1}) =$$
$$-(1302.600 \pm 1.500) + (7.844 \pm 0.163) = \mathbf{-(1294.756 \pm 1.509)}\ \text{kJ·mol}^{-1},$$

$$S_m^\circ(H_3PO_4, aq) = S_m^\circ(H_2PO_4^-) + \Delta_r S_m^\circ(\text{V.1}) =$$
$$(92.500 \pm 1.500) + (67.516 \pm 0.555) = \mathbf{(160.016 \pm 1.599)}\ \text{J·K}^{-1}\text{·mol}^{-1},$$

which compare well with $\Delta_f H_m^\circ(H_3PO_4, aq) = -1295.14$ kJ·mol^{-1} and $S_m^\circ(H_3PO_4, aq) = 158.506$ J·K^{-1}·mol^{-1}, derived independently from the Pitzer treatment mentioned in Section V.1.1, and are our selected data.

The corresponding partial molar volume data are summarised in the Appendix A entry for [1982LAR/ZEE].

V.1.4 Second dissociation constant of phosphoric acid

$$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-} \qquad \text{(V.3)}$$

Grenthe *et al*. [1992GRE/FUG] selected $\log_{10}K_2^\circ(\text{(V.3)}, 298.15\ \text{K}) = -(7.212 \pm 0.013)$ from the CODATA values in [1989COX/WAG], but with a reduced uncertainty. They noted that their selected value was consistent with that recommended by Martell and Smith [1982MAR/SMI], – (7.199 ± 0.002). Two potentiometric studies from the Bates school [1943BAT/ACR], [1945BAT/ACR] formed the foundation for most reviews of the thermodynamics of this equilibrium, including that of Martell and Smith. In their potentiometric experiments Bates and Acree used various ratios of Na^+ and K^+ in mixtures of $H_2PO_4^-$ and HPO_4^{2-} in the presence of 0.005 mol·kg^{-1} Cl^- to create two large datasets, which extended to $I_m = 0.6$ from 273.15 to 333.15 K. The SIT treatment was applied to these data with the caveat in their case that the Debye-Hückel extended terms are not strictly ion interaction parameters because they apply to mixtures of cations and anions with no swamping electrolyte. Nevertheless, no systematic deviations in these fits with respect to ionic strength or temperature were observed. The $\log_{10}K_2^\circ$ values so derived are given in Table V-10 and are seen to be completely

compatible with those obtained from the other works and with those from two more recent modelling approaches by Partanen and Covington [2005PAR/COV], [2005PAR/COV2]. These authors [2005PAR/COV2] applied their Hückel activity coefficient model for Reaction (V.3) based on the data in [1943BAT/ACR] and [1945BAT/ACR].

Table V-10: Values of $\log_{10}K_2^\circ$ and the corresponding $\Delta\varepsilon$ at 298.15 K from SIT fits (uncertainties are 2σ).

Ionic medium	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$-\log_{10}K_2^\circ$ SIT	I_m max /mol·kg^{-1}	$-\log_{10}K_2^\circ$ reported	Reference
NaCl	– (0.029 ± 0.023)	(0.208 ± 0.079)	(7.200 ± 0.005)	0.6	(7.198 ± 0.001)	[1943BAT/ACR]
NaCl	– (0.026 ± 0.008) [(a)]	(0.231 ± 0.018) [(a)]	(7.204 ± 0.008)	0.6		[1945BAT/ACR]
KCl	– (0.157 ± 0.064) [(a)]	(0.28 ± 0.33) [(a)]	7.198 (fixed)	1.0		[1974MES/BAE]
NaCl	– (0.056 ± 0.008)	(0.102 ± 0.012)	7.199 (fixed)	6.0		[1989HER/FER]
KCl	– (0.061 ± 0.007) [(a)]	– (0.278 ± 0.043) [(a)]	7.145	1.03	(7.200 ± 0.002)	[1991DAN/DER]
NaCl	– (0.083 ± 0.017) [(a)]	(0.122 ± 0.102) [(a)]	7.184	1.03		
NaCl	(0.018 ± 0.003)	(0.092 ± 0.007)	(7.198 ± 0.002)	2.0	7.199	[2005PAR/COV]
KCl	(0.081 ± 0.011)	(0.233 ± 0.004)	(7.197 ± 0.002)	2.0	7.199	[1943BAT/ACR]
NaCl	(0.04 ± 0.01)	– (0.008 ± 0.010)			(7.212 ± 0.013)	NEA currently
KCl	(0.11 ± 0.01)	(0.007 ± 0.002)				selected

(a) Obtained over a range of temperature assuming these values are independent of temperature.

Application of the SIT treatment to the results from the individual studies shown in Table V-10 showed that two of them are clearly divergent (*e.g.* [1974MES/BAE] and [1989HER/FER]) as discussed in detail in the corresponding Appendix A entries.

Results taken from the literature are given in Table V-11, some of which could be evaluated using the SIT. In addition, values retained for inclusion in the overall data evaluation are shown in columns 5 and 6 with uncertainties estimated in this review.

Table V-11: Summary of available values of $\log_{10}K_2^\circ$ (V.3) and corresponding thermodynamic quantities with uncertainties estimated in this review at 2σ.

T/K	$-\log_{10}K_2^\circ$ SIT	$-\log_{10}K_2^\circ$ reported	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} reported	$-\log_{10}K_2^\circ$ retained	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} retained	Method	Reference
291.15		7.227		(7.23 ± 0.03)		pot	[1929BJE/UNM]
298.15		7.207		(7.21 ± 0.03)			
310.15		7.165		(7.17 ± 0.03)			
293.15		7.218		(7.22 ± 0.03)		pot	[1933NIM]
298.15		7.206		(7.21 ± 0.03)			
303.15		7.197		(7.20 ± 0.03)			
308.15		7.192		(7.19 ± 0.03)			
313.15		7.189		(7.19 ± 0.03)			
318.15		7.189		(7.19 ± 0.04)			
323.15		7.191		(7.19 ± 0.04)			
298.15			(3.35 ± 0.33)		(3.35 ± 0.33)	cal	[1937PIT]
273.15	(7.318 ± 0.007)	(7.314 ± 0.001)		(7.32 ± 0.05)		pot	[1943BAT/ACR]
278.15	(7.284 ± 0.006)	(7.281 ± 0.001)		(7.28 ± 0.04)			
283.15	(7.256 ± 0.006)	(7.254 ± 0.001)		(7.26 ± 0.03)			
288.15	(7.233 ± 0.007)	(7.230 ± 0.001)		(7.23 ± 0.02)			
293.15	(7.215 ± 0.006)	(7.213 ± 0.001)		(7.22 ± 0.02)			
298.15	(7.200 ± 0.005)	(7.198 ± 0.001)		(7.20 ± 0.02)			
303.15	(7.195 ± 0.007)	(7.190 ± 0.001)		(7.20 ± 0.02)			
308.15	(7.188 ± 0.006)	(7.185 ± 0.002)		(7.19 ± 0.02)			
313.15	(7.188 ± 0.006)	(7.182 ± 0.002)		(7.19 ± 0.02)			
318.15	(7.186 ± 0.007)	(7.181 ± 0.002)		(7.19 ± 0.03)			
323.15	(7.188 ± 0.006)	(7.184 ± 0.002)		(7.19 ± 0.03)			
328.15	(7.191 ± 0.007)	(7.188 ± 0.002)		(7.19 ± 0.04)			
333.15	(7.198 ± 0.007)	(7.196 ± 0.002)		(7.20 ± 0.05)			
273.15	(7.310 ± 0.007)	7.313		(7.31 ± 0.05)		pot	[1945BAT/ACR]
278.15	(7.277 ± 0.005)	7.282		(7.28 ± 0.04)			
283.15	(7.251 ± 0.007)	7.254		(7.25 ± 0.03)			

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Table V-11 (continued)

T/K	$-\log_{10}K_2^\circ$ SIT	$-\log_{10}K_2^\circ$ reported	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} reported	$-\log_{10}K_2^\circ$ retained	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} retained	Method	Reference
288.15	(7.227 ± 0.005)	7.232		(7.23 ± 0.02)			
293.15	(7.208 ± 0.004)	7.213		(7.21 ± 0.02)			
298.15	(7.184 ± 0.004)	7.197		(7.20 ± 0.02)			
303.15	(7.183 ± 0.004)	7.189		(7.18 ± 0.02)			
308.15	(7.178 ± 0.006)	7.185		(7.18 ± 0.02)			
313.15	(7.175 ± 0.005)	7.180		(7.18 ± 0.02)			
318.15	(7.175 ± 0.007)	7.181		(7.18 ± 0.03)			
323.15	(7.178 ± 0.006)	7.182		(7.18 ± 0.03)			
328.15	(7.183 ± 0.008)	7.187		(7.18 ± 0.04)			
333.15	(7.189 ± 0.007)	7.193		(7.19 ± 0.05)			
278.15		7.280		(7.28 ± 0.05)		pot	[1958GRZ]
283.15		7.253		(7.25 ± 0.04)			
288.15		7.231		(7.23 ± 0.04)			
293.15		7.213		(7.21 ± 0.04)			
298.15		7.200		(7.20 ± 0.04)			
303.15		7.190		(7.19 ± 0.04)			
308.15		7.183		(7.18 ± 0.04)			
313.15		7.178		(7.18 ± 0.04)			
318.15		7.176		(7.18 ± 0.04)			
323.15		7.176		(7.18 ± 0.05)			
298.15			3.77		(3.35 ± 1.20)	cal	[1962CHR/IZA]
298.15			(2.5 ± 1.3)		(3.35 ± 0.90)	cal	[1966IRA/TAU]
298.15			(4.70 ± 0.10) [a]			cal	[1981VAC/SAB]
298.15		(7.199 ± 0.002)	(3.77 ± 0.42)			est	[1982MAR/SMI]
298.15		(7.200 ± 0.002)		(7.20 ± 0.04)		pot	[1991DAN/DER]
298.15		(7.199 ± 0.004)	(4.07 ± 0.27) [b]		(4.07 ± 1.20)	cal	[1993VAS/KOC2]
298.15		(6.81 ± 0.01) [c]	(5.12 ± 0.03) [c]			cal, pot	[1998FUK/TAK]
298.15		7.198	3.6			est	[2002GOL/KIS]

(Continued on next page)

Table V-11 (continued)

T /K	$-\log_{10} K_2^\circ$ SIT	$-\log_{10} K_2^\circ$ reported	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} reported	$-\log_{10} K_2^\circ$ retained	$\Delta_r H_{m,2}^\circ$ /kJ·mol^{-1} retained	Method	Reference
298.15		(7.17 ± 0.04)				est	[2003RAP/ZUL]
298.15	(7.198 ± 0.002)	7.199				est	[2005PAR/COV]
298.15	(7.197 ± 0.002)					est	

(a) I_c = 0.15 (NaCl)
(b) A value of $\Delta_r C_{p,m,2}^\circ = -(160 \pm 40)$ J·K^{-1}·mol^{-1} was given.
(c) I_c = 0.10 (KCl) and a value of $\Delta_r C_{p,m,2}^\circ = -(187 \pm 3)$ J·K^{-1}·mol^{-1} was given.

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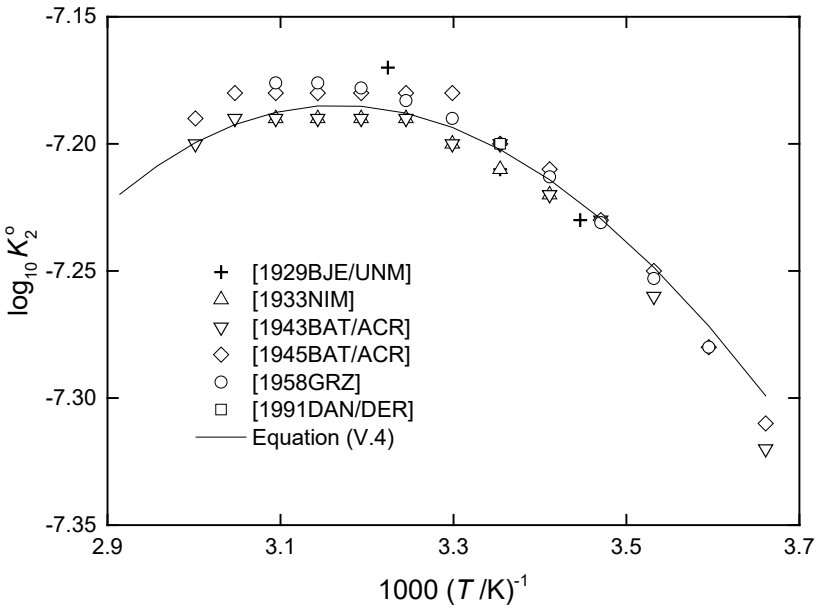
Results from the studies shown in Figure V-7 were fitted by a weighted regression which included a constant $\Delta_r C^\circ_{p,m,2}$ term resulting in:

$$\log_{10} K_2^\circ (\text{V.3}) = (62.86692 \pm 5.83042) - (3274.319 \pm 26.006)(T/\text{K})^{-1} - (10.37051 \pm 0.87008)\ln(T/\text{K}) \quad \text{(V.4)}$$

Figure V-7: Plot of the temperature dependence of $\log_{10} K_2^\circ$ (V.3).



The thermodynamic quantities obtained with this fit at 298.15 K are:

$$\log_{10} K_2^\circ ((\text{V.3}), 298.15\ \text{K}) = -(7.202 \pm 0.003),$$

$$\Delta_r G^\circ_{m,2} ((\text{V.3}), 298.15\ \text{K}) = (41.110 \pm 0.015)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H^\circ_{m,2} ((\text{V.3}), 298.15\ \text{K}) = (3.491 \pm 0.130)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^\circ_{m,2} ((\text{V.3}), 298.15\ \text{K}) = -(126.175 \pm 0.439)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C^\circ_{p,m,2} ((\text{V.3}), 298.15\ \text{K}) = -(198.542 \pm 32.600)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

which agree well with those in [1992GRE/FUG] (CODATA): $\log_{10} K_2^\circ = -(7.212 \pm 0.013)$; $\Delta_r G^\circ_{m,2}$ (V.3) = (41.166 ± 0.074) kJ·mol^{-1}; $\Delta_r H^\circ_{m,2}$ (V.3) = (3.600 ± 1.000) kJ·mol^{-1}; $\Delta_r S^\circ_{m,2}$ (V.3) = – (126.000 ± 3.363) J·K^{-1}·mol^{-1}.

By way of further comparison, Martell and Smith [1982MAR/SMI] reported $\Delta_r H^\circ_{m,2}$ = (3.77 ± 0.42) kJ·mol^{-1}; Latimer *et al.* [1938LAT/PIT] estimated $\Delta_r S^\circ_{m,2}$ = – 126.8 J·K^{-1}·mol^{-1}; and Larsen *et al.* [1982LAR/ZEE] reported – 220 J·K^{-1}·mol^{-1}

for $\Delta_r C^o_{p,m,2}$ based on their flow calorimetric measurements, as well as a value for $\Delta \overline{V}^o_2$ of – 26.1 $cm^3 \cdot mol^{-1}$. Interestingly, if $\Delta_r H^o_{m,2}$ were assumed to be independent of temperature, then $\log_{10} K^o_2 = -(7.212 \pm 0.004)$ at 298.15 K and $\Delta_r H^o_{m,2} = (3.35 \pm 0.25)$ $kJ \cdot mol^{-1}$.

Covington *et al*. [1985COV/BAT] provided two specific combinations of aqueous sodium and potassium phosphates which give reference pH values of 6.865 (0.025 $mol \cdot kg^{-1}$ Na_2HPO_4 + 0.025 $mol \cdot kg^{-1}$ KH_2PO_4) and 7.413 (0.03043 $mol \cdot kg^{-1}$ Na_2HPO_4 + 0.008695 $mol \cdot kg^{-1}$ KH_2PO_4) at 298.15 K. Utilising the SIT based on the value of $\log_{10} K^o_2$ (V.3) = – (7.202 ± 0.003) and the appropriate ion interaction coefficients for Na^+ and K^+ with $H_2PO_4^-$ and HPO_4^{2-} in Table V-7, the reference pH value would be 6.868 and 7.411, respectively, in excellent agreement.

V.1.5 Third dissociation constant of phosphoric acid

$$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-} \qquad \text{(V.5)}$$

Grenthe *et al*. [1992GRE/FUG] selected $\log_{10} K^o_3$ ((V.5), 298.15 K) = – (12.35 ± 0.03) from an average of the values reported in [1929BJE/UNM] and [1961VAN/QUI] that is the same as those recommended by Martell and Smith [1982MAR/SMI], – (12.35 ± 0.02), and later by Goldberg *et al*. [2002GOL/KIS], – 12.35. Other studies now considered are those by Hershey *et al*. [1989HER/FER] who carried out potentiometric measurements at 298.15 K to I_m = 6 (NaCl) and Daniele *et al*. [1991DAN/DER] who performed similar experiments over a narrower range of ionic strength. Application of the SIT treatment to both datasets derived for KCl solutions at 298.15 K yielded the curve shown in Figure V-8. The SIT ion interaction coefficients obtained from the individual fits of the $\log_{10} K_{m,3}$ (V.5) data are given in Table V-12. Other potentiometric measurements at I_m = 3.31 (KCl) were not considered further due to the high phosphate concentrations employed [1974FER/GAL]. Also, the dissociation constants for all three dissociation steps in [1969SAL/HAK] deviated significantly from other reliable studies and were therefore rejected.

Table V-12: Values of $\log_{10} K^o_3$ (V.5) and the corresponding $\Delta\varepsilon$ at 298.15 K from SIT fits (uncertainties are 2σ).

Ionic medium	$\Delta\varepsilon_1$ /$kg \cdot mol^{-1}$	$\Delta\varepsilon_2$ /$kg \cdot mol^{-1}$	$-\log_{10} K^o_3$ SIT	I_m max /$mol \cdot kg^{-1}$	$-\log_{10} K^o_3$ reported	Reference
NaCl	– (0.071 ± 0.017)	(0.124 ± 0.025)	12.35 [(a)]	6.0	12.35 [(a)]	[1989HER/FER]
KCl	(0.132 ± 0.085) [(b)]	(0.39 ± 0.50) [(b)]	(12.34 ± 0.02)	1.03	(12.33 ± 0.08)	[1991DAN/DER]
KCl	(0.234 ± 0.014)	(0.272 ± 0.082)	(12.32 ± 0.01)			
NaCl	– (0.116 ± 0.035) [(b)]	(0.41 ± 0.20) [(b)]	(12.34 ± 0.02)			
NaCl	– (0.094 ± 0.015)	(0.396 ± 0.090)	(12.32 ± 0.01)			
KCl	(0.15 ± 0.02)	– (0.05 ± 0.03)	(12.34 ± 0.02)			This review
NaCl	(0.00 ± 0.02)	(0.06 ± 0.04)	(12.34 ± 0.02)			

(a) This value had to be fixed in the regression.

(b) Obtained over a range of temperatures assuming that the ion interaction parameters are independent of temperature.

Figure V-8: Application of the SIT treatment for the molal third stepwise dissociation constant of phosphoric acid at 298.15 K in NaCl solutions.

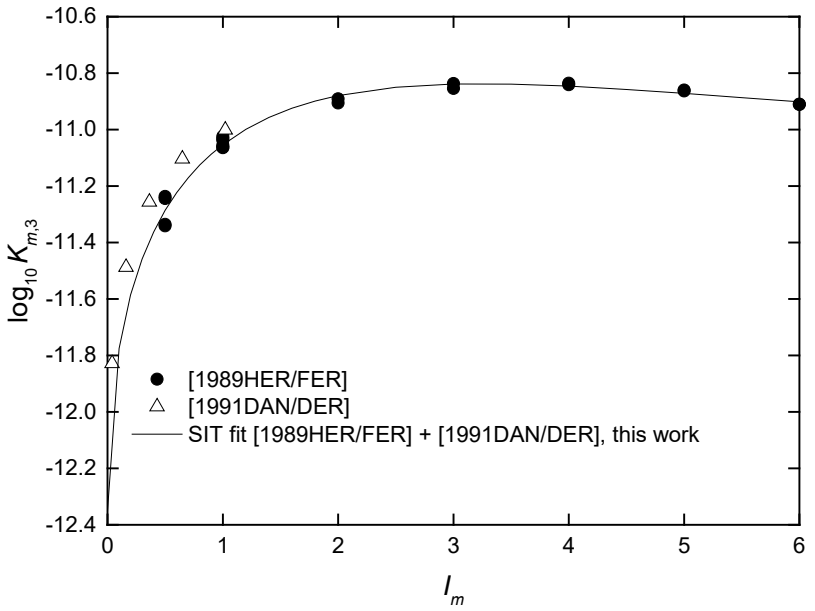


Table V-13 summarises the available values of $\log_{10}K_3^{\circ}$(V.5) and $\Delta_r H_{m,3}^{\circ}$((V.5), 298.15 K) and those accepted for the final assessment with uncertainties imposed by the reviewer. The $\Delta_r H_m^{\circ}$((V.5), 298.15 K) value attributed to Daniele *et al*. of 27 kJ·mol^{-1} was discounted by Goldberg *et al*. [2002GOL/KIS] in favour of (16.0 ± 0.5) kJ·mol^{-1} based on an average of the existing calorimetric data with $\Delta_r C_{p,m}^{\circ}$((V.5), 298.15 K) = – 242 J·K^{-1}·mol^{-1} [1982LAR/ZEE]. Grenthe *et al*. [1992GRE/FUG] accepted the $\Delta_r H_m^{\circ}$((V.5), 298.15 K) value of (14.6 ± 3.8) kJ·mol^{-1} advocated by Martell and Smith [1982MAR/SMI] with this reduced uncertainty, over those given in [1938LAT/PIT] and [1980GHO/GHO]. Pitzer [1937PIT] reported this exact value from his calorimetry experiments with a stated "conservative" uncertainty outweighing the value published soon after [1938LAT/PIT]. Our assessment of the potentiometric data reported from 283.15 to 323.15 K by Ghosh *et al*. [1980GHO/GHO] found them to be very scattered and unreliable because the tabulated hydrolysis constants could not be reproduced.

Table V-13: Summary of available values of $\log_{10}K_3^\circ$ and corresponding thermodynamic quantities for Reaction (V.5) with uncertainties estimated in this review at 2σ.

T /K	$-\log_{10}K_3^\circ$ reported	$\Delta_r H_{m,3}^\circ$ /kJ·mol^{-1} reported	$-\log_{10}K_3^\circ$ retained	$\Delta_r H_{m,3}^\circ$ /kJ·mol^{-1} retained	Method	Reference
291.15	12.465				pot	[1929BJE/UNM]
298.15	12.325		(12.33 ± 0.02)			
310.15	12.180					
298.15		(14.6 ± 4.1)		(14.6 ± 3.8)	cal	[1937PIT]
298.15		16.8 (a)				[1938LAT/PIT]
298.15	(12.375 ± 0.010)		(12.38 ± 0.03)		sp	[1961VAN/QUI]
298.15	(12.39 ± 0.03)	(18.1 ± 0.8)	(12.39 ± 0.20)	(18.1 ± 7.0)	cal	[1965HAN/CHR]
298.15	12.387	18.8			cal	[1965PAP/TOR]
298.15	(12.39 ± 0.03)	(17.57 ± 1.30)	(12.39 ± 0.20)	(17.6 ± 7.0)	cal	[1966CHR/IZA]
298.15		(11.3 ± 1.3)		(11.3 ± 5.0)	cal	[1966IRA/TAU]
298.15	(11.962 ± 0.008)	5.175			pot	[1980GHO/GHO]
298.15		(18.6 ± 0.2) (b)			cal	[1981VAC/SAB]
293.15	(12.33 ± 0.04)	(27 ± 8)	(12.33 ± 0.08)		pot	[1991DAN/DER]
298.15		13.70		(13.7 ± 6.0)	cal	[1994VAS/KOC]
298.15	12.35	16	(12.35 ± 0.03)		est	[2002GOL/KIS]
298.15	(12.31 ± 0.03)		(12.31 ± 0.03)		est	[2003RAP/ZUL]

(a) Derived from the reported value of $\Delta_r S_{m,3}^\circ = -180$ J·K^{-1}·mol^{-1}.
(b) $I_c = 0.15$ (NaCl).

Weighted averages of the retained values in Table V-13 at 298.15 K produced the thermodynamic quantities given in Table V-14 that compare very closely with those in [1992GRE/FUG] so that at the present time the latter can be retained.

Table V-14: Summary of thermodynamic data for Reaction (V.5) at 298.15 K.

$\log_{10}K_3^\circ$	$\Delta_r G_{m,3}^\circ$ /kJ·mol^{-1}	$\Delta_r H_{m,3}^\circ$ /kJ·mol^{-1}	$\Delta_r S_{m,3}^\circ$ /J·K^{-1}·mol^{-1}	Reference
– (12.350 ± 0.030)	(70.494 ± 0.171)	(14.600 ± 3.800)	– (187.470 ± 12.758)	[1992GRE/FUG]
– (12.340 ± 0.016)	(70.437 ± 0.091)	(14.463 ± 3.000)	– (187.739 ± 10.067)	This review

Larsen *et al*. [1982LAR/ZEE] also reported values for $\Delta_r C_{p,m}^\circ$(V.5) = – 242 J·K^{-1}·mol^{-1} and $\Delta\overline{V}_3^\circ$(V.5) = – 24.0 cm^3·mol^{-1} at 298.15 K.

Ion pairing to form KPO_4^{2-} has been reported [1994KOC/DUS] at 298.15 K based on calorimetric data, but ion pairing was not treated specifically in the above SIT analysis of potentiometric results in KCl solutions [1991DAN/DER] that were compatible with those in NaCl solutions.

V.1.6 Dissociation constants of polyphosphoric acids

In the absence of papers dealing with the thermodynamics of dissociation of polyphosphoric acids since the review by Grenthe *et al.* [1992GRE/FUG], their values (see pp. 388-389 of their text) are accepted by this review.

V.1.7 Summary of thermodynamic quantities

The following values are recommended at this time for the mononuclear phosphate species because small differences found between the values determined in this review for all the monophosphate species except H_3PO_4(aq) and those currently recommended by the NEA TDB can be ignored until a comprehensive review is made of the thermodynamics of all the phosphate complexes. The differences in $\Delta_f H_m^\circ(298.15\,\mathrm{K})$ and $S_m^\circ(298.15\,\mathrm{K})$ for H_3PO_4(aq) are larger and the values from this review are preferred.

Table V-15: Summary of thermodynamic quantities retained and selected in this review for the individual aqueous phosphate species.

Species	$\Delta_f G_m^\circ(298.15\,\mathrm{K})$ /kJ·mol^{-1}	$\Delta_f H_m^\circ(298.15\,\mathrm{K})$ /kJ·mol^{-1}	$S_m^\circ(298.15\,\mathrm{K})$ /J·K^{-1}·mol^{-1}	Reference
PO_4^{3-}	**– (1025.491 ± 1.576)**	**– (1284.400 ± 4.085)**	**– (220.970 ± 12.846)**	[1992GRE/FUG]
HPO_4^{2-}	**– (1095.985 ± 1.567)**	**– (1299.000 ± 1.500)**	**– (33.500 ± 1.500)**	[1989COX/WAG]
$H_2PO_4^-$	**– (1137.152 ±1.567)**	**– (1302.600 ± 1.500)**	**(92.500 ± 1.500)**	[1989COX/WAG]
H_3PO_4(aq)	**– (1149.438 ± 1.567)**	**– (1294.756 ± 1.509)**	**(160.016 ± 1.599)**	This review

Chapter V.2

H_2CO_3(aq) and CO_2(aq)

V.2.1 Dissociation constants of carbonic acid

In order to evaluate accurately the magnesium/calcium carbonate solubility data reported for relatively wide temperature (273-573 K), CO_2 pressure (0-40 MPa) and ionic strength (0-6 mol·kg^{-1}) ranges, knowledge of thermodynamic data for carbonic acid dissociation under the same conditions is of fundamental importance. Despite the large number of equilibrium data reported for different background electrolytes, a complete set with T and I_m dependencies is available only for NaCl and KCl solutions (several data are reported for $NaClO_4$ medium, too, but most of these refer only to 298.15 K). The vast majority of data reported for the KCl background electrolyte were determined in [1993HE/MOR]. However, the pK values reported by He and Morse in NaCl solutions deviate considerably from those in other reports at several temperatures. Therefore, none of their data were considered in this review. As a consequence, reliable sets of pK_m values of carbonic acid, including their T and I_m dependencies, are available only for a NaCl background electrolyte.

Speciation models for the (CO_2-H_2O-NaCl) system have been already developed in [2007LI/DUA], [2007MIL/HUA] and [2012VIS/VAN]. The experimental ionic strength dependence of the pK values was not included in the models in [2007LI/DUA] and [2012VIS/VAN], nevertheless, the former was able to predict the ionic strength dependence. Although the ionic strength dependence is considered in [2007MIL/HUA], it is based on the Pitzer formalism. On the other hand, the examples related to carbonic acid pKs discussed by Grenthe *et al.* [1997GRE/PLY2] are very useful to introduce the SIT formalism, but it is not a complete model and many details of the calculations are not included. Therefore, for the purposes of this review, we developed an SIT-based equilibrium model for carbonic acid speciation.

In aqueous solutions of carbon dioxide the following three equilibria should be considered:

$$CO_2(g) \rightleftharpoons CO_2(aq) \tag{V.6}$$

$$CO_2(aq) + H_2O(l) \rightleftharpoons HCO_3^- + H^+ \tag{V.7}$$

$$HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ \tag{V.8}$$

In fact, Equation (V.7) is the sum of two equilibria:

$$CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \quad \text{(V.9)}$$

and

$$H_2CO_3(aq) \rightleftharpoons HCO_3^- + H^+ \quad \text{(V.10)}$$

Since the Equilibrium (V.9) is strongly shifted to the left (K~1/600), only ~ 0.2 % of CO_2(aq) is transformed to H_2CO_3(aq) [1983PAL/ELD]. As a consequence, the equilibrium constant for Reaction (V.7) is considerably smaller than that for Equation (V.10), *i.e.*, than one would expect from the chemical structure of H_2CO_3. For the temperature dependence of the CO_2(g) solubility constant in aqueous solutions Reaction (V.6), we adopted the equation derived in [2012VIS/VAN], which is based on the large dataset compiled by Crovetto [1991CRO], using an equation similar to that applied in [2003FER/ALV].

$$\ln(55.508/p_{1v}K_m) = -9.14122/T_r + 2.8192\tau^{0.355}/T_r + 11.28516T_r^{-0.41}\exp(\tau) - 0.8066$$

where K_m is the solubility constant of CO_2(g) in mol·kg^{-1}·bar^{-1}, p_{1v} is the saturated vapour pressure of water (bar), T_r is the reduced temperature (T_r = (T/K)/(T_c/K)), T_c is the critical temperature of water (647.096 K), and $\tau = (1 - T_r)$.

V.2.2 The dissociation constants of carbonic acid at 298.15 K, and the formation of $NaCO_3^-$ and $NaHCO_3$(aq) ion pairs

Several references report equilibrium data for the $NaCO_3^-$ and $NaHCO_3$(aq) ion pairs [1961GAR/THO], [1962GAR/THO], [1970BUT/HUS], [1970NAK], [1971NAK2], [1974PYT/HAW], [1999CAP/BUC], [2006CRE/STE], [2013STE/BEN], using individual activity coefficients and employing a high-precision sodium ion-selective electrode. Nevertheless, only the existence of $NaCO_3^-$ is supported by dielectric relaxation and Raman spectroscopic data [1999CAP/BUC], [2000SIP/BOL]. The interactions are rather weak (at 298.15 K: $\log_{10}K^\circ(NaCO_3^-) = 1.15$, $\log_{10}K^\circ(NaHCO_3(aq)) = 0.26$ [2006CRE/STE]), so that all these studies are complicated by the inherent difficulties related to the determination of small stability constants ($K < 100$). On the other hand, weak interactions between oppositely charged ions can be incorporated into the ion interaction coefficients. However, the TDB-recommended single (I_m independent) ion interaction parameters $\varepsilon(Na^+, HCO_3^-) = 0.00$ kg·mol^{-1}, $\varepsilon(Na^+, CO_3^{2-}) = -0.08$ kg·mol^{-1} are clearly inappropriate to reproduce the ionic-strength dependence of carbonic acid pKs (where pK_1 = – $\log_{10}K_2$, Eq.(V.7), and pK_2 = – $\log_{10}K_1$, Eq.(V.8)) at 298.15 K (see Figure V-9 and Figure 10 (dashed lines). The non-linear dependence of (pK_1 + 2D + $\log_{10}a_{H_2O}$) and (pK_2 + 4D) values (where D is defined by the Debye-Hückel term, see Chapter B.1) as a

function of ionic strength could relate either to weak ion pairing, or to notable deviation from the widely accepted (and TDB-recommended) $Ba_j = 1.5$ kg$^{1/2}$·mol$^{1/2}$ value in the denominator of the Debye-Hückel term [2005POW/BRO]. On the other hand, the experimental data can be reproduced within the experimental uncertainties by SIT using ionic-strength dependent ($\varepsilon = \varepsilon_1 + \varepsilon_2 \log_{10} I_m$) ion interaction coefficients, *i.e.* ionic-strength dependent $\Delta\varepsilon$ values ($\Delta\varepsilon = \Delta\varepsilon_1 + \Delta\varepsilon_2 \log_{10} I_m$), without the assumption of ion-pair formation. The non-linear regression using the data depicted in Figure V-9 and Figure V-10 (except for those reported in [1993HE/MOR]) resulted in the following $\Delta\varepsilon$ values for Equations (V.7) and (V.8):

$$\Delta\varepsilon_1(\text{V.7}) = -\ (0.017 \pm 0.002)\ \text{kg·mol}^{-1},$$

$$\Delta\varepsilon_2(\text{V.7}) = (0.098 \pm 0.003)\ \text{kg·mol}^{-1},$$

and

$$\Delta\varepsilon_1(\text{V.8}) = -\ (0.035 \pm 0.003)\ \text{kg·mol}^{-1},$$

$$\Delta\varepsilon_2(\text{V.8}) = (0.145 \pm 0.005)\ \text{kg·mol}^{-1}.$$

Considering that $\Delta\varepsilon(\text{V.7}) = \varepsilon(H^+, Cl^-) + \varepsilon(Na^+, HCO_3^-) - \varepsilon(CO_2(aq), Na^+ + Cl^-)$ and $\Delta\varepsilon(\text{V.8}) = \varepsilon(H^+, Cl^-) + \varepsilon(Na^+, CO_3^{2-}) - \varepsilon(Na^+, HCO_3^-)$, and that $\varepsilon(H^+,Cl^-) = 0.12$ kg·mol^{-1} (Table B-4), $\varepsilon(CO_2(aq), Na^+ + Cl^-) = 0.083$ kg·mol^{-1} [1997GRE/PLY2], it is possible to calculate the selected individual ion interaction coefficients:

$$\varepsilon_1(Na^+, HCO_3^-) = \mathbf{-\ (0.054 \pm 0.010)}\ \text{kg·mol}^{-1},$$

$$\varepsilon_2(Na^+, HCO_3^-) = \mathbf{(0.098 \pm 0.010)}\ \text{kg·mol}^{-1},$$

and

$$\varepsilon_1(Na^+, CO_3^{2-}) = \mathbf{-\ (0.209 \pm 0.010)}\ \text{kg·mol}^{-1},$$

$$\varepsilon_2(Na^+, CO_3^{2-}) = \mathbf{(0.243 \pm 0.010)}\ \text{kg·mol}^{-1}.$$

Figure V-9: Ionic-strength dependence of the first dissociation constant of carbonic acid at 298.15 K. The data from [1993HE/MOR] were not considered in the fit.

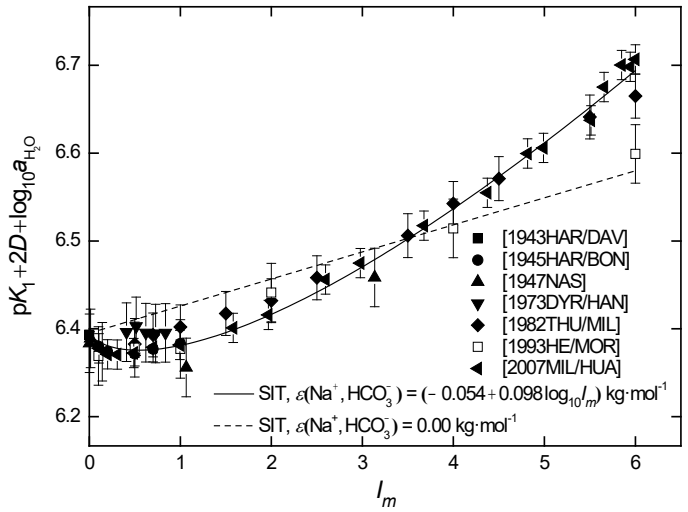
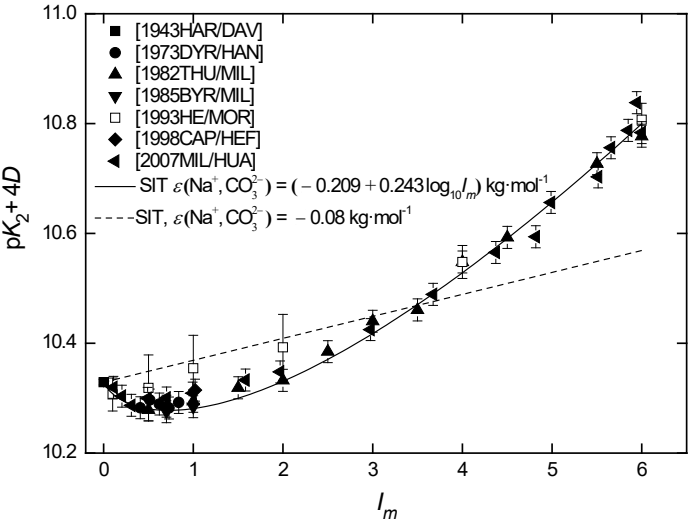


Figure V-10: Ionic-strength dependence of the second dissociation constant of carbonic acid at 298.15 K. The data from [1993HE/MOR] were not considered in the fit.



During this evaluation pK_1^o and pK_2^o were constrained to the NEA TDB recommended values,

$$pK_1^o \text{ ((V.7), 298.15 K)} = (6.354 \pm 0.020),$$

$$pK_2^o \text{ ((V.7), 298.15 K)} = (10.329 \pm 0.020),$$

but their refinement would result only in marginal differences.

In order to verify the validity of the above I_m-dependent ε values, the mean activity coefficients of $NaHCO_3$ and Na_2CO_3 calculated by the SIT are compared to the experimental values. There are two datasets available for the mean activity coefficients of $NaHCO_3$ solutions. Sarbar *et al.* [1982SAR/COV2] applied the isopiestic technique, while Pitzer and Peiper [1980PIT/PEI] used the potential-difference data of Harned *et al.* [1943HAR/DAV], [1945HAR/BON]. The two datasets are somewhat different (Figure V-11), the maximal deviation ($\Delta \log_{10}\gamma_{\pm} \sim 0.04$) is in the $I_m = 0.3$-0.6 region, reflecting the inherent complexity of this system. The mean activity coefficients calculated by the SIT using the above-determined ionic-strength-dependent, ion interaction coefficients lay between the two experimental datasets (Figure V-11, solid line). On the other hand, the TDB recommended ionic-strength independent $\varepsilon(Na^+, HCO_3^-) = 0.00\ kg \cdot mol^{-1}$ value is clearly inappropriate to reproduce the experimental data at $I_m > 0.5$ (dashed line in Figure V-11).

The two independent and reliable datasets for the mean activity coefficients of Na_2CO_3 solutions [1979ROB/MAC] and [1982PEI/PIT] show better agreement (Figure V-12), although Robinson and Macaskili [1979ROB/MAC] did not correct their data for hydrolysis. Considering the whole ionic-strength range ($I_m = 0$-9), the experimentally determined mean activity coefficients can be better predicted using the ionic-strength dependent ion interaction coefficients (solid line in Figure V-12), as compared to the TDB recommended $\varepsilon(Na^+, CO_3^{2-}) = -0.08\ kg \cdot mol^{-1}$ value (dashed line in Figure V-12).

As a consequence, although the existence of $NaCO_3^-$ or $NaHCO_3$(aq) ion pairs in aqueous solutions is likely, some important solution properties (ionic-strength dependence of carbonic acid pK_ms, mean activity coefficients of $NaHCO_3$ and Na_2CO_3 solutions) can be well reproduced by ionic-strength-dependent ion interaction coefficients, without considering ion association.

Figure V-11: $\log_{10}\gamma_{\pm}$ values as a function of $NaHCO_3$ concentration at 298.15 K.

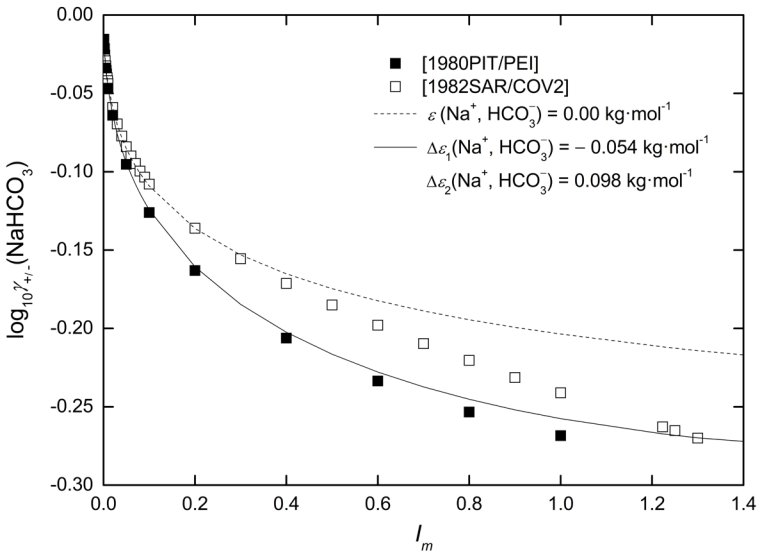
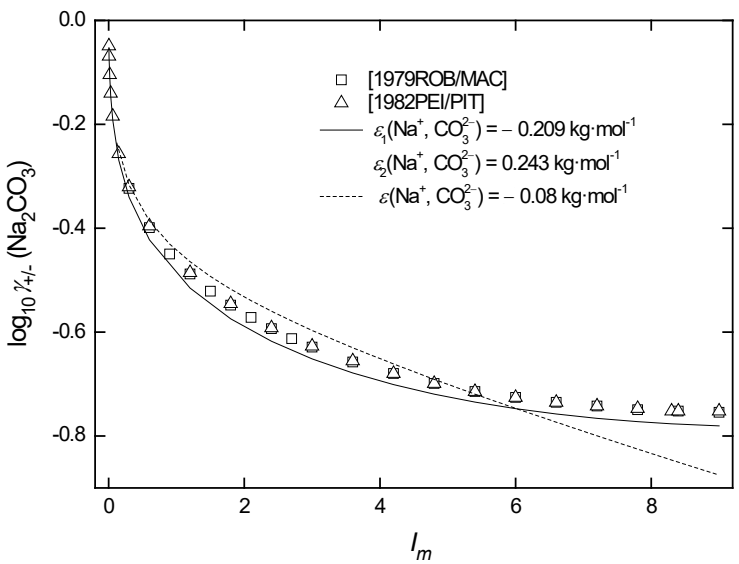


Figure V-12: $\log_{10}\gamma_{\pm}$ values as a function of Na_2CO_3 concentration at 298.15 K.



V.2.3 Temperature and pressure dependencies of the dissociation constants of carbonic acid

As mentioned above, a complete T and I_m dependence of carbonic acid pK values is available only for an NaCl background electrolyte. In this medium 376 data have been reported for pK_1 ((273.15-568.15 K), I_m = 0-6, [1943HAR/DAV], [1945HAR/BON], [1947NAS], [1963RYZ], [1973DYR/HAN], [1975REA2], [1982PAT/SLO], [1982THU/MIL], [1993HE/MOR], [1998PAR/KIM], [2007MIL/HUA], [2013STE/BEN] and [2014STE/BEN]) and 286 for pK_2 (273.15-523.15 K, I_m = 0-6, [1941HAR/SCH], [1947NAS], [1954CUT/STR], [1963RYZ], [1971NAK2], [1973DYR/HAN], [1975REA2], [1982THU/MIL], [1984PAT/BUS], [1993HE/MOR], [1999PAR/KIM], [2007MIL/HUA], [2013STE/BEN] and [2014STE/BEN]). The pK_1 and pK_2 values determined in [2013STE/BEN] and [2014STE/BEN] at identical temperatures agree well with each other, thus only the more comprehensive dataset of [2013STE/BEN] were used in our evaluation.

The non-linear ionic-strength dependence of the data depicted in Figure V-13 and Figure V-14 may indicate the existence of $NaCO_3^-$ or $NaHCO_3$(aq) ion pairs at all temperatures. The experimental data can be well reproduced by the SIT using ionic-strength-dependent $\Delta\varepsilon$ values (due to the lack of TDB approved temperature dependence of $\varepsilon(H^+,Cl^-)$ and $\varepsilon(CO_2(aq), Na^+ + Cl^-) = \log_{10}\gamma_{CO_2}/I_m$, in the first phase of our evaluation only temperature dependent $\Delta\varepsilon$ values, for Equation (V.7) and (V.8), were obtained).

Figure V-13: Ionic-strength dependence of the first dissociation constant of carbonic acid at 273.15 K (A), 323.15 K (B), 373.15 K (C) and 473.15 K (D). The experimental data reported in [1993HE/MOR] were not considered.

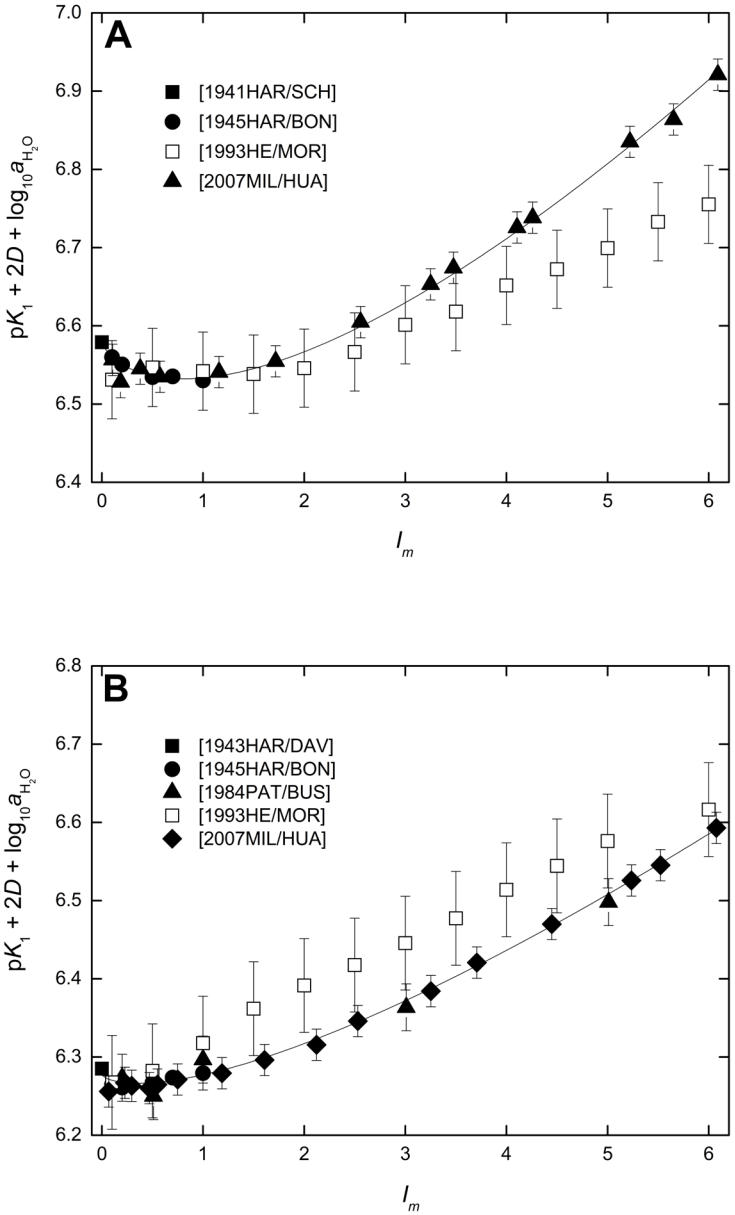
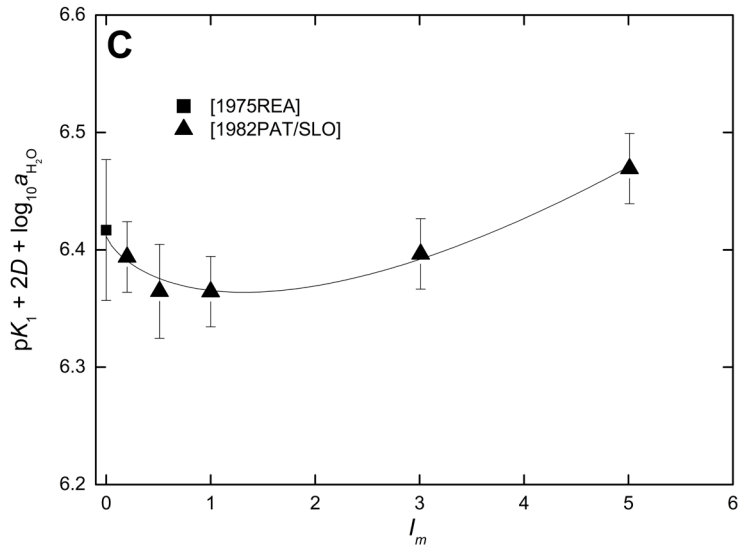


Figure V-13 (cont.): Ionic-strength dependence of the first dissociation constant of carbonic acid at 273.15 K (A), 323.15 K (B), 373.15 K (C) and 473.15 K (D). The experimental data reported in [1993HE/MOR] were not considered



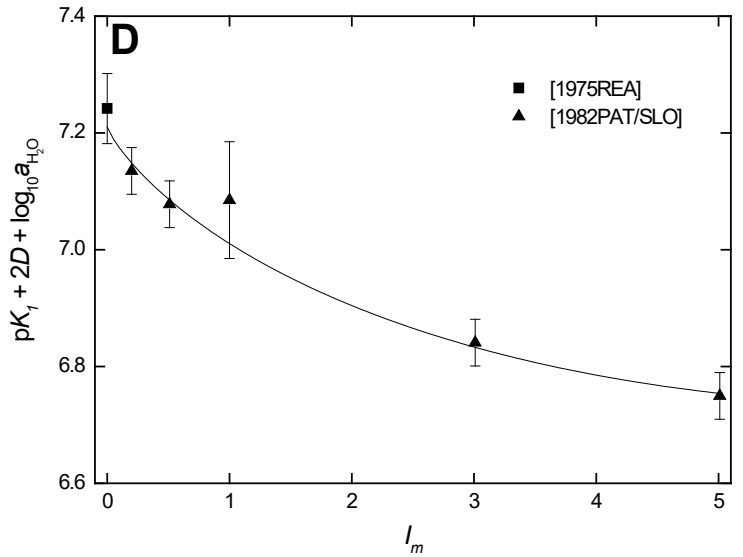


Figure V-14: Ionic-strength dependence of the second dissociation constant of carbonic acid at 273.15 K (A), 323.15 K (B), 373.15 K (C) and 473.15 K (D). The experimental data reported in [1993HE/MOR] were not considered.

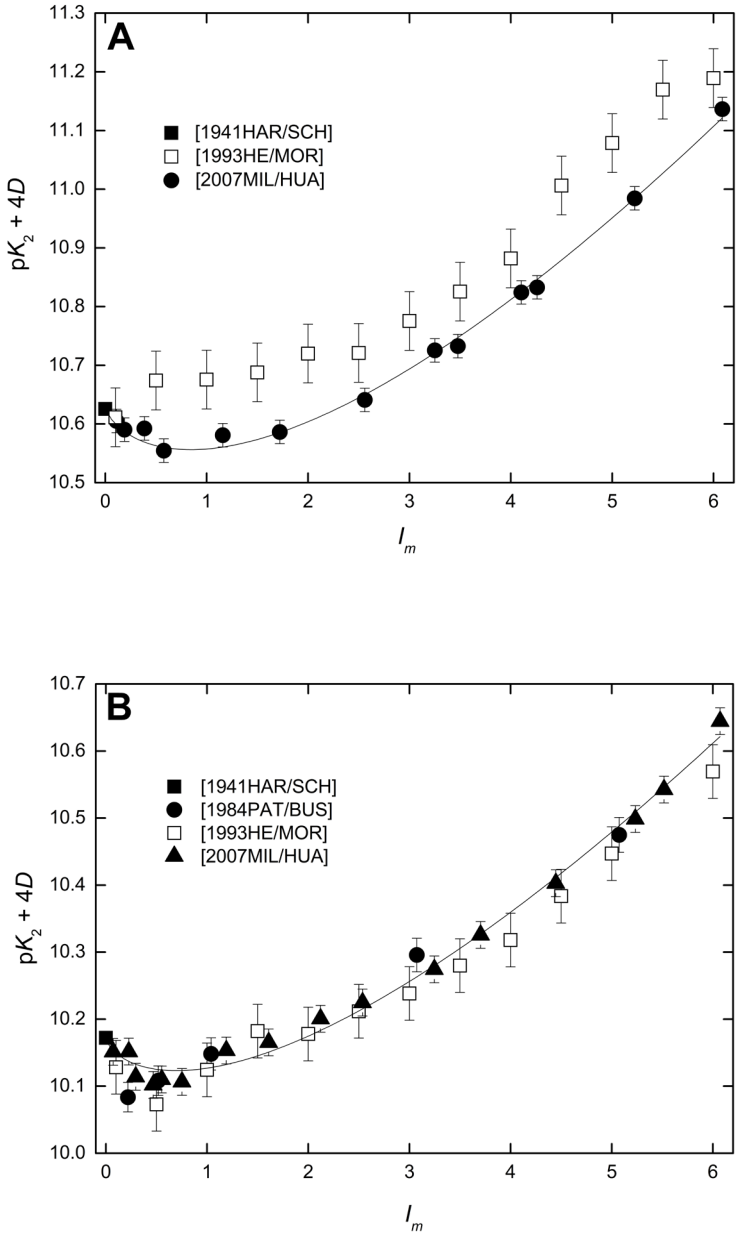
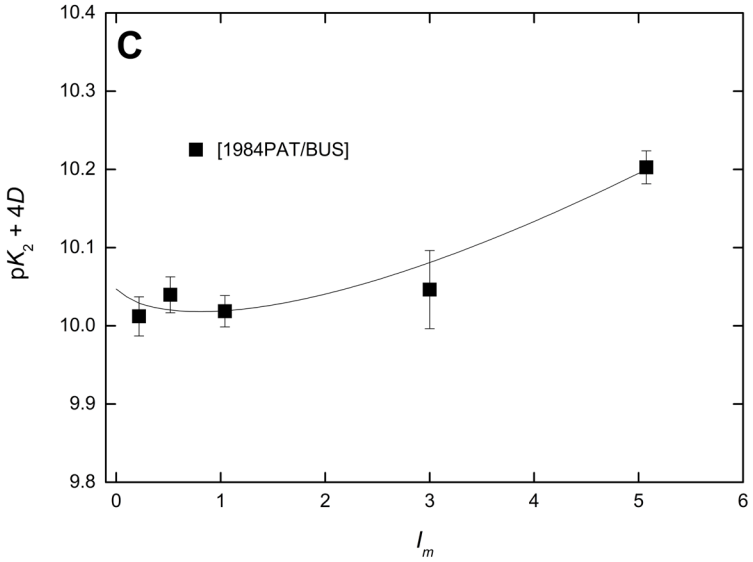
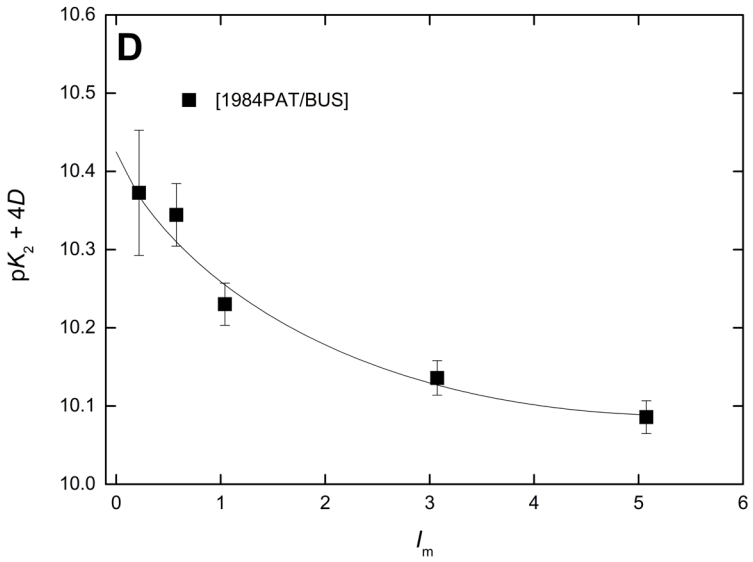


Figure V-14 (cont.): Ionic-strength dependence of the second dissociation constant of carbonic acid at 273.15 K (A), 323.15 K (B), 373.15 K (C) and 473.15 K (D). The experimental data reported in [1993HE/MOR] were not considered.





The thermodynamic dissociation constants (pK_1° and pK_2°) determined from the ionic-strength dependence at different temperatures are depicted in Figure V-15. These figures also include the data reported in [1954CUT/STR], [1963RYZ], [1971NAK2], [1975REA2], [1998PAR/KIM], [1999PAR/KIM] and [2013STE/BEN], since in these papers only thermodynamic constants have been reported. As can be seen, the pK_2° values from different sources are scattered between 373.15 and 523.15 K. In this region only four references are available. Due to the relatively high uncertainties of the data reported in [1963RYZ] and [1999PAR/KIM], these constants were not considered in our compilation (see the appropriate entries in Appendix A). In this way, above 100 °C our evaluation for the temperature dependence of pK_2° is based on the data of [1984PAT/BUS] and [2013STE/BEN]. In [1984PAT/BUS] at each temperature stoichiometric constants have been reported between 0.2-5.0 mol·kg^{-1} NaCl, which were re-evaluated for the purposes of this review by the SIT approach (see *e.g.* Figure V-15 C and D). Since this re-evaluation resulted in lower uncertainties of the thermodynamic constants than those reported in [2013STE/BEN], the fitted curve is closer to the data derived from [1984PAT/BUS]. The experimentally determined pK° values have been fitted by the equation

$$pK^\circ = -(\log_{10} K^\circ) = a + b\,(T/\mathrm{K})^{-1} + c\log_{10}(T/\mathrm{K}) + d\,(T/\mathrm{K}) + e\,(T/\mathrm{K})^{-2} \qquad \text{(V.11)}$$

(see solid lines in Figure V-15), which is the same as that used in [1982PLU/BUS], [2007LI/DUA] and [2012VIS/VAN]. The unconstrained regression (see dashed lines in Figure V-15) resulted in the following parameters:

pK_1° (V.7): a = 322.007, b = – 19334.0, c = – 115.213, d = 0.05823, e = 1.50351 × 10^6;

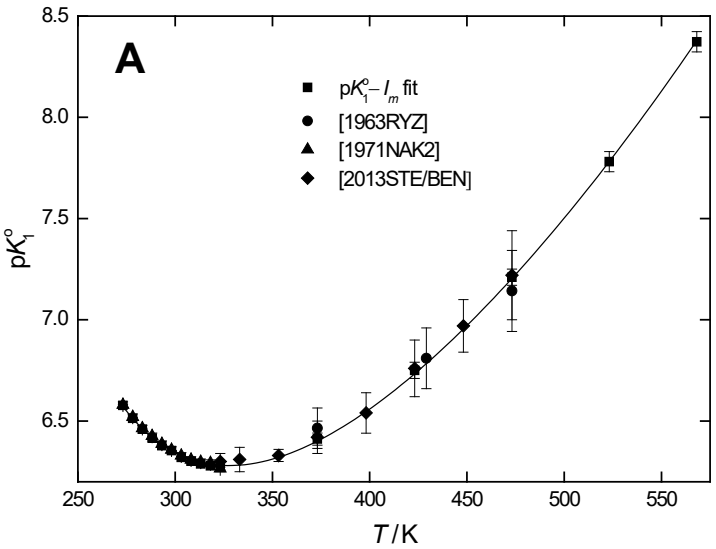
pK_2° (V.8): a = 61.5762, b = – 4191.31, c = – 19.4526, d = 0.01524, e = 569164.

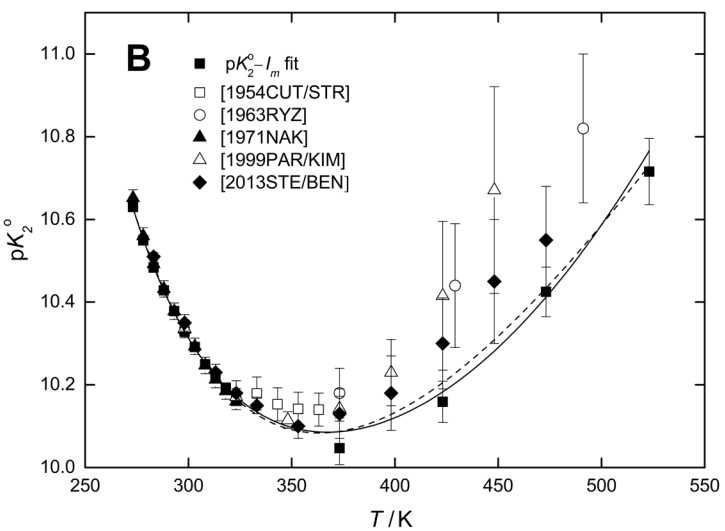
However, the reaction enthalpies for the first and second dissociation reactions of carbonic acid calculated from the above parameters ($\Delta_r H_m^\circ$((V.7), 298.15 K) = 9.451 kJ·mol^{-1} and $\Delta_r H_m^\circ$((V.8), 298.15 K) = 15.139 kJ·mol^{-1}) are somewhat higher than the corresponding CODATA values ($\Delta_r H_m^\circ$((V.7), 298.15 K) = (9.155 ± 0.063) kJ·mol^{-1} and $\Delta_r H_m^\circ$((V.8), 298.15 K) = (14.698 ± 0.105) kJ·mol^{-1}), which are based on the calorimetric data of Berg and Vanderzee [1978BER/VAN], [1978BER/VAN2]. If our data evaluation is constrained to the CODATA (and NEA TDB) recommended reaction enthalpies, somewhat different parameter sets are calculated and accepted (see solid lines in Figure V-15):

$$pK_1^\circ(\mathrm{V.7}) = 291.5972 - 16744.549\,(T/\mathrm{K})^{-1} - 105.2550\log_{10}(T/\mathrm{K}) + 0.056656\,(T/\mathrm{K}) + 1286540/(T/\mathrm{K})^2 \qquad \text{(V.12)}$$

$$pK_2^\circ(\mathrm{V.8}) = 629.9596 - 35481.965\,(T/\mathrm{K})^{-1} - 224.9503\log_{10}(T/\mathrm{K}) + 0.098513\,(T/\mathrm{K}) + 2367190/(T/\mathrm{K})^2 \qquad \text{(V.13)}$$

Figure V-15: Temperature dependence of the pK_1^o (A) and pK_2^o (B) values of carbonic acid. The solid (constrained) and dashed (unconstrained) lines are the best fits using Equation (V.11) with the parameters immediately above. The data represented by open symbols were not considered during the evaluation.





In order to empower the present SIT model, we examined the temperature dependence of the $\Delta\varepsilon_1$ and $\Delta\varepsilon_2$ values (Figure V-16). The experimental $\Delta\varepsilon$ values can be well fitted by the equation

$$\Delta\varepsilon = a + b\,(T/\mathrm{K})^{-1} + c\log_{10}(T/\mathrm{K}) + d\,(T/\mathrm{K}) + e(T/\mathrm{K})^{-2} \qquad \text{(V.14)}$$

(see solid lines in Figure V-16). The non-linear regression using the data depicted in Figure V-16 resulted in the following parameters:

pK_1(V.7), $\Delta\varepsilon_1$: a = 27.95825, b = – 1191.70103, c = – 9.908093, d = 1.82682 × 10^{-3}, e = 0;

pK_1(V.7), $\Delta\varepsilon_2$: a = – 17.64058, b = 803.80771, c = 6.21147, d = – 1.11618 × 10^{-3}, e = 0;

pK_2(V.8), $\Delta\varepsilon_1$: a = – 183.33257, b = 9412.61093, c = 69.8379548, d = – 0.0342612, e = – 522573.997;

pK_2(V.8), $\Delta\varepsilon_2$: a = 803.49636, b = – 42186.72714, c = – 293.412025, d = 0.1273697, e = 2330398.655.

Using Equations (V.12), (V.13) and (V.14) it is now possible to predict both pK_m values of carbonic acid using the SIT at T = 273-600 K and I_m = 0-6 (NaCl) providing a self-consistent dataset for the calculation of carbonic acid pKs. However, the evaluation of magnesium/calcium carbonate solubility data required knowledge of individual $\varepsilon(\mathrm{Na^+}, \mathrm{HCO_3^-})$ and $\varepsilon(\mathrm{Na^+}, \mathrm{CO_3^{2-}})$ values at T = 273-600 K. Therefore, we derived $\varepsilon(\mathrm{Na^+}, \mathrm{HCO_3^-})$ and $\varepsilon(\mathrm{Na^+}, \mathrm{CO_3^{2-}})$ from the above $\Delta\varepsilon$ values at all temperatures, using $\varepsilon(\mathrm{CO_2(aq)}, \mathrm{Na^+} + \mathrm{Cl^-}) = \log_{10}\gamma_{\mathrm{CO_2}}/I_m \sim 2\lambda_{\mathrm{Na^+\text{-}CO_{2(aq)}}} = 2\,(-12.0230 + 576.930(T/\mathrm{K})^{-1} + 1.78993\ln(T/\mathrm{K}))$ [2007MIL/HUA] and $\varepsilon(\mathrm{H^+},\mathrm{Cl^-}) = 0.12 - 0.00045((T/\mathrm{K}) - 298.15)$ [2001PRE/GAM]. The $\varepsilon_{1/2}(\mathrm{Na^+}, \mathrm{HCO_3^-})$ and $\varepsilon_{1/2}(\mathrm{Na^+}, \mathrm{CO_3^{2-}})$ values calculated in this way are depicted in Figure V-17, and were fitted by Equation (V.15)

$$\varepsilon_{1/2} = a + b\,(T/\mathrm{K})^{-1} + c\log_{10}(T/\mathrm{K}) + d\,(T/\mathrm{K}) \qquad \text{(V.15)}$$

(see solid lines in Figure V-17). The non-linear regression resulted in the following parameters:

$\varepsilon_1(\mathrm{Na^+}, \mathrm{HCO_3^-})$: a = 21.6440, b = – 820.238, c = – 8.04727, d = 3.30350 × 10^{-3};

$\varepsilon_2(\mathrm{Na^+}, \mathrm{HCO_3^-})$: a = – 21.7592, b = 927.862, c = 7.81874, d = – 2.03733 × 10^{-3};

$\varepsilon_1(\mathrm{Na^+}, \mathrm{CO_3^{2-}})$: a = – 47.87072, b = 884.70998, c = 20.11645, d = 0.0170407;

$\varepsilon_2(\mathrm{Na^+}, \mathrm{CO_3^{2-}})$: a = 127.34221, b = – 3180.60697, c = – 51.45016, d = 0.0365654.

Figure V-16: Temperature dependence of the $\Delta\varepsilon_1$ and $\Delta\varepsilon_2$ values related to the first (A) and second (B) dissociation constants of carbonic acid.

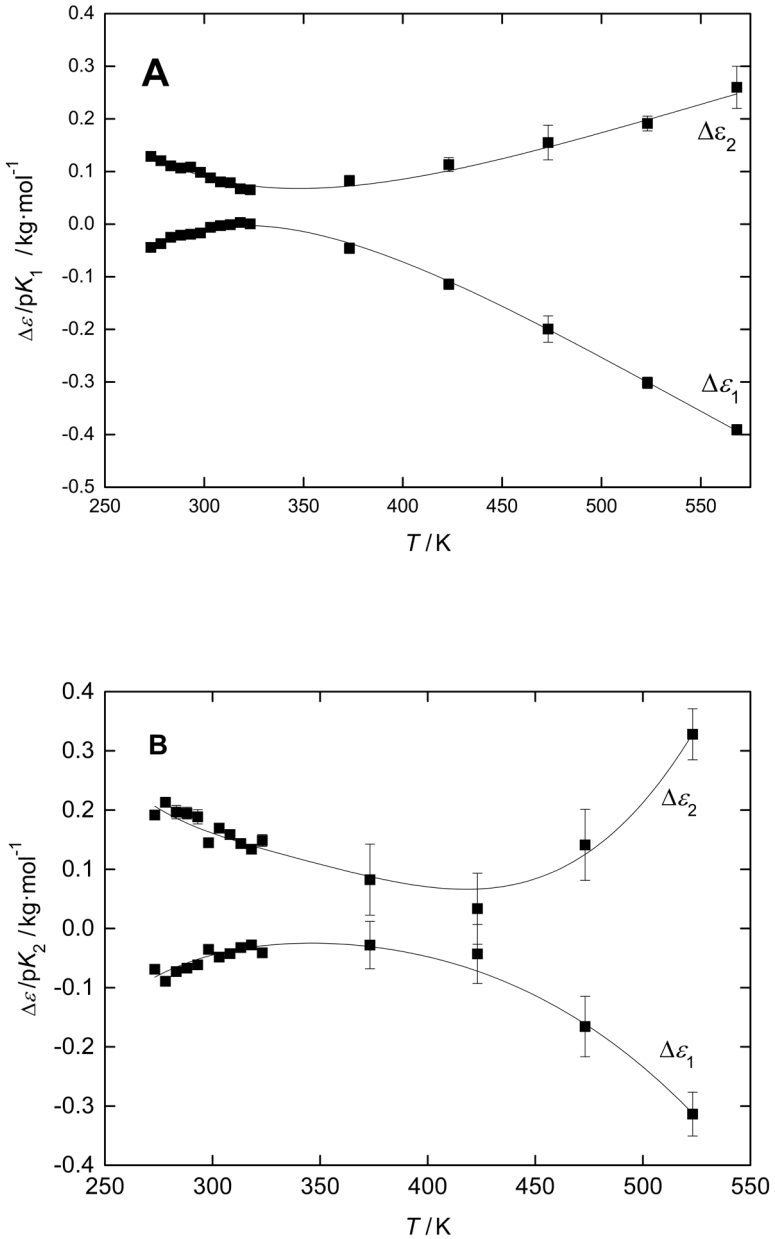
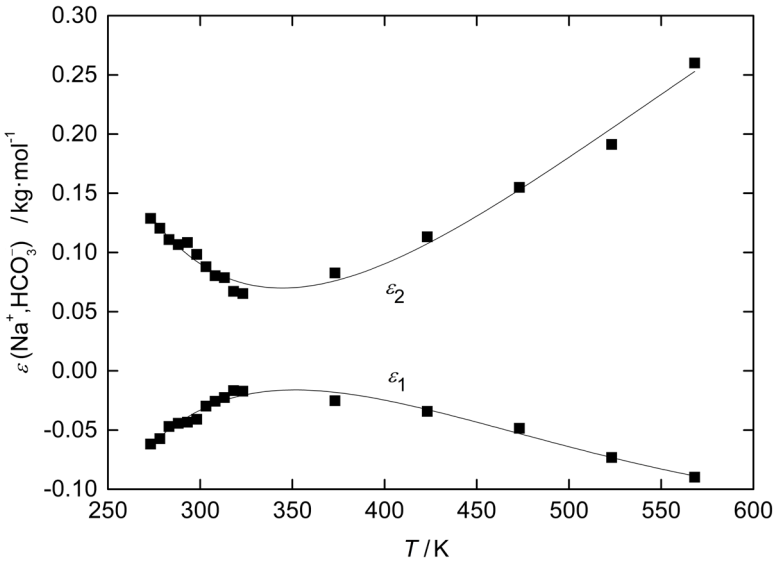
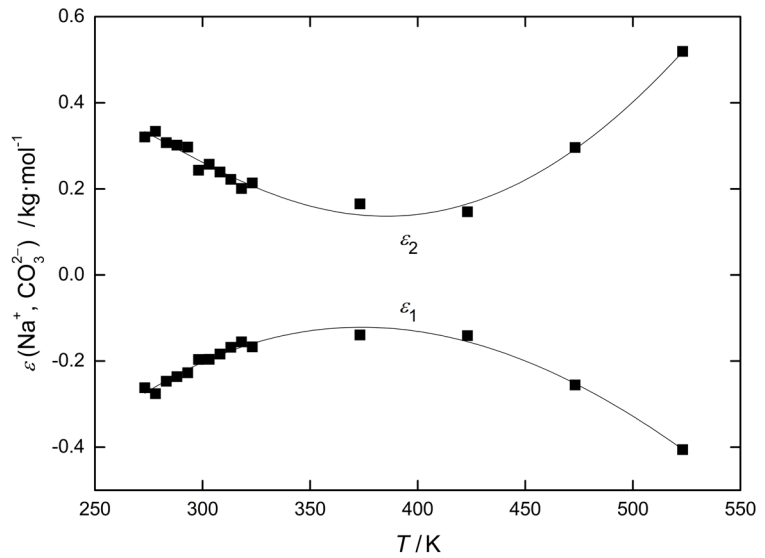


Figure V-17: Temperature dependence of $\varepsilon_{1/2}(Na^+, HCO_3^-)$ and $\varepsilon_{1/2}(Na^+, CO_3^{2-})$.





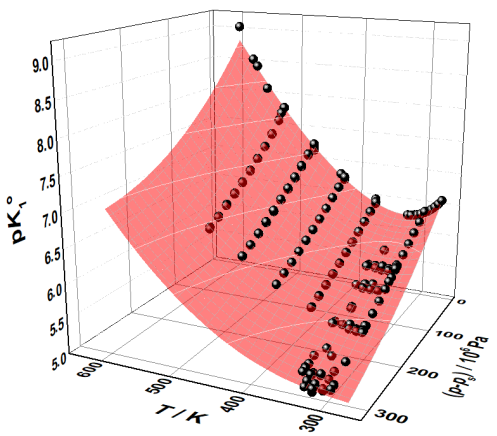
Since Ellis [1959ELL2], Read [1975REA2] and Che *et al.* [1992CHE/GIL] reported the temperature and pressure dependencies of the first dissociation constant of carbonic acid over relatively wide ranges, it is possible to expand Equation (V.11) with a pressure-dependent component [2007LI/DUA] valid up to 300 MPa

$$\begin{aligned}\log_{10}K_1^{\circ}\,(\text{V.6}) = {} & a + b\,(T/\text{K})^{-1} + c\log_{10}(T/\text{K})^{-1} + d\,(T/\text{K}) + e\,(T/\text{K})^{-2} + \\ & (f/(T/\text{K}) + g\,(T/\text{K})^{-2} + h\log_{10}(T/\text{K})/(T_c/\text{K}))\,((p-p_s)/\text{MPa}) + \\ & (i\,(T/\text{K})^{-1} + j\,(T/\text{K})^{-2} + k\log_{10}(T/\text{K})/(T_c/\text{K}))\,((p-p_s)/\text{MPa})^2 \qquad \text{(V.16)}\end{aligned}$$

where p is the actual pressure in MPa, $p_s = 0.1$ MPa below 373.15 K and p_s the saturation pressure of water in MPa at temperatures above 373.15 K. The parameters *a-e* have been already determined (see above the constrained fit), thus only the parameters *f-k* were optimised using the dataset depicted in Figure V-18. The non-linear regression resulted in the following constants: $f = -22.2506$, $g = 169.84087$, $h = 0.01314$, $i = 0.05413$, $j = -9.15731$, $k = -1.27712 \times 10^{-4}$.

Apparently, no data are available for the pressure dependence of $\text{p}K_2^{\circ}$.

Figure V-18: Temperature and pressure dependencies of $\text{p}K_1^{\circ}$ of carbonic acid. Black spheres are the thermodynamic protonation constants determined from the ionic-strength dependence (see also Figure V-15) and the experimental data from [1959ELL2], [1975REA2] and [1992CHE/GIL], the fitted surface is in red.



Chapter V.3

SiO_2(s) and SiO_2(aq) species

V.3.1 Silica polymorphs

V.3.1.1 Introduction

There is a large number of polymorphs of silica—Vieillard [1986VIE] lists 11, many of which exist in several closely related structures. However, it is not clear if all of these are pure SiO_2 phases. Until the 1950's, tridymite was considered to be the stable phase of silica at 0.1 MPa from 1143 to 1743 K, as suggested in the pioneering study of the phase relationships by Fenner [1913FEN]. However, as discussed in detail in Section V.3.1.4, studies by Flörke [1955FLO], [1956FLO] and by Holmquist [1958HOL2], [1961HOL] have thrown considerable doubt on this.

In this review, we shall consider only quartz, cristobalite, tridymite and vitreous silica in detail, with a short note on the amorphous phase. Each of the first four phases has second-order transitions but, for convenience, these transitions are treated here as first-order with small enthalpies of transition.

V.3.1.2 SiO_2(quar)

Quartz is the polymorph stable *ca.* to 1200 K, and exists in two very closely related hexagonal structures: α-quartz is stable up to *ca.* 847 K, where it undergoes a second-order transition to the β form.

There is, of course, extensive literature on the structure, transformation and thermodynamic data for quartz; only the more important and reliable studies can be quoted here.

V.3.1.2.1 Structures

There is a voluminous literature on the detailed structures and their interrelationships, the details of which do not concern this review. We have taken the structural data from the review by Schnurre *et al.* [2004SCH/GRO]:

SiO_2(α-quar), hexagonal cell, space group $P3_221$,
a = 4.912 Å, c = 5.402 Å at 299 K;

SiO_2 (β-quar), hexagonal cell, space group $P6_221$,
a = 5.020 Å, c = 5.523 Å, at 873 K.

V.3.1.2.2 SiO_2 (α-quar)

V.3.1.2.2.1 Heat capacity and entropy

There is very good agreement in the low-temperature data for α-quartz, except for the early, less accurate data of [1936AND], as shown in Table V-16. The low-temperature heat capacities accepted in [1989GRO/STO], JANAF [1985CHA/DAV], [1989COX/WAG] and [1991GUR/VEY] (the latter two are identical), are based primarily on data by Westrum, supported by the consistent data of Gurevich and Khlyustov [1979GUR/KHL]. Unfortunately, Westrum's detailed experimental data have not been published, although a related paper by Westrum on the heat capacities of two samples of irradiated quartz [1956WES] does give a coarse graph of these unirradiated data; a similar graph is also given in the paper by Leadbetter and Morrison [1963LEA/MOR] examining the very low-temperature specific heats of the silica phases. Moreover, the values of Westrum's smoothed data at rounded temperatures from 10 to 700 K can be calculated from Table I in the paper by Lord and Morrow [1957LOR/MOR], which gives the difference between their calculated values and Westrum's values.

Table V-16 shows the relevant experimental values of $C^{\circ}_{p,\mathrm{m}}$(298.15 K) and S°_{m}(298.15 K), including those from Westrum's study as quoted by various authors. We see that these are very consistent apart from the early, less accurate, measurements by Anderson [1936AND].

Since, as noted below, we are accepting the high temperature $C^{\circ}_{p,\mathrm{m}}$ values assessed in [1989GRO/STO], we have also selected their assessed values for $C^{\circ}_{p,\mathrm{m}}$ (with a minor change, see below) and S°_{m} at 298.15 K with conservative uncertainty limits; the latter is identical with the current TDB/CODATA value, from [1989COX/WAG].

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{SiO_2},\ \alpha\text{-quar},\ 298.15\ \mathrm{K}) = \mathbf{(44.50 \pm 0.20)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{SiO_2},\ \alpha\text{-quar},\ 298.15\ \mathrm{K}) = \mathbf{(41.460 \pm 0.200)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

Table V-16: Data for SiO_2(α-quar) at 298.15 K.

Reference	$C^{\circ}_{p,\mathrm{m}}$(298.15 K) /J·K^{-1}·mol^{-1}	S°_{m}(298.15 K) /J·K^{-1}·mol^{-1}
Values based on Westrum's study		
JANAF[1985CHA/DAV]	44.589	(41.463 ± 0.130)
[1989COX/WAG]	(44.602 ± 0.300)	(41.460 ± 0.200)
[1989GRO/STO]	44.50	41.46
[1991GUR/VEY]	44.602	(41.460 ± 0.200)

(Continued on next page)

Table V-16 (continued)

Reference	$C^{o}_{p,m}$(298.15 K) /J·K^{-1}·mol^{-1}	S^{o}_{m}(298.15 K) /J·K^{-1}·mol^{-1}
	Other measurements	
[1936AND]	(44.4 ± 0.3) [(a)]	(42.09 ± 0.13)
[1979GUR/KHL]	(44.43 ± 0.03)	(41.43 ± 0.02)

(a) Extrapolated value, data extend only to 296.3 K

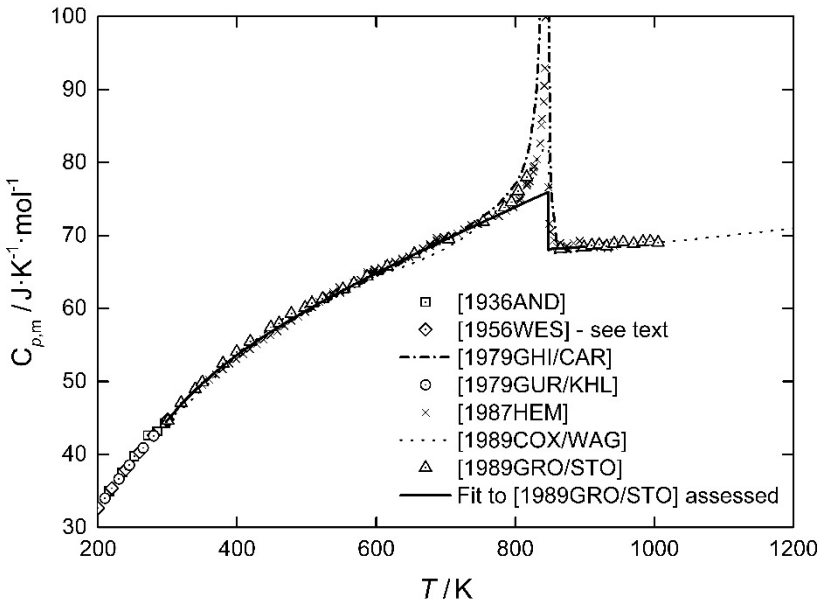
The three later detailed studies of the high temperature heat capacity in [1979GHI/CAR], [1987HEM] and [1989GRO/STO] (see the quite full Appendix A entries for these) show fairly clearly that the related assessments in [1989COX/WAG] and [1991GUR/VEY] do not give an entirely satisfactory description of the $C^{o}_{p,m}$ of α-quartz between 600 and 750 K. (Note incidentally that the $C^{o}_{p,m}$ coefficients for α-quartz given in Table 16.1 in [1991GUR/VEY] are those for the slightly different $C^{o}_{p,m}$ values given in the original Russian version of this assessment [1979GLU/GUR]). The excellent paper by Grønvold *et al.* [1989GRO/STO] gives not only detailed $C^{o}_{p,m}$ measurements, but also an equally detailed and thorough assessment of all the experimental heat capacity and enthalpy data for both α- and β-quartz. Since we cannot improve on their assessment, and the $C^{o}_{p,m}$ measurements in the more recent study by Akaogi *et al.* [1995AKA/YUS] (from (250-675) K, see their Fig. 1) are in good agreement with those assessed by Grønvold *et al.* [1989GRO/STO], these latter values have been accepted by this review.

Figure V-19 shows the differences more clearly; the later values in [1995AKA/YUS] were only reported graphically, so are not included in the figure.

The heat capacity data up to 780 K, just below the transition point (see below) have been fitted by the normal four-term equation, with a constraint of $C^{o}_{p,m}$ (218.15 K) = 44.50 J·K^{-1}·mol^{-1}, to give:

$$C^{o}_{p,m}(SiO_2, \alpha\text{-quar}, T)/\text{J·K}^{-1}\text{·mol}^{-1} = \mathbf{46.53598} + \mathbf{3.32117 \times 10^{-2}}(T/\text{K}) + \mathbf{3.88114 \times 10^{-6}}(T/\text{K})^2 - \mathbf{1.091883 \times 10^{6}}(T/\text{K})^{-2} \quad (298.15\text{-}780\,\text{K}).$$

Figure V-19: The heat capacity of SiO_2(quar).



V.3.1.2.2.2 Transition temperature

As noted, the transition from α- to β-quartz is second-order, the heat capacity increasing to very high values between 847 and 848 K ([1979GHI/CAR], [1987HEM], [1989GRO/STO]). However, in common with other assessments, we shall represent it by a first-order transition, the high heat capacities being taken into account by introducing a small enthalpy of transition. It is clear that the precise transition temperature depends slightly on the purity, crystallite size and thermal history of the sample used; Grønvold *et al.* [1989GRO/STO] give a table of 20 measurements, all of which, except the earliest, lie within the range (847 ± 3) K. As for other data, we have accepted the value from their thorough analysis with 847.3 K for T_{trs}. The adjusted enthalpy of transition is considered in the next section.

V.3.1.2.2.3 Enthalpy of formation

The recent literature does not reveal any information that would lead to a revision of the value for the standard enthalpy of formation of α-quartz selected by CODATA [1989COX/WAG], based on two sets of concordant data, namely:

- the difference between the enthalpies of reaction of Si(cr) and SiO_2(α-quar) with F_2(g) by Wise *et al.* [1963WIS/MAR], leading

these authors to $-(910.94 \pm 1.42)$ kJ·mol^{-1} while CODATA quote $\Delta_f H^\circ_m$ (SiO_2, α-quar, 298.15 K) = $-(911.1 \pm 1.4)$ kJ·mol^{-1} from this source.

- the enthalpy of combustion by Good *et al.* [1964GOO/LAC] of Si(cr) in an oxygen-bomb calorimeter in the presence of aqueous HF (see the relevant Appendix A entry for details of the technique), which, combined with the later measurements by Kilday and Prosen [1973KIL/PRO] on the enthalpy of solution of SiO_2(α-quar) in aqueous HF, yield $\Delta_f H^\circ_m$ (SiO_2, α-quar, 298.15 K) = $-(910.4 \pm 1.4)$ kJ·mol^{-1}.

  The value adopted by CODATA, [1989COX/WAG], $\Delta_f H^\circ_m$ (SiO_2, α-quar, 298.15 K) = $-(910.7 \pm 1.0)$ kJ·mol^{-1}, is identical to that adopted in [1991GUR/VEY]. Therefore, this review maintains and selects

$$\Delta_f H^\circ_m (SiO_2, \text{α-quar, 298.15 K}) = \mathbf{-(910.7 \pm 1.0)}\ \text{kJ·mol}^{-1}.$$

V.3.1.2.3 SiO_2(β-quar)

V.3.1.2.3.1 Heat capacity and entropy

The heat capacities for β-quartz are also taken from the comprehensive assessment in [1989GRO/STO], based on four heat capacity studies (including their own) and nine enthalpy-drop measurements; the data from three other more discordant enthalpy measurements were rejected. Their values from 860 to 1650 K have been fitted by the four-term expression:

$$C^\circ_{p,m}(SiO_2, \beta\text{-quar}, T)/\text{J·K}^{-1}\text{·mol}^{-1} = \mathbf{74.89141 - 1.037879 \times 10^{-2}}\,(T/\text{K}) + \mathbf{6.375226 \times 10^{-6}}\,(T/\text{K})^2 - \mathbf{1.819987 \times 10^{6}}\,(T/\text{K})^{-2} \qquad (860\text{-}1650\,\text{K}).$$

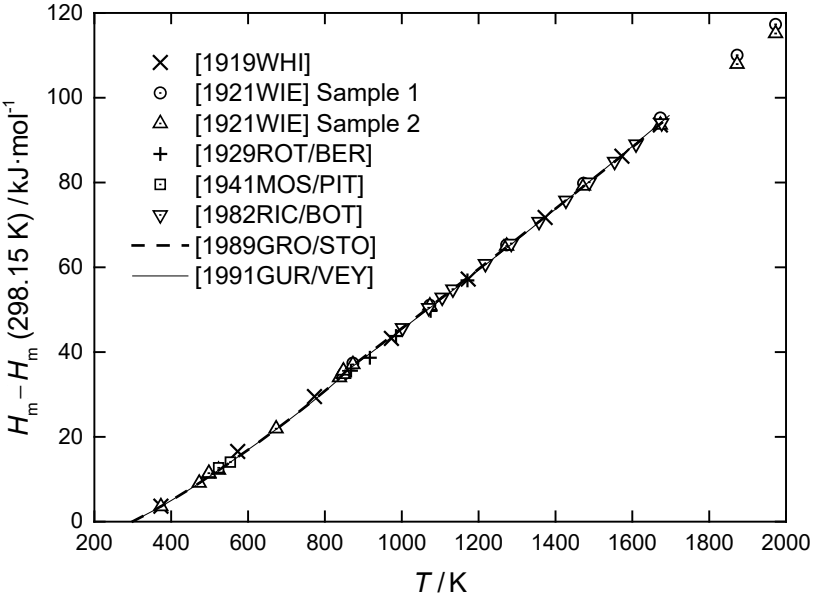
As noted, the second-order transition will, for convenience, be modelled as a first-order transition at 847.3 K, by assuming the $C^\circ_{p,m}$ equations for α- and β-quartz extend up to and from 847.3 K, and introducing an artificial enthalpy of transition of $\Delta_{trs} H_m$ = 742 J·mol^{-1}, the difference between the values of (H(847.3 K) – H(298.15 K)) calculated from the two fits of the $C^\circ_{p,m}$ and enthalpy equations.

A comparison of the enthalpy data for quartz is shown in Figure V-20.

This approximation implies there will be small differences in the Gibbs energy function $(G^\circ_m - H^\circ_m(298.15\,\text{K}))/T$ between the tabulated data in [1989GRO/STO] and our representation between 760 and 860 K, but these are quite negligible (less than 0.015 J·K^{-1}·mol^{-1}), compared to the overall uncertainty of 0.2 J·K^{-1}·mol^{-1} in S°_m (298.15 K) and thus in $(G^\circ_m - H^\circ_m(298.15\,\text{K}))/T$.

These data correspond to $H_m(SiO_2, \beta\text{-quar}, 1000\,\text{K}) - H_m(SiO_2, \alpha\text{-quar}, 298.15\,\text{K}) = 45.598$ kJ·mol^{-1}, in excellent agreement with the consensus value of experimental values (45.5 ± 0.2) kJ·mol^{-1} given by Hosieni *et al.* [1985HOS/HOW].

Figure V-20: Enthalpy data for SiO_2(quar).



V.3.1.3 SiO_2(cris)

Cristobalite is the stable polymorph crystallising from the melt, at *ca.* 2000 K. It exists in two closely related structures with a second-order transition from the tetragonal α form to the cubic β form at *ca.* 540 K. The relative stabilities of cristobalite and tridymite at ambient pressure in the temperature range from *ca.* 1140 K to *ca.* 1740 K is still not established unambiguously (see the discussion under tridymite in Section V.3.1.4). However as noted there, in this review, tridymite will be taken to be metastable in this region, so that cristobalite will be taken to be stable from 1309 K (see below) to the melting point. As for quartz, only the more important and reliable of the experimental data and assessments are quoted here.

V.3.1.3.1 Structures

We have taken the structural data from the review in [2004SCH/GRO]:

SiO_2(α-cris), tetragonal cell, space group $P4_12_12$,
a = 4.9877 Å, c = 6.9679 Å, at 473 K;

SiO_2(β-cris), cubic cell, space group $Fd\bar{3}m$,
a = 7.166 Å, at 583 K.

V.3.1.3.2 SiO_2(α-cris)

V.3.1.3.2.1 Heat capacity and entropy

The low-temperature data for α-cristobalite, are taken from unpublished data by Westrum, as quoted in [1982RIC/BOT], who deduced these values from the values of $C^{\circ}_{p,\mathrm{m}}$(α-quar) and the difference of ($C^{\circ}_{p,\mathrm{m}}$(α-cris) – $C^{\circ}_{p,\mathrm{m}}$(α-quar)) plotted in [1956WES]. The assessments by Gurvich *et al.* [1991GUR/VEY] and JANAF [1985CHA/DAV], who may have had access to Westrum's detailed data, give slightly larger values. The early, less accurate and less complete $C^{\circ}_{p,\mathrm{m}}$ data in [1936AND] (23 measurements from 55 to 297 K) give slightly lower values. The uncertainty in $C^{\circ}_{p,\mathrm{m}}$ is estimated by this review. This review selects the following heat capacity and entropy values:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{SiO_2}, \text{α-cris}, 298.15\ \mathrm{K}) = \mathbf{(44.885 \pm 0.300)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{SiO_2}, \text{α-cris}, 298.15\ \mathrm{K}) = \mathbf{(43.363 \pm 0.150)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

The data for the high temperature $C^{\circ}_{p,\mathrm{m}}$ of cristobalite are not well-defined, since there seem to be notable inconsistencies between the enthalpy-drop measurements and the heat capacity measurements. The selected values in [1991GUR/VEY] seem to be based solely on the enthalpy data of Mosesman and Pitzer [1941MOS/PIT] and the resultant heat capacity has a very shallow maximum at 404 K followed by a shallow minimum at 415 K. Instead, for α-cristobalite, we have accepted the heat capacity values selected in the thorough assessment by Richet *et al.* [1982RIC/BOT], which are based on the data of White [1919WHI] (373-1673 K), Westrum [1956WES] (260-300 K) and Leadbetter and Smith**,** [1976LEA/SMI] (316-653 K). The data of Wietzel [1921WIE] (373-523 K) and Mosesman and Pitzer [1941MOS/PIT] (367-494 K) diverge from the selected values by 2.7 % and 0.7 % respectively and were not included in the fitting.

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{SiO_2}, \text{α-cris}, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{33.350} + \mathbf{5.9066 \times 10^{-2}}(T/\mathrm{K}) - \mathbf{5.4004 \times 10^{5}}(T/\mathrm{K})^{-2} \quad (298.15\text{-}540\ \mathrm{K}).$$

V.3.1.3.2.2 Transition temperature

The α-β transition temperature of cristobalite is not well-defined, since it is clearly dependent of the type of complicated ordering in the crystal, as shown by Hill and Roy [1958HIL/ROY]. Richet *et al.* [1982RIC/BOT] give a table of numerous experimental inversion temperatures varying from 498 K measured on cooling and 544 K upon heating. We have accepted the value of (540 ± 15) K given in [1958HIL/ROY] (consistent with the sharp DSC peak at 535 K found by Thompson and Wennemer [1979THO/WEN]) for the most stable form of α-cristobalite (but with a much larger uncertainty), rather than the lower temperature of 525 K chosen in [1982RIC/BOT].

As for quartz, the second-order transition will, for convenience, be modelled as a first-order transition at 540 K, and introducing an artificial enthalpy of transition of $\Delta_{trs}H_m = 1.280\ kJ{\cdot}mol^{-1}$, the difference between the values of (H_m(540 K) – H_m(298.15 K)) calculated from the integrated forms of the two $C^o_{p,m}$ equations.

V.3.1.3.3 SiO_2(β-cris)

The heat capacities for β-cristobalite are also taken from the comprehensive assessment in [1982RIC/BOT], based on the heat capacity study by Leadbetter and Smith [1976LEA/SMI], (532-653) K, and the enthalpy-drop studies by White [1919WHI] (573-1673) K, and their own enthalpy data (1008-1834) K; the enthalpy data by Wietzel [1921WIE], and Mosesman and Pitzer [1941MOS/PIT] were given zero weight; they differ from the selected values by less than 2.1 and 1.3 % respectively.

$$C^o_{p,m}(SiO_2, \beta\text{-cris}, T)/J{\cdot}K^{-1}{\cdot}mol^{-1} = \mathbf{66.930} + \mathbf{4.8429 \times 10^{-3}}(T/K) - \mathbf{2.5419 \times 10^{6}}(T/K)^{-2}$$
(540-2000 K).

V.3.1.3.4 Enthalpy of formation

The enthalpy of formation of α-cristobalite has been obtained from the difference between the enthalpy of solution of α-quartz and α-cristobalite in aqueous hydrofluoric acid and in high temperature melts. A summary of the literature data is given in Table V-17.

Table V-17: Literature values for the enthalpy of the reaction SiO_2(α-quar) ⇌ SiO_2(α-cris), from the differences in enthalpies of dissolution.

Reference	Solvent and temperature	$\Delta_r H^o_m$ (298.15 K) /kJ·mol^{-1}	Comments
[1921WIE]	35 % HF, room T	(7.24 ± 0.30)	Uncertainty covering only the scatter of the results, as estimated by this review
[1949ROT/TRO]	20.6 % HF, 350 K	(7.15 ± 0.25) at 350 K	As reported
		(7.15 ± 0.25) at 298.15 K	Correction by this review to 298.15 K is 0.003 kJ·mol^{-1}
[1952HUM/KIN2]	20.1 % HF, 346.85 K	(3.89 ± 0.38)	Correction to 298.15 K is negligible, see above
[1953KRA]	20 % HF, 347.85 K	(2.64 ± 0.63)	Correction to 298.15 K is negligible, see above. Uncertainty estimated by this review
[1959HUM/SCH]	39 % HF, 299.65 K	(7.53 ± 0.96)	Correction to 298.15 K is negligible
[1967HOL/KLE]	High temperature melt [(a)], 970 K	(1.9 ± 0.6) at 970 K	As reported
		(2.7 ± 0.6) at 298.15 K	Correction by this review to 298.15 K is 0.813 kJ·mol^{-1}

(Continued on next page)

Table V-17 (continued)

Reference	Solvent and temperature	$\Delta_r H_m^o$ (298.15 K) /kJ·mol^{-1}	Comments
[1980NAV/HON]	High temperature melt [(b)], 985 K	(0.91 ± 0.18) at 985 K	As reported
		(1.73 ± 0.30) at 298.15 K	Correction by this review to 298.15 K is 0.819 kJ·mol^{-1}

(a) Approximate composition: $9PbO \cdot 3CdO \cdot 4B_2O_3$
(b) $2PbO \cdot B_2O_3$ melt

There is an Appendix A entry for each of the references listed in this table, in which in particular, the limitations of some of the early results have been outlined. The scatter in the data obtained from the dissolution in HF media is particularly large, but the obvious underestimation of the uncertainties in some of these papers is not sufficient to explain this scatter. It remains, as noted in [1991GUR/VEY], that these results are obtained from the small difference of much larger values of about 150 kJ·mol^{-1}, whereas the enthalpies of dissolution in a high temperature melt in the experiments of Holm and Kleppa [1967HOL/KLE] and Navrotsky *et al*. [1980NAV/HON] are much smaller (*ca.* 15 kJ·mol^{-1}), and the corrections to 298.15 K are still relatively small (*ca.* 0.8 kJ·mol^{-1}).

However, a more precise value for the enthalpy of formation can be obtained from a consideration of the temperature at which quartz transforms to cristobalite. Although this is not known very precisely, the $\Delta_f G_m^o$ curves of these polymorphs intersect at such a small angle that the resultant uncertainty in the differences in $\Delta_r H_m^o$ (298.15 K) is very small. As noted in the Appendix A entry for [1956FLO], Flörke found that the anode section of samples of electrolysed tridymite contained quartz at 1323 K but cristobalite at 1373 K. These results are consistent with the later work of Holmquist, [1961HOL], who found that, when quartz was heated with sodium chloride *in vacuo*, cristobalite started to appear at 1323 K, but not at 1273 K. The temperature at which quartz transforms to cristobalite is thus taken to be (1323 ± 50) K. With our selected data for the thermal properties of these phases, this corresponds to a value of $\Delta_r H_m^o$ (cristobalite-quartz) = (3.0 ± 0.1) kJ·mol^{-1} at 298.15 K, consistent with the data reported by Kracek [1953KRA], and Holm and Kleppa [1967HOL/KLE] within their uncertainties (and close to that of Navrotsky *et al.* [1980NAV/HON]). This value corresponds to:

$$\Delta_f H_m^o (SiO_2, \alpha\text{-cris}, 298.15\ K) = \mathbf{-\ (907.7 \pm 1.1)}\ kJ \cdot mol^{-1}.$$

which is our selected value. With this value, the calculated temperature at which cristobalite becomes stable is 1309 K. The selected value is very close to that adopted in [1991GUR/VEY], – (908.0 ± 1.2) kJ·mol^{-1}.

V.3.1.4 SiO_2(trid)

V.3.1.4.1 Introduction and structures

Tridymite is a very complex phase and many of its properties are still surprisingly poorly defined: the number of structures and their detailed ordering, the thermodynamic data and even its stability in the pure SiO_2 system, are not well-established.

There are numerous forms of tridymite with slight differences in the complex ordering within the crystal, which we shall not discuss in any detail; in their assessments/reviews of the phase stability in the Si-O system, Wriedt [1990WRI] and Schnurre *et al.* [2004SCH/GRO] list nine forms. But only two or three of these are commonly encountered and are given the labels of S-I, S-II and S-VI, a nomenclature proposed by Sosman [1965SOS], widely but not universally accepted. Wriedt [1990WRI] discusses the correlation of these phases with other nomenclatures.

The first two, which are low-temperature forms, are certainly metastable with respect to α-quartz, and, as discussed below, the high temperature form S-VI is very probably metastable with respect to quartz and cristobalite, though easily and normally stabilised by impurities, particularly alkali metal ions. The structures of these phases are given in Table V-18.

Table V-18: Structure of tridymite phases.

Phase	Symmetry and Space Group	Lattice parameters /Å	*T* /K	Reference
Tridymite S-I	Monoclinic *Cc*	*a* = 18.494 *b* = 4.991 *c* = 23.758 *β* = 105.79 °	295	[1990WRI] [2004SCH/GRO]
Tridymite S-II	Orthorhombic [(a)]	*a* = 10.04 *b* = 17.28 *c* = 8.20	348-388	[1990WRI] [2004SCH/GRO]
Tridymite S-VI	Hexagonal $P6_3/mmc$	*a* = 5.03 *c* = 8.22	Indefinite, but above 436 K	[1926GIB]
		a = 5.05-5.06 *c* = 8.24-8.30	*ca.* 523 K	[1964SAT]

(a) Check references for additional details on symmetry and space group.

Until the 1950's, tridymite S-VI was thought to be the stable phase of silica at normal pressure from 1143 to 1743 K. However, the work of Flörke [1955FLO], [1956FLO] and of Holmquist [1958HOL2], [1961HOL] provides strong evidence that this is not the case, and that quartz transforms directly to cristobalite at a temperature between 1273 and 1373 K as discussed more fully in Section V.3.1.3.4 on cristobalite. The relevant papers are summarised briefly here, with fuller details in the relevant Appendix A entries, which also quote a number of further references not mentioned here.

Flörke [1955FLO] found that he was only able to obtain tridymite by the use of a mineraliser such as Na_2O, K_2O or CaO, and flame spectrometric analyses showed that the tridymite prepared in this way always contained a fraction of a percent of foreign ions, such as Na^+, K^+ or Ca^{2+}.

Subsequently, Flörke [1956FLO] also showed that after a pellet of tridymite was electrolysed for (8-20) days at (1473-1623) K, cristobalite was formed in the anode section of the sample, where the impurity levels were less than 0.05 %, while the cathode section contained only tridymite. A later paper [1961FLO] gives more detail of these experiments, including XRD diagrams showing the presence of cristobalite in the anode, but not the cathode, section. These findings were confirmed by Holmquist [1958HOL2], who found that in a similar experiment held at 1623 K, the anode section was completely converted to pure 'snow-white' cristobalite in as short time as 12 hours.

Shortly after, Holmquist [1961HOL] found further convincing evidence that impurities are required to stabilise tridymite. When quartz was heated with 2 % alkali oxide, tridymite formed directly at 1145 to 1175 K with Na_2O and K_2O, and above 1278 K with Li_2O; in each case cristobalite was formed as an intermediate phase. Tridymite was also formed in the presence of the alkali silicates, carbonates and chlorides when heated in air; in all these cases, reaction occurs to form the silicates and or oxides. But, significantly, when quartz was heated with sodium chloride *in vacuo*, (when Na_2O or Na_2SiO_3 are not formed) tridymite did not form but cristobalite started to appear at 1323 K, but not at 1273 K.

Cyprès [1962CYP] also reports experiments on the transformation of quartz at high temperature both in the absence and presence of 5 % Na_2O. In a sample heated to 1623 K for 24 hours without Na_2O, XRD showed the presence of 83 % quartz and 17 % cristobalite, the latter being formed within the crystals. However, in a sample heated to 1523 K for 11 hours in the presence of 5 % Na_2CO_3, the quartz had completely disappeared, leaving a mixture of cristobalite and tridymite, the latter forming on the exterior of the grains.

The case for tridymite being stable has been made by Hill and Roy, [1958HIL/ROY2] and Wriedt [1990WRI]. Hill and Roy prepared tridymite without any solid or liquid mineralisers by heating a range of very pure silicas or transistor-grade Si in Pt containers with H_2O or D_2O to (1113-1513) K (at pressures of (3-23) MPa). This tridymite was certainly free from any appreciable amount of alkali ions, but Hill and Roy [1958HIL/ROY2] indicate that their tridymite products could have contained up to 3 mol % H_2O or D_2O, and Holmquist [1958HOL2] and others have pointed out that the H_3O^+ ion, which is approximately the same size as K^+, could act in the same way to stabilise tridymite. Wriedt [1990WRI] presented a clear summary of all the experiments discussed above, but noted that the reported range of stability of tridymite in numerous publications is always close to (1143-1743) K, independent of the mineraliser used. He suggests that if impurities are required to stabilise tridymite, one would expect a more

or less continuous range of temperatures at which it is formed. Since this is not the case, he makes the choice (his terminology) that tridymite is indeed stable from 1143 to1743 K at ambient pressures, but does not give any explanation of the results of the electrolysis experiments of both Flörke and Holmquist.

This review finds that the evidence of the two electrolysis studies, and those of Holmquist [1961HOL] and Cyprès [1962CYP], that tridymite is only stable in the presence of impurities, more convincing than that in [1958HIL/ROY2] and [1990WRI] to the contrary, and the data in this review reflect this position, which was also adopted in other assessments, such as JANAF [1985CHA/DAV], [1991GUR/VEY] and [2004SCH/GRO], but not in the reviews of Wriedt [1990WRI] (see above), Hallstedt [1992HAL] and Swamy *et al.* [1994SWA/SAX]. In fact, the authors of the last two assessments of the Si-O system did not discuss this problem, accepting the traditional view without question.

V.3.1.4.2 SiO_2(trid S-I)

V.3.1.4.2.1 Heat capacity and entropy

There is only one measurement of the low-temperature heat capacity of tridymite, by Anderson [1936AND], for a sample prepared at 1523 K, using a considerable amount (14 wt %) of Na_2WO_4 as a flux, and which after a second purification contained 99.46 % SiO_2. Anderson's measurements only ranged from 54.2 to 294.9 K. He reported a standard entropy at 298.15 K of (43.93 ± 0.42) $J \cdot K^{-1} \cdot mol^{-1}$, obtained by extrapolation to 0 K with a Debye function and graphical integration. Gurvich *et al.* [1991GUR/VEY] improved the extrapolation down to 0 K, and calculated a slightly lower value of (43.51 ± 0.80) $J \cdot K^{-1} \cdot mol^{-1}$, essentially the same as that adopted in [1982WAG/EVA], 43.50 $J \cdot K^{-1} \cdot mol^{-1}$, and slightly higher than the value of 43.34 $J \cdot K^{-1} \cdot mol^{-1}$ given in [1941MOS/PIT], which they admit is a somewhat arbitrary re-calculation.

The rounded value from [1991GUR/VEY] is selected here, as is their heat capacity at 298.15 K:

$$C^{\circ}_{p,\mathrm{m}}(SiO_2, \text{trid S-I}, 298.15\ \mathrm{K}) = \mathbf{(44.642 \pm 0.500)}\ J \cdot K^{-1} \cdot mol^{-1},$$

$$S^{\circ}_{\mathrm{m}}(SiO_2, \text{trid S-I}, 298.15\ \mathrm{K}) = \mathbf{(43.50 \pm 0.80)}\ J \cdot K^{-1} \cdot mol^{-1}.$$

The thermal properties of tridymites above 298.15 K are very poorly defined, in part due to the many polymorphs formed depending on the very complicated ordering behaviour. Although there seem to be no experimental data above 840 K, a number of assessments give data up to 2000 K, based on the (reasonable) assumption that the heat capacities of cristobalite and tridymite are very similar.

There are four papers on the thermal properties above 298.15 K: Mosesman and Pitzer [1941MOS/PIT] measured the enthalpy content up to 625 K, Sinel'nikov

[1956SIN] gives a small plot of the heat capacity up to *ca.* 500 K, together with details of the enthalpies of transition, while Thompson and Wennemer [1979THO/WEN] made measurements of $C_{p,\mathrm{m}}$ at intervals of 2 K between the temperatures of the two transitions, (360-500) K, and thereafter at 10 K intervals up to 840 K. These measurements are reasonably consistent, except that the heat capacities in [1956SIN] below 370 K seem slightly high (see Figure V-21 below), as is also indicated by the enthalpy data of [1941MOS/PIT]. Shahid and Glasser [1970SHA/GLA] also give plots of $C_{p,\mathrm{m}}$ up to 523 K, measured in a dynamic calorimeter, but since they state these values are accurate only to within 6-8 %, they were not included in the fitting.

V.3.1.4.2.2 Transition data

There is, in general, reasonable agreement in the three studies [1941MOS/PIT], [1956SIN] and [1970SHA/GLA] on the temperatures and enthalpies of the transitions in the tridymites just above ambient temperature, as shown in Table V-19, though some of the detailed conclusions differ. The enthalpies measured by Sabatier [1957SAB] are much smaller, probably because his tridymite sample was prepared at the low temperature of 1223 K, and thus was not well-ordered, and may have contained unconverted quartz or cristobalite. The status of the third transition is not clear. Mosesman and Pitzer [1941MOS/PIT] found this to be sharp transition, in the studies of Shahid and Glasser [1970SHA/GLA] it was a faint, broad peak, while Thompson and Wennemer [1979THO/WEN] found it in mixed (tridymite + cristobalite) samples, but not in cristobalite-free tridymite; Sinel'nikov [1956SIN] and Sabatier [1957SAB] did not observe it at all. The indications are therefore that this effect is probably due to the presence of cristobalite in the samples used, as suggested in [1956SIN] and [1979THO/WEN].

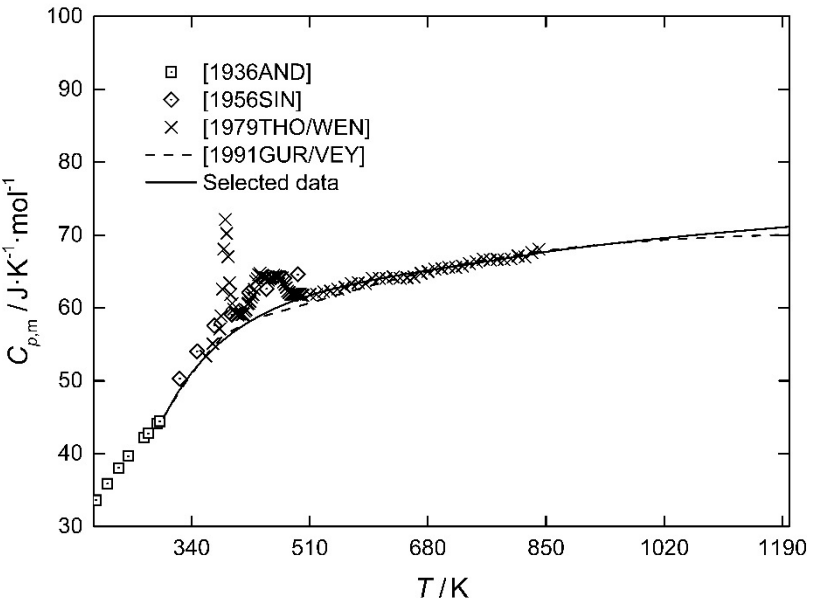
Table V-19: Transition data for tridymite.

Reference	T /K	$\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ /J·mol^{-1}	T /K	$\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ /J·mol^{-1}	T /K	$\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ /J·mol^{-1}
[1941MOS/PIT]	390	(293 ± 40)	436	(167 ± 40)	498	(188 ± 40)
[1956SIN]	388	(272 ± 20)	433	(145 ± 20)		not observed
[1957SAB]	388	108	438	58		not observed
[1970SHA/GLA]	390	—	436	broad peak	503-523	faint broad peak
[1978NUK/NAK]	383	—	423	—	463	
[1979THO/WEN]	390	Sharp	436-470	broad peak		none within experimental uncertainty

In view of the very small enthalpies of transition and uncertainties in the compositions of the samples, many of which were made with a large excess of flux/mineraliser, we shall select a simplified transition-free representation of tridymite (*cf.* the data used in [2004SCH/GRO]), with a heat capacity fitted from the enthalpy data in [1941MOS/PIT] and the heat capacities in [1956SIN] and [1979THO/WEN], excluding the high values around the various transitions. These values are compared

with the experimental data in Figure V-21; the mean deviation from the enthalpy data of [1941MOS/PIT] is 0.99 %.

Figure V-21: High-temperature heat capacity of SiO_2(trid).



$$C^{\circ}_{p,\mathrm{m}}(\mathrm{SiO_2,trid},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{87.48693} - \mathbf{4.427262\times10^{-2}}(T/\mathrm{K}) + \mathbf{3.020355\times10^{-5}}(T/\mathrm{K})^2 - \mathbf{2.873920\times10^{6}}(T/\mathrm{K})^{-2} \quad (298.15\text{-}850)\ \mathrm{K}.$$

This equation should not be used above (850-900) K, where the calculated values of $C_{p,\mathrm{m}}$ are probably too high. However, in order to check the instability of tridymite with respect to quartz and cristobalite above 850 K, we have used, but do not select, tridymite $C_{p,\mathrm{m}}$ data very close to those of cristobalite in this temperature range.

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{SiO_2,trid},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 67.10126 + 4.8429\times10^{-3}(T/\mathrm{K}) - 2.5419\times10^{6}(T/\mathrm{K})^{-2} \quad (850\text{-}1650)\ \mathrm{K}.$$

V.3.1.4.2.3 Enthalpy of formation

The direct experimental data leading to the enthalpy of formation of tridymite are quite scanty and rest on the difference between the enthalpy of solution of α-quartz and tridymite (S-I) in aqueous hydrofluoric acid.

As discussed in Appendix A, Hummel and Schwiete [1959HUM/SCH] report a value of (5.52 ± 0.63) kJ·mol^{-1} at 299.65 K for the enthalpy of the reaction

$$SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(\text{trid S-I}) \qquad \text{(V.17)}$$

from dissolutions in 39.0 % HF at 299.65 K. These authors mention, for the same reaction, a value of (1.5 ± 0.5) kJ·mol^{-1} attributed to Troitzsch [1936TRO]. Unfortunately, the latter document is a thesis not available to this review. This value of 1.5 kJ·mol^{-1} is also listed in [1991GUR/VEY], as originating from the same group ([1932ROT/BEC], [1949ROT/TRO] and [1936TRO]). However, neither of the papers [1932ROT/BEC] and [1949ROT/TRO] mentions data on tridymite. In any case, the characterisation of the samples and the experimental details are far more explicit in [1959HUM/SCH] than in [1949ROT/TRO] for reactions involving other forms of SiO_2. This, coupled with the unavailability of [1936TRO], leads us to give a lower weight to the difference of 1.5 kJ·mol^{-1} in the evaluation of the Reaction (V.17) and the value indicated by the enthalpy of solution measurements is thus:

$$\Delta_f H^\circ_m(SiO_2, \text{trid S-I}, 298.15\ \text{K}) = -(906.3 \pm 2.5)\ \text{kJ·mol}^{-1},$$

keeping conservative uncertainty limits. This value is 0.5 kJ·mol^{-1} more negative than that obtained using the lower limit of the result in [1959HUM/SCH] for the quartz-to-tridymite transition.

Another, equally uncertain, value can be estimated from the Gibbs energy of the reaction SiO_2(β-quar) $\rightleftharpoons$ SiO_2(trid S-VI) at 1150 K, the temperature at which the presence of a small (< 1 %) impurity from the mineraliser stabilises tridymite.

Table V-20 gives the equilibrium constant for this reaction, as a function of $\Delta_f H^\circ_m(SiO_2, \text{trid}, 298.15\ \text{K})$; K, is of course, the activity of SiO_2 in the impure tridymite when the reaction occurs.

Table V-20: Data for the reaction SiO_2(β-quar) $\rightleftharpoons$ SiO_2(trid S-VI).

$\Delta_f H^\circ_m$ (SiO_2, trid S-I, 298.15 K) /kJ·mol^{-1}	$\Delta_r G^\circ_m$ (Reaction (V.17), 1150 K)/kJ·mol^{-1}	K
– 908.0	0.292	0.970
– 907.5	0.792	0.921
– 907.0	1.292	0.874
– 906.9	1.392	0.865
– 906.3	1.992	0.812
– 906.0	2.292	0.787
– 905.5	2.792	0.747

Since there is only a small impurity content (probably at most 1 %) in the tridymite formed, so that the activity of SiO_2 is close to 1.0, this table suggests that the value $\Delta_f H^\circ_m$(tridymite, 298.15 K) is close to – 907.5 kJ·mol^{-1} and that the enthalpy of Reaction (V.17) measured by Hummel and Schwiete [1959HUM/SCH] (which corresponds to $\Delta_f H^\circ_m$(tridymite, 298.15 K) = – 906.3 kJ·mol^{-1}) is somewhat too large (as was their corresponding value for cristobalite, see Section V.3.1.3.4), and that a

value of $\Delta_f H^o_m(SiO_2,$ trid S-I, 298.15 K) near – 907.5 kJ·mol^{-1} may be more appropriate.

Our final selected value for the enthalpy of formation is:

$$\Delta_f H^o_m(SiO_2, \text{trid S-I}, 298.15\ K) = \mathbf{-\ (906.9 \pm 2.5)}\ kJ{\cdot}mol^{-1},$$

which also keeps tridymite less stable than cristobalite at low temperatures. The uncertainty is estimated conservatively by this review, based on the values for the other polymorphs. This value for the enthalpy of formation differs somewhat from that selected in [1991GUR/VEY], – (905.2 ± 1.5) kJ·mol^{-1}.

V.3.1.5 SiO_2(vit)

We have used the terminology SiO_2(vit) for the phase often described in the literature as vitreous quartz. As discussed in some detail in Hetherington *et al.* [1965HET/JAC] and Brückner [1970BRU], most samples of SiO_2(vit) contain small amounts of both OH, Cl and metallic ions, (possibly as much as 1 mol %) depending on the method of manufacture; even the purest available material usually contains 0.4 ppm OH and 200 ppm Cl by mass. This, together with effects of the temperature of preparation, means that there are differences in some of the thermodynamic properties, especially at very low temperatures, depending on the sample.

V.3.1.5.1 Heat capacity and entropy

Because of its common use to contain other materials in enthalpy-content and heat capacity measurements, many such studies also report the heat capacity of vitreous silica, for which there is thus a wealth of data—Schnurre e*t al.* [2004SCH/GRO] give a non-exhaustive list of 26 studies. Both they and Richet *et al.* [1982RIC/BOT] have analysed many of these data, and we have accepted the thorough analysis of the heat capacity given by the latter authors, since as shown in Figure V-22, the data given by [2004SCH/GRO] do not merge smoothly with the low-temperature data. Only the more important studies on which the assessment of [1982RIC/BOT] are based are mentioned below.

The preferred low-temperature data are those of Westrum [1956WES] (indirectly obtained from [1957LOR/MOR]—see Section V.3.1.2.2.1) and Turdakin (values given in [1967TAR]), which agree to within 1 %. Above 298.15 K, the data are based primarily on the studies of Fischer [1931FIS], ($H_m - H_m$(273.15 K)), White [1919WHI] ($H_m - H_m$(273.15 K), (573-1173) K), Shmidt and Maksimov, [1979SHM/MAK] ($C^o_{p.m}$, 294-621 K), and their own work, [1982RIC/BOT] ($H_m - H_m$(273.15 K), 1007-1837 K). However, the data from all the remaining 17 studies differed by less than 1 % from their assessed data, except for three early studies which deviate by up to 2 %. We have re-fitted the data from the $C^o_{p.m}$ equation used in

[1982RIC/BOT] up to 1400 K, which included a $T^{1/2}$ term, to the usual four-term expression, with negligible differences.

The value of S°_{m}(298.15 K) is also based on that given in [1982RIC/BOT]; we note that these authors refer to data for the amorphous phase, although their sample was, in fact, vitreous silica (see relevant Appendix A). However, the entropy of SiO_2(vit) is not zero at 0 K, and they, like the authors of other assessments, derived their entropy values by equating the Gibbs energies of cristobalite and vitreous (regarded as supercooled liquid SiO_2) at 1999 K, the melting point of cristobalite. Although our selected entropy and heat capacity values of cristobalite are the same as those in [1982RIC/BOT], our selected values for the enthalpies of formation of both cristobalite and vitreous SiO_2 differ slightly from their values. Thus, to maintain the equality of cristobalite and vitreous SiO_2 at 1999 K, the entropy of vitreous SiO_2 has been adjusted slightly (by 0.105 $J \cdot K^{-1} \cdot mol^{-1}$).

Our selected values are thus:

$$C^{\circ}_{p,m}(SiO_2, \text{vit}, 298.15\ K) = \mathbf{(44.061 \pm 0.800)}\ J \cdot K^{-1} \cdot mol^{-1},$$

$$S^{\circ}_{m}(SiO_2, \text{vit}, 298.15\ K) = \mathbf{(48.17 \pm 1.00)}\ J \cdot K^{-1} \cdot mol^{-1},$$

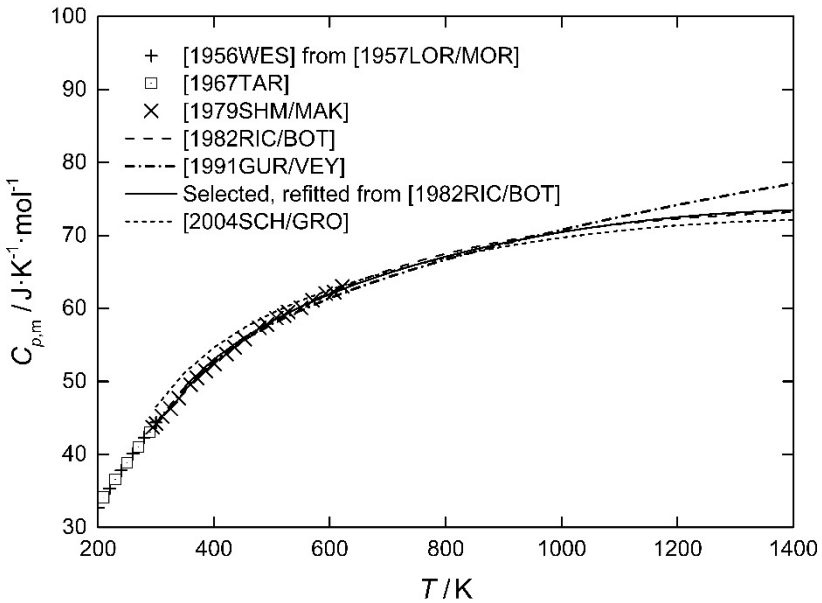
$$C^{\circ}_{p,m}(SiO_2, \text{vit}, T)/J \cdot K^{-1} \cdot mol^{-1} = \mathbf{48.8375} + \mathbf{3.4958 \times 10^{-2}}(T/K) - \mathbf{1.20830 \times 10^{-5}}(T/K)^2 - \mathbf{1.255648 \times 10^{6}}(T/K)^{-2} \quad (298.15\text{-}1480\ K),$$

$$C^{\circ}_{p,m}(SiO_2, \text{vit}, T)/J \cdot K^{-1} \cdot mol^{-1} = \mathbf{81.373} \quad (1480\text{-}2000\ K),$$

where the uncertainties have been conservatively estimated by this review. The discontinuity in the heat capacity at 1480 K is due to a glass transition [1982RIC/BOT].

Some of the experimental data and selected values (including, for comparison, those in [1991GUR/VEY]) are shown in Figure V-22.

Figure V-22: Heat capacity of SiO_2(vit).



V.3.1.5.2 Enthalpy of formation

The enthalpy of formation of vitreous silica has been obtained from the difference between the enthalpy of various reactions for α-quartz and vitreous silica, summarised in Table V-21.

Table V-21: Data for the enthalpy change of reaction SiO_2(α-quar) ⇌ SiO_2(vit).

Reference	Reaction type and temperature	$\Delta_r H_m$ /kJ·mol^{-1}	Comments
[1912MUL]	Dissolution in 20 % HF, (290-291) K	(9.25 ± 2.55)	As reported
[1921WIE]	Dissolution in 35 % HF and fuming HF, ambient temperature	(9.49 ± 0.30)	Average value, with uncertainty covering only the scatter of the reported results, as estimated by this review
[1949ROT/TRO]	Dissolution in 20.6 % HF, 350 K	(12.47 ± 0.29) at 350 K	As reported
		(12.50 ± 0.30) at 298.15 K	Correction by this review to 298.15 K is 0.029 kJ·mol^{-1}
[1953KRA]	Dissolution from 298.15 K into 20 % HF, 347.85 K	(9.12 ± 0.54) at 298.15 K	No correction required; uncertainty estimated by this review

(Continued on next page)

Table V-21 (continued)

Reference	Reaction type and temperature	$\Delta_r H_m$ /kJ·mol^{-1}	Comments
[1959HUM/SCH]	Dissolution in 39 % HF, 299.65 K	(9.50 ± 0.84) at 299.65 K	As reported
[1963WIS/MAR]	Fluorine bomb combustion, 298.15 K	(7.468 ± 0.293) at 298.15 K	As reported
[1967HOL/KLE]	Dissolution in $9PbO{\cdot}3CdO{\cdot}4B_2O_3$ melt, 970 K	(5.9 ± 0.6) at 970 K	As reported
		(9.0 ± 0.6) at 298.15 K	As corrected to 298.15 K by the authors
		(9.0 ± 0.6) at 298.15 K	Correction by this review to 298.15 K is 2.38 kJ·mol^{-1}
[1970TAK/YOS]	Dissolution in 5 % HF, 298.15 K	9.6 at 298.15 K	As reported
[1973KIL/PRO]	Dissolution in 24.4 % HF, 353.15 K	(9.749 ± 0.293) at 353.15 K	As reported
		(9.78 ± 0.29) at 298.15 K	Correction by this review to 298.15 K is 0.029 kJ·mol^{-1}
[1980NAV/HON]	Dissolution in $2PbO{\cdot}B_2O_3$ melt, 985 K	(7.00 ± 0.24) at 985 K	As reported
		(9.34 ± 0.24) at 298.15 K	Correction by this review to 298.15 K is 2.34 kJ·mol^{-1}

There are Appendix A entries for all these references, which include, in particular, notes on the limitations affecting some of the early results. The results of Holm and Kleppa [1967HOL/KLE] and Navrotsky *et al.* [1980NAV/HON], obtained from the dissolution of the two SiO_2 forms in a high temperature melt, are not derived from the difference of large numbers, unlike the other results, and are, potentially, the most precise. The dissolution data in aqueous HF from the careful work in [1973KIL/PRO] and from Hummel and Schwiete [1959HUM/SCH] agree well with those in [1967HOL/KLE] and [1980NAV/HON]. It should be noted, however, that, in the case of the reaction $SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(\alpha\text{-cris})$ the results in [1959HUM/SCH] were far from the selected value. The uncertainty affecting the value reported by Takashi and Yoshio [1970TAK/YOS] is hard to evaluate in view of the lack of experimental details concerning, in particular, the identification of the phases. The fluorine bomb reaction data of Wise *et al.* [1963WIS/MAR] yield a distinctly smaller value than those of other authors, for no obvious reason.

Under these circumstances, we adopt for the reaction:

$$SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(\text{vit}),$$

the value of $\Delta_r H_m^o(298.15\ \text{K}) = (9.15 \pm 0.60)$ kJ·mol^{-1}, and thus select the rounded value

$$\Delta_f H^o_m(SiO_2, \text{vit}, 298.15\ K) = \mathbf{-\ (901.6 \pm 1.2)}\ kJ{\cdot}mol^{-1},$$

where the uncertainties have arbitrarily been increased in view of the fairly wide range of experimental values.

These selected values for the vitreous phase are very close to those recommended by [1991GUR/VEY], who adopted $\Delta_f H^o_m(SiO_2, \text{vit}, 298.15\ K) = -(901.4 \pm 1.1)\ kJ{\cdot}mol^{-1}$, and are in almost complete agreement with the assessment by Richet *et al.* [1982RIC/BOT] which was appreciably more comprehensive than that in [1991GUR/VEY].

V.3.1.6 SiO_2(am)

We are not able to select any definite thermodynamic data for amorphous silica, owing to the variability of the properties of this phase, depending on its impurity content and particle size, but we note briefly here some data for the enthalpy and Gibbs energy of formation at 298.15 K. The calorimetric data for reaction,

$$SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(am) \qquad (V.18)$$

obtained by dissolutions in HF solutions are summarised in Table V-22; the details of these determinations are given in the relevant Appendix A entries.

Table V-22: Experimental data for the enthalpy of Reaction (V.18).

Reference	$\Delta_r H_m$ (298.15 K) /kJ·mol^{-1}	Solvent and temperature
[1949ROT/TRO]	*ca.* 16.3	20.6 % HF, 350 K
[1959HUM/SCH]	(14.27 ± 0.59)	39 % HF, 299.65 K, extremely fine particulates

The difference in these enthalpy data contrasts sharply with the very consistent data for the solubility of this phase described in Section V.3.2.2.2 which give:

$$\Delta_f G^o_m(SiO_2, am, 298.15\ K) = -(850.287 \pm 1.022)\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^o_m(SiO_2, am, 298.15\ K) = -(902.854 \pm 1.259)\ kJ{\cdot}mol^{-1}$$

and

$$S^o_m(SiO_2, am, 298.15\ K) = (47.651 \pm 2.604)\ J{\cdot}K^{-1}{\cdot}mol^{-1},$$

and a difference of $(7.85 \pm 0.76)\ kJ{\cdot}mol^{-1}$ between the enthalpies of formation of SiO_2(α-quar) and SiO_2(am).

The reasons for the difference in the data from the enthalpy of solution and solubility studies are not clear, but may be related to different pre-treatments (if any) of the samples, such as water-washing. This certainly played a role in gibbsite experiments, for example, where acid pre-wash seemed essential to remove fine particulates and impurities.

If data are required for the amorphous phase, we strongly recommend using the results from the very consistent solubility experiments listed above and fully discussed in Section V.3.2.2. It is worth noting that the value for S^{o}_{m} (SiO_2, am, 298.15 K) = (47.651 ± 2.604) $J \cdot K^{-1} \cdot mol^{-1}$ is not statistically different from that selected (see Section V.3.1.5.1) for S^{o}_{m} (SiO_2, vit, 298.15 K) = (48.17 ± 1.00) $J \cdot K^{-1} \cdot mol^{-1}$, indicating that our reported data do not allow us to distinguish unambiguously between the vitreous and the amorphous species.

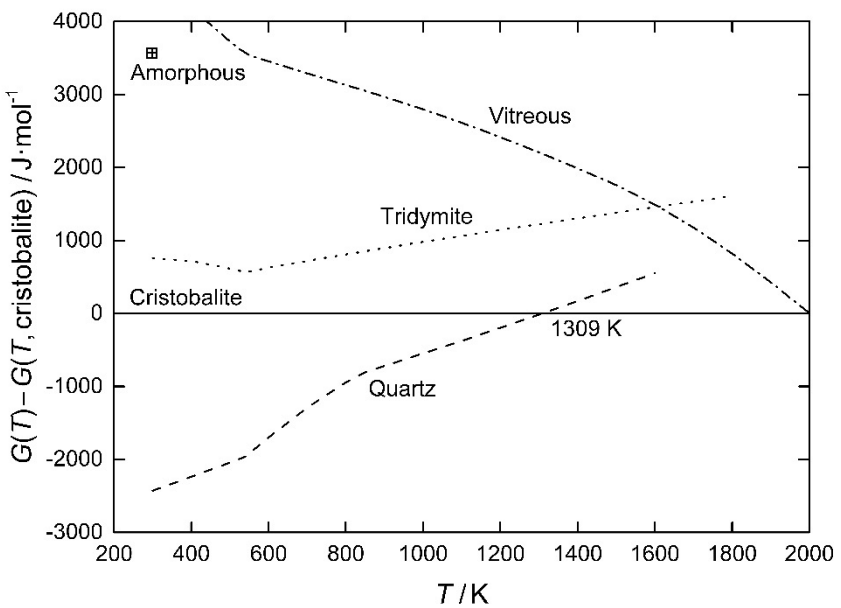
V.3.1.7 Summary

The following table (Table V-23) summarises the selected data at 298.15 K for the silica polymorphs. The Gibbs energies of formation have been calculated internally through the Gibbs-Helmholtz relation. The stabilities of the silica phases relative to cristobalite are shown in Figure V-23.

Table V-23: Summary of selected values at 298.15 K.

Phase	$\Delta_f G^{o}_{m}$ /kJ·mol^{-1}	$\Delta_f H^{o}_{m}$ /kJ·mol^{-1}	S^{o}_{m} /J·K^{-1}·mol^{-1}
SiO_2(α-quar)	**– (856.287 ± 1.002)**	**– (910.700 ± 1.000)**	**(41.460 ± 0.200)**
SiO_2(α-cris)	**– (853.854 ± 1.101)**	**– (907.700 ± 1.100)**	**(43.363 ± 0.150)**
SiO_2(trid S-I)	**– (853.095 ± 2.511)**	**– (906.900 ± 2.500)**	**(43.500 ± 0.800)**
SiO_2(vit)	**– (849.188 ± 1.237)**	**– (901.600 ± 1.200)**	**(48.170 ± 1.000)**

Figure V-23: The relative stabilities of the silica phases.



V.3.2 Si-O-H

V.3.2.1 Aqueous silicon tetrahydroxide

Quartz is one of the most abundant minerals found on earth. Consequently, geochemists in particular have carried out numerous experimental and modelling [1977WAL/HEL] studies of this mineral and amorphous silica ($SiO_2 \cdot xH_2O$) over a wide range of temperature, pH and ionic strength. Practical considerations as well as the slow rate of dissolution of crystalline quartz have meant that most studies have concentrated on temperatures well in excess of 298 K (*e.g.*, Rimstidt [1997RIM] allowed 13.5 years with weekly agitation to achieve equilibration of crushed quartz with water at 294 K). The generalised equation applicable to the dissolution of both crystalline quartz and amorphous silica in water is:

$$SiO_2(s) + 2H_2O(l) \rightleftharpoons Si(OH)_4(aq) \qquad \text{(V.19)}$$

As indicated, experimental studies of quartz solubility in water have been reported over wide ranges of temperature and pressure: Kennedy, 433-883 K [1950KEN]; Fournier, 357-413 K [1960FOU]; Kitahara, 433-573 K [1960KIT]; van Lier *et al*., 333-373 K [1960LIE/BRU]; Morey *et al*., 342-413 K [1962MOR/FOU]; Siever, 398-454.5 K [1962SIE]; Anderson and Burnham, 873-1173 K [1967AND/BUR], Crerar and Anderson, 398-498 K [1971CRE/AND]; Hemley *et al*., 473-773 K [1980HEM/MON]; Saccocia and Seyfried, 573-673 K [1990SAC/SEY]; von Damm *et al*., 473-733 K [1991DAM/BIS]; and Rimstidt, 294-363 K [1997RIM]. The kinetics of quartz dissolution and precipitation were studied by: Bohlmann *et al.*, 298.15 K [1980BOH/MES]; Rimstidt and Barnes, 291-573 K [1980RIM/BAR]; and Mitra and Rimstidt, 298-373 K [2009MIT/RIM]. Modelling studies in the temperature range 373 to 523 K show general agreement, but at temperatures below 373 K and above 523 K significant discrepancies exist: Greenberg, 473 K [1957GRE]; Walther and Helgeson, 298-873 K [1977WAL/HEL]; Fournier and Potter, 298-1173 K [1982FOU/POT]; Eikenberg, 298-363 K [1990EIK]; Stefánsson, 273-623 K [2001STE]; Gerya *et al.*, 293-573 K [2005GER/MAR]; and Plyasunov, 298-2000 K [2011PLY], [2012PLY]. This review has set a somewhat arbitrary limit of 473 K on the experimental results that are included in our assessments. This choice justifies setting $\Delta_r C^o_{p,m}$((V.19), 298.15 K) = 0 $J \cdot K^{-1} \cdot mol^{-1}$, as the discussion below in Section V.3.2.2.1 will show.

There have also been numerous studies made of the solubility of amorphous silica and although no thermodynamic constants are selected here for this metastable phase, the excellent reproducibility of the data even at high temperature (583 K) attests to its prolonged lifetime. This longevity and potential importance of amorphous silica in the interpretation of cement-water interactions justifies the treatment of solubility data in Section V.3.2.2.2. The experimental studies and ranges of temperature considered are: Lenher and Merrill, 298-363 K [1917LEN/MER]; Hitchen, 401-468 K [1935HIT];

Alexander *et al.*, 298.15 K [1954ALE/HES]; Krauskopf, 298.15 K [1956KRA3]; Greenberg and Price, 298.15-308.15 K [1957GRE/PRI]; Elmer and Nordberg, 308-368 K [1958ELM/NOR]; Siever, 298.15 K [1962SIE]; Morey *et al.*, 298.15 K [1964MOR/FOU]; Marshall, 298-473 K [1980MAR]; Chen and Marshall, 373-473 K [1982CHE/MAR]; Gallup, 298-473 K [1998GAL]; and Gunnarsson and Arnórsson, 281-473 K [2000GUN/ARN].

V.3.2.2 $Si(OH)_4(aq)$

V.3.2.2.1 Solubility of α-quartz

Grenthe *et al.* [1992GRE/FUG] derived the thermodynamic properties for $Si(OH)_4(aq)$ from the quartz solubility results reported by Morey *et al.* [1962MOR/FOU] from 298 to 513 K with expanded uncertainties that were adopted by the NEA TDB Project:

$$\Delta_f G^\circ_m (Si(OH)_4, aq, 298.15\ K) = -(1307.735 \pm 1.156)\ kJ\cdot mol^{-1},$$

$$\Delta_f H^\circ_m (Si(OH)_4, aq, 298.15\ K) = -(1456.960 \pm 3.163)\ kJ\cdot mol^{-1},$$

$$S^\circ_m (Si(OH)_4, aq, 298.15\ K) = (189.973 \pm 11.296)\ J\cdot K^{-1}\cdot mol^{-1}.$$

In their analysis, Grenthe *et al.* accepted the argument that the higher solubility reported by van Lier *et al.* [1960LIE/BRU] may have resulted from the greater surface energy of the fine-grained quartz used in that study. A rule-of-thumb for judging the quality of solubility experiments is that lower solubilities generally are considered more reliable due to crystallinity, grain size and surface property considerations. Subsequently, Rimstidt's [1997RIM] definitive study yielded higher quartz solubilities that are in good agreement with data reported by van Lier *et al*. [1960LIE/BRU] giving rise to a more negative value of the Gibbs energy of formation of $Si(OH)_4(aq)$ now accepted by others, *e.g.*, Rimstidt [1997RIM], Mitra and Rimstidt [2009MIT/RIM], and Stefánsson [2001STE].

A summary of the available solubility constants for α-quartz (Reaction (V.19)) is given in Table V-24 where the upper temperature limit was set to 473 K. Figure V-24 shows the dependence of $\log_{10} K^\circ_{s,4}$ on $(T/K)^{-1}$. The results reported in pure water were treated with activity coefficients set to unity due primarily to the absence of ionic species and the low concentrations of dissolved silica.

Table V-24: Molal solubilities of α-quartz and logarithms of the solubility constants with experimental uncertainties estimated by the reviewers (2σ).

T /K	p /MPa [(a)]	Medium	m_{Si}	$\log_{10} K^\circ_{s,4}$	Reference
433	0.62	water	1.17×10^{-3}	– 2.932 [(b)]	[1950KEN]
455	1.11		2.83×10^{-3}	– 2.548 [(b)]	
473	1.55		4.00×10^{-3}	– 2.398 [(b)]	
357	0.1	water	4.16×10^{-4}	– 3.381 [(b)]	[1960FOU]

(Continued on next page)

Table V-24 (continued)

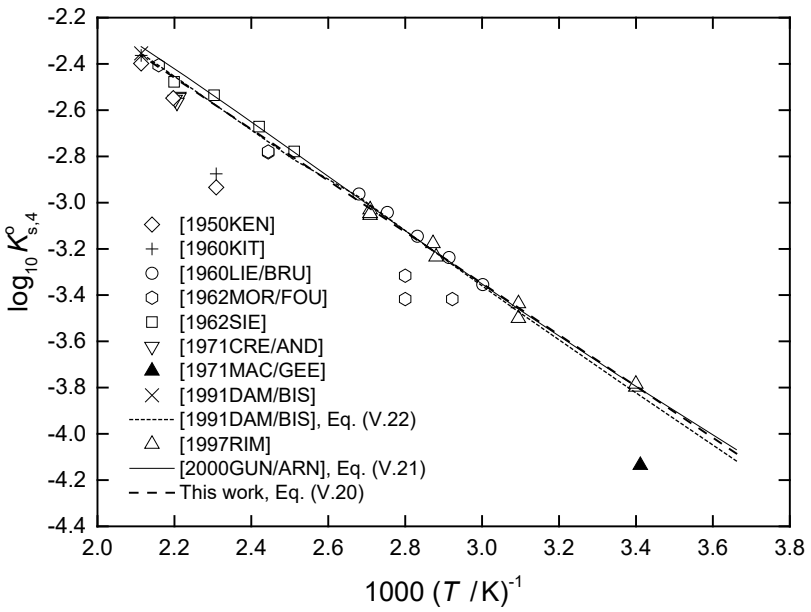
T /K	p /MPa [(a)]	Medium	m_{Si}	$\log_{10} K^\circ_{s,4}$	Reference
438	0.70		2.75×10^{-3}	– (2.561 ± 0.035)	
433	0.61	water	1.33×10^{-3}	– 2.876 [(a)]	[1960KIT]
473	1.52		4.33×10^{-3}	– (2.364 ± 0.035)	
333	0.1	water	4.42×10^{-4}	– (3.355 ± 0.025)	[1960LIE/BRU]
343	0.1		5.79×10^{-4}	– (3.237 ± 0.025)	
353	0.1		7.15×10^{-4}	– (3.146 ± 0.025)	
363	0.1		9.06×10^{-4}	– (3.043 ± 0.025)	
373	0.1		1.09×10^{-3}	– (2.963 ± 0.025)	
363	0.1	10^{-4} c_{NaOH}	9.20×10^{-4}	– (3.036 ± 0.025)	
342	0.1	water	3.83×10^{-4}	– 3.417 [(b)]	[1962MOR/FOU]
357	0.1		4.83×10^{-4}	– 3.316 [(b)]	
357	0.1		3.83×10^{-4}	– 3.417 [(b)]	
409	0.32		1.65×10^{-3}	– (2.783 ± 0.030)	
409	0.32		1.66×10^{-3}	– (2.779 ± 0.030)	
463	1.26		3.93×10^{-3}	– (2.406 ± 0.035)	
398	0.23	water	1.66×10^{-3}	– (2.779 ± 0.030)	[1962SIE]
413	0.37		2.13×10^{-3}	– (2.672 ± 0.030)	
434	0.63	pH = 8.5 KOH [(c)]	2.91×10^{-3}	– (2.536 ± 0.030)	
454.6	1.04		3.33×10^{-3}	– (2.478 ± 0.030)	
452	0.98	water	2.88×10^{-3}	– 2.541 [(b)]	[1971CRE/AND]
452	0.98		2.85×10^{-3}	– 2.546 [(b)]	
453	1.00		2.71×10^{-3}	– 2.567 [(b)]	
454	1.03		2.79×10^{-3}	– 2.554 [(b)]	
293	0.1	sea water	7.3×10^{-5}	– 4.14 [(b)]	[1971MAC/GEE]
473	1.55	water	4.42×10^{-3}	– (2.355 ± 0.030)	[1991DAM/BIS]
294	0.1	water	1.60×10^{-4}	– (3.797 ± 0.050)	[1997RIM]
294	0.1		1.64×10^{-4}	– (3.784 ± 0.050)	
323	0.1		3.16×10^{-4}	– (3.500 ± 0.025)	
323	0.1		3.66×10^{-4}	– (3.437 ± 0.025)	
348	0.1		6.66×10^{-4}	– (3.177 ± 0.020)	
347	0.1		5.83×10^{-4}	– (3.235 ± 0.020)	
369	0.1		8.99×10^{-4}	– (3.046 ± 0.020)	
369	0.1		8.82×10^{-4}	– (3.055 ± 0.020)	
369	0.1		9.32×10^{-4}	– (3.031 ± 0.020)	
369	0.1		8.99×10^{-4}	– (3.046 ± 0.020)	

(a) von Damm *et al*. [1991DAM/BIS] showed that small pressure increases up to *ca.* 1.5 MPa would cause an increase in $\log_{10} K^\circ_{s,4}$ of < 0.005.

(b) These solutions appear to be under-saturated so that these data were not included in the final overall analysis.

(c) pH was adjusted with KOH at room temperature before starting the experiment.

Figure V-24: The dependence of $\log_{10}K^{\circ}_{s,4}$ (or $\log_{10}m_{Si}$) for the solubility of α-quartz on $(T/\mathrm{K})^{-1}$.



A weighted regression of the data selected from Table V-24 resulted in the following equation, noting that inclusion of an additional temperature term was not justifiable statistically:

$$\log_{10}K^{\circ}_{s,4}(\mathrm{SiO_2},\alpha\text{-quar}) = -(0.0133296 \pm 0.0525632) - (1112.05 \pm 19.37)(T/\mathrm{K})^{-1} \quad \text{(V.20)}$$

Equation (V.20) gives the selected values:

$$\log_{10}K^{\circ}_{s,4}(\mathrm{SiO_2},\ \alpha\text{-quar},\ 298.15\ \mathrm{K}) = \mathbf{-\ (3.743 \pm 0.027)},$$

$$\Delta_r G^{\circ}_m((\mathrm{V.19}),\ 298.15\ \mathrm{K}) = \mathbf{(21.366 \pm 0.154)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H^{\circ}_m((\mathrm{V.19}),\ 298.15\ \mathrm{K}) = \mathbf{(21.290 \pm 0.727)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r S^{\circ}_m((\mathrm{V.19}),\ 298.15\ \mathrm{K}) = \mathbf{-\ (0.255 \pm 2.492)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

where the uncertainties are 2σ. The solubility of α-quartz at 298.15 K is therefore calculated to be $(1.81 \pm 0.10) \times 10^{-4}$ mol·kg^{-1}. Differences between these thermodynamic quantities and those proposed by Stefánsson [2001STE], and Gunnarsson and Arnórsson [2000GUN/ARN] are due at least in part to the more comprehensive datasets treated by these authors. Grenthe *et al.* [1992GRE/FUG]

accepted values of $\log_{10}K^\circ_{s,4}(SiO_2, \alpha\text{-quar}, 298.15\ K) = -(4.0 \pm 0.1)$ and $\Delta_r H^\circ_m((V.19), 298.15\ K) = (25.4 \pm 3.0)\ kJ\cdot mol^{-1}$ from [1962MOR/FOU] and hence a solubility of $1 \times 10^{-4}\ mol\cdot kg^{-1}$.

The standard molal thermodynamic quantities of formation are shown in Table V-25. The values selected by this review are shown in bold.

Table V-25: Standard molal thermodynamic quantities of formation for the species involved in the dissolution of α-quartz and the hydrolysis of $Si(OH)_4$(aq) at 298.15 K compared to those in the current TDB.

$\Delta_f G^\circ_m$ /kJ·mol^{-1}	$\Delta_f H^\circ_m$ /kJ·mol^{-1}	S°_m /J·K^{-1}·mol^{-1}	$C^\circ_{p,m}$ /J·K^{-1}·mol^{-1}	Reference
SiO_2(α-quar)				
– (856.287 ± 1.002)	**– (910.700 ± 1.000)**	**(41.460 ± 0.200)**	**(44.500 ± 0.200)**	This review
– (856.287 ± 1.002)	– (910.700 ± 1.000)	(41.460 ± 0.200)	(44.602 ± 0.300)	NEA currently selected
H_2O(l)				
– (237.140 ± 0.041)	– (285.830 ± 0.040)	(69.950 ± 0.030)	(75.351 ± 0.080)	NEA/CODATA
$Si(OH)_4$(aq)				
– (1309.201 ± 1.017)	**– (1461.070 ± 1.239)**	**(181.105 ± 2.501)**	**(195.000 ± 3.000)** [(a)]	This review
– (1307.735 ± 1.156)	– (1456.960 ± 3.163)	(189.973 ± 11.296)		NEA currently selected
$SiO(OH)_3^-$				
– (1253.236 ± 1.034)	**– (1431.753 ± 1.352)**	**(91.727 ± 3.153)**		This review
– (1251.740 ± 1.162)	– (1431.360 ± 3.743)	(88.024 ± 13.144)		NEA currently selected
$SiO_2(OH)_2^{2-}$				
– (1177.088 ± 1.140)	**– (1386.070 ± 15.051)**	**– (10.453 ± 50.402)**		This review
– (1175.651 ± 1.265)	– (1381.960 ± 15.330)	– (1.488 ± 51.592)		NEA currently selected

(a) This value was calculated by assigning $\Delta_r C^\circ_{p,m}((V.19), 298.15\ K)$ a value of $(0 \pm 3)\ J\cdot K^{-1}\cdot mol^{-1}$ as dictated by Eq. (V.20).

Gunnarsson and Arnórsson [2000GUN/ARN] took solubility data for crystalline α-quartz from Rimstidt [1997RIM] and the literature values cited therein, to derive the following equation valid from 273 to 583 K at a pressure of 0.1 MPa to 373 K and p_{sat} above this temperature:

$$\log_{10}K^\circ_{s,4}(SiO_2, \alpha\text{-quar}) = -34.188 + 197.47(T/K)^{-1} + 12.245\log_{10}(T/K) - (5.851\times10^{-6})(T/K)^2 \quad (V.21)$$

Von Damm *et al*. [1991DAM/BIS] had previously derived a more complicated expression to fit their quartz solubility results in synthetic seawater (~ 0.5 mol·kg^{-1} NaCl) along with a critically-evaluated set of quartz solubility data in dilute aqueous solutions, from 318.15 to 1173 K and pressures from < 0.1 to 986 MPa:

$$\log_{10}K^{\circ}_{s,4}(\mathrm{SiO_2}, \alpha\text{-quar}) = -1.011 - 983.078(T/\mathrm{K})^{-1} + 0.0017691(T/\mathrm{K}) + 0.77976\log_{10}(\rho_s) + 0.01732(p/\mathrm{MPa})(T/\mathrm{K}) \quad \text{(V.22)}$$

where ρ_s is the solution density (g·cm^{-3}). This equation gives results compatible with those from Equation (V.21) and indicates that $\log_{10}K^{\circ}_{s,4}$ (SiO_2, α-quar) increases by less than 0.1 from p_{sat} to 50 MPa at temperatures below 598 K. Fournier and Potter [1982FOU/POT] provided a model for the solubility of quartz over even greater ranges of pressure and temperatures (to about 1 GPa and 1173 K, respectively). Shi *et al.* [2019SHI/MAO] provided a density model applicable to 1273 K and 2000 MPa for the solubility of quartz involving a large number of experimental studies in aqueous NaCl and CO_2 solutions. Most of these studies were made at conditions which exceed those of interest to this review. Their model is compatible with the results in [1997RIM]. For these reasons this model was not considered further by this review.

V.3.2.2.2 Solubility of amorphous silica

The term amorphous silica encompasses a range of metastable solids, many of which have been studied experimentally in attempts to quantify their solubilities. In general, a surprisingly common solubility was found that is approximately ten times higher than that of α-quartz with, as expected, much shorter equilibration times that enable experiments to be conducted conveniently at low temperatures. Notable among these studies is the work by Alexander *et al.* [1954ALE/HES] who found no difference in the solubility of powdered silica, colloidal or freshly prepared silica gel with $m_{Si(OH)_4(aq)} = (2.16 \pm 0.17) \times 10^{-3}$ at 298.15 K. Greenberg and Price [1957GRE/PRI] determined the solubility of colloidal silica (surface area 750 m^2·g^{-1}) to be 1.8×10^{-3} mol·dm^{-3} at 298.15 K (*cf.* 1.91×10^{-3} mol·kg^{-1} [1964MOR/FOU] using two commercial silica gels and silica glass) with an equilibration time of less than two days compared to 1000 days for initially supersaturated solutions. As a strong indicator of the relative stability of SiO_2(am), Gunnarsson and Arnórsson [2000GUN/ARN] were able to reverse the approach to equilibrium in their definitive experiments to 583 K. Marshall [1980MAR] also conducted highly reproducible amorphous silica experiments to 573 K as a function of ionic strength ($NaNO_3$) that required the solid phase to remain unchanged for considerable time periods during thermal and chemical equilibration. Marshall's results [1980MAR], [1980MAR/WAR] provided an opportunity to apply the SIT treatment which gave consistent and reasonable values of $\log_{10}K^{\circ}_{s,4}$ and $\Delta\varepsilon$ as demonstrated below. A summary of the available solubility constants for amorphous silica (Reaction (V.19)) is given in Table V-26 and shown in Figure V-25 as a function of $(T/\mathrm{K})^{-1}$.

Table V-26: Experimental and re-calculated solubility constants for amorphous silica with experimental uncertainties estimated by the reviewers (2σ).

T /K	p /MPa	Medium	m_{Si}	$\log_{10} K^{\circ}_{s,4}$ (SiO_2, am)	Reference
273	0.1	water	2.7×10^{-3}	2.57 [(a)]	[1917LEN/MER]
363			7.1×10^{-3}	– 2.15 [(a)]	
401	2.54	water	8.24×10^{-3}	– (2.08 ± 0.03)	[1935HIT]
401	2.54		8.19×10^{-3}	– (2.09 ± 0.03)	
403	2.70		8.83×10^{-3}	– (2.05 ± 0.03)	
403	2.70		8.66×10^{-3}	– (2.06 ± 0.05)	
432	6.03		1.12×10^{-2}	– (1.95 ± 0.05)	
432	6.03		1.10×10^{-2}	– (1.96 ± 0.05)	
441	7.54		1.18×10^{-2}	– (1.93 ± 0.06)	
443	7.91		1.24×10^{-2}	– (1.91 ± 0.06)	
468	13.98		1.50×10^{-2}	– (1.82 ± 0.08)	
468	13.98		1.50×10^{-2}	– (1.82 ± 0.08)	
298	0.1	water	2.2×10^{-3}	– (2.66 ± 0.03)	[1954ALE/HES]
298	0.1	water	1.8×10^{-3}	– (2.74 ± 0.03)	[1957GRE/PRI]
308	0.1	HNO_3(sln)	2.66×10^{-3}	– (2.58 ± 0.03)	[1958ELM/NOR]
338	0.1		4.33×10^{-3}	– (2.36 ± 0.03)	
368	0.1		7.75×10^{-3}	– 2.11 [(a)]	
298	0.1	water	2.33×10^{-3}	– 2.63 [(a,b)]	[1962SIE]
298	0.1	water	1.93×10^{-3}	– (2.72 ± 0.03)	[1964MOR/FOU]
298	0.1	$NaNO_3$(sln)	2.10×10^{-3}	– (2.68 ± 0.02) [(c)]	[1980MAR]
373	0.1		7.03×10^{-3}	– (2.15 ± 0.09) [(c)]	
423	0.48		1.06×10^{-2}	– (1.98 ± 0.05) [(c)]	
473	1.55		1.52×10^{-2}	– (1.82 ± 0.05) [(c)]	
298	0.1	KCl(sln)	1.99×10^{-3}	– (2.70 ± 0.02) [(c)]	[1980MAR/WAR]
		KNO_3(sln)	2.06×10^{-3}	– (2.69 ± 0.02) [(c)]	
		NaCl(sln)	2.08×10^{-3}	– (2.68 ± 0.02) [(c)]	
		LiCl(sln)	2.01×10^{-3}	– (2.70 ± 0.02) [(c)]	
		$LiNO_3$(sln)	1.98×10^{-3}	– (2.70 ± 0.02) [(c)]	
373	0.1	water	6.40×10^{-3}	– (2.19 ± 0.04)	[1982CHE/MAR]
423	0.48		1.03×10^{-2}	– (1.99 ± 0.04)	
473	1.55		1.61×10^{-2}	– (1.79 ± 0.04)	
298	0.1	water	2.00×10^{-3}	– (2.70 ± 0.04)	[1987DAN/SCH]
298	0.1	water	1.91×10^{-3}	– (2.72 ± 0.04)	[1998GAL]
373	0.1		6.24×10^{-3}	– (2.20 ± 0.03)	
423	0.48		1.03×10^{-2}	– (1.99 ± 0.03)	
473	1.55		1.58×10^{-2}	– (1.80 ± 0.03)	
281	0.1	water	1.27×10^{-3}	– (2.90 ± 0.06)	[2000GUN/ARN]
281			1.41×10^{-3}	– (2.85 ± 0.06)	
293			1.71×10^{-3}	– (2.77 ± 0.04)	
293			2.20×10^{-3}	– (2.66 ± 0.04)	
323			2.75×10^{-3}	– (2.56 ± 0.04)	
324			2.83×10^{-3}	– (2.55 ± 0.04)	
324			2.93×10^{-3}	– (2.53 ± 0.04)	
362			5.29×10^{-3}	– (2.28 ± 0.03)	
363			5.48×10^{-3}	– (2.26 ± 0.03)	
403	0.27		8.28×10^{-3}	– (2.08 ± 0.03)	
423	0.48		1.01×10^{-2}	– (2.00 ± 0.03)	
423	0.48		1.01×10^{-2}	– (2.00 ± 0.03)	

(Continued on next page)

Table V-26 (continued)

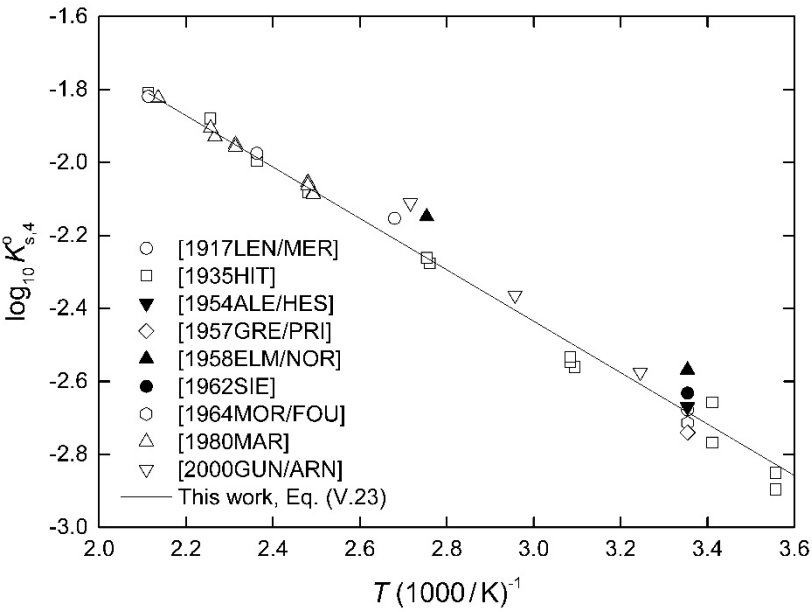
T /K	p /MPa	Medium	m_{Si}	$\log_{10} K^{o}_{s,4}$ (SiO_2, am)	Reference
443	0.79		1.32×10^{-2}	$-(1.88 \pm 0.03)$	
473	1.55		1.55×10^{-2}	$-(1.81 \pm 0.03)$	

(a) These values were rejected from the final fitting process.
(b) This value is an average of three datasets for silica gel without taking the value for opal into consideration.
(c) These values were obtained from a SIT treatment of the reported solubilities in these solutions.

A weighted regression of the data selected in Table V-26 resulted in the following equation (see Figure V-25):

$$\log_{10} K^{o}_{s,4}(SiO_2, am) = -(0.336466 \pm 0.013877) - (702.2267 \pm 5.5668)(T/K)^{-1} \quad (V.23)$$

Figure V-25: Dependence of $\log_{10} K^{o}_{s,4}$ (SiO_2, am) or $\log_{10} m_{Si}$ on $(T/K)^{-1}$.



For the apparent equilibrium, Equation (V.23) gives: $\log_{10} K^{o}_{s,4}$ (SiO_2, am, 298.15 K) = $-(2.692 \pm 0.009)$, $\Delta_r G^{o}_m$ ((V.19), 298.15 K) = (15.366 ± 0.054) kJ·mol^{-1}; $\Delta_r H^{o}_m$ ((V.19), 298.15 K) = (13.444 ± 0.209) kJ·mol^{-1}; $\Delta_r S^{o}_m$ ((V.19), 298.15 K) = $-(6.446 \pm 0.724)$ J·K^{-1}·mol^{-1}, where the uncertainties are 2σ. Consequently, the solubility equilibrium of amorphous silica at 298.15 K is $(2.03 \pm 0.04) \times 10^{-3}$ mol·kg^{-1}.

When these data are combined with data for $Si(OH)_4$(aq) derived above, the following values for SiO_2(am) can be calculated:

$$\Delta_f G^o_m (SiO_2, am, 298.15\ K) = -(850.287 \pm 1.022)\ kJ\cdot mol^{-1},$$

$$\Delta_f H^o_m (SiO_2, am, 298.15\ K) = -(902.854 \pm 1.259)\ kJ\cdot mol^{-1},$$

$$S^o_m (SiO_2, am, 298.15\ K) = (47.651 \pm 2.604)\ J\cdot K^{-1}\cdot mol^{-1}.$$

and hence a difference of (7.85 ± 0.76) kJ·mol^{-1} between the enthalpies of formation of quartz and amorphous silica, where the uncertainty is calculated from the difference in the enthalpies of solution. Section V.3.1.6 has a further short discussion of the data for SiO_2(am). As expected, these data for SiO_2(am) are close to those derived for SiO_2(vit) in Section V.3.1.5.

As alluded to above, the SIT treatment could be applied to the results of Marshall [1980MAR] giving rise to the quantities shown in Table V-27. His results at 523 and 573 K were ignored, and those at lower temperatures were truncated to ionic strengths < 3.5 mol·kg^{-1} ($NaNO_3$). The SIT plots show good linearity and illustrate that the values of $\log_{10} K^o_{s,4}(SiO_2, am)$ were in reasonable agreement with those determined directly in highly dilute solutions by other researchers.

Table V-27: Results of application of the SIT to solubility data for amorphous silica in the presence of 1:1 salts where the pressure was 0.101 MPa at 298.15 and 373.15 K, and the saturation vapour pressure of water at 423.15 and 473.15 K.

T /K	Electrolyte	m_{MeX} max.	$\Delta\varepsilon$ /kg·mol^{-1}	$\varepsilon(Si(OH)_4(aq), Me^+, X^-)$ /kg·mol^{-1}	Reference
298.15	$NaNO_3$	3.36	(0.053 ± 0.008)	(0.05 ± 0.01)	[1980MAR]
373.15		2.12	(0.046 ± 0.039)	(0.05 ± 0.04)	
423.15		3.31	(0.029 ± 0.014)	(0.03 ± 0.01)	
473.15		3.31	(0.016 ± 0.012)	(0.02 ± 0.01)	
298.15	KCl	4.81	$-(0.011 \pm 0.002)$	$-(0.011 \pm 0.002)$	[1980MAR/WAR]
298.15	KNO_3	3.76	$-(0.002 \pm 0.003)$	$-(0.002 \pm 0.003)$	
298.15	NaCl	4.98	(0.043 ± 0.003)	(0.043 ± 0.003)	
298.15	LiCl	4.36	(0.084 ± 0.006)	(0.084 ± 0.006)	
298.15	$LiNO_3$	4.57	(0.084 ± 0.004)	(0.084 ± 0.004)	

Taking the single value for $m_{Si} = (1.360 \pm 0.012) \times 10^{-3}$ mol·kg^{-1} in 1.0515 mol·kg^{-1} ($NaClO_4$) [1968JOR] and $\log_{10} K^o_{s,4}(SiO_2, am, 298.15\ K) = -(2.696 \pm 0.016)$, then $\Delta\varepsilon = \varepsilon(Si(OH)_4(aq), Na^+ + ClO_4^-) = (0.13 \pm 0.04)\ kg\cdot mol^{-1}$ where the uncertainty was increased because only one data point was measured.

From the solubility study of amorphous SiO_2 in $NaNO_3$ solutions by Marshall [1980MAR], the SIT treatment gave a value of

$$\varepsilon(Si(OH)_4(aq), Na^+ + NO_3^-) = \mathbf{(0.05 \pm 0.01)}\ kg\cdot mol^{-1}.$$

V.3.2.3 $SiO(OH)_3^-$

Grenthe *et al.* [1992GRE/FUG] applied the SIT to treat potentiometric data from [1977BUS/MES], [1981SJO/NOR] and [1983SJO/HAG] obtained at 298.15 K in NaCl and $NaClO_4$ media to extract the dissociation constants for the following equilibria:

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \quad \text{(V.24)}$$

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \quad \text{(V.25)}$$

The values so obtained are $\log_{10}K^\circ$ ((V.24), 298.15 K) = – (9.81 ± 0.02) and $\log_{10}\beta^\circ$ ((V.25), 298.15 K) = – (23.14 ± 0.09) with an average value of $\Delta\varepsilon$ = (0.04 ± 0.02) kg·mol^{-1} in these media. As an aside, Grenthe *et al.* [1992GRE/FUG] accepted the view of Baes and Mesmer [1976BAE/MES] that the conductance measurements of Ryzhenko and Khitarov [1968RYZ/KHI] are flawed so that their dissociation constants for Reactions (V.24) and (V.25) should be ignored. This opinion is upheld in this review.

An SIT treatment was carried out incorporating $\log_{10}K_m$ (V.26) values available in $NaClO_4$ solutions at 298.15 K [1959LAG], [1967BIL/ING], [1983SJO/HAG] to I_m = 2.086. This treatment resulted in $\log_{10}K^\circ$ ((V.26), 298.15 K) = (4.197 ± 0.030) and $\Delta\varepsilon$ (V.26) = – (0.113 ± 0.014) kg·mol^{-1}. Busey and Mesmer [1977BUS/MES] measured the dissociation constant for Reaction (V.24) in 0-5 mol·kg^{-1} NaCl solutions from 333.15 to 573.15 K. Accepting the $\log_{10}K^\circ$ (V.26) values derived from the SIT treatment of the data in [1977BUS/MES] at 333.15-473.15 K and the value of $\log_{10}K_m$ (V.26) = (4.27 ± 0.02) at I_m = 0.608 (NaCl) and 298.15 K calculated from Sjöberg *et al*. [1981SJO/NOR], and constraining the fit to $\log_{10}K^\circ$ ((V.26), 298 K) = 4.197, produced Equation (V.27).

$$Si(OH)_4(aq) + OH^- \rightleftharpoons SiO(OH)_3^- + H_2O(l) \quad \text{(V.26)}$$

$$\log_{10}K^\circ(\text{V.26}) = -(0.44554 \pm 0.04766) + (1384.09 \pm 14.32)(T/\text{K})^{-1} \quad \text{(V.27)}$$

The results of this regression are compared with other estimates in Table V-28 and are shown in Figure V-26.

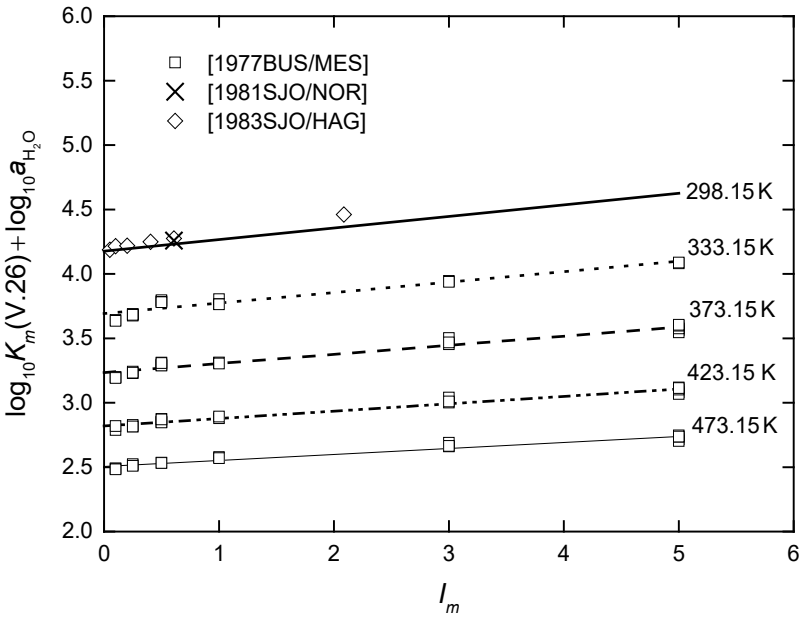
Table V-28: Comparison of our results with those from different sources in the literature for Reaction (V.26) converted to (2σ) uncertainties.

Quantity	[1959LAG] /[1967BIL/ING]	[1977BUS/MES][(a)]	[1992GRE/FUG]	From Eq. (V.27)
Ionic medium	$NaClO_4$	NaCl	NaCl/$NaClO_4$	$NaClO_4$
$\log_{10}K^\circ$	(4.19 ± 0.01)	(4.168 ± 0.022)	(4.19 ± 0.03)	(4.197 ± 0.030)
$\Delta_r H_m^\circ$ /kJ·mol^{-1}		– (30.21 ± 0.075)	– (30.21 ± 2.0)	– (26.50 ± 0.54)
$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}		– (21.5 ± 2.2)	– (21 ± 12)	– (8.5 ± 1.8)
$\Delta_r C_{p,m}^\circ$ /J·K^{-1}·mol^{-1}		(49.4 ± 7.3)		
$\Delta\varepsilon$ /kg·mol^{-1}	– (0.13 ± 0.02)		– (0.07 ± 0.05)	– (0.11 ± 0.01)

(a) These quantities were reported by Busey and Mesmer [1977BUS/MES] from an empirical fit of their data, whereas the other values in this table were obtained from application of the SIT.

The significant discrepancy in the values of $\Delta_r H_m^o$ (V.26) can be traced largely to the inclusion by Busey and Mesmer of a heat capacity term that could not be justified for the more limited dataset treated here, truncated at 473 K.

Figure V-26: Plot of apparent $\log_{10} K_m$ (V.26) *versus* ionic strength (the lines correspond to a SIT model for the non-ideality).



With the addition of the corresponding properties for water dissociation ($H_2O(l) \rightleftharpoons H^+ + OH^-$) (*viz.*, $\Delta_r G_m^o$ ((V.26), 298.15 K) = (79.920 ± 0.083) kJ·mol^{-1}, $\Delta_r H_m^o$ ((V.26), 298.15 K) = (55.815 ± 0.057) kJ·mol^{-1}), the selected standard thermodynamic quantities for Reaction (V.24) at 298.15 K are therefore:

$$\log_{10} K^o \text{ ((V.24), 298.15 K)} = \mathbf{-\ (9.804 \pm 0.033)},$$

$$\Delta_r G_m^o \text{ ((V.24), 298.15 K)} = \mathbf{(55.965 \pm 0.188)} \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((V.24), 298.15 K)} = \mathbf{(29.317 \pm 0.541)} \text{ kJ·mol}^{-1},$$

$$\Delta_r S_m^o \text{ ((V.24), 298.15 K)} = \mathbf{-\ (89.378 \pm 1.921)} \text{ J·K}^{-1}\text{·mol}^{-1}.$$

From the $\Delta\varepsilon$ (V.26) value in NaCl medium extrapolated to 298.15 K (see Appendix A entry [1977BUS/MES]), it follows that:

$\varepsilon(Na^+, SiO(OH)_3^-) = \Delta\varepsilon$ (V.26) – (0.09 ± 0.05) + $\varepsilon(Na^+, OH^-)$ +
$\varepsilon(Si(OH)_4(aq), Na^+ + Cl^-)$ = – (0.09 ± 0.05) + (0.05 ± 0.01) + (0.04 ± 0.01)
= (0.00 ± 0.05) kg·mol^{-1}
(*cf.* $\varepsilon(Na^+, SiO(OH)_3^-)$ = – (0.08 ± 0.03) kg·mol^{-1} [1992GRE/FUG]).

For the reaction in the $NaClO_4$ medium at 298.15 K in Table V-29 $\Delta\varepsilon$(V.26) = – (0.11 ± 0.01) kg·mol^{-1}, $\varepsilon(Si(OH)_4(aq), Na^+ + ClO_4^-)$ = – $\Delta\varepsilon$ (V.26) – $\varepsilon(Na^+, OH^-)$ + $\varepsilon(Na^+, SiO(OH)_3^-)$ = (0.11 ± 0.01) – (0.04 ± 0.01) + (0.00 ± 0.05) = (0.07 ± 0.05) kg·mol^{-1}. From this value of $\varepsilon(Si(OH)_4(aq), Na^+ + ClO_4^-)$ and that of (0.13 ± 0.04) kg·mol^{-1} [1968JOR] it follows that the weighted average of

$$\varepsilon(Si(OH)_4(aq), Na^+ + ClO_4^-) = \mathbf{(0.10 \pm 0.08)}\ \text{kg·mol}^{-1},$$

which is our selected value.

V.3.2.4 $SiO_2(OH)_2^{2-}$

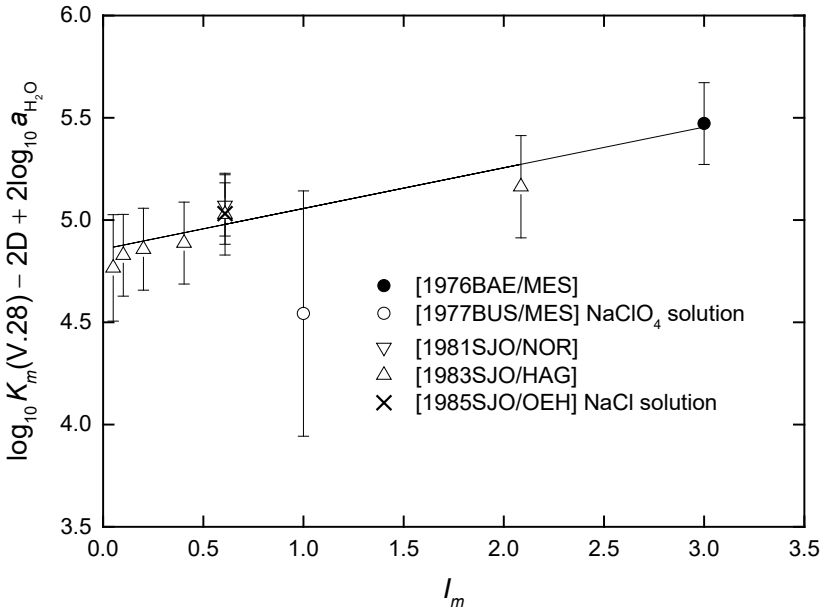
Potentiometry was used exclusively to extract the second hydrolysis constant of $Si(OH)_4(aq)$ represented by Equation (V.28) [1959LAG], [1977BUS/MES], [1981SJO/NOR], [1983SJO/HAG] and [1985SJO/OEH]. The accuracy of these results was affected adversely by the dominant presence of $SiO(OH)_3^-$ and the rapid onset of formation of polysilicates. Indeed, Busey and Mesmer [1977BUS/MES] presented different values of $\log_{10}\beta_m$ (V.28), determined in 1 mol·kg^{-1} NaCl solutions at 333.15 K, depending on which polysilicate species was thought to coexist.

$$Si(OH)_4(aq) + 2OH^- \rightleftharpoons SiO_2(OH)_2^{2-} + 2H_2O(l) \qquad \text{(V.28)}$$

Grenthe *et al.* [1992GRE/FUG] reviewed all of these results in terms of the dissociation Reaction (V.29) and used the SIT to produce the value of $\log_{10}\beta^\circ$ ((V.29), 298.15 K) = – (23.14 ± 0.09). The current review also relied on the SIT treatment but applied it to Reaction (V.28). Extrapolation of the higher temperature results of Busey and Mesmer [1977BUS/MES] resulted in a very uncertain value of $\log_{10}\beta^\circ$ ((V.28), 298.15 K) = (4.54 ± 0.60), which had little effect on the weighted SIT regression. The remaining values, all measured at 298.15 K, were taken from the original papers, including the result favoured by Baes and Mesmer [1976BAE/MES] in 3 mol·kg^{-1} $NaClO_4$ solution taken from [1959LAG]. Grenthe *et al.* also incorporated this value into their treatment of the results obtained in NaCl solutions. This seems reasonable in view of the larger uncertainties in the individual values that were adopted here from the SIT treatment by Grenthe *et al*. The results of the SIT treatment are shown in Figure V-27.

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \qquad \text{(V.29)}$$

Figure V-27: An SIT plot of $\log_{10} K_m\,(\text{V.28}) - 2D + 2\log_{10} a_{H_2O}$ *versus* I_m.



The selected value of the logarithm of the hydrolysis constant obtained from the SIT treatment is

$$\log_{10}\beta^{\circ}\,((\text{V.28}),\,298.15\ \text{K}) = (4.857 \pm 0.082),$$

$$\text{or}\ \log_{10}\beta^{\circ}\,((\text{V.29}),\,298.15\ \text{K}) = \mathbf{-\,(23.145 \pm 0.090)},$$

which is virtually identical to that proposed by Grenthe *et al*. with a slightly smaller uncertainty.

The enthalpy of reaction given by Grenthe *et al.* of $\Delta_r H_m^{\circ}\,((\text{V.29}),\,298.15\ \text{K}) = (75 \pm 15)\ \text{kJ·mol}^{-1}$ is accepted without comment. The resulting selected thermodynamic values are:

$$\Delta_r G_m^{\circ}(\,(\text{V.29}),\,298.15\ \text{K}) = \mathbf{(132.113 \pm 0.514)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^{\circ}(\,(\text{V.29}),\,298.15\ \text{K}) = \mathbf{(75.000 \pm 15.000)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^{\circ}(\,(\text{V.29}),\,298.15\ \text{K}) = \mathbf{-\,(191.558 \pm 50.340)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

A value of $\Delta\varepsilon\,(\text{V.28}) = -\,(0.20 \pm 0.07)\ \text{kg·mol}^{-1}$ in NaCl medium was obtained from the SIT treatment and led to: $\varepsilon(Na^+, SiO_2(OH)_2^{2-}) =$

$\Delta\varepsilon + 2\varepsilon(Na^+, OH^-) + \varepsilon(Si(OH)_4(aq), Na^+ + Cl^-) = -(0.20 \pm 0.07) + 2\ (0.04 \pm 0.01) + (0.05 \pm 0.01) = -(0.07 \pm 0.07)$ kg·mol^{-1} (*cf.* $-(0.10 \pm 0.07)$ kg·mol^{-1} [1992GRE/FUG]).

Therefore, the accepted values at 298.15 K are:

$$\varepsilon(Na^+, SiO(OH)_3^-) = \mathbf{-\ (0.00 \pm 0.05)}\ \text{kg·mol}^{-1},$$

$$\varepsilon(Na^+, SiO_2(OH)_2^{2-}) = \mathbf{-\ (0.07 \pm 0.07)}\ \text{kg·mol}^{-1}.$$

V.3.2.5 Polysilicates

Busey and Mesmer [1977BUS/MES] interpreted their potentiometric results, obtained in a 1 mol·kg^{-1} NaCl solution from 373.15 to 561 K with Si(IV) concentrations ranging from 0.005 to 0.05 mol·kg^{-1} in terms of one additional polysilicate species. They found a slight statistical advantage in including the $Si_4O_6(OH)_6^{2-}$ species (Reaction (V.30)) rather than $Si_3O_6(OH)_3^{3-}$, $Si_5O_9(OH)_4^{2-}$ or $Si_6O_{10}(OH)_7^{3-}$, but considered the results too ambiguous to assign thermodynamic quantities to this polysilicate species. Conversely, Lagerström [1959LAG] speculated that eight polysilicates were present in 3 mol·kg^{-1} $NaClO_4$ at 298.15 K during his potentiometric titrations. Felmy *et al*. [2001FEL/CHO] cast doubt on the significance of Reaction (V.30).

$$4Si(OH)_4(aq) + 2OH^- \rightleftharpoons Si_4O_6(OH)_6^{2-} + 6H_2O(l) \qquad \text{(V.30)}$$

Clearly, controversy exists as to the nature and stability of such multinuclear species. Grenthe *et al.* [1992GRE/FUG] based their analysis on the detailed potentiometric and ^{29}Si NMR study conducted by Sjöberg *et al.* [1985SJO/OEH] also with consideration of results from [1977BUS/MES], [1981SJO/NOR] and [1983SJO/HAG]. The prevailing experimental conditions in the 1985 study were $I_c = 0.6$ (NaCl); $c_{Si} = (3\text{-}48) \times 10^{-3}$, $2.5 \le pH_c \le 12.2$ (potentiometric titrations), $10.35 \le pH_c \le 12.02$ (NMR measurements) and the temperature was constant at 298.15 K. The reaction scheme for the formation of polysilicates proposed by Sjöberg *et al.* [1985SJO/OEH] involved Reactions (V.31) to (V.36) (Reactions (V.24) and (V.25) were included in their treatment with the equilibrium constant for the former being taken from [1983SJO/HAG]). However, Felmy *et al.* [2001FEL/CHO] identified another four polysilicates that should be significant species under these conditions as well as those prevailing in their more sophisticated NMR and potentiometric experiments.

$$2Si(OH)_4(aq) \rightleftharpoons Si_2O_2(OH)_5^- + H^+ + H_2O(l) \qquad \text{(V.31)}$$

$$2Si(OH)_4(aq) \rightleftharpoons Si_2O_3(OH)_4^{2-} + 2H^+ + H_2O(l) \qquad \text{(V.32)}$$

$$3Si(OH)_4(aq) \rightleftharpoons Si_3O_6(OH)_3^{3-}(cyclic) + 3H^+ + 3H_2O(l) \qquad \text{(V.33)}$$

$$3Si(OH)_4(aq) \rightleftharpoons Si_3O_5(OH)_5^{3-}(linear) + 3H^+ + 2H_2O(l) \qquad \text{(V.34)}$$

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_7(OH)_5^{3-} + 3H^+ + 4H_2O(l) \qquad \text{(V.35)}$$

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_8(OH)_4^{4-}(cyclic) + 4H^+ + 4H_2O(l) \qquad (V.36)$$

The NMR measurements reported by Sjöberg *et al.* [1985SJO/OEH] gave a unique perspective on Reaction (V.37), because analysis of the potentiometric data alone could not distinguish between Reactions (V.33) and (V.34).

$$Si_3O_6(OH)_3^{3-}\ (cyclic) + H_2O(l) \rightleftharpoons Si_3O_5(OH)_5^{3-}\ (linear) \qquad (V.37)$$

Grenthe *et al*. [1992GRE/FUG] accepted the speciation scheme and formation constants in [1985SJO/OEH] and utilised SIT equations to extrapolate the formation constants to $I = 0$. The lack of the prerequisite ion interaction coefficients necessitated using surrogate values from the aqueous sodium phosphate system tabulated in the TDB NEA:

$$\varepsilon(Na^+, Si_2O_2(OH)_5^-) \approx \varepsilon(Na^+, H_2PO_4^-) = -(0.08 \pm 0.04)\ kg{\cdot}mol^{-1};$$

$$\varepsilon(Na^+, Si_2O_3(OH)_4^{2-}) \approx \varepsilon(Na^+, HPO_4^{2-}) = -(0.15 \pm 0.06)\ kg{\cdot}mol^{-1};$$

$$\varepsilon(Na^+, Si_3O_6(OH)_3^{3-}) = \varepsilon(Na^+, Si_3O_5(OH)_5^{3-}) = \varepsilon(Na^+, Si_4O_7(OH)_5^{3-}) \approx$$
$$\approx \varepsilon(Na^+, PO_4^{3-}) = -(0.25 \pm 0.03)\ kg{\cdot}mol^{-1};$$

$$\varepsilon(Na^+, Si_4O_8(OH)_4^{4-}) \approx \varepsilon(Na^+, P_2O_7^{4-}) = -(0.26 \pm 0.05)\ kg{\cdot}mol^{-1};$$

and

$$\varepsilon(H^+, Cl^-) = (0.12 \pm 0.01)\ kg{\cdot}mol^{-1}.$$

In an attempt to test this approach, a second route involving the equivalent hydrolysis reactions (see Appendix A entry for [1985SJO/OEH]) was invoked in this review to reduce the magnitude of the activity coefficient approximation, which necessitated also including the value $\varepsilon(Na^+, OH^-) = (0.04 \pm 0.01)\ kg{\cdot}mol^{-1}$. The proposed species and their reported molal formation constants are shown in Table V-29 (taken from Table 4 in [1985SJO/OEH]).

Table V-29: Summary of the formation constants from [1985SJO/OEH] and [2001FEL/CHO] at 298.15 K with uncertainties adjusted to 2σ.

Species formed	$\log_{10}\beta_m$ $I_m = 0.608$	$\log_{10}\beta^\circ$ Indirect [a]	$\log_{10}\beta^\circ$ Direct [b]	$\log_{10}\beta^\circ$ [1992GRE/FUG]	$\log_{10}\beta^\circ$ [2001FEL/CHO]
$SiO(OH)_3^-$	– 9.467 [c]		– (9.81 ± 0.11)	– (9.81 ± 0.02)	– 9.82 [d]
$SiO_2(OH)_2^{2-}$	– (22.11 ± 0.01)		– (23.14 ± 0.08)	– (23.14 ± 0.09)	– 23.27 [d]
$Si_2O_3(OH)_4^{2-}$	– (17.99 ± 0.04)	– 18.85	– 19.05	– (19.0 ± 0.3)	– 19.4
$Si_2O_5(OH)_5^-$	– (7.74 ± 0.06)	– 8.02	– 8.03	– (8.1 ± 0.3)	– 8.50
$Si_3O_6(OH)_3^{3-}$ cyclic	– (26.42 ± 0.06)	– 28.29	– 28.57	– (28.6 ± 0.3)	– 29.3
$Si_3O_5(OH)_5^{3-}$ linear	– (25.39 ± 0.04	– 27.25	– 27.54	– (27.5 ± 0.3)	– 29.4
$Si_4O_7(OH)_5^{3-}$ cyclic	– (23.42 ± 0.04)	– 25.30	– 25.58	– (25.5 ± 0.3)	

(Continued on next page)

Table V-29 (continued)

Species formed	$\log_{10}\beta_m$ $I_m = 0.608$	$\log_{10}\beta^\circ$ Indirect [a]	$\log_{10}\beta^\circ$ Direct [b]	$\log_{10}\beta^\circ$ [1992GRE/FUG]	$\log_{10}\beta^\circ$ [2001FEL/CHO]
$Si_4O_7(OH)_6^{4-}$ linear					– 39.1
$Si_4O_8(OH)_4^{4-}$ cyclic	– (32.80 ± 0.06)	– 35.98	– 36.36	– (36.3 ± 0.5)	– 39.2
$Si_5O_{10}(OH)_5^{5-}$ [e]	≤ – 41.5				
$Si_4O_8(OH)_4^{4-}$ sub					– 39.1
$Si_4O_6(OH)_6^{2-}$					– 15.6
$Si_6O_{15}^{6-}$ prismatic					– 61.8

(a) Obtained indirectly *via* the SIT treatment of the hydrolysis reactions.
(b) Repeating the direct approach taken by Grenthe *et al.* [1992GRE/FUG].
(c) Taken from [1981SJO/NOR].
(d) Taken from [1986HER/MIL].
(e) The exact stoichiometry was not given by Sjöberg *et al.* [1985SJO/OEH] and is only given here as one of a number of possibilities.

Felmy *et al*. [2001FEL/CHO] took an alternative approach by utilising the Pitzer ion interaction model whereby ion interaction parameters for the polysilicates were estimated from known values for like-charged anions to treat existing experimental data to $I_c \leq 3$. These authors also reported results of molecular dynamics calculations, which included interatomic potentials for $NaOH\text{-}H_2O\text{-}SiO_2$ solutions. In addition to Reactions (V.31)-(V.34) and (V.36), their analysis strongly supported inclusion of the following four reactions (where sub represents "substituted tetramer"):

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_7(OH)_6^{4-}(linear) + 4H^+ + 3H_2O(l) \quad (V.38)$$

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_8(OH)_4^{4-}(sub) + 4H^+ + 4H_2O(l) \quad (V.39)$$

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_6(OH)_6^{2-} + 2H^+ + 4H_2O(l) \quad (V.40)$$

$$6Si(OH)_4(aq) \rightleftharpoons Si_6O_{15}^{6-}(prismatic) + 6H^+ + 9H_2O(l) \quad (V.41)$$

The formation constants derived by Felmy *et al.* are also given in Table V-29 while noting that inclusion of the $Si_4O_6(OH)_6^{2-}$ species gave only a minor improvement in their fit.

Considering the results available prior to those in [2001FEL/CHO], it is apparent from the values of $\log_{10}\beta^\circ$ ((V.31)-(V.36)) in Table V-29 that there are only relatively minor differences within the uncertainties assigned by Grenthe *et al.*, resulting from the different calculation routes taken, perhaps with the exception of that for the cyclic $Si_4O_8(OH)_4^{4-}$, which was therefore given a larger uncertainty. It is noteworthy that Baes and Mesmer [1976BAE/MES] tabulated a value of $\log_{10}\beta_m$ ((V.36), 298.15 K) of – 32.48 for this reaction derived from their treatment of a more restricted reaction scheme than that proposed by Lagerström [1959LAG] in

3 mol·kg^{-1} $NaClO_4$. Admittedly, it is even more tenuous to apply the same activity coefficient correction to this high ionic strength value, but, nevertheless, the result is $\log_{10}\beta^{o}$ ((V.36), 298.15 K) ≈ -35.7, which is not unreasonably different from that derived for a much lower ionic strength NaCl medium. At this point it must be acknowledged that the ion interaction parameters for the $Na^{+}-H_nPO_4^{(3-n)-}$ system used as a surrogate are definitely ionic strength dependent, but within the large uncertainties introduced by adopting these surrogate values, this correction can be ignored at $I_m = 0.608$ for the data in [1985SJO/OEH].

It is beyond dispute that the ^{29}Si NMR results of Felmy *et al.* [2001FEL/CHO], while confirming for the most part the earlier NMR work [1985SJO/OEH] (which could not account for all the silicon in solution) have established the importance of four additional polysilicate species in ≤ 0.1 mol·kg^{-1} Si(IV) solutions at ambient temperatures. Their analysis also showed the importance of the protonated dimer in Lagerström's [1959LAG] potentiometric data. The Pitzer ion interaction model appears to describe adequately the behaviour of Si(IV) solutions to $I_c < 3$ as evidenced by its prediction of SiO_2(am) solubility. However, application of the Pitzer model also relied on the use of known ion interaction coefficients of surrogate anions in the absence of such data for the polysilicates. This review recommends cautious use of the polysilicate formation constants derived by Felmy *et al*. and given in Table V-29 noting that their assignment of speciation was supported in a more recent study, which combined ^{29}Si NMR measurements and sophisticated NMR spectral simulations [2006CHO/FEL].

The $\Delta_f G_m^{o}$ values for the silicate and polysilicate anions derived from the SIT and Pitzer ion interaction treatments described above (columns 4 and 6, respectively, in Table V-29) and the selected value of $\Delta_f G_m^{o}(Si(OH)_4, aq, 298.15\ K) = -(1309.201 \pm 1.017)$ kJ·mol^{-1}, are compared in Table V-30 with those from [1992GRE/FUG]. The uncertainties selected by Grenthe *et al.* have been rounded up to give a more realistic indicator of their accuracy, but those for the data in [2001FEL/CHO] are unavailable at this time.

Table V-30: Summary of the Gibbs energies of formation of the silicate and polysilicates at 298.15 K. Values selected by this review are highlighted in bold.

Species formed	$\Delta_f G_m^{o}$ /kJ·mol^{-1} [1992GRE/FUG]	$\Delta_f G_m^{o}$ /kJ·mol^{-1} re-evaluation	$\Delta_f G_m^{o}$ /kJ·mol^{-1} [2001FEL/CHO]
$Si(OH)_4$(aq)	– (1307.735 ± 1.156)	**– (1309.201 ± 1.017)**	
$SiO(OH)_3^-$	– (1251.740 ± 1.162)	**– (1253.236 ± 1.034)**	– 1253.1
$SiO_2(OH)_2^{2-}$	– (1175.651 ± 1.265)	**– (1177.088 ± 1.140)**	– 1176.4
$Si_2O_3(OH)_4^{2-}$	– (2269.878 ± 2.878)	**– (2272.500 ± 3.000)**	– 2270.5
$Si_2O_2(OH)_5^-$	– (2332.096 ± 2.878)	**– (2335.400 ± 3.000)**	– 2332.7
$Si_3O_6(OH)_3^{3-}$ cyclic	– (3048.536 ± 3.870)	**– (3053.100 ± 4.000)**	– 3048.9
$Si_3O_5(OH)_5^{3-}$ linear	– (3291.955 ± 3.869)	**– (3296.100 ± 4.000)**	– 3285.5
$Si_4O_7(OH)_5^{3-}$	– (4136.826 ± 4.934)	**– (4142.200 ± 5.000)**	
$Si_4O_8(OH)_4^{4-}$ cyclic	– (4075.179 ± 5.437)	**– (4080.700 ± 6.000)**	– 4064.5

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Table V-30 (continued)

Species formed	$\Delta_f G_m^o$ /kJ·mol^{-1} [1992GRE/FUG]	$\Delta_f G_m^o$ /kJ·mol^{-1} re-evaluation	$\Delta_f G_m^o$ /kJ·mol^{-1} [2001FEL/CHO]
$Si_4O_7(OH)_6^{4-}$ linear			– 4302.2
$Si_4O_8(OH)_4^{4-}$ sub			– 4065.1
$Si_4O_6(OH)_6^{2-}$			– 4199.2
$Si_6O_{15}^{6-}$ prismatic			– 5368.2

The $\Delta_f G_m^o$ values given in column 3 of Table V-30 are our new selected values but those in column 4 may prove to be more reliable in the future.

Chapter V.4

Aqueous borate species

V.4.1 Monoborate species

V.4.1.1 Boric acid and the borate anion

The current NEA TDB selected auxiliary data for $B(OH)_3(aq)$ are the CODATA values [1989COX/WAG]:

$$\Delta_f G^o_m(B(OH)_3, aq, 298.15\ K) = \mathbf{-\ (969.268 \pm 0.820)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^o_m(B(OH)_3, aq, 298.15\ K) = \mathbf{-\ (1072.800 \pm 0.800)}\ kJ{\cdot}mol^{-1},$$

$$S^o_m(B(OH)_3, aq, 298.15\ K) = \mathbf{(162.400 \pm 0.600)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

State-of-the-art dielectric relaxation experiments involving NaOH and $NaB(OH)_4$ solutions should readily detect contact Na^+-$B(OH)_4^-$ ion pairs as noted by Buchner *et al*. [1999BUC/HEF]. Therefore, the lack of a detectable solute relaxation process indicated that the ion-pair lifetimes must be similar or even shorter than the water-relaxation time. Ion association of alkali metal cations with borate anions is not treated explicitly in this review. Note that a more recent conductivity study [2016ARC/FER] identified sodium and potassium borate ion-pair formation to near supercritical temperatures, but we believe these species are sufficiently weak at ambient conditions (*i.e.*, $\log_{10} K_1^o \sim -0.2$ at 298.15 K) that they can be assimilated in the SIT activity-coefficient models.

The value of $\Delta_r H^o_m$ for the hydrolysis reaction,

$$B(OH)_3(aq) + OH^- \rightleftharpoons B(OH)_4^- \qquad \text{(V.42)}$$

was derived from the enthalpies of solution of $B(OH)_3(cr)$ in water and sodium hydroxide solutions measured at various ionic strengths between 288 and 330 K [1982DEV/YEF2]. Their value of $\Delta_r H^o_m$ ((V.42), 298.15 K) = – (42.468 ± 0.084) kJ·mol^{-1} agrees reasonably well with that suggested in a review [2000PAL/BEN], *viz.* $\Delta_r H^o_m$ ((V.42), 298.15 K) = – (42.7 ± 0.2) kJ·mol^{-1}.

The hydrolysis of boric acid was studied by Mesmer *et al.* [1972MES/BAE] with a hydrogen-electrode concentration cell over the temperature range 323 to 563 K as a function of ionic strength up to 1.02 mol·kg^{-1} (KCl), while the molality of boric

acid was kept ≤ 0.02 mol·kg^{-1}. For re-treatment with the SIT, the upper temperature was limited to 473 K. The isocoulombic nature of Reaction (V.42) eliminated the need to include the Debye-Hückel expression. Much earlier, Owen and King [1943OWE/KIN] had used a Harned-type cell for the same purpose from 278.15 to 328.15 K at ionic strengths from 0.02 to 3.0 mol·kg^{-1} (NaCl), while the boric acid molality was varied from 0.005 to 0.03 mol·kg^{-1} at each ionic strength. In view of the fact that $\varepsilon(Na^+, B(OH)_4^-)$ is listed by the NEA TDB as being ionic strength dependent, two SIT treatments were implemented (see [1972MES/BAE] in Appendix A). The resulting $\log_{10} K_1^\circ$ values differed by less than the combined uncertainties.

Manov *et al*. [1944MAN/DEL] treated their potentiometric data obtained with a Harned-type cell in a similar manner to that applied in [1934OWE] and [1935OWE]. They chose to treat the acid dissociation equilibrium (Reaction (V.43))

$$B(OH)_3(aq) + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+ \qquad \text{(V.43)}$$

Their reported average values of $\log_{10} K_1^\circ$ (V.43) were accepted for inclusion in the overall data treatment after conversion to $\log_{10} K_1^\circ$ (V.42) and are shown in Table V-31. Two potentiometric studies [1973DYR/HAN], [1996KHA/ISS] gave disparate results that could not be rationalised simply on the basis of activity coefficient models; perhaps their solutions were corrupted by uptake of atmospheric CO_2. Another study [1992JUM/ABD] used circular reasoning to establish an artificially reasonable value of $\log_{10} K_1^\circ$ (V.43) and was, therefore, rejected.

Table V-31: Summary of data for the hydrolysis of boric acid according to Reaction (V.42) obtained from experimental studies, reviews and re-evaluations where the uncertainties (2σ) are derived from application of the SIT model over the listed temperature ranges (units: $\Delta_r H_m^o$, kJ·mol^{-1}; $\Delta_r S_m^o$, J·K^{-1}·mol^{-1}; $\Delta_r C_{p,m}^o$, J·K^{-1}·mol^{-1}).

T /K	Ionic medium	I_m max	$\log_{10} K_1^o$ reported [(a)]	$\log_{10} K_1^o$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	$\log_{10} K_1^o$ Equation (V.44)	Method	Reference
283.15	NaCl	0.097	(5.154 ± *0.020*)			5.161	pot	[1934OWE]
288.15			(5.019 ± *0.020*)			5.024		
293.15			(4.888 ± *0.020*)			4.891		
298.15			(4.765 ± *0.020*)			4.764		
303.15			(4.642 ± *0.020*)			4.642		
313.15			(4.412 ± *0.020*)			4.411		
323.15			(4.196 ± *0.020*)			4.197		
278.15	KI	0.01	(5.294 ± *0.015*)			5.305	pot	[1935OWE]
283.15			(5.154 ± *0.015*)			5.161		
288.15			(5.020 ± *0.015*)			5.024		
293.15			(4.889 ± *0.015*)			4.891		
298.15			(4.764 ± *0.015*)			4.764		
303.15			(4.641 ± *0.015*)			4.642		
308.15			(4.524 ± *0.015*)			4.524		
313.15			(4.412 ± *0.015*)			4.411		
278.15	NaCl	3.00	(5.296 ± *0.020*)	(5.311 ± 0.020)	(0.140 ± 0.004)	5.305	pot	[1943OWE/KIN]
283.15			(5.153 ± *0.020*)	(5.163 ± 0.020)	(0.138 ± 0.013)	5.161		
288.15			(5.018 ± *0.015*)	(5.028 ± 0.014)	(0.134 ± 0.010)	5.024		
293.15			(4.888 ± *0.015*)	(4.895 ± 0.013)	(0.131 ± 0.009)	4.891		
298.15			(4.764 ± *0.015*) $\Delta_r H_m^o = -41.82$ $\Delta_r S_m^o = -49.2$ $\Delta_r C_{p,m}^o = 177$	(4.768 ± 0.012)	(0.130 ± 0.008)	4.764		

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Table V-31 (continued)

T /K	Ionic medium	I_m max	$\log_{10} K_1^o$ reported [a]	$\log_{10} K_1^o$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	$\log_{10} K_1^o$ Equation (V.44)	Method	Reference
303.15			(4.642 ± *0.015*)	(4.646 ± 0.011)	(0.128 ± 0.007)	4.642		
308.15			(4.525 ± *0.015*)	(4.527 ± 0.010)	(0.128 ± 0.007)	4.524		
313.15			(4.412 ± *0.015*)	(4.414 ± 0.009)	(0.128 ± 0.006)	4.411		
318.15			(4.303 ± *0.015*)	(4.304 ± 0.007)	(0.127 ± 0.005)	4.302		
323.15			(4.195 ± *0.015*)	(4.197 ± 0.006)	(0.128 ± 0.004)	4.197		
328.15			(4.092 ± *0.015*)	(4.094 ± 0.006)	(0.127 ± 0.004)	4.096		
273.15	NaCl, $NaB(OH)_4$	0.053	(5.439 ± *0.015*)			5.454	pot	[1944MAN/DEL]
278.15			(5.297 ± *0.015*)			5.305		
283.15			(5.156 ± *0.015*)			5.161		
288.15			(5.021 ± *0.015*)			5.024		
293.15			(4.890 ± *0.015*)			4.891		
298.15			(4.767 ± *0.015*)			4.764		
303.15			(4.644 ± *0.015*)			4.642		
308.15			(4.528 ± *0.015*)			4.524		
313.15			(4.416 ± *0.015*)			4.411		
318.15			(4.306 ± *0.015*)			4.302		
323.15			(4.199 ± *0.015*)			4.197		
328.15			(4.098 ± *0.015*)			4.096		
333.15			(4.003 ± *0.015*)			3.998		
323	KCl	1.03	(4.188 ± *0.020*)	(4.206 ± 0.015)	− (0.098 ± 0.030)	4.197	pot	[1972MES/BAE]
373			(3.314 ± *0.020*)	(3.335 ± 0.008)	− (0.091 ± 0.014)	3.331		
423 [b]			(2.697 ± *0.020*)	(2.712 ± 0.008)	− (0.078 ± 0.012)	2.719		
473 [b]			(2.267 ± *0.020*)	(2.282 ± 0.010)	− (0.063 ± 0.016)	2.279		

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Table V-31 (continued)

T /K	Ionic medium	I_m max	$\log_{10}K_1^o$ reported [a]	$\log_{10}K_1^o$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	$\log_{10}K_1^o$ Equation (V.44)	Method	Reference
→ 298.15 [d]			4.76 $\Delta_r H_m^o = -(42.3 \pm 0.3)$ $\Delta_r S_m^o = -(51 \pm 9)$ $\Delta_r C_{p,m}^o = (28.0 \pm 1.1)$			4.756	est	
298.15	NaOH	I_c 2.5	$\Delta_r H_m^o = -(42.47 \pm 0.08)$ (4.757 ± 0.015)				cal	[1982DEV/YEF2]
298.15			$\Delta_r H_m^o = -(42.7 \pm 0.2)$ $\Delta_r S_m^o = -(51.1 \pm 0.8)$ $\Delta_r C_{p,m}^o = (28 \pm 7)$			4.756	est	[2000PAL/BEN]
298.15		< 0.1	(4.92 ± 0.02) [c]				con	[2016ARC/FER]

(a) The experimental uncertainties in the reported values of $\log_{10}K_1^o$ are shown in italics because they were estimated by this review and in all cases exceeded the statistical uncertainties given in the original papers.
(b) At or slightly greater than saturation vapour pressure.
(c) Measurements were made at 20 MPa up to 623 K.
(d) → 298.15: values were extrapolated to 298.15 K.

The best weighted fit of the $\log_{10}K_1^\circ$ values derived at low ionic strengths in [1934OWE], [1935OWE] and [1944MAN/DEL], and those resulting from SIT treatments of the data to high ionic strengths in [1943OWE/KIN] and [1972MES/BAE], and the $\Delta_r H_m^\circ$ value in [1982DEV/YEF2] gave Equation (V.44) as illustrated in Figure V-28.

$$\log_{10}K_1^\circ(\text{V.42}) = -(4.93117 \pm 0.06247) + (2548.59 \pm 10.98)\,(T/\text{K})^{-1} + (3.8466 \pm 0.0873)\times 10^{-3}\,(T/\text{K}) \qquad \text{(V.44)}$$

The resulting thermodynamic quantities derived from Equation (V.44) at 298.15 K are: $\log_{10}K_1^\circ$ ((V.42), 298.15 K) = (4.764 ± 0.077), $\Delta_r G_m^\circ$ ((V.42), 298.15 K) = – (27.191 ± 0.440) kJ·mol^{-1}, $\Delta_r H_m^\circ$ ((V.42), 298.15 K) = – (42.246 ± 0.257) kJ·mol^{-1}, $\Delta_r S_m^\circ$ ((V.42), 298.15 K) = – (50.494 ± 1.556) J·K^{-1}·mol^{-1}, $\Delta_r C_{p,m}^\circ$ ((V.42), 298.15 K) = (43.913 ± 0.996) J·K^{-1}·mol^{-1}. The value of $\Delta_r C_{p,m}^\circ$ ((V.42), 298.15 K) differs notably from the heat capacity measurements described in the Appendix A entry for [1988HOV], $\Delta_r C_{p,m}^\circ$ ((V.42), 298.15 K) = (24 ± 3) J·K^{-1}·mol^{-1}. Equation (V.44) leads to $\log_{10}K_1^\circ$ ((V.43), 298.15 K) = – (9.24 ± 0.06).

To investigate the difference when the temperature-independent value of $\Delta_r C_{p,m}^\circ$ ((V.42), 298.15 K) = 24 J·K^{-1}·mol^{-1} [1988HOV], an unweighted fit of the same data set excluding this $\Delta_r C_{p,m}^\circ$ ((V.42), 298.15 K) and $\Delta_r H_m^\circ$ ((V.42), 298.15 K) values gave the following equation:

$$\log_{10}K_1^\circ(\text{V.42}) = -(3.9168 \pm 0.0044) + (2405.13 \pm 27.54)\,(T/\text{K})^{-1} + (2.1023 \pm 0.2628)\times 10^{-3}\,(T/\text{K}) \qquad \text{(V.45)}$$

The resulting and accepted thermodynamic quantities at 298.15 K are

$$\log_{10}K_1^\circ\text{((V.42), 298.15 K)} = \mathbf{(4.777 \pm 0.121)},$$

$$\Delta_r G_m^\circ\text{((V.42), 298.15 K)} = \mathbf{-(27.267 \pm 0.692)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^\circ\text{((V.42), 298.15 K)} = \mathbf{-(42.468 \pm 0.691)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^\circ\text{((V.42), 298.15 K)} = \mathbf{-(50.986 \pm 3.001)}\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C_{p,m}^\circ\text{((V.42), 298.15 K)} = \mathbf{(24.000 \pm 3.000)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

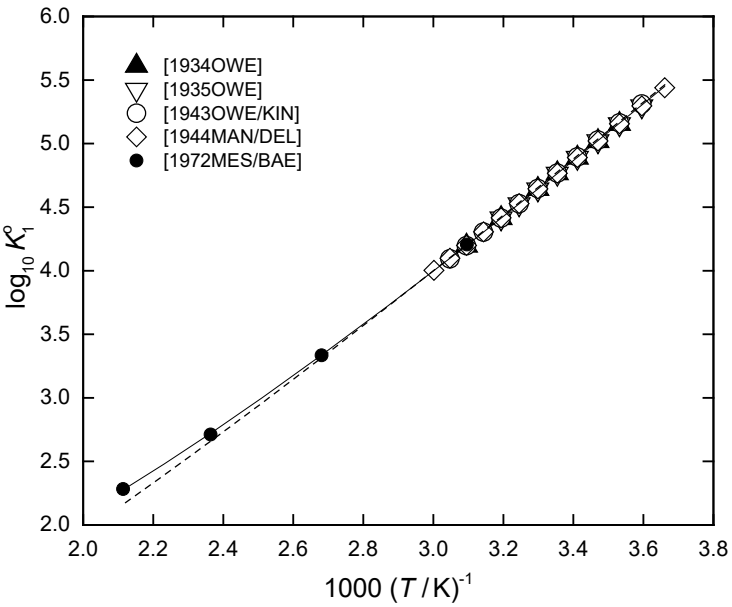
Similar thermodynamic quantities were obtained by Mesmer *et al.* [1972MES/BAE], who extrapolated their high-temperature potentiometric results in KCl media to 298.15 K, as well as by interpolation of the low-temperature data by Owen and King [1943OWE/KIN]. From this assessment and consideration of the CODATA values for $B(OH)_3(aq)$ and OH^-, this review selects:

$$\Delta_f G_m^\circ(B(OH)_4^-, 298.15\ \text{K}) = \mathbf{-(1153.755 \pm 1.075)}\ \text{kJ·mol}^{-1},$$

$$\Delta_f H_m^\circ(B(OH)_4^-, 298.15\ \text{K}) = \mathbf{-(1345.283 \pm 1.058)}\ \text{kJ·mol}^{-1},$$

$$S_m^\circ(B(OH)_4^-, 298.15\ \text{K}) = \mathbf{(100.514 \pm 3.067)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

Figure V-28: Temperature dependence of $\log_{10} K_1^\circ$ (V.42) where the solid and dashed lines were generated by Equation (V.44) and Equation (V.45).



After completion of this review, Arcis *et al.* [2017ARC/FER] reported the ionisation constant of boric acid from 298 to 623 K based on conductivity and innovative Raman measurements. They demonstrated that Equation (V.44) represents their data very well over their entire temperature range.

V.4.1.1.1 Ion interaction coefficients involving the borate anion

From an SIT treatment of the $\log_{10} K_1$ (V.42) values in KCl solutions [1972MES/BAE] between 323 and 473 K with $I_m \leq 1.0$, a simple extrapolation of the $\Delta\varepsilon$ value gives $\Delta\varepsilon = -(0.12 \pm 0.03)$ kg·mol^{-1} at 298.15 K with no detectable dependence on I_m. It follows that if the ion interactions with $B(OH)_3$(aq) can be ignored:

$$\varepsilon(K^+, B(OH)_4^-) = \Delta\varepsilon + \varepsilon(K^+, OH^-) = -(0.12 \pm 0.03) + (0.09 \pm 0.01)$$
$$= -\mathbf{(0.03 \pm 0.03)}\,\text{kg·mol}^{-1}.$$

From an SIT treatment of the $\log_{10}K_1$ (V.42) values in NaCl solutions [1943OWE/KIN] between 278.15 and 328.15 K, an average $\Delta\varepsilon$ value of $-(0.13 \pm 0.01)$ kg·mol^{-1} was obtained and hence:

$$\varepsilon(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = \Delta\varepsilon + \varepsilon(\mathrm{Na^+}, \mathrm{OH^-}) = -(0.13 \pm 0.01) + (0.04 \pm 0.01)$$
$$= \mathbf{-(0.09 \pm 0.01)}\ \mathrm{kg\cdot mol^{-1}}.$$

at 298.15 K.

This compares to $\varepsilon(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = -(0.07 \pm 0.05)$ kg·mol^{-1} in the current NEA TDB listing, where $\varepsilon(\mathrm{Na^+}, \mathrm{B(OH)_4^-})$ was also considered to be ionic strength dependent. For the latter situation, we derive:

$$\varepsilon_1(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = \mathbf{-(0.156 \pm 0.001)}\ \mathrm{kg\cdot mol^{-1}},$$

$$\varepsilon_2(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = \mathbf{(0.053 \pm 0.001)}\ \mathrm{kg\cdot mol^{-1}}$$

at 298.15 K, which are our selected values, compared to the current selected values:

$$\varepsilon_1(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = -(0.092 \pm 0.002)\ \mathrm{kg\cdot mol^{-1}},$$

$$\varepsilon_2(\mathrm{Na^+}, \mathrm{B(OH)_4^-}) = (0.103 \pm 0.005)\ \mathrm{kg\cdot mol^{-1}}.$$

These revised ion interaction coefficients are the recommended values.

V.4.2 Polyborate species

The opinion generally accepted in the literature is that polyborates tend to be insignificant at concentrations ≤ 0.025 mol·dm^{-3}. However, some disagreement existed in the past between the sets of results from two laboratories as to the composition of the polyborates anions formed, while those from a third source [1970SPE], have been discredited, because the borate concentration was not varied so that these results are therefore ignored in this review. The Swedish group published a series of papers based on potentiometric titrations performed at 298.15 K over a wide range of borate concentrations [1957ING/LAG], [1962ING], [1963ING], [1963ING2] whereas Mesmer *et al.* [1972MES/BAE] later considered 14 possible boron species in analyses of their potentiometric results obtained from 323.15 to 473.15 K at boron concentrations from 0.03 to 0.6 mol·kg^{-1}. These data are all summarised in Table V-32. It must be noted that the abbreviation used by Mesmer *et al*. was *x*:*y* for the species $\mathrm{B}_x(\mathrm{OH})_{3x+y}^{y-}$, whereas in this review these species are referred to as *y*, *z*, *x*, $\mathrm{B}_x\mathrm{O}_y(\mathrm{OH})_z^{(3x-2y-z)}$. From these two main sources, the pertinent equilibria for the formation of polyborate anions are:

$$2\mathrm{B(OH)_3(aq)} + \mathrm{OH^-} \rightleftharpoons \mathrm{B_2O_3OH^-} + 3\mathrm{H_2O(l)}\ (\text{or } \mathrm{B_2(OH)_7^-}) \qquad \text{(V.46)}$$

$$2\mathrm{B(OH)_3(aq)} + \mathrm{OH^-} \rightleftharpoons \mathrm{B_2O(OH)_5^-} + \mathrm{H_2O(l)}\ (\text{or } \mathrm{B_2(OH)_7^-}) \qquad \text{(V.47)}$$

$$3\mathrm{B(OH)_3(aq)} + \mathrm{OH^-} \rightleftharpoons \mathrm{B_3O_3(OH)_4^-} + 3\mathrm{H_2O(l)}\ (\text{or } \mathrm{B_3(OH)_{10}^-}) \qquad \text{(V.48)}$$

$$3B(OH)_3(aq) + 2OH^- \rightleftharpoons B_3O_3(OH)_5^{2-} + 3H_2O(l) \text{ (or } B_3(OH)_{11}^{2-}) \quad \text{(V.49)}$$

$$4B(OH)_3(aq) + 2OH^- \rightleftharpoons B_4O_5(OH)_4^{2-} + 5H_2O(l) \text{ (or } B_4(OH)_{14}^{2-}) \quad \text{(V.50)}$$

$$5B(OH)_3(aq) + OH^- \rightleftharpoons B_5O_5(OH)_6^- + 5H_2O(l) \text{ (or } B_5(OH)_{16}^-) \quad \text{(V.51)}$$

$$5B(OH)_3(aq) + 3OH^- \rightleftharpoons B_5O_5(OH)_8^{3-} + 5H_2O(l) \text{ (or } B_5(OH)_{18}^{3-}) \quad \text{(V.52)}$$

$$5B(OH)_3(aq) + OH^- \rightleftharpoons B_5O_6(OH)_4^- + 6H_2O(l) \text{ (or } B_5(OH)_{16}^-) \quad \text{(V.53)}$$

$$5B(OH)_3(aq) + 3OH^- \rightleftharpoons B_5O_6(OH)_6^{3-} + 6H_2O(l) \text{ (or } B_5(OH)_{18}^{3-}) \quad \text{(V.54)}$$

All of these studies included the hydrolysis Reaction (V.42). The species $B_3O_3(OH)_4^-$ was found in all the fitting schemes. Conversely, inclusion of the species $B_2O_3OH^-$ was claimed [1972MES/BAE] to give a much better fit at high temperatures, but it was not included in any of the schemes preferred in the Swedish studies conducted at 298.15 K. Baes and Mesmer suggested the presence of a third polyborate species, either $B_4O_5(OH)_4^{2-}$ or $B_5O_5(OH)_8^{3-}$, in as much as it improved the quality of fit slightly so that two alternative reaction schemes were presented. The structure of the $B_3O_3(OH)_4^-$ species was proposed [1957ING/LAG] to be a six-membered ring, which is consistent with the five rules suggested for the structures of crystalline hydrated polyborates [1960EDW/ROS].

Raman spectroscopic measurements at 298.15 K and *ab-initio* calculations were used to identify the stoichiometry and structure of aqueous polyborates [2011ZHO/FAN]. These calculations demonstrated the existence of six species in solution: $B(OH)_3(aq)$, $B(OH)_4^-$, $B_3O_3(OH)_4^-$, $B_3O_3(OH)_5^{2-}$, $B_4O_5(OH)_4^{2-}$ and $B_5O_6(OH)_4^-$ which were also identified in [2016MIR/WAG], except for $B_4O_5(OH)_4^{2-}$. No formation constants were given. Interestingly, it was suggested that the $B_2O_3OH^-$ species proposed in a potentiometric study by Mesmer and Baes [1972MES/BAE] cannot be observed by Raman because the depolymerisation/polymerization rate is too fast to be detected by spectroscopy. However, more recently Applegarth *et al.* [2017APP/PYE] suggested this species may be responsible for a weak shoulder observed at ~ 773 cm^{-1}. Miao *et al.* [2016MIA/FAN] carried out ^{11}B NMR and EXAFS measurements of aqueous $RbB_5O_6(OH)_4$ solutions and $Rb[B_5O_6(OH)_4]\cdot 2H_2O$(cr), whose purity and crystallinity were confirmed by TGA and XRD. They came to the same conclusions concerning the dominant aqueous species present at 298 and 333 K as in Zhou *et al.* [2011ZHO/FAN] five years earlier based on Raman spectroscopic measurements and *ab-initio* calculations. Note that Applegarth *et al.* [2017APP/PYE] tentatively assigned a weak shoulder at ~ 773 cm^{-1} found in all the solutions and decreasing slightly with temperature, to $B_2O_3OH^-$.

Table V-32: Summary of the experimental speciation and formation constant data for borate and polyborate anions reported in the literature at 298.15 K after conversion to molal units and where necessary to the all anionic forms of the reactions.

Ionic medium	$c_{B\,max}$	I_m	Reaction	Species	$\log_{10}K_m$	Method	Reference
$NaClO_4$	0.6	3.50	(V.43)	$B(OH)_4^-$	(5.15 ± 0.06)	pot	[1957ING/LAG]
			(V.48)	$B_3O_3(OH)_4^-$	(7.18 ± 0.10)		
			(V.49)	$B_3O_3(OH)_5^{2-}$	(11.73 ± 0.20)		
$NaClO_4$	0.2	0.101	(V.43)	$B(OH)_4^-$	(4.80 ± 0.06)	pot	[1962ING]
		3.50			(5.15 ± 0.06)		
		0.101	(V.48)	$B_3O_3(OH)_4^-$	(6.48 ± 0.04)		
		3.50			(7.11 ± 0.04)		
		0.101	(V.51)	$B_5O_5(OH)_6^-$	(6.99 ± 0.02)		
		3.50			(7.27 ± 0.02)		
$NaClO_4$ + $NaB(OH)_4$	3	3.50	(V.43)	$B(OH)_4^-$	5.24	pot [(a)]	[1963ING3]
			(V.49)	$B_3O_3(OH)_5^{2-}$	11.0		
			(V.50)	$B_4O_5(OH)_4^{2-}$	13.6		
$NaClO_4$ [(b)]	0.6	3.50	(V.43)	$B(OH)_4^-$	(5.20 ± 0.06)	pot	[1963ING2]
			(V.48)	$B_3O_3(OH)_4^-$	(7.32 ± 0.08)		
			(V.49)	$B_3O_3(OH)_5^{2-}$	(11.73 ± 0.19)		
			(V.50)	$B_4O_5(OH)_4^{2-}$	(13.38 ± 0.11)		
			(V.51)	$B_5O_5(OH)_6^-$	7.4		
LiBr	0.6	3.27	(V.43)	$B(OH)_4^-$	(5.23 ± 0.06)		[1963ING2]
			(V.48)	$B_3O_3(OH)_4^-$	(7.24 ± 0.07)		
			(V.49)	$B_3O_3(OH)_5^{2-}$	(11.83 ± 0.13)		
			(V.50)	$B_4O_5(OH)_4^{2-}$	(13.32 ± 0.12)		
NaBr	0.6	3.27	(V.43)	$B(OH)_4^-$	(5.23 ± 0.06)		[1963ING2]
			(V.48)	$B_3O_3(OH)_4^-$	(7.18 ± 0.07)		
			(V.49)	$B_3O_3(OH)_5^{2-}$	(11.89 ± 0.14)		
			(V.50)	$B_4O_5(OH)_4^{2-}$	(13.30 ± 0.11)		
KBr	0.6	3.39	(V.43)	$B(OH)_4^-$	(5.07 ± 0.06)		[1963ING2]
			(V.48)	$B_3O_3(OH)_4^-$	(7.11 ± 0.08)		
			(V.49)	$B_3O_3(OH)_5^{2-}$	12.8		
			(V.50)	$B_4O_5(OH)_4^{2-}$	(13.32 ± 0.10)		
	0.93		(V.43)	$B(OH)_4^-$	5.0	^{11}B NMR	[1967MOM/NAC]
			(V.48)	$B_3O_3(OH)_4^-$	(6.92 ± 0.22)		
			(V.50)	$B_5O_5(OH)_4^-$	(7.05 ± 0.15)		
KCl [(c)]	0.6	1.00	(V.47)	$B_2O_3OH^-$	(5.95 ± 0.06) [(d)]	pot	[1972MES/BAE]
					(6.02 ± 0.04) [(e)]		
			(V.48)	$B_3O_3(OH)_4^-$	(7.319 ± 0.002) [(d)]		
					(7.378 ± 0.003) [(e)]		
			(V.50)	$B_4O_5(OH)_4^{2-}$	(13.29 ± 0.07) [(e)]		
			(V.52)	$B_5O_5(OH)_8^{3-}$	(19.20 ± 0.08) [(e)]		
	0.4		(V.43)	$B(OH)_4^-$	5.39	^{11}B NMR	[1983SAL]
			(V.48)	$B_3O_3(OH)_4^-$	6.44		

(Continued on next page)

Table V-32 (continued)

Ionic medium	$c_{B\,max}$	I_m	Reaction	Species	$\log_{10}K_m$	Method	Reference
			(V.50)	$B_4O_5(OH)_4^{2-}$	11.83		
			(V.53)	$B_5O_6(OH)_4^-$	7.03		
			(V.43)	$B(OH)_4^-$		RAM/DFT	[2011ZHO/FAN]
			(V.48)	$B_3O_3(OH)_4^-$			
			(V.49)	$B_3O_3(OH)_5^{2-}$			
			(V.50)	$B_4O_5(OH)_4^{2-}$			
			(V.53)	$B_5O_6(OH)_4^-$			
				$B(OH)_4^-$		^{11}B NMR / EXAFS	[2016MIA/FAN]
				$B_3O_3(OH)_4^-$			
				$B_3O_3(OH)_5^{2-}$			
				$B_4O_5(OH)_4^{2-}$			
				$B_5O_6(OH)_4^-$			
				$B(OH)_4^-$		Raman	[2017APP/PYE]
				$B_2O(OH)_5^-$			
				$B_3O_3(OH)_4^-$			
				$B_4O_5(OH)_4^{2-}$			
				$B_5O_6(OH)_4^-$			
$NaClO_4$	0.643	0.223	7.2.2	$B_3O_3(OH)_4^-$	6.659	Raman	[2019SAS/ARC][f]

(a) A so-called "self-medium" method was employed such that $NaClO_4$ was added to produce $c_{Na^+} \approx 3$; the conversion to molal units was based on the density of a pure $NaClO_4$ solution [1985SOH/NOV] and the formation constants tabulated have a higher degree of uncertainty.
(b) Re-treatment of the data in [1957ING/LAG].
(c) The formation constants were extrapolated from the measured temperature range of 323 to 473 K.
(d) According to Scheme 1 in [1972MES/BAE].
(e) According to Scheme 2 in [1972MES/BAE].
(f) According to fitted Model 2 (20 MPa) in [2019SAS/ARC].

It is noteworthy that high boron concentrations were required to observe the polyborate species potentiometrically with the possible exception of the $B_3O_3(OH)_4^-$ species. This situation was most pronounced in the study by Mesmer *et al.* [1972MES/BAE], where the ionic strength was lower, as well as in study by Ingraham [1963ING] who utilised the self-medium method. Surprisingly, the latter study did not identify the $B_3O_3(OH)_4^-$ species at all. The high boron concentrations used in all of these investigations tend to invalidate the concept of a supporting electrolyte and would also lead to significant changes in ionic strength during the titrations. Therefore, the available data in the form of {Z or $\bar{n}$ (molality of hydroxide ions missing from solution/molality of total boron)) *versus* c_{H^+} or c_{OH^-} } in [1962ING] and [1963ING2] were re-treated by limiting the molality of boron in solution to $< 0.1 \times I_c$. The results are given in Table V-33.

Table V-33: Re-evaluation of the potentiometric results at 298.15 K; limiting total boron concentrations $\leq 0.1 \times I_m$ with conversion of the reported values from molar to molal units [1985SOH/NOV], assuming the densities of the inert electrolyte could be applied (the uncertainties are statistical in nature and indicate the goodness of fit in each scheme).

Ionic medium	I_m	Reaction	Species	Reported $\log_{10}K_m$	Scheme 1 $\log_{10}K_m$	Scheme 2 $\log_{10}K_m$	Reference
$NaClO_4$	3.50	(V.43)	$B(OH)_4^-$	(5.15 ± 0.06)	(5.14 ± 0.06)	(5.13 ± 0.01)	[1962ING]
		(V.48)	$B_3O_3(OH)_4^-$	(7.11 ± 0.04)	(7.18 ± 0.01)	(7.04 ± 0.02)	
		(V.50)	$B_4O_5(OH)_4^{2-}$ (a)			(13.90 ± 0.09)	
$NaClO_4$	3.50	(V.43)	$B(OH)_4^-$	(5.20 ± 0.02)	(5.16 ± 0.09)	(5.14 ± 0.01)	[1963ING2]
		(V.48)	$B_3O_3(OH)_4^-$	(7.32 ± 0.08)	(7.28 ± 0.05)	(7.01 ± 0.06)	
		(V.50)	$B_4O_5(OH)_4^{2-}$	(13.38 ± 0.11)		(13.31 ± 0.05)	

(a) This species was not chosen by the author.

The value of $\log_{10}K_m$ (V.43) at $I_m = 3.50$ ($NaClO_4$) using the appropriate ion interaction coefficients is (5.25 ± 0.60) which is slightly higher than those given in Table V-33. By limiting the total boron concentrations to levels consistent with having a constant ionic medium, only one polyborate species (*viz.* $B_3O_3(OH)_4^-$) with possibly a second polyborate species (such as $B_4O_5(OH)_4^{2-}$) is/are required to obtain consistent values of $\log_{10}K_m$ (V.48). A mean value of $\log_{10}K_m$ (V.48) = (7.14 ± 0.25) at 298.15 K and $I_m = 3.50$ ($NaClO_4$) is proposed, which is in close accord with that reported by Ingri [1962ING] (see Table V-33), probably reflecting its dominating presence at moderate boron concentrations. Application of the SIT to this equilibrium is simplified by the absence of the Debye-Hückel term:

$$\log_{10}\beta^{\circ}(\text{V.48}) = \log_{10}\beta_m(\text{V.48}) + 3\log_{10}a_{H_2O} + \Delta\varepsilon I_m .$$

The approximation is made that $\varepsilon(Na^+, B(OH)_4^-) \approx \varepsilon(Na^+, B_3O_3(OH)_4^-)$ = $(-(0.092 \pm 0.002) + (0.103 \pm 0.005)\log_{10}I_m)$ kg·mol^{-1} due to the high ionic strengths involved. As a result, $\log_{10}K_m$ ((V.48), 298.15 K) = (6.71 ± 0.30) where the uncertainty is approximate. Consequently, $\Delta_f G_m^{\circ}(B_3O_3(OH)_4^-, 298.15\,K) = -(2391.905 \pm 3.000)$ kJ·mol^{-1} and $\log_{10}\beta^{\circ}$ ((V.55), 298.15 K) = – (7.29 ± 0.30).

$$3\,B(OH)_3(aq) \rightleftharpoons B_3O_3(OH)_4^- + H^+ + 2\,H_2O(l) \qquad \text{(V.55)}$$

The rest of this section is devoted to the references found during the review process.

Very recently, Sasidharanpillai *et al.* [2019SAS/ARC] reported solvent-corrected reduced isotropic Raman spectra of aqueous borate solutions obtained from perpendicular and parallel polarization measurements in a titanium flow cell from 298.15 to 573.15 K at 20 MPa using perchlorate as an internal standard. The spectra obtained at low sodium:boron stoichiometric molal ratios of 0 to 0.254 showed bands at 880, 747, 615 and 532 cm^{-1}, consistent with the species $B(OH)_3(aq)$, $B(OH)_4^-$,

$B_3O_3(OH)_4^-$ and $B_5O_6(OH)_4^-$, respectively. The $B_3O_3(OH)_4^-$ anion was dominant over the whole range of conditions, thermodynamic formation constants of which calculated from the peak areas agree with those reported by Mesmer *et al.* [1972MES/BAE] from 323.15 to 423.15 K, except for the $\log_{10}K$(V.48) value of (2.259 ± 0.060), which is significantly larger than the value extrapolated from the earlier result. These new results are also summarised in Table V-32.

All of these studies included Reaction (V.43) with a hydrolysis constant similar or identical to that selected in this review. Perhaps the most important feature of these recent studies is the use of varied techniques such as advanced Raman spectroscopy, flow-through conductance, NMR and EXAFS to identify the important polyborates in solution. Peter Tremaine's group [2017APP/PYE], [2019SAS/ARC], [2019FER/ARC] used Raman and conductance measurements to high temperatures to produce the following equilibrium constants, all at 298.15 K. For the sake of comparison and because they are interconnected, the values estimated by Bassett are also listed.

Reaction	**Boron species**	**$\log_{10}K$**	**Source**	**$\log_{10}K$ [1980BAS]**
(V.43)	$B(OH)_4^-$	(4.775 ± 0.065)	[2013WAN/KOS]	4.76
		(4.764 ± 0.077)	Equation (V.44)	
(V.46)	$B_2O(OH)_5^-$	4.408	[2013WAN/KOS]	
		(4.65 ± 0.05)	[2019FER/ARC]	4.69
(V.47)	$B_3O_3(OH)_4^-$	6.662	[2013WAN/KOS]	
		6.622	[2013WAN/KOS]	
			[2019SAS/ARC]	
		(6.74 ± 0.01)	[2019FER/ARC]	6.69
(V.50)	$B_4O_5(OH)_4^{2-}$	11.747	[2013WAN/KOS]	12.94
(V.54)	$B_5O_6(OH)_6^{3-}$	15.126	[2013WAN/KOS]	

Finally, from a comparison of all the data sets shown above it would appear that the $B_3O_3(OH)_4^-$ species exists and the equilibrium constants assigned to it are in reasonable agreement. Conversely the formation constant for the dimer from Tremaine's work is in stark disagreement with that proposed by the Palmer *et al.* model, especially at high temperatures. This review favors the values suggested by the Tremaine group who made their own high-temperature measurements while incorporating the estimated values of Bassett [1980BAS], which not unknown to this review until after their evaluation was completed.

Chapter V.5

Aluminium species

V.5.1 Solid aluminium compounds

V.5.1.1 Aluminium oxide, Al_2O_3(cr)

V.5.1.1.1 Structure

The hexagonal structure of aluminium oxide, corundum, space group $R\bar{3}c$, was refined by Newnham and de Haan [1962NEW/HAA] who reported a = (4.759 ± 0.001) Å, c = (12.991 ± 0.005) Å (1σ) at room temperature for synthetic single crystals. Several concurring data were reported in the literature. The lattice parameters for single crystal synthetic corundum are compiled as a = (4.7540 ± 0.0005) Å, c = (12.9820 ± 0.0006) Å in the Handbook of Mineralogy [1997ANT/BID].

V.5.1.1.2 Heat capacity and entropy

Since α-alumina (corundum) is stable to high temperatures and is readily available in a very pure form (synthetic sapphire), it was designated by the Calorimetry Conference in 1948 as a suitable material for a standard for heat capacity measurements from 0 to 1600 K. In the following years the US National Bureau of Standards (NBS) made an extensive series of precise measurements of the heat capacity and enthalpy content from *ca.* 10 to 1170 K on this so-called Calorimetry Conference material and of α-alumina standard reference material (SRM720) that had the same origin (Linde Air Product Company) and was fabricated by the same Verneuil flame fusion process [1971DIT/DOU].

These studies are summarised in Table V-34, further details being given in the relevant Appendix A entries. Ditmars *et al.* [1982DIT/ISH] analysed all these NBS measurements and gave the recommended values for the thermal properties of this standard reference material.

Table V-34: Measurements at US National Bureau of Standards.

Reference	Temperature range /K	Experimental technique
[1947GIN/COR2]	323-1172	Drop calorimetry
[1953GIN/FUR]	14-1173	Adiabatic and drop calorimetry
[1956FUR/DOU]	13-1170	Adiabatic and drop calorimetry
[1957WES/GIN]	*ca.* 310-690	Adiabatic calorimetry
[1958WES/GIN]	312-689	Drop calorimetry
[1982DIT/ISH]	1173-2257	Adiabatic receiving calorimetry
	10-2250	Assessment

Other low-temperature measurements on the same standard sample were made by Kerr *et al.* [1950KER/JOH] (50 measurements, 20 to 295 K), Fugate and Swenson [1969FUG/SWE] (40 measurements, 2 to 25 K) and Chang [1977CHA], (160 measurements, 8.6 to 371.3 K) with excellent agreement. The values from a more recent study by Brunet *et al.* [2004BRU/MOR] (*ca.* 200 measurements, 34.25 to 299.2 K) on a sample of commercial high-purity alumina (but not synthetic sapphire) are less precise, gave very similar heat capacities, but a slightly higher derived S_m (298.15 K) of (51.20 ± 0.75) J·K^{-1}·mol^{-1} (in the text) or (51.37 ± 0.75) J·K^{-1}·mol^{-1} (in a figure), possibly due to the smaller particle size of the material used.

Archer [1993ARC] very thoroughly re-analysed all the data for synthetic sapphire, including those from the more recent studies listed in Table V-35, taking into account various sources of experimental error in the differing experimental techniques. He made a more sophisticated fit to all the data from 0 to 2400 K, using 14 cubic splines and has provided a detailed table of $C_p(T)$, $H(T) - H(0\ \mathrm{K})$ and S_m at closely spaced intervals over this temperature range.

Unfortunately, this table does not include the values at 298.15 K but these have been calculated by fitting the assessed heat capacities from 260 to 320 K, the relevant cubic spline range, by a cubic equation, from which these values have been calculated. These $C^o_{p,m}$ and S^o_m values differ only negligibly (0.02 J·K^{-1}·mol^{-1}) from the existing CODATA/TDB values, so the latter have been retained for this review,

$$C^o_{p,m}(Al_2O_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(79.033 \pm 0.100)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^o_m(Al_2O_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(50.920 \pm 0.100)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

and where the uncertainty limits have been conservatively doubled from those suggested by Archer.

Table V-35: Recent studies on synthetic sapphire in Archer's assessment [1993ARC].

Reference	Temperature range /K	Experimental technique
[1966MAR/SNO]	300-475	Adiabatic calorimetry
[1967GRO]	298-1048	Adiabatic calorimetry
[1968LEA]	298-773	Adiabatic calorimetry

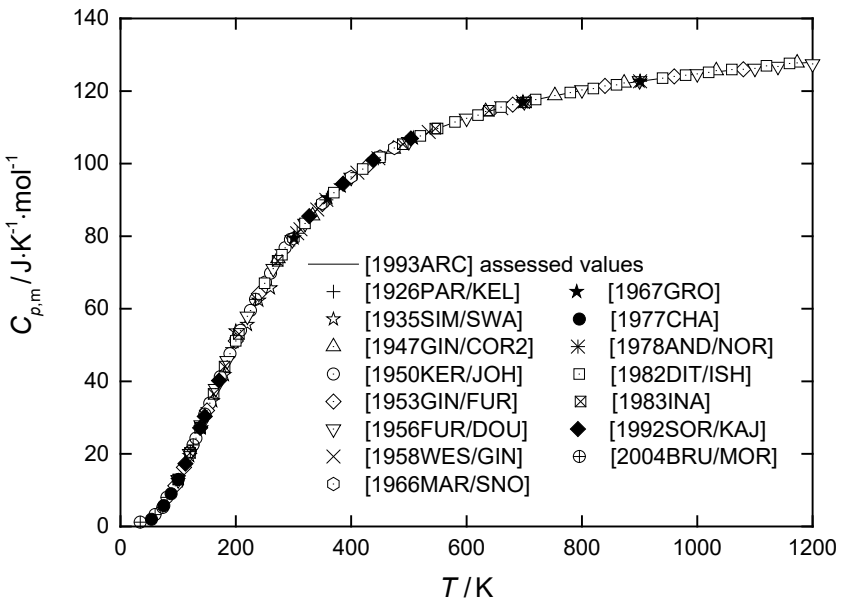
(Continued on next page)

Table V-35 (continued)

Reference	Temperature range /K	Experimental technique
[1971DIT/DOU]	273-1173	Bunsen ice calorimetry
[1977CHA]	8-375	Adiabatic calorimetry
[1978AND/NOR]	300-550	Adiabatic calorimetry
[1983INA]	70-700	Adiabatic calorimetry
[1992SOR/KAJ]	60-505	Adiabatic calorimetry

Selected points from all these studies are shown in Figure V-29, together with those of three earlier low-temperature studies by Parks and Kelley [1926PAR/KEL] (12 measurements from 91 to 291 K), Simon and Swain [1935SIM/SWA] (17 measurements from 30 to 280 K). This figure illustrates the excellent consistency of all the data—the less precise data from [1935SIM/SWA] are the only slight deviating points[1].

Figure V-29: Heat capacity of Al_2O_3(cr).



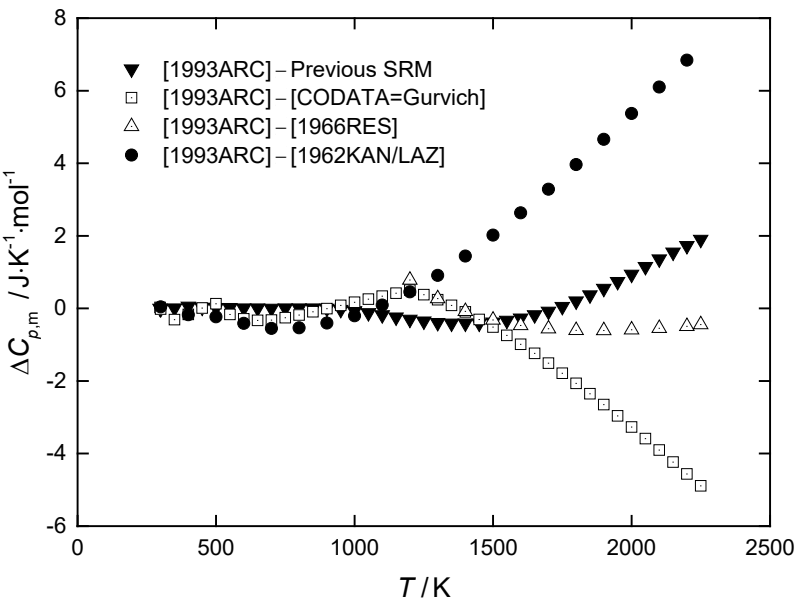
Above 1800 K Archer's revised heat capacities are slightly larger (by 0.2 to 1.2 %) than those given earlier for the synthetic sapphire Standard Reference Material by Ditmars *et al*. [1982DIT/ISH]. More importantly, it should be noted that above 1500 K they are distinctly and increasingly smaller (by 0.4 to 3.4 %) than the CODATA values given by Cox *et al.* in [1989COX/WAG]. The latter authors do not indicate the

[1] No conversion to the ITS-90 scale has been applied through the book since changes are negligible compared to the combined uncertainties in these data.

source of their high-temperature $C_{p,\mathrm{m}}$ data for alumina, but we note that they are identical to those given by Gurvich *et al.* in [1994GUR/VEY], although the source of these is not entirely clear. Gurvich *et al.* indicate that in this temperature range their heat capacities are based solely on the enthalpy data of Kantor *et al.* [1962KAN/LAZ]. However, the $C_{p,\mathrm{m}}$ values in [1994GUR/VEY] (derived from a linear equation in T) are not identical with those given by Kantor *et al.,* which are calculated from a three-term equation with a term involving T^{-2}. Above 1500 K they are appreciably larger, by as much as 4.56 $J{\cdot}K^{-1}{\cdot}mol^{-1}$ at 2200 K, whereas the $C_{p,\mathrm{m}}$ values of Kantor *et al.* are close to Archer's [1993ARC], (although they were not included in Archer's fitting, since Kantor *et al.* give very few details of their alumina sample). These differences are shown more clearly in the Figure V-30 below. This Figure also includes the differences for the high-temperature data extrapolated by Reshetnikov [1966RES] from data below 370 K, see the details in Appendix A.

We therefore conclude that the above 1200 K the $C_{p,\mathrm{m}}$ values in Archer's re-analysis [1993ARC] are much to be preferred to those in [1994GUR/VEY] and the CODATA volume [1989COX/WAG].

Figure V-30: Differences in the data for $C_{p,\mathrm{m}}(Al_2O_3, \mathrm{cr})$.



Since we cannot improve on Archer's comprehensive and thorough treatment [1993ARC], and these data are now (presumably) the recommended heat capacities for

the NIST standard reference alumina, his assessed values have been re-fitted, with negligible differences, by the following three usual four-term equations for this review.

$$C_{p,\mathrm{m}}(\mathrm{Al_2O_3, cr}, T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{54.13155} + \mathbf{0.1807542}\,(T/\mathrm{K}) - \mathbf{1.28383\times10^{-4}}\,(T/\mathrm{K})^2 - \mathbf{1.56257\times10^{6}}\,(T/\mathrm{K})^{-2} \quad (298.15\text{-}501.2\,\mathrm{K});$$

$$= \mathbf{113.9367} + \mathbf{1.902946\times10^{-2}}\,(T/\mathrm{K}) - \mathbf{4.186664\times10^{-6}}\,(T/\mathrm{K})^2 - \mathbf{4.061287\times10^{6}}\,(T/\mathrm{K})^{-2} \quad (501.2\text{-}1162\,\mathrm{K});$$

$$= \mathbf{117.909} + \mathbf{1.119230\times10^{-2}}\,(T/\mathrm{K}) - \mathbf{4.688480\times10^{-7}}\,(T/\mathrm{K})^2 - \mathbf{3.906959\times10^{6}}\,(T/\mathrm{K})^{-2} \quad (1162\text{-}2250\,\mathrm{K});$$

V.5.1.1.3 Enthalpy of formation

The value $\Delta_\mathrm{f}H^\mathrm{o}_\mathrm{m}(\mathrm{Al_2O_3, cr}, 298.15\ \mathrm{K}) = -(1675.7 \pm 1.3)\ \mathrm{kJ\cdot mol^{-1}}$ selected by both CODATA [1989COX/WAG] and Gurvich *et al.* [1994GUR/VEY] rests essentially on the nearly identical oxygen-bomb calorimetry results of Holley and Huber [1951HOL/HUB] and Mah [1957MAH] that are the most careful studies on the topic —see the Appendix A entries. The bomb-calorimetric data of Snyder and Seltz [1945SNY/SEL], yield a value of $-(1669.66 \pm 1.05)\ \mathrm{kJ\cdot mol^{-1}}$, after our correction to the 1981 atomic weights; Gurvich *et al.* [1994GUR/VEY] list $-1670\ \mathrm{kJ\cdot mol^{-1}}$ without giving uncertainty limits. We suggest that the uncertainty limits reported in [1945SNY/SEL] may have been underestimated, in particular, by a potentially improper estimation of the superficial oxide layer on the samples, or by an inadequate evaluation of the small amounts of unburned aluminium. Schneider and Gattow [1954SCH/GAT3], using the same technique, suggest a value of $-(1682 \pm 8)\ \mathrm{kJ\cdot mol^{-1}}$, the large uncertainty limits resulting from corrections for incomplete combustion and the presence of other alumina phases. In earlier years, many results [1901BER], [1919WAR/WIT], [1924MOO/PAR], [1929ROT/MUL], [1934MEI/ROT], [1940ROT/WOL2], [1942ROT/WIR], arising from combustion techniques were reported, with results varying considerably, sometimes more than two percent between successive values by the same group. The causes of this were varied: insufficient purity or characterisation of the samples, inadequate evaluation of incomplete reactions, reaction with the support materials, formation of extraneous phases in the combustion process. For these reasons, these results will not be discussed individually.

The values of $\Delta_\mathrm{f}H^\mathrm{o}_\mathrm{m}(\mathrm{Al_2O_3, cr}, 298.15\ \mathrm{K})$ that can be deduced from electromotive force measurements at high temperature [1933THO/SEY], [1933TRE/TER], [1957MAS/REV], [1965REY], [1968THO], [1976STE/HAU] and [1977GHO/KAY] yield values scattered over about 25 $\mathrm{kJ\cdot mol^{-1}}$, generally less negative than the calorimetric values, with relatively large uncertainty limits. Although they were all considered by this review, they will not be further discussed here.

In absence of more recent information, we agree with [1989COX/WAG] and [1994GUR/VEY] (the latter reproducing the earlier Russian assessment by Glushko *et al.* [1981GLU/MED]) in using the data of Holley and Huber

[1951HOL/HUB] and of Mah [1957MAH] as our basis for the adopted value of $\Delta_f H_m^o(Al_2O_3, cr, 298.15\ K)$. However, we note that the results in [1951HOL/HUB], corrected by this review to the 1981 atomic weights, yield $\Delta_f H_m^o(Al_2O_3, cr, 298.15\ K) = -(1675.1 \pm 1.3)\ kJ \cdot mol^{-1}$, but that [1989COX/WAG] gives $-(1675.7 \pm 1.3)\ kJ \cdot mol^{-1}$. The results of Mah [1957MAH] yield $-(1675.61 \pm 1.05)\ kJ \cdot mol^{-1}$ (the correction to 1981 atomic weights being negligible), while [1989COX/WAG] gives $-(1675.7 \pm 1.0)\ kJ \cdot mol^{-1}$. In addition, we note that in the NBS assessment by Wagman *et al.* [1982WAG/EVA] (data assessed in 1965), the value of $-1675.7\ kJ \cdot mol^{-1}$ is also given.

Having mentioned these small discrepancies, but taking into account the far-reaching implications of a departure from the CODATA assessment for a compound as important as corundum, this review adopts:

$$\Delta_f H_m^o(Al_2O_3, cr, 298.15\ K) = \mathbf{-(1675.7 \pm 1.3)}\ kJ \cdot mol^{-1}.$$

V.5.1.2 $Al(OH)_3$(cr)

There are four polymorphs of $Al(OH)_3$(cr), gibbsite (also known as hydrargillite), bayerite, doyleite, and nordstrandite; only gibbsite and, in part, bayerite are considered in this review.

V.5.1.2.1 $Al(OH)_3$(gibb)

V.5.1.2.1.1 Structure

Gibbsite has a sheet structure of double layers of OH ions with Al ions occupying two-thirds of the octahedral interstices within the layers. The monoclinic structure, originally determined by Megaw [1934MEG] using a single crystal from Langesundsfjord, Norway, was refined by Saalfeld and Wedde [1974SAA/WED], using a crystal from the same location, to give the parameters $a = (8.684 \pm 0.001)$, $b = (5.078 \pm 0.001)$, $c = (9.736 \pm 0.002)$ Å, $\beta = (94.54 \pm 0.02)$ °, space group $P2_1/n$ at room temperature. More recently Balan *et al.* [2006BAL/LAZ] have obtained similar values from a first-principles calculation of the structure.

V.5.1.2.1.2 Heat capacity and entropy

The first measurements of the low-temperature $C_{p,m}$ of gibbsite were made from 52.8 to 296.5 K by Shomate and Cook [1946SHO/COO], but Hemingway *et al.* [1977HEM/ROB3] have made a more detailed study on a high-purity sample using an adiabatic calorimeter from 13.4 to 379 K, supplemented by DSC measurements from 340 to 480 K. The earlier measurements [1946SHO/COO] are higher than those of Hemingway *et al.,* by as much as *ca.* 7 % at 52.8 K, but decreasing to *ca.* 2 % at 200-300 K. The data from the later measurements [1977HEM/ROB3] have been

preferred since they extend to appreciably lower temperatures. The derived heat capacity and entropy selected values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Al(OH)_3},\ \mathrm{gibb},\ 298.15\ \mathrm{K}) = \mathbf{(91.70 \pm 0.30)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$$

and

$$S^{\circ}_{\mathrm{m}}(\mathrm{Al(OH)_3},\ \mathrm{gibb},\ 298.15\ \mathrm{K}) = \mathbf{(68.44 \pm 0.30)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$$

where the uncertainties have been increased from the conservative values given by Hemingway *et al.*

The tabulated heat capacities above 298.15 K in [1977HEM/ROB3] have been fitted by the equation:

$$C_{p,\mathrm{m}}(\mathrm{Al(OH)_3},\mathrm{gibb},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{77.39936} + \mathbf{9.87237\times 10^{-2}}\,(T/\mathrm{K}) + \mathbf{5.99677\times 10^{-5}}\,(T/\mathrm{K})^2 - \mathbf{1.819167\times 10^{6}}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}500\,\mathrm{K}).$$

V.5.1.2.1.3 Enthalpy of formation

Given the lack of reactivity of gibbsite, hydrofluoric acid solution calorimetry has played an essential role in the evaluation of the enthalpy of formation of this compound. The results obtained by this technique and by calorimetry of thermal decomposition are listed in Table V-36. All the references cited in that Table have been discussed in Appendix A.

The reported values are in general agreement with the exception of those given by Barany and Kelley [1961BAR/KEL] who obtain a value noticeably less negative than the other calorimetric results. The reason for this discrepancy is not entirely clear. However, Hemingway and Robie [1977HEM/ROB] allude to the great hygroscopicity of aluminium chloride hexahydrate, which was contained in gelatine capsules in the work of Barany and Kelley. Also, there could have been inadequacies in the evaluation of the enthalpy of transfer of water between the media used. Under these circumstances, the enthalpy value reported by Barany and Kelley is given for information only.

Table V-36: Experimental data for the calorimetric determination of the standard enthalpy of formation of gibbsite.

Reference	Method	$\Delta_f H^{\circ}_{\mathrm{m}}$ (Al(OH)$_3$, gibb, 298.15 K) / kJ·mol^{-1}	
		As reported	As re-evaluated by this review
[1961BAR/KEL]	HF solution calorimetry (gibbsite, $AlCl_3{\cdot}6H_2O$)	– (1281.9 ± 0.7)	For information only
[1970GRO/CHR]	HF solution calorimetry (gibbsite, $AlCl_3$)	– (1294.24 ± 1.41)	– (1294.15 ± 1.45)

(Continued on next page)

Table V-36 (continued)

Reference	Method	$\Delta_f H_m^o$ ($Al(OH)_3$, gibb, 298.15 K) / kJ·mol^{-1}	
		As reported	As re-evaluated by this review
[1977HEM/ROB]	HF solution calorimetry (gibbsite, Al)	– (1293.13 ± 1.19)	– (1293.12 ± 2.42)
[1992JOH/TAS]	HF solution calorimetry (gibbsite, Al)	– (1294.9 ± 1.2)	– (1294.92 ± 1.17)
[1996CHE/ZEN2]	Thermal decomposition (gibbsite to γ-Al_2O_3)	– (1293.18 ± 1.20)	– (1293.23 ± 1.71)

In the study of Gross *et al.* [1970GRO/CHR], the enthalpy of formation of gibbsite was obtained by comparing the enthalpy of solution of the pair aluminium chloride and water with that of gibbsite and hydrochloric acid. As detailed in Appendix A, combination of these two dissolution processes leads to:

$$2AlCl_3(cr, 298.15\ K) + 6H_2O(l, 298.15\ K) \rightleftharpoons 2Al(OH)_3(gibb, 298.15\ K) + 6HCl\ (in\ HCl\ 3.617\ H_2O, 298.15\ K) \quad (V.56)$$

with $\Delta_r H_m^o$ ((V.56), 298.15 K) = – (369.44 ± 2.36) kJ·mol^{-1}.

With $\Delta_f H_m^o(AlCl_3, 298.15\ K)$ = – (706.26 ± 0.70) kJ·mol^{-1} previously determined by the same group [1970GRO/HAY2], the authors reported $\Delta_f H_m^o(Al(OH)_3, gibb, 298.15\ K)$ = – (1294.24 ± 1.41) kJ·mol^{-1}. Noting that correction to the 1981 atomic weights would be negligible and using CODATA accepted values for $\Delta_f H_m^o(H_2O, l, 298.15\ K)$ = – (285.830 ± 0.040) kJ·mol^{-1}, and of $\Delta_f H_m^o(HCl·3.617H_2O, 298.15\ K)$ = – (151.441 ± 0.150) kJ·mol^{-1} (interpolated from the enthalpies of dilution given by Parker, [1965PAR]) this review obtains $\Delta_f H_m^o(Al(OH)_3, gibb, 298.15\ K)$ = – (1294.15 ± 1.45) kJ·mol^{-1} from this very careful work. It may be noted that the value for $\Delta_f H_m^o(AlCl_3, 298.15\ K)$ = – (706.26 ± 0.70) kJ·mol^{-1} used in [1970GRO/CHR] is in good agreement with that obtained earlier by Coughlin [1958COU], – (705.30 ± 0.84) kJ·mol^{-1}.

As detailed in the relevant Appendix A, Hemingway and Robie [1977HEM/ROB] used the following reaction sequence at 303.3 K:

$$Al(cr) + (1849\,HF + 8163\,H_2O)(sln) \rightleftharpoons (AlF_3 + 1846\,HF + 8163\,H_2O)(sln) + 3/2\,H_2(g) \quad (V.57)$$

$$3H_2O(l) + (AlF_3 + 1846\,HF + 8163\,H_2O)(sln) \rightleftharpoons (AlF_3 + 1846\,HF + 8166\,H_2O)(sln) \quad (V.58)$$

$$Al(OH)_3(cr) + (1849\,HF + 8163\,H_2O)(sln) \rightleftharpoons (AlF_3 + 1846\,HF + 8166\,H_2O)(sln) \quad (V.59)$$

from which they obtained

$$Al(cr) + 3H_2O(l) \rightleftharpoons Al(OH)_3(cr) + 3/2\,H_2(g) \quad (V.60)$$

The uncertainties in the values given by the authors for Reactions (V.57), (V.58) and (V.59) relate to the 1σ level, with a variable number of experiments. We have thus re-calculated their data and adjusted the uncertainty to the 95 % confidence interval. Thus at 303.3 K $\Delta_r H^o_m$ (V.60) = – (595.195 ± 2.384) – (0.622 ± 0.018) + (159.572 ± 0.367) = – (436.245 ± 2.412) kJ·mol^{-1}. Accepting as such a correction of 0.615 kJ·mol^{-1} given by the authors to bring this value to 298.15 K and using the CODATA value for the enthalpy of formation of water, we obtain $\Delta_f H^o_m$ (Al(OH)$_3$, gibb, 298.15 K) = – (1293.12 ± 2.42) kJ·mol^{-1} from that source.

The results of Gross *et al.* [1970GRO/CHR] and Hemingway and Robie [1977HEM/ROB] were the basis for the selection of $\Delta_f H^o_m$ (Al(OH)$_3$, gibb, 298.15 K) = – (1293.5 ± 1.2) kJ·mol^{-1} by Gurvich *et al.* [1994GUR/VEY], who also took into account a value of $\Delta_f H^o_m$ (Al(OH)$_3$, gibb, 298.15 K) = – (1293.3 ± 1.3) kJ·mol^{-1} from Khodakovsky *et al.* [1980KHO/KAT], which is a broad assessment of equilibria in the Al_2O_3-H_2O system.

Two other relevant papers have been published since then.

The study of Johnson *et al.* [1992JOH/TAS] involved the following reactions:

$$Al(cr) + 2218.39\,HF \cdot 7632.87\,H_2O \rightleftharpoons 3/2\,H_2(g) + AlF_3 \cdot 2215.39\,HF \cdot 7632.87\,H_2O \quad (V.61)$$

$$Al(OH)_3(cr) + 2218.39\,HF \cdot 7629.87\,H_2O \rightleftharpoons AlF_3 \cdot 2215.39\,HF \cdot 7632.87\,H_2O \quad (V.62)$$

$$2218.39\,HF \cdot 7629.87H_2O + 3H_2O(l) \rightleftharpoons 2218.39\,HF \cdot 7632.87H_2O \quad (V.63)$$

$$3H_2(g) + 3/2\,O_2(g) \rightleftharpoons 3H_2O(l) \quad (V.64)$$

with $\Delta_r H^o_m$ (V.61) = – (602.93 ± 1.13) kJ·mol^{-1}, $\Delta_r H^o_m$ (V.62) = – (166.76 ± 0.19) kJ·mol^{-1}, $\Delta_r H^o_m$ (V.63) = – (1.26 ± 0.20) kJ·mol^{-1} from Johnson *et al.* [1973JOH/SMI] and $\Delta_r H^o_m$ (V.64) = – (857.49 ± 0.13) kJ·mol^{-1} from CODATA.

Combination of these reactions leads to:

$$Al(cr) + 3H_2(g) + 3/2O_2(g) \rightleftharpoons Al(OH)_3(cr) \quad (V.65)$$

$\Delta_f H^o_m$ (V.65) = – (1294.92 ± 1.17) kJ·mol^{-1}.

More recently, Chen and Zeng [1996CHE/ZEN2] obtained the standard enthalpy of formation of gibbsite from the enthalpy change for its decomposition to γ-alumina. For Reaction (V.66) at 958 K

$$Al(OH)_3(gibb, 298.15\ K) \rightleftharpoons (1/2Al_2O_3(cr, \gamma) + 3/2H_2O(g), 958\ K) \quad (V.66)$$

they obtained, on the basis of three experiments, a value of $\Delta_r H^o_m$ (V.66) = (176.72 ± 0.90) kJ·mol^{-1}, the uncertainty limits being increased by this review to reflect the 95 % confidence interval. As noted in the Appendix A entry, our

correction of this value to 298.15 K using the heat capacities of γ-Al_2O_3 from Chen *et al.* [1995CHE/ZEN2] and H_2O(g) from CODATA [1989COX/WAG] is (74.664 ± 1.000) kJ·mol^{-1}, which gives $\Delta_r H_m^o$ ((V.66), 298.15 K) = (102.06 ± 1.35) kJ·mol^{-1}.

For the enthalpy of formation of γ-Al_2O_3 required to complete the calculation, we have for consistency used the value of $\Delta_f H_m^o$(Al_2O_3, cr, γ, 298.15 K) = – (1656.86 ± 2.10) kJ·mol^{-1} re-calculated from measurements by the same team [1995CHE/ZEN2]—see the Appendix A review for that paper. This is consistent with the value measured by Yokokawa and Kleppa [1964YOK/KLE2] using drop calorimetry, which we calculate gives $\Delta_f H_m^o$(Al_2O_3, γ, 298.15 K) = – (1657.2 ± 1.6) kJ·mol^{-1}. We note also that the value selected in [1994GUR/VEY], $\Delta_f H_m^o$(Al_2O_3, cr, γ, 298.15 K) = – (1657 ± 5) kJ·mol^{-1}, encompasses the more precise value determined later [1995CHE/ZEN2].

Thus, with the CODATA value for $\Delta_f H_m^o$(H_2O, g, 298.15 K), $\Delta_f H_m^o$(Al(OH)$_3$, gibb, 298.15 K) is found to be – (1293.23 ± 1.71) kJ·mol^{-1} based on the results of Chen and co-workers.

Based on the calculated weighted and unweighted means, – 1294.17 and – 1293.86 kJ·mol^{-1}, of our re-calculated values in Table V-36, the best calorimetric value for

$$\Delta_f H_m^o(\mathrm{Al(OH)_3}, \text{gibb}, 298.15\ \mathrm{K}) = \mathbf{-\ (1294.0 \pm 1.5)}\ \mathrm{kJ\cdot mol^{-1}},$$

and this is our selected value.

As noted in Section V.5.2.3 these data are completely consistent with the solubility data for gibbsite.

V.5.1.2.2 Al(OH)$_3$(β-baye)

V.5.1.2.2.1 Structure

The first refinement of the monoclinic structure of bayerite using X-ray diffraction by Rothbauer *et al.* [1967ROT/ZIG] led to the assignment of space group $P2_1/a$. However, later neutron-diffraction measurements by authors [1978ZIG/JOS] using deuterated bayerite showed that the refinement of the positions of the hydrogen atoms led to space group $P2_1/n$, with a = (5.0626 ± 0.0020) Å, b = (8.6719 ± 0.0020) Å, c = (9.4254 ± 0.0030) Å and β = (90.26 ± 0.01) ° at room temperature. These lattice parameters corresponded to those reported by Rothbauer *et al.* [1967ROT/ZIG], with the expected doubling of the c value. The structure of bayerite is thus very similar to that of gibbsite.

V.5.1.2.2.2 Heat capacity and entropy

We have not found any published data for the heat capacity of bayerite, so no values of $C^{o}_{p,m}$ and S^{o}_{m} are selected, but a value of S^{o}_{m} (298.15 K) = (79.780 ± 3.525) J·K^{-1}·mol^{-1}, based on the thorough analysis of the solubility data for bayerite in Section V.5.2.4 is recommended. We note that this is noticeably larger than that of gibbsite, 68.44 J·K^{-1}·mol^{-1}.

V.5.1.2.2.3 Enthalpy of formation

To date, there has been no direct calorimetric determination of the enthalpy of formation of bayerite, but there are a number of data, besides solubility investigations, linking the thermochemistry of bayerite to that of gibbsite.

Fricke and Wullhorst [1932FRI/WUL] have reported the enthalpies of solution of both bayerite and gibbsite (hydrargillite) in 11.59 and 12.00 % HF. For bayerite, two samples were used. These gave average values of – (158.32 ± 1.26) kJ·mol^{-1} (based on six measurements) and – (157.03 ± 0.54) kJ·mol^{-1} (based on five measurements in 11.59 % HF only); the uncertainty for the latter value was calculated by this review. For gibbsite the average value of – (153.09 ± 0.84) kJ·mol^{-1} was reported based on nine measurements. Unfortunately, the temperature at which these experiments were carried out was not reported.

From their results, Fricke and Wullhorst reported a value of – 5.02 kJ·mol^{-1} for $\Delta_r H_m$ (V.67):

$$Al(OH)_3(baye) \rightleftharpoons Al(OH)_3(gibb) \qquad (V.67)$$

The authors indicate, in a footnote, that, in their laboratory, a value of – 5.44 kJ·mol^{-1} was obtained for $\Delta_r H_m$ (V.67) from dissolutions in 20 % HF.

This review thus considers that a value of – (5.2 ± 2.0) kJ·mol^{-1} can be taken as the "best" result for $\Delta_r H_m$ (V.67) from the data of Fricke and Wullhorst.

Besides the solubility data that are treated fully in Section V.5.2.4 of the present review, several comparative studies of the thermal decomposition of bayerite and gibbsite have been reported [1932FRI/SEV], [1955EYR/GOT], [1957MIC], [1969MUK/TAK] and [1972STR/HEN] that give rather scattered values for the enthalpy of Reaction (V.67) above. A few details concerning these studies are given in the evaluation of the thermochemical properties of bayerite, gibbsite and other aluminium species by Apps *et al.* [1988APP/NEI]. We concur with the opinion expressed by the latter authors that such studies only permit a rough estimate of the enthalpy of transition for Reaction (V.67) above, since the thermal decomposition of different bayerite and gibbsite samples rarely result in identical decomposition products.

Under these circumstances, this review uses the value $\Delta_r H_m$ ((V.67), 298.15 K) = – (5.2 ± 2.0) kJ·mol^{-1} from Fricke and Wullhorst in the

evaluation of $\Delta_f H_m^o$ (Al(OH)$_3$, β-baye, 298.15 K) in conjunction with $\Delta_f H_m^o$ (Al(OH)$_3$, α-gibb, 298.15 K) = – (1294.0 ± 1.5) kJ·mol^{-1}, leading to $\Delta_f H_m^o$ (Al(OH)$_3$, β-baye, 298.15 K) = – (1288.8 ± 2.5) kJ·mol^{-1}.

This value is in good agreement with that obtained in this review (Section V.5.2.4) from a full analysis of the solubility data of bayerite – (1289.164 ± 4.453) kJ·mol^{-1}. We therefore select a value of:

$$\Delta_f H_m^o \text{(Al(OH)}_3\text{, β-baye, 298.15 K)} = \mathbf{-\ (1289.0 \pm 2.5)}\ \text{kJ·mol}^{-1}.$$

V.5.2 Discussion of data selection for aqueous aluminium species

V.5.2.1 Aqueous Al^{3+}

Fratiello *et al*. [1968FRA/LEE] concluded from proton nuclear magnetic resonance (^{1}H NMR) measurements that Al^{3+} and Mg^{2+} are too strongly hydrated to allow ion pairing with chloride ions.

We see no reason to doubt the CODATA [1989COX/WAG] value for the enthalpy of formation of Al^{3+}, $\Delta_f H_m^o$ (Al^{3+}, 298.15 K) = – (538.4 ± 1.5) kJ·mol^{-1}, which (see note 85 in [1989COX/WAG]) was based on the concordant measurements of the enthalpies of solution of Al(cr), $AlCl_3$(cr) and $AlCl_3 \cdot 6H_2O$(cr) in aqueous HCl by Coughlin [1958COU] and of the enthalpy of solution of $AlCl_3$(cr) in water by Krivtsov *et al.* [1971KRI/ROS]. These gave $\Delta_f H_m^o$ (Al^{3+}, 298.15 K) = – (538.3 ± 2.0) and – (538.5 ± 1.3) kJ·mol^{-1} respectively; we note that Krivtsov *et al.* utilised the value for enthalpy of formation of $AlCl_3$(cr) obtained by Coughlin, so these determinations are not entirely independent. More recently Wenming Zeng *et al.* [1994WEN/QIY2] derived the concordant value of – (538.25 ± 1.00) kJ·mol^{-1} from calorimetric measurements of aluminium in HCl solutions.

However, the CODATA value for the standard entropy S_m^o (Al^{3+}, 298.15 K) = – (325 ± 10) J·K^{-1}·mol^{-1} is far less certain. This is based on the two values of – (332 ± 7) J·K^{-1}·mol^{-1} from the earlier review of the solubility of gibbsite by Parks [1972PAR] and – (313 ± 6) J·K^{-1}·mol^{-1} from the solubility of caesium alum $CsAl(SO_4)_2$ by Latimer and Greensfelder [1928LAT/GRE]. The former value is in good agreement with the value derived from the more comprehensive evaluation of the solubility of gibbsite in the current review, – (334.0 ± 2.0) J·K^{-1}·mol^{-1} (Section V.5.2.3), but there are some serious doubts about the determination by Latimer and Greensfelder.

These authors measured the low-temperature heat capacity of caesium alum, $CsAl(SO_4)_2 \cdot 12H_2O$(cr) from 18.7 to 297.8 K to give S_m^o ($CsAl(SO_4)_2 \cdot 12H_2O$, cr, 298.15 K) = 163.85 cal·K^{-1}·mol^{-1} = 685.55 J·K^{-1}·mol^{-1}. Kelley and King [1961KEL/KIN] re-interpreted these data to give a value of (682.0 ± 20.9) J·K^{-1}·mol^{-1},

whereas a further re-examination by Hemingway and Robie [1977HEM/ROB2] indicated a value of (679.4 ± 17.0) $J \cdot K^{-1} \cdot mol^{-1}$. The latter authors also indicated the possibilities of a low temperature transition in the alum (as in the similar potassium and ammonium alums) and the presence of some disorder in the alum structure (leading to a non-zero entropy at 0 K).

Latimer and Greensfelder [1928LAT/GRE] also reported, without any experimental detail whatsoever, their measurement of the enthalpy of solution of caesium alum, 13.56 $kcal \cdot mol^{-1}$ (56.74 $kJ \cdot mol^{-1}$); we suggest that their estimated uncertainty, ~ 25 $cal \cdot mol^{-1}$ (105 $J \cdot mol^{-1}$) is probably too small, especially as this enthalpy of solution is noticeably different from that subsequently measured by Zalogina and Selivanova [1969ZAL/SEL], 14.9 $kcal \cdot mol^{-1}$ (62.3 $kJ \cdot mol^{-1}$).

Latimer and Greensfelder then combined these data with the Gibbs energy of solution of caesium alum calculated from the solubility measured by Berkeley [1904BER] (and confirmed by others), and the activity coefficient of the salt, derived from their own freezing-point measurements, to calculate the standard entropy of Al^{3+} to be – 296.6 $J \cdot K^{-1} \cdot mol^{-1}$. This derivation of $S^o_m(Al^{3+}, 298.15\ K)$ was repeated by the CODATA team [1989COX/WAG] using modern data for the constituent species to give $S^o_m(Al^{3+}, 298.15\ K) = -(313 \pm 6)\ J \cdot K^{-1} \cdot mol^{-1}$, a calculation we have confirmed, although the uncertainty may have been underestimated. However, in view of the lack of experimental detail in [1928LAT/GRE], and the considerable uncertainties in the data for the enthalpy of solution and standard entropy of caesium alum, this review like Hemingway and Robie [1977HEM/ROB2], does not have great confidence in [1928LAT/GRE] and the value of $S^o_m(Al^{3+}, 298.15\ K)$ derived from it.

Wolery and Jove-Colon [2017WOL/JOV] corrected S^o_m values for Al^{3+} from [1995POK/HEL] and [2001TAG/SCH]. Their reported S^o_m value of – 338.1 $J \cdot K^{-1} \cdot mol^{-1}$ appears to be a correction for [1995POK/HEL] —see Table 24 in [2017WOL/JOV]. The previously selected value by the NEA is – (325 ± 10) $J \cdot K^{-1} \cdot mol^{-1}$. Other values were reported in [1997SHO/SAS], [2001BEN/PAL] and [2001TAG/SCH].

We therefore suggest that the entropy of Al^{3+} be based on the consistent values for the solubility of gibbsite, using the values from the current review, which is the most comprehensive and up-to-date assessment of all the relevant studies, and incorporates a constraint to be consistent with the calorimetric enthalpies of formation of Al^{3+} and gibbsite.

From the assessed value for the solubility of gibbsite given in Section V.5.2.3.1, the entropy change for the reaction $\alpha\text{-}Al(OH)_3(cr) + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l)$ was found to be $\Delta_r S^o_m(298.15\ K) = -(192.544 \pm 7.353)\ J \cdot K^{-1} \cdot mol^{-1}$. With the value of $S^o_m(Al(OH)_3, gibb)$ selected in this review (68.44 $J \cdot K^{-1} \cdot mol^{-1}$), this gives $S^o_m(Al^{3+}, 298.15\ K) = -(334.0 \pm 2.0)\ J \cdot K^{-1} \cdot mol^{-1}$. Our selected value is thus:

$$S^o_m(Al^{3+}, 298.15\ K) = \mathbf{-\ (334.000 \pm 2.000)}\ J \cdot K^{-1} \cdot mol^{-1}.$$

The SIT treatment by Lemire *et al.* [2013LEM/BER] of the osmotic coefficient data for $AlCl_3$(sln) at 298.15 K in [1941MAS], [1949ROB/STO], [1955HAR/GAR] and [2002APE/KOR2] give a value of $\varepsilon(Al^{3+}, Cl^-) = (0.34 \pm 0.02)$ kg·mol^{-1}, when the molality of $AlCl_3$ is limited to ≤ 0.9 mol·kg^{-1}. This is virtually identical to the value previously recommended by the NEA (presumably taken from [1988CIA]) of $\varepsilon(Al^{3+}, Cl^-) = (0.33 \pm 0.02)$ kg·mol^{-1}, which was based solely on the tabulated osmotic coefficients in [1949ROB/STO], restricted also to $m_{Al} \leq 0.9$ mol·kg^{-1}. The available data actually extend to 1.8 mol·kg^{-1} and are well represented by a two-term epsilon fit $\varepsilon_1 + \varepsilon_2 \log_{10}(I_m)$ where $\varepsilon_1 = (0.34 \pm 0.04)$ kg·mol^{-1} and $\varepsilon_2 = (0.40 \pm 0.04)$ kg·mol^{-1}, although this function does not extrapolate well to the osmotic coefficient value at a much higher molality, *i.e.*, $m_{sat} = 3.43$ [2002APE/KOR2]. By way of comparison, vapour-pressure measurements by Pearce and Blackman [1935PEA/BLA] yield a value of $\varepsilon(Al^{3+}, NO_3^-) = (0.25 \pm 0.02)$ kg·mol^{-1} at 298.15 K to relatively high concentrations, whereas isopiestic measurements cut off at $I_m = 1$ yield $\varepsilon(Al^{3+}, ClO_4^-) = (0.52 \pm 0.01)$ kg·mol^{-1} [1970KAL].

V.5.2.2 Hydrolysis of aqueous aluminium(III)

V.5.2.2.1 The first hydrolysis constant of Al^{3+}

Swaddle *et al.* [2005SWA/ROS] produced surprising kinetic evidence that "$AlOH^{2+}$" has in fact a five-coordinate configuration rather than being octahedral as was generally believed. The ramifications of this structural change resulting from hydrolysis are not immediately apparent but may signal a different thermodynamic hydrolysis behaviour compared to other trivalent cations.

The first hydrolysis constant of Al^{3+} represented by Reaction (V.68) has been measured by potentiometry, solubility and electrospray mass spectrometry, although the former provided the most accurate thermodynamic data. Nevertheless, experimental problems arise because polynuclear hydrolysed ions are readily formed in the pH range < 5 at room temperature and < 3.5 by 343 K, aluminium molalities must be limited $< 10^{-3}$ mol·kg^{-1} to avoid their formation. From an ^{17}O-NMR study, Nordin *et al.* [1998NOR/SUL] established that the rapid rate of water exchange between hydrated aluminium ions and the bulk aqueous solution ensures that equilibrium is reached readily within the time frame of a potentiometric titration in the determination of ${}^*K_1$ (V.68). All the relevant thermodynamic experimental data are listed in Table V-37.

$$Al^{3+} + H_2O(l) \rightleftharpoons AlOH^{2+} + H^+ \qquad \text{(V.68)}$$

Table V-37: Values of $\log_{10}{}^{*}K_1^{\circ}$ (V.68) as cited in the literature and/or recalculated with the SIT limiting the temperature to ≤ 473.15 K.

T /K	Method	$\log_{10}{}^{*}K_1^{\circ}$ reported	References	$\log_{10}{}^{*}K_1^{\circ}$ SIT [a]	$\Delta\varepsilon$ [a] /kg·mol^{-1}	Comments
288.15	pot	– (5.28 ± 0.02)	[1954SCH/TAY]	– (5.28 ± 0.20)		($AlCl_3$, KCl) $c_{Al} \leq 0.01$; $c_K = 0.001$
293.15		– (5.15 ± 0.02)		– (5.15 ± 0.20)		
298.15		– (4.98 ± 0.02)		– (4.99 ± 0.20)	0	$0.001 \leq c_{Al} \leq 0.0005$
298.15	pot	– (5.02 ± 0.03)	[1956KUB]	– (4.93 ± 0.25)	– (0.31 ± 0.06)	I_c = 0.003-1.055 ($NaClO_4$)
298.15	pot	– (5.02 ± 0.04)	[1963FRI/PEE]	– (4.97 ± 0.20)		($AlCl_3$, $Al(ClO_4)_3$, Na_2CO_3) $10^{-5} \leq c_{Al} \leq 0.01$
298.15	sp	$\log_{10}{}^{*}K_{c,1}$ – 4.89	[1969NAZ/NEV]			I_c = 0.1 ($NaClO_4$) $c_{Al} \approx 2 \times 10^{-5}$
298.15	pot	– (4.99 ± 0.01) $\Delta_r H^{\circ}_{m,1}$ (11.4 ± 2.1) [c]	[1971VOL/PAV]	– (5.00 ± 0.20)	– 0.3 [b]	$I_c \approx 0.0025$-0.07 (NaCl, NaBr, $NaNO_3$, $NaClO_4$, Na_2SO_4)
298.15	pot	$\log_{10}{}^{*}K_{c,1}$ – (5.52 ± 0.04)	[1981OHM/FOR]	$\log_{10}{}^{*}K_{m,1}$ – (5.51 ± 0.12)		$I_c \approx 0.6$ (NaCl) $0.0025 \leq c_{Al} \leq 0.02$
423	sol	– 2.6	[1983KUY/MAL]			$I_m \approx 0.1$ ($NaClO_4$)
473		– 1.9				
298.15	pot	– 5.00 $\Delta_r H^{\circ}_{m,1}$ 49.4 [c]	[1984COU/MIC]			$I_m \approx 0.1$ (KNO_3) $c_{Al} = 2 \times 10^{-4}$ 293.15-343.15 K
298	nmr	– (4.93 ± 0.08)	[1985AKI/ELD]			$0.001 \leq c_{Al} \leq 0.2$
298.15	pot	$\log_{10}{}^{*}K_{c,1}$ – (5.41 ± 0.01)	[1985BRO/SYL]	– (4.99 ± 0.10) [d]		$I_c \approx 0.1$ ($NaNO_3$) $c_{Al} \leq 0.1$
298.15	pot	– (4.86 ± 0.03)	[1988HED]	– (4.81 ± 0.20)	– (0.12 ± 0.10)	I_c = 0.6-3.0 (NaCl)
		– (4.86 ± 0.04)		– (4.80 ± 0.20)	– (0.11 ± 0.04)	I_c = 0.6-3.0 ($NaClO_4$)

(Continued on next page)

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Table V-37 (continued)

T/K	Method	$\log_{10}{}^{*}K_1^{\circ}$ reported	References	$\log_{10}{}^{*}K_1^{\circ}$ SIT [(a)]	$\Delta\varepsilon$ [(a)]/kg·mol^{-1}	Comments
423	sol	– (2.1 ± 0.1)	[1993BOU/KNA]	– (2.1 ± 0.5)		I_m < 0.15 (various solutes)
473		– (1.1 ± 0.1)		– (1.1 ± 0.5)		
443	sol	– (1.76 ± 0.32)	[1993CAS/DAN]	– (1.8 ± 0.3)		I_m ≈ 0.01 (HCl + NaCl)
473		– (1.38 ± 0.32)		– (1.4 ± 0.3)		
298.15	pot	– (4.95 ± 0.03) $\Delta_r H^{\circ}_{m,1}$ (55.4 ± 1.0) [(c)]	[1993PAL/WES]	– (4.88 ± 0.10)	– (0.30 ± 0.11)	I_m = 0.1-5.0 (NaCl) m_{Al} ≤ 0.001
323.15		– (4.20 ± 0.03)		– (4.12 ± 0.20)	– (0.15 ± 0.01)	
347.95		– (3.56 ± 0.03)		– (3.47 ± 0.20)	(0.02 ± 0.08)	
372.95		– (3.00 ± 0.03)		– (2.95 ± 0.20)	(0.10 ± 0.08)	
397.95		– (2.51 ± 0.03)		– (2.41 ± 0.20)	(0.16 ± 0.08)	

(a) Estimated (2σ) uncertainties by this review.
(b) The value of $\Delta\varepsilon$ was fixed from [1993PAL/WES].
(c) Units are kJ·mol^{-1}.
(d) Assuming $\Delta\varepsilon$ = – 0.17 kg·mol^{-1} (see below).

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Only four potentiometric studies were carried out as a function of ionic strength that allowed the first hydrolysis constant of Al^{3+} to be treated with the SIT according to Equation (V.69) [1956KUB], [1971VOL/PAV], [1988HED], [1993PAL/WES]. Some early research used batch potentiometric measurements of individual dilute solutions where the only hydrolysis product was assumed quite reasonably to be $AlOH^{2+}$ and activity coefficients were virtually unity [1954SCH/TAY], [1963FRI/PEE]. Öhman *et al.* [1981OHM/FOR] performed careful titrations at a single condition, *viz.*, $I_c = 0.6$ (NaCl, 298.15 K), over a wide range of aluminium concentrations with the main aim of establishing the stoichiometry and stability of high-order hydroxidoaluminium (III) species. They found that the stoichiometry of such species could not be determined unambiguously from potentiometric titrations alone, but their value of $\log_{10}{}^{*}K_1$ is incorporated into the current SIT treatment as illustrated in Figure V-31. High-temperature solubility studies, the results of which are truncated at 473 K in Table V-37 produced divergent results [1983KUY/MAL], [1993BOU/KNA], [1993CAS/DAN]. Results for the latter two studies are accepted by this review, but are assigned larger uncertainties.

$$\log_{10}{}^{*}K_{m,1} + 4D - \log_{10} a_{H_2O} = \log_{10}{}^{*}K_1^{o} - \Delta\varepsilon I_m \qquad \text{(V.69)}$$

Application of the SIT to the results at 298.15 K to $I_m \leq 1.0$ and 3.2 (NaCl) from [1954SCH/TAY], [1956KUB], [1963FRI/PEE], [1971VOL/PAV], [1981OHM/FOR], [1988HED] and [1993PAL/WES] yielded: $\log_{10}{}^{*}K_1^{o} = -(4.90 \pm 0.06)$ and $\Delta\varepsilon = -(0.17 \pm 0.04)$ kg·mol^{-1}, and $\log_{10}{}^{*}K_1^{o} = -(4.92 \pm 0.07)$ with $\Delta\varepsilon = (\Delta\varepsilon_1 + \Delta\varepsilon_2 \log_{10}(I_m))$, $\Delta\varepsilon_1 = -(0.27 \pm 0.09)$ and $\Delta\varepsilon_2 = (0.24 \pm 0.16)$ kg·mol^{-1} (see Figure V-31). Thus, two approaches are available to calculate the ion interaction coefficient:

(1) $\varepsilon(AlOH^{2+}, Cl^-) = \Delta\varepsilon + \varepsilon(Al^{3+}, Cl^-) - \varepsilon(H^+, Cl^-) =$
$-(0.17 \pm 0.04) + (0.33 \pm 0.02) - (0.12 \pm 0.01) = (0.04 \pm 0.05)$ kg·mol^{-1}

and

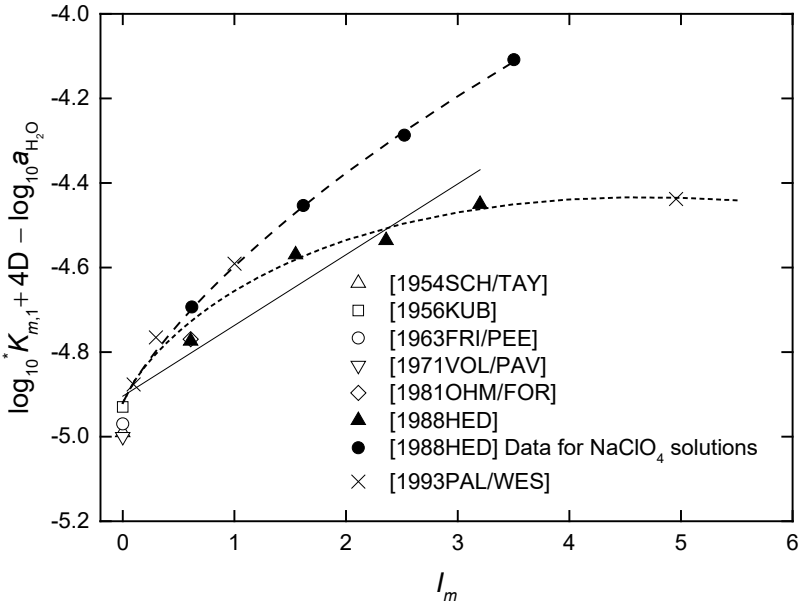
(2) $\varepsilon(AlOH^{2+}, Cl^-) = -(0.27 \pm 0.09) - (0.24 \pm 0.16)\log_{10}(I_m)$ +
$((0.34 \pm 0.04) + (0.40 \pm 0.04)\log_{10}(I_m))$ $- (0.12 \pm 0.01) =$
$(-(0.05 \pm 0.10) + (0.64 \pm 0.16)\log_{10}(I_m))$ kg·mol^{-1}.

Figure V-31 shows that the latter fit extrapolates very well to the value at $I_m = 4.95$ [1993PAL/WES] that was not included in the regression, because it was considered to be outside of the general range of applicability of the SIT. A value of $\varepsilon(AlOH^{2+}, Cl^-) = 0.09$ kg·mol^{-1} was previously recommended by the NEA TDB [2003GUI/FAN].

From the four values of $\log_{10}{}^{*}K_{m,1}$ reported in $NaClO_4$ solutions from $I_m = 0.62$ to 3.50 at 298.15 K [1988HED], an SIT fit with $\log_{10}{}^{*}K_1^{o}$ constrained at

– 4.92, yielded the ion interaction coefficients, $\Delta\varepsilon_1 = -(0.33 \pm 0.01)$ kg·mol^{-1} and $\Delta\varepsilon_2 = (0.17 \pm 0.02)$ kg·mol^{-1} (see Figure V-31).

Figure V-31: Plot of the first hydrolysis constant of Al^{3+} at 298.15 K in NaCl solutions where the solid and short-dashed lines were generated assuming that $\Delta\varepsilon$ was independent and dependent on I_m, respectively. The solid circles represent data in $NaClO_4$ solutions [1988HED] with the long-dashed curve coming from a constrained SIT fit.



In the absence of calorimetric data, a simple weighted regression of the values of $\log_{10}{}^{*}K_1^{\circ}$ in column 5 of Table V-37 as a function of (T/K) gives:

$$\log_{10}{}^{*}K_1^{\circ}\,(\text{V.68}) = (4.988827 \pm 0.132089) - (2954.321 \pm 39.383)(T/\mathrm{K})^{-1} \quad \text{(V.70)}$$

which yielded our selected values:

$$\log_{10}{}^{*}K_1^{\circ}\,((\text{V.68}),\ 298.15\ \mathrm{K}) = \mathbf{-\ (4.920 \pm 0.070)},$$

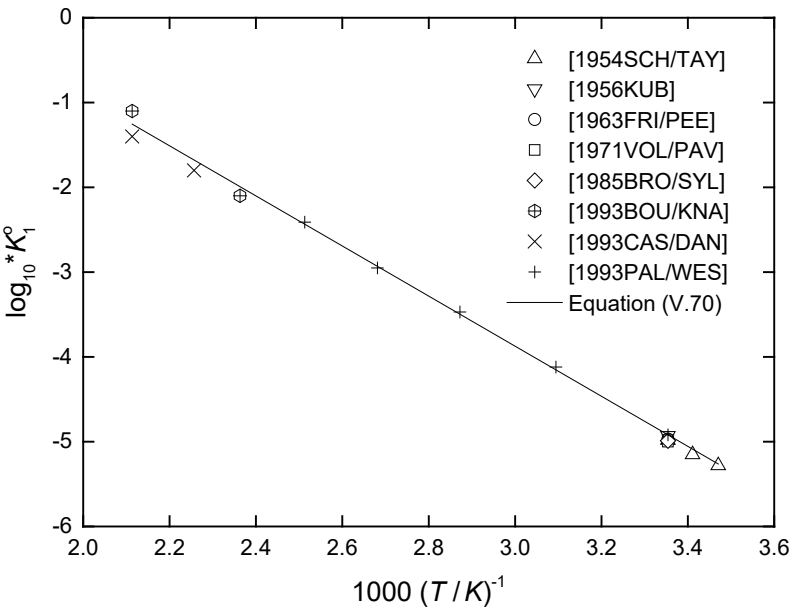
$$\Delta_r G_m^{\circ}\,((\text{V.68}),\ 298.15\ \mathrm{K}) = \mathbf{(28.084 \pm 0.400)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H_m^{\circ}\,((\text{V.68}),\ 298.15\ \mathrm{K}) = \mathbf{(56.560 \pm 1.478)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r S_m^{\circ}\,((\text{V.68}),\ 298.15\ \mathrm{K}) = \mathbf{(95.509 \pm 5.136)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

Palmer and Wesolowski [1993PAL/WES] reported $\Delta_r H_m^o$ ((V.68), 298.15 K) = (55.4 ± 0.7) kJ·mol^{-1} and (49.8 ± 2.1) kJ·mol^{-1} was recommended by Baes and Mesmer [1976BAE/MES]. Palmer and Wesolowski also treated these data in terms of the Pitzer ion interaction model taking the single electrolyte parameters for NaCl, HCl and $AlCl_3$ from [1984PIT/PEI], [1987HOL/BUS] "model 1" and [1992PAL/WES], respectively. The resulting value of $\log_{10}{}^*K_1^o$ ((V.68), 298.15 K) was found to be – 4.91, *cf.* – 5.00 estimated by Khodakovsky *et al*. [1980KHO/KAT].

Figure V-32: Plot of $\log_{10}{}^*K_1^o$ (V.68) *versus* 1000(T /K)$^{-1}$.



Whereas the thermodynamic values for the formation of most of the mononuclear aqueous Al species were derived from gibbsite solubility data, those for $AlOH^{2+}$ were derived predominantly from the potentiometric measurements summarised in Table V-37 for the hydrolysis reaction and from the corresponding properties of Al^{3+} (see Section V.5.2.3.1), *i.e.*, a combination of Equations (V.69) and (V.70). This review selects:

$$\Delta_f G_m^o (AlOH^{2+}, 298.15\ K) = \mathbf{-\ (697.880 \pm 1.663)}\ kJ\cdot mol^{-1},$$

$$\Delta_f H_m^o (AlOH^{2+}, 298.15\ K) = \mathbf{-\ (767.670 \pm 2.106)}\ kJ\cdot mol^{-1},$$

$$S_m^o (AlOH^{2+}, 298.15\ K) = \mathbf{-\ (168.541 \pm 5.512)}\ J\cdot K^{-1}\cdot mol^{-1}.$$

The bulk of previous assessments were derived from solubility measurements and these results are listed in Table V- 38.

Table V- 38: Thermodynamic quantities derived for the $AlOH^{2+}$ species at 298.15 K.

$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}	Reference
– (696.0 ± 2.5)	– (769.7 ± 3.7)	– (181.5 ± 8.0)	[2001BEN/PAL]
– (697.9 ± 1.7)	– (767.7 ± 1.9)	(168.5 ± 5.5)	This review

V.5.2.2.2 The second hydrolysis constant of Al^{3+}

Kubota [1956KUB] could not derive consistent $\log_{10}{}^{*}K_c$ (V.71) values from his potentiometric titration results as a function of aluminium concentration and ionic strength, but rather suggested the formation of the dimer, $Al_2(OH)_2^{4+}$, although the evidence was not totally convincing. ^{27}Al NMR spectroscopic measurements carried out in the pH range 2.00-5.20 and total Al(III) concentrations in the range (1-10) × 10^{-6} mol·dm^{-3}, were unable to distinguish between $AlOH^{2+}$ and $Al(OH)_2^+$ solely on the basis of either chemical shift or line-width measurements [1995FAU/LAB]. On the other hand, Baes and Mesmer [1976BAE/MES] considered the novel spectrophotometric results of Nazarenko and Nevskaya [1969NAZ/NEV] that involved low aluminium concentrations (*ca*. 2 × 10^{-5} mol·dm^{-3} in 0.1 mol·dm^{-3} $NaClO_4$) to be the most reliable investigation of Reaction (V.71) with $\log_{10}{}^{*}K_c = -5.43$ at 298.15 K. Despite "possible complications" in this approach, Baes and Mesmer recommended a value of $\log_{10}{}^{*}\beta_c$ ((V.72), 298.15 K) = – 9.9 at $I_c = 0.1$ ($NaClO_4$). Brown *et al.* [1985BRO/SYL] reported detailed potentiometric titrations of Al^{3+} solutions and proposed a similar value of $\log_{10}{}^{*}K_c$ ((V.71), 298.15 K) = – 5.41 at $I_c = 0.1$($NaNO_3$).

$$AlOH^{2+} + H_2O(l) \rightleftharpoons Al(OH)_2^+ + H^+ \qquad \text{(V.71)}$$

$$Al^{3+} + 2H_2O(l) \rightleftharpoons Al(OH)_2^+ + 2H^+ \qquad \text{(V.72)}$$

Table V-39 summarises the available data for $\log_{10}{}^{*}K_2^o$ ((V.71), T/K) restricted to $T \leq 473.15$ K, but neglecting the results from Verdes [1990VER] that are consistently higher than those from other studies by a factor of *ca*. 0.8 $\log_{10}{}^{*}K_2^o$. The increasing tendency for hydrolysis to occur with increasing temperature amplifies the significance of the high-temperature data.

The paper [2001BEN/PAL] contains hydrolysis constants derived from new experimental solubility data for boehmite from 373.15 to 573.15 K that were combined with existing hydrolysis data at lower temperatures to provide best estimates over a wide range of temperature and ionic strength. However, these combined data were not included in the current data treatment because past emphasis on the NEA TDB assessments has been to provide the most accurate thermodynamic data at 298.15 K. Therefore, these high-temperature results for a phase that this review has not targeted were omitted to place more weight on the low-temperature data.

Table V-39: Values of $\log_{10}{}^{*}K_2^{\circ}$ ((V.71), T/K) obtained by application of the SIT to temperatures ≤ 473.15 K with the 2σ uncertainties assigned by this review.

T/K	Method	$\log_{10}{}^{*}K_2^{\circ}$ reported	References	$\log_{10}{}^{*}K_2^{\circ}$ retained	Comments
298.15	sp	$\log_{10}{}^{*}K_{c,2}$ – 5.43	[1969NAZ/NEV]	– (5.60 ± 0.10)	I_c = 0.1 ($NaClO_4$)
423	sol [(a)]	– 2.60	[1983KUY/MAL]		I_m = 0.1 ($NaClO_4$)
473		– 1.88			
298.15	pot	$\log_{10}{}^{*}K_{c,2}$ – (5.26 ± 0.02)	[1985BRO/SYL]	– (5.58 ± 0.10)	I_c = 0.1 ($NaNO_3$)
423	sol [(a)]	– 2.7	[1993BOU/KNA]	– (2.7 ± 0.5)	I_m < 0.15 (variable)
473		– 2.3		– (2.3 ± 0.5)	
363	sol [(a)]	– 4.37	[1993CAS/DAN]	– (4.4 ± 0.3)	I_m ≤ 0.01 (variable)
443		– 2.82		– (2.8 ± 0.4)	
473		– 2.59		– (2.6 ± 0.4)	
323.15	sol [(b)]	– (5.15 ± 0.12) [(b)]	[1994WES/PAL]	– (5.15 ± 0.20)	I_m = 0.1 (NaCl)
373	sol [(a)]	– (3.76 ± 0.20)	[2001BEN/PAL]	– (3.8 ± 0.2)	$I_m \rightarrow 0.0$
398		– (3.27 ± 0.20)		– (3.3 ± 0.2)	
423		– (2.81 ± 0.20)		– (2.8 ± 0.2)	
448		– (2.38 ± 0.20)		– (2.4 ± 0.2)	
473		– (2.06 ± 0.20)		– (2.1 ± 0.2)	

(a) Solubility experiments were conducted with crystalline boehmite as the solid phase.
(b) Solubility experiments were conducted with crystalline gibbsite as the solid phase

A weighted regression of the values of $\log_{10}{}^{*}K_2^{\circ}$ in Table V-39 as a function of $(T/\text{K})^{-1}$, depicted in Figure V-33, gives:

$$\log_{10}{}^{*}K_2^{\circ}\,(\text{V.71})) = (3.68290 \pm 0.23871) - (2774.763 \pm 80.130)(T/\text{K})^{-1} \quad \text{(V.73)}$$

which yielded:

$$\log_{10}{}^{*}K_2^{\circ}\,((\text{V.71}), 298.15\ \text{K}) = -(5.614 \pm 0.101),$$

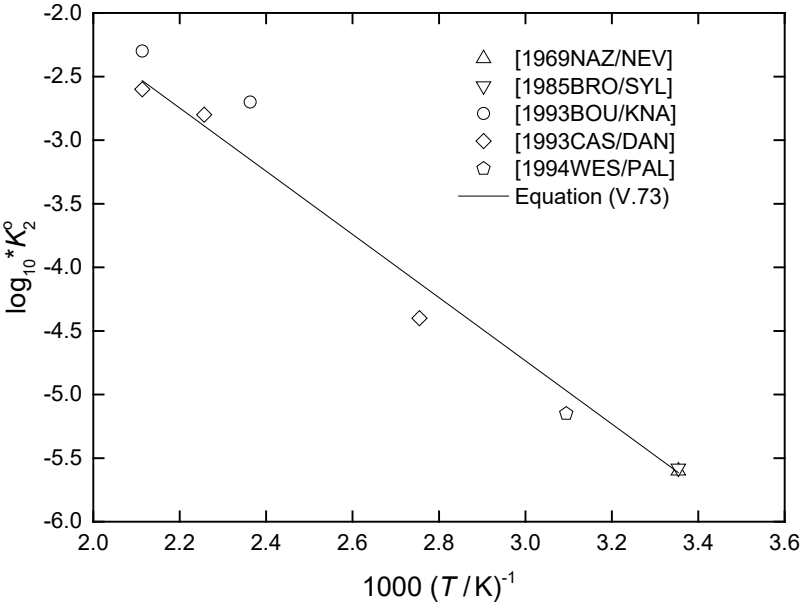
$$\Delta_r G_m^{\circ}\,((\text{V.71}), 298.15\ \text{K}) = (32.097 \pm 0.574)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^{\circ}\,((\text{V.71}), 298.15\ \text{K}) = (53.122 \pm 3.007)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^{\circ}\,((\text{V.71}), 298.15\ \text{K}) = (70.518 \pm 10.268)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

From the Nagra/PSI chemical thermodynamic database 01/01, Hummel *et al*. [2002HUM/BER] proposed a $\log_{10}{}^{*}\beta_2^{\circ}\,((\text{V.72}), 298.15\ \text{K})$ value of – 10.594 (*cf*. – (10.53 ± 0.12)) (Equations (V.70) and (V.73)) and – 10.55 from an SIT treatment of the single $\log_{10}{}^{*}\beta_c^{\circ}\,((\text{V.72}), 298.15\ \text{K})$ value in [1976BAE/MES].

Figure V-33: Plot of $\log_{10}{}^{*}K_2^{\circ}$ (V.71) *versus* $1000(T/\mathrm{K})^{-1}$.



V.5.2.2.3 The third hydrolysis constant of Al^{3+}

The narrow stability field of $Al(OH)_3(aq)$ ($6 \leq pH \leq 7.5$ at 298.15 K [1976BAE/MES]) coupled with the extremely low solubility of aluminium(III) hydroxides/oxides at low temperatures precluded potentiometry as a viable method to derive the formation constant of this species according to Reactions (V.74) or (V.75). Indeed, solubility studies of boehmite, AlO(OH), at elevated temperatures, where hydrolysis and solubility are enhanced, provided the best means of obtaining thermodynamic data, apart from a unique spectrophotometric investigation at 298.15 K [1969NAZ/NEV], also cited in Section V.5.2.2.2. Table V-40 summarises the available data for $\log_{10}{}^{*}K_3^{\circ}$ ((V.74), T/K) restricted to ≤ 473.15 K, again neglecting the discordant results from Verdes [1990VER].

$$Al(OH)_2^{+} + H_2O(l) \rightleftharpoons Al(OH)_3(aq) + H^{+} \qquad (V.74)$$

$$Al^{3+} + 3H_2O(l) \rightleftharpoons Al(OH)_3(aq) + 3H^{+} \qquad (V.75)$$

Table V-40: Values of $\log_{10}{}^{*}K_3^{\circ}$ ((V.74), T /K) obtained by application of the SIT to temperatures ≤ 473.15 K with 2σ uncertainties assigned by this review.

T /K	Method	$\log_{10}{}^{*}K_3^{\circ}$ reported	References	$\log_{10}{}^{*}K_3^{\circ}$ retained	Comments
298.15	sp	$\log_{10}{}^{*}K_{c,3}$ – 5.86	[1969NAZ/NEV]	– (5.86 ± 0.30)	I_c = 0.1 ($NaClO_4$)
423	sol [(a)]	– 3.91	[1983KUY/MAL]		I_m = 0.1 ($NaClO_4$)
473		– 3.35			
423	sol [(a)]	– 3.0	[1993BOU/KNA]	– (3.0 ± 0.7)	I_m < 0.2 (variable)
473		– 2.7		– (2.7 ± 0.7)	
363	sol [(a)]	– (4.6 ± 0.3)	[1993CAS/DAN]	– (4.6 ± 0.4)	I_m ≤ 0.01 (variable)
443		– (3.0 ± 0.3)		– (3.0 ± 0.5)	
473		– (2.7 ± 0.3)		– (2.7 ± 0.5)	

(a) Solubility experiments were conducted with crystalline boehmite as the solid phase.

A weighted regression of the limited number of $\log_{10}{}^{*}K_3^{\circ}$ values in Table V-40 as a function of (T /K) with $\log_{10}{}^{*}K_3^{\circ}$ fixed at – 5.86 at 298.15 K [1969NAZ/NEV] gives:

$$\log_{10}{}^{*}K_3^{\circ}\,(\text{V.74}) = (2.7708 \pm 0.3650) - (2573.29 \pm 130.25)(T/\text{K})^{-1} \qquad \text{(V.76)}$$

which yielded:

$$\log_{10}{}^{*}K_3^{\circ}\,((\text{V.74}),\ 298.15\ \text{K}) = -(5.860 \pm 0.300),$$

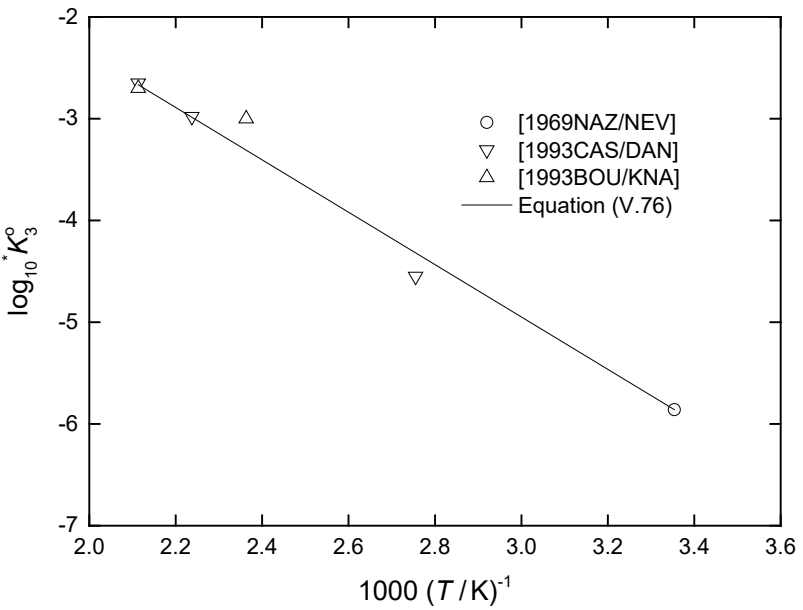
$$\Delta_r G_m^{\circ}\,((\text{V.74}),\ 298.15\ \text{K}) = (33.449 \pm 1.712)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^{\circ}\,((\text{V.74}),\ 298.15\ \text{K}) = (49.265 \pm 3.270)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^{\circ}\,((\text{V.74}),\ 298.15\ \text{K}) = (53.047 \pm 17.377)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

Figure V-34 shows the dependence of $\log_{10}{}^{*}K_3^{\circ}$ on (T /K)$^{-1}$. Combining the stepwise hydrolysis constants gives: $\log_{10}{}^{*}\beta_3^{\circ}\,((\text{V.75}),\ 298.15\ \text{K}) = -(16.39 \pm 0.23)$. This compares with the following values derived from various models: – 16.57 [1980KHO/KAT], – 16.94 [1990NOR/PLU], – 16.43 [1995POK/HEL], – 15.58 [1997SHO/SAS], and – 16.432 [2002HUM/BER].

Figure V-34: Plot of $\log_{10}{}^{*}K_3^{\circ}$ (V.74) *versus* $1000(T/\mathrm{K})^{-1}$.



V.5.2.2.4 The fourth hydrolysis constant of Al^{3+}

$$Al^{3+} + 4H_2O(l) \rightleftharpoons Al(OH)_4^- + 4H^+ \qquad \text{(V.77)}$$

The inability to obtain direct thermodynamic measurements of this hydrolysis equilibrium forced us to combine the gibbsite solubility equilibria in acidic (Reaction (V.78)) and basic (Reaction (V.85)) solutions where Al^{3+} and $Al(OH)_4^-$ dominate. This combination, together with the data for the ionisation of water at 298.15 K gave:

$$\log_{10}{}^{*}\beta_4^{\circ}\ ((\text{V.77}),\ 298.15\ \text{K}) = -\ (22.949 \pm 0.106),$$

$$\Delta_r G_m^{\circ}\ ((\text{V.77}),\ 298.15\ \text{K}) = (130.994 \pm 0.500)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^{\circ}\ ((\text{V.77}),\ 298.15\ \text{K}) = (182.702 \pm 2.410)\ \text{kJ mol}^{-1},$$

$$\Delta_r S_m^{\circ}\ ((\text{V.77}),\ 298.15\ \text{K}) = (173.430 \pm 3.940)\ \text{J·K}^{-1}\ \text{mol}^{-1}.$$

From their calorimetric measurements and estimates of the hydrolysis constants taken from the literature, Wu *et al.* [1997WU/DOB] estimated an enthalpy of reaction equivalent to $\Delta_r H_m^{\circ}\ ((\text{V.77}),\ 298.15\ \text{K}) = 179.9\ \text{kJ·mol}^{-1}$. Caiani *et al.* [1989CAI/CON] determined a value of $C_{p,m}^{\circ}(NaAl(OH)_4, aq, 298.15\ \text{K}) =$

74 $J \cdot K^{-1} \cdot mol^{-1}$ by extrapolation of their calorimetric data available from 323.15 to 523.15 K. This led to a value of $\Delta_r C^o_{p,m}$ ((V.77), 298.15 K) = – 139 $J \cdot K^{-1} \cdot mol^{-1}$.

V.5.2.2.5 Higher-order hydrolysed aqueous Al(III) species

Despite the relatively numerous studies made of this complex system, none of the constants derived from them are accepted by this review. The hydrolysis of Al^{3+} was studied from 335.65 to 422.95 K in 1 $mol \cdot kg^{-1}$ KCl in a hydrogen-electrode concentration cell to elucidate the stoichiometries of the polyaluminium hydrolysed species and determine their formation constants as a function of temperature by employing relatively high concentrations of Al(III), 0.0045-0.045 $mol \cdot kg^{-1}$ [1971MES/BAE]. The temperature range was limited at the low end by the slow kinetics of formation of some of these species. Three schemes were presented that included different combinations of these species with the preferred one involving, $Al_2(OH)_2^{4+}$, $Al_3(OH)_4^{5+}$ and $Al_{14}(OH)_{34}^{8+}$, although these authors [1976BAE/MES] later re-evaluated the data and replaced the latter cation by $Al_{13}O_4(OH)_{24}^{7+}$. The revised formation constants are given in Table V-41 together with those derived by others.

Table V-41: Values of the molal hydrolysis constants of Al^{3+} to form multinuclear aluminium(III) species.

T /K	Method	I_m	Medium	$\log_{10}{}^*K$	Reference
$Al_2(OH)_2^{4+}$					
298.15	pot	1.05	$NaClO_4$	– 8.0	[1976BAE/MES] [(a)]
298.15	pot	1.05	$NaClO_4$	– 8.0	[1976BAE/MES] [(b)]
335.65	pot	1.00	KCl	– (5.90 ± 0.04)	[1976BAE/MES] [(c)]
372.75				– (4.81 ± 0.15)	
397.95				– (3.95 ± 0.07)	
298.15		0.0		– 7.7	
298.15	pot	3.33	$NaNO_3$	– (7.41 ± 0.21)	[1991MIL/BUG]
298.15	^{27}Al NMR	0.01	$NaClO_4$	– 4.2	[1995FAU/LAB] [1997OHM]
298.15	pot	2.84	$NaClO_4$	– (6.9 ± 0.5)	[1998BUS/BUR]
298.15	pot	3.20	NaCl	– (7.59 ± 0.03)	[2015MAR/ARA]
$Al_2(OH)_4^{2+}$					
298.15	pot	3.33	$NaNO_3$	– (16.27 ± 0.13)	[1991MIL/BUG]
$Al_3(OH)_4^{5+}$					
298.15	pot	1.05	$NaClO_4$	– 13.47	[1976BAE/MES] [(a)]
335.65	pot	1.00	KCl	– (10.74 ± 0.04)	[1976BAE/MES] [(c)]
372.75				– (8.20 ± 0.14)	
397.95				– (7.01 ± 0.9)	
298.15		0.0		– 13.94	
298.15	pot	0.608	NaCl	– (13.56 ± 0.02)	[1981OHM/FOR]
298.15	pot	0.101	$NaNO_3$	– (13.12 ± 0.01)	[1985BRO/SYL]
298.15	pot	3.20	NaCl	– (13.90 ± 0.01)	[1987HED/SJO]
298.15	pot	3.33	$NaNO_3$	– (13.15 ± 0.08)	[1991MIL/BUG]
298.15	pot	2.84	$NaClO_4$	– (12.2 ± 0.2)	[1998BUS/BUR]
298.15	pot	3.20	NaCl	– (14.30 ± 0.04)	[2015MAR/ARA]

(Continued on next page)

Table V-41(continued)

T /K	Method	I_m	Medium	$\log_{10}{}^*K$	Reference
$Al_{13}O_4(OH)_{22}^{9+}$					
298.15	pot	3.20	NaCl	– (104.17 ± 0.05)	[2015MAR/ARA]
$Al_{13}O_4(OH)_{24}^{7+}$					
298.15	pot	1.05	$NaClO_4$	– 104.81	[1976BAE/MES] (a)
298.15	pot	0.01	$NaClO_4$	– 100.4	[1976BAE/MES] (d)
335.65	pot	1.00	KCl	– 64.3	[1976BAE/MES] (c)
372.75				– 52.9	
397.95				– 43.9	
298.15		0.0		– 98.73	
298.15	pot	0.608	NaCl	– (109.1 ± 0.12)	[1981OHM/FOR]
298.15	pot	3.20	NaCl	– (112.79 ± 0.13)	[1987HED/SJO]
298.15	pot	3.20	NaCl	– (113.20 ± 0.06)	[2015MAR/ARA]
$Al_{13}O_5(OH)_{24}^{5+}$					
298.15	pot	3.20	NaCl	– (123.0 ± 0.2)	[2015MAR/ARA]

(a) Re-calculated from data supplied by Aveston [1965AVE].
(b) Re-calculated from data reported by Kubota [1956KUB].
(c) Re-calculated for scheme II preferred in their earlier study [1971MES/BAE].
(d) Re-calculated from data reported by Smith and Hem [1972SMI/HEM].

The species $Al_2(OH)_4^{2+}$ and $Al_{13}O_4(OH)_{24}^{7+}$ are difficult to identify uniquely from potentiometric titration data. Baes and Mesmer [1976BAE/MES] favoured these species in their scheme because their structures had been identified in the solid state, *viz*., $Al_2(OH)_2(H_2O)_8(SO_4)_2 \cdot 2H_2O(cr)$, and other basic sulfates and selenates. They declared that the choice of the two lower species, $Al_2(OH)_4^{2+}$ and $Al_3(OH)_4^{5+}$, over the alternative combination of $Al_2(OH)_3^{3+}$ (Al_2) and $Al_3(OH)_3^{6+}$ (Al_3), was based mainly on the fit parameters and the known existence of the $M_2(OH)_4^{2+}$ species in other metal systems. However, Öhman *et al.* [1981OHM/FOR] did not identify $Al_2(OH)_4^{2+}$ from an analysis of their detailed potentiometric titration data at 298.15 K. It is known that the symmetrical Keggin ion, $Al_{13}O_4(OH)_{24}^{7+}$ (Al_{13}), is slow to form and decompose, thereby complicating analyses of experimental data obtained at moderate temperatures and high aluminium concentrations. Finally, these higher-order aqueous species all eventually become less stable with increasing temperature.

Bradley and Hanna [1994BRA/HAN] combined ^{27}Al and ^{23}Na MAS NMR (Magic Angle Spinning Nuclear Magnetic Resonance) and powder X-ray diffraction studies of freeze-dried sodium aluminate solutions to suggest that in the solid state, apart from the dominant tetrahedral anion $Al(OH)_4^-$, the dimeric species, $(Al(OH)_3)_2O^{2-}$, and polymeric species $Al(OH)_2(OAl)_2^{3+}$ and $Al(OH)(OAl)_3^{5+}$ also exist in relative amounts corresponding to the Al:OH ratio in solution. A review by Sipos [2009SIP] suggested that in synthetic homogeneous Bayer liquors (*i.e.*, strongly alkaline solutions) the dominant ions in solution are simply, Na^+, OH^-, $Al(OH)_4^-$, a dimer with the probable formula $Al_2O(OH)_6^{2-}$, and ion pairs thereof up to 373 K. No definitive evidence was found for the presence of other higher-order mononuclear Al

species, such as $Al(OH)_5^{2-}$ and $Al(OH)_6^{3-}$, and polynuclear Al(III) species, such as hexameric cations. Interestingly, Watling *et al.* [1999WAT/SIP] claimed that Raman and IR spectra of solid aluminate compounds could not be used to deduce unambiguously the structure of aluminate species in solution.

Electrospray mass spectrometric measurements of aqueous aluminium solutions (1 s-14 d) were reported by Sarpola [2007SAR] as a function of pH (3.2-10), concentration 0.001 to 0.1 mol·dm^{-3} and reaction time (1 s-14 d). The main species detected were Al_2, Al_3 and Al_{13} at these conditions.

Yang *et al*. [2010YAN/QIA] presented a mechanism for the formation of $Al_{30}O_8(OH)_{56}(H_2O)_{26}^{18+}$ (Al_{30}) using density functional theory and kinetic analyses of ^{27}Al-NMR experimental results. This is the largest aggregate characterised to date. It was isolated previously in the solid state as a sulfate salt. The possible mechanisms of formation of Al_{30} differ but all invoke Al_{13} as an intermediate.

^{27}Al NMR of aqueous Al species was pioneered by Akitt *et al*. [1969AKI/GRE], [1969AKI/GRE2]. The main tetrahedral, octahedral and Keggin structures of aluminium were investigated subsequently in [1989AKI/ELD2]. The major species detected were: Al^{3+}, $AlOH^{2+}$, $Al(OH)_2^+$ and/or $Al_2(OH)_x^{6-x}$ or $Al_2O(OH)_6^{2-}$ [1970MOO/EVA], $Al_{13}O_4(OH)_{24}(H_2O)_{12}^{7+}$ and $Al(OH)_4^-$. It should be mentioned that a more recent NMR study by Faust *et al.* [1995FAU/LAB] predicted a much larger formation constant for $Al_2(OH)_4^{2+}$, but this finding was later brought into question [1997OHM].

Dielectric relaxation was used to investigate the structure of concentrated aqueous alkaline aluminate solutions at total aluminium concentrations of 1 to 6 mol·dm^{-3} and total sodium ion concentrations of 8.33 mol·dm^{-3} at 298.15 K over a range of frequencies [2002BUC/SIP]. These results indicated the presence of solvent-shared ion pairs, [Na^+ $(HO)_3$Al-O-Al$(OH)_3^{2-}$], although even with this sophisticated method the exact nature of higher-order Al(III) species in solution remained indefinite. For very concentrated alkaline aluminate solutions, supporting Raman spectral evidence exists that the only identifiable high-order Al(III) species is the dimer $Al_2O(OH)_6^{2-}$, which Sipos *et al*. [2006SIP/MAY] suggested may exist as $[(HO)_3Al\text{-}O\text{-}Al(OH)_3]^{2-}$.

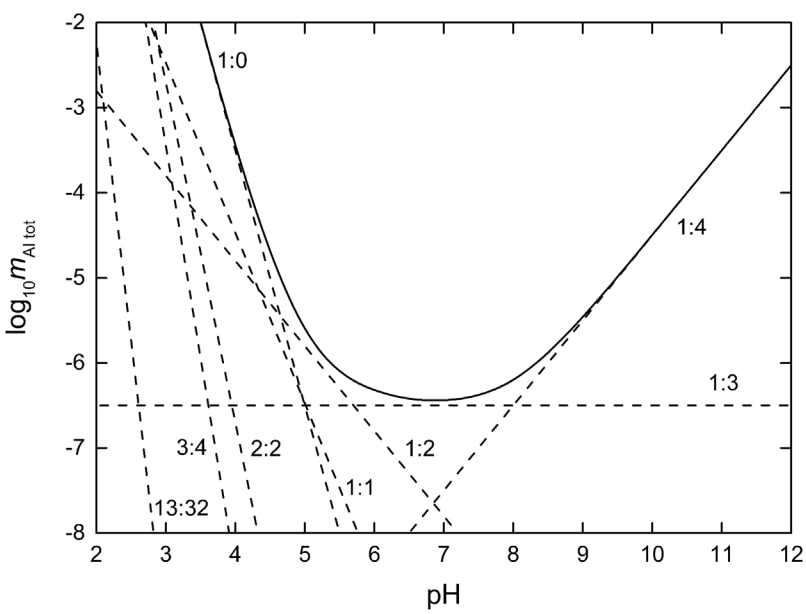
V.5.2.3 Solubility data for gibbsite or hydrargillite, α-Al(OH)$_3$

Figure V-35 was taken from [1976BAE/MES] to provide an impression of the solubility profile of gibbsite as a function of pH_m at 298.15 K while also illustrating the nature of aqueous species that can be encountered at low temperatures.

In general the experimental uncertainties in the reported solubility constants of gibbsite in acidic to near neutral pH conditions are large and often severely under-estimated. These uncertainties arise because of the following problems: slow dissolution kinetics near ambient temperature, particularly at low pH and low ionic strengths;

the presence of fine particles; also according to Hemingway *et al*. [1989HEM/KIT] the presence of chloride ions and variations in the techniques used by different authors (although more recent results tend to show that adequate pre-treatment of gibbsite eliminated this problem as originally proposed by Bloom and Weaver [1982BLO/WEA]); the presence of complexing pH buffers [1979MAY/HEL], [1990WES/PAL]; and conversion to boehmite above *ca*. 353.15 K. The persistence of the metastable phase (bayerite) and to a lesser extent boehmite or pseudo-boehmite over a wide temperature range were also complicating factors in the interpretation of solubility results, particularly in earlier studies where detailed characterisation of the solid phase before and after completion of the experiments was either impossible or not attempted [1971SMI/MIS2]. In this regard, Verdes *et al*. [1992VER/GOU] referred to Kittrick [1966KIT2] who postulated that an $NaAlO_2$(s) impurity existed in gibbsite prepared by the Bayer process that can lead to weakly basic solutions becoming rapidly supersaturated, resulting in precipitation of bayerite. Indeed, in initial tests with untreated gibbsite, Verdes *et al*. found that at high solid to solution loadings and low pH, higher solubility quotients were obtained consistent with the effects of a soluble impurity being present in the commercial grade gibbsite.

Figure V-35: The solubility curve of gibbsite at 298.15 K and 0.1 MPa ($I_m = 0$) taken from [1976BAE/MES] where the Al(III) species are defined by their Al:OH ratios.



V.5.2.3.1 First solubility constant in acidic media

There is reasonable agreement amongst the values of $\log_{10}{}^{*}K^{\circ}_{s,0}$ (V.78) for the dissolution of α-$Al(OH)_3$ [2] in moderately acidic solutions as evident from the values summarised in Table V-42.

$$\alpha\text{-}Al(OH)_3 + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l) \qquad (V.78)$$

Slow kinetics of dissolution at low temperatures and low ionic strengths probably contributed to spurious results in [1962FRI/PEE] and [1980HIT/MES]. Kittrick [1966KIT2] was perhaps the first to point out the need to pre-treat gibbsite particles, preferably with dilute acid, to remove fines and unwanted surface phases. He later proposed a novel "immiscible displacement" method that appears to have allowed equilibrium to be reached in several days rather than many months at ambient conditions [1980KIT]. Most solubility studies carried out after 1966 used various pre-treatments and careful characterisations of the solids at least before the experiments were undertaken. Moreover, in later studies, equilibrium was often approached from over- and under-saturation to ensure the reversibility of the measurements.

Palmer and Wesolowski [1992PAL/WES] made a global empirical fit of their 77 $\log_{10}{}^{*}K_{s,0}$ values with six adjustable parameters and $\Delta_r C^{\circ}_{p,m}((V.78), T)$ values from [1986BAR/HEP] and [1986HOV/TRE]. This fit yielded the following values at 298.15 K: $\log_{10}{}^{*}K^{\circ}_{s,0} = (7.74 \pm 0.04)$, $\Delta_r G^{\circ}_m = -(44.2 \pm 0.2)$ kJ·mol^{-1}, $\Delta_r H^{\circ}_m = -(105.3 \pm 2.0)$ kJ·mol^{-1}, $\Delta_r S^{\circ}_m = -(205 \pm 6)$ J·K^{-1}·mol^{-1} and $\Delta_r C^{\circ}_{p,m} = (13 \pm 8)$ J·K^{-1}·mol^{-1}. In addition, a Pitzer ion interaction treatment was described using known single electrolyte coefficients and only four adjustable parameters, *viz*., $\ln{}^{*}K_{s,0} = p_1 + p_2(T/\mathrm{K})^{-1}$, ${}^{s}\theta_{AlNa} = p_3$ and $\psi_{AlNaCl} = p_4$. The latter two mixing parameters were found to be similar in magnitude to those of the analogous La^{3+}-Na^+-Cl^--$H_2O(l)$ system. However, a further complication was incurred depending on whether the higher-order electrostatic mixing terms, ${}^{E}\theta_{AlNa}$ and ${}^{E}\theta_{AlH}$, taken from the literature were included or not [1975PIT]. Inclusion of these terms led to $\log_{10}{}^{*}K^{\circ}_{s,0} = 7.65$ at 298.15 K (*cf*. 7.612 derived by Christov *et al*. from their Pitzer treatment [2007CHR/DIC]), whereas neglecting these terms gave $\log_{10}{}^{*}K^{\circ}_{s,0} = 7.76$, which is virtually identical to that obtained from the empirical fit given above.

Table V-42: Values of $\log_{10}{}^{*}K^{\circ}_{s,0}$ ($Al(OH)_3$, gibb) cited in the literature and where possible re-calculated by the SIT using either a constant $\Delta\varepsilon$ or concentration-dependent, $\Delta\varepsilon = \Delta\varepsilon_1 + \Delta\varepsilon_2 \log_{10}(I_m)$. Uncertainties were assigned by this review.

T /K	$\log_{10}{}^{*}K^{\circ}_{s,0}$ reported	References	$\log_{10}{}^{*}K^{\circ}_{s,0}$ re-calculated	$\Delta\varepsilon$ /kg·mol^{-1}	Comments
298.15	9.04	[1958GAY/THO]			m_{HClO_4} (initial) $\leq 1.82 \times 10^{-3}$
298.15	8.5	[1962FRI/PEE]			$I_m = (0.03\text{-}60) \times 10^{-3}$

(Continued on next page)

[2] Gurvich *et al.* referred to gibbsite as "the monoclinic γ-modification" [1994GUR/VEY].

Table V-42 (continued)

T /K	$\log_{10}{}^{*}K^{\circ}_{s,0}$ reported	References	$\log_{10}{}^{*}K^{\circ}_{s,0}$ re-calculated	$\Delta\varepsilon$ /kg·mol^{-1}	Comments
296.15-298.15	(7.97 ± 0.12)	[1966KIT2]	(7.95 ± 0.12)		I_m = (1.3-5.55) × 10^{-3}
288.15	(8.59 ± 0.04)	[1974SIN]	(8.50 ± 0.12)		I_m = (2.2-3.0) × 10^{-2}
298.15	(8.04 ± 0.04)		(7.92 ± 0.12)		
308.15	(7.52 ± 0.04)		(7.41 ± 0.12)		
298.15	(8.11 ± 0.06)	[1979MAY/HEL]			I_m = (0.1-60.0) × 10^{-3} ≤ 48 % hydrolysed
333.15	$\log_{10}{}^{*}K_{s,0}$ (7.20 ± 0.15)	[1980HIT/MES]			I_m = 1.0 (NaCl)
298	$\log_{10}{}^{*}K_{s,0}$ (7.98 ± 0.06)	[1980KIT]	(7.86 ± 0.20)		I_c = 2 × 10^{-3} $AlCl_3$
295	(7.71 ± 0.06)	[1982BLO/WEA]			
298.15	(7.76 ± 0.14) (7.78 ± 0.14) [a]	[1988PER/KIT]	(7.83 ± 0.13)		[a] corrected, see Appendix A I_m = (0.3-3.3) × 10^{-2}
273.15 283.15 298.15	9.62 8.95 8.05	[1989BOU/GRI]			Re-calculated from [1981BOU2]
353.15	(5.00 ± 0.08)	[1992NAG/LAS]	(4.95 ± 0.12)		I_m = (1.2-4.0) × 10^{-3} ≤ 37 % hydrolysed
303.15	(7.43 ± 0.06) 7.34 [f]	[1992PAL/WES]	(7.33 ± 0.10) 7.64	(0.19 ± 0.08) [c] (0.14 ± 0.04) [d] – (0.20 ± 0.06) [e]	[c] $I_m \leq 1$ (NaCl) [d] $\Delta\varepsilon_1$ [e] $\Delta\varepsilon_2$
323.15	(6.31 ± 0.05) 6.19 [f]		(6.24 ± 0.10) 6.25	(0.17 ± 0.09) [c]	[f] derived from the Pitzer model
343.15	(5.32 ± 0.07) 5.18 [f]		(5.26 ± 0.10) 5.28	(0.08 ± 0.04) [c]	
298	(8.02 ± 0.03)	[1994SU/HAR]			gibbsite + imogolite ($Al_2SiO_3(OH)_4$) mixture, $I_c \approx 0.01$
298.15	(7.83 ± 0.03)	[1996MOG/GAN]			kinetic study c_{HClO_4} and $c_{Al} < 10^{-3}$
298.15	7.612	[2007CHR/DIC]			derived from a Pitzer model

Application of the SIT treatment to the available $\log_{10}{}^{*}K_{s,0}$ values according to Equation (V.79) resulted in the values listed in Table V-42. In the case of the data in [1992PAL/WES], where ionic strengths up to *ca.* 5 mol·kg^{-1} were reported, an alternative treatment was necessary with $\Delta\varepsilon = \Delta\varepsilon_1 + \Delta\varepsilon_2 \log_{10}(I_m)$.

$$\log_{10}{}^{*}K_{s,0} - 6D + 3\log_{10} a_{H_2O} = \log_{10}{}^{*}K^{\circ}_{s,0} - \Delta\varepsilon I_m \qquad \text{(V.79)}$$

The results of a weighted linear regression, Equation (V.80), of the $\log_{10}{}^{*}K^{\circ}_{s,0}$ values in the fourth column of Table V-42 with the value of $\Delta_r H^{\circ}_m$ (V.78) set to comply with $\Delta_f H^{\circ}_m$ ($Al(OH)_3$, gibb, 298.15 K) = – (1294.000 ± 1.500) kJ·mol^{-1} recommended by this review, as well as $\Delta_f H^{\circ}_m$ (Al^{3+}, aq, 298.15 K) = – (538.400 ± 1.500) kJ·mol^{-1} and

$\Delta_f H_m^o$ (H_2O, l, 298.15 K) = – (285.830 ± 0.040) kJ·mol^{-1} [1989COX/WAG] are shown in Figure V-36.

$$\log_{10} {}^{*}K_{s,0}^{o} = -(10.057 \pm 0.098) + (5322.053)(T/\mathrm{K})^{-1} \quad \text{(V.80)}$$

The resulting thermodynamic quantities of reaction at 298.15 K are:

$$\log_{10} {}^{*}K_{s,0}^{o}\ ((\text{V.78}), 298.15\ \mathrm{K}) = (7.793 \pm 0.098),$$

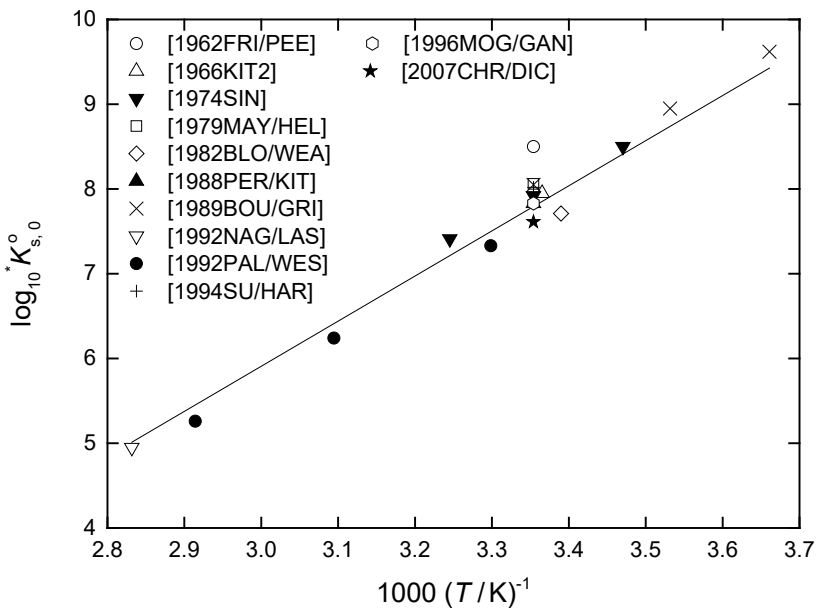
$$\Delta_r G_m^o\ ((\text{V.78}), 298.15\ \mathrm{K}) = -(44.483 \pm 0.558)\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H_m^o\ ((\text{V.78}), 298.15\ \mathrm{K}) = -(101.890 \pm 2.120)\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r S_m^o\ ((\text{V.78}), 298.15\ \mathrm{K}) = -(192.544 \pm 7.353)\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

The known heat capacities for this reaction [1986HOV/TRE], [1977HEM/ROB3] and [1972KEL2] were not included in the fitting process, but assuming that $\Delta_r C_{p,m}^o$ ((V.78), 298.15 K) is only *ca*. 1.3 J·K^{-1}·mol^{-1} [2002HUM/BER], this assumption has an insignificant effect on the fit. In their review of aluminium aqueous chemistry, Khodakovsky *et al*. [1980KHO/KAT] derived the following quantities at 298.15 K converted to the acidic form: $\log_{10} {}^{*}K_{s,0}^{o}$ (V.78) = (8.14 ± 0.30) and $\Delta_r H_m^o$ = – (102.03 ± 1.29) kJ·mol^{-1}.

Figure V-36: Plot of $\log_{10} {}^{*}K_{s,0}^{o}$ values for gibbsite solubility *versus* $(T/\mathrm{K})^{-1}$; the values reported in [1962FRI/PEE], [1982BLO/WEA], [1989BOU/GRI], [1994SU/HAR] and [2007CHR/DIC] were excluded from the fit (see Appendix A entries).



V.5.2.3.2 Second solubility constant of gibbsite in acidic media

Potentiometric hydrolysis studies provided a better route to determine the stability of $AlOH^{2+}$ rather than from the solubility studies (see Section V.5.2.2.1).

$$\alpha\text{-}Al(OH)_3 + 2H^+ \rightleftharpoons AlOH^{2+} + 2H_2O(l) \quad \text{(V.81)}$$

The appropriate SIT expression to describe this equilibrium is:

$$\log_{10} {}^{*}K_{s,1} - 2D + 2\log_{10} a_{H_2O} = \log_{10} {}^{*}K^{o}_{s,1} - \Delta\varepsilon I_m \quad \text{(V.82)}$$

Table V-43: Values of $\log_{10} {}^{*}K_{s,1}$ as cited in the literature and re-calculated using SIT.

T /K	$\log_{10} {}^{*}K_{s,1}$	I_m	Medium	References	$\log_{10} {}^{*}K^{o}_{s,1}$ re-calculated	Comments
298.15	$\log_{10} {}^{*}K^{o}_{s,1}$ 3.12	0.011	HNO_3	[1979MAY/HEL]	3.11	
298.15		0.101	$NaNO_3$	[1985BRO/SYL]	(2.89 ± 0.10) [(a)]	From their value of ${}^{*}K_{c,1}$ and $K^{o}_{s,0}$ [1992PAL/WES]
323.15	2.32 [(a)]	0.1	NaCl	[1994WES/PAL]	(2.11 ± 0.22)	$\Delta\varepsilon$ from [1993PAL/WES

(a) Re-calculated in this review as (2.30 ± 0.22).

Wesolowski and Palmer [1994WES/PAL] reported a value of $\log_{10} {}^{*}K_{s,1} = (3.04 \pm 0.04)$ at $I_m = 0.1$ (NaCl) and 323.15 K from gibbsite solubility data, but their choice of speciation is questioned in this review. May *et al*. [1979MAY/HEL] reported $\log_{10} {}^{*}K^{o}_{s,1} = 3.12$ at 298.15 K from their gibbsite solubility study, but this value and those for $\log_{10} K^{o}_{s,3}$ and $\log_{10} {}^{*}K^{o}_{s,4}$ have been criticised [1990WES/PAL], because of the effect of complexation of the aluminate anion by the pH buffer, Bis-Tris (Bis(2-hydroxyethyl)amino-tris(hydroxymethyl)methane), used by May *et al*. in the intermediate pH range.

A combination of the thermodynamic quantities for Reactions (V.68) and (V.78) provided the best estimate of the corresponding quantities for Reaction (V.81).

$$\log_{10} {}^{*}K^{o}_{s,1} \text{((V.81), 298.15 K)} = (2.873 \pm 0.120),$$

$$\Delta_r G^{o}_{m} \text{((V.81), 298.15 K)} = -(16.399 \pm 0.687) \text{ kJ·mol}^{-1},$$

$$\Delta_r H^{o}_{m} \text{((V.81), 298.15 K)} = -(45.330 \pm 2.584) \text{ kJ·mol}^{-1},$$

$$\Delta_r S^{o}_{m} \text{((V.81), 298.15 K)} = -(97.035 \pm 5.298) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

Khodakovsky *et al*. [1980KHO/KAT] derived the following quantities at 298.15 K converted to the acidic form of Reaction (V.81): $\log_{10} {}^{*}K^{o}_{s,1} = 3.14$ and $\Delta_r H^{o}_{m} = -55.19$ kJ·mol^{-1}.

V.5.2.3.3 Third solubility constant of gibbsite

$$\alpha\text{-}Al(OH)_3 + H^+ \rightleftharpoons Al(OH)_2^+ + H_2O(l) \quad \text{(V.83)}$$

The $Al(OH)_2^+$ species becomes insignificant with increasing temperature and ionic strength until the critical temperature of water is approached [1976BAE/MES].

A combination of the thermodynamic quantities for Reactions (V.71) and (V.81) provided the best estimate of the corresponding quantities for Reaction (V.83), which this review selects:

$$\log_{10} {}^{*}K^{\circ}_{s,2}\,((V.83), 298.15\ K) = \mathbf{-\ (2.741 \pm 0.152)},$$

$$\Delta_r G^{\circ}_m\,((V.83), 298.15\ K) = \mathbf{(15.644 \pm 0.868)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_r H^{\circ}_m\,((V.83), 298.15\ K) = \mathbf{(2.224 \pm 5.604)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_r S^{\circ}_m\,((V.83), 298.15\ K) = \mathbf{-\ (45.011 \pm 19.020)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

Khodakovsky *et al*. [1980KHO/KAT] derived the following quantities at 298.15 K converted to the form of Reaction (V.83): $\log_{10} {}^{*}K^{\circ}_{s,2} = -2.3$ and $\Delta_r H^{\circ}_m = 22.2\ kJ{\cdot}mol^{-1}$.

The recommended values for $Al(OH)_2^+$ given in bold in Table V-44 were derived from the solubility equilibria given immediately above and are compared with previous estimates in the following table.

Table V- 44: Thermodynamic quantities derived for the $Al(OH)_2^+$ species at 298.15 K.

$\Delta_f G^{\circ}_m$ /kJ·mol^{-1}	$\Delta_f H^{\circ}_m$ /kJ·mol^{-1}	S°_m /J·K^{-1}·mol^{-1}	Reference
– (900.2 ± 2.3)	– (998.3 ± 3.1)	– (30.6 ± 3.0)	[2001BEN/PAL]
– (902.991 ± 1.736)	**– (1005.946 ± 5.801)**	**– (46.521 ± 19.022)**	This review

V.5.2.3.4 Fourth solubility constant of gibbsite

$$\alpha\text{-}Al(OH)_3 \rightleftharpoons Al(OH)_3(aq) \qquad (V.84)$$

Khodakovsky *et al*. [1980KHO/KAT] derived the following quantities at 298.15 K: $\log_{10} K^{\circ}_{s,3} = -8.43$ and $\Delta_r H^{\circ}_m = 63.9\ kJ{\cdot}mol^{-1}$. A re-analysis of the solubility data in [1994WES/PAL] yields: $\log_{10} K^{\circ}_{s,3} \approx \log_{10} K_{s,3} = -8.3$ at 323.15 K.

A combination of the thermodynamic quantities for Reactions (V.71) and (V.83) provides the best estimate of the corresponding quantities for Reaction (V.84), which this review selects:

$$\log_{10} K^{\circ}_{s,3}\,((V.84), 298.15\ K) = \mathbf{-\ (8.601 \pm 0.336)},$$

$$\Delta_r G^{\circ}_m\,((V.84), 298.15\ K) = \mathbf{(49.093 \pm 1.919)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_r H^{\circ}_m\,((V.84), 298.15\ K) = \mathbf{(51.489 \pm 6.488)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_r S^{\circ}_m\,((V.84), 298.15\ K) = \mathbf{(8.036 \pm 22.693)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

The recommended values for $Al(OH)_3(aq)$ given in bold in Table V-45 were derived exclusively from the solubility equilibria given immediately above and are compared with previous estimates in the following table.

Table V- 45: Thermodynamic quantities derived for the $Al(OH)_3(aq)$ species at 298.15 K.

$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}	Reference
– (1110.4 ± 4.1)	– (1270.7 ± 6.5)	– (5.7 ± 7.0)	[2001BEN/PAL]
– (1106.682 ± 2.438)	**– (1242.511 ± 6.659)**	**(76.476 ± 22.695)**	This review

V.5.2.3.5 Fifth solubility constant of gibbsite in basic media

The dissolution and precipitation reactions of gibbsite in alkaline solutions have been widely studied mainly because of their importance to the Bayer process. Kinetics of dissolution are significantly faster than in acidic media (conversely, faster at low ionic strengths but slower in acidic solutions). The solution speciation is constant over a wide pH range involving only the tetrahedral aluminate anion. The solubility constants are generally reported in terms of Reaction (V.85) because the hydroxide ion concentration can generally be determined accurately from the mass balance equation:

$$\alpha\text{-}Al(OH)_3 + OH^- \rightleftharpoons Al(OH)_4^- \qquad (V.85)$$

A pioneering solubility study of gibbsite [1930FRI/JUC] provided early insight into the behaviour of the aluminate anion by extending the NaOH(sln) concentration to 20.6 mol·kg^{-1}, although these early investigations by Fricke and co-workers [1928FRI], [1929FRI3] and [1930FRI/JUC] were hampered by the lack of precise analytical methods and techniques to characterise thoroughly the solid phase. However, they provided reasonable solubility constants for gibbsite in strongly basic solutions. Russell *et al.* [1955RUS/EDW] later provided a large dataset of reliable solubilities for gibbsite and boehmite over wide ranges of temperature and ionic strength (NaOH). Much later, Wesolowski [1992WES] performed a large number of batch experiments in NaCl/NaOH solutions from 279.55 to 353.15 K at ionic strengths from 0.01 to 5 mol·kg^{-1}, where the dominant presence of NaCl as the supporting electrolyte allowed these results to be treated by the SIT according to Equation (V.86). Moreover, some of these experiments were conducted with initially supersaturated solutions so that the approach to equilibrium was reversed. Wesolowski used both an empirical and the Pitzer ion interaction approach to reconcile his data and those from previous studies, [1955RUS/EDW] in particular. The results of application of the SIT to the data in NaCl/NaOH and KOH solutions are shown in Figure V-37 and Figure V-38.

$$\log_{10} K_{s,4} = \log_{10} K_{s,4}^o - \Delta\varepsilon I_m \qquad (V.86)$$

A value of $\log_{10} K_{s,4}^o = -(1.138 \pm 0.036)$ at 298.15 K resulted from the weighted regression of the data in NaCl dominated solutions [1992WES] shown in Figure V-37 with $\Delta\varepsilon = -(0.039 \pm 0.014)$ kg·mol^{-1}. The corresponding value calculated by Wesolowski using the Pitzer ion interaction approach to treat a much larger data set was $\log_{10} K_{s,4}^o = -(1.143 \pm 0.004)$ (2σ).

Figure V-37: Plot of $\log_{10}K_{s,4}$ *versus* I_m for gibbsite solubility data in an aqueous NaCl medium [1992WES]. Solid lines result from isothermal fits; dashed lines result from a global fit with a temperature independent value of $\Delta\varepsilon = -0.039$ kg·mol^{-1}.

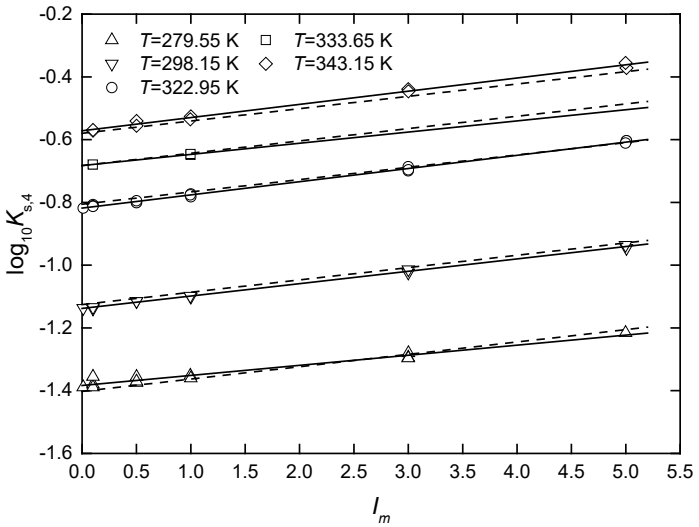
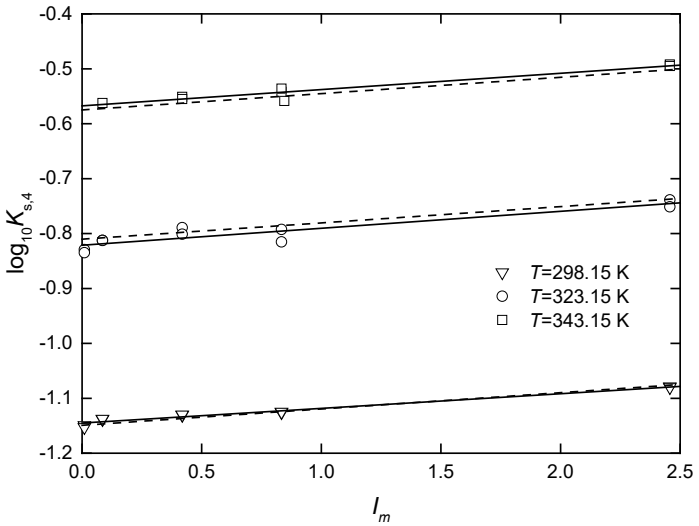


Figure V-38: Plot of $\log_{10}K_{s,4}$ *versus* I_m for gibbsite solubility data in an aqueous KOH medium [1992WES]. Solid lines result from isothermal fits; dashed lines result from a global fit with $\Delta\varepsilon = -0.030$ kg·mol^{-1}.



The values of $\log_{10}K^{\circ}_{s,4}$ from all the relevant studies are given in Table V-46.

Table V-46: Values of $\log_{10} K^{\circ}_{s,4}$ cited in the literature and recalculated using the SIT treatment where the uncertainties from the SIT were estimated by this review (2σ).

T /K	I_m (max)	$-\log_{10} K^{\circ}_{s,4}$ reported	$-\log_{10} K^{\circ}_{s,4}$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	$-\log_{10} K^{\circ}_{s,4}$ SIT	References
303.15	4 (NaOH)				(1.16 ± 0.18) [a, b]	[1928FRI]
303.15	7 (NaOH)		(0.98 ± 0.11)	(0.056 ± 0.020)	(0.98 ± 0.11) [a]	[1930FRI/JUC]
347.15	5.1 (NaOH)		(0.53 ± 0.11)	(0.064 ± 0.011)	(0.53 ± 0.11)	[1951FUL/GIN]
352.15	5.2		(0.48 ± 0.11)		(0.48 ± 0.11)	
357.15	5.3		(0.42 ± 0.11)		(0.42 ± 0.11)	
362.15	5.3		(0.37 ± 0.11)		(0.37 ± 0.11)	
298	?	1.94				[1952KUZ]
313.15	7.9 (NaOH)		(1.01 ± 0.04)	(0.060 ± 0.008)	(1.03 ± 0.06) [a]	[1955RUS/EDW]
333.15	4.1		(0.72 ± 0.04)	(0.053 ± 0.008)	(0.73 ± 0.04) [a]	
343.15	11.1		(0.61 ± 0.04)	(0.061 ± 0.004)	(0.61 ± 0.04) [a]	
353.15	4.1		(0.44 ± 0.04)	(0.055 ± 0.004)	(0.46 ± 0.04) [a]	
363.15	4.2		(0.34 ± 0.04)	(0.059 ± 0.002)	(0.34 ± 0.04) [a]	
313.15	4.9 (NaOH)	(0.90 ± 0.03)	(0.89 ± 0.10)	(0.058 ± 0.010)	(0.90 ± 0.10) [a]	[1963IKK/OKA]
343.15	5.1	(0.47 ± 0.02)	(0.46 ± 0.10)	(0.041 ± 0.014)	(0.52 ± 0.10) [a]	
333	5 (NaOH)		(0.64 ± 0.08)	(0.038 ± 0.021)	(0.72 ± 0.10) [a]	[1964LYA/KHO]
	7.8 (NaCl/OH)		(0.69 ± 0.05)	(0.048 ± 0.010)	(0.76 ± 0.10) [a]	
298.15	7.4 (NaOH)	1.27	(1.24 ± 0.10)	(0.071 ± 0.002)	(1.21 ± 0.10) [a, b]	[1970SZI/BER]
308.15	8.0	1.05	(1.02 ± 0.10)	(0.061 ± 0.002)	(1.01 ± 0.10) [a]	
323.15	7.6	0.78	(0.76 ± 0.10)	(0.061 ± 0.005)	(0.77 ± 0.10) [a]	
333.15	7.5	0.63	(0.63 ± 0.10)	(0.062 ± 0.003)	(0.63 ± 0.10) [a]	
294	0.05 (NH_4Cl/OH)	1.26			(1.24 ± 0.06) [c]	[1992VER/GOU]
323	0.025	0.76			(0.74 ± 0.08) [c]	
343	0.025	0.46			(0.46 ± 0.12) [c]	

(Continued on next page)

Table V-46 (continued)

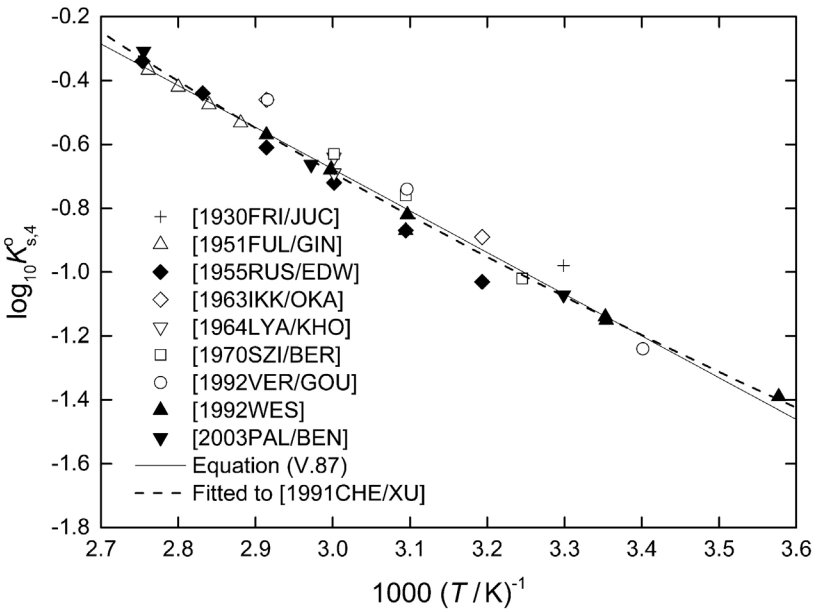
T/K	I_m (max)	$-\log_{10}K^{\circ}_{s,4}$ reported	$-\log_{10}K^{\circ}_{s,4}$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	$-\log_{10}K^{\circ}_{s,4}$ SIT	References
279.55	5 (NaCl/OH)	1.39 (d)	(1.38 ± 0.04)	(0.032 ± 0.004)	(1.39 ± 0.04) (c)	[1992WES]
298.25		1.14 (d)	(1.14 ± 0.04)	(0.039 ± 0.001)	(1.14 ± 0.04) (c)	
322.95		0.82 (d)	(0.82 ± 0.04)	(0.042 ± 0.001)	(0.82 ± 0.04) (c)	
333.65		0.68 (d)	(0.68 ± 0.03)	(0.036 ± 0.004)	(0.68 ± 0.03) (c)	
343.15		0.56 (d)	(0.57 ± 0.03)	(0.042 ± 0.002)	(0.56 ± 0.03) (c)	
298.15	2.5 (KOH)	1.14 (d)	(1.15 ± 0.06)	(0.027 ± 0.004)	(1.15 ± 0.06)	
323.15		0.81 (d)	(0.82 ± 0.05)	(0.031 ± 0.008)	(0.81 ± 0.12)	
343.15		0.56 (d)	(0.57 ± 0.05)	(0.030 ± 0.006)	(0.57 ± 0.05)	
303.15	5 ($NaNO_3$)		(1.06 ± 0.06)	(0.014 ± 0.006)	(1.06 ± 0.06)	[2016BEN/HIL]
336.45			(0.66 ± 0.12)	0.014	(0.66 ± 0.12)	
362.95			(0.30 ± 0.05)	(0.015 ± 0.004)	(0.30 ± 0.05)	

(a) $\Delta\varepsilon$ was fixed at – 0.06 kg·mol^{-1}, based on an average of the $\Delta\varepsilon$ values in NaOH media [1951FUL/GIN], [1955RUS/EDW].
(b) These values were not included in the overall data analysis for reasons outlined in the Appendix A entries.
(c) $\Delta\varepsilon$ was fixed at – (0.039 ± 0.002) kg·mol^{-1}, based on a weighted regression of all the data in NaCl/NaOH solutions [1992WES].
(d) The individual values were not cited and therefore they were derived from the global fitting Equation (12) in [1992WES], which incorporated data from other sources.

A weighted linear regression, Equation (V.87), of the $\log_{10}K^{\circ}_{s,4}$ values in the sixth column of Table V-46 as a function of the reciprocal temperature depicted in Figure V-39.

$$\log_{10}K^{\circ}_{s,4}(\mathrm{Al(OH)_3, gibb}) = (3.22456 \pm 0.09243) - (1305.655 \pm 30.216)(T/\mathrm{K})^{-1} \quad \text{(V.87)}$$

Figure V-39: Temperature dependence of $\log_{10}K^{\circ}_{s,4}$ for gibbsite where the solid line was generated by Equation (V.87) and the dashed curve by an equation with an additional term such that $\Delta_r C^{\circ}_{p,m}$ values approximated those given in [1991CHE/XU] (see below).



Equation (V.87) gave the following thermodynamic quantities at 298.15 K:

$$\log_{10}K^{\circ}_{s,4}((\text{V.85}), 298.15\ \mathrm{K}) = -(1.155 \pm 0.022),$$

$$\Delta_r G^{\circ}_m((\text{V.85}), 298.15\ \mathrm{K}) = (6.591 \pm 0.127)\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H^{\circ}_m((\text{V.85}), 298.15\ \mathrm{K}) = (24.997 \pm 1.134)\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r S^{\circ}_m((\text{V.85}), 298.15\ \mathrm{K}) = (61.734 \pm 3.827)\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

or for the corresponding Reaction (V.88) using the thermodynamic properties for the ionisation of water [1992GRE/FUG]. Khodakovsky *et al*. [1980KHO/KAT] derived the following quantities at 298.15 K for Reaction (V.85): $\log_{10}K^{\circ}_{s,4} = -1.29$ and $\Delta_r H^{\circ}_m = 25.72\ \mathrm{kJ\cdot mol^{-1}}$.

$$\alpha\text{-Al(OH)}_3 + \text{H}_2\text{O(l)} \rightleftharpoons \text{Al(OH)}_4^- + \text{H}^+ \quad \text{(V.88)}$$

$$\log_{10}{}^{*}K^{\circ}_{s,4}\ ((\text{V.88}),\ 298.15\ \text{K}) = \mathbf{-\ (15.156 \pm 0.040)},$$

$$\Delta_r G^{\circ}_m\ ((\text{V.88}),\ 298.15\ \text{K}) = \mathbf{(86.511 \pm 0.228)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H^{\circ}_m\ ((\text{V.88}),\ 298.15\ \text{K}) = \mathbf{(80.812 \pm 1.140)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^{\circ}_m\ ((\text{V.88}),\ 298.15\ \text{K}) = \mathbf{-\ (19.115 \pm 3.900)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

The reaction values above are selected by this review. Taking the thermodynamic quantities for the formation of gibbsite from the current review (Section V.5.1.2.1.2 and Section V.5.1.2.1.3) and those for hydroxide ion from the CODATA values in the NEA TDB, our corresponding selected values for the aluminate anion are:

$$\Delta_f G^{\circ}_m(\text{Al(OH)}_4^-,\ 298.15\ \text{K}) = \mathbf{-\ (1306.404 \pm 1.521)}\ \text{kJ·mol}^{-1},$$

$$\Delta_f H^{\circ}_m(\text{Al(OH)}_4^-,\ 298.15\ \text{K}) = \mathbf{-\ (1499.018 \pm 1.884)}\ \text{kJ·mol}^{-1},$$

$$S^{\circ}_m(\text{Al(OH)}_4^-,\ 298.15\ \text{K}) = \mathbf{(119.275 \pm 3.911)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

The above value for $\Delta_f G^{\circ}_m(\text{Al(OH)}_4^-, 298.15\ \text{K})$ is in line with that from Hemingway *et al.* [1978HEM/ROB], – (1305 ± 1) kJ·mol^{-1}, based on the calorimetric and solubility data available at that time.

Based on calorimetric measurements summarised in Table V-48, Chen *et al.* [1991CHE/XU] determined the corresponding values for Al(OH)_4^- at 298.15 K: $\Delta_f G^{\circ}_m = -1305.00$ kJ·mol^{-1}, $\Delta_f H^{\circ}_m = -1502.77$ kJ·mol^{-1}, $S^{\circ}_m = 101.49$ J·K^{-1}·mol^{-1} and $C^{\circ}_{p,m} = 89.6$ J·K^{-1}·mol^{-1}. Wenming Zeng *et al.* [1994WEN/QIY] combined their calorimetric measurements of the enthalpies of reaction of Al(cr) with aqueous NaOH at 298.15 K and that of Chen *et al.* to give their recommended value: $\Delta_f H^{\circ}_m(\text{Al(OH)}_4^-, \text{aq}, 298.15\ \text{K}) = -(1503.61 \pm 1.00)$ kJ·mol^{-1}, *cf.* – 1497.41 kJ·mol^{-1} assessed by Khodakovsky *et al.* [1980KHO/KAT]. Hovey and Helper [1988HOV/HEP] reported values of $\Delta_f G^{\circ}_m(\text{Al(OH)}_4^-, \text{aq}, 298.15\ \text{K}) = -1305.6$ kJ·mol^{-1} and $S^{\circ}_m(\text{Al(OH)}_4^-, \text{aq}, 298.15\ \text{K}) = 111.3$ J·K^{-1}·mol^{-1} based on their calorimetric measurements extrapolated to standard state conditions using the HKF correlations [1981HEL/KIR].

For the sake of consistency, the thermodynamic properties are given in the following table.

Table V- 47: Thermodynamic quantities derived for the Al(OH)_4^- species at 298.15 K.

$\Delta_f G^{\circ}_m$ /kJ·mol^{-1}	$\Delta_f H^{\circ}_m$ /kJ·mol^{-1}	S°_m /J·K^{-1}·mol^{-1}	Reference
– (1305.7 ± 4.1)	– (1503.0 ± 3.5)	(103.7 ± 6.0)	[2001BEN/PAL]
– (1306.4 ± 1.5)	– (1499.0 ± 1.9)	(119.3 ± 3.9)	This review

Table V-48: Values of $\Delta_r H^o_m$ and $\Delta_r C^o_{p,m}$ taken from Table 3 in [1991CHE/XU] and values [(a)] derived from Equation (V.89).

T/K	$\Delta_r H^o_m$ /kJ·mol^{-1}	$\Delta_r H^o_m$ [(a)] /kJ·mol^{-1}	$\Delta_r C^o_{p,m}$ /J·K^{-1}·mol^{-1}	$\Delta_r C^o_{p,m}$ [(a)] /J·K^{-1}·mol^{-1}
373.25	30.07	30.52	116.2	112.2
386.05	31.56	31.98	116.9	116.0
398.75	33.06	33.49	117.6	119.9
411.75	34.59	35.06	118.3	122.8
423.45	35.97	36.53	118.9	127.2

The dashed curve in Figure V-39 resulted from adding an additional temperature term constrained to fit the $\Delta_r C^o_{p,m}$ values given in Table V-48 to produce the following equation:

$$\log_{10} K^o_{s,4} = -(1.8031 \pm 0.0818) - (500.648 \pm 26.766)(T/\text{K})^{-1} + 7.848 \times 10^{-3}(T/\text{K}) \quad \text{(V.89)}$$

which yielded the following thermodynamic quantities at 298.15 K, $\log_{10} K^o_{s,4} = -(1.142 \pm 0.020)$, $\Delta_r G^o_m$ (V.85) = (6.519 ± 0.115) kJ·mol^{-1}, $\Delta_r H^o_m$ (V.85) = (22.941 ± 1.004) kJ·mol^{-1}, $\Delta_r S^o_m$ (V.85) = (55.080 ± 3.071) J·K^{-1}·mol^{-1} and $\Delta_r C^o_{p,m}$ (V.85) = 89.593 J·K^{-1}·mol^{-1}.

Assuming that $\Delta\varepsilon = -(0.039 \pm 0.014)$ kg·mol^{-1} at 298.15 K when NaCl is the dominant electrolyte in solution [1992WES], then,

$$\varepsilon(\text{Na}^+, \text{Al(OH)}_4^-) = \Delta\varepsilon + \varepsilon(\text{Na}^+, \text{OH}^-) = -(0.039 \pm 0.014) + (0.04 \pm 0.01) = \mathbf{(0.00 \pm 0.02)}\ \text{kg·mol}^{-1}.$$

The corresponding treatment of all the data obtained in KOH solutions gave a value of $\Delta\varepsilon = -(0.030 \pm 0.005)$ kg·mol^{-1} in good agreement with the value obtained from the 298.15 K data only, *viz*., $-(0.027 \pm 0.003)$ kg·mol^{-1}. Thus,

$$\varepsilon(\text{K}^+, \text{Al(OH)}_4^-) = \Delta\varepsilon + \varepsilon(\text{K}^+, \text{OH}^-) = \mathbf{(0.06 \pm 0.01)}\ \text{kg·mol}^{-1}$$

at 298.15 K, since $\varepsilon(\text{K}^+,\text{OH}^-) = (0.09 \pm 0.01)$ kg·mol^{-1} [1992GRE/FUG]. After completion of this review an ^{27}Al NMR study [2018GRA/DEM] appeared, which identified the formation of solvent-separated $\text{Al(OH)}_4^- \ldots \text{Na}^+$ ion pairs, but not contact ion pairs, during the crystallisation of gibbsite with implications to the sluggish kinetics of this process.

V.5.2.4 Solubility data for bayerite

Rodgers *et al*. [1991ROD/GRE] characterised rare deposits of mainly bayerite on Raoul Island in the Pacific Ocean and on Campbell Island in the sub-Antarctic. More commonly, it is apparent that bayerite forms preferentially (although not always in pure form) to the other aluminium oxide/hydroxide phases at low temperatures in the

absence of pre-existing seed crystals of gibbsite [1970SCH/ROB], [1997REN/FRA], [2003PAL/BEN] or on clean aluminium metal surfaces or on certain clays.

There have been relatively few studies of bayerite solubility and generally the results are quite disparate and confined almost exclusively to basic solutions (Reaction (V.90)) with the exception of [1981GAM/MER]. Moreover, the kinetics of dissolution are sufficiently slow that few thermodynamic data are available at ambient conditions. Apps *et al*. [1988APP/NEI] tabulated results from some of the earlier works which were often published only with the results shown in the form of small figures. These data were generally given in $g \cdot dm^{-3}$ or similar units which required developing a function of solution density *versus* solution composition and temperature. The available data are summarised in Table V-49.

$$Al(OH)_3(\beta\text{-baye}) + OH^- \rightleftharpoons Al(OH)_4^- \qquad (V.90)$$

Table V-49: Values of $\log_{10} K^\circ_{s,4}$ (V.90) for bayerite as cited in the literature and re-calculated using the SIT.

T /K	I_m (max.)	$-\log_{10} K^\circ_{s,4}$ reported	$-\log_{10} K^\circ_{s,4}$ SIT	$-\Delta\varepsilon$ /kg·mol^{-1}	References
303.15	4.00 (NaOH)		(1.00 ± 0.18) (1.06 ± 0.24)	0.06 [(a)]	[1928FRI] [(b)]
303.15	3.98 (NaOH)		(1.01 ± 0.24)	0.06 [(a)]	[1929FRI] [(b)]
303.15	7.70 (NaOH)		(0.98 ± 0.11)	(0.056 ± 0.020)	[1930FRI/JUC]
298.15	4.22 (NaOH)		(1.21 ± 0.20)	0.06 [(a)]	[1950HER/STI]
301.65	4.22 (NaOH)		(1.14 ± 0.20)		
313.15	1.17 (NaOH)	0.65	(0.70 ± 0.14)	0.06 [(a)]	[1955RUS/EDW]
323.15	1.17 (NaOH)	0.54	(0.56 ± 0.14)		
333.15	1.17 (NaOH)	0.43	(0.48 ± 0.13)		
343.15	1.15 (NaOH)	0.33	(0.38 ± 0.13)		
353.15	1.16 (NaOH)	0.23	(0.27 ± 0.13)		
363.15	1.16 (NaOH)	0.15	(0.17 ± 0.14)		
293.15	7.89 (NaOH)		(0.92 ± 0.16)	0.06 [(a)]	[1964CHR]
313.15	4.61 (NaOH)		(0.81 ± 0.18)	0.06	[1973LYA/KRY]
323.15	5.04 (NaOH)		(0.67 ± 0.18)		
333.15	5.51 (NaOH)		(0.56 ± 0.18)		
294	0.06 (NH_4Cl/OH)	0.355	(0.34 ± 0.12)	0.039 [(a)]	[1992VER/GOU]
323	0.06 (NH_4Cl/OH)	0.017	(0.01 ± 0.12)		
338	0.05 (NH_4Cl/OH)	– 0.136	– (0.08 ± 0.12)		
303.15	5 ($NaNO_3$)	0.853 [(c)]	(0.85 ± 0.10)	(0.020 ± 0.004)	[2016BEN/HIL]
336.15		0.493 [(c)]	(0.52 ± 0.10)	(0.021 ± 0.007)	
363.05		0.215 [(c)]	(0.22 ± 0.10)	0.021	

(a) Fixed at this value based on the $\Delta\varepsilon$ obtained from the more accurate and abundant data for gibbsite.
(b) These data were taken from graphs in [1988APP/NEI].
(c) These values were derived from a global Pitzer ion interaction treatment.

The results of the early studies [1928FRI], [1929FRI], [1930FRI/JUC] and [1950HER/STI] were rejected in our analysis, because the purity of the solid phase could not be established and these results are more consistent with gibbsite being the solubility controlling phase (Figure V-40). Sanjuan and Michard [1987SAN/MIC] reported a value of $\log_{10}{}^{*}K^{\circ}_{s,4} = -13.4$ at 323.15 K obtained from oversaturation in the initial presence of gibbsite, but neither was the attainment of equilibrium confirmed nor was the new phase precipitated identified adequately [1990SCH/STR], [1991HEM]. Verdes *et al*. [1992VER/GOU] measured the solubility of bayerite from 293 to 353 K by conducting dissolution and precipitation experiments. Despite the internal concordance of these results, Figure V-40 shows that their reported $\log_{10}K^{\circ}_{s,4}$ values are systematically higher than all the others, leaving one to speculate that some other higher energy solid phase controlled the solubility in their experiments. The results from [2003PAL/BEN] and [2016BEN/HIL] for gibbsite are also shown in Figure V-40 because these experiments were run concurrently with those for the more labile bayerite and the agreement with the comprehensive results for gibbsite [1992WES] supports the attainment of equilibrium in the bayerite experiments which were not reversed.

Application of the SIT treatment according to Equation (V.91) yielded the $\log_{10}K^{\circ}_{s,4}$(V.90) values given in Table V-49, and a regression (the uncertainties were assigned by this review) gave Equation (V.92).

$$\log_{10}K_{s,4}\,(\text{V.90}) = \log_{10}K^{\circ}_{s,4}\,(\text{V.90}) - \Delta\varepsilon I_m \qquad \text{(V.91)}$$

$$\log_{10}K^{\circ}_{s,4}\,(\text{V.90}) = (2.722539 \pm 0.327758) - (1053.057 \pm 0.108)(T/\text{K})^{-1} \qquad \text{(V.92)}$$

which corresponds to $\Delta_r H^{\circ}_m\,(\text{V.90}) = (20.161 \pm 4.036)$ kJ·mol^{-1}, and hence to $\Delta_f H^{\circ}_m\,(\text{Al(OH)}_3\text{, baye, 298.15 K}) = -(1289.164 \pm 4.453)$ kJ·mol^{-1}.

This is in excellent agreement with the calorimetric value of $-(1288.8 \pm 2.5)$ kJ·mol^{-1} derived in Section V.5.1.2.2.3, where a final value of

$$\Delta_f H^{\circ}_m\,(\text{Al(OH)}_3\text{, baye, 298.15 K}) = \mathbf{-(1289.0 \pm 2.5)}\ \text{kJ·mol}^{-1}$$

is selected.

We have therefore re-fitted the data for $\log_{10}K^{\circ}_{s,4}$(V.90) values given in Table V-49, with a weighted regression with the value of $\Delta_f H^{\circ}_m$(V.90) constrained to 19.997 kJ·mol^{-1}, corresponding to $\Delta_f H^{\circ}_m\,(\text{Al(OH)}_3\text{, baye, 298.15 K}) = -(1289.0 \pm 2.5)$ kJ·mol^{-1}. This gives the preferred equation shown in Figure V-40:

$$\log_{10}K^{\circ}_{s,4}\,(\text{Al(OH)}_3\text{, baye}) = (2.6322 \pm 0.01338) - (1044.510 \pm 19.290)(T/\text{K})^{-1} \qquad \text{(V.93)}$$

The resulting thermodynamic parameters at 298.15 K where the uncertainties are 2σ:

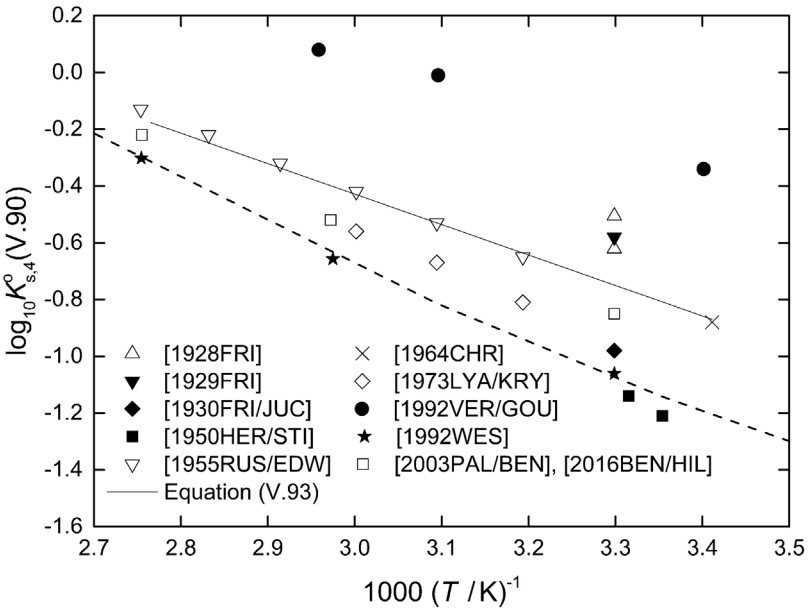
$$\log_{10}K^{\circ}_{s,4}\,((\text{V.90}), 298.15\ \text{K}) = -(0.871 \pm 0.026),$$

$$\Delta_r G_m^o \text{ ((V.90), 298.15 K)} = (4.972 \pm 1.484) \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((V.90), 298.15 K)} = 19.997 \text{ kJ·mol}^{-1},$$

$$\Delta_r S_m^o \text{ ((V.90), 298.15 K)} = 50.394 \text{ J·K}^{-1}\text{·mol}^{-1}.$$

Figure V-40: Temperature dependence of $\log_{10} K_{s,4}^o$ (V.90) for bayerite where the open symbols represent data which were retained and solid symbols represent data which were not included in the regression, Equation (V.93).



The corresponding quantities for Reaction (V.94) at 298.15 K are:

$$\text{Al(OH)}_3\text{(β-baye)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{Al(OH)}_4^- + \text{H}^+ \qquad \text{(V.94)}$$

$$\log_{10}{}^* K_{s,4}^o \text{ ((V.94), 298.15 K)} = -(14.872 \pm 0.029),$$

$$\Delta_r G_m^o \text{ ((V.94), 298.15 K)} = (84.892 \pm 1.486) \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((V.94), 298.15 K)} = (75.812 \pm 2.501) \text{ kJ·mol}^{-1},$$

$$\Delta_r S_m^o \text{ ((V.94), 298.15 K)} = -(30.454 \pm 9.757) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

These thermodynamic values for bayerite solubility are recommended rather than selected because of the inherent instability of bayerite over extended periods of time. The derived values for the thermodynamic properties of bayerite at 298.15 K are:

$$\Delta_f G_m^o \text{ (Al(OH)}_3\text{, baye)} = -(1154.156 \pm 2.709) \text{ kJ·mol}^{-1},$$

$$\Delta_f H_m^o(\text{Al(OH)}_3, \text{baye}) = \mathbf{-\ (1289.000 \pm 2.500)}\ \text{kJ·mol}^{-1},$$

$$S_m^o(\text{Al(OH)}_3, \text{baye}) = (79.777 \pm 10.496)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

where only the value of $\Delta_f H_m^o(\text{Al(OH)}_3, \text{baye})$ is selected due to its excellent agreement with calorimetric measurements, as explained above. We note that this value for $S_m^o(\text{Al(OH)}_3, \text{baye})$ is distinctly larger than that measured for gibbsite, $(68.44 \pm 0.30)\ \text{J·K}^{-1}\text{·mol}^{-1}$.

Hemingway *et al*. [1989HEM/KIT] implied that the principal reason for differences in the reported solubilities of this and other aluminium oxide/hydroxide solid phases was related to problems associated with the different experimental techniques used by various authors rather than to any pre-treatment procedures of the solids as proposed previously [1982BLO/WEA]. However, the general agreement between the various solubility studies involving gibbsite and bayerite, in basic solutions in particular, tends to refute the claim by Hemingway *et al*.

Finally, bayerite (BET surface area $8.8\ \text{m}^2\text{·g}^{-3}$) solubility experiments were carried out by Gamsjäger and Meralla [1981GAM/MER] who reported that equilibrium was attained after one week at 363 K and $I_m = 1.0$ ($HClO_4$, $NaClO_4$). The final pH was in the approximate range of 2.2 to 2.7 and a mean value for $\log_{10}{}^{*}K_{s,0}$ ((V.95), 363.15 K) of (5.43 ± 0.10) was obtained.

$$\text{Al(OH)}_3\text{(baye)} + 3\text{H}^+ \rightleftharpoons \text{Al}^{3+} + 3\text{H}_2\text{O(l)} \qquad \text{(V.95)}$$

An extended Debye-Hückel relationship was used to estimate an approximate value for $\log_{10}{}^{*}K_{s,0}^o$ ((V.95), 363.15 K) of 4.12.

V.5.2.5 Summary of thermodynamic data

Table V-50 gives a summary of the selected thermodynamic data for aluminium(III) species.

Table V-50: Summary of selected thermodynamic quantities determined at 0.1 MPa and 298.15 K from our assessment.

Species	$\Delta_f G_m^o$ (298.15 K) /kJ·mol^{-1}	$\Delta_f H_m^o$ (298.15 K) /kJ·mol^{-1}	S_m^o (298.15 K) /J·K^{-1}·mol^{-1}	Sections
Al^{3+}	**– (488.824 ± 1.614)**	**– (538.400 ± 1.500)**	**– (334.000 ± 2.000)**	V.5.2.1
$AlOH^{2+}$	**– (697.880 ± 1.663)**	**– (767.670 ± 2.106)**	**– (168.541 ± 5.512)**	V.5.2.2.1
$Al(OH)_2^+$	**– (902.991 ± 1.736)**	**– (1005.946 ± 5.801)**	**– (46.521 ± 19.022)**	V.5.2.2.2
$Al(OH)_3$(aq)	**– (1106.682 ± 2.438)**	**– (1242.511 ± 6.659)**	**(76.476 ± 22.695)**	V.5.2.2.3
$Al(OH)_3$(gibb)	**– (1155.775 ± 1.503)**	**– (1294.000 ± 1.500)**	**(68.440 ± 0.300)**	V.5.1.2.1
$Al(OH)_3$(baye)	– (1154.156 ± 2.709)[(a)]	**– (1289.000 ± 2.500)**	(79.777 ± 10.496)[(a)]	V.5.1.2.2
$Al(OH)_4^-$	**– (1306.404 ± 1.511)**	**– (1499.018 ± 1.884)**	**(119.275 ± 3.869)**	V.5.2.3.5

(a) Values recommended but not selected (see Section V.5.2.4)

The ion interaction coefficients derived in this review are summarised in Table V-51 and are the recommended values.

Table V-51: Summary of selected ion interaction coefficients for the aqueous Al(III) species at 298.15 K.

Interactions	Values derived in this review kg·mol^{-1}	Current NEA TDB values kg·mol^{-1}
$\varepsilon(Al^{3+}, Cl^-)$	**(0.34 ± 0.02)** [(a)]	(0.33 ± 0.02)
$\varepsilon(Al^{3+},NO_3^-)$	**(0.25 ± 0.02)**	
$\varepsilon(Al^{3+},ClO_4^-)$	**(0.52 ± 0.01)**	
$\varepsilon(AlOH^{2+},Cl^-)$	**(0.040 ± 0.051)** [(b)]	0.09
$\varepsilon(Na^+,Al(OH)_4^-)$	**(0.00 ± 0.02)**	
$\varepsilon(K^+,Al(OH)_4^-)$	**(0.06 ± 0.01)**	

(a) For $I_m > 0.9$, ε_1 = **(0.34 ± 0.04)** and ε_2 = **(0.40 ± 0.04)** kg·mol^{-1}.
(b) For $I_m > 1.0$, ε_1 = **– (0.05 ± 0.06)** and ε_2 = **(0.64 ± 0.06)** kg·mol^{-1}.

Chapter V.6

Mg and Ca species

V.6.1 Magnesium species

V.6.1.1 Magnesium solid species

V.6.1.1.1 MgO(cr)

V.6.1.1.1.1 Structure

Magnesium oxide has the fcc NaCl structure, space group $Fm\bar{3}m$, with $a = (4.2110 \pm 0.0012)$ Å at 296 K as given by Hazen [1976HAZ] on synthetic single crystals. These results are in agreement with the more recent data in [1997DUB/SAX] who extended the thermal expansion data of [1976HAZ] (from 77 to 1313 K and up to 2.3 MPa) to the melting point of periclase at atmospheric pressure, (3098 ± 40) K.

V.6.1.1.1.2 Heat capacity and entropy

The low-temperature heat capacity has been studied by Giauque and Archibald [1937GIA/ARC] (20-301 K), Barron *et al.* [1959BAR/BER] (10-270 K), and Gmelin [1969GME2] (4-320 K). The last two measurements are in excellent agreement, as shown in Figure V-41, but those of [1937GIA/ARC] are slightly higher, presumably because they used a microcrystalline sample. The CODATA assessment [1989COX/WAG] was based predominantly on the data of Barron *et al.* [1959BAR/BER], which are the most precise, and the CODATA values have been accepted for this review. Thus:

$$C^{\circ}_{p,\mathrm{m}}\,(\mathrm{MgO,\ cr,\ 298.15\ K}) = \mathbf{(37.237 \pm 0.200)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

and

$$S^{\circ}_{\mathrm{m}}\,(\mathrm{MgO,\ cr,\ 298.15\ K}) = \mathbf{(26.95 \pm 0.15)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

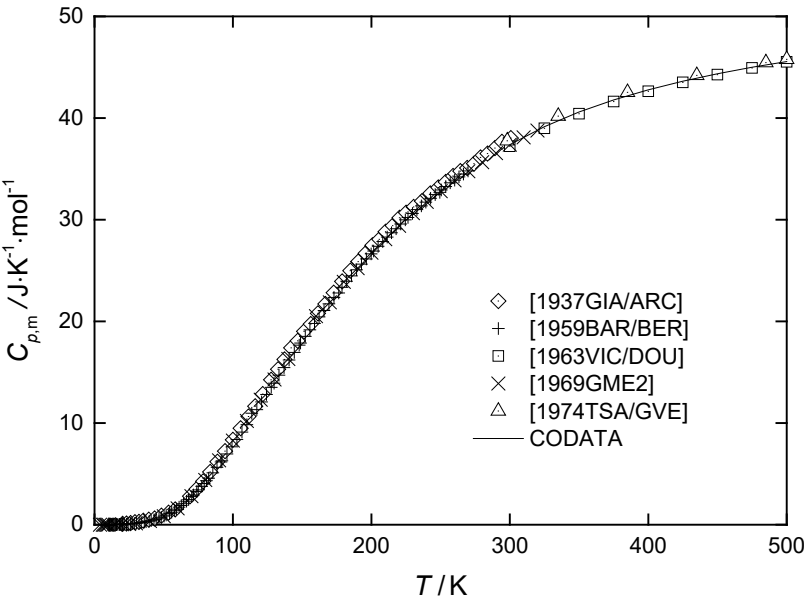
The enthalpy content of MgO(cr) has been studied by Victor and Douglas [1963VIC/DOU], (373-1173) K and Tsagareishvili and Gvelesiani [1974TSA/GVE] (370-1591) K, again with good agreement. Gurvich *et al.* [1994GUR/VEY] also mention measurements from 1182 to 2488 K by Kantor and Fomichev [1969KAN/FOM], but these are contained in a Russian book which we have not

been able to access. The source of the selected values in [1989COX/WAG] is not described, but they are identical to the data in [1994GUR/VEY], which are based on the data of Victor and Douglas [1963VIC/DOU], and Tsagareishvili and Gvelesiani [1974TSA/GVE], and have been accepted by this review.

The data up to 1000 K can be represented with adequate precision by the equation:

$$C_{p,\mathrm{m}}(\mathrm{MgO,cr},T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{46.374} + \mathbf{7.0518\times 10^{-3}}(T/\mathrm{K}) - \mathbf{1.57771\times 10^{-6}}(T/\mathrm{K})^2 - \mathbf{9.8661\times 10^{5}}(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1000\,\mathrm{K}).$$

Figure V-41: Heat capacity of MgO(cr).



V.6.1.1.1.3 Enthalpy of formation

Apart from two questionable high-temperature studies (see below), the recent literature does not reveal any information that would lead to a revision of the value for the enthalpy of formation of MgO(cr) selected by CODATA [1989COX/WAG], based principally on the calorimetric measurements of Shomate and Huffman [1943SHO/HUF], and Vasil'ev *et al*. [1981VAS/VAS2]. This value is identical to that adopted by Gurvich *et al.* in [1994GUR/VEY]. Therefore, this review maintains and selects:

$$\Delta_{\mathrm{f}}H^{\mathrm{o}}_{\mathrm{m}}\,(\mathrm{MgO,\ cr,\ 298.15\ K}) = \mathbf{-\ (601.6 \pm 0.3)}\ \mathrm{kJ\cdot mol^{-1}}.$$

The enthalpy of transition from the macro- to the microcrystalline form of MgO was selected in [1994GUR/VEY] as (3.9 ± 0.3) kJ·mol^{-1} based "on differences between the enthalpies of dissolution of these modifications in hydrochloric acid" obtained as 3.7 kJ·mol^{-1} using the results of Giauque and Archibald [1937GIA/ARC] for microcrystals and of Shomate and Huffman [1943SHO/HUF] for macrocrystals (although we do not understand the details of their derivation), and as 4.0 kJ·mol^{-1} in [1938TAY/WEL]. The value listed in [1994GUR/VEY] for $\Delta_f H^o_m$(MgO, microcr, 298.15 K) = – (597.7 ± 0.4) kJ·mol^{-1} is given here for information only.

There are two apparently markedly discrepant values of the enthalpy of formation of MgO(cr) from high-temperature equilibria. Gourishankar *et al*. [1993GOU/RAN] studied the free vaporisation of MgO(cr) from 1830 to 2070 K and Ono *et al*. [1993ONO/NAK] studied the equilibrium of MgO(cr) and Nb(l) at (923 to 1173) K. However, these have been discredited by the potential-difference study by Jacob and Varghese [1996JAC/VAR], who showed that the difference ($\Delta_f H^o_m$(MgO, cr) – $\Delta_f H^o_m$(CaO, cr)) in the range 950 to 1280 K is consistent with the CODATA selections — see the Appendix A entries for these three references for further comments on the uncertainties in the discrepant studies, both in the experiments themselves and the subsidiary data required.

V.6.1.1.2 $Mg(OH)_2$(cr) (brucite)

The experimental data for $Mg(OH)_2$(cr) have been discussed in some detail in [1994GUR/VEY].

V.6.1.1.2.1 Structure

Magnesium hydroxide (found in nature as the mineral brucite) possesses hexagonal symmetry, CdI_2-type (space group $P\bar{3}m1$) with lattice parameters a = (3.142 ± 0.001) Å and c = (4.766 ± 0.002) Å, as given in [1967ZIG/ROT] from neutron diffraction studies on powder material and on a naturally occurring single crystal.

V.6.1.1.2.2 Heat capacity and entropy

Giauque and Archibald [1937GIA/ARC] measured the low-temperature heat capacity of a sample of $Mg(OH)_2$(cr) whose only significant impurity was 0.1 mol % KOH from 22 to 320 K. Gurvich *et al*. [1994GUR/VEY] have adopted these data, taking S_m (22 K) = 0.17 J·K^{-1}·mol^{-1}; their assessed value of the standard entropy (the same as in [1982WAG/EVA]) is slightly greater than that given in [1937GIA/ARC] (corrected to modern atomic weights) and is our selected value:

$$S^o_m(Mg(OH)_2, cr, 298.15\ K) = \mathbf{(63.18 \pm 0.40)}\ J\cdot K^{-1}\cdot mol^{-1}.$$

The enthalpy increments of $Mg(OH)_2$(cr) have been measured by King and Ferrante [1975KIN/FER] from 350 to 699 K, using a sample containing 1.23 mol %

impurities; their consistent data merge well with the low-temperature heat capacities around 300 K and a simultaneous fit of both data sets, with a constraint of $C_{p,\mathrm{m}}(300\ \mathrm{K})$ = 77.252 J·K^{-1}·mol^{-1} [1937GIA/ARC] gives the equation:

$$C_{p,\mathrm{m}}(\mathrm{Mg(OH)_2}, \mathrm{cr}, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{93.3928} + \mathbf{3.0121 \times 10^{-2}}\,(T/\mathrm{K}) - \mathbf{2.2659 \times 10^{6}}\,(T/\mathrm{K})^{-2} \quad (298.15\text{-}600\ \mathrm{K})$$

and thus we select:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Mg(OH)_2}, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(76.9 \pm 0.3)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

Lashchenko and Kompanskii [1935LAS/KOM] had earlier made similar measurements from 393 to 945 K. However, their sample contained more than 3.5 % impurities, given as mainly Fe_2O_3 and Cr_2O_3. Typically, only the measurements below *ca.* 700 K relate to the undecomposed hydroxide. Although these measurements agree reasonably with those in [1975KIN/FER], they are more scattered, and were not used in the fitting.

V.6.1.1.2.3 Enthalpy of formation

The value selected in [1994GUR/VEY] for the enthalpy of formation of $Mg(OH)_2$(cr) results from five very concordant results on the enthalpies of solution of MgO(cr) and $Mg(OH)_2$(cr) in hydrochloric acid ([1937GIA/ARC], [1937WEL/TAY], [1938TAY/WEL], [1948TOR/SAH]) or hydrofluoric acid [1956BEN]. After correction, where needed, for the enthalpy of transition from the macro- to the microcrystalline form of MgO (see Section V.6.1.1.1.3), these five individual results were given in [1994GUR/VEY] as – (925.03 ± 1.00), – (924.32 ± 0.35), – (924.32 ± 0.66), – (924.35 ± 0.30) and – (924.32 ± 0.60) kJ·mol^{-1}, and their weighted average was selected as $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Mg(OH)_2}, \mathrm{cr}, 298.15\ \mathrm{K})$ = – (924.35 ± 0.30) kJ·mol^{-1}. A re-culation by this review of the results of Bennington [1956BEN] obtained from dissolution of MgO(cr) and $Mg(OH)_2$(cr) in hydrofluoric acid (see the relevant Appendix A entry) by this author without much discussion yields $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Mg(OH)_2}, \mathrm{cr}, 298.15\ \mathrm{K})$ = – (925.421 ± 0.678) kJ·mol^{-1} in marginal agreement with the value of – (924.32 ± 0.60) kJ·mol^{-1} extrapolated by Gurvich *et al.* [1994GUR/VEY].

Gurvich *et al*. [1994GUR/VEY] also reference without details two older results from similar dissolutions in hydrochloric acid media, [1928ROT/CHA] and [1935FRI/LUK], which our examination shows yield respectively, – (920.5 ± 2.0) and – (925.3 ± 1.0) kJ·mol^{-1}, and other values of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Mg(OH)_2}, \mathrm{cr}, 298.15\ \mathrm{K})$ = – (926.7 ± 5.0) kJ·mol^{-1}, from the enthalpies of interaction of MgO(cr) and $Mg(OH)_2$(cr) with sodium peroxide [1915MIX] and – (925.6 ± 4.0) kJ·mol^{-1} from the calorimetric determination in [1986MUL/PRA] of the enthalpy of Reaction (V.96).

$$\mathrm{Mg(OH)_2(cr)} \rightleftharpoons \mathrm{MgO(cr) + H_2O(g)} \qquad \text{(V.96)}$$

Gurvich *et al*. [1994GUR/VEY] also cite 15 studies on the equilibrium of Reaction (V.96) and noted the very large scatter in the results; however the more detailed measurements in [1954ZHA/KAD], [1937GIA/ARC], [1956KEN2] and [1957ROY/ROY] gave values of – 927, – 927, – 927 and – 922 kJ·mol^{-1}, respectively, for $\Delta_f H^\circ_m$(Mg(OH)$_2$, cr, 298.15 K), which agree with their adopted value within their uncertainties.

However, it is clear from a comparison of the texts in the Russian [1981GLU/GUR] and English [1994GUR/VEY] versions of this assessment that all the values quoted from the latter are actually based on $\Delta_f H^\circ_m$(MgO, cr, 298.15 K) = – 601.5 kJ·mol^{-1}, the value selected in the original Russian edition of their text [1981GLU/GUR], rather than – 601.6 kJ·mol^{-1}, the CODATA value actually selected in [1994GUR/VEY].

We have not found any more recent experimental results that would lead to a further revision of the standard enthalpy of formation of Mg(OH)$_2$(cr) and, consequently, after examining in detail the important references quoted above, we adopt the value of:

$$\Delta_f H^\circ_m \text{(Mg(OH)}_2\text{, cr, 298.15 K)} = \mathbf{-\ (924.45 \pm 0.30)}\ \text{kJ·mol}^{-1},$$

which is 0.1 kJ·mol^{-1} more negative than that selected in [1994GUR/VEY].

V.6.1.1.3 Magnesium chloride, MgCl$_2$(cr)

V.6.1.1.3.1 Structure

Magnesium chloride has the hexagonal $CdCl_2$ structure, space group $R\bar{3}m$ [1963WYC] up to its melting point at 987 K. From powder X-ray diffractometry Partin and O'Keeffe [1991PAR/OKE] reported a = (3.6363 ± 0.0010) Å and c = (17.6663 ± 0.0005) Å, without specifying precisely the temperature that would have been warranted by such a precision, compared with the less precise values given in [1984DOR], a = (3.640 ± 0.004) Å and c = (17.673 ± 0.015) Å. In any case, these results are to be preferred to the somewhat divergent data assessed by Wyckoff [1963WYC], a = 3.596 Å and c = 17.589 Å.

A β-modification of MgCl$_2$(cr), obtained by dehydration of the hexahydrate using $SOCl_2$ at 351 K, was reported by Bassi and Polato [1982BAS/POL] as being trigonal, space group $P\bar{3}m1$, with lattice parameters a = (3.641 ± 0.003) Å and c = (5.927 ± 0.006) Å. As the thermodynamic relationship between the two MgCl$_2$ phases is not established, we are giving these values for information only. All the data discussed below thus will pertain to the hexagonal phase with a $CdCl_2$ structure.

V.6.1.1.3.2 Heat capacity and entropy

Kelley and Moore [1943KEL/MOO] measured the low-temperature heat capacity from 53.6 to 295.4 K of a sample of MgCl$_2$(cr) containing 0.2 % MgO and extrapolated these

data down to 0 K using a sum of Debye and Einstein functions. Moore [1943MOO] made enthalpy measurements from 494 to 1428 K (above the melting point of 987 K) of the same sample, while later Holm and Grønvold [1973HOL/GRO] studied the enthalpy content just above and below the melting point. The two sets of data for $MgCl_2$(cr) are in excellent agreement and although the derived heat capacities of $MgCl_2$(l) are very similar in the two studies, Holm and Grønvold [1973HOL/GRO] derived a slightly smaller enthalpy of fusion.

Gurvich *et al*. [1994GUR/VEY] combined all these data to give the equation

$$C_{p,\mathrm{m}}(\mathrm{MgCl_2}, \mathrm{cr}, T)\ /\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{75.655} + \mathbf{9.975 \times 10^{-3}}(T/\mathrm{K}) - \mathbf{6.440 \times 10^{5}}(T/\mathrm{K})^{-2} \quad (298.15\text{-}987\ \mathrm{K})$$

which we have accepted. Thus:

$$C^{\circ}_{p,\mathrm{m}}\ (\mathrm{MgCl_2},\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(71.384 \pm 0.200)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

The selected standard entropy:

$$S^{\circ}_{\mathrm{m}}\ (\mathrm{MgCl_2},\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(89.62 \pm 0.80)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

is that given in [1994GUR/VEY]; this is very slightly smaller than that given in [1943KEL/MOO], presumably due to a slightly different extrapolation below 53 K —see also a similar, but larger, difference in $S^{\circ}_{\mathrm{m}}\ (\mathrm{CaCl_2},\ \mathrm{cr})$ in Section V.6.2.1.3.2.

V.6.1.1.3.3 Enthalpy of formation

A series of` rather concordant solution calorimetric data involving Mg(cr), $MgCl_2$(cr) and MgO(cr) yielded a good basis for the selection in [1994GUR/VEY] of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\ (\mathrm{MgCl_2},\ \mathrm{cr},\ 298.15\ \mathrm{K})$. The enthalpy of solution in hydrochloric acid of Mg(cr) measured by Shomate and Huffman [1943SHO/HUF] and of $MgCl_2$(cr) by Shin and Criss [1979SHI/CRI] under the same concentration conditions, yield a value of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\ (\mathrm{MgCl_2},\ \mathrm{cr},\ 298.15\ \mathrm{K}) = -(644.28 \pm 0.69)\ \mathrm{kJ{\cdot}mol^{-1}}$. The value obtained in [1943SHO/HUF], based on their own data on the dissolution of the metal and of $MgCl_2$(cr) was discarded as it leads to a value for the enthalpy of formation of $MgCl_2$(cr) that is more positive by *ca.* 3 $\mathrm{kJ{\cdot}mol^{-1}}$, possibly due to an MgO impurity in their $MgCl_2$(cr) sample.

Several other authors reported the enthalpy of solution of $MgCl_2$(cr) in water at rather low concentrations that were easily corrected to infinite dilution, to give the values of $-(157.49 \pm 0.21)\ \mathrm{kJ{\cdot}mol^{-1}}$ [1974VOR/UMY], $-(155.84 \pm 0.20)\ \mathrm{kJ{\cdot}mol^{-1}}$ [1979SHI/CRI] and $-(158.59 \pm 0.28)\ \mathrm{kJ{\cdot}mol^{-1}}$ [1984MON/ALE] at infinite dilution, while Greyson and Snell [1969GRE/SNE] obtained $-(154.93 \pm 0.17)\ \mathrm{kJ{\cdot}mol^{-1}}$ for a concentration of 0.015 molal. Noting that these four results diverge more than expected from the reproducibility of the measurements and ascribing the differences to varying purities of the samples, Gurvich *et al*. [1994GUR/VEY] selected an actual value of $-(157.0 \pm 1.1)\ \mathrm{kJ{\cdot}mol^{-1}}$ for the dissolution of $MgCl_2$(cr) at infinite dilution.

Combination of this with the CODATA value $\Delta_f H^o_m(Mg^{2+}, aq, 298.15\ K) = -(467.0 \pm 0.6)$ kJ·mol^{-1} yields $\Delta_f H^o_m(MgCl_2, cr, 298.15\ K) = -(644.2 \pm 1.3)$ kJ·mol^{-1}.

Monaenkova *et al*. [1971MON/PAS] also reported enthalpies of solution of Mg(cr) in $HClO_4$ solutions that they extrapolated to infinite dilution to obtain $\Delta_f H^o_m(Mg^{2+}, aq, 298.15\ K) = -(468.86 \pm 075)$ kJ·mol^{-1}. They also made use of the enthalpy of solution of MgO(cr) in $(HClO_4, 450H_2O)$, $-(151.08 \pm 0.50)$ kJ·mol^{-1}, previously determined in [1960VOR/PRI] and extrapolated to infinite dilution. These results, when combined with the enthalpy of solution of $MgCl_2$(cr) at infinite dilution, $-(157.49 \pm 0.21)$ kJ·mol^{-1} from [1974VOR/UMY], lead from two partially dependent cycles to $\Delta_f H^o_m(MgCl_2, cr, 298.15\ K) = -(644.8 \pm 1.3)$ kJ·mol^{-1}. We have examined in great detail the references mentioned above (see comments in the relevant Appendix A entries) and see no reason to alter the values re-culated in [1994GUR/VEY] that are the basis for their selection.

Gurvich *et al*. [1994GUR/VEY] also mention that the literature contains a number of references, (previously assembled in [1979GLU/MED]) to the dissolution of Mg(cr) and $MgCl_2$(cr). However, the relevant data are less accurate, and the concentrations of the final solutions arising from the dissolution of the metal and the dichloride were different, so that the data in these references are difficult to use for the present purposes.

Gurvich *et al*. [1994GUR/VEY] also cite a dozen references that yield, with a lower precision, the enthalpy of formation of $MgCl_2$(cr) from various equilibria and which centre on $-(642 \pm 3)$ kJ·mol^{-1}.

Under these circumstances and in the absence of more recent results, we select and adopt the value assessed in [1994GUR/VEY]:

$$\Delta_f H^o_m(MgCl_2, cr, 298.15\ K) = \mathbf{-(644.3 \pm 0.7)}\ \text{kJ·mol}^{-1}.$$

Use of this value with that selected for $S^o_m(MgCl_2, cr, 298.15\ K)$ in the preceding Section V.6.1.1.3.2 yields the selected

$$\Delta_f G^o_m(MgCl_2, cr, 298.15\ K) = \mathbf{-(594.768 \pm 0.740)}\ \text{kJ·mol}^{-1}.$$

V.6.1.1.4 Magnesium sulfate, $MgSO_4$(cr)

V.6.1.1.4.1 Structure

There are three polymorphs of $MgSO_4$, two structurally similar orthorhombic α and β forms and a γ phase that can be obtained above 1283 K. We shall adopt the nomenclature and structural data given by Fortes and Wood [2007FOR/WOO], whereby the more common form derived by dehydration of the hydrates is called β, with an orthorhombic structure, space group *Pbnm* with $a = (4.7460 \pm 0.0001)$ Å, $b = (8.5831 \pm 0.0001)$ Å, $c = (6.7093 \pm 0.0001)$ Å at 300 K. The α-form has the space group *Cmcm* with $a = (5.1747 \pm 0.0001)$, $b = (7.8756 \pm 0.0001)$, $c = (6.4952 \pm 0.0001)$ Å at 300 K

[2007FOR/WOO]. It is obtained by dissolution of MgO(cr) in H_2SO_4(aq), or from the β-form at 868 K [1972YAM/KAT]. The structure of the γ-phase formed at (1283 ± 10) K noted by Dewing and Richardson [1959DEW/RIC] is not known, although Du [2000DU] indicates that it is monoclinic, without quoting a reference. Note that this designation of the phases is different from that used by some other authors. The enthalpy data in [1980KO/DAU] (see below) strongly indicate that the α-form is the more stable at 298.15 K.

V.6.1.1.4.2 Heat capacity and entropy

Moore and Kelley [1942MOO/KEL] measured the low-temperature heat capacity from 53 to 295 K of a sample of $MgSO_4$(cr) containing 0.1-0.15 % MgO. This was prepared by dehydration of the heptahydrate at 673 K for two hours, which according to [2007FOR/WOO] gives the β-form. Moore and Kelley [1942MOO/KEL] do not give any structural data, so we have indeed assumed their data refer to the β-phase, but with an increased uncertainty. They extrapolated the heat capacity from 53 K down to 0 K using a sum of a Debye and three Einstein functions.

There seem to be no measurements of the enthalpy content of $MgSO_4$(cr) above room temperature. In the assessments by Chase *et al.* [1985CHA/DAV] and Gurvich *et al.* [1994GUR/VEY] the authors indicated that the heat capacity was estimated from $C_{p,\mathrm{m}}$ ($CaSO_4$ + MgO – CaO), although the tabulated values are actually noticeably different from those derived from this expression at higher temperatures. No values are selected for this review.

The selected values of the standard heat capacity and entropy are those given in [1994GUR/VEY], based on the data in [1942MOO/KEL]:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{MgSO_4}, \beta, 298.15\ \mathrm{K}) = \mathbf{(96.4 \pm 0.8)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{MgSO_4}, \beta, 298.15\ \mathrm{K}) = \mathbf{(91.6 \pm 1.0)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

There are no other data on the low-temperature heat capacity of $MgSO_4$(cr), so we have not been able to select any values for $C^{\circ}_{p,\mathrm{m}}$ and S°_{m} of the α-$MgSO_4$ polymorph, although since the structures of the α and β polymorphs are similar, they are not likely to be very different from the above values. We note, though, that Gurvich *et al.* [1994GUR/VEY] give the above values for the phase (which, unlike us, they call α) as that with an enthalpy of formation of – (1288.8 ± 0.6) kJ·mol^{-1}, which may not be correct.

V.6.1.1.4.3 Enthalpy of formation

The solution calorimetric determinations of $\Delta_f H^{\circ}_{\mathrm{m}}(\mathrm{MgSO_4})$, mainly dating from the 19$^{\mathrm{th}}$ century ([1871FAV/VAL], [1882BER/ILO], [1883THO], [1883BER/ILO], [1885PIC], [1886PIC]) are subject to uncertainties that cannot now be eliminated since, in these early

studies, the preparative methods for the $MgSO_4$ samples are either missing or lacking detail. These data will thus not be considered further.

In fact, the only fully meaningful measurements of $\Delta_f H^\circ_m(MgSO_4)$ based on solution calorimetry are those of Ko and Daut [1980KO/DAU], since these authors unambiguously determined the enthalpy of formation of well-characterised α-$MgSO_4$ and β-$MgSO_4$. The preparative techniques used in [1980KO/DAU] were similar to those employed by the authors who characterised structurally the two phases for the first time, namely Rentzeperis and Soldatos [1958REN/SOL] for α-$MgSO_4$ (obtained by heating a solution of MgO in H_2SO_4 to complete elimination of excess H_2SO_4), and Coing-Boyat [1962COI] for β-$MgSO_4$ (obtained by desiccation to 773 K of a neutral solution of $MgSO_4$). Ko and Daut [1980KO/DAU] measured the enthalpies of solution of MgO(cr) and well-characterised $MgSO_4$ (α and β) in 4.36 molal HCl solution. As noted in Appendix A, when corrected by using the CODATA [1989COX/WAG] values for the auxiliary data instead of those from [1968WAG/EVA] used by Ko and Daut, these data give $\Delta_f H^\circ_m(MgSO_4, \alpha, 298.15\ K) = -(1288.64 \pm 0.61)\ kJ{\cdot}mol^{-1}$ and $\Delta_f H^\circ_m(MgSO_4, \beta, 298.15\ K) = -(1284.80 \pm 0.61)\ kJ{\cdot}mol^{-1}$. After rounding, these are our selected values:

$$\Delta_f H^\circ_m(MgSO_4, \alpha, 298.15\ K) = \mathbf{-\ (1288.6 \pm 0.6)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^\circ_m(MgSO_4, \beta, 298.15\ K) = \mathbf{-\ (1284.8 \pm 0.6)}\ kJ{\cdot}mol^{-1}.$$

The value for $\Delta_f H^\circ_m(MgSO_4, \alpha, 298.15\ K)$ is very close to that selected in [1994GUR/VEY] ($-(1288.8 \pm 0.6)\ kJ{\cdot}mol^{-1}$), who also report, for information only, equilibrium measurements on the decomposition of $MgSO_4$(cr) to MgO(cr) + SO_2(g) + 1/2O_2(g)) ([1925MAR], [1959DEW/RIC] and [1977LAU/CUB]), and to MgO(cr) + SO_3(g) [1977LAU/CUB]. The enthalpies of formation of $\Delta_f H^\circ_m(MgSO_4, 298.15\ K)$ derived by Gurvich *et al.* from these studies based on third-law treatments, – 1290 to – 1300 $kJ{\cdot}mol^{-1}$ are somewhat more negative than our selected value, but with large uncertainties (up to 18 $kJ{\cdot}mol^{-1}$ for [1959DEW/RIC]), possibly related to errors in the estimated heat capacities of $MgSO_4$ (cr and l) necessarily used.

V.6.1.1.5 Magnesium carbonate, $MgCO_3$(cr) (magnesite)

It should be noted that, due to the low stability of magnesite and its slow kinetics of crystallisation at low temperature, several magnesium hydrated carbonates, hydroxidocarbonates and related minerals are well known compounds, *e.g.*, see papers from the CRNS research centre, Géosciences Environnement Toulouse (GET), France, by P. Bénézeth and coworkers. Such compounds have been the topic of extensive studies in recent years. Unfortunately, such minerals fall well outside the scope of the present study.

V.6.1.1.5.1 Structure

Magnesium carbonate has the hexagonal calcite structure, up to its melting point at 1263 K, space group $R\bar{3}c$, with a = (4.633 ± 0.002) Å, c = (15.016 ± 0.003) Å, as given by Graf [1961GRA] for a synthetic material at (299 ± 3) K. This author gave a broad discussion of the structure of many rhombohedral carbonates (including calcite and dolomite). Concordant values, a = (4.635 ± 0.002) Å, c = (15.019 ± 0.003) Å at 297 K, were given in [1985MAR/REE] for a natural sample (containing 0.8 at % Mn, 0.7 at % Fe and 0.1 at % Ca). These authors reported structure refinement of magnesite up to 773 K and made similar measurements for calcite up to 1073 K.

V.6.1.1.5.2 Heat capacity and entropy

There are three studies of the low-temperature heat capacity of $MgCO_3$(cr). Anderson [1934AND] measured the low-temperature heat capacity from 56 to 292 K of a naturally-occurring sample of $MgCO_3$(cr) containing 9.0 % $CaCO_3$ and 1.3 % $FeCO_3$ (not just the 0.3 % $CaCO_3$ given in [1994GUR/VEY]). After correction for these impurities, the heat capacity was extrapolated from 56 K down to 0 K using a sum of two Debye and one Einstein functions, and also graphically, giving standard entropies of (65.7 ± 0.8) and (66.1 ± 0.8) J·K^{-1}·mol^{-1} respectively. Hemingway and Robie have studied two samples of magnesite. In [1977HEM/ROB3] they made measurements from 12.7 to 384 K on fine crystals of a very pure synthetic material (see Appendix A for details), obtaining S°_{m} (298.15 K) = (65.09 ± 0.13) J·K^{-1}·mol^{-1}. Subsequently in [1994ROB/HEM] they studied large glassy clear magnesite crystals from Brazil which microprobe analysis indicated contained 0.5 % $FeCO_3$, 0.3 % $CaCO_3$ and 0.1 % $MnCO_3$ (mole percentages). Measurements were made by adiabatic calorimetry from 8 to 343 K and from 339 to 676 K by DSC, with good agreement between the two techniques in the overlap region. The calculated standard entropy from this study was 64.98 J·K^{-1}·mol^{-1}; no uncertainty limits were given. In view of the high purity of the sample and the careful investigation, the data in [1977HEM/ROB3] are considered by this review to be the most accurate, and with a minor change in $C^{\circ}_{p,m}$, are the selected values.

The only other high-temperature enthalpy measurements are the unpublished data up to 753 K by Shomate (documented by Kelley in [1960KEL]) on an unknown sample.

A comparison of these data is shown in Figure V-42 together with the assessed data in [1994GUR/VEY]. As noted in the Appendix A entry for [1994ROB/HEM], there are grounds for thinking that they were not very confident of their DSC measurements, and for this review we have therefore selected the values assessed in [1994GUR/VEY], which are a fit to the high-temperature enthalpy data of Shomate. The heat capacity equation is:

$$C^{\circ}_{p,m}(MgCO_3, cr, T)\ /J\cdot K^{-1}\cdot mol^{-1} = \mathbf{73.335} + \mathbf{6.399\times10^{-2}}(T/K) - \mathbf{1.4495\times10^{6}}(T/K)^{-2} \qquad (298.15\text{-}800\,K)$$

with the selected standard values at 298.15 K:

$$C^{\circ}_{p,\mathrm{m}}\,(\mathrm{MgCO_3,\ cr,\ 298.15\ K}) = \mathbf{(76.108 \pm 0.760)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

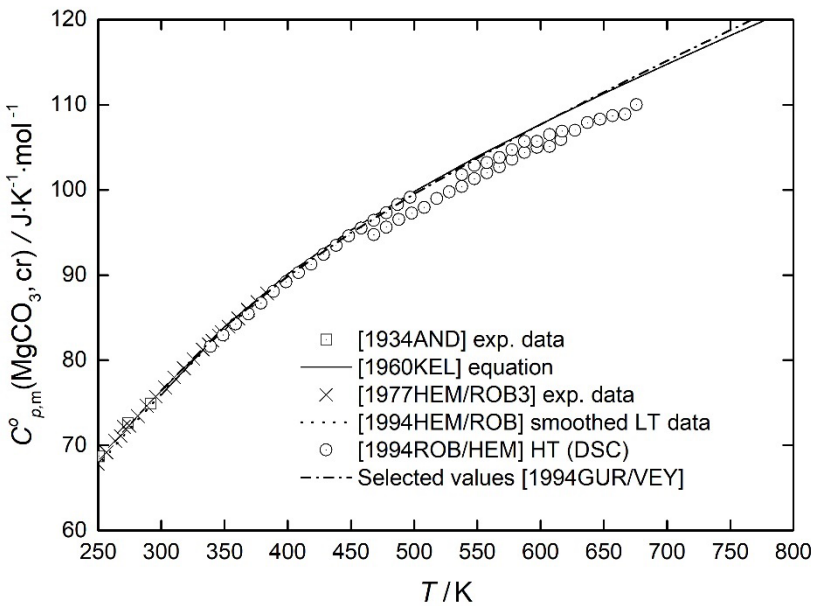
and

$$S^{\circ}_{\mathrm{m}}\,(\mathrm{MgCO_3,\ cr,\ 298.15\ K}) = \mathbf{(65.1 \pm 0.6)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

where the entropy is the rounded value from the optimisation described in Section V.6.1.1.6, which is the same as the experimental value obtained by Hemingway and Robie [1977HEM/ROB3].

We note with interest that Gurvich *et al*. [1994GUR/VEY] adopted this value of S°_{m} (298.15 K), identical to that obtained in [1977HEM/ROB3], although the only study mentioned in their text is that of Anderson [1934AND], which gives a noticeably higher value.

Figure V-42: Heat capacity of $MgCO_3$(cr) ($J{\cdot}K^{-1}{\cdot}mol^{-1}$).



V.6.1.1.5.3 Enthalpy of formation

There are a number of studies aimed at measuring $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{MgCO_3,\ cr})$ by combustion calorimetry, by calorimetric dissolution at near-ambient and at high temperatures, and by measuring the decomposition pressures, which are summarised in Table V-52.

Unfortunately, many of these studies suffer from notable uncertainties, mainly due to one or more of the following factors:

- The use of impure magnesite, either natural minerals or inadequate removal of hydroxidocarbonate from synthetic material.
- Crude procedures and/or complex corrections.
- Lack of experimental heat capacity measurements of magnesite above 675 K.
- For the high-temperature, high-pressure studies of the decomposition the $p\Delta V$ contribution to the Gibbs energy is not negligible but not all the requisite data on the variation of the molar volumes of magnesite and periclase with T and p are available.

Table V-52: Experimental data on the enthalpy of formation of $MgCO_3$(cr).

Reference	Method	No. of Exp. Points	Derived $\Delta_f H^\circ_m$ (298.15 K) /kJ·mol^{-1}	Comments
[1913MAR/SIM]	Decomposition 670-782 K	23	– (1113.85 ± 5.04) †	See Appendix A Possible small contribution from KNO_3 ---------- **† Included in optimisation**
[1941ROT]	Combustion	6	– (1089.7 ± 2.20) For information only	Crude experimental procedure; complex corrections ---------- Not included in optimisation
[1941ROT]	Soln. calorimetry	Not given	– 1104.7 For information only	Value cited as such from earlier unpublished results ---------- Not included in optimisation
[1953CRE/ALL]	Decomposition 500-700 K	9	– (1098.8 ± 2.7) For information only	Product contained CO_2(g) from impure magnesite ---------- Not included in optimisation
[1954KAP/STA]	Soln. calorimetry	5	– (1093.5 ± 25.0) For information only	Sample was probably hydrated hydroxidocarbonate ---------- Not included in optimisation
[1955HAR/TUT]	Decomposition 923-1123 K	5	– (1112.04 ± 1.03)	Uncertainties in $C_{p,m}$ and $p\Delta V$ contribution to $\Delta_r G_m$ ---------- *Included in Combined high temperature decomposition*

(Continued on next page)

Table V-52 (continued)

Reference	Method	No. of Exp. Points	Derived $\Delta_f H_m^o$ (298.15 K) /kJ·mol^{-1}	Comments
[1958FOL/KLI]	DTA	Not given	− (1111.8 ± 10.0) For information only	Too few details to elaborate on the cited value ---------- Not included in the optimisation
[1958FOL/KLI]	High-temperature calorimetry	average of 5-6 points is given	− (1107.6 ± 10.0) For information only	Values corrected to 298.15 K ---------- Not included in the optimisation
[1964THO/CUN]	High-temperature calorimetry 923 K	Not given	– (1112.14 ± 3.00)	Uncertainties in High-temperature $C_{p,m}(MgCO_3)$ ---------- *Included in Combined high temperature decomposition*
[1968JOH/MET]	Decomposition 980-1040 K	2	– (1112.08 ± 1.54)	Uncertainties in $C_{p,m}$ and $p\Delta V$ contribution to $\Delta_r G_m$ ---------- *Included in Combined high temperature decomposition*
[1993CHA/NAV]	High-temperature calorimetry 977 K	19	− (1112.0 ± 1.5) † Uncertainties increased by 50 % from 2σ	Confirmed by the results of [1995CHA/NAV], see Appendix A ---------- † **Included in optimisation**
[1995KOZ/NEW]	Decomposition 988 K	1	– (1111.57 ± 0.46) Uncertainty just that of uncertainty in *p*	Uncertainties in $C_{p,m}$ and $p\Delta V$ contribution to $\Delta_r G_m$ ---------- *Included in Combined high temperature decomposition*

The uncertainties for the decomposition reactions are 2σ statistical only, while most of the uncertainties in the calorimetric values are estimated by this review.

The nature of the uncertainties is noted briefly in Table V-52, but more details of the problems, both experimental and calculational, in extracting the values of $\Delta_f H_m^o$ ($MgCO_3$, 298.15 K) are given in the full Appendix A entries for these studies. The review by Langmuir [1965LAN] also analysed in some detail those studies which were available at the time.

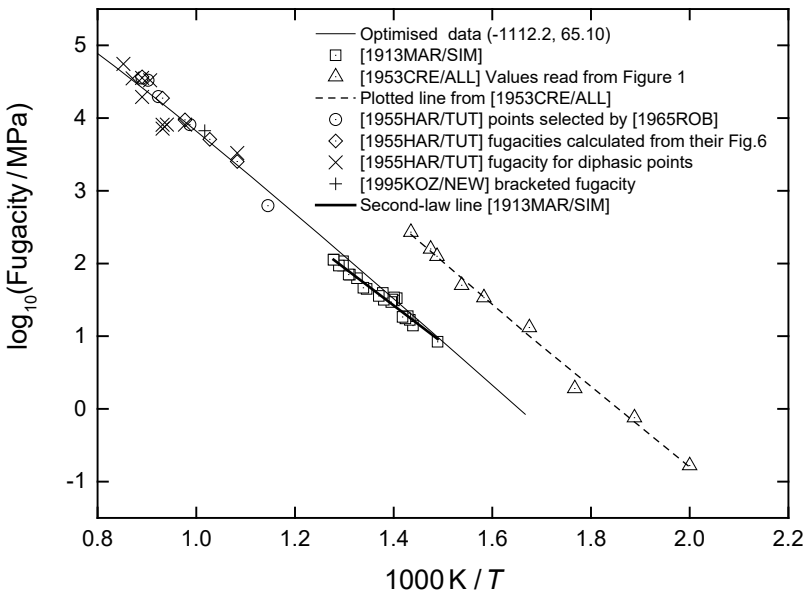
A selection of the more reliable of these values has been used, together with the solubility data discussed in Section V.6.1.2.6 in a simple optimisation of the 298.15 K data for magnesite and $S^{o}(Mg^{2+})$, as described in Section V.6.1.1.6. Three values for the enthalpy of formation were included in the optimisation: the two values marked with † in the above Table, and the mean value of $-(1111.90 \pm 2.50)$ kJ·mol^{-1} derived from the more uncertain decomposition data (references in italic).

As noted in Section V.6.1.1.6, the optimisation gives $-(1112.180 \pm 1.30)$ kJ·mol^{-1} for the enthalpy of formation at 298.15 K, which, after rounding is our selected value, with an increased uncertainty:

$$\Delta_f H^{o}_m(MgCO_3, cr, 298.15\ K) = \mathbf{-(1112.200 \pm 1.300)}\ kJ \cdot mol^{-1}.$$

Figure V-43 compares the calculated decomposition pressure of magnesite (plotted as the fugacity) with the experimental data, showing reasonable agreement with the more reliable data considering the uncertainties in the latter.

Figure V-43: Decomposition pressures of $MgCO_3$(cr).



V.6.1.1.6 Optimisation of some data for Mg species

From our assessments of the data on the solubilities of brucite and magnesite, and from the decomposition pressures of magnesite, it became clear that an appreciably better fit of these data could be achieved with a slightly different value for $S^{o}_m(Mg^{2+}, 298.15\ K)$

from that given by CODATA [1989COX/WAG] and currently used in the TDB reviews, – (137.0 ± 4.0) J·K^{-1}·mol^{-1}. We have therefore made a simple optimisation of the inter-related values for: one value of the Gibbs energy at 298.15 K of the reaction,

$$MgCO_3(cr) \rightleftharpoons Mg^{2+} + CO_3^{2-} \quad \text{(V.97)}$$

three values of $\Delta_f H_m^o$ ($MgCO_3$, cr, 298.15 K), one of S_m^o ($MgCO_3$, cr, 298.15 K), and three values S_m^o (Mg^{2+}, 298.15 K), using the values in Table V-53. The solubility of magnesite ($MgCO_3$(cr)) is discussed in detail in Section V.6.1.2.6.

This table also gives the final rounded optimised values which are therefore our selected values with slightly increased uncertainties.

Table V-53: Input data used in the optimisation of the data for some Mg reactions and species.

No.	Property	Initial value	Optimised value ± 2 sigma	Reference
	$MgCO_3(cr) \rightleftharpoons Mg^{2+} + CO_3^{2-}$			
1	$\Delta_r G_m^o$ /kJ·mol^{-1}	(44.952 ± 1.700)	(44.841 ± 0.932)	$MgCO_3$ solubility see Section V.6.1.2.6
	$MgCO_3$(cr)			
2	$\Delta_f H_m^o$ /kJ·mol^{-1}	– (1113.85 ± 5.00)	– (1112.180 ± 1.300)	[1913MAR/SIM]
3	$\Delta_f H_m^o$ /kJ·mol^{-1}	– (1112.0 ± 1.5)		[1993CHA/NAV]
4	$\Delta_f H_m^o$ /kJ·mol^{-1}	– (1111.90 ± 2.50)		Combined less-precise high-temperature decomposition
5	S_m^o /J·K^{-1}·mol^{-1}	(65.09 ± 0.13)	(65.090 ± 0.600)	[1977HEM/ROB3]
	Mg^{2+}			
6	S_m^o /J·K^{-1}·mol^{-1}	– (135.02 ± 7.60)	– (135.931 ± 1.700)	$Mg(OH)_2$ solubility see Section V.6.1.2.2
7	S_m^o /J·K^{-1}·mol^{-1}	– (136.9 ± 5.0)		[1989COX/WAG], (note 130c), based on $MgSO_4 \cdot 6H_2O$
8	S_m^o /J·K^{-1}·mol^{-1}	– (134.5 ± 6.0)		[1989COX/WAG], (note 130e) based on $MgCl_2 \cdot 4H_2O$ and $MgCl_2 \cdot 6H_2O$

Selected values:

$$S_m^o(Mg^{2+}, 298.15\ K) = \mathbf{-\ (135.9 \pm 1.7)}\ J \cdot K^{-1} \cdot mol^{-1},$$

$$\Delta_f H_m^o(MgCO_3, cr, 298.15\ K) = \mathbf{-\ (1112.2 \pm 1.3)}\ kJ \cdot mol^{-1},$$

$$S_m^o(MgCO_3, cr, 298.15\ K) = \mathbf{(65.1 \pm 0.6)}\ J \cdot K^{-1} \cdot mol^{-1},$$

giving

$$\Delta_f G^o_m (MgCO_3, cr, 298.15\ K) = \mathbf{-\ (1028.409 \pm 1.313)}\ kJ \cdot mol^{-1}.$$

V.6.1.2 Magnesium aqueous species

V.6.1.2.1 Aqueous magnesium(II)

In their comprehensive review, Ohtaki and Radnai [1993OHT/RAD] specifically mentioned the magnesium cation as being "one of the most widely investigated ions by the X-ray diffraction method and computer simulations" and that these results are "fairly" concordant [1970BOL/GER], [1975BUL/IRI], [1979CAM/LIC], [1993OHT/RAD], [1996MAR/GLU], [1998PAV/SIE], [1998PYE/RUD]. The distribution of hydration numbers is very narrow at six per metal centre with Mg^{2+}-OH_2 lengths of 200 pm. Ohtaki and Radnai cited ample NMR supporting evidence for the structure of the hydrated magnesium(II) cation. Raman spectroscopic measurements of aqueous magnesium perchlorate, chloride and sulfate solutions (3.0, 3.3 and 3.05 $mol \cdot kg^{-1}$, respectively) from 298.15 to 398.15 K by Pye and Rudolph [1998PYE/RUD] were also interpreted as demonstrating that the octahedral $Mg(H_2O)_6^{2+}$ cation is thermodynamically stable in perchlorate and chloride solutions over this range of conditions, *i.e.*, there was no indication of contact ion-pair formation, although it was apparent that a solvent-separated ion pair existed in sulfate solutions.

As noted in Section V.6.1.2.6, it became clear that a better fit of the data for the solubilities of $Mg(OH)_2$(cr) (brucite) and $MgCO_3$(cr) (magnesite) and the data for the entropy and enthalpy of formation (calculated from the calorimetric and decomposition pressure data) of magnesite could be achieved with a small change in the value of $S^o_m(Mg^{2+}, aq, 298.15\ K)$ from the value assessed by CODATA [1989COX/WAG], – (137 ± 4) $J \cdot K^{-1} \cdot mol^{-1}$. No change is made to the enthalpy of formation:

$$\Delta_f H^o_m (Mg^{2+}, aq, 298.15\ K) = \mathbf{-\ (467.0 \pm 0.6)}\ kJ \cdot mol^{-1}.$$

Cox *et al.* [1989COX/WAG] based their choice for $S^o_m(Mg^{2+}, aq, 298.15\ K)$ on three values: – (137.1 ± 4.0) $J \cdot K^{-1} \cdot mol^{-1}$ based on data for the solubility of $Mg(OH)_2$(cr) by Hostetler [1963HOS] and data for $Mg(OH)_2$(cr) and $\Delta_f H^o_m(Mg^{2+}, aq, 298.15\ K)$, – (136.9 ± 5.0) $J \cdot K^{-1} \cdot mol^{-1}$ from the solubility and relevant data for $MgSO_4 \cdot 6H_2O$(cr), and – (134.5 ± 6.0) $J \cdot K^{-1} \cdot mol^{-1}$ based on the solubility of $MgCl_2 \cdot 6H_2O$(cr).

The first value has clearly been overtaken by the thorough analysis of all the relevant data for the solubility of brucite given in Section V.6.1.2.2, which indicates a slightly more positive value of $S^o_m(Mg^{2+}, aq, 298.15\ K) = -\ (135.02 \pm 7.60)\ J \cdot K^{-1} \cdot mol^{-1}$.

As described in Section V.6.1.1.6, this latter value and the second and third values noted in [1989COX/WAG] were included in the simple optimisation of the relevant data giving the accepted value of

$$S_m^o(Mg^{2+}, aq, 298.15\ K) = \mathbf{-\ (135.900 \pm 1.700)}\ J\cdot K^{-1}\cdot mol^{-1},$$

and thus we obtain:

$$\Delta_f G_m^o(Mg^{2+}, aq, 298.15\ K) = \mathbf{-\ (455.703 \pm 0.786)}\ kJ\cdot mol^{-1}.$$

The optimised $\Delta_f G_m^o(Mg^{2+})$ value derived by Miron *et al.* [2017MIR/WAG] based on Gibbs energies of formation of other magnesium bearing phases is – 456.005 $kJ\cdot mol^{-1}$ at 298.15 K.

V.6.1.2.2 Solubility of magnesium(II) hydroxide (brucite)

Hostetler [1963HOS], in an extensive series of experiments, provided probably the first valuable estimates of the solubility of brucite, $Mg(OH)_2(cr)$, and the extent of hydrolysis of Mg^{2+} that prevailed for many years. Mention was made of potential contamination by CO_2 that could have contributed to the disparity amongst earlier results. An electron microscope was used to monitor grain size as a function of time and established that grains larger than 1 μm provided constant solubility attributable to crystalline brucite. This limiting grain size is now generally accepted as the minimum for most solubility studies of crystalline solids. A subsequent study from the same laboratory by McGee and Hostetler [1977MCG/HOS] improved the accuracy of the solubility constants considerably. By taking samples over time (for up to 13 days compared to one set of samples taken after five days [1963HOS]) McGee and Hostetler alleviated a concern of the reviewers that equilibrium was not fully achieved in the earlier study.

Baes and Mesmer [1976BAE/MES] had to base their selected value of the solubility product of brucite (Equation (V.98)) on the work of Hostetler [1963HOS], but questioned the corresponding value for Equation (V.98) of $\log_{10}{}^*K_{s,0} = 16.58$ in 3 $mol\cdot dm^{-3}$ ($NaClO_4$) solution reported by Horn [1969HOR], because their treatment of the pH dependence of the solubility led to an unreasonably high value of $\log_{10}{}^*K_1^o$ (V.98), *i.e.*, – 9.64, *cf.* our value of – (11.59 ± 0.12) at 298.15 K.

$$Mg(OH)_2(cr) \rightleftharpoons Mg^{2+} + 2OH^- \qquad (V.98)$$

$$Mg(OH)_2(cr) + 2H^+ \rightleftharpoons Mg^{2+} + 2H_2O(l) \qquad (V.99)$$

More recently, Altmaier *et al*. [2003ALT/MET] performed a limited number of experiments on the solubility equilibria of brucite whereby they resolved apparent minor differences reported in some earlier studies, particularly with reference to the established Harvie, Møller and Weare (HMW) database [1984HAR/MOL] and the recommended values from the NEA database. Their measurements were made in an argon-filled glove box at room temperature (295 ± 2) K for periods of several days to 14 months with relatively low solid to liquid ratios, *viz*., 0.3 g/0.2 dm^{-3} of solution. The pH_m of the samples taken over time initially showed a slight decrease in pH_m for a month, but reached constant values after 50 to 350 days indicating that equilibrium had been achieved. A decrease observed at high $MgCl_2$ molalities from 2.67 to 5.15 $mol\cdot kg^{-1}$ was shown by XRD analysis to be due to the transformation of brucite to an anhydrous magnesium

oxychloride. Significant hydrolysis of Mg^{2+} could be ruled out from the results of Palmer and Wesolowski [1997PAL/WES]. The experimental results are listed in Table V-54, where the activity coefficients were calculated from published Pitzer ion interaction parameters in [1984HAR/MOL]. Altmaier *et al*. presented possible reasons for finding that $m_{Mg^{2+}} > 2m_{OH^-}$ although no definitive explanation was offered other than assuming that magnesium-containing impurities were responsible.

Table V-54: Composition of solutions in equilibrium [2003ALT/MET] with $Mg(OH)_2$(cr) and the molal solubility constants based on the Pitzer treatment of activity coefficients using parameters from [1984HAR/MOL] at 295 K and on the SIT treatment by this review. The values in italics indicate that the fit was empirical, because of the absence of a supporting electrolyte.

Initial m_{MgCl_2}	Initial m_{NaCl}	$-\log_{10} m_{H^+}$	$-\log_{10} m_{Mg^{2+}}$	Pitzer $\log_{10}{}^{*}K^{o}_{s,0}$	SIT $\log_{10}{}^{*}K^{o}_{s,0}$
0	0	10.34	3.55	17.10	*17.10*
0.01	0	9.67	2.02	17.19	*17.18*
0.02	0	9.54	1.73	17.17	*17.17*
0.05	0	9.32	1.30 (a)	17.10	*17.09*
0.10	0	9.20	1.00 (a)	17.10	*17.09*
0.25	0	9.09	0.60 (a)	17.17	*17.18*
1.00	0	8.97	0.00 (a)	17.21	*17.40*
2.67	0	9.00	– 0.43 (a)	17.22	
0	0.1	10.33	3.38	17.04	17.05
0	0.5	10.38	3.28	17.07	17.08
0	1.0	10.45	3.30	17.08	17.10
0	5.61	10.72	3.33	16.92	17.02
0.01	0.50	9.74	1.99	17.08	17.09
0.03	0.50	9.49	1.53	17.03	17.03
0.05	0.50	9.38	1.31	17.02	17.02
0.01	5.0	10.04	1.94	17.03	17.12
0.03	5.0	9.79	1.50	16.96	17.05
0.05	5.0	9.70	1.30	16.98	17.06

(a) $m_{Mg^{2+}}$ was not measured but was assumed to be equal to the initial solution concentration.

The mean $\log_{10}{}^{*}K_{s,0}$ (V.99) values from both sets of experiments (*i.e.*, without[1] and with NaCl added) were given as (17.16 ± 0.10) and (17.01 ± 0.10) (2σ), respectively, at 295 K with an overall average value of (17.1 ± 0.2). Altmaier *et al*. calculated $\log_{10}{}^{*}K^{o}_{s,0}$ (V.99) = 16.30 at 298.15 K from the values of $\Delta_f G^o_m$, $\Delta_f H^o_m$ and S^o_m at 298.15 K for brucite selected by Helgeson *et al*. [1978HEL/DEL]. The Helgeson *et al*. analysis was based on high-pressure, high-temperature data for the equilibrium between $Mg(OH)_2$(cr) and MgO(cr) in combination with calorimetric data for MgO(cr) and auxiliary data for the standard molal volume, heat capacity and entropy of brucite. This discrepant solubility constant was thought to originate primarily from the choice of the

[1] Although the absence of a supporting electrolyte invalidates application of the SIT in this case, the results of this fit are provided for an empirical and perhaps fortuitous agreement with that obtained in the presence of excess supporting electrolyte.

standard entropy of brucite ($S^{o}_{298.15} = 63.14$ J·K^{-1}·mol^{-1}) [1968ROB/WAL] which apparently stemmed from a collection of entropy data [1961KEL/KIN] with an assigned uncertainty of ± 0.21 J·K^{-1}·mol^{-1} and from experimental heat capacities for $Mg(OH)_2$(cr) from 323.15 to 373.15 K calculated by Giauque and Archibald [1937GIA/ARC].

Brown *et al*. [1996BRO/DRU] precipitated brucite *in situ* by addition of an NaOH(sln) titrant to an Mg^{2+} solution in NaCl media within a well-stirred hydrogen-electrode concentration cell at 333.15, 373.15, 423.15 and 473.15 K. Whereas the precipitation of metal hydroxides in the absence of seed crystals of the desired phase normally leads to the formation of amorphous and/or very fine grained material, this precipitate was characterised by XRD as being crystalline brucite and back titrations gave identical results from under-saturation. The pH_m of the mixture was monitored continuously and was used as an indicator of equilibrium having been attained. Königsberger *et al*. [1999KON/KON] combined solubility data [1977MCG/HOS], [1996BRO/DRU] with CODATA values for Mg^{2+} and H_2O(l) [1989COX/WAG] to give $\Delta_f G^{o}_{m}$ ($Mg(OH)_2$, cr, 298.15 K) = – 831.9 kJ·mol^{-1}, $\Delta_f H^{o}_{m}$ ($Mg(OH)_2$, cr, 298.15 K) = – 926.7 kJ·mol^{-1}, and S^{o}_{m} ($Mg(OH)_2$, cr, 298.15 K) = 50.6 J·K^{-1}·mol^{-1}. Königsberger *et al*. proposed that the precipitation experiments of Brown *et al*. [1996BRO/DRU] could have generated smaller less-crystalline brucite particles with higher surface energies. However, Altmaier *et al*. [2003ALT/MET] were able to refute this suggestion because they reported a similar $\log_{10} {}^{*}K^{o}_{s,0}$ value of (17.1 ± 0.2) for Reaction (V.99) for well-crystallised brucite (0.3-1.0 μm) for which the excess Gibbs surface energy contribution was known to be negligible [1963HOS]. From this they derived $\Delta_f G^{o}_{m}$ ($Mg(OH)_2$, cr, 298.15 K) = – 832.1 kJ·mol^{-1} using CODATA [1989COX/WAG] auxiliary values. Furthermore, batch experiments performed by Xiong [2008XIO] produced crystalline brucite (0.63 to 23 μm) by precipitation from homogeneous $MgCl_2$ solutions at ambient temperature with identical solubilities to those obtained from their dissolution experiments. Hostettler had originally asserted that differences between the results of all previous solubility studies could be attributed to particle size effects.

Table V-55: Summary of experimental data, reported logarithms of the molar solubility products and solubility constants of $Mg(OH)_2$(cr) according to Equations (V.98) and (V.99) where reported uncertainties are assumed to be 1σ. Only individual sets of measurements are shown when sufficient supporting electrolyte was present to permit application of SIT.

T/K	Ionic medium	I_c max	$-\log_{10}{}^{*}K_s^{o}$ reported	$-\log_{10}{}^{*}K_s^{o}$ re-calculated	$-\log_{10}{}^{*}K_{s,0}^{o}$ (2σ) re-calculated	Method	Reference
291.15	NaOH + $MgCl_2$	0.3 (a)	(10.93 ± 0.08)			sol, pot	[1925GJA]
298	$KCl + MgCl_2$	1.6	(11.15 ± 0.03) (b)	(11.15 ± 0.25)	(16.85 ± 0.24)	sol	[1963HOS]
298	$KCl + MgCl_2$	0.4	(11.16 ± 0.03)	(11.16 ± 0.12)	(16.85 ± 0.12)		
298.15	$NaClO_4$	3.0	(16.58 ± 0.01)		(16.51 ± 0.04)	sol	[1969HOR]
298.15	$MgCl_2$	0.075	(10.86 ± 0.03)			sol	[1973LIU/NAN]
283.15	$Mg(OH)_2$	0.0036	(10.91 ± 0.18)	(10.87 ± 0.22)	(18.24 ± 0.22)	sol	[1977MCG/HOS]
298.15		0.0030	(10.92 ± 0.06)	(10.94 ± 0.05)	(17.07 ± 0.05)		
313.15		0.0012	(10.90 ± 0.12)	(10.91 ± 0.13)	(16.15 ± 0.13)		
328.15		0.0012	(10.87 ± 0.15)	(10.92 ± 0.14)	(15.36 ± 0.14)		
343.15		0.0006	(10.96 ± 0.16)	(11.03 ± 0.18)	(14.58 ± 0.18)		
363.15		0.0008	(11.06 ± 0.16)	(11.13 ± 0.17)	(13.71 ± 0.17)		
333.15	$NaCl + MgCl_2$	I_m = 0.1,1.0	15.07		(15.05 ± 0.21)	pot, sol	[1996BRO/DRU]
373.15			13.18		(12.98 ± 0.21)		
423.15			11.33		(11.33 ± 0.11)		
473.15			9.87		(9.88 ± 0.11)		
298.15	$KCl + MgCl_2$				(16.95 ± 0.16) (c) (16.84 ± 0.05) (c)	est	[2003ALT/MET]
295.15	$MgCl_2$	$I_m \leq 3.0$	(17.16 ± 0.10)		(17.12 ± 0.04)	sol	
295.15	$NaCl + MgCl_2$	$I_m \leq 5.15$	(17.01 ± 0.10)		(17.06 ± 0.05)		
295.65	$NaCl + MgCl_2$	$I_m \leq 4.41$	(17.05 ± 0.20)		(17.04 ± 0.07)	sol	[2008XIO]

(a) The first measurement at $I_c \approx 1.5$ was neglected here.
(b) This value was accepted by Baes and Mesmer [1976BAE/MES].
(c) Re-calculation of the data from first and third datasets in [1963HOS].

Application of the SIT to treat the ionic-strength dependence of the brucite dissolution equilibrium (Equation(V.99)) gave:

$$\log_{10}{}^{*}K_{s,0} - 2D + 2\log_{10} a_{H_2O} = \log_{10}{}^{*}K^{\circ}_{s,0} - \Delta\varepsilon I_m \qquad \text{(V.100)}$$

An SIT treatment of the data in [2003ALT/MET] resulted in the following values for the NaCl-free (see footnote (a) in Table V-55) and NaCl-containing solutions: $\log_{10}{}^{*}K^{\circ}_{s,0} = (17.12 \pm 0.04)$ and (17.06 ± 0.05); $\Delta\varepsilon$ (apparent) = – (0.092 ± 0.036) and $\Delta\varepsilon = -(0.081 \pm 0.010)$ kg·mol^{-1}, respectively, at 295 K (*cf.* $\Delta\varepsilon = -(0.05 \pm 0.02)$ kg·mol^{-1}) derived from the ion interaction terms recommended by the NEA). The uncertainties quoted are 2σ of the statistical fits. The corresponding average value for $\log_{10}{}^{*}K^{\circ}_{s,0}$ reported by Altmaier *et al*. obtained from the Pitzer treatment was (17.1 ± 0.2), identical to that obtained from the SIT, the latter having fewer parameters. Moreover, SIT treatments (see discussion below) of two other datasets gave very similar values, *viz*. [1977MCG/HOS] using low-ionic-strength < 0.01 data resulted in $\log_{10}{}^{*}K^{\circ}_{s,0} = (17.07 \pm 0.05)$ at 298.15 K, whereas values from [1996BRO/DRU] at = 0.1 and 1.0, extrapolated from data from 333.15 to 473.15 K, gave $\log_{10}{}^{*}K^{\circ}_{s,0} = 17.01$ extrapolated from data from 333.15 to 473.15 K [1996BRO/DRU]. Finally, an SIT regression of the solubility data at 296.65 K obtained from under-saturation in NaCl solutions to $I_m = 4.41$ resulted in a compatible value of $\log_{10}{}^{*}K^{\circ}_{s,0} = (17.06 \pm 0.07)$, but a distinctly different $\Delta\varepsilon = (0.48 \pm 0.04)$ kg·mol^{-1} (2σ) presumably due in part to a difference in activity coefficient models [2008XIO]. The combined datasets [1977MCG/HOS], [1996BRO/DRU], [2003ALT/MET], [2008XIO] were regressed with the following equation providing the most adequate fit:

$$\log_{10}{}^{*}K^{\circ}_{s,0}\,(\text{V.99}) = -(3.0430 \pm 0.202) + (5978.97 \pm 76.45)\,(T/\text{K})^{-1} \qquad \text{(V.101)}$$

Equation (V.101) gives:

$$\log_{10}{}^{*}K^{\circ}_{s,0}\,((\text{V.99}), 298.15\ \text{K}) = (17.011 \pm 0.100),$$

$$\Delta_r G^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = -(97.100 \pm 0.569)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = -(114.467 \pm 2.831)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = -(58.249 \pm 9.685)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C^{\circ}_{p,m}\,((\text{V.99}), 298.15\ \text{K}) = (0 \pm 11)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

These values are in excellent agreement with the calculated values for $Mg(OH)_2$(cr), based on the more precise calorimetric data, which gave:

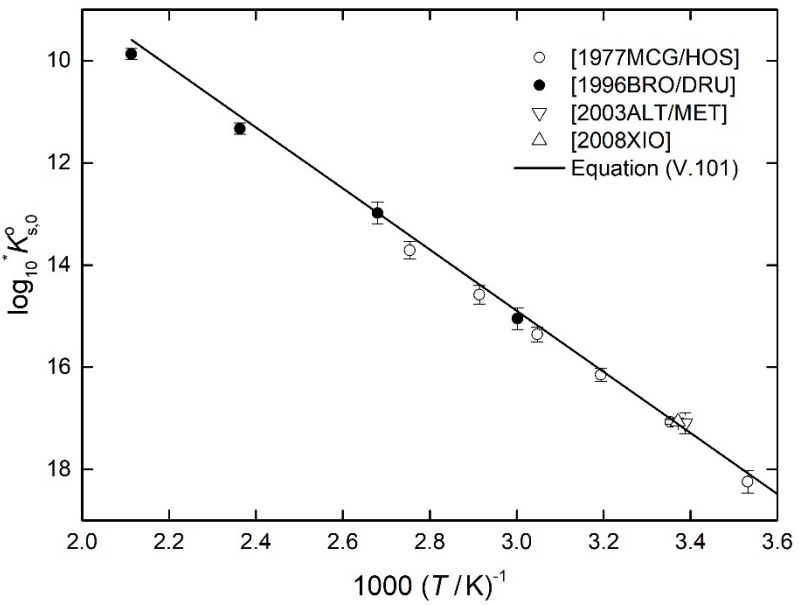
$$\Delta_r G^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = -(96.565 \pm 0.854)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = -(114.210 \pm 0.675)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^{\circ}_m\,((\text{V.99}), 298.15\ \text{K}) = \quad (59.180 \pm 4.020)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

The data in [1982WAG/EVA] correspond to $\Delta_r {}^{*}H_m^{\circ}$ ((V.99), 298.15 K) = – 114.0 kJ·mol^{-1} which was adopted also in [1995ROB/HEM]. Table V-58 summarises the thermodynamic quantities derived for the formation of Mg^{2+}. The effect of adding a constant heat capacity of reaction term was within the inherent uncertainty of the solubility measurements that were small compared to data for most other metal hydroxides/oxides. Neither version of the regression falls within the range of the two major assessments by CODATA and NIST, and therefore do not solve the dilemma discussed in detail by Wolery [2005WOL] (updated as Appendix A of Wolery and Jove-Colon [2007WOL/JOV]) as to why the values of $\Delta_r {}^{*}G_m^{\circ}$ ((V.99), 298.15 K) selected in these assessments disagree to the extent that they do, other than perhaps to suggest that experimental uncertainties are of the same order of magnitude.

Figure V-44: Plot $\log_{10} {}^{*}K_{s,0}^{\circ}$ for $Mg(OH)_2$(cr) as function of $(T/K)^{-1}$.



The effect of chloride ion pairing was ignored in these studies as it has been in all other brucite solubility investigations carried out in chloride media.

The second system studied by Altmaier *et al.* [2003ALT/MET] involved the equilibrium:

$$Mg_2(OH)_3Cl \cdot 4H_2O(cr) + 3H^+ \rightleftharpoons 2Mg^{2+} + Cl^- + 7H_2O(l) \qquad (V.102)$$

which, although outside the scope of this review, is mentioned briefly here for the sake of completeness. According to the authors this hydrated hydroxido-chlorido phase is

formed at room temperature only at magnesium molalities > 1.7 mol·kg^{-1} (see Figure 5 in this paper). The activity coefficients at these high ionic strengths were estimated from the Pitzer ion interaction model. The mean $\log_{10} {}^{*}K^{\circ}_{s,0}$ ((V.102), 295 K) value from six independent experiments was given as (26.0 ± 0.2) at (295 ± 2) K.

V.6.1.2.3 Hydrolysis of aqueous magnesium(II)

V.6.1.2.3.1 Monohydroxidomagnesium(II) complex, $MgOH^+$

The first hydrolysis equilibrium of magnesium ion is:

$$Mg^{2+} + H_2O(l) \rightleftharpoons MgOH^+ + H^+ \qquad \text{(V.103)}$$

Experimental difficulties incurred in studies of this kind are well documented, but in the case of magnesium, a particular problem emerged in that after only a minor amount of hydrolysis had taken place, precipitation of $Mg(OH)_2(s)$ occurred. The onset of precipitation was fortunately very rapid and could be observed by an abrupt change in pH, see for example [1996BRO/DRU]. In a subsequent potentiometric study in the same laboratory, Palmer and Wesolowski [1997PAL/WES], using carefully selected magnesium(II) and base titrant concentrations, showed that values of the first hydrolysis constant in NaCl solutions could be determined. The resulting value extrapolated to 298.15 K is in good agreement with those obtained by McGee and Hostetler [1975MCG/HOS] from their potentiometric investigation and from an earlier similar study [1948STO/DAV], which was limited to only two titrations. All the available hydrolysis data are summarised Table V-56.

Table V-56: Summary of experimental data and hydrolysis constants according to Reaction (V.103), where # is the number of data points collected.

T /K	#	Ionic medium	I	$-\log_{10}K_1^\circ$ reported	$-\log_{10}{}^*K_1^\circ$ reported	$-\log_{10}{}^*K_1^\circ$ Re-calculated	Method	Reference
291.15	15	$MgCl_2$ + NaOH	$I_{c\,(max)}$ 1.5		(11.61 ± 0.46)	(11.68 ± 0.33)	pot	[1925GJA]
298.15	18	$MgCl_2$ + $Ba(OH)_2$	I_m 0.092		11.42	(11.44 ± 0.13)	pot	[1948STO/DAV]
298.15	6	$MgCl_2$ + KCl	$I_{m\,(max)}$ 0.400	(2.80 ± 0.14)		(11.13 ± 0.16)	sol	[1963HOS]
298.15	4	$MgCl_2$ + KCl + $Ba(OH)_2$		(2.60 ± 0.25)		(11.46 ± 0.26)	pot	
298.15	2	$MgCl_2$ + NaCl	I_m 4.24		(12.31 ± 0.06)	(12.41 ± 0.15)	pot	[1963LEW]
	3	$MgCl_2$ + NaCl	I_m 4.56		(12.17 ± 0.06)	(12.44 ± 0.16)		
	2	$MgCl_2$	I_m 4.68		(12.19 ± 0.12)	(12.45 ± 0.16)		
298.15	12	$NaClO_4$	I_m 3.48		(9.56 ± 0.07)	(9.64 ± 0.20)	sol	[1969HOR]
298.15	4	$MgCl_2$ + $Ba(OH)_2$	$I_{m\,(max)}$ 0.076	(2.55 ± 0.05)		(11.61 ± 0.19)	pot	[1973LIU/NAN]
298.15	5	$MgCl_2$ + $Ba(OH)_2$	$I_{m\,(max)}$ 0.0015	(2.55 ± 0.05)			con	
283.15	2	$MgCl_2$ + $Ba(OH)_2$	$I_{c\,(max)}$ 0.12	(2.18 ± 0.08) [(b)]		(12.35 ± 0.10)	pot	[1975MCG/HOS]
298.15	5			(2.20 ± 0.05) [(b)]		(11.79 ± 0.06)		
313.15	3			(2.29 ± 0.03) [(b)]		(11.21 ± 0.06)		
328.15	3			(2.37 ± 0.03) [(b)]		(10.74 ± 0.06)		
343.15	2			(2.45 ± 0.04) [(b)]		(10.28 ± 0.10)		
363.15	5			(2.54 ± 0.09) [(b)]		(9.82 ± 0.12)		
298.15						(11.44 ± 0.10)	est	[1976BAE/MES][(a)]

(Continued on next page)

Table V-56 (continued)

T /K	#	Ionic medium	I	$-\log_{10}K_1^o$ reported	$-\log_{10}{}^*K_1^o$ reported	$-\log_{10}{}^*K_1^o$ Re-calculated	Method	Reference
274.15	6	NaCl	$I_{m\ (max)}$ 5.12		11.92 (c)	(12.58 ± 0.11)	pot	[1997PAL/WES]
297.65	9				11.73 (c)	(11.61 ± 0.09)		
323.05	6				10.73 (c)	(10.63 ± 0.10)		
348.05	6				9.92 (c)	(9.89 ± 0.12)		
372.95	8				9.22 (c)	(9.06 ± 0.05)		
422.95	5				8.06 (c)	(8.10 ± 0.06)		
472.95	5				7.14 (c)	(7.20 ± 0.06)		
523.05	7				6.40 (c)	(6.60 ± 0.04)		

(a) Derived from data in [1948STO/DAV] and [1963HOS]; this value was also accepted without comment in [1990NOR/PLU].
(b) Uncertainties were given as (2σ).
(c) Smoothed values at p_{sat} taken from Table V in [1997PAL/WES].

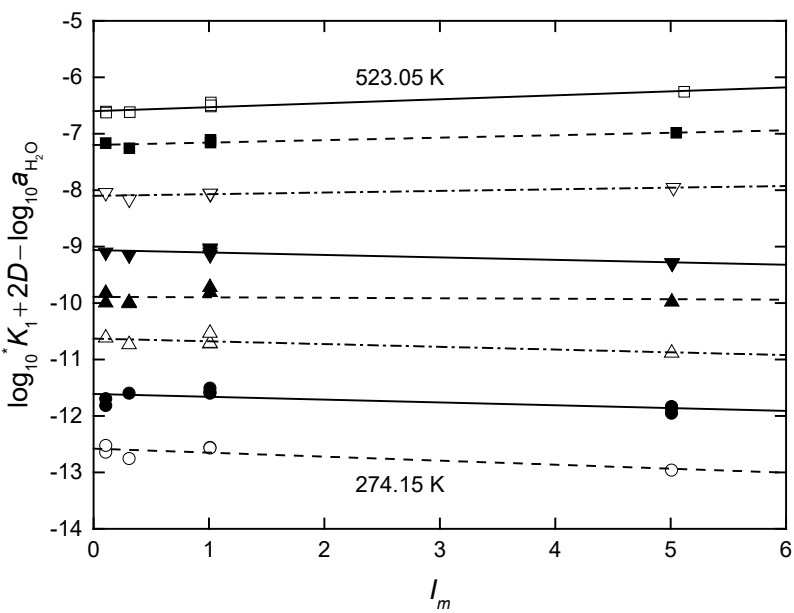
The SIT Equation (V.104) applied to the results presented in the potentiometric study [1997PAL/WES] of Mg^{2+} hydrolysis from 274.15 to 523.15 K at ionic strengths of 0.11, 0.31, 1.01, and 5.0 mol·kg^{-1} (NaCl) yielded the parameters listed in Table V-57. The values of $\log_{10}{}^{*}K_1^{\circ}$ thus obtained are in good agreement with those reported originally by the authors who used the Pitzer ion interaction model; the larger uncertainty probably resulting from the fewer adjustable parameters being allowed by the SIT. The fits are illustrated in Figure V-45, noting that these linear plots extend to an ionic strength of 5 mol·kg^{-1} which is higher than normally expected for equilibria involving 2:1 electrolytes.

$$\log_{10}{}^{*}K_{m,1} + 2D - \log_{10} a_{H_2O} = \log_{10}{}^{*}K_1^{\circ} - \Delta\varepsilon I_m \qquad \text{(V.104)}$$

Table V-57: Summary of the application of Equation (V.104) to the hydrolysis data in [1997PAL/WES].

T/K	p_{sat} /MPa	$-\log_{10}{}^{*}K_1^{\circ}$	$\Delta\varepsilon$ /kg·mol^{-1}	T/K	p_{sat} /MPa	$-\log_{10}{}^{*}K_1^{\circ}$	$\Delta\varepsilon$ /kg·mol^{-1}
274.15	0.1	(12.58 ± 0.11)	(0.071 ± 0.052)	372.95	0.1	(9.06 ± 0.05)	(0.043 ± 0.021)
297.65	0.1	(11.61 ± 0.09)	(0.050 ± 0.025)	422.95	0.474	(8.10 ± 0.06)	– (0.029 ± 0.026)
323.05	0.1	(10.63 ± 0.10)	(0.048 ± 0.045)	472.95	1.548	(7.20 ± 0.06)	– (0.044 ± 0.026)
348.05	0.1	(9.89 ± 0.12)	(0.009 ± 0.058)	523.05	3.970	(6.60 ± 0.04)	– (0.070 ± 0.022)

Figure V-45: The SIT treatment of the data in [1997PAL/WES], the results of which are summarised in Table V-57.



Equation (V.105) resulted from a weighted linear regression of the re-culated $\log_{10}{}^{*}K_1^{\circ}$(V.103) values listed in Table V-56 from [1925GJA], [1948STO/DAV], [1963HOS], [1975MCG/HOS] and [1997PAL/WES] *versus* $(T/\mathrm{K})^{-1}$. These results are shown in Figure V-46.

$$\log_{10}{}^{*}K_1^{\circ}(\text{V.103}) = (0.129 \pm 0.296) - (3494.2 \pm 88.1)(T/\mathrm{K})^{-1} \qquad \text{(V.105)}$$

The resulting selected value at 298.15 K is,

$$\log_{10}{}^{*}K_1^{\circ}((\text{V.103}), 298.15\ \mathrm{K}) = \mathbf{-\ (11.591 \pm 0.117)}.$$

The corresponding selected thermodynamic parameters are:

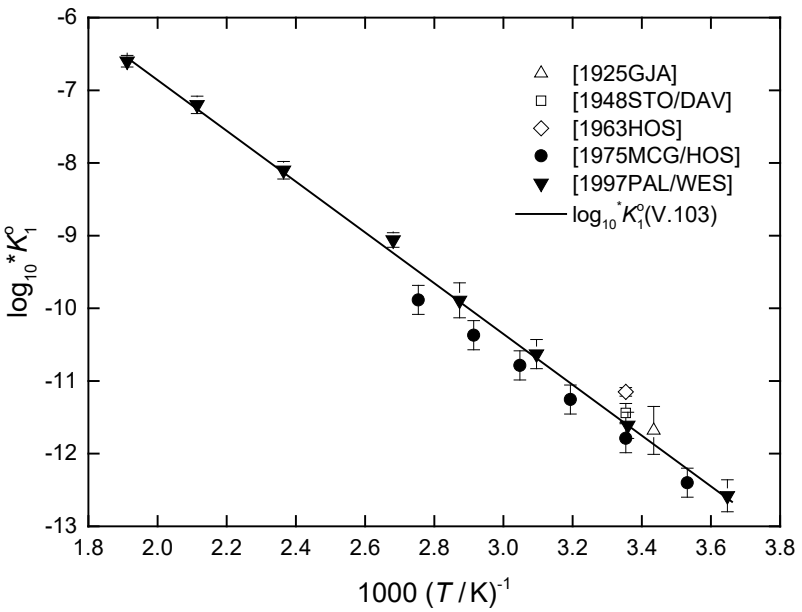
$$\Delta_r G_{m,1}^{\circ}((\text{V.103}), 298.15\ \mathrm{K}) = \mathbf{(66.160 \pm 0.666)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H_{m,1}^{\circ}((\text{V.103}), 298.15\ \mathrm{K}) = \mathbf{(66.896 \pm 3.373)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r S_{m,1}^{\circ}((\text{V.103}), 298.15\ \mathrm{K}) = \mathbf{(2.469 \pm 11.532)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

Accepting that $\Delta_r C_{p,m}^{\circ}((\text{V.103}), 298.15\ \mathrm{K}) = 0$, it follows that $\Delta_r C_{p,m}^{\circ}(\mathrm{MgOH^+}) = 0.0 + 75.3 - 73.8 = 1.5\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$ (uncertainties unknown).

Figure V-46: Plot of the temperature dependence of $\log_{10}{}^{*}K_1^{\circ}$(V.103) derived from application of the SIT.



The ion interaction parameters listed in Table V-57 decreased monotonically with temperature and the interpolated $\Delta\varepsilon$ value at 298.15 K resulted in:

$$\varepsilon(MgOH^+, Cl^-) = \Delta\varepsilon - \varepsilon(H^+,Cl^-) + \varepsilon(Mg^{2+},Cl^-)$$

$$= (0.050 \pm 0.025) - (0.12 \pm 0.01) + (0.19 \pm 0.02) = \mathbf{(0.12 \pm 0.03)}\ kg{\cdot}mol^{-1},$$

which is our selected value.

The corresponding selected molal thermodynamic properties of $MgOH^+$ are summarised in Table V-58.

Table V-58: Molal standard thermodynamic quantities of formation for the system Mg-OH-H_2O at 298.15 K.

Quantity	$Mg(OH)_2$(cr)	H_2O(l)	OH^-	Mg^{2+}	$MgOH^+$	Reference
$\Delta_f G^o_m$ /kJ·mol^{-1}		– (237.140 ± 0.041)	– (157.220 ± 0.072)	– (455.375 ± 1.335)		NEA currently selected
					– 626.03	[2017MIR/WAG]
	– (833.418 ± 0.324)			**– (455.703 ± 0.786)**	**– (626.683 ± 1.031)**	This review
$\Delta_f H^o_m$ /kJ·mol^{-1}		– (285.830 ± 0.040)	– (230.015 ± 0.040)	– (467.000 ± 0.600)		NEA currently selected
	– (924.450 ± 0.300)			**– (467.0 ± 0.6)**	**– (685.934 ± 3.426)**	This review
S^o_m /J·K^{-1}·mol^{-1}		(69.950 ± 0.030)	– (10.90 ± 0.20)	– (137.000 ± 4.000)		NEA currently selected
	(63.180 ± 0.400)			**– (135.9 ± 1.7)**	**– (63.481 ± 11.657)**	This review
$C^o_{p,m}$ /J·K^{-1}·mol^{-1}		(75.351 ± 0.080)				NEA currently selected
	(76.900 ± 0.300)			**(74 ± 11)**		This review

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V.6.1.2.3.2 Higher-order hydroxidomagnesium(II) complexes

Potentiometric titrations of solutions containing high magnesium concentrations (1.000, 1.250 and 1.500 mol·dm^{-3}) with and without added NaCl were performed by Lewis [1963LEW] at 298.15 K. Earlier, Stock and Davies [1948STO/DAV] had used a barium hydroxide titrant to mitigate formation of precipitates. Nevertheless, back titrations produced reproducible hysteresis, which additional turbidity experiments indicated could be due to temporary formation of colloids. Gjalbaek [1925GJA] had reported a similar hysteresis phenomenon during titrations of magnesium chloride solutions. Lewis offered two speciation schemes based on an analysis of his potentiometric curves using the LETAGROP routine [1961DYR/ING], [1962ING/SIL], and a less rigorous graphical method. The first involved only $MgOH^+$ species while the second involved Reactions (V.106) and (V.107). These values are listed in Table V-59. Re-analysis by Baes and Mesmer [1976BAE/MES] led them to dispute this speciation scheme, replacing Mg_2OH^{3+} by $MgOH^+$, because the hydrolysis constant thus obtained for this mononuclear ion was compatible with the constant derived in dilute magnesium chloride solutions. They also included results from [1948STO/DAV] and [1963HOS] to derive the ionic strength dependence of $\log_{10}{}^{*}K_{m,1}$ based on the ion interaction model proposed by Pitzer and Brewer [1961LEW/RAN] and data from Lewis only, to obtain the ion interaction term for the 4:4 species with chloride ion. Baes and Mesmer selected a value of $\log_{10}{}^{*}\beta_{4,4} = -39.71$ from these very limited high ionic strength data with the extrapolated value being very sensitive to the approximated ratio of activity coefficients for chloride and hydroxide ions in these media. No hydrolysis constant values for formation of these multinuclear magnesium(II) species are recommended in this review.

$$2Mg^{2+} + H_2O(l) \rightleftharpoons Mg_2OH^{3+} + H^+ \qquad \text{(V.106)}$$

and

$$4Mg^{2+} + 4H_2O(l) \rightleftharpoons Mg_4(OH)_4^{4+} + 4H^+ \qquad \text{(V.107)}$$

Table V-59: Summary of experimental data at 298.15 K converted to molal units and the reported hydrolysis constants according to Reactions (V.106) and (V.107) at 298.15 K.

No. of Experiments	Ionic medium	I_m	$-\log_{10}{}^{*}K_{m,1}$	$-\log_{10}{}^{*}\beta_{m,1,2}$	$-\log_{10}{}^{*}\beta_{m,4,4}$	Method	Reference
2	NaCl + $MgCl_2$	4.24	12.24	(12.33 ± 0.06)	(39.80 ± 0.02) 39.77	pot est	[1963LEW] [1976BAE/MES]
3	NaCl + $MgCl_2$	4.56	12.25	(12.30 ± 0.06)	(39.77 ± 0.02) 39.77	pot est	[1963LEW] [1976BAE/MES]
2	$MgCl_2$	4.68	(12.25 ± 0.12) 12.25	(12.38 ± 0.12)	(39.74 ± 0.03) 39.72	pot est	[1963LEW] [1976BAE/MES]

V.6.1.2.4 Aqueous magnesium(II) chloride

V.6.1.2.4.1 Ion-association of aqueous magnesium chloride

$$Mg^{2+} + Cl^{-} \rightleftharpoons MgCl^{+} \quad \text{(V.108)}$$

Wahab *et al.* [2005WAH/MAH] summarised the results of studies which involved a variety of experimental approaches to identify ion association in $Mg(NO_3)_2$ and $Mg(CH_3COO)_2$ solutions. Their discussion emphasised the disparity between the findings of various researchers not in only observing widely different degrees of weak ion association, but also in claiming different modes of association (*e.g.*, contact, solvent-separated ion pairs) and mono- *versus* bi-dentate modes of interaction. This applies to interactions with nitrate and most other monovalent inorganic anions, carboxylic anions being the exception where ion pairing is readily identifiable even at ambient temperatures. For example, Zhang *et al.* [2004ZHA/CHO] employed Raman spectroscopy to observe the onset of contact ion-pair formation in *ca*. 9.25 mol·kg^{-1} $Mg(NO_3)_2$ solutions contained in water droplets because there was insufficient water to fully hydrate the Mg^{2+} cation. It is apparent, at least in the cited examples, that Raman spectroscopy identifies apparent ion pairing more consistently, even when other techniques do not [1990ALV/SOU], [1998PYE/RUD] [2003RUD/IRM]. Nevertheless, no significant ion association was detected in even 5.9 mol·kg^{-1} $MgCl_2$ solutions to 473 K in the detailed Raman work of Rudolph and co-workers [1998PYE/RUD] [2003RUD/IRM]. Note that the arguments presented here apply even more forcibly to ion association in $CaCl_2$(sln) and as discussed in Section V.6.2.2.2.1.

It must be acknowledged that at high temperatures, hydrolysis and ion association of highly concentrated solutions of $MgCl_2$(sln) eventually become significant. Wang *et al*. [1998WAN/PIT] provided an excellent critique of the papers dealing with the formation of $MgCl^+$ ion pairs. In brief, mention was made of the ion-pair association constants derived from enthalpies of dilution and mixing of $MgCl_2$ solutions by Gillespie *et al*. [1992GIL/OSC] from 523 to 598 K that were considered by Rard and Clegg [1997RAR/CLE] to be compromised by covariance in their fitting procedures. Density measurements of $MgCl_2$ solutions over a wider range of temperature and higher pressures in [1997OBS/MAJ] seemed to confirm the values reported by Gillespie *et al*. However, Wang *et al*. [1998WAN/PIT] pointed out that the interpretation of the data in [1992GIL/OSC] was based on the assumption that the activity coefficient of $MgCl^+$ could be set equal to that of NaCl(aq), while the mean activity coefficient of $MgCl_2$(sln) was taken from the isopiestic data of Holmes *et al*. [1978HOL/BAE] which Wang *et al*. claimed were first criticised in [1986EMO/VOI], and then superseded by more consistent results from the same group [1996HOL/MES]. Solubility measurements [1990SAC/SEY] of talc-quartz mixtures from 573 to 673 K (50 MPa) were interpreted by taking the formation constants of $MgCl^+$ ion pairs into account in this chloride-rich, mixed-electrolyte medium, but again the values derived for these weak interactions are known to be highly sensitive to the choice of activity coefficient models. Electrical conductivity

data for $MgCl_2$ and $CaCl_2$ solutions obtained from 298 to 873 K [1982FRA/MAR] at pressures up to 400 MPa yielded 2:1 formation constants which are considerably lower than those mentioned above with the difference becoming larger with increasing temperature, but did establish the significance of $MgCl^+$ at high temperatures and pressures [1981FRA/POP]. The global model developed by Wang *et al*. [1998WAN/PIT] could not assign specific ion-pair formation constants to this system largely because of interference from the ion association of HCl(aq), although the comment was made that "The dilution enthalpy measurements of Simonson *et al*.[12] (where reference 12 in [1998WAN/PIT] refers to unpublished data) in 0.02 mol·kg^{-1} HCl solution are essentially consistent with the assumption of complete dissociation of $MgCl_2$, extending to low molalities." Thus, even under conditions where ion pairing must occur, there is no consensus amongst these high-temperature studies as to the magnitude of their formation constants. Concordant, enthalpy of dilution measurements of 0.1 to 2.0 mol·kg^{-1} $MgCl_2$ solutions from 298.15 to 353.15 K were also interpreted without consideration of ion pairing [1975SNI/MAN].

Potentiometric (Ag|AgCl *versus* glass electrodes) and vapour-pressure measurements by Havel and Högfeldt [1973HAV/HOG] of Mg^{2+} in 3 mol·dm^{-3} ionic strength ($NaClO_4$) solutions containing NaCl, NaBr and NaSCN at 298.15 K yielded the following values of $\log_{10}K_{c,1}$ (Reaction (V.108)): $-(0.98 \pm 0.19)$, $-(1.45 \pm 0.26)$ and $-(0.91 \pm 0.19)$, respectively. However, the Mg^{2+} concentration reached almost 1.5 mol·dm^{-3} in some cases while the concentration of the anion of interest increased to *ca*. 1 mol·dm^{-3}. Havel and Högfeldt made the further gross assumption that the activity coefficient quotient associated with this equilibrium was constant and virtually the same in the three systems. Similarly, other potentiometric studies utilising chloride-sensitive (ise-Cl) and Ag|AgCl electrodes *versus* calomel electrodes [1973HAV/HOG], [1975SUC/CAD], [1978ELG/WED], [1978JOH/PYT], [1982MAJ/STU], [1998ROB/STE] employed high magnesium and/or chloride concentrations relative to the prevailing ionic strength such that the variation of activity coefficients could not be estimated accurately: K_1^o values were reported from 0.1 to 3.1 (see Table V-60).

Conductance measurements of dilute $MgCl_2$(sln) were made by Fisher and Fox [1977FIS/FOX] at 298.15 K as a function of pressure to *ca*. 200 MPa. They provided two sets of association constants calculated at two estimated distances-of-closest-approach (*r*). Pethybridge and Taba [1980PET/TAB] made special reference to fitting conductance data for asymmetrical electrolytes such as $MgCl_2$, suggesting that pre-setting the distance parameter to the Bjerrum distance would allow the values of Λ_0 and K_1^o to be determined from a two-parameter regression. Their values of K_1^o for the formation of the $MgCl^+$ ion pair varied from 1.55 to 4.44 for r = 500 to 600 pm depending on the values of the molar ionic conductance assigned to the ion pair.

Table V-60: Experimental conditions and reported $MgCl^+$ ion-pair formation constants for the Reaction (V.108)

T/K	Ionic medium	$c_{Cl^-(max)}$ /mol·dm^{-3}	$c_{Mg^{2+}(tot)}$ /mol·dm^{-3}	K_1^o	Method	Reference
272.55	$KClO_4$ or $KClO_3 + MgCl_2$	< 0.516	< 0.258	8.1	cry	[1959KEN2]
298.15	3 mol·dm^{-3} $NaClO_4$	1	1.5 (max)	$K_{c,1} = (0.10 \pm 0.04)$	pot vp [a]	[1973HAV/HOG]
298.15	1 mol·kg^{-1} $NaNO_3$	m_{Cl^-} 0.001	$m_{Mg\ total}$ 0.1/0.2	$K_{m,1} = (0.8 \pm 0.4)$	pot	[1975SUC/CAD]
298.15 [b]	$MgCl_2$	0.02 (max)	0.01 (max)	0.23-0.43	con	[1977FIS/FOX]
298.15	$I_c = 0.7$ $MgCl_2 + NaCl$ NaF(minor)			$K_{c,1} = (0.34 \pm 0.02)$	pot	[1978ELG/WED]
298.15	$I_c = 0.37$-0.75 $MgCl_2 + HCl$	0.820	≤ 0.163	(1.8 ± 0.3)	pot	[1978JOH/PYT]
298.15		0.0014- 00052	0.0007-00026	1.55-4.44	con	[1980PET/TAB][c]
298.15	$MgCl_2$	$m_{Cl^-} \leq 0.01$	$m_{Mg^{2+}} \leq 0.005$	$10^{-4.86}$	con	[1982FRA/MAR]
298.15	$I_m = 1.0$ $Mg(NO_3)_2 + MgNO_3$	m_{Cl^-} 0.001	$m_{Mg^{2+}}$ 0.05-0.3	$K_{m,1} = (0.77 \pm 0.23)$ $\Delta_r H_1 = 2.3$ kJ·mol^{-1}	pot	[1982MAJ/STU]
298.15				(3.7 ± 0.7)	est	[1987ROB/RIG]
298.15				0.22	est [d]	[1996TIK/FAW]
298.15				0.73 $\Delta_r H_1^o = 1.54$ kJ·mol^{-1}	est	[1997SVE/SHO]
298.15	$I_c = 0.17$-0.94 $Mg(NO_3)_2 + NaCl$, $NaNO_3$	0.054-0.16	0.0013-0.072	(3.1 ± 0.2)	pot	[1998ROB/STE]

(a) vp, vapour pressure measurements.
(b) Other measurements were made at pressures up to 202.7 MPa.
(c) Data taken from [1934SHE/BRO].
(d) MSA (Mean Spherical Approximation) calculations.

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All of the experimental evidence supports the conclusion that no distinct value for the $MgCl^+$ ion-pair formation constant can be assigned and that this interaction can be neglected when modelling the thermodynamic properties of $MgCl_2$ solutions to moderate concentrations (*e.g.*, to at least 3 $mol \cdot kg^{-1}$) at near ambient conditions.

V.6.1.2.5 Aqueous magnesium(II) sulfate

Reardon and Beckie [1987REA/BEC] used the thermodynamic data for the Mg(II) and Fe(II) sulfate-water system at 298.15 K in [1974OYK/BAL] to establish Pitzer parameters for the aqueous $FeSO_4$ system. Libus *et al*. [1980LIB/SAD] measured the osmotic coefficients of (Mg, Mn, Co, Ni, Cu, Zn)SO_4, although Lemire *et al*. [2013LEM/BER] noted that *"*The unexpected difference between the osmotic coefficient data for $FeSO_4$ and all the other $MeSO_4$ (except $MgSO_4$, which lies in between, and might be suitable for an SIT treatment) may suggest that either the iron data and/or all the other transition metal data are of questionable quality.". Lemire *et al*. did not resolve these ambiguities.

Potential difference measurements [1997MAL/ZAM] yielded activity coefficients for $MeSO_4$ (Me = Mg, Ca, Sr and Zn) solutions at 298.15 K over a very wide concentration range based on a Pitzer ion interaction model which did not take ion association into account.

Magnesium sulfate heptahydrate (Epsom salt) is highly soluble as are the other less stable hydrates with anhydrous magnesium sulfate being hygroscopic. Duplicate independent measurements by Rard and Miller [1981RAR/MIL] established that the solubility of $MgSO_4 \cdot 7H_2O$ at 298.15 K is (3.0211 ± 0.0022) $mol \cdot kg^{-1}$ after 6 to 15 day equilibrations. This value is in good agreement with the values of 3.026 and 3.015 $mol \cdot kg^{-1}$ reported previously [1965LIN] and [1978FIL/ANT].

V.6.1.2.5.1 Ion association of aqueous magnesium(II) sulfate

Pitzer and Mayorga [1974PIT/MAY] made the following statement which is fundamental to the approach taken in this review, *viz*. "While we have noted that the thermodynamic properties of 2-2 electrolytes do not require the assumption of a discrete chemical species of ion pairs, we recognise that spectroscopic and ultrasonic measurements do indicate a discrete species of inner-shell ion pairs in some cases. Thus, for some purposes it is clear that ion association must be recognised. In many cases, however, the proportion of inner-shell ion pairs is never large, and it may be a satisfactory approximation for thermodynamic purposes to ignore the distinction between inner-shell and solvent-separated ion pairs."

Rudolph *et al*. [2003RUD/IRM] presented Raman spectroscopic evidence for the formation of an $MgOSO_3$(aq) contact ion pair and detected another ν_1-SO_4^{2-} mode at 982.2 cm^{-1} which they assigned to a solvent-shared ion pair. Moreover, at high Mg^{2+}: SO_4^{2-} concentration ratios, another ν_1-SO_4^{2-} mode was observed at 1005 cm^{-1},

which was tentatively assigned to an $MgSO_4$ ion triplet [1985ARC/WOO] at concentrations > 1 mol·dm^{-3} at ambient conditions. Prior to these spectroscopic studies, there existed some skepticism as to the existence of higher-order species, claiming that they were fitting artifacts [1981RAR/MIL], [1997MAL/ZAM]. Nevertheless, dielectric relaxation spectroscopy [2004BUC/CHE] provided definitive evidence for the coexistence of three 2:2 ion-pair types (*i.e.*, contact (CIP), solvent-shared (SIP) and double-solvent-separated (2SIP) ion pairs). Moreover, there has been an extraordinary number of quantitative measurements of the formation of the 2:2 ion pair according to Reaction (V.109) which may or may not (see [2004BUC/CHE]) include contributions from all three ion-pair types. This singular latter approach was also taken in the analysis of the ion pair association constants in $NiSO_4$(sln) [2005GAM/BUG]. The results of the $MgSO_4$ ion pairing measurements are summarised in Table V-57. The values cited from the literature generally have uncertainties of (1σ) whereas those re-calculated using the SIT have (2σ) uncertainties. From the results listed it is clear that a broad range of experimental techniques have been employed over the last 100 years: conductance, [1932MON/DAV], [1938DAV2], [1973KAT], [1977PET/TAB], [1979FIS/FOX], [1980PET/TAB], [1986SIN/DAH], [2000BIA/DUJ], [2000TSI/MOL], [2002TOM/BES], [2006HUA/PAP], [2007BES/HAU]; freezing-point depression, [1955BRO/PRU], [1971ISO]; potentiometry, [1958NAI/NAN2], [1968KES/PYT], [1970LAR] [1973POW], [1978ELG/WED], [1978EMA/FAR], [1980EMA/LIN], [1983PRA/GHO], [2001KRA/HEF]; calorimetry, [1969IZA/EAT], [1970LAR], [1975LEU/MIL], [1975SNI/MAN], [1977WOO/HEP], [1980JOH/PER], [1981WAC/RIE], [1983MAY/WOO]; vapour-pressure osmometry, [1975YOK/YAM]; spectrophotometry, [1975YOK/YAM2], [1978EMA/FAR], [1980EMA/LIN]; a critical coagulation concentration (ccc) method, [1978KAL/KRZ]; spectroscopic, [2003RUD/IRM]; and dielectric relaxation, [2004BUC/CHE], [2006AKI/ROH]. There are actually fewer modelling studies than for $CaSO_4$(sln), although ironically there are significantly fewer experimental results for the latter. All of the diverse experimental data for both systems involved at least binary mixed electrolytes with the exception of the conductance experiments where only dilute $MgSO_4$ solutions (*i.e.*, 0.0001 to 0.01 mol·dm^{-3}) were investigated. Conductance measurements provided the most precise and accurate formation constants for Reaction (V.109) in these dilute solutions that could be adequately treated using an SIT compatible Lee-Wheaton equation. Re-treatment of the remaining data at moderate ionic strengths (*e.g.*, 0.1 to 0.2 mol·kg^{-1}) relied on the SIT Debye-Hückel equation solely to estimate the mean ionic activity coefficients because specific ion interaction contributions were negligible compared to the experimental uncertainties in K_1 (V.109) (*e.g.*, Appendix A for [1955BRO/PRU]).

$$Mg^{2+} + SO_4^{2-} \rightleftharpoons MgSO_4(aq) \qquad \text{(V.109)}$$

Table V-61: Experimental and re-calculated formation constants for $MgSO_4$(aq) and other thermodynamic quantities (units: $\Delta_r H_m^o$, kJ·mol^{-1}; $\Delta_r S_m^o$, J·K^{-1}·mol^{-1}; $\Delta_r C_{p,m}^o$, J·K^{-1}·mol^{-1}; $\Delta_r V_m^o$, cm^3·mol^{-1}).

Method	T /K	Measured concentration range	Reported value	Re-calculated SIT value	Derived value	Reference
con	291.15	c_{MgSO_4} : 0.0001-0.0010	$\log_{10} K_1^o = 2.201$			[1932MON/DAV]
con	291.15	c_{MgSO_4} : 0.0005-0.0050	$\log_{10} K_1^o = (2.107 \pm 0.017)$	$\log_{10} K_1^o = (2.15 \pm 0.13)$	$\log_{10} K_1^o = 2.30$	[1912NOY/FAL] [1938DAV2]
fpt	273.15	m_{MgSO_4} : 0.0061-0.0321	$\log_{10} K_1^o = 1.98$-2.39	$\log_{10} K_1^o = (2.18 \pm 0.05)$	$\log_{10} K_1^o = 2.23$	[1955BRO/PRU]
pot	273.15	I_m (HCl): 0.018-0.076	$\log_{10} K_1^o = (1.96 \pm 0.04)$	$\log_{10} K_1^o = (2.15 \pm 0.20)$	$\log_{10} K_1^o = 2.23$	[1958NAI/NAN2]
	293.15	I_m (HCl): 0.017-0.022	$\log_{10} K_1^o = (2.20 \pm 0.03)$	$\log_{10} K_1^o = (2.40 \pm 0.10)$	$\log_{10} K_1^o = 2.30$	
	298.15	I_m (HCl): 0.016-0.102	$\log_{10} K_1^o = (2.25 \pm 0.02)$	$\log_{10} K_1^o = (2.42 \pm 0.10)$	$\log_{10} K_1^o = 2.32$	
	303.15	I_m (HCl): 0.047-0.095	$\log_{10} K_1^o = (2.35 \pm 0.01)$	$\log_{10} K_1^o = (2.42 \pm 0.15)$	$\log_{10} K_1^o = 2.34$	
	308.15	I_m (HCl): 0.016-0.093	$\log_{10} K_1^o = (2.39 \pm 0.02)$	$\log_{10} K_1^o = (2.53 \pm 0.15)$	$\log_{10} K_1^o = 2.35$	
	313.15	I_m (HCl): 0.044-0.091	$\log_{10} K_1^o = (2.45 \pm 0.02)$	$\log_{10} K_1^o = (2.52 \pm 0.16)$	$\log_{10} K_1^o = 2.37$	
	318.15	I_m (HCl): 0.015-0.087	$\log_{10} K_1^o = (2.49 \pm 0.05)$ $\Delta_r H_1^o = (20.3 \pm 1.7)$	$\log_{10} K_1^o = (2.62 \pm 0.13)$	$\log_{10} K_1^o = 2.39$ $\Delta_r H_1^o = 5.76$	
pot	298.15	I_m (NaCl+$MgCl_2$): 0.670	$\log_{10} K_{c,1}^o = (1.01 \pm 0.02)$ [(a)]			[1968KES/PYT]
cal	298.15	→ 0	$\log_{10} K_1^o = (2.23 \pm 0.07)$ [(b)] $\Delta_r H_1^o = (2.1 \pm 0.1)$			[1969IZA/EAT]
cal	298.15	→ 0	$\log_{10} K_1^o$ (2.31 ± 0.08) [(b)] $\Delta_r H_1^o = (5.3 \pm 0.8)$ $\Delta_r S_m^o = (62 \pm 2)$	$\Delta_r H_1^o = (5.3 \pm 0.8)$	$\Delta_r H_1^o = 5.76$	[1970LAR] [1930LAN/MON], [1931LAN/STR], [1931LAN/STR2] [1933LAN/MON], [1956LAN/MIE], [1959LAN]
fpt	273.15	m_{MgSO_4} : 0.006-0.021	$\log_{10} K_1^o = 1.89$ -2.35	$\log_{10} K_1^o = (2.20 \pm 0.05)$	$\log_{10} K_1^o = 2.23$	[1971ISO]
est	273.15	0	$\log_{10} K_1^o = 2.33$			[1972PIT]
	298.15	0	$\log_{10} K_1^o = 2.38$			

(Continued on next page)

Table V-61 (continued)

Method	T /K	Measured concentration range	Reported value	Re-calculated SIT value	Derived value	Reference
est	298.15	0	$\Delta_r H_1^o$ = (5.7 ± 0.2)	$\Delta_r H_1^o$ = (5.7 ± 0.4)	$\Delta_r H_1^o$ = 5.76	[1973POW] [1969IZA/EAT]
con	273.15	c_{MgSO_4} : 0.0002-0.0030	$\log_{10} K_1^o$ = (2.013 ± 0.004)	$\log_{10} K_1^o$ = (2.24 ± 0.04)	$\log_{10} K_1^o$ = 2.23	[1973KAT]
	278.15		$\log_{10} K_1^o$ = (2.033 ± 0.004)	$\log_{10} K_1^o$ = (2.25 ± 0.04)	$\log_{10} K_1^o$ = 2.25	
	283.15		$\log_{10} K_1^o$ = (2.064 ± 0.007)	$\log_{10} K_1^o$ = (2.26 ± 0.04)	$\log_{10} K_1^o$ = 2.27	
	288.15		$\log_{10} K_1^o$ = (2.083 ± 0.007)	$\log_{10} K_1^o$ =(2.28 ± 0.038)	$\log_{10} K_1^o$ = 2.29	
	293.15		$\log_{10} K_1^o$ = (2.111 ± 0.010)	$\log_{10} K_1^o$ = (2.30 ± 0.04)	$\log_{10} K_1^o$ = 2.31	
	298.15		$\log_{10} K_1^o$ = (2.130 ± 0.006)	$\log_{10} K_1^o$ = (2.31 ± 0.04)	$\log_{10} K_1^o$ = 2.33	
	303.15		$\log_{10} K_1^o$ = (2.167 ± 0.009)	$\log_{10} K_1^o$ = (2.33 ± 0.04)	$\log_{10} K_1^o$ = 2.34	
	308.15		$\log_{10} K_1^o$ = (2.179 ± 0.009)	$\log_{10} K_1^o$ = (2.35 ± 0.04)	$\log_{10} K_1^o$ = 2.36	
	313.15		$\log_{10} K_1^o$ = (2.223 ± 0.010)	$\log_{10} K_1^o$ = (2.39 ± 0.03)	$\log_{10} K_1^o$ = 2.37	
	318.15		$\log_{10} K_1^o$ = (2.243 ± 0.007)	$\log_{10} K_1^o$ = (2.41 ± 0.04)	$\log_{10} K_1^o$ = 2.39	
cal	303.15	m_{MgSO_4} : 0.1-1.0	$\log_{10} K_1^o$ = 2.250 [(b)] $\Delta_r H_1^o$ = 4.81-5.69		$\Delta_r H_1^o$ = 5.76	[1975LEU/MIL]
vp	298.15	m_{MgSO_4} : 0.004-0.149	$\log_{10} K_1^o$ = 1.86 - 2.29	$\log_{10} K_1^o$ = (2.03 ± 0.07)	$\log_{10} K_1^o$ = 2.33	[1975YOK/YAM]
sp	298.15	c_{MgSO_4} : 0.00 0.008	$\log_{10} K_1^o$ = (2.26 ± 0.02)			[1975YOK/YAM2]
con	298.15	c_{MgSO_4} : 0.000137-?	$\log_{10} K_1^o$ = 2.225	$\log_{10} K_1^o$ = (2.32 ± 0.08)	$\log_{10} K_1^o$ = 2.33	[1977PET/TAB]
	323.15	0.119	$\log_{10} K_1^o$ = 2.318	$\log_{10} K_1^o$ = (2.40 ± 0.08)	$\log_{10} K_1^o$ = 2.40	
est	298.15		$\log_{10} K_1^o$ = 2.30 [(b)] $\Delta_r H_1^o$ = 5.0 [(b)] $\Delta_r C_{p\,m}^o$ = 125			[1977WOO/HEP]
pot	298.15	I_c(NaCl + $NaClO_4$): 0.7	$\log_{10} K_{c,1}^o$ = (0.79 ± 0.01) [(a)]			[1978ELG/WED]
pot,sp	288.15	I_c($MgSO_4$ + NaCl): 0.7	$\log_{10} K_1^o$ = (2.206 ± 0.016)			[1978EMA/FAR]

(Continued on next page)

Table V-61 (continued)

Method	T/K	Measured concentration range	Reported value	Re-calculated SIT value	Derived value	Reference
	298.15	I_c($MgSO_4$ + NaCl): 0.7	$\log_{10} K_1^o = (2.232 \pm 0.025)$ $\Delta_r H_1^o = 4.81$	$\log_{10} K_1^o = (2.36 \pm 0.04)$	$\log_{10} K_1^o = 2.33$	
	308.15	I_c($MgSO_4$ + NaCl): 0.7	$\log_{10} K_1^o = (2.262 \pm 0.047)$	$\log_{10} K_1^o = (2.38 \pm 0.04)$	$\log_{10} K_1^o = 2.36$	
ccc	298.15	I_c($Mg(NO_3)_2$ + K_2SO_4) ≤0.062	$\log_{10} K_1^o = (2.20 \pm 0.11)$			[1978KAL/KRZ]
con	288.15	c_{MgSO_4} : (0.0005-0.1)	$\log_{10} K_1^o = 2.161$ $\Delta_r V_m^o = -10.8$	$\log_{10} K_1^o = (2.13 \pm 0.10)$	$\log_{10} K_1^o = 2.29$	[1979FIS/FOX]
	293.15		$\log_{10} K_1^o = 2.223$ $\Delta_r V_m^o = -9.3$	$\log_{10} K_1^o = (2.24 \pm 0.19)$	$\log_{10} K_1^o = 2.31$	
			$\log_{10} K_1^o = 2.207$ $\Delta_r V_m^o = -9.0$	$\log_{10} K_1^o = (2.22 \pm 0.21)$	$\log_{10} K_1^o = 2.33$	
pot, sp	298.15	I_c($MgSO_4$ + $MgCl_2$ + $(CH_3)_4NCl$): (0.076-1.0)	$\log_{10} K_1^o = (2.45 \pm 0.03)$			[1980EMA/LIN]
cal	298.15	I_c (LiCl + NaCl + $NaClO_4$ + $MgCl_2$ + $MgSO_4$) = 0.75	$\log_{10} K_1^o = 0.48 - 1.08$ [(b)] $\Delta_r H_1^o = 0.7\text{-}3.3$			[1980JOH/PER]
con	298.15	$c_{MgSO_4} \leq 0.0040$	$\log_{10} K_1^o = 2.241$			[1980PET/TAB]
cal	298.15	m_{MgSO_4} : 0.0005-0.01	$\log_{10} K_1^o = (2.207 \pm 0.001)$ [(b)] $\Delta_r H_1^o = (5.776 \pm 0.002)$		$\Delta_r H_1^o = 5.76$	[1981WAC/RIE]
pot	278.15	I_c($MgSO_4$ + H_2SO_4) ≤ 0.05	$\log_{10} K_1^o = (3.04 \pm 0.08)$			[1983PRA/GHO]
	288.15		$\log_{10} K_1^o = (2.96 \pm 0.01)$			
	298.15		$\log_{10} K_1^o = (2.86 \pm 0.03)$			
	308.15		$\log_{10} K_1^o = (2.80 \pm 0.02)$ $\Delta_r H_1^o = (12.9 \pm 0.3)$			
est	298.15	I_m ($MgSO_4$) ≤ 0.3	$\log_{10} K_1^o = 2.102$ $\Delta_r H_1^o = 5.972$ $\Delta_r C_{p,m}^o = 203$			[1985ARC/WOO]
con	298.15	c_{MgSO_4} : 0.001-0.05	$\log_{10} K_1^o = 2.19$			[1986SIN/DAH]

(Continued on next page)

Table V-61 (continued)

Method	T/K	Measured concentration range	Reported value	Re-calculated SIT value	Derived value	Reference
est drs	298.15	m_{MgSO_4} : 0.2-1.2	$\log_{10} K_1^o = 2.215$			[1996TIK/FAW] [1992BAR/HET]
con	298.15	c_{MgSO_4} : 0.00013-0.0053	$\log_{10} K_1^o = 2.239$ [(c)]			[2000BIA/DUJ] [1977PET/TAB]
con	293.15	c_{MgSO_4} : 0.00029-0.00294	$\log_{10} K_1^o = (2.215 \pm 0.003)$	$\log_{10} K_1^o = (2.40 \pm 0.05)$	$\log_{10} K_1^o = 2.31$	[2000TSI/MOL]
pot	298.15	I_c(CsCl)= 0.25-6.0	$\log_{10} K_1^o = (2.38 \pm 0.03)$	$\log_{10} K_1^o = (2.36 \pm 0.08)$	$\log_{10} K_1^o = 2.33$	[2001KRA/HEF]
		I_c(Me_4NCl) = 1.0	$\log_{10} K_{c,1}^o = (0.98 \pm 0.02)$			
con	278.15	c_{MgSO_4} : 0.00026-1.8009	$\log_{10} K_1^o = 2.160$ $\Delta_r H_1^o = 6.63$	$\log_{10} K_1^o = (2.26 \pm 0.04)$	$\log_{10} K_1^o = 2.25$	[2002TOM/BES]
	283.15		$\log_{10} K_1^o = 2.167$	$\log_{10} K_1^o = (2.27 \pm 0.04)$	$\log_{10} K_1^o = 2.27$	
	288.15		$\log_{10} K_1^o = 2.172$	$\log_{10} K_1^o = (2.28 \pm 0.04)$	$\log_{10} K_1^o = 2.29$	
	293.15		$\log_{10} K_1^o = 2.182$	$\log_{10} K_1^o = (2.29 \pm 0.04)$	$\log_{10} K_1^o = 2.31$	
	298.15		$\log_{10} K_1^o = 2.196$	$\log_{10} K_1^o = (2.31 \pm 0.04)$	$\log_{10} K_1^o = 2.33$	
	303.15		$\log_{10} K_1^o = 2.219$	$\log_{10} K_1^o = (2.33 \pm 0.04)$	$\log_{10} K_1^o = 2.34$	
	308.15		$\log_{10} K_1^o = 2.249$	$\log_{10} K_1^o = (2.35 \pm 0.03)$	$\log_{10} K_1^o = 2.35$	
drs	298.15	m_{MgSO_4} : 0.017-2.24	$\log_{10} K_1^o = (2.22 \pm 0.03)$			[2004BUC/CHE]
drs	298.15	m_{MgSO_4} : 0.017-2.24	$\log_{10} K_1^o = (2.22 \pm 0.08)$ $\Delta_r H_1^o = (13.3 \pm 0.2)$ $\Delta_r H_{n\,m}^o = (614 \pm 20)$			[2006AKI/ROH]

(a) Value derived assuming sodium sulfate ion pairing.
(b) Value taken from other sources;
(c) Result from the Lee-Wheaton equation.

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For the evaluations of the conductance data with the Lee-Wheaton equation, the following sources were used: the viscosity of water was taken from [1978KES/SOK]; and the dielectric constant of water from [1979BRA/PIT]. Bianchi *et al*. [2000BIA/DUJ] used the conductance data in [1977PET/TAB] taking the example of $MgSO_4$ solutions to test the performance of various conductivity equations (*e.g.*, Fuoss-Hsia-Fernández-Prini (FHFP) [1967FUO/HSI], [1969FER], Lee-Wheaton (LW) [1987BIA/COR], Mean Spherical Approximation (MSA) [1995TUR/BLU], [1955LEI] and an abridged MSA equation (MSAp)) for symmetrical electrolytes. They showed the utility of the Lee-Wheaton equation at concentrations of less than *ca*. 0.005 $mol \cdot dm^{-3}$. Fisher and Fox [1979FIS/FOX] reported conductance data for $MgSO_4$ and $CaSO_4$, from 0.101 to 202.65 MPa within the temperature range of 283.15 to 303.15 K using a Pyrex cell with a sliding Teflon piston. This cell design limited the accuracy of these measurements compared to those with conventional glass cells [1973KAT], [2002TOM/BES], which gave the most accurate equilibrium constants of all the techniques employed.

The calorimetric data were generally consistent with those from the most recent study [1981WAC/RIE], which essentially anchor the enthalpy of dilution due to the high precision of this work.

Equation (V.110) resulted from a weighted linear regression of selected $\log_{10} K_1^\circ$ and $\Delta_r H_m^\circ$ values taken from column 5 of Table V-61 *versus* $(T/\mathrm{K})^{-1}$ and is illustrated in Figure V-47.

$$\log_{10} K_1^\circ (\text{V.109}) = (3.3314 \pm 0.0600) - (300.906 \pm 17.546)(T/\mathrm{K})^{-1} \quad (\text{V.110})$$

Hence, K_1°((V.109), 298.15 K) = (210.0 ± 7.6) which is the value accepted by this review. The resulting thermodynamic parameters at 298.15 K are accepted:

$$\log_{10} K_1^\circ ((\text{V.109}), 298.15\ \mathrm{K}) = \mathbf{(2.322 \pm 0.016)},$$

$$\Delta_r G_m^\circ ((\text{V.109}), 298.15\ \mathrm{K}) = \mathbf{-\ (13.255 \pm 0.090)}\ \mathrm{kJ \cdot mol^{-1}},$$

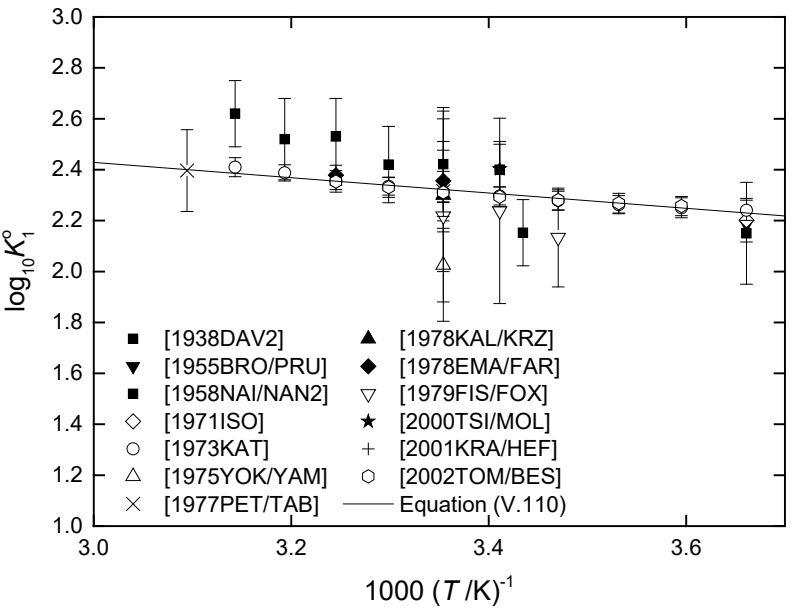
$$\Delta_r H_m^\circ ((\text{V.109}), 298.15\ \mathrm{K}) = \mathbf{(5.761 \pm 0.658)}\ \mathrm{kJ \cdot mol^{-1}},$$

$$\Delta_r S_m^\circ ((\text{V.109}), 298.15\ \mathrm{K}) = \mathbf{(63.780 \pm 2.227)}\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}},$$

where the uncertainties are 2σ.

Only one study [2001KRA/HEF] was found in which the ionic strength was varied (from 0.25 to 4.86 $mol \cdot kg^{-1}$; the highest ionic strength value was ignored here), using an "inert" salt (CsCl), although it appears likely that sulfate may have contributed significantly to the ionic strength, especially at the two lowest I_m values. Application of the SIT to the remaining three values yielded K_1°((V.109), 298.15 K) = (231 ± 7) and $\Delta\varepsilon = -(0.011 \pm 0.002)\ \mathrm{kg \cdot mol^{-1}}$ from which, given $\varepsilon(Mg^{2+}, Cl^-) = (0.19 \pm 0.02)\ \mathrm{kg \cdot mol^{-1}}$, a value of $\varepsilon(Cs^+, SO_4^{2-}) = -(0.20 \pm 0.02)\ \mathrm{kg \cdot mol^{-1}}$ was derived, *cf.* $\varepsilon(K^+, SO_4^{2-}) = -(0.06 \pm 0.02)\ \mathrm{kg \cdot mol^{-1}}$.

Figure V-47: Plot of $\log_{10}K_1^\circ$ for the formation of $MgSO_4$(aq) *versus* $(T/K)^{-1}$ where the solid line was generated from Equation (V.110).



The near equality of the thermodynamic quantities for the formation of sulfate ion pairs with Mg^{2+}, Zn^{2+}, Cd^{2+}, Ca^{2+}, Cu^{2+} and Ni^{2+} tends to confirm Larson's hypothesis [1970LAR] that the sulfate anion replaces two waters of hydration in the primary solvation shell of these cations rather than forming solvent-separated ion pairs, at the very least for the non-transition metal ions, Mg^{2+} and Ca^{2+}.

Gamsjäger *et al*. [2005GAM/BUG] used the weighted average of potentiometric and conductance data to derive $K_1^\circ(298.15\ K) = (226.1 \pm 15.2)$ for the formation of $NiSO_4$(aq) while calorimetric data gave a selected value of $\Delta_r H_m^\circ = (5.66 \pm 0.81)\ kJ{\cdot}mol^{-1}$ at 298.15 K. Lemire *et al*. [2013LEM/BER] more recently selected the value of $K_1^\circ(298.15\ K) = (275 \pm 19)$ for the formation of $FeSO_4$(aq) with $\Delta_r H_m^\circ = (8.4 \pm 6.2)\ kJ{\cdot}mol^{-1}$ at 298.15 K. Not surprisingly, high-temperature conductance measurements (398.15-548.15 K) [2011MAD/TRE] clearly established that Equation (V.110) severely underestimates the ion association constant of $NiSO_4$(sln) and $MgSO_4$(sln) at these temperatures if the contribution of $\Delta_r C_{p,m}^\circ$ is ignored.

Based on the thermodynamic quantities derived from Equation (V.110) and the optimised auxiliary data for Mg^{2+} and SO_4^{2-} selected by this review, the following quantities were calculated and are selected by this review:

$$\Delta_f G^o_m(MgSO_4, aq, 298.15\ K) = \mathbf{-\ (1212.962 \pm 0.895)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^o_m(MgSO_4, aq, 298.15\ K) = \mathbf{-\ (1370.579 \pm 0.976)}\ kJ{\cdot}mol^{-1},$$

$$S^o_m(MgSO_4, aq, 298.15\ K) = \mathbf{-\ (53.620 \pm 2.830)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

We note that whereas this value for $S^o_m(MgSO_4, aq, 298.15\ K)$ is compatible with that proposed by Glushko *et al.* [1981GLU/MED], – (53.68 ± 3.60) $J{\cdot}K^{-1}{\cdot}mol^{-1}$, it is significantly smaller than found for other divalent metal sulfates, including $CaSO_4(aq)$.

Higher-order magnesium sulfate ion pairs such as $Mg_2SO_4^+$ and $Mg_3(SO_4)_3(aq)$ were proposed by Archer and Wood [1985ARC/WOO] based on their modelling of thermodynamic data (*i.e.*, freezing point depression data, isopiestic data, vapour-pressure osmometry and enthalpy of dilution measurements) available for the $MgSO_4$(sln) system from 273 to 423 K, extending to ionic strengths of 3 $mol{\cdot}kg^{-1}$. Their findings must be considered as being conjectural until confirmed by Raman spectroscopy or dielectric relaxation, for example.

V.6.1.2.6 The solubility of magnesite ($MgCO_3$(cr))

In spite of its wide-spread geological occurrence, only a few papers report experimental data for the aqueous solubility of magnesite. According to Christ and Hostetler [1970CHR/HOS] "few carbonate mineral solubilities have proven more difficult to investigate experimentally than magnesite". As a consequence, the reported solubility products at 298.15 K differ by more than four orders of magnitude (see Figure V-48). This is partly due to the very slow rate of magnesite dissolution, which results in serious difficulties for the measurement of its low-temperature solubility (despite the fact that under all conditions magnesite is the most stable carbonate phase forming in the system MgO-CO_2-H_2O). Among the 21 available references, 16 papers ([1867WAG], [1881ENG/VIL], [1892LUB], [1901CAM/BRI], [1915WEL], [1932BAR], [1935HAL/RIT], [1954YAN], [1960BER/BOR], [1964HAL/TAS], [1966HAL/TAS], [1969HOR], [1969RIE], [1970CHR/HOS], [1986KIT/PER], [2011BEN/SAL]) reported primary data for the so-called open systems, and seven papers ([1898TRE/REU], [1903CAM/SEI], [1915GOT], [1932BAR], [1932LEI], [1962MOR], [2011BEN/SAL]) for the closed (to CO_2) systems. Since different methods were used by the authors to derive solubility products, the reported solubility data were re-evaluated for the purposes of this review, considering the formation of Mg^{2+}, $MgOH^+$ and $MgCO_3(aq)$ as aqueous species (using their thermodynamic data selected in this review). The thermodynamic models used here for the open and closed systems are described in detail in [2012VIS/VAN]. However, the activity coefficients were calculated using the NEA TDB recommended SIT approach. In our calculations the temperature-dependent $\varepsilon(Na^+, CO_3^{2-})$ and $\varepsilon(Na^+, HCO_3^-)$ coefficients obtained in Chapter V.2 and $\varepsilon(Mg^{2+}, HCO_3^-) = -(0.06 \pm 0.05)\ kg{\cdot}mol^{-1}$ (Section V.6.1.2.7) were used; other values were taken from Table B-4. The temperature dependence of latter ε(i,j) parameters were taken from [2001PRE/GAM]. Some ion interaction coefficients,

$\varepsilon(MgOH^+, CO_3^{2-})$, $\varepsilon(MgOH^+, HCO_3^-)$ and the temperature dependence of $\varepsilon(Mg^{2+}, HCO_3^-)$, are unknown, but as the concentrations of these species are very low, they do not affect the results markedly. The evaluation of magnesite, calcite, aragonite, gypsum and anhydrite solubilities required knowledge of the temperature dependence of auto-ionisation constant of water $K^{\circ}_{H_2O}$ and a_{H_2O} (in NaCl solution), which were taken from [2006BAN/LVO] and [2007MIL/HUA], respectively.

The reported high solubility for $MgCO_3$ in [1881ENG/VIL] suggests that Engel and Ville probably studied a hydrated $MgCO_3$ (nesquehonite or landsfordite) rather than magnesite. In some cases ([1892LUB], [1964HAL/TAS]), re-evaluation was not possible, due to lacking experimental data for the CO_2 pressure. The solubilities by Leick [1932LEI] were obtained in boiling water, which resulted in non-equilibrium systems. In addition, Leick stated that a possible transformation to brucite might have occurred during the measurements. A similar observation was reported by Morey [1962MOR] above 403 K. One of the data points in [1932BAR] refers to precipitated $MgCO_3$, which has a considerably higher solubility than magnesite, so this datum was rejected too. The solubility determined in [1954YAN] at 328.15 K deviates from the expected value, therefore it was rejected (Table V-62). Some old references ([1898TRE/REU], [1903CAM/SEI], [1915GOT], [1932BAR]) reported much higher than expected solubilities of magnesite in a closed system, partly due to possible external CO_2 uptake. Therefore, the data in these references were also rejected. The solubility of magnesite was also determined in presence of different salts (NaCl, $NaClO_4$, $NaNO_3$, Na_2SO_4, Na_2CO_3, $MgCl_2$) by several authors, both in open ([1903CAM/SEI], [1915WEL]) and closed ([1903CAM/SEI], [1915GOT], [1932LEI], [1969HOR], [1969RIE], [2011BEN/SAL]) systems. In open systems only the solubility data reported in [1915WEL] for 0.469 mol·dm^{-3} NaCl solution were accepted. Due to the experimental difficulties related to the systems closed to CO_2, only the data reported in [2011BEN/SAL] were accepted, because Bénézeth *et al*. accurately determined the pH, $c(Mg^{2+})_{tot}$ and total dissolved inorganic carbon. Since these authors determined the solubility at a constant (0.1 mol·dm^{-3}) NaCl concentration, an evaluation of ionic-strength dependence was not possible, but the calculated $\log_{10}K^{\circ}_{s,0}$ values were extrapolated to zero ionic strength using the SIT approach. The solubility data determined in 3 mol·dm^{-3} $NaClO_4$ ([1969HOR], [1969RIE]) seem to be correct, but the solubility constants extrapolated to $I_m = 0$ are much lower than expected; therefore these data were also rejected.

Re-evaluation of the experimental data resulted in 37 accepted $\log_{10}K^{\circ}_{s,0}$ values for Reaction (V.111), from eight different sources (21 data points from [2011BEN/SAL]), in the temperature range 291 to 473 K (Table V-62, half-filled symbols in Figure V-48). An unconstrained regression of the data resulted in Equation (V.112).

$$MgCO_3(cr) \rightleftharpoons Mg^{2+} + CO_3^{2-} \qquad \text{(V.111)}$$

$$\log_{10}K^{\circ}_{sp} = (7.68236 \pm 3.62397) - (0.03341255 \pm 0.00502635)(T/\text{K}) - (1609.5086 \pm 639.0336)(T/\text{K})^{-1} \qquad \text{(V.112)}$$

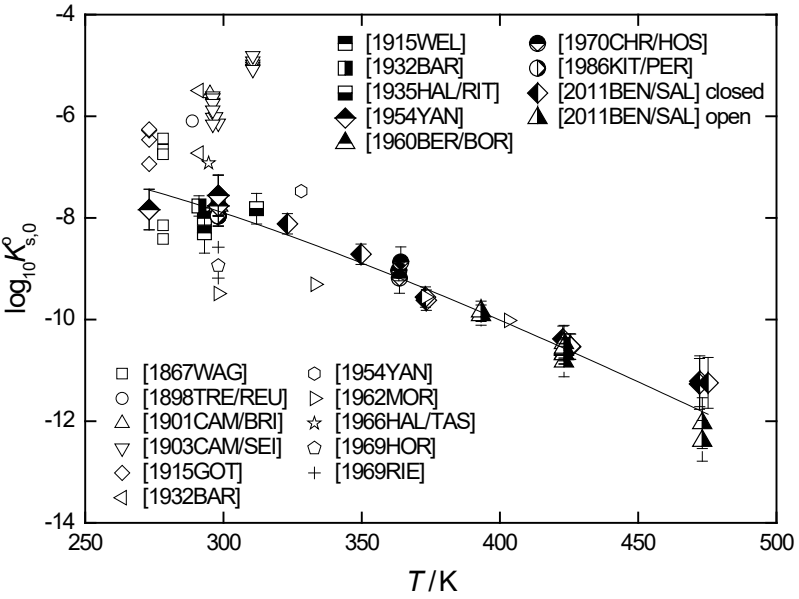
Table V-62: References with accepted solubility data for $MgCO_3$(cr).

Reference	Experimental conditions	Reported solubility mol·kg^{-1}	Re-calculated $\log_{10}K^\circ_{s,0}$
[1915WEL]	T = 293.15 K, p_{CO_2} = 0.029 kPa,	0.00074	– (8.007 ± 0.200)
	T = 293.15 K, p_{CO_2} = 0.029 kPa, 0.469 mol·dm^{-3} NaCl	0.0012	– (8.293 ± 0.400)
[1932BAR]	T = 291.15 K, p_{CO_2} = 0.031 kPa	0.00095	– (7.766 ± 0.200)
[1935HAL/RIT]	T = 298.15 K, p_{CO_2} = 0.0968 MPa,	0.01657	– (7.737 ± 0.200)
	T = 311.95 K, p_{CO_2} = 0.0945 MPa	0.01298	– (7.819 ± 0.300)
[1954YAN]	T = 273.15 K, p_{CO_2} = 0.101 MPa	0.02252	– (7.835 ± 0.400)
	T = 298.15 K, p_{CO_2} = 0.0101 MPa	0.01650	– (7.761 ± 0.200)
	T = 298.15 K, p_{CO_2} = 0.0012 MPa	0.00165	– (7.560 ± 0.400)
	T = 328.15 K, p_{CO_2} = 0.101 MPa	0.01559	– 7.473 (rejected)
[1960BER/BOR]	T = 298.15 K, p_{CO_2} = 0.100 MPa	0.01649	– (7.757 ± 0.200)
[1970CHR/HOS]	T = 364.15 K, p_{CO_2} = 0.0316 MPa	0.00198	– (8.869 ± 0.300)
	T = 364.45 K, p_{CO_2} = 0.00278 MPa	0.00095	– (9.036 ± 0.200)
	T = 364.65 K, p_{CO_2} = 0.0312 MPa	0.00174	– (9.183 ± 0.300)
[1986KIT/PER]	T = 298.15 K, p_{CO_2} = 0.034333 kPa	0.000763	– (7.965 ± 0.200)
	T = 298.15 K, p_{CO_2} = 0.034333 kPa	0.000766	– (7.960 ± 0.200)
	T = 298.15 K, p_{CO_2} = 0.034333 kPa	0.000764	– (7.963 ± 0.200)
[2011BEN/SAL]	I_c = 0.1 (NaCl)		
	T = 393.15 K, p_{CO_2} = 2.5 MPa	0.00586	– (9.912 ± 0.200)
	T = 393.15 K, p_{CO_2} = 1.5 MPa	0.00525	– (9.839 ± 0.200)
	T = 423.15 K, p_{CO_2} = 1.6 MPa	0.00306	– (10.463 ± 0.350)
	T = 423.15 K, p_{CO_2} = 3 MPa	0.00320	– (10.669 ± 0.200)
	T = 423.15 K, p_{CO_2} = 3 MPa	0.00282	– (10.826 ± 0.300)
	T = 423.15 K, p_{CO_2} = 3 MPa	0.00317	– (10.680 ± 0.200)
	T = 423.15 K, p_{CO_2} = 1.5 MPa	0.00270	– (10.597 ± 0.200)
	T = 423.15 K, p_{CO_2} = 1.5 MPa	0.00274	– (10.576 ± 0.200)
	T = 423.15 K, p_{CO_2} = 1.3 MPa	0.00266	– (10.554 ± 0.200)
	T = 473.15 K, p_{CO_2} = 3 MPa	0.00081	– (12.377 ± 0.400)
	T = 473.15 K, p_{CO_2} = 1.5 MPa	0.00086	– (12.026 ± 0.500)
	T = 472.25 K, closed system	0.00116	– (11.212 ± 0.500)
	T = 422.75 K, closed system	0.00141	– (10.380 ± 0.250)
	T = 373.35 K, closed system	0.00179	– (9.619 ± 0.200)
	T = 472.15 K, closed system	0.00132	– (11.266 ± 0.500)
	T = 425.55 K, closed system	0.00145	– (10.535 ± 0.250)
	T = 373.05 K, closed system	0.00184	– (9.558 ± 0.200)
	T = 425.35 K, closed system	0.00148	– (10.527 ± 0.250)
	T = 349.75 K, closed system	0.00207	– (8.714 ± 0.200)
	T = 323.15 K, closed system	0.00242	– (8.116 ± 0.200)
	T = 475.25 K, closed system	0.00135	– (11.246 ± 0.500)

Equation (V.112) yields the following thermodynamic parameters for Reaction (V.111): $\Delta_r G^\circ_m$ ((V.111), 298.15 K) = (45.04 ± 1.70) kJ·mol^{-1}, $\Delta_r H^\circ_m$ ((V.111), 298.15 K) = – (27.26 ± 2.50) kJ·mol^{-1}, $\Delta_r S^\circ_m$ ((V.111), 298.15 K) = – (242.5 ± 10.0) J·K^{-1}·mol^{-1} and $\Delta_r C^\circ_{p,m}$ ((V.111), 298.15 K) = – (389.8 ± 10.0) J·K^{-1}·mol^{-1}. However, the low-temperature heat capacity measurements noted in Section V.6.1.1.5.2 indicate that the standard entropy of magnesite is close to

S_m^o ($MgCO_3$, cr, magnesite, 298.15 K) = (65.09 ± 0.80) J·K^{-1}·mol^{-1}, which corresponds to $\Delta_r S_m^o$ ((V.111), 298.15 K) = – (252.09 ± 1.62) J·K^{-1}·mol^{-1}.

Figure V-48: Temperature dependence of the re-evaluated solubility product of magnesite. The selected thermodynamic parameters are based on the accepted values (half-filled symbols).



The regression constrained to this value resulted in Equation (V.113) (see solid line in Figure V-48):

$$\log_{10} K_{s,0}^o = (4.99512 \pm 0.91989) - (3.04590 \pm 0.15427)\times 10^{-2}\,(T/\mathrm{K}) - (1129.681 \pm 129.321)(T/\mathrm{K})^{-1} \qquad \text{(V.113)}$$

leading to the following values at 298.15 K:

$$\log_{10} K_{s,0}^o \text{((V.111), 298.15 K)} = -(7.875 \pm 0.150),$$

$$\Delta_r G_m^o \text{((V.111), 298.15 K)} = (44.952 \pm 1.700)\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r H_m^o \text{((V.111), 298.15 K)} = -(30.209 \pm 2.000)\ \mathrm{kJ\cdot mol^{-1}}.$$

This value of $\Delta_r G_m^o$ has been used as one of the input values in the final optimisation of the data for $MgCO_3$(cr) and Mg^{2+} described in Section V.6.1.1.6, which leads to the final selected values for $MgCO_3$(cr):

$$\Delta_f H_m^o(MgCO_3, cr, 298.15\ K) = \mathbf{-\ (1112.200 \pm 1.300)}\ kJ\cdot mol^{-1},$$

$$S_m^o(MgCO_3, cr, 298.15\ K) = \mathbf{(65.100 \pm 0.600)}\ J\cdot K^{-1}\cdot mol^{-1},$$

giving: $\Delta_f G_m^o(MgCO_3, cr, 298.15\ K) = -(1028.409 \pm 1.313)\ kJ\cdot mol^{-1}$, and with the optimised value of $S_m^o(Mg^{2+}, 298.15\ K) = -(135.900 \pm 1.700)\ J\cdot K^{-1}\cdot mol^{-1}$:

$$\log_{10} K_{sp}^o((V.111), 298.15\ K) = -(7.850 \pm 0.229),$$

$$\Delta_r G_m^o((V.111), 298.15\ K) = (44.806 \pm 1.579)\ kJ\cdot mol^{-1},$$

$$\Delta_r H_m^o((V.111), 298.15\ K) = -(30.030 \pm 1.453)\ kJ\cdot mol^{-1},$$

$$\Delta_r S_m^o((V.111), 298.15\ K) = -(251.000 \pm 7.197)\ J\cdot K^{-1}\cdot mol^{-1}.$$

This value of $\log_{10} K_{s,0}^o$ is slightly smaller than those predicted in [2011BEN/SAL] and [2012VIS/VAN] ($-(7.80 \pm 0.30)$ and -7.78, respectively), partly due to the larger number of experimental data considered here. The optimised values of $\Delta_r G_m^o$, $\Delta_r H_m^o$ and $\Delta_r S_m^o$ of Reaction (V.111) are in good agreement with those predicted by Model 2 in the recent compilation by de Visser *et al.* [2012VIS/VAN] using the Pitzer model ($\Delta_r G_m^o = 44.42\ kJ\cdot mol^{-1}$, $\Delta_r H_m^o = -30.54\ kJ\cdot mol^{-1}$, ($\Delta_r S_m^o$ (V.97), 298.15 K) $= -251.43\ J\cdot K^{-1}\cdot mol^{-1}$) (this model uses the stability constant for the $MgCO_3$(aq) ion pair, and Pitzer parameters for $Mg(HCO_3)^+$ and $Mg(HCO_3)_2$(aq); owing to the insignificant contribution of $Mg(HCO_3)_2$(aq), it was not considered in the current review). In addition, the selected values for $MgCO_3$(cr) agree well within the uncertainty range with those selected by Robie and Hemingway (p. 271 of [1995ROB/HEM]): $\Delta_f G_m^o = -(1029.5 \pm 1.4)\ kJ\cdot mol^{-1}$, $\Delta_f H_m^o = -(1113.3 \pm 1.3)\ kJ\cdot mol^{-1}$, $S_m^o = (65.09 \pm 0.10)\ J\cdot K^{-1}\cdot mol^{-1}$.

V.6.1.2.7 Aqueous carbonato complexes of Mg^{2+}

The interaction of Mg^{2+} with HCO_3^-/CO_3^{2-} ions is of great importance for calculations of both magnesium speciation in natural waters and the solubility of magnesite. The available references reporting explicit equilibrium constants for the Mg^{2+}-HCO_3^-/CO_3^{2-} ion pairs are listed in Table V-63. The majority of these studies report the existence of two ion pairs (Reactions (V.114) and (V.115)).

$$Mg^{2+} + HCO_3^- \rightleftharpoons MgHCO_3^+ \qquad (V.114)$$

$$Mg^{2+} + CO_3^{2-} \rightleftharpoons MgCO_3(aq) \qquad (V.115)$$

The formation of $Mg(HCO_3)_2$(aq) and $Mg_2CO_3^{2+}$ complexes reported in [1977RIE/GAM] and [1974PYT/HAW] was not confirmed in other studies.

Most of the studies listed in Table V-63 report thermodynamic constants for the ion-pair (complex) formation using activity coefficients calculated by the Debye-Hückel equation (with A, B and a_i values taken from various literature sources). A correct re-evaluation of these data by SIT is not possible, due to unpublished experimental data and/or to the high number of strongly correlating unknown parameters. Nevertheless, in

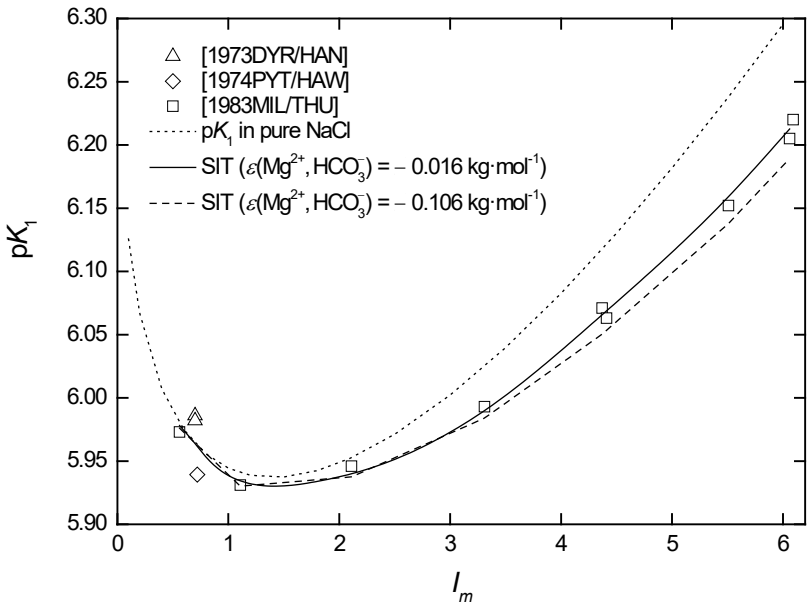
most cases SIT-related methods were used, generally at low ionic strength and some of these papers are the most reliable publications on this topic (with a large number of data points). Thus several reported values can be accepted (depending on the precision of the work and on the similarity with other studies), which allow a reliable temperature dependence of the thermodynamic equilibrium constants to be established.

However, far fewer data are available to determine a reliable ionic-strength dependence of the above equilibria [1960RAA], [1973DYR/HAN], [1974PYT/HAW], [1975LIN/ATK], [1977RIE/GAM], [1983MIL/THU], [1993HE/MOR]. Among these the carbonic acid pK_m values determined in (Na,Mg)Cl media [1973DYR/HAN], [1974PYT/HAW], [1983MIL/THU], [1993HE/MOR]) are the most valuable, since the comparison of stoichiometric pK_m values of carbonic acid determined in NaCl and (Na,Mg)Cl media can be used to quantify the Mg^{2+}- HCO_3^-/CO_3^{2-} ion pairing interactions [1973DYR/HAN], [1974PYT/HAW], [1983MIL/THU].

Although the existence of $MgHCO_3^+$ ion pair was vindicated in a detailed IR study [2017STE/LEM], the interaction between Mg^{2+} and HCO_3^- is rather weak ($\log_{10}K^\circ \sim 1$ at 298.15 K, Table V-63). In such cases the distinction between ion-pair formation and activity coefficient variations is difficult to establish; therefore, it is more appropriate to take into account this interaction by the corresponding ion interaction coefficients (see [1983MIL/THU] and [1985PIT/OLS]).

In the presence of Mg^{2+} the first dissociation constant of carbonic acid differs only slightly in NaCl and (Na,Mg)Cl media ($\Delta pK_1 < 0.1$, [1973DYR/HAN], [1974PYT/HAW], [1983MIL/THU]), confirming the rather weak Mg^{2+}- HCO_3^- interaction. Therefore, in this review the SIT formalism has been used to interpret the slight changes of pK_1 values in the mixed (Na,Mg)Cl electrolyte (as compared to the pure NaCl media), using the expression $\log_{10}\gamma(HCO_3^-) = -D + \varepsilon(Na^+, HCO_3^-)m_{Na} + \varepsilon(Mg^{2+}, HCO_3^-)m_{Mg}$. The results of the SIT analysis are depicted in Figure V-49, which confirms that the $\varepsilon(Mg^{2+}, HCO_3^-)$ ion interaction coefficient is appropriate to account for the weak Mg^{2+}- HCO_3^- interaction. Using the ionic-strength dependent $\varepsilon(Na^+, HCO_3^-)$ selected in this review, the SIT analysis resulted in $\varepsilon(Mg^{2+}, HCO_3^-) = -(0.016 \pm 0.030)$ kg·mol^{-1} at 298.15 K.

Figure V-49: SIT treatment of the carbonic acid pK_1 values reported in [1973DYR/HAN], [1974PYT/HAW], [1983MIL/THU] for the mixed (Na,Mg)Cl media at 298.15 K, without considering the formation of the weak $MgHCO_3^+$ ion pair (dashed line pK_1 values are calculated by $\varepsilon(Mg^{2+}, HCO_3^-) = -0.106$ kg·mol^{-1} (see the text), dotted line shows the pK_1 values in pure NaCl medium selected in this review).



It should be noted, however, that this calculation neglects the changes in the activity coefficient of CO_2(aq) and the water activity in the presence of Mg^{2+} (although the molal concentration of $MgCl_2$ during these measurements was always less than 5, 9 and 15 % of the concentration of NaCl in [1983MIL/THU], [1973DYR/HAN] and [1974PYT/HAW], respectively.

Table V-63: References with explicit equilibrium constants for the Mg^{2+}-HCO_3^-/CO_3^{2-} ion pairs.

Reference	T /K	Method	I_m range	$\log_{10}K$ reported	$\log_{10}K$ re-calulated	$\log_{10}K$ accepted	Notes
$Mg^{2+} + HCO_3^- \rightleftharpoons MgHCO_3^+$							
[1941GRE]	295.15	pH, sol	0.15 mol·dm^{-3} (KCl)	− 0.77			
[1963HOS2]	298.15	pH	0-0.07 ($MgCl_2$)				
			→ 0	(0.95 ± 0.10)			
[1971NAK]	298.15	pH, ise-Mg^{2+}	0.01-0.06 mol·dm^{-3} ($MgCl_2$)				
			→ 0	1.23			
[1973DYR/HAN]	298.15	pH	0.7 mol·dm^{-3} (NaCl)	0.02			
[1972MAR/VAS]	295.15	ise-Mg^{2+}		1.27			
[1974MAR/VAS]	333.15			1.64			
	343.15			1.77			
	353.15			1.82			
	363.15			1.94			
	371.15			2.01			
[1974PYT/HAW]	298.15	pH	0.72 (NaCl,$MgCl_2$)	0.21			
[1975LIN/ATK]	273.15	pH	0.15 (Na,Mg)Cl	0.87			
	283.15		0.15 (Na,Mg)Cl	0.61			
	293.15		0.15 (Na,Mg)Cl	0.37			
			0.35 (Na,Mg)Cl	0.37			
			0.64 (Na,Mg)Cl	0.30			
			→ 0 (Na,Mg)Cl	0.85			
	303.15		0.15 (Na,Mg)Cl	0.40			
[1977RIE/GAM]	298.15	pH	3.5 ($NaClO_4$)	0.16			
[1977SIE/HOS]		pH	0.07-0.55 (K,Mg)Cl				
	283.15		→ 0	(1.051 ± 0.018)			
	298.15		→ 0	(1.066 ± 0.012)			
	313.15		→ 0	(1.108 ± 0.006)			
	328.15		→ 0	(1.160 ± 0.011)			

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Table V-63 (continued)

Reference	T /K	Method	I_m range	$\log_{10}K$ reported	$\log_{10}K$ re-calculated	$\log_{10}K$ accepted	Notes
	343.15		$\rightarrow 0$	(1.230 ± 0.017)			
	363.15		$\rightarrow 0$	(1.337 ± 0.007)			
[1981MAR/GUY]		sol	? ($Ca(HCO_3)_2$)				
	278.15		$\rightarrow 0$	(1.23 ± 0.03)			
	288.15		$\rightarrow 0$	(1.23 ± 0.03)			
	298.15		$\rightarrow 0$	(1.23 ± 0.03)			
	303.15		$\rightarrow 0$	(1.24 ± 0.03)			
	313.15		$\rightarrow 0$	(1.24 ± 0.03)			
	323.15		$\rightarrow 0$	(1.25 ± 0.03)			
[1985GUY/DOR]		sol	? ($Ca(HCO_3)_2$)				
	298.15		$\rightarrow 0$	1.23			
[2014STE/BEN]			0.1 m NaCl + 0-0.115 m $MgCl_2$ + 0.0-0.0022 m $NaHCO_3$ + 0.0-0.01 m Na_2CO_3				
	283.15	pH	$\rightarrow 0$	(1.14 ± 0.08)			
	298.15	pH	$\rightarrow 0$	(1.09 ± 0.07)			
	313.15	pH	$\rightarrow 0$	(1.09 ± 0.05)			
	333.15	pH	$\rightarrow 0$	(1.15 ± 0.06)			
	353.15	pH(sp)	$\rightarrow 0$	(1.31 ± 0.08)			
	373.15	pH(sp)	$\rightarrow 0$	(1.40 ± 0.15)			
	398.15	pH(sp)	$\rightarrow 0$	(1.55 ± 0.15)			
	423.15	pH(sp)	$\rightarrow 0$	(1.75 ± 0.20)			
[2017STE/LEM]			0.0-0.15 m $NaHCO_3$ + 0.0-0.05 m Na_2CO_3 + 0-0.16 m $MgCl_2$				
	298.15	pH(IR)	$\rightarrow 0$	(1.12 ± 0.11)			

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Table V-63 (continued)

Reference	T /K	Method	I_m range	$\log_{10}K$ reported	$\log_{10}K$ re-calculated	$\log_{10}K$ accepted	Notes
$Mg^{2+} + CO_3^{2-} \rightleftharpoons MgCO_3(aq)$							
[1941GRE]	295.15	pH, sol	0.15 mol·dm^{-3} (KCl)	2.37			
[1960RAA]	293.15	Sp	0.1 mol·dm^{-3} (NaCl)	2.15	(2.15 ± 0.20)		
			→ 0		(3.10 ± 0.20)		
[1961GAR/THO]	298.15	pH	0-0.1($MgCl_2$)				
			→ 0	3.40			Re-calculated in
			→ 0	2.99		(2.99 ± 0.15)	[1974REA/LAN]
[1971NAK]	298.15	pH, ise-Mg^{2+}	0.01-0.06 mol·dm^{-3} ($MgCl_2$)	3.24			
[1973DYR/HAN]	298.15	pH	0.7 mol·dm^{-3} (NaCl, $MgCl_2$)	1.51	(1.42 ± 0.15)	(1.42 ± 0.15)	Re-evaluated by the SIT method together with the data reported in [1974PYT/HAW] and [1983MIL/THU]
			→ 0		(3.01 ± 0.15)	(3.00 ± 0.10)	
[1973LAR/SOL]		pH	0.003-0.032 ? (K,Mg)ClO_4				
	278.15		→ 0	2.81		(2.81 ± 0.15)	
	288.15		→ 0	2.82		(2.82 ± 0.15)	
	298.15		→ 0	2.90		(2.90 ± 0.15)	
[1974PYT/HAW]	298.15	pH	0.72 mol·kg^{-1} (NaCl, $MgCl_2$)	2.05	1.24	(1.24 ± 0.20)	Re-evaluated by the SIT method together with the data reported in [1973DYR/HAN] and [1983MIL/THU]
			→ 0		(2.82 ± 0.20)	(3.00 ± 0.10)	
[1974REA/LAN]		pH	< 0.07 ($MgCl_2$/K_2CO_3)				
	283.15		→ 0				
	298.15		→ 0	(2.79 ± 0.10)		(2.79 ± 0.10)	
	314.15		→ 0	(2.88 ± 0.05)		(2.88 ± 0.10)	
	324.65		→ 0	(3.03 ± 0.07)		(3.03 ± 0.10)	
				(3.17 ± 0.08)		(3.17 ± 0.10)	

(Continued on next page)

Table V-63 (continued)

Reference	T /K	Method	I_m range	$\log_{10}K$ reported	$\log_{10}K$ re-calculated	$\log_{10}K$ accepted	Notes
[1975LIN/ATK]	273.15	pH	0.15 (Na,Mg)Cl	2.56			
	283.15		0.15 (Na,Mg)Cl	2.53			
	293.15		0.15 (Na,Mg)Cl	2.35			
			0.35 (Na,Mg)Cl	2.11			
			0.64 (Na,Mg)Cl	1.95			
			→ 0 (Na,Mg)Cl	3.26			
	303.15		0.15 (Na,Mg)Cl	2.22			
[1977SIE/HOS2]		pH	0.11-0.2 (K,Mg)Cl				
	283.15		→ 0	(2.89 ± 0.019)		(2.89 ± 0.10)	
	298.15		→ 0	(2.98 ± 0.028)		(2.98 ± 0.10)	
	313.15		→ 0	(3.07 ± 0.021)		(3.07 ± 0.10)	
	328.15		→ 0	(3.18 ± 0.026)		(3.18 ± 0.10)	
	343.15		→ 0	(3.28 ± 0.042)		(3.28 ± 0.10)	
	363.15		→ 0	(3.41 ± 0.067)		(3.41 ± 0.10)	
[1977RIE/GAM]	298.15	pH	3.5 ($NaClO_4$)	1.73			
[1981MAR/GUY]		sol					
	278.15		→ 0	2.98			
	288.15		→ 0	3.15			
	298.15		→ 0	3.32			
	303.15		→ 0	3.41			
	313.15		→ 0	3.56			
	323.15		→ 0	3.72			
[1983MIL/THU]	298.15	pH	0.56 mol·kg^{-1} ($NaCl/MgCl_2$)				Calculated by the
			1.11	1.43	1.49	(1.49 ± 0.10)	Pitzer formalism (see
			2.11	1.22	1.34	(1.34 ± 0.10)	text). Re-evaluated
			3.31	1.03	1.21	(1.21 ± 0.10)	by the SIT method
			4.37	0.89	1.19	(1.19 ± 0.10)	together with the data
			4.41	0.97	1.25	(1.25 ± 0.10)	reported in
			5.52	0.92	1.21	(1.21 ± 0.10)	[1973DYR/HAN]
			6.06	1.07	1.32	(1.32 ± 0.10)	and
							[1974PYT/HAW]

(Continued on next page)

Table V-63 (continued)

Reference	T /K	Method	I_m range	$\log_{10}K$ reported	$\log_{10}K$ re-calculated	$\log_{10}K$ accepted	Notes
			6.09	1.03	1.29	(1.29 ± 0.10)	
			→ 0	1.10	1.35	(1.35 ± 0.10)	
				3.00	3.00	(3.00 ± 0.10)	
[1985GUY/DOR]	298.15	sol	→ 0	3.32			
[2014STE/BEN]			0.1 m NaCl + 0-0.115 m $MgCl_2$ + 0.0-0.0022 m $NaHCO_3$ + 0.0-0.01 m Na_2CO_3				
	283.15	pH	→ 0	(2.86 ± 0.04)		(2.86 ± 0.10)	
	298.15	pH	→ 0	(3.01 ± 0.07)		(3.01 ± 0.10)	
	313.15	pH	→ 0	(3.11 ± 0.06)		(3.11 ± 0.10)	
	333.15	pH	→ 0	(3.24 ± 0.05)		(3.24 ± 0.10)	
	353.15	pH(sp)	→ 0	(3.37 ± 0.08)		(3.37 ± 0.10)	
	373.15	pH(sp)	→ 0	(3.48 ± 0.15)		(3.48 ± 0.15)	
[2017STE/LEM]			0-0.15 m $NaHCO_3$ + 0.0-0.05 m Na_2CO_3 + 0-0.16 m $MgCl_2$				
	298.15	pH(IR)	→ 0	(2.98 ± 0.06)		(2.98 ± 0.10)	

The Pitzer parameters of the Mg^{2+}- HCO_3^- interaction have been determined at 298.15 K in [1983MIL/THU] ($\beta^0 = 0.019$ kg·mol^{-1}, $\beta^1 = 0.58$ kg·mol^{-1}), in [1985PIT/OLS] ($\beta^0 = 0.033$ kg·mol^{-1}, $\beta^1 = 0.85$ kg·mol^{-1}), and in [1993HE/MOR] ($\beta^0 = 0.03$ kg·mol^{-1}, $\beta^1 = 0.80$ kg·mol^{-1}). Based on the above β^0 values, $\varepsilon(Mg^{2+}, HCO_3^-) = -(0.106 \pm 0.040)$ kg·mol^{-1} can be estimated [1997GRE/PLY]. Using the latter value, the calculated first dissociation constants for a (Na,Mg)Cl medium are identical within the experimental errors to the measured ones, especially up to 3 mol·kg^{-1} (NaCl) (see dashed line in Figure V-49). Therefore, in this review the average of above two values is selected for

$$\varepsilon(Mg^{2+}, HCO_3^-) = \mathbf{-\ (0.06 \pm 0.05)}\ \text{kg·mol}^{-1}.$$

The $\log_{10}K^\circ$(V.115) values listed in Table V-63 for the $MgCO_3$(aq) ion pair suggest a much stronger interaction. Indeed, the presence of magnesium results in a much greater effect on the second dissociation constant of carbonic acid ($0.2 < \Delta pK_2 < 0.9$ [1983MIL/THU]), which cannot be explained by the changes in the activity coefficients.

Assuming $MgCO_3$(aq) is the only complex species in the solution, the stoichiometric pK_2 values determined in NaCl and (Na,Mg)Cl media at identical ionic strengths are related to K_m(V.116).

$$K_m(\text{V.115}) = (K_{2,(\text{Na,Mg})\text{Cl}} / K_{2,\text{NaCl}} - 1) / m(Mg^{2+})_{\text{free}} \qquad \text{(V.116)}$$

Considering that at $pH = pK_{2,(\text{Na,Mg})\text{Cl}}$ the concentration of HCO_3^- can be expressed as $m(HCO_3^-) = m(CO_3^{2-}) + m(MgCO_3(aq))$, one can calculate $m(Mg^{2+})_{\text{free}}$, the concentration of uncomplexed Mg^{2+}, and thus K_m(V.115) (see Table V-63).

It should be noted that Equation (V.116) neglects the changes in the activity coefficient of H^+ in the presence of Mg^{2+} (as does the SIT model), but if the concentration of $MgCl_2$ is low compared to that of NaCl, this effect should be small. Millero and Thurmond [1983MIL/THU] treated their data, and those reported in [1973DYR/HAN] and [1974PYT/HAW], using the Pitzer formalism, by calculating the total or experimental activity coefficients. However, they used an erroneous equation for the NaCl concentration dependence of $\ln\gamma(CO_2, aq)$ ($\ln\gamma(CO_2, aq) = 0.2421m - 0.0106m^2$) for the interpretation of their potentiometric data [1997GRE/PLY2]. This error is likely to have been propagated to the constant K_m(V.115), too. This may be the reason why the stoichiometric K_m(V.115) values calculated in [1983MIL/THU] are somewhat smaller than those re-calculated in this review (see Table V-63), but the values extrapolated to zero ionic strength are identical. The stoichiometric constants K_m(V.115) re-calculated in this review based on the data reported in [1983MIL/THU], [1973DYR/HAN] and [1974PYT/HAW] are depicted in Figure V-50. Since the concentration of magnesium was different for every experimental point, instead of the classical SIT treatment, it is more appropriate to calculate the thermodynamic constants point by point, using the following activity coefficients:

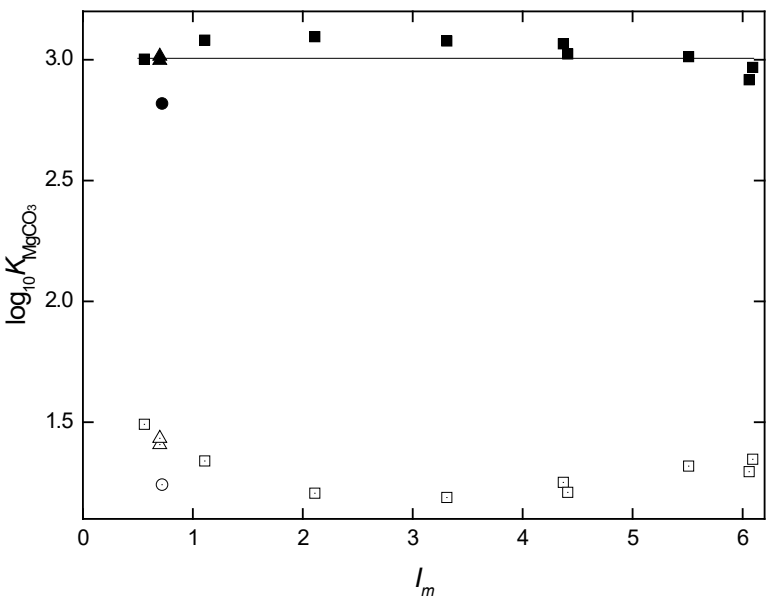
$$\log_{10}\gamma(Mg^{2+}) = -4D + \varepsilon(Mg^{2+}, HCO_3^-)\, m(HCO_3^-) + \varepsilon(Mg^{2+}, Cl^-) m(Cl^-)\,,$$

$$\log_{10}\gamma(CO_3^{2-}) = -4D + \varepsilon(Na^+, CO_3^{2-})\, m(Na^+)\,,$$

and

$$\log_{10}\gamma(MgCO_3(aq)) = \varepsilon(MgCO_3(aq), Na^+ + Cl^-)\, m(NaCl)\,.$$

Figure V-50: Stoichiometric (open symbols) and thermodynamic (filled symbols) stability constants of $MgCO_3$(aq) as a function of ionic strength (NaCl) re-calculated using SIT (squares: [1983MIL/THU], circles: [1973DYR/HAN] and triangle: [1974PYT/HAW]).



Using the ion interaction coefficients selected in this review, the SIT analysis resulted in $\log_{10}K^{\circ}$(V.115) = (3.00 ± 0.10) and $\varepsilon(MgCO_3(aq), Na^+ + Cl^-)$ = (0.073 ± 0.030) kg·mol^{-1}. These values are close to those determined by Millero and Thurmond [1983MIL/THU] $\log_{10}K^{\circ}$(V.115) = (3.000 ± 0.024) and $\log_{10}\gamma(MgCO_3(aq))$ = 0.056$m(Na^+ + Cl^-)$.

He and Morse also reported ionic strength and temperature dependencies of carbonic acid pK_m values in mixed (Na,Mg)Cl medium [1993HE/MOR]. In principle, these data would allow similar treatment as described above. However, the comparison of the pK_m values determined by these authors in pure NaCl media with the selected ones

reflects notable deviations, and the calculated values are also contradictory. This clearly indicates that the data reported in [1993HE/MOR] have high uncertainties.

Based on the seven independent values listed in Table V-63:

$$\log_{10}K^{\circ}((V.115), 298.15\ K) = (2.966 \pm 0.041).$$

The above $\log_{10}K^{\circ}((V.115), 298.15\ K)$ value and those accepted at other temperatures (see Table V-63) are plotted as a function of temperature in Figure V-51. The temperature dependence of $\log_{10}K^{\circ}(V.115)$ can be described by the following simple equation:

$$\log_{10}K^{\circ}(V.115) = (5.43685 \pm 0.08422) - (736.70941 \pm 25.26770)(T/K)^{-1} \quad (V.117)$$

From the above equation the following thermodynamic parameters can be calculated and are accepted by this review:

$$\log_{10}K^{\circ}((V.115), 298.15\ K) = \mathbf{(2.966 \pm 0.169)},$$

$$\Delta_r G_m^{\circ}((V.115), 298.15\ K) = \mathbf{-(16.930 \pm 0.965)}\ kJ\cdot mol^{-1},$$

$$\Delta_r H_m^{\circ}((V.115), 298.15\ K) = \mathbf{(14.104 \pm 0.484)}\ kJ\cdot mol^{-1},$$

$$\Delta_r S_m^{\circ}((V.115), 298.15\ K) = \mathbf{(104.089 \pm 3.621)}\ J\cdot K^{-1}\cdot mol^{-1}.$$

These values are in excellent agreement with those predicted in [2014STE/BEN] ($\Delta_r G_m^{\circ} = -17.1$ kJ·mol^{-1}, $\Delta_r H_m^{\circ} = 13.8$ kJ·mol^{-1}, $\Delta_r S_m^{\circ} = 103.5$ J·K^{-1}·mol^{-1}), and in [1974REA/LAN] ($\Delta_r G_m^{\circ} = -16.44$ kJ·mol^{-1}, $\Delta_r H_m^{\circ} = 13.51$ kJ·mol^{-1}, $\Delta_r S_m^{\circ} = 100.42$ J·K^{-1}·mol^{-1}), but show relatively significant deviations from the values calculated using the data listed in [2012VIS/VAN] ($\Delta_r G_m^{\circ} = -16.69$ kJ·mol^{-1}, $\Delta_r H_m^{\circ} = 11.45$ kJ·mol^{-1}, $\Delta_r S_m^{\circ} = 94.4$ J·K^{-1}·mol^{-1}). Combining the above selected thermodynamic parameters with those listed in Table III-1 for Mg^{2+} and CO_3^{2-}, the following values are calculated and selected:

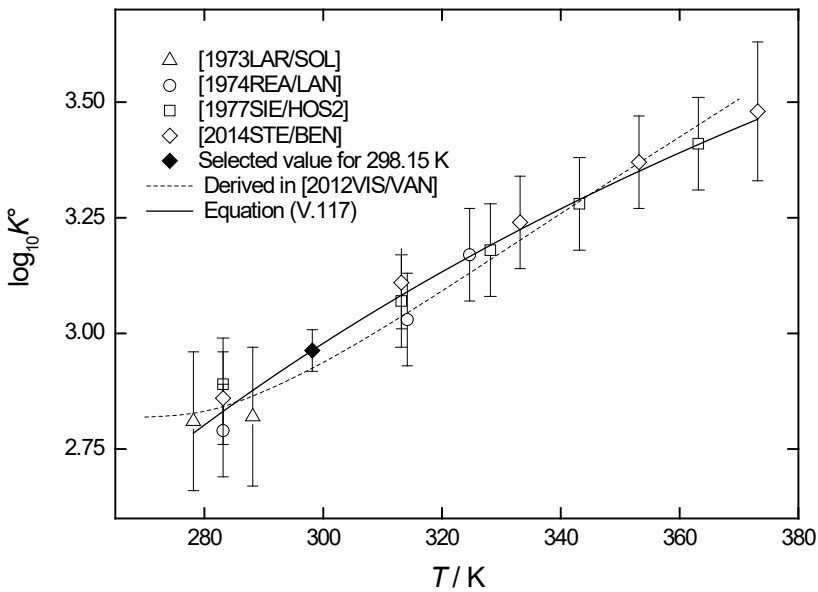
$$\Delta_f G_m^{\circ}(MgCO_3, aq, 298.15\ K) = \mathbf{-(1000.533 \pm 1.304)}\ kJ\cdot mol^{-1},$$

$$\Delta_f H_m^{\circ}(MgCO_3, aq, 298.15\ K) = \mathbf{-(1128.126 \pm 0.810)}\ kJ\cdot mol^{-1},$$

$$S_m^{\circ}(MgCO_3, aq, 298.15\ K) = \mathbf{-(81.811 \pm 4.123)}\ J\cdot K^{-1}\cdot mol^{-1}.$$

Considering the slightly different values used for the formation data of Mg^{2+} and CO_3^{2-} ions, these parameters compare well with those reported in [2014STE/BEN] ($\Delta_f G_m^{\circ} = -(1000.3 \pm 1.1)$ kJ·mol^{-1}, $\Delta_f H_m^{\circ} = -(1127.0 \pm 1.1)$ kJ·mol^{-1}, $S_m^{\circ} = -(79.4 \pm 6.0)$ J·K^{-1}·mol^{-1}). On the other hand, the above values are considerably different from those in [1997SVE/SHO] ($\Delta_f G_m^{\circ} = -998.97$ kJ·mol^{-1}, $\Delta_f H_m^{\circ} = -1132.07$ kJ·mol^{-1}, $S_m^{\circ} = -100.4$ J·K^{-1}·mol^{-1}). In a more recent evaluation, a nearly identical $\Delta_f G_m^{\circ} = -1000.992$ kJ·mol^{-1} but quite different S_m° – 102.57 J·K^{-1}·mol^{-1}, have been reported for $MgCO_3$(aq) [2017MIR/WAG].

Figure V-51: Temperature dependence of the thermodynamic equilibrium constants of $MgCO_3(aq)$.



V.6.2 Calcium species

V.6.2.1 Solid calcium compounds

V.6.2.1.1 CaO(cr)

V.6.2.1.1.1 Structure

Calcium oxide has the fcc NaCl structure, space group $Fm\bar{3}m$, with $a = (4.813 \pm 0.002)$ Å at 298 K, as obtained by Fiquet *et al.* [1999FIQ/RIC], from three concordant sets of data using samples held on refractory metal wires (Pt-10 % Rh, Ir, Re). A room temperature value of $a = 4.8105$ Å was assessed by Wyckoff [1963WYC], as a result of several earlier results. Fiquet *et al.* [1999FIQ/RIC] also give the parameter at temperatures up to 3073 K.

V.6.2.1.1.2 Heat capacity and entropy

As noted in [1989COX/WAG] and [1994GUR/VEY], the most accurate measurements of the low-temperature heat capacity are by Gmelin [1966GME2] (1 to 65 K) and [1969GME2] (1.2 to 300 K). We have accepted the values selected in [1989COX/WAG] which are based on these data:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaO, cr, 298.15\ K}) = \mathbf{(42.049 \pm 0.400)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

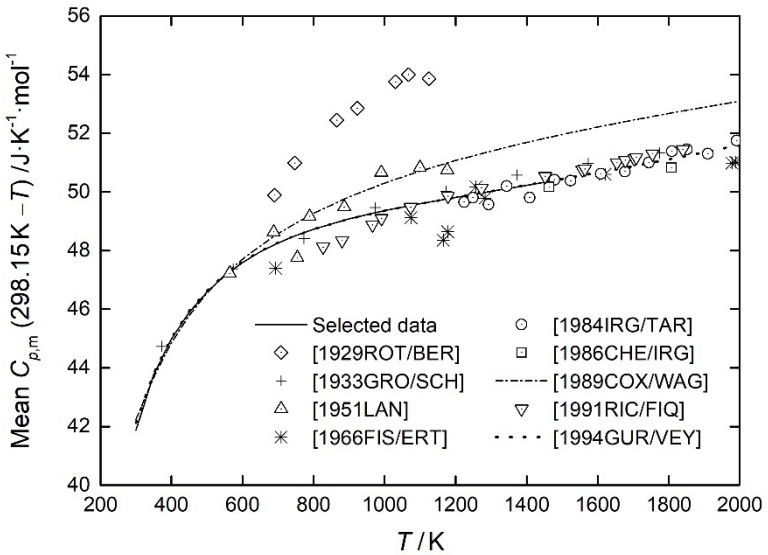
$$S^{\circ}_{\mathrm{m}}(\mathrm{CaO, cr, 298.15\ K}) = \mathbf{(38.1 \pm 0.4)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

For the high-temperature enthalpy content of CaO(cr), we have adopted the values assessed by Gurvich *et al.* [1994GUR/VEY], rather than the noticeably higher values given by CODATA, since Gurvich *et al.* included the more recent data by Irgashov *et al.* [1984IRG/TAR] (1224 to 2542 K) and Chekhovskoi *et al.* [1986CHE/IRG] (1463 to 3086 K), which were not available at the time of the CODATA assessment. There are a number of other studies of the high-temperature enthalpy, whose data are compared with the selected values in Figure V-52.

At the lower temperatures the selected values data can be represented by the equation:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaO, cr}, T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{57.801} - \mathbf{1.08436\times 10^{-2}}(T/\mathrm{K}) + \mathbf{5.34523\times 10^{-6}}(T/\mathrm{K})^2 - \mathbf{1.155008\times 10^{6}}(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1500\ \mathrm{K}).$$

Figure V-52: Mean heat capacity (298.15 K – T) for CaO(cr).



V.6.2.1.1.3 Enthalpy of formation

Apart some questionable high-temperature studies (see below), the recent literature does not reveal any information that would lead to a revision of the value for the enthalpy of formation of CaO(cr) selected by CODATA [1989COX/WAG], derived principally from measurements of the enthalpy of combustion of Ca(cr) in O_2(g) by Huber and Holley [1956HUB/HOL] discussed in the relevant Appendix A.

Cox *et al.* [1989COX/WAG] indicated that measurements of the enthalpy of solution of Ca(cr) and CaO(cr) in aqueous acids (detailed in [1981GLU/GUR]) lead to values of lower accuracy. However, the analysis in [1994GUR/VEY] of these measurements leads to a value of the standard enthalpy of formation of – 635.5 kJ·mol^{-1}, (with no uncertainty), well within the uncertainty of the CODATA value, which is also that adopted in [1994GUR/VEY].

This review therefore maintains and selects:

$$\Delta_f H^\circ_m(\text{CaO, cr, 298.15 K}) = \mathbf{-\ (634.920 \pm 0.900)}\ \text{kJ·mol}^{-1}.$$

Subsequent to the CODATA assessment, several high-temperature equilibrium studies have reported data (mainly also involving CaC_2(cr)) which were interpreted by their authors to give markedly discrepant values of the enthalpies of formation of CaO(cr) (and also MgO(cr)), although the data for CaC_2(cr) have much larger uncertainties. These experiments are summarised in Table V-64.

Table V-64: Summary of discordant studies on $\Delta_f H^\circ_m$(CaO, cr).

Reference	Experimental study	Comments
[1968SWI]	p(CO, g) over CaC_2-CaO-C in CaO-SiO_2-Al_2O_3 slags, 1983 K	$\Delta_f G^\circ_m$ (CaC_2, cr) uncertain; CaC_2 solubility in slag?
[1970RAI/GRE]	$CaO(cr) + C(cr) \rightleftharpoons Ca(g) + CO(g)$, 1429-1525 K	Not inconsistent with the CODATA value within the large uncertainty
[1972EDM/TAY]	p(CO, g) over CaC_2,CaO-C in CaO-CaF_2 slags, 1693-1823 K	Unknown solubilities of CaC_2 and CaO in the slag
[1989WAK/SAN]	Ca-Ag(l) + CaO-saturated slag in graphite, 1570-1831 K	$\Delta_f G^\circ_m$ (CaC_2, cr) uncertain;
[1991CHR/GRI]	p(CO, g) over CaC_2, CaO, C 1753-1973 K	$\Delta_f G^\circ_m$ (CaC_2, cr) uncertain
[1993GOU/RAN]	$CaO(cr) \rightleftharpoons Ca(g) + O(g)$ in graphite susceptor, 1830-2072 K	Probable lowering of p(O,O_2) due to use of a graphite susceptor

However, these studies have been discredited by the potential-difference study in [1996JAC/VAR] (see Appendix A entry), which showed that the difference $(\Delta_f H^\circ_m(\text{MgO}) - \Delta_f H^\circ_m(\text{CaO}))$ in the range 950 to 1280 K is consistent with the CODATA selections. This paper also includes an excellent critical survey of the discordant studies, which is the basis of the experimental uncertainties, and those in the

subsidiary data required, listed in Table V-64. The Appendix A entries for all these references give further comments on these uncertainties.

It seems strange that most of the studies involving CaC_2(cr) attribute their discordant results to an error in $\Delta_f H^o_m$ (CaO, cr) when the uncertainties in the data for CaC_2(cr) are appreciably greater than those in lime—we estimate the uncertainty in $\Delta_f H^o_m$ (CaC_2, cr) alone to be ± 15 kJ·mol^{-1}. Ono *et al.* [1992ONO/KOB] have remeasured $\Delta_f G^o_m$ (CaC_2, cr) by equilibrating Ca-CaC_2 mixtures with a thin sheet of Fe held in a Fe crucible at 1223 to 1673 K, using the known activities of C in Fe. However, Jacob and Varghese [1996JAC/VAR] pointed out uncertainties even in these measurements, for example the unknown solubility of carbon in Ca(l).

Moreover, there are two more recent high-temperature studies which are in agreement with the assessed data: the study in [1982SAT/SAS] (see Appendix A) on the evaporation of CaO(cr) to Ca(g) and O(g) from 1773 to 2173 K, using a tungsten rather than a graphite heater, and that on the reaction of CaO(cr) with aluminium [1991SRI/JAC].

An overview of the more reliable high-temperature data is given in the Appendix A entry for [1996JAC/VAR].

Given the excellent consistency in the calorimetric data for $\Delta_f H^o_m$ (CaO, cr) and the uncertainties in these more difficult high-temperature experiments and the subsidiary data required for their interpretation, we do not think the discordant experiments, despite their apparent relative consistency, provide sufficient grounds for seriously questioning the CODATA value for this.

V.6.2.1.2 $Ca(OH)_2$(cr) (portlandite)

The experimental data for $Ca(OH)_2$(cr) have been discussed in some detail in [1994GUR/VEY].

V.6.2.1.2.1 Structure

Calcium hydroxide (found in nature as the mineral portlandite), possesses hexagonal symmetry, CdI_2-type (space group $P\bar{3}m1$) with lattice parameters a = (3.592 ± 0.001) and c = (4.906 ± 0.001) Å, as given in [1957BUS/LEV] from neutron-diffraction studies on synthetic crystals grown from interdiffusing NaOH and $CaCl_2$ solutions. More recently, in a study of the thermal motion of the hydrogen atoms in the compound, Desgranges *et al.* [1993DES/GRE] reported a = (3.589 ± 0.008) and c = (4.911 ± 0.014) Å at room temperature, using similarly prepared single crystals. Although Wyckoff [1963WYC] selected rather precise values, a = 3.5844 and c = 4.8962 Å, the many values reported in the literature show quite a variation. Therefore, the conservative uncertainties reported in [1993DES/GRE] may be taken as covering the range of reported data.

V.6.2.1.2.2 Heat capacity and entropy

The assessments in [1987GAR/PAR] and [1994GUR/VEY] adopt the same values for the heat capacity of $Ca(OH)_2$(cr), based on the low-temperature measurements by Hatton *et al.* [1959HAT/HIL], who made measurements from 19 to 330 K on a sample containing 99.75 % $Ca(OH)_2$(cr), the residue being $CaCO_3$(cr) and H_2O. The heat capacities were extrapolated from 19 to 0 K by assuming a constant value of $C^{\circ}_{p,\mathrm{m}}/T^3$. The resulting selected values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca(OH)_2, cr, 298.15\ K}) = \mathbf{(87.488 \pm 0.100)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{Ca(OH)_2, cr, 298.15\ K}) = \mathbf{(83.4 \pm 0.4)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

Kobayashi [1950KOB] measured the heat capacity from 314 to 670 K, and Lashchenko *et al.* [1935LAS/KOM] from 386 to 1317 K, with only those values up to *ca.* 780 K referring to the undecomposed hydroxide. We have accepted the fit to all these data given in [1987GAR/PAR] and [1994GUR/VEY]:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca(OH)_2}, \mathrm{cr}, T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{89.264} + \mathbf{3.3112\times 10^{-2}}\,(T/\mathrm{K}) - \mathbf{1.0355\times 10^{6}}\,(T/\mathrm{K})^{-2} \quad (298.15\text{-}700\ \mathrm{K}).$$

V.6.2.1.2.3 Enthalpy of formation

The basis for the selection of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Ca(OH)_2, cr, 298.15\ K})$ in [1994GUR/VEY] (involving the CODATA-recommended values for Ca^{2+} and CaO(cr)) is a series of consistent solution-calorimetric data involving Ca(cr), CaO(cr) and $Ca(OH)_2$(cr).

From the early studies in [1883THO] and [1897BER], carried out at 291 K, and those in [1928ROT/CHA], carried out at 323 K, on the difference in heat of dissolution in dilute hydrochloric acid of CaO(cr) and $Ca(OH)_2$(cr), Gurvich *et al.* [1994GUR/VEY] calculated values of -985.9, -984.1 and $-986.5\ \mathrm{kJ\cdot mol^{-1}}$ respectively, for $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Ca(OH)_2, cr, 298.15\ K})$, without indicating uncertainty limits. We note that Garvin *et al.* [1987GAR/PAR] assessed an uncertainty of $\pm 2.09\ \mathrm{kJ\cdot mol^{-1}}$ for the reported experimental data in [1883THO] and [1897BER] and $\pm 0.42\ \mathrm{kJ\cdot mol^{-1}}$ to those in [1928ROT/CHA].

Gurvich *et al.* [1994GUR/VEY] also calculated, from the difference of the enthalpies of solution of CaO(cr) and $Ca(OH)_2$(cr), the following values for $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{Ca(OH)_2, cr, 298.15\ K})$: $-(986.1 \pm 1.5)\ \mathrm{kJ\cdot mol^{-1}}$ based on the results in [1929THO/BRO] and [1930THO/BRO] (see Appendix A for [1930THO/BRO2]), $-(986.1 \pm 1.5)\ \mathrm{kJ\cdot mol^{-1}}$ based on the results in [1934SCH/HEY] and [1935SCH/PRA], and $-(985.9 \pm 1.0)\ \mathrm{kJ\cdot mol^{-1}}$ based on the studies in [1937WEL/TAY] and [1938TAY/WEL] (see the respective Appendix A entries). Garvin *et al.* [1987GAR/PAR] list an uncertainty of $\pm 0.42\ \mathrm{kJ\cdot mol^{-1}}$ for each of these three sets of reported experimental data.

Measurements in [1965HOP/WUL] (see Appendix A entry) of the enthalpy of solution of $Ca(OH)_2$(cr) in water were interpreted by Gurvich *et al.* [1994GUR/VEY] to give $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.2 ± 1.5) kJ·mol^{-1}. We note, however, that Garvin *et al.* [1987GAR/PAR] (their reaction 76) also take into account the calorimetric measurements of the enthalpy of solution of $Ca(OH)_2$(cr) in dilute HCl by these authors, and take larger uncertainty limits of ± 1.67 kJ·mol^{-1} when using the all of these results, noting that the $Ca(OH)_2$(cr) sample (certified reagent) was not further characterised. This leads to $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (984.79 ± 1.95) kJ·mol^{-1}.

In their evaluation Gurvich *et al.* [1994GUR/VEY] also use results on the enthalpy of hydration of CaO(cr) by Thorvaldson *et al.* [1930THO/BRO2] (see Appendix A) and by Taylor and Wells [1938TAY/WEL] (see Appendix A) that lead them to $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.9 ± 1.5) and – (985.5 ± 2.5) kJ·mol^{-1}, respectively. The larger uncertainty on the latter value may be due to the fact that Taylor and Wells [1938TAY/WEL] did not report any details of their measurements (although in their evaluation Garvin *et al.* [1987GAR/PAR] credited these results with a very high weighting).

Gurvich *et al.* [1994GUR/VEY] also considered the high-temperature calorimetric study by Müller and Prasser [1986MUL/PRA] (see Appendix A), not quoted in [1987GAR/PAR], of the decomposition of $Ca(OH)_2$(cr) to CaO(cr) and H_2O(g) and report from these experiments $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.6 ± 2.7) kJ·mol^{-1}.

Gurvich *et al.* [1994GUR/VEY] finally selected $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.9 ± 1.0) kJ·mol^{-1}, the same value as that selected in [1981GLU/GUR]. They note also a series of other studies of lower accuracy that yield values close to their selected value. These are two equilibrium studies on the dissociation of $Ca(OH)_2$(cr) in [1932TAM/SIO], [1953BER/RAS] (given zero weight in [1987GAR/PAR]), and [1957HAL/MOO], yielding $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.0 ± 2.0), – (988.3 ± 4.0) and – (985.0 ± 2.0) kJ·mol^{-1}, respectively, and data on the solubility of $Ca(OH)_2$(cr) in water in [1960GRE/COP] and [1967YEA/MAR], which give $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (983.7 ± 2.0) kJ·mol^{-1}.

The more comprehensive assessment of the solubility of $Ca(OH)_2$(cr) in this review in Section V.6.2.2.1.1 corresponds to $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.157 ± 1.008) kJ·mol^{-1} and our selected value is thus:

$$\Delta_f H^\circ_m (Ca(OH)_2, \text{cr}, 298.15\ \text{K}) = \mathbf{-\ (985.160 \pm 1.000)}\ \text{kJ·mol}^{-1},$$

which is the same as that assessed in [1987GAR/PAR] based on a large network of reactions involving many calcium compounds and is consistent with that selected in [1994GUR/VEY].

V.6.2.1.3 $CaCl_2$(cr)

V.6.2.1.3.1 Structure

Calcium chloride has an orthorhombic (distorted rutile) structure up to its melting point at 1045 K, space group *Pnnm*, with $a = 6.24$ Å, $b = 6.43$ Å $c = 4.20$ Å according to Bever *et al.* [1935BEV/NIE] and $a = 6.22$ Å, $b = 6.42$ Å $c = 4.15$ Å according to Döll and Klemm, [1939DOL/KLE].

V.6.2.1.3.2 Heat capacity and entropy

Kelley and Moore [1943KEL/MOO] measured the low-temperature heat capacity from 53 to 295 K of a sample of $CaCl_2$(cr) containing 0.59 % $MgCl_2$(cr), for which corrections were made. We have accepted the extrapolation down to 0 K used in [1987GAR/PAR] and [1994GUR/VEY], based on a comparison with $SrCl_2$(cr) and $BaCl_2$(cr), rather than that used in [1943KEL/MOO], which gives an unreasonably high value of S(54 K). Garvin *et al.* [1987GAR/PAR] combined these low-temperature data with the enthalpy measurements of Moore [1943MOO] from 404 to 1667 K (above the melting point of 1045 K). Their resultant heat capacities can be represented with sufficient accuracy by the equation:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaCl_2,cr},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{87.9697} - \mathbf{2.98445\times10^{-2}}\,(T/\mathrm{K}) + \mathbf{2.83591\times10^{-5}}\,(T/\mathrm{K})^2 - \mathbf{7.76527\times10^{5}}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1045\,\mathrm{K}).$$

The standard values, as calculated in [1987GAR/PAR] are:

$$C^{\circ}_{p,\mathrm{m}}\,(\mathrm{CaCl_2,\ cr,\ 298.15\ K}) = \mathbf{(72.857 \pm 0.400)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(\mathrm{CaCl_2,\ cr,\ 298.15\ K}) = \mathbf{(108.368 \pm 4.000)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

which this review selects. The (unwarranted) number of decimal digits are given to maintain consistency with other data.

V.6.2.1.3.3 Enthalpy of formation

The basis for the selection of $\Delta_f H^{\circ}_{\mathrm{m}}\,(\mathrm{CaCl_2,\ cr,\ 298.15\ K})$ in [1994GUR/VEY] (using the CODATA accepted values for Ca^{2+} and CaO(cr)) is a series of consistent solution-calorimetric data involving Ca(cr), CaO(cr) and $CaCl_2$(cr), although the authors mention in passing a report [1951SIE] of the combustion enthalpy of Ca(cr) in Cl_2(g) that leads to the value – 786 kJ·mol^{-1}, diverging by *ca.* 10 kJ·mol^{-1} from the other values. That study is not mentioned in the assessment by Garvin *et al.* [1987GAR/PAR].

The study of Ehrlich *et al.* [1963EHR/PEI] (see Appendix A entry), involving the dissolution enthalpy of Ca(cr) and $CaCl_2$(cr) in (HCl, 500 H_2O) led Gurvich *et al.* [1994GUR/VEY] to a value of – (799.3 ± 5.0) kJ·mol^{-1}. However, this study was partly discredited in [1987GAR/PAR], as they had reservations about the value used for the enthalpy of dissolution of the metal, and was not used in their fitting. Based mainly on the dissolution enthalpy in CaO(cr) in hydrochloric acid by Thorvaldson *et al.*

[1929THO/BRO], Torgeson and Shomate [1947TOR/SHO] (not used in [1987GAR/PAR]), Wells and Taylor [1937WEL/TAY] (see Appendix A) and Taylor and Wells [1938TAY/WEL] (see Appendix A entry), Gurvich *et al.* [1994GUR/VEY] calculated $\Delta_f H^o_m(CaCl_2, cr, 298.15\ K) = -(794.9 \pm 2.0)\ kJ \cdot mol^{-1}$. Using the measurements by Greyson and Snell [1969GRE/SNE] of the enthalpy of dissolution of $CaCl_2$(cr) in water (see Appendix A entry), together with other less accurate values that are mainly in agreement with these (see Appendix A entry), Gurvich *et al.* [1994GUR/VEY] obtained an extrapolated enthalpy of dissolution at infinite dilution of $-(81.4 \pm 0.3)\ kJ \cdot mol^{-1}$, and calculated $\Delta_f H^o_m(CaCl_2, cr, 298.15\ K) = -(795.9 \pm 1.0)\ kJ \cdot mol^{-1}$ (although our calculation gives $-(795.76 \pm 1.04)\ kJ \cdot mol^{-1}$). They suggest that this study gives the most accurate value. [1994GUR/VEY] finally select $\Delta_f H^o_m(CaCl_2, cr, 298.15\ K) = -(795.8 \pm 1.0)\ kJ \cdot mol^{-1}$.

Using essentially the same dissolution data but their much bigger network of reactions involving the calcium compounds, Garvin *et al.* [1987GAR/PAR] selected:

$$\Delta_f H^o_m(CaCl_2, cr, 298.15\ K) = \mathbf{-(795.39 \pm 1.00)}\ kJ \cdot mol^{-1}.$$

This is also the value selected by this review, in view of the absence of more recent data.

V.6.2.1.4 $CaSO_4$(cr) (anhydrite)

V.6.2.1.4.1 Structure

Anhydrous calcium sulfate (found in nature as the mineral anhydrite) obtained by dehydrating the dihydrate (gypsum) or the hemihydrate at 1173 to 1273 K has been known for many years to have an orthorhombic structure. Using X-ray and neutron diffraction, Hawthorne and Ferguson [1975HAW/FER] and Kirfel and Will [1980KIR/WIL] reported, respectively, $a = (6.993 \pm 0.002)$, $b = (6.995 \pm 0.002)$, $c = (6.245 \pm 0.001)$ Å and $a = (7.006 \pm 0.001)$, $b = (6.998 \pm 0.001)$, $c = (6.245 \pm 0.001)$ Å, space group *Amma*, while Morikawa *et al.* [1975MOR/MIN], from X-ray diffraction, give $a = (6.992 \pm 0.001)$, $b = (6.999 \pm 0.001)$, $c = (6.240 \pm 0.001)$ Å, space group *Bmmb*, equivalent to *Amma* with the *a* and *b* axes interchanged.

The orthorhombic form of $CaSO_4$ is accepted as the standard form, until its transition point at (1473 ± 10) K [1994GUR/VEY], at which it transforms to a cubic face-centred species with $a = 7.8$ Å as given by [1952FLO2], using high-temperature XRD.

Anhydrite and the material obtained by high-temperature treatment have a low solubility in water. According to Kelley *et al.* [1941KEL/SOU] and Kelley and King [1961KEL/KIN], when the α- and β-sulfate hemihydrates are dehydrated at very moderate temperatures two “soluble” or more reactive (especially towards re-hydration) species were obtained, called α- and β-soluble anhydrite. More recent studies in [1984LAG/ARM] and [1995BEZ/NON] also report the progressive thermal decomposition of gypsum towards a lower hydrate, to “soluble” anhydrite, designated as

γ-$CaSO_4$, and then as insoluble anhydrite. In this review, we shall only consider the data for the stable anhydrite mineral.

V.6.2.1.4.2 Heat capacity and entropy

There are two studies of the low-temperature heat capacity. Kelley *et al.* [1941KEL/SOU] made 23 measurements from 50 to 296 K on a sample of $CaSO_4$(cr) containing 0.09 % $CaSO_4 \cdot 2H_2O$; the data were extrapolated down to 0 K using a sum of one Debye and two Einstein functions. More recently Robie *et al.* [1989ROB/RUS] made more extensive measurements from 9 to 366 K on a sample containing the equivalent of 0.09 % K_2O. These data were consistently 0.4 to 1.4 % higher than the earlier values and have been preferred by this review in view of the much closer approach to 0 K, and the greater number of measurements. They give the standard values of:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaSO_4, cr, 298.15\ K}) = \mathbf{(101.2 \pm 0.2)}\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$$

and

$$S^{\circ}_{\mathrm{m}}(\mathrm{CaSO_4, cr, 298.15\ K}) = \mathbf{(107.4 \pm 0.2)}\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$$

which are the selected values.

Robie *et al.* [1989ROB/RUS] also made the only reliable measurements on the high-temperature heat capacity, from 339 to 990 K. These were obtained using DSC, so are more scattered than those from the low-temperature data, but agreed well with these in the overlap region. Other measurements were made by Kopp [1865KOP] from 291 to 319 K, and by Lashchenko and Kompanskii, [1928LAS/KOM] from 293 to 1393 K. The latter data were accepted by Kelley [1960KEL] whose equation is plotted in Figure A-22 in the Appendix A entry for [1989ROB/RUS]. However, as noted by Gurvich *et al.* [1994GUR/VEY], they are rather scattered and lead to unrealistically high $C^{\circ}_{p,\mathrm{m}}$ values at the higher temperatures, and have not been used in this review.

We have fitted the high-temperature data of Robie *et al.* [1989ROB/RUS] with a constraint of $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K}) = 101.2\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$ to give the equation:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaSO_4}, \mathrm{cr}, T)/\mathrm{J \cdot K^{-1} \cdot mol^{-1}} = \mathbf{98.9745} + \mathbf{7.67322 \times 10^{-2}}\,(T/\mathrm{K}) + \mathbf{1.514915 \times 10^{-5}}\,(T/\mathrm{K})^2 - \mathbf{1.71613 \times 10^{6}}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1000\ \mathrm{K}).$$

V.6.2.1.4.3 Enthalpy of formation

For the evaluation of $\Delta_f H^{\circ}_{\mathrm{m}}(\mathrm{CaSO_4, cr, 298.15\ K})$, data based on solution calorimetry, solubility measurements and high-temperature equilibria are available.

Gurvich *et al.* [1994GUR/VEY] listed explicitly those values they calculated from these three types of sources which they considered to be the most accurate. They also gave a number of references that permit the calculation of the enthalpy of formation of $CaSO_4$(cr), but were not considered further because either the material used lack characterisation or because the measurements are less accurate. *In fine*, however, the

value they adopt for $\Delta_f H_m^o(CaSO_4, cr, 298.15\ K)$ is based on three concordant sets of calorimetric data for the reaction:

$$CaSO_4(cr) + 2H_2O(l) \rightleftharpoons CaSO_4 \cdot 2H_2O(cr) \qquad (V.118)$$

from the measurements of Newman and Wells [1938NEW/WEL], $-(16.70 \pm 0.10)$ kJ·mol^{-1}, Southard [1940SOU], $-(16.90 \pm 0.10)$ kJ·mol^{-1} and Levy [1948LEV] $-(16.90 \pm 0.10)$ kJ·mol^{-1} of the dissolution of $CaSO_4$(cr) and $CaSO_4 \cdot 2H_2O$(cr) in dilute HCl, rejecting the discordant value from a similar determination by de Forcrand [1906FOR]. However, in our view, the value given by Levy [1948LEV] is merely an assessment of values from other authors rather than an independent measurement. The mean of the other two values gives $\Delta_f H_m^o((V.118), 298.15\ K) = -(16.80 \pm 0.10)$ kJ·mol^{-1}, which with our selected value for $\Delta_f H_m^o(CaSO_4 \cdot 2H_2O, cr, 298.15\ K)$, $-(2023.0 \pm 1.3)$ kJ·mol^{-1}, gives $\Delta_f H_m^o(CaSO_4, cr, 298.15\ K) = -(1434.54 \pm 1.30)$ kJ·mol^{-1} from these dissolution data.

In their evaluation of $\Delta_f H_m^o(CaSO_4, cr, 298.15\ K)$, Garvin *et al.* [1987GAR/PAR], in contrast to [1994GUR/VEY], did not mention the solution calorimetric data of [1906FOR], [1938NEW/WEL], [1940SOU] and [1948LEV] but involve their own adopted value for $\Delta_f H_m^o(CaSO_4 \cdot 2H_2O, cr, 298.15\ K) = -(2022.92 \pm 1.10)$ kJ·mol^{-1}, the enthalpy of solution measurements on $CaSO_4 \cdot 2H_2O$(cr) by Lange and Monheim [1930LAN/MON] and by Wallace [1946WAL] and the summary of the dehydration data given by Kelley *et al.* [1941KEL/SOU]. They also account for the entire literature on solubility data and high-temperature equilibria involving calcium sulfate and insert them in their whole network of CODATA-compatible data on calcium compounds, leading to their selected value of $\Delta_f H_m^o(CaSO_4, cr, 298.15\ K) = -(1434.508 \pm 1.300)$ kJ·mol^{-1}.

The detailed re-assessment of the solubility data for anhydrite by this review in Section V.6.2.2.5.4 gives (with the fitting constrained to reproduce the standard entropy of anhydrite of 107.4 J·K^{-1}·mol^{-1}) a value of $\Delta_f H_m^o(CaSO_4, cr, 298.15\ K) = -(1434.465 \pm 2.000)$ kJ·mol^{-1}.

Giving additional emphasis to the value assessed by [1987GAR/PAR], our selected value for the enthalpy of formation of anhydrite from these three very consistent values is:

$$\Delta_f H_m^o(CaSO_4, cr, 298.15\ K) = -(1434.51 \pm 1.30)\ kJ \cdot mol^{-1}.$$

This gives a temperature of 314.1 K for decomposition of gypsum to anhydrite at a pressure of 0.1 MPa, in agreement with the experimental values of (315 ± 2) K, see Section V.6.2.2.5.6.

V.6.2.1.5 $CaSO_4 \cdot 2H_2O$(cr) (gypsum)

V.6.2.1.5.1 Structure

The crystal structure of gypsum (often called selenite when in the form of very large pure crystals) was determined by Wooster [1936WOO] as monoclinic, space group *C*2/*c*, with four molecules per unit cell, using single crystal X-ray diffraction. All subsequent publications confirmed this structure, but the reported lattice parameters vary considerably, and some of these studies transformed the atomic positions into space group *I*2/*a*. This was the case of the neutron powder diffraction study of Schofield *et al.* [1996SCH/KNI] who reported the thermal expansion of deuterated gypsum between 4.2 and 320 K. More recently, Comodi *et al.* [2008COM/NAZ] reported a single crystal X-ray study of the high-pressure behaviour of a natural sample of gypsum up to 5 GPa. These authors indicated that the sample maintained its crystal structure up to (4.0 ± 0.1) GPa. At ambient pressure, the lattice parameters were given as a = (6.277 ± 0.002) Å, b = (15.181 ± 0.006) Å, c = (5.672 ± 0.002) Å and β = (114.11 ± 0.02) °.

V.6.2.1.5.2 Heat capacity and entropy

There are three reasonably concordant measurements of the low-temperature heat capacity of gypsum. Latimer *et al.* [1933LAT/HIC] measured the heat capacity of two samples, one with a crystallite size of *ca.* 1 μm, (studied down to 18.7 K) and one derived from a natural mineral, broken into pieces of *ca.* 1 cm size for which measurements were made from 62 to 274 K. There were small differences in the heat capacities of these materials, but the authors report the values $C^{\circ}_{p,\mathrm{m}}$ (298.15 K) = (187.4 ± 0.8) J·K^{-1}·mol^{-1}, S°_{m} (298.15 K) = (194.0 ± 0.8) J·K^{-1}·mol^{-1}. Shortly after, Kelley *et al.* [1941KEL/SOU] studied two natural samples of $CaSO_4 \cdot 2H_2O$(cr) from *ca.* 53 to 296 K for a selenite mineral, but only from 93.6 to 297.5 K for a gypsum mineral, with concordant results. More recently Robie *et al.* [1989ROB/RUS] measured the heat capacity of a very pure selenite mineral from 9 to 323.51 K (which is just above the decomposition temperature of gypsum). The quoted heat capacities and entropies at 298.15 K are given in Table V-65.

Table V-65: Experimental data for gypsum.

	Gypsum	
Reference	$C^{\circ}_{p,\mathrm{m}}$ (298.15 K) /J·K^{-1}·mol^{-1}	S°_{m} (298.15 K) /J·K^{-1}·mol^{-1}
[1933LAT/HIC]	(187.4 ± 0.8)	(194.0 ± 0.8)
[1941KEL/SOU]	183.7	(194.1 ± 1.70)
[1989ROB/RUS]	(187.2 ± 0.2)	(193.8 ± 0.3)

It is seen that these data are, on the whole quite consistent, and our selected values are those of Robie *et al.*:

$$C^{\circ}_{p,\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K}) = \mathbf{(187.2 \pm 0.2)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K}) = \mathbf{(193.8 \pm 0.3)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

Gypsum decomposes to anhydrite and water around 315 K (*ca.* 314.1 K with our selected data), so no further values for the heat capacity above 298.15 K are selected. If any values are required for the short range above 298.15 K, the linear extrapolation of the low-temperature data suggested by Robie *et al.* [1989ROB/RUS] can be used:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaSO_4{\cdot}2H_2O, cr}, T)\,/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = 76.885 + 0.37(T/\mathrm{K}) \qquad (298.15\text{-}325\,\mathrm{K}).$$

V.6.2.1.5.3 Enthalpy of formation

In their evaluation of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K})$, Garvin *et al.* [1987GAR/PAR] accounted for the entire literature on the enthalpy of solution, solubility data and high-temperature equilibria involving gypsum and inserted them in their whole network of CODATA-compatible data on calcium compounds, leading to their selected value of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K}) = -\,(2022.92 \pm 1.10)\ \mathrm{kJ{\cdot}mol^{-1}}$.

For the enthalpy of solution of gypsum at infinite dilution according to Reaction (V.119) at 298.15 K

$$\mathrm{CaSO_4{\cdot}2H_2O(cr)} \rightleftharpoons \mathrm{Ca^{2+} + SO_4^{2-} + 2H_2O(l)} \qquad \text{(V.119)}$$

Garvin *et al.* [1987GAR/PAR] made use of the values $-\,(1.142 \pm 0.209)\ \mathrm{kJ{\cdot}mol^{-1}}$ and $-\,(1.130 \pm 0.209)\ \mathrm{kJ{\cdot}mol^{-1}}$ they re-evaluated from the enthalpies of dilution of Lange and Monheim [1930LAN/MON] and Wallace [1946WAL], respectively. These values were also used by CODATA [1989COX/WAG] (see their note 135i on the assessment of $S^{\circ}_{\mathrm{m}}\,(\mathrm{Ca^{2+},\ aq})$). They lead to $-\,(2022.86 \pm 1.10)\ \mathrm{kJ{\cdot}mol^{-1}}$ and $-\,(2022.87 \pm 1.10)\ \mathrm{kJ{\cdot}mol^{-1}}$ for $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K})$, in total agreement with the rest of the literature data assessed in [1987GAR/PAR].

The detailed re-assessment of the solubility data for gypsum by this review in Section V.6.2.2.5 gives, with the fitting constrained to reproduce the standard entropy of gypsum of $193.8\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$, the very similar value of $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K}) = -\,(2023.095 \pm 1.500)\ \mathrm{kJ{\cdot}mol^{-1}}$.

From these very consistent assessments, we select:

$$\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(\mathrm{CaSO_4{\cdot}2H_2O,\ cr,\ 298.15\ K}) = \mathbf{-\,(2023.0 \pm 1.3)}\ \mathrm{kJ{\cdot}mol^{-1}}$$

for the standard enthalpy of formation of gypsum.

This, when combined with our selected values for anhydrite, gives a temperature of 314.1 K for the decomposition of gypsum to anhydrite at a pressure of 0.1 MPa, consistent with the experimental values, detailed in Section V.6.2.2.5.6.

V.6.2.1.6 $Ca_3(PO_4)_2$(cr)

V.6.2.1.6.1 Structure

This compound has three polymorphs stable at ambient pressure: the low-temperature β form, (rhombohedral, space group $R3c$, $a = (10.4352 \pm 0.0002)$ Å, $c = (37.4029 \pm 0.0005)$ Å in the hexagonal setting [2003YAS/SAK]), stable up to *ca.* 1383 K, the monoclinic α-form, stable up to *ca.* 1745 K, and the α′-form, stable up to the melting point [1959NUR/WEL], 1969FIX/HEY]. There is also a γ-polymorph stable at higher pressures [1987SUG/TOK]. We shall deal only with β-$Ca_3(PO_4)_2$ in any detail, but some further mention is also made of α-$Ca_3(PO_4)_2$.

V.6.2.1.6.2 Heat capacity and entropy

Southard and Milner [1935SOU/MIL] measured the low-temperature $C^{\circ}_{p,\mathrm{m}}$ of both forms from *ca.* 15.3 to 297.5 K (286.6 K for the α-form), and Britske and Veselovskii [1937BRI/VES] the high-temperature enthalpy contents at 11 temperatures from 871 to 1571 K, using a drop calorimeter. As noted in Appendix A, some of the experimental details in the latter paper are far from clear. The authors indicated a β → α transition temperature of (1373 ± 10) K, with an approximate enthalpy of $\Delta_{\mathrm{trs}}H^{\circ} = 19$ kJ·mol^{-1}. This temperature is slightly lower than those derived from phase diagram studies ([1959NUR/WEL], [1969FIX/HEY]) and we suggest a value of (1383 ± 10) K to cover the range of values. We have combined these studies by re-fitting the high-temperature data for the β-phase (see Figure V-53 below), subject to the constraint of $C^{\circ}_{p,\mathrm{m}}$ ($Ca_3(PO_4)_2$, cr, β, 300 K) = 229.116 J·K^{-1}·mol^{-1} as given by the more precise measurements in [1935SOU/MIL]. The resulting equation for $C^{\circ}_{p,\mathrm{m}}$ is:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_3(PO_4)_2, cr, \beta}, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{208.755} + \mathbf{0.159058}(T/\mathrm{K}) - \mathbf{2.46211{\times}10^{6}}(T/\mathrm{K})^{-2} \quad (298.15\text{-}1383\,\mathrm{K})$$

from which we derive and select:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_3(PO_4)_2, cr, \beta}, 298.15\ \mathrm{K}) = \mathbf{(228.481 \pm 1.600)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

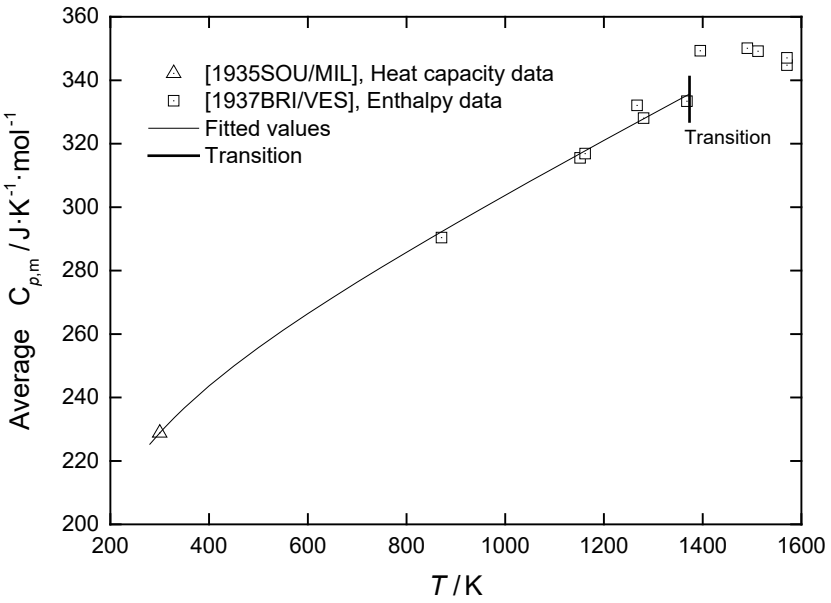
We also adopt from [1935SOU/MIL]:

$$S^{\circ}_{\mathrm{m}}(\mathrm{Ca_3(PO_4)_2, cr, \beta}, 298.15\ \mathrm{K}) = \mathbf{(236.0 \pm 1.6)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

and give, for information S°_{m} ($Ca_3(PO_4)_2$, cr, α, 298.15 K) = (240.9 ± 0.6) J·K^{-1}·mol^{-1} as obtained by these authors.

We shall not deal further with the α-phase here.

Figure V-53: Mean heat capacity of β-$Ca_3(PO_4)_2$.



V.6.2.1.6.3 Enthalpy of formation

For their evaluation of $\Delta_f H^\circ_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – 4120.82 kJ·mol^{-1}, Glushko *et al.* [1979GLU/MED], cite four references: [1963JAC], [1963MEA/RIC], [1967COR/OUW] and [1972KAN/UME].

Glushko *et al.* did not take into consideration the results of Smirnova *et al.* [1962SMI/ILL] that concern the enthalpy of formation of tricalcium phosphate (α- and β-modifications), as well as those of hydroxylapatite and fluorapatite, by solution calorimetry. As mentioned in the relevant Appendix A, this review believes that the results of Smirnova *et al.* on calcium phosphate should only be considered as informative in view of the very limited purity (estimated to 90 %) of the material used.

The results obtained by Jacques [1963JAC] that concern beta tricalcium phosphate, hydroxylapatite and fluorapatite have been analysed in considerable detail by this review (see the relevant Appendix A entry). Although it was virtually impossible to retrace exactly the various calculations made by Jacques, as no detailed information was given on some of the auxiliary values used in the cycles, in particular the precise concentrations of the nitric and phosphoric acid solutions, we have re-calculated the various enthalpies of formation, using CODATA or CODATA-compatible values.

Two values for $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) were obtained using two distinct but related cycles: – (4127.140 ± 5.230) kJ·mol^{-1} and – (4130.591 ± 5.778) kJ·mol^{-1}, yielding, with a 2σ uncertainty, a weighted mean of – (4128.691 ± 7.755) kJ·mol^{-1}.

However, in view of the reservations expressed in the examination of the results of Jacques, we feel that an uncertainty interval of ± 12.0 kJ·mol^{-1} would more adequately represent these results, thus, $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4128.7 ± 12.0) kJ·mol^{-1}.

The study of Meadowcroft and Richardson [1963MEA/RIC] concerns the measurement, at 308.15 K, of the enthalpies of solution in aqueous 6 molar HCl of a series of crystalline and glassy phosphates, of which only $Ca_3(PO_4)_2$(cr) is of interest to this review. The results have been analysed in considerable detail as discussed in the relevant Appendix A entry. Among the weaknesses of this paper, one should mention that there was no clear identification of the phase composition of the $Ca_3(PO_4)_2$ used in the dissolutions and the fact that the calorimetric results were not reported individually, their uncertainty being given only as "full spread of results".

The thermodynamic cycle used to obtain $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, 298.15 K) involved P_2O_5(cr), $CaCO_3$(cr) and CO_2(g). In view of the above uncertainties, the experimental results were assumed to be valid at 298.15 K. In the re-calculation of the results we made use of $\Delta_f H^o_m$ ($CaCO_3$, calc, 298.15 K) = – (1207.5 ± 1.2) kJ·mol^{-1} from this review, $\Delta_f H^o_m$ (CO_2, g, 298.15 K) = – (393.51 ± 0.13) kJ·mol^{-1} from CODATA, and $\Delta_f H^o_m$ (P_2O_5, cr, 298.15 K) = – (1505.05 ± 1.60) kJ·mol^{-1} from the thorough assessment by Gurvich *et al.* [1989GUR/VEY]. This is the value compatible with the CODATA phosphate data and is fully up-to-date since a recent review on phosphorus species [2012JUN/HUD] did not reveal new studies.

We thus obtain from this source $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4118.5 ± 10.0) kJ·mol^{-1}; the uncertainty limits are believed to cover both the uncertainties in the identity of the calcium phosphate phase used and on its impurity level.

The work on Cordfunke and Ouweltjes [1967COR/OUW] concerning the determination by solution calorimetry of the standard enthalpies of formation of Ag_3PO_4(cr) and Ag_3AsO_4(cr) made use of thermodynamic cycles that involved CaO(cr), $Ca_3(PO_4)_2$(cr) and $Ca_3(AsO_4)_2$(cr). However, this paper did not produce any new data on $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr), as a literature value was used in the cycle. Consequently, this paper should not really have been given in [1979GLU/MED] in their reference list for tricalcium phosphate.

Reference was also given in [1979GLU/MED] to the study of Kanazawa and Umegaki [1972KAN/UME] that concerns the heat effects upon dissolution of $Ca_3(PO_4)_2$ (cr, β), fluorapatite and various calcium phosphate compounds in hydrochloric acid and citric acid at various concentrations, and the kinetics of these reactions.

However, this review fails to see how meaningful standard enthalpy of formation values could have been extracted from the results presented.

More recently, Ben Abdelkader *et al.* [1999BEN/BEN] reported a study on the preparation, characterisation and determination of the enthalpy of formation of β-$Ca_3(PO_4)_2$ (and of $Mg_3(PO_4)_2$ that is not relevant to this review), based on solution calorimetry in HNO_3, $5.36H_2O$(sln).The relevant Appendix A gives details of the characterisation of the samples and of the calorimetric procedure. The enthalpy of solution in the medium of $Ca(NO_3)_2 \cdot 4H_2O$(cr), and (H_3PO_4, $0.756H_2O$), involved in the cycle, were taken, without discussion, from the thesis of Ben Cherifa [1988BEN] that was not available to this review. The values taken for the standard enthalpies of formation of these two species were the same as those in [1982WAG/EVA]. Ben Abdelkader *et al.* reported $\Delta_f H^\circ_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – 4056 kJ·mol^{-1} without any indication of uncertainty limits. They noted, without further comments, that this result was some 60 kJ·mol^{-1} less negative than literature values. They also observed that a very large discrepancy in the same direction was affecting their result on $Mg_3(PO_4)_2$(cr). The use of the values accepted by CODATA for the phosphoric acid species would shift the result in [1999BEN/BEN] to $\Delta_f H^\circ_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) ≈ – 4070 kJ·mol^{-1}, a value still unacceptably too positive. Under these circumstances, the results of Ben Abdelkader *et al.* [1999BEN/BEN] are given for information only.

Thus only the results from the studies of Jacques [1963JAC], $\Delta_f H^\circ_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4128.7 ± 12.0) kJ·mol^{-1} and of Meadowcroft and Richardson [1963MEA/RIC] – (4118.5 ± 10.0) kJ·mol^{-1} deserve to be taken into consideration.

A straight average of these two relatively consistent results leads to the value selected in this review:

$$\Delta_f H^\circ_m (Ca_3(PO_4)_2, \text{cr}, \beta, 298.15\ \text{K}) = \mathbf{-\ (4123.6 \pm 12.0)}\ \text{kJ·mol}^{-1},$$

keeping the larger uncertainty limits. It is very regrettable that the literature lacks more precise data for such an important compound.

V.6.2.1.7 $Ca(H_2PO_4)_2 \cdot H_2O$(cr)

This compound, $Ca(H_2PO_4)_2 \cdot H_2O$(cr), is the only calcium phosphate hydrate briefly considered in this review, as its standard enthalpy of formation is needed in the evaluation of the results of Gottschal [1958GOT] on the enthalpy of formation of hydroxylapatite and fluorapatite.

V.6.2.1.7.1 Structure

Based on goniometric measurement, Haushofer [1883HAU] reported some 140 years ago the crystal morphology of $Ca(H_2PO_4)_2 \cdot H_2O$(cr) as triclinic pinacoidal. Single crystal X-ray data were reported, in general agreement, by Smith *et al.*

[1955SMI/LEH] and by MacLennan and Beevers [1956MAC/BEE]. The triclinic $P\bar{1}$ structure was refined by Dickens and Bowen [1971DIC/BOW] who reported, at 298.15 K, a = (5.6261 ± 0.0005) Å, b = (11.889 ± 0.002) Å, c = (6.4731 ± 0.0008) Å, α = (98.633 ± 0.006) °, β = (118.262 ± 0.006) ° and γ = (83.344 ± 0.006) ° with two molecules per unit cell. This X-ray diffraction study also indicated the approximate positions of the hydrogen atoms. The hydrogen bonding in $Ca(H_2PO_4)_2{\cdot}H_2O$(cr) was more precisely determined by Schroeder *et al.* [1975SCH/PRI]. The positional parameters did not differ significantly from the refinements of Dickens and Bowen but a few thermal parameters differed by three or four standard deviations.

V.6.2.1.7.2 Heat capacity and entropy

The low-temperature heat capacity was measured from 10 to 345 K by Egan and Wakefield [1956EGA/WAK] and the entropy at 298.15 K calculated by graphical integration, the extrapolation to 0 K being made with a Debye function with a Debye temperature of 133 K. With the tiny correction to 1981 molecular weights, the calculated values at 298.15 K are our selected values:

$$C^{\circ}_{p,\mathrm{m}}\,(Ca(H_2PO_4)_2{\cdot}H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(258.84 \pm 0.25)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(Ca(H_2PO_4)_2{\cdot}H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(259.84 \pm 0.42)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

where the authors' conservative estimate of the accuracy of their heat capacities (± 0.05 %) has been doubled.

V.6.2.1.7.3 Enthalpy of formation

There are rather few studies on the enthalpy of formation of this compound. Egan *et al.* [1956EGA/WAK] made use of Reaction (V.120):

$$2(H_3PO_4, \mathrm{in}\,1.0055\,H_2O)(\mathrm{sln}) + Ca(OH)_2\,(\mathrm{cr}) \rightleftharpoons Ca(H_2PO_4)_2{\cdot}H_2O(\mathrm{cr}) + H_2O(\mathrm{l}) \quad \text{(V.120)}$$

by measuring the difference between the sums of the heats of solution of the products and of the reactants in 4 molal hydrochloric acid at 298.15 K under such conditions that the ionic concentrations of the solutions of products and reactants were identical. Details concerning the purity of the reagents, the calorimetric procedure and the experimental results are given in the relevant Appendix A entry.

A re-calculation by this review of the results with CODATA and CODATA-compatible values (including that for (H_3PO_4, in 100 H_2O)(sln)) leads to:

$$\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}\,(Ca(H_2PO_4)_2{\cdot}H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{-\,(3423.405 \pm 2.295)}\ \mathrm{kJ{\cdot}mol^{-1}}$$

and this is our selected value.

Although the anhydrous species, $Ca(H_2PO_4)_2$(cr), is not treated in detail in the present review, it may be useful to mention and comment on the results of Egan *et al.* [1956EGA/WAK] for this species. They obtained, for Reaction (V.121):

$$2(H_3PO_4, \text{ in } 1.0055H_2O)(sln) + Ca(OH)_2(cr) \rightleftharpoons Ca(H_2PO_4)_2(cr) + 2H_2O(l) \qquad (V.121)$$

in a process similar to that used for the monohydrate. Our re-calculation yields:

$$\Delta_f H^o_m (Ca(H_2PO_4)_2, cr, 298.15 \text{ K}) = -(3118.440 \pm 2.302) \text{ kJ·mol}^{-1}.$$

These results yield the enthalpy of dehydration according to Reaction (V.122):

$$Ca(H_2PO_4)_2 \cdot H_2O(cr) \rightleftharpoons Ca(H_2PO_4)_2(cr) + H_2O(l) \qquad (V.122)$$

as $\Delta_f H^o_m$((V.122), 298.15 K) = (19.135 ± 0.542) kJ·mol^{-1}, keeping uncertainty limits that avoid redundancy. Egan *et al.* [1956EGA/WAK] report for Reaction (V.122), a value of 10.293 kJ·mol^{-1}, likely due, as noted in Appendix A, to the use in the calculations (but not in their text) of a positive rather than negative enthalpy of solution of $Ca(H_2PO_4)_2$(cr) of the same magnitude. The larger value for Reaction (V.122) is fully consistent with the values given in Wagman *et al.* [1982WAG/EVA]. In their early calorimetric study of calcium phosphates, Matignon and Séon [1932MAT/SEO] (see Appendix A entry) report, from the dissolution of the monohydrated and anhydrous salts in dilute hydrochloric acid, a value of 20.7 kJ·mol^{-1} for the dehydration process, in good agreement with the larger value above.

It should be mentioned that Sorokin [1941SOR] reported the vapour pressure of $Ca(H_2PO_4)_2 \cdot H_2O$(cr) at temperatures in the range 313 to 363 K and calculated, for the dehydration process at 363 K, a value of 7.63 kJ·mol^{-1}. Unfortunately the pressures observed in these measurements were different when approaching equilibrium from higher and lower pressures, indicating that equilibrium may not have been reached. Moreover, a satisfactory calculation of these results at 298.15 K is not possible owing to the lack of heat capacity data for $Ca(H_2PO_4)_2$(cr).

V.6.2.1.8 $CaCO_3$ (cr)

V.6.2.1.8.1 Structure and polymorphism

Calcium carbonate is found in nature in three modifications, calcite, aragonite and the far less common vaterite. Calcite, taken as the standard state, is the only equilibrium modification up to the melting point at 1603 K. It displays a hexagonal structure, space group $R\bar{3}c$ with a = 4.9899 Å, c = 17.064 Å, as given by Graf [1961GRA] at 299 K for a synthetic material. Graf also gives a broad discussion of the structure of many isomorphous carbonates (including magnesium carbonate and dolomite). Concordant values, a = (4.988 ± 0.001) Å, c = (17.061 ± 0.001) Å at 297 K, are given by Markgraf and Reeder [1985MAR/REE] for a natural sample (containing 0.2 at % Mg). These authors reported a structural refinement of calcite up to 1173 K and made similar measurements for magnesite up to 773 K.

Aragonite is orthorhombic, space group *Pmcn*. Based on single crystal X-ray data on a mineral sample containing 0.2 at % Sr, 0.05 at % Pb and 0.04 at % Mg, de Villiers [1971VIL] reported a = (4.9614 ± 0.0003), b = (7.9671 ± 0.0004) and

c = (5.7404 ± 0.0004) Å at 294 K, while, in a nearly simultaneous paper, Dal Negro and Ungaretti [1971NEG/UNG] give a = (4.9616 ± 0.0002), b = (7.9705 ± 0.0006) and c = (5.7394 ± 0.0004) Å using the same technique, on a natural sample also, with a Ca content of 99.9 % of the theoretical amount.

At ambient pressure aragonite is metastable with respect to calcite at all temperatures up the decomposition temperature.

Since vaterite is not found in natural carbonate sediments (Lippmann [1973LIP]) we shall not deal in any detail with the thermodynamic properties of this polymorph, but only summarise the details of the structure. It is also relevant to note that Wolf *et al.* [2000WOL/KON] have recently suggested that traces of water are needed to stabilise this phase.

It is reasonably well-established that vaterite possesses a hexagonal structure, but the detailed space group, dimensions of the unit cell, and ordering of the carbonate ions are still uncertain. From X-ray and optical data on single crystals grown by ionic diffusion in aqueous solutions Kamhi [1963KAM] inferred a space group $P6_3/mmc$, with a prominent pseudo-cell with a' = (4.13 ± 0.01), c' = (8.49 ± 0.02) Å, and Z = 2. However, the presence of weak superstructure reflections, apparently due to the partial ordering of the carbonate ions, suggested a true cell rotated 30 ° from the pseudo-cell, with $a = \sqrt{3}\,a'$ = (7.15 ± 0.02), $c = 2c'$ = (16.98 ± 0.04) Å , and Z = 12 but details of the structure were not given.

Meyer [1959MEY] first reported an orthorhombic structure, space group *Pbnm,* but in a later paper [1969MEY], he proposed a hexagonal structure, with space group and lattice parameters consistent with the results of Kamhi [1963KAM], but with a different carbonate site symmetry, and noted some evidence for a true cell with $c = 3c'$ and Z = 18. An idealised structure, space group $P6_322$, was developed by Lippmann [1973LIP], based on the same structure of $YbBO_3$ in [1966BRA/GRA].

V.6.2.1.9 $CaCO_3$(calc)

V.6.2.1.9.1 Heat capacity and entropy

There are three extensive measurements of the low-temperature $C^{\circ}_{p,\mathrm{m}}$ ($CaCO_3$, calc), by Anderson [1934AND2] (from 57 to 296 K), Staveley and Linford [1969STA/LIN] (11 to 303 K) and Nan *et al.* [2007NAN/SHI] (80 to 390 K). The data from the first two studies agree reasonably well, but as noted in Appendix A, the $C^{\circ}_{p,\mathrm{m}}$ values from the most recent study by Nan *et al.* [2007NAN/SHI] are noticeably higher than the earlier values at lower temperatures, but are in very good agreement at 298.15 K. Simon and Swain [1935SIM/SWA] also made ten measurements from 15 to 80 K which are 2-3 % higher than those in [1934AND2] and [1969STA/LIN]. Because of the much bigger temperature range in the study by Staveley and Linford [1969STA/LIN], their values at 298.15 K have been selected, despite the regrettable lack of analysis of the mineral sample studied.

The standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaCO_3}, \text{calc}, 298.15\ \mathrm{K}) = \mathbf{(83.472 \pm 0.420)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

and

$$S^{\circ}_{\mathrm{m}}(\mathrm{CaCO_3}, \text{calc}, 298.15\ \mathrm{K}) = \mathbf{(91.710 \pm 0.210)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

where the uncertainties are taken from Garvin *et al.* [1987GAR/PAR], who selected the same values.

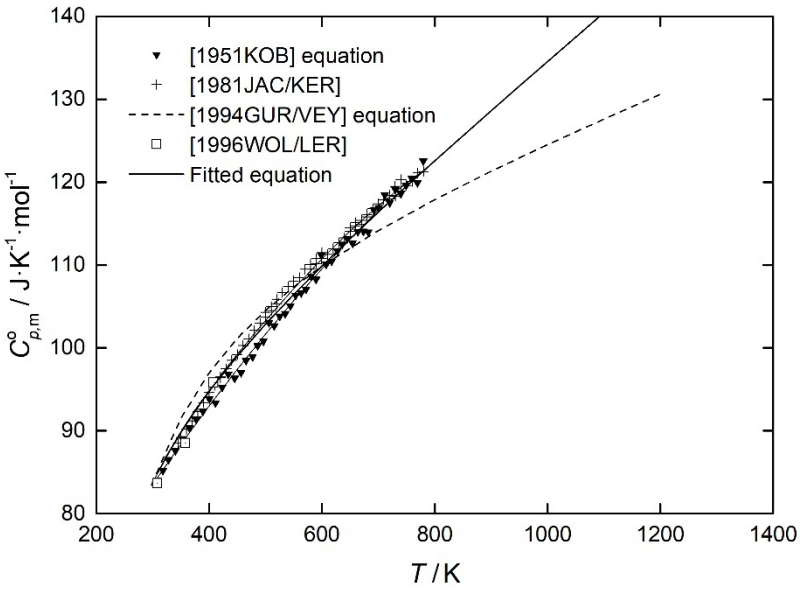
There are two measurements of the enthalpy content of calcite, by Magnus [1913MAG] (a single determination at 659 K) and von Gronow and Schweite, [1933GRO/SCH] (eight points up to 1173 K) and two detailed studies of the high-temperature $C^{\circ}_{p,\mathrm{m}}$, using DSC, by Kobayashi [1951KOB] (48 points from 308 to 780 K) and Jacobs *et al.* [1981JAC/KER] (61 points from 345 to 775 K), plus several DSC measurements by Nan *et al.* [2007NAN/SHI] up to 390 K. These measurements are all reasonably consistent, although those in [1933GRO/SCH] above 1000 K seem slightly low. They have been fitted by the following equation, with the constraint of $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K}) = (83.472)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$.

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaCO_3}, \text{calc}, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{78.78075} + \mathbf{5.68682 \times 10^{-2}}\,(T/\mathrm{K}) - \mathbf{1.09019 \times 10^{6}}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1200\ \mathrm{K}).$$

The four values given more recently by Wolf *et al.* [2000WOL/KON] are slightly lower, but well within their uncertainty ($\pm$ 0.6 $\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$); the similar values measured by DSC by the same team [1996WOL/LER] are also consistent within their larger uncertainties (1.1 to 2.2 $\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$).

The $C^{\circ}_{p,\mathrm{m}}$ values given by the equation given in [1994GUR/VEY] (and the similar data given earlier in [1960KEL]) are distinctly lower above 700 K, but we note that that in this temperature region, the heat capacities of calcite given by both these assessments are lower than those of $\mathrm{MgCO_3}$(cr), which seems anomalous. Figure V-54 summarises the heat capacity data.

Figure V-54: High-temperature heat capacity of calcite.



V.6.2.1.9.2 Enthalpy of formation

Our knowledge on the standard enthalpy of formation of calcite is based on determinations of the enthalpies of solution of CaO(cr) and calcite in hydrochloric acid, the analysis of solubility measurements of calcite and the examination of the equilibrium of decomposition of calcite to CaO(cr) and CO_2(g).

The solution-calorimetric measurements used by Gurvich *et al.* [1994GUR/VEY] in their evaluation of $\Delta_f H^{\circ}_m$ ($CaCO_3$, cr, 298.15 K), together with the values obtained, are shown below in Table V-66.

Table V-66: Solution calorimetric results assessed by Gurvich *et al.* [1994GUR/VEY] in the evaluation of $\Delta_f H^{\circ}_m$ ($CaCO_3$, calc, 298.15 K).

Reference	Solvent	$\Delta_f H^{\circ}_m$ (298.15 K) /kJ·mol^{-1}
[1925BAC] [(a)], [1925BAC2]	1 molar HCl(sln), saturated with CO_2(g)	– 1206.8
[1928ROT/CHA]	2 molar HCl (2.25m) at 323 K	– (1206.4 ± 1.0)
[1934BIL/ROH] [(b)]	30 % HCl(sln), 363 K	– (1207.0 ± 1.0)
[1941ROT/BER] [(c)]	2.42 molar HCl(sln)	– (1205.6 ± 1.0)

(a) See Appendix A, covering also [1925BAC2]
(b) No information was given on the purity of the calcite (presumably single crystal) used. This reference is not listed in [1987GAR/PAR].
(c) These results were discarded in [1987GAR/PAR] as being largely deviant.

Gurvich *et al*. [1994GUR/VEY] do not mention the solution data of Hull and Turnbull [1973HUL/TUR] who reported the enthalpy of solution of calcite in CO_2-saturated (at 0.098 MPa) 0.1 mol·dm^{-3} HCl (0.01 mole $CaCO_3$ in 525 g 0.1 mol·dm^{-3} HCl) as – (14.121 ± 0.063) kJ·mol^{-1}, after a correction of – 1.36 kJ·mol^{-1} for the saturation of water of the evolved CO_2. This result was used in the evaluation by Garvin *et al.* [1987GAR/PAR] with increased uncertainty limits of ± 0.209 kJ·mol^{-1}.

Gurvich *et al*. [1994GUR/VEY] deduced the value $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1206.5 \pm 2.0)$ kJ·mol^{-1} from the evaluation by Kelley and Anderson [1935KEL/AND] of the literature of the thermodynamics of metal carbonates (in particular their solubilities), and the value of $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1207.3 \pm 1.5)$ kJ·mol^{-1} from the study of Jacobson and Langmuir [1974JAC/LAN] that was centred on the solubility of calcite between 273.15 and 323.15 K, taking into account the formation of $CaCO_3$(aq) and $CaHCO_3^+$. The evaluation in [1935KEL/AND] was not, *per se*, mentioned by Garvin *et al.* [1987GAR/PAR] who however made full use of the study by Jacobson and Langmuir [1974JAC/LAN], which does take into account all the references involving calcite in Kelley and Anderson [1935KEL/AND]. In addition, Garvin *et al.* [1987GAR/PAR] made use of the more recent study by Plummer and Busenberg [1982PLU/BUS] (not mentioned in [1994GUR/VEY] in which all the references predate 1981) who measured the solubility of calcite, aragonite and vaterite in CO_2-H_2O solutions between 273 and 363 K and developed a model involving the species $CaCO_3$(aq) and $CaHCO_3^+$. From this source, Garvin *et al.* [1987GAR/PAR] (their reaction 337) obtained $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1208.6 \pm 1.6)$ kJ·mol^{-1}, a value somewhat more negative than those obtained from earlier solubility studies.

More recently Chai and Navrotsky [1993CHA/NAV] reported a drop-calorimetry study from which we calculate (see relevant Appendix A entry) a value of – (178.794 ± 2.000) kJ·mol^{-1} for the enthalpy of formation of calcite from CaO(cr) and CO_2(g), which corresponds to $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1207.2 \pm 2.2)$ kJ·mol^{-1}.

As a result of their assessment, Gurvich *et al.* [1994GUR/VEY] recommended $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1206.6 \pm 1.0)$ kJ·mol^{-1}, while Garvin *et al.* [1987GAR/PAR] selected $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) = -(1207.586 \pm 1.000)$ kJ·mol^{-1}. It is not unlikely that the slightly more negative value than (but overlapping with) that selected in [1994GUR/VEY] is due to a different appreciation and treatment of the data arising from solubility determinations. We also note that the uncertainties adopted in these two assessments may be somewhat low given the uncertainties by CODATA for $\Delta_f H^o_m(Ca^{2+}, 298.15\ K)$ and $\Delta_f H^o_m(CO_3^{2-}, \text{aq}, 298.15\ K)$, respectively 1.0 and 0.25 kJ·mol^{-1}.

The literature on the solubility of calcite has been extensively analysed in the present review (see Section V.6.2.2.4.2), leading to $\Delta_f H^o_m(CaCO_3, \text{calc}, 298.15\ K) =$

– (1207.288 ± 1.500) kJ·mol^{-1}, in good agreement with the value selected by Garvin *et al.* [1987GAR/PAR]. Taking into account both calorimetry and solubility data this review selects the value:

$$\Delta_f H_m^o(CaCO_3, \text{calc}, 298.15\ \text{K}) = \mathbf{-\ (1207.5 \pm 1.2)}\ \text{kJ·mol}^{-1},$$

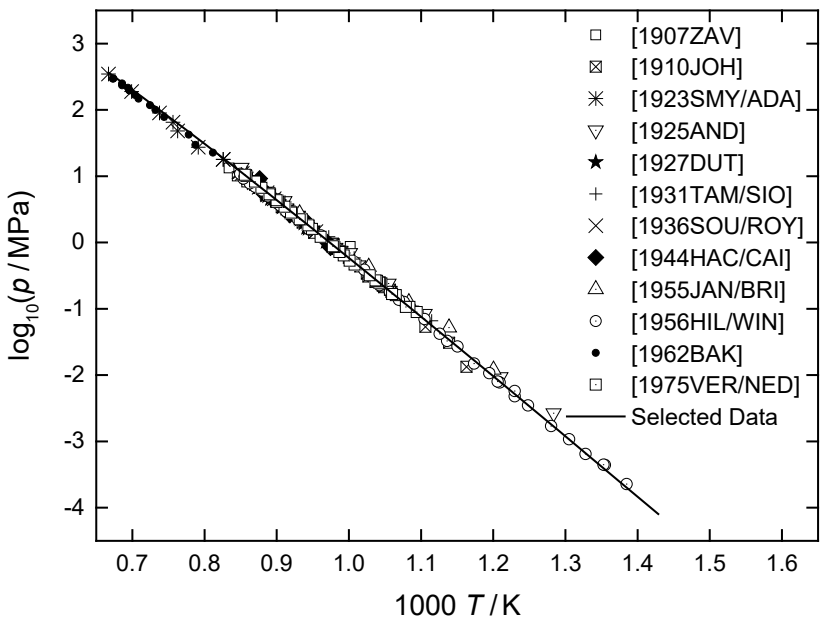
keeping conservative uncertainty limits.

There are at least 11 studies on the equilibrium for the decomposition reaction:

$$CaCO_3(\text{calc}) \rightleftharpoons CaO(\text{cr}) + CO_2(\text{g}) \qquad \text{(V.123)}$$

from which the stability of calcite can be derived; both [1994GUR/VEY] and [1987GAR/PAR] included these in their assessments. Figure V-55 shows that these studies are all in excellent agreement with the values of CaO(cr) and $CaCO_3$(calc) selected in this review. More recently, Forbes and Radha [2011FOR/RAD] (see Appendix A) reported dissolution and water-adsorption calorimetric studies of the surface energy of calcite, using a bulk sample and four commercial nanophase samples. These results, which were aimed at understanding the effect of particle size on the stability of the calcium carbonate phases, do not contain any information that would affect the above assessments of the standard enthalpy of formation of calcite.

Figure V-55: Decomposition pressure of $CaCO_3$(calc).



V.6.2.1.10 $CaCO_3$(arag)

V.6.2.1.10.1 Heat capacity and entropy

Two of the three studies of the low-temperature $C^{\circ}_{p,\mathrm{m}}$ of calcite also studied the low-temperature $C^{\circ}_{p,\mathrm{m}}$ of aragonite: Anderson [1934AND2] (from 57 to 296 K) and Staveley and Linford [1969STA/LIN] (22 to 291 K). As for calcite, the data are in very good agreement and the value of the standard entropy is taken from the latter more extensive study in [1969STA/LIN]. The earlier measurements by Günther [1916GUN] from 23-56 K are 25-40 % lower than the consistent later measurements and have been ignored.

The selected standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaCO_3}, \text{arag}, 298.15\ \mathrm{K}) = \mathbf{(82.320 \pm 0.200)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

and

$$S^{\circ}_{\mathrm{m}}(\mathrm{CaCO_3}, \text{arag}, 298.15\ \mathrm{K}) = \mathbf{(87.990 \pm 0.210)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

where the uncertainty of S°_{m} is taken from the fitting of [1987GAR/PAR], who selected the same value.

The heat capacity above 298.15 K has been measured by Kobayashi [1951KOB2] from 303 to 735 K in a heat conduction calorimeter; as noted in Appendix A, only the values up to 573 K refer to pure aragonite. More recently, Wolf *et al.* [1996WOL/LER] have also reported three measurements at 308, 358 and 408 K, though these have higher uncertainties. All these values have been fitted by the following equation:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaO_3}, \text{arag}, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{79.4282} + \mathbf{4.9320 \times 10^{-2}}\,(T/\mathrm{K}) - \mathbf{1.05010 \times 10^{6}}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}600\ \mathrm{K}).$$

V.6.2.1.10.2 Enthalpy of formation

The value for the standard enthalpy of formation of aragonite stems from a limited number of determinations of the difference between the enthalpy of solution of calcite and aragonite, the analysis of the transition between the two crystallographic forms and a larger number of solubility determinations.

Gurvich *et al.* [1994GUR/VEY] do not give any quantitative data on aragonite, but in the earlier Russian edition [1981GLU/GUR] they mention the value $\Delta_{\mathrm{f}}H^{\circ}_{\mathrm{m}}(\mathrm{CaCO_3}, \text{arag}, 298.15\ \mathrm{K}) = -1206.8\ \mathrm{kJ{\cdot}mol^{-1}}$, based on the calorimetric results of Bäckström [1925BAC] (see Appendix A entry) and [1925BAC2] that yield – 0.172 and – 0.234 $\mathrm{kJ{\cdot}mol^{-1}}$, respectively, for the $\Delta_{trs}H^{\circ}_{\mathrm{m}}$ (298.15 K) for the reaction:

$$\mathrm{CaCO_3}(\text{calc}, 298.15\ \mathrm{K}) \rightleftharpoons \mathrm{CaCO_3}(\text{arag}, 298.15\mathrm{K}) \qquad \text{(V.124)}$$

and of [1928ROT/CHA] that yield, for Reaction (V.124), a value of $\Delta_{trs}H_m$ (323 K) = – (0.226 ± 0.008) kJ·mol^{-1}. These three studies were taken into account in the assessment in [1987GAR/PAR].

But it should be noted that there seems to be a small inconsistency (perhaps due to rounding) in the treatment of the data for the calcite-aragonite transition in the assessment by Garvin *et al*., since they seem to have fixed S^o_m (298.15 K) for both calcite and aragonite (their Reactions 334 and 336) and also $\Delta_{trs}G^o_m$ (298.15 K) = 0.929 kJ·mol^{-1} (their Reaction 375), so $\Delta_{trs}H^o_m$ (298.15 K) should also be fixed to – 0.180 kJ·mol^{-1}, whereas the values in their Table 6.2.1 for these quantities are 0.920 and – 0.190 kJ·mol^{-1} respectively.

The results for Reaction (V.124) from Roth and Chall [1928ROT/CHA] were corrected (following Kelley [1960KEL]) to 298.15 K as $\Delta_rH^o_m$ ((V.124), 298.15 K) = – (0.176 ± 0.084) kJ·mol^{-1}.

Kobayashi [1951KOB2] derived a rather uncertain value of the enthalpy calcite-aragonite transition from an analysis of the $C^o_{p,m}$ curves for aragonite in the region where the transition to calcite occurs fairly rapidly (573-673 K) and derived a value of $\Delta_rH^o_m$ ((V.124), 298.15 K) = (0.121 ± 0.096) kJ·mol^{-1}, but these results were given zero weight by Garvin *et al.* (their Reaction 378).

The transition was also studied based on decomposition pressures between 302 and 350 K by Jamieson [1953JAM] and used in [1987GAR/PAR].

The comprehensive analysis of the solubility data of aragonite given in Section V.6.2.2.4.3 gives an enthalpy of formation of $\Delta_fH^o_m$ ($CaCO_3$, arag, 298.15 K) = – (1207.538 ± 2.000) kJ·mol^{-1}, 0.250 kJ·mol^{-1} more negative than the value for calcite from the similar solubility analysis.

Garvin *et al.* [1987GAR/PAR] finally obtained a value of $\Delta_rH^o_m$ ((V.124), 298.15 K) = – (0.190 ± 0.458) kJ·mol^{-1} and selected $\Delta_fH^o_m$ ($CaCO_3$, arag, 298.15 K) = – (1207.776 ± 1.100) kJ·mol^{-1}.

More recently, Königsberger *et al.* [1989KON/BUG] obtained, from a potentiometric study, further values for the thermodynamic data associated with Reaction (V.124) in the temperature range 278 to 318 K, obtaining at 298.15 K: $\Delta_rG^o_m$ = (0.830 ± 0.024) kJ·mol^{-1}, $\Delta_rH^o_m$ = – (0.540 ± 0.060) kJ·mol^{-1} and $\Delta_rS^o_m$ = – (4.60 ± 0.21) J·K^{-1}·mol^{-1}. The latter value is noticeably more negative than that derived from the selected calorimetric data, – (3.72 ± 0.32) J·K^{-1}·mol^{-1}, and the data in [1989KON/BUG] were re-fitted by this review with this constraint of $\Delta_rS^o_m$ = – 3.72 J·K^{-1}·mol^{-1}, which gave the following values at 298.15 K: $\Delta_rG^o_m$ = (0.834 ± 0.097) kJ·mol^{-1}, $\Delta_rH^o_m$ = – (0.275 ± 0.035) kJ·mol^{-1}, where the uncertainties are only those of the fitting.

The same group [1996WOL/LER] (see Appendix A entry) measured calorimetrically the thermodynamic quantities relative to the aragonite-calcite transition and proposed an optimised value of $\Delta_r H_m^o = -(0.440 \pm 0.050)$ kJ·mol^{-1} for Reaction (V.124) at 298.15 K.

Wolf *et al.* [1996WOL/LER] also reported the enthalpy change upon transition at high temperature as $\Delta_r H_m^o$ (V.124) $= -(0.403 \pm 0.008)$ kJ·mol^{-1} for $T = 720$-742 K. All these data for the values of $\Delta_r H_m^o$, $\Delta_r S_m^o$ and $\Delta_r G_m^o$ of Reaction (V.124) are summarised in Table V-67.

This selected value for $\Delta_f H_m^o(CaCO_3, \text{arag}, 298.15\ K) =$ **$-(1207.700 \pm 1.300)$** kJ·mol^{-1} is close to that selected by Garvin *et al*.

Table V-67: Values at 298.15 K for the transition, $CaCO_3(cr, \text{calcite}) \rightleftharpoons CaCO_3(\text{arag})$. Values in italics were constraints. Selected values are in bold.

Reference	Species for the transition calcite = aragonite	$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}
[1925BAC] (see Appendix A)	aragonite – calcite		– 0.172	
[1925BAC2] (see Appendix A)	aragonite – calcite		– 0.234	
[1928ROT/CHA]	aragonite – calcite		– (0.176 ± 0.084)	
[1987GAR/PAR] (calculated from Table 6.2.1)	aragonite – calcite	(0.920 ± 1.487)	– (0.190 ± 1.486)	*– (3.720 ± 0.320)*
[1987GAR/PAR] (reaction 375)	aragonite – calcite	*(0.929 ± 0.05)*		
[1989KON/BUG] (from potential difference—21 pts from 288.15 to 318.15 K)	aragonite – calcite	(0.830 ± 0.020)	– (0.540 ± 0.060)	– (4.600 ± 0.210)
[1989KON/BUG] (with constrained $\Delta_r S^{(a)}$)	aragonite – calcite	(0.834 ± 0.097)	– (0.275 ± 0.035)	*– (3.720 ± 0.320)*
[1996WOL/LER] (calorimetric—see Appendix A)	aragonite – calcite		– (0.100 ± 0.540) – (0.210 ± 0.460) – (0.050 ± 0.200)	
[1996WOL/LER] + [1989KON/BUG] (optimised data—see Appendix A of [1996WOL/LER])	aragonite – calcite	(0.840 ± 0.020)	– (0.440 ± 0.050)	– (4.300 ± 0.200)
Solubility (this review) (aragonite–calcite)	aragonite – calcite	0.859	– 0.250	*– (3.720 ± 0.320)*

(Continued on next page)

Table V-67 (continued)

Reference	Species for the transition calcite = aragonite	$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}
Selected value	calcite	***– (1128.983 ± 1.208)***	***– (1207.500 ± 1.200)***	***(91.710 ± 0.210)***
Selected value	aragonite	***– (1128.074 ± 1.307)***	***– (1207.700 ± 1.300)***	***(87.990 ± 0.210)***
	aragonite – calcite	(0.909 ± 0.221)	– (0.200 ± 0.200)	*– (3.720 ± 0.320)*

(a) The uncertainty in the constrained fit of the [1989KON/BUG] data is just the statistical value from the spread of values and does not include any experimental uncertainty.

It may be noted that, somewhat unusually, the greater stability of calcite compared to aragonite at all temperatures is due to the higher entropy of calcite rather than a more negative enthalpy of formation.

V.6.2.2 Calcium aqueous species

V.6.2.2.1 Aqueous calcium(II)

V.6.2.2.1.1 Solubility of calcium(II) hydroxide

The advantage of studying this system is that hydrolysis takes place at high pH, so that this parameter, which is normally difficult to control and more to the point quantify, can be determined with accuracy from simple mass-balance equations. Thus, virtually all of the solubility results were treated using the dissolution equilibrium in the form of Reaction (V.125) rather than the acidic form, Reaction (V.126). The formation constants and enthalpies of reaction for Reaction (V.125) reported in the literature are given in Table V-68 together with the molal values derived in this review and the results of SIT treatments where applicable. In virtually all the cases cited in Table V-68, hydrolysis of Ca^{2+} to form $CaOH^+$ was taken into account. An example of the magnitude this correction is presented in the first entry in Table V-68 showing its effect on $\log_{10} K_s^o$.

$$Ca(OH)_2(cr) \rightleftharpoons Ca^{2+} + 2OH^- \quad \text{(V.125)}$$

$$Ca(OH)_2(cr) + 2H^+ \rightleftharpoons Ca^{2+} + 2H_2O(l) \quad \text{(V.126)}$$

Table V-68: Summary of data for the solubility of calcium hydroxide according to Reactions (V.125) and (V.126) obtained from experimental studies or from reviews and re-evaluations. The quantities shown in italics were not corrected for the formation of $CaOH^+$. (units: $\Delta_r H_m^o$, kJ·mol^{-1}).

T/K	Ionic medium	I_m [(a)] max	$-\log_{10} K_s^o$ reported	$-\log_{10} K_s^o$ (V.125) SIT	$\Delta\varepsilon$ /kg·mol^{-1}	$\log_{10}{}^* K_{s,0}^o$ (V.126)	Reference
298.15	NaCl	3.82		(*5.023 ± 0.010*) (5.220 ± 0.018)	(*0.191 ± 0.004*) (0.184 ± 0.010)	(*22.98 ± 0.02*) (22.78 ± 0.03)	[1931JOH/GRO]
	KCl	2.93		(5.237 ± 0.006)	(0.226 ± 0.004)	(4.448 ± 0.02)	
	CsCl	3.60		(5.241 ± 0.010)	(0.242 ± 0.006)	(22.76 ± 0.02)	
	LiCl	3.92		(5.273 ± 0.006)	– (0.130 ± 0.006) (0.188 ± 0.010) [(b)]	(22.73 ± 0.02)	
	$NaClO_4$	3.61		(5.240 ± 0.002)	(0.321 ± 0.001)	(22.76 ± 0.02)	
	$NaNO_3$	3.86		(5.216 ± 0.016)	(0.113 ± 0.008)	(22.79 ± 0.03)	
293.15	self	0.0526		5.188			[1956BAT/BOW]
298.15		0.0505		5.237			
303.15		0.0486		5.284			
298.15	0		5.07 $\Delta_r H_s^o$ – 13.8			22.93	[1960GRE/COP]
298.15	$Ca(OH)_2$ + $CaCl_2$ + HCl	< 0.06	5.203 $\Delta_r H_s^o$ – (18.2 ± 0.2)	5.222 $\Delta_r H_s^o$ – 17.87[c]			[1965HOP/WUL]
273.65	$NaNO_3$	2.83		(4.888 ± 0.020)	(0.051 ± 0.018)	(24.96 ± 0.03)	[1967YEA/MAR]
298.15		2.88	5.03 $\Delta_r H_s^o$ – 12.7	(5.232 ± 0.008) $\Delta_r H_s^o$ – (26.2 ± 0.1)	(0.107 ± 0.008)	(22.77 ± 0.02)	
323.15		2.82		(5.481 ± 0.012)	(0.113 ± 0.010)	(21.07 ± 0.02)	
373.15		2.36		(6.200 ± 0.012)	(0.093 ± 0.010)	(18.33 ± 0.02)	
298.15		0	(5.30 ± 0.26) [(c)]			(22.70 ± 0.26)	[1987GAR/PAR]

(Continued on next page)

Table V-68 (continued)

T/K	Ionic medium	I_m (a) max	$-\log_{10}K_s^o$ reported	$-\log_{10}K_s^o$ (V.125) SIT	$\Delta\varepsilon$ /kg·mol^{-1}	$\log_{10}{}^*K_{s,0}^o$ (V.126)	Reference
373.15		→0	$\log_{10}{}^*K_{s,0}^o$				[1991SEE/SEY]
50 MPa			18.15				
473.15			$\log_{10}{}^*K_{s,0}^o$				
50 MPa			14.40				
573.15			$\log_{10}{}^*K_{s,0}^o$				
50 MPa			11.89				
623.15			$\log_{10}{}^*K_{s,0}^o$				
50 MPa			10.86				
298.15	KOH	→0	(5.19 ± 0.04) (d)				[1995DUC/REA]

(a) Results at $I_m > 4$ were excluded from the SIT treatment.
(b) This value corresponds to ε_2 in the term, $\varepsilon_2 \log_{10} I_m$, noting that $\varepsilon(Li^+, OH^-)$ is ionic strength dependent.
(c) Obtained from the thermodynamic data tabulated by the authors.
(d) Obtained from a Pitzer treatment of solubility data from this study and those presented in the 15 earlier datasets.

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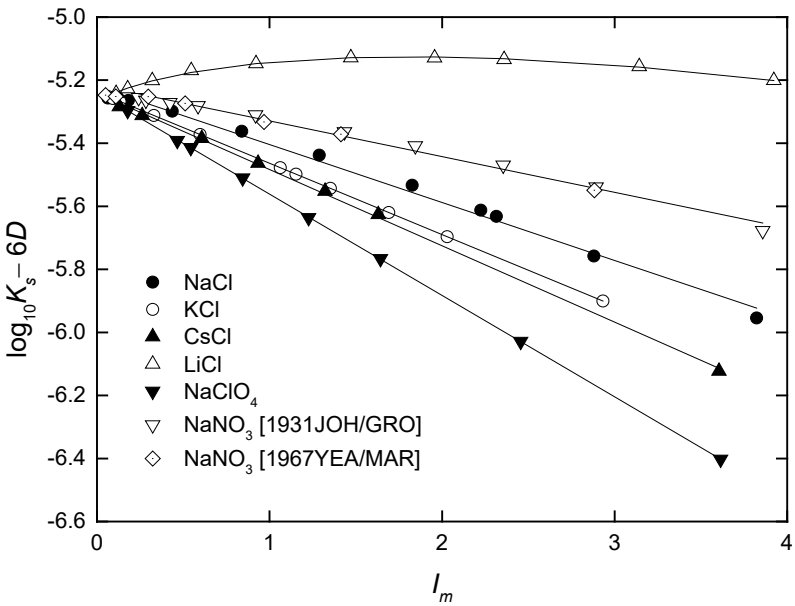
Figure V-56 shows the application of the SIT treatment to the comprehensive data sets from [1931JOH/GRO] and [1967YEA/MAR] at 298.15 K that yielded the $\log_{10}K_s^o$ (V.125) and $\Delta\varepsilon$ (V.125) values given in Table V-68. For the case of the $NaNO_3$ solutions, the mean values are: $\log_{10}K_s^o$ (V.125) = – (5.225 ± 0.010) and $\Delta\varepsilon$ (V.125) = (0.110 ± 0.006) kg·mol^{-1} (2σ). It is interesting to note that the careful study by Bates and Bower [1956BAT/BOW], which involved multiple independently synthesised samples of $Ca(OH)_2$(cr) with equilibration times of up to 6 months in highly dilute solutions, produced a virtually identical value of $\log_{10}K_s^o$ (V.125) = – 5.237.

Figure V-56: SIT plots of the solubility constant of $Ca(OH)_2$(cr) (Reaction (V.125)) at 298.15 K in the following selected ionic media: NaCl, KCl, CsCl, LiCl, $NaClO_4$, $NaNO_3$ [1931JOH/GRO], and $NaNO_3$ [1967YEA/MAR].



From the SIT treatment, values for $\varepsilon(M^+, OH^-)$ can be derived, where M^+= Na^+, K^+, Cs^+ and Li^+:

$$\varepsilon(Ca^{2+}, X^-) = (\Delta\varepsilon - 2\varepsilon(M^+, OH^-))\ \text{kg·mol}^{-1} \qquad \text{(V.127)}$$

Table V-69 gives the derived ion interaction coefficients $\varepsilon(Ca^{2+}, X^-)$ compared to the previously published values. No explanation can be given for the poor agreement obtained for $\varepsilon(Ca^{2+}, Cl^-)$ in the case of NaCl as the supporting electrolyte and the unsatisfactory correlation for KCl and LiCl solutions.

Table V-69: Summary of the treatment of SIT ion interaction coefficients (kg·mol^{-1}) at 298.15 K according to Equation (V.127).

Salt *NX*	$\varepsilon(Ca^{2+},X^-)$ NEA currently selected	$\varepsilon(M^+,OH^-)$ NEA currently selected	$\Delta\varepsilon$ exp.	$\varepsilon(Ca^{2+},X^-)$ calc.
LiCl	(0.14 ± 0.01)	– (0.039 ± 0.002) [(a)] (0.072 ± 0.006) [(b)]	– (0.130 ± 0.006) (0.188 ± 0.010)	
NaCl	(0.14 ± 0.01) [(c)]	(0.04 ± 0.01)	(0.184 ± 0.010)	(0.10 ± 0.02) (X = Cl^-)
$NaClO_4$	(0.27 ± 0.03)	(0.04 ± 0.01)	(0.321 ± 0.001)	(0.24 ± 0.02) (X = ClO_4^-)
$NaNO_3$	(0.02 ± 0.01)	(0.04 ± 0.01)	(0.110 ± 0.006)	(0.03 ± 0.01) (X = NO_3^-)
KCl	(0.14 ± 0.01)	(0.09 ± 0.01)	(0.226 ± 0.004)	(0.05 ± 0.02) (X = Cl^-)
CsCl	(0.14 ± 0.01)	(0.05 ± 0.01) [(d)]	(0.242 ± 0.006)	(0.14 ± 0.01) (X = Cl^-)

(a) $\Delta\varepsilon_1$.

(b) $\Delta\varepsilon_2$ where $\Delta\varepsilon = (\Delta\varepsilon_1 + \Delta\varepsilon_2 \log_{10} I_m)$.

(c) This value was verified by treating the more recent smoothed osmotic coefficients for $CaCl_2$ solutions which were generated in [1997RAR/CLE] by application of the extended Pitzer equation to a large collection of critically evaluated osmotic coefficient data.

(d) $\varepsilon(Cs^+,OH^-)$ was derived from the smoothed osmotic coefficient data in [1965ROB/STO].

A weighted regression of the $\log_{10}K_s^o$ (V.125) values derived from the SIT ion interaction model as a simple function of $(T/K)^{-1}$, as shown in Figure V-57, including the $\Delta_r H_s^o$ (V.125) value in [1965HOP/WUL], gave rise to Equation (V.128) which is also illustrated in Figure V-57 and compared to other fits and the primary data.

$$\log_{10}K_s^o\,(\text{V.125}) = -(8.3682 \pm 0.0086) + (933.576 \pm 2.564)(T/\text{K})^{-1} \qquad (\text{V.128})$$

Application of this equation led to following thermodynamic quantities at 298.15 K: $\log_{10} K_s^o$ ((V.125), 298.15 K) = – (5.237 ± 0.049), $\Delta_r G_s^o$ = (29.893 ± 0.280) kJ·mol^{-1}, $\Delta_r H_s^o$ = – (17.873 ± 0.096) kJ·mol^{-1}, $\Delta_r S_s^o$ = – (160.208 ± 0.993) J·K^{-1}·mol^{-1}, *cf.* CODATA recommended values [1987GAR/PAR]: $\Delta_r G_s^o$ = (30.3 ± 1.5) kJ·mol^{-1}, $\Delta_r H_s^o$ = – (17.87 ± 1.4) kJ·mol^{-1}, and $\Delta_r S_s^o$ = – (161.4 ± 1.1) J·K^{-1}·mol^{-1} (all 2σ).

Seewald and Seyfried [1991SEE/SEY] studied the solubility of $Ca(OH)_2$(cr) from 373 to 625 K at a fixed pressure of 50 MPa using a flexible gold bag containing either pure water or dilute acetic acid, 0.001 to 0.01 mol·kg^{-1}. These experimental conditions are well beyond the scope of the current review, but it is of interest that a simple extrapolation of their values of $\log_{10}{}^*K_{s,0}^o$ (V.126) given in Table V-68 to 298.15 K (50 MPa) resulted in $\log_{10}{}^*K_{s,0}^o$ (V.126) = 22.73 and $\Delta_r{}^*H_s^o$ (V.126) = – 129.4 kJ·mol^{-1}. The corresponding values derived from Equation (V.128) and the thermodynamics of dissociation of water at 298.15 K are:

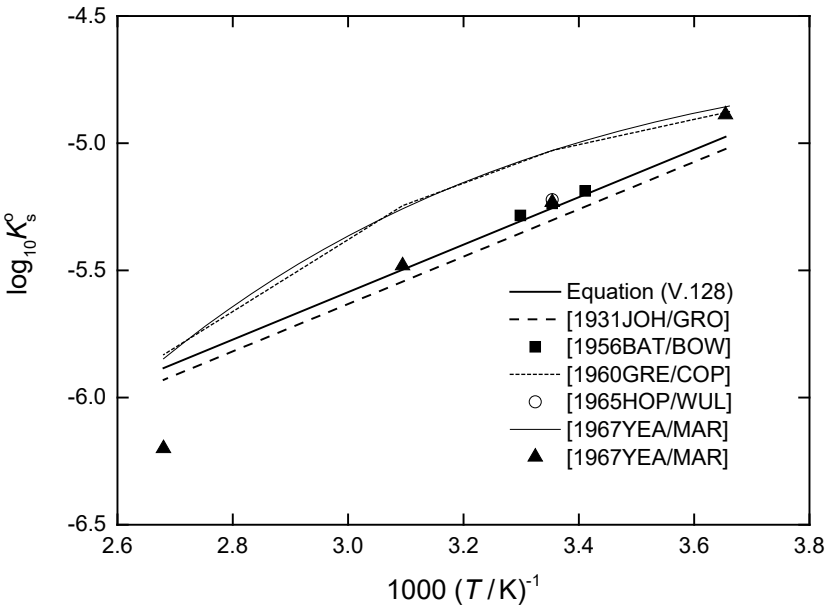
$$\log_{10}{}^*K_{s,0}^o\,(\text{V.126}) = (22.765 \pm 0.051),$$

$$\Delta_r{}^*H_s^o\,(\text{V.126}) = -(129.503 \pm 0.112)\ \text{kJ·mol}^{-1}.$$

The selected values are very slightly different: $\log_{10}K^o$ ((V.126), 298.15 K) = (22.694 ± 0.260) and $\Delta_r H_m^o$ ((V.126), 298.15 K) =

– (129.500 ± 1.633) kJ·mol^{-1} since the selected value for $\Delta_f H^\circ_m(Ca(OH)_2, cr, 298.15\ K)$ = – (985.160 ± 1.000) kJ·mol^{-1} is very slightly different from that derived from the solubility data alone, – (985.157 ± 1.008) kJ·mol^{-1} (Section V.6.2.2.1.1).

Figure V-57: Temperature dependence of $\log_{10} K^\circ_s$ (V.125).



V.6.2.2.1.2 Hydrolysis of aqueous Ca^{2+}

Felmy *et al*. [1998FEL/DIX] used both macroscopic thermodynamic models and molecular modelling based on the density functional theory to verify that the aqueous calcium ion is surrounded by six water molecules in keeping with the later findings of an EXAFS study [2003FUL/HEA]. Moreover, no delocalisation of a hydroxide group from the calcium ion was observed, suggesting the absence of hydrolysed species.

Most studies typically reported dissociation constants for $CaOH^+$ according to Reaction (V.129), because they were measured in basic solutions, rather than as hydrolysis constants of Ca^{2+}, Reaction (V.130), knowledge of which is more directly desirable for modelling purposes. Guiomar *et al*. [1998GUI/LIT] widely deemed as invalid the assumption that the dissociation of $Ca(OH)_2$(aq) is incomplete. However, their methodology in re-analysing the data in [1959BAT/BOW] is considered in this review to be flawed (see Appendix A entry for details).

$$CaOH^+ \rightleftharpoons Ca^{2+} + OH^- \qquad \text{(V.129)}$$

$$Ca^{2+} + H_2O(l) \rightleftharpoons CaOH^+ + H^+ \quad \text{(V.130)}$$

The Davies equation [1962DAV] was commonly employed to estimate activity coefficients generally with multiple extended terms (*e.g.*, [1959BAT/BOW] being the most cited), often resulting in different $\log_{10}K_1^o$ (V.129) values being given (*e.g.*, – 1.13 to – 1.45). These discussions generally invoked the premise that $\gamma_{OH^-} \approx \gamma_{Cl^-}$, where chloride was the most common other anion in these systems. A further assumption was made that the ion interaction coefficients for Ca^{2+} and $CaOH^+$ were the same for both anions.

Table V-70 summarises all of the data found in the literature and includes results of re-assessments carried out in this review (the 2σ uncertainties in the $\log_{10}{}^*K_1^o$ values were estimated by this review). No thermodynamic data for the formation of $CaOH^+$ could be extracted from the copious solubility/potentiometric work of Bates and Bower [1956BAT/BOW], because their experiments were conducted in NaCl solutions with no added NaOH. Conversely, in the presence of $Ca(OH)_2$(cr), $CaOH^+$ can be the dominant form of calcium in solution [1991SEE/SEY]. A major concern is that working at high pH requires the strict avoidance of exposure of the solution to CO_2(g) because its uptake is very rapid, being first order in respect to hydroxide ion concentration, leading to formation of stable calcium carbonate species. Moreover, potentiometric techniques [1954GIM/MON], [1959BAT/BOW], [1972MAR/VAS] are generally difficult to implement due to the high concentration of free hydroxide ion required to achieve measurable hydrolysis, whereas other methods can generally rely on mass-balance considerations to determine the free hydroxide ion concentration. Kinetic methods [1949BEL/PRU], [1950BEL/WAI], [1956BEL/PAN] provided an indirect, and generally less accurate, measure of the hydrolysis equilibrium by relying on the catalytic effect of free hydroxide ions on either the depolymerisation of diacetone alcohol or its reaction with nitroethane in solutions of calcium hydroxide, whereby $CaOH^+$ was assumed to have little to no effect on the reaction rate.

Table V-70: Summary of experimental data, dissociation and hydrolysis constants according to Reactions (V.129) and (V.130), respectively (units: $\Delta_r H_m^o$, kJ·mol^{-1}).

T /K	Ionic medium	I_m max	$-\log_{10} K_1^o$ (reported)	$-\log_{10} K_1^o$ (re-calculated)	$-\log_{10}{}^* K_1^o$ (re-calculated)	Method	Reference
298.15	$NaOH + Ca(IO_3)_2 + NaCl$	0.15	(1.51 ± 0.06)	(1.47 ± 0.22)	(12.53 ± 0.31)	sol	[1938DAV],[1936KIL2]
298.15	$KOH + NaOH + CaCl_2$	0.0457	(1.30 ± 0.04)	(1.28 ± 0.08)	(12.72 ± 0.08)	kin	[1949BEL/PRU]
273.15	$NaOH + CaCl_2 + Ca(IO_3)_2$	0.076, .295	1.38, 1.24	(1.38 ± 0.05) [(a)]	(13.57 ± 0.10)	kin, sol [(a)]	[1950BEL/WAI]
298.15	$Ca(OH)_2 + Ca(IO_3)_2$	0.067	1.30	(1.37 ± 0.02)	(12.63 ± 0.08)	sol	[1951DAV/HOY]
273.15	$KOH + Ca(IO_3)_2$	0.0761	1.37	(1.36 ± 0.11)	(13.59 ± 0.20)	sol	[1953BEL/GEO]
298.15		0.0779	1.40	(1.46 ± 0.04)	(12.54 ± 0.08)		
313.15		0.1026	1.48	(1.57 ± 0.05)	(11.98 ± 0.10)		
288.15	$NaOH + CaCl_2$	0.050	(1.366 ± 0.020)	(1.352 ± 0.002)	(12.99 ± 0.06)	pot	[1954GIM/MON]
298.15		0.080	(1.363 ± 0.015)	(1.50 ± 0.04)	(12.50 ± 0.08)		
308.15		0.087	1.39	(1.62 ± 0.07)	(12.07 ± 0.14)		
298.15	$NaOH + CaCl_2$	0.12	(1.46 ± 0.06)	(1.47 ± 0.06)	(12.53 ± 0.11)	kin	[1956BEL/PAN]
273.15	$KCl + CaCl_2$	0.074		(1.14 ± 0.04)	(13.81 ± 0.08)	pot	[1959BAT/BOW]
283.15		0.077	1.12	(1.04 ± 0.12)	(13.50 ± 0.12)		
288.15		0.076		(1.17 ± 0.12)	(13.17 ± 0.12)		
293.15		0.075		(1.15 ± 0.07)	(13.02 ± 0.07)		
298.15		0.075	1.13-1.15	(1.21 ± 0.09)	(12.79 ± 0.09)		
303.15		0.076		(1.26 ± 0.13)	(12.58 ± 0.13)		
308.15		0.075		(1.30 ± 0.11)	(12.39 ± 0.11)		
313.15		0.074	1.39	(1.38 ± 0.15)	(12.16 ± 0.15)		
298.15	$Ca(OH)_2$	0.06	$\Delta_r H_m^o$ − (8.3 ± 0.2)		$\Delta_r^* H_m^o$ (62.0 ± 1.7)	cal	[1965HOP/WUL]

(Continued on next page)

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Table V-70 (continued)

T /K	Ionic medium	I_m max	$-\log_{10}K_1^o$ (reported)	$-\log_{10}K_1^o$ (re-calculated)	$-\log_{10}{}^*K_1^o$ (re-calculated)	Method	Reference
333.15	$Ca(NO_3)_2$	0.001	2.80	2.90	10.13	pot	[1972MAR/VAS]
343.15			3.06	3.17	9.64		
353.15			3.44	3.62	8.99		
363.15			3.43	3.68	8.75		
371.15			3.88	3.99	8.31		
298.15			(1.15 ± 0.02) $\Delta_r H_m^o$ – (10.5 ± 2.9)		(12.85 ± 0.02) (b)	ass	[1976BAE/MES] re-treatment of [1959BAT/BOW]
298.15	Na/KOH + Na/KCl	4	1.06			ass	[1984HAR/MOL] re-treatment of [1977MCG/HOS]
298.15					12.78	ass	[1990NOR/PLU]
373 50MPa	pure H_2O + CH_3COOH	0.01			10.04 (b)	sol	[1991SEE/SEY]
473					8.20 (b)		
573					6.88 (b)		
623					6.35 (b)		
298.15	KOH + NaOH	3.86 3.63	1.34			sol	[1995DUC/REA]
298.15	NaOH	5	1.17			sol	[1998FEL/DIX]
298.15			1.45			ass	[1998GUI/LIT] re-treatment of [1959BAT/BOW]

(a) Only the solubility data were re-analysed due to the lower ionic strengths employed.
(b) The values were reported in the original papers.

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Solubility studies involving either $Ca(IO_3)_2(s)$ as the controlling phase [1938DAV], [1950BEL/WAI], [1951DAV/HOY], [1953BEL/GEO] or those involving $Ca(OH)_2(cr)$ directly [1956BAT/BOW], [1965HOP/WUL], [1991SEE/SEY], [1995DUC/REA], [1998FEL/DIX] have been more commonly employed. Davies [1938DAV] re-analysed measurements made by Kilde [1936KIL2] of the solubility of calcium iodate in dilute NaOH solutions at 298.15 K taking into account the formation of $CaIO_3^+$ and $NaIO_3(aq)$ ion pairs to determine the dissociation constant of $CaOH^+$. Experiments of this kind are discussed here in more detail because they showed reasonable consistency (except ironically for the treatment in [1938DAV]). This consistency suggests that the otherwise preferred data from Bates and Bower [1959BAT/BOW] may in fact be less reliable than heretofore believed. Kilde's measurements were repeated [1950BEL/WAI], [1951DAV/HOY], [1953BEL/GEO] from 273.15 to 313.15 K, although Bell's group also carried out some independent kinetic measurements. Re-treatment of these solubility results using the SIT involved re-fitting the solubility constant data for $Ca(IO_3)_2(s)$ and those associated with dissociation of the $CaIO_3^+$ ion pair. The proposed $KIO_3(aq)$ and $NaIO_3(aq)$ ion pairs were excluded from our re-treatment procedure which led to the results summarised in Table V-71. Bell and George [1953BEL/GEO] noted that the stable solid phase at 273.15 and 298.15 K is $Ca(IO_3)_2{\cdot}6H_2O(cr)$, whereas by 313.15 K, $Ca(IO_3)_2{\cdot}H_2O(cr)$ is more stable.

Table V-71: Summary of dissociation and solubility constants reported and re-calculated from solubility studies involving $Ca(IO_3)_2(s)$ with statistically based uncertainties at the 2σ level.

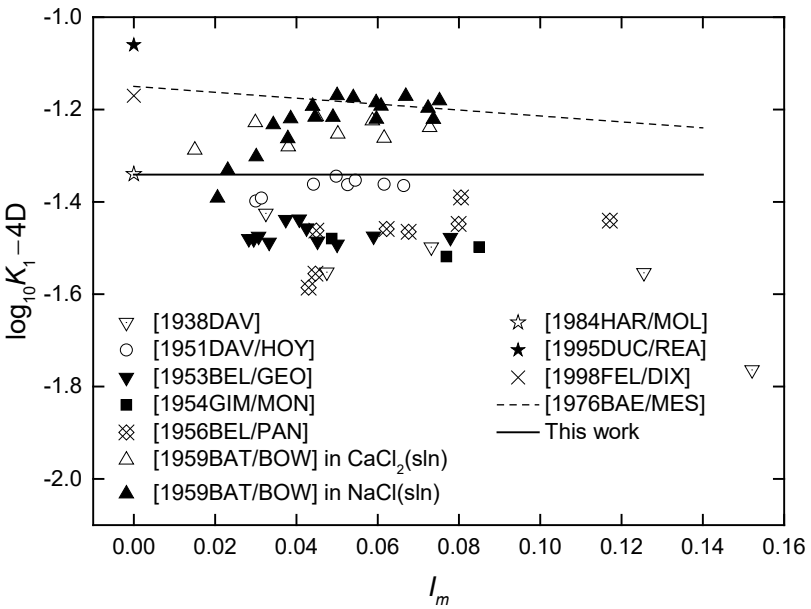
	Temperature			Reference
	273.15 K	298.15 K	313.15 K	
$K_1^\circ(CaIO_3^+)$	0.115	0.113	0.115	[1951DAV/HOY]
$K_s^\circ(Ca(IO_3)_2,s)$	2.921×10^{-8}			[1950BEL/WAI]
	$(3.00 \pm 0.06) \times 10^{-8}$ (a)			Re-calculated
	2.859×10^{-8}	7.119×10^{-7} (a)	2.437×10^{-6} (a)	[1953BEL/GEO]
	2.88×10^{-8}	7.170×10^{-7}	2.485×10^{-6}	Re-calculated (b)

(a) Accepted values.
(b) These values were based on only one measurement reported in the absence of added NaOH.

There exist several compilations [1990NOR/PLU], reviews [1976BAE/MES], [1984HAR/MOL], [1995DUC/REA], [1998FEL/DIX], [1998GUI/LIT], and assessments [1987GAR/PAR] (CODATA) of the thermodynamics of Ca^{2+} hydrolysis. Nordstrom *et al*. [1990NOR/PLU] compiled a database for geochemical modelling purposes that included a value $\log_{10}{}^*K_1^\circ$ ((V.130), 298.15 K) of – 12.78 for the hydrolysis of Ca^{2+} although its origin is difficult to trace. The CODATA assessment [1987GAR/PAR] provides an estimate of $\Delta_f G_m^\circ$ ($CaOH^+$, 298.15 K) presumably based solely on the results of Bates and Bower [1959BAT/BOW]. Baes and Mesmer [1976BAE/MES] give a brief summary and treatment of the available thermodynamic data associated with calcium(II) hydrolysis at 298.15 K and acknowledge that these hydrolysis constants had been

determined with only limited accuracy (see also the Appendix A entry for [1959BAT/BOW]). Nevertheless, they provided a function based on the Pitzer and Brewer [1961LEW/RAN] ion interaction activity coefficient model for $\log_{10}K_1$ (V.129) with the coefficient in the extended term varying depending on the range of ionic strength of interest. Three independent Pitzer ion interaction models have been applied to the $CaOH^+$ dissociation equilibrium, [1984HAR/MOL], [1995DUC/REA], [1998FEL/DIX] resulting in the $\log_{10}K_1$ (V.129) values shown in Figure V-58, where the last assessment benefited from having additional $Ca(OH)_2$(cr) solubility data to high ionic strengths. It should be noted that data from [1956BAT/BOW] and [1959BAT/BOW] were heavily favoured in the construction of these models, whereas the results illustrated in Figure V-58 indicate surprisingly that data from other sources are significantly more consistent. Certainly, the established Pitzer ion interaction approach provides a viable alternative to resolve the activity coefficient question. However, no sensitivity analyses were carried out in these studies and hence the uncertainties in these extrapolations are not known. The solid line in Figure V-58 corresponds to $\log_{10}K_1^\circ = -(1.34 \pm 0.06)$ at 298.15 K (2σ statistical uncertainty) and was obtained from a weighted regression, Equation (V.131), of all the available $\log_{10}K_1$ data at low ionic strengths.

Figure V-58: The ionic strength dependence of the logarithm of the dissociation constant of $CaOH^+$ (Reaction (V.129)) at 298.15 K.



The experimental conditions employed in [1991SEE/SEY] are beyond the scope of the current review. Nevertheless, they provided support for the choice of the enthalpy of reaction for the hydrolysis of calcium, noting that pressure should not affect this equilibrium significantly at least up to 373 K. The potentiometric results in [1972MAR/VAS] were thought to be unreliable and were also not included in the final data analysis routine. The heats of dissolution of $Ca(OH)_2$(s) in water were measured at 298.15 K [1965HOP/WUL]. Hopkins and Wulff used an equation for the ionic strength dependence of the first dissociation constant of $CaOH^+$ proposed in [1959BAT/BOW] that yielded a value $\log_{10}K_1^\circ$((V.129), 298.15 K) = – 1.222 and $\Delta_r H_m^\circ$((V.129), 298.15 K) = – (8.4 ± 1.7) kJ·mol^{-1}. However, in [1953BEL/GEO] and [1954GIM/MON] more negative values for the dissociation constant were used, which in combination with the lower $\Delta_r H_m^\circ$((V.129), 298.15 K) value led to similar values for $\Delta_r S_m^\circ$((V.129), 298.15 K).

Equation (V.131) resulted from a weighted linear regression of the $\log_{10}{}^*K_1^\circ$ values in Table V-70 as a function of $(T/\mathrm{K})^{-1}$ shown in Figure V-59.

$$\log_{10}{}^*K_1^\circ(\text{V.130}) = -(1.5608 \pm 0.3669) - (3309.54 \pm 107.21)(T/\mathrm{K})^{-1} \quad \text{(V.131)}$$

The resulting thermodynamic parameters at 298.15 K are:

$$\log_{10}{}^*K_1^\circ((\text{V.130}), 298.15\ \mathrm{K}) = \mathbf{-\ (12.661 \pm 0.514)},$$

$$\Delta_r{}^*G_m^\circ((\text{V.130}), 298.15\ \mathrm{K}) = \mathbf{(72.269 \pm 2.932)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$\Delta_r{}^*H_m^\circ((\text{V.130}), 298.15\ \mathrm{K}) = \mathbf{(63.360 \pm 4.000)}\ \mathrm{kJ\cdot mol^{-1}},$$

and $\Delta_r{}^*S_m^\circ((\text{V.130}), 298.15\ \mathrm{K}) = \mathbf{-\ (29.884 \pm 16.635)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$ (2σ).

The corresponding $\log_{10}K_1^\circ$(V.129) value at 298.15 K is – (1.34 ± 0.06).

Taking the thermodynamic properties of Ca^{2+} and H_2O(l) from the auxiliary data in Table IV-1, the corresponding values for the formation of $CaOH^+$ are:

$$\Delta_f G_m^\circ(CaOH^+, 298.15\ \mathrm{K}) = \mathbf{-\ (717.676 \pm 3.115)}\ \mathrm{kJ\cdot mol^{-1}},$$

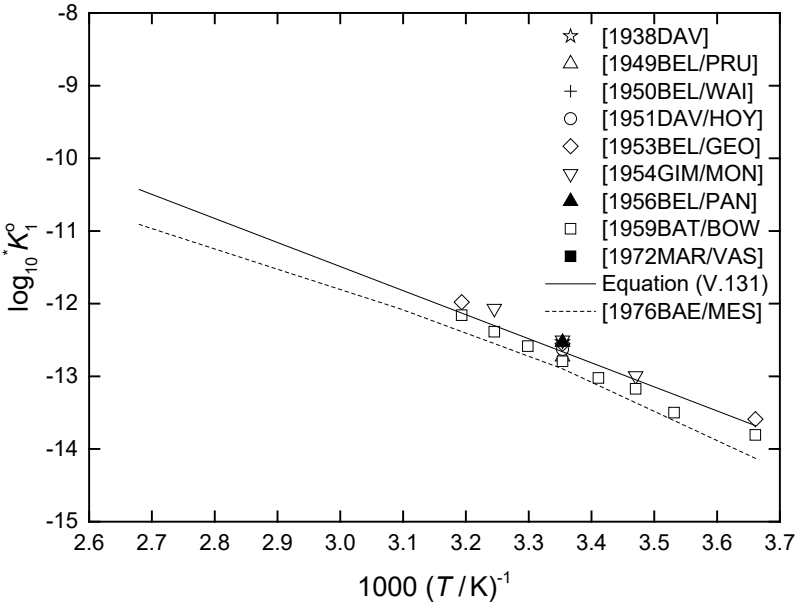
$$\Delta_f H_m^\circ(CaOH^+, 298.15\ \mathrm{K}) = \mathbf{-\ (765.470 \pm 4.123)}\ \mathrm{kJ\cdot mol^{-1}},$$

$$S_m^\circ(CaOH^+, 298.15\ \mathrm{K}) = \mathbf{-\ (16.134 \pm 16.665)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

and these are our selected values.

Garvin *et al.* [1987GAR/PAR] (p.181) give a value of $\Delta_f G_m^\circ(CaOH^+, 298.15\ \mathrm{K}) = -(717.0 \pm 1.4)\ \mathrm{kJ\cdot mol^{-1}}$, apparently based solely on [1959BAT/BOW] (their reference (p.262) to [1956BAT/BOW] is presumably an error) which leads to $\log_{10}{}^*K_1^\circ$(V.129) = – 1.221.

Figure V-59: Temperature dependence of $\log_{10}{}^{*}K_1^{\circ}$ (V.130).



V.6.2.2.2 Aqueous calcium(II) chloride

V.6.2.2.2.1 Ion association in aqueous calcium(II) chloride

The structure of $CaCl_2$ solutions was determined by X-ray diffraction [1976LIC/PIC]. A coordination number of six with octahedral symmetry was proposed for both Cl^- and Ca^{2+} ions in solution, based on analysis of the correlation functions. The Ca-OH_2 distance was given as 240-245 pm. More significantly perhaps, no ion-ion interaction was detected even in 4 mol·dm^{-3} solutions at 291 K. X-ray, neutron diffraction and XAFS measurements [1987END/CUM], [1993OHT/RAD], [2003FUL/HEA] at ionic strengths between 1 and 14 mol·kg^{-1} did not reveal any evidence of direct contact between the calcium and chloride ions, suggesting that a contact ion pair would represent less than a few percent of the total calcium in solution. Fulton *et al*. [2006FUL/CHE] used XAFS spectroscopy to determine the hydration structure of Ca^{2+} and ion pairing between Ca^{2+} and Cl^- ions at 523.15 K (14 MPa) and 673.15 K (35 MPa) in 1 mol·kg^{-1} $CaCl_2$ solutions. K-edges were used to identify the hydration numbers in the extended XAFS regions. On average, the hydration shell of Ca^{2+} contained 3.2 water molecules and 1.8 chloride ions giving a total coordination number of (5 ± 1). These authors did not quantify the fractions of the ion pairs in solution at these conditions.

Based on the need to consider ion pairing explicitly in the treatment of the thermodynamic properties of $CaCl_2$ solutions, it was proposed [1987PHU/PIT] that a contact ion pair is formed at concentrations above 5 mol·kg^{-1} at ambient temperatures, whereas Rard and Clegg [1997RAR/CLE] estimated the minimum threshold chloride concentration for formation of a contact ion pair to be 3 mol·kg^{-1}. The larger ionic radius for the calcium ion compared to that of the magnesium ion ensures that the ion-ion interactions in chloride solutions are weaker than those of magnesium and in Section V.6.1.2.3.1 even the latter were taken to be insignificant at ambient conditions at moderate chloride ion concentrations.

A limited number of studies are available concerning formation of a $CaCl^+$ ion pair at the ambient conditions, [1975ELG/WED], [1975SUC/CAD], [1978JOH/PYT], [1982MAJ/STU], [1987ROB/RIG], [1994ROB/GIA], [1998ROB/STE], [2005MEN/WOO]. Most of the studies dealing with speciation in $CaCl_2$ solutions reported formation of a monochlorido ion pair, $CaCl^+$, according to Reaction (V.132) whereas at very high chloride concentrations or at elevated temperatures, some studies also proposed formation of the neutral $CaCl_2$(aq) species [1982FRA/MAR], [1987END/CUM], [1989WIL/SEW], [1992GIL/OSC], [1993OHT/RAD], [1994HOL/BUS], [1997RAR/CLE], [2000SAV], [2001OSC/GIL]. Holmes *et al*. [1994HOL/BUS] found that at temperatures greater than 523 K explicit consideration of ion association was required to describe the thermodynamic properties of 7.26 mol·kg^{-1} $CaCl_2$ solutions up to 40 MPa. Indeed, Oscarson *et al*. [2001OSC/GIL] found it was necessary to include the presence of $CaCl_2$(aq) ion pairs to interpret their heat of dilution calorimetric results obtained from 573 to 623 K at 10.3 to 17.6 MPa, respectively.

$$Ca^{2+} + Cl^- \rightleftharpoons CaCl^+ \qquad \text{(V.132)}$$

The formation constant at finite ionic strength is defined by:

$$K_1(\text{V.132}) = m_{CaCl^+} / m_{Ca^{2+}}\, m_{Cl^-} = K_1^{\circ}(\text{V.132})\, \gamma_{Ca^{2+}} \gamma_{Cl^-} / \gamma_{CaCl^+} \qquad \text{(V.133)}$$

Different ionic media were employed in the experimental studies, but in no cases were both the total calcium and chloride concentrations kept systematically below 10 % of the prevailing ionic strength. Most of the experimental data were collected using electrochemical cells [1975SUC/CAD], [1978JOH/PYT], [1982MAJ/STU], [1998ROB/STE] with chloride sensitive electrodes (ISE) or Ag|AgCl electrodes. The K_1(V.132) values in Table V-72 are very scattered, which can be attributed to the fact that conventional measurement techniques, such as potentiometry, are relatively insensitive to weak ionic interactions in dilute solutions and require high concentrations of the anion to achieve a measurable degree of “apparent” ion pairing.

Table V-72: Summary of the experimental conditions and $CaCl^+$ molar formation constants for Reaction (V.132) at 298.15 K taken from a selected number of references (units: $\Delta_r H_1$, kJ·mol^{-1}).

$K_{c,1}$	$c_{Cl^-(max)}$	$c_{Ca^{2+}}$	Ionic medium	Method	Reference
1.2	0.7	0.032-0.045	I_c = 0.7 Synthetic seawater	sol	[1975ELG/WED]
K_m 0.72	m_{Cl^-} 0.001	$m_{Ca^{2+}}$ 0.1-0.2	I_m = 1.0 $NaNO_3$	pot	[1975SUC/CAD]
K_1^o (2.3 ± 0.1)	0.78	0.03-0.10	I_c=0.37-0.74 $CaCl_2$ + HCl	pot	[1978JOH/PYT]
K_m 0.54-0.98 $\Delta_r H_1$ 4.91	m_{Cl^-} 0.001	$m_{Ca^{2+}}$ 0.05-0.3	I_m =1.0 $NaNO_3$	pot	[1982MAJ/STU]
K_1^o $10^{-5.4}$	m_{Cl^-} 0.002-0.010	$m_{Ca^{2+}}$ 0.001-0.005	$CaCl_2$	con	[1982FRA/MAR]
K_1^o (2.5 ± 0.6)				est	[1987ROB/RIG]
K_1^o (2.7 ± 0.5)		0.0095-0.014	$Ca(NO_3)_2$ + NaCl + $(CH_3)_4NCl$	pot	[1994ROB/GIA]
K_1^o 0.51 $\Delta_r H_1^o$ 4.67				est	[1997SVE/SHO]
K_1^o (2.6 ± 0.9)	0.06-0.23	0.0015-0.12	I_c = 0.18-0.94 $Ca(NO_3)_2$ + NaCl + $NaNO_3$	pot	[1998ROB/STE]

A number of studies [1975ELG/WED], [1987ROB/RIG], [1997RAR/CLE], [1997SVE/SHO], [2005MEN/WOO] estimated the activity coefficients using various forms of the extended Debye-Hückel approach or the HFK correlations or the Pitzer ion interaction model. The value of K_1^o (V.132) depends significantly on the choice of the activity-coefficient model.

The potentiometric measurements reported in [1978JOH/PYT] for $CaCl_2$(sln) in the presence of HCl as a background electrolyte were referred to by Rard and Clegg [1997RAR/CLE] as being from a manuscript "in preparation", which has apparently

never been published, although the original authors did report the solution compositions from which the values of K_1 (V.132) at 298.15 K could be extracted. An extensive dataset was collected by Johnson and Pytkowicz, but the sole anion was chloride.

Šucha *et al.* [1975SUC/CAD] presented one of the few studies where the composition of the ionic medium was held constant, albeit with higher metal ion concentrations than could satisfy the inert supporting electrolyte criterion (the empirical rule is for the contribution to the ionic strength of ions other than the supporting electrolyte should not exceed 10% [1997ALL/BAN]). These authors used both ISE and AgCl|Ag electrodes to monitor the free chloride molality. However, the resulting formation constants were so small and the uncertainties so large as to put the validity of the constants in question. The lack of original experimental results is unfortunate, but noting that for $K_{m,1}$ (V.132) = 1 at a total chloride molality of 10^{-3} mol·kg^{-1} and $m_{Ca^{2+}} \sim 0.1$ would imply a value of $m_{CaCl^+} < 10^{-4}$ mol·kg^{-1}, which would in turn result in a < 10 % change in the measured chloride molality and hence a small but measurable potential signal. Similar results using the same potentiometric ISE technique were reported in [1982MAJ/STU] at an ionic strength of 1 mol·kg^{-1} ($NaNO_3$), but high chloride concentrations between 0.05 and 0.2 mol·kg^{-1} were employed. A value of $K_{m,1}$ (V.132) = 0.68 kg·mol^{-1} at 298.15 K was reported. In [1975ELG/WED] a value of $K_{c,1}$ (V.132) = 1.2 dm^3·mol^{-1} was derived from measurements of the solubility of gypsum in simulated sea water, I_c = 0.7 mol·dm^{-3}. Continuing their work with synthetic sea water, Elgquist and Wedborg used an ISE, and reported the same value of $K_{c,1}$ (V.132) based on a fully speciated model in which the ion pairs $NaSO_4^-$, NaCl(aq), KCl(aq), $MgCl_2$(aq), $NaClO_4$(aq), $Mg(ClO_4)_2$(aq) and Na_2SO_4(aq) were included, while the formation constants for $MgCl^+$, $CaCl^+$ and those of other major species were treated as fitting parameters. Solving such a complex series of equilibria based on one experimental technique is dubious at best especially in the absence of an uncertainty analysis.

De Robertis *et al*. [1987ROB/RIG], [1994ROB/GIA], [1998ROB/STE] wrote a series of papers which reported values of $\log_{10} K_{c,1}$ (V.132) based on potentiometric titrations using an ISE (calcium and chloride) in conjunction with a glass reference electrode. Raw experimental data were not provided. In all cases, high calcium and/or chloride concentrations were employed relative to the ionic strength. Rard and Clegg [1997RAR/CLE] raised the concern that the ISE may not be sensitive exclusively to free chloride ions so that other ions in solution may interfere.

Mėndez De Leo and Wood [2005MEN/WOO] used their flow-through high-temperature (348.0-522.5 K) conductivity cell to determine the first association constants of Ca^{2+} with chloride and acetate ions. The association constants were derived from the TBBK conductance model (Turq-Blum-Bernard-Kunz equation) [1995TUR/BLU] and were corrected for hydrolysis of Ca^{2+} with activity coefficients calculated using the mean spherical approximation. Implicit in their data treatment was the extra-thermodynamic assumption that $\lambda_{Cl^-} = \lambda_{CaCl^+} = \lambda_{CaOH^+}$ which was required to obtain the single ion molar conductivities, but this seems to be an unusual assumption.

These authors also re-calculated the experimental results at ambient temperature presented in [1978JOH/PYT] and [1982MAJ/STU] using a Debye-Hückel model in the former case to calculate the activity coefficients with an additional linear term in ionic strength. They used the Pitzer model to treat the data (presumably obtained by private communication) at $I_m = 1$ in [1982MAJ/STU] and arrived at $K_1^o(\text{V.132}) = 6.61$ at 298.15 K which is considerably higher than most of the other values in Table V-72.

The thermodynamic properties of aqueous $CaCl_2$ solutions have been extensively reviewed in [1997RAR/CLE]. However, these authors did not carry out comparative analyses of the actual values of the formation constants nor did they attempt to obtain its value at infinite dilution. Recognising the considerable inconsistency of the low-temperature formation constants, Rard and Clegg pointed out a number of experimental difficulties incurred in the determination of K_1^o(V.132) at concentrations above 5 mol·kg^{-1}, where such a weak ion pair would become significant if only due to the lack of sufficient water for hydration of the ions. They correctly identified the very significant impact that the various activity coefficient models have on the association constants reported by different research groups. Rard and Clegg speculated that a more reliable value of K_1^o(V.132) at ambient conditions could be obtained by extrapolating measurements made at high temperatures. While this suggestion is appealing, analysis of the conductance data of Frantz and Marshall [1982FRA/MAR] shows that their extrapolation from elevated temperatures (to 873 K) and pressures (to almost 400 MPa) produced formation constants four orders of magnitude lower that those generally reported at ambient conditions. Gillespie *et al*. [1992GIL/OSC] suggested a value of $K_1^o(\text{V.132}) = 0.0025$ at 298 K based on a fit of their titration calorimetric data measured from 523 to 598 K. However, it has been pointed out (see Appendix A entry) that the values of formation constants and the enthalpies of reaction derived by Gillespie *et al*. from their calorimetric data are not unique and are highly covariant.

The results available in the literature confirm that it is impossible to assign a definitive value to $K_1^o(CaCl^+)$(V.132) at ambient conditions and indeed the thermodynamic properties of $CaCl_2$ solutions can be modelled effectively to moderate concentrations without taking ion pairing into account.

Savenko [2000SAV] even suggested the existence of $CaCl_2$(aq) ion pairs at 298.15 K. A stability constant ($\beta_2^o = 0.35$) for $CaCl_2$(aq) formation was derived from a dubious linear relationship with solubility that involved a key and unrealistic value for $\beta_2^o(Ca(NO_3)_2(aq))$. Several high-temperature investigations produced β_2^o values based on conductance experiments: [1982FRA/MAR] at extreme conditions (to 873 K and 400 MPa) and AgCl(cr) solubility experiments [1989WIL/SEW] from 423 to 633 K. Not surprisingly, there are challenging technical difficulties with these experiments, and the relevant assumptions and activity coefficient treatments generally appear to be naive. These difficulties are discussed in the corresponding Appendix A entries. It is significant that the β_2^o values obtained from these two studies by extrapolation to 298 K are three orders of magnitude apart (*viz*., $10^{-5.4}$ and $10^{-2.1}$, respectively).

In conclusion, the experimental data presented in Table V-72 embody the typical range of issues encountered in attempts to quantify weak ion association without the benefit of any supporting spectroscopic evidence. Garvin *et al.* [1987GAR/PAR] make no mention of $CaCl^+$ or $CaCl_2(aq)$ species in their CODATA Thermodynamic Tables so that the consensus appears to be that these ion pairs can be ignored in treatments of the thermodynamic properties of $CaCl_2(sln)$ at ambient conditions to moderate concentrations, as is the case for $MgCl_2$ solutions.

V.6.2.2.3 Aqueous calcium(II) sulfate

V.6.2.2.3.1 Ion association of aqueous calcium(II) sulfate

The range of experimental techniques employed to study Reaction (V.134) is similar to those reported for the formation of $MgSO_4(aq)$ although fewer experimental studies have been reported, *viz*., solubility [1953BEL/GEO], [1966MAR/SLU], [1969YEA/MAR], [1975ELG/WED], [1976REA/LAN]; freezing-point depression [1955BRO/PRU]; calorimetry [1969IZA/EAT], [1970LAR]; [1981WAC/RIE]; potentiometry [1967NAK/RAS], [1973AIN], [1978EMA/FAR], [1979ELG/WED], [1997MAL/ZAM]; vapour-pressure osmometry [1975YOK/YAM]; spectrophotometry [1975YOK/YAM2]; conductivity [1932MON/DAV], [1979FIS/FOX]; compared to the number of modelling or data re-treatments in [1962GAR/THO], [1970GAR/GLU], [1972PIT], [1973POW], [1976REA/LAN], [1981TUR/WHI], [1987WEA2], [1990NOR/PLU], [1997SVE/SHO]. The Pitzer ion interaction treatments invoked in [1974PIT/MAY], [1997MAL/ZAM] and [2001MAR] were conducted at a level of precision that did not require consideration of ion pairing to fit the available data satisfactorily. Weare [1987WEA2] gave a particularly elegant discussion of application of the Pitzer ion interaction treatment to cases where a specific ion interaction approach is desirable with the introduction of the β^2 interaction parameter or where ion-association equilibria must be included explicitly, citing the $CaSO_4(sln)$ system specifically. Whereas the former is favoured slightly at ambient temperatures, the latter has to be used at higher temperatures where ion association becomes substantial. The results of these experimental studies, including those where such data were re-treated, are summarised in Table V-73.

$$Ca^{2+} + SO_4^{2-} \rightleftharpoons CaSO_4(aq) \qquad \text{(V.134)}$$

Table V-73: Formation constants for $CaSO_4$(aq) and other thermodynamic quantities (units: $\Delta_r H^o_m$, kJ·mol^{-1}; $\Delta_r S^o_m$, J·K^{-1}·mol^{-1}; $\Delta_r C^o_{p,m}$, J·K^{-1}·mol^{-1}; $\Delta_r V^o_m$, cm^3·mol^{-1}) with 2σ uncertainties.

Method	T/K	Medium	Reported $\log_{10} K^o_1$	Re-calculated SIT $\log_{10} K^o_1$	Derived Eq.(V.135) $\log_{10} K^o_1$	Reference
con	291.15	c_{CaSO_4} 0.0001-0.001	(2.28 ± 0.08)	(1.99 ± 0.20)	2.32	[1932MON/DAV] [1905LAN] [1919HAR/PAI]
sol	273.15	$c_{K_2SO_4}$ 0.0059-0.024	2.22 [a]	(2.20 ± 0.06)	2.24	[1953BEL/GEO]
	298.15	$c_{K_2SO_4}$ 0.0051-0.020	2.31 [a]	(2.24 ± 0.08)	2.35	
	313.15	$c_{K_2SO_4}$ 0.0031-0.014	2.39 [a]	(2.33 ± 0.05)	2.41	
fpt [b]	273.15	m_{CaSO_4} 0.004-0.011	2.27-2.48	(2.41 ± 0.07)	2.24	[1955BRO/PRU]
sol	298.15	m_{NaCl}, m_{CaSO_4} 0-1	(1.9 ± 0.4)			[1966MAR/SLU]
pot	298.15	→ 0 I_c = 0.045-0.179 $CaSO_4 + Na_2SO_4 +$ NaCl	(2.27 ± 0.03)	(2.27 ± 0.10)	2.35	[1967NAK/RAS]
cal	298.15	→ 0	(2.43 ± 0.03) $\Delta_r H^o_m$ (3.39 ± 0.04) $\Delta_r S^o_m$ (57.3 ± 0.04)			[1969IZA/EAT]

(Continued on next page)

Table V-73 (continued)

Method	T /K	Medium	Reported $\log_{10}K_1^\circ$	Re-calculated SIT $\log_{10}K_1^\circ$	Derived Eq.(V.135) $\log_{10}K_1^\circ$	Reference
sol	273.65	$\rightarrow 0$	1.93			[1969YEA/MAR]
		I_m = 0.23-6.02	$\Delta_r H_m^\circ$			
		Na_2SO_4, $NaNO_3$	8.08			
			$\Delta_r S_m^\circ$			
			66.1			
			$\Delta_r C_p^\circ$			
			142			
	298.15		2.03			
est	298.15		2.41			[1970GAR/GLU]
cal	298.15	$\rightarrow 0$	(2.31 ± 0.08)		$\Delta_r H_m^\circ$	[1970LAR]
			$\Delta_r H_m^\circ$		6.88	
			(6.28 ± 0.83)			
			$\Delta_r S_m^\circ$			
			(65.3 ± 2.1)			
est	273.15	$\rightarrow 0$	2.44			[1972PIT]
	298.15	$\rightarrow 0$	2.50			
pot	298.15	$\rightarrow 0$	(2.31 ± 0.01)	(2.42 ± 0.02)	2.35	[1973AIN]
		$I_m \approx 0.0125$	$\Delta_r H_m^\circ$			
		$CaCl_2 + CaSO_4$	(10 ± 2)			
			$\Delta_r S_m^\circ$			
			(80 ± 7)			
			$\Delta_r C_{p,m}^\circ$			
			(260 ± 230)			
	313.15		(2.38 ± 0.01)	(2.43 ± 0.02)	2.41	
	323.15		(2.43 ± 0.01)	(2.42 ± 0.02)	2.44	

(Continued on next page)

Table V-73 (continued)

Method	T /K	Medium	Reported $\log_{10}K_1^o$	Re-calculated SIT $\log_{10}K_1^o$	Derived Eq.(V.135) $\log_{10}K_1^o$	Reference
est	298.15		2.31			[1973POW]
			$\Delta_r H_m^o$ (7.8 ± 0.2)	$\Delta_r H_m^o$ (7.83 ± 0.16)	$\Delta_r H_m^o$ 6.88	Re-calculation of [1969IZA/EAT]
sol	298.15	I_c = 0.7 $CaCl_2$ + Na_2SO_4 + NaCl	$\log_{10} K_{c,1}$ 1.49			[1975ELG/WED]
vp	298.15	m_{CaSO_4} 0.003-0.015	2.07-2.34	(2.12 ± 0.16)	2.35	[1975YOK/YAM]
sp	298.15	c_{CaSO_4} 0.002-0.008	1.78-2.30			[1975YOK/YAM2]
est, sol	298.15	I_m < 0.6 variable	(2.31 ± 0.04)			[1976REA/LAN]
pot, sp	288.15	→ 0 $I_m \leq$ 0.0403 $CaSO_4$, NaCl	(2.24 ± 0.04)	(2.32 ± 0.03)	2.31	[1978EMA/FAR]
	298.15	→ 0 $I_m \leq$ 0.0207 $CaSO_4$ + NaCl	(2.29 ± 0.02) $\Delta_r H_m^o$ 9.20 $\Delta_r S_m^o$ 75.8	(2.33 ± 0.03)	2.35	
	308.15	→ 0 $I_m \leq$ 0.0201 $CaSO_4$ + NaCl	(2.34 ± 0.04)	(2.42 ± 0.03)	2.39	
pot	298.15	I_c = 0.7 $CaCl_2$ + Na_2SO_4 + NaCl	$\log_{10} K_{c,1}$ 1.40			[1979ELG/WED]

(Continued on next page)

Table V-73 (continued)

Method	T/K	Medium	Reported $\log_{10}K_1^\circ$	Re-calculated SIT $\log_{10}K_1^\circ$	Derived Eq.(V.135) $\log_{10}K_1^\circ$	Reference
con	383.15	$c_{CaSO_4} < 0.021$	2.39 $\Delta_r V_m^\circ$ 11.6	(2.50 ± 0.08)	2.28	[1979FIS/FOX]
	388.15		2.34 $\Delta_r V_m^\circ$ 11.6	(2.42 ± 0.21)	2.31	
	293.15		2.33 $\Delta_r V_m^\circ$ 11.6	(2.40 ± 0.05)	2.33	
	298.15		2.34 $\Delta_r V_m^\circ$ 11.6	(2.39 ± 0.08)	2.35	
	303.15		2.33 $\Delta_r V_m^\circ$ 12.0	(2.40 ± 0.10)	2.37	
cal	298.15	m_{CaSO_4} 0.0005-0.01	(2.283 ± 0.001) $\Delta_r H_m^\circ$ (6.665 ± 0.004)		$\Delta_r H_m^\circ$ 6.88	[1981WAC/RIE]
est	298.15	→ 0	2.11 $\Delta_r H_m^\circ$ 5.44 $\Delta_r C_{p,m}^\circ$ 196 $\Delta_r V_m^\circ$ 8.9			[1997SVE/SHO]

(a) Originally reported as ion-pair dissociation constants.
(b) Freezing point measurements.

The solubility product of $CaSO_4 \cdot 2H_2O$(cr) obtained from data at 298.15 K in [1969YEA/MAR], $\log_{10} K_s^\circ = -(4.325 \pm 0.004)$ and $\Delta\varepsilon = (0.040 \pm 0.004)$ kg·mol^{-1} from a less than strict application of the SIT was not considered further in this review, but rather it casts doubt on the validity of the ion-pair constants derived from that study in mixed $Na_2SO_4 + NaNO_3$ electrolyte solutions especially using a simple empirical activity-coefficient equation. An interesting cell design was used to determine the dissociation constant of the $CaSO_4$(aq) ion pair in aqueous $CaCl_2 + CaSO_4$ solutions at 298.15, 313.15 and 323.15 K [1973AIN]. Application of the SIT to these data obtained at relatively low ionic strengths yielded reasonable equilibrium constants which differed from those reported by the author indicating the bias imposed by the activity coefficient model applied.

The osmotic coefficients of magnesium and calcium sulfates measured by vapour-pressure osmometry [1975YOK/YAM] at 298.15 K were considered by Rard and Miller [1981RAR/MIL2] to be suspect due to the uncertain nature of these measurements and the scatter in the values of their association constants.

Emara *et al*. [1978EMA/FAR] utilised a cation-sensitive electrode to quantify magnesium and calcium sulfate ion pairing at 288.15, 298.15 and 308.15 K, followed two years later by similar work conducted exclusively at 298.15 K [1980EMA/LIN]. Attempts were made to minimise the liquid-junction potential, but as in most other studies reported in this section, the solutions contained mixed electrolytes, in this case, $CaSO_4 + NaCl$, at comparable concentrations. However, the limited data tabulated were at relatively low ionic strengths such that activity coefficients could be estimated solely with the Debye-Hückel expression (see the Appendix A entry for [1953BEL/GEO] for more details). In this regard, the potentiometric/solubility study by Nakayama and Rasnick [1967NAK/RAS] at 298.15 K and $I_m < 0.2$ indicated that the activity coefficients were inadequately represented by this expression to this ionic strength (higher than were employed in other studies), so that an additional empirical correction, linear in ionic strength, was required to give internally consistent equilibrium constants.

Equation (V.135) results from a weighted regression of selected $\log_{10} K_1^\circ$ (V.134) and $\Delta_r H_m^\circ$ (V.134) values (listed in Table V-73, column 5) *versus* $(T/\text{K})^{-1}$. Figure V-60 illustrates the goodness of this fit noting that the enthalpy of reaction was determined largely from the calorimetric data.

$$\log_{10} K_1^\circ(\text{V.134}) = (3.5533 \pm 0.0349) - (359.31 \pm 8.55)(T/\text{K})^{-1} \quad \text{(V.135)}$$

Hence, $K_1^\circ((\text{V.134}), 298.15\ \text{K}) = (223 \pm 21)$ is the value selected by this review.

The resulting selected thermodynamic parameters at 298.15 K are:

$$\log_{10} K_1^\circ((\text{V.134}), 298.15\ \text{K}) = \mathbf{(2.348 \pm 0.045)},$$

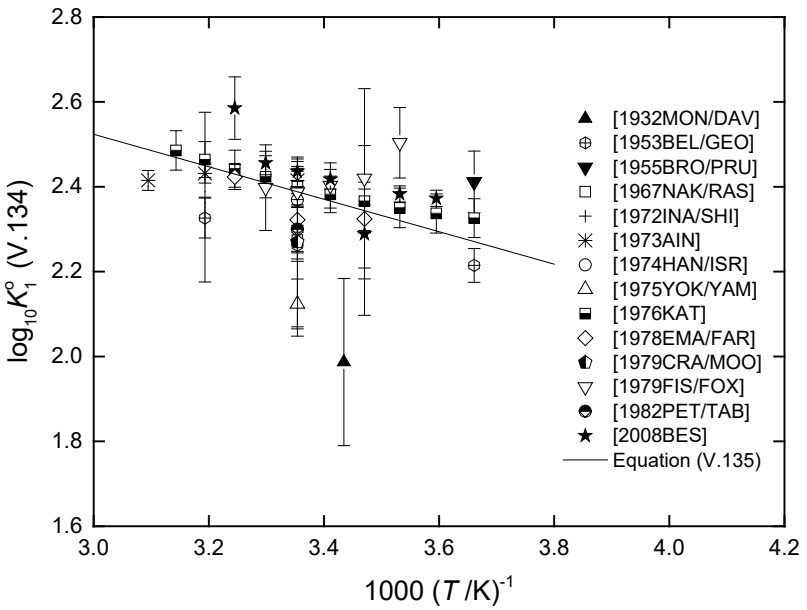
$$\Delta_r G_m^\circ((\text{V.134}), 298.15\ \text{K}) = \mathbf{-(13.403 \pm 0.258)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^\circ((\text{V.134}), 298.15\ \text{K}) = \mathbf{(6.879 \pm 0.321)}\ \text{kJ·mol}^{-1},$$

and $\Delta_r S^o_m$ ((V.134), 298.15 K) = **(68.028 ± 1.381)** J·K^{-1}·mol^{-1} (2σ).

For comparison, for the magnesium analogue (Section V.6.1.2.5.1) K^o_1 = (211.3 ± 5.4) for the formation of $MgSO_4$(aq) and $\Delta_r H^o_m$ = (5.78 ± 0.10) kJ·mol^{-1} at 298.15 K. Gamsjäger *et al*. [2005GAM/BUG] selected K^o_1 = (226.1 ± 15.2) for the formation of $NiSO_4$(aq) and $\Delta_r H^o_m$ = (5.66 ± 0.81) kJ·mol^{-1} at 298.15 K. Lemire *et al*. [2013LEM/BER] selected K^o_1 (298.15 K) = (275 ± 19) for the formation of $FeSO_4$(aq) with $\Delta_r H^o_m$ = (8.4 ± 6.2) kJ·mol^{-1} at 298.15 K.

Figure V-60: Plot of $\log_{10} K^o_1$ (V.134) for the formation of $CaSO_4$(aq) *versus* $(T/\mathrm{K})^{-1}$.



Pitzer [1972PIT] analysed an array of experimental data such as heats of dilution, osmotic coefficients and cell potentials of MSO_4 solutions from 273.15 to 298.15 K. He estimated the ion-pair formation constants including values of K^o_2 which referred to both MSO_4(aq) and $M(SO_4)^{2-}_2$, which were indistinguishable. The formation of higher-order sulfate ion pairs is still controversial and the relatively small values for K^o_2 may be an artefact of the fitting procedure. Based on the quality of the fits, Pitzer also concluded that assigning an activity coefficient of unity for the neutral ion pair was justified. Reardon and Langmuir's [1976REA/LAN] treatment of previous gypsum solubility data and consideration of available $CaSO_4$ ion-pair constants led them to conclude that assuming the activity coefficient of the $CaSO_4$(aq) ion pair is unity is incorrect. However, their treatment also included consideration of ion pairing between

univalent cations and sulfate and relied on the extra-thermodynamic assumption that $\gamma_{K^+} = \gamma_{Cl^-} = \gamma_{\pm KCl}$. Therefore, this review prefers the more conservative conclusion (also adopted for the magnesium analogue) of Pitzer and others that $\gamma_{CaSO_4(aq)} = 1$, also noting that $\gamma_{NiSO_4(aq)}$ was assumed to be unity by Gamsjäger *et al*. [2005GAM/BUG].

Taking the thermodynamic properties of Ca^{2+} and SO_4^{2-} from the auxiliary data in Table IV-1, the corresponding values for the formation (2σ) of $CaSO_4$(aq) are:

$$\Delta_f G^o_m(CaSO_4, aq, 298.15\ K) = \mathbf{-\ (1310.213 \pm 1.159)}\ kJ\cdot mol^{-1},$$

$$\Delta_f H^o_m(CaSO_4, aq, 298.15\ K) = \mathbf{-\ (1445.461 \pm 1.124)}\ kJ\cdot mol^{-1},$$

$$S^o_m(CaSO_4, aq, 298.15\ K) = \mathbf{(30.328 \pm 1.751)}\ J\cdot K^{-1}\cdot mol^{-1},$$

and these are our selected values.

Note that these values vary somewhat from the CODATA evaluation by Garvin *et al*. [1987GAR/PAR] who reported: $\Delta_f G^o_m(CaSO_4, aq, 298.15\ K) = -(1311.548 \pm 0.738)\ kJ\cdot mol^{-1}$, $\Delta_f H^o_m(CaSO_4, aq, 298.15\ K) = -(1448.951 \pm 0.818)\ kJ\cdot mol^{-1}$ and $S^o_m(CaSO_4, aq, 298.15\ K) = (22.423 \pm 3.716)\ J\cdot K^{-1}\cdot mol^{-1}$.

V.6.2.2.4 Aqueous calcium(II) carbonate

V.6.2.2.4.1 Aqueous carbonato complexes of Ca^{2+}

In spite of the great geochemical and industrial interest, only a limited number of experimental data have been reported for the aqueous interaction of Ca^{2+} with HCO_3^-/CO_3^{2-} ions, especially at non-zero ionic strength. The available references reporting explicit equilibrium constants for the Ca^{2+}- HCO_3^-/CO_3^{2-} ion pairs are listed in Table V-74. These studies describe the existence of two ion pairs (Equations (V.136) and (V.137)).

$$Ca^{2+} + HCO_3^- \rightleftharpoons CaHCO_3^+ \quad \text{(V.136)}$$

$$Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3(aq) \quad \text{(V.137)}$$

There was a debate in the literature concerning the formation the $CaHCO_3^+$ ion pair. However, some recent publications seemingly resolved the conflicting experimental observations (‘*analysis of anhydrous calcium carbonate solubility determinations from literature shows that the data can be split up into two internally consistent but mutually inconsistent data sets. The first is consistent with the occurrence of a stable* $CaHCO_3^+$ *ion pair, while the second is inconsistent. It is assumed that the latter data set was obtained with charged calcite particles, and that this surface charge results in an increased solubility constant at low* $CO_2(g)$ *pressure’* [2003VIS/VAN], [2005VIS/VAN]).

Although, the formation of a $CaHCO_3^+$ ion pair has a significant effect on the p_{CO_2} dependence of calcium carbonate solubility [1982PLU/BUS], the data listed

in Table V-74 suggest a rather weak interaction. According to [1985PIT/OLS] it is more appropriate to take this weak interaction into account by the introducing ion interaction coefficients $\varepsilon(Ca^{2+}, HCO_3^-)$, similar to the Mg^{2+}- HCO_3^- interaction. Although He and Morse determined the first pK of carbonic acid in (Na, Ca) Cl media [1993HE/MOR], the relatively high uncertainties in these data (see Appendix A) do not allow the calculation of $\varepsilon(Ca^{2+}, HCO_3^-)$ in the same way as was described in Section V.6.1.2.7. On the other hand, several references report the Pitzer parameters of Ca^{2+}- HCO_3^- interaction: $\beta^0 = 0.4$ kg·mol^{-1} and $\beta^1 = 2.977$ kg·mol^{-1} [1984HAR/MOL], $\beta^0 = 0.28$ kg·mol^{-1}, $\beta^1 = 0.3$ kg·mol^{-1} [1985PIT/OLS], $\beta^0 = 0.20$ kg·mol^{-1}, $\beta^1 = 0.30$ kg·mol^{-1} [1993HE/MOR]. Among these data, those reported in [1985PIT/OLS], based on electrochemical measurements with a Harned cell are the most accurate. From their values,

$$\varepsilon(Ca^{2+}, HCO_3^-) = \mathbf{(0.113 \pm 0.030)}\ \text{kg·mol}^{-1}$$

can be estimated [1997GRE/PLY]. In this review this value is selected and used in the further evaluations.

As mentioned above, very limited data are available for the stability constant of the $CaCO_3$(aq) ion pair at non-zero ionic strength (Table V-74). Greenwald reported $\log_{10} K_{CaCO_3} = 3.0$ at $I_m = 0.15$ KCl [1941GRE]. Since this value is too high compared to other data, and several questionable assumptions were made [1974JAC/LAN], it was not considered further. The second dissociation constant of carbonic acid in (Na,Ca)Cl medium was reported in [1974PYT/HAW] and [1995RUS/CHE]. Since in both cases only $I_m = 0.72$ was studied, an evaluation of ionic strength dependence was not possible. Therefore, $\varepsilon(CaCO_3(aq), Na^+ + Cl^-) = 0$ kg·mol^{-1} was used in this review, analogous to [1993HE/MOR] and [2012VIS/VAN]. The stoichiometric and thermodynamic constants based on these data (see Table V-74) were calculated as described in Section V.6.1.2.5.

Table V-74: References with explicit equilibrium constants for the Ca^{2+}– HCO_3^- / CO_3^{2-} ion pairs.

Reference	T /K	method	I_m range	$\log_{10}K_m$ /reported	$\log_{10}K_m$ /re-calculated	$\log_{10}K_m$ /accepted	Notes
$Ca^{2+} + HCO_3^- \rightleftharpoons CaHCO_3^+$							
[1941GRE]	295.15	pH, sol	0.15 [(a)] KCl	0.81			
[1973DYR/HAN]	298.15	pH	0.7 [(a)] NaCl	0.02			estimated
[1974JAC/LAN]		cond	0.002-0.02 $Ca(HCO_3)_2$				
	288.15		→ 0	0.88			
	298.15		→ 0	0.99			
	308.15		→ 0	1.16			
	318.15		→ 0	1.29			
[1972MAR/VAS]		ise-Ca^{2+}	$Ca(HCO_3)_2$ + 0-0.15 [(a)] NaCl				
[1974MAR/VAS]			→ 0				
	295.15		→ 0	1.27			
	333.15		→ 0	1.64			
	343.15		→ 0	1.77			
	353.15		→ 0	1.82			
	363.15		→ 0	1.94			
	371.15			2.01			
[1974PYT/HAW]	298.15	pH	0.72 (NaCl, $CaCl_2$)	0.29			
[1981BAU]		pH, cal	? $KHCO_3$/$CaCl_2$				
	383.15		→ 0	1.23			
	298.15		→ 0	1.23			
	313.15		→ 0	1.26			
	328.15		→ 0	1.31			
	343.15		→ 0	1.38			
	363.15		→ 0	1.50			

(Continued on next page)

Table V-74 (continued)

Reference	T /K	method	I_m range	$\log_{10}K_m$ /reported	$\log_{10}K_m$ /re-calculated	$\log_{10}K_m$ /accepted	Notes
[1982PLU/BUS]		pH, sol	0-0.016 $Ca(HCO_3)_2$				
	277.65		$\rightarrow 0$				
	277.65		$\rightarrow 0$	(0.91 ± 0.06)			
	288.25		$\rightarrow 0$	(0.97 ± 0.06)			
	298.15		$\rightarrow 0$	(1.14 ± 0.07)			
	308.15		$\rightarrow 0$	(1.17 ± 0.07)			
	318.15		$\rightarrow 0$	(1.21 ± 0.10)			
	338.15		$\rightarrow 0$	(1.21 ± 0.08)			
	353.15		$\rightarrow 0$	(1.27 ± 0.08)			
	363.15		$\rightarrow 0$	(1.35 ± 0.10)			
[1983GUY/DOR]		sol	? $Ca(HCO_3)_2$				
	298.15		$\rightarrow 0$	1.14			
[1990DOR/MAR]		sol	? $Ca(HCO_3)_2$				
	278.15		$\rightarrow 0$	1.03			
	288.15		$\rightarrow 0$	1.10			
	298.15		$\rightarrow 0$	1.14			
	308.15		$\rightarrow 0$	1.19			
	318.15		$\rightarrow 0$	1.23			
	328.15		$\rightarrow 0$	1.26			
	338.15		$\rightarrow 0$	1.29			
	348.15		$\rightarrow 0$	1.32			
$Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3(aq)$							
[1941GRE]	295.15	pH, sol	0.15 [(a)] KCl	3.0			
[1962GAR/THO]	298.15	pH	$\rightarrow 0$	3.2		(3.2 ± 0.2)	
[1968NAK]	298.15	pH, ise-Ca^{2+}	$\rightarrow 0$	4.48			
[1973DYR/HAN]	298.15	pH	0.7 [(a)] NaCl	1.51			estimated

(Continued on next page)

Table V-74 (continued)

Reference	T /K	method	I_m range	$\log_{10}K_m$ /reported	$\log_{10}K_m$ /re-calculated	$\log_{10}K_m$ /accepted	Notes
[1974PYT/HAW]	298.15	pH	0.72 (NaCl, $CaCl_2$)	2.21	1.49	(1.49 ± 0.20)	
					3.10	(3.10 ± 0.20)	
[1970LAF]	298.15	sol	→ 0	(3.1 ± 0.3)		(3.1 ± 0.3)	calculated from literature solubility data
[1972MAR/VAS]		ise-Ca^{2+}	$Ca(HCO_3)_2$ + 0-0.15 NaCl				
[1974MAR/VAS]	295.15		→ 0	4.39			
	333.15		→ 0	5.34			
	343.15		→ 0	5.55			
	353.15		→ 0	5.74			
	363.15		→ 0	5.82			
	371.15		→ 0	6.00			
[1974REA/LAN]		pH	0-0.02 $MgCl_2/K_2CO_3$				
	282.65		→ 0	(3.04 ± 0.04)		(3.04 ± 0.10)	
	298.15		→ 0	(3.15 ± 0.08)		(3.15 ± 0.10)	
	312.15		→ 0	(3.34 ± 0.07)		(3.34 ± 0.10)	
[1982PLU/BUS]		pH	0-0.02 $K_2CO_3/CaCl_2$				
	278.65		→ 0	(3.13 ± 0.02)		(3.13 ± 0.02)	
	298.15		→ 0	(3.20 ± 0.07)		(3.20 ± 0.07)	
	313.15		→ 0	(3.42 ± 0.11)		(3.42 ± 0.11)	
	333.15		→ 0	(3.63 ± 0.16)		(3.63 ± 0.16)	
	353.15		→ 0	(3.92 ± 0.18)		(3.92 ± 0.18)	
[1983GUY/DOR]		sol	? $Ca(HCO_3)_2$				
	298.15		→ 0	4.44			
[1973LAR/SOL]		pH	0.003-0.032 $(K,Mg)ClO_4$				
	278.15		→ 0	3.17		(3.17 ± 0.15)	
	288.15		→ 0	3.17		(3.17 ± 0.15)	
	298.15		→ 0	3.22		(3.22 ± 0.15)	

(Continued on next page)

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Table V-74 (continued)

Reference	T /K	method	I_m range	$\log_{10}K_m$ /reported	$\log_{10}K_m$ /re-calculated	$\log_{10}K_m$ /accepted	Notes
[1990DOR/MAR]		sol	? $Ca(HCO_3)_2$				
	278.15		$\rightarrow 0$	4.15			
	288.15		$\rightarrow 0$	4.19			
	298.15		$\rightarrow 0$	4.31			
	308.15		$\rightarrow 0$	4.27			
	318.15		$\rightarrow 0$	4.35			
	328.15		$\rightarrow 0$	4.45			
	338.15		$\rightarrow 0$	4.45			
	348.15		$\rightarrow 0$	4.53			
[1995RUS/CHE]	298.15	pH	0.718 NaCl		1.58	(1.58 ± 0.25)	
			$\rightarrow 0$		3.19	(3.19 ± 0.25)	
[2007BYC/BEN]		sol	? NaCl				
	393.15		$\rightarrow 0$	(4.40 ± 0.03)		(4.40 ± 0.20)	
	413.15		$\rightarrow 0$	(5.16 ± 0.16)		(5.16 ± 0.60)	
	433.15		$\rightarrow 0$	(5.00 ± 0.10)		(5.00 ± 0.30)	

(a) Ionic strength given in molar units.

On the other hand, several references reported stability constants of the $CaCO_3$(aq) ion pair at zero ionic strength (Table V-74). All these data were determined at low ionic strength using the SIT-related extrapolation to zero ionic strength. The stability constants reported by Nakayama [1968NAK] and Martynova *et al*. [1972MAR/VAS], [1974MAR/VAS] are too high as compared with the accepted data. Besides, their experimental details have been criticised in [1974JAC/LAN] and [1982PLU/BUS]; therefore, these data were not considered further. The data reported in [1983GUY/DOR] and [1990DOR/MAR] were also discarded due to large deviations from the expected values. Based on the seven independent values of $\log_{10}K^\circ$ (V.137) at 298.15 K, listed in Table V-74, the following value is selected in this review:

$$\log_{10}K^\circ\,((\text{V.137}),\,298.15\ \text{K}) = \mathbf{(3.166 \pm 0.100)}.$$

The temperature dependence of $\log_{10}K^\circ$ (V.137) was derived from the above selected value and from those accepted for $T \neq 298.15$ K (see Table V-74). The constants reported in [2007BYC/BEN] extended considerably the temperature range of experimentally determined $\log_{10}K^\circ$ (V.137) values, but were not considered in the earlier compilations. However, only limited experimental details are reported in this reference and the constants reported for 413 K do not fit with the expected trend (see dashed line in Figure V-61 for the predicted temperature dependence in [2012VIS/VAN]). Therefore, the uncertainties in these data were increased, especially that of the constant at 413 K. The accepted $\log_{10}K^\circ$ (V.137) values as a function of temperature are depicted in Figure V-61 and can be best described by Equation (V.138).

$$\log_{10}K^\circ\,(\text{V.137}) = -308.46387 - 0.05225(T/\text{K}) + 9908.87522(T/\text{K})^{-1} + 118.80558\log_{10}(T/\text{K}) \qquad \text{(V.138)}$$

From the above data the following thermodynamic parameters can be calculated and selected[2]:

$$\Delta_r G_m^\circ\,((\text{V.137}),\,298.15\ \text{K}) = \mathbf{-\,(18.089 \pm 0.570)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^\circ\,((\text{V.137}),\,298.15\ \text{K}) = \mathbf{(15.890 \pm 1.500)}\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^\circ\,((\text{V.137}),\,298.15\ \text{K}) = \mathbf{(113.966 \pm 5.382)}\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

These values compare relatively well with those reported in [1974REA/LAN] ($\Delta_r G_m^\circ = -(18.01 \pm 0.46)$ kJ·mol^{-1}, $\Delta_r H_m^\circ = (16.84 \pm 3.35)$ kJ·mol^{-1}, $\Delta_r S_m^\circ = (116.73 \pm 11.30)$ J·K^{-1}·mol^{-1}), and with the data calculated using the parameters

[2] During the review process one of our reviewers made us aware of the recently published [2020BYC/BEN], which reports refined values for $\log_{10}K$ (V.137) at 413.15 K compared to the previous publication [2007BYC/BEN]. Although, the refined new value at 413.15 K ($\log_{10}K$ (V.137) = 4.75) is considerably different from the earlier one (reported in [2007BYC/BEN]), in our assessment we increased the uncertainty of this (old) data point, and thus the fitted curve gives a value close to 4.75. Some new solubility data for calcite at 100-160 °C and 1-50 bar p_{CO_2} are also reported in [2020BYC/BEN].

listed in [2012VIS/VAN] ($\Delta_r G_m^o = -18.22$ kJ·mol^{-1}, $\Delta_r H_m^o = 17.66$ kJ·mol^{-1}, $\Delta_r S_m^o = 120.32$ J·K^{-1}·mol^{-1}).

Combining our calculated values with the recommended parameters for Ca^{2+} and CO_3^{2-}, the following thermodynamic quantities have been selected:

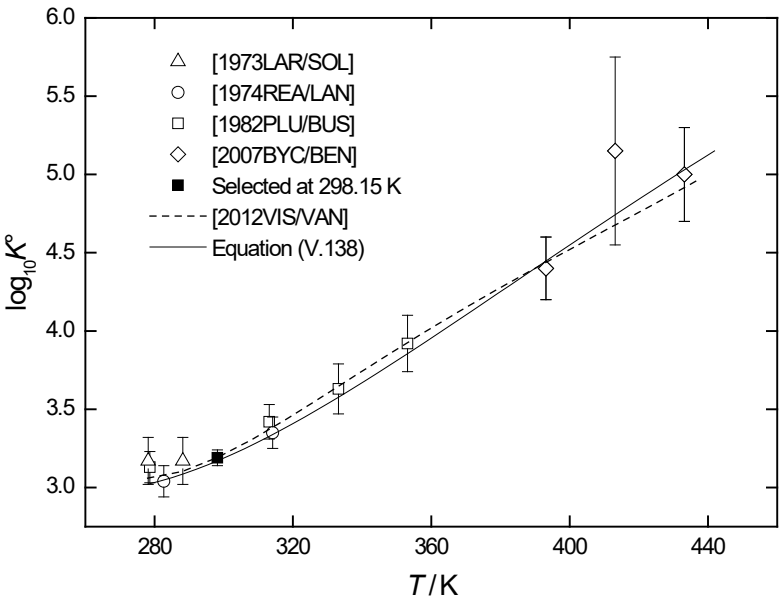
$$\Delta_f G_m^o (CaCO_3, aq, 298.15\ K) = \mathbf{-\ (1098.795 \pm 1.257)}\ kJ \cdot mol^{-1},$$

$$\Delta_f H_m^o (CaCO_3, aq, 298.15\ K) = \mathbf{-\ (1202.340 \pm 1.820)}\ kJ \cdot mol^{-1},$$

$$S_m^o (CaCO_3, aq, 298.15\ K) = \mathbf{(7.766 \pm 5.565)}\ J \cdot K^{-1} \cdot mol^{-1}.$$

Despite the slightly different values used for the formation data of Ca^{2+} and CO_3^{2-} ions, these parameters compare well with those reported in [1997SVE/SHO] ($\Delta_f G_m^o = -1099.764$ kJ·mol^{-1}, $\Delta_f H_m^o = -1202.440$ kJ·mol^{-1}, $S_m^o = 10.46$ J·K^{-1}·mol^{-1}) and in [2017MIR/WAG] ($\Delta_f G_m^o = -1099.084$ kJ·mol^{-1}, $S_m^o = 20.11$ J·K^{-1}·mol^{-1}).

Figure V-61: The temperature dependence of the equilibrium formation constants of $CaCO_3$(aq) (dashed line was calculated from the temperature dependence derived in [2012VIS/VAN]).



V.6.2.2.4.2 The solubility of calcite

Calcite, the most common and stable form of natural calcium carbonate, $CaCO_3$(cr), is one of the most widely distributed minerals in the earth's crust and is extensively used in

a number of industrial processes. In accordance with its exceptional geological and industrial importance, a remarkably large number of papers have reported solubility data in open or closed systems both in pure water and in different electrolyte solutions (HCl, NaCl, KCl, NH_4Cl, NH_4NO_3, $MgCl_2$, $CaCl_2$). Among these electrolyte solutions only NaCl is considered in the present review, due to the number of experimental data and its general importance. Since different methods were used by the authors to derive solubility products, the reported solubility data were re-evaluated for the purposes of this review, considering the formation of only Ca^{2+}, $CaOH^+$ and $CaCO_3(aq)$ as aqueous species.

As for magnesite and aragonite, the thermodynamic models used here for the evaluation of calcite solubility in open and closed systems are described in [2012VIS/VAN], but the activity coefficients were calculated using the NEA TDB recommended SIT approach. In our calculations, the equilibrium constants ($\log_{10}K_{CaOH^+}$, $\log_{10}K_{CaCO_3}$) and ion interaction coefficients ($\varepsilon(Na^+, CO_3^{2-})$, $\varepsilon(Na^+, HCO_3^-)$, $\varepsilon(Ca^{2+}, HCO_3^-)$, $\varepsilon(Ca^{2+}, Cl^-)$) selected in the present review were used. Other ε(i,j) parameters and their temperature dependencies were taken from [2001PRE/GAM]. Some ion interaction coefficients $\varepsilon(CaOH^+, CO_3^{2-})$, $\varepsilon(CaOH^+, HCO_3^-)$ and the temperature dependence of $\varepsilon(Ca^{2+}, HCO_3^-)$ are unknown, although the concentrations of these species are very low, so this does not affect the results markedly.

The solubility of calcite is defined by the Reaction (V.139):

$$CaCO_3(calc) \rightleftharpoons Ca^{2+} + CO_3^{2-} \qquad \text{(V.139)}$$

For the open system (calcite-H_2O-CO_2) 37 references were considered for evaluation [1869COS], [1872SCH], [1888ENG3], [1910VET], [1911MCC/SMI], [1916CAV], [1917SEY/LLO], [1921BAC], [1923MIT2], [1924HAE2], [1929FRE/JOH], [1932BAR], [1935HAL/RIT], [1945SHT], [1945SHT/FRO], [1952MIL2], [1952SHT/FRO], [1958TRO], [1959ELL], [1960BER/BOR], [1960YAN], [1962SEG/HOL], [1963MAL], [1964PIC], [1964SIP/GLO], [1968NAK], [1971STC], [1973TUR], [1982PLU/BUS], [1983SAS/MOR], [1989WOL/BRE], [2004LOO/PAS] as well as [1909LEA/SEN], [1912EHL/HEM], [1912KEN], [1924KUG/SHO], [1960YAN], [1971STC]. The experimental data reported in the last six references were rejected. In [1912KEN], [1960YAN] and [1971STC] the reported solubilities were too low, whereas in [1912EHL/HEM] and [1924KUG/SHO] they were too high compared to those reported in other references, while the solubility data in [1909LEA/SEN] are highly scattered. Most of the data reported in [1945SHT], [1945SHT/FRO] and [1952MIL2] were also neglected, since too high and low solubilities were determined, respectively. Several outlying points were discarded from the other sources. Finally, among the 900 reported solubility data, 604 were accepted (see Figure V-62).

Ten references reported 292 data points for calcite solubility in the open system (calcite-H_2O-CO_2) based on pH measurements [1922KOL], [1965GRE/BAS], [1971STC], [1972NAK], [1973PIC], [1973TUR], [1974JAC/LAN], [1983SAS/MOR],

[1983GUY/DOR], [1990DOR/MAR]. The experimental data reported in [1922KOL] and [1971STC], as well as four other outliers were discarded. The remaining 279 data points were accepted (Figure V-63).

Eight references reported solubility data in the (calcite-H_2O-CO_2-NaCl) system [1912EHL/HEM], [1929FRE/JOH], [1945SHT], [1945SHT/FRO], [1952SHT/FRO], [1963ELL2], [1965AKI/LAG] and [1989WOL/BRE]. As mentioned above, the solubilities reported in [1912EHL/HEM] are too high, and all their data are rejected. The solubilities reported in [1989WOL/BRE] and the re-evaluated solubility products as a function of ionic strength are depicted in Figure V-64. The solubility passes through a maximum at around 2.0 mol·dm^{-3} (NaCl). The re-evaluated $\log_{10}K^{\circ}_{s,0}$ values are close (± 0.1) to those determined in pure aqueous solutions up to I_m = 2-2.5 (NaCl), but considerably higher values were calculated at higher ionic strengths. Besides the possible experimental errors, these deviations indicate that the validity range of the present SIT treatment is I_m = 0-2.5 (NaCl). Therefore, the solubility products calculated for I_m > 2.0 (NaCl) (I_m > 1.0 (NaCl) for [1945SHT] and [1952SHT/FRO]) were rejected, *i.e.*, among the 123 reported solubility data, 84 were accepted (Figure V-63).

Figure V-62: Temperature dependence of the re-evaluated solubility product of calcite in the open system (calcite-H_2O-CO_2).

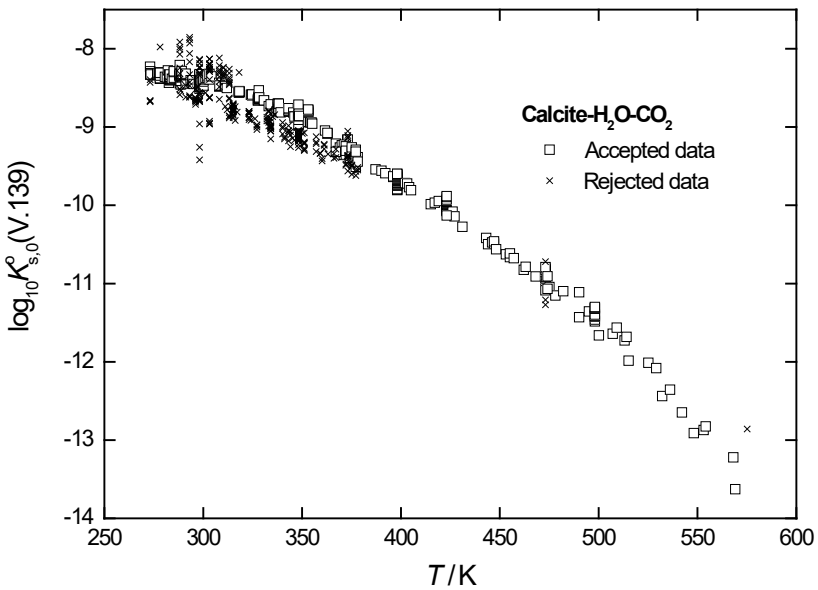


Figure V-63: Temperature dependence of the re-evaluated solubility product of calcite in (calcite-H_2O-CO_2), based on pH measurements, and in (calcite-H_2O-CO_2-NaCl).

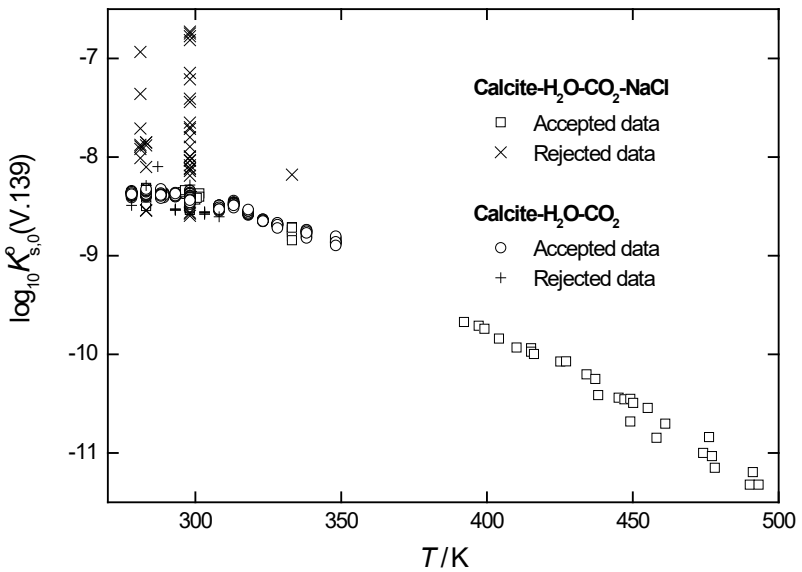
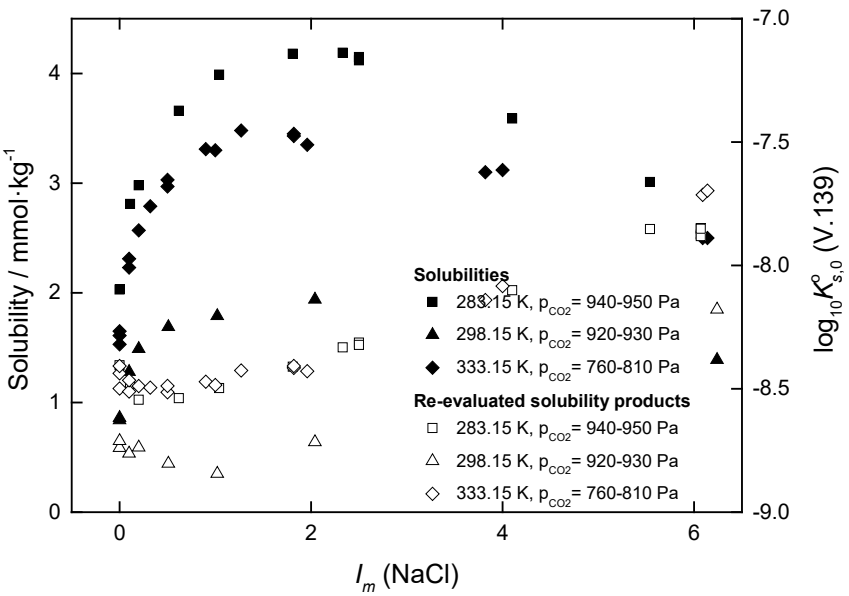
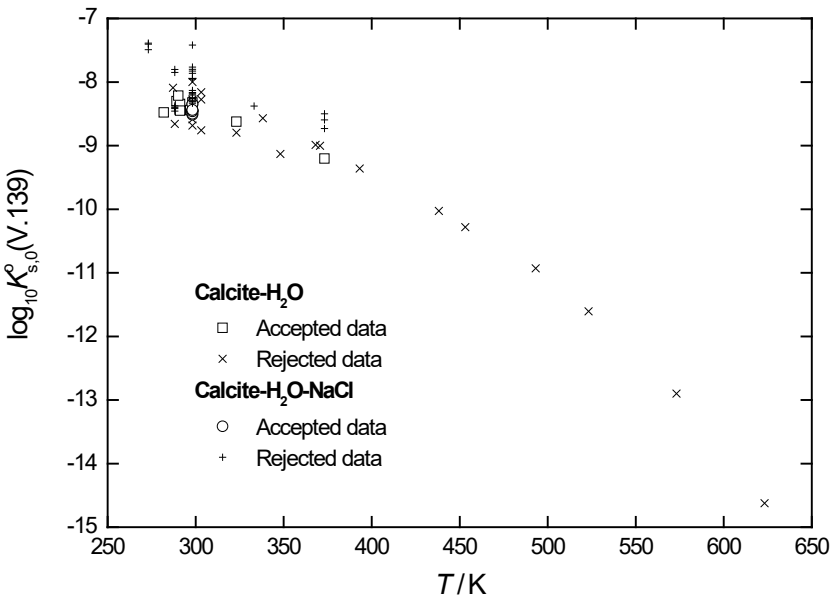


Figure V-64: Solubilities and re-evaluated solubility products in (calcite + H_2O + CO_2 + NaCl) as a function of I_m (data from [1989WOL/BRE]).



For the (calcite-H_2O) system, 13 references reported solubility data. Due to the well-known experimental difficulties related to closed systems (*e.g.*, CO_2 contamination) only ten experimental data reported in [1872SCH], [1893HOL], [1906BLA/NOV], [1912KEN], [1912TIL/HEU], [1923ASK] and [1925STU] were accepted (Figure V-65). The other sources reported lower [1929PAU/STE], [2002KON/NAN] or higher [1922KOL], [1962MOR] solubility than expected, while considerably scattered data are reported in [1932BAR].

Figure V-65: Temperature dependence of the re-evaluated solubility product of calcite in the (calcite-H_2O) systems.



Six references are available for calcite solubility in the (calcite-H_2O-NaCl) system. Most of these report considerably higher solubilities than expected, probably due to unwanted CO_2 uptake [1915GOT], [1916WAR/KOU], [1923ASK], [1932LEI]. Although the re-evaluated solubility products based on the data in [1905CAN/GOG] up to $I_m \leq 2$ are in the expected range, the system cannot be considered as a truly closed one (see Appendix A entry). Consequently, only ten data points reported in [1984MIL/MIL] up to $I_m < 3$ can be accepted (Figure V-65). From the solubility data measured at higher ionic strength ($I_m \geq 3$) considerably higher $\log_{10} K^{\circ}_{s,0}$ values were calculated, as described above for the (calcite-H_2O-CO_2-NaCl) system (Figure V-64). This observation illustrates again that the currently used set of ion interaction coefficients is valid only up to $I_m = 2.0$, which is the usual validity range of the SIT approach.

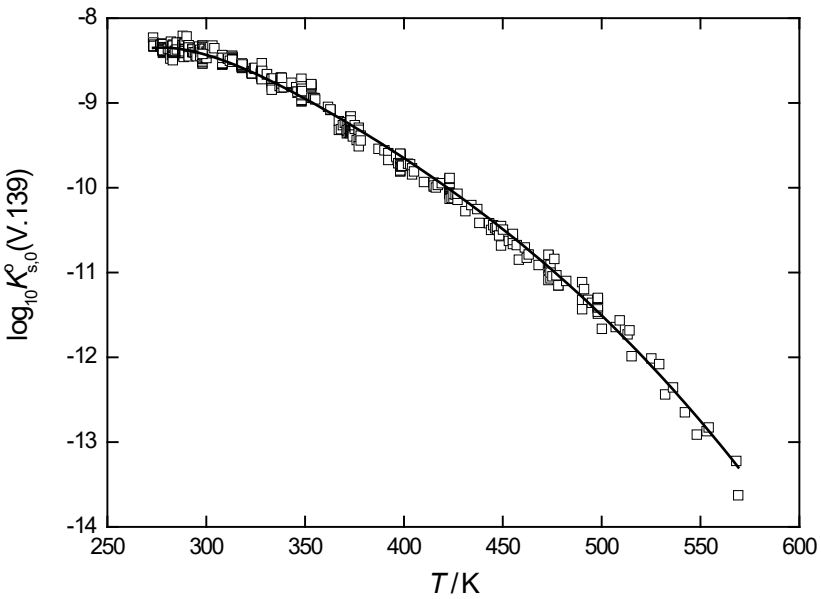
Accordingly, all together 988 data points were accepted among them 164 data refer to 298.15 K. From the latter data set:

$$\log_{10} K^{\circ}_{s,0}\,((\text{V.139}),\ 298.15\ \text{K}) = -\,(8.421 \pm 0.100),$$

can be calculated for Reaction (V.139).

All the 988 accepted data points were used to derive the temperature dependence of the calcite solubility product ($\log_{10} K^{\circ}_{s,0}$ for Reaction (V.139)) in the temperature range 273.15-575.15 K. See Figure V-66.

Figure V-66: Temperature dependence of the re-evaluated solubility product of calcite (Reaction (V.139)) accepted for both open and closed systems. The solid line is based on Equation (V.141).



Regression of the data depicted in Figure V-66 was first constrained to $\Delta_r G^{\circ}_m\,((\text{V.139}),\ 298.15\ \text{K}) = (48.065 \pm 0.571)\ \text{kJ·mol}^{-1}$, which is based on $\log_{10} K^{\circ}_{s,0}\,((\text{V.139}),\ 298.15\ \text{K}) = -\,(8.421 \pm 0.100)$ calculated from the 144 experimental data accepted for 298.15 K. In this way the regression resulted in Equation (V.140):

$$\log_{10} K^{\circ}_{s,0}\,(\text{V.139}) = -1485.679 + 83982.618(T/\text{K})^{-1} + 535.1952\log_{10}(T/\text{K}) - 0.23477(T/\text{K}) - 5220526.8(T/\text{K})^{-2} \qquad (\text{V.140})$$

The above parameters correspond to the following thermodynamic quantities for Equation (V.139) at 298.15 K: $\Delta_r G^{\circ}_m\,((\text{V.139}),\ 298.15\ \text{K}) = (48.065 \pm 0.571)\ \text{kJ·mol}^{-1}$,

$\Delta_r H_m^o$ ((V.139), 298.15 K) = – (10.211 ± 1.000) kJ·mol^{-1} and $\Delta_r S_m^o$ ((V.139), 298.15 K) = – (195.460 ± 5.000) J·K^{-1}·mol^{-1}. The latter value would lead to S_m^o ($CaCO_3$, calc, 298.15 K) = (89.257 ± 5.000) J·K^{-1}·mol^{-1}. However, a considerably more precise value S_m^o ($CaCO_3$, calc, 298.15 K) = (91.710 ± 0.210) J·K^{-1}·mol^{-1} has been selected in this review (see Section V.6.2.1.9.1), which corresponds to $\Delta_r S_m^o$ ((V.139), 298.15 K) = – (197.910 ± 1.430) J·K^{-1}·mol^{-1}. Therefore, regression of the data in Figure V-66 has been doubly constrained, on one hand to $\Delta_r G_m^o$ ((V.139), 298.15 K) = (48.065 ± 0.571) kJ·mol^{-1}, and on the other hand to the latter $\Delta_r S_m^o$ ((V.139), 298.15 K) value. This doubly constrained regression resulted in Equation (V.141) (see solid line in Figure V-66).

$$\begin{aligned}\log_{10} K_{s,0}^o\,(\text{V.139}) = &-1528.838 + 85733.361(T/\text{K})^{-1} \\ &+ 551.4571\log_{10}(T/\text{K}) - 0.24285(T/\text{K}) \\ &- 5268934.6(T/\text{K})^{-2}\end{aligned} \qquad \text{(V.141)}$$

In this way the following thermodynamic quantities have been calculated:

$$\Delta_r G_m^o\,((\text{V.139}), 298.15\ \text{K}) = (48.065 \pm 0.571)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H_m^o\,((\text{V.139}), 298.15\ \text{K}) = -\,(10.942 \pm 1.000)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^o\,((\text{V.139}), 298.15\ \text{K}) = -\,(197.910 \pm 3.862)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

These values agree very well with those calculated in [2012VIS/VAN2] for Model 2, where Pitzer parameters (not the stability constant) were used for the $CaHCO_3^-$ ion pair. The above thermodynamic parameters, combined with those recommended for Ca^{2+} and CO_3^{2-}, correspond to:

$$\Delta_f G_m^o\,(CaCO_3, \text{calc}, 298.15\ \text{K}) = -\,(1128.771 \pm 1.500)\ \text{kJ·mol}^{-1},$$

$$\Delta_f H_m^o\,(CaCO_3, \text{calc}, 298.15\ \text{K}) = -\,(1207.288 \pm 1.500)\ \text{kJ·mol}^{-1}.$$

The above values compare well with those selected in this review, which include several other experimental sources (see Table V-67), as well as with those selected by Robie and Hemingway ($\Delta_f G_m^o$ = – (1128.5 ± 1.5) kJ·mol^{-1}, $\Delta_f H_m^o$ = – (1207.4 ± 1.3) kJ·mol^{-1}, S_m^o = (91.7 ± 0.20) J·K^{-1}·mol^{-1}, [1995ROB/HEM]) and by Villafáfila García *et al.* using the Extended UNIQUAC model ($\Delta_f G_m^o$ = – 1129.057 kJ·mol^{-1}, $\Delta_f H_m^o$ = – 1207.4 kJ·mol^{-1}, [2006VIL/THO]).

V.6.2.2.4.3 The solubility of aragonite ($CaCO_3$(cr))

Among the three $CaCO_3$ minerals, which are metastable relative to calcite (ikaite, vaterite, aragonite), only the latter is considered in this review. Despite its wide ranging geological and biological occurrences, only a few papers report experimental data for the aqueous solubility of aragonite, such as for the open system (aragonite-H_2O-CO_2): [1921BAC], [1961YAN/RAS], [1973HUL/TUR], [1976BER], [1982PLU/BUS] and [1983SAS/MOR]. Synthetic aragonite, prepared by the authors, has been also studied in [1961YAN/RAS]. They noted the presence of calcite in their compound, and the reported

solubilities are close to those of calcite. Therefore, these data were neglected in our evaluation. Since different methods were used by the authors to derive the solubility products, the reported solubility data were re-evaluated for the purposes of this review using the SIT approach, considering only the aqueous species Ca^{2+}, $CaOH^{+}$ and $CaCO_3(aq)$.

The solubility of aragonite has been also studied in an open system in presence of $MgCl_2$ [1974SIE/HOS], $CaCl_2$ [1983SAS/MOR] and artificial sea water [1965MAC], [1976BER], [1980MOR/MUC], [1980COO/KEP]. Since these electrolytes are beyond the scope of this review, the reported data were not considered. Browman and Hastings studied the aragonite solubility at 311 K in open systems in the presence of 25×10^{-3} mol·dm^{-3} $NaHCO_3$ and $(0\text{-}200) \times 10^{-3}$ mol·dm^{-3} NaCl [1937BRO/HAS]. Although the re-evaluation of the experimental data is not possible (the applied partial pressure of CO_2 was not reported; they merely stated "CO_2 was present in the gas phase in an amount such that the pH at equilibrium varied between 7.2 and 7.6"), the extrapolation of the reported solubility constants to zero ionic strength using the SIT approach resulted in comparable $\log_{10}K^{\circ}_{s,0}$ values to those reported by others. Therefore, the extrapolated value has been accepted with increased uncertainty ($\log_{10}K^{\circ}_{s,0} = -(8.41 \pm 0.12)$).

Experimental solubility data have been reported in pure water and in the closed system (aragonite-H_2O) in [1893KOH/ROS], [1912KEN], [1916WAR/KOU] and [2002KON/NAN]. The solubility determined in [1916WAR/KOU] was considerably higher, whereas that in [2002KON/NAN] at 348.15 K was notably lower than expected, and consequently these data were neglected. Due to the well-known experimental difficulties related to closed systems, relatively high uncertainties (± 0.12) have been assigned to the re-evaluated $\log_{10}K^{\circ}_{s,0}$ values calculated from the remaining accepted solubility data. Kendall reported the formation of some calcite crystals at 373 K during his experiments [1912KEN], therefore the uncertainty of solubility product at this temperature was increased to ± 0.20. Only a single paper reports solubility data in the (aragonite-H_2O-NaCl) system [1916WAR/KOU]. The re-evaluation of this datum resulted in a higher solubility than expected, therefore it was neglected.

The solubility in closed systems has been also studied in the presence of artificial sea water [1980COO/KEP2], [1989MUC/CAN], in (evaporated) supratidal brines [1988OOM/KAN] and in HCl solution [1983SAS/MOR]. These data were not considered further in the present review. The system studied by Weyl [1959WEY] is neither open nor closed, so in the absence of data for p_{CO2} at the equilibrium, calculation of $\log_{10}K^{\circ}_{s,0}$ using the reported solubility data was not possible.

Table V-75: References with accepted solubility data for aragonite ($CaCO_3$(cr)).

Reference	Experimental conditions	Reported solubility	Re-calculated $\log_{10} K^{o}_{s,0}$
[1921BAC]	T = 282.15 K, p_{CO2} = 0.1036 MPa	0.01462 mol·kg^{-1}	– (8.147 ± 0.100)
	T = 298.15 K, p_{CO2} = 0.1015 MPa	0.0107 mol·kg^{-1}	– (8.240 ± 0.060)
	T = 308.15 K, p_{CO2} = 0.0992 MPa	0.00882 mol·kg^{-1}	– (8.316 ± 0.060)
[1937BRO/HAS]	T = 311.15 K, p_{CO2} = ? MPa, 25 mmol·dm^{-3} ($NaHCO_3$) and 0-200 mmol·dm^{-3} (NaCl)	0.00017-0.00081 mmol·dm^{-3}	– (8.41 ± 0.12)
[1973HUL/TUR]	T = 298.15 K, p_{CO2} = 0.0983 MPa,	0.01035 mol·kg^{-1}	– (8.263 ± 0.060)
[1976BER] [(a)]	T = 278.15 K, p_{CO2} = 0.0098/0.0981 MPa	0.00607-0.01509 mol·kg^{-1}	– (8.185 ± 0.030)
	T = 298.15 K, p_{CO2} = 0.0003/0.0098/0.0981 MPa	0.00126-0.0104 mol·kg^{-1}	– (8.277 ± 0.030)
[1982PLU/BUS] [(a)]	T = 273.85 K, p_{CO2} = ~ 0.0982 MPa	~ 0.01618 mol·kg^{-1}	– (8.168 ± 0.020)
	T = 277.05 K, p_{CO2} = ~ 0.0003-0.0304 MPa	0.0017-0.097 mol·kg^{-1}	– (8.193 ± 0.020)
	T = 278.05 K, p_{CO2} = ~ 0.0993 MPa	0.015-0.0153 mol·kg^{-1}	– (8.164 ± 0.030)
	T = 288.25 K, p_{CO2} = 0.00003-0.0973 MPa	0.0006-0.0125 mol·kg^{-1}	– (8.217 ± 0.030)
	T = 298.15 K, p_{CO2} = 0.00003-0.0963 MPa	0.00055-0.0106 mol·kg^{-1}	– (8.284 ± 0.030)
	T = 308.15 K, p_{CO2} = 0.0003-0.0942 MPa	0.001-0.00882 mol·kg^{-1}	– (8.323 ± 0.030)
	T = 318.05 K, p_{CO2} = 0.0003-0.0902 MPa	0.0008-0.00727 mol·kg^{-1}	– (8.423 ± 0.040)
	T = 328.15 K, p_{CO2} = 0.0253-0.0851 MPa	0.0036-0.00577 mol·kg^{-1}	– (8.527 ± 0.040)
	T = 338.15 K, p_{CO2} = 0.0002-0.0760 MPa	0.00028 mol·kg^{-1}	– (8.659 ± 0.040)
	T = 353.15 K, p_{CO2} = 0.0002-0.0527 MPa	0.0047 mol·kg^{-1}	– (8.829 ± 0.040)
	T = 363.15 K, p_{CO2} = 0.0001-0.0304 MPa	0.0004-0.00313 mol·kg^{-1} 0.00028-0.0021 mol·kg^{-1}	– (8.970 ± 0.040)
[1983SAS/MOR] [(a)]	T = 298.15 K, p_{CO2} = 0.0983 MPa	0.011075 mol·kg^{-1}	– (8.186 ± 0.080)
[1893KOH/ROS]	T = 291.15 K	0.00015 mol·kg^{-1}	– (8.192 ± 0.120)
[1912KEN]	T = 298.15 K	0.0001531 mol·kg^{-1}	– (8.254 ± 0.120)
	T = 323.15 K	0.0001635 mol·kg^{-1}	– (8.546 ± 0.120)
	T = 373.15 K	0.0001983 mol·kg^{-1}	– (9.129 ± 0.120)
[2002KON/NAN]	T = 298.15 K	0.00014 mol·kg^{-1}	– (8.342 ± 0.120)
	T = 323.15 K	0.00016 mol·kg^{-1}	– (8.569 ± 0.120)

(a) 4-39 data points at each temperature

Among the 229 accepted solubility constants for aragonite 205 are from [1982PLU/BUS] and 10 are from [1976BER]. Although the data reported in [1982PLU/BUS] are highly reliable, their data obviously would dominate the re-evaluation. Since at a given temperature 6-39 independent solubility data were reported in [1982PLU/BUS], in our evaluation of temperature dependence of $\log_{10} K^{\circ}_{s,0}$ only an average value at each temperature was used (similar to the data reported in [1976BER] and [1983SAS/MOR]). In this way 25 data points (Table V-75) were used to derive the temperature dependence of the $\log_{10} K^{\circ}_{s,0}$ for Reaction (V.142). The unconstrained regression resulted in Equation (V.143) (see the solid line in Figure V-67).

$$CaCO_3(arag) \rightleftharpoons Ca^{2+} + CO_3^{2-} \qquad \text{(V.142)}$$

$$\log_{10} K^{\circ}_{s,0}(\text{V.142}) = (8.93572 \pm 1.10879) - (0.0319812 \pm 0.0017879)(T/\text{K}) - (2286.87492 \pm 170.74324)(T/\text{K})^{-1} \qquad \text{(V.143)}$$

The refined parameters of Equation (V.143) led to the following thermodynamic quantities for Reaction (V.142) at 298.15 K:

$$\log_{10} K^{\circ}_{s,0}((\text{V.142}), 298.15\ \text{K}) = -(8.263 \pm 0.030),$$

$$\Delta_r G^{\circ}_m((\text{V.142}), 298.15\ \text{K}) = (47.165 \pm 0.500)\ \text{kJ·mol}^{-1},$$

$$\Delta_r H^{\circ}_m((\text{V.142}), 298.15\ \text{K}) = -(10.684 \pm 1.000)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^{\circ}_m((\text{V.142}), 298.15\ \text{K}) = -(194.030 \pm 1.500)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C^{\circ}_{p,m}((\text{V.142}), 298.15\ \text{K}) = -(365.1 \pm 19.0)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

These values are slightly different from those determined by Plummer and Busenberg (*e.g.*, $\log_{10} K^{\circ}_{s,0} = -(8.334 \pm 0.020)$ based on their own dataset and considering the formation of the $CaHCO_3^+$ ion pair, but are close to those calculated in [2012VIS/VAN2] for Model 2 (the formation of $CaHCO_3^+$ ion pair is neglected) using the Pitzer approach. From the above $\Delta_r S^{\circ}_m((\text{V.142}), 298.15\ \text{K})$ value, and using the recommended S°_m parameters for Ca^{2+} and CO_3^{2-}, $S^{\circ}_m(CaCO_3, \text{arag}, 298.15\ \text{K}) = (87.83 \pm 5.00)\ \text{J·K}^{-1}\text{·mol}^{-1}$ can be calculated.

However,

$$S^{\circ}_m(CaCO_3, \text{arag}, 298.15\ \text{K}) = (87.99 \pm 0.21)\ \text{J·K}^{-1}\text{·mol}^{-1}$$

is selected in this review based on the more precise low-temperature heat capacity measurements, which corresponds to $\Delta_r S^{\circ}_m((\text{V.142}), 298.15\ \text{K}) = -(194.19 \pm 1.43)\ \text{J·K}^{-1}\text{·mol}^{-1}$. Since latter value is more reliable, we re-calculated our data using constrained regression, which resulted in Equation (V.144).

$$\log_{10} K^{\circ}_{s,0}(\text{V.142}) = (8.83346 \pm 0.78682) - (0.031824 \pm 0.001319)(T/\text{K}) - (2270.48754 \pm 116.57557)(T/\text{K})^{-1} \qquad \text{(V.144)}$$

The parameters of Equation (V.144) correspond to the following thermodynamic quantities:

$$\log_{10} K^{\circ}_{s,0}\ ((V.142),\ 298.15\ K) = -\ (8.270 \pm 0.030),$$

$$\Delta_r G^{\circ}_m\ ((V.142),\ 298.15\ K) = (47.206 \pm 0.171)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r H^{\circ}_m\ ((V.142),\ 298.15\ K) = -\ (10.692 \pm 1.000)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r S^{\circ}_m\ ((V.142),\ 298.15\ K) = -\ (194.191 \pm 3.403)\ J{\cdot}K^{-1}{\cdot}mol^{-1},$$

$$\Delta_r C^{\circ}_{p,m}\ ((V.142),\ 298.15\ K) = -\ (363.300 \pm 19.000)\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

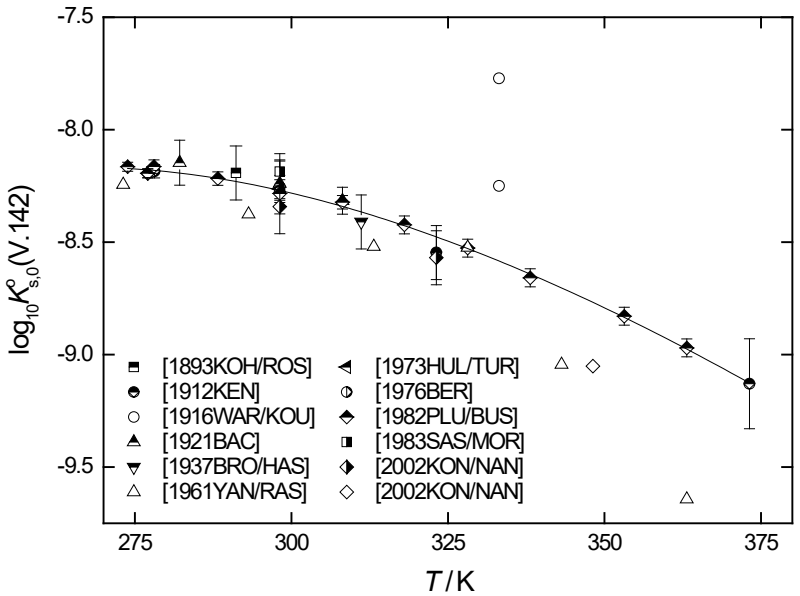
Combining the above thermodynamic parameters with those recommended for Ca^{2+} and CO_3^{2-}, the following values are calculated from our solubility assessment:

$$\Delta_f G^{\circ}_m\ (CaCO_3,\ arag,\ 298.15\ K) = -\ (1127.912 \pm 2.000)\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^{\circ}_m\ (CaCO_3,\ arag,\ 298.15\ K) = -\ (1207.538 \pm 2.000)\ kJ{\cdot}mol^{-1}.$$

The above values compare well with those selected in this review, which include several other experimental sources (see Table V-67), and with those selected by Robie and Hemingway: $\Delta_f G^{\circ}_m = -\ (1127.4 \pm 1.5)\ kJ{\cdot}mol^{-1}$, $\Delta_f H^{\circ}_m = -\ (1207.4 \pm 1.4)\ kJ{\cdot}mol^{-1}$ (see p. 25 of [1995ROB/HEM]).

Figure V-67: Temperature dependence of the re-evaluated solubility product of aragonite. The solid line (Equation (V.144)) and the selected thermodynamic parameters are based on the accepted values (half-filled symbols).



V.6.2.2.5 The solubility of gypsum and anhydrite

Among the three forms of $CaSO_4 \cdot nH_2O$(cr) (anhydrite (n = 0), hemihydrate (n = 0.5) and dihydrate (n = 2)), the hemihydrate is metastable at all temperatures. Therefore, in this review only the dihydrate (gypsum) and anhydrite are considered. Although several earlier papers report thermodynamic models to describe experimental solubilities of gypsum and anhydrite, most of them are based on the Pitzer model (*e.g.*, [1980HAR/WEA], [1984MEI/ROS], [1985LAN/MEL], [1988MOL], [1988PLU/PAR], [1990RAJ/ATK], [2011LI/DUA], [2017DAI/KAN] and [2018KRU]), others neglected the formation of the $CaSO_4$(aq) species [1966MAR/SLU]. To our knowledge a comprehensive, SIT-based model is not available in the literature.

V.6.2.2.5.1 The solubility of gypsum at 298.15 K

The dissolution of gypsum is governed by the following two equilibria:

$$CaSO_4 \cdot 2H_2O(cr) \rightleftharpoons Ca^{2+} + SO_4^{2-} + 2H_2O \qquad \text{(V.145)}$$

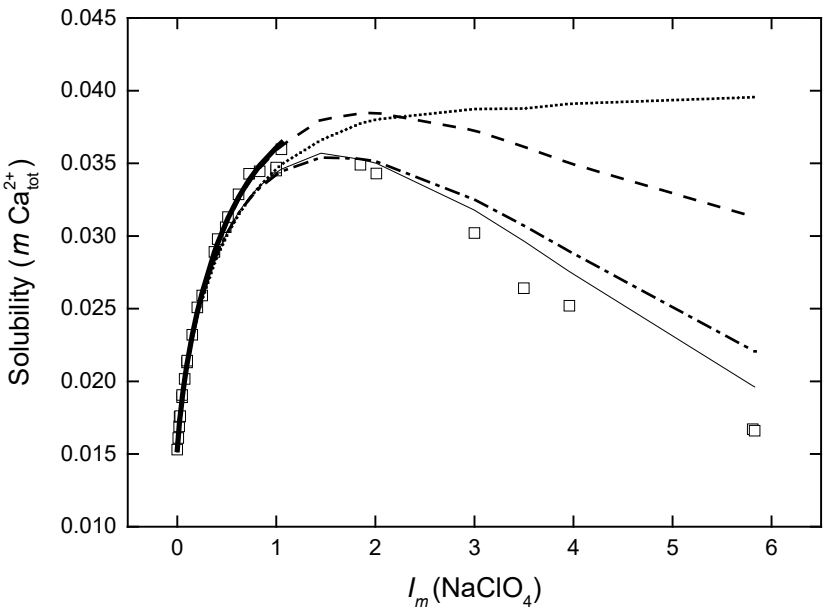
$$Ca^{2+} + SO_4^{2-} \rightleftharpoons CaSO_4(aq) \qquad \text{(V.146)}$$

There is a large amount of data in the literature for the solubility of gypsum at 298.15 K both in pure water and in different electrolyte solutions. In pure water, the reported solubilities ([1902HUL/ALL], [1935HIL/YAN], [1952SHT/FRO], [1956MAD/SWA],[1956MAD/SWA2],[1961BOC],[1963RZA/RUS],[1966MAR/SLU], [1966POW/FAB], [1967NAK/RAS], [1968BLO/WAT], [1968MAN/KRY], [1969TAN], [1978CUL/LAT], [1980INN/RAB], [2013WAN/ZEN]) agree well with each other, and their average value is (0.0153 ± 0.0002) mol·dm^{-3}. From these data $\log_{10} K^{\circ}_{s,0}$ ((V.145), 298.15 K) = – (4.626 ± 0.002) can be calculated. Due to the practical importance of calcium sulfate dissolution and precipitation in several environmental, geological and industrial processes, the solubility of gypsum has been studied in several electrolyte solutions, such as NaCl, $NaClO_4$, $NaNO_3$, Na_2SO_4, HCl, H_2SO_4, MSO_4 (where M = Ca^{2+}, Mg^{2+}, Mn^{2+}, Zn^{2+}) and $Fe_2(SO_4)_3$), $MgCl_2$, $CaCl_2$ and sea water (see for example [1968MAR/SLU], [1973BRI], [1981ROG], [1984MEI/ROS], [1988MOL], [2007AZI/PAP], [2010VIS/PAZ] and references therein). Among these electrolyte solutions only NaCl and $NaClO_4$ are considered in the present review, due to the number of available experimental data and their general importance.

Four papers report solubilities in $NaClO_4$ solutions [1958NIL/REN], [1971NAK3], [1973KAL/YEA] and [1973KAL/YEA2]. The experimental data are shown in Figure V-68. Since the data from different sources show good agreement, all experimental points were treated together. The solubility passes through a relatively sharp maximum at around 1.5 mol·dm^{-3} ($NaClO_4$). The decrease of solubility at high electrolyte concentrations correlates with the hydration ability of the electrolyte [2003FRE/VOI]. The simplest SIT model, using $\varepsilon(CaSO_4(aq), NaClO_4(aq)) =$ 0 kg·mol^{-1}, $\varepsilon(Ca^{2+}, ClO_4^-) = (0.27 \pm 0.03)$ kg·mol^{-1} (Table B-4) and $\varepsilon(Na^+, SO_4^{2-}) =$ – (0.12 ± 0.06) kg·mol^{-1} (Table B-5), is able to reproduce the solubility data only up to

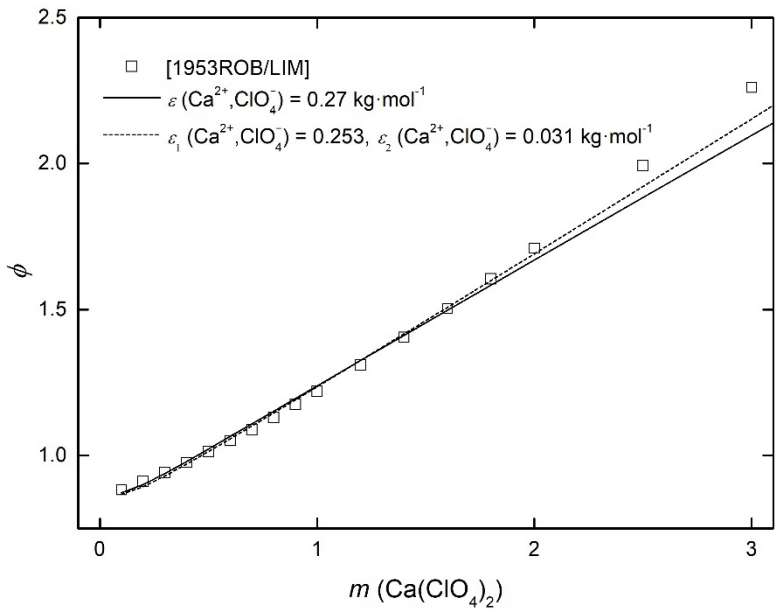
I_m = 0.25 (dotted line in Figure V-68). A somewhat better fit can be obtained using ionic strength dependent Na^+-SO_4^{2-} ion interaction coefficients ($\varepsilon_1(Na^+, SO_4^{2-})$ = – (0.184 ± 0.002) kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-})$ = (0.139 ± 0.006) kg·mol^{-1}, NEA TDB values (Table B-6), dashed line in Figure V-68) and allowing $\varepsilon(CaSO_4(aq), Na^+ + Cl^-)$ to have a non-zero value (dotted-dashed line in Figure V-68). Although, in the latter case the calculated $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = (0.235 ± 0.040) kg·mol^{-1} is an anomalously high value, the agreement with the experimental data is still poor at higher ionic strength.

Figure V-68: The solubility of gypsum in aqueous $NaClO_4$ solutions at 298.15 K. Experimental data are from [1958NIL/REN], [1971NAK3], [1973KAL/YEA] and [1973KAL/YEA2], lines were calculated using different ion interaction coefficients (see also the text): **Dotted line**: $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = 0 kg·mol^{-1}, $\varepsilon(Ca^{2+}, ClO_4^-)$ = 0.27 kg·mol^{-1}, $\varepsilon(Na^+, SO_4^{2-})$ = – 0.12 kg·mol^{-1}; **Dashed line**: $\varepsilon_1(Na^+, SO_4^{2-})$ = – 0.184 kg·mol^{-1},$\varepsilon_2(Na^+, SO_4^{2-})$ = 0.139 kg·mol^{-1}, $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = 0.235 kg·mol^{-1}; **Dotted-dashed line**: $\varepsilon_1(Na^+, SO_4^{2-})$ = – 0.188 kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-})$ = 0.151 kg·mol^{-1}, $\varepsilon_1(Ca^{2+}, ClO_4^-)$ = 0.243 kg·mol^{-1}, $\varepsilon_2(Ca^{2+}, ClO_4^-)$ = 0.045 kg·mol^{-1}; $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = 0 kg·mol^{-1}; **Thin solid line**: $\varepsilon_1(Na^+, SO_4^{2-})$ = – 0.188 kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-})$ = 0.151 kg·mol^{-1}, $\varepsilon_1(Ca^{2+}, ClO_4^-)$ = 0.243 kg·mol^{-1}, $\varepsilon_2(Ca^{2+}, ClO_4^-)$ = 0.045 kg·mol^{-1}, $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = 0.27 kg·mol^{-1}; **Thick solid line (accepted)**: $\varepsilon_1(Na^+, SO_4^{2-})$ = – 0.184 kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-})$ = 0.139 kg·mol^{-1}, $\varepsilon(Ca^{2+}, ClO_4^-)$ = 0.27 kg·mol^{-1}, $\varepsilon(CaSO_4(aq), NaClO_4(aq))$ = 0.000 kg·mol^{-1}.



Therefore, we used the available osmotic coefficient data to derive ionic strength dependent $\varepsilon(Na^+\text{-}SO_4^{2-})$ [2000RAR/CLE] and $\varepsilon(Ca^{2+}\text{-}ClO_4^-)$ [1953ROB/LIM] ion interaction coefficients valid up to $I_m = 6$, since solubility data for gypsum and anhydrite are reported between $I_m = 0$-6. The above-mentioned NEA TDB recommended $\varepsilon_1(Na^+, SO_4^{2-})$ and $\varepsilon_2(Na^+, SO_4^{2-})$ values nearly perfectly reproduce the osmotic coefficients of Na_2SO_4 solutions [2000RAR/CLE] up to $I_m = 3.0$, but slightly different ion interaction coefficients $\varepsilon_1(Na^+, SO_4^{2-}) = -(0.188 \pm 0.002)$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) = (0.151 \pm 0.006)$ kg·mol^{-1} are more appropriate for a wider ionic strength range ($I_m = 0$-6.0). Similarly, the NEA TDB recommended $\varepsilon(Ca^{2+}, ClO_4^-) = (0.27 \pm 0.03)$ kg·mol^{-1} is suitable up to $I_m = 3$, but $\varepsilon_1(Ca^{2+}, ClO_4^-) = (0.253 \pm 0.020)$ kg·mol^{-1} and $\varepsilon_2(Ca^{2+}, ClO_4^-) = (0.031 \pm 0.020)$ kg·mol^{-1} are somewhat better between $I_m = 0$-6.0 (Figure V-69).

Figure V-69: Plot of experimental and calculated osmotic coefficient *versus* molality of aqueous $Ca(ClO_4)_2$ solutions at 298.15 K.



However, even using these ion interaction coefficients and allowing $\varepsilon(CaSO_4(aq), Na^+ + Cl^-)$ to be different from zero ($\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = 0.27$ kg·mol^{-1}, thin solid line in Figure V-68), the ionic strength dependence of gypsum solubility cannot be reproduced correctly. Therefore, only the solubility data up to $I_m = 1.2$ were considered in our evaluation. Using the NEA TDB recommended ion interaction coefficients ($(\varepsilon_1(Na^+, SO_4^{2-}) = -(0.184 \pm 0.002)$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) =$

(0.139 ± 0.006) kg·mol^{-1}, $\varepsilon(Ca^{2+}, ClO_4^-) = (0.27 \pm 0.03)$ kg·mol^{-1}) and $\varepsilon(CaSO_4(aq), NaClO_4(aq)) = 0.000$ kg·mol^{-1}, $\log_{10} K^\circ_{s,0}$ ((V.145), 298.15 K) = – (4.629 ± 0.008) can be calculated (thick solid line in Figure V-68) from the solubility data determined in $NaClO_4$ solutions.

The solubility of gypsum in NaCl solutions ([1952SHT/FRO], [1956MAD/SWA2], [1961BOC], [1966MAR/SLU], [1966POW/FAB], [1967NAK/RAS], [1968BLO/WAT], [1969TAN], [1971NAK3], [1973MEL/MOS]) also passes through a maximum at around 2.5-3.0 mol·kg^{-1} NaCl (Figure V-70). The simplest SIT model using $\varepsilon(Ca^{2+}, Cl^-) = (0.14 \pm 0.01)$ kg·mol^{-1} (a value confirmed in Table B-4), $\varepsilon(Na^+, SO_4^{2-}) = – (0.12 \pm 0.06)$ kg·mol^{-1} (Table B-5) and $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = 0$ kg·mol^{-1} shows notable deviation from the experimental data ($\log_{10} K^\circ_{s,0}$ (V.145) = – 4.625, see dotted line in Figure V-70). A somewhat better fit can be obtained using ionic strength dependent ion interaction coefficients ($\varepsilon_1(Na^+, SO_4^{2-}) = – (0.184 \pm 0.002)$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) = (0.139 \pm 0.006)$ kg·mol^{-1}, TDB values), which results in $\log_{10} K^\circ_{s,0}$ (V.145) = – (4.621 ± 0.010) and $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = (0.085 \pm 0.020)$ kg·mol^{-1} (see dashed line in Figure V-70). Nevertheless, the agreement with the experimental data is still poor at $I_m > 0.3$. Therefore, we re-evaluated the osmotic coefficients reported earlier for $CaCl_2$ solutions ([1977STA/NUT], [1985ANA/ATK], [1997RAR/CLE]) in order to derive ionic strength dependent ion interaction coefficients valid between I_m = 0-6.0 (Figure V-71).

Figure V-70: The solubility of gypsum in aqueous NaCl solutions at 298.15 K. Experimental data are from [1952SHT/FRO], [1956MAD/SWA2], [1961BOC], [1963RZA/RUS], [1966MAR/SLU], [1966POW/FAB], [1967NAK/RAS], [1968BLO/WAT], [1969TAN], [1971NAK], [1973MEL/MOS], lines were calculated using different ion interaction coefficients (see also the text); **Dotted line**: $\varepsilon(Ca^{2+}, Cl^-) = 0.14$ kg·mol^{-1}, $\varepsilon(Na^+, SO_4^{2-}) = -0.12$ kg·mol^{-1}, $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = 0$ kg·mol^{-1}; **Dashed line**: $\varepsilon(Ca^{2+}, Cl^-) = 0.14$ kg·mol^{-1}, $\varepsilon_1(Na^+, SO_4^{2-}) = -0.184$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) = 0.139$ kg·mol^{-1}, $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = (0.085 \pm 0.020)$ kg·mol^{-1}; **Solid line**: $\varepsilon_1(Ca^{2+}, Cl^-) = 0.117$ kg·mol^{-1}, $\varepsilon_2(Ca^{2+}, Cl^-) = 0.072$ kg·mol^{-1}, $\varepsilon_1(Na^+, SO_4^{2-}) = -0.188$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) = 0.151$ kg·mol^{-1}, $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = 0.015$ kg·mol^{-1}.

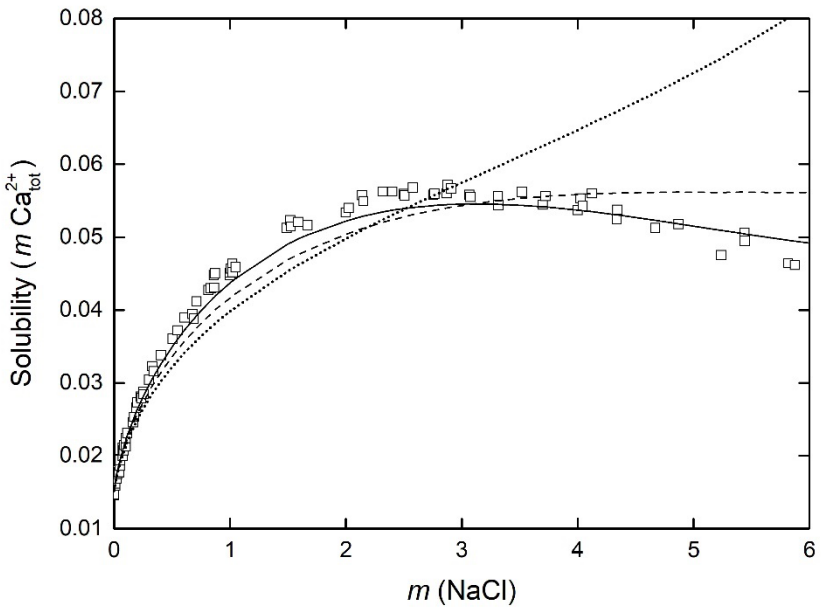
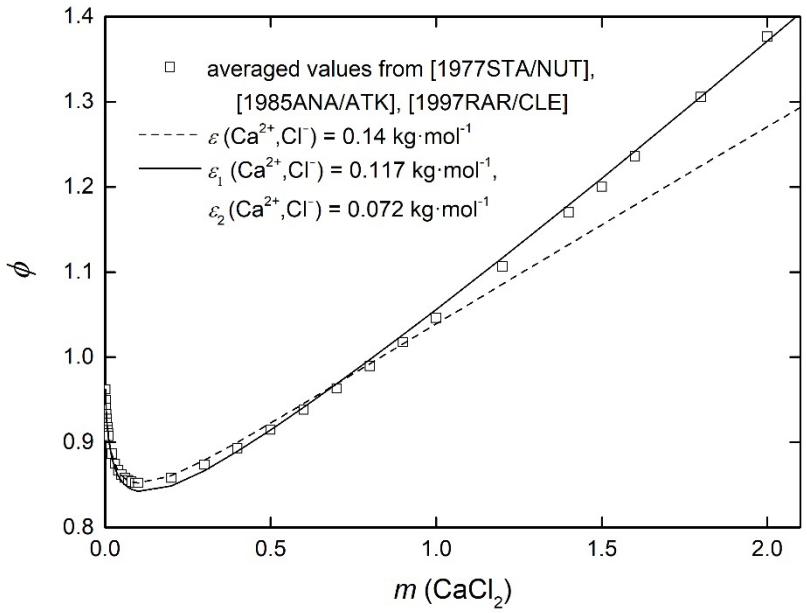


Figure V-71: Plot of osmotic coefficient *versus* molality of aqueous $CaCl_2$ solutions at 298.15 K.



The TDB recommended single epsilon value, $\varepsilon(Ca^{2+}, Cl^-) = 0.14$ kg·mol^{-1} results in a good fit to the experimental data up to $I_m = 3$ (dashed line in Figure V-71). However, at $I_m > 3$ the ionic strength dependent ion interaction coefficients

$$\varepsilon_1(Ca^{2+}, Cl^-) = \mathbf{(0.117 \pm 0.01)}\ \text{kg·mol}^{-1}$$

and

$$\varepsilon_2(Ca^{2+}, Cl^-) = \mathbf{(0.072 \pm 0.01)}\ \text{kg·mol}^{-1}$$

(see solid line in Figure V-71) are clearly more accurate. Consequently, the latter values can be recommended when experimental data are available at high (up to $I_m = 6$) ionic strengths. Using these ionic strength dependent values and $\varepsilon_1(Na^+, SO_4^{2-}) = -(0.188 \pm 0.002)$ kg·mol^{-1}, $\varepsilon_2(Na^+, SO_4^{2-}) = (0.151 \pm 0.006)$ kg·mol^{-1} (see above), gypsum solubility in NaCl medium can be well reproduced (solid line in Figure V-70), although the agreement between 1 and 3 mol·kg^{-1} (NaCl) is not perfect. The resulting parameters are $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = (0.015 \pm 0.020)$ kg·mol^{-1} and $\log_{10} K^\circ_{s,0}$ (V.145) $= -(4.618 \pm 0.020)$.

Considering all of the above-mentioned data, the best value for the thermodynamic solubility constant of gypsum in water at 298.15 K is:

$$\log_{10} K^{\circ}_{s,0}\,((\text{V.145}),\ 298.15\ \text{K}) = -\,(4.626 \pm 0.020).$$

Obviously, the value of $\log_{10} K^{\circ}_{s,0}$ strongly depends on the ion interaction model used. The above value agrees relatively well with those determined by using the Pitzer model: – 4.598 [1978CUL/LAT], – 4.590 [1985LAN/MEL], – 4.597 [1990RAJ/ATK], – 4.577 [2018KRU] but differs considerably from those reported in the less comprehensive studies in [1966MAR/SLU] (– 4.373, which neglected the formation of $CaSO_4$(aq)), or [1966POW/FAB] (– 4.359, which neglected that a_{H_2O} differs from 1.0).

V.6.2.2.5.2 The solubility of gypsum between 273.15 and 383.15 K

A number of papers report experimental data for the solubility of gypsum at temperatures other than 298.15 K both in pure water ([1901CAM], 1902HUL/ALL], [1910MEL], [1934HIL], [1935HIL/YAN], [1938POS], [1955ANS/BRE], [1956MAD/SWA], [1957KYD/DRU], [1960GRO2], [1963RZA/RUS], [1966OST/MET], [1968MAN/KRY], [1978CUL/LAT], [1980INN/RAB], [2010YUA/WAN], [2013WAN/ZEN]), and in NaCl solutions ([1901CAM], [1943FOR], [1956MAD/SWA2], [1961BOC], [1964MAR/SLU], [1965ZEN], [1966MAR/SLU], [1966OST/MET], [1966POW/FAB], [1970GLE/HAM]).

The interpretation of these data is complicated by the gypsum-anhydrite equilibrium.

$$CaSO_4 \cdot 2H_2O(\text{s, gypsum}) \rightleftharpoons CaSO_4(\text{s, anhydrite}) + 2H_2O(\text{l}) \quad (\text{V.147})$$

Discussion of the gypsum-anhydrite transition temperature has a long history, starting with the pioneering work of Van't Hoff [1903HOF/ARM]. However, the slow crystallisation kinetics of anhydrite in aqueous solutions at temperatures below 343 K result in considerable uncertainty in the gypsum-anhydrite conversion temperature in water. Consequently, 100 years after the Van't Hoff's paper, Freyer and Voigt could give only best estimates, which varied between 315 and 333 K [2003FRE/VOI]. The advantage of the slow kinetics is that gypsum can exist under metastable conditions well above the transition temperature. Hence, solubility data have been reported up to 383 K (at higher temperatures gypsum may be converted into the hemihydrate). On the other hand, the gypsum-anhydrite transition temperature decreases with increasing concentration of unlike salts (*e.g.*, NaCl, *cf.* Section V.6.2.2.5.6). Although the transformation is very slow in NaCl solutions, Ostroff reported dehydration of gypsum within ten days at 363.65 K and concentrations higher than 2 mol·dm^{-3} (NaCl) [1964OST2]. Therefore, the data points corresponding to $T \geq 363$ K and c(NaCl) > 2 mol·dm^{-3} were neglected in our re-evaluation.

In order to re-evaluate the solubility data reported for $T \neq 298.15$ K, the temperature dependence of $\varepsilon(Na^+, SO_4^{2-})$ and $\varepsilon(Ca^{2+}, Cl^-)$ must be known. In [2000RAR/CLE] the activity and osmotic coefficients of Na_2SO_4 solutions (0.05-3.0 mol·kg^{-1}) are reported between 273 and 398 K. Based on these data the

following temperature dependencies of $\varepsilon_1(Na^+, SO_4^{2-})$ and $\varepsilon_2(Na^+, SO_4^{2-})$ can be calculated.

$$\varepsilon_1(Na^+,SO_4^{2-}) = 77.08577 - 1692.24305(T/K)^{-1} - 30.6237\log_{10}(T/K) + 0.01861996(T/K) - 122023.562(T/K)^{-2}$$

and

$$\varepsilon_2(Na^+,SO_4^{2-}) = 131.00442 - 9487.35992(T/K)^{-1} - 44.92032\log_{10}(T/K) + 0.0126761(T/K) + 741368.842(T/K)^{-2}.$$

The Ca^{2+}-Cl^- I_m dependent ion interaction coefficients values valid between 273.15 and 373.15 K are derived from the osmotic coefficients of $CaCl_2$ solutions reported in [1977STA/NUT], [1985ANA/ATK] and [1997RAR/CLE]. At higher temperatures $\varepsilon_1(Ca^{2+},Cl^-)$ and $\varepsilon_2(Ca^{2+},Cl^-)$ are simultaneously refined with $\log_{10}K^\circ_{s,0}$ (V.145), using the solubility of anhydrite in NaCl medium reported in [1967TEM/ROD] and [2004FRE/VOI] between 423.15 and 598.15 K (during this latter evaluation $\varepsilon(CaSO_4(aq), Na^+ + Cl^-) = 0$ is used, see later).

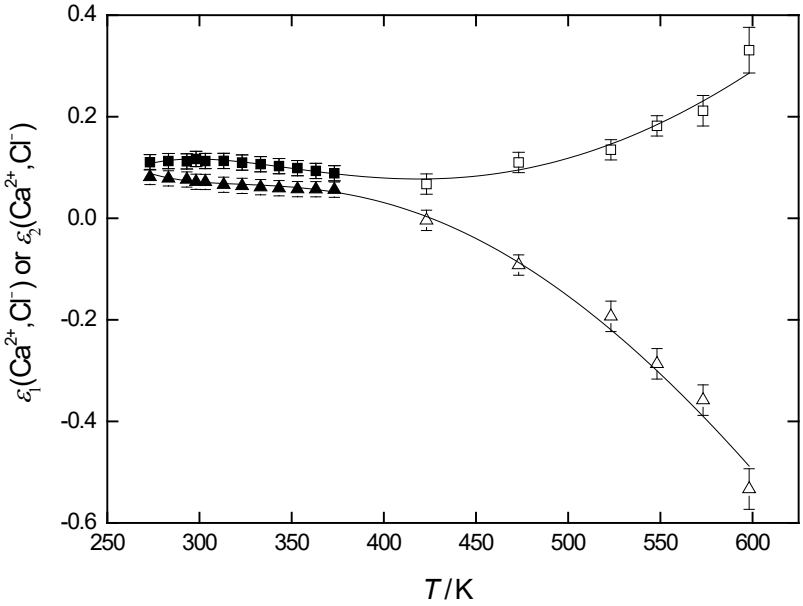
The $\varepsilon_1(Ca^{2+},Cl^-)$ and $\varepsilon_2(Ca^{2+},Cl^-)$ values calculated in this way are depicted in Figure V-72, and show a clear non-linear temperature dependence. Based on these data the following temperature dependencies of $\varepsilon_1(Ca^{2+},Cl^-)$ and $\varepsilon_2(Ca^{2+},Cl^-)$ can be derived.

$$\varepsilon_1(Ca^+,Cl^-) = 55.225273 - 1134.393275(T/K)^{-1} - 22.237578\log_{10}(T/K) + 0.0148726(T/K) - 63502.500(T/K)^{-2}$$

and

$$\varepsilon_2(Ca^+,Cl^-) = -3.092055 - 2079.45359(T/K)^{-1} + 4.2072175\log_{10}(T/K) - 0.0105602(T/K) + 255727.200(T/K)^{-2}.$$

Figure V-72: $\varepsilon_1(Ca^{2+}, Cl^-)$ (squares) and $\varepsilon_2(Ca^{2+},Cl^-)$ (triangles) values as a function of temperature derived from the osmotic coefficients of $CaCl_2$ solutions (filled symbols, ■: ε_1, ▲: ε_2) or anhydrite solubility (open symbols, □: ε_1, △: ε_2).



Some representative solubility curves, depicted in Figure V-73, indicate that in NaCl solutions the solubility is nearly independent of temperature. Between 273.15 and 373.15 K, $\varepsilon(CaSO_4(aq), Na^+ + Cl^-)$ was simultaneously refined with $\log_{10}K^\circ_{s,0}$ (V.145) in NaCl medium and it turned out that the activity coefficient of $CaSO_4(aq)$ was close to unity, as expected. The values of $\varepsilon(CaSO_4(aq), Na^+ + Cl^-)$ are scattered around zero (0.01 ± 0.04) independently of the temperature.

The $\log_{10}K^\circ_{s,0}$ (V.145) values determined either directly from the solubilities in pure water, or extrapolated from solubilities measured in NaCl solutions are presented in Figure V-74. As can be seen, the $\log_{10}K^\circ_{s,0}$ (V.145) values obtained from different sources are generally within $\pm$ 0.03 $\log_{10}$ units at a given temperature. In the cases of greater deviation, outliers have been identified (*e.g.*, at 288.15 and 355.15 K reported in [1901CAM], at 308.15 and 338.3 K reported in [1965ZEN] and [1970GLE/HAM], respectively). An unconstrained regression of the remaining 120 data points resulted in the equation:

$$\log_{10}K^\circ_{s,0}(\text{V.145}) = 5.2776174 - 0.0170046\,(T/\text{K}) - 1441.171438\,(T/\text{K})^{-1},$$

and hence

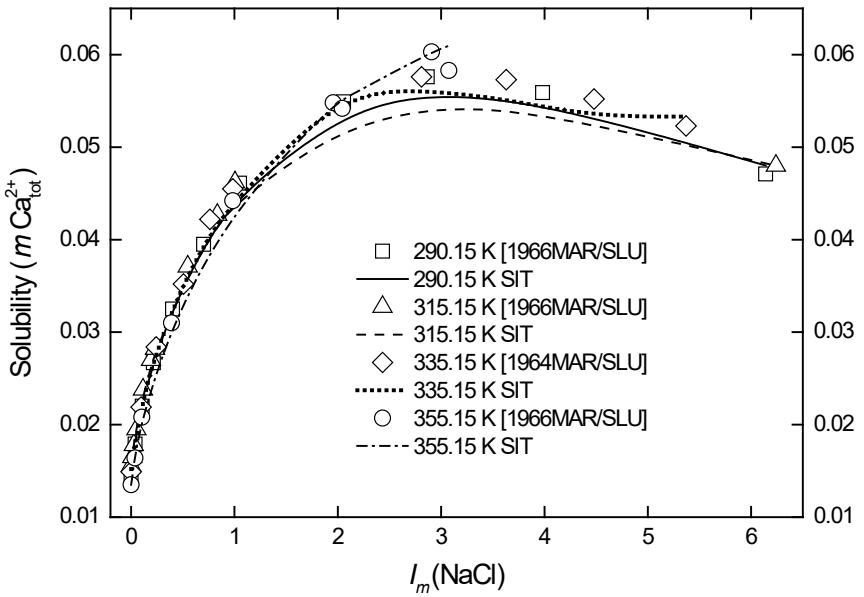
$$\Delta_r G^o_m \text{ ((V.145), 298.15 K)} = (26.405 \pm 0.300) \text{ kJ·mol}^{-1},$$

$$\Delta_r H^o_m \text{ ((V.145), 298.15 K)} = -(1.348 \pm 0.600) \text{ kJ·mol}^{-1},$$

$$\Delta_r S^o_m \text{ ((V.145), 298.15 K)} = -(93.086 \pm 1.500) \text{ J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C^o_{p,m} \text{ ((V.145), 298.15 K)} = -(194.125 \pm 6.800) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

Figure V-73: The solubility of gypsum in aqueous NaCl solutions at different temperatures.



However, based on calorimetric data, the value of S^o_m ($CaSO_4 \cdot 2H_2O$, cr, 298.15 K) = (193.80 ± 0.50) J·K^{-1}·mol^{-1} has been selected in this review, which implies $\Delta_r S^o_m$ ((V.145), 298.15 K) = – (91.60 ± 1.00) J·K^{-1}·mol^{-1}. Using the latter value, the constrained regression of the selected experimental data resulted in Equation (V.148) (see solid line in Figure V-74).

$$\log_{10} K^o_{s,0} \text{ (V.145)} = 6.08130767 - 0.01822217(T/\text{K}) - 1572.555564(T/\text{K})^{-1} \quad \text{(V.148)}$$

Our fitted curve is between the two recently published temperature dependencies ([2017DAI/KAN] and [2018KRU]), which were based on the Pitzer formalism.

The parameters of the above equation led to the following thermodynamic quantities for Reaction (V.145) at 298.15 K:

$$\Delta_r G_m^o \text{ ((V.145), 298.15 K)} = (26.405 \pm 0.144) \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((V.145), 298.15 K)} = -(0.905 \pm 0.600) \text{ kJ·mol}^{-1},$$

$$\Delta_r S_m^o \text{ ((V.145), 298.15 K)} = -(91.598 \pm 2.048) \text{ J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C_{p,m}^o \text{ ((V.145), 298.15 K)} = -(208.025 \pm 6.780) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

These values differ slightly from those reported in [1990RAJ/ATK] ($\Delta_r G_m^o$ ((V.145), 298.15 K) = 26.24 kJ·mol^{-1}, $\Delta_r H_m^o$ ((V.145), 298.15 K) = – 0.58 kJ·mol^{-1}, $\Delta_r S_m^o$ ((V.145), 298.15 K) = – 89.3 J·K^{-1}·mol^{-1} and $\Delta_r C_{p,m}^o$ ((V.145), 298.15 K) = – 332.7 J·K^{-1}·mol^{-1}), probably due to the different approach and more comprehensive dataset used for the present re-evaluation. Based on the above thermodynamic data, the corresponding values for the formation of $CaSO_4 \cdot 2H_2O$(cr) from these solubility measurements are:

$$\Delta_f G_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = -(1797.495 \pm 1.000) \text{ kJ·mol}^{-1},$$

$$\Delta_f H_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = -(2023.095 \pm 1.500) \text{ kJ·mol}^{-1},$$

$$S_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = (193.800 \pm 0.500) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

Our selected values for $\Delta_f G_m^o$ and $\Delta_f H_m^o$, including consideration of the calorimetric data for the enthalpy of formation and entropy of gypsum are nearly identical:

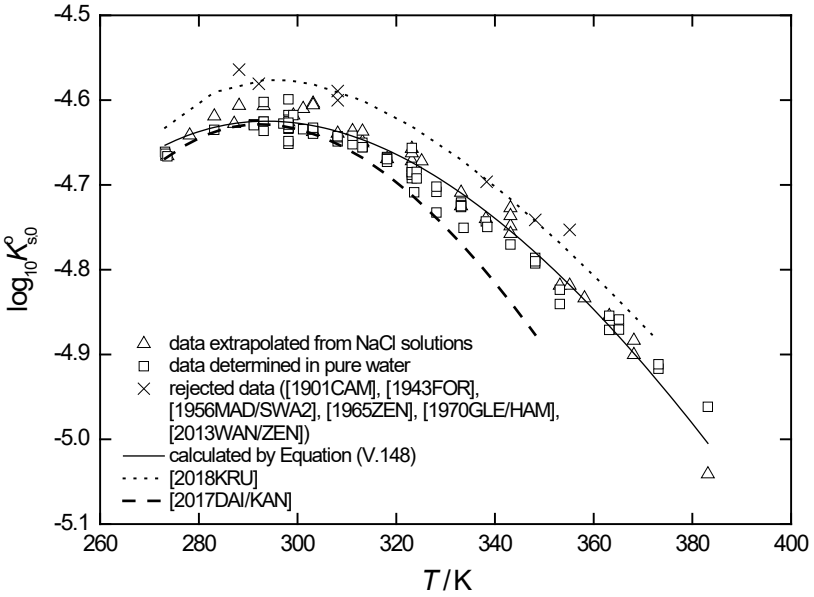
$$\Delta_f G_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = \mathbf{-(1797.402 \pm 1.309)} \text{ kJ·mol}^{-1},$$

$$\Delta_f H_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = \mathbf{-(2023.000 \pm 1.300)} \text{ kJ·mol}^{-1},$$

$$S_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = \mathbf{(193.800 \pm 0.300)} \text{ J·K}^{-1}\text{·mol}^{-1}.$$

These values are very close to the CODATA values in Table 6.2.1 of [1987GAR/PAR]: $\Delta_f G_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = -(1797.361 \pm 1.100)$ kJ·mol^{-1}, $\Delta_f H_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = -(2022.92 \pm 1.10)$ kJ·mol^{-1}, $S_m^o (CaSO_4 \cdot 2H_2O, \text{cr}, 298.15 \text{ K}) = (193.93 \pm 2.10)$ J·K^{-1}·mol^{-1}. However, it should be noted that the values for $\Delta_f G_m^o (SO_4^{2-}, 298.15 \text{ K})$ and $S_m^o (SO_4^{2-}, 298.15 \text{ K})$ (but not $\Delta_f H_m^o (SO_4^{2-}, 298.15 \text{ K})$) used in the assessment by Garvin *et al.* [1987GAR/PAR] (see their Table 7.1.1) are slightly different from those in the later CODATA volume [1989COX/WAG] which are used here and in previous NEA TDB volumes.

Figure V-74: The solubility constant of gypsum as a function of temperature. Lines calculated from the temperature dependence derived in [2017DAI/KAN] (dashed) and [2018KRU] (dotted).



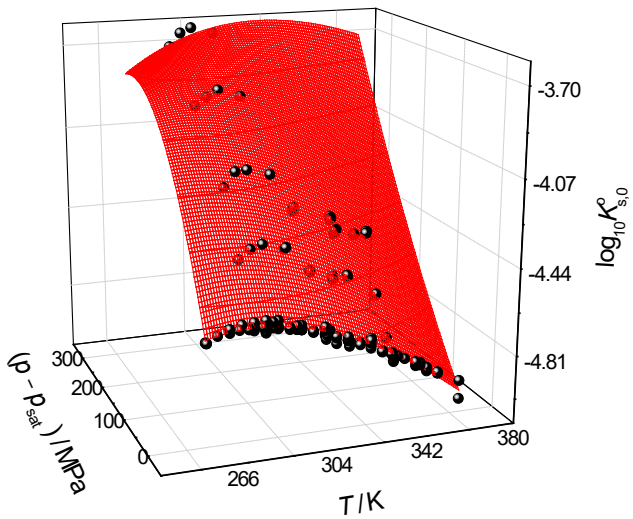
V.6.2.2.5.3 The pressure dependence of gypsum solubility

Only two references have been found on the pressure dependence of gypsum solubility [1968MAN/KRY] and [1973BLO/DIC]. As can be seen from Figure V-75, the solubility increases with increasing pressure. Although only a limited number of data is available, the pressure dependence of $\log_{10} K^{\circ}_{s,0}$ (V.145) can be best described by following equation:

$$\begin{aligned}\log_{10} K^{\circ}_{s,0}\,(\text{V.145}) = {} & a + b\,(T/\text{K}) + c\,(T/\text{K})^{-1} \\ & + (d\,(T/\text{K}) + e\,(T/\text{K})^{-1})(p - p_{\text{sat}}) \\ & + (f\,(T/\text{K}) + g\,(T/\text{K})^{-1})(p - p_{\text{sat}})^2 \end{aligned} \qquad \text{(V.149)}$$

where p is the actual pressure (in MPa), $p_{\text{sat}} = 0.1$ MPa below 373.15 K and p_{sat} is the saturation pressure of water in MPa at temperatures above 373.15 K. The constants a, b and c are identical to those in Equation (V.148), and $d = -5.96148 \times 10^{-6}$, $e = 2.18326$, $f = 6.16874 \times 10^{-8}$, $g = -7.8300 \times 10^{-3}$. We note that similar temperature and pressure dependent model has been developed in [2017DAI/KAN], using the Pitzer formalism. In our calculations, we used the equilibrium constant for $CaSO_4$(aq) selected in the present review.

Figure V-75: The pressure dependence of $\log_{10} K^{\circ}_{s,0}$ (V.145).



V.6.2.2.5.4 The solubility of anhydrite

As for gypsum, a considerable number of papers report experimental data for the solubility of anhydrite at different temperatures both in pure water ([1910MEL], [1929PAR/WHI], [1934HIL], [1935HIL/YAN], [1937HIL], [1938POS], [1950BOO/BID], [1955ANS/BRE], [1960GRO2], [1966MAN2], [1967TEM/ROD], [1980INN/RAB], [2013WAN/ZEN]), and in NaCl solutions ([1903ANS], [1955ANS/BRE], [1956MAD/SWA2], [1961BOC], [1964MAR/SLU], [1967TEM/ROD], [1968FUR/GLU], [1970GLE/HAM], [2004FRE/VOI]). Although, anhydrite solubility was studied in aqueous solutions of several other electrolytes (for example Na_2SO_4, $NaNO_3$, $CaCl_2$, $MgCl_2$ [1967TEM/ROD], or sea water [1968MAR/SLU]), these are beyond the scope of this review.

The dissolution of anhydrite is governed by the Equilibria (V.146) and (V.150)

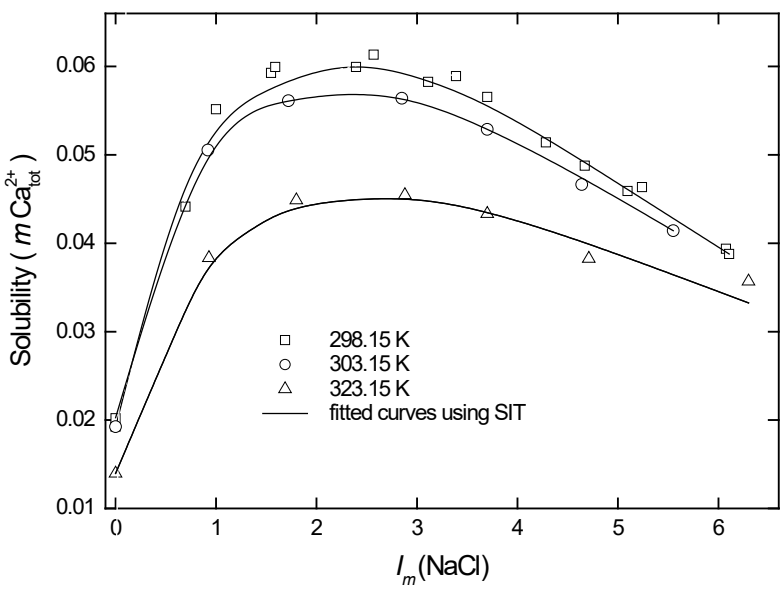
$$CaSO_4(cr) \rightleftharpoons Ca^{2+} + SO_4^{2-} \qquad \text{(V.150)}$$

As mentioned above, crystallisation of anhydrite is very slow below 343 K, even in presence of anhydrite seed crystals. Therefore, approaching true solubility equilibrium from both under- and supersaturation, is very difficult to achieve between 273 and 343 K. Nevertheless, several authors reported experimental solubility data within this temperature range. Hill approached equilibrium from both under- and supersaturated solutions [1937HIL], and at 338 K observed a 12.6 % difference between the two

concentrations after four months equilibration (which means a 0.065 difference in $\log_{10} K^{\circ}_{s,0}$ (V.150)). Surprisingly, the difference at lower temperatures was smaller (0.026 and 0.007 $\log_{10}$ units, respectively, at 318 and 308 K). On the other hand, the data reported by Innorta *et al.* [1980INN/RAB] indicated nearly identical solubility in under- and supersaturated solutions after a week equilibration between 313 and 338 K. Other authors ([1955ANS/BRE], [1956MAD/SWA2], [1961BOC], [1966MAN2], [1968FUR/GLU], [1970GLE/HAM], [2013WAN/ZEN]) reported constant solubility, approaching from undersaturated solutions below 343 K after 2-25 days (except for d'Anselme who observed stable solubility after three hours [1903ANS], and Posnak who at some temperatures measured the solubility after 1-2 years equilibration [1938POS]). Obviously, most of these data underestimate anhydrite solubility. A further complication arises from the nature of the anhydrite studied. Hill reported [1934HIL] that gypsum, dehydrated over a Bunsen burner flame at temperatures of 873 to 1073 K gave an anhydrite which gave erratic solubility results, probably due to the poorly crystallised, polymorphic form of the mineral.

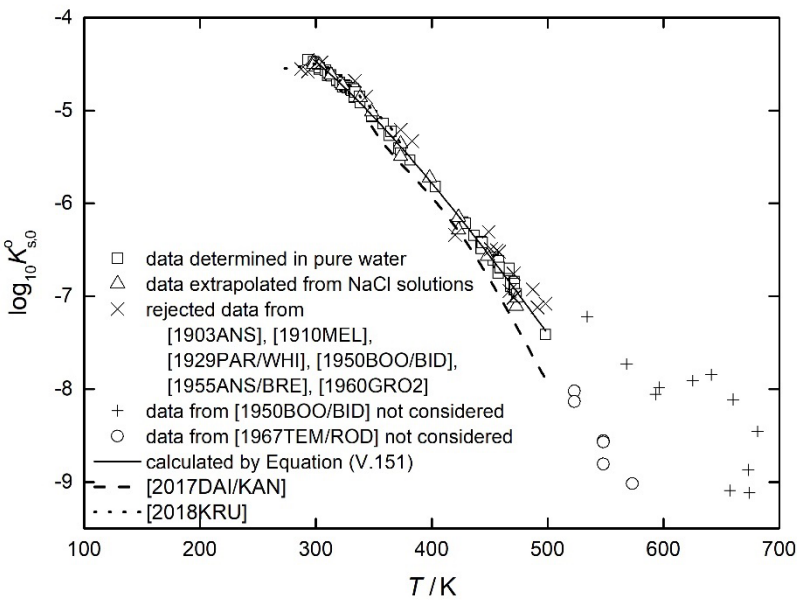
As in the case of gypsum, the addition of unlike ions, *e.g.*, NaCl (Figure V-73), increases anhydrite solubility which passes through a maximum around 2-3 mol·kg^{-1} (NaCl). However, in contrast to gypsum, the maximum solubility continuously decreases with temperature (Figure V-76).

Figure V-76: The solubility of anhydrite in aqueous NaCl solutions at different temperatures (experimental data are from [1956MAD/SWA2] and [1961BOC]).



The $\log_{10}K^{\circ}_{s,0}$ (V.150) values, determined either directly from the solubility in pure water or extrapolated from solubilities in NaCl solutions, are depicted in Figure V-77. The $\log_{10}K^{\circ}_{s,0}$ (V.150) values calculated from the data in [1903ANS] and [1950BOO/BID] were rejected (see Appendix A entry). Some further points ([1910MEL], [1929PAR/WHI], [1955ANS/BRE]) were also rejected due to large deviations from the other data. The $\log_{10}K^{\circ}_{s,0}$ (V.150) values calculated for $T > 500$ K from the data listed in [1967TEM/ROD] were not considered, because this temperature range is beyond the scope of the present review (although these values seem to follow the trend observed up to 500 K).

Figure V-77: The solubility constant of anhydrite as a function of temperature. Dashed and dotted lines were calculated from the temperature dependence derived in [2017DAI/KAN] and [2018KRU], respectively.



The unconstrained regression of the remaining 89 data points (with lower weights between 293 and 343 K) resulted in the equation:

$$\log_{10}K^{\circ}_{s,0}\,(\text{V.150}) = 4.237185 - 0.020041\,(T/\text{K}) - 803.925044\,(T/\text{K})^{-1}$$

for the temperature dependence of $\log_{10}K^{\circ}_{s,0}$ (V.150) and thus the following thermodynamic quantities for Reaction (V.150) at 298.15 K:

$$\log_{10} K^{\circ}_{s,0}\,((V.150),\ 298.15\ K) = -\,(4.434 \pm 0.030),$$

$$\Delta_r G^{\circ}_m\,((V.150),\ 298.15\ K) = (25.312 \pm 0.500)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r H^{\circ}_m\,((V.150),\ 298.15\ K) = -\,(18.716 \pm 1.000)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r S^{\circ}_m\,((V.150),\ 298.15\ K) = -\,(147.669 \pm 1.500)\ J{\cdot}K^{-1}{\cdot}mol^{-1},$$

$$\Delta_r C^{\circ}_{p,m}\,((V.150),\ 298.15\ K) = -\,(228.800 \pm 10.000)\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

However, based on calorimetric data the value of $S^{\circ}_m(CaSO_4,\ cr,\ 298.15\ K) = (107.4 \pm 0.2)\ J{\cdot}K^{-1}{\cdot}mol^{-1}$ has been selected in this review, which implies $\Delta_r S^{\circ}_m((V.150),\ 298.15\ K) = -\,(145.1 \pm 1.1)\ J{\cdot}K^{-1}{\cdot}mol^{-1}$. Using the latter value, a constrained regression of the selected experimental data resulted in Equation (V.151) (see solid line in Figure V-77).

$$\log_{10} K^{\circ}_{s,0}(V.150) = 4.7844025 - 0.02073362\,(T/K) - 909.409936\,(T/K)^{-1} \qquad (V.151)$$

Again, our fitted curve is between (at least in the overlapping region) the two recently published temperature dependencies ([2017DAI/KAN] and [2018KRU]), which were based on the Pitzer formalism. The refined parameters of Equation (V.151) lead to the following thermodynamic quantities for Reaction (V.150) at 298.15 K:

$$\log_{10} K^{\circ}_{s,0}\,((V.150),\ 298.15\ K) = -\,(4.448 \pm 0.030),$$

$$\Delta_r G^{\circ}_m\,((V.150),\ 298.15\ K) = (25.386 \pm 0.171)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r H^{\circ}_m\,((V.150),\ 298.15\ K) = -\,(17.875 \pm 1.000)\ kJ{\cdot}mol^{-1},$$

$$\Delta_r S^{\circ}_m\,((V.150),\ 298.15\ K) = -\,(145.098 \pm 3.403)\ J{\cdot}K^{-1}{\cdot}mol^{-1},$$

$$\Delta_r C^{\circ}_{p,m}\,((V.150),\ 298.15\ K) = -\,(236.700 \pm 4.000)\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

These values are somewhat different from those reported in [1990RAJ/ATK] ($\Delta_r G^{\circ}_m((V.150),\ 298.15\ K) = 24.32\ kJ{\cdot}mol^{-1}$, $\Delta_r H^{\circ}_m((V.150),\ 298.15\ K) = -\,17.66\ kJ{\cdot}mol^{-1}$, and $\Delta_r S^{\circ}_m((V.150),\ 298.15\ K) = -\,139.7\ J{\cdot}K^{-1}{\cdot}mol^{-1}$ and $\Delta_r C^{\circ}_{p,m}((V.150),\ 298.15\ K) = -\,396.8\ J{\cdot}K^{-1}{\cdot}mol^{-1}$).

Based on the thermodynamic data derived from Equation (V.151), the corresponding values for the formation of $CaSO_4$(cr) are:

$$\Delta_f G^{\circ}_m\,(CaSO_4,\ cr,\ 298.15\ K) = -\,(1322.196 \pm 1.500)\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^{\circ}_m\,(CaSO_4,\ cr,\ 298.15\ K) = -\,(1434.465 \pm 2.000)\ kJ{\cdot}mol^{-1},$$

$$S^{\circ}_m\,(CaSO_4,\ cr,\ 298.15\ K) = (107.400 \pm 0.200)\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

Our selected values for $\Delta_f G^{\circ}_m$ and $\Delta_f H^{\circ}_m$, including consideration of the calorimetric data for the enthalpy of formation and entropy of anhydrite are nearly identical:

$$\Delta_f G^\circ_m (CaSO_4, cr, 298.15\ K) = \mathbf{-\ (1322.242 \pm 1.307)}\ kJ{\cdot}mol^{-1},$$

$$\Delta_f H^\circ_m (CaSO_4, cr, 298.15\ K) = \mathbf{-\ (1434.510 \pm 1.300)}\ kJ{\cdot}mol^{-1},$$

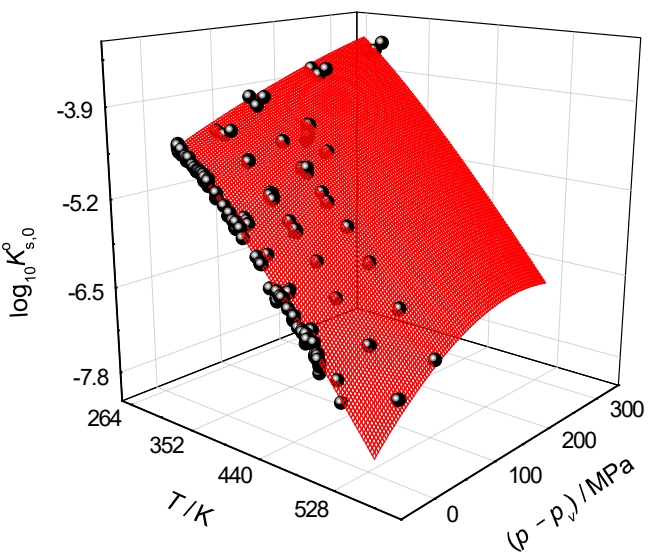
$$S^\circ_m (CaSO_4, cr, 298.15\ K) = \mathbf{(107.400 \pm 0.200)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

The values given in Table 6.2.1 of [1987GAR/PAR] are very similar (– (1321.98 ± 1.20) kJ·mol^{-1}, – (1434.508 ± 1.300) kJ·mol^{-1} and (106.525 ± 1.500) J·K^{-1}·mol^{-1}, respectively). However, note the slightly different data for the sulfate ion used in that assessment—see above. Their selected value for the standard entropy was based on the earlier work in [1961KEL/KIN].

V.6.2.2.5.5 The pressure dependence of anhydrite solubility

Three references report the pressure dependence of anhydrite solubility [1963DIC/BLO], [1966MAN2] and [1969BLO/DIC]. Like gypsum, the solubility increases with increasing pressure (Figure V-78). The pressure dependent $\log_{10} K^\circ_{s,0}$ (V.150) values were fitted by Equation (V.149) (see Section V.6.2.2.5.3). The constants a, b and c are identical with those in Equation (V.151), while $d = 2.31518 \times 10^{-5}$, $e = -0.58177$, $f = -5.52063 \times 10^{-8}$, $g = 3.7900 \times 10^{-3}$.

Figure V-78: The pressure dependence of $\log_{10} K^\circ_{s,0}$ (V.150).



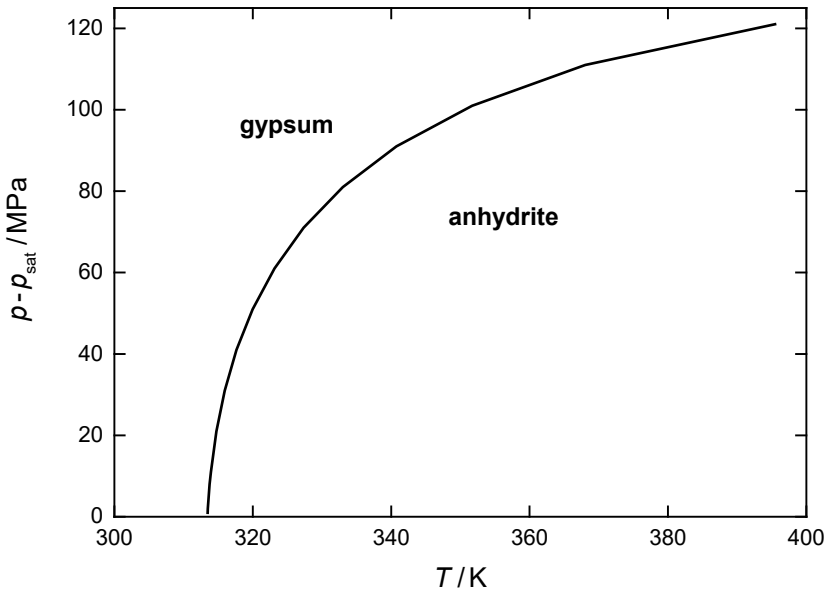
V.6.2.2.5.6 The gypsum-anhydrite phase transition

Based on the present dataset (Equations (V.148), (V.149), (V.151)), gypsum is the stable phase in pure water below 316.88 K at 0.1 MPa (although the slow kinetics of anhydrite crystallisation allow the formation of metastable phases both below and above this temperature). The gypsum-anhydrite transition temperature has been extensively studied due to its relevance to the evolution of evaporitic deposits. The reported values are spread between 311.65 and 339.15 K. Our value (316.9 ± 2 K) is within the uncertainty range of many earlier reports (*e.g.* [1937HIL], [1938POS], [1955ANS/BRE], [2007CHA/PUH] and [2018KRU]). On the other hand, it differs from that suggested in other evaluations (311.65 [1929PAR/WHI], 322.15 K [1988MOL], 322.65 K [1980INN/RAB], 328.65 K [1977KNA/GAN], 329.15 K [1973BLO/DIC], 331.15 K [1967HAR], 333.05 K [1990RAJ/ATK]). The main reasons for the considerable ambiguity of the gypsum-anhydrite transition temperature are (i) the very slow dissolution/growth kinetics of anhydrite at temperatures lower than 343 K, (ii) the small variation of gypsum solubility between 311 and 339 K, and (iii) partly the different ion interaction models used. Due to the first of above reasons, the transition temperature determined from solubility studies may be only the lower limit of the corresponding temperature range. At the same time, Freyer and Voigt [2003FRE/VOI] questioned some experimental details in those two papers which reported experimentally determined transition temperature higher than 328 K ([1967HAR], [1977KNA/GAN]), and they concluded that only a relatively wide temperature range (313-333 K) could be given for the gypsum-anhydrite transition temperature with high certainty.

As mentioned above, the addition of NaCl increases both gypsum and anhydrite solubilities, which pass through maxima around 2-3 $mol \cdot dm^{-3}$ (NaCl) (Figure V-73 and Figure V-76). However, the effect of NaCl above the maximum solubility is more pronounced in the case of anhydrite. Therefore, at 298 and 313 K the solubility curves of gypsum and anhydrite intersect at NaCl molalities of ~ 4.5 and ~ 3.0 $mol \cdot kg^{-1}$, respectively. Above 312 K at any practical NaCl concentration, anhydrite is the stable phase.

Increasing pressure results in higher solubilities of both gypsum and anhydrite, but this effect is more pronounced on the anhydrite solubility; therefore, below 313.47 K gypsum is the stable phase at any pressure. Above this temperature and at low pressure, anhydrite is the stable phase, but at higher pressures gypsum becomes the stable phase even well above 313.47 K (Figure V-79).

Figure V-79: The stability regions of gypsum and anhydrite as a function of temperature and pressure (the calculation is based on Equations (V.148), (V.149), and (V.151), p is the actual pressure, $p_{sat} = 0.1$ MPa below 373.15 K and p_{sat} is the saturation pressure of water in MPa at temperatures above 373.15 K).



V.6.3 Dolomite, $CaMg(CO_3)_2$(cr)

V.6.3.1 Structure

Dolomite has a rhombohedral structure, space group $R\bar{3}$ with alternating layers of Ca and Mg atoms. The lattice parameters vary slightly, depending on the precise Ca/Mg ratio, the degree of ordering (and for natural samples, the presence of impurities). Goldsmith and Graf [1958GOL/GRA] measured the lattice parameters of five selected natural dolomites, with $CaCO_3$ contents varying from 49.4 to 50.6 mol %, with impurity levels of 0.6 mol % (except for one with 3.2 mol % Fe) Their interpolated lattice parameters for the ideal dolomite composition were $a = 4.8079$ Å and $c = 16.010$ Å. In a later paper [1961GOL/GRA] the same authors also measured the lattice parameters of an ordered synthetic dolomite with the ideal composition to be $a = 4.8067$ Å and $c = 16.025$ Å. All these are, as expected, near the middle of the ranges of parameters reported in the voluminous literature on this subject ($a = 4.803$-4.827 Å and $c = 15.984$-16.137 Å).

V.6.3.2 Heat capacity and entropy

The only measurements of the heat capacity of dolomite are two studies at the U.S. Geological Survey. Stout and Robie [1963STO/ROB] measured the low-temperature heat capacity from 11.6 to 301.1 K in an adiabatic calorimeter, and these data were extended to 620 K by Krupka *et al.* [1985KRU/HEM] using DSC (see Appendix A entry). These merge smoothly with the low-temperature data of Stout and Robie [1963STO/ROB], as shown in Figure V-80 below, and these consistent data are the basis of our selected values, after a tiny correction for the molecular weight.

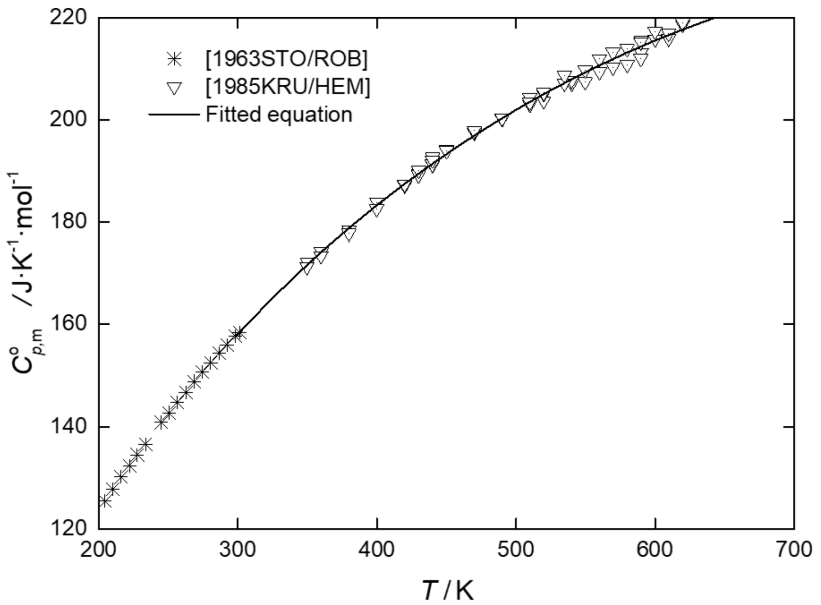
$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaMg(CO_3)_2},\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(157.51 \pm 0.80)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{CaMg(CO_3)_2},\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(155.17 \pm 0.80)}\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{CaMg(CO_3)_2},\mathrm{cr},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{82.79486} + \mathbf{0.347996}\,(T/\mathrm{K}) - \mathbf{2.03776\times10^{-4}}\,(T/\mathrm{K})^2 - \mathbf{9.71209\times10^{5}}\,(T/\mathrm{K})^{-2} \quad (298.15\text{-}650\,\mathrm{K}).$$

The uncertainties in the data at 298.15 K have been doubled from those suggested by Stout and Robie [1963STO/ROB], since below 35 K, the measured heat capacities of dolomite were less than those of calcite which seems irregular, as noted in Appendix A.

Figure V-80: Heat capacity of dolomite.



V.6.3.3 Enthalpy and Gibbs energy of formation

Our direct knowledge of the enthalpy of formation of dolomite rests essentially on enthalpies of solution and solubility studies. Values from electrochemical and thermal decomposition experiments are considered to be less reliable.

The early results of Mitchell [1923MIT], [1923MIT2] on the enthalpy of solution and on decomposition pressure, which were the basis of the selection of $\Delta_f H_m^o(CaMg(CO_3)_2, cr, 298.15\ K) = -(2326.3)\ kJ\cdot mol^{-1}$ by Rossini *et al.* [1952ROS/WAG] and Wagman *et al.* [1982WAG/EVA] (data assessed in 1968), were criticised by Bäckström [1924BAC], and Mitchell's subsequent data on the solubility [1923MIT2] are also criticised in the Section V.6.3.5 on the solubility of dolomite.

Navrotsky and Capobianco [1987NAV/CAP] compared the enthalpies of solution at 358 K of both a natural ordered dolomite and a synthetically disordered sample of the same origin, with the enthalpies of solution of synthetic mixtures magnesium and calcium carbonates. The reaction medium was 1.00 $mol\cdot dm^{-3}$ HCl solution saturated in NaCl at room temperature and containing 0.75 $mol\cdot dm^{-3}$ $CaCl_2$, and 0.25 molar $MgCl_2$. Details of the analysis of the materials and of the operating procedure are given in the relevant Appendix A.

Navrotsky and Capobianco reported their results as averages of at least six identical measurements; we have doubled their reported 1σ uncertainty. Given the complexity of the medium (sln), no attempt was made to identify solutes.

The details of the enthalpies of the individual reactions are also given in Appendix A, together with the correction to 298.15 K. From these, the enthalpy of formation of dolomite from binary carbonates:

$$CaCO_3(calc) + MgCO_3(cr) \rightleftharpoons CaMg(CO_3)_2(cr) \qquad (V.152)$$

is obtained as:

$$\Delta_f H_m^o((V.152), \text{ordered compound}, 298.15\ K) = -(11.60 \pm 1.02)\ kJ\cdot mol^{-1},$$

$$\Delta_f H_m^o((V.152), \text{disordered compound}, 298.15\ K) = (2.34 \pm 1.28)\ kJ\cdot mol^{-1},$$

where the corrections for the ordered and disordered compounds are assumed to be the same.

Using $\Delta_f H_m^o(MgCO_3, cr, 298.15\ K) = -(1112.2 \pm 1.3)\ kJ\cdot mol^{-1}$ and $\Delta_f H_m^o(CaCO_3, calc, 298.15\ K) = -(1207.5 \pm 1.2)\ kJ\cdot mol^{-1}$ selected in this review we obtain for the ordered compound $\Delta_f H_m^o(CaMg(CO_3)_2, cr, 298.15\ K) = -(2331.500 \pm 1.866)\ kJ\cdot mol^{-1}$.

In a similar approach, Chai and Navrotsky [1993CHA/NAV] compared the enthalpy of solution of dolomite with that of a stoichiometric mixture of magnesium and calcium carbonates in lead borate ($2PbO\cdot B_2O_3$) solvent at 977 K, using a natural dolomite sample of the same origin (Euguy, Spain) as in the study [1987NAV/CAP] just described

and also a synthetic sample. Details of the sample characterisation and of the high-temperature drop-solution calorimetric procedure are given in the relevant Appendix A. The results were reported as averages (with 2σ uncertainty) of a large number of similar measurements. Given the complexity of the medium (designated sln*), no attempt was made to identify the solute species. Thus, for the dissolution of mechanical mixture of binary carbonates:

$$[(CaCO_3(cr) + MgCO_3(cr)), 298.15\ K] + (sln^*, 977\ K) \rightleftharpoons (CaO + MgO)(sln^*, 977\ K) + 2CO_2(g, 977\ K) \quad (V.153)$$

the reported value from seven measurements was:

$$\Delta_r H^\circ_m\ (V.153) = (373.7 \pm 1.9)\ kJ \cdot mol^{-1}.$$

For the dissolution of dolomite:

$$CaMg(CO_3)_2(cr, 298.15\ K) + (sln^*, 977\ K) \rightleftharpoons (CaO + MgO)(sln^*, 977\ K) + 2CO_2(g, 977\ K) \quad (V.154)$$

the values were:

$\Delta_r H^\circ_m$ ((V.154), natural sample) = (383.7 ± 1.1) kJ·mol^{-1} from 6 measurements and

$\Delta_r H^\circ_m$ ((V.154), synthetic sample) = (385.8 ± 1.1) kJ·mol^{-1} from 16 measurements.

Chai and Navrotsky noted that the limited agreement between these two values could not be due to a potential disordering in the synthetic sample, (when the value would have been smaller than that for the natural sample), nor to differences in stoichiometries, and therefore adopted $\Delta_r H^\circ_m$ (V.154) = (384.8 ± 1.6) kJ·mol^{-1} as the best value. This was supported by later measurements of the same laboratory [1995CHA/NAV]—see Appendix A entry.

Combination of the value for Reactions (V.153) and (V.154) gives for the enthalpy of formation of dolomite from the binary carbonates:

$$CaCO_3(calc) + MgCO_3(cr) \rightleftharpoons CaMg(CO_3)_2(cr) \quad (V.155)$$

$$\Delta_r H^\circ_m\ ((V.155), 298.15\ K) = -(11.1 \pm 2.5)\ kJ \cdot mol^{-1}.$$

This value is in good agreement with (albeit less precise than) the value of $-(11.60 \pm 1.02)$ kJ·mol^{-1} obtained in the same laboratory [1987NAV/CAP] as described above and leads to $\Delta_r H^\circ_m$ ($CaMg(CO_3)_2$, cr, 298.15 K) = – (2331.00 ± 2.95) kJ·mol^{-1}.

Hemingway and Robie [1994HEM/ROB] reported a solution calorimetry study of the enthalpy of formation of natural dolomite through a cycle involving the dissolution (at 300.15 K) in 4 mol·dm^{-3} HCl of this compound, of CaO(cr) and of MgO(cr). The experimental study was carried out in 1973, some 20 years before the final publication and had previously been referred to as “Robie and Hemingway, unpublished data, (1978)”

in a number of papers.

For the reaction:

$$CaMg(CO_3)_2(cr) + (178HCl + 2276H_2O)(sln) \rightleftharpoons (CaCl_2 + MgCl_2 + 174HCl + 2278H_2O)(sln) + 2CO_2(g) \quad \text{(V.156)}$$

the value $\Delta_r H_m^o$ ((V.156), 300.15 K) = – (47.08 ± 0.35) kJ·mol^{-1} was obtained after correction for the CO_2 that remained dissolved (93 %) in the solution in the actual experiment. The authors refer to [1973HEM/ROB] for the discussion of this correction.

Also used in the cycle are the reactions:

$$MgO(cr) + (178HCl + 2276H_2O)(sln) \rightleftharpoons (MgCl_2 + 176HCl + 2277H_2O)(sln) \quad \text{(V.157)}$$

$\Delta_r H_m^o$ ((V.157), 300.15 K) = – (150.30 ± 0.08) kJ·mol^{-1}, from [1973HEM/ROB] for the dissolution under very similar conditions, after correction for a small difference in temperature (from 303.15 to 300.15 K) and:

$$CaO(cr) + (MgCl_2 + 176HCl + 2277H_2O)(sln) \rightleftharpoons (CaCl_2 + MgCl_2 + 174HCl + 2278H_2O)(sln) \quad \text{(V.158)}$$

$\Delta_r H_m^o$ ((V.158), 300.15 K) = – (197.72 ± 0.34) kJ·mol^{-1}.

These give for the reaction:

$$CaO(cr) + MgO(cr) + 2CO_2(g) \rightleftharpoons CaMg(CO_3)_2(cr) \quad \text{(V.159)}$$

$\Delta_r H_m^o$ ((V.159), 300.15 K) = – (300.94 ± 0.49) kJ·mol^{-1}.

Using the data from [1978ROB/HEM], Robie and Hemingway corrected the latter result to 298.15 K and calculated the standard enthalpy and Gibbs energy of formation, and solubility constant of dolomite. With the $C_{p,m}^o$ values accepted in this review, we calculate $\Delta_r H_m^o$ ((V.159), 298.15 K) = – (300.95 ± 0.49) kJ·mol^{-1}. With CODATA values for the reactants, we calculate $\Delta_f H_m^o$ ($CaMg(CO_3)_2$, cr, 298.15 K) = – (2324.49 ± 1.10) kJ·mol^{-1}.

Rock *et al.* [2001ROC/MAN] reported data for the Gibbs energy of dolomite at 298.15 K, based on electrochemical cell measurements and also on lattice energy calculations. The cell used was:

$$Cd(Hg)(\text{two phase}) \mid CdCO_3(cr), Ca_{0.50}Mg_{0.50}CO_3(cr) \mid CaCl_2(aq, m_1),$$

$$MgCl_2(aq, m_2) \mid Hg_2Cl_2(cr) \mid Hg(l).$$

where Cd(Hg) represents a saturated cadmium-mercury amalgam and m_1, m_2, are respectively the molalities of calcium and magnesium in the aqueous medium. As noted in detail in the relevant Appendix A, this study leads to $\Delta_f G_m^o$ ($CaMg(CO_3)_2$, cr) = – (2144.82 ± 2.90) kJ·mol^{-1}, *ca.* 20 kJ·mol^{-1} more positive than that derived from the

three consistent calorimetric studies discussed above and is not considered in our final selection of the data for dolomite. We note that Rock *et al.* [2001ROC/MAN] also refer to their unpublished value for $\Delta_f G_m^o$ ($MgCO_3$, cr) by a similar electrochemical study, which is also *ca.* 20 kJ·mol^{-1} more positive than the well-established value assessed in this review.

V.6.3.4 Decomposition pressures

There are a number of studies of the decomposition pressures of dolomite, such as those by Graf and Goldsmith [1955GRA/GOL], Harker and Tuttle [1955HAR/TUT] and Goldsmith and Heard [1961GOL/HEA], as described in Appendix A.

As for the decomposition of magnesite, at lower temperatures equilibrium is established only after many hours, and in addition, the products are difficult to establish unequivocally. Since magnesite has an appreciably higher decomposition pressure than calcite, the general reaction can be expressed by the simplified equation:

$$CaMg(CO_3)_2(cr) \rightleftharpoons CaMg_{1-x}(CO_3)_{2-x}(cr) + xMgO(cr) + xCO_2(g)$$

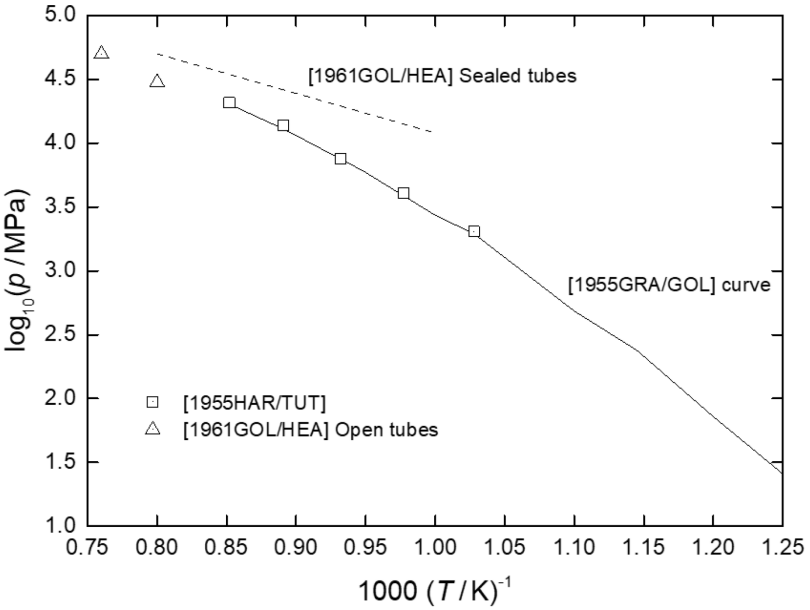
to give a calcium-rich mixed carbonate and MgO. Since x varies with temperature the decomposition pressures do not follow a linear $\log_{10}(p)$ *vs.* $1/T$ relation at higher temperatures and in reality the MgO may also contain small quantities of CaO. In addition, partial disordering of the dolomite occurs at temperatures above *ca*. 1273 K [1961GOL/HEA]. With these complications, and the unknown activities of $CaCO_3$ and $MgCO_3$ in the mixed carbonate, it is not possible to derive reliable enthalpy or Gibbs energies of formation of dolomite from these studies, as noted by Harker and Tuttle [1955HAR/TUT] and Graf and Goldsmith [1955GRA/GOL] (see also below). For this review, we have therefore limited ourselves to a simple comparison of the decomposition pressures from the more important studies.

Graf and Goldsmith [1955GRA/GOL] measured the decomposition pressures at temperatures up to 1136 K and pressures of 138 MPa, and Harker and Tuttle up to 1223 K and pressures of 275 MPa [1955HAR/TUT] with excellent agreement. Goldsmith and Heard in their more extended investigation of the Mg-Ca-CO_3 system [1961GOL/HEA] obtained data in both open and sealed tubes at higher temperatures and pressures. The data in the open tubes agreed well with an extrapolation of the earlier data but the decomposition pressures in sealed tubes were appreciably greater, for an unknown reason. These data are summarised in Figure V-81.

It should be noted that despite the uncertainties in the compositions of the products noted above, Stout and Robie, in a final section of their paper on the heat capacity of dolomite noted above [1963STO/ROB], made approximate calculations of the enthalpy of the reverse of Reaction (V.152) based on the decomposition data of Graf and Goldsmith [1955GRA/GOL], using a number of estimated quantities and simplifying assumptions. These calculations would indicate a relatively constant value of

(12.3 ± 1.0) kJ·mol^{-1} over the temperature range 830 to 1136 K, consistent with the more negative values for the enthalpy of formation of dolomite.

Figure V-81: Dissociation pressure of dolomite.



V.6.3.5 Solubility of dolomite

The reported values of solubility products of stoichiometric or ordered dolomite ($CaMg(CO_3)_2(cr)$) at 298.15 K range between $\log_{10}K_s^o$ (V.160) = – 15.0 to – 19.4 (see Table V-76). The large variation of data is likely due to the slow dissolution rate and even slower crystallisation rate of ordered dolomite [2000SHE/BAR], [2001POK/SCH]. The rates of both processes increase with the temperature. Siebert and Hostetler estimated *ca.* two months to reach equilibrium saturation for dolomite at 363.15 K, while *ca.* 15 years would be needed at 298.15 K [1970SIE/HOS]. Accordingly, although ordered dolomite can be readily synthesised under hydrothermal conditions, its formation under 373.15 K is extremely slow [2001POK/SCH], which is also confirmed by its absence in modern marine depositional environments.

$$CaMg(CO_3)_2(cr) \rightleftharpoons Ca^{2+} + Mg^{2+} + 2CO_3^{2-} \qquad \text{(V.160)}$$

As Sherman and Barak [2000SHE/BAR] noted, most data listed in Table V-76 are unsatisfactory due to one or more experimental problems: (i) insufficient equilibration times [1923MIT2], [1960GAR/THO]; (ii) closed systems with respect to CO_2

[1962MOR]; (iii) inconsistency between measured data at different $p(CO_2)$ [1923MIT2], or between pH and HCO_3^- concentration [1950YAN]; (iv) the *in situ* groundwater measurements are hampered by questions of attainment of equilibrium with respect to dolomite [1963HSU], [1964BAR/BAC]. The solubilities reported by Mitchell [1923MIT2] at 298.15 K after 150 hours equilibration are probably erroneous, since they indicate a strong dependence of $\log_{10}K^\circ_{s,0}$ on CO_2 pressure. The dissolution experiments performed by Garrels *et al.* [1960GAR/THO] in pure water at 298.15 K and 0.1 MPa $p(CO_2)$ indicated considerably different solubility for coarsely and finely ground dolomite, which may indicate non-equilibrium concentrations in the former case. The continuous flow reaction tube used by Morey resulted in undersaturated solutions, at least at lower temperatures, similar to his experiments with magnesite [1962MOR]. Among the data for 298.15 K in Table V-76, those of Sherman and Barak are the most reliable [2000SHE/BAR]. Their solubility study was performed at 298.15 K in Ca(II)-Mg(II)-HCO_3^-/CO_3^{2-} solutions for nearly two years. Based on statistical inference by comparison of alkalinity in unseeded and dolomite-seeded solutions, Sherman and Barak determined $\log_{10}K^\circ_s$ (V.160) = – (17.2 ± 0.2). This is in good agreement with the value that can be calculated from the thermodynamic properties of dolomite determined by Hemingway and Robie [1994HEM/ROB], $\log_{10}K^\circ_s$ (V.160)= – (17.09 ± 0.38), using HC1 solution calorimetry (see Section V.6.3.3). Recently, Bénézeth *et al*. performed a detailed study on the solubility of dolomite between 323.15 and 525.65 K in 0.1 mol·kg^{-1} NaCl solutions using a hydrogen-electrode concentration cell [2018BEN/BER]. Although this study was performed in a closed system with respect to CO_2, the authors accurately and simultaneously determined the pH, $[Mg^{2+}]_{tot}$, $[Ca^{2+}]_{tot}$ and the total dissolved inorganic carbon (TDIC) concentrations. In this review we accept the $\log_{10}K^\circ_s$ (V.160) values reported in [2000SHE/BAR] and [2018BEN/BER] (see Table V-76). An unconstrained regression of these data resulted in Equation (V.161) .

$$\log_{10}K^\circ_s\,(\text{V.160}) = (16.63917 \pm 3.87939) - (0.067767 \pm 0.0048780)(T/\text{K}) - (4058.17 \pm 756.67)(T/\text{K})^{-1} \qquad \text{(V.161)}$$

The above parameters correspond to the following thermodynamic quantities for Equation (V.160) at 298.15 K: $\log_{10}K^\circ_{s,0}$ ((V.160), 298.15 K) = – (17.177 ± 0.200), $\Delta_r G^\circ_m$ ((V.160), 298.15 K) = (98.046 ± 1.142) kJ·mol^{-1}, $\Delta_r H^\circ_m$ ((V.160), 298.15 K) = – (37.636 ± 1.500) kJ·mol^{-1} and $\Delta_r S^\circ_m$ ((V.160), 298.15 K) = – (455.081 ± 6.323) J·K^{-1}·mol^{-1}. The latter value would lead to S°_m ($CaMg(CO_3)_2$, cr, 298.15 K) = (162.981 ± 6.684) J·K^{-1}·mol^{-1}. However, a considerably more precise value S°_m ($CaMg(CO_3)_2$, cr, 298.15 K) = (155.170 ± 0.800) J·K^{-1}·mol^{-1} has been selected in this review (see Section V.6.3.2), which corresponds to $\Delta_r S^\circ_m$ ((V.160), 298.15 K) = – (447.27 ± 2.154) J·K^{-1}·mol^{-1}. Therefore, regression of the accepted data was constrained to the latter $\Delta_r S^\circ_m$ ((V.160), 298.15 K) value. This constrained regression resulted in Equation (V.162) (solid line in Figure V-82).

$$\log_{10}K^\circ_s\,(\text{V.160}) = (18.1594 \pm 0.9175) - (0.069632 \pm 0.001538)(T/\text{K}) - (4359.635 \pm 120.405)(T/\text{K})^{-1} \qquad \text{(V.162)}$$

In this way the following thermodynamic quantities can be calculated:

$$\log_{10} K_s^o \text{ ((V.160), 298.15 K)} = -(17.224 \pm 0.200),$$

$$\Delta_r G_m^o \text{ ((V.160), 298.15 K)} = (98.314 \pm 1.142) \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((V.160), 298.15 K)} = -(35.039 \pm 1.500) \text{ kJ·mol}^{-1},$$

$$\Delta_r S_m^o \text{ ((V.160), 298.15 K)} = -(447.270 \pm 2.427) \text{ J·K}^{-1}\text{·mol}^{-1}.$$

The above thermodynamic parameters, combined with those recommended for Mg^{2+}, Ca^{2+} and CO_3^{2-}, correspond to:

$$\Delta_f G_m^o (CaMg(CO_3)_2, \text{cr}, 298.15 \text{ K}) = -(2162.742 \pm 1.778) \text{ kJ·mol}^{-1} \text{ and}$$

$$\Delta_f H_m^o (CaMg(CO_3)_2, \text{cr}, 298.15 \text{ K}) = -(2325.421 \pm 1.933) \text{ kJ·mol}^{-1}.$$

These values compare well with those selected in this review (see the next chapter), and are within the uncertainty range of those selected by Hemingway and Robie [1994HEM/ROB]: $\Delta_f G_m^o = -(2161.7 \pm 1.1)$ kJ·mol^{-1}, $\Delta_f H_m^o = -(2324.5 \pm 1.1)$ kJ·mol^{-1}. There are slightly larger deviations from the values of Bénézeth *et al.* ($\Delta_f G_m^o = -(2160.9 \pm 2.0)$ kJ·mol^{-1}, $\Delta_f H_m^o = -(2323.1 \pm 2.0)$ kJ·mol^{-1}, [2018BEN/BER]), which are partly due to the different values used for the formation data of Mg^{2+}, Ca^{2+} and CO_3^{2-} ions.

Figure V- 82: Solubility constants of dolomite as a function of temperature

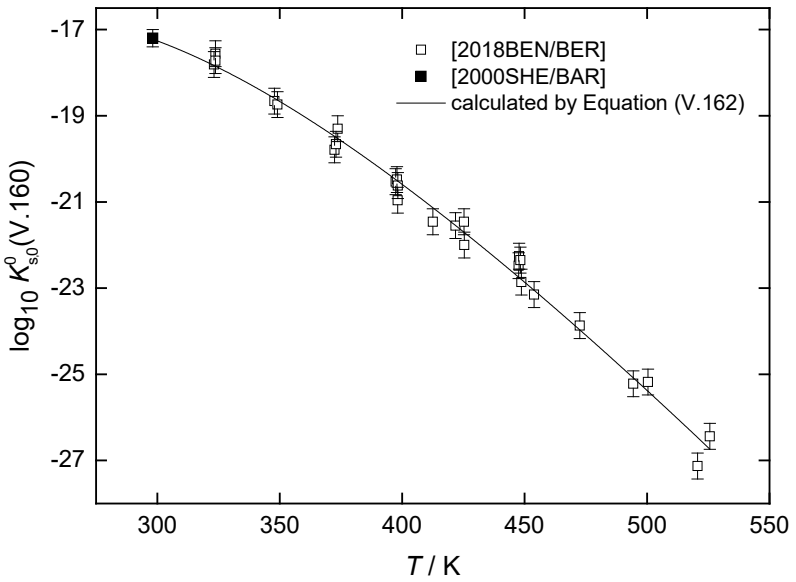


Table V-76: Solubility data reported for dolomite.

Reference	Experimental conditions	Reported solubility (m)			Reported $\log_{10} K_s^o$	Accepted $\log_{10} K_s^o$
		Mg(II)	Ca(II) + Mg(II)	Ca(II)		
[1923MIT2]	T = 298.15 K, p_{CO_2} = 0.0253 MPa,		0.0048		– 17.68[a]	
	T = 298.15 K, p_{CO_2} = 0.0507 MPa,		0.0081		– 17.13[a]	
	T = 298.15 K, p_{CO_2} = 0.076 MPa,		0.0171		– 15.81[a]	
	T = 298.15 K, p_{CO_2} = 0.1013 MPa,		0.0282		– 14.72[a]	
[1950YAN]	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa, pH = 5.7	0.00329	0.00651	0.00322	– 17.8[b], – 18.4[c]	
	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa, 0.086 m NaCl		0.00742		– 18.5[b], – 19.3[c]	
	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa, 0.346 m NaCl		0.01326			
[1954YAN]	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa,	0.00348	0.00723	0.00375	– 18.2[d]	
[1954YAN2]	T = 273.15 K, p_{CO_2} ~ 0.1013 MPa		0.0108			
	T = 298.15 K, p_{CO_2} = 121.6 Pa,	0.00102	0.00142	0.0004		
	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa,		0.00609			
	T = 328.15 K, p_{CO_2} ~ 0.1013 MPa,		0.00458			
	T = 343.15 K, p_{CO_2} ~ 0.1013 MPa,		0.00722			
[1959KRA]	T = 298.15 K, at 35, 45 and 60 ‰ salinity, atmospheric CO_2				– 16.82, –17.2[c]	
[1960GAR/THO]	T = 298.15 K, p_{CO_2} ~ 0.1013 MPa, pH = 5.68 (coarse);				– 19.33	
	T = 298.15 K, p_{CO_2} 0.1013 MPa, pH = 6.21 (fine ground)				– 16.5[c]	

(Continued on next page)

Table V-76 (continued)

Reference	Experimental conditions	Reported solubility (m)		Reported $\log_{10} K_s^o$	Accepted $\log_{10} K_s^o$
[1962MOR]	T = 298.15 K	0.000036	0.00004		
	T = 323.15 K	0.000066	0.000061		
	T = 353.15 K	0.000088	0.000087		
	T = 383.15 K	0.000096	0.00011		
	T = 403.15 K	0.000094	0.00013		
	T = 423.15 K	0.000072	0.00015		
[1962HAL]	T = 294.15 K, p_{CO_2} = 0.1013 MPa		0.0092		– 17.0, – 17.7[c]
			pH = 6.02		– 16.6, – 17.5[c]
[1963HSU]	Based on the composition of ground waters in dolomitic limestones, T ~ 298.15 K			– 16.7,– 17.1[c]	
[1964BAR/BAC]	Based on the composition of ground waters in dolomitic limestones			– (16.67 ± 0.15)	
[1969RIE]	T = 298.15 K, p_{CO_2} = 0.0924 MPa, 3 M $NaClO_4$	0.0031-0.0508 m between pH 5.2-5.8		– 18.04 ± 0.02)[e]	
	T = 323.45 K, p_{CO_2} = 0.0843 MPa, 3 M $NaClO_4$	0.0023-0.0513 m between pH 5-5.7		– (17.3 ± 0.04)[e]	
[1970SIE/HOS]	T = 298.15 K, p_{CO_2} = 0.1013 MPa			[f]	
	T = 363.15 K, p_{CO_2} = 0.1013 MPa			– (17.4 ± 0.1)	
[2000SHE/BAR]	T = 298.15 K, p_{CO_2} = 0.0952 MPa			– (17.2 ± 0.2)	– (17.2 ± 0.2)

(Continued on next page)

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Table V-76 (continued)

Reference	Experimental conditions	Reported solubility (m)			Reported $\log_{10} K_s^o$	Accepted $\log_{10} K_s^o$
[2018BEN/BER]		$[Mg^{2+}]_{tot}$ (m)	$[Ca^{2+}]_{tot}$ (m)	TDIC (m) (g)		
	323.15 K, 0.1 m NaCl	0.0010965	0.0015996	0.002125	– (17.81 ± 0.30)	– (17.81 ± 0.30)
	323.75 K, 0.1 m NaCl	0.0015346	0.0015205	0.002957	– (17.56 ± 0.30)	– (17.56 ± 0.30)
	323.75 K, 0.1 m NaCl	0.0016827	0.0010209	0.002522	– (17.72 ± 0.30)	– (17.72 ± 0.30)
	347.85 K, 0.1 m NaCl	0.0009705	0.0013213	0.001632	– (18.66 ± 0.30)	– (18.66 ± 0.30)
	349.05 K, 0.1 m NaCl	0.0011885	0.0013032	0.002044	– (18.74 ± 0.30)	– (18.74 ± 0.30)
	372.35 K, 0.1 m NaCl	0.0008072	0.0012942	0.001106	– (19.79 ± 0.30)	– (19.79 ± 0.30)
	372.95 K, 0.1 m NaCl	0.0010186	0.0011429	0.001454	– (19.66 ± 0.30)	– (19.66 ± 0.30)
	373.65 K, 0.1 m NaCl	0.0010864	0.0013243	0.002225	– (19.30 ± 0.30)	– (19.30 ± 0.30)
	397.35 K, 0.1 m NaCl	0.0008318	0.0012134	0.000919	– (20.53 ± 0.30)	– (20.53 ± 0.30)
	397.85 K, 0.1 m NaCl	0.0007345	0.0010023	0.000781	– (20.48 ± 0.30)	– (20.48 ± 0.30)
	398.15 K, 0.1 m NaCl	0.0006668	0.0010375	0.000691	– (20.96 ± 0.30)	– (20.96 ± 0.30)
	398.25 K, 0.1 m NaCl	0.0008356	0.0010691	0.001061	– (20.62 ± 0.30)	– (20.62 ± 0.30)
	412.55 K, 0.1 m NaCl	0.0007962	0.0013932	0.001478	– (21.46 ± 0.30)	– (21.46 ± 0.30)
	421.75 K, 0.1 m NaCl	0.0007228	0.0010023	0.000755	– (21.55 ± 0.30)	– (21.55 ± 0.30)
	425.25 K, 0.1 m NaCl	0.0007161	0.0009840	0.001072	– (21.46 ± 0.30)	– (21.46 ± 0.30)
	425.35 K, 0.1 m NaCl	0.0007464	0.0009705	0.000984	– (22.00 ± 0.30)	– (22.00 ± 0.30)
	447.55 K, 0.1 m NaCl	0.0006012	0.0010471	0.000414	– (22.48 ± 0.30)	– (22.48 ± 0.30)
	447.75 K, 0.1 m NaCl	0.0006138	0.0009419	0.000525	– (22.26 ± 0.30)	– (22.26 ± 0.30)
	448.65 K, 0.1 m NaCl	0.0005794	0.0009354	0.000413	– (22.86 ± 0.30)	– (22.86 ± 0.30)
	448.25 K, 0.1 m NaCl	0.0006471	0.0009441	0.000658	– (22.35 ± 0.30)	– (22.35 ± 0.30)
	453.85 K, 0.1 m NaCl	0.0006577	0.0014191	0.000988	– (23.15 ± 0.30)	– (23.15 ± 0.30)
	472.55 K, 0.1 m NaCl	0.0005768	0.0011324	0.000349	– (23.87 ± 0.30)	– (23.87 ± 0.30)
	494.35 K, 0.1 m NaCl	0.0005395	0.0012823	0.000698	– (25.22 ± 0.30)	– (25.22 ± 0.30)
	500.35 K, 0.1 m NaCl	0.0002992	0.0012503	0.000423	– (25.18 ± 0.30)	– (25.18 ± 0.30)
	520.65 K, 0.1 m NaCl	0.0001030	0.0015101	0.000348	– (27.13 ± 0.30)	– (27.13 ± 0.30)
	525.65 K, 0.1 m NaCl	0.0004699	0.0011695	0.000779	– (26.44 ± 0.30)	– (26.44 ± 0.30)

(a) re-calculated in [1935HAL/RIT]
(b) re-calculated in [1962HAL]
(c) re-calculated in [2000SHE/BAR]
(d) re-calculated in [1994HEM/ROB]
(e) $\log_{10} K_s$
(f) time estimated to reach equilibrium is ~ 15 years
(g) TDIC = total dissolved inorganic carbon

V.6.3.6 Selected values for dolomite

As noted in Section V.6.3.2, the selected value for the standard entropy of dolomite is

$$S^\circ_m(CaMg(CO_3)_2, cr, 298.15\ K) = \mathbf{(155.17 \pm 0.80)}\ J\cdot K^{-1}\cdot mol^{-1}.$$

Our selection of the enthalpy of formation is based on the values in Table V-77.

Table V-77: Experimental data for $\Delta_f H^\circ_m(CaMg(CO_3)_2, cr, 298.15\ K)$.

Reference	Method	$\Delta_f H^\circ_m(298.15\ K)$/kJ·mol^{-1}
[1987NAV/CAP]	Calorimetry from carbonates dissolved in HCl-NaCl (sln)	– (2331.50 ± 1.87)
[1993CHA/NAV]	High temperature calorimetry from carbonates	– (2331.00 ± 2.95)
[1994HEM/ROB]	Calorimetry from oxides dissolved in HCl-NaCl (sln)	– (2324.49 ± 1.10)
This review	Solubility assessment	– (2325.40 ± 1.13)

Our selected value for ordered dolomite is then the rounded weighted mean of these four values, with an increased uncertainty:

$$\Delta_f H^\circ_m(CaMg(CO_3)_2, cr, 298.15\ K) = \mathbf{-\ (2326.2 \pm 2.0)}\ kJ\cdot mol^{-1},$$

which corresponds to

$$\Delta_f G^\circ_m(CaMg(CO_3)_2, cr, 298.15\ K) = \mathbf{-\ (2163.402 \pm 2.019)}\ kJ\cdot mol^{-1},$$

and a solubility of

$$\log_{10} K^\circ_{s,0}(298.15\ K) = -\ (17.339 \pm 0.438),$$

consistent with the value of – (17.224 ± 0.200) derived from the solubility measurements.

Chapter V.7

Na and K species

V.7.1 Sodium carbonate, Na_2CO_3 (cr)

V.7.1.1 Structure and polymorphism

The polymorphism of Na_2CO_3(cr) has been the subject of many studies, initially with conflicting conclusions. Some of the early authors *e.g.*, Makarov and Shul'gina [1940MAK/SHU], suggested there were at least three polymorphic transitions, some extending over a noticeable temperature range, but it is now clear that some of these were due either to partial decomposition to Na_2O(cr) and CO_2(g), or to the presence of impurities, particularly KCl(cr), in the samples used. Later work in an atmosphere of CO_2(g) on high-purity materials by Reisman and co-workers [1958REI], [1958REI/HOL], [1959REI], indicates clearly that there are only two polymorphic transitions above 298.15 K, at *ca.* 630 and 750 K. Arakcheeva and Chapuis [2005ARA/CHA] give a detailed summary of the complicated crystallographic aspects of the relevant phases which do not concern this review.

The polymorph stable at 298.15 K has an incommensurate monoclinic structure, space group $C2/m$, with parameters $a = (8.91 \pm 0.01)$ Å, $b = (5.24 \pm 0.01)$ Å, $c = (6.035 \pm 0.015)$ Å, $\gamma = (101.3 \pm 0.1)$ ° [1976AAL/HOL], [2002ZUB/PUS], [2003DUS/CHA], [2010ARA/BIN]. At *ca.* (628 ± 5) K, this transforms to another monoclinic structure, space group $C2/m$, with parameters $a = 8.99$ Å, $b = 5.2$ Å, $c = 6.26$ Å, $\gamma = 98.87$ ° at 645 K [1995SWA/DOV]. At (754 ± 5) K, this in turn transforms to a hexagonal phase, stable up to the melting point at (1129 ± 2) K. The parameters of this hexagonal phase, space group $P6_3/mmc$ are $a = 5.208$ Å, $c = 6.454$ Å at 756 K [1995SWA/DOV].

In addition, below *ca.* 170 K, there is a commensurate monoclinic structure in pure Na_2CO_3, space group $C2/m$, with parameters $a = (8.898 \pm 0.007)$ Å, $b = (5.237 \pm 0.005)$ Å, $c = (5.996 \pm 0.005)$ Å, $\gamma = (101.87 \pm 0.08)$ ° at 110 K [2003DUS/CHA]; de Pater and Helmholdt [1979PAT/HEL] reported the transition temperature to be 130 K, and Arakcheeva *et al.* [2010ARA/BIN] did not find any transition above 120 K in a natural natrite crystal.

V.7.1.2 Heat capacity and entropy

There are two measurements of the low-temperature $C_p(Na_2CO_3)$, by Anderson [1933AND] from *ca.* 56 to 290 K and Waterfield *et al.* [1968WAT/LIN] from *ca.* 10 to 300 K, with agreement to within 1 %. Because of the much bigger temperature range in the study by Waterfield *et al.*, their values at 298.15 K have been selected.

The standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(Na_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(112.30 \pm 0.50)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

and

$$S^{\circ}_{\mathrm{m}}(Na_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(135.00 \pm 0.40)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

The data for the high-temperature heat capacity are less well-defined since, as noted above, the transitions can extend over appreciable temperature ranges and some measurements are compromised by partial decomposition. The measurements of $H_{\mathrm{m}}(T) - H_{\mathrm{m}}(298.15\ \mathrm{K})$ summarised in Table V-78 are plotted in Figure V-83, together with the assessed values in [1982GLU/GUR]. Since the latter give an excellent representation of the rather scattered data, we have accepted these, with small adjustments to the transition temperatures to take account of more recent studies.

Table V-78: Summary of the studies of the high-temperature enthalpy of Na_2CO_3(cr).

Reference	Number of measurements	Temperature range /K
[1952MAY]	13	410-1470
[1956POP/GIN]	10	476-871
[1963JAN/NEU]	28	*ca.* 600-1200
[1964ROL/REC3]	22	645-1322

The selected values are thus:

$$C^{\circ}_{p,\mathrm{m}}(Na_2CO_3, \mathrm{cr}, \alpha, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{64.118} + \mathbf{0.161604}\,(T/\mathrm{K}) \qquad (298.15\text{-}628\ \mathrm{K})$$

Transition at (628 ± 5) K; $\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ = 0.8 kJ·mol^{-1},

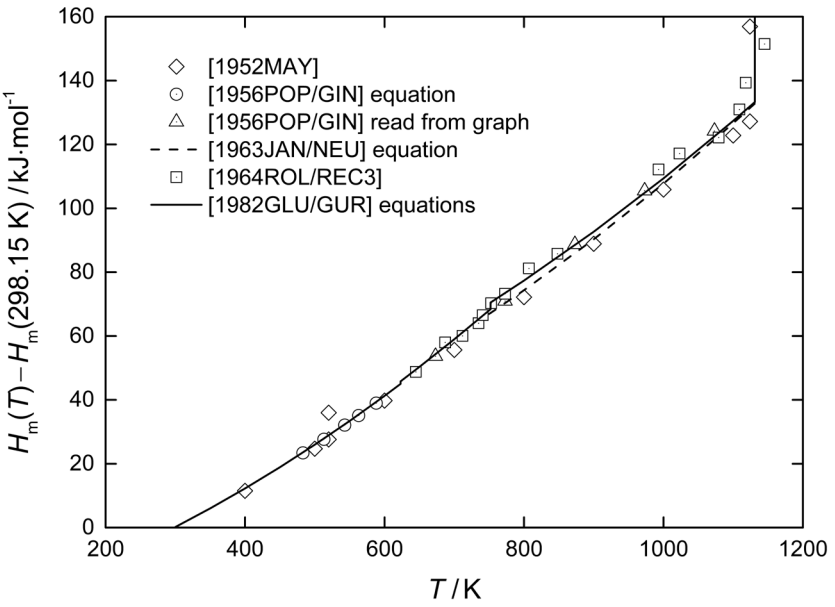
$$C^{\circ}_{p,\mathrm{m}}(Na_2CO_3, \mathrm{cr}, \beta, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{64.118} + \mathbf{0.161604}\,(T/\mathrm{K}) \qquad (628\text{-}754\ \mathrm{K})$$

Transition at (754 ± 5) K; $\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ = 2.1 kJ·mol^{-1},

$$C^{\circ}_{p,\mathrm{m}}(Na_2CO_3, \mathrm{cr}, \gamma, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{36.836} + \mathbf{0.136800}\,(T/\mathrm{K}) \qquad (754\text{-}1129\ \mathrm{K})$$

Melting at (1129 ± 2) K; $\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ = 28.0 kJ·mol^{-1}.

Figure V-83: High-temperature enthalpy of Na_2CO_3(cr).



V.7.1.3 Enthalpy of formation

The most accurate measurements by Berg and Vanderzee [1978BER/VAN] of the dissolution at 298.15 K of Na_2CO_3(cr) and CO_2(g) in excess NaOH solution are the basis for the selection by this review of the standard enthalpy of formation of Na_2CO_3(cr), in agreement with the assessment of Glushko *et al.* [1982GLU/GUR].

Using the extrapolation procedure to infinite dilution, given in detail in [1978BER/VAN2], Berg and Vanderzee [1978BER/VAN] report

$$Na_2CO_3(cr) \rightleftharpoons 2Na^+ + CO_3^{2-} \qquad \text{(V.163)}$$

$$\Delta_r H_m^\circ \text{ ((V.163), 298.15 K)} = -(26.656 \pm 0.126)\ \text{kJ·mol}^{-1}.$$

Berg and Vanderzee [1978BER/VAN] also indicate that their extrapolation procedure is "somewhat" similar to that used earlier by Rupert *et al.* [1965RUP/HOP] (see relevant Appendix A entry) for concentrations ranging between 0.007 to 0.075 mol·kg^{-1}. This extrapolation procedure used by Berg and Vanderzee is compatible with extrapolations used in this review.

Rupert *et al.* [1965RUP/HOP] reported:

$$\Delta_r H_m^\circ \text{ ((V.163), 298.15 K)} = -(26.61 \pm 0.33)\ \text{kJ·mol}^{-1}.$$

The other studies on the enthalpy of solution of Na_2CO_3(cr) are detailed in Table V-79.

Table V-79: Summary of literature values used by [1982GLU/GUR] for the enthalpy of solution of Na_2CO_3(cr).

Reference	Final Molal Concentration	T /K	$\Delta_{sol}H_m$ /kJ·mol^{-1}	Extrapolated $\Delta_{sol}H^o_m$ (298.15 K) /kJ·mol^{-1} [1982GLU/GUR]
[1928MAT/NAK]	0.055	298.15	– (23.458 ± 0.092)	– 26.35
[1932MAT/KAM]	0.055	298.15	– (23.727 ± 0.034)	
[1931SWA/ALT]	0.05	303.15	– (27.154 ± 0.081) (a)	– 27.10 (b)
[1948TOR]	2.6×10^{-4}	298	– (22.150 ± 0.063)	– 26.36
[1968WAT/LIN]	7.5×10^{-3}	298.15	– (22.681 ± 0.084)	– 26.26

(a) Uncertainty estimated by this review;
(b) This review does not understand this extrapolated value.

All of these measurements involved final solutions of molality near or below about 0.1 and are consistent with the values obtained by Berg and Vanderzee [1978BER/VAN] and Rupert *et al.* [1965RUP/HOP].

In their assessment, Glushko *et al.* [1982GLU/GUR] also referenced early dissolution measurements leading to higher final concentrations and to various reactions involving Na_2CO_3 in solution and in the solid state. Since all these yield far less precise values of the enthalpy of formation of this compound, they are not considered further here.

Noting that the early results listed in Table V-79 are somewhat scattered, and in the absence of details on the extrapolation procedure to infinite dilution used by Glushko *et al.* [1982GLU/GUR], this review will rely only on the results of Berg and Vanderzee [1978BER/VAN] for Reaction (V.163). Using the CODATA values $\Delta_f H^o_m(Na^+, 298.15\ K) = -(240.340 \pm 0.060)$ kJ·mol^{-1} and $\Delta_f H^o_m(CO_3^{2-}, 298.15\ K) = -(675.23 \pm 0.25)$ kJ·mol^{-1} this review obtains and selects the value:

$$\Delta_f H^o_m(Na_2CO_3, cr, 298.15\ K) = \mathbf{-(1129.25 \pm 0.30)}\ \text{kJ·mol}^{-1}.$$

V.7.2 Sodium hydrogen carbonate, $NaHCO_3$(cr)

V.7.2.1 Structure and polymorphism

At 298.15 K $NaHCO_3$ has a monoclinic structure, space group $P2_1/c$, with parameters $a = (7.475 \pm 0.010)$ Å, $b = (9.686 \pm 0.010)$ Å, $c = (3.481 \pm 0.010)$ Å, $\beta = 111.85$ ° [1962SAS/SCH], confirmed by Sharma [1965SHA]. These agree within the uncertainties with the earlier data of Zachariasen [1933ZAC], except for the angle β, for which he gave 112.42 °.

V.7.2.2 Heat capacity and entropy

The only measurements of $C_p(NaHCO_3)$ are by Anderson [1933AND] from *ca.* 56 to 290 K and by Kobayashi [1952KOB] from 300.3 to 385.5 K. The latter forms a consistent continuation of the low-temperature data, but the C_p values above 360 K rise very steeply, presumably due to the onset of decomposition. We have accepted the standard entropy derived in [1933AND] and the linear $C^{\circ}_{p,m}$ equation given by Kobayashi [1952KOB]:

$$C^{\circ}_{p,m}(NaHCO_3, cr, T)/J\cdot K^{-1}\cdot mol^{-1} = \mathbf{45.898} + \mathbf{0.14108}(T/K) \qquad (298.15\text{-}360\,K)\ .$$

The standard values selected at 298.15 K are:

$$C^{\circ}_{p,m}(NaHCO_3, cr, 298.15\ K) = \mathbf{(87.96 \pm 0.50)}\ J\cdot K^{-1}\cdot mol^{-1}$$

and

$$S^{\circ}_{m}(NaHCO_3, cr, 298.15\ K) = \mathbf{(102.0 \pm 1.0)}\ J\cdot K^{-1}\cdot mol^{-1}$$

where the uncertainties are estimated by this review.

As announced in a brief note added in proof, Berg and Vanderzee [1978BER/VAN] (see the relevant Appendix A entry) suggested the value $S^{\circ}_{m}(NaHCO_3, cr, 298.15\ K) = (102.36 \pm 0.80)\ J\cdot K^{-1}\cdot mol^{-1}$, insignificantly different from the value selected above. This result was given by Vanderzee [1982VAN] in the frame of a study that concerned dehydration and decomposition studies of various sodium carbonate hydrates and related salts.

V.7.2.3 Enthalpy of formation

The most accurate measurements by Berg and Vanderzee [1978BER/VAN] of the dissolution of $NaHCO_3$(cr) in excess NaOH solution are the basis for the selection of the standard enthalpy of formation of $NaHCO_3$(cr) by this review. The studies by Berthelot and Thomsen dating back to the nineteenth century have too many uncertainties, such as the purity and lack of control of the CO_2 content of their apparatus, to be considered when there are more reliable modern data.

Rupert and Hopkins [1965RUP/HOP] reported without any experimental detail an equation giving the enthalpies of solution of $NaHCO_3$(cr) as a function of the square root of the molality, to obtain, at infinite dilution, a value of $(18.71 \pm 0.21)\ kJ\cdot mol^{-1}$. Under these circumstances, this value will not be discussed further, although we note its excellent agreement with the value obtained by Berg and Vanderzee [1978BER/VAN] and given below (Reaction (V.164)).

We have only been able to identify two other relatively modern calorimetric studies relevant to the enthalpy of solution of $NaHCO_3$(cr).

Perreu [1940PER] reported the dissolution of the salt in water from saturation down to *ca.* 0.22 $mol\cdot kg^{-1}$ at *ca.* 291 K and, probably from a graphical extrapolation,

gave a value of 19.16 kJ·mol^{-1} at infinite dilution. Although this value is in reasonable agreement with that obtained by Berg and Vanderzee [1978BER/VAN] for Reaction (V.164) below, we mention it here for information only, in view of the lack of experimental details in [1940PER].

The other relevant study is that of Torgeson [1948TOR] in his determination of the enthalpy of formation of trona, $NaHCO_3 \cdot Na_2CO_3 \cdot 2H_2O$. However, the only relevant data he gave was for the enthalpy of solution of $NaHCO_3$ into a solution of Na_2CO_3, so no value for $\Delta_f H^\circ_m(NaHCO_3)$ can be derived from this study.

Using the extrapolation procedure to infinite dilution, given in detail in [1978BER/VAN], Berg and Vanderzee [1978BER/VAN2] reported

$$NaHCO_3(cr) \rightleftharpoons Na^+ + HCO_3^- \qquad (V.164)$$

$$\Delta_r H^\circ_m((V.164), 298.15\ K) = (18.719 \pm 0.126)\ kJ \cdot mol^{-1}.$$

With the 1973 CODATA values for the standard enthalpies of formation of Na^+, OH^- and $CO_2(g)$, they reported $\Delta_f H^\circ_m(NaHCO_3, cr, 298.15\ K) = -(948.952 \pm 0.209)\ kJ \cdot mol^{-1}$.

With the CODATA enthalpies of formation accepted in this review, we re-calculate this to give:

$$\Delta_f H^\circ_m(NaHCO_3, cr, 298.15\ K) = -(948.989 \pm 0.244)\ kJ \cdot mol^{-1}.$$

Our selected value is rounded from this:

$$\Delta_f H^\circ_m(NaHCO_3, cr, 298.15\ K) = \mathbf{-(949.0 \pm 0.2)}\ kJ \cdot mol^{-1}.$$

This value is essentially the same as that selected by Glushko *et al.* [1981GLU/MED], $-(949.08 \pm 0.25)\ kJ \cdot mol^{-1}$ and close that selected by NBS [1982WAG/EVA], $-950.8\ kJ \cdot mol^{-1}$, and in agreement with the enthalpy of decomposition of $NaHCO_3$ at 473 K measured by Thomasson and Cunningham [1964THO/CUN], which after correction to 298.15 K, gives $\Delta_f H^\circ_m(NaHCO_3, cr, 298.15\ K) = -(949.07 \pm 0.69)\ kJ \cdot mol^{-1}$—see the Appendix A entry for this study. We also note the measurements of the decomposition pressure by Lescoeur [1892LES] and Caven and Sands [1911CAV/SAN] according to Reaction (V.165):

$$NaHCO_3(cr) \rightleftharpoons 1/2Na_2CO_3(cr) + 1/2H_2O(g) + 1/2CO_2(g) \qquad (V.165)$$

However, their measured pressures are noticeably [1892LES] or considerably [1914CAV/SAN] larger than the calculated values, and as detailed in the Appendix A entries for these studies, a third-law treatment of their decomposition pressures using our selected entropies leads to values of $\Delta_r H^\circ_m((V.165), 298.15\ K) = (65.35 \pm 0.51)\ kJ \cdot mol^{-1}$ [1892LES] or $(62.321 \pm 0.245)\ kJ \cdot mol^{-1}$ [1914CAV/SAN] compared to $(66.742 \pm 0.248)\ kJ \cdot mol^{-1}$ from our selected values. The reason for these discrepancies is not clear, see the Appendix A entries.

V.7.3 Potassium carbonate, K_2CO_3(cr)

V.7.3.1 Structure and polymorphism

As for Na_2CO_3(cr), some early studies, *e.g.*, [1940MAK/SHU] and [1973PAP], suggested there were two polymorphic transitions in K_2CO_3(cr), but later work on high-purity materials in an atmosphere of CO_2(g) indicates clearly that there is only a single transition above 298.15 K, at (693 ± 5) K, followed by melting at (1173 ± 2) K, [1958REI], [1959REI], [1973SCH/LEV].

The polymorph stable at 298.15 K has a monoclinic structure, space group $P2_1/c$, with parameters a = (5.64 ± 0.01) Å, b = (9.82 ± 0.02) Å, c = (6.88 ± 0.01) Å, β = (98.8 ± 0.01) ° [1973GAT/LLO], [1998IDE/RIC]. At (693 ± 5) K, this transforms to a hexagonal phase, stable up to the melting point at (1173 ± 2) K. The parameters of this hexagonal phase are a = 5.70 Å, c = 7.17 Å at 723 K [1973SCH/LEV].

V.7.3.2 Heat capacity and entropy

The only measurement of the low-temperature $C_p(K_2CO_3)$ are by Stull *et al.* [1970STU/HIL] from *ca.* 15 K to *ca.* 310 K; only the values at round temperatures from 16 to 298.15 K were tabulated in this paper. The standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(K_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(114.43 \pm 1.20)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

and

$$S^{\circ}_{\mathrm{m}}(K_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = \mathbf{(155.50 \pm 1.60)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

and these are our selected values.

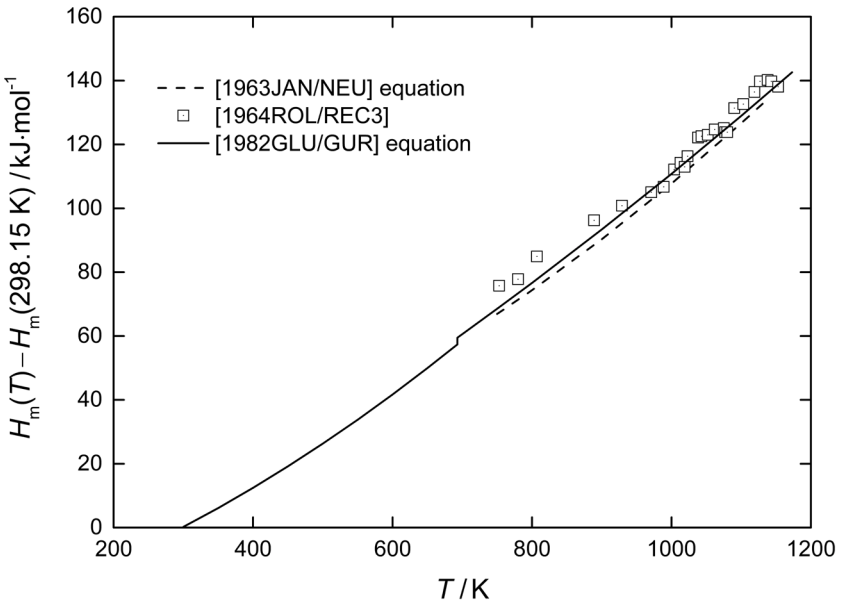
The data for the high-temperature heat capacity are also ill-defined; the heat capacities given (in the form of an extended abstract) by Ginzburg *et al.* [1971GIN/GUB] (298.15 to 1273 K) obviously suffer from problems of interference due to the transitions, but are broadly in agreement with the consistent enthalpy measurements of Janz *et al.* [1963JAN/NEU] (630 to 1250 K) and Rolin and Recapet [1964ROL/REC3] (753 to 1306 K). The assessed values of Glushko *et al.* [1982GLU/GUR] give a good representation of these rather scattered data (Figure V-84) and since these authors presumably had access to the detailed data in [1971GIN/GUB], which we do not have, we have accepted the data assessed in [1982GLU/GUR]. The selected values are thus:

$$C^{\circ}_{p,\mathrm{m}}(K_2CO_3, \mathrm{cr}, \alpha, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{67.798} + \mathbf{0.156403}\,(T/\mathrm{K}) \qquad (298.15\text{-}693\ \mathrm{K}),$$

$$C^{\circ}_{p,\mathrm{m}}(K_2CO_3, \mathrm{cr}, \beta, T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{108.922} + \mathbf{0.068852}\,(T/\mathrm{K}) \qquad (693\text{-}1173\ \mathrm{K}).$$

Transition at (693 ± 5) K; $\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ = 2.1 kJ·mol^{-1}. Melting at (1173 ± 2) K; $\Delta_{\mathrm{trs}}H_{\mathrm{m}}$ = 30.0 kJ·mol^{-1}.

Figure V-84: High-temperature enthalpy of K_2CO_3(cr).



V.7.3.3 Enthalpy of formation

The basis for the enthalpy of formation adopted in this review is the enthalpy of solution of K_2CO_3(cr) in water at 298.15 K by Benjamin [1962BEN] to reach final concentrations between 0.0042 and 0.1630 mol·kg^{-1} (11 determinations). Using the method suggested by Rupert *et al.* [1965RUP/HOP] the data were corrected for hydrolysis and extrapolated to infinite dilution in the assessment of Glushko *et al*. [1982GLU/GUR] who, for Reaction (V.166):

$$K_2CO_3(cr) \rightleftharpoons 2K^+ + CO_3^{2-} \qquad \text{(V.166)}$$

give $\Delta_r H_m^o$ ((V.166), 298.15 K) = – (28.0 ± 2.0) kJ·mol^{-1}.

The older measurements of the enthalpy of solution of K_2CO_3(cr) in water at higher final concentrations [1875BER5], [1882THO], [1883BER/ILO] and also the measurements of the enthalpies of a series of reactions involving solutions of K_2CO_3 [1873BER], [1888MUL] lead to less precise values, since they may have suffered, as noted by Benjamin [1962BEN] and by JANAF [1985CHA/DAV], from the incomplete elimination of CO_2(g) from the medium. Indeed, Benjamin [1962BEN] (see the relevant Appendix A) gives experimental evidence on the effect of the presence of CO_2(g). For these reasons, those early data will not be discussed further in this review.

Similarly, high-temperature equilibrium reactions involving K_2CO_3(cr) do not lead to reliable values of the enthalpy of formation of K_2CO_3(cr) owing to the complicated nature of the reaction products and the lack of data for the activity coefficients of the components in the solid solutions formed.

Thus, from Reaction (V.166) above, this review selects:

$$\Delta_f H^o_m(K_2CO_3, \text{cr}, 298.15\ \text{K}) = \mathbf{-\ (1151.5 \pm 2.0)}\ \text{kJ·mol}^{-1}.$$

in agreement with the selection of Glushko *et al.* [1982GLU/GUR].

The selected thermodynamic data for K_2CO_3(cr) are also in excellent agreement with the NIST-JANAF selections [1998CHA].

V.7.4 Potassium hydrogen carbonate, $KHCO_3$(cr)

V.7.4.1 Structure and polymorphism

At 298.15 K $KHCO_3$ has a monoclinic structure, space group $P2_1/a$, with parameters $a = (15.17 \pm 0.01)$ Å, $b = (5.63 \pm 0.04)$ Å, $c = (3.71 \pm 0.03)$ Å, $\beta = (104.63 \pm 0.03)$ °, at room temperature [1954NIT/TOM], [1974THO/TEL], but at 318 K it transforms to a closely related monoclinic structure, space group $C2/m$, with parameters $a = (15.194 \pm 0.002)$ Å, $b = (5.640 \pm 0.001)$ Å, $c = (3.734 \pm 0.001)$ Å, $\beta = (104.86 \pm 0.01)$ ° at 353 K, [1986HAU], [1990KAS/YAM]. The DSC measurements of [2001LEE/KIM] indicated another transition at 155 K, and decomposition starting around 373 K.

V.7.4.2 Heat capacity and entropy

We have not been able to find any measurements of the heat capacity of $KHCO_3$(cr), above or below 298.15 K. Kelley and Anderson [1935KEL/AND] suggest an estimated value of $S^o_m(298.15\ \text{K}) = 108.8\ \text{J·K}^{-1}\text{·mol}^{-1}$, while the NBS tables [1982WAG/EVA] give a value of $S^o_m(298.15\ \text{K}) = 115.5\ \text{J·K}^{-1}\text{·mol}^{-1}$, whose source is not clear, but is possibly from an estimated difference between the standard entropies of MX(cr), M = Na and K, which is *ca.* $(13 \pm 7)\ \text{J·K}^{-1}\text{·mol}^{-1}$. No values for $S^o_m(298.15\ \text{K})$ or $C_{p,m}$ are selected for this review, but a value of $S^o_m(298.15\ \text{K}) = (115 \pm 7)\ \text{J·K}^{-1}\text{·mol}^{-1}$ can be used if one is required.

V.7.4.3 Enthalpy of formation

Recent literature does not reveal any modern experimental results on the enthalpy of formation of $KHCO_3$(cr). The successive assessments by the NBS [1952ROS/WAG] and [1982WAG/EVA] giving respectively $\Delta_f H^o_m(KHCO_3, \text{cr}, 298.15\ \text{K}) = -959.4\ \text{kJ·mol}^{-1}$ and $\Delta_f H^o_m(KHCO_3, \text{cr}, 298.15\ \text{K}) = -963.2\ \text{kJ·mol}^{-1}$, and that of

Glushko *et al.* [1981GLU/MED], yielding $\Delta_f H^o_m(KHCO_3, cr, 298.15\ K) = -961.9\ kJ{\cdot}mol^{-1}$, are all based on literature data that are more than hundred years old, [1873BER], [1875BER5], [1888MUL], [1909FOR] and [1914CAV/SAN].

Using the values for $\Delta_f H^o_m(K^+, aq, 298.15\ K)$ and $\Delta_f H^o_m(HCO_3^-, aq, 298.15\ K)$ found in the relevant assessments, we obtain, for the dissolution reaction,

$$KHCO_3(cr) \rightleftharpoons K^+ + HCO_3^- \qquad (V.167)$$

$\Delta_{sol} H^o_m((V.167), 298.15\ K)$, the values of 17.08, 18.83 and 18.367 $kJ{\cdot}mol^{-1}$, from, respectively, [1952ROS/WAG], [1982WAG/EVA] and [1981GLU/MED], indicating, at least, reasonable consistency in the analysis of the early experimental data.

We do not feel in a position to improve the earlier assessments of these, in part because of the very limited amount of available experimental information in the original publications that could lead to a proper extrapolation of the data to standard conditions. Taking, for Reaction (V.167), the average value, 18.60 $kJ{\cdot}mol^{-1}$, from the concordant analyses in [1982WAG/EVA] and [1981GLU/MED] and using the CODATA [1989COX/WAG] values, $\Delta_f H^o_m(K^+, aq, 298.15\ K) = -(252.14 \pm 0.08)\ kJ{\cdot}mol^{-1}$ and $\Delta_f H^o_m(HCO_3^-, aq, 298.15\ K) = -(689.93 \pm 0.20)\ kJ{\cdot}mol^{-1}$, we obtain $\Delta_f H^o_m(KHCO_3, cr, 298.15\ K) = -(960.67 \pm 0.22)\ kJ{\cdot}mol^{-1}$, disregarding for the moment the uncertainty in the value on the standard enthalpy of solution of $KHCO_3$(cr). However, in view of the uncertainties in the procedures used to extrapolate the early results to infinite dilution, and also those resulting from the experimental limitations of the very old studies, this review suggests, rather arbitrarily, a much larger uncertainty limit and selects the value:

$$\Delta_f H^o_m(KHCO_3, cr, 298.15\ K) = \mathbf{-(960.7 \pm 2.0)}\ kJ{\cdot}mol^{-1}.$$

As for $NaHCO_3$, both Lescoeur [1892LES], and Caven and Sands [1914CAV/SAN] have measured the decomposition pressures of $KHCO_3$(cr) according to the Reaction (V.168):

$$KHCO_3(cr) \rightleftharpoons 1/2K_2CO_3(cr) + 1/2H_2O(g) + 1/2CO_2(g) \qquad (V.168)$$

Their data are compared in Figure A-2 in the Appendix A entry for [1914CAV/SAN]. As noted there, both of these studies lead to unreasonably large values for $S^o_m(KHCO_3, cr, 298.15\ K)$, and their data have not been considered further for this review.

Chapter V.8

Minerals

V.8.1 Magnesium minerals

We have examined the data for two hydrated magnesium silicates found as minerals: chrysotile and sepiolite.

V.8.1.1 Chrysotile, $Mg_3Si_2O_5(OH)_4 = 3MgO{\cdot}2SiO_2{\cdot}2H_2O$

Chrysotile, together with antigorite and lizardite, form the three principal minerals of the serpentine group. As noted by Rinaudo *et al*. [2003RIN/GAS] they have essentially the same chemical composition, $Mg_3Si_2O_5(OH)_4$, but significantly different structures, related to variations in the detailed arrangement of the sheets of SiO_4 tetrahedra and $MgO_2(OH)_4$ octahedra. We shall only discuss the data for chrysotile in this review.

The most relevant determination of the structure is probably that by Falini *et al*. [2004FAL/FOR], who prepared nanotubes of the very pure material by the interaction of silica with an aqueous solution of $MgCl_2$ with a Si/Mg mole ratio of 0.68, at a pH of 13.0 (by the addition of OH solution), followed by hydrothermal treatment at 8.2 MPa for 24 hours. This synthetic chrysotile had a monoclinic structure, *Cc* space group, with a = 5.340 Å, b = 9.241 Å, and c = 14.689 Å, β = 93.66 °.

V.8.1.1.1 Heat capacity and entropy

King *et al*. [1967KIN/BAR] measured the heat capacities of both chrysotile and its structurally-related polymorph antigorite from *ca.* 53 to 297 K (*ca.* 30 measurements for each), with very similar results. However, the (natural) sample of chrysotile used was rather impure (see Appendix A for details) and for the heat capacity measurements was taken to contain 94 % chrysotile and 6 % impurities, including 1.7 % excess water (ice) for which corrections were made. The heat capacities were extrapolated to 0 K using the sum of one Debye and three Einstein functions. The standard entropy of chrysotile at 298.15 K was calculated to be:

$$S^{\circ}_{m}(Mg_3SiO_2O_5(OH)_4, \text{cr}, 298.15\ \text{K}) = \mathbf{(221.3 \pm 1.5)}\ \text{J}{\cdot}\text{K}^{-1}{\cdot}\text{mol}^{-1},$$

which is our selected value.

The value of $C^{\circ}_{p,m}$ (298.15 K) was not given by King *et al*. but extrapolation of

the data just below 298.15 K indicates a value of:

$$C^{\circ}_{p,\mathrm{m}}(Mg_3SiO_2O_5(OH)_4, \text{cr}, 298.15\ \text{K}) = \mathbf{(273.7 \pm 1.5)}\ \text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1},$$

which is again our selected value.

King *et al*. did not measure the heat capacity of chrysotile above 298.15 K because neither of their samples was dense enough. However, they did measure that of the polymorph antigorite by drop calorimetry from 405.5 to 847.9 K, and fitted their data by an equation corresponding to:

$$C^{\circ}_{p,\mathrm{m}}(\text{antigorite}, T)/\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} = 317.23 + 0.1323\,(T/\text{K}) - 7.3555\times10^{6}\,(T/\text{K})^{-2},$$

and since the heat capacities of chrysotile and antigorite below 298.15 K differ by less than 1.2 %, this expression, slightly modified to reproduce the value given above for $C^{\circ}_{p,\mathrm{m}}(Mg_3SiO_2O_5(OH)_4, \text{cr}, 298.15\ \text{K})$ can be adopted for chrysotile without undue error. Thus,

$$C^{\circ}_{p,\mathrm{m}}(Mg_3SiO_2O_5(OH)_4, T)/\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} = \mathbf{317.00} + \mathbf{0.1323}\,(T/\text{K}) - \mathbf{7.3555\times10^{6}}\,(T/\text{K})^{-2}$$
(298.15 to 800 K).

Kumzerov and colleagues measured the heat capacity of chrysotile samples from 6 to 80 K [2005KUM/PAR] and from 80 to 300 K [2011KUM/KAR], but as noted in the relevant short Appendix A entries, in both papers the results are only presented in small graphs from which no precise data could be derived.

V.8.1.1.2 Enthalpy of formation

Hydrofluoric acid solution calorimetry and high-temperature melt solution calorimetry provided direct experimental information on the enthalpy of formation of this compound through the use, essentially, of non-stoichiometric compounds.

In the frame of a broader study on mineral reactions (see the relevant Appendix A entries), Bennington [1956BEN] used two natural fibrous chrysotile samples containing, as major impurities (in wt %), 0.66 FeO, 0.64 Al_2O_3 and 0.07 MnO for the Gila County (AZ) sample, and 0.27 Al_2O_3, 1.68 Fe_2O_3, 0.36 FeO, 0.06 MnO and 0.08 NiO for the Thedford (Ontario) sample. Only optical (no diffraction) data were given. The solution medium (sln) was 20.1 % HF at 354.15 K. Corrected for impurities, the values for the dissolution of the two samples were in good agreement. For Reaction (V.169):

$$Mg_3Si_2O_5(OH)_4(\text{cr}, 298.15\,\text{K}) + 18\text{HF}(\text{sln}, 354.15\,\text{K}) \rightleftharpoons$$
$$3MgF_2(\text{s}, 354.15\,\text{K}) + 9H_2O(\text{sln}, 354.15\,\text{K}) + 2H_2SiF_6(\text{sln}, 354.15\,\text{K}) \quad \text{(V.169)}$$

Bennington obtained as a best value, $\Delta_r H^{\circ}_{\mathrm{m}}(\text{V.169}) = -(550.606 \pm 1.264)$ kJ·mol^{-1} at 354.15 K. His thermodynamic cycle involved MgO(cr), $Mg(OH)_2$(cr),

SiO_2(α-quar) and H_2O(l), and values for the relevant reactions were taken from the nearly simultaneous study of Weeks [1956WEE2].

The enthalpy of formation of chrysotile from its constituent oxides according to Reaction (V.170):

$$3MgO(cr, 298.15\ K) + 2SiO_2(\alpha\text{-quar}, 298.15\ K) + 2H_2O(l, 298.15\ K) \rightleftharpoons Mg_3Si_2O_5(OH)_4(cr, 298.15\ K) \quad (V.170)$$

was obtained by the author as $\Delta_r H^\circ_m$ (V.170) = – (165.410 ± 1.464) kJ·mol^{-1}, for which this review prefers a larger uncertainty, ± 2.510 kJ·mol^{-1} by analogy with the better documented and more detailed subsequent study of King *et al*. [1967KIN/BAR] who used a similar cycle (see below). From Reaction (V.170), using CODATA values, this review calculates $\Delta_f H^\circ_m(Mg_3Si_2O_5(OH)_4, cr, 298.15\ K)$ = – (4363.27 ± 3.33) kJ·mol^{-1} from this work.

King *et al*. [1967KIN/BAR] used a mixture of 10.5 wt % HF and 10.5 wt % HCl at 346.85 K as solution medium for the dissolution of a natural sample (identified as sample 2) that contained, as major impurities, 0.23 wt % Al_2O_3, 0.40 wt % Fe_2O_3, 0.06 wt % FeO, 0.25 wt % CaO and 0.24 wt % CO_2 (presumably as a carbonate). Details of the analyses of the compounds involved in this study were also given.

Their thermodynamic cycle, involving the dissolution of MgO(cr), SiO_2(α-quar) and H_2O(l), is noted in detail in the relevant Appendix A entry. Combination of these reactions led also to Reaction (V.170) above, with $\Delta_r H^\circ_m$ ((V.170), 298.15 K) = – (166.774 ± 2.510) kJ·mol^{-1}; the reported uncertainty, which this review accepts, contains allowances for uncertainties in the solution measurements, the calibrations and the corrections for impurities. Using CODATA values [1989COX/WAG], this review calculates for Reaction (V.170) from this extremely careful work:

$$\Delta_f H^\circ_m(Mg_3Si_2O_5(OH)_4, cr, 298.15\ K) = -(4364.0 \pm 3.400)\ kJ\cdot mol^{-1}.$$

Ogorodova *et al*. [2006OGO/KIS] made use of high-temperature melt ($2PbO\cdot B_2O_3$ at 973 K) solution calorimetry to obtain the enthalpy of formation of a natural sample of chrysotile-asbestos (Eastern Sayan Mountains, Russia) analysed as $(Mg_{2.96}Ca_{0.04})(Al_{0.05}Si_{1.94}Fe^{3+}_{0.02})O_{5.03}(OH)_{3.91}$ and of a synthetic sample analysed as $Mg_{2.96}Si_{2.02}O_5(OH)_4\cdot 0.66H_2O$. Further details concerning the samples and the calorimetric procedure are described in the relevant Appendix A entry. The enthalpies of formation at 298.15 K of the chrysotile samples from the constituent oxides were obtained by the authors from the heat contents and enthalpies of solution of the constituent oxides taken from earlier data, either from their group (some of them estimated) or from others. These references are listed in Appendix A. The authors obtained:

$$\Delta_f H_m^\circ((Mg_{2.96}Ca_{0.04})(Al_{0.05}Si_{1.94}Fe^{3+}_{0.02})O_{5.03}(OH)_{3.91}, \text{cr}, 298.15\,\text{K}) = -(4338 \pm 17)\ \text{kJ·mol}^{-1}$$

and

$$\Delta_f H_m^\circ(Mg_{2.96}Si_{2.02}O_5(OH)_4 \cdot 0.66H_2O, \text{cr}, 298.15\,\text{K}) = -(4473.7 \pm 9.3)\ \text{kJ·mol}^{-1}.$$

In a further publication [2007OGO/KIS], concerning the system $(Mg, Ni)_3Si_2O_5(OH)_4$(cr), that is beyond the scope of the present review, Ogorodova *et al*. list $\Delta_f H_m^\circ(Mg_{2.96}Si_{2.02}O_5(OH)_4 \cdot 0.66H_2O, \text{cr}, 298.15\ \text{K}) = -(4479.3 \pm 9.9)\ \text{kJ·mol}^{-1}$, using the same experimental techniques as those in [2006OGO/KIS], but the values are only in marginal agreement. The authors do not give any additional details.

Using a procedure that is not entirely clear to this review, Ogorodova *et al*. [2007OGO/KIS] reported for the stoichiometric composition $\Delta_f H_m^\circ(Mg_3Si_2O_5(OH)_4, \text{cr}, 298.15\ \text{K}) = -(4353 \pm 17)\ \text{kJ·mol}^{-1}$ from the data on the natural sample and $\Delta_f H_m^\circ(Mg_3Si_2O_5(OH)_4, \text{cr}, 298.15\ \text{K}) = -(4291 \pm 26)\ \text{kJ·mol}^{-1}$ from the data on the synthetic sample. The higher stability of the natural sample, attested also in the DTA and TGA measurements by a 100 K higher temperature of decomposition, was ascribed by the authors to structural differences. Although their value for the natural sample is not incompatible with the results of Bennington [1956BEN] and King *et al*. [1967KIN/BAR], we prefer to give the results of Ogorodova *et al*. [2006OGO/KIS] for information only.

Hostetler and Christ [1968HOS/CHR] measured the solubility of chrysotile in water at 363 K, but the composition of the equilibrium solution contained much more Mg than that corresponding to congruent dissolution. Nevertheless, the authors assumed congruent dissolution and used the compositions to calculate an equilibrium solubility of $\log_{10}K_m\,(363\ \text{K}) = -(49.2 \pm 0.5)$. As noted in Appendix A, Bricker and Nesbitt [1973BRI/NES] subsequently suggested that chrysotile in fact dissolves incongruently to form talc ($Mg_3Si_4O_{10}(OH)_2$(cr)) and that the solution composition was actually controlled by equilibrium with talc. Indeed, on this basis, they used their estimated $\Delta_f G_m^\circ$(talc) to re-interpret the data in [1968HOS/CHR] to calculate $\Delta_f G_m^\circ(Mg_3SiO_2O_5(OH)_4, 298.15\ \text{K}) = -4025\ \text{kJ·mol}^{-1}$ in tolerable agreement with our selected value of $-(4036.371 \pm 3.431)\ \text{kJ·mol}^{-1}$ (see below), in view of the uncertainties in the calculation in [1973BRI/NES]. Because of these various uncertainties in the derivation of thermodynamic data from their results, the data of Hostetler and Christ [1968HOS/CHR] are not considered further in this review, as noted in Appendix A.

The straight average of the results of Bennington [1956BEN] and King *et al*. [1967KIN/BAR] is $-(4363.95 \pm 3.33)\ \text{kJ·mol}^{-1}$ and our selected value for the enthalpy of formation of chrysotile is rounded to give:

$$\Delta_f H_m^\circ(Mg_3Si_2O_5(OH)_4, \text{cr}, 298.15\ \text{K}) = \mathbf{-(4364.0 \pm 3.4)}\ \text{kJ·mol}^{-1}$$

and thus

$\Delta_f G^o_m$ ($Mg_3Si_2O_5(OH)_4$, cr, 298.15 K) = **– (4036.371 ± 3.431)** kJ·mol^{-1},

and these are our selected values.

An additional source of thermodynamic values for minerals of petrological interest can be found in the broad study of Holland and Powell [2011HOL/POW].

V.8.1.2 Sepiolite, $Mg_4Si_6O_{15}(OH)_2{\cdot}nH_2O = 4MgO{\cdot}6SiO_2{\cdot}(n+1)H_2O$

V.8.1.2.1 Structure and general survey

There are extensive deposits of sepiolite (also known as meerschaum), with a variable water content, and of course containing small quantities of other elements, particularly Al and Fe. The variable water content is a function of the temperature and water content of the atmosphere, and as noted by Post *et al.* [2007POS/BIS], there is general agreement that the zeolitic water is lost at ambient temperature under vacuum or on heating to *ca.* 415 K and the bound water is lost in two stages between 525 and 875 K while the OH groups are retained up to *ca*. 1075 K.

There are many studies of the structure, with lattice parameters varying with the impurity content, but with mean values of a = 13.40 Å, b = 27.02 Å, c = 5.28 Å for the orthorhombic space group *Pncn* and a formula with nominally 8 Mg and about 15 H_2O. Recently, in a publication that deserves special mention and also summarises earlier data, Sánchez del Río *et al.* [2011SAN/GAR] studied 20 samples of sepiolite of known compositions at 298 K (for hydrated ambient material) and three of these at 498 K (for the zeolitically dehydrated material). All showed reasonable agreement with the above crystallographic data. The same year, Giustetto *et al.* [2011GIU/LEV], in a study of a natural material corresponding to the composition $(Mg_{7.33}Al_{0.34}Fe_{0.10})(Si_{11.92}Al_{0.08})K_{0.07}Ca_{0.01}O_{32}$ and containing 15.44H_2O per unit cell, reported for the same orthorhombic space group *Pncn*, a = (13.395 ± 0.002) Å, b = (27.016 ± 0.003) Å and c = (5.2765 ± 0.0006) Å.

V.8.1.2.2 Heat capacity and entropy

We have not been able to find any experimental data on the heat capacity and entropy of sepiolite. In their study of the solubility of sepiolite (see below), Christ and Hostetler [1973CHR/HOS] estimated the standard entropy of a sepiolite with the composition $2MgO{\cdot}3SiO_2{\cdot}3.5H_2O$ ($Mg_2Si_3O_{7.5}(OH){\cdot}3H_2O$) simply from the sum of the standard entropies of the constituent oxides. However, it is not clear whether the contribution of H_2O should be that of ice or liquid water (see comment in Appendix A), or a mixture of these, and Christ and Hostetler estimated two values: 315.5 J·K^{-1}·mol^{-1} if all 3.5 moles of water are considered to behave like ice but 372.0 J·K^{-1}·mol^{-1} if the 2 moles of zeolitic H_2O are regarded as liquid water. Besides, it is also not clear how good an estimate this value is, since a similar calculation for chrysotile gives a value about 9 % greater than the experimental value, even if the lower H_2O contribution is taken. Ogorodova

et al. [2014OGO/KIS] made similar calculations for $Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O$(cr) and $Mg_4Si_6O_{15}(OH)_2$(cr), which we also do not regard as reliable, see Appendix A.

No reliable values of the heat capacity or entropy can be selected for this review.

V.8.1.2.3 Enthalpy of formation

There is only one study of the enthalpy of formation of sepiolites, *viz.* the aforementioned paper by Ogorodova *et al*. [2014OGO/KIS], which gives values of the enthalpies of formation of the two sepiolites, $Mg_4Si_6O_{15}(OH)_2$(cr) and $Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O$(cr), obtained by drop calorimetry into a $2PbO\text{-}B_2O_3$ melt at 973 K.

However, as noted in the extensive Appendix A entry, they give so few details of their actual experimental results and intermediate calculations that we cannot endorse their calculated enthalpies of formation, $\Delta_f H^o_m(Mg_4Si_6O_{15}(OH)_2, cr, 298.15\ K) = -(8213.0 \pm 10.5)\ kJ{\cdot}mol^{-1}$ and $\Delta_f H^o_m(Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O, cr, 298.15\ K) = -(9386.5 \pm 14.0)\ kJ{\cdot}mol^{-1}$, which are given for information only.

As noted above Christ and Hostetler [1973CHR/HOS] have studied the solubility of a well-crystallised sample of sepiolite at three temperatures (324, 343 and 363 K). However, the only analyses of their material were by X-ray diffraction and electron-microscopy, and although the ratio of the Si/Mg concentrations in the solution was close to 1.5, as expected for congruent dissolution, the water content of the mineral was not accurately defined. Nevertheless, assuming the composition of the sample was $Mg_4Si_6O_{15}(OH)_2{\cdot}3.5H_2O$, Christ and Hostetler calculated the following $\log_{10}K_{sp}$ values for their well-crystallised sample: $-(38.8 \pm 0.3)$ at 324 K, $-(37.5 \pm 0.3)$ at 343 K, and $-(37.2 \pm 0.2)$ at 363 K.

However, because of a number of uncertainties in their calculations noted in Appendix A, these data are mentioned here for information only.

Thus, unfortunately no reliable data for sepiolite can be recommended in this review.

V.8.2 Calcium minerals: two hydrated borates (colemanite, inyoite) and OH- and F-apatites

V.8.2.1 Colemanite, $CaB_3O_4(OH)_3{\cdot}H_2O$(cr)

V.8.2.1.1 Structure

The monoclinic structure of colemanite, first determined by Christ [1953CHR], has been refined by Burns and Hawthorne [1993BUR/HAW]. The lattice

parameters at room temperature are $a = (8.712 \pm 0.002)$ Å, $b = (11.247 \pm 0.003)$ Å, $c = (6.091 \pm 0.001)$ Å, $\beta = (110.12 \pm 0.02)°$, space group $P2_1/a$. It undergoes a small crystallographic transition to the less symmetrical $P2_1$ structure from 238 to 282 K, [1960PER/BLO], [1980GUR/SEM].

V.8.2.1.2 Heat capacity and entropy

Gurevich *et al.* [1980GUR/SEM] have measured the low-temperature heat capacity of an essentially stoichiometric sample of colemanite ($Ca_{1.01}B_{2..99}O_4(OH)_3 \cdot 0.99H_2O$) from 12.53 to 313.83 K, and found the values $C^{\circ}_{p,m}$ (298.15 K) = (217.06 ± 0.21) $J \cdot K^{-1} \cdot mol^{-1}$ and S°_{m} (298.15 K) = (191.84 ± 0.20) $J \cdot K^{-1} \cdot mol^{-1}$, and the rounded values of these, with larger uncertainties, are the selected values for the stoichiometric composition:

$$C^{\circ}_{p,m}(CaB_3O_4(OH)_3 \cdot H_2O, \text{cr}, 298.15\ \text{K}) = \mathbf{(217.1 \pm 1.0)}\ J \cdot K^{-1} \cdot mol^{-1},$$

$$S^{\circ}_{m}(CaB_3O_4(OH)_3 \cdot H_2O, \text{cr}, 298.15\ \text{K}) = \mathbf{(191.8 \pm 1.5)}\ J \cdot K^{-1} \cdot mol^{-1}.$$

V.8.2.1.3 Enthalpy of formation

Gurevich *et al.* [1976GUR/SOK] measured the enthalpies of solution of analysed colemanite and inyoite samples (see Section V.8.2.2.3) in 1 $mol \cdot dm^{-3}$ HCl and reported the enthalpies of formation of the compounds with the theoretical stoichiometric composition.

However, as described with more detail in Appendix A, the information furnished by the authors is not sufficient to decide whether the theoretical or actual composition of the samples were used in the calculation of the enthalpies of solution or if and how a correction to theoretical composition was made at some stage. Besides, the reviewers found problems in understanding the statistical approach for the treatment of the data. A re-calculation by the review, as described in Appendix A, leads to:

$$\Delta_f H^{\circ}_{m}(CaB_3O_4(OH)_3 \cdot H_2O, 298.15\ \text{K}) = -(3470.52 \pm 2.29)\ kJ \cdot mol^{-1},$$

assuming that the given enthalpy of solution concerns the stoichiometric compound. The same calculation (using, of course, the only available experimental enthalpy of solution of the compound) assuming the non-stoichiometric composition yields $\Delta_f H^{\circ}_{m}(CaO_{0.97}B_{3.02}O_4(OH)_3 \cdot 0.87H_2O, 298.15\ \text{K}) = -(3421.36 \pm 2.29)\ kJ \cdot mol^{-1}$.

In view of these uncertainties, the review recommends the value:

$$\Delta_f H^{\circ}_{m}(CaB_3O_4(OH)_3 \cdot H_2O, \text{cr}, 298.15\ \text{K}) = \mathbf{-(3470 \pm 20)}\ kJ \cdot mol^{-1}$$

from this, the only experimental source.

V.8.2.2 Inyoite, $CaB_3O_3(OH)_5{\cdot}4H_2O$(cr)

V.8.2.2.1 Structure

The monoclinic structure of inyoite, first determined by Christ [1953CHR2], was refined by Clark [1959CLA]. The lattice parameters at room temperature are a = (10.63 ± 0.02) Å, b = (12.06 ± 0.02) Å, c = (8.405 ± 0.020) Å, β = (114.03 ± 0.08) °, space group $P2_1/a$.

V.8.2.2.2 Heat capacity and entropy

Gurevich *et al.* [1980GUR/SEM] have measured the low-temperature heat capacity of a sample of inyoite from 11.54 to 306.16 K, in two different calorimeters, with reasonable agreement in the range of the overlapping measurements. However, the analysed composition of their material was $Ca_{1.01}B_{2.99}O_3(OH)_5{\cdot}3.46H_2O$, distinctly deficient in water with respect to the ideal stoichiometric composition $CaB_3O_3(OH)_5{\cdot}4H_2O$. For this water-deficient sample, they found the following values:

$$C^{\circ}_{p,\mathrm{m}}\,(Ca_{1.01}B_{2.99}O_3\,(OH)_5{\cdot}3.46H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = (338.6 \pm 0.9)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(Ca_{1.01}B_{2.99}O_3\,(OH)_5{\cdot}3.46H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = (332.3 \pm 0.8)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

We can only estimate values for the heat capacity and entropy of the stoichiometric mineral based on these results. For a number of well-characterised hydrated solids (carbonates, chlorides, nitrates and sulfates of Ca, Mg and Na), the mean contributions of one mole of water of hydration to $C^{\circ}_{p,\mathrm{m}}$ and S°_{m} at 298.15 K are (41.3 ± 2.6) and (42.9 ± 2.1) $\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ respectively. We therefore select for stoichiometric inyoite:

$$C^{\circ}_{p,\mathrm{m}}\,(CaB_3O_3\,(OH)_5{\cdot}4H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(361.0 \pm 5.0)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(CaB_3O_3\,(OH)_5{\cdot}4H_2O,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = \mathbf{(355.5 \pm 5.0)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

introducing rather large uncertainty limits to account for other sources of experimental errors.

V.8.2.2.3 Enthalpy of formation

Gurevich *et al.* [1976GUR/SOK] measured the enthalpies of solution of analysed inyoite and colemanite (see Section V.8.2.1.3) samples in 1 $\mathrm{mol{\cdot}dm^{-3}}$ HCl and reported the enthalpies of formation of the compounds with the theoretical stoichiometric composition.

However, as described with more detail in Section V.8.2.1.3 and in Appendix A, the reviewers encountered difficulties in understanding the treatment of the data in [1976GUR/SOK]. In addition, in the case of inyoite, the composition of the actual samples was reported as $Ca_{1.02}B_{2.99}O_3(OH)_5{\cdot}3.94H_2O$. The reviewers note that a

minor correction imposed by electroneutrality requirements leads to either $Ca_{1.02}B_{2.9866}O_3(OH)_5 \cdot 3.94H_2O$ or $Ca_{1.015}B_{2.99}O_3(OH)_5 \cdot 3.94H_2O$.

Calculations similar to those for colemanite (see Section V.8.2.1.3) give:

$$\Delta_f H^o_m (CaB_3O_3(OH)_5 \cdot 4H_2O, \text{cr}, 298.15\ \text{K}) = -(4648.01 \pm 2.31)\ \text{kJ} \cdot \text{mol}^{-1},$$

assuming stoichiometric composition and either:

$$\Delta_f H^o_m (Ca_{1.02}B_{2.9866}O_3(OH)_5 \cdot 3.94H_2O, \text{cr}, 298.15\ \text{K}) = -(4638.81 \pm 2.31)\ \text{kJ} \cdot \text{mol}^{-1}$$

or

$$\Delta_f H^o_m (Ca_{1.015}B_{2.99}O_3(OH)_5 \cdot 3.94H_2O, \text{cr}, 298.15\ \text{K}) = -(4636.86 \pm 2.31)\ \text{kJ} \cdot \text{mol}^{-1}.$$

In view of this situation, the review selects:

$$\Delta_f H^o_m (CaB_3O_3(OH)_5 \cdot 4H_2O, \text{cr}, 298.15\ \text{K}) = \mathbf{-(4648 \pm 12)}\ \text{kJ} \cdot \text{mol}^{-1},$$

from this, the only experimental source.

V.8.2.3 Hydroxylapatite, $Ca_5(PO_4)_3OH$(cr)

V.8.2.3.1 Structure

We shall use the simpler formula $Ca_5(PO_4)_3OH$, rather than the doubled formula used in many publications.

Hydroxylapatite was originally thought to have a hexagonal crystal structure, space group $P6_3/m$, with $a = 9.417$ Å, $c = 6.875$ Å at ambient temperatures but there is now convincing evidence that the structure at 298.15 K is very slightly distorted from hexagonal to the monoclinic space group $P2_1/b$ with $a = 9.421$ Å, $b = 2a$, $c = 6.881$ Å, $\gamma = 120$ °. This structure was originally proposed by Elliott *et al.* [1973ELL/MAC] and Van Rees *et al.* [1973REE/MEN] but their studies are not entirely conclusive, since they did not give any chemical analyses for their samples, and small deviations from stoichiometric and/or small amounts of impurities can clearly affect the structure (though probably to the hexagonal form). However, the study by Suda *et al.* [1995SUD/YAS] using analysed samples, shows convincingly that hydroxylapatite does indeed have a slightly monoclinic distortion at room temperature, but transforms reversibly to the hexagonal structure at (480 ± 5) K, with the very small enthalpy of transition of (315.0 ± 12.5) $J \cdot mol^{-1}$. The transition temperature is in good agreement with those found by Van Rees *et al.* [1973REE/MEN], 484 K (from the disappearance of birefringence in hydroxylapatite crystals mounted on a hot-stage microscope) and (474.5 ± 5.0) K by Sharpataya *et al.* [1995SHA/FED] from DSC measurements, although the latter authors give a smaller enthalpy of transition of (187.5 ± 7.5) $J \cdot mol^{-1}$. The difference in the two structures is thought to be the statistical disordering of the columns of OH groups in the hexagonal phase. More details of these studies are given in Appendix A.

V.8.2.3.2 Heat capacity and entropy

There are two studies of the low-temperature heat capacity. The more extensive results are those by Egan *et al.* [1951EGA/WAK], who reported measurements from 11 K to room temperature, with a tabulation at round values of T up 298.16 K. The sample used was a high-purity synthetic sample, but no structural data are reported, so it is unclear whether it was the low-temperature monoclinic phase or the high-temperature hexagonal phase.

More recently, Palkin *et al.* [1991PAL/KUZ] reported measurements on a hexagonal form of hydroxylapatite from 58.5 to 305.18 K. The heat capacities of Egan *et al.* are lower than those of Palkin *et al.* by 1.8 to 2.5 % up to 100 K, but increasingly higher by 1.3 to 2.0 % from 150 to 300 K. Sharpataya *et al.* also made measurements at temperatures down to 299.2 K [1995SHA/FED]. In order to maintain a consistent treatment of the data we have accepted the value of $C^{\circ}_{p,\mathrm{m}}$ (298.15 K) derived from the assessment of the data for the monoclinic phase up to 480 K —see below. This is midway between the $C^{\circ}_{p,\mathrm{m}}$ (298.15 K) values in [1951EGA/WAK] (385.1 ± 2.0) J·K^{-1}·mol^{-1} and in [1991PAL/KUZ] (376.9 ± 2.0) J·K^{-1}·mol^{-1}, where both the uncertainties have been increased owing to the uncertainties in the both studies. We have similarly taken the average of the entropies obtained in these two studies (390.4 ± 2.0) and (388.4 ± 2.0) J·K^{-1}·mol^{-1} respectively. This review selects:

$$S^{\circ}_{\mathrm{m}}(\mathrm{Ca_5(PO_4)_3OH}, \text{cr, monoclinic, 298.15 K}) = \mathbf{(389.4 \pm 2.0)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

and

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_5(PO_4)_3OH}, \text{cr, monoclinic, 298.15 K}) = \mathbf{(381.15 \pm 3.00)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

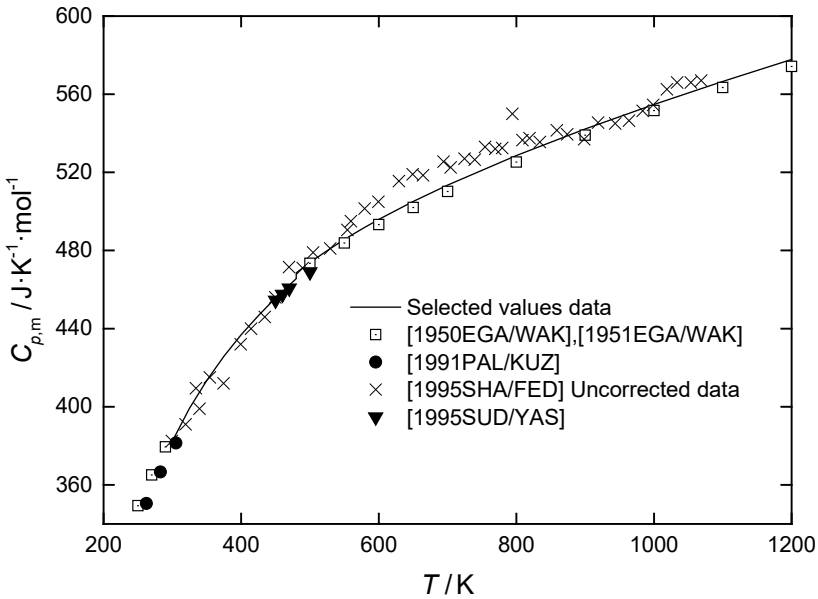
Egan *et al.* [1950EGA/WAK] had slightly earlier also measured the enthalpy content of two samples of hydroxylapatite from 473.4 to 1472.5 K, and more recently Sharpataya *et al.* [1995SHA/FED] have measured the heat capacity from 300 to 1070 K by DSC. In their study of the transition around 480 K Suda *et al.* [1995SUD/YAS] also reported graphically their DSC measurements of the heat capacity from 440 to 510 K. All these data have been fitted into two ranges, from 298.15 to 480 K for the monoclinic phase and from 480 to 1200 K for the hexagonal phase, with the results shown in Figure V-85. This shows that the uncorrected values in [1995SHA/FED] are rather scattered—unfortunately the authors do not report their individual corrected values, which are presumably smoother. The two values of the heat capacity at the transition temperature are, as expected, very similar: 465.60 J·K^{-1}·mol^{-1} for the monoclinic phase and 468.57 J·K^{-1}·mol^{-1} for the hexagonal phase. For the small enthalpy of transition, we have taken a rounded mean of the rather different values given by Sharpataya *et al.* [1995SHA/FED], (0.188 ± 0.008) kJ·mol^{-1} and Suda *et al.* [1995SUD/YAS], (0.315 ± 0.013) kJ·mol^{-1}, with a considerably increased uncertainty to cover the two values.

$$C_{p,\mathrm{m}}^{\circ}(\mathrm{Ca_5(PO_4)_3OH, cr, monoclinic}, T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{410.119} + \mathbf{0.182626}(T/\mathrm{K}) - \mathbf{7.41505\times 10^{6}}(T/\mathrm{K})^{-2} \quad (298.15 \text{ to } 480 \text{ K}),$$

$$\Delta_{\mathrm{trs}}H_{\mathrm{m}}(480\ \mathrm{K}) = (0.250 \pm 0.075)\ \mathrm{kJ\cdot mol^{-1}},$$

$$C_{p,\mathrm{m}}^{\circ}(\mathrm{Ca_5(PO_4)_3OH, cr, hexagonal}, T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = \mathbf{49.0467} + \mathbf{0.124878}(T/\mathrm{K}) - \mathbf{1.04128\times 10^{-5}}(T/\mathrm{K})^{2} - \mathbf{8.75879\times 10^{6}}(T/\mathrm{K})^{-2} \quad (480 \text{ to } 1200 \text{ K}).$$

Figure V-85: Heat capacity of hydroxylapatite $Ca_5(PO_4)_3OH$(cr).



More recently, Cruz *et al.* [2005CRU/CAN] reported *ab initio* studies for the hexagonal form of hydroxylapatite (and also fluorapatite), and the monoclinic form [2006CRU/CAN] of hydroxylapatite—see the short relevant Appendix A of [2005CRU/CAN]. These calculations yielded values for $C_{p,\mathrm{m}}^{\circ}(298.15\ \mathrm{K})$, with substantial uncertainties, but owing to the general uncertainties in such *ab initio* calculations, these values have not been utilised for this Review and this study is mentioned here for information only.

V.8.2.3.3 Enthalpy of formation

The enthalpy of formation of hydroxylapatite rests on the determination of its enthalpy of solution in various nitric and hydrochloric acid media, using varied thermodynamic cycles.

As detailed and discussed in the relevant Appendix A, Gottschal [1958GOT] reported measurements of calorimetric dissolutions of hydroxylapatite in 20 and 30 % HNO_3. The calorimetric results were given for each medium as averages of two single determinations only. The purity of the samples was declared as 98.5 %. Gottschal described the calorimetric Reaction (V.171) as:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + (14HNO_3 + H_2O)(sln) \rightleftharpoons 7Ca(NO_3)_2(sln) + 3Ca(H_2PO_4)_2 \cdot H_2O(cr) \quad (V.171)$$

He also reported the similar reaction scheme for his determination of the enthalpy of solution of fluorapatite. The scheme is thus resting on the enthalpy of formation of $Ca(H_2PO_4)_2 \cdot H_2O$(cr). However, the validity of this process was strongly criticised by Jacques [1963JAC]. In view of this, this review gives, for information only, a CODATA-compatible value of $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6770 \pm 15)\ kJ \cdot mol^{-1}$ from this study, keeping conservative uncertainty limits from these results.

Smirnova *et al.* [1962SMI/ILL] reported the dissolution of hydroxylapatite, fluorapatite and $Ca_3(PO_4)_2$(cr, α and β) in *ca.* 10 % HCl (given as HCl, 18.3 H_2O). Details on the samples and their limited purity, and the calorimetric procedure are given the relevant Appendix A. Three dissolutions of a synthetic hydroxylapatite sample were reported according to Reaction (V.172):

$$Ca_5(PO_4)_3OH(cr) + 10HCl(sln) \rightleftharpoons (5CaCl_2 + 3H_3PO_4 + H_2O)(sln) \quad (V.172)$$

for which this review re-calculated $\Delta_r H^\circ_m(V.172) = -(156.971 \pm 12.350)\ kJ \cdot mol^{-1}$, noting that these uncertainty limits arise from the dispersion of the three experimental results only and the added uncertainty due to the limited purity of the sample could amount to a further 6.0 $kJ \cdot mol^{-1}$. As we were unable to properly trace the auxiliary values used by the authors in their evaluation of the enthalpies of formation of the hydroxylapatite from Reaction (V.172), we have resorted to using CODATA-compatible values of the enthalpies of formation of the relevant species in just 18.3H_2O, realising that such a procedure would not include any interaction between the products in solution. We thus obtained $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6724.7 \pm 15.3)\ kJ \cdot mol^{-1}$, keeping conservative uncertainty limits imposed by the limitations of this study. For the same formula, the authors obtained – 6719.5 $kJ \cdot mol^{-1}$.

The enthalpy of solution data reported by Jacques [1963JAC] for hydroxylapatite concerns also beta tricalcium phosphate and fluorapatite and have been analysed in considerable detail by this review. As mentioned in the relevant Appendix A, it was virtually impossible to retrace exactly the various calculations made by Jacques for the lack of detailed information on some of the auxiliary values he used in the thermodynamic cycles. We have thus re-calculated the various enthalpies of formation using CODATA or CODATA-compatible values, starting from the experimental enthalpies of solution given by Jacques. The value

$\Delta_f H^o_m$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K) = – (6741.5 ± 12.2) kJ·mol^{-1} is based on $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4128.691 ± 7.749) kJ·mol^{-1} also obtained from the result of Jacques [1963JAC].

The enthalpies of solution of hydroxylapatite and fluorapatite in dilute HCl were reported by Craig and Rootare [1974CRA/ROO]. As detailed in the relevant Appendix A, there is a large scatter of the data (several tens of kJ·mol^{-1}) between samples of different origins that is not satisfactorily explained. Under these conditions, the results published by Craig and Rootare are mentioned for information only and will not be considered further.

As discussed in detail in the relevant Appendix A, Jemal *et al.* [1995JEM/BEN] reported the enthalpy of formation of hydroxylapatite and fluorapatite from solution calorimetry at 298.15 K in nitric acid (HNO_3, 35.35H_2O). In fact, these calorimetric results concerning the two compounds were already reported by Ben Cherifa *et al.* [1991BEN/SOM] of the same group, but with far less detail (see Appendix A). Under these conditions, only the discussion in [1995JEM/BEN] will be considered further.

Eight measurements of enthalpies of solution of hydroxylapatite were reported, but it was not possible to reproduce the average value given by the authors, as they mention a weighted average without giving the uncertainties of the individual measurements. We thus calculated, as a straight average $\Delta_{sol} H^o_m$ ($Ca_{10}(PO_4)_6(OH)_2$, cr, in (HNO_3, 35.35H_2O), 298.15 K) = – (364.157 ± 2.660) kJ·mol^{-1} (2σ), or for the formulation used in the present text, $\Delta_{sol} H^o_m$ ($Ca_5(PO_4)_3OH$, cr, in (HNO_3, 35.35H_2O), 298.15 K) = – (182.079 ± 1.330) kJ·mol^{-1} (2σ). The cycle used by the authors to yield the enthalpy of formation of the compound is given in detail in the Appendix A entry using CODATA or CODATA-compatible auxiliary values.

We thus obtain, from the data in [1995JEM/BEN], $\Delta_f H^o_m$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K) = – (6682.0 ± 9.2) kJ·mol^{-1} a value in reasonable agreement with that, – 6652.5 kJ·mol^{-1}, given by the authors without uncertainty limits, noting that the major difference arises from the use of different sets of auxiliary data, in particular for phosphoric acid.

Krivtsov *et al.* [1997KRI/ORL] obtained the enthalpy of formation of hydroxylapatite $Ca_{10}(PO_4)_6(OH)_2$(cr) at 298.15 K from the enthalpy of solution of this compound, of $CaCl_2$(cr) and of β-$Ca_3(PO_4)_2$ in 14.04 % HCl. In the relevant Appendix A details are given on the instrumentation, the identification of the compounds involved, and the experimental thermodynamic cycle leading to the process:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 2HCl(g) \rightleftharpoons CaCl_2(cr) + 3\ \beta\text{-}Ca_3(PO_4)_2 + 2H_2O(l) \qquad (V.173)$$

for which this review re-calculates, for a 95 % confidence interval,

$$\Delta_r H^\circ_m((V.173), 298.15\ K) = -(35.638 \pm 2.852)\ kJ{\cdot}mol^{-1}.$$

Use of $\Delta_f H^\circ_m(H_2O, l, 298.15\ K) = -(285.830 \pm 0.040)$ and $\Delta_f H^\circ_m(HCl, g, 298.15\ K) = -(92.310 \pm 0.100)\ kJ{\cdot}mol^{-1}$ from CODATA [1989COX/WAG], $\Delta_f H^\circ_m(CaCl_2, cr, 298.15\ K) = -(795.39 \pm 1.00)\ kJ{\cdot}mol^{-1}$ from [1987GAR/PAR] and $\Delta_f H^\circ_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = -(4123.6 \pm 12.0)\ kJ{\cdot}mol^{-1}$ selected by this review, leads to $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6758.8 \pm 18.1)\ kJ{\cdot}mol^{-1}$.

Cruz *et al.* [2005CRU/MIN] reported the enthalpy of formation of hydroxylapatite from dissolution of the compound in two media, (HCl, 553.41H_2O) and (HNO_3, 35.35H_2O). These experiments are analysed and discussed in considerable detail in the relevant Appendix A.

From the dissolutions in HCl, the authors derive the enthalpy of formation of $Ca_{10}(PO_4)_6(OH)_2$(cr) from Reaction (V.174):

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 20(HCl, 553.41H_2O)(sln) + 18H_2O(l) \rightleftharpoons 10CaCl_2{\cdot}2H_2O(cr) + 6(H_3PO_4, 0.77H_2O) \quad (V.174)$$

for which they report $\Delta_f H^\circ_m(V.174) = (237.1 \pm 10.6)\ kJ{\cdot}mol^{-1}$.

However, this review noted an inconsistency in the reaction sequence used by the authors, without being able to discover its origin. Using the selected value for $\Delta_f H^\circ_m(CaCl_2{\cdot}2H_2O,\ cr, 298.15\ K) = -(1404.000 \pm 1.800)\ kJ{\cdot}mol^{-1}$ from [1987GAR/PAR] we have thus resorted to evaluate $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K)$ from Reaction (V.174) as written by the authors. This led to $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(13482.0 \pm 19.0)\ kJ{\cdot}mol^{-1}$ and thus for the smaller formula used in this review $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6741.0 \pm 15.0)\ kJ{\cdot}mol^{-1}$, where the uncertainty has been increased substantially to allow for the unbalanced reaction scheme noted in Appendix A. Cruz *et al.* [2005CRU/MIN] reported $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(13430 \pm 81)\ kJ{\cdot}mol^{-1}$.

This review did not make use of the results of Cruz *et al.* [2005CRU/MIN] obtained from HNO_3 media, as, in this case, we are even less certain about the stoichiometries in the cycle involved. From that medium, the authors reported $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(13398 \pm 11)\ kJ{\cdot}mol^{-1}$, in quite limited agreement with their own value from the dissolutions in HCl.

The final values taken into consideration by this review for the evaluation of $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K)$ are summarised in Table V-80 noting that the value we obtained from the results of Jemal *et al.* [1995JEM/BEN] is about 60 $kJ{\cdot}mol^{-1}$ less negative than that which results from the average of the other four sources and that a similar bias is observed in the case of fluorapatite (see Section V.8.2.4.3).

Table V-80: Values taken into consideration for the evaluation of $\Delta_f H^o_m$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K).

Reference	$\Delta_f H^o_m$ (298.15 K) /kJ·mol^{-1}	Comments
[1962SMI/ILL]	– (6724.7 ± 15.3)	
[1963JAC]	– (6741.5 ± 12.2)	
[1995JEM/BEN]	– (6682.0 ± 9.2)	To be discarded from average; see text
[1997KRI/ORL]	– (6758.8 ± 18.1)	
[2005CRU/MIN]	– (6741.0 ± 15.0)	

Thus, from a weighted average of these four acceptable sources this review obtains and selects:

$$\Delta_f H^o_m (Ca_5(PO_4)_3OH, cr, 298.15\ K) = \mathbf{-\ (6740.4 \pm 15.0)}\ kJ{\cdot}mol^{-1},$$

keeping conservative uncertainty limits.

For information only, we note (see the relevant Appendix A entry) that Orlovskii and Ionov [1996ORL/ION2], using a structure-thermochemical model they had proposed earlier for phosphorus compounds, reported a calculated value of $\Delta_f H_m$ ($Ca_5(PO_4)_3OH$, cr, 0 K) = – 6744 kJ·mol^{-1}, quite consistent with our selected value.

V.8.2.4 Fluorapatite, $Ca_5(PO_4)_3F$(cr)

V.8.2.4.1 Structure

Fluorapatite has a hexagonal crystal structure, space group $P6_3/m$ with a = (9.397 ± 0.003) Å, c = (6.878 ± 0.002) Å as given by Hughes *et al.* [1989HUG/CAM] for a natural sample. We shall use the simpler formula $Ca_5(PO_4)_3F$, rather than the doubled formula used in many publications.

V.8.2.4.2 Heat capacity and entropy

There are three measurements of the heat capacity. Egan and Wakefield [1951EGA/WAK2] measured the heat capacity of fluorapatite from *ca.* 13 to *ca.* 300 K and the enthalpy content from 474.6 to 1581.7 K at intervals of *ca.* 100 K, using an adiabatic calorimeter. More recently Dachs *et al.* [2010DAC/HAR] measured the heat capacities of small (mg) samples of fluorapatite over the temperature range 5 to 764 K using relaxation calorimetry and differential scanning calorimetry (DSC). Their low temperature values of $C^o_{p,m}$ agree very well with the adiabatic calorimetric measurements in [1951EGA/WAK2] up to about *ca.* 150 K, but are increasingly smaller above this, being 1.7 % lower at 298.15 K.

The third study is by Meis *et al.* [2000MEI/GAL], who made DSC measurements of $C^{\circ}_{p,\mathrm{m}}$ from *ca.* 300 K to 1000 K and also calculated $C^{\circ}_{p,\mathrm{m}}$ from a force field model. However as noted in Appendix A, the two small graphs in which the results are given are not entirely consistent, and their experimental values (reported to have an uncertainty of *ca.* 3 %) are consistently *ca.* 4 % greater than those derived from the two consistent measurements noted above. These data were not included in our fitted heat capacities. The data at 298.15 K from the first two studies are compared in Table V-81.

Table V-81: Heat capacity and entropy values of fluorapatite.

Reference	$C^{\circ}_{p,\mathrm{m}}(298.15\,\mathrm{K})$ /J·K^{-1}·mol^{-1}	$S^{\circ}_{\mathrm{m}}(298.15\,\mathrm{K})$ /J·K^{-1}·mol^{-1}
[1951EGA/WAK2]	(376.0 ± 0.8)	(388.1 ± 0.8)
[2010DAC/HAR]	(369.6 ± 1.1)	(383.2 ± 1.5)

Figure V-86 shows the smoothed experimental data as reported by in these three studies, and three fits, two to each of the individual studies (with $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K})$ constrained to the appropriate value from the low-temperature measurements) and one to the combined data, with no constraint on $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K})$. For the selected values, we have taken the $C^{\circ}_{p,\mathrm{m}}$ values from the combined fit, and a mean value of $S^{\circ}_{\mathrm{m}}(298.15\ \mathrm{K})$ with an increased uncertainty, but restrict the range of validity to 1200 K.

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,\ cr,\ 298.15\ K}) = \mathbf{(373.0 \pm 3.5)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

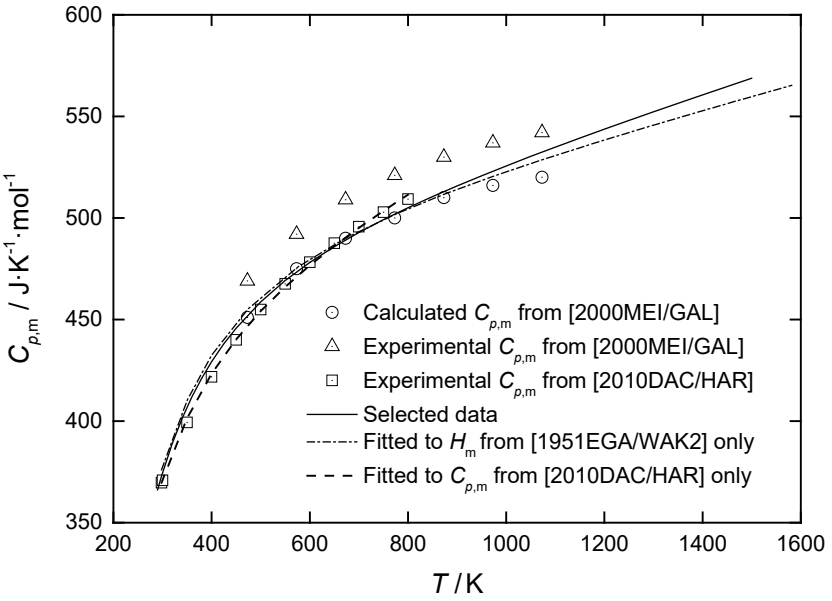
$$S^{\circ}_{\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,\ cr,\ 298.15\ K}) = \mathbf{(385.6 \pm 2.5)}\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,cr},T)/\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = \mathbf{459.7936} + \mathbf{7.5546\times10^{-2}}(T/\mathrm{K}) - \mathbf{9.72048\times10^{6}}(T/\mathrm{K})^{-2} \qquad (298.15\ \text{to}\ 1200\ \mathrm{K}).$$

This Figure also includes both the calculated and experimental data from the study of Meis *et al.* [2000MEI/GAL], and shows that their calculated values agree rather well with the selected values, at least up to 873 K.

As noted in Section V.8.2.3, Cruz *et al.* [2005CRU/CAN] reported *ab initio* studies for the hexagonal form of fluorapatite, which yielded a value for $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K})$, with a substantial uncertainty, and this value has not been utilised for this review.

Figure V-86: Heat capacity of $Ca_5(PO_4)_3F$(cr).



V.8.2.4.3 Enthalpy of formation

The enthalpy of formation of fluorapatite rests on the determination of its enthalpy of solution in nitric and hydrochloric acid and in a molten salt medium, using varied thermodynamic cycles.

As detailed and discussed in the relevant Appendix A, Gottschal [1958GOT] reported measurements of calorimetric dissolutions of fluorapatite in 20 and 30 % HNO_3. The calorimetric results were given for each medium as averages of two single determinations only. The purity of the samples was declared as 97 %. Gottschal described the calorimetric reaction as Reaction (V.175):

$$\begin{aligned} &Ca_{10}(PO_4)_6F_2(cr) + (14HNO_3 + H_2O)(sln) \rightleftharpoons \\ &(7Ca(NO_3)_2 + 2HF)(sln) + 3Ca(H_2PO_4)_2 \cdot H_2O(cr) \end{aligned} \qquad \text{(V.175)}$$

As in the case of the hydroxylapatite (see Section V.8.2.3.3), the validity of this process was strongly criticised by Jacques [1963JAC]. In view of this we have calculated, for information only, a CODATA compatible value of: $\Delta_f H_m^o(Ca_5(PO_4)_3F, cr, 298.15\ K) = -(6841 \pm 15)$ kJ·mol^{-1}, keeping conservative uncertainty limits on those results.

Smirnova *et al.* [1962SMI/ILL] reported the dissolution of fluorapatite, hydroxylapatite and $Ca_3(PO_4)_2$(cr, α and β) in *ca.* 10 % HCl (given as HCl, 18.3H_2O). Details on the samples and their limited purity, and the calorimetric procedure are given in the relevant Appendix A. Four measurements of the enthalpy of Reaction (V.176) for a synthetic sample and three for a natural sample, according to Reaction (V.176):

$$Ca_5(PO_4)_3F(cr) + 10HCl(sln) \rightleftharpoons (5CaCl_2 + 3H_3PO_4 + HF)(sln) \quad (V.176)$$

were reported, for which this review re-calculated as $\Delta_r H^\circ_m$ (V.176) = – (70.596 ± 4.960) kJ·mol^{-1}, with uncertainty limits arising only from the dispersion of the seven experimental results; the added uncertainty due to the limited purity of the two samples could amount to a further 6.0 kJ·mol^{-1}. As we were unable to properly trace the auxiliary values used by the authors in their evaluation of the enthalpies of formation of the fluorapatite from Reaction (V.176), as was the case for hydroxylapatite, we have resorted to using the CODATA-compatible values of the enthalpies of formation of the relevant species just in 18.3H_2O, realising that such a procedure would not include any interaction between the products in solution. We thus obtained $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) = – (6837.0 ± 15.0) kJ·mol^{-1}, keeping conservative uncertainty limits imposed by the limitations of this study. For the same formula, the authors obtained – 6838.8 kJ·mol^{-1}.

The enthalpy of solution data reported by Jacques [1963JAC] for fluorapatite concerns also β-tricalcium phosphate and hydroxylapatite. They have been analysed in considerable detail by this review. As mentioned in the relevant Appendix A, it was virtually impossible to retrace exactly the various calculations made by Jacques for the lack of detailed information on some of the auxiliary values he used in the thermodynamic cycles. We have thus re-calculated the various enthalpies of formation using CODATA or CODATA-compatible values, starting from the experimental enthalpies of solution given by Jacques. Three values for $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) were obtained using distinct but related cycles: – (6891.703 ± 11.851), – (6890.922 ± 6.613) and – (6897.878 ± 7.565) kJ·mol^{-1}. The weighted mean of these three values gives $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) = – (6893.6 ± 9.2) kJ·mol^{-1} from this study.

The enthalpies of solution of fluorapatite and hydroxylapatite in dilute HCl were reported by Craig and Rootare [1974CRA/ROO]. As detailed in the relevant Appendix A, in view of the general context of the reporting of the data in this publication, and also because of the criticism expressed by this Review on the stoichiometry of the fluorapatite sample, we shall consider these results reported for fluorapatite for information only.

Westrich and Navrotsky [1981WES/NAV] have used oxide melt solution calorimetry with lead borate solvent (2PbO·B_2O_3) at 985 K to determine the enthalpy of

formation of fluorapatite from its component species, CaF_2(cr)[1] and β-$Ca_3(PO_4)_2$. As described in the relevant Appendix A, in order to enhance the rate of dissolution of the component oxides and of the fluorapatite, they used AlF_3 and SiO_2 (separately) as fluxes of which the enthalpy of solution in the same medium had been determined earlier. The enthalpy of formation (at 985 K) of $Ca_5(PO_4)_3F$(cr) from its constituents, according to Reaction (V.177)

$$0.5CaF_2(cr) + 1.5Ca_3(PO_4)_2(cr) \rightleftharpoons Ca_5(PO_4)_3F(cr) \qquad (V.177)$$

were calculated by this review to be: $\Delta_r H_m$ (V.177) = – (74.06 ± 18.04) kJ·mol^{-1}, from the data with the AlF_3(cr) diluent, and $\Delta_r H_m$ (V.177) = – (71.78 ± 9.91) kJ·mol^{-1}, from the data with the SiO_2(cr) diluent, while the authors reported values (also for 2σ uncertainty) of – (74.27 ± 17.90) kJ·mol^{-1}, and – (73.09 ± 6.86) kJ·mol^{-1}, respectively. The reason for these small differences could not be identified by this review, as it is not entirely clear what values Westrich and Navrotsky used in their calculation for the enthalpies of solution of CaF_2(cr) and $Ca_3(PO_4)_2$(cr).

Under these conditions, this review used in the calculations the average value $\Delta_r H_m$ (V.177) = – (73.3 ± 10.0) kJ·mol^{-1}, with uncertainty limits covering adequately the two sets of results. Using the heat capacity functions accepted in this review, we re-calculate $\Delta_r H^o_m$ ((V.177), 298.15 K) = – 60.721 kJ·mol^{-1} and thus with $\Delta_f H^o_m$ (CaF_2, cr, 298.15 K) = – (1228.004 ± 2.000) kJ·mol^{-1} [1987GAR/PAR] (that this review will have to formally adopt) and $\Delta_f H^o_m$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4123.6 ± 12.0) kJ·mol^{-1} from this review, we find from this study:

$$\Delta_f H^o_m (Ca_5(PO_4)_3F, cr, 298.15\ K) = – (6860.1 \pm 20.6)\ kJ \cdot mol^{-1}.$$

As noted in Section V.8.2.3.3, the brief details of measurements of the enthalpy of formation of fluorapatite by Ben Cherifa *et al*. [1991BEN/SOM] were expanded by Jemal *et al.* [1995JEM/BEN], and only the discussion in [1995JEM/BEN] will be considered further here. Seven measurements of the enthalpy of solution of fluorapatite were reported, but it was not possible to reproduce the average value given by the authors, as they mention a weighted average without giving the uncertainties of the individual measurements. We thus calculated, as a straight average, for the formulation used in the present text, $\Delta_{sol} H^o_m$ ($Ca_5(PO_4)_3F$, cr, in (HNO_3, 35.35H_2O), 298.15 K) = – (98.061 ± 0.823) kJ·mol^{-1} (2σ). The cycle then used by the authors to yield the enthalpy of formation of the compound is given in detail in the Appendix A, and our calculations were made using CODATA or CODATA-compatible auxiliary values. We thus obtain, from the data in [1995JEM/BEN], $\Delta_f H^o_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) = – (6804.3 ± 9.2) kJ·mol^{-1}.

[1] This compound was not part of those subjected to our thermodynamic assessment and only its enthalpy of formation is mentioned in Table IV-1.

This value is in reasonable agreement with that, – 6772.5 kJ·mol^{-1}, given by the authors without any uncertainty, noting that the major difference arises from the use of different sets of auxiliary data, in particular for phosphoric acid.

In conclusion, Table V-82 summarises the values taken into consideration by this review for the final evaluation of $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K), noting that the value obtained from the results of Jemal *et al.* [1995JEM/BEN] is about 70 kJ·mol^{-1} less negative than that which results from the average of the other three sources as is the case for their measurements on hydroxylapatite (see Section V.8.2.3.3).

Table V-82: Values taken into consideration for the evaluation of $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K).

Reference	$\Delta_f H^\circ_m$ (298.15 K) /kJ·mol^{-1}	Comments
[1962SMI/ILL]	– (6837.0 ± 15.0)	
[1963JAC]	– (6893.6 ± 9.2)	
[1981WES/NAV]	– (6860.1 ± 20.6)	
[1995JEM/BEN]	– (6804.3 ± 9.2)	To be discarded from average; see text above

Thus, from a weighted average of these three sources this review obtains and selects:

$$\Delta_f H^\circ_m (Ca_5(PO_4)_3F, \text{cr}, 298.15\ \text{K}) = \mathbf{-\ (6875.8 \pm 20.0)}\ \text{kJ·mol}^{-1},$$

keeping conservative uncertainty limits.

V.8.3 Na mineral: borax

V.8.3.1 Borax, $Na_2B_4O_7{\cdot}10H_2O$(cr)

We shall use the traditional formula for borax, rather than the structurally more correct $Na_2B_4O_5(OH)_4{\cdot}8H_2O$—see below. As will be apparent from the following sections, there are few definitive data on the thermochemistry of borax.

V.8.3.1.1 Structure

There is good agreement between a number of studies of the crystal structure, all of which indicate a monoclinic structure, space group *C*2/*c,* and we quote only the parameters given by the neutron diffraction study by Levy and Lisensky [1978LEV/LIS] on a re-crystallised sample at 296.7 K: a = 11.885 Å, b = 10.654 Å, c = 12.206 Å. β = 106.623 °. This, and other studies, clearly show the presence of $B_4O_5(OH)_4$ groups in the crystal.

V.8.3.1.2 Heat capacity and entropy

The only heat capacity data we have been able to locate are those from the early work of Kopp [1865KOP]. This is a long, beautifully written paper, with full details (including engravings of the apparatus) of measurements on a large number (> 100) of elements and compounds. The method is essentially just drop calorimetry, the sample, contained in a glass ampoule, being dropped into water and the change in temperature in both being measured. All the details of the temperatures and masses and $C^{o}_{p,m}$ of the empty capsule are given.

Table V-83: Enthalpy drop measurements by Kopp [1865KOP].

T_1 /K	T_2 /K	T_3 /K	T_4 /K	m /g	C^{o}_{p} /J·K^{-1}·g^{-1}
324.05	291.85	288.58	291.58	3.38	1.619
323.45	291.55	288.30	291.28	3.38	1.623
322.25	291.65	288.48	291.31	3.38	1.594
322.65	291.95	288.76	291.60	3.38	1.602

where T_1 and T_2 are the initial and final temperatures of the sample, and T_3 and T_4 are those of water and m the mass of the sample. The mean C^{o}_{p} for 292 to 323 K is thus (1.610 ± 0.013) J·K^{-1}·g^{-1} = (614.0 ± 5.0) J·K^{-1}·mol^{-1}. We assume this is close to the value of C^{o}_{p}(307.5 K), and thus select the value:

$$C^{o}_{p,m}(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K) = \mathbf{(608 \pm 6)}\ J{\cdot}K^{-1}{\cdot}mol^{-1}.$$

where the ratio of $C^{o}_{p,m}$(298.15 K) / $C^{o}_{p,m}$(307.5 K) is taken to be 0.991, the same as that calculated from a short extrapolation of the data for $Na_2CO_3{\cdot}10H_2O$(cr) [1968WAT/LIN]. Wagman *et al.* [1982WAG/EVA] give $C^{o}_{p,m}$(borax, 298.15 K) = 615 J·K^{-1}·mol^{-1}; no value is given in [1981GLU/MED].

Wagman *et al.* [1982WAG/EVA] also give the values of S^{o}_{m}(298.15 K) = 586 J·K^{-1}·mol^{-1}, $\Delta_f G^{o}_{m}$(298.15 K) = – 5516 kJ·mol^{-1}, the source for which is unclear. The latter value is close to that given by Bassett [1977BAS] – 5518.1 kJ·mol^{-1}, see below.

No values for S^{o}_{m}(298.15 K) or $\Delta_f G^{o}_{m}$(298.15 K) are selected in this review.

V.8.3.1.3 Enthalpy of formation

The early results of Favre and Valson [1873FAV/VAL] and Thomsen [1883THO] (excerpts) concerned enthalpies of solution in a narrow concentration range of a large number of salts, in an effort to rationalise enthalpies of hydration. These results are, of course, of historical interest only.

More recently, Scholle and Szmigielská [1965SCH/SZM] measured the integral enthalpies of solution in water at different concentrations for the deca- and

pentahydrate of disodium tetraborate, and for the anhydrous salt. Measurements at 318 K were reported for a final concentration of 2.5 % of salt. For the process:

$$Na_2B_4O_7(cr) + 10H_2O(l) \rightleftharpoons Na_2B_4O_7 \cdot 10H_2O(cr) \quad (V.178)$$

which is the only one of interest to this study, the authors reported an enthalpy effect of – 161.0 cal·(g anhydrous salt)$^{-1}$ or – 135.54 kJ·mol^{-1}. For the evaluation of the enthalpy of formation of borax from this study, we have used $\Delta_f H^\circ_m(Na_2B_4O_7, cr, 298.15\ K) = -(3290.1 \pm 5.0)$ kJ·mol^{-1}, which is the average of the assessments by Glushko *et al.* [1981GLU/MED], – (3289.04 ± 5.02) kJ·mol^{-1}, and Wagman *et al.* [1982WAG/EVA], – 3291.1 kJ·mol^{-1}, in view of the fact that $\Delta_f H^\circ_m(Na_2B_4O_7, cr, 298.15\ K)$ is not being re-evaluated in the present review. With this, we obtain, from the results in [1965SCH/SZM], $\Delta_f H^\circ_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -(6283.94 \pm 10.00)$ kJ·mol^{-1} (the uncertainty limits being ours), a value that should be taken for information purposes only in view of the uncertainties incurred in its derivation, but it is consistent with our selected value below. The heat capacities of borax solutions are also reported in this paper [1965SCH/SZM].

In an evaluation of boron systems in thermal waters, Bassett [1977BAS], citing Scholle and Szmigielská [1965SCH/SZM], gives (in his Table 3.1) $\Delta_f H^\circ_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -6089.054$ kJ·mol^{-1}, a likely misprint for – 6289.054 kJ·mol^{-1} in view of his selected value (in his Table 3.8) for $\Delta_f G^\circ_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -5518.1$ kJ·mol^{-1}.

Vasil'ev *et al.* [1984VAS/KOC] reported the enthalpy of formation of borax from very careful measurements of the enthalpy of solution at 298.15 K of the salt in HCl, NaOH and water, using an isothermally jacketed calorimeter. All the experimental results were extrapolated to zero ionic strength using a Debye-Hückel type relation. More details on experimental conditions are given in the relevant Appendix A entry.

For the dissolution in HCl according to Reaction (V.179):

$$Na_2B_4O_7 \cdot 10H_2O(cr) + 2H^+ \rightleftharpoons 4B(OH)_3(aq) + 2Na^+ + 5H_2O(l) \quad (V.179)$$

the authors reported the infinite dilution value $\Delta_{sol} H^\circ_m(V.179) = (86.06 \pm 1.10)$ kJ·mol^{-1}. Accepting this value and making use of CODATA values [1989COX/WAG], this review obtains $\Delta_f H^\circ_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -(6287.09 \pm 3.39)$ kJ·mol^{-1} from this scheme.

For the dissolution in NaOH solutions according to Reaction (V.180):

$$Na_2B_4O_7 \cdot 10H_2O(cr) + 2OH^- \rightleftharpoons 4B(OH)_4^- + 2Na^+ + 3H_2O(l) \quad (V.180)$$

Vasil'ev *et al.* obtained the infinite dilution value $\Delta_{sol} H^\circ_m(V.180) = (31.05 \pm 1.60)$ kJ·mol^{-1}. With $\Delta_f H^\circ_m(B(OH)_4^-, aq, 298.15\ K) = -(1345.283 \pm 0.805)$ kJ·mol^{-1} obtained in this review and otherwise CODATA values, we calculate from Reaction (V.180) $\Delta_f H^\circ_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -(6290.32 \pm 3.60)$ kJ·mol^{-1}.

For the dissolution in water according to Reaction (V.181) at infinite dilution:

$$Na_2B_4O_7{\cdot}10H_2O(cr) \rightleftharpoons 2B(OH)_4^- + 2B(OH)_3(aq) + 2Na^+ + 3H_2O(l) \quad (V.181)$$

the value $\Delta_{sol}H_m^o(V.181)$ = (117.1 ± 1.1) kJ·mol^{-1} was reported.

Use of $\Delta_f H_m^o(B(OH)_4^-, aq, 298.15\ K)$ = – (1345.283 ± 0.805) kJ·mol^{-1} (see above) and otherwise of CODATA values leads to $\Delta_f H_m^o(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K)$ = – (6291.44 ± 2.53) kJ·mol^{-1}.

As this review finds no reason to prefer one of the above three excellently consistent sets of data, their simple average, $\Delta_f H_m^o(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K)$ = – (6289.62 ± 3.40) kJ·mol^{-1} is recommended as the best value from the data in [1984VAS/KOC], keeping uncertainty limits that take into account that the main sources of error are common to the three schemes.

In their assessment of the sodium borate data Glushko *et al.* [1981GLU/MED] recommended $\Delta_f H_m^o(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K)$ = – (6285.2 ± 2.1) kJ·mol^{-1}. Among their cited literature—which we have thoroughly checked—they included a publication with the same four authors as those of [1984VAS/KOC], which was due to appear in *Zhur. Neorg. Khim.* (1982). We have been unable to locate this paper, and suspect that it may have been a preliminary unpublished version of Vasil'ev *et al.* [1984VAS/KOC] in the same journal.

Given the lack of detail of all the other assessments, this review thus accepts the mean of the three values obtained by Vasil'ev *et al.* [1984VAS/KOC] for the selected value for the enthalpy of formation of borax:

$$\Delta_f H_m^o(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K) = \mathbf{-\ (6289.6 \pm 3.4)}\ kJ{\cdot}mol^{-1}.$$

As noted above, no value for $\Delta_f G_m^o(Na_2B_4O_7{\cdot}10H_2O, cr, 298.15\ K)$ can be selected in this review.

Part 4

Appendices

Appendix A

Discussion of selected references

[1865KOP]

This is a long, beautifully written paper, with full details (including engravings of the apparatus) of measurements on a large number (> 100) of elements and compounds; the data for borax are given on page 150 of the article. The method essentially uses a drop calorimeter, dropping the sample contained in a glass ampoule into water, and measuring the change in temperature in both. All the details of the temperatures and masses and C_p of the empty capsule are given.

Naphtha was involved in some of the experiments, it is not immediately clear how or why. For borax, there were four drops (see Table A-1), where t_1 and t_2 are the initial and final temperatures of the sample, t_3 and t_4 are those of the water and m is the mass of the sample. The mean C_p for 292-323 K is thus (0.385 ± 0.003) cal·K^{-1}·g^{-1} = 614 J·K^{-1}·mol^{-1}.

Table A-1: Enthalpy drop measurements for borax.

t_1/°C	t_2/°C	t_3/°C	t_4/°C	m/g	C_p/cal·K^{-1}·g^{-1}
50.9	18.7	15.43	18.43	3.38	0.387
50.3	18.4	15.15	18.13	3.38	0.388
49.1	18.5	15.33	18.16	3.38	0.381
49.5	18.8	15.61	18.45	3.38	0.383

For comparison, Kopp's four measured C_p values for the anhydrous $Na_2B_4O_7$(cr) are also mentioned. These give a value of 0.227 cal·K^{-1}·g^{-1} as a mean C_p from *ca.* 290 to 320 K. This gives 191.1 J·K^{-1}·mol^{-1} compared with $C^{\circ}_{p,\mathrm{m}}$(298.15 K) = 186.77 J·K^{-1}·mol^{-1} in [1982WAG/EVA] and (186.77 ± 0.63) J·K^{-1}·mol^{-1} in [1981GLU/MED].

[1867WAG]

The solubility of synthetic magnesite (no indication about crystallinity) was determined at 5 °C and at $p(CO_2)$ = 1-6 bar. No analysis of the material used was given. However, the reported solubilities are inconsistent: between $p(CO_2)$ = 2-3 and 5-6 bar, the solubility increased considerably, but between $p(CO_2)$ = 1-2, 3-4 and 4-5 bar the increase was found to be very limited. Therefore, these data were rejected.

[1892LES]

Lescoeur measured the decomposition pressures of $NaHCO_3$(cr) and $KHCO_3$(cr) and some related compounds, but only these two species are of interest to this review.

$NaHCO_3$(cr) was prepared by passing CO_2(g) into a saturated solution of Na_2CO_3 and drying the resulting precipitate in a current of dry CO_2(g). Analysis showed it contained somewhat more carbon and somewhat less hydrogen (analysed as CO_2 and H_2O, respectively) than theoretical. Six measurements of the dissociation pressure were made from 328 to 373 K. The measured pressures are slightly greater than those calculated from the selected values, a third-law processing giving $\Delta_r H^o_m$ ((A.1), 298.15 K) = (65.35 ± 0.51) kJ·mol^{-1}, compared to (66.742 ± 0.248) kJ·mol^{-1} from our selected values.

$$NaHCO_3(cr) \rightleftharpoons \tfrac{1}{2}Na_2CO_3(cr) + \tfrac{1}{2}CO_2(g) + \tfrac{1}{2}H_2O(g) \qquad (A.1)$$

$KHCO_3$(cr) was prepared in a similar manner to $NaHCO_3$(cr), and analysis showed it also contained somewhat more carbon and somewhat less hydrogen than theoretical. Six measurements of the dissociation pressure were made from 85 to 127 °C. No entropy or heat capacity data are available for $KHCO_3$ for a third-law processing but the following values for Reaction (A.2) are obtained from a least squares fit to log(p) as a linear function of (T/K)$^{-1}$:

$$KHCO_3(cr) \rightleftharpoons \tfrac{1}{2}K_2CO_3(cr) + \tfrac{1}{2}CO_2(g) + \tfrac{1}{2}H_2O(g) \qquad (A.2)$$

$\Delta_r H^o_m$ ((A.2), 376 K) = (57.75 ± 4.02) kJ·mol^{-1} and

$\Delta_r S^o_m$ ((A.2), 376 K) = (133.64 ± 10.34) J·K^{-1}·mol^{-1}.

Judging from the very small heat capacity change for Reaction (A.1), the values for 298.15 K will be only slightly different. With our selected values for the remaining species, these values correspond to $\Delta_f H^o_m$($KHCO_3$, cr, 298.15 K) = – (951.2 ± 4.1) kJ·mol^{-1} and S^o_m($KHCO_3$, cr, 298.15 K) = (145.4 ± 10.4) J·K^{-1}·mol^{-1}. The former value is distinctly less negative than our selected value, – (960.7 ± 2.0) kJ·mol^{-1}, while the latter value is unreasonably large compared to that for $NaHCO_3$(cr), (102.2 ± 1.0) J·K^{-1}·mol^{-1}.

The data from this study are not considered further in this review.

[1905CAN/GOG]

The solubilities of $CaCO_3$(cr) (polymorph not specified, but based on its solubility it was probably calcite) and $BaCO_3$(cr) in NaCl and KCl solutions (I_m ~ 1-4.3) were determined at 13-18 °C. Calcium carbonate suspended in a solution of NaCl or KCl was flushed continuously for 98 days with CO_2-free air. The CO_2(g) concentration released

during the solubilisation of the carbonate was measured by absorption in a KOH solution.

Since a considerable part of the dissolved carbonate was flushed out from the solution as CO_2(g), the system cannot be considered as a true closed one. Therefore, these data were not considered in our evaluation.

[1911CAV/SAN]

Caven and Sand measured the dissociation pressures of $NaHCO_3$(cr) and $Na_2CO_3 \cdot H_2O$(cr) at temperatures from *ca.* 313 to 388 K for the former and *ca.* 303 to 373 K for the latter. Only the data for $NaHCO_3$(cr) are of interest to the current review.

The pressures were measured by balancing those in the decomposition reaction with a measured pressure of air in a mercury U-tube. Two experimental difficulties were acknowledged by the authors, *viz* ensuring that there was no absorption of either CO_2(g) or H_2O(g) during the measurements, and the slow reaction, perhaps due to the formation of impermeable crusts on the decomposing $NaHCO_3$. No details were given of the sample of sodium bicarbonate used in the experiments, except that it was dried in a desiccator before being mixed with 'ignited' sodium carbonate for the experiments. In a later paper on the similar decomposition of $KHCO_3$(cr) [1914CAV/SAN], the authors reported two analyses of the gas from the decomposition of $NaHCO_3$(cr) at 374.7 and 375 K, where the CO_2/H_2O ratios were 1.02 and 0.98 respectively.

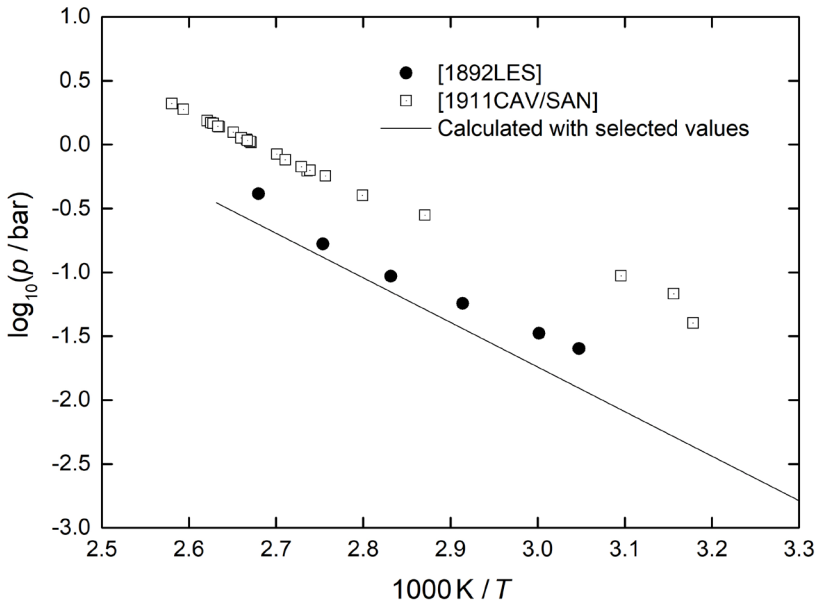
The results of *ca.* 30 experiments at temperatures from *ca.* 313 to 388 K were tabulated, taken where the pressure was both rising and falling. The results are not convincingly consistent, even after excluding four data points where the observed pressures differ from the smoothed values by factors of 3 to 4. The authors fitted their data by a linear equation in $(T/\mathrm{K})^{-1}$, corresponding to an enthalpy of Reaction (A.3) of $\Delta_r H^\circ_m$ (A.3) = 63.95 kJ·mol^{-1}. Strictly this would apply to the mid temperature of the measurements, 351 K, but the change in heat capacity for this reaction is so small that this may be taken to be the same at 298.15 K.

$$NaHCO_3(cr) \rightleftharpoons \tfrac{1}{2}Na_2CO_3(cr) + \tfrac{1}{2}CO_2(g) + \tfrac{1}{2}H_2O(g) \qquad (A.3)$$

A third-law fit to the data using our selected values of the entropies of this reaction is slightly different giving $\Delta_r H^\circ_m$ (A.3) = (62.321 ± 0.245) kJ·mol^{-1}. Both these values of $\Delta_r H^\circ_m$ (A.3) are noticeably different from that calculated from our selected values for $NaHCO_3$(cr) and Na_2CO_3(cr), $\Delta_r H^\circ_m$ ((A.3), 298.15 K) = (66.742 ± 0.248) kJ·mol^{-1}, consistent with the measured pressures being a factor of *ca.* 3 higher than the calculated pressures. The reason for this discrepancy is not clear, but may have been due to an impurity in the $NaHCO_3$(cr) used.

Figure A-1 shows a plot of the experimental data as compared with the earlier determination by Lescoeur [1892LES] and those calculated from the selected values.

Figure A-1: Decomposition pressure of $NaHCO_3$(cr).



[1912MUL]

This paper reports the enthalpies of solution in aqueous hydrofluoric acid of crystalline quartz, quartz glass powder, hydrated silicic acid and various silicates. Specific heat determinations of aqueous hydrofluoric acid at various concentrations were also reported. We shall only deal here with the few results concerning quartz and quartz glass.

The calorimeter was described in some detail and reactions were carried at 290-291 K. The platinum reaction vessel contained about 470 mL aqueous HF (20 % for the results described here) and the samples were of the order of one gram. Prior to use the samples were dried in an oven for one hour at 393 K. Little information was given on the purity of the starting materials. To obtain a sufficient rate of dissolution of crystalline quartz, the author used the finest particles obtained by sedimentation of a material ground for hours in agate. The sample was shown to contain about 1 % foreign material, as shown by fuming with HF. The enthalpy of solution was given as – 29.98 kcal·mol^{-1}, with an indicated inaccuracy of 2 % at the most. The quartz glass sample was also pulverised in agate and its enthalpy of solution was given as – (32.14 ± 0.105) kcal·mol^{-1} as an average of two concordant results.

From his results, [1912MUL] reports the value of 2.21 kcal·mol^{-1} or 9.25 kJ·mol^{-1} for the enthalpy of the transformation $SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(\text{vit})$.

No uncertainty limits are given for this value but, from the above, a minimal value of ± 0.61 kcal·mol^{-1} (± 2.55 kJ·mol^{-1}) is assumed here.

[1913MAG]

Magnus reported the enthalpy differences (as mean specific heats) of twelve inorganic solids between specified high temperatures and varying near-ambient temperatures, using a drop calorimeter. Those of interest to this review are: MgO(cr), (four temperatures, 334, 415, 560 and (663 ± 3) K), CaO(cr) (two temperatures, (559 ± 1) K, (661 ± 3) K and $CaCO_3$(calc), one point at (659 ± 2) K). The only indication of purity of the samples is that they were chemically pure, (manufacturer's guaranteed purity). These data points have all been used in selecting the data for the relevant compounds.

[1913MAR/SIM]

Measurement of the dissociation pressure of $MgCO_3$(cr) is very difficult because both the forward and backward reactions are extremely slow, and the magnesite must be very pure to avoid the formation of basic carbonates. For example, with completely dry carbonate at 300-400 °C Marc and Šimek found that equilibrium was not attained in weeks, and dry MgO could be left in contact with CO_2(g) at 100 K below the dissociation temperature for days without any CO_2(g) being taken up. However, both the forward and backward reactions are catalysed by water. For these reasons, attempts to determine the *P*-*T* curve of $MgCO_3$(cr) by the static method failed, even in the presence of H_2O or KNO_3 as catalysts. No better results were obtained by passing dry CO_2-free air over the heated solid. Satisfactory results were finally obtained by heating $MgCO_3$ in a closed evacuated tube with the addition of a large excess of KNO_3, which served to prevent the MgO taking up more CO_2 on cooling; the free CO_2 was determined by pumping out the gas and absorption as usual. The full results, both of the many unsuccessful runs and of the 24 successful experiments at temperatures from 671.5 to 782.3 K and CO_2 pressures of 0.835 to 11.21 bar are tabulated and plotted (the corrections for the non-ideality of CO_2(g) are insignificant at these pressures). Most of the values obtained were reasonably self-consistent, and these were fitted by the authors by an equation equivalent to $\log_{10}(p/\text{bar}) = -47.271 + 16.70 \log_{10}(T/\text{K})$.

The authors measured the pressures generated by heating KNO_3 alone, and found that these were negligible in the temperature range of their experiments (their Table 5). However, we note that later measurements by Centnerszwer [1930CEN], using a KNO_3 of unspecified purity, found much higher decomposition pressures, so

there may have been a small contribution from this in the pressures measured by Marc and Šimek.

With our selected data for the entropies and heat capacities of $MgCO_3$(cr), the calculated value of the third-law enthalpy of decomposition of magnesite according to Equation (A.4) is – (118.74 ± 2.11) kJ·mol^{-1}, where the uncertainty is statistical (2σ) only. This uncertainty has been increased to 5.00 to allow for the possible contribution of KNO_3(g) to the measured pressures.

$$MgCO_3(cr) \rightleftharpoons MgO(cr) + CO_2(g) \quad (A.4)$$

This corresponds to a value of $\Delta_f H^o_m(MgCO_3, cr, 298.15\ K)$ = – (1113.85 ± 5.04) kJ·mol^{-1}.

[1914CAV/SAN]

Caven and Sand measured the dissociation pressures of $KHCO_3$(cr), $RbHCO_3$(cr) and $CsHCO_3$(cr) at temperatures from *ca.* 333 to 443 K. Only the data for $KHCO_3$(cr) from 336.9 to 429.2 K are of interest to the current review.

As in the earlier paper by the same authors on $NaHCO_3$ [1911CAV/SAN], the pressures were measured by balancing those in the decomposition reaction with a measured pressure of air in a mercury U-tube. The composition of the gas was measured at three temperatures from 418 to 424 K showing CO_2/H_2O ratios of 1.28, 1.07 and 0.98, indicating, in the authors' view, that the dissociation products were those corresponding to the reaction:

$$KHCO_3(cr) \rightleftharpoons ½K_2CO_3(cr) + ½CO_2(g) + ½H_2O(g) \quad (A.5)$$

The authors also report two similar analyses of the gas from the decomposition of $NaHCO_3$(cr) at 374.7 and 375.0 K, where the CO_2/H_2O ratios were 1.02 and 0.98 respectively.

No details were given of the sample of potassium bicarbonate used in the experiments. The results of 18 experiments at temperatures from *ca.* 349 to 429 K were tabulated, taken where the pressure was both rising and falling. The pressures are very consistent with the linear relation:

$$\log_{10}(p\ /\text{bar}) = (8.08570 \pm 0.13933) - (3474.336 \pm 41.623)\ (T\ /\text{K})^{-1}$$

corresponding to the values:

$$\Delta_r H^o_m\ (A.5) = (66.514 \pm 1.144)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S^o_m\ (A.5) = (154.800 \pm 2.667)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

where all the uncertainties are twice the standard deviation.

These values strictly relate to the mid-temperature of the experiments, 389 K. There are no heat capacity data available for $KHCO_3$(cr) to make the correction to 298.15 K, but presumably the heat capacity change will be small, as for the corresponding decomposition of $NaHCO_3$(cr), so they will not be greatly different at 298.15 K. As noted in the main text, these would imply the values of:

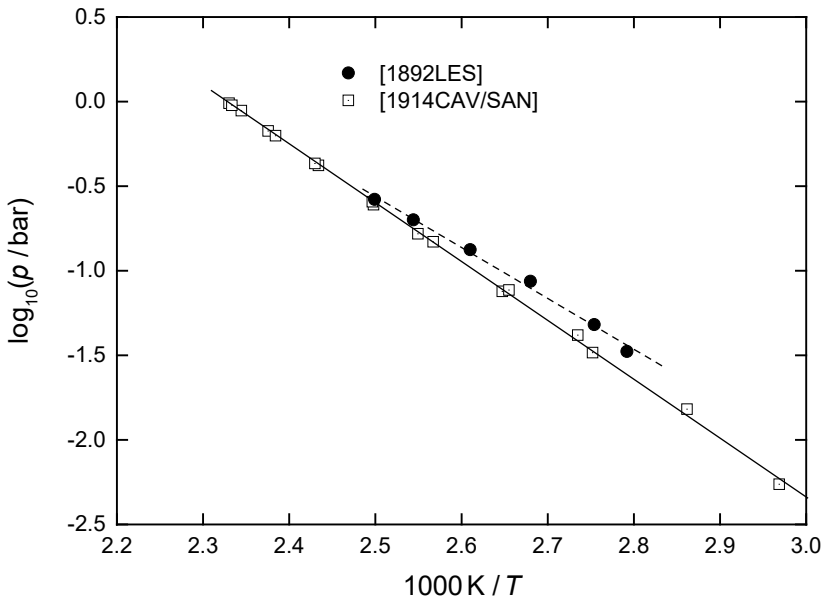
$$\Delta_f H^o_m (KHCO_3, cr, 298.15\ K) = -(959.9 \pm 1.5)\ kJ \cdot mol^{-1} \text{ and}$$

$$S^o_m (KHCO_3, cr, 298.15\ K) = (124.3 \pm 2.7)\ J \cdot K^{-1} \cdot mol^{-1}.$$

The former value is in excellent agreement with our selected value of $-(960.7 \pm 2.0)$ kJ·mol^{-1}, but the latter is probably too large compared to the value of (102.0 ± 1.0) J·K^{-1}·mol^{-1} for the standard entropy of $NaHCO_3$(cr).

Figure A-2 shows a plot of the experimental data as compared with the earlier determination by Lescoeur [1892LES].

Figure A-2: Decomposition pressure of $KHCO_3$(cr).



[1915MIX]

This paper concerns the determination by bomb calorimetry of the enthalpies of formation and of hydration of a number of compounds, of which only brucite is of relevance for this review. The fusion medium used was sodium peroxide, to which

sulfur or lampblack were added as combustible substances. Although the results were given, it is not entirely clear to this review how the thermal effects due to sulfur or lampblack were deducted from the overall heat effects. It was, thus, assumed that the results given by the author had been corrected, and were accepted unchanged.

The sample of brucite was described as being of good crystallinity, containing the theoretical quantity of water, 0.3 % of FeO and traces of Mn and Ca.

For the reaction:

$$Mg(OH)_2(cr) + Na_2O(l) \rightleftharpoons MgO(cr) + 2NaOH(l) \qquad (A.6)$$

three measurements with added sulfur yielded – 476, – 438 and – 456 cal·g$(Mg(OH)_2)^{-1}$ for $\Delta_r H^o$ (A.6), while the single measurement using lampblack, gave – 461 cal·g$(Mg(OH)_2)^{-1}$ with a mean of – 458 cal·g$(Mg(OH)_2)^{-1}$, corresponding to $\Delta_r H^o_m$ (A.6) = – 26.7 kcal·mol^{-1} (or – 111.71 kJ·mol^{-1}).

The heat effect corresponding to the combination of Na_2O with water according to:

$$Na_2O(l) + H_2O(l) \rightleftharpoons 2\ NaOH(l) \qquad (A.7)$$

was taken by the author as $\Delta_r H^o_m$ (A.7) = – 36.1 kcal·mol^{-1} (or – 151.04 kJ·mol^{-1}), from a combination of his own results and an examination of literature data.

Thus, for the dehydration reaction:

$$Mg(OH)_2(cr) \rightleftharpoons MgO(cr) + H_2O(l) \qquad (A.8)$$

the author obtained $\Delta_r H^o_m$ (A.8) = 9.4 kcal·mol^{-1} (or 39.33 kJ·mol^{-1}).

There is no indication in the paper of the temperature at which these measurements were made, and as no temperature corrections were given in the description of the processes, this review takes the same value for 298.15 K, with increased uncertainty limits, thus:

$$\Delta_r H^o_m((A.8), 298.15\ K) = (39.3 \pm 5.0)\ kJ \cdot mol^{-1}.$$

This value agrees reasonably well with that calculated from the selected data – (37.020 ± 0.425) kJ·mol^{-1}.

[1915WEL]

The solubility of magnesite (and nesquehonite) at 20 °C in air-saturated ($p(CO_2)$ = 2.9 × 10^{-4} bar) pure water and in 27.2 g·dm^{-3} (0.469 mol·kg^{-1}) NaCl solution was studied. Air was filtered through cotton and distilled water before being bubbled through suspensions of the solid in thermostatted containers for 36-61 days. The carbonate and bicarbonate concentrations were determined by titration with $NaHSO_4$, first with phenolphthalein indicator, then with methyl orange. The solubilities measured

for magnesite were 0.00074 and 0.0012 mol·dm^{-3} Mg^{2+} in pure water and in 0.469 mol·kg^{-1} NaCl, respectively.

The magnesite studied was described by the author as "amorphous natural magnesium carbonate" and originated from Placer County, Colorado (USA). Since the solubilities obtained are somewhat lower than expected, it is very likely that the material was not amorphous but microcrystalline.

[1917LEN/MER]

There had been at least fourteen studies of the solubility of silica in various forms prior to this study which involved amorphous silica in water as well as HCl and H_2SO_4 solutions at 25 and 90 °C. The solubility of SiO_2(am) decreased with increasing acid concentrations which were measured up to 40 and 70 wt % for the HCl and H_2SO_4 solutions, respectively. The solubility of gelatinous silica in conductivity water appeared to reach constant values of 0.0081 g·(50 cm^3)$^{-1}$ at 25 °C after 8-13 days and (0.0214 ± 0.0003) g·(50 cm^3)$^{-1}$ (mean of six values, 2σ) at 90 °C after 24 hours. Conversion to molality units gives 2.7×10^{-3} and 7.1×10^{-3} mol·kg^{-1}, respectively. The value at 25 °C compares to 2.2×10^{-3} mol·kg^{-1} reported by Kennedy [1950KEN] as an average for three different amorphous silicas over longer equilibration periods. The results of these earlier studies tend to show that relatively consistent solubilities could be determined for amorphous silica over long periods of time regardless of the source of the material or the amount.

[1919WHI]

This paper describes a very careful investigation of the enthalpy differences ($H(T) - H(273.15\ \mathrm{K})$) of 14 silicates, including quartz, cristobalite and vitreous silica. There is a very thorough consideration of the possible experimental errors, both systematic and random, including the temperature scale and uniformity of the temperatures in the high-temperature furnace. However, there are few details of the samples used, except that the quartz was a natural mineral, the silica glass was "prepared from quartz of practically the same composition" and the cristobalite was made by "crystallising the silica glass"; no impurity levels are given.

The results are given as values of ($H(T) - H(273.15\ \mathrm{K})$) at round temperatures of T from 373 to 1373 K for quartz, to 1673 K for cristobalite and to 1173 K for silica glass. Since these data were included in the assessments for these polymorphs that we have accepted, they are not considered in any more detail here.

[1921BAC]

The solubilities of Iceland spar (calcite) and synthetic aragonite in water have been determined in open systems ($p(CO_2)$ = 0.99-1.04 bar) at 9, 25 and 35 °C. Aragonite was prepared according to a modified method of Bourgeois [1886BOU]: less than the calculated amount of urea was added to a 1 mol·dm^{-3} $CaCl_2$ solution saturated with CO_2 in a closed tube. The solution was heated to 140 °C in a sealed tube for about two hours. At this temperature the hydrolysis of urea resulted in the formation of ammonium carbonate at low concentrations, which avoided local supersaturation. Aragonite prepared in this way contained 0.5 % calcite. Washed CO_2 gas was passed through the reaction bulb, which contained 5 g of aragonite or 60 g of calcite. The total gas pressure was adjusted to 1.04 bar. The vapour pressure of water was taken into account. Equilibrium was attained within 40-140 hours. Concentrations were determined either from the alkalinity using an HCl solution or from the electrical conductivity.

The presence of small amounts of calcite in the aragonite probably has a negligible effect. The solubility data were obtained from both supersaturated and undersaturated solutions. A notable difference between the aragonite solubilities determined from the under- or supersaturated solutions was found at 9 °C. Therefore, an uncertainty of ± 0.10 has been assigned to the $\log_{10}K_s^o(CaCO_3, \text{arag})$ value – (8.147 ± 0.100) calculated for this temperature. The reported data fit very well to the selected values.

[1921WIE]

This paper is concerned with the determination of the enthalpies of transformation of α-quartz to cristobalite, vitreous silica and chalcedony from the difference in the enthalpies of solution of the various forms in aqueous hydrofluoric acid, and the measurement of the heat capacities of these forms through their stability range up to the melting point of cristobalite. The results on chalcedony will not be discussed here, as this crystalline form is outside the scope of the present review.

Two samples of α-quartz were used: one of natural origin (Carrara) and one from the firm Heraeus. The two samples of vitreous silica were from Heraeus, Hanau, and from the Deutsche-English Quartzschmelze (Berlin-Pankow). Wietzel declares an impurity level smaller than 0.5 % for all samples.

The two cristobalite samples were obtained from precipitated silica from the firm Kahlbaum and heated, respectively, at 1373 K for 68 hours and at 1873 K for 20 minutes. For reaction calorimetry, the materials were pulverised in agate mills for many hours, so as to obtain samples with a dissolution time (*ca.* 15 minutes for 0.7 g amounts) compatible with the operation of the calorimeter. The size of the powder granules was not indicated.

The reaction calorimeter (a platinum vessel) and its mode of operation were briefly described. For α-quartz and the vitreous phase, dissolutions were performed in 35 % HF and in fuming HF (4-5 reactions for each material), while only 35 % HF was used for cristobalite (two measurements).

This review could not link the limited scatter of the enthalpies of solution with differences in the treatment of the samples prior to dissolution.

The author reports his calorimetric results without formally giving overall uncertainties that cannot be evaluated at this stage. Consequently, the uncertainties in the values of the enthalpies of transformation given below cover only the scatter of the data.

Table A-2: Enthalpies of transformation

Reaction	$\Delta_t H_m^o$ /kJ·mol^{-1}
α-quartz = cristobalite	(7.24 ± 0.30)
α-quartz = vitreous silica	(9.49 ± 0.30)

A high-temperature drop-calorimeter was used for heat capacity measurements, using samples of the same materials that were studied in the enthalpy of reaction experiments. The samples, sealed in Ag or Pt, were dropped into a copper tube inside a Dewar vessel filled with water at 273.15 K. For the two quartz samples a total of 23 measurements were made, for cristobalite 23 measurements and for vitreous silica, 12, in each case to a maximum temperature of 1973 K. The results of all these data are reported only as the mean C_p between 273.15 K and round temperatures up to 1973.15 K.

The enthalpy content values lie appreciably (up to 3.7 %) above the assessed values of Richet *et al.* [1982RIC/BOT], so were given zero weight in that fitting.

Two different studies were also made of the melting point of cristobalite. In the first, cristobalite was heated in a platinum container in an iridium furnace at (1943-1983) K and quenched to observe the formation of tiny crystals of cristobalite, from which a value of T(fus) = 1970 K was derived (no uncertainty given). In the second, small SiO_2-glass fragments were heated for a very short time to 1873 K to form a thin film of cristobalite. These were subsequently heated to higher temperatures and quenched. The disappearance of the cristobalite film and a rounding of the small glass fragments indicated a melting point of 1967 K. The author thus recommended a value of (1969 ± 10) K for the melting point. This value is significantly lower than other measurements. [1994GUR/VEY] recommend a value of (1996 ± 5) K.

[1925BAC]

The heats of solution of calcite and aragonite in 1 molar hydrochloric acid at (298.15 ± 0.10) K were determined under two different sets of conditions, in pure acid and in acid saturated with carbon dioxide. As the measurements in pure acid were only used to demonstrate the need to saturate the acid with CO_2 in view of the importance of the enthalpy of solution of CO_2, only results in CO_2 saturated media will be mentioned here.

The preparation of the aragonite and synthetic calcite samples was briefly described. A sample of Iceland spar was also used. The Dewar type calorimeter and its operation were described. Ten-gram calcium carbonate samples were dissolved in 0.800 dm^3 of 1 molar HCl.

The author reports an enthalpy of solution of – 3.245 kcal·mol^{-1} from the average of one determination with synthetic calcite and one with Iceland spar, in agreement, and an enthalpy of solution of – 3.204 kcal·mol^{-1} from the average of two determinations with the aragonite sample. These data lead to $\Delta_r H^\circ_m(298.15\ \mathrm{K}) =$ – 0.041 kcal·mol^{-1} (or – 0.172 kJ·mol^{-1}) for the calcite to aragonite transformation at (298.15 ± 0.10) K.

In a subsequent publication centred on the enthalpy of formation of calcite from its dissolution in comparable conditions in 1 molar HCl and that of CaO in the same medium, [1925BAC2], Bäckström reported an additional determination of the enthalpy of solution of calcite, yielding – 3.290 kcal·mol^{-1}. In this latter publication, the author adopted the value of – 3.260 kcal·mol^{-1} as the best value, leading to $\Delta_{trs} H^\circ_m(298.15\ \mathrm{K}) =$ – 0.056 kcal·mol^{-1} = – 0.234 kJ·mol^{-1} for the calcite to aragonite transition. Bäckström also reported – 46.200 kcal·mol^{-1} for the enthalpy of solution of CaO(cr) in the same conditions, from the weighted average of three concordant measurements. From these results, the author obtained for reaction:

$$\mathrm{CaCO_3(calc)} \rightleftharpoons \mathrm{CaO(cr) + CO_2(g)} \qquad \text{(A.9)}$$

a value $\Delta_r H^\circ_m((\text{A.9}), 298.15\ \mathrm{K}) =$ – (42.60 ± 0.20) kcal·mol^{-1} (or – (178.24 ± 0.84) kJ·mol^{-1}), after introduction of a correction of 0.330 kcal·mol^{-1} (or 1.381 kJ·mol^{-1}) for the saturation by water vapour of the evolving CO_2(g).

[1925GJA]

These brucite solubility experiments were performed in electrochemical concentration cells fitted with hydrogen electrodes. The magnesium content of the solutions was determined from the cell potential corrected for the liquid-junction “diffusion” potential by an undeclared method. Six solubility measurements were tabulated in which the added $MgCl_2$ molarity was varied (0.5, 0.1, 0.02, 0.004. 0.0008 and 0 mol·dm^{-3}). Excluding the first measurement in their Table 6, the mean of the logarithm of the

solubility product was calculated in this review to be – (10.93 ± 0.08) (2σ), *cf.* – 10.5 calculated from a provisional fit of more recent data. Similarly, from 15 of the 18 electrochemical measurements listed in Table 9, the logarithm of the mean hydrolysis constant converted to the acidic form was calculated to be – (11.61 ± 0.46) (2σ), *cf.* – 11.87 calculated from a provisional fit of more recent data. The experimental limitations include the lack of information on the purity, particle size and crystallinity of the solid magnesium hydroxide used in the solubility experiments. Equilibration times were not given. More important is the use of a cell with liquid junction that contained different solutes in each compartment so that calculation of the liquid-junction potential, particularly at the higher $MgCl_2$ concentrations, is difficult. From a data treatment point of view, the use of the simple cubic concentration term to estimate activity coefficients is inadequate by today's standards. For these reasons no attempt was made to re-treat these data.

[1928FRI]

The results of solubility experiments involving uncharacterised crystalline bayerite and gibbsite are summarised in Table A-3 along with results of re-calculations of the solubility constants in molal units.

Table A-3: Summary of bayerite experimental data at 30 °C (after three days equilibration) with the solution densities calculated iteratively from the density function converted from that in [2006REY/BER].

Solid	$c_{Al_2O_3}$ /g·dm^{-3}	c_{NaOH} /mol·dm^{-3}	ρ_{calc} /g·cm^{-3}	m_{Na^+} /mol·kg^{-1}	m_{OH^-}	$m_{Al(OH)_4^-}$	$\log_{10}K_{s,4}$
bayerite A	28.9	3.75	1.150	4.004	3.398	0.605	– 0.75
	29.8	3.75	1.154	4.003	3.379	0.624	– 0.73
	26.1	3.75	1.147	4.006	3.459	0.547	– 0.80
bayerite B	22.4	3.75	1.137	4.010	3.540	0.470	– 0.88
	26.4	3.75	1.147	4.006	3.453	0.553	– 0.80
	26.4	3.75	1.147	4.006	3.453	0.553	– 0.80
gibbsite	19.8	3.75	1.130	4.013	3.597	0.412	– 0.94
	19.8	3.75	1.130	4.013	3.604	0.409	– 0.95

Application of the SIT treatment with $\Delta\varepsilon$ fixed at – 0.039 kg·mol^{-1} [1992WES] yielded: bayerite A, $\log_{10}K^\circ_{s,4} = -(0.60 \pm 0.18)$; bayerite B, $\log_{10}K^\circ_{s,4} = -(0.66 \pm 0.24)$; gibbsite, $\log_{10}K^\circ_{s,4} = -(0.78 \pm 0.18)$; (2σ) at 30 °C.

[1929BJE/UNM]

Potentiometric measurements were made using a cell with liquid junction fitted with $Pt|H_2$ and $HgCl_2|Hg$ electrodes. Ionic strengths for the phosphate system were between 0.1 and 0.18 mol·dm^{-3}. Activity coefficients were approximated with an extended Debye-Hückel equation. Stepwise dissociation constants for H_3PO_4(aq) were determined at 18, 25 and 37 °C. Grenthe *et al.* [1992GRE/FUG] included the value of $\log_{10}K_3^\circ = -(12.35 \pm 0.03)$ at 25 °C in their evaluation. The current review includes all nine values reported by Bjerrum and Unmack in the databases for the three acid dissociation equilibria of H_3PO_4(aq).

[1929FRI3]

The results of solubility experiments involving uncharacterised bayerite are given in Table A-4. Application of the SIT treatment with $\Delta\varepsilon$ fixed at -0.039 kg·mol^{-1} [1992WES] yielded: $\log_{10}K_{s,4}^\circ = -(0.62 \pm 0.24)$ (2σ) at 30 °C.

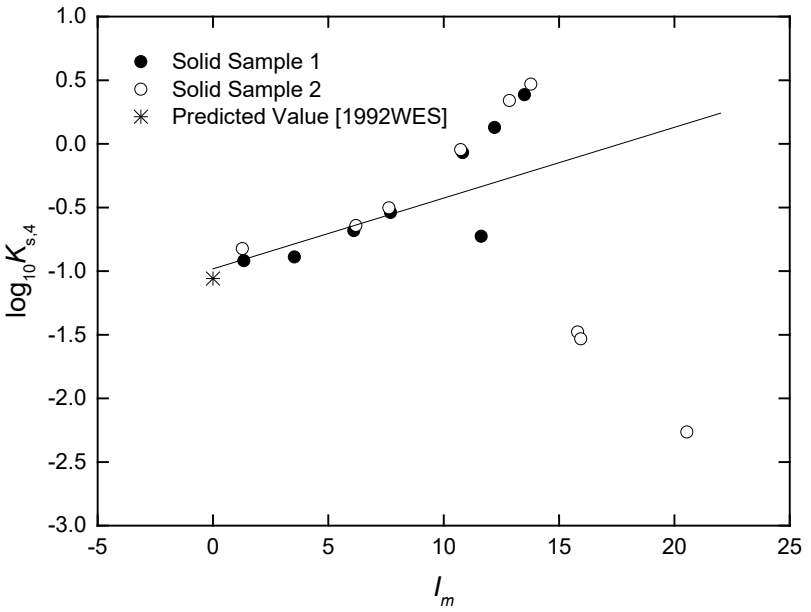
Table A-4: Summary of experimental data from a bayerite solubility experiment conducted at 30 °C with solution densities calculated from a density function converted from that of [2006REY/BER]. (The measured concentrations in columns 1 and 2 were taken from a figure in [1929FRI] by Apps *et al.* [1988APP/NEI]).

$c_{Al_2O_3}$ /g·dm^{-3}	c_{NaOH} /mol·dm^{-3}	ρ_{calc} /g·cm^{-3}	m_{Na^+} /mol·kg^{-1}	m_{OH^-}	$m_{Al(OH)_4^-}$	$\log_{10}K_{s,4}$
27.5	3.73	1.150	3.981	3.405	0.576	– 0.77

[1930FRI/JUC]

Slow precipitation produced gibbsite (labelled as XXI in this paper), whereas more rapid precipitation produced a "somewhat younger solid XX" with an XRD pattern, more consistent with bayerite. These measurements covered a wide range of NaOH concentration to the extent that sodium aluminate began to precipitate after the aluminium concentration reached a maximum. SiO_2 concentrations were found to be < 0.1 % but no mention was made of how CO_2 was excluded during the handling of the reaction mixtures. A sample of their data was re-treated here for $Al(OH)_3$(s) equilibrated for six weeks at 30 °C. The limited data fit shown in Figure A-3 was based on a cut-off at $I_m < 7$ corresponding to the expected limit of the SIT and resulted in $\log_{10}K_{s,4}^\circ = -(0.98 \pm 0.11)$ and $\Delta\varepsilon = -(0.056 \pm 0.020)$ kg·mol^{-1}, (2σ statistical uncertainties only) in reasonable agreement with $\log_{10}K_{s,4}^\circ = -1.058$ [1992WES] at 30 °C.

Figure A-3: A plot of $\log_{10}K_{s,4}$ *versus* I_m at 30 °C; the asterisk indicates the value of $\log_{10}K_{s,4}$ from [1992WES].



The results at 30 °C show reasonable agreement for the two solid phases despite the comment by Fricke and Jucaitis that bayerite could have been present in the XX sample, noting that the presence of silica is known to promote the formation of bayerite at low temperatures. Finally, the apparent formation or deposition of sodium aluminate as the solubility-limiting phase apparently occurs abruptly near $I_m \sim 14$, although only after considerable over-saturation with respect to gibbsite.

[1930THO/BRO2]

This paper, in which a determination of the heat of hydration of calcium oxide is reported, is the third of a series. The first paper [1929THO/BRO] concerned the heat of solution of calcium oxide in hydrochloric acid and the second [1930THO/BRO], the heat of solution of calcium hydroxide, both in (HCl, $200H_2O$). The comments below thus include the results of the first two papers of the series to the extent of their involvement with the third [1930THO/BRO2].

The adiabatic calorimeter used has been described in [1929THO/BRO], together with details on qualification and calibration experiments. Reference to earlier

literature is given for more details. For the experiments described in the three papers, 500 to 600 g of solvent (water or dilute hydrochloric acid) were used.

The CaO(cr) samples for calorimetry were obtained by freshly igniting (1073-1473 K) in platinum crucibles various calcium carbonate and calcium oxalate samples resulting from several preparative schemes. In contrast with this great care in the preparative schemes, no detailed analysis of the resulting samples was given.

Two techniques were used [1930THO/BRO] for the preparation of $Ca(OH)_2$(cr), namely: a) prolonged exposure of CaO(cr) to water vapour at 294 K in an evacuated vessel; b) exposure of CaO(cr) to saturated steam in an autoclave at 423 K for 24 hours or more.

The samples prepared by method a) contained a 11-15 % water in excess of stoichiometry, that could be removed down to a few tenths of a percent by placing them over CaO(cr) in vacuum. The samples prepared by method b) were *almost* of theoretical composition after drying over CaO(cr). The morphology and reactivity of the two types of sample were different, although the X-ray diffraction patterns were reported as identical.

As a result of 11 determinations of the enthalpy of solution of 3.6 to 4.5 g samples of CaO(cr) in 500.3 g saturated lime water, an average value of -276.8 cal·(g CaO)$^{-1}$ was reported at 293 K [1930THO/BRO2], corrected to -276.3 cal·(g CaO)$^{-1}$ (or -64.83 kJ·(mol CaO)$^{-1}$), by introduction of a correction (variable with the sample history) due to supersaturation of the calorimetric solutions. This question of supersaturation is also discussed in [1930THO/BRO].

In [1930THO/BRO2] Thorvaldson *et al*. also make use of the enthalpy of solution of CaO(cr) in (HCl, 200H_2O), -829.2 cal·(g CaO)$^{-1}$ (although the original reports by [1929THO/BRO] mentions -828.9 cal·(g CaO)$^{-1}$), and of $Ca(OH)_2$(cr) in the same medium, -550.3 cal·(g CaO)$^{-1}$ as an average from three different sets of data (each one with 5-9 determinations), after a correction of -0.5 to -3 cal·(g CaO)$^{-1}$ for the enthalpy of wetting of the $Ca(OH)_2$(cr) samples. The range of this measured enthalpy of wetting reflects differences between the samples of different origin [1930THO/BRO].

These figures lead to -278.9 cal·(g CaO)$^{-1}$ (or -65.44 kJ·(mol CaO)$^{-1}$) for the enthalpy of hydration of CaO(cr), a value that is 0.61 kJ·(mol CaO)$^{-1}$ more negative than that obtained from the dissolution of CaO(cr) in saturated lime water.

On the basis of microscopic examinations, the authors suggested that the discrepancy may be due to differences in the reactivity and surface energy of the CaO(cr) and the $Ca(OH)_2$(cr).

[1931JOH/GRO]

Johnston and Grove were careful to prepare pure, crystalline calcium hydroxide particles and proposed that the scatter in previously published results was due to the presence of fine particles or even colloids. Earlier work had also shown that it was necessary to work at partial CO_2 pressures of $< 10^{-13}$ atm although the presence of some $CaCO_3$(s) would not necessarily jeopardise the experiments due to its much lower solubility. A special glass apparatus was constructed which allowed the solution to be driven back and forth through a packed column of $Ca(OH)_2$(cr) at (25.00 ± 0.02) °C. These experiments were conducted as a function of ionic strength in the presence of excess NaCl, KCl, CsCl, LiCl, $BaCl_2$, $SrCl_2$, $NaClO_3$, $NaClO_4$, NaBr, KBr, $NaNO_3$, $NaNO_2$, NaI and $NaC_2H_3O_2$. The equilibrium in question is given by Reaction (A.10):

$$Ca(OH)_2(cr) \rightleftharpoons Ca^{2+} + 2OH^- \quad \text{(A.10)}$$

$$\log_{10} K_{s,0} - 6D = \log_{10} K^{\circ}_{s,0} - \Delta\varepsilon I_m \quad \text{(A.11)}$$

Application of the SIT according to Equation (A.11) was possible because the supporting electrolyte was maintained in sufficient excess over that of $Ca(OH)_2$(sln) except at the lowest concentrations. A correction for hydrolysis was applied; the value of $\log_{10} K^{\circ}_1$ for the formation of $CaOH^+$ was taken as – 1.34. However, the corresponding value of $\Delta\varepsilon$ had to be derived from the formation of $CaOH^+$ and the SIT-based value for the dissociation of water in NaCl medium, such that the hydrolysis correction applied strictly only to that medium. The mean corrected value was $\log_{10} K^{\circ}_{s,0} = -(5.24 \pm 0.04)$, (2σ). The mean solubility of $Ca(OH)_2$(cr) was (0.0114 ± 0.0003) mol·kg^{-1} at 25 °C and is identical to that obtained from a SIT treatment of the data in [1967YEA/MAR] and very close to that proposed in the compilation in [1987GAR/PAR]. The measured $\Delta\varepsilon$ value in NaCl media was (0.184 ± 0.010) kg·mol^{-1} which compares favourably with $\Delta\varepsilon$ = (0.22 ± 0.02) kg·mol^{-1} derived from the NEA TDB recommended values for the NaCl solutions.

[1932BAR]

The solubility of natural calcite was determined (at 15.5, 25 and 30 °C, in aqueous solutions with $p(CO_2)$ = 0, 0.00031, 0.958, 1 and 2 bar, as well as natural and freshly precipitated magnesite (at 18 °C) in CO_2-free and air-saturated ($p(CO_2)$ = 0.00031 bar) solutions).

No clear analytical method is mentioned in the paper. The solubilities of freshly precipitated $MgCO_3$ in both solvents and the natural magnesite in CO_2-free aqueous solution are too high as compared to the selected values. Therefore, only the solubility of natural magnesite in air-saturated water (0.08 g·dm^{-3} $MgCO_3$ or 0.00095 mol·dm^{-3}) is accepted. On the other hand, the reported calcite solubilities in open systems are considerably lower than the selected values, while those reported for

the closed systems are considerably scattered. Therefore, no data for calcite solubility are accepted for this review.

[1932MAT/SEO]

This early study of the dissolution at room temperature of several hydrated and anhydrous calcium phosphates in dilute hydrochloric acid should, for its most part, be considered of historical interest only, given the lack of information as to the analysis of the samples and the precise concentration of the species dissolved. Nevertheless, it is worth noting that, from the reported enthalpies of dissolution of $Ca(H_2PO_4)_2{\cdot}H_2O$(cr) and of the corresponding anhydrous species in 0.5 molar HCl at an unspecified temperature (but presumably room temperature), Matignon and Séon give for Reaction (A.12):

$$Ca(H_2PO_4)_2{\cdot}H_2O(cr) \rightleftharpoons Ca(H_2PO_4)_2(cr) + H_2O(l) \qquad (A.12)$$

a value of 20.7 kJ·mol^{-1}, in reasonable agreement, considering the circumstances, with that calculated by this review from the results of Egan *et al.* [1956EGA/WAK] (19.135 ± 0.542 kJ·mol^{-1}).

[1932MON/DAV]

No new conductance results were reported and a modified Onsager equation was applied for incomplete dissociation of 2:2 electrolytes, including $CaSO_4$(sln) and $MgSO_4$(sln). For the former, Money and Davies treated two datasets at 18 °C from [1905LAN] and [1919HAR/PAI], and proposed a mean value for the association constant at infinite dilution of 189 dm^3·mol^{-1}. Re-treatment with an SIT compatible Lee-Wheaton equation [1978LEE/WHE2] of seven of the eight molar conductance results gave $K_1^{\circ}(CaSO_4, aq)$ = (97 ± 22) dm^3·mol^{-1}; this 2σ statistical uncertainty was doubled to ± 44 because the raw data came from an unknown source. This smaller value attests to the sensitivity of the ion association constant to the conductance equation chosen to represent the data. Values for the formation constant of $MgSO_4$(aq) and $NiSO_4$(aq) are given as 159 and 250 dm^3·mol^{-1}, respectively. The NEA selected value for the latter is (226.1 ± 15.2) dm^3·mol^{-1} [2005GAM/BUG]. The re-calculated limiting molar conductance for $CaSO_4$(sln) is 243.3 S·m^2·mol^{-1}.

[1933AND]

Anderson measured the low-temperature heat capacity from *ca.* 56 to 290 K of sodium carbonate and bicarbonate in an adiabatic calorimeter. The materials were commercial samples, which unspecified analyses showed contained respectively < 0.1 % and *ca.* 0.2 % impurities. No corrections were made.

The detailed measurements (17 for each material) are tabulated, and the heat capacities were extrapolated from 56 K down to 0 K using a sum of two Debye and one Einstein functions and also graphically, giving standard entropies of (136.0 ± 2.5) and 138.1 $J \cdot K^{-1} \cdot mol^{-1}$ respectively for the Na_2CO_3(cr) and (102.1 ± 1.7) and 102.1 $J \cdot K^{-1} \cdot mol^{-1}$ for $NaHCO_3$(cr). No explicit data are given for $C^{o}_{p,m}$ (298.15 K), but the values read from the graph gives values of 110.1 and 87.6 $J \cdot K^{-1} \cdot mol^{-1}$.

Their heat capacities and derived entropy values for Na_2CO_3(cr) agree to within 2 % of the later measurements of [1969STA/LIN] but because of much closer approach to 0 K of the latter, their data have been preferred in this review.

[1933GRO/SCH]

The authors reported the enthalpy differences (as mean specific heats) of the title compounds (and also $CaCO_3$(cr)) between specified high temperatures and 273.15 K, using a drop calorimeter. Those compounds of interest to this review are: CaO(cr), (eight temperatures up to 1773 K), Al_2O_3(cr) (seven temperatures up to 1773 K) and $CaCO_3$(calc), (eight temperatures to 1173 K). The only indication of purity of the samples is that they were chemically pure, (manufacturer's guarantee). Before use, the completely crystalline corundum was heated to 1800 °C, the CaO(cr) (prepared by decomposition of highly pure $CaCO_3$(cr)) was annealed at 1500 °C for a long period. No details of the sample of $CaCO_3$(cr) are given, and it was assumed by the review to be pure calcite.

For the measurements, 1-3 g samples contained in a thin Pt capsule, were heated in a Ni-Cr or Pt-Rh furnace and dropped into a copper-water calorimeter. Each measurement was repeated five times.

The results are in good agreement with most other data, and were included in the assessment by [1994GUR/VEY], whose data we have selected for this review.

[1933LAT/HIC]

The heat capacities of two samples of gypsum ($CaSO_4 \cdot 2H_2O$, cr) and one sample of $BaSO_4$(cr) were measured at low temperatures. The C_p against $\log_{10}(T)$ plots of the molal heat capacities of the two samples of gypsum differing in crystal surface by a factor of roughly 10^6 differed slightly by amounts thought by the authors to be outside the limits of experimental error.

A commercial sample of $CaSO_4 \cdot 2H_2O$(cr) was intimately stirred with water, filtered, dried and heated to 520 °C in a muffle furnace. The water content was 20.85 % (theor. 20.93 %) and unspecified analysis showed that it contained only a trace of

sodium, but no carbonate or barium. The crystallite size was approximately 1 μm. About 50 measurements were made on this sample from 18.7 to 302.1 K.

The natural gypsum was obtained from a large clear single crystal (500 g) which had a water content of 20.88 %. Most of the material used was bigger than 1 cm and all fine powder was excluded. About 30 measurements were made on this sample from 62.4 to 274.1 K; measurements were not extended below 60 K since the values at the lower temperatures in this range agreed closely with those from the smaller crystals.

The full data for both samples were tabulated. For the calculation of entropy the data were extended to 0 K by a T^3 extrapolation with a Debye temperature of 167.2 K, followed by graphical integration, which gave S°_m ($CaSO_4 \cdot 2H_2O$, cr, 298.15 K) = 46.36 $cal \cdot K^{-1} \cdot mol^{-1}$ with an unspecified uncertainty of ± 0.05 %. We therefore derive a value of S°_m ($CaSO_4 \cdot 2H_2O$, cr, 298.15 K) = (194.0 ± 0.8) $J \cdot K^{-1} \cdot mol^{-1}$ from this study. The value of $C^\circ_{p,m}$ (298.15 K) was not given, but by interpolation from the tabulated values near 298.15 K is calculated to be (44.8 ± 0.2) $cal \cdot K^{-1} \cdot mol^{-1}$ = (187.4 ± 0.8) $J \cdot K^{-1} \cdot mol^{-1}$.

[1933NIM]

This early potentiometric study of the second dissociation constant of phosphoric acid utilised a Harned cell from 25 to 50 (± 0.01) °C in 5 °C intervals. The ionic strength (NaCl) was < 0.167 $mol \cdot kg^{-1}$. Eight measurements were reported for mixtures of NaH_2PO_4 and Na_2HPO_4, and a second series of eight measurements were made with a different ratio of KH_2PO_4 and NaH_2PO_4 at each temperature. A simple Debye-Hückel equation was employed to estimate activity coefficients. The average values of $\log_{10} K^\circ_2$ from 25 to 50 °C are: – 7.218, – 7.206, – 7.197, – 7.192, – 7.189, – 7.189 and – 7.192, respectively. The standard deviation between the two sets of values is < 0.001. The value at 25 °C agrees quite well with – 7.206 [2005PAR/COV2], – 7.198 [1943BAT/ACR] and – 7.212 [1992GRE/FUG]. The agreement improved with increasing temperature.

[1934AND]

The author measured the low-temperature heat capacity at 14 temperatures from 56 to 292 K of a naturally-occurring (Norwegian) sample of $MgCO_3$(cr) containing 9.0 % $CaCO_3$ and 1.3 % $FeCO_3$ (not just the 0.3 % $CaCO_3$ given in [1994GUR/VEY]). After correction for these impurities, the heat capacity was extrapolated from 56 K down to 0 K using a sum of two Debye and one Einstein functions and also graphically, giving standard entropies of (65.7 ± 0.8) and (66.1 ± 0.8) $J \cdot K^{-1} \cdot mol^{-1}$ respectively.

Their heat capacities (and hence entropy values) are slightly higher than the consistent measurements of [1977HEM/ROB3] and [1994ROB/HEM] using much purer magnesite samples and the later data have been preferred in this review.

[1934AND2]

The author measured the low-temperature heat capacity from 57 K to 296 K of two naturally-occurring (Utah) samples of $CaCO_3$(calc) and one sample of aragonite (California). One sample of calcite was a coarse powder, 99.97 % pure, the second was comminuted to a fine powder (*ca.* 5 μm size). The latter was less pure (98.656 %), and corrections were made for the presence of 0.762 % $Ca(OH)_2$ and 0.618 % H_2O. The aragonite was > 99.9 % pure and no corrections were made.

For the calcite samples, there was no discernible difference in the heat capacities of the two samples at the lower temperatures, but the values for the fine sample were progressively slightly higher at the higher temperatures. The heat capacity was extrapolated from 57 K down to 0 K using a sum of one Debye and two Einstein functions and also graphically, giving standard entropies at 298.15 K of (93.05 ± 1.67) and 93.30 $J \cdot K^{-1} \cdot mol^{-1}$ respectively for the coarse sample, while graphical integration gave 93.72 $J \cdot K^{-1} \cdot mol^{-1}$ for the fine sample.

For the aragonite sample, 20 measurements were made from 54 K to 291 K, and the heat capacities were extrapolated to 0 K using the sum two Debye and one Einstein functions, giving standard entropies at 298.15 K of (88.62 ± 1.67) and 88.95 $J \cdot K^{-1} \cdot mol^{-1}$ respectively.

Their heat capacities and derived entropy values for both substances agree to within 1 % of the later measurements in [1969STA/LIN] but because of much closer approach to 0 K of the latter, their data have been preferred in this review. The C_p values from the most recent study [2007NAN/SHI] are noticeably higher than the earlier values at lower temperatures, but in very good agreement at 298.15 K.

[1934OWE]

The technique and data treatment used in this potentiometric investigation of the acid dissociation of boric acid from this laboratory were described in [1935OWE] and [1944MAN/DEL]. The temperature was varied from 10 to 50 °C while the ionic strength was ≤ 0.1 $mol \cdot kg^{-1}$ ($m_{H_4BO_4^-} + m_{NaCl}$). The resulting acid dissociation constants are in excellent agreement with those from a number of subsequent studies.

[1935HAL/RIT]

The aqueous solubilities of natural magnesite (Kraubath, Obersteiermark, Ca-free) were determined at 25 °C and 38.8 °C and that of calcite (Iceland) at 38.8 °C, all at $p(CO_2) \sim 1$ bar. The solids were kept in suspension for 6-30 days by passing a stream of CO_2(g) through the thermostatted reaction vessel. After equilibrium was reached, 200 mL samples were titrated with H_2SO_4 to determine the alkalinity of the solution. The calcium content was determined by precipitation of $CaSO_4$. The solubilities of magnesite at 25 °C ($p(CO_2)$ = 0.955 bar) and 38.8 °C ($p(CO_2)$ = 0.933 bar) were 0.01657 and 0.01298 mol·dm^{-3}, respectively, and that of calcite at 38.8 °C ($p(CO_2)$ = 0.918 bar) was 0.00703 mol·dm^{-3}. These values agree relatively well with the selected solubilities.

[1935HIT]

Hitchen gives an incredibly detailed description of the experimental method and of the large capacity pressure vessel employed in this study of amorphous silica solubility in water to 336 °C. Data are tabulated from 128 °C and are reproduced in Table A-5, although Figure 8 in this paper indicated that two additional values were derived at 25 and 90 °C. Generally, 24 to 48 hours was considered adequate to attain equilibrium.

Table A-5: Solubility of amorphous silica in water taken from Table I of [1935HIT], limited to temperatures < 200 °C.

t /°C	p_{sat} /MPa	Equilibration time /hours	Reported solubility/weight % SiO_2	Solubility /mol·kg^{-1}
128	2.54	36	0.0495	8.24×10^{-3}
128	2.54	60	0.0492	8.19×10^{-3}
130	2.70	48	0.0530	8.83×10^{-3}
130	2.70	72	0.0520	8.66×10^{-3}
159	6.03	24	0.0673	1.12×10^{-2}
159	6.03	48	0.0662	1.10×10^{-2}
168	7.54	24	0.0706	1.18×10^{-2}
170	7.91	72	0.0746	1.24×10^{-2}
195	13.98	24	0.0900	1.50×10^{-2}
195	13.98	48	0.0903	1.50×10^{-2}

These results yielded solubility constants which are virtually equivalent to infinite dilution values because of the low concentrations of the neutral $Si(OH)_4$(aq). Regression of these data as a function of (T /K) gave:

$$\log_{10} K^{o}_{s,4} = -(0.321 \pm 0.047) - (704.82 \pm 20.30)(T/\mathrm{K})^{-1}$$

with $r^2 = 0.994$. The value of m_{Si} extrapolated to 25 °C is 2.07×10^{-3} which compares to 2.33×10^{-3} [1962SIE].

[1935KEL/AND]

Kelley and Anderson briefly summarised the data for carbonates and bicarbonates available in 1935, and discussed some implications, particularly on their decomposition. The only interest to this review is their estimated value for the standard entropy of $KHCO_3$(cr), for which there are still no experimental data. Their value of S^{o}_{m} ($KHCO_3$, cr, 298.15 K) = 108.8 $J \cdot K^{-1} \cdot mol^{-1}$, from 'certain analogies involving sodium and potassium compounds' is smaller than the 115.5 $J \cdot K^{-1} \cdot mol^{-1}$ given in [1982WAG/EVA], whose source is not clear.

[1935LAS/KOM]

The authors measured the enthalpy content of a brucite ($Mg(OH)_2$) mineral from 393 to 945 K by drop calorimetry. However, their sample contained more than 3.5 % impurities, given as unspecified sesquioxides 3.29 %, silica 0.19 % with traces of carbonates and Mn oxides. It is not clear whether any corrections were made for these impurities.

Although the sample was contained in a sealed Pt tube, only the nine measurements below 700 K relate to the undecomposed hydroxide. The measured enthalpies agree reasonably with those of [1975KIN/FER], but they are more scattered, and were not used in the C_p fitting.

The authors also measured the heat capacity of $Ca(OH)_2$(cr). As mentioned in Section V.6.2.1.2.2, the fit of these data by [1987GAR/PAR] and [1994GUR/VEY] was accepted by this review.

[1935OWE]

This potentiometric investigation of the acid dissociation of boric acid utilised the same cell fitted with hydrogen and Ag|AgCl (and two Ag|AgI) electrodes as used elsewhere [1934OWE], [1943OWE/KIN]. The temperature was varied from 5 to 40 °C at 5° intervals while the ionic strength was 0.01 $mol \cdot kg^{-1}$ ($m_{H_3BO_3(aq)} = m_{H_4BO_4^-} = m_{KCl}$). The activity coefficient ratio of chloride (iodide) to borate was considered to be unity for the specific conditions chosen in this study. The dissociation constants tabulated for the Ag|AgI reference electrode are in excellent agreement with those reported in

[1943OWE/KIN] where NaCl was the supporting electrolyte up to molalities of 3 mol·kg^{-1}.

[1935PEA/BLA]

Vapour-pressure measurements were made of aluminium and calcium nitrate solutions at 25 °C using an apparatus described previously [1932PEA/NEL]. These measurements at selected molalities extended to 8.3601 (saturation) mol·kg^{-1} for $Ca(NO_3)_2$(sln) and from 0.0 to 3.1607 (saturation) mol·kg^{-1} for $Al(NO_3)_3$(sln). Partial molar volumes were also given at each molality. An SIT treatment resulted in a value of $\varepsilon(Al^{3+},NO_3^-) = (0.25 \pm 0.02)$ kg·mol^{-1} over the entire range of I_m with the caveat that the lower ionic strength data, *i.e.*, $I_m \leq 6$, are very scattered.

[1935SOU/MIL]

Southard and Milner measured the low-temperature C_p of α and β-$Ca_3(PO_4)_2$ (the β-polymorph is the stable form at 298.15 K and up to *ca.* 1383 K). We shall not deal with the α-form in any detail in this review, but some details are included here for completeness.

The α-form was prepared from $Ca(OH)_2$(cr) and H_3PO_4(sln) and heated to 1673 K for one hour and quickly cooled; it contained < 0.1 % impurities. The β-form was prepared from Na_3PO_4(cr) and $Ca(NO_3)_2$(cr) at 1673 K, then 1173 K for one hour, followed by slow cooling. The diffraction pattern was merely reported to be 'that of the β-phase'. However, chemical analysis showed 2.1 mol % Na_3PO_4 and 0.1 % SiO_2. Approximate corrections were made for these, but since the C_p of Na_3PO_4(cr) was not known, values were estimated from those for α-$Ca_3(PO_4)_2$, Na_2CO_3 and $CaCO_3$, but no details are given. This correction leads to an uncertainty of ± 0.5% in the C_p values of this phase.

Southard and Milner reported values of S(α, 298.16 K) and S(β, 298.16 K) in calories which correspond to:

$$S^\circ_m(Ca_3(PO_4)_2, \text{cr}, \alpha, 298.15\ \text{K}) = (240.9 \pm 0.6)\ \text{J·K}^{-1}\text{·mol}^{-1} \text{ and}$$

$$S^\circ_m(Ca_3(PO_4)_2, \text{cr}, \beta, 298.15\ \text{K}) = (236.0 \pm 1.6)\ \text{J·K}^{-1}\text{·mol}^{-1}.$$

No values of $C^\circ_{p,m}$ (298.15 K) were given, only the uncorrected experimental data up to 286.63 K for α and 297.48 K for β, plus the corrected C_p values at round temperatures, including 280 and 300 K.

As noted in Section V.6.2.1.6.2, these data have been combined with the high-temperature values obtained by Britske and Veselovskii [1937BRI/VES]

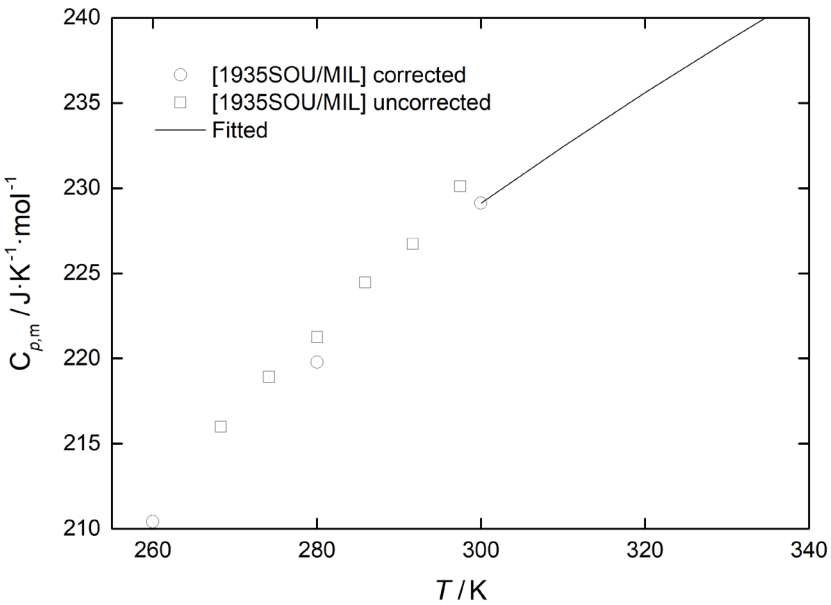
with the constraint of $C^{o}_{p,m}$ ($Ca_3(PO_4)_2$, cr, β, 300 K) = 229.116 J·K^{-1}·mol^{-1} from [1935SOU/MIL].

The resulting $C^{o}_{p,m}$ equation gives:

$$C^{o}_{p,m}\ (Ca_3(PO_4)_2,\ cr,\ \beta,\ 298.15K) = (228.481 \pm 1.600)\ J{\cdot}K^{-1}{\cdot}mol^{-1}$$

where the uncertainty has been estimated by the review. The details of the fit around 300 K are shown in Figure A-4.

Figure A-4: The heat capacity of β-$Ca_3(PO_4)_2$ /J·K^{-1}·mol^{-1}.



[1936AND]

The heat capacities of quartz, cristobalite and tridymite were measured from *ca.* 54 K to *ca.* 295 K and the standard entropies at 298.15 K calculated both by graphical integration and by fitting the heat capacities to the sum of Debye and Einstein functions. The quartz was a purified natural sample, containing 0.07 % impurities. The cristobalite sample was obtained by inductively heating a different quartz sample in a graphite tube for 90 minutes at 1600 °C; petrographic analysis showed essentially complete conversion to cristobalite (99.99 % purity). The tridymite was prepared by heating a purified quartz with 14 wt % of Na_2WO_4 as a flux in a SiC furnace for four hours at 1250 °C. After washing with a solution of hydrated ammonia, reheating and rewashing, analyses showed it contained 99.46 % SiO_2. Unspecified petrographic examination

indicated complete conversion to tridymite, though it is not clear how reliable such examinations are for tridymite, since the phase composition of individual grains depends on the grain size; we note also that in his work on the polymorphic transformations of tridymite, Sinel'nikov [1956SIN] indicated that considerable doubt has been expressed in the literature as to the reliability of such analyses.

The detailed values for 20-25 runs for each polymorph are tabulated from which the reproducibility near 298 K was judged to be *ca.* 1 %. The values of S^{o}_{m} (298.15 K) for quartz, cristobalite and tridymite obtained by graphical integration (the author's preferred method) were respectively 42.09, 43.26 and 43.93 $J \cdot K^{-1} \cdot mol^{-1}$.

[1937BRI/VES]

The authors have studied the enthalpy content of $Ca_3(PO_4)_2$(cr) at 11 temperatures between 871 and 1571 K (spanning the $\beta \rightarrow \alpha$ transition temperature), using a drop calorimeter. However, the temperature of the bath into which the samples were dropped is not stated unambiguously, either in this paper, or related ones; internal evidence suggests, however that it was 298.15 K, and the data have been processed for this value. Some other experimental details are also unclear. The results are reported as the average C_p. From the enthalpy-drop measurements, the transition to the α-form was observed to occur between 1368 and 1398 K, with an associated enthalpy change of *ca.* 19 $kJ \cdot mol^{-1}$. A more precise value of T_{trs} = (1373 ± 10) K was found by slow cooling of a sample with an embedded thermocouple; this is slightly lower than the values (1383 and 1393 K) found in phase diagram studies ([1959NUR/WEL], [1969FIX/HEY]).

The data for the low-temperature β-form have been re-fitted subject to the constraint of $C_{p,m}(Ca_3(PO_4)_2, cr, \beta, 300\ K) = 229.116\ J \cdot K^{-1} \cdot mol^{-1}$ given by [1935SOU/MIL], in their low-temperature study, resulting in the equation:

$$C_{p,m}(Ca_3(PO_4)_2, cr, \beta, T)\ /J \cdot K^{-1} \cdot mol^{-1} = 208.755 + 0.159058(T/K) - 2.46211 \times 10^{6}(T/K)^{-2}$$
(298.15-1383 K)

from which we derive:

$$C^{o}_{p,m}(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = (228.48 \pm 1.60)\ J \cdot K^{-1} \cdot mol^{-1}$$

where the uncertainty has been estimated by the review.

Somewhat surprisingly, the experimental average C_p(298.15 K – T) for the α-phase is essentially constant (and may even decrease slightly) from 1395 to 1571 K, which would imply that the actual C_p of this phase decreases as the temperature increases. However, the relatively slow transition at *ca.* 1373 K will mean these values for the α-phase have considerable uncertainty. A comparison of the fitted and experimental values is shown in Figure V-53 (Section V.6.2.1.6.2).

[1937BRO/HAS]

In this paper the solubility of aragonite was determined at 38 °C and at an ionic strength equal to that of blood serum (I_m = 0.012-0.23 *i.e.*, 0.01-0.025 mol·dm^{-3} $NaHCO_3$ and 0-0.2 mol·dm^{-3} NaCl). After equilibration (20-40 hours) solutions were centrifuged and the following properties of the supernatant solution were determined: pH (colorimetrically by phenol red), dissolved CO_2 (volumetrically, based on the method of Van Slyke and Neil [1924SLY/NEI]) and dissolved Ca^{2+} (according to the method described by Clark and Collip [1925CLA/COL]).

Browman and Hastings studied an open system, but the partial pressure of CO_2 was not reported (CO_2 was present in the gas phase in an amount such that the pH at equilibrium varied between 7.2 and 7.6). Therefore, re-evaluation of the solubility data is not possible. However, the extrapolation of the reported $\log_{10} K_s$ values to zero ionic strength, using SIT, resulted in a reasonable value. Moreover, the solubility ratio aragonite/calcite determined by the authors is also close to that expected. Therefore, the extrapolated value of $\log_{10} K_s^\circ$ for the reaction $CaCO_3(arag) \rightleftharpoons Ca^{2+} + CO_3^{2-}$ has been accepted with a relatively large uncertainty ($\log_{10} K_{s,0}^\circ = -(8.41 \pm 0.12)$).

[1937GIA/ARC]

As indicated in its title, the ultimate purpose of this study was to obtain an accurate value of the entropy of water by the third law of thermodynamics. This analysis of the paper, however, will be restricted to the thermodynamics of MgO(microcr) and $Mg(OH)_2$(cr).

$Mg(OH)_2$(cr) was prepared by heating $MgCl_2 \cdot 6H_2O$, KOH and water to 210 °C, cooled slowly so that the soluble residue dissolved away. After the residual $Mg(OH)_2$(cr) was dried at 100 °C for two hours, analysis showed it contained 0.15 % K and less than 0.005 % phosphate (from the KOH used). The MgO(cr) was prepared from this $Mg(OH)_2$(cr) by heating to 300 °C *in vacuo* for 10 days, followed by a short excursion to 350 °C, but it still contained *ca.* 5 % $Mg(OH)_2$(cr), for which corrections were made. For the stated purpose of the authors, this was the correct material to use, rather than a macro-crystalline sample, and they note later in the paper that their heat capacities of MgO(cr) (and thus also the calculated entropy) are indeed higher than those from a sample prepared from fused MgO. The heat capacities were measured in a conventional adiabatic calorimeter from *ca.* 20 to 300 K, using *ca.* 173 g $Mg(OH)_2$ and *ca.* 118 g MgO.

The calculated entropies, after correction to modern atomic weights, are S_m°(MgO, microcrystalline, 298.15 K) = (27.85 ± 0.15) J·K^{-1}·mol^{-1} and S_m°($Mg(OH)_2$, cr, 298.15 K) = (63.12 ± 0.15) J·K^{-1}·mol^{-1}. The former is noticeably larger than the selected value for MgO(macrocrystalline), (26.95 ± 0.15) J·K^{-1}·mol^{-1}. The selected value for S_m°($Mg(OH)_2$, cr, 298.15 K), based on the assessment by

Gurvich *et al.* [1994GUR/VEY], is (63.18 ± 0.40) $J \cdot K^{-1} \cdot mol^{-1}$, due presumably to a very slightly different extrapolation below 20 K.

The authors also measured the enthalpy of the reaction:

$$Mg(OH)_2(cr) \rightleftharpoons MgO(microcr) + H_2O(l) \qquad (A.13)$$

at 298.15 K by dissolution of the solids in 1.000 molar HCl solution.

For $Mg(OH)_2$, 5.000 g samples were dissolved in 0.9 dm^3 HCl(sln); for MgO the required weight (3.5386 g) (taking into account the $Mg(OH)_2$ impurity) to produce an identical final solution were dissolved in the 0.9 dm^3 HCl medium, to which the appropriate amount of water (1.461 cm^3) had been added. The 0.22 mol % potassium impurity in the $Mg(OH)_2$ was assumed to remain as KOH when the $Mg(OH)_2$ was converted to oxide, in which case, its presence is irrelevant, since the final solutions are identical.

From three concordant experiments on each compound and after adding 4 $cal \cdot mol^{-1}$ for the partial molar heat content of water in the medium, the authors obtained the average for Reaction (A.13) as 9.739 $kcal \cdot mol^{-1}$. With the use of IUPAC 1981 atomic weights, this value becomes:

$$\Delta_r H^\circ_m \text{ ((A.13), 298.15 K)} = 9.736 \text{ kcal} \cdot \text{mol}^{-1} = 40.735 \text{ kJ} \cdot \text{mol}^{-1}.$$

No uncertainty is given, but is estimated by this review to be ± 0.216 $kJ \cdot mol^{-1}$ (95 % confidence), from the scatter of values for the individual determinations, using a value of 4.30 for the Student-t factor.

It should be emphasised that this value refers to microcrystalline MgO. According to Gurvich *et al.* [1994GUR/VEY], the difference between the enthalpies of formation of the micro- and macrocrystalline forms is 3.9 $kJ \cdot mol^{-1}$.

[1937PIT]

From this early batch calorimetric study of a number of electrolytes, the enthalpies of neutralisation of $H_3PO_4(aq)$, $H_2PO_4^-$ and HPO_4^{2-} were reported as: – (8.159 ± 0.335), (3.35 ± 0.33), and (14.64 ± 2.09) $kJ \cdot mol^{-1}$ at 25 °C. Using the enthalpy of acid dissociation of $H_3PO_4(aq)$ reported by Nims [1933NIM] from potentiometric experiments, Pitzer gave a "best value" for the former of – (7.87 ± 0.34) $kJ \cdot mol^{-1}$.

[1937WEL/TAY]

This paper describes methods (both new and from the literature) for determining the degree of hydration of magnesia in hydrated dolomitic limes and lime putties, and the factors influencing the rate of hydration of magnesia. In this context, the enthalpies of

solution in hydrochloric acid of MgO, $Mg(OH)_2$, CaO, $Ca(OH)_2$, and $CaCO_3$ were measured. We shall limit the discussion of this paper to these latter determinations.

Samples of about 1.5 g were dissolved in 0.690 dm^3 of 2 molar HCl, in a conventional calorimeter yielding rather dilute solutions. The results were obtained at 298.15 K. The paper gives few details on the calculational procedures, but it will be assumed that they are the same as in [1938TAY/WEL], as these authors indicate that Wells *et al.* [1937WEL/TAY] give preliminary results.

Results concerning magnesium species

The magnesium oxide used in the heats of solution was obtained by dehydrating at temperatures ranging between 1323 to 1373 K a sample of $Mg(OH)_2$ prepared from MgO(cr) of analytical purity, free of any water-soluble substances, of chlorides and of carbonates, the only impurities being traces of iron and aluminium oxides.

Before being used for enthalpy of solution determinations, the $Mg(OH)_2$(cr) sample was calcined at 1023 K for several hours, re-hydrated in an autoclave for two hours at 8.6 bar, then oven-dried at 378 K for one hour.

Four dissolution experiments of each MgO(cr) and $Mg(OH)_2$(cr) samples yielded, respectively, the average values of – 150.247 kJ·mol^{-1} and – 112.550 kJ·mol^{-1}. Using these results, Gurvich *et al.* [1994GUR/VEY] reported from this source $\Delta_f H^o_m$ ($Mg(OH)_2$, cr, 298.15 K) = – (925.03 ± 1.00) kJ·mol^{-1}. We concur with their calculations, noting, as in the main text, that for $\Delta_f H^o_m$ (MgO, cr, 298.15 K) they used the value – (601.5 ± 0.3) kJ·mol^{-1} and not the CODATA value of – (601.6 ± 0.3) kJ·mol^{-1}. The increased uncertainty limits of ± 1.00 kJ·mol^{-1} from those of ± 0.4 kJ·mol^{-1} recommended by [1937WEL/TAY] are fully justified by the lack of details on the crystallinity of the MgO(cr) samples used.

Results concerning calcium species

The calcium carbonate used in the preparation of CaO(cr) and $Ca(OH)_2$(cr) was of analytical grade, completely soluble in hydrochloric acid and free of chlorides, silicates and sulfates. It contained only traces of aluminium and iron oxides. Just prior to calorimetry, the calcium oxide was obtained by calcining the carbonate for four hours at 1323 to 1373 K. In the preparation of calcium hydroxide, the carbonate was calcined for several hours at 1223 K, hydrated in an autoclave for two hours at 8.6 bar, then oven-dried at 378 K for one hour (as in the preparation of the magnesium hydroxide). These conditions are nearly identical with those used in [1938TAY/WEL].

Three dissolution experiments of each of CaO(cr) and $Ca(OH)_2$(cr) yielded, respectively, the average values of – 195.644 kJ·mol^{-1} and – 130.666 kJ·mol^{-1} for the enthalpies of dissolution. These values are in excellent agreement with those from the comparable determinations in 0.277 and 2.085 molar HCl given in [1938TAY/WEL] and were used as such by Garvin *et al.* [1987GAR/PAR], together with several other

reactions, in their evaluation of the relation between the enthalpies of formation of CaO(cr) and $Ca(OH)_2$(cr).

The enthalpies of solution of $CaCO_3$(cr) in CO_2-saturated and CO_2-unsaturated 0.277 and 2.085 molar HCl, were, respectively, – 13.891 kJ·mol^{-1} and – 24.184 kJ·mol^{-1}. As noted by the authors, the data obtained from CO_2-unsaturated solution vary with the experimental conditions (since the amount of escaping CO_2(g) varies) and are of no general use. In any case, these results were listed for information only by Garvin *et al.* [1987GAR/PAR] (their reaction 360), as they lead to a value for the difference between the enthalpies of formation of CaO(cr) and $Ca(OH)_2$(cr) which differs by about 3 kJ·mol^{-1} from other consistent measurements.

[1938DAV]

The dissociation constants of $CaOH^+$ were determined from measurements by Kilde [1936KIL2] at 25 °C of the solubility of calcium iodate in NaOH solutions, NaOH and NaCl ternary solutions, and solutions of NaOH and $Ca(OH)_2$, presumably at 25 °C and ionic strengths from 0.03 to 0.15 mol·dm^{-3}. Davies assumed the formation of $CaOH^+$, $NaIO_3$(aq), $Ca(IO_3)_2$(aq) and $CaIO_3^+$ ion pairs, which were ignored by Kilde. Activity coefficients were accounted for by the expression $\log_{10}\gamma_{\pm} = -2.02\sqrt{I} + 2I$. The dissociation constants exhibited significant scatter with an average value, ignoring obvious discrepant values, of (0.031 ± 0.004). Davies did not characterise or analyse the solid phase and used the term "harmonised" to describe the derivation of the ion-pair concentrations in solution, without any further explanation. It is assumed that Davies carried out a simultaneous minimisation of the equilibrium concentrations in solution, making assumptions about the ionisation constants of the $CaOH^+$, $NaIO_3$(aq), $Ca(IO_3)_2$(aq) and $CaIO_3^+$ ion pairs based on results in [1936KIL2]. Re-treatment of these results using $K_s^\circ(Ca(IO_3)_2(s)) = 7.112 \times 10^{-7}$ and $K_1^\circ(CaIO_3^+) = 0.13$ [1938DAV] at 25 °C, but neglecting the $NaIO_3$(aq) ion pair, yielded the results shown in Table A-6 taken from Table II in this paper.

Table A-6: Dissociation constants for $CaOH^+$ at 25 °C re-calculated from the solubility data.

c_{NaOH}	c_{Ca}	c_{NaCl}	$10^3 c_{Ca^{2+}}$	$10^4 c_{CaIO_3^+}$	$10^3 c_{CaOH^+}$	I_c	K_1°
0.010	0.0084	0	7.27	4.68	0.66	0.0329	0.0526
0.025	0.0094	0	6.72	4.38	2.25	0.0478	0.0314
0.025	0.0114	0.100	8.03	4.36	2.93	0.1525	0.0191
0.050	0.0105	0	6.51	4.18	3.57	0.0735	0.0343
0.050	0.0114	0.050	7.24	4.21	3.74	0.1259	0.0302

Despite the observed high concentration of $CaIO_3^+$ in the first experiment, all of these results are retained. The mean value of $K_1^\circ(CaOH) = (0.034 \pm 0.024)$ (2σ) or $\log_{10}{}^*K_1^\circ = -(12.53 \pm 0.31)$ is in agreement with the bulk of the later studies but its large uncertainty provided little weight to the overall fit of the data.

[1938DAV2]

Davies re-examined his previous treatment [1927DAV] of early conductance data of Kohlrausch cited in [1912NOY/FAL] to extract ion-pair dissociation constants and molar conductance values of $MgSO_4$(sln), among other electrolytes, at 18 °C. The original Onsager equation was replaced by the expression:

$$\Lambda_0 - \Lambda_c = b\sqrt{I_c} / (1 + \sqrt{I_c})$$

and gave an average value of $K_c^\circ = (128 \pm 5)$ from the four highest molarities (0.0005 to 0.005) mol·dm^{-3} with $\Lambda_0 = 112.7$ S·cm^2·mol^{-1} and $b = 202.6$. Table A-7 gives the experimental data and re-analysis results taking the dielectric constant from [1979BRA/PIT] and the viscosity of water from a polynomial fit of the data in [1978KES/SOK] from 0 to 70 °C. The modified Lee-Wheaton equation [1978LEE/WHE2] gave consistent results at the higher concentrations, *viz*., $K^\circ = (142 \pm 42)$ (2σ) with $\Lambda_0 = 118.7$ S·cm^2·mol^{-1} for a Bjerrum critical distance of 1.4182 nm.

Table A-7: Concentration of $MgSO_4$ and molar conductance at 18 °C and calculated association constants.

$10^4 c$ /mol·dm^{-3}	$\Lambda_{exp.}$ /S·cm^2·mol^{-1}	K_c° (reported)	$\Lambda_{calc.}$ /S·cm^2·mol^{-1} (Lee Wheaton)	K_c° (Lee Wheaton re-calculated)
1.0	109.80			
2.0	108.01			
5.0	104.15	132	104.92	165
10	99.85	133	99.48	132
20	94.12	132	94.12	126
50	84.53	123	84.53	145

Despite the age of the original experiments attributed to Noyes and Falk [1912NOY/FAL], the re-calculated results are in good agreement with the majority of results garnered over the next 100 years and are accepted by this review.

[1938NEW/WEL]

Through the measure of the enthalpy of solution at 298.15 K in 2.09 molal HCl of gypsum, calcium sulfate hemihydrate, natural anhydrite and anhydrous $CaSO_4$ samples obtained by heating gypsum at increasing temperatures up to 1373 K, the authors

obtained the enthalpies of dehydration of gypsum to the other calcium sulfate samples. This review is only interested in the dehydration of gypsum to anhydrite.

The solution calorimeter used was described and its mode of operation was detailed. Its charge was 640.0 g of acid solution and the samples were of size equivalent to 2.0 g of $CaSO_4 \cdot 2H_2O$(cr).

Two methods were used for the preparation of the gypsum samples: crystallisation from a supersaturated hemihydrate in water and precipitation from a mixture of approximately equal volumes of 10 % solutions of reagent quality calcium chloride and sulfuric acid. Seven preparations were made and the authors indicated that the preparation (not unambiguously identified) used for further investigation had a loss on ignition of 20.98 % (compared to a theoretical value of 20.93 %) and that no foreign material was identified upon microscopic investigation.

A sample of natural anhydrite (Midland, CA) containing SiO_2, Al_2O_3, Fe_2O_3, and MgO for a total of no more than 0.5 % and showing no foreign material by microscopy was used in the overall study. However, because of the presence of these impurities, and although the enthalpies of solution of the natural sample and of the purer dehydration product were in agreement within *ca.* 1 %, only the $CaSO_4$(cr) obtained from dehydration of gypsum at 1273 K was used for the evaluation of the reaction:

$$CaSO_4(cr) + 2H_2O(l) \rightleftharpoons CaSO_4 \cdot 2H_2O(cr) \qquad (A.14)$$

The authors report the value of $-(3.99 \pm 0.02)$ kcal·mol^{-1} or $-(16.7 \pm 0.1)$ kJ·mol^{-1} for $\Delta_r H^\circ_m$ (298.15 K) for Reaction (A.14) above for a sample of anhydrite heated to 1273 K, including a small correction for the transfer of H_2O(l) to the medium used for the dissolution reaction.

[1938TAY/WEL]

This paper deals with several related topics.

- An experimental determination of the enthalpy of solution of various samples of MgO(cr) in hydrochloric acid as a function of their temperature of preparation. This variation is related to the increase in particle size as the temperature of ignition is increased.

- An experimental determination of the enthalpy of solution of natural brucite and synthetic $Mg(OH)_2$(cr).

- An experimental determination of the enthalpy of solution of CaO(cr) and $Ca(OH)_2$(cr).

- A brief account of the calorimetric determination of the enthalpy of the direct hydration of CaO(cr) to $Ca(OH)_2$ (cr) in saturated lime water.

• A discussion of the literature results for the enthalpy of hydration of MgO(cr) to $Mg(OH)_2$(cr) and of CaO(cr) to $Ca(OH)_2$(cr).

The calorimeter used was described in a previous publication [1938NEW/WEL] from the same group and details of the calorimetric procedure were given, including correction for differences in the concentration of the final solutions and in the actual temperature during dissolution, together with the method of estimating uncertainties on repeated determinations.

Results concerning magnesium species

The magnesium oxide used in the determination of the heats of solution was obtained by dehydrating, at temperatures ranging between 723 K and 1698 K, a sample of $Mg(OH)_2$(cr) prepared from MgO(cr) of analytical purity, free of any water-soluble substances, chlorides and carbonates, the only impurities being traces of iron and aluminium oxides.

Before being used for heats of solution determinations, the $Mg(OH)_2$(cr) sample was calcined at 1023 K for several hours, re-hydrated in an autoclave for two hours at 8.3 bar then oven-dried at 378 K for one hour. The natural brucite sample was used as such. It was free of silica, entirely soluble in hydrochloric acid and contained only traces of calcium, iron and aluminium oxide. For the dissolution of MgO(cr), five sets of results were reported with values varying smoothly with the $Mg(OH)_2$(cr) dehydration temperature. For the sample prepared by dehydration for 2.5 hours at 723 K, the enthalpy of dissolution of 1 mole MgO(cr) into 17.630 kg of 2.085 molar HCl was $-(153.43 \pm 0.29)$ kJ·mol^{-1}, while for the sample treated at 1698 K, the value of $-(149.45 \pm 0.13)$ kJ·mol^{-1} was obtained. These values included all the appropriate corrections, including, for the sample prepared at low temperature, one for the presence of unreacted $Mg(OH)_2$(cr). The authors also showed that calcining a sample above 1698 K would have had essentially no further effect on the heat of dissolution. Moreover, they reported that a ground sample of fused MgO (periclase) would not dissolve entirely even when stirred for three hours at 25 °C with 4.355 molar HCl in the calorimeter.

For the dissolution of brucite and synthetic $Mg(OH)_2$(cr) in the same medium, values of $-(111.46 \pm 0.04)$ kJ·mol^{-1} and $-(112.47 \pm 0.13)$ kJ·mol^{-1}, respectively, were obtained for $\Delta_r H^\circ_m$.

After accounting for the fact that MgO samples fired at the lowest temperature were microcrystalline (see main text Section V.6.1.1.3) with a correction of 4.0 kJ·mol^{-1}, Gurvich *et al.* [1994GUR/VEY] reported from the data in [1938TAY/WEL], a value of $\Delta_f H^\circ_m(Mg(OH)_2, cr, 298.15\ K) = -(924.32 \pm 0.66)$ kJ·mol^{-1}, based essentially on the results from the synthetic material with conservative uncertainty limits. We note that, in these calculations, a value of $\Delta_f H^\circ_m(MgO, cr, 298.15\ K) = -(601.5 \pm 0.3)$ kJ·mol^{-1} was used.

Results concerning calcium species

The calcium carbonate used in the preparation of CaO(cr) and $Ca(OH)_2$(cr) was of analytical grade, completely soluble in hydrochloric acid and free of chlorides, silicates and sulfates. It contained only traces of aluminium and iron oxides. Just prior to the measurement, the calcium oxide was prepared by calcining the carbonate for four hours at 1323 to 1373 K. In the preparation of calcium hydroxide, the carbonate was calcined for several hours at 1223 K, hydrated in an autoclave for two hours at 8.3 bar, then oven-dried at 378 K for one hour (as in the preparation of the magnesium hydroxide). The enthalpies of solution of both CaO(cr) and $Ca(OH)_2$(cr) were determined using two different acid concentrations.

The dissolution of CaO(cr) and $Ca(OH)_2$(cr), respectively, into 28.3 kg of 2.085 molar HCl yielded values of $-(195.73 \pm 0.25)$ kJ·mol^{-1} and $-(130.67 \pm 0.17)$ kJ·mol^{-1}. The dissolution of 1 mole of CaO(cr) and $Ca(OH)_2$(cr), respectively, into 27.42 kg of 0.277 molar HCl yielded, after all corrections, values of $-(193.64 \pm 0.29)$ kJ·mol^{-1} and $-(128.45 \pm 0.24)$ kJ·mol^{-1}. These values are in excellent agreement with those derived from the comparable determinations in 2 molar HCl given by the same authors [1937WEL/TAY].

As mentioned above, the enthalpy of hydration of CaO(cr) to $Ca(OH)_2$ was also determined by direct hydration of the oxide in saturated $Ca(OH)_2$(sln). Two experiments yielded – 15.38 and – 15.47 kcal·mol^{-1} that were averaged by the authors as $-(15.43 \pm 0.09)$ kcal·mol^{-1} ($-(64.56 \pm 0.38)$ kJ·mol^{-1}). No details were given.

All the above results on the calcium species were fully taken into account by Garvin *et al.* [1987GAR/PAR] in their evaluation of the thermochemical properties of calcium compounds. Therefore, we shall not consider this topic further here.

[1940MAK/SHU]

The solidus-liquidus in the Na_2CO_3-K_2CO_3 system was investigated by the differential method. The only details of interest to this review are the reported transition temperatures of the components, *viz.* 629, 759 and 891 K for Na_2CO_3 and 523, 701 and 895 K for K_2CO_3.

However, most of these transitions have been attributed by Reisman and his colleagues [1958REI], [1958REI/HOL], [1959REI] to the presence of the corresponding monoxides, formed by decomposition (or impurity KCl) since the DTA was carried out in air rather than CO_2(g).

[1940SOU]

The enthalpies of hydration to gypsum ($CaSO_4 \cdot 2H_2O$) of calcium sulfates in various degrees of hydration were measured either by direct hydration in a calorimeter for the species "soluble" anhydrite ($CaSO_4$) and metastable or unstable hydrates, or indirectly through the enthalpies of solution in hydrochloric acid. The first technique concerned species that were not within the scope of the present review. The latter technique was used for species that could not be hydrated rapidly enough: this was the case of natural anhydrite, the only species of concern to this review. Its enthalpy of solution was thus compared to that of gypsum.

The solution calorimeter built for this study and its mode of operation were described. Its charge was 1862 g of 2.031 molar HCl solution and the samples were of size equivalent to 6 g of $CaSO_4 \cdot 2H_2O$(cr). It was operated at (298.15 ± 0.10) K.

The natural anhydrite was from Ludwig (Nevada). After grinding to pass 100-mesh screen, washing with hydrochloric acid and drying it was shown to be free from water, carbonates, sesquioxides and chlorides. It contained 1.15 % material resistant to attack by the hydrochloric acid solution.

The microcrystalline gypsum used for this study was from the Perkins deposit (Alaska). It was used as such after grinding to pass a 100-mesh screen. Its analysis was (in mass %): H_2O, 20.75 (theor. 20.93); $CaCO_3$, 0.28; M_2O_3, 0.05; insoluble 0.17.

Corrections in the enthalpies of solution were made for impurities, in particular for the small amount of calcium carbonate present in the Alaskan gypsum. After accounting for a small correction for the transfer of 2 H_2O(l) molecules to the medium used for the dissolution reaction, the enthalpy value reported by Southard for the hydration of natural anhydrite to gypsum according to Reaction (A.15):

$$CaSO_4(cr) + 2H_2O(l) \rightleftharpoons CaSO_4 \cdot 2H_2O(cr) \qquad (A.15)$$

was – (4040 ± 20) cal·mol^{-1} or – (16.903 ± 0.084) kJ·mol^{-1}.

Dissolution of selenite crystals from Gerlach, Nevada gave a value identical within 2 parts in 5000 to that obtained for the Alaskan gypsum sample.

[1941HAR/SCH]

By means of potential-difference measurements using the cell Pt,H_2|Na_2CO_3(m_1), $NaHCO_3$(m_2),NaCl(m_3)|AgCl-Ag, the ionisation constant of HCO_3^- was determined at 5° intervals from 0 to 50 °C. Based on these data, thermodynamic parameters for the dissociation of HCO_3^- ($\Delta_r G_m^\circ$, $\Delta_r H_m^\circ$, $\Delta_r S_m^\circ$, $\Delta_r C_{p,m}^\circ$) were also calculated. For the extrapolation to zero ionic strength the Debye-Hückel theory was used.

Harned and Scholes used rather a narrow and low ionic strength range (I_m = 0.02-0.16). The ratio of the molality of NaCl to the sum of carbonate and hydrogencarbonate ions was kept at ~ 0.5. Nevertheless, this dataset is considered in the literature (and by the present reviewers) as one of the most reliable. Indeed, the reported data agree very well with those determined 66 years later in [2007MIL/HUA].

[1941KEL/SOU]

The authors measured the low-temperature heat capacity of $CaSO_4$(cr) at 23 temperatures from 50 to 296 K on a sample of a natural mineral containing 0.09 % gypsum and extrapolated the values down to 0 K using a sum of one Debye and two Einstein functions. Integration of the heat capacities gave for the standard entropy S^o_m ($CaSO_4$, 298.15 K) = (106.7 ± 1.7) $J \cdot K^{-1} \cdot mol^{-1}$. Measurements were also made on two samples of $CaSO_4 \cdot 2H_2O$(cr), (24 on a selenite mineral (from 53 to 295.1 K), and nine on a gypsum sample (from 93.6 to 297.5 K)). The calculated standard entropy S^o_m ($CaSO_4 \cdot 2H_2O$, cr, 298.15 K) was (194.1 ± 1.7) $J \cdot K^{-1} \cdot mol^{-1}$.

The measured C_p values (and thus the calculated entropy) of anhydrite are slightly smaller than the more extensive measurements of Robie *et al.* [1989ROB/RUS], whose calculated value of S^o_m (298.15 K) is (107.4 ± 0.2) $J \cdot K^{-1} \cdot mol^{-1}$. For gypsum, the two sets of values are very similar up to *ca.* 200 K, but those of Kelley *et al.* are slightly smaller above this temperature; the standard entropy of (194.1 ± 1.7) $J \cdot K^{-1} \cdot mol^{-1}$ is essentially the same as the more precise value of (193.8 ± 0.3) $J \cdot K^{-1} \cdot mol^{-1}$ in [1989ROB/RUS]. The values from more extensive recent data of Robie *et al.* [1989ROB/RUS] have been preferred for both anhydrite and gypsum for this review.

The paper also discusses other studies of the thermodynamic properties of gypsum and its dehydration products, but these have been overtaken by the comprehensive assessment by Garvin *et al*. [1987GAR/PAR].

[1941MOS/PIT]

The authors present the detailed results of about 30 measurements of the high-temperature enthalpy content for each of cristobalite (367-622) K and tridymite (366-624) K. They mention that a few measurements were also made on quartz, but these are not documented. The samples (*ca.* 6 g) of quartz, cristobalite, and tridymite were from the same materials used by Anderson [1936AND] for his low-temperature heat capacity study, with impurity levels, as reported by him, of 0.07 % and 0.01 % and 0.54 %, respectively. The results of about 30 measurements from *ca.* (377-623) K are tabulated for each polymorph; the authors remark that within the uncertainty of their measurements, the enthalpy contents are the same for cristobalite and tridymite. The

data for cristobalite from this study are about 1 % higher than those derived from the assessment in [1982RIC/BOT] (in which they were given zero weight).

The temperatures and enthalpies of transition in cristobalite and tridymite are also discussed and tabulated. For cristobalite, a broad transition was observed from 498 to 535 K, with an enthalpy of (1300 ± 80) J·mol^{-1}, and for tridymite, three transitions at *ca.* 390 K (rapid, $\Delta_{trs}H_m = (293 \pm 40)$ J·mol^{-1}), *ca.* 436 K (rapid, $\Delta_{trs}H_m = (167 \pm 40)$ J·mol^{-1}) and 498 K (sharp and rapid, $\Delta_{trs}H_m = (188 \pm 40)$ J·mol^{-1}). These values are in good accord with later work.

The authors then combined their data with other then-existing data to tabulate the properties of quartz, cristobalite, and tridymite up to 2000 K. Since there were (and indeed still are) no data on tridymite at high temperatures, the heat capacities of tridymite were assumed to be close to those of cristobalite, as measured by [1919WHI] and [1921WIE]. A further constraint was applied, based on the then current assumption that cristobalite and tridymite are in equilibrium at 1743 K. As noted in Section V.3.1.4.1 this is not accepted by this review, so the details of the thermal functions given by [1941MOS/PIT] are not discussed further.

[1941ROT]

This paper concerns the determination of the enthalpy of formation of $PbCO_3$(cerusite) and $BaCO_3$(witherite) by solution calorimetry and of $MgCO_3$(magnesite) by combustion calorimetry. The authors cite also unpublished results on solution calorimetry of magnesite.

Only the results on magnesite are of relevance to the present review. The author noted that they could not obtain precise results on natural magnesite using bomb calorimetry and, therefore, used synthetic $MgCO_3$ (Merck, *pro analysi*) in the present study. The technique consisted in burning in a paraffin oil flame the sample (a few hundred mg) contained in a cellophane crucible: when the cellophane burned the carbonate fell through the flame and the resulting MgO collected below in a quartz dish. After correction of the very important thermal effect due to the combustion of the paraffin oil and other organic materials, the value $\Delta_r H_m$(A.16) = (94.6 ± 2.1) kJ·mol^{-1} ((22.6 ± 0.5) kcal·mol^{-1}), for Reaction (A.16):

$$MgCO_3(cr) \rightleftharpoons MgO(cr) + CO_2(g) \qquad (A.16)$$

is given by the author for 1 atm and 293 K, as average of six determinations.

Used as such, with CODATA values for the enthalpies of formation of MgO(cr) and CO_2(g) this result leads to $\Delta_f H_m^\circ$ ($MgCO_3$, cr, 298.15 K) = $-(1089.7 \pm 2.2)$ kJ·mol^{-1}. In view of the rather crude nature of the experimental procedure and, more especially, of the importance and complexity of the correction

terms involved in the calculations, it appears that the uncertainty of these results may be underestimated by an order of magnitude.

In the same paper, Roth also mentions unpublished results obtained with Müller-Mangold on the enthalpy of solution of magnesite in concentrated hydrochloric acid at 263 K. A value of 109.6 kJ·mol^{-1} (26.2 kcal·mol^{-1}) is given for Reaction (A.16). Taken as such, this value leads to $\Delta_f H^o_m$ ($MgCO_3$, cr, 298.15 K) = – (1104.7) kJ·mol^{-1}. It should be noted that Roth expressed scepticism about his solution calorimetry results, in comparison to his bomb calorimetry data mentioned above. In any case, in absence of detailed information, the solution calorimetric results of Roth are given for information only.

[1942MOO/KEL]

Moore and Kelley measured the low-temperature heat capacity of $MgSO_4$(cr). Their sample was prepared by dehydrating $MgSO_4 \cdot 7H_2O$(cr) at temperatures up to 400 °C for two hours; analysis showed it contained 0.15 % MgO. However, there was no structural analysis of the sample, so it is not clear which polymorph of $MgSO_4$(cr) was studied; this method of preparation normally gives the β-phase. The heat capacity was measured at 29 temperatures from 53 to 295 K. Extrapolations down to 0 K were made using a sum of one Debye and three Einstein functions.

Integration of the heat capacities gave the following value for the standard entropy:

$$S^o_m (MgSO_4, cr, \beta, 298.15\ K) = (91.6 \pm 0.8)\ J \cdot K^{-1} \cdot mol^{-1}.$$

The correction for modern atomic weights is negligible, and this is the selected value, with a slightly larger uncertainty of ± 1.0 J·K^{-1}·mol^{-1}.

[1943BAT/ACR]

This potentiometric study reported the dissociation constant of $H_2PO_4^-$ from 0 to 60 °C in a cell with hydrogen and Ag|AgCl electrodes. The ionic strength was varied to 0.6 mol·kg^{-1} (NaCl) with nine measurements at each temperature. An extended Debye-Hückel equation was employed and the average $\log_{10} K^o_2$ values obtained from five independent series of measurements at 5 °C intervals are: – (7.314 ± 0.001), – (7.281 ± 0.001), – (7.254 ± 0.001), – (7.230 ± 0.001), – (7.213 ± 0.001), – (7.198 ± 0.001), – (7.190 ± 0.001), – (7.185 ± 0.002), – (7.182 ± 0.002), – (7.181 ± 0.002), – (7.184 ± 0.002), – (7.188 ± 0.002), – (7.196 ± 0.002), respectively. Bates and Acree proposed the following equation to fit their data:

$$\log_{10} K^o_2 = 5.3541 - 1979.5\ (T/K)^{-1} - 0.019840(T/K) \qquad (A.17)$$

which led to the following thermodynamic quantities at 25 °C: $\log_{10}K_2^\circ = -(7.198 \pm 0.001)$, $\Delta_r G_m^\circ = 41.093$ kJ·mol^{-1}, $\Delta_r H_m^\circ = 4.129$ kJ·mol^{-1}, $\Delta_r S_m^\circ = -123.8$ J·K^{-1}·mol^{-1}, $\Delta_r C_{p,m}^\circ = -226.3$ J·K^{-1}·mol^{-1}. Table A-8 contains previous estimates of the value of $\log_{10}K_2^\circ$ taken from this paper.

Table A-8: Values of $\log_{10}K_2^\circ$ at 25 °C taken from earlier literature where LJ refers to the liquid-junction potential.

Reference	Method	$-\log_{10}K_2^\circ$
[1909ABB/BRA]	con	6.69
[1914MIC/GAR]	pot (with LJ)	7.07
[1921MIC/KRU]	pot (with LJ)	7.11
[1924PRI/WAR]	pot (with LJ)	7.17
[1927COH]	pot (with LJ)	7.17
[1929BJE/UNM]	pot (with LJ)	7.207
[1933NIM]	pot (without LJ)	7.206
[1934GUG/SCH]	pot (with LJ)	7.228
[1939SHI]	pot (with LJ)	7.02
[1943BAT/ACR]	pot (without LJ)	7.198

We note that [2005PAR/COV2] contains tables of the solution compositions and $\log_{10}K_{m,2}$ values for the experiments cited in [1943BAT/ACR] and [1945BAT/ACR]. Application of an isothermal SIT treatment, ignoring the absence of a swamping inert electrolyte and assuming that the difference in the ion interaction parameters with Na^+ and K^+ is small, gives the parameters in Table A-9.

Table A-9: Values of $\log_{10}K_2^\circ$ and the corresponding $\Delta\varepsilon$ from SIT fits (uncertainties are 2σ).

t /°C	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$\log_{10}K_2^\circ$	R Fit factor	$\log_{10}K_2^\circ$ reported
0	−(0.049 ± 0.027)	(0.284 ± 0.094)	−(7.318 ± 0.007)	0.9529	−(7.314 ± 0.002)
5	−(0.045 ± 0.027)	(0.252 ± 0.091)	−(7.284 ± 0.006)	0.9454	−(7.281 ± 0.002)
10	−(0.040 ± 0.027)	(0.234 ± 0.092)	−(7.256 ± 0.006)	0.9346	−(7.254 ± 0.002)
15	−(0.034 ± 0.029)	(0.234 ± 0.097)	−(7.233 ± 0.007)	0.9212	−(7.230 ± 0.002)
20	−(0.028 ± 0.026)	(0.235 ± 0.088)	−(7.215 ± 0.006)	0.9269	−(7.213 ± 0.002)
25	−(0.029 ± 0.023)	(0.208 ± 0.079)	−(7.200 ± 0.005)	0.9313	−(7.198 ± 0.002)
30	−(0.018 ± 0.027)	(0.233 ± 0.093)	−(7.195 ± 0.007)	0.9047	−(7.190 ± 0.003)
35	−(0.027 ± 0.023)	(0.214 ± 0.079)	−(7.188 ± 0.006)	0.9317	−(7.185 ± 0.003)
40	−(0.015 ± 0.025)	(0.230 ± 0.088)	−(7.188 ± 0.006)	0.9077	−(7.182 ± 0.003)
45	−(0.013 ± 0.027)	(0.236 ± 0.093)	−(7.186 ± 0.007)	0.8970	−(7.181 ± 0.003)
50	−(0.018 ± 0.025)	(0.216 ± 0.086)	−(7.188 ± 0.006)	0.9077	−(7.184 ± 0.003)
55	−(0.013 ± 0.027)	(0.214 ± 0.094)	−(7.191 ± 0.007)	0.8818	−(7.188 ± 0.004)
60	−(0.013 ± 0.028)	(0.208 ± 0.097)	−(7.198 ± 0.007)	0.8686	−(7.196 ± 0.004)

[1943KEL/MOO]

Kelley and Moore measured the low-temperature heat capacity of $MgCl_2$(cr) and $CaCl_2$(cr). Their sample of $MgCl_2$(cr) was prepared by heating $MgCl_2 \cdot NH_4Cl \cdot 6H_2O$(cr) at temperatures up to 600 °C until no more HCl(g) was evolved; analysis, however, showed that it still contained 0.2 % MgO. $CaCl_2$(cr) was prepared from the hexahydrate by heating firstly *in vacuo* to 130 °C, followed by HCl(g) at temperatures up to 740 °C; it contained 0.59 % $MgCl_2$ impurity.

The heat capacities of each material were measured at twenty temperatures from 53 to 295 K Extrapolations down to 0 K were made using sums of Debye and Einstein functions. Integration of the heat capacities gave the following values for the standard entropies: S^o_m ($MgCl_2$, cr, 298.15 K) = (89.52 ± 0.80) $J \cdot K^{-1} \cdot mol^{-1}$ and S^o_m ($CaCl_2$, cr, 298.15 K) = (113.80 ± 1.67) $J \cdot K^{-1} \cdot mol^{-1}$, where we have applied small corrections for modern atomic weights. We note that both Garvin *et al.* [1987GAR/PAR] (for $CaCl_2$) and Gurvich *et al.* [1994GUR/VEY] used somewhat different extrapolations of the data to 0 K, and preferred the slightly larger value of (89.62 ± 0.80) $J \cdot K^{-1} \cdot mol^{-1}$ for S^o_m ($MgCl_2$, cr, 298.15 K), and the much smaller value of (108.40 ± 4.00) $J \cdot K^{-1} \cdot mol^{-1}$ for S^o_m ($CaCl_2$, cr, 298.15 K). We have adopted the data from [1994GUR/VEY] as relevant data for $MgCl_2$(cr) and those from [1987GAR/PAR] for $CaCl_2$(cr)—see Sections V.6.1.1.3.2 and V.6.2.1.3.2.

[1943MOO]

The enthalpy content of the same samples of $MgCl_2$(cr) and $CaCl_2$(cr) as those used in the related low-temperature study by [1943KEL/MOO] were measured from 494 to 1428 K (21 points) and 404 to 1667 K (20 points) respectively; in both cases, the measurements extended well above the melting points. These data merge smoothly with the low-temperature measurements of the same samples by [1943KEL/MOO].

For $MgCl_2$(cr), their data also agree excellently with the later measurements in [1973HOL/GRO] around the melting point, and although the derived heat capacities of $MgCl_2$(l) are very similar in the two studies, Holm and Grønvold obtained a slightly smaller enthalpy of fusion.

[1943OWE/KIN]

This potentiometric investigation of the hydrolysis of boric acid Reaction (A.18) used a cell with hydrogen and Ag|AgCl electrodes and no liquid junction. The temperature was varied in 5 ° intervals from 5 to 55 °C. The ionic strength was varied from 0.02 to 3.0 $mol \cdot kg^{-1}$ (NaCl) and the boric acid molality from 0.005 to 0.03 $mol \cdot kg^{-1}$. Dissociation constants were reported, converted here to hydrolysis constants:

$$H_3BO_3(aq) + OH^- \rightleftharpoons H_4BO_4^- \quad (A.18)$$

Activity coefficients were calculated using an extended Debye-Hückel expression where the extended terms were the empirical functions: $b_1 m_{NaCl}$, $b_2(m_{NaCl})^2$ and $b_3 \log_{10}(1+b_4 m_{NaCl})$. These multiple terms reflect the complexity of the variation of the activity coefficients that could have been minimised by converting to the isocoulombic form of Reaction (A.18). The ion interaction parameters listed in Table A-10 are treated as follows by ignoring the interactions of the neutral H_3BO_3(aq) species with Na^+ and Cl^-:

$\varepsilon(Na^+, H_4BO_4^-) = \Delta\varepsilon + \varepsilon(Na^+, OH^-) = -(0.13 \pm 0.01) + (0.04 \pm 0.01) = -(0.09 \pm 0.01)$ kg·mol^{-1} at 25 °C, *cf.* $\varepsilon(Na^+, H_4BO_4^-) = -(0.07 \pm 0.05)$ kg·mol^{-1} according to the current NEA TDB listing.

Table A-10: Summary of the reported hydrolysis constants of boric acid and those obtained from application of the SIT treatment with ionic-strength independent and dependent approaches. The uncertainties are at the 2σ level.

t /°C	$\log_{10} K_1^\circ$ (A.18)	$-\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	R Fit factor	$\log_{10} K_1^\circ$ reported
5	(5.31 ± 0.020)	(0.140 ± 0.014)		0.994	5.296
	(5.288 ± 0.001)	(0.182 ± 0.001)	(0.085 ± 0.003)	1.000	
10	(5.163 ± 0.020)	(0.138 ± 0.013)		0.994	5.153
	(5.140 ± 0.006)	(0.180 ± 0.008)	(0.086 ± 0.015)	1.000	
15	(5.028 ± 0.014)	(0.134 ± 0.010)		0.996	5.018
	(5.011 ± 0.002)	(0.166 ± 0.002)	(0.065 ± 0.004)	1.000	
20	(4.895 ± 0.013)	(0.131 ± 0.009)		0.997	4.888
	(4.880 ± 0.003)	(0.159 ± 0.004)	(0.058 ± 0.006)	1.000	
25	(4.768 ± 0.012)	(0.130 ± 0.008)		0.997	4.764
	(4.754 ± 0.011)	(0.156 ± 0.001)	(0.053 ± 0.001)	1.000	
30	(4.646 ± 0.011)	(0.128 ± 0.007)		0.998	4.642
	(4.634 ± 0.001)	(0.151 ± 0.001)	(0.047 ± 0.001)	1.000	
35	(4.527 ± 0.001)	(0.128 ± 0.007)		0.998	4.525
	(4.516 ± 0.002)	(0.149 ± 0.003)	(0.043 ± 0.005)	1.000	
40	(4.414 ± 0.009)	(0.128 ± 0.006)		0.998	4.412
	(4.403 ± 0.001)	(0.147 ± 0.002)	(0.038 ± 0.003)	1.000	
45	(4.304 ± 0.007)	(0.127 ± 0.005)		0.999	4.303
	(4.295 ± 0.002)	(0.143 ± 0.003)	(0.032 ± 0.005)	1.000	
50	(4.197 ± 0.006)	(0.128 ± 0.004)		0.999	4.195
	(4.190 ± 0.003)	(0.140 ± 0.004)	(0.025 ± 0.008)	1.000	
55	(4.094 ± 0.006)	(0.127 ± 0.004)		0.999	4.092
	(4.088 ± 0.003)	(0.139 ± 0.005)	(0.025 ± 0.009)	1.000	

The SIT treatment of the $\log_{10}K_1^o$ values in Table A-10 derived with ionic-strength independent ion interaction terms yielded at 25 °C: $\log_{10}K_1^o = (4.770 \pm 0.002)$, $\Delta_r G_m^o = -(27.227 \pm 0.010)$ kJ·mol^{-1}, $\Delta_r H_m^o = -(42.365 \pm 0.181)$ kJ·mol^{-1}, $\Delta_r S_m^o = -(50.774 \pm 0.594)$ J·K^{-1}·mol^{-1}, and $\Delta_r C_{p,m}^o = 0$ J·K^{-1}·mol^{-1}. Owen and King reported the following thermodynamic quantities converted here to those for Reaction (A.18) at 25 °C: $\Delta_r G_m^o = -27.214$ kJ·mol^{-1}, $\Delta_r H_m^o = -41.815$ kJ·mol^{-1}, $\Delta_r S_m^o = -49.15$ J·K^{-1}·mol^{-1} and $\Delta_r C_{p,m}^o = -24.2$ J·K^{-1}·mol^{-1}, where the $\Delta_r C_{p,m}^o$ for the dissociation of water was taken from [1975OLO/HEP].

[1943SHO/HUF]

This is a calorimetric study of the dissolution in *ca.* 1 molar HCl of Mg(cr) and of the title compounds leading to their enthalpies of formation. Only the data involving MgO(cr) and $MgCl_2$(cr) will be discussed here.

The magnesium metal used was made by carbon reduction of magnesium oxide, followed by vacuum sublimation. The total impurities were stated to be less than 0.01 %, excepting surface oxidation. The samples for calorimetry were taken from the interior of a large ingot and their exposure to air was reduced to a minimum.

Magnesium oxide was made by igniting in a silica flask magnesium hydroxide obtained from reagent-quality magnesium chloride and ammonium hydroxide, followed by a heating for 100 hours to a temperature gradually raised to 1273 K. As noted in [1994GUR/VEY], this temperature is sufficient to obtain macrocrystalline samples. Analysis showed 60.36 % Mg (calculated 60.32 %) and 99.90 % MgO by titration.

A sample of $MgCl_2$(cr) prepared by Kelley and Moore [1943KEL/MOO] and used by these authors for specific heat measurements was treated before use by heating in a silica flask at 873 K in a stream of dry hydrogen chloride. Analysis showed 74.27 mass % Cl (calculated 74.46 %), 25.65 mass % Mg (calculated 25.54 %), and 0.15 mass % MgO.

The calorimeter used has been described in more detail in the literature [1940SOU]. The equations for the dissolution of the metal and of the compounds corresponded to the reaction (at 298.15 K) of one mole of substance in (77.2 HCl, 4190 H_2O). The actual weights of the compounds employed are equivalent to the average weight of the metal samples, 0.5725 g.

The enthalpy of solution of Mg(cr) according to:

$$Mg(cr) + 2H^+(sln) \rightleftharpoons Mg^{2+}(sln) + H_2(g) \qquad (A.19)$$

was reported as $\Delta_{sol}H_m$ (A.19) = $-(111.322 \pm 0.041)$ kcal·mol^{-1} or $-(465.771 \pm 0.172)$ kJ·mol^{-1} (95 % confidence), as an average of six measurements, after correction for the vaporisation of water by the (assumed 100 %) hydrogen evolution.

Use of 1981 atomic weights yields $\Delta_{sol}H^{o}_{m}$(A.19) = – (465.48 ± 0.17) kJ·mol^{-1}. This determination was a key input in the CODATA assessment [1989COX/WAG] of $\Delta_{f}H^{o}_{m}(Mg^{2+}, aq, 298.15\ K)$.

The enthalpy of solution of MgO(cr) according to:

$$MgO(cr) + 2H^{+}(sln) \rightleftharpoons Mg^{2+}(sln) + H_2O(sln) \qquad (A.20)$$

was reported as $\Delta_{sol}H^{o}_{m}$(A.20) = – (35.799 ± 0.021) kcal·mol^{-1} or – (149.783 ± 0.088) kJ·mol^{-1} (95 % confidence), as an average of five measurements, after correction for the enthalpy of dilution of the water formed in the reaction. These two determinations were key results in the adoption of the identical value for $\Delta_{f}H^{o}_{m}$(MgO, cr, 298.15 K) by CODATA [1989COX/WAG] and Gurvich *et al.* [1994GUR/VEY].

The enthalpy of solution of $MgCl_2$(cr) according to:

$$MgCl_2(cr) + (sln) \rightleftharpoons [Mg^{2+} + 2Cl^{-}](sln) \qquad (A.21)$$

was reported as $\Delta_{sol}H_{m}$(A.21) = – (36.566 ± 0.018) kcal·mol^{-1} or – (152.992 ± 0.075) kJ·mol^{-1} (95 % confidence), as an average of seven measurements, after a correction (74 cal) for the presence of 0.15 % magnesium oxide in the sample. These results led to a value of $\Delta_{f}H^{o}_{m}(MgCl_2, cr, 298.15\ K)$ about 3 kJ·mol^{-1} less negative than the values obtained from several other solution calorimetric determinations. It appears reasonable to assume, as suggested in [1979SHI/CRI], that this discrepancy is due to the presence of more than 0.15 % MgO(cr) in the magnesium chloride sample of [1943SHO/HUF]. The data of [1943SHO/HUF] for $\Delta_{sol}H^{o}_{m}(MgCl_2, cr)$ were also rejected by Cox *et al.* [1989COX/WAG] in their assessment of $\Delta_{f}H^{o}_{m}(Mg^{2+}, aq, 298.15\ K)$.

[1944MAN/DEL]

This potentiometric investigation of the hydrolysis of boric acid (Reaction (A.18)) utilised the same type of cell with no liquid junction potential as used by Owen *et al.* [1935OWE], [1943OWE/KIN]. The temperature was varied in five degree intervals from 0 to 60 °C. Solutions contained equimolal mixtures of boric acid, borate and sodium chloride at ionic strengths to 0.053 mol·kg^{-1}. Two sources of borate where used, crystalline borax and fused borax, with no discernible differences being observed. Dissociation constants were converted in this review to hydrolysis constants according to Equation (A.18).

The treatment of these potentiometric data was similar to that applied in [1935OWE] and [1943OWE/KIN]. From the sum of Equation (A.22), which defines the activity of hydrogen ion, and Equation (A.23), which is essentially the Nernst equation for the Harned cell, Equation (A.24) was derived to determine the $\log_{10}K^{o}_{1}$ (ionisation) value:

$$a_{H^+} = {}^{*}K^{o}_{1} m_{B(OH)_3(aq)} / m_{B(OH)_4^-} \gamma_{B(OH)_4^-} \qquad (A.22)$$

$$E = E^{\circ} - k \log_{10}({}^{*}K_1^{\circ} m_{B(OH)_3(aq)} m_{Cl^-} \gamma_{Cl^-} / m_{B(OH)_4^-} \gamma_{B(OH)_4^-}) \quad (A.23)$$

$$-\log_{10} K_1^{\circ} = (E - E^{\circ})/k + \log_{10}(m_{B(OH)_3(aq)} / m_{B(OH)_4^-}) + \log_{10} m_{Cl^-} + \log_{10}(\gamma_{Cl^-} / \gamma_{B(OH)_4^-}) \quad (A.24)$$

where $k = \ln(10)RT/F$. The extended Debye-Hückel equation was considered adequate to describe the activity coefficients of the anions such that their ratio (denoted as β^* in this paper) was approximated by the linear variation of $\log_{10} K_1^{\circ}$ (ionisation) with temperature. The maximum value of β^* was 0.19 such that $\beta^* I_m$ contributed ≤ 0.010 to $\log_{10} K_1^{\circ}$ (ionisation). The average values of $\log_{10} K_1^{\circ}$ (ionisation) listed in Table 4 of this publication are accepted for inclusion in the overall data treatment after conversion to $\log_{10} K_1^{\circ}$ (A.18), which are shown in Table V-31 in Section V.4.1.1 of the main text. These results are in excellent agreement with those in [1935OWE], [1943OWE/KIN] and [1972MES/BAE].

[1945BAT/ACR]

The experimental details are outlined in their previous paper [1943BAT/ACR] and as in that case, the $\log_{10} K_{m,2}$ values are also reported in [2005PAR/COV2]. However, in this 1945 study, solutions were prepared from five different stock solutions with varying KH_2PO_4 to Na_2HPO_4 concentration ratios and diluted to the desired ionic strength, which ranged from 0.00466 to 0.45 mol·kg^{-1}. Measurements were made at five degree intervals from 25 to 50 °C and the average $\Delta\varepsilon_1$ and $\Delta\varepsilon_2$ values over this temperature range are: – (0.050 ± 0.012) and (0.03 ± 0.07) kg·mol^{-1}, respectively.

[1945HAR/BON]

By means of potential-difference measurements using the so-called Harned cell (Pt, H_2, CO_2|$NaHCO_3$(m_1), NaCl(m_2), CO_2(m_3)|AgCl-Ag) the ionisation constant of H_2CO_3(aq) was determined between 0 and 50 °C in NaCl solutions, extending the data reported by Harned and Davis [1943HAR/DAV] up to a concentration of 1 mol·dm^{-3}(NaCl). The original data were obtained using various CO_2-H_2 mixtures as the gas phase. Measurements were made with five constant NaCl concentrations (0.1, 0.2, 0.5, 0.7 and 1.0 mol·dm^{-3}). At each fixed salt concentration (m_2), six different sodium hydrogencarbonate concentrations (c_1 ~ 0.0005, 0.0007, 0.01, 0.015, 0.02 and 0.03 mol·dm^{-3}) were used. Based on these data thermodynamic parameters for the dissociation of HCO_3^- ($\Delta_r H_m^{\circ}$, $\Delta_r S_m^{\circ}$, $\Delta_r C_{p,m}^{\circ}$) were also calculated.

The vapour pressure of water over the sodium chloride solutions was taken into account for the calculation of $p(CO_2)/p(H_2)$ in the cell. For the extrapolation to zero ionic strength the Debye-Hückel theory was used. The reported set of pK_1 values is considered in the literature (and by the reviewers) to be one of the most reliable.

[1945SNY/SEL]

Full details were given of the isothermal bomb calorimeter (335 cm^3) and the operating procedure used in this study. The oxygen used was purified from hydrogen and hydrocarbons by oxidation by palladised asbestos and carbon monoxide oxidised with hopcalite as catalyst. Water and carbon dioxide were then removed by "suitable" absorbents.

The calorimeter was operated under 40 bar O_2, and was calibrated by combustion of standard benzoic acid (NBS sample 39f). The authors specifically expressed in detail their efforts to minimise potential differences between calibration and Al combustion runs. They also commented that methods and practices in use at the NBS were used. The aluminium foil used was from Aluminum Company of America. The metallic impurities, totalling 0.006 %, consisted of Si, Fe, Cu, Mg, Ca and Na. The carbon content was estimated to 0.001 %; nitrogen was not determined. The estimated thickness of the oxide film on the aluminium samples corresponded to an oxide content of 0.01 % and was corrected for in the calculations. High-grade zircon crucibles, lined with a heavy layer of pure Al_2O_3 were used. Several crucibles were used in achieving the fifteen reported experiments. Combustion was quantitative in all but two cases, for which combustion reached 99.94 and 99.90 % (as determined and corrected for by recovery of a small piece of unburned foil). XRD of the combustion product showed it to be corundum. The weight (*in vacuo*, to the nearest 10 μg) of the samples was adjusted between 1.38 and 1.39 g. Combustion results were individually corrected and reported (2σ confidence interval) at 296.58 K and yielded an average value of (30.8681 ± 0.0111) $kJ \cdot g^{-1}$.

After correcting for the effect of preassure on the enthalpy of $O_2(g)$, neglecting the very small correction to 298.15 K and taking into account the 0.01 % of aluminium in the superficial layer of the samples, the authors obtain $\Delta_f H^\circ_m$ (Al_2O_3, cr, 298.15 K) = – (1669.31 ± 1.05) $kJ \cdot mol^{-1}$.

The authors used 26.97 for the atomic weight of aluminium. Correction to the 1981 atomic weights leads to – (1669.66 ± 1.05) $kJ \cdot mol^{-1}$.

[1945STO2]

The osmotic and activity coefficients of NaH_2PO_4 and KH_2PO_4 were obtained using KCl(sln) as the reference at 25 °C. The concentrations ranged from 0.1262 to 6.585 $mol \cdot kg^{-1}$ and 0.0993 to 1.820 $mol \cdot kg^{-1}$, respectively. These osmotic coefficients were included in an SIT treatment to derive the corresponding ion interaction coefficients.

[1946SHO/COO]

The authors measured the heat capacity of gibbsite from 52.8 to 296.5 K (29 measurements) in an adiabatic calorimeter, and also the enthalpy content from 322.2 to 423.5 K (five measurements) in a drop calorimeter. Similar measurements that we have not analysed further were made on a sample of $Al_2O_3 \cdot H_2O$. The sample of gibbsite was prepared by dissolving Al wire in 0.2 molar NaOH, and purified by washing with 4 molar HCl (to remove the Fe impurity present in the original Al wire) and dried at 140 °C. Chemical analysis showed it contained slightly more Al (65.62 wt %) than the theoretical 65.35 wt % and the XRD pattern was that of gibbsite. For the calculation of the standard entropy, the heat capacities were extrapolated to 0 K using the sum of one Debye and two Einstein functions.

The data for all the measurements are tabulated, together with fitted values of the enthalpy content from 325 to 425 K.

The calculated values at 298.15 K were:

$$C^{\circ}_{p,\mathrm{m}}\,(\mathrm{Al(OH)_3,\ gibb,\ 298.15\ K}) = (93.07 \pm 0.42)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(\mathrm{Al(OH)_3,\ gibb,\ 298.15\ K}) = (70.10 \pm 0.42)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}.$$

The authors found difficulties in the heat content measurements, owing to the evolution of water which did not recombine with the residual alumina, and indicate that their measurements above 300 K may be inaccurate by a few per cent.

As noted in the main text, the later measurements of Hemingway *et al.* [1977HEM/ROB3], using a high-purity sample, and extending down to 13.4 K, have been preferred to the data of Shomate and Cook.

[1947GIN/COR2]

A detailed description is given of a drop calorimeter for the measurement of the enthalpy content at high temperatures, consisting of a furnace and ice calorimeter and detailed results for a sample of aluminium oxide (corundum) in the range 323-1172 K are given.

The 8.0406 g sample was taken from a polished rod of synthetic sapphire. The detailed results of a total of 31 measurements at nine temperatures between 323 and 1173 K are tabulated. The deviations of the experimental values of $(H(T) - H(273.15\ \mathrm{K}))$ from a smoothed curve (no equations given) were less than 0.05 %. The derived values of specific heat and enthalpy at round temperatures from 0 to 900 °C are given.

These results are in excellent agreement with the later more detailed study [1956FUR/DOU].

[1948LEV]

In this very general paper Levy re-assessed the thermodynamic relations between gypsum ($CaSO_4 \cdot 2H_2O$(cr)), anhydrite ($CaSO_4$ (cr)) and the intermediary hydrates. The author essentially refers to the studies of Kelley *et al.* [1941KEL/SOU]. The crystalline structure of the various species and their structural relations are also described and briefly discussed. Values for entropy of the compounds at 298.15 K are also given, as well as the enthalpies of hydration at 298.15 K.

The values given for the standard entropies of the only species of interest to this review, gypsum and anhydrite, (46.4 ± 0.4) cal·K^{-1}·mol^{-1} and (25.5 ± 0.4) cal·K^{-1}·mol^{-1}, respectively, are those given by Kelley *et al.* [1941KEL/SOU]. The value for the enthalpy of hydration of anhydrite according to Reaction (A.25):

$$CaSO_4(cr) + 2H_2O(l) \rightleftharpoons CaSO_4 \cdot 2H_2O(cr) \qquad (A.25)$$

is $-(4.030 \pm 0.020)$ kcal·mol^{-1} as given by Levy, while Southard [1940SOU] and Kelley *et al.* [1941KEL/SOU] report $-(4.040 \pm 0.020)$ kcal·mol^{-1}.

[1948STO/DAV]

The second dissociation constant of magnesium hydroxide was determined from pH measurements with a glass electrode and Ag|AgCl reference electrode standardised against "a series of standard buffers". The temperature was maintained at (25.00 ± 0.01) °C. Apparently only two titrations were made of an $MgCl_2$ solution, prepared *in situ* by the addition of the solid salt to a known mass of conductivity water. A standard $Ba(OH)_2$ titrant was chosen primarily to inhibit the formation of high localised hydroxide concentrations that could lead to premature precipitation of $Mg(OH)_2$(s). Stock and Davies were careful to minimise CO_2 contamination in their titration vessel, although a correction for this minor impurity was still made to each dataset. As recognised subsequently in [2003ALT/MET] and other papers, these early results are quite reliable. Although the ionic medium was predominantly $MgCl_2$(sln), thereby strictly invalidating the use of the SIT treatment, its concentration was constant and the ionic strength was relatively low. Therefore, the activity coefficients approximated by the SIT Debye-Hückel expression, resulted in the logarithm of the second dissociation constant of $Mg(OH)_2$(aq) , $-(11.44 \pm 0.13)$, for the two titrations.

[1948TOR]

In the course of this study that was primarily devoted to the thermochemistry of trona, $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$(cr), a compound that falls outside the interest of the present review, the enthalpies of solution in water of Na_2CO_3(cr), and of $NaHCO_3$(cr) in Na_2CO_3 solutions were reported at (298.15 ± 0.1) K.

Prior to use, reagent grade sodium bicarbonate of unspecified origin was treated with tank carbon dioxide for several hours at room temperature. A "virtually 100 %" purity was reported from titration with standard hydrochloric acid. Sodium carbonate was prepared from reagent grade sodium bicarbonate.

For Na_2CO_3(cr), three experiments were reported: each time; 2.814 g (0.02655 mole) were dissolved in 1814.9 g H_2O, leading to a 0.01463 molal solution. The reported values, converted from kcal·mol^{-1}, for the enthalpies of solution were – 22.213, – 22.112 and – 22.125 kJ·mol^{-1}, from which the author selected – (22.150 ± 0.063) kJ·mol^{-1}.

In the case of $NaHCO_3$(cr) three experiments were also reported: each time, 2.230 g (0.02653 mole) were dissolved in the final solution resulting from the dissolution of Na_2CO_3. The reported values, converted from kcal·mol^{-1}, for the enthalpies of solution were 15.690, 15.707 and 15.715 kJ·mol^{-1}, from which the author selected (15.703 ± 0.017) kJ·mol^{-1}.

[1948TOR/SAH]

This study principally concerns the determination of the enthalpy of formation of Mg_2SiO_4(cr), $MgSiO_3$(cr) and $CaSiO_3$(cr), compounds that are outside the scope of the present review. However, in their study, the authors found it necessary to measure the enthalpy of solution of $Mg(OH)_2$(cr) in 1.000 molar HCl. Their sample was prepared in a manner similar to that described in [1938TAY/WEL] (see relevant Appendix A entry).

For the reaction:

$$Mg(OH)_2(cr) + 2H^+ \rightleftharpoons Mg^{2+} + 2H_2O(l) \qquad (A.26)$$

six values were reported that included a small correction (35 cal) for MgO(cr) impurity in the sample and for the dilution by the water formed in the reaction (10 cal), leading to an average value $\Delta_r H^\circ_m$ ((A.26), 298.15 K) = – (26.950 ± 0.010) kcal·mol^{-1} (– (112.758 ± 0.042) kJ·mol^{-1}).

The calorimeter used for these determinations was that used by [1943SHO/HUF] in their determination of the enthalpy of formation of MgO(cr) and other magnesium compounds, and the experimental conditions were reported as virtually identical.

For the reaction:

$$MgO(cr) + 2H^+ \rightleftharpoons Mg^{2+} + H_2O(l) \qquad (A.27)$$

Shomate and Huffman [1943SHO/HUF] had reported five very concordant values, each of them being also corrected for the dilution by the water formed in the

reaction (5 cal), leading to an average value $\Delta_r H_m^o$ ((A.27), 298.15 K) = – (35.799 ± 0.021) kcal·mol^{-1} (– (149.783 ± 0.088) kJ·mol^{-1}).

From these, the enthalpy of dehydration of $Mg(OH)_2$(cr):

$$Mg(OH)_2(cr) \rightleftharpoons MgO(cr) + H_2O(l) \quad \text{(A.28)}$$

is obtained as $\Delta_r H_m^o$ ((A.28), 298.15 K) = (149.783 ± 0.088) – (112.758 ± 0.042) = (37.025 ± 0.098) kJ·mol^{-1}.

[1949BEL/PRU]

Bell and Prue derived the dissociation constant of $CaOH^+$ at 298.15 K from a comparison of the rates of depolymerisation of diacetone alcohol in $Ca(OH)_2$ solutions with those in KOH and NaOH solutions. The reaction is catalysed by free hydroxide ions. They assumed that $CaOH^+$ had no catalytic effect and that the concentration of free hydroxide ions could be calculated using a constant catalytic coefficient averaged over the same range of the total hydroxide concentrations for the alkali metal hydroxide and $Ca(OH)_2$ solutions. The ionic strength was varied from 0.02 to 0.05 mol·dm^{-3}, while the c_{Ca} concentration ranged from 0.007 to 0.14. Measurements were also made in mixed 0.025 mol·dm^{-3} NaOH, 0.175 mol·dm^{-3} NaCl, $CaCl_2$ solutions at an ionic strength of 0.2 mol·dm^{-3} with concentrations of $CaCl_2$ between 7×10^{-4} and 3×10^{-3} mol·dm^{-3}. Activity coefficients were described using the Debye-Hückel expression with $Ba = 1$. The resulting dissociation constants did not show any trend with ionic strength with an average scatter less than the experimental error in the kinetic measurements of ± 0.5 %. Bell and Prue reported values of 0.051 mol·dm^{-3} for the dissociation constant in the $Ca(OH)_2$(aq) catalysed solutions and 0.050 mol·dm^{-3} in the NaOH, NaCl and $CaCl_2$ solutions. The ionic strengths were so low that $\Delta\varepsilon \approx 0$ in the SIT approach. The average value $\log_{10} K_1^o = -(1.28 \pm 0.08)$ obtained from the SIT is consistent with other literature values and resulted in $\log_{10}{}^* K_1^o = -(12.72 \pm 0.08)$ for the equivalent hydrolysis reaction. There is an error in the tabulated $K_{c,1}$ values and a typographical error in the authors' Equation (2); the exponent of ionic strength in the denominator of the second term should be 1/2 not 1/3.

[1949KON/LEB]

Konopik and Leberl discussed the treatment of potentiometric data for very weak acids, including boric acid, in very general terms. Potentiometric titrations were presumably made at 20 °C of "concentrated" boric acid solutions. Values for the first and second dissociation constants were proposed and an estimate was made of the third constant. Neither the composition nor the concentrations used were given and no mention was made of multinuclear species. These results were not considered further in this review.

[1949ROB/STO]

A synopsis of this paper was given in [2013LEM/BER].

[1949ROT/TRO]

This paper concerns determinations of the enthalpies of solution in concentrated acids of α-quartz, β-cristobalite, quartz glass and amorphous silicic acid that we shall consider here and of a few silicates (Fe_2SiO_4, $CaSiO_3$, Na_2SiO_3, *etc.*) that are outside the interest of the present review, as are also, the measurements on the specific heat and setting heats of cements.

A description of the 300 cm^3 silver calorimeter with gold insert suitable for working in concentrated HF and in HCl-HF mixtures around 350 K is given. All calorimetry samples were used as "finely pulverised" material. In all instances, except where mentioned below, the samples were dissolved at 350 K in 20.6 % aqueous HF solution. The results were corrected for the water content. All results, $\Delta_{sol}H_m$, were reported without any indication of the number of experiments.

For α-quartz, Roth and Troitzsch indicate having used "pure quartz sand" containing 0.64 % water. It yielded $\Delta_{sol}H_m = -(33.51 \pm 0.04)$ kcal·mol^{-1} or $-(140.21 \pm 0.17)$ kJ·mol^{-1}.

β-cristobalite was obtained by treatment of amorphous silica for seventy hours at 1373 K, followed by a brief digestion in 40 % HF at room temperature to remove the remaining particles of silicic acid, and dried. The sample was checked by XRD. It gave $\Delta_{sol}H_m = -(35.22 \pm 0.05)$ kcal·mol^{-1} or $-(147.36 \pm 0.21)$ kJ·mol^{-1}.

An impure "cristobalite" sample obtained by [1932ROT/BEC] in the oxidation of Si in an oxygen bomb and containing 0.86 % water was also used which gave a value of $\Delta_{sol}H_m = -(36.17 \pm 0.11)$ kcal·mol^{-1} or $-(151.35 \pm 0.46)$ kJ·mol^{-1}.

A sample of quartz glass containing *ca.* 0.25 % water was also used which yielded $\Delta_{sol}H_m = -(36.49 \pm 0.05)$ kcal·mol^{-1} or $-(152.67 \pm 0.21)$ kJ·mol^{-1}.

Amorphous (no XRD lines) silicic acid was obtained by hydrolysis of $SiCl_4$, followed by washing the resulting gel until free from Cl and drying for five hours at 673 K. The sample contained 0.99 % water. The reported solution value was $\Delta_{sol}H_m = -(37.17 \pm 0.09)$ kcal·mol^{-1} or $-(155.52 \pm 0.38)$ kJ·mol^{-1}.

A similar preparation of silicic acid containing 2.33 % water was also used, leading to $\Delta_{sol}H_m = -(36.04 \pm 0.11)$ kcal·mol^{-1} or $-(150.79 \pm 0.46)$ kJ·mol^{-1}. However, as this sample was dissolved at 294 K in 10.3 % HF, comparison with the other results is not immediately feasible. We note, nevertheless, that in a cycle involving Fe_2SiO_4, [1949ROT/TRO] make use of the value -546.8 cal·g^{-1}, or -32.85 kcal·mol^{-1}, for the enthalpy of solution of quartz at 295 K in 10.3 % HF. In the case of amorphous silicic

acid, the authors also mention using the extrapolation to zero water content of the enthalpies of solution at about 293 K.

Based on their data [1949ROT/TRO] finally report the following enthalpies of transformation at about 293 K.

Table A-11: Enthalpies of transformation of α-quartz to β-cristobalite, quartz glass and amorphous silicic acid.

Reaction	$\Delta_r H_m$ /kJ·mol^{-1}
α-quartz ⇌ β-cristobalite	(7.15 ± 0.25)
α-quartz ⇌ quartz glass	(12.47 ± 0.29)
α-quartz ⇌ amorphous silicic acid	*ca.* 16.3

[1950BEL/WAI]

From kinetic measurements of the alkaline hydrolysis of carbethoxymethyltriethylammonium iodide, Bell and Waind derived values for the dissociation constant of $CaOH^+$ at 0 °C. As in [1949BEL/PRU] and [1956BEL/PAN], it was accepted that only free hydroxide ions act as catalysts for the alkaline hydrolysis reaction, allowing the dissociation constant of $CaOH^+$ to be determined by comparison of the reaction rates using mixtures of NaOH(sln) and $CaCl_2$(sln). The concentration of $CaCl_2$ was 0.0726 mol·dm^{-3} while that of NaOH was varied between 0.003 and 0.006 mol·dm^{-3} with I_c reaching 0.295. The dissociation constants were derived by also assuming formation of $NaIO_3(aq)$ and $CaIO_3^+$ ion pairs. Values for these two dissociation constants were taken from conductivity measurements in [1938DAV]. Activity coefficients were defined by the Davies equation with the extended term equal to either zero or $-0.2\ I_c$. The dissociation constants showed no significant trend with ionic strength and mean values of $K_1^o(CaOH^+)$ were reported as (0.046 ± 0.003) and (0.059 ± 0.002) for the mixtures of NaOH(sln) and $CaCl_2$(sln), respectively.

Bell and Waind also measured the solubility of calcium iodate in water and dilute NaOH solutions at 0 °C following the procedure in [1938DAV]. The concentration of NaOH was varied between 0.02 and 0.07 mol·dm^{-3}, and the solubility of $Ca(IO_3)_2$(cr) varied from 0.003 to 0.004 mol·dm^{-3} such that $I_c \leq 0.077$. Re-calculation of the solubility data in Table II of [1950BEL/WAI], ignoring formation of $NaIO_3$(aq) ion pairs and either including or excluding those of $CaIO_3^+$ and simply using the SIT Debye-Hückel expression to account for activity coefficients, led to a solubility product, $K_s^o(Ca(IO_3)_2(s))$ at 0 °C of (3.00 ± 0.06) × 10^{-8} (2σ) (assuming an ion-pair dissociation constant for $CaIO_3^+$ of 0.115 [1953BEL/GEO]), *cf.* 2.921 × 10^{-8} proposed by the authors and 2.86 × 10^{-8} [1953BEL/GEO]. From these measurements we derived a mean value of K_1^o = (0.044 ± 0.005) mol·dm^{-3} (2σ), *cf.* (0.044 ± 0.057) proposed by the authors using the Davies equation with either no extension term or with a $-0.2\ I_c$ extension. Minor contributions of the two ion pairs and the differences in the Debye-Hückel models could

readily account for these differences. The $CaIO_3^+$ ion pair only represented *ca*. 2 % of the total calcium in solution. The corresponding value $\log_{10}{}^{*}K_1^\circ = -(13.57 \pm 0.10)$ is in good agreement with other results.

[1950EGA/WAK]

The authors measured the enthalpy content of two samples of hydroxylapatite from 473.4 to 1472.5 K at intervals of *ca.* 100 K, using an adiabatic calorimeter. One sample contained 0.3 % Al_2O_3(cr) impurity, the second 0.18 % Cl; no significant difference was observed in the results from the two samples. The XRD patterns of both were said to agree with the then-current ASTM data (presumably for the hexagonal structure), but with the addition of two weak lines. These were attributed to the presence of a small amount of chlorapatite, although, as pointed out by Suda *et al.* [1995SUD/YAS], could possibly have been observed because [1950EGA/WAK] had, in fact, prepared the more stable monoclinic phase. The experimental data show no indication of the transition to the hexagonal phase at *ca.* 480 K, which is not surprising, since Suda *et al.* [1995SUD/YAS] and Sharpataya *et al.* [1995SHA/FED] have shown that this has a very small enthalpy change—see Section V.8.2.3.1.

The measured enthalpy differences were fitted in [1950EGA/WAK] by an equation corresponding to a three-term C_p equation. However, their tabulated values of the thermal function at round temperatures show some appreciable differences from the values calculated from this equation. We have therefore re-fitted their data above 480 K (27 points with tiny corrections to conform with modern atomic weights) with those from the DSC measurements of C_p in [1995SHA/FED] and [1995SUD/YAS] to give the following equation for the hexagonal polymer:

$$C_{p,\mathrm{m}}(Ca_5(PO_4)_3OH, \text{cr, hexagonal}, T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 449.0467 + 0.124878(T/\mathrm{K}) - 1.04128\times10^{-5}(T/\mathrm{K})^2 - 8.75879\times10^{6} \quad (480 \text{ to } 1200\ \mathrm{K})$$

Figure V-85 in Section V.8.2.3.2 of the main text shows a plot of these data for the heat capacity.

[1950HER/STI]

Herrmann and Stipetić conducted two bayerite solubility experiments which are summarised in Table A-12, as well as the results of application of the SIT treatment which was constrained by fixing $\Delta\varepsilon = -(0.052 \pm 0.037)\ \mathrm{kg\cdot mol^{-1}}$ obtained from the analogous gibbsite solubility data in the same NaOH(sln) medium over a wider temperature range.

Table A-12: Summary of bayerite solubility experiments with molalities calculated using the density function based on [2006REY/BER].

t /°C	$c_{Al_2O_3}$ /g·dm^{-3}	c_{Na_2O} /g·dm^{-3}	ρ_{calc} /g·cm^{-3}	I_m /mol·kg^{-1}	m_{OH^-} /mol·kg^{-1}	$m_{Al(OH)_4^-}$ /mol·kg^{-1}	$\log_{10}K_{s,4}$	$\log_{10}K^\circ_{s,4}$ SIT
25.0	20.20	123.95	1.167	4.22	3.81	0.419	− 0.96	− (0.74 ± 0.24)
28.5	23.50	123.95	1.166	4.22	3.73	0.487	− 0.89	− (0.67 ± 0.24)

[1950KEN]

The solubility of quartz in water was measured to 610 °C and pressures to 175 MPa. The experimental method involved equilibration of pre-weighed quartz plates with water in pre-oxidised stainless steel bombs, which were then rapidly quenched whereupon the plates were removed, washed and re-weighed to determine the amount of quartz dissolved. Equilibration periods ranged from two hours to 10 days. Of the 294 measurements made only three which fell below 200 °C, were examined in this review (see Table A-13). The $\log_{10}K^\circ_{s,4}$ values were derived assuming that the activity coefficient of $Si(OH)_4(aq)$ is unity. The first two values probably represent an under-saturated condition, noting that the autoclaves were not rocked and the surface area of the quartz plates must have been small.

Table A-13: Solubility measurements of α-quartz.

t /°C	p_{sat} /MPa	Equilibration time /hours	Reported solubility weight % SiO_2	Solubility mol·kg^{-1}
160	0.62	144	0.007	1.17×10^{-3}
182	1.11	120	0.017	2.83×10^{-3}
200	1.55	230	0.024	4.00×10^{-3}

[1950KER/JOH]

The authors measured the low-temperature heat capacity of synthetic sapphire from 20.2 to 294.9 K (50 measurements) in an adiabatic calorimeter. The sapphire was from the same manufacturer as that used in the numerous studies at the US National Bureau of Standards (NBS), though from a different batch. No analysis for impurities was made, being assumed that it was of the same high purity (> 99.98 wt %) as the NBS sample. The individual results are tabulated together with smoothed values of the thermal functions at round temperatures up to 300 K. For the calculation of the standard entropy, the heat capacity was extrapolated to 0 K by a T^3 extrapolation, using a Debye temperature of 571 K. The derived values of $C^\circ_{p,m}$ and S°_m at 298.15 K were 79.87 and 50.90 J·K^{-1}·mol^{-1} compared to the CODATA values of (79.033 ± 0.200) and (50.92 ± 0.10) J·K^{-1}·mol^{-1}, based on the later more detailed studies by Furukawa and Douglas [1956FUR/DOU] (13-300 K) and Fugate and Swenson [1969FUG/SWE] (2-25 K).

[1950KOB]

The heat capacity of $Ca(OH)_2$(cr) was measured from 314 to 670 K, using a heat conduction calorimeter. Decomposition of the sample was avoided by having a dish of $Ca(OH)_2$(cr) in bottom of the calorimeter at a higher temperature than the sample. The sample was prepared by mixing $Ba(OH)_2$ and $CaCl_2$ solutions and dried at 150 °C for 10 hours, care being taken to exclude CO_2(g). Analysis showed it to be 99.91 % pure, the residue being assumed to be absorbed water; no corrections were made to the heat capacities. Two sample sizes were used, one of 8.3 g and one of 9.3 g; the results of the two samples agreed to within 1 %. Detailed results of the 35 measurements are tabulated. The author fitted his results to the equation:

$$C_{p,\mathrm{m}}(\mathrm{Ca(OH)_2}, \mathrm{cr}, T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 59.329 + 1.3397\times10^{-2}(T/\mathrm{K}) - 9.079\times10^{-5}(T/\mathrm{K})^2 \quad (360\text{-}670\ \mathrm{K})$$

As noted elsewhere, these data merge smoothly with the later low-temperature measurements of Hatton *et al.* [1959HAT/HIL], and the selected data are a re-fit in [1994GUR/VEY] of these experimental results and (with a lesser weighting) those of Lashchenko and Kompanskii [1935LAS/KOM] with a constraint of $C^{\circ}_{p,\mathrm{m}}$ ($Ca(OH)_2$, cr, 298.15 K) = 87.488 $\mathrm{J\cdot K^{-1}\cdot mol^{-1}}$, as given in [1959HAT/HIL].

[1951BAT]

Bates employed a Harned-type cell to measure the potential difference in aqueous mixtures of KH_2PO_4 (x m) and HCl ($(1-x)$ m) at 0 to 60 (± 0.005) °C. The difficulty imposed by trying to measure the dissociation constant of a moderately strong acid was somewhat circumvented by applying two experimental strategies. One involved increasing the value of x while still maintaining a stable pH buffer mixture and then estimating the activity coefficient independently using a Debye-Hückel expression. The second method involved substituting a weak acid, such as formic acid, for HCl, and adding KCl to establish the chloride molality. The ionic strength attained was < 0.4 $\mathrm{mol\cdot kg^{-1}}$. The average $\log_{10}K^{\circ}_1$ value for phosphoric acid dissociation from both methods was – 2.148 with $\Delta_r H^{\circ}_{\mathrm{m},1} = -7.650\ \mathrm{kJ\cdot mol^{-1}}$ and $\Delta_r S^{\circ}_{\mathrm{m},1} = -66.8\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$ at 25.0 °C, *cf.* $\Delta_r H^{\circ}_{\mathrm{m},1} = -8.160\ \mathrm{kJ\cdot mol^{-1}}$ [1937PIT]. Nims [1934NIM] used a similar potentiometric technique and obtained different $\log_{10}K^{\circ}_1$ values, but similar $\Delta_r H^{\circ}_{\mathrm{m},1}$ values.

[1951DAV/HOY]

This note reported the dissociation constants of $CaOH^+$ based on measurements of the solubility of calcium iodate in the presence of $Ca(OH)_2$(sln) at 25 °C and ionic strengths from 0.022 to 0.067 $\mathrm{mol\cdot dm^{-3}}$. These constants were derived by assuming formation of $CaOH^+$ and $CaIO_3^+$ ion pairs. Davies and Hoyle took the dissociation constant of

$CaIO_3^+$ to be 0.13 mol·dm^{-3} and the molar solubility product of $CaIO_3(s)$ to be 7.119×10^{-7} [1938DAV]. Activity coefficients were accounted for using the Davies equation with a $0.2I$ extension term. The dissociation constants are consistent and independent of the ionic strength with a mean value of 0.049 mol·dm^{-3}; the authors excluded two data points at the highest ionic strengths to obtain a mean of 0.050 mol·dm^{-3}. They commented that neglect of any contribution of $CaOH^+$ to the catalysis of the hydrolysis of diacetone alcohol in [1949BEL/PRU] led to a virtually identical value of 0.052. Other factors such as the choice of activity coefficient equation, range of concentrations studied and the inherent accuracy of the experimental procedure could readily overshadow this small difference. Re-calculation using the infinite dilution solubility product, 7.119×10^{-7}, and a $CaIO_3^+$ dissociation constant of 0.13 mol·dm^{-3}, gave an average K_1° value for the dissociation of $CaOH^+$ of (0.0425 ± 0.0039) (2σ) for all eight measurements in which $Ca(OH)_2(sln)$ was present, including the two highest I_c values of 0.0616 and 0.0664. This corresponds to $\log_{10} K_1^\circ = -(1.372 \pm 0.020)$, which shows the good precision of these measurements. Neglecting the formation of $CaIO_3^+$ ion pairs led to lower K_1° values which exhibited a distinct monotonic trend to higher values with increasing ionic strength. Therefore, inclusion of the $CaIO_3^+$ ion pair is vindicated.

[1951EGA/WAK]

The authors measured the heat capacity of hydroxylapatite from 11 K to *ca.* 300 K, using an adiabatic calorimeter. The analysed CaO and P_2O_5 contents of the sample differed from the stoichiometric values by only a few hundredths of a percent and no corrections were made for impurities. The individual measurements were not reported, only values at round temperatures from 15 to 298.16 K. For the calculation of entropy, the data were extrapolated from 13.18 K to 0 K using a Debye temperature of 95 K. This is a high-quality experimental study; the changes in the values for the conversion to modern atomic weights are negligible.

The derived values at 298.15 K are:

$$C_{p,\mathrm{m}}^\circ(Ca_5(PO_4)_3OH, \mathrm{cr}, 298.15\ \mathrm{K}) = (385.05 \pm 1.50)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S_\mathrm{m}^\circ(Ca_5(PO_4)_3OH, \mathrm{cr}, 298.15\ \mathrm{K}) = (390.43 \pm 1.00)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

where the uncertainties are estimated by the review.

[1951EGA/WAK2]

Egan *et al.* measured the heat capacity of fluorapatite from *ca.* 13 to *ca.* 300 K and the enthalpy content from 474.6 to 1581.7 K at intervals of *ca.* 100 K, using an adiabatic calorimeter. The sample, prepared by heating $Ca_3(PO_4)_2$ with CaF_2 for 30 min at

1370 °C, contained an estimated 0.1 % unreacted CaF_2(cr) impurity, but no other impurities were detected in the XRD pattern. The measured enthalpy differences were fitted by the authors to an equation corresponding to a three-term C_p equation, constrained to reproduce their $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K})$ exactly. However, the tabulated values of the thermal function at round temperatures show some appreciable differences from the values calculated from this equation. We have therefore re-fitted these enthalpy data (22 points) together with the high-temperature C_p measurements in [2010DAC/HAR], with a constraint of $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K}) = (375.99)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ from their own low-temperature measurements to give the equation:

$$C_{p,\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,\ cr}, T)\ /\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = 469.872 + 6.2877\times10^{-2}(T/\mathrm{K}) - 1.00116\times10^{7}(T/\mathrm{K})^{-2} \qquad (298.15\ \mathrm{K\ to\ 1600\ K})$$

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,\ cr},\ 298.15\ \mathrm{K}) = (376.0 \pm 0.8)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

The entropy:

$$S^{\circ}_{\mathrm{m}}(\mathrm{Ca_5(PO_4)_3F,\ cr},\ 298.15\ \mathrm{K}) = (388.1 \pm 0.8)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$$

was calculated by the authors, using a Debye extrapolation (with $\theta_D = 112$ K) below 13.32 K.

For the final selected values, these data have been combined with the more recent measurements in [2010DAC/HAR] (see Section V.8.2.4.2).

[1951FUL/GIN]

The equilibrium studied here is:

$$\mathrm{Al(OH)_3(\alpha\text{-gibb}) + OH^- \rightleftharpoons Al(OH)_4^-} \qquad (A.29)$$

The data were limited by the reviewer to 74, 79, 84 and 89 °C which represents the limit of validity of the density equation created by the reviewer, but more significantly, gibbsite dehydrates to boehmite at higher temperatures. The experimental and analytical methods were mentioned briefly with such general comments as the individual reaction vessels were stirred in the lower temperature range and rocked in the higher range. It appears that each result was obtained from five independent measurements. Solution samples were removed after five days' equilibration (filtration was not mentioned) which is remarkable because these measurements extended to 200 °C where conversion to boehmite should have been quite rapid, although the solubility should reflect that of the more soluble solid phase so that presumably sufficient gibbsite remained to control the overall solubility. Despite the high aluminate molalities reported, application of the SIT treatment to the equation $\log_{10}K_{\mathrm{s},4} = a + b(T/\mathrm{K})^{-1} - cI_m$ yielded values of $\log_{10}K^{\circ}_{\mathrm{s},4}$ in good agreement with those from many previous and subsequent publications.

[1951HOL/HUB]

The 360 cm^3 Parr oxygen bomb calorimeter used in this study was fully described, with its peripherals, together with its mode of operation under 25 bar O_2 (99 %). Before use, the oxygen was thoroughly purified from residual combustible gases and nitrogen. Calibration was made electrically and by combustion of standard benzoic acid (NBS sample 39f). The two methods gave substantial agreement. Metals for combustion were weighed to 0.1 mg and weights were reported *in vacuo*. The completeness of the combustion of the metals was verified by treating the combustion products with 6 molar HCl and measuring the amount of hydrogen evolved from any unreacted metal present. All calorimetric final average values were reported for the 2σ confidence level.

Combustion of magnesium and standard enthalpy of formation of MgO(cr, 298.15 K)

Doubly distilled magnesium was used, with a chief metallic impurity, present to less than 0.01 %; the overall purity of the Mg was at least 99.98 %. Analysis for nitrogen yielded a content of 0.004 %. The superficial oxygen layer was removed with dilute hydrochloric acid and the cleaned metal was kept for a few hours in dry oxygen at 25 bar without gaining as much as 0.1 mg in weight. Combustion was achieved on sintered MgO. The initial temperature of the bomb was 297.75 K and the average final temperature was 299.05 K. Fifteen runs were made using 0.42 to 0.61 g Mg. The percentage burned ranged between 99.78 and 99.99 %. Taking into account the uncertainty in the energy equivalent, the final average for the heat of combustion of magnesium is (24.667 ± 0.020) $kJ \cdot g^{-1}$. After a very small correction to 298.15 K and after correcting for the pressure dependence of the enthalpy of $O_2(g)$, the authors reported $\Delta_f H^\circ_m$(MgO, cr, 298.15 K) $= -(601.23 \pm 0.49)$ $kJ \cdot mol^{-1}$. Correction to the 1981 atomic weights leads to $-(601.00 \pm 0.49)$ $kJ \cdot mol^{-1}$. This is in good agreement with the value selected in CODATA, [1989COX/WAG] $-(601.60 \pm 0.30)$ $kJ \cdot mol^{-1}$, based essentially on enthalpy of solution measurements since, as these latter authors comment, Holley and Huber did not establish that the product of their combustion was macrocrystalline.

Combustion of aluminium and standard enthalpy of formation of Al_2O_3(cr, 298.15 K)

The aluminium used (Aluminum Company of America) was nominally 99.9968 % Al, mainly as 0.13 mm sheets. Analysis of a portion of the sheet yielded 0.002 % N, 0.03 % C and 0.01 % O detected impurities. Difficulties in the ignition of the samples were fully overcome. Combustion was achieved on an alumina disk surrounded by a covered alumina cylinder, itself provided with holes for easy oxygen access. The initial temperature was 297.75 K and the average final temperature was 299.15 K. Fifteen runs were made, using 0.33 to 0.49 g Al. The percentage burned ranged between 99.36 and 99.98 %. There was no mention that the structural identity the alumina formed was verified. Taking into account the uncertainty in the energy equivalent,

the final average for the heat of combustion of aluminium is (30.976 ± 0.024) kJ·g^{-1}. After corrections similar to those for magnesium above, the authors obtain $\Delta_f H^o_m$ (Al_2O_3, cr, 298.15 K) = – (1674.8 ± 1.3) kJ·mol^{-1}. Correction to the 1981 atomic weights leads to – (1675.1 ± 1.3) kJ·mol^{-1}. This value was important in the adoption by CODATA of $\Delta_f H^o_m$ (Al_2O_3, cr, 298.15 K), although as noted in the main text they indicate a value of – (1675.7 ± 1.3) kJ·mol^{-1}.

[1951KOB]

The heat capacity of $CaCO_3$(calc) was measured from 308 to 780 K in a heat conduction calorimeter. The sample was prepared by mixing solutions of $CaCl_2$ and $(NH_4)_2CO_3$ at 20 °C, with continuous stirring for two hours, followed by drying at 200 °C. Microscopic examination showed only rhombic crystals; the lime content measured after decomposition at 800 °C was 100.1 % of the theoretical.

Three sets of measurements (48 in all) were made with *ca.* 13 g of calcite contained in a silver ampoule, and the full results are tabulated. These results merge well with the low-temperature data of Anderson [1934AND2] and Staveley and Linford [1969STA/LIN] but are slightly lower than the more recent measurements of Jacobs *et al.* [1981JAC/KER] below 700 K.

Using these and other relevant data, the author showed that calculated decomposition pressures of the reaction $CaCO_3(calc) \rightleftharpoons CaO(cr) + CO_2(g)$ are in good agreement with the experimental data, particularly the pressures measured by Tamaru *et al.* [1931TAM/SIO].

[1951KOB2]

The heat capacity of $CaCO_3$(arag) was measured from 303 to 735 K in a heat conduction calorimeter; as noted below, only the values up to 573 K refer to pure aragonite. The sample was prepared by the method described in [1916JOH/MER], *viz.* very slow precipitation (0.5 g·h^{-1}) by dropping of solutions of $CaCl_2$ and K_2CO_3 into 1 dm^3 of water at 85-90 °C, with vigorous stirring. The aragonite was filtered and dried at 150 °C. The density was measured to be 2.93 kg·dm^{-3}, that of aragonite and the lime content, measured after decomposition at 800 °C, was 100.1 % of the theoretical. No other characterisation was reported.

Three sets of measurements (51 in all) were made with *ca.* 13 g of aragonite contained in a silver capsule, and the full results are tabulated. From 573-673 K, the C_p values showed anomalous humps due to the gradual transition of the sample into calcite, and above 673 K, the results were the same as those for calcite [1951KOB]. The author fitted the data up to 580 K by a quadratic function of T, but we note that there is a typographical error in the printed coefficient of T^2, which should clearly be

-3.31×10^{-5} cal·mol^{-1}·K^{-3} rather than -3.11×10^{-5} cal·mol^{-1}·K^{-3}. The measured values merge well with the low-temperature data in [1934AND2] and [1969STA/LIN], and are in good agreement with the three values given in [1996WOL/LER] at higher temperatures.

Kobayashi also derived a rather uncertain value of the enthalpy calcite-aragonite transition from an analysis of the above-mentioned humps in the C_p curves for aragonite in the region where the transition to calcite occurs fairly rapidly (573-673 K) and derived a value of $\Delta_r H^\circ_m$ ((A.30), 298.15 K) = (0.121 ± 0.096) kJ·mol^{-1}.

$$\mathrm{CaCO_3(calc)} \rightleftharpoons \mathrm{CaCO_3(arag)} \qquad \text{(A.30)}$$

We note, however that these results were given zero weight in the assessment of Garvin *et al.* [1987GAR/PAR] (their Reaction 378).

[1951LAN]

The high-temperature heat contents of the title compounds were measured using drop calorimetry. Only the data for CaO(cr) are relevant for the present review.

The CaO(cr) was obtained by heating $CaCO_3$(cr) to 800 °C in the platinum capsule that was used to contain the oxide for the measurement; the only detail of the composition and purity given is that the test for CO_2 was negative. Eight measurements were made, at temperatures from 564 to 1176 K. The results were fitted by an enthalpy equation which corresponds to the heat capacity equation:

$$C_{p,\mathrm{m}}(\mathrm{CaO,cr},T)/\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 49.62 - 4.519\times10^{-3}(T/\mathrm{K}) - 6.945\times10^{6}(T/\mathrm{K})^{-2} \quad (298.15\text{-}1200\ \mathrm{K}).$$

We note a typographical error in the paper in the first coefficient of the corresponding enthalpy equation, which should be 11.86 cal·K^{-1}·mol^{-1} and not 11.68 cal·K^{-1}·mol^{-1}.

As shown in Figure V-52 in Section V.6.2.1.1.2, above 600 K, the results of this study are distinctly higher than the consensus of other measurements (except those by [1929ROT/BER]) and were given a low weighting by [1994GUR/VEY] in their assessment of C_p(CaO, cr), which are our selected data. This paper also reports the dissociation pressures of the title carbonates, but these are not relevant to the current review.

[1952HUM/KIN2]

A combination of combustion and solution calorimetry was used to obtain the enthalpies of formation of alpha quartz and alpha cristobalite from the elements.

The combustion of 99.81 % pure, fully analysed silicon under 40 atm O_2(g) was carried out in a previously described apparatus [1951HUM]. The combustion led to two products: the major portion (*ca.* 87 %) remained in the crucible and was identified by XRD as alpha cristobalite. The remaining part, found on the walls of the bomb, was a fine silica powder giving no XRD pattern. Its enthalpy of solution in 20.1 % hydrofluoric acid at 346.85 K was reported as – (35.41 ± 0.03) kcal·mol^{-1} (average of three determinations). This value was used, in conjunction with the enthalpy of solution of the cristobalite (obtained in the combustion), that was reported as – (34.22 ± 0.03) kcal·mol^{-1} (average of six determinations) under the same conditions, in order to calculate the standard enthalpy of formation of alpha-quartz that was reported as – (879.73 ± 1.13) kJ·mol^{-1} at 298.15 K. However, the experimental procedure and analytical techniques used in [1952HUM/KIN2] were criticised in [1956GOL] and the difficulties encountered in the oxygen combustion calorimetry of silicon were outlined in later papers [1963WIS/MAR], [1964GOO/LAC] that were, with [1973KIL/PRO], the basis of the CODATA selection [1989COX/WAG]:

$$\Delta_f H^\circ_m (\alpha\text{-quar}, 298.15\ \text{K}) = -(910.7 \pm 1.00)\ \text{kJ·mol}^{-1}.$$

Under these circumstances, the results given in [1952HUM/KIN2] for alpha quartz will not be considered further.

Nevertheless, Humphrey and King [1952HUM/KIN2], citing an earlier paper [1951KIN], give, from the average of seven concordant determinations, the value of – (33.290 ± 0.080) kcal·mol^{-1} for the enthalpy of solution of alpha quartz under the same conditions of temperature and medium as that reported above for their alpha cristobalite, and deduce for Reaction (A.31):

$$\text{SiO}_2(\alpha\text{-quar}, 298.15\ \text{K}) \rightleftharpoons \text{SiO}_2(\text{cris}, 298.15\ \text{K}) \qquad \text{(A.31)}$$

$\Delta_r H^\circ_m$ ((A.31), 298.15 K) = (3.89 ± 0.38) kJ·mol^{-1}.

This latter value has been included in the values considered in the present review.

[1952KOB]

The heat capacities of Na_2CO_3(cr) and $NaHCO_3$(cr) were measured in a heat conduction calorimeter from 320 to 487 K and 300 to 385.5 K respectively. $NaHCO_3$(cr) was first prepared from a solution of re-crystallised Na_2CO_3(cr) and CO_2(g), and anhydrous Na_2CO_3(cr) was prepared from this by heating to 773 K for one hour. The only analysis reported is that the carbonate content of the $NaHCO_3$(cr) was 99.98 % of the theoretical value.

Two separate samples of *ca.* 16 g of each compound were used in the measurements (23 for Na_2CO_3(cr) and 15 for $NaHCO_3$(cr)); the detailed results are given for both compounds. Kobayashi fitted his values using the equations:

$$C_{p,\mathrm{m}}(\mathrm{Na_2CO_3}, \mathrm{cr}, T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 58.493 + 0.21552(T/\mathrm{K}) + 1.3075\times10^{-4}(T/\mathrm{K})^2 \quad (300\text{-}500\ \mathrm{K});$$

$$C_{p,\mathrm{m}}(\mathrm{NaHCO_3}, \mathrm{cr}, T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 45.898 + 0.14108(T/\mathrm{K}) \quad (300\text{-}360\ \mathrm{K}).$$

The heat capacities of Na_2CO_3(cr) at the lowest temperatures agree well with the low-temperature values measured by Anderson [1933AND] and Waterfield *et al.* [1968WAT/LIN] around 300 K, but the data at higher temperatures are appreciably lower than the direct measurements in [1951POP/GAL] and those derived from the numerous subsequent enthalpy measurements.

The data for $NaHCO_3$(cr) are a consistent continuation of the low-temperature C_p data of Anderson [1933AND], but the values above 360 K rise very steeply, presumably due to the onset of decomposition.

For unknown reasons, this paper is not quoted in either of the assessments by JANAF [1985CHA/DAV] or Glushko *et al.* [1982GLU/GUR].

[1952MAY]

This paper gives the results of drop-calorimetric measurements of the enthalpy of Na_2S(cr), Na_2SO_4(cr) and Na_2CO_3(cr); only the last are of interest to this review. No details are given of the apparatus used, nor of the sample preparation or analyses, and the results are presented only as a smoothed value of $H(T) - H(298.15\ \mathrm{K})$ at round temperatures from 400 to 1500 K. Despite the lack of detail, the results agree well with other data, the enthalpy values falling slightly below the selected values.

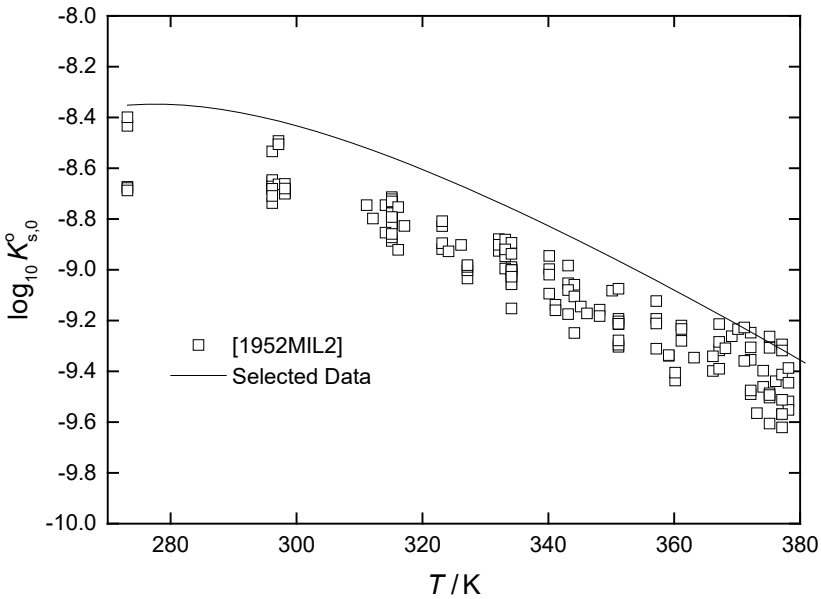
[1952MIL2]

The solubility of three different types of $CaCO_3$ (optical grade calcite, Chihuahua, Mexico; Solenhofen limestone and *Venus mercenaria* shell) was studied in distilled water, 0.5 $\mathrm{mol\cdot dm^{-3}}$ NaCl solution, and in standard sea water at temperatures ranging from 0 to 105 °C, and at CO_2-pressures ranging from 1 to 100 bar. A steel bomb was partially filled with a known volume of water into which a relatively large crystal (plate) of $CaCO_3$ was placed. The bomb was inserted in a furnace at a regulated temperature and connected to a CO_2(g) source. After 8-50 hours of equilibration the plate was removed, rinsed with distilled water, dried on a hot plate, cooled in a desiccator and weighed (mass losses were 0.001-0.025 g).

In this review only the 165 solubility data reported for optical grade calcite in distilled water were considered. However, the solubility products calculated from these data are 0.2 to 0.3 $\log_{10}$ units smaller (more negative) than the selected values (see

Figure A-5). Segnit *et al.* [1962SEG/HOL] repeated the experiments of Miller, and they found that that some re-precipitation takes place on the calcite crystals during quenching of the run to room temperature and pressure (especially above $p(CO_2)$ = 10 bar) and that is probably the reason for the low solubilities reported by Miller. Plummer and Busenberg in [1982PLU/BUS] also noted that metal impurities in solution (from corrosion of the steel bombs) strongly inhibit the rate of calcite dissolution even at trace levels. Consequently, these data were rejected.

Figure A-5: Solubility of optical grade calcite as a function of temperature.

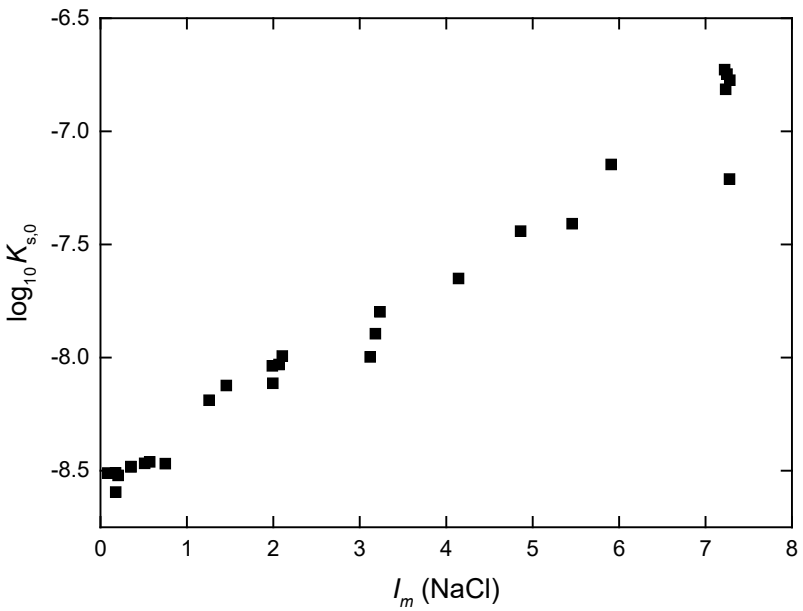


[1952SHT/FRO]

The solubility of calcite (natural Iceland spar) was measured at 25 °C in 200 cm^3 NaCl solutions (I_m = 0-6.2) containing 10 g calcite through which CO_2 ($p(CO_2) \sim 0.00125$, 0.06, 0.1, 0.54, 0.96 bar) was bubbled. The equilibration lasted 1-2 months. Equilibrium with calcite was approached from both under- and supersaturation by changing the temperature (from 20 to 25 °C). Dissolved Ca^{2+} was determined by gravimetry as an oxalate precipitate. $\Sigma m\,(HCO_3^-)$ was determined by titration with HCl in the presence of methyl orange. Assays for Ca^{2+} and HCO_3^- were made for a mutual check and to confirm each other. The partial pressure of CO_2 was determined using the $Ba(OH)_2$ method.

Our re-evaluations of the experimental data of related publications dealing with {calcite + H_2O + CO_2 + NaCl} or {calcite + H_2O + NaCl} systems ([1984MIL/MIL], [1989WOL/BRE]) indicate that reliable solubility products can be obtained up to I_m < 2-3, which is probably related to the validity range of the SIT treatment used in this review. In the case of the present paper, the re-evaluated $\log_{10}K^{\circ}_{s,0}$ values are close to those selected in pure aqueous solutions only up to I_m = 1 (NaCl), but considerably higher values were calculated at higher ionic strengths. Consequently, the experimental data obtained in this paper at I_m > 1.0 (NaCl) were rejected.

Figure A-6: The re-evaluated $\log_{10}K_{s,0}$ ($CaCO_3$, calc) values at 298.15 K based on the data listed in [1952SHT/FRO] as a function of NaCl molality.



[1953BEL/GEO]

Among an array of similar solubility measurements involving thallous iodates, a more limited number were made of the solubility of $Ca(IO_3)_2$ over ranges of K_2SO_4 concentration at 0, 25 and 40 °C. The dissociation constants of $CaOH^+$ were also determined from similar solubility measurements in KOH solutions at concentrations ranging from 0.005 to 0.07 mol·dm^{-3} and ionic strengths from 0.006 to 0.09 mol·dm^{-3}. No supporting electrolyte was added. A flow-through packed column was used. Experiments over long periods of time showed no change in solubility (< 0.2 %) indicating that the solid was thermodynamically stable for the duration of their

experiments. Bell and George used the Davies equation to estimate activity coefficients with an extension term of $0.2I_c$ and also considered the formation of $CaIO_3^+$ ion pairs with dissociation constants of 0.15, 0.13 and 0.115 mol·dm^{-3} at 0, 25 and 40 °C, respectively [1950BEL/WAI].

$CaSO_4$(aq)

The average values of the dissociation constant of $CaSO_4$(aq) were reported as 0.0060, 0.0049 and 0.0041 mol·dm^{-3}. Conversion of the reported molar to molal quantities involved a correction of < 0.5 %. Bell and George reported the following thermodynamic quantities converted here to those for the association reaction: $\Delta_r H_m^o =$ (3.15 ± 0.03) kcal·mol^{-1} or (13.2 ± 0.1) kJ·mol^{-1}; $\Delta_r S_m^o =$ (16.1 ± 0.8) cal·K^{-1}·mol^{-1} or (67 ± 3) J·K^{-1}·mol^{-1}. Average values of K_1^o were calculated in this review by applying the SIT. The contribution of $\Delta\varepsilon = \varepsilon(CaSO_4^+, IO_3^-) - \varepsilon(Ca^{2+}, IO_3^-) - \varepsilon(K^+, SO_4^{2-})$ was approximated by:

$$\varepsilon(MX^+, ClO_4^-) - \varepsilon(Ca^{2+}, ClO_4^-) - \varepsilon(K^+, SO_4^{2-}) =$$
$$(0.23 \pm 0.07) - (0.27 \pm 0.03) + (0.06 \pm 0.02) = (0.02 \pm 0.08) \text{ kg·mol}^{-1}.$$

In view of the low ionic strengths employed and the uncertainty in $\Delta\varepsilon$, its contribution to $\log_{10} K_1^o$ is < 0.002. The contributions from possible $CaIO_3^+$ and KSO_4^- ion pairs were ignored so that the resulting values of $K_1^o(CaSO_4(aq))$ are (160 ± 24), (183 ± 8) and (212 ± 24) mol·kg^{-1} at 0, 25 and 40 °C, respectively, where the uncertainties are statistically based (2σ) (*cf.* 167, 205 and 244 mol·dm^{-3} reported by the authors). The solubility constants of $Ca(IO_3)_2$(s) at 0, 25 and 40 °C are given as 2.859×10^{-8}, 7.119×10^{-7} and 2.437×10^{-6} mol^2·dm^{-6}, respectively, bearing in mind that the authors stated at 40 °C the stable solid phase was $Ca(IO_3)_2 \cdot H_2O$(cr) rather than $Ca(IO_3)_2 \cdot 6H_2O$(cr). In the absence of added sulfate or hydroxide, the corresponding solubility constants corrected for solution non-ideality and for formation of $CaIO_3^+$ are calculated by the reviewer to be 2.88×10^{-8}, 7.170×10^{-7} and 2.485×10^{-6} mol^2·dm^{-6}, ignoring ion pairing of K^+ and IO_3^-. The K_s^o value at 0 °C was measured earlier by Bell and Waind [1950BEL/WAI] at four NaOH concentrations to be 2.921×10^{-8} mol^2·dm^{-6} which is re-calculated to be $(3.00 \pm 0.06) \times 10^{-8}$ mol^2·dm^{-6}.

$CaOH^+$

The dissociation constants of $CaOH^+$ showed no trend with the ionic strength leading the authors to quote the following mean values (the 2σ uncertainties are added by the reviewer): (0.043 ± 0.006), (0.040 ± 0.004) and (0.033 ± 0.004) mol·dm^{-3} at 40 °C after assuming formation of $CaOH^+$, and KIO_3(aq) and $CaIO_3^+$ species in solution. The two latter dissociation constants are derived from conductivity measurements. The authors reported the following thermodynamic quantities: $\Delta_r H_m^o(298.15\text{ K}) = -(1.19 \pm 0.25)$ kcal·mol^{-1} or $-(5.0 \pm 1.1)$ kJ·mol^{-1}, and $\Delta_r S_m^o(298.15\text{ K}) = -(10.4 \pm 0.8)$ cal·K^{-1}·mol^{-1} or $-(43.5 \pm 3.4)$ J·K^{-1}·mol^{-1}. Considering the experimental uncertainties (1σ), these values are close to those reported in [1954GIM/MON]. Re-calculations as described above for the sulfate system were made while accepting contributions from $CaIO_3^+$ ion

pairs but ignoring those suggested for $KIO_3(aq)$ ion pairs, led to average values of $K_1^o(CaOH^+)$ of (0.044 ± 0.011), (0.035 ± 0.003) and (0.027 ± 0.003) mol·kg^{-1} (2σ) at 0, 25 and 40 °C, respectively.

[1953CRE/ALL]

This paper aimed to measure equilibrium pressures of $CO_2(g)$ over $MgCO_3$ in the temperature range from *ca.* 500 to 700 K.

The equipment and operating procedures were briefly described. An "ordinary" glass reaction vessel (volume 1 cm^3) was used; the possible reaction with magnesite at the reaction temperatures was not even mentioned.

Natural magnesite (particle size less than 0.1 mm) from the Zillertal region (Austria) was used. Unfortunately, no analysis was given, even though it was recognised by the authors that the sample contained impurities (see immediately below). The gases formed in the decomposition reaction were subjected to a rather primitive analysis, mainly by cooling in solid CO_2 and liquid air, which showed that 5-10 % remained uncondensed at liquid air temperature (*ca*. 78 K). It was also shown that the residual gases reacted positively to a test for CO(g) using $PdCl_2$, so it was assumed that an impurity in the magnesite reduced CO_2 to CO. This conclusion was supported by the fact that the amount of residual gases was independent of the period of outgassing, duration of the experiment and degree of decomposition of the magnesite.

This review notes that it is not uncommon to find organic impurities in natural magnesites that could have been responsible for the partial reduction of the evolved $CO_2(g)$ in these experiments. The authors indicate that their reported results were corrected for the presence of residual gases, although no details were given. The results for three representative temperatures (600, 670 and 693 K) were tabulated, but a plot of $\log_{10}(p/\text{atm})$ *versus* $(T/\text{K})^{-1}$ contained nine experimental points ranging from 500 to 700 K. More disturbingly, the average of the tabulated values at 600 K, $\log_{10}(p/\text{atm}) = -0.3$ differs markedly from that plotted on their Figure, $\log_{10}(p/\text{atm}) = ca.\ 0.0$.

The reported pressures are about a factor of 10 higher than those in other studies, and with the thermal functions selected in this review, a third-law treatment gives a mean enthalpy of decomposition of (103.7 ± 2.7) kJ·mol^{-1} at 298.15 K, leading to $\Delta_f H^o(MgCO_3, \text{cr}, 298.15\ \text{K}) = -(1098.8 \pm 2.7)$ kJ·mol^{-1} from these questionable results.

In a related paper, Cremer *et al.* [1953CRE/ALL2] report results on the kinetics of decomposition of magnesite, where they failed to obtain reproducible results with their sample of natural magnesite from Zillertal, but found rates of decomposition with pure synthetic magnesite that are much higher than indicated in other studies, all of which note that the decomposition is extremely slow. This information, together with

the criticisms expressed above, leads this review to the opinion that the results of [1953CRE/ALL] should be mentioned for information only, and they have not been included in our assessments.

[1953GIN/FUR]

As part of a detailed program at the US National Bureau of Standards to provide materials and the data for heat capacity standards for a range of temperatures, this paper describes further results of measurements on aluminium oxide (corundum) and also on four other materials of no immediate interest to this review.

For corundum, measurements were made from 14 to 1173 K, using calorimeters and techniques which have been described in other papers from the same laboratory. The sample was from the same batch of synthetic sapphire used in the studies of [1947GIN/COR2] and [1956FUR/DOU]; spectrographic analysis showed the presence of 0.01-0.02 wt % impurities, predominantly silica.

The detailed experimental results are not given in the paper, only smoothed values of C_p and ($H(T) - H(0\ \mathrm{K})$) at round values of T from 0 to 1200 K. As shown in Figure V-29 in Section V.5.1.1.2 of the main text, these results are entirely consistent with numerous similar studies.

[1953KRA]

This section of an annual report deals with the enthalpies of solution of various oxides from 298.15 K into 20 % aqueous HF at 347.85 K. Only the experimental data on quartz, cristobalite and vitreous quartz are of interest to this review. No indication is given in this paper as to the calorimeter used, nor as to the origin or the purity of the samples. The results of the enthalpies of solutions are given as $-(33.30 \pm 0.04)$, $-(33.93 \pm 0.03)$ and $-(35.48 \pm 0.02)$ kcal·mol^{-1} for dissolution of quartz, cristobalite and vitreous quartz, respectively. The statistical meaning of these uncertainties cannot be evaluated here as the details of the measurements were not given.

From these results, using conservative uncertainty limits, this review estimates for the two reactions:

$$\mathrm{SiO_2(\alpha\text{-}quar) \rightleftharpoons SiO_2(cris)} \qquad \text{(A.32)}$$

$$\mathrm{SiO_2(\alpha\text{-}quar) \rightleftharpoons SiO_2(vit)} \qquad \text{(A.33)}$$

$\Delta_r H^\circ_m$ ((A.32), 298.15 K) = (0.63 ± 0.15) kcal·mol^{-1} or (2.64 ± 0.63) kJ·mol^{-1} and

$\Delta_r H^\circ_m$ ((A.33), 298.15 K) = (2.18 ± 0.13) kcal·mol^{-1} or (9.12 ± 0.54) kJ·mol^{-1}.

[1954ALE/HES]

The solubility of amorphous silica was measured in water and NaOH solutions at 25 °C. No difference was found when powdered silica, colloidal or freshly prepared silica gel were used. Equilibration times of up to six months were reported. The solubility of silica was generally constant from pH 5 to 8 at (0.012 to 0.014 wt %) which corresponds to $m_{\mathrm{Si(OH)_4\,(aq)}} = (2.16 \pm 0.17) \times 10^{-3}$ for amorphous SiO_2. The pH dependence at higher pH was reported to be consistent with a $\log_{10}K_c$(A.34) value of *ca.* 4.2. Activity coefficients were not mentioned.

$$\mathrm{Si(OH)_4\,(aq) + OH^- \rightleftharpoons SiO(OH)_3^- + H_2O(l)} \qquad \text{(A.34)}$$

The resolution of the spectrophotometric analytical method employed was limited for low solubilities observed in near neutral pH solutions, although the result is compatible with accepted values and is retained with an increased uncertainty, $\log_{10}K_{s,4} = -(2.67 \pm 0.12)$ at 25 °C. Their observation that the solubility of "amorphous" silica is independent of the source of the material is consistent with the findings of other investigators.

[1954GIM/MON]

The dissociation constants of a number of alkali and alkaline earth hydroxides were reported based on potential-difference measurements at 15, 25 and 35 °C using a hydrogen electrode in conjunction with an Ag|AgCl reference electrode. The E^o values for the reference electrode were taken from Harned and Ehlers [1932HAR/EHL], *viz.*, 0.22847, 0.22239 and 0.21563 V, respectively. The appropriate Nernst expression is:

$$\log_{10}c_{\mathrm{OH^-}} = F(E - E^o)/2.303RT + \log_{10}(c_{\mathrm{Cl^-}}K_w^o) \qquad \text{(A.35)}$$

For the case of $CaOH^+$, three solutions of $CaCl_2$ and $Ca(OH)_2$ were used with stoichiometric concentrations of m_1 and m_2, respectively, given in Table A-14 after application of an average minor correction of 1.0033 to convert from molar to molal units [1985SOH/NOV]. Ion-pair formation between Ca^{2+} and Cl^- was justifiably ignored. Activity coefficients were described using a modified Davies equation with the *Ba* term of unity and the non-electrostatic term fixed at 0.2*I*. The dissociation constants of $CaOH^+$, given in Table A-14, were taken from Table 3 in [1954GIM/MON]: $\Delta_r H_m^o(298.15\ \mathrm{K}) = -5.23\ \mathrm{kJ\cdot mol^{-1}}$ and $\Delta_r S_m^o(298.15\ \mathrm{K}) = -47.23\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$.

The raw data were re-calculated (see lower half of Table A-14) using $\log_{10}K_{H_2O}^o$ of – 14.346, – 14.001 and – 13.688 at 15, 25 and 35 °C, respectively. The SIT Debye-Hückel equation was used to estimate the activity coefficients with no ion interaction term, because $I_m \leq 0.0874$. The average values of $\log_{10}K_1^o$ are $-(1.352 \pm 0.002)$, $-(1.499 \pm 0.038)$ and $-(1.622 \pm 0.071)$ at 15, 25 and 35 °C, respectively, corresponding to $\log_{10}{}^*K_1^o$ values of $-(12.994 \pm 0.003)$, $-(12.502 \pm 0.038)$ and $-(12.066 \pm 0.071)$.

Table A-14: Experimental data and dissociation constants for $CaOH^+$ reported and derived from the SIT.

$10^2 m_1$	$10^2 m_2$	E/V 15 °C	E/V 25 °C	E/V 35 °C	K_1^o 15 °C	K_1^o 25 °C	K_1^o 35 °C
1.184	0.576	1.0271_5	1.0278	1.0283_5	0.047	0.043	0.308 [(a)]
1.472	1.585	1.0457_5	1.0469	1.0483_5	0.047	0.042	0.040
1.883	1.396	1.0401	1.0373_5	1.0384	0.044	0.045	0.042
		I_m/15 °C	I_m/25 °C	I_m/35 °C	K_1^o /15 °C	K_1^o /25 °C	K_1^o /35 °C
		0.0495	0.0487	0.0479	0.0450	0.0331	0.0256
SIT		0.0803	0.0769	0.0737	0.0454	0.0303	0.0217
		0.0874	0.0850	0.0828	0.0430	0.0318	0.0245

(a) This is presumably a typographical error.

[1954KAP/STA]

This paper is concerned with the determination of the enthalpies of formation of synthetic $MgCO_3$(cr) and $BaCO_3$(cr) by the dissolution of the compounds in 1 molar HCl (CO_2-saturated) at 298.15 K. The authors discuss also their earlier results on the thermochemistry of the other alkaline earth carbonates, as well as regularities within the series and extrapolate their results to $RaCO_3$(cr). Only the results concerning $MgCO_3$(cr) will be discussed here.

The calorimeter chamber was a Pyrex glass beaker of 450 cm^3 capacity. The equipment was described in detail in an earlier publication by the same group [1943KAP/KLO]. Modifications from that earlier publication are described, together with the operating procedure including calibration.

As attempts to dissolve a pyrite-contaminated natural magnesite samples failed, owing to its low reactivity, the authors resorted to synthesising the compound, using two methods that both involved precipitation of a "basic magnesium carbonate" at 413 to 423 K under 1 atm CO_2. In one case, the amorphous precipitate was heated with urea to promote crystallisation and the urea was removed by decantation and washing; the crystalline precipitate was dried at 383 K. For the second preparation, the "basic magnesium carbonate" was heated for five hours at 413 K in dry CO_2, following by washing and then drying to constant weight in CO_2 at the same temperature. The product of the second preparation was reported as crystalline from optical and X-ray techniques with a purity of 99.7 %.

However, criticism has been voiced as to the purity of the samples obtained, in particular by Langmuir [1965LAN] who noted that it was highly unlikely that pure magnesite was obtained at temperatures as low as 413 K under only one atmosphere of CO_2(g). In fact, Langmuir [1965LAN] indicated that all his attempts to reproduce the experimental conditions used by the authors yielded pure hydromagnesite (hydrated hydroxidocarbonate), as identified by XRD.

In their thermodynamic cycle leading to the enthalpy of formation of $MgCO_3$(cr), Kapustinsky *et al.* [1954KAP/STA] also used MgO(cr). The material at their disposal, of undefined origin, was calcined at 1373 K. The analytical result (gravimetry of the pyrophosphate) indicated a magnesium content of 60.4 % (theoretical, 60.3 %).

The authors obtained the average values of:

$$\Delta_r H_m(\text{A.36}) = -(53.56 \pm 0.84)\ \text{kJ·mol}^{-1}\ (-(12.80 \pm 0.20)\ \text{kcal·mol}^{-1})$$

(six listed determinations with samples of *ca.*1 g) for the dissolution of their magnesite sample according to Reaction (A.36):

$$MgCO_3(cr) + 2H^+(\text{in 1 molar HCl}) \rightleftharpoons \{Mg^{2+} + H_2O\}(\text{1 molar HCl}) + CO_2(g) \qquad \text{(A.36)}$$

and $\Delta_r H_m(\text{A.37}) = -(151.96 \pm 0.96)$ kJ·mol^{-1} ($-(36.32 \pm 0.23)$ kcal·mol^{-1}) (five listed determinations with samples of *ca.* 0.4 g) for that of their MgO(cr) according to Reaction (A.37):

$$MgO(cr) + 2H^+(\text{1 molar HCl}) \rightleftharpoons \{Mg^{2+} + H_2O\}(\text{1 molar HCl}) \qquad \text{(A.37)}$$

As all reactions were made using *ca.* 400 cm^3 of solution, the magnesium content of the two resulting series were thus reasonably similar. Combination with the CODATA values of $\Delta_f H^\circ_m(CO_2, \text{g}, 298.15\ \text{K}) = -(393.51 \pm 0.13)$ kJ·mol^{-1} and $\Delta_f H^\circ_m(\text{MgO, cr}, 298.15\ \text{K}) = -(601.60 \pm 0.30)$ kJ·mol^{-1}, also accepted in this review, leads to $\Delta_f H^\circ_m(MgCO_3, \text{cr}, 298.15\ \text{K}) = -(1093.5 \pm 1.3)$ kJ·mol^{-1}.

Nevertheless, in view of the reservations made above as to the identity of the magnesite samples used by Kapustinky *et al.* [1954KAP/STA], this review suggests an uncertainty of ± 25 kJ·mol^{-1} for these results which are therefore given for information only, as noted in the main text.

[1954SCA/BRE]

Osmotic coefficients of NaH_2PO_4, KH_2SO_4, Na_2HPO_4, K_2HPO_4, Na_3PO_4 and K_3PO_4 were measured relative to a NaCl(sln) standard at 25 °C. The corresponding arsenates were also investigated. All six sets of osmotic coefficient data were considered to be reliable and were accepted for inclusion in the final SIT analyses. Robinson and Stokes [1965ROB/STO] incorporated these osmotic coefficients into their Appendix which is the most common source of data used by Ciavatta [1980CIA] [1988CIA] to apply the SIT treatment.

[1954SCH/GAT3]

Schneider and Gattow measured the enthalpy of combustion of aluminium, at an oxygen pressure of 25 bar, using a bomb calorimeter described previously.

The aluminium used was indicated to be 99.99 % pure, in the form of fine filings wrapped in cellophane. Eight experiments were reported, with aluminium samples ranging from 0.1 to 0.2 g. Combustion in porcelain crucibles was far from complete, ranging from 45 to 92 %; corrections were made for the amount of unreacted aluminium, which was determined eudiometrically.

An average value of $-(838.06 \pm 2.92)$ kJ·(mol Al)$^{-1}$ was reported (2σ uncertainties) for the heat of combustion. The authors tried to identify and quantify crystallographically as best as they could other alumina modifications(s) (amounting up to 25 %) in the combustion products. They came to the conclusion that a correction of at most 10 kJ·(mol Al_2O_3)$^{-1}$ would account for these impurities.

Finally, they suggested $\Delta_f H^\circ_m(Al_2O_3, cr, 298.15\ K) = -(1682 \pm 8)$ kJ·mol^{-1}, a value that this review will consider only as indicative.

[1954SCH/TAY]

Schofield and Taylor employed a batch potentiometric method to determine the first hydrolysis constant of Al^{3+} with $AlCl_3$ concentrations varying from 2×10^{-5} to 10^{-2} mol·kg^{-1}. The pH was given on the conventional activity scale. A simple Debye-Hückel expression was used to extrapolate the hydrolysis constants to $I = 0$. As in a later study [1963FRI/PEE], which used essentially the same approach except that the solutions were slightly neutralised by addition of Na_2CO_3, it was assumed that the concentration of $AlOH^{2+}$ was equal to that of H^+. Re-calculation of the experimental data tabulated in this paper using the SIT, gave consistent hydrolysis constant values of $\log_{10}{}^*K_1^\circ$ for aluminium concentrations in the range 5×10^{-5} to 10^{-3} mol·kg^{-1} when data for the lowest and two highest concentrations were ignored. This resulted in an average value of $\log_{10}{}^*K_1^\circ = -(4.99 \pm 0.20)$ at 298.15 K.

[1955BRO/PRU]

Freezing-point depression, θ, measurements at 0 °C (± 0.0001 °C precision) were made for the solutes, KCl, $CuSO_4$, $ZnSO_4$, $MgSO_4$, $CaSO_4$, $CoSO_4$, and $NiSO_4$. Data for $NiSO_4$(sln) were treated with the SIT [2005GAM/BUG] and it was concluded that the value of the ion-pair association constant so obtained was highly dependent on the concentration range treated, which was limited to < 0.033 mol·kg^{-1} in that review. Molal ion-pair association constants were calculated by Brown and Prue for various assumed distances of closest approach in the denominator of the Debye-Hückel expression and

resulted in a range of predicted K_1^o values from 95 to 244 $dm^3 \cdot mol^{-1}$ for $MgSO_4(aq)$ and for $CaSO_4(aq)$ 185 to 303 $dm^3 \cdot mol^{-1}$. In re-treating their results, either the value of α (the degree of dissociation of the ion pair) or the association constant was varied at each molality, m, until the osmotic coefficient, Φ, or θ, matched the experimental value. The real ionic strength, activity coefficient and ultimately the ion-pair association constant were derived in each iteration loop. Table A-15 and Table A-16 summarise the results of the procedure for $MgSO_4(aq)$ and $CaSO_4(aq)$, respectively.

Table A-15: Summary of the experimental and SIT-based values of the formation constants for $MgSO_4(aq)$ at 0 °C.

$10^3 m$ /mol·kg^{-1}	θ /°C	Φ	α	I_m /mol·kg^{-1}	γ	K_1^o
6.10	0.0176	0.7798	0.8069	0.0197	0.2277	139
9.67	0.0272	0.7602	0.7518	0.0291	0.2667	155
15.23	0.0406	0.7205	0.7240	0.0441	0.3137	147
19.35	0.0505	0.7054	0.6997	0.0542	0.3388	151
26.50	0.0672	0.6854	0.6679	0.0708	0.3735	157
28.30	0.0715	0.6828	0.6590	0.0746	0.3805	160
32.10	0.0803	0.6761	0.6447	0.0828	0.3947	164

The average value of K_1^o for the formation of $MgSO_4(aq)$ is (153 ± 17) $kg \cdot mol^{-1}$ with a 2σ statistical uncertainty.

Table A-16: Summary of the experimental and SIT based values of the formation constants for $CaSO_4(aq)$ at 0 °C.

$10^3 m$ /mol·kg^{-1}	θ /°C	Φ	α	I_m /mol·kg^{-1}	γ	K_1^o
4.18	0.0124	0.7974	0.7608	0.0127	0.6465	237
4.87	0.0141	0.7781	0.7479	0.0146	0.6299	234
5.26	0.0152	0.7762	0.7386	0.0155	0.6219	236
5.60	0.0160	0.7672	0.7366	0.0165	0.6145	230
6.25	0.0176	0.7548	0.7220	0.0180	0.6031	231
6.70	0.0190	0.7612	0.7092	0.0190	0.5965	243
7.14	0.0200	0.7549	0.6916	0.0198	0.5916	260
7.42	0.0206	0.7460	0.6905	0.0205	0.5869	254
7.82	0.0218	0.7498	0.6769	0.0212	0.5827	266
9.15	0.0249	0.7318	0.6536	0.0239	0.5668	276
9.37	0.0256	0.7352	0.6519	0.0244	0.5640	275
9.52	0.0257	0.7266	0.6502	0.0248	0.5623	275
10.46	0.0282	0.7240	0.6404	0.0268	0.5519	275
10.92	0.0293	0.7206	0.6286	0.0275	0.5487	286
11.00	0.0294	0.7197	0.6210	0.0273	0.5494	296

The average value of K_1^o for the formation of $CaSO_4(aq)$ is (258 ± 43) $kg \cdot mol^{-1}$ with a 2σ uncertainty. The limited range of concentration accessible resulted in small measurable perturbations of the freezing point of these solutions such that the experimental uncertainty of the ion-pair association constant is larger than for the other

metal sulfates. The average value of K_1° for $MgSO_4$(aq) is in very good agreement with that determined subsequently [1971ISO] by the same method and is retained in this review. There is a trend in the ion-pair association constant for $CaSO_4$(aq) with concentration that may reflect the small change in the freezing point at the lower temperatures in particular.

[1955FLO]

The notable variation of a number of physical properties (optical properties, transition temperature) of cristobalite and tridymite suggests that these phases are frequently structurally inhomogeneous, and XRD indicates that this variation is due to one-dimensional disorder within the crystal. A cristobalite prepared from pure silica glass at 1650 °C was found to be the most ordered. Dilatometry studies were recommended as the best means of studying the disorder, and an interesting series of dilatometric curves for cristobalite and tridymite samples prepared at various temperatures (and thus with varying disorder) is given.

The author was only able to obtain tridymite by the use of a mineraliser such as Na_2O, K_2O or CaO, and flame spectrometric studies showed that the tridymite prepared in this way always contains a fraction of a percent of foreign ions, such as Na^+, K^+ or Ca^{2+}. He therefore suggests that tridymite is not a stable phase in the pure SiO_2 unary system, and that quartz converts directly to cristobalite at a temperature probably between 1000 and 1100 °C. See also [1957EIT] for a review and discussion of this paper.

[1955GRA/GOL]

Graf and Goldsmith measured the decomposition pressures of dolomite at temperatures from 618 to 1136 K and pressures up to 138 MPa. The dolomite was a natural sample from Brazil, containing less than 0.5 mol % Fe and Mn carbonates, and with a Ca:Mg ratio of 1:1, as determined by wet and spectrographic analysis (no details given). About 50 runs were made, with heating times of 1.5 hours at the highest temperature and 1630 hours at the lowest. As in other studies the products after heating were examined by XRD to establish whether decomposition had occurred. As a check on the results dolomite was synthesised in its derived stability field, but no details of these experiments are given. The authors also studied the equilibrium between dolomite and two magnesian calcites of different compositions, which do not concern this review.

As noted in the main text, these results of this study are in excellent agreement with those obtained by Harker and Tuttle [1955HAR/TUT].

[1955HAR/GAR]

A synopsis of this paper appeared in [2013LEM/BER].

[1955HAR/TUT]

The authors measured the decomposition of magnesite, calcite and dolomite at temperatures up to 1173 K and CO_2 pressures up to 280 MPa (for magnesite).

This study illustrates clearly the difficulties and uncertainties in the determination of the dissociation pressure of magnesite: the very slow reaction rate for both the forward and backward reactions, even at temperatures above 1000 K and the corresponding high pressures of more than 2500 bar (and thus the need to consider the non-ideality of CO_2(g) and volume changes in MgO(cr) and $MgCO_3$(cr) in processing the data), the lack of data on the thermodynamic properties of magnesite above 700 K and the possible interference from even small quantities of impurities in the reactants.

Both the forward and backward reactions were studied. The only description of the magnesite and periclase used is that they were commercial analytical reagents, with no further details of their purity.

The apparatus consisted of a nichrome furnace which could be raised and lowered to surround the pressure vessels (of unspecified material); the pressures were measured on a commercial gauge and the temperatures by unspecified thermocouples. For a run, a mixture of both the initial and final materials was wrapped in gold foil and placed in the pressure vessel. At the end of the run, the temperature and pressure were lowered simultaneously to minimise any back reactions, and the remaining products were examined by optical and X-ray analyses.

Measurements on magnesite were made from 873 to 1173 K, at pressures from from 60 to 2800 bar. Owing to the difficulties involved in the measurements, these were not of the highest precision; the uncertainties in T were estimated to be ± 7 K and those in the pressures, to be ± 1 bar for $p < 35$ bar, ± 3.5 bar for $35 < p/\text{bar} < 70$ and ± 35 bar for $p > 70$ bar. In addition, it was of course impossible to establish whether equilibrium had been achieved during the experiments, so the results are just correlated with the presence or absence of MgO and $MgCO_3$ after the run. The phases found to be present in 54 experiments in the range noted above are tabulated, and a curve is drawn indicating the univariant pressure as a function of T. However, for many temperatures, the correspondence between the table and the curve is far from clear, since the Figure 6 in their paper includes only about a half of the 54 tabulated points. Moreover, at some pressures, the bracketing temperatures (whether periclase or magnesite is present) are different by 20 or even 50 K.

In using these results, we have read the values of the pressures at 923 to 1123 K from the univariant curve presented by the authors; this also seems to be the procedure followed by Robie [1965ROB] in his analysis of these data.

In analysing these data to obtain the enthalpy of dissociation of magnesite, it is necessary to include corrections for the non-ideality of CO_2(g) at the high pressures involved (for which we have used the data in [1988SPY/REE]) and to include a relatively small term to allow for the Gibbs energy involved in the changes of the molar volumes of MgO(cr) and $MgCO_3$(cr) with temperature and pressure—see Equation 17 of [1965ROB].

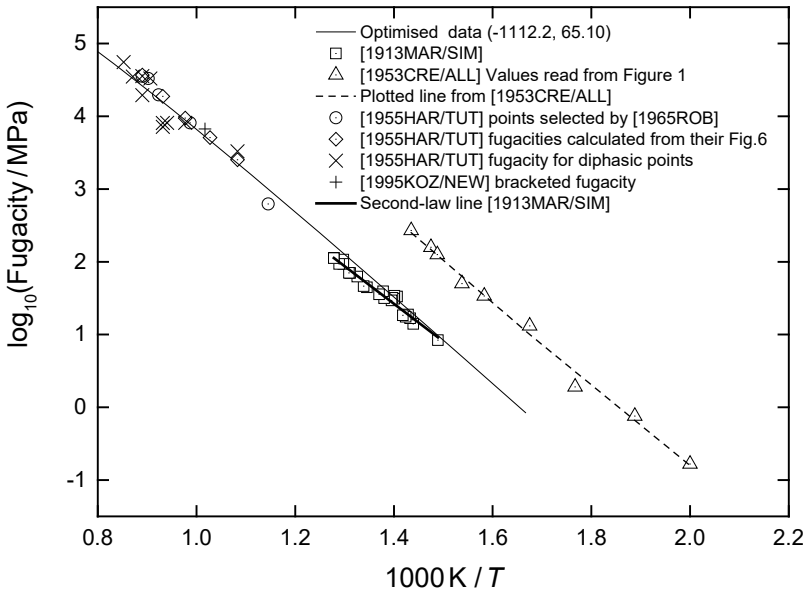
Of course, the full data for these molar volume corrections are not available (especially for magnesite above its decomposition pressure at ambient pressure, *ca.* 675 K). We have therefore approximated this integral using separate data on the T and p dependence of the molar volumes of MgO ([1995REE/GOE], [2008JAC/HOL]) and $MgCO_3$ ([1985MAR/REE], [1999FIQ/REY], [2002FIQ/GUY]), by calculating the molar volumes at the relevant temperature and the mean pressure between 1 bar and the equilibrium pressure. This is a probably better approximation than that used by Robie [1965ROB], who just took ΔV_m at 298.15 K.

The derived value of $\Delta_r H_m^o(298.15\ \mathrm{K})$ for the decomposition of magnesite (Eq.(A.38)):

$$MgCO_3(cr) \rightleftharpoons MgO(cr) + CO_2(g) \qquad (A.38)$$

is $-(116.93 \pm 0.98)\ \mathrm{kJ \cdot mol^{-1}}$, where the 2σ uncertainty does not include any experimental or data analysis uncertainties, which are much larger. However, in view of all these problems/difficulties with the data by these authors and the lack of data for their processing, we have given this value a low weight in our final analysis of the data for $MgCO_3$(cr), although this value does, in fact, agree excellently with similar high-pressure studies in [1968JOH/MET] and [1995KOZ/NEW] and overlaps within the statistical uncertainties with those from the lower-pressure study in [1913MAR/SIM], $-(118.74 \pm 2.11)\ \mathrm{kJ \cdot mol^{-1}}$, which suffers from fewer uncertainties in the experimental values and their analysis. A comparison of these results with other experimental and the assessed data is shown in Figure A-7. Note that the "Optimised data" corresponds to the selected $\Delta_f H_m^o$ and S_m^o for $MgCO_3$(cr), as explained through Section V.6.1.1.5.

Figure A-7: Decomposition pressure of $MgCO_3$(cr).



[1955RUS/EDW]

Density measurements were made with a pycnometer for the more viscous $NaAl(OH)_4$ + NaOH solutions and with a platinum cylinder of known mass and volume for the less viscous solutions. No further details of these measurements were provided to allow a critique of the methodology and they are probably reliable to ± 0.001 $g \cdot cm^{-3}$ at best. Therefore, these results were not included in the fitting equation [2006REY/BER] for predicting the density of $NaAl(OH)_4$ + NaOH solutions, noting that the reported concentration units were $g \cdot dm^{-3}$. In fact, the densities determined by Russell *et al*. showed a strong positive bias of up to 9 % compared with the predicted values [2006REY/BER]. Fortunately, for calculations of the values for $\log_{10} K_{s,4}$, corresponding to Reaction (A.39), the density terms for conversion to molal units cancel out.

$$Al(OH)_3(cr) + OH^- \rightleftharpoons Al(OH)_4^- \quad \text{(A.39)}$$

Only the gibbsite solubility data at 40 °C (264 hours), 50 °C (168 hours), 60 °C (168 hours), 70 °C (96 hours), 80 and 90 °C were re-treated because of concerns that the trihydroxides dehydrate to boehmite at higher temperatures. Furthermore, the solution samples were withdrawn from pressure vessels so that the assumption that no change in composition occurred during quenching becomes more tenuous with increasing

experimental temperature. An unweighted SIT fit of the $\log_{10}K^{\circ}_{s,4}$ data for gibbsite as a function of temperature and ionic strength resulted in Equation (A.40) (1σ):

$$\log_{10}K_{s,4}(\mathrm{Al(OH)_3, gibb}) = (3.864 \pm 0.064) - (1529.76 \pm 21.71)(T/\mathrm{K})^{-1} + (0.0613 \pm 0.0013)\, I_m \quad \text{(A.40)}$$

This fit included all the data to I_m=11.09 with the implied temperature-independent value of $\Delta\varepsilon$(A.39) = – (0.061 ± 0.003) kg·mol^{-1} (2σ), which is influenced strongly by the high ionic strength results at 40 and 70 °C. This and the subsequent SIT fits apply to a self-medium of Na^+-$Al(OH)_4^-$-OH^--H_2O so that $\Delta\varepsilon$ is a composite quantity. Contributions from $Al(OH)_3$(aq) can be ignored due to low stability of this species, but the existence of dimeric Al(III) species were ignored with less confidence. Any ion pairing involving Na^+ is accommodated in the activity-coefficient expressions. Finally, these data for gibbsite were later included in a Pitzer ion interaction treatment of the combined literature data [1992WES]. Despite these assumptions, the simple linear fit according to interactions involving only singly charged anions gave a good representation of the available data in this paper, especially considering the techniques that were available to them at that time. The value of $\log_{10}K^{\circ}_{s,4}$ ($Al(OH)_3$, gibb) from Equation (A.40) corrected to 25 °C is – (1.27 ± 0.19) (2σ), *cf.* the mean value of – (1.14 ± 0.06) [1992WES]. Considering the equation, $\varepsilon(Na^+, Al(OH)_4^-) = \Delta\varepsilon$(A.39) + $\varepsilon(Na^+, OH^-)$, where $\varepsilon(Na^+, OH^-)$ = (0.04 ± 0.01) kg·mol^{-1}, then $\varepsilon(Na^+, Al(OH)_4^-)$ = (0.02 ± 0.02) kg·mol^{-1} at 25 °C, *cf.* $\varepsilon(Na^+, Al(OH)_4^-)$ = (0.02 ± 0.01) kg·mol^{-1} in mainly NaCl + NaOH solutions in [1992WES]. Finally, the value of $\Delta_r H^{\circ}_m$((A.39), 298.15 K) = (29.3 ± 0.8) kJ·mol^{-1} (2σ) is distinctly different from (23.74 ± 0.06) kJ·mol^{-1} obtained by Wesolowski [1992WES].

A simultaneous unweighted SIT fit of the $(\log_{10}K_{s,4} - \log_{10}a_{H_2O})$ data for boehmite was made to $I_m \leq 10$, as a function of temperature (80 to 170 °C) and ionic strength, resulting in Equation (A.41) (1σ):

$$\log_{10}K_{s,4}(\mathrm{boehmite}) = (2.088 \pm 0.064) - (1039.01 \pm 25.29)(T/\mathrm{K})^{-1} + (0.080 \pm 0.001) I_m \quad \text{(A.41)}$$

Only the bayerite (γ-$Al(OH)_3$) solubility data at 40 °C (148 hours), 50 °C (120 hours), 60 °C (96 hours), 70 °C (48 hours), 80 °C (16 hours), and 90 °C (5.5 hours) were re-treated. An unweighted fit of the $\log_{10}K_{s,4}$ data for bayerite as a function of temperature is represented in Equation (A.42). The value of $\log_{10}K^{\circ}_{s,4}$ ($Al(OH)_3$, baye) at 25 °C is – (0.81 ± 0.22) where the large uncertainty (2σ) is due to the paucity of data and the larger experimental uncertainty of the bayerite data.

$$\log_{10}K_{s,4}(\mathrm{Al(OH)_3, baye}) = (3.171 \pm 0.146) - (1186.1 \pm 48.8)(T/\mathrm{K})^{-1} + (0.046 \pm 0.022) I_m \quad \text{(A.42)}$$

The results of applying the SIT are given in Table A-17 and are compared to those predicted by the authors.

Table A-17: Summary of solubility results at 25 °C with 2σ uncertainties.

Solid	$\log_{10}K^{\circ}_{s,4}$ /SIT	$\Delta_r H^{\circ}_m$ /kJ·mol^{-1} /SIT	$\log_{10}{}^{*}K^{\circ}_{s,4}$ /SIT	$\log_{10}K^{\circ}_{s,4}$ [1955RUS/EDW]	$\Delta_r H^{\circ}_m$ /kJ·mol^{-1} [1955RUS/EDW]
Gibbsite	– (1.3 ± 0.3)	(29.5 ± 1.2)	– 15.29	– 1.27	30.7
Bayerite	– (0.9 ± 0.4)	(22.7 ± 3.7)	– 14.9	– 0.83	21.9
Boehmite	– (1.4 ± 0.2)	(19.8 ± 0.4)	– 15.39	– 1.27	19.9

The mean value of the "fitting factor" $\Delta\dot{\varepsilon}$(A.39) = – (0.070 ± 0.002) kg·mol^{-1} (2σ) accommodates all the data to ionic strengths of 10 mol·kg^{-1} from 40 to 170 °C.

[1956BAT/BOW]

The solubilities of five independently prepared samples of $Ca(OH)_2$(cr) were measured after they had been initially ignited at 1000 °C for 45 minutes then exposed to boiling water, filtered and dried at 110 °C. The initial average solubility was (0.02037 ± 0.00011) mol·kg^{-1} at 25 °C. After an additional month the average of four samples was (0.02032 ± 0.00006) mol·kg^{-1}, and after six months the average of two samples was 0.02022 mol·kg^{-1}. The one-month average was accepted by Bates *et al*. as representing true thermodynamic equilibrium. The solubilities at 20 and 30 °C were reported as 0.0211 and 0.0196 mol·kg^{-1}, respectively. Thus, given that $K^{\circ}_{s,0} = 4m^3_{sat}\gamma^3$ then without making a correction for hydrolysis $\log_{10}K^{\circ}_{s,0}$ = – 4.987, – 5.038 and – 5.083 at 20, 25 and 30 °C, respectively. However, hydrolysis is significant in these solutions as the results in Table A-18 indicate.

Table A-18: Evaluation of the $Ca(OH)_2$(cr) solubility data with the hydrolysis corrections based on the selected values in this report.

t /°C	Solubility mol·kg^{-1}	$\log_{10}K_{H_2O}$	I_m	$m_{Ca^{2+}}$	$10^3 m_{CaOH^+}$	$\log_{10}K^{\circ}_{s,0}$
20	0.0211	– 13.990	0.0526	0.0157	5.37	– 5.188
25	0.0203	– 13.822	0.0505	0.0151	5.21	– 5.237
30	0.0196	– 13.663	0.0486	0.0145	5.11	– 5.284

Note that the value of $\log_{10}K^{\circ}_{s,0}$ = – 5.237 at 25 °C in Table A-18 is in excellent agreement with those obtained from SIT treatments of the solubility results in [1931JOH/GRO], namely a mean $\log_{10}K^{\circ}_{s}$ = – (5.24 ± 0.04) (2σ), and $\log_{10}K^{\circ}_{s}$ = (5.232 ± 0.008) [1967YEA/MAR]. This is a very rigorous publication dealing with pH measurements of $Ca(OH)_2$ solutions, especially in regards to the number of experiments and their quality.

[1956BEL/PAN]

Values of the dissociation constants of $CaOH^+$, TlOH(aq) and $BaOH^+$ at 25 °C were derived from kinetic measurements in an adiabatic calorimeter. The proposed mechanism involved the rate-determining step, $kc_{C_2H_5NO_2}c_{OH^-}^{1/2}$. This "thermal maximum method" was applied previously by Bell and co-workers [1949BEL/PRU], [1950BEL/WAI] in which it was assumed that only free hydroxide ions are involved in the reaction, allowing the dissociation constant of $CaOH^+$ to be determined by a comparison of the reaction rates in mixtures of NaOH and NaCl, and NaOH and $CaCl_2$. The $CaCl_2$ concentration was varied between 0.02 and 0.06 mol·dm^{-3}, whereas the NaOH concentration was varied between 0.02 and 0.03 mol·dm^{-3}. Activity coefficients were described using a simple Debye-Hückel expression with $Ba = 1$. Bell and Panckhurst reported a value of (0.035 ± 0.005) mol·dm^{-3} for the dissociation constant of $CaOH^+$ at 25 °C. The experimental data presented in Table 2 in [1956BEL/PAN] were re-calculated using the SIT treatment with $\Delta\varepsilon = 0$ and the results are shown in Table A-19. Higher ionic strengths were reached in this study compared to those from most other studies of this equilibrium and yet it is significant that the second-order rate constants for the reaction in the absence of $CaCl_2$ gave a constant value of (5.83 ± 0.011) dm^3·mol^{-1}·s^{-1} to $I_c = 0.062$ and showed no trend within a larger scatter to $I_c = 0.9$ (NaCl). The reported and SIT values of K_1° in Table A-19 are virtually identical with an average value $K_1^\circ = (0.0337 \pm 0.0047)$ mol·kg^{-1} (2σ).

Table A-19: Stoichiometric molal concentrations[(a)], the fraction of free hydroxide ion (α), real ionic strength and the dissociation constant of $CaOH^+$ at 25 °C.

m_{NaOH} stoi.	m_{CaCl_2} stoi.	α	K_1°	I_m	α	K_1° /SIT
reported				re-calculated		
0.0262	0.0198	0.86	0.036	0.0449	0.8491	0.0345
0.0301	0.0198	0.84	0.030	0.0447	0.8319	0.0278
0.0250	0.0198	0.83	0.028	0.0430	0.8148	0.0260
0.0280	0.0294	0.82	0.036	0.0622	0.8062	0.0348
0.0206	0.0395	0.80	0.042	0.0805	0.7804	0.0407
0.0224	0.0330	0.79	0.035	0.0676	0.7804	0.0343
0.0231	0.0395	0.78	0.036	0.0799	0.7633	0.0357
0.0204	0.0593	0.72	0.036	0.1171	0.7033	0.0362

(a) Only a trivial correction was necessary to convert from molar to molal concentrations [1985SOH/NOV].

[1956BEN]

In the frame of a broader study of the effects of pressure and shearing stress on mineral reactions, that are beyond the scope of the present review, Bennington [1956BEN] presented new experimental data leading to the enthalpy of formation of talc (that is not part of the charter of the present review) and serpentine (with nominal formula $Mg_3Si_2O_5(OH)_4$) from their component binary oxides. Following the reference to this study by King *et al.* [1967KIN/BAR] from the same laboratory, we take the latter

material to be chrysotile. Brief results leading to the enthalpy of formation of brucite are also presented. The hydrofluoric solution calorimeter used is a version derived from earlier designs at the U.S. Bureau of Mines [1948TOR/SAH], [1951KIN], [1954KEL/TOD].

Two natural fibrous chrysotile samples were used, containing, as major impurities (in wt %), 0.66 FeO, 0.64 Al_2O_3 and 0.07 MnO for the Gila County (AZ) sample, and 0.27 Al_2O_3, 1.68 Fe_2O_3, 0.36 FeO, 0.06 MnO and 0.08 NiO for the Thedford (Ontario) sample. Only optical (not diffraction) data were given. For the brucite used, the authors only mention that no impurities were visible under microscope after prolonged grinding (five hours) in a power mortar. In the calorimetric procedure, the samples at 298.15 K were dropped into the solution medium used (20.1 % HF at 354.15 K). The thermodynamic cycle leading to the enthalpy of formation of chrysotile involved MgO(cr), $Mg(OH)_2$(cr), SiO_2(α-quar) and H_2O(l). This cycle is analogous to that used later by King *et al.* [1967KIN/BAR] and detailed in the relevant Appendix A entry. Five dissolution runs were given for each chrysotile sample leading, after correction for impurities, to the reported averages of $-(474.92 \pm 1.40)$ cal·g^{-1} (Gila sample) and $-(476.41 \pm 1.09)$ cal·g^{-1} (Thedford sample), in good agreement. In their thermodynamic cycle, the author used, for Reaction (A.43):

$$\mathrm{Mg_3Si_2O_5(OH)_4(cr, 298.15\ K) + 18HF(sln, 354.15\ K) \rightleftharpoons 3MgF_2(s, 354.15\ K) + 9H_2O(sln, 354.15\ K) + 2H_2SiF_6(sln, 354.15\ K)} \quad \text{(A.43)}$$

a value of $-(550.606 \pm 1.264)$ kJ·mol^{-1}, this value and all others to follow being converted from the originally reported values in kcal. A straight average of the two sets, using the same atomic weights leads this review to a value 1.00 kJ more negative. We have, however, kept the value used by the author as he may have had reasons to bias his final value towards one set of results. For the other dissolution reactions involved in the cycle the author indicates that he relied on the results of Weeks [1956WEE2]. The full reference to this work, not given in [1956BEN], is included in the bibliography.

The enthalpy of formation of chrysotile from its constituent oxides according to Reaction (A.44) is then given by the author as:

$$\mathrm{3MgO(cr, 298.15\ K) + 2SiO_2(\alpha\text{-quar}, 298.15\ K) + 2H_2O(l, 298.15\ K) \rightleftharpoons Mg_3Si_2O_5(OH)_4(cr, 298.15\ K)} \quad \text{(A.44)}$$

$\Delta_r H^\circ_m$((A.44), 298.15 K) = $-(165.410 \pm 1.464)$ kJ·mol^{-1}. As this review is of the opinion that these uncertainties are somewhat optimistic in view of the complexity of the cycle used and of the measurements carried out, we have taken more realistic uncertainties, namely ± 2.510 kJ·mol^{-1} by analogy with the later work of King *et al.* [1967KIN/BAR]—see the relevant Appendix A entry.

From Reaction (A.44), using the CODATA values [1989COX/WAG], this review calculates, from this work, $\Delta_f H_m^o$ ($Mg_3Si_2O_5(OH)_4$, cr, 298.15 K) = – (165.410 ± 2.510) – 2 (910.7 ± 1.00) – 3 (601.60 ± 0.30) – 2 (285.830 ± 0.040) = – (4363.3 ± 3.3) kJ·mol^{-1}. This is in excellent agreement with the value measured subsequently in the same laboratory by King *et al.* [1967KIN/BAR], which we re-calculate to be – (4364.63 ± 3.33) kJ·mol^{-1}.

The determination of the enthalpy of formation of brucite $Mg(OH)_2$(cr) from magnesium oxide and water, according to Reaction (A.45):

$$MgO(cr, 298.15\ K) + H_2O(l, 298.15\ K) \rightleftharpoons Mg(OH)_2(cr, 298.15\ K) \quad (A.45)$$

was also obtained from the following reactions:

$$MgO(cr, 298.15\ K) + 2HF(sln, 354.15\ K) \rightleftharpoons MgF_2(s, 354.15\ K) + H_2O(sln, 354.15\ K) \quad (A.46)$$

with $\Delta_r H_m$ (A.46) = – (144.507 ± 0.460) kJ·mol^{-1},

$$H_2O(l, 298.15\ K) \rightleftharpoons H_2O(sln, 354.15\ K) \quad (A.47)$$

with $\Delta_r H_m$ (A.47) = (2.301 ± 0.054) kJ·mol^{-1},

$$Mg(OH)_2(cr, 298.15\ K) + 2HF(sln, 354.15\ K) \rightleftharpoons MgF_2(s, 354.15\ K) + H_2O(sln, 354.15\ K) \quad (A.48)$$

with $\Delta_r H_m$ (A.48) = – (104.215 ± 0.397) kJ·mol^{-1}.

Thus $\Delta_r H_m$ (A.45) = – (37.991 ± 0.607) kJ·mol^{-1}.

This result was not discussed by Bennington.

Using the CODATA values [1989COX/WAG], $\Delta_f H_m^o$ (MgO, cr, 298.15 K) = – (601.60 ± 0.30) kJ·mol^{-1} and $\Delta_f H_m^o$ (H_2O, l, 298.15 K) = – (285.830 ± 0.040) kJ·mol^{-1}, this review calculates $\Delta_f H_m^o$ ($Mg(OH)_2$, cr, 298.15 K) = – (925.421 ± 0.678) kJ·mol^{-1}. This value, rounded to – (925.42 ± 0.60) kJ·mol^{-1}, was included in the assessment of Gurvich *et al.* [1994GUR/VEY], that is the basis of the value of $\Delta_f H_m^o$ ($Mg(OH)_2$, cr, 298.15 K) selected in this review.

[1956EGA/WAK]

This paper is concerned with low-temperature heat capacity measurements of monocalcium phosphate monohydrate, $Ca(H_2PO_4)_2 \cdot H_2O$(cr), and, based on solution calorimetry, the enthalpy of formation of this salt and of its anhydrous form, $Ca(H_2PO_4)_2$(cr). The preparative methods of both salts were given. Based on petrographic, X-ray and spectrographic examination, the monohydrate was crystallographically pure; its chemical analysis was reported as 22.45 % CaO

(theor., 22.25 %) and 56.20 % P_2O_5(theor., 56.31 %). The anhydrous compound was shown to be a pure crystallographic phase, based on microscopic examination; its chemical analysis was reported as 23.94 % CaO (theor., 23.95 %) and 60.63 % P_2O_5 (theor., 60.65 %).

The other species used in the solution calorimetry were also well characterised. The average of six chemical analyses of the specially prepared calcium hydroxide corresponded to 100.027 % $Ca(OH)_2$(cr) and microscopic examination showed the presence of insignificant amounts of calcium oxide as cores in the hydroxide crystals. Phosphoric acid hemihydrate, $2H_3PO_4 \cdot H_2O$, was crystallised from reagent grade phosphoric acid and dissolved in distilled water to form an acid containing 84.40 % H_3PO_4 (H_3PO_4 in $1.0055H_2O$).

The phosphate content of the solution was checked by quadruplicate determination as magnesium pyrophosphate. The low-temperature calorimeter used was described previously by the same authors [1951EGA/WAK] and used without modifications. The Dewar-type solution calorimeter (0.875 dm^{-3}) was similar to the equipment used by Southard [1940SOU]; its features and mode of operation were given in great detail in the present paper.

For the heat capacity determination (10.065 to *ca.* 345 K), 68.4939 g $Ca(H_2PO_4)_2 \cdot H_2O$(cr) were used. This corresponds to 0.271726 mol, corrected for 1981 atomic weights, rather than the reported 0.271706 mol. The experimental heat capacity curve showed a small sharp peak at 273 K corresponding to 0.0308 cal that was positively identified as being due to a small amount of water that arose from the decomposition of a trace amount of the hydrate during the soldering of the calorimeter chamber. A correction was made for this impurity, although it was in fact negligible.

The heat capacities were tabulated at 5 K intervals from 10 to 345 K, and the entropy at 298.15 K was calculated by graphical integration, the extrapolation to 0 K being made with a Debye function with a Debye temperature of 133 K. With the tiny correction to 1981 atomic weights, the calculated values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(Ca(H_2PO_4)_2 \cdot H_2O, \mathrm{cr}, 298.15\ \mathrm{K}) = (258.84 \pm 0.25)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(Ca(H_2PO_4)_2 \cdot H_2O, \mathrm{cr}, 298.15\ \mathrm{K}) = (259.84 \pm 0.42)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}},$$

where the authors' conservative estimate of the accuracy of their heat capacities (± 0.05 %) has been doubled.

The enthalpy of formation of the hydrate is based on the use of Reaction (A.49):

$$2(H_3PO_4, \mathrm{in}\, 1.0055\, H_2O)(\mathrm{sln}) + Ca(OH)_2(\mathrm{cr}) \rightleftharpoons Ca(H_2PO_4)_2 \cdot H_2O(\mathrm{cr}) + H_2O(\mathrm{l}) \quad \text{(A.49)}$$

through the measure of the difference between the sums of the enthalpies of solution of the products and the sum of the enthalpies of solution of the reactants in 900 g of

4 molal hydrochloric acid at 298.15 K (within a few hundredths of a degree) with suitable additions so that the ionic concentrations of the final solutions of the products and reactants of Reaction (A.49) were identical.

The following enthalpies of solution (re-calculated here with 95 % confidence limits) were obtained:

H_3PO_4, in 1.0055H_2O(l) (3-7 g samples, 4 determinations): – (7.443 ± 0.039) kJ·mol^{-1},

$Ca(OH)_2$(cr) (1-2g samples, 5 determinations): – (131.047 ± 0.508) kJ·mol^{-1},

H_2O(l) (1-10g samples, 5 determinations):– (0.2713 ± 0.0063) kJ·mol^{-1},

(larger uncertainty given by the authors),

$Ca(H_2PO_4)_2{\cdot}H_2O$(cr) (5-9 g samples, 6 determinations): (14.435 ± 0.025) kJ·mol^{-1}.

Substitution in Reaction (A.49) of these molar enthalpies leads to:

$$\Delta_r H^\circ_m \text{ ((A.49), 298.15 K)} = -(160.097 \pm 0.515) \text{ kJ·mol}^{-1}.$$

In the re-evaluation of $\Delta_f H^\circ_m$ ($Ca(H_2PO_4)_2{\cdot}H_2O$, cr, 298.15 K), we used: $\Delta_f H^\circ_m$ (H_2O, l, 298.15 K) = – (285.830 ± 0.040) kJ·mol^{-1}, $\Delta_f H^\circ_m$ ($Ca(OH)_2$, cr, 298.15 K) = – (985.16 ± 1.00) kJ·mol^{-1} from CODATA and this review, and of the CODATA-compatible $\Delta_f H^\circ_m$ ((H_3PO_4, in 1.0055H_2O), sln, 298.15 K) obtained as follows. We have accepted, for consistency reasons, the value of – 8.297 kJ·mol^{-1} for the enthalpy of transfer of H_3PO_4(l) to (H_3PO_4, in 1.0055H_2O) listed by Egan *et al.*[1956EGA/WAK] as their own unpublished data. Then, making use of $\Delta_f H^\circ_m$ (H_3PO_4, l) = – 1266.9 kJ·mol^{-1} from [1982WAG/EVA], corrected by – 6.792 kJ·mol^{-1} to obtain a value compatible with the CODATA $\Delta_f H^\circ_m$ (H_3PO_4, in 100H_2O) = – (1296.2 ± 1.0) kJ·mol^{-1}, we obtain:

$$\Delta_f H^\circ_m \text{ ((H_3PO_4, in 1.0055}H_2O\text{), 298.15 K)} = -(1281.989 \pm 1.000) \text{ kJ·mol}^{-1}.$$

With these, we obtain $\Delta_f H^\circ_m$ ($Ca(H_2PO_4)_2{\cdot}H_2O$, cr, 298.15 K) = – (3423.405 ± 2.295) kJ·mol^{-1}.

For the anhydrous species, $Ca(H_2PO_4)_2$(cr), the enthalpy of Reaction (A.50):

$$2(H_3PO_4, \text{in} 1.0055\,H_2O)(\text{sln}) + Ca(OH)_2(\text{cr}) \rightleftharpoons Ca(H_2PO_4)_2(\text{cr}) + 2H_2O(\text{l}) \qquad \text{(A.50)}$$

was determined similarly.

Re-calculation of the enthalpy of solution of $Ca(H_2PO_4)_2$(cr) (4-9 g samples, six determinations, 4 molal HCl solution medium to which was added the stoichiometric amount of water indicated by Reaction (A.50)), yielded – (4.428 ± 0.171) kJ·mol^{-1}, with 95 % confidence limits. We then obtain $\Delta_r H^\circ_m$ ((A.50), 298.15 K) = – (140.962 ± 0.542) kJ·mol^{-1}. We note that the authors' value for this (– 35.770 kcal·mol^{-1} = – 149.661 kJ·mol^{-1}) corresponds to a positive rather than negative enthalpy of solution of $Ca(H_2PO_4)_2$(cr) of the same magnitude. We have assumed that this was due to a neglect

of the negative sign during the calculation rather than a spurious addition of one during the publication of the paper.

With these values, we calculate:

$$\Delta_f H^\circ_m (Ca(H_2PO_4), cr, 298.15\ K) = -(3118.440 \pm 2.302)\ kJ \cdot mol^{-1}.$$

These results yield the enthalpy of dehydration according to Reaction (A.51):

$$Ca(H_2PO_4)_2 \cdot H_2O(cr) \rightleftharpoons Ca(H_2PO_4)_2(cr) + H_2O(l) \qquad (A.51)$$

$\Delta_r H^\circ_m$ ((A.51), 298.15 K) = (19.135 ± 0.542) kJ·mol^{-1}, keeping uncertainty limits that avoid redundancy.

Because of the difference in the enthalpy of solution of $Ca(H_2PO_4)_2$(cr) noted above, the value for Reaction (A.51) is appreciably different from that given by [1956EGA/WAK], (10.293 kJ·mol^{-1}), but is fully consistent with the values given in [1982WAG/EVA].

[1956FLO]

This paper is one of a number by Flörke investigating the stability of tridymite at normal pressure in the absence of impurities. A pellet of finely-powdered tridymite was heated to temperatures of (1200-1350) °C between Pt electrodes with an applied voltage of 250 V, for 8 to 20 days. These temperatures are within the previously accepted range of stability of tridymite. In a typical experiment at 1300 °C, the current increased to *ca.* 1 A during the first day, and fell slowly to 0.2 A during the course of 18 days. The pellet was then divided into anode and cathode portions, which were analysed microscopically, chemically and by XRD. In all the experiments cristobalite was formed in the anode section, which contained less than 0.05 % impurities, while the cathode section was still tridymite. These experiments strongly suggest that tridymite is not stable at 1 bar in the absence of impurities, and that quartz transforms directly to cristobalite.

Further similar experiments were thus carried out at lower temperatures to try to determine the temperature of this transition. Following electrolysis at 1050 °C, the anode section was found to contain quartz rather than cristobalite, but at 1100 °C, there was no evidence for the formation of quartz. Flörke thus suggests the transition quartz-cristobalite temperature is *ca*. 1050 °C, which agrees well with the subsequent findings by Holmquist [1961HOL].

[1956FUR/DOU]

In 1948, α-alumina (corundum) was designated by the Calorimetry Conference as a suitable material for a standard for heat capacity measurements from 10 to 1800 K, and

this paper reports the results of very accurate measurements from 13-1170 K on a sample of the very pure so-called Calorimetry Conference alumina available from the US National Bureau of Standards.

The sample used was a colourless synthetic sapphire and spectrographic analysis showed the purity to be between 99.98 and 99.99 % by weight. The only impurities were 0.005 % Si, 0.005 % Fe and 0.002 % Cr. It was estimated that these impurities would not affect the measured heat capacity by more than 0.02 % in the temperature range studied.

An adiabatic calorimeter was used from 13 to 380 K and a drop method was used with a Bunsen ice calorimeter from 273 to 1700 K. The full details of these calorimeters are given.

For the low-temperature measurements two series of runs were made, with samples of 226 g (from 13-120 K and 280-380 K, 97 experiments) and 252 g (from 80-380 K, 84 experiments), with good agreement in the overlap region. For the drop-calorimetric study, 47 runs were made with a 16.3 g sample. The full results of all these runs and of the empty container are tabulated with deviation plots. The deviations of the experimental data from a smoothed curve were less than 0.1 $J \cdot K^{-1} \cdot mol^{-1}$. The smoothed values of C_p from all these runs are tabulated at closely spaced intervals. As shown in Figure V-29 in Section V.5.1.1.2 of the main text, these authors' data agree excellently with all the eight earlier and subsequent studies, with the exception of the early, less precise, data of Simon and Swain [1935SIM/SWA]. These data ([1956FUR/DOU]), together with those of [1969FUG/SWE] for the range 2-25 K, form the basis of the CODATA selection for alumina.

[1956HUB/HOL]

These very careful oxygen-bomb-calorimetric measurements are the basis for the adoption by CODATA [1989COX/WAG] of the standard enthalpy of formation of CaO(cr). The instrument used and its calibration procedure were described by Holley and Huber [1951HOL/HUB]. The bomb, with a volume of 360 cm^3, was operated at a pressure of 50 bar as preliminary experiments showed that at lower pressures the amount of unburnt material was quite large (up to 7 % under 25 bar).

The high-purity calcium produced in the same laboratory was analysed with the following results (in wt %): Mg, 0.01 %; C, 0.011 %; H, 0.025 %, O, 0.02 %; N, 0.001 %. No other metallic impurities were present in amounts greater than 0.01 %. X-ray diffraction showed only lines of metallic calcium.

Six itemised combustion experiments with weighed samples of calcium chunks ranging from approximately 1 to 5 g were reported, giving an average specific energy of combustion of (15.816 ± 0.014) $kJ \cdot g^{-1} (2\sigma)$. Combustion varied from 99.49 to 99.96 %

of completion. The initial temperature was 298.05 K and the average final temperature was 299.25 K. The corrections for impurities (including that for the presence of the 0.008 g magnesium wire used as fuse) were fully detailed. The absence of CaO_2 in the combustion product was ascertained to a sensitivity of 0.0005 g CaO_2. X-ray diffraction of the combustion product showed only lines of CaO(cr). The final corrected value for the heat of combustion of calcium metal in oxygen at 50 bar pressure at 298.05 K was (15.8065 ± 0.022) kJ·g^{-1}(2σ) or ΔU(298.05K) = – (633.52 ± 0.89) kJ·mol^{-1}. The authors noted that the correction of this value to 298.15 K is less than the uncertainty in the result—the correction is indeed entirely negligible (*ca.* 0.14 J·mol^{-1}). After correcting for the deviation of oxygen from the perfect gas law and converting from energy to enthalpy effect, the authors report $\Delta_f H^\circ_m$(CaO, cr, 298.15 K) = – (635.09 ± 0.89) kJ·mol^{-1}.

The value adopted by CODATA [1989COX/WAG] and by [1989GUR/VEY] (and selected by this review) is very slightly different $\Delta_f H^\circ_m$(CaO, cr, 298.15 K) = – (634.92 ± 0.90) kJ·mol^{-1}, presumably due to the use of more modern subsidiary data.

[1956KRA3]

This early semi-quantitative study of the solubility of amorphous silica reported on a large number of experiments in sea water, but generally with relatively short equilibration times (maximum of 95 days). Krauskopf concluded that the solubility of amorphous silica at 25 °C is between 100 and 140 ppm, which encompasses the value of 115 ppm reported in [1964MOR/FOU]. No thermodynamic data were extracted from this paper.

[1956POP/GIN]

The high-temperature enthalpies of the title compounds were determined by conventional drop calorimetry; only the data for Na_2CO_3(cr) are of interest for this review. The only details given of the sample of Na_2CO_3(cr) was that it was “chemically pure”. Sixty three measurements were made at 20 different temperatures, the sample being held in Pt crucibles; it is not stated whether these were open or sealed. It should be noted, however, that later authors (for example Janz *et al.* [1961JAN/LOR2], [1963JAN/NEU]) indicate that Na_2CO_3 reacts with Pt at high temperatures, and used Au-Pd liners in their crucible. Two major transitions were noted, one between 713 and 733 K and melting between 1113 and 1133 K. These transition data are consistent with those found by adiabatic calorimetry in the same laboratory [1951POP/GAL]. The data are presented only in the form of graphs and derived C_p equations for two ranges for the solid, 476 to 592 K and 737 to 1112 K and for that for the liquid, 1134 to 1380 K. No C_p values could be derived for the other temperature ranges of the solid, owing to the transformation (at *ca.* 628 K).

The data from this study are included in Figure V-83 in Section V.7.1.2 of the main text, which shows they are consistent with later studies and the selected data.

[1956SIN]

Samples of tridymite were prepared by heating quartz grains of a uniform size (*ca*. 100 μm) with a range of mineralisers: Na_2WO_4, Li_2CO_3, Na_2CO_3 and K_2CO_3 in Pt crucibles at 1300 to 1320 °C; a vast excess (300 % by weight) of Na_2WO_4 was used, but modest amounts (3 % by weight) of the carbonates.

The transition temperatures, measured by highly sensitive DTA, were only slightly dependent on the mineraliser used for the preparation, but were appreciably different on heating and cooling, being respectively (388 ± 2) and (345 ± 2) K for the first transition and (433 ± 2) and (422 ± 2) K for the second. There was no indication of a third transition found by other authors, and the author attributes their results to the presence of residual cristobalite in their samples.

The heat capacities and enthalpies of the samples were measured from *ca*. 300 to 500 K using a vacuum adiabatic calorimeter, but were only reported in the form of a small graph. The enthalpies of transition are reported to be (272 ± 20) and (145 ± 20) $J \cdot mol^{-1}$ for the transitions at *ca.* 388 K and 433 K, respectively.

[1956WES]

The author studied the heat capacity of quartz after irradiation at *ca.* 35 °C by neutrons in the CP 5 reactor of Argonne National Laboratory at two integrated damaging fluxes, 2.5×10^{19} and 7.7×10^{19} $n \cdot cm^{-2}$. (The damaging flux was defined as the upper two decades of epithermal neutron energies that are significant in generating structural damage).

The data for the irradiated samples are of no immediate interest to this review, but the author does give some comparisons with his data for unirradiated quartz, which are otherwise unpublished. Unfortunately, these are only in the form of a quite coarse graph with a non-linear temperature axis. However, Westrum does give his measured (presumably smoothed) value for unirradiated quartz, $C^{o}_{p,m}(SiO_2, \alpha\text{-quar}, 298.15\ K) = 44.593\ J \cdot K^{-1} \cdot mol^{-1}$, but with no uncertainty.

A separate portion of the Brazilian quartz sample was converted to cristobalite which contained < 0.5 % quartz and < 1.0 % tridymite, but no details of the treatment used for the conversion are given. The heat capacity of this cristobalite was also measured, but again, these measurements are only reported on the coarse graph mentioned above.

[1957GRE/PRI]

The solubility of colloidal silica (surface area 750 $m^2 \cdot g^{-1}$) was measured at 25 and 35 °C with relatively low solid loadings of 1 g·(100 cm^3 solution)$^{-1}$. The sampling method was not mentioned and separating the colloidal material from the solution would have been difficult especially without high-speed centrifuges. The required equilibration times were apparently ≤ two days. The solubility of colloidal SiO_2 is given as (1.8×10^{-3}) mol·dm^{-3} at 25 °C (*cf.* (1.93×10^{-3} mol·kg^{-1} [1964MOR/FOU] for amorphous SiO_2). The latter value was only attained after 1000 days.

[1957HAR]

This experimental study of oxide growth on pure aluminium metal surfaces (cold rolled and electropolished) in contact with pure water (*ca*. 1 g in 23 dm^3 at 0 °C and 1 bar O_2 pressure at each temperature) yielded the following properties: below 60 to 70 °C, an amorphous layer was formed rapidly, followed by the formation of boehmite, orthorhombic γ-AlOOH (defined by others as pseudoboehmite), followed by the formation of a monoclinic outer (solubility controlling) bayerite layer. Aluminium ions escaping to the liquid are apparently re-precipitated as bayerite to form the outer layer. Above 70 °C the outer bayerite layer does not form.

[1957ING/LAG]

Ingri *et al.* conducted titrations at seven borate concentrations between 0.01 and 0.6 mol·dm^{-3} in a cell fitted with hydrogen and Ag|AgCl electrodes in solutions of 3 mol·dm^{-3} $NaClO_4$ at (25.0 ± 0.1) °C. Polyborate anions are referred to as ($3x–2y–z$), x species, $B_xO_y(OH)_z^{(3x-2y-z)}$. The equilibria which gave the best fits to the data are:

$$B(OH)_3(aq) + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+ \quad \text{(A.52)}$$

$$3B(OH)_3(aq) \rightleftharpoons B_3O_3(OH)_4^- + H^+ + 2H_2O(l) \quad \text{(A.53)}$$

$$3B(OH)_3(aq) \rightleftharpoons B_3O_3(OH)_5^{2-} + 2H^+ + H_2O(l) \quad \text{(A.54)}$$

$$4B(OH)_3(aq) \rightleftharpoons B_4O_4(OH)_6^{2-} + 2H^+ + 2H_2O(l) \quad \text{(A.55)}$$

The following formation constants were proposed where the fourth constant was presented as an alternative to the third although its inclusion gave a poorer quality fit: $\log_{10}{}^*K_{c,1}$ (A.52) = – (9.00 ± 0.05), $\log_{10}{}^*\beta_{c,1,3}$ (A.53) = – (6.84 ± 0.10) and $\log_{10}{}^*\beta_{c,2,3}$ (A.54)= – (15.44 ± 0.20) or $\log_{10}{}^*\beta_{c,2,4}$ (A.55) = – (15.71 ± 0.20). A pK_c value for the dissociation of water at I_c = 3 ($NaClO_4$) and 25 °C was measured at (14.22 ± 0.02) and is believed to have been used in this and subsequent papers to derive the molarity of hydroxide ion from the measured hydrogen ion concentrations in their titrations of boric acid. Mesmer *et al.* [1972MES/BAE] proposed

a slightly different speciation scheme based on their potentiometric results in I_m = 1.0 (KCl), *viz*., 1,1, 1,2, 1,3 and (2,4 or 3,5). Both articles support the formation of the 1,3 species which is considered to have a six membered ring of bridging B and O atoms with four OH groups bonded to the three B atoms. A discussion of the rules governing the structures of the polyborates proposed by Edwards and Ross [1960EDW/ROS] is given in the synopsis to [1972MES/BAE].

[1957LOR/MOR]

The heat capacity of α-quartz has been computed from (10-700) K, and that of vitreous silica from (10-900) K, from the Debye temperatures obtained from a T^3 relation below 10 K for the translational degrees of freedom, and recent infrared and Raman data for the other frequencies. $C_p - C_V$ was negligible for vitreous silica and was computed from compressibility and thermal-expansion data for α-quartz. The agreement with C_p measurements is excellent for α-quartz (within 1 % above 60 K) but is less satisfactory for vitreous silica because of inadequate knowledge of the distribution of frequencies below 500 cm^{-1}.

This paper is also very useful in that it allows us to derive the otherwise unpublished low-temperature heat capacities of [1956WES] at round temperatures up to 300 K, since the authors give their own calculated values and the differences from Westrum's smoothed experimental values.

[1957MAH]

This oxygen-bomb-calorimetric study was conducted in an instrument that was previously described in detail by Humphrey [1951HUM]. The bomb had a volume of 355 cm^3 and was operated under 30 bar O_2. Calibration was made by combustion of standard benzoic acid (NBS sample 39 g). Only the results concerning the combustion of aluminium will be discussed here.

Fine lathe turnings from an aluminium bar (99.998 % purity), were used for the combustions. Samples showed no oxidation under bomb conditions before ignition.

Fused alumina (alundum) disks were used to hold the samples. The percentage completion of combustion (99.14 to 99.66 %) was determined by weighing the total combustion products. It was verified by XRD that the combustion product on the disk corresponded to corundum with no evidence of any other modification of alumina, though such modifications appeared on the small amounts of wall deposits.

Combustion results were reported (2σ confidence interval) at 303.15 K as average values of seven runs, using an average Al mass of 0.40359 g, and yielded an observed energy change under bomb conditions of (30.9729 ± 0.0117) $kJ \cdot g^{-1}$.

After correction to a constant pressure process and to 298.15 K, the authors report $\Delta_f H_m^\circ(Al_2O_3, cr, 298.15\ K) = -(1675.61 \pm 1.05)\ kJ \cdot mol^{-1}$. The correction to the 1981 atomic weights is negligible.

[1957SAB]

The temperatures and enthalpies of the transitions in tridymite and cristobalite were measured by DTA.

Tridymite was prepared by heating quartz in molten Na_2WO_4 at 950 °C for one week; the purity was checked by XRD, with no further details. Cristobalite was obtained by devitrification of melted silica at 1540 °C for 48 hours; its purity was checked from the XRD and density of 2.322 $g \cdot cm^{-3}$ at 20 °C. The tridymite was found to have transitions at 115 °C and 165 °C, with enthalpy changes of 0.43 $cal \cdot g^{-1}$ (108 $J \cdot mol^{-1}$) and 0.23 $cal \cdot g^{-1}$ (58 $J \cdot mol^{-1}$); these are the mean values of five measurements for each. The corresponding values for the sharp transition in cristobalite were 270 °C and 4.36 $cal \cdot g^{-1}$ (1095 $J \cdot mol^{-1}$).

The author also found that the enthalpies of transition were proportional to the corresponding volume changes.

[1958ELM/NOR]

The solubility of porous glass and silica gel were reported in nitric acid solutions at 36 and 65 °C (porous glass) and 95 °C (silica gel). Dissolved silica was analysed using molybdate as the colorimetric reagent and the weight-loss method was also employed in the case of the porous glass experiments. The solubility of both materials decreased linearly with the molality of HNO_3 (actually plotted against normality) over the range 0.001 to 10.2 $mol \cdot dm^{-3}$. The rate of dissolution increased with increasing acid concentration with equilibrium only being established after 23 days in 0.001 $mol \cdot dm^{-3}$ solutions at 36 °C from under-saturation. Conversely, equilibrium was approached from over-saturated solutions when silica gel was used at 95 °C where seven to nine days were required to reach an apparent equilibrium condition. These results cannot be interpreted in terms of formation a specific cationic Si species; the effects of ionic strength and acidity could not be uncoupled. Elmer and Nordberg gave empirical fits of $\log_{10} c_{Si}$ *versus* molality, extrapolation of which gave c_{Si} values equivalent to 2.66×10^{-3}, 4.33×10^{-3} and 7.75×10^{-3} $mol \cdot dm^{-3}$ at $c_{HNO_3} = 0$ at 36, 65 and 95 °C, respectively. A preliminary comparison with the more internally consistent, higher temperature values of Hitchen [1935HIT] suggests that the experiments from over-saturation at 95 °C did not reach equilibrium, whereas those at lower temperatures may have reached a steady-state with respect to the metastable SiO_2(s).

[1958FOL/KLI]

This paper reports results, without details, of measurements by differential thermal analysis and by high-temperature calorimetry of the thermal effects associated with the dissociation of various minerals and related materials listed in Table A-20. The two techniques used are described but, in the case of the DTA, there is little or no information as to the expected accuracy, even under future improved conditions.

The calorimeter comprised a small furnace immersed in a well-insulated copper vase containing 200 cm^3 water at room temperature. The reaction heat was obtained as the difference between the electrical energy input to heat the sample at the decomposition temperature and the amount of energy returned to the water surrounding upon cooling.

Little is said about the precise chemical composition or the purity of the natural materials. The authors ascribe the obvious disagreement between the results by the two techniques for all the materials, except for magnesium carbonate, to difficulties in properly controlling the water release in hydrous materials, especially in DTA. Under these circumstances, only the results for magnesium carbonate, which do not involve water release, will be considered further.

Table A-20: Materials studied and measured heats of dissociation

Material	Measured heat /cal·g^{-1}	
	DTA	Calorimetry
Hydrargillite (nat., Istria) (gibbsite)	220	80 to 10
Diaspore (nat., Banská Bela)		73
Boehmite (nat, from bauxite of Halimba-Tamáskút)		68
Kaolinite (nat., Zettlitz)	106	98
Magnesium carbonate (synthetic, *pro analysi*)	320	319
Gypsum (synthetic, *pro analysi*)	170	84

In the case of calorimetry, the sample size is given as "0.7 to 0.8 g" and the maximal error in the individual determinations is stated as "2 to 3 cal". The results are given as average of five or six determinations. With a reported heat of 319 cal·g^{-1}, the measured heat, for a sample of 0.7 g, would have been 223 cal with an uncertainty of, say, ± 2.5 cal, corresponding to ± 1.1 %. This corresponds to (26.90 ± 0.30) kcal·mol^{-1} or (112.5 ± 1.3) kJ·mol^{-1} for Reaction (A.56) at room temperature, assumed here to be 298 K:

$$MgCO_3(cr) \rightleftharpoons MgO(cr) + CO_2(g) \qquad (A.56)$$

With CODATA values for $\Delta_f H^\circ_m(CO_2, g, 298.15\ K) = -(393.51 \pm 0.13)$ kJ·mol^{-1} and $\Delta_f H^\circ_m(MgO, cr, 298.15\ K) = -(601.60 \pm 0.30)$ kJ·mol^{-1}, also accepted in this review, one obtains $\Delta_f H^\circ_m(MgCO_3, cr, 298.15\ K) = -(1107.6 \pm 1.3)$ kJ·mol^{-1}.

Nevertheless, in view of the lack of detailed information about the details and accuracy of the calorimetric technique used by [1958FOL/KLI], and the well-known slow kinetics of the decomposition, this review suggests the uncertainty be increased to at least ± 10 kJ·mol^{-1} for these results.

The authors' value of 320 cal·g^{-1} = 112.9 kJ·mol^{-1} for the enthalpy of dissociation of magnesite measured by DTA at an unspecified temperature, but presumably at least 700 K, corresponds to a standard enthalpy of formation of – (1111.8 ± 10.0) kJ·mol^{-1} (where the uncertainty is estimated by this review), close to our selected value of – (1112. 2 ± 1.3) kJ·mol^{-1}.

[1958GOT]

The enthalpies of formation of hydroxyl-, fluor- and chlorapatite have been obtained from measurement of the enthalpies of solution of these compounds in 20 and 30 % HNO_3 at (298.15 ± 1.00) K, using an air-jacketed calorimeter, described in detail in the publication. The weight of the samples ranged between 1.5 and 2.4 g and the weight of the solvent between 6.8 and 7.6 g. The compounds were prepared according to methods described in the literature. From X-ray and chemical analyses (results not given), the compounds were described as being of "moderate to high purity", *viz.* approximately 98.5 % for the hydroxylapatite and 97 % for the fluorapatite. The data on the chlorapatite are beyond the scope of this review and will not be discussed here.

According to the author, the measured calorimetric reactions correspond to:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + (14HNO_3 + H_2O)(sln) \rightleftharpoons 7Ca(NO_3)_2(sln) + 3Ca(H_2PO_4)_2 \cdot H_2O(cr) \quad \text{(A.57)}$$

with $\Delta_r H^\circ_m$ (A.57) = – 291.62 and – 301.25 kJ·mol^{-1}

and

$$Ca_{10}(PO_4)_6F_2(cr) + (14HNO_3 + H_2O)(sln) \rightleftharpoons (7Ca(NO_3)_2 + 2HF)(sln) + 3Ca(H_2PO_4)_2 \cdot H_2O(cr) \quad \text{(A.58)}$$

with $\Delta_r H^\circ_m$ (A.58) = – 223.01 and – 228.86 kJ·mol^{-1} for the 20 % and 30 % HNO_3 solution media, respectively.

These results are given as averages of two single determinations only, spanning a range between 2.5 and 5.8 kJ·mol^{-1}.

We believe that uncertainty limits of ± 25 kJ·mol^{-1} should be accepted for Reactions (A.57) and (A.58), in view of the moderate purity of the samples and the reported range for the experimental determinations, without taking into consideration concerns expressed below on the validity of the reaction scheme.

From his X-ray and undefined chemical analyses, the author asserted that the solid final product of the calorimetric reactions is "entirely" $Ca(H_2PO_4)_2 \cdot H_2O$(cr). It should be noted that this very simple scheme has been questioned by Jacques [1963JAC] who did not find any precipitated product such as $Ca(H_2PO_4)_2 \cdot H_2O$(cr) in repeated attempts to reproduce these reactions, although $Ca(H_2PO_4)_2 \cdot H_2O$(s) was found upon evaporation of the solutions. Jacques also noted that, as written, Reactions (A.57) and (A.58) do not make any allowance for interactions between calcium nitrate, hydrogen fluoride and calcium phosphates in solution.

Without questioning, at this stage, the accuracy of the scheme, we have attempted to estimate how the results of the author arising from Reaction (A.57) and Reaction (A.58) could be converted to CODATA compatible values. We note that for $\Delta_f H^\circ_m(Ca(H_2PO_4)_2 \cdot H_2O, cr, 298.15\ K)$, the author uses – 3417.574 kJ·mol^{-1} from [1956EGA/WAK], while the value selected by this review is – (3423.405 ± 2.295) kJ·mol^{-1}. For the other terms of Reaction (A.57) and Reaction (A.58) involving H_2O(sln), HNO_3(sln), $Ca(NO_3)_2$(sln) and HF(sln) (the latter for Reaction (A.58) only), we have assumed that the author used the auxiliary data from Rossini *et al*. [1952ROS/WAG] (his reference 5), rather than his reference 4 ([1956EGA/WAK]) as he quotes. However, we were unable to understand unambiguously the selection of some values by the author. Under these conditions, in the conversion of the data to CODATA compatible values, besides taking into account the case of $\Delta_f H^\circ_m(Ca(H_2PO_4)_2 \cdot H_2O, cr, 298.15\ K)$ above, we have limited ourselves to the most important differences from [1952ROS/WAG].

In particular, $\Delta_f H^\circ_m(NO_3^-, 298.15\ K)$ and $\Delta_f H^\circ_m(F^-, 298.15\ K)$ are given as – 206.572 kJ·mol^{-1} and – 329.113 kJ·mol^{-1} in [1952ROS/WAG], while the CODATA values are – (206.85 ± 0.40) kJ·mol^{-1} and – (335.35 ± 0.65) kJ·mol^{-1} respectively.

For $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K)$, the author reported – 13518.5 kJ·mol^{-1}. Using the above corrections, we obtain, for the simpler composition, a CODATA compatible value of $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6770 \pm 15)$ kJ·mol^{-1}, keeping conservative uncertainty limits.

Similarly, for fluorapatite Gottschal's value of $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K) = -13648.2$ kJ·mol^{-1} becomes a CODATA compatible value of $\Delta_f H^\circ_m(Ca_5(PO_4)_3F, cr, 298.15\ K) = -(6841 \pm 15)$ kJ·mol^{-1}, again with conservative uncertainty limits.

However, even these CODATA compatible values, for both hydroxylapatite and fluorapatite, derived from the data of Gottschal, must be considered as informative only, in view of the criticism expressed about the validity of the reaction schemes mentioned above.

[1958HIL/ROY]

The authors studied by DTA the transition temperatures of about twenty samples of cristobalite prepared from various forms of silica (quartz, silica gel and silicic acid) at temperatures from 1175 to 1600 °C.

It is demonstrated experimentally that even in the purest laboratory silicas available the temperature of the α-β inversion in cristobalite is variable and depends on the temperature and length of heat treatment of the starting material. It is shown that this variability is an index of the ordering achieved in the cristobalite structure, quite independent of the presence of impurities. There is no specific order characteristic of a particular temperature; the completely ordered so-called 3C stacking of cristobalite is the most stable cristobalite throughout the temperature range and has an inversion temperature of 267 °C (540 K). All disordered cristobalites will tend toward the 3C 267 °C cristobalite with time.

[1958HIL/ROY2]

Hill and Roy attempted to address the problem as to whether tridymite is a stable phase in the pure silica system by heating a range of very pure silica-based materials and also transistor grade silicon to (1113-1513) K in gold or platinum capsules with only H_2O or D_2O and no other mineralisers, but at pressures from (30-230) bar. The following starting materials, with their impurity levels, were used:

- Silicic acid— < 0.4 % non-volatile matter with HF and < 0.02 % of other materials tested for.
- SiO_2 glass— < 0.1 % (manufacturer's analysis).
- SiO_2 gel—prepared by hydrolysis of ethyl orthosilicate; only ppm of inorganic impurities.
- Cristobalite, prepared from the above gel—no additional analysis.
- Transistor grade Si— <1 ppm impurities

After preparation, the products were examined by powder XRD at room temperature and high temperature, and by DTA to determine their inversion temperatures. About 100 runs were carried out and the brief results from about 50 are tabulated. Tridymite and cristobalite were found in samples heated over the whole temperature range, with quartz also present in some samples heated below 1323 K, but not above this temperature.

Two forms of tridymite, which they called tridymite-M (for metastable) and tridymite-S (for stable), were found from different heat treatment conditions; they are related to each other by a very sluggish reconstructive monotropic transformation. The

M-phase may be transformed to the S-phase by prolonged heat treatment. Each form has its characteristic X-ray powder diffraction pattern and also a different DTA pattern.

The authors did consider the possible presence of H_2O or D_2O incorporation into their samples, but were able only to show by IR analysis that this was < *ca.* 3 mol %, although "the loss in weight at 1573 K was not greater than 0.1 %". We take this to be 0.1 wt %, corresponding to 0.3 mol % H_2O.

The authors conclude from these experiments that tridymite is a stable phase in the SiO_2 system, but as discussed in the text (Section V.3.1.4.1 on tridymite), other authors have pointed out that they have not shown unambiguously that small amounts of H_2O and D_2O are not incorporated into the lattice. We also note that, strictly, their results apply only at the high pressures of the experiments.

The paper also includes a discussion of the detailed XRD patterns, ordering structure and forms of the transition in the tridymites found, which are not relevant for the purposes of this review. We note also that Flörke [1961FLO] suggests that the XRD in this paper leads to unsatisfactory results, since the authors did not use single crystals.

[1958HOL2]

This paper discusses briefly the various possible transformations of silica, together with the papers of Flörke which suggest that tridymite is stabilised by impurities, and reports an experiment on the electrolysis of tridymite at high temperature, similar to that reported by Flörke in [1956FLO].

A sample of silica brick was fired enough times to convert all the silica to tridymite, and a 1 cm^3 of this sample was placed between spring-loaded Pt electrodes at 1350 °C. After 12 hours electrolysis with a 200 V potential, it was found that the anode section was 'snow-white', while the cathode section was greyish and had partially melted. XRD showed only cristobalite in the anode section, but tridymite, with some cristobalite, in the cathode section. No such changes were observed when an A.C. potential was applied.

This experiment therefore supports the view that tridymite is not a stable phase in the pure SiO_2 system.

[1958NAI/NAN]

A cell without liquid junction was used to monitor the pH of $ZnSO_4$-HCl and $MgSO_4$-HCl solutions. Iteration in I_m and activity coefficient, which was determined from the Davies expression with the linear ionic strength term set at 0.2, yielded the desired ion-pair formation constants. The temperatures investigated were 0, 20, 25, 30, 35, 40 and 45 °C. Small differences are noted between their calculated ionic strength and formation

constants at 25 °C and those simply calculated from mass balance of data provided in their Table 1. An average value of $\log_{10}K_2^o(H_2SO_4(aq))$ for the second dissociation constant of H_2SO_4(aq) determined by this review is $-(1.955 \pm 0.010)$ (2σ), *cf.* – 1.96 [1958NAI/NAN2] and $-(1.98 \pm 0.008)$ [1992GRE/FUG]. It had been noted previously, as discussed in [1973KAT] and [2005GAM/BUG], that the final value of $\log_{10}K_1^o(MSO_4(aq))$ is highly dependent on the activity coefficient model used and the value of $\log_{10}K_2^o(H_2SO_4(aq))$, where even differences in the latter of 0.02 can lead to differences of 20 % in K_1^o [2005GAM/BUG]. Therefore, a re-examination of these data using the SIT model for non-ideality was carried out using the NEA recommended value for $\log_{10}K_2^o(H_2SO_4(aq))$. The formation constants were calculated using the ion interaction coefficient values: $\varepsilon(Mg^{2+},Cl^-) = (0.19 \pm 0.02)$ kg·mol^{-1}, $\varepsilon(H^+,Cl^-) = (0.12 \pm 0.01)$ kg·mol^{-1} and $\varepsilon(H^+,HSO_4^-) = (0.01 \pm 0.02)$ kg·mol^{-1}. The resulting average value of K_1^o at 25 °C is (263 ± 70) (2σ). For the data at the other temperatures, the SIT approach yielded $K_1^o = (141 \pm 65)$ kg·mol^{-1} at 0 °C, (254 ± 57) at 20 °C, (266 ± 91) at 30 °C, (338 ± 117) at 35 °C, (327 ± 120) at 40 °C and (416 ± 124) kg·mol^{-1} at 45 °C. The enthalpy of reaction reported by Nair and Nancollas, *viz*., 20.3 kJ·mol^{-1}, is approximately a factor of four larger than the consensus of more reliable calorimetric data.

[1958NAI/NAN2]

Nair and Nancollas derived thermodynamic quantities for the dissociation of HSO_4^- from their potentiometric measurements with a Harned-type cell at 0, 5, 15, 25, 35, and 45 °C in H_2SO_4-HCl mixtures: $K_2^o = 0.0110$ mol·kg^{-1} at 25 °C (note that the molal dissociation constants in their Table 2 should be divided by 100), $\Delta_r G_{m,2}^o = 11.17$ kJ·mol^{-1}, $\Delta_r H_{m,2}^o = -(23.4 \pm 0.8)$ kJ·mol^{-1} and $\Delta_r S_{m,2}^o = -116$ J·K^{-1}·mol^{-1}. The value of $\Delta_r G_{m,2}^o$ given in [1992GRE/FUG] is (11.302 ± 0.285) kJ·mol^{-1}.

[1958REI]

The melting point of K_2CO_3(cr) in a CO_2(g) atmosphere was determined by DTA to be (1174 ± 1) K (12 determinations), and a small anomaly was observed in K_2CO_3(cr) at (695 ± 1) K. The material used was said to be of > 99.9 % purity, but no analyses are given. Before use, the samples were heated to constant mass in a CO_2(g) atmosphere and cooled from the melt to improve crystallinity. When heated in air, additional transitions were observed around 526 and 640 K, close to the temperatures reported in [1940MAK/SHU], but these are probably due to formation of K_2O(cr), since they disappear after the samples were allowed pick up moisture, forming KOH(cr).

Although the abstract of the paper indicates that the author also measured the melting point of Na_2CO_3(cr) to be (1127 ± 1) K, there is no further mention of this in

the paper. The author also reported a transition in K_2CO_3(cr) at the same temperature, 659 K, as in his study of the Na_2CO_3-K_2CO_3 phase diagram [1959REI].

[1958REI/HOL]

The only interest in this paper for the present review lies in the authors' re-examination of the polymorphic behaviour of Na_2CO_3(cr), using DTA in a CO_2(g) atmosphere to inhibit decomposition. The samples of high-purity Na_2CO_3 (no analysis given) were contained in small Au-Pd crucibles, a material which, unlike Pt or pure Au, is inert to Na_2CO_3 at high temperatures. Two sluggish transitions were observed at 628 and 758 K, but none in the region of 873 K, as reported by Makarov and Shul'gina, [1940MAK/SHU]. In a subsequent paper [1959REI], this was attributed to the probable presence of KCl(cr) as an impurity in the material used in [1940MAK/SHU].

[1958WES/GIN]

This paper contains mainly a detailed description of a new drop calorimeter, for enthalpy precise measurements accurate to 0.1 %, for heat capacity measurements of solids and liquids over the temperature range 300 to 870 K. Sixteen measurements, from 311.9 to 689 K, all replicated, were made on samples of the same synthetic sapphire (containing less than 0.02 wt % impurities) as used in other studies at US National Bureau of Standards, and the detailed results are tabulated. These heat capacities agreed to within 0.1 % of other values derived from the enthalpy measurements in the same laboratory.

[1959BAR/BER]

This is a careful study of the C_p of a macrocrystalline sample of MgO(cr) from 9.8 to 269.8 K. The sample was prepared by lightly crushing a single crystal to give smaller crystals of *ca.* 5 mm size, which were heated in a vacuum at 150 °C for 18 hours prior to use. The full results of the 125 measurements are given.

The measured C_p values are noticeably greater than those obtained in [1937GIA/ARC] on a microcrystalline sample prepared from $Mg(OH)_2$(cr) and the authors discuss the reasons for this in some detail.

The low-temperature values for MgO(cr) in the CODATA assessment [1989COX/WAG] were based predominantly on this study and these values have been accepted by the review:

$$C^{\circ}_{p,\mathrm{m}}(\mathrm{MgO, cr, 298.15\ K}) = (37.237 \pm 0.200)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}\ \text{and}$$

$$S^{\circ}_{\mathrm{m}}(\mathrm{MgO, cr, 298.15\ K}) = (26.95 \pm 0.15)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

[1959BAT/BOW]

Dissociation constants of $CaOH^+$ were derived at eight temperatures (0 to 40 °C) from potentiometric measurements with hydrogen *versus* Ag|AgCl reference electrodes. The concentrations of $Ca(OH)_2$ varied between 0.002 and 0.02 $mol \cdot kg^{-1}$ with similar concentrations of either KCl or $CaCl_2$. Bates *et al*. pointed out that values of the dissociation constants are sensitive to the choice of $k = \log_{10}\{\gamma(Cl^-)/\gamma(OH^-)\}$ required to calculate the molality of OH^- in solution. The authors calculated three values of $k = 0$, -0.3 and -1.0. The ionic strengths used were generally low ($I_m < 0.08$), and both the standard Davies equation and a Debye-Hückel equation with a variable ion-size parameter between 0 and 8 Å were used. The SIT calculations in this review used the standard potentials for the Ag|AgCl electrode calculated from the equation in [1954BAT/BOW].

The SIT results at 25 °C are illustrated in Figure A-8 which shows that the scatter in the data is larger than the difference between the two datasets (KCl *versus* $CaCl_2$). The average of all these data is $-(1.21 \pm 0.09)$, compared to -1.15 and -1.13 suggested in [1959BAT/BOW] for the KCl and $CaCl_2$ solutions, respectively, for $k = 0$, *cf*. $\log_{10}K_1^\circ = -1.27$ [1987GAR/PAR]. Furthermore, the re-calculated results in Table A-21 are consistent with those from other studies where the same concerns regarding activity coefficients exist.

Figure A-8: Values of $\log_{10}K_1^\circ$ for $CaOH^+$ derived from data collected in KCl and $CaCl_2$ solutions as a function of I_m at 25 °C (average value indicated by the solid line).

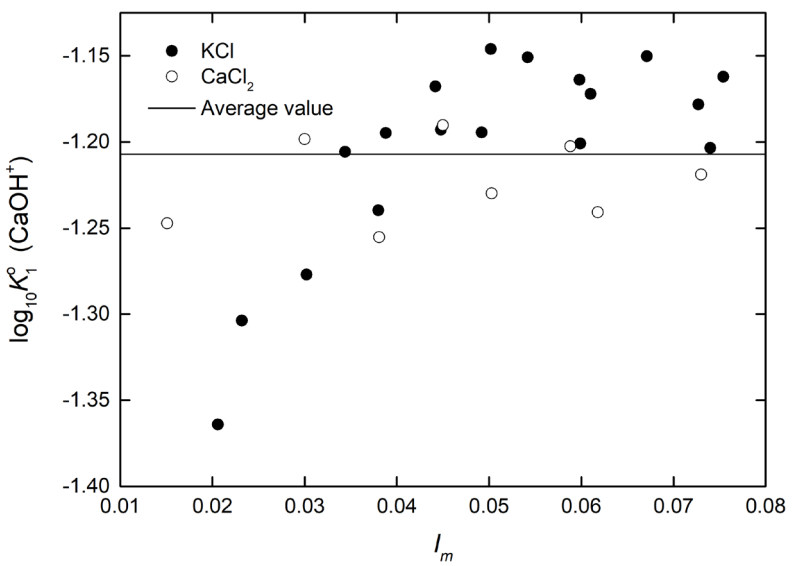


Table A-21: SIT calculated dissociation constants for $CaOH^+$ and the corresponding hydrolysis constants with statistical uncertainties at the 2σ level.

t /°C	$-\log_{10}K_1^\circ$ re-calculated	$-\log_{10}{}^*K_1^\circ$ re-calculated	I_m /max	t /°C	$-\log_{10}K_1^\circ$ re-calculated	$-\log_{10}{}^*K_1^\circ$ re-calculated	I_m /max
10	(1.035 ± 0.121)	(13.499 ± 0.121)	0.077	30	(1.255 ± 0.128)	(12.584 ± 0.128)	0.075
15	(1.173 ± 0.122)	(13.173 ± 0.122)	0.076	35	(1.302 ± 0.107)	(12.386 ± 0.107)	0.075
20	(1.147 ± 0.073)	(13.022 ± 0.073)	0.075	40	(1.384 ± 0.147)	(12.160 ± 0.147)	0.074
25	(1.207 ± 0.093)	(12.794 ± 0.093)	0.075				

[1959DEW/RIC]

The authors studied the decomposition of $MgSO_4$(cr) and $CaSO_4$(cr) between 900 and 1365 °C by DTA in flowing mixtures of SO_2(g) and O_2(g). Very small specimens of a few hundred mg were used. It should be noted that there are rather few experimental details in the paper; there was no characterisation of the samples used, a DTA plot is given only for the $CaSO_4$ experiments, and there are no details of the gas compositions used. A correction was made for the presence of SO_3(g), but again, with no details of the data used. The results of the decomposition are only given as graphs of log ($p_{SO2} \times p_{O2}$½) *versus* $(T/\mathrm{K})^{-1}$, with, it was stated, good reproducibility between the data taken on heating and cooling.

Peaks were observed on the DTA plots due to polymorphic transitions at (1283 ± 10) K for $MgSO_4$ and (1483 ± 2) K for $CaSO_4$, and the Gibbs energies of the decomposition reactions:

$$\mathrm{MSO_4(cr) \rightleftharpoons MO(cr) + SO_2(g) + ½\ O_2(g)} \qquad \text{(A.59)}$$

were given as:

$$\Delta_r G_m(\text{A.59}) = 364760 - 251.88\ (T/\mathrm{K})\ \mathrm{J\cdot mol^{-1}}\ (1173\text{-}1423\ \mathrm{K})\ \text{for M = Mg}$$

and

$$\Delta_r G_m(\text{A.59}) = 464424 - 236.90\ (T/\mathrm{K})\ \mathrm{J\cdot mol^{-1}}\ (1483\text{-}1638\ \mathrm{K})\ \text{for M = Ca.}$$

The reproducibility of the data corresponds to *ca.* ± 2900 $\mathrm{J\cdot mol^{-1}}$ for Mg and ± 800 $\mathrm{J\cdot mol^{-1}}$ for Ca.

Third-law calculations by Gurvich *et al.* [1994GUR/VEY] gave $\Delta_f H_m^\circ(\mathrm{MSO_4, cr, 298.15\ K})$ = – (1290 ± 18) $\mathrm{kJ\cdot mol^{-1}}$ for M = Mg and – (1439 ± 15) $\mathrm{kJ\cdot mol^{-1}}$ for M = Ca, the large uncertainty being due to the lack of experimental heat capacities and enthalpy of transition data (especially for $MgSO_4$(cr)). Given this large uncertainty, the good agreement with both the values selected in this review may be somewhat fortuitous.

[1959ELL]

The solubility of calcite (Cumberland, England) crystals in distilled water was measured between 100 and 300 °C at CO_2 pressures between 1 and 40 bar. The solubilities were determined in a stainless steel bomb of ~ 100 cm^3 capacity, using a single natural clear calcite crystal of approximately 2 g mass. The weighed crystal was contained in a wire cage attached to the inside of the bomb-head. The bomb was placed up-side down in an electric tube furnace and the water and carbon dioxide were allowed to equilibrate for 24 hours with the calcite crystal exposed to the gas phase. After this time, the furnace was inverted so that the liquid phase in the bomb came into contact with the calcite. The time allowed for equilibrium between solid and solution was about 24 hours at 300 °C, and up to a week at 100 °C. At the end of a run the furnace was again inverted, and with the crystal in the gas phase, the bomb was removed from the furnace and quenched in cold water. The crystal was again weighed to obtain the amount dissolved.

The method used by Ellis eliminated the difficulties experienced by Miller [1952MIL2] in quenching the runs to room temperature and pressure. Therefore, the solubilities agree well with those reported in [1962SEG/HOL], [1963MAL] and [1964SIP/GLO] on which our selected data above 100 °C are based.

[1959ELL2]

The first dissociation constant of carbonic acid has been determined for pressures up to 3000 atm in the temperature range 25-65 °C by conductivity measurements of 0.0001-0.1 m HCl, KCl and $KHCO_3$ solutions.

The measurements were made in a stainless-steel pressure vessel (25 cm^3) into which the conductivity cell (Teflon cylinder of 7 cm^3) containing the solution was placed. A tight-fitting Teflon piston supporting the electrodes separated the solution in the cell from medicinal paraffin used to transmit pressure within the pressure vessel. The electrode assembly in the cell consisted of platinum foil electrodes tightly bound by fine platinum wire to the arms of a U-shaped Pyrex glass piece. A thick platinum lead from one electrode supported the assembly from the Teflon piston, and a lead of fine wire from the other electrode to the piston allowed for the slight play caused by the contraction of Teflon under pressure. The cell therefore incorporated the advantages of the sliding piston design and almost constant spacing of the electrodes by glass. The platinum in the cell was lightly coated with platinum black. The reported pressures were estimated to be accurate within ± 0.5 %. The pressure vessel and a paraffin reservoir were heated in an oil-bath controlled to ± 0.05 °C. The mass of the vessel damped out temperature variations in the bath. About 2 hours were required for the apparatus to reach initial thermal equilibrium, and a further 30 minutes after a change of pressure for equilibrium to be regained. The resistances were measured with a capacity-compensated non-inductive Wheatstone bridge. Corrections to the conductances were made for the

resistance of lead wires and the change in cell constant with pressure. The latter was 0.1 % change per 1000 atm. Measurements of the cell resistance were taken again at 1 atm pressure at the end of each run.

At a given temperature, the increasing pressure results in somewhat higher K_1 values owing largely to the increased hydration of carbon dioxide to H_2CO_3 at high pressure (the fraction of carbon dioxide in solution present as H_2CO_3 rises from 0.259 % at 1 atm to ~ 2.3 % at 3000 atm). The author noted that no significant amount of carbon dioxide was lost from the cell during the measurements. The activity coefficients of CO_2(aq) and the undissociated H_2CO_3 were taken as unity at all pressures and temperatures. The mean activity coefficient of the H^+ and HCO_3^- ions were obtained from the Debye-Hückel equation. The thermodynamic dissociation constants determined by the author agree well with those evaluated in this review ($\Delta p K_1^\circ = 0.02\text{-}0.03$).

[1959HAT/HIL]

The authors measured the low-temperature heat capacity of $Ca(OH)_2$(cr). Their sample was prepared by calcining 'Analar'-grade $CaCO_3$(cr) at an unspecified temperature and reaction with the stoichiometric amount of water. Analysis showed it contained 99.75 % $Ca(OH)_2$, the residue being $CaCO_3$ and water; no corrections were made to the heat capacities. The heat capacity was measured at 92 temperatures from 19 to 330 K, using a 39 g sample. Extrapolation down to 0 K was made assuming C_p was proportional to T^3, using the data from 19-30 K to define the proportionality constant. The uncertainties in C_p were estimated to be 0.2-0.3 % above 50 K and 1 % below 50 K. The data around 300 K merge smoothly with the high-temperature data of [1950KOB].

Integration of the heat capacities gave the following standard values:

$$C_{p,\mathrm{m}}^\circ (\mathrm{Ca(OH)_2, cr, 298.15\ K}) = (87.488 \pm 0.100)\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$S_\mathrm{m}^\circ (\mathrm{Ca(OH)_2, cr, 298.15\ K}) = (83.4 \pm 0.4)\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

These values are the selected values for this review.

[1959HUM/SCH]

This paper concerns the determination by solution calorimetry of the enthalpies of formation and of fusion of the compounds in the system Na_2O-SiO_2. The influence on the enthalpy of solution of the various experimental conditions (acid concentration, mode of preparation of the compounds, *etc.*) was discussed with the added purpose of defining the best conditions for the calorimetric determinations.

This review will only discuss the results concerning the modifications of silica.

Use was made of a new solution calorimeter using thermoelectric temperature measurement. The reaction vessel made of platinum, thus allowing measurements with hydrofluoric acid solutions, was centred in a Dewar vessel, containing water covered with paraffin. The samples (50-200 mg) were wrapped in fine Japanese silk paper before weighing and introduction in the calorimeter.

The samples (two preparations) for the determination with quartz were obtained by grinding in an agate mill 99.9 % pure rock crystal, followed by sedimentation. The *ca.* (2.5-5) μm fractions were then treated with dilute hydrofluoric acid and dried to 723 K. XRD of the samples ready for calorimetry showed only diffraction lines belonging to quartz. For the dissolution of 60 to 120 mg samples in 39.0 % HF at 299.65 K, the authors report a value of $-(141.84 \pm 0.38)$ kJ·mol^{-1} as an average of 13 determinations covering the two preparations that gave indistinguishable results. We note that the authors designate these samples as β-quartz. They also give the values of $-(134.85 \pm 0.42)$ kJ·mol^{-1} and $-(132.13 \pm 0.42)$ kJ·mol^{-1} for the dissolution, at the same temperature, in, respectively, 19.5 and 9.8 % HF.

The quartz glass (99.9 % SiO_2) obtained from Heraeus, Hanau, had particle sizes smaller than 5 μm and was fully amorphous as seen by XRD. For the dissolution of 50 to 120 mg samples in 39.0 % HF at 299.65 K, the authors report a value of $-(151.34 \pm 0.46)$ kJ·mol^{-1} as an average of 8 determinations and $-(143.97 \pm 0.08)$ kJ·mol^{-1} for the dissolution, at the same temperature, in 19.5 % HF.

The tridymite sample used (which the authors designate as γ-tridymite) was obtained by treatment at 1583 K of amorphous silicic acid (Aerosil from Degussa) to which 2 % K_2CO_3 was added. After washing treatments, the sample was shown to contain 0.15 % K_2O. After grinding, the fraction of size smaller than 2.5 μm was shown by XRD to be free of quartz and cristobalite, but in view of the broadness and intensity of the diffraction lines, was assumed to contain 50 % of amorphous material. A treatment with 5 % HF was needed to eliminate these amorphous particles and to restore the expected line sharpness and intensity of the tridymite. For the dissolution of this tridymite in 39.0 % HF at 299.65 K, the authors report a value of $-(147.36 \pm 0.25)$ kJ·mol^{-1}, as an average of three determinations.

The cristobalite sample was prepared by treating a (20-60) μm fraction of rock crystal at 1973 K for four hours in a Tammann furnace. A fraction of particles smaller than 2.5 μm prepared as for the other SiO_2 samples was used for calorimetry. For the dissolution of the tridymite obtained in this way in 39.0 % HF at 299.65 K, the authors report a value of $-(149.37 \pm 0.59)$ kJ·mol^{-1} as an average of eight determinations.

Measurements were carried out with amorphous silicic acid (Aerosil from Degussa). Upon calcination to 723 K, the sample was shown to have lost 0.95 % by weight that was assumed to be water. After correction for this water content, the results of the dissolution at 299.65 K of the samples in 39.0, 19.5 and 9.8 % HF, respectively,

were $-(156.11 \pm 0.21)$, $-(149.08 \pm 0.25)$ and $-(146.36 \pm 0.33)$ kJ·mol^{-1}, as averages of three determinations in each medium.

From the above results, the authors report the following enthalpy changes for the various transformations at 299.65 K, mostly from dissolutions in 39.0 % HF.

Table A-22: Enthalpies of transformation of α-quartz into other silicon dioxide species at 299.65 K.

Reaction	$\Delta_r H_m$ /kJ·mol^{-1}
SiO_2(α-quar) ⇌ SiO_2 (vit)	(9.50 ± 0.84)
SiO_2(α-quar) ⇌ SiO_2(trid S-I)	(5.52 ± 0.63)
SiO_2(α-quar) ⇌ SiO_2 (β-cris)	(7.53 ± 0.96)
SiO_2(α-quar) ⇌ SiO_2(am)	(14.27 ± 0.59)

The authors also compare their data with similar literature values.

[1959KEN2]

Ion association of the chloride salts of Cu(II), Zn(II), Ni(II) and Mg(II) in the supporting electrolytes $KClO_3$ and $KClO_4$ was quantified from freezing-point-depression measurements. A Debye-Hückel expression was used to estimate activity coefficients with a variable term in the denominator. The ionic strength was based on the supporting electrolyte alone after extrapolation of the molality of the metal chloride to zero (in the case of $MgCl_2$(sln) its molality reached 0.258 and 0.0485 mol·kg^{-1}, in the two media, respectively). Nevertheless, the ionic strengths exceeded the applicability of the Debye Hückel equation so that the resulting value of $K_1^\circ = 8.1$ kg·mol^{-1} at 0 °C for the formation of $MgCl^+$ contains unknown uncertainties and is much larger than expected based on the results of similar studies.

[1959LAG]

Potentiometric titrations of silicic acid solutions were carried out in 0.5 mol·dm^{-3} (25 and 50 °C) and 3 mol·dm^{-3} (25 °C) $NaClO_4$ solutions using hydrogen and Ag|AgCl electrodes with a salt bridge. The Jena-Geräte glass titration vessel was pre-treated with alkali in order to inhibit Si release. The silicic acid concentration was increased to relatively high values, 0.0025 to 0.080 mol·dm^{-3} in 0.5 mol·dm^{-3} $NaClO_4$ and 0.005 to 0.20 mol·dm^{-3} in 3 mol·dm^{-3} $NaClO_4$ with the express purpose of observing the formation of polysilicate species. The equilibria for the formation of the two mononuclear species are represented by Reactions (A.60) and (A.61):

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \quad (A.60)$$

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \quad (A.61)$$

An unweighted SIT treatment of the reported $\log_{10}K_m$ (A.60) values after conversion to molal units yielded: $\log_{10}{}^{*}K^{\circ}$ (A.60) = – (9.81 ± 0.01) (*cf.* – (9.84 ± 0.02) derived by Lagerström) and $\Delta\varepsilon$ = (0.046 ± 0.010) kg·mol^{-1} (all corrected to 2σ). The former value corresponds to $\log_{10}K^{\circ}$ = (4.19 ± 0.01) and compares very favourably with (4.14 ± 0.05) derived from an SIT treatment of the potentiometric results in [1977BUS/MES]. Eight equilibrium constants for the formation of polynuclear Si(IV) species (25 °C) and six at 50 °C are given in Table 5 of this paper at I_c = 3. No uncertainties are listed and this reviewer suggests that the covariance between these multiple constants prevents any further consideration of these values. Subsequent studies carried out by Ingri and co-workers [1985SJO/OEH] produced a more limited range of species for which ambiguities still exist as to their stoichiometry and stability.

[1959REI]

Reisman studied the Na_2CO_3-K_2CO_3 phase diagram using DTA and XRD techniques; only the transition data for the pure components are of interest to the present review. The high-purity materials were dried at 450 K in Au-Pd crucibles in a CO_2(g) atmosphere before use. Transitions were observed at 634 and 762 K for Na_2CO_3(cr) and 695 K for K_2CO_3(cr); the melting points were 1127 and 1174 K respectively.

These data are very similar, though not identical, to those reported by the same team in a related paper [1958REI/HOL].

[1960BER/BOR]

The solubilities of synthetic magnesite (prepared by calcining hydromagnesite at 300-350 °C and under $p(CO_2)$ = 70-80 bar), calcite (Iceland spar from the deposits of Maidantal, Kazakh. SSR) and nesquehonite were studied. The solubility was examined in specially prepared vessels with capacities of 700-800 cm^3 having oil seals and tubes for the introduction of CO_2(g) at a pressure of about 1 bar. The suspensions were agitated at 298.15 K for 3-8 months until equilibrium was established. The dissolved metal ions were determined gravimetrically (Mg^{2+}: as pyrophosphate, Ca^{2+}: as oxalate) and complexometrically using Trilon. Total alkalinity was measured by acid titration with 0.1 mol·dm^{-3} HCl (methyl orange indicator). The measured solubilities of synthetic magnesite and calcite at 298.15 K and at $p(CO_2)$ = 0.987 bar were 0.0165 and 0.0090 mol·dm^{-3}, respectively.

The solid phases were examined optically and thermographically (calcite and magnesite did not alter during the long equilibration). The solubility of synthetic magnesite is given as 0.0165 mol·dm^{-3} in their Table 1, but 0.0156 mol·dm^{-3} is given in the text (in both the Russian and English versions). The former value is accepted in this review, since in their Table 1 the value 0.0165 mol·dm^{-3} is repeated twice, and this

value is practically identical with those determined in [1935HAL/RIT] and [1954YAN] under nearly identical conditions.

[1960EDW/ROS]

Edwards and Ross began by stating that determination of the exact structures of polyborates is difficult due largely to the fact that boron exhibits both trigonal and tetrahedral configurations, and because there is rapid conversion between the various structures. They presented the following postulates that govern the principles of bonding in polyborates based on information gained from XRD and NMR studies: the basic structure is a six-membered ring with alternate boron and oxygen atoms; it is particularly noteworthy that in solutions containing boric acid and borate anions, the species $B_3O_3(OH)_4^-$ is predicted to be stable and the species $B_3O_3(OH)_5^{2-}$ is also likely to have a significant presence; the existence of a long-chain polymer with units of $B_3O_4(OH)_2^-$ is also considered probable; the two trimers are stabilised by the presence of small cations such as Mg^{2+} and Ca^{2+}. Other workers have proposed that at high concentrations $B_5O_6(OH)_5^{2-}$ and $B_5O_6(OH)_6^{3-}$ also may exist based on the NMR evidence.

[1960FOU]

This short abstract was followed by an extensive paper two years later [1962MOR/FOU], although the latter publication did not cite this abstract. Fournier noted that equilibrium with quartz could not be achieved at room temperature using over-saturated solutions of silica, even with respect to the precipitation of silica gel. Equilibrium with respect to quartz was only achieved in a pressure vessel containing pure water from under- and over-saturation after various equilibration periods at elevated temperatures and saturation-vapour pressure or at 100 MPa. At the former pressures, solubilities were reported at 84 °C (25 ppm), 165 °C (165 ppm) and 240 °C (435 ppm). Only the first two values are considered here. The equivalent molal solubilities are 4.16×10^{-4} and 2.75×10^{-3} mol·kg^{-1} at 84 and 165 °C, respectively. Interestingly, the former value is virtually identical to those given in [1962MOR/FOU], but both are lower by a factor of *ca*. two than those accepted in this review. The author claimed to have obtained identical results when equilibrium was approached from under- and over-saturation, and therefore the lower solubilities cannot be simply rationalised by inadequate equilibration periods. On the other hand, the value at 165 °C is in excellent agreement with other reliable data including those reported at 136 and 190 °C by Morey *et al*. [1962MOR/FOU].

[1960GRE/COP]

Existing experimental data were processed to determine the solubility product of $Ca(OH)_2$(cr) [1923MEL], [1924HAS/CAL], [1931JOH/GRO], [1934BAS], [1942DSC/MOS], [1950FAR] and [1956BAT/BOW], and the hydrolysis constant, K_1°, of Ca^{2+} which was taken to be 0.071 mol·kg^{-1} at 25 °C [1959BAT/BOW]. Activity coefficients were estimated either from an expression attributed to Hitchcock [1928HIT] or from the conventional Debye-Hückel expression. Note that average pK_{sp} values were given incorrectly in their Table 1 rather than those calculated from the individual K_{sp} values. Greenberg and Copeland derived Equation (A.62) with strong weighting applied to the data of Bates *et al*. [1956BAT/BOW] and recommended the value $\log_{10} K_{s,0}^\circ = -5.07$ at 25 °C, although Equation (A.62) gives $\log_{10} K_s^\circ = -5.029$; presumably the discrepancy is due to rounding errors. Their value of $\Delta_r H_s^\circ$ derived from Equation (A.62) is – 13.8 kJ·mol^{-1}. There is remarkable agreement between the temperature dependence of $\log_{10} K_{s,0}^\circ$ predicted from Equation (A.62) and that from a four-term equation derived in [1967YEA/MAR].

$$\ln K_{s,0}^\circ = 17.67 \ln(T/\mathrm{K}) - 0.078(T/\mathrm{K}) - 89.0 \qquad \text{(A.62)}$$

[1960KIT]

Measurements were made of the solubility of quartz from 140 to 500 °C at saturation vapour pressure up to the critical temperature of water and then at elevated pressures at supercritical conditions. The weight-loss method using "quartz blocks" was employed so that the accuracy of the low-temperature (low-solubility) results was poor. Only the first two data points at 160 and 200 °C were considered for which the solubilities converted from weight % SiO_2 to mol·kg^{-1} were 1.3×10^{-3} (0.61 MPa) and 4.3×10^{-3} (1.52 MPa). Comparison with other high-temperature data shows clearly that 120 hours' equilibration time at 160 °C was inadequate to achieve equilibrium, whereas the solubility value at 200 °C is in excellent agreement with those reported by other investigators.

[1960LIE/BRU]

Van Lier *et al*. used crushed Brazilian quartz, which was also used almost 40 years later in the excellent Rimstidt study [1997RIM], and also found higher solubilities than in many of the interim studies. These measurements were carried out at 60, 70, 80, 90 and 100 °C in 0.01 mol·dm^{-3} NaCl solutions, but only those at 60 °C could be re-treated by this review and a mean solubility of $(4.8 \pm 0.5) \times 10^{-4}$ mol·kg^{-1} was calculated which compares favourably with $(4.5 \pm 0.5) \times 10^{-4}$ mol·kg^{-1} [1997RIM], neglecting the effect of NaCl on the activity coefficient of $Si(OH)_4$(aq). Solubilities at the other temperatures

are in excellent agreement with regressed data in [1991DAM/BIS], [1997RIM] and [2000GUN/ARN] corresponding to the equilibrium:

$$SiO_2(s) + 2\,H_2O(l) \rightleftharpoons Si(OH)_4(aq)$$

where the pressure above 100 °C was p_{sat}. A maximum temperature limit of 200 °C was imposed in keeping with the policy of this review. Their expression used to fit these data took the form:

$$\log_{10} m_{Si} = -8.476 - 485.24(T/\mathrm{K})^{-1} + 3.068\log_{10}(T/\mathrm{K}) \\ -2.268\times10^{-6}(T/\mathrm{K})^2 \qquad \text{(A.63)}$$

An experiment is summarised in Table I of this paper involving the dissolution of quartz in 0.12 mol·dm^{-3} NaOH at 25 °C that gave constant silica concentrations, which were determined by colorimetric and gravimetric analyses, after only 17 days and consequently this solution was severely under-saturated. However, four quartz solubility results given in Table V of this paper at initial hydroxide concentrations of 10^{-4}, 10^{-3}, 10^{-2} and 0.1 mol·kg^{-1} where the final NaOH concentration was measured in the first two cases (presumably by titration), led to a mean re-calculated $\log_{10}K_{s,1}$ value of (3.35 ± 0.21) (2σ), *cf.* (3.3 ± 0.2) (2σ) derived from the $\log_{10}{}^{*}K_{s,1}$ values in their Table V. The equilibration times were not given. Finally, experiments were reported in their Table IV on the solubility of quartz in water and a 0.01 mol·dm^{-3} borax solution. The reported solubility of quartz was rejected by Grenthe *et al*. [1992GRE/FUG] in favour of the lower value reported by Morey *et al*. [1962MOR/FOU] who suggested that the higher solubility may have resulted from the fine-grained quartz used by van Lier *et al.* having a higher surface energy. More recent data [1997RIM] tend to refute this claim.

[1961BAR/KEL]

Using hydrofluoric acid solution calorimetry, the authors measured the enthalpies of formation of three aluminosilicates and of gibbsite. Our comments will be restricted to gibbsite, as the aluminosilicates are not within the topic of the present review.

The calorimeter used had been described in the literature. The instrument was operated at 346.85 K, using 940.1 g of 20.10 wt % of reagent grade HF. All reagents involved in the thermodynamic cycle were dropped in the calorimeter from 298.15 K. The gibbsite sample supplied by J. C. Kennedy (UCLA) contained, by weight, 65.20 % alumina and 34.80 % water, as compared with the theoretical 65.35 and 34.65 %. Hydrochloric acid and aluminium chloride hexahydrate, reagent grade, were also used in the thermodynamic cycles. The aluminium chloride was analysed as containing 11.18 % Al and 44.09 % Cl (theoretical: 11.17 % Al and 44.06 % Cl). The authors described in great detail the reactions they measured themselves, *i.e.*, the dissolution of gibbsite and of $AlCl_3{\cdot}6H_2O$, and the solution of H_2O and 4 molar HCl and ascertained

the closing of the thermodynamic cycle. To derive the enthalpy of formation of gibbsite, they had recourse to the data of Coughlin [1958COU] who, in the frame of his work on the enthalpy of formation of aluminium chloride, reported the enthalpy of solution of aluminium in 4.360 molal HCl.

As a result of their study, Barany and Kelly [1961BAR/KEL] reported $\Delta_f H_m^o$ (Al(OH)$_3$, gibb, 298.15 K) = – (1281.9 ± 0.7) kJ·mol^{-1}, a value some 11 kJ·mol^{-1} less negative than other calorimetric results. The reason for this discrepancy is not entirely clear. However, Hemingway and Robie [1977HEM/ROB] allude to the great hygroscopicity of aluminium chloride hexahydrate that was contained in gelatin capsules in the work of Barany and Kelley. Also, there could have been inadequacies in the evaluation of the enthalpy of transfer of water between the media used.

Under these circumstances, the value reported by Barany and Kelley for $\Delta_f H_m^o$ (Al(OH)$_3$, gibb, 298.15 K) is only given for information by this review.

[1961FLO]

Flörke found that when very pure samples of silica prepared from the hydrolysis of $SiCl_4$ or $Si(C_2H_5)_4$ are heated above 1373 K for 10 days they transformed into a mixture of quartz and cristobalite, and that the quartz was still present after a further 50 days annealing; no tridymite was found. Moreover, when a sample of pure Brazilian quartz was heated above 1373 K, it transformed at a measurable rate into cristobalite. This paper also has further details of the author's experiment on the electrolysis of tridymite described in [1956FLO] (see Appendix A), including good X-ray patterns of the anode and cathode sections after electrolysis, showing clearly the presence of cristobalite at the anode, but none at cathode. From these two studies, the author concluded that tridymite is not a stable phase of silica at ambient pressure in the absence of impurities.

[1961GOL/HEA]

This paper is mainly a study of the phase relations in the $MgCO_3$-$CaCO_3$ system, which do not concern this review, but it reports in an Annex measurements of the decomposition pressures of four different dolomites, at higher pressures (100, 300 and 500 MPa) than those obtained in the earlier study by the same team in [1955GRA/GOL]. Runs were made using both open and sealed tubes. The data in open tubes agreed well with the extrapolation of the curve given by Graf and Goldsmith [1955GRA/GOL], but in sealed tubes the decomposition pressures at a given temperature were much higher than those in open tubes, for unknown reasons, as shown in Figure V-81 in Section V.6.3.4 of the main text.

[1961HOL]

Holmquist heated quartz with alkali oxides, chlorides, carbonates and silicates to investigate the conditions under which tridymite becomes a stable phase. When quartz was heated with 2 % alkali oxide, tridymite formed directly at 872 to 898 °C with Na_2O, at 883 to 902 °C with K_2O, and above 1005 °C with Li_2O; in each case cristobalite was formed as an intermediate phase. Tridymite was also formed in the presence of the alkali silicates and carbonates and the chlorides when heated in air; in all these cases, reaction occurs to form the silicates and or oxides. But, significantly, when quartz was heated with sodium chloride *in vacuo*, tridymite did not form but cristobalite started to appear at 1050 °C, but not at 1000 °C. These results strongly suggest that quartz and cristobalite are the only stable crystalline phases of pure silica, and that some alkali impurity is required to stabilise or at least form tridymite. These results are consistent with the earlier work by Flörke [1955FLO], [1956FLO], which also indicate that tridymite is almost certainly metastable at ambient pressures.

From these results and the electrolysis experiments of [1956FLO] and [1958HOL2], Holmquist suggests the temperature at which quartz transforms to cristobalite in the absence of impurities is (1025 ± 25) °C. It is noted above that it is formed at lower temperatures, but only in the presence of alkali oxides.

[1961VAN/QUI]

Vanderzee and Quist used a comparative spectrophotometric method to determine the dissociation constant of HPO_4^{2-} at 25 °C. The indicator disodium 3-carboxy-4-hydroxy-4'-(4-sulfophenylazo)azobenzene was used to compare the pH of $HCO_3^- + CO_3^{2-}$ and $HPO_4^{2-} + PO_4^{3-}$ buffers. A value of $\log_{10} K_3^\circ = -(12.375 \pm 0.010)$ for HPO_4^{2-} was reported based on $\log_{10} K_2^\circ = -10.329$ for HCO_3^-, which is identical to the value selected by the NEA. An ion-size parameter *a* of 4.8 Å was chosen for the extended Debye-Hückel equation to best represent activity coefficients in these solutions which varied in ionic strength from 0.0149 to 0.507 mol·dm^{-3}. The value of $\log_{10} K_3^\circ = -(12.35 \pm 0.02)$ for HPO_4^{2-} selected in [1992GRE/FUG] is the average of this value of $-(12.375 \pm 0.010)$ and -12.325 from [1929BJE/UNM], and both values are retained for consideration here.

[1961YAN/RAS]

In this study the solubilities of $CaCO_3$-$MgCO_3$-H_2O ternary (and binary) systems have been determined at 0, 20, 40, 55, 70 and 90 °C, under ~ 1 bar CO_2. The mixtures were stirred continuously in a thermostat for several days, although they showed little change in composition after 7-8 hours. After 1-2 days the liquid was sampled periodically, and the total alkalinity, the calcium and magnesium contents as well as the density of the

solution were determined. The solids were analysed microscopically, in part chemically and thermogravimetrically.

The aragonite solubility determined by Yanat'eva and Rassonskaya is very close to that of calcite. This is probably due to the presence of calcite in the aragonite samples. The aragonite was synthesised by the authors by slowly pouring aqueous $CaCl_2$ solution into a hot $(NH_4)_2CO_3$ solution. They noted the presence of individual crystals of rhombohedral calcite among the spheroidal crystals of aragonite. Since the reported solubility is considerably lower than expected, the derived corresponding solubility products were rejected from our evaluation.

[1962BEN]

In this study, a stirred vacuum-flask calorimeter was used to determine the enthalpy of solution in water of anhydrous K_2CO_3(cr) at 298.15 K between concentrations of 0.004 and 0.16 molal. The internal consistency of the results was given as ± 0.4 % and it was mentioned that the (unreported) enthalpy of solution of sodium chloride studied in the same apparatus agreed with that found by Benson and Benson [1955BEN/BEN], whose results were in agreement with the best literature data.

Granular potassium carbonate (Baker analysed reagent of 100.0 % assay), dried at 383 K before use, was employed and the water was free of CO_2 and inorganic ions. No other analytical details were given.

The results of the individual enthalpies of solution, $\Delta_{sln}H$, were listed as given in Table A-23 below.

Table A-23: Enthalpies of solution of K_2CO_3(cr) at 298.15 K.

Molality of solution	$\Delta_s H_m$ /kJ·mol^{-1}	Molality of solution	$\Delta_s H_m$ /kJ·mol^{-1}
0.0042	– 21.757	0.0598	– 29.196
0.0099	– 25.221	0.0952	– 29.225
0.0171	– 26.970	0.1014	– 29.598
0.0312	– 27.677	0.1398	– 29.623
0.0480	– 29.087	0.1630	– 29.568
0.0518	– 28.752	0.0507 (a)	– 33.401 initial value – 38.367 final value (after 30min) See text

(a) CO_2 present in the solution at near saturation.

The authors suggest that the difference between their results and the values assessed by Rossini *et al.* [1952ROS/WAG] may be due to the presence of CO_2 in the water used in some the early determinations on which the latter assessment was based,

resulting in an extraneous heat from the formation of HCO_3^- (*cf.* the last entry in Table A-23).

[1962CHR/IZA]

Thermometric titrations were carried out at (25.00 ± 0.05) °C using a thermistor as the temperature sensor with a detection limit of ± 0.02 °C to 1 % accuracy. Christensen and Izatt derived an enthalpy of acid dissociation, $\Delta_r H^\circ_{m,2}(298.15\ \mathrm{K})$, for $H_2PO_4^-$ of 0.9 kcal·mol^{-1} or 3.77 kJ·mol^{-1}.

[1962CYP]

After a rather general overview on the various polymorphs of silica, this paper gives an excellent summary of Flörke's electrolysis experiment [1956FLO], [1961FLO] (including XRD patterns of anode and cathode taken from [1961FLO]); these two papers indicate that foreign ions are necessary for the stabilisation of tridymite. Cyprès also gives micrographs of the work of his own laboratory on the transformation of quartz at high temperature both in the absence and presence of 5 % Na_2O. In a sample heated to 1350 °C for 24 hours without Na_2O, XRD showed the presence of 83 % quartz and 17 % cristobalite, the latter being formed within the crystals. However, in a sample heated to 1250 °C for 11 hours in the presence of 5 % Na_2CO_3, the quartz had completely disappeared, leaving a mixture of cristobalite and tridymite, the latter forming on the exterior of the grains.

The author also showed, using radioactive ^{45}CaO, that when CaO or $CaSiO_3$ is used as a mineraliser, Ca^{2+} ions are incorporated into both the tridymite and (to a much smaller extent) the cristobalite formed when a silica sample was heated in a marble crucible under pressure at 1350 °C for 60 hours, as described in more detail in [1960KEY/CYP]. It is therefore concluded that for a CaO mineraliser, at least, the rapid increase in the rate of conversion to both cristobalite and tridymite is due to the incorporation of small amounts of Ca^{2+} ions into the silica lattice.

[1962FRI/PEE]

Solubility experiments were initiated by adding 250 mg of synthetic gibbsite, which had been carefully pre-treated, to 50 cm^3 of solution. Seven experiments were initiated with no gibbsite present, but with excess Al^{3+} added. The mixtures were allowed to equilibrate at 25 °C with intermittent shaking for one to three months. The solid particles were then separated by high-speed centrifugation. The pH was measured at 25 °C with the glass electrode calibrated against commercial buffers. The experimental results are listed in Table A-24.

Table A-24: Summary of experimental data and solubility constants at 25 °C.

Time /months	Gibbsite present	pH	$\log_{10} c_{Al}$	$10^3 I_m$	$\log_{10}{}^{*}K_{s,0}$	$\log_{10}{}^{*}K^{\circ}_{s,0}$	% Al^{3+} (a)
3	no	3.70	– 2.00	0.0598	9.09	8.17	97.8
3	no	4.10	– 3.00	0.00580	9.25	8.90	90.1
3	no	4.48	– 4.00	0.00055	9.30	9.19	73.6
3	no	4.98	– 5.00	0.00005	9.56	9.52	41.5
3	yes	3.70	– 2.00	0.05980	9.09	8.17	97.8
3	yes	3.96	– 3.00	0.00591	8.84	8.49	92.7
3	yes	4.21	– 4.08	0.00052	8.47	8.36	84.1
3	yes	4.72	– 5.59	0.00003	8.33	8.30	57.9
3	yes	2.11 (b)	– 5.13	0.00783	1.20	0.80	99.9
3	yes	3.95 (b)	– 5.35	0.00014	6.45	6.39	90.1
3	yes	4.08 (b)	– 5.49	0.00010	6.69	6.64	87.1
1	yes	3.08	– 2.00	0.0609	7.23	6.32	99.5
1	yes	3.33	– 3.01	0.00630	6.97	6.61	98.2
1	yes	3.95	– 4.02	0.00066	7.79	7.66	90.8
1	yes	4.26	– 5.02	0.00011	7.67	7.62	81.4
1	yes	3.91	– 2.02	0.05660	9.69	8.50	96.4
1	yes	4.01	– 3.05	0.00524	8.94	8.61	91.7
1	yes	4.23	– 4.20	0.00041	8.41	8.31	83.3
1	yes	4.58	– 5.46	0.00004	8.10	8.07	66.3
1	no	3.95	– 2.02	0.05650	9.81	8.92	96.0
1	no	4.24	– 3.02	0.00541	9.63	9.30	86.6
1	no	4.54	– 4.05	0.00048	9.42	9.31	70.5
1	no	4.88	– 5.48	0.00003	8.84	8.81	47.6

(a) % Al^{3+} represents the mole percentage of Al^{3+} in solution after correcting for the formation of $AlOH^{2+}$.

(b) In these three cases no HCl, NaOH or $AlCl_3$ was present in solution initially so that equilibrium was approached from under-saturation unequivocally, whereas in all other cases $AlCl_3$(aq) was present initially and the c_{Al} decreased or remained virtually constant indicating these solutions were initially over-saturated to generally small extents.

Frink and Peech reported an overall value for $\log_{10} K^{\circ}_{s}$ of 33.5 (*i.e.* $\log_{10}{}^{*}K^{\circ}_{s,0}$ = 8.5). Kittrick [1966KIT] argued that the presence of fine-grained material in the gibbsite synthesised by Frink and Peech led to the higher solubilities. Bloom and Weaver [1982BLO/WEA] later re-examined the gibbsite used in both former studies and established the existence of a fine-grained, poorly crystallised surface phase in both cases, although the $\log_{10}{}^{*}K^{\circ}_{s,0}$ values reported by Kittrick proved to be compatible with those obtained from subsequent reliable studies. It would appear from the $\log_{10}{}^{*}K^{\circ}_{s,0}$ values listed in Table A-24 that inadequate equilibration time was the main reason for the wide range in these results that is particularly noticeable in the three experiments conducted from extreme under-saturation and those performed in the absence of added gibbsite seed crystals where high apparent $\log_{10}{}^{*}K^{\circ}_{s,0}$ values were calculated. No reliable thermodynamic data could be extracted from these results.

[1962ING]

A very detailed potentiometric study was conducted of borate solutions (0.01 to 0.2 mol·dm^{-3}) at 0.1 and 3 mol·dm^{-3} ionic strength ($NaClO_4$) and 25.0 °C in a cell fitted with hydrogen and Ag|AgCl electrodes. The LETAGROP regression program [1962ING/SIL] was used to determine the following formation constants (this paper like its predecessor [1957ING/LAG] uses the designation (3x–2y–z), x for the species $B_xO_y(OH)_z^{(3x-2y-z)}$): at $I_c = 0.1$, $\log_{10}{}^*K_{c,1}$ (A.64) = – (8.98 ± 0.01), $\log_{10}{}^*\beta_{c,1,3}$ (A.65) = – (7.29 ± 0.02) and $\log_{10}{}^*\beta_{c,1,5}$ (A.66) = – (6.77 ± 0.01); and at $I_c = 1.0$, $\log_{10}{}^*K_{c,1}$ (A.64) = – (9.00 ± 0.02), $\log_{10}{}^*\beta_{c,1,3}$ (A.65) = – (6.91 ± 0.03) and $\log_{10}{}^*\beta_{c,1,5}$ (A.66) = – (6.62 ± 0.16).

$$B(OH)_3(aq) + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+ \quad \text{(A.64)}$$

$$3B(OH)_3(aq) \rightleftharpoons B_3O_3(OH)_4^- + H^+ + 2H_2O(l) \quad \text{(A.65)}$$

$$5B(OH)_3(aq) \rightleftharpoons B_5O_6(OH)_4^- + H^+ + 5H_2O(l) \quad \text{(A.66)}$$

[1962ING/SIL]

This paper draws attention to computer programs which were designed to treat equilibria for the formation of polynuclear species in solution. Of these early algorithms, the LETAGROP generalised least-squares program [1962ING/SIL] has enjoyed continued success. No new thermodynamic data were contained in this paper.

[1962MOR/FOU]

The solubility of quartz was measured in pure water from under-saturation from 69 to 240 °C ($SiO_2(cr) + 2\ H_2O(l) \rightleftharpoons Si(OH)_4(aq)$) at saturation vapour pressure. Additional measurements made at 1000 bar are not considered here. The reported solubilities are shown in Table A-25, limited to temperatures < 200 °C. The values at room temperature, 69 and 84 °C, indicate that equilibrium was not reached. The results at 136 and 190 °C are in good agreement with those from other studies such as [1991DAM/BIS], also indicating that the colorimetric analytical method used was reliable. The values of $\log_{10}K^\circ_{s,4}$ in Table A-25 were estimated by the reviewer.

Table A-25: Solubility measurements of α-quartz in water, where the reported ppm refer to $SiO_2(aq)$.

t /°C	Equilibration time /days	Reported solubility /ppm	Solubility /mol·kg^{-1}	$\log_{10}K^\circ_{s,4}$
69	116	23	3.83×10^{-4}	– 3.42
84	18	29	4.83×10^{-4}	– 3.32
84	8	23	3.83×10^{-4}	– 3.42

(Continued on next page)

Table A-25 (continued)

t /°C	Equilibration time /days	Reported solubility /ppm	Solubility /mol·kg^{-1}	$\log_{10} K^{o}_{s,4}$
136	4	99	1.65×10^{-3}	– 2.78
136	3	100	1.66×10^{-3}	– 2.78
190	5	236	3.93×10^{-3}	– 2.41
Room temperature [(a)]	530	(6.4 ± 1)	1.07×10^{-4}	– 3.98

(a) The reaction mixture was contained in polypropylene bottles which were rotated continuously.

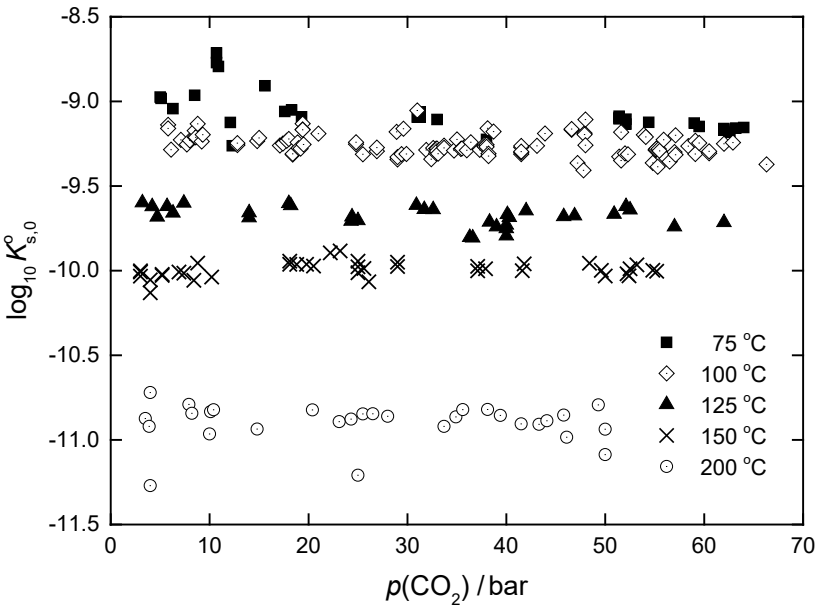
[1962SEG/HOL]

The solubility of calcite (Iceland spar) crystals in distilled water was measured at 75, 100, 125, 150 and 200 °C at CO_2 pressures between 2 and 60 bar. The bomb was filled with calcite crystals and doubly distilled water. Air was removed and the bomb was then filled with CO_2(g). The bomb was placed in an oil bath and allowed to heat up to the desired temperature. After 48 hours, the solution was sampled through a sampling valve at the pressure and temperature of the experiment. The weighed sample of the solution was acidified, buffered to a pH of 10.5 and calcium was titrated with EDTA using Eriochrome Black T as an indicator. Equilibrium was obtained both from over- and undersaturation.

Segnit *et al*. mentioned that at 75 °C and to a lesser extent at 100 °C difficulties were encountered in re-precipitating $CaCO_3$ upon release of CO_2 pressure (supersaturation approach). Indeed, the calculated $\log_{10} K^{o}_{s,0}$ ($CaCO_3$, calc) values are somewhat more scattered at 75 and 100 °C than at higher temperature (see Figure A-9), but the values show no pressure dependence, and after eliminating some outliers, the high number of data determine correct averages.

These values agree well with those calculated using the data in [1959ELL], [1963ELL2], [1963MAL] and [1964SIP/GLO] on which our selected data above 100 °C are based.

Figure A-9: The calculated $\log_{10} K^{\circ}_{s,0}$ ($CaCO_3$, calc) values based on the solubility data reported in [1962SEG/HOL] as a function of the partial pressure of CO_2.



[1962SIE]

The solubility of α-quartz was measured from under-saturation from 125 to 181.5 °C (SiO_2(s) + 2 H_2O(l) ⇌ $Si(OH)_4$(aq)) in sealed pressure vessels. Apparently analyses of the higher temperature solutions were made by weight loss of the quartz crystals, whereas the lower solubilities were measured spectrophotometrically. Siever reported the solubility of amorphous silica at 25 °C as 2.33×10^{-3} mol·kg^{-1}.

[1962SMI/ILL]

We have not been able to understand many of the final calculations in this rather poorly written paper concerning the dissolution of $Ca_3(PO_4)_2$(cr, α and β), fluorapatite and hydroxylapatite in *ca.* 10 % HCl (given as HCl, 18.3 H_2O). A 3 dm^3 adiabatic calorimeter, briefly described, was used. The preparation of the samples and their analytical data were given.

For β-$Ca_3(PO_4)_2$, although the elemental analytical results (unspecified techniques) appeared reasonably satisfactory (found: CaO, 53.95 %; P_2O_5, 45.60 %, calculated: CaO, 54.19 %; P_2O_5, 45.81 %), and X-ray examination did not reveal

extraneous phases, optical measurements revealed the presence of 2.5-5 % of a second phase with a refractive index near that of hydroxylapatite. At that stage the sample was heated to 1000 °C in a stream of nitrogen. The authors also made undefined solubility measurements in citrate solutions, presumably to distinguish between phases, as apatites (among other phases) are far less soluble in the test conditions, than calcium phosphate. The heat treatment of the β-$Ca_3(PO_4)_2$ sample increased the solubility in citrate to *ca.* 90 % of the P_2O_5 found by analysis. In view of the lack of transparency of these results, this review conservatively concludes that the purity of the sample is of the order of 90 %. The α-$Ca_3(PO_4)_2$) sample was obtained by heating the β-$Ca_3(PO_4)_2$ sample to 1200 °C in dry nitrogen. X-ray and optical examination revealed a homogeneous product, with *ca.* 97 % of the phosphate content soluble by the citrate test (see above).

Hydroxylapatite was prepared according to a method described in the literature; X-ray examination and optical measurements revealed no extraneous phases. The analytical results (found: CaO, 54.30 %; P_2O_5, 41.89 %; calculated: CaO, 55.82 %; P_2O_5, 42.39 %), however, revealed a deficit of Ca and P of *ca* 2 % with regard to stoichiometry. The citrate test indicated that 3 % of the phosphate content found by analysis was soluble, and thus probably not from an apatite phase.

A synthetic and a natural (Kola Peninsula) sample of fluorapatite were used. The analytical results of the synthetic sample (found: CaO, 56.49 %; P_2O_5, 42.48 %; F, 3.73 %—calculated: CaO, 55.59 %; P_2O_5, 42.22 %; F, 3.77 %) revealed a 2 % excess of CaO relative to stoichiometry, while the fluorine content was close to the expected value. X-ray examination and optical measurements revealed no extraneous phase. The preparation was reported to contain no citrate soluble phosphate species. The analytical results (found: CaO, 52.50 %; P_2O_5, 41.50 %; F, not given) of the natural sample (selected under a lens) revealed a more serious deficit with respect to stoichiometry, close to 6 % for Ca.

Samples for calorimetry were in the 3 g range for calcium phosphate and hydroxylapatite and in the 3-5 g range for fluorapatite. A quantity of (1000 ± 1) g of (HCl, 18.3 H_2O) solution (hereunder represented as (sln)) was used for each experiment, leading to relatively dilute solutions, in the range 0.01 to 0.03 molar, of the solutes. Calorimetric experiments were carried out in the temperature range 288 to 293 K.

No uncertainty limits were given for the individual determinations of the enthalpies of solution nor for their average. This review has therefore estimated uncertainties compatible with the limited purity of the compounds (see above) and the limited number of reported experiments. The values given below in $kJ \cdot mol^{-1}$ are converted from $cal \cdot g^{-1}$ given in the paper.

For the enthalpies of solution of α- and β-$Ca_3(PO_4)_2$ only two individual values for each were reported, corresponding to Reaction (A.67):

$$Ca_3(PO_4)_2(cr,\ \alpha \text{ or } \beta) + 6HCl(sln) \rightleftharpoons (3\ CaCl_2 + 2H_3PO_4)(sln) \quad (A.67)$$

For this we obtain $\Delta_r H_m$((A.67), α phase) = – (115.608 ± 6.760) kJ·mol^{-1}; those uncertainty limits cover the 95 % confidence limits and are much larger than those (± 3.0 kJ·mol^{-1}) arising from the reported purity of the sample. For the β phase, we obtain $\Delta_r H_m$((A.67), β phase) = – (94.513 ± 9.500) kJ·mol^{-1}, the uncertainty arising essentially from the low purity of the sample (see above). These results lead to (21 ± 12) kJ·mol^{-1} for the difference in the enthalpy of formation of the alpha and beta polymorphs of $Ca_3(PO_4)_2$(cr).

We were unable to properly trace the auxiliary values used by the authors in their evaluation of the enthalpies of formation of tricalcium phosphate from Reaction (A.67) above, nor of hydroxylapatite and fluorapatite, from Reactions (A.68) and (A.69) below. Although this procedure will not include any interaction between the products in solution, we have resorted to use just the CODATA-compatible values of the enthalpies of formation of the relevant species in 18.3 H_2O, namely, $\Delta_f H_m^\circ$ ($CaCl_2$, in 18.3 H_2O) = – 869.339 kJ·mol^{-1}, $\Delta_f H_m^\circ$ (H_3PO_4, in 18.3 H_2O) = – 1294.622 kJ·mol^{-1}, $\Delta_f H_m^\circ$ (HCl, in 18.3 H_2O) = – 163.487 kJ·mol^{-1}, $\Delta_f H_m^\circ$ (HF, in 18.3 H_2O) = – 321.872 kJ·mol^{-1}, and $\Delta_f H_m^\circ$ (H_2O, in HCl·18.3H_2O) = – 285.979 kJ·mol^{-1}. As the enthalpy of formation of β-$Ca_3(PO_4)_2$ is being assessed in the present review (see Section V.6.2.1.6.3), we have re-calculated its enthalpy of formation from the results of the authors for Reaction (A.67) above and obtain $\Delta_f H_m^\circ$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4121.8 ± 10.4) kJ·mol^{-1}, keeping conservative uncertainty limits. The authors reported $\Delta_f H_m^\circ$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – 4112.9 kJ·mol^{-1}.

For the enthalpy of solution of hydroxylapatite (three measurements), according to:

$$Ca_5(PO_4)_3OH(cr) + 10HCl(sln) \rightleftharpoons (5\ CaCl_2 + 3\ H_3PO_4 + H_2O)(sln) \quad (A.68)$$

this review obtains $\Delta_r H_m^\circ$ (A.68) = – (156.971 ± 12.350) kJ·mol^{-1}, with uncertainty limits from the dispersion of the three experimental results only; the added uncertainty due to the limited purity of the sample could amount to a further 6.0 kJ·mol^{-1}.

For the enthalpy of solution of fluorapatite (four measurements on the synthetic sample and three on the natural sample), according to:

$$Ca_5(PO_4)_3F(cr) + 10HCl(sln) \rightleftharpoons (5\ CaCl_2 + 3\ H_3PO_4 + HF)(sln) \quad (A.69)$$

the results for the two samples that cannot be statistically distinguished, and their average gives $\Delta_r H_m^\circ$ (A.69) = – (70.596 ± 4.960) kJ·mol^{-1}, with uncertainty limits arising from the dispersion of the seven experimental results only; the added uncertainty due to the limited purity of the two samples could amount to about 6.0 kJ·mol^{-1}.

For $\Delta_f H^\circ_m$ ($Ca_{10}(PO_4)_6(OH)_2$, cr, 298.15 K), the authors reported a value of – 13439.0 kJ·mol^{-1}, or $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K) = – 6719.5 kJ·mol^{-1} for the simpler composition. Using the above corrections, we obtain a CODATA compatible value of $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K) = – (6724.7 ± 15.3) kJ·mol^{-1}, keeping conservative uncertainty limits.

The authors reported a value of $\Delta_f H^\circ_m$ ($Ca_{10}(PO_4)_6F_2$, cr, 298.15 K) = – 13677.5 kJ·mol^{-1} or $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) = – 6838.7 kJ·mol^{-1} for fluorapatite. The corrections by this review yields a CODATA compatible value of $\Delta_f H^\circ_m$ ($Ca_5(PO_4)_3F$, cr, 298.15 K) = – (6837.0 ± 15.0) kJ·mol^{-1}, again with conservative uncertainty limits.

In view of the reservations mentioned above about the purity of the samples and the lack of consideration for the potential interactions between the species brought in solution, this review considers that the values of the enthalpies of formation for β-tricalcium phosphate, hydroxylapatite and fluorapatite, re-calculated from the results Smirnova *et al.* [1962SMI/ILL], should be taken for information only, although our re-calculated enthalpy of formation of β-$Ca_3(PO_4)_2$, – (4121.8 ± 10.4) kJ·mol^{-1}, from their experiments (see the discussion of Reaction (A.67) above) is in fact in good agreement with our selected value – (4123.6 ± 12.0) kJ·mol^{-1}.

[1963EHR/PEI]

The enthalpies of solution of calcium, strontium, barium metals, their anhydrous chlorides, bromides and iodides, as well as their corresponding hydrogenohalides were obtained at 298.15 K from dissolution calorimetry in dilute solutions (0.1 mol·dm^{-3}) of the corresponding acids so as to obtain a final concentration of 2500 mol H_2O per mol of salt. The calorimeter used was described in detail. We shall restrict the present discussion to the results concerning Ca(cr) and $CaCl_2$(cr).

The anhydrous halides were obtained by treating the anhydrous salts with the corresponding acids at high temperature. Although this process frequently leads to hydroxide or oxide contamination in the case of $CaCl_2$ (see [1969GRE/SNE]), the reported analytical results for $CaCl_2$ (Cl = 63.88 % (calc. 63.89 %) and Ca = 36.15 % (calc. 36.11 %)) were excellent. The metals were distilled under vacuum before use to free them of oxides, nitrides and alkali metals. For the metal, complexometric analysis yielded a purity of 99.7 % for calcium, but hydrogen determination indicated only 99.3 % free metal.

Calorimetric results were reported without any details as the mean of 4-5 individual measurements. The enthalpy of dissolution of the metal was reported as – 546.0 kJ·mol^{-1} and that of the chloride as – 79.66 kJ·mol^{-1}. In view of the analytical results on the metal sample, the results on the metal were listed for information only in [1987GAR/PAR].

[1963ELL2]

The solubility of calcite was determined at 12 bar CO_2 pressure in 0.2, 0.5, and 1.0 mol·kg^{-1} NaCl solutions for temperatures between 120 and 320 °C. The partial pressure of CO_2 was calculated from the solubility of the gas in the salt solution at the particular temperature. The loss of weight of the calcite crystals was measured. The solubility of calcite in natural hydrothermal solutions was found to be significantly higher than in salt solutions of comparable ionic strength. Therefore, complexing of calcium with silica was suspected. The reported data considerably extend the available temperature range for {calcite-H_2O-CO_2-NaCl} system. The calculated $\log_{10} K^{\circ}_{s,0}$ values agree well with other accepted data.

[1963FRI/PEE]

The first hydrolysis constant of Al^{3+} was measured using a glass electrode in conjunction with calomel and Ag|AgCl reference electrodes at 25 °C. Both cells with and without liquid junctions were used. Individual solutions were made up of either $AlCl_3$ and Na_2CO_3 or $Al(ClO_4)_3$ and Na_2CO_3 with aluminium concentrations varying from 10^{-5} to 10^{-2} mol·kg^{-1}. A simple Debye-Hückel expression was used to extrapolate the hydrolysis constants to $I = 0$. Results from the former mixture were considered by Frink and Peech to be more reliable and yielded an average value of $\log_{10} {}^{*}K^{\circ}_{1} = -(5.02 \pm 0.04)$. Note that this and other potentiometric studies ignore any contribution from $Al(OH)^{+}_{2}$.

[1963HOS]

Hostetler used precipitation and dissolution experiments at 25.0 °C under one atmosphere nitrogen gas to determine the solubility product of brucite and the first hydrolysis constant of Mg^{2+}. Mention was made of possible CO_2 contamination in some earlier studies that could have contributed to the wide disparity among previous studies. An electron microscope was used to monitor grain size as a function of time and established that sizes greater than 1 μm provided a constant solubility attributable to crystalline brucite. This limiting grain size as is now generally accepted as a rule of thumb for most solid phases. Interestingly, the author used a palladium-junction reference electrode in conjunction with a glass electrode to monitor pH over long periods of time in order to eliminate contamination of the reaction solutions by KCl diffusing from conventional reference electrodes. The ionic strength was maintained > 0.01 mol·kg^{-1} with additions of KCl in order for the palladium wire to give reliable readings. No mention was made of the pH calibration method employed, but pH values are given on the activity scale. The brucite was shown by XRD to be well crystallised before and after the dissolution experiments. The solubility products were calculated using empirical activity coefficients for Mg^{2+} after allowing for hydrolysis. The mean

$\log_{10}K_s^o$ value is – (11.05 ± 0.16) (2σ) although the author reported – (11.15 ± 0.03) (1σ) and suggested that equilibrium was achieved in five days. The dissolution equilibrium is represented by:

$$Mg(OH)_2(cr) + 2H^+ \rightleftharpoons Mg^{2+} + 2H_2O(l) \quad (A.70)$$

Application of the SIT to Reaction (A.70) gave $\log_{10}{}^*K_{s,0}^o$ = (16.85 ± 0.24) (2σ) or $\log_{10}K_s^o$ = – (11.15 ± 0.25) in agreement with that proposed later in [1976BAE/MES]. A second set of experiments lasting three days with constant bubbling of N_2 to reduce CO_2 contamination produced a mean $\log_{10}K_s^o$ value of – (10.62 ± 0.45) (2σ), but these solutions apparently remained over-saturated with respect to brucite and this result was not considered further by this review. A third set of complicated solubility experiments was reported in which $Mg(OH)_2$(s) was added to HCl solutions under strong agitation whereupon after one day the suspensions were boiled with reflux before being returned to the thermostat. This procedure was repeated until the pH remained constant for a day. XRD patterns were the same for the original and final solids; this was one of the earliest studies where the solid phase was monitored during the experiments. A mean $\log_{10}K_s^o$ value of – 11.16 was reported, *cf.* – (11.16 ± 0.12) obtained with the SIT treatment. Baes and Mesmer [1976BAE/MES] used an ion interaction treatment of the first and third datasets to give $\log_{10}{}^*K_{s,0}^o$ = (16.85 ± 0.05) and $\Delta\varepsilon$ = (0.201 ± 0.145) kg·mol^{-1}, noting that only one experiment was conducted at relatively high I_m. Altmaier *et al.* [2003ALT/MET] re-calculated the following values for these three sets of dissolution experiments: $\log_{10}{}^*K_{s,0}^o$ = (16.95 ± 0.16), (17.26 ± 0.10) and (16.84 ± 0.05 (2σ)).

Hostettler was able to estimate values for the hydrolysis constant represented by Reaction (A.71) only from the third dataset, but acknowledged that solubility experiments were not the best way to obtain such data which rely on mass and charge-balance equations, where the latter in turn is based on the pH measurements.

$$Mg^{2+} + OH^- \rightleftharpoons MgOH^+ \quad (A.71)$$

The mean value of $\log_{10}K_1^o$ reported was (2.80 ± 0.14). Re-calculation using the SIT gave $\log_{10}{}^*K_1^o$ = – (11.13 ± 0.16) which is almost 0.5 larger than the mean value calculated from more recent data. In order to address the hydrolysis equilibrium more directly, four titrations of $MgCl_2$ solutions, two containing added KCl, with a $Ba(OH)_2$ solution as titrant were reported. The ionic strength was ≤ 0.135 mol·kg^{-1}, but varied considerably during each titration due to the absence of a supporting electrolyte. Re-treatment of the combined datasets (26 points) with the SIT yielded $\log_{10}{}^*K_1^o$ = – (11.46 ± 0.26) (2σ), *cf.* – (11.40 ± 0.25) suggested by Hostetler and – (11.44 ± 0.1) recommended by Baes and Mesmer [1976BAE/MES] who set their ion interaction contribution term to zero in this case.

[1963JAC]

The enthalpies of solution in nitric acid of fluorapatite $Ca_{10}(PO_4)_6F_2$(cr) and hydroxylapatite $Ca_{10}(PO_4)_6(OH)_2$(cr) were measured and compared with the enthalpies of solution in the same medium of stoichiometrically identical mixtures of their component calcium oxide, fluoride, hydroxide and phosphate. From these, the standard enthalpies of formation of the two apatites were obtained. In each case, three thermodynamic cycles have been utilised. The procedure aimed at eliminating the need for detailed knowledge of the products of the solution if the relative activities of the solutes are kept constant in all experiments. The author also obtained solution calorimetric results leading to the enthalpy of formation of β-$Ca_3(PO_4)_2$.

The β-$Ca_3(PO_4)_2$ was obtained by heating laboratory-prepared hydrated calcium phosphate at 1023 K for 12 hours. Calcium hydroxide was prepared by precipitation from air-free solutions of "Analar" calcium nitrate by carbonate-free 10 % sodium hydroxide solution. The precipitate was washed with water, dried at 453 K and stored in a desiccator. Calcium oxide was prepared by heating the calcium hydroxide (as prepared above) at 1173 K to constant weight. No further characterisations were given for these compounds.

The fluorapatite sample was from a Canadian mine. Its chemical analysis revealed the presence of impurities (Al_2O_3, Fe_2O_3 and MgO) totalling less than 1.7 % of the sample weight and were accounted as "dead" weight in the final calculations, as they were not expected, according to the author, to have an appreciable influence on the observed heats of reaction. However, the fluorine content is given as 2.2 %, identical to an erroneously stated theoretical content that, in fact, should be 3.77 %, as already noted in [1981WES/NAV].

Properly treated human teeth were used the source of hydroxylapatite. X-ray powder analysis revealed a very pure apatite structure with only traces of free tricalcium phosphate and 0.17 % CaF_2. Allowance was made for the latter compound in the measured heats of solution. The dissolution medium was 31 % nitric acid (250 cm^3 for each experiment). Three to four g of the apatites were used in determinations made in triplicate, although only the average values (corrected, "as far as possible" for impurities) were reported.

The author gives no information on the calorimeter used, saying only that the procedure for the measurements was that of Gottschal [1958GOT] and the reactions were carried at 298.15 K. No undissolved material was found, apart from traces of silica and alumina for the fluorapatite. The author indicates that in the calculation of the enthalpies of formation from the enthalpies of solution, the enthalpies of formation of β-$Ca_3(PO_4)_2$(cr), CaF_2(cr), CaO(cr) and $Ca(OH)_2$(cr) from Rossini *et al.* [1952ROS/WAG] were used. The data of Egan and Luff [1961EGA/LUF] were used for the enthalpies of formation of phosphoric acid solutions.

It was virtually impossible to retrace exactly the various calculations made by Jacques, as no detailed information was given on some of the auxiliary values used in the cycles, in particular for the nitric and phosphoric acid solutions—the latter was merely described as having density of 1.75, presumably in g·cm^{-3}. We have, thus, re-calculated the various enthalpies of formation, using CODATA or CODATA-compatible values, taking as starting points the ten experimental enthalpies of solution given by Jacques. To ensure a better transparency with the original paper, we have kept hereunder the same sequence and used also calorie units. In the schematic representation below "HNO_3" represents 31 % HNO_3 and "sln" the product of the reaction under consideration. Thus:

Table A-26: List of reactions used by Jacques [1963JAC].

Reaction		$\Delta_r H_m$ /kcal·mol^{-1}
$Ca_9(PO_4)_6\cdot CaF_2(cr) + HNO_3 \rightleftharpoons sln$	(A.72)	– (49.6 ± 0.84)
$3Ca_3(PO_4)_2(cr) + CaF_2(cr) + HNO_3 \rightleftharpoons sln$	(A.73)	– (90.07 ± 0.54)
$2H_3PO_4 + HNO_3 \rightleftharpoons sln$	(A.74)	– (5.84 ± 0.06)
$CaF_2(cr) + 9Ca(OH)_2(cr) + (HNO_3)^* \rightleftharpoons sln$	(A.75) [(a)]	– (310.8 ± 1.5)
$CaF_2(cr) + 9CaO(cr) + (HNO_3)^* \rightleftharpoons sln$	(A.76) [(a)]	– (451.6 ± 2.5)
$Ca_3(PO_4)_2(cr) + HNO_3 \rightleftharpoons sln$	(A.77)	– (31.55 ± 0.74)
$Ca_{10}(PO_4)_6(OH)_2(cr) + HNO_3 \rightleftharpoons sln$	(A.78)	– (98.6 ± 1.12)
$3Ca_3(PO_4)_2(cr) + Ca(OH)_2(cr) + HNO_3 \rightleftharpoons sln$	(A.79)	– (125.3 ± 1.38)
$3Ca(OH)_2(cr) + (HNO_3)^* \rightleftharpoons sln$	(A.80) [(a)]	– (104.88 ± 0.51)
$3CaO(cr) + (HNO_3)^* \rightleftharpoons sln$	(A.81) [(a)]	− (151.53 ± 0.84)

(a) $(HNO_3)^*$ is the solution resulting from Reaction (A.74), so as to properly close the relevant cycle.

For the calculations below, we have used the values in Table A-27:

Table A-27: Auxiliary data used in the re-calculation of the results of [1963JAC].

Species	$\Delta_f H^\circ_m$ (298.15 K) /kJ·mol^{-1}	$\Delta_f H^\circ_m$ (298.15 K)/kcal·mol^{-1}	Source
CaO(cr)	– (634.92 ± 0.90)	– (151.750 ± 0.215)	CODATA
$Ca(OH)_2$(cr)	– (985.16 ± 1.00)	– (235.458 ± 0.239)	This review
CaF_2(cr)	– (1228.004 ± 2.000)	− (293.500 ± 0.478)	[1987GAR/PAR]
H_2O in (HNO_3,7.130H_2O)	– (286.323 ± 0.030)	– (68.433 ± 0.007)	(a)
H_3PO_4, 0.5937H_2O	– (1279.182 ± 1.016)	– (305.732 ± 0.243)	(b)

(a) Calculated from an interpolation of the enthalpies of formation of HNO_3(sln) given in [1982WAG/EVA].

(b) For the H_3PO_4 solution, we have accepted the value of 1.75 g·cm^{-3} for the density, as given in Table I of [1963JAC]; this, with the density data of Söhnel and Novotný [1985SOH/NOV], corresponds to a concentration of 90.159 wt % H_3PO_4 and thus to $H_3PO_4\cdot 0.5937H_2O$. The enthalpies of formation of H_3PO_4 solutions given in [1982WAG/EVA] were then corrected to have $\Delta_f H_m(H_3PO_4,100\ H_2O)$ = – (1296.2 ± 1.0) kJ·mol^{-1}, the basis of the CODATA phosphate values, and interpolated to give $\Delta_f H_m(H_3PO_4,0.5937H_2O)$ = – (1279.182 ± 1.016) kJ·mol^{-1}, where the small increase in uncertainty is due to the uncertainty in the density, taken to be ± 0.005 g·cm^{-3}, just from the two significant digits of the given value of 1.75 g cm^{-3}.

The value of $\Delta_f H^o_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K)$ was obtained using two distinct but related cycles:

A combination of Reactions (A.74), (A.77) and (A.80) above yields for:

$$3Ca(OH)_2(cr) + 2(H_3PO_4, 0.5937H_2O) \rightleftharpoons Ca_3(PO_4)_2(cr) + 6H_2O(\text{in } HNO_3, 7.130H_2O)$$

$\Delta_r H^o_m = -(79.17 \pm 0.90)$ kcal and, hence, $\Delta_f H^o_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = -(986.410 \pm 1.250)$ kcal·mol^{-1}.

Similarly, Reactions (A.74), (A.77) and (A.81) yield for:

$$3CaO(cr) + 2(H_3PO_4, 0.5937H_2O) \rightleftharpoons Ca_3(PO_4)_2(cr) + 3H_2O(\text{in } HNO_3, 7.130H_2O)$$

$\Delta_r H^o_m = -(125.82 \pm 1.12)$ kcal and, hence, $\Delta_f H^o_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = -(987.235 \pm 1.381)$ kcal·mol^{-1}.

The weighted mean of these values gives $\Delta_f H^o_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = -(986.781 \pm 1.852)$ kcal·mol^{-1} or $-(4128.691 \pm 7.749)$ kJ·mol^{-1}.

For reasons of internal consistency, we have used this value with its uncertainty in the calculations related to the apatite results obtained by Jacques. However, given the comments above, we believe that a value of ± 12.0 kJ·mol^{-1} would better represent the overall uncertainty of this result.

For hydroxylapatite, subtraction of the value for Reaction (A.78) from that of Reaction (A.79) yields $\Delta_r H^o_m = -(26.70 \pm 1.78)$ kcal and use of $\Delta_f H^o_m(Ca_3(PO_4)_2, cr, \beta, 298.15\ K)$ above gives $\Delta_f H^o_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(3222.501 \pm 5.839)$ kcal·mol^{-1} or $-(13482.94 \pm 24.43)$ kJ·mol^{-1}.

Three distinct but related cycles were used to obtain $\Delta_f H^o_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K)$. Subtraction of the value for Reaction (A.72) above from that for Reaction (A.73) yields $\Delta_r H^o_m = -(40.47 \pm 1.00)$ kcal and, hence, $\Delta_f H^o_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K) = -(3294.313 \pm 5.665)$ kcal·mol^{-1}. Similarly, combination of the values for Reactions (A.72), (A.74) and (A.75) yields $\Delta_r H^o_m = -(278.72 \pm 1.73)$ kcal and, hence, $\Delta_f H^o_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K) = -(3293.940 \pm 3.161)$ kcal·mol^{-1}. Finally, combination of the values for Reactions (A.72), (A.74) and (A.76) yields $\Delta_r H^o_m = -(419.52 \pm 2.64)$ kcal and, hence, $\Delta_f H^o_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K) = -(3297.265 \pm 3.616)$ kcal·mol^{-1}.

The weighted mean of these three values gives $\Delta_f H^o_m(Ca_{10}(PO_4)_6F_2, cr, 298.15\ K) = -(3295.215 \pm 4.375)$ kcal·mol^{-1} or $-(13787.18 \pm 18.31)$ kJ·mol^{-1} from this study.

As noted in the main text, the values for the enthalpies of formation of the two apatites from this study were included in those used to select the final values for these species.

[1963IKK/OKA]

Measurements were made of the specific gravity and viscosity of a series of sodium aluminate, sodium hydroxide solutions utilising pycnometers and capillary-type viscometers over ranges of temperature (293.15-353.15 K) and composition. Solubility experiments were made from 313 to 403 K, but there was no description of the solid phase used. The lack of precision of all these measurements meant that no reliable thermodynamic data could be extracted.

[1963ING2]

This potentiometric study of the formation of polyborates used the same cell described in [1957ING/LAG] with I_c = 3 mol·dm^{-3} ($NaClO_4$, LiBr, KBr and NaBr) at 25.0 °C. The borate concentration was varied from 0.01 to 0.4 mol·dm^{-3}. The LETAGROP regression program [1962ING/SIL] was used to determine formation constants for the optimal polyborate species. The following five equilibria were presented.

$$B(OH)_3(aq) + OH^- \rightleftharpoons B(OH)_4^- \quad (A.82)$$

$$3B(OH)_3(aq) + OH^- \rightleftharpoons B_3O_3(OH)_4^- + 3H_2O(l) \quad (A.83)$$

$$4B(OH)_3(aq) + 2OH^- \rightleftharpoons B_4O_5(OH)_4^{2-} + 5H_2O(l) \quad (A.84)$$

$$3B(OH)_3(aq) + 2OH^- \rightleftharpoons B_3O_3(OH)_5^{2-} + 3H_2O(l) \quad (A.85)$$

$$5B(OH)_3(aq) + OH^- \rightleftharpoons B_5O_6(OH)_4^- + 6H_2O(l) \quad (A.86)$$

The molar formation constants given in Ingri's Table 2 were converted to molal values [1985SOH/NOV] and are shown in Table A-28 as the weighted averages from three or four titrations. There are two typographical errors in Table 1 of this paper, "B = 0.50" should read "B = 0.05" and the 5th entry in this titration has an invalid value for Z.

Table A-28: Logarithms of the weighted average formation constants according to Reactions (A.82)-(A.86) where (3σ) statistical uncertainties were given by the author.

Medium	$\log_{10}K_{m\,1}$	$\log_{10}\beta_{m\,1\,3}$	$\log_{10}\beta_{m\,2\,4}$	$\log_{10}\beta_{m\,2\,3}$	$\log_{10}\beta_{m\,1\,5}$
3.50 mol·kg^{-1}($NaClO_4$)	(5.20 ± 0.02)	(7.32 ± 0.08)	(13.38 ± 0.11)	(11.73 ± 0.19)	7.4 [(b)]
3.27 mol·kg^{-1}(LiBr)	(5.23 ± 0.01)	(7.24 ± 0.07)	(13.32 ± 0.12)	(11.83 ± 0.13)	
3.27 mol·kg^{-1}(NaBr)	(5.23 ± 0.01)	(7.18 ± 0.07)	(13.30 ± 0.11)	(11.89 ± 0.14)	
3.39 mol·kg^{-1}(KBr)	(5.07 ± 0.01)	(7.11 ± 0.08)	(13.23 ± 0.10)	12.8 [(a)]	

(a) Only a single value was reported.

(b) The uncertainty in only one of four values was quoted and this species, which was only observed in $NaClO_4$ solutions, was considered to "probably" occur at high borate concentrations.

[1963ING3]

This potentiometric study is one of a series of investigations of the formation of polyborates using a cell fitted with hydrogen and Ag|AgCl electrodes. High borate

concentrations were used (*i.e.*, 0.6 to 3 mol·dm^{-3}) at 25.0 °C and the total sodium molality was maintained at *ca*. 3 mol·dm^{-3} by addition of $NaClO_4$. This experimental approach was termed the "self-medium" method. The LETAGROP regression program [1962ING/SIL] was again used to determine the optimum speciation and respective formation constants. The following three equilibria were proposed that best represented the data over the entire range of boron concentrations:

$$B(OH)_4^- \rightleftharpoons B(OH)_3(aq) + OH^- \quad \text{(A.87)}$$

$$3B(OH)_4^- \rightleftharpoons B_3O_3(OH)_5^{2-} + OH^- + 3H_2O(l) \quad \text{(A.88)}$$

$$4B(OH)_4^- \rightleftharpoons B_4O_5(OH)_4^{2-} + 2OH^- + 5H_2O(l) \quad \text{(A.89)}$$

The following values were reported: $\log_{10} K^{\circ}_{c,1}$ (A.87) = – (5.31 ± 0.02), $\log_{10} \beta^{\circ}_{c,2,3}$ (A.88) = – (4.64 ± 0.19) and $\log_{10} \beta^{\circ}_{c,2,4}$ (A.89) = – (7.34 ± 0.11) (3σ). There is a distinct decreasing trend in the values of $\log_{10} \beta^{\circ}_{c,2,3}$ (A.88) with increasing borate concentration. Moreover, their value of $\log_{10} \beta^{\circ}_{c,2,3}$ (A.88) is almost 2 $\log_{10}$ less than reported in [1957ING/LAG] where the dominant electrolyte was $NaClO_4$.

[1963LEA/MOR]

The authors have calculated the very low-temperature C_p (up to 20 K) of quartz, cristobalite and vitreous silica from the low-frequency distribution of the crystal, plus a few superimposed peaks. The data, in the form of graphs of C_p *vs.* T^2, are compared with experimental data.

[1963LEW]

Lewis studied the hydrolysis of Mg^{2+} potentiometrically (glass and hydrogen electrodes *versus* an Ag|AgCl electrode, although the latter generally gave unsatisfactory results) at (25.0 ± 0.1) °C and a total chloride concentration of 3.0 mol·dm^{-3} at high magnesium concentrations (1.000, 1.250 and 1.500 mol·dm^{-3} with additional NaCl added when needed). Back titrations (*i.e.*, an acidic titrant was used) produced reproducible hysteresis. Turbidity experiments indicated at least temporary formation of colloids upon addition of titrant that could have led to the observed hysteresis. Gjaldbaek [1925GJA] reported similar hysteresis during titrations of $Mg(OH)_2$ solutions. Two speciation schemes were obtained from the LETAGROP routine and a less rigorous graphical method:

$$Mg^{2+} + H_2O(l) \rightleftharpoons MgOH^+ + H^+ \quad \text{(A.90)}$$

or

$$2Mg^{2+} + H_2O(l) \rightleftharpoons Mg_2OH^{3+} + H^+ \quad \text{(A.91)}$$

and

$$4Mg^{2+} + 4H_2O(l) \rightleftharpoons Mg_4(OH)_4^{4+} + 4H^+ \quad (A.92)$$

The titration results are summarised in Table A-29 after conversion to molal quantities and compared with results obtained at low magnesium concentrations in NaCl solutions at the same ionic strengths [1996BRO/DRU]. Comparison of these three datasets does not lead to an unambiguous choice between the two equilibrium schemes within the given experimental uncertainties, which are presumed to be at the 1σ level in this paper.

Table A-29: Molal hydrolysis constants [(a)] for Mg^{2+} from the LETAGROP program (the corresponding values obtained graphically were significantly different) at 25 °C.

Medium /mol·kg^{-1}	I_m	$\log_{10}{}^*K_{m,1}$	$\log_{10}{}^*\beta_{m,2,1}$	$\log_{10}{}^*\beta_{m,4,4}$
$MgCl_2$ (1.06) + NaCl (1.06)	4.24	– (12.31 ± 0.06) – (12.41 ± 0.15) [(b)]	– (12.33 ± 0.06)	– (39.80 ± 0.02)
$MgCl_2$ (1.34) + NaCl (0.54)	4.56	– (12.17 ± 0.06) – (12.44 ± 0.16) [(b)]	– (12.30 ± 0.06)	– (39.77 ± 0.02)
$MgCl_2$ (1.56)	4.68	– (12.19 ± 0.12) – (12.45 ± 0.16) [(b)]	– (12.38 ± 0.12)	– (39.74 ± 0.03)

(a) Converted from molar to molal units using densities calculated from [1985SOH/NOV], [2010QIB/ABU].
(b) Calculated from the SIT parameters in [1996BRO/DRU] for NaCl media with the uncertainties at the 2σ level.

This appears to have been a well thought-out experimental program potentially providing rare hydrolysis constants for multinuclear magnesium species, but the results are nevertheless ambiguous. Baes and Mesmer [1976BAE/MES] also reviewed Lewis' results and disputed the speciation scheme, replacing Mg_2OH^{3+} by $MgOH^+$, because the hydrolysis constant was compatible with the constant derived in dilute $MgCl_2$ solutions (see the Appendix A entry [1976BAE/MES]). Both schemes predict $Mg_4(OH)_4^{4+}$ would dominate at high Mg(II) concentrations.

[1963MEA/RIC]

This paper concerns the measurement, at 308.15 K, of the enthalpies of solution in 6 molar HCl of a series of crystalline and glassy phosphates with the principal aim to obtain the enthalpies of formation from the component oxides, evaluation of values in terms of crystal energies and heats of mixing of different glasses. Of the compounds studied, only ($Ca_3(PO_4)_2$, cr) is of interest to this review. The compound used was said to be reagent material, 98 % pure, with less than 0.2 % Mg. It was ignited for eight hours at 1473 K, well above the stability range for the beta form. The authors do not report any verification of the structure of their sample, although they mention in passing the β-form in the discussion of their results. The calcium carbonate involved in the cycle of interest was said to be analytical reagent grade dried at an elevated temperature. The crystalline phosphorus pentoxide was made from commercial oxide by distillation.

The solution calorimeter used, previously described in the literature, had a capacity of 800 cm^3 and the final concentration of the solute was less than 0.005 $g \cdot cm^{-3}$. Samples were introduced in the chamber in gelatine capsules. For reactions involving carbonates, the solvent was saturated with CO_2 before use and a steady stream of CO_2 was passed through the solution during the experiment. Corrections were made for the heat of vaporisation of water and HCl into the CO_2 released. The calorimetric results were not given individually: only mention was made that three measurements were carried out. The uncertainty of the reported measurement was only given as "full spread of results".

The authors write the reactions of interest as:

$$Ca_3(PO_4)_2(cr) + 6HCl(sln) \rightleftharpoons (2H_3PO_4 + 3CaCl_2)(sln) \quad (A.93)$$

with $\Delta_r H_m$ (A.93) = – (111.29 ± 1.67) kJ·mol^{-1}

$$P_2O_5(cr) + 3H_2O(sln) \rightleftharpoons 2H_3PO_4(sln) \quad (A.94)$$

with $\Delta_r H_m$ (A.94) = – (209.83 ± 4.18) kJ·mol^{-1} and

$$3CaCO_3(cr) + 6HCl(sln) \rightleftharpoons (3H_2O + 3CaCl_2)(sln) + 3CO_2(g) \quad (A.95)$$

with $\Delta_r H_m$ (A.95) = – 3(24.31 ± 0.42) kJ·mol^{-1}

Combination of these reactions gives:

$$P_2O_5(cr) + 3CaCO_3(cr) \rightleftharpoons Ca_3(PO_4)_2(cr) + 3CO_2(g) \quad (A.96)$$

and $\Delta_r H_m$ (A.96) = – (171.47 ± 4.67) kJ·mol^{-1}.

In closing the cycle, Reaction (A.96), it was assumed that the small amounts of solutes had no influence on the enthalpy of the solvent, since no information in this regard was given by the authors. In view of the uncertainties affecting the enthalpy values for the above reactions at 308.15 K, we shall assume that these values are valid for 298.15 K. To evaluate $\Delta_f H_m^\circ$ ($Ca_3(PO_4)_2$, cr, 298.15 K), we made use of $\Delta_f H_m^\circ$ ($CaCO_3$, calc, 298.15 K) = – (1207.5 ± 1.2) kJ·mol^{-1} from this review (see Section V.6.2.1.9.2), $\Delta_f H_m^\circ$ (CO_2, g, 298.15 K) = – (393.51 ± 0.13) kJ·mol^{-1} from CODATA, and $\Delta_f H_m^\circ$ (P_2O_5, cr, 298.15 K) = – (1505.05 ± 1.60) kJ·mol^{-1} from the thorough assessment by Gurvich *et al.* [1989GUR/VEY]. This value is fully up-to-date, (a recent review on phosphorus species [2012JUN/HUD] does not reveal new studies) and is fully compatible with the CODATA value for $\Delta_f H_m^\circ$ (H_3PO_4, in 100H_2O) = – (1296.2 ± 1.0) kJ·mol^{-1}. We thus obtain $\Delta_f H_m^\circ$ ($Ca_3(PO_4)_2$, cr, 298.15 K) = – (4118.5 ± 6.1) kJ·mol^{-1} from this work.

However, there is no clear evidence as to the phase composition of the $Ca_3(PO_4)_2$ used in the dissolutions, and since these α–β transitions are particularly sluggish [2014MAT/RAD], we have increased the uncertainty of the above result. The difference in the enthalpies of formation of the α and β polymorphs of $Ca_3(PO_4)_2$(cr) is

not well-defined—it may be as large as (19-21) kJ·mol^{-1} ([1937BRI/VES], [1962SMI/ILL], with adopted values as different as 10.9 kJ·mol^{-1} [1982WAG/EVA] and 4.2 kJ·mol^{-1} [1979GLU/MED]). Our selected value from this source is thus – (4118.5 ± 10.0) kJ·mol^{-1}, the increased uncertainty also covering additional doubt about the impurity level in the calcite used.

[1963RYZ]

The conductances of aqueous solutions of CO_2 were measured at 100, 156 and 218 °C and at pressures approximately equal to that of the saturated vapour over the solutions. Using the results of similar measurements for Na_2CO_3 and $NaHCO_3$ solutions reported in a previous paper [1963KHI/RYZ], Ryzhenko calculated the true dissociation constants of carbonic acid (K_1 and K_2) at the given temperatures.

The dissociation constants for carbonic acid were used to calculate the equilibrium hydrolysis concentrations of HCO_3^-, CO_3^{2-}, and OH^- ions in solutions of alkali carbonate and bicarbonate. The results indicated that as the temperature rises from 100 °C, the degree of hydrolysis increases and carbonate and bicarbonate solutions become more alkaline.

The reported values of the first dissociation constant agree well with those of other reports. However, the pK_2 values deviate by up to 0.4 $\log_{10}$ units from those reported in [1984PAT/BUS], which are considered in this review as one of the most precise values at high temperatures. The reasons for these deviations are not clear, but are, at least in part, related to the propagation of errors due to the complicated calculation of pK_2, and to the assumption that the equivalent conductances of Na_2SO_4 and the hypothetical (unhydrolysed) Na_2CO_3 solutions are identical.

Due to the large uncertainties assigned to the constants reported in [1963RYZ], they were not considered in our evaluation.

[1963STO/ROB]

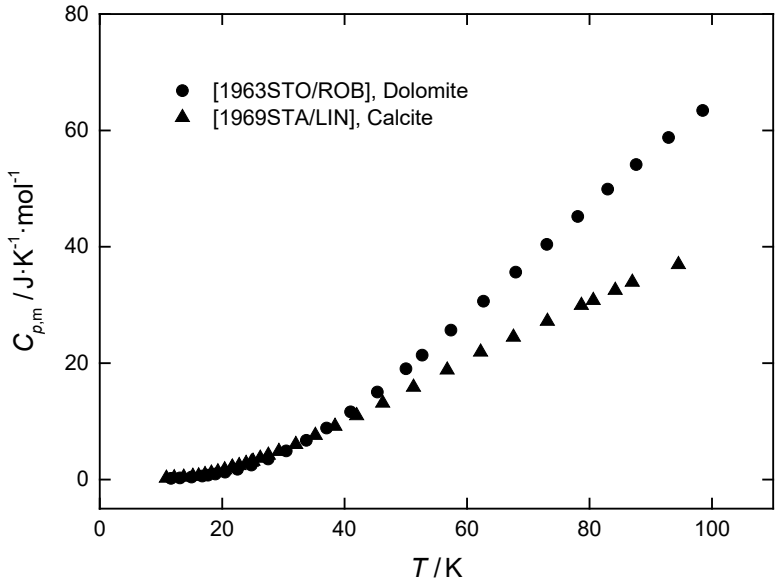
The authors measured the heat capacity of dolomite from 11.6 to 301.1 K using an adiabatic calorimeter. Since it is difficult to prepare a synthetic sample of well-ordered dolomite (the reaction of $CaCO_3$ and $MgCO_3$ is very slow at temperature below the decomposition temperature of $MgCO_3$), a purified natural mineral from Binnental, Switzerland was used. The surface layers of cleaned original crystals (0.5 to 3.0 cm) were scraped off, and the residual crystals washed three times in boiling distilled water, dried at 100 °C and crushed. A detailed chemical analysis showed that the sample contained a slight excess of both Mg and Ca with respect to pure dolomite, with also small quantities (< 0.10 wt %) of Fe, Mn and Sr. The latter elements, and the excess Ca were assumed to be present as carbonates, but the Mg as brucite $Mg(OH)_2$, for reasons

which are not entirely clear. Suitable small corrections for these impurities were applied to the measured heat capacities. XRD examination showed only the sharp lines of well-ordered dolomite.

The results of about 70 runs are tabulated, together with smoothed values of C_p, S and $H(T) - H(0\ \mathrm{K})$ at round temperatures from 10 to 300 K; it is not stated how the heat capacities below 11.6 K were extrapolated to 0 K. An unusual feature of the results is that below 35 K, the measured heat capacity of dolomite is less than that of calcite—see Figure A-10 below. The tabulated values of C_p and S at 298.15 K are (157.53 ± 0.42) and $(155.18 \pm 0.42)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$, respectively. However, with the small correction to 1981 atomic weight values, these become (157.51 ± 0.80) and $(155.17 \pm 0.80)\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$, where the uncertainties have been increased because of the unusual behaviour below 35 K. These are the selected values.

Stout and Robie also made approximate calculations of the enthalpy of the reaction $\mathrm{CaMg(CO_3)_2(cr) \rightleftharpoons MgCO_3(cr) + CaCO_3(calc)}$ based on the decomposition data of Graf and Goldsmith [1955GRA/GOL], using a number of estimated quantities and simplifying assumptions. These calculations would indicate a relatively constant value of $(12.3 \pm 1.0)\ \mathrm{kJ \cdot mol^{-1}}$ over the temperature range (830 to 1136) K, but this value is not considered further in this review.

Figure A-10: Heat capacity of dolomite and calcite.



[1963VIC/DOU]

The enthalpy content $H(T) - H(273.15\ \mathrm{K})$ of fused MgO(cr) and sintered BeO(cr) was measured up to 1173 K using a Bunsen ice-calorimeter. The clear, colourless MgO(cr) sample contained 99.9 weight % MgO, the principal impurities being CaO and Fe_2O_3 and the estimated uncertainty in the measurements 0.25 %. The data (nine measurements, each duplicated) were fitted to the equation corresponding to the C_p expression:

$$C_{p,\mathrm{m}}(\mathrm{MgO,cr},T)\ /\mathrm{J\cdot K^{-1}\cdot mol^{-1}} = 44.940 + 1.0190\times10^{-2}\,(T/\mathrm{K}) - 2.9099\times10^{-6}\,(T/\mathrm{K})^2 - 9.46216\times10^{5}\,(T/\mathrm{K})^{-2} \qquad (298.15\text{-}1000)\ \mathrm{K}$$

The data merge smoothly with the low-temperature measurements of Gmelin [1969GME2] and together with the data in [1974TSA/GVE] were the main basis of the high-temperature C_p data selected by CODATA [1989COX/WAG].

[1963WIS/MAR]

In this very thorough paper, Wise *et al.* reported the enthalpies of formation at 298.15 K of SiF_4(g), SiO_2(α-quar) and SiO_2(vit). The high-pressure fluorine bomb calorimetric equipment and its mode of operation had been detailed in earlier publications of the same group. In the present study, amounts of 400-500 mg of silicon and 1.0-1.2 g of each form of silica were used. All experiments were repeated seven or more times. As neither forms of silica reacted spontaneously with fluorine, powdered silicon had to be added for the combustion of the two silica forms. The only experimental differences (fluorine pressure and geometrical arrangement) were dictated by the differences in behaviour and reactivity of the two polymorphs under fluorine attack. In particular, coupons of vitreous silica were used, while α-quartz had to be reduced to granules of 60-100 mesh size. Infrared spectrometric analysis of the combustion gases showed only bands due to SiF_4 and none to Si_2F_6 or to oxyfluorides. Mass-spectrometric analyses showed only SiF_4, plus O_2 (after silica combustions), and small amounts of N_2 (impurity in the fluorine) and of CF_4 (carbon impurity in the silicon samples).

The two values reported in this very thorough paper, namely $\Delta_f H^\circ_\mathrm{m}(\mathrm{SiF_4,\ g,\ 298.15\ K}) = -(385.98 \pm 0.19)\ \mathrm{kcal\cdot mol^{-1}}$ or $-(1614.94 \pm 0.79)\ \mathrm{kJ\cdot mol^{-1}}$ and $\Delta_f H^\circ_\mathrm{m}(\alpha\text{-quartz, 298.15 K}) = -(217.72 \pm 0.34)\ \mathrm{kcal\cdot mol^{-1}}$ or $-(910.94 \pm 1.42)\ \mathrm{kJ\cdot mol^{-1}}$, were essential in the adoption by CODATA [1989COX/WAG] of the standard enthalpies of formation of these compounds. As we have no reason to question the CODATA assessment, we shall not discuss further the results on the enthalpies of formation of SiF_4(g) and of α-quartz.

However, [1963WIS/MAR] also reported $\Delta_f H^\circ_\mathrm{m}(\mathrm{SiO_2,\ vit,\ 298.15\ K}) = -(215.94 \pm 0.31)\ \mathrm{kcal\cdot mol^{-1}}$ or $-(903.49 \pm 1.30)\ \mathrm{kJ\cdot mol^{-1}}$ based on the reaction of vitreous silica with fluorine under analogous conditions as for α-quartz. The only

experimental differences (fluorine pressure and geometrical arrangement) were dictated by the differences in behaviour and reactivity of the two polymorphs under fluorine attack. In particular, coupons of vitreous silica were used, while α-quartz had to be reduced to granules of 60-100 mesh size. The vitreous silica samples were obtained from Englehard Industries and from Corning Glass Works. No metallic impurities were detected by spectrographic analysis in either of these samples and the absence of XRD lines showed them to be amorphous.

Based on seven determinations, the authors report for the combustion Reaction (A.97):

$$SiO_2(vit) + 2F_2(g) \rightleftharpoons SiF_4(g) + O_2(g) \quad (A.97)$$

the average value, $\Delta_r U°((A.97), 298.15\ K) = -(2830.0 \pm 0.4)$ cal·(g SiO_2)$^{-1}$ (1σ). For the same number of experiments with α-quartz (Reaction (A.98)):

$$SiO_2(\alpha\text{-quar}) + 2F_2(g) \rightleftharpoons SiF_4(g) + O_2(g) \quad (A.98)$$

they obtained the average value $\Delta_r U°((A.98), 298.15\ K) = -(2800.3 \pm 1.1)$ cal·(g SiO_2)$^{-1}$.

From these values, this review calculates for the transformation according to Reaction (A.99):

$$SiO_2(\alpha\text{-quar}) \rightleftharpoons SiO_2(vit) \quad (A.99)$$

$\Delta_r U°((A.99), 298.15\ K) = (1.785 \pm 0.070)$ kcal·mol^{-1} or (7.468 ± 0.293) kJ·mol^{-1}.

In their abstract of the paper, the authors give this difference as (1.78 ± 0.46) kcal·mol^{-1}.

In their discussion, [1963WIS/MAR] note that their results resolve apparent discrepancies between thermochemical and equilibrium values for the Gibbs energy of formation of silica that were previously noted [1961CHI], but that unresolved discrepancies still existed in the enthalpy of formation of silicon carbide.

[1964CHR]

The concentrations in these bayerite solubility experiments at 20 °C were given in g·dm^{-3} of Na_2O and Al_2O_3, and were converted here to molalities using the modified equation in [2006REY/BER]. Application of the SIT treatment to a restricted, scattered dataset yielded: $\log_{10} K^{o}_{s,4} = -(0.88 \pm 0.16)$ with $\Delta\varepsilon$ fixed at $-(0.052 \pm 0.037)$ kg·mol^{-1}.

[1964GOO/LAC]

This study concerns the development of an original use of an existing rotating-bomb combustion calorimeter for its application to the determination of the enthalpy of formation of α-quartz and of organosilicon compounds. It consisted of submitting for combustion, mixtures of the compound to be investigated with a fluorine donating substance, which in the case of the combustion of silicon was polymeric vinylidene fluoride, $(CH_2CF_2)_n$. We shall not discuss here the combustion of hexamethyldisiloxane that was also the topic of this publication. The high-purity silicon used (50 to 75 μm particle size) (courtesy of Bell Telephone Laboratories, Murray Hill, NJ) was estimated to have, after proper treatment, a surface coating of SiO_2 50 Å thick that was accounted for in the combustion results. The vinylidene trifluoride (Pennsalt Chemical Corp.) was shown to deliver in the combustion experiments, an amount of HF corresponding to the theoretical formula of the polymer. Thus, for the combustions, the amounts of silicon and vinylidene fluoride used in the mixtures were such that the polymer would deliver 7.4 atoms of fluorine per atom of silicon. Two types of combustions were made under 30 bar O_2(g) in the presence of water or of concentrated HF solutions so as to realise the necessary conditions for Reactions (A.100) and (A.101) below:

$$\mathrm{Si\,(cr,\alpha) + O_2(g) + (7.444HF, 209.729H_2O)(sln) \rightleftharpoons (H_2SiF_6,\ 1.444HF, 211.729H_2O)(sln)} \qquad \text{(A.100)}$$

and

$$\mathrm{Si(cr,\alpha) + O_2(g) + (47.157HF,\ 172.657H_2O)(sln) \rightleftharpoons (H_2SiF_6, 41.157\ HF, 174.657H_2O)(sln)} \qquad \text{(A.101)}$$

Only the experiments (nine and eight for Reactions (A.100) and (A.101), respectively) that showed the absence of carbon or unburned silicon were taken into account. Because vinylidene fluoride contributes a large amount of the combustion energy delivered in these experiments, an accurate value of its enthalpy of combustion was determined in a separate series of ten combustions.

After conversion of energy values to enthalpy values, the enthalpies of combustion were reported as $\Delta_r H^\circ_m$ ((A.100), 298.15 K) = – (249.89 ± 0.25) kcal·mol^{-1} or – (1045.54 ± 1.05) kJ·mol^{-1}, $\Delta_r H^\circ_m$ ((A.101), 298.15 K) = – (250.30 ± 0.25) kcal·mol^{-1} or – (1047.26 ± 1.05) kJ·mol^{-1}, the uncertainty intervals representing twice the standard deviation of the means. Combining their result for Reaction (A.101) with the results of King [1951KIN] for the enthalpy of solution of α-quartz in aqueous HF, Good *et al.* reported $\Delta_f H^\circ_m$ (SiO_2, cr, α, 298.15 K) = – (217.5 ± 0.5) kcal·mol^{-1} or – (910.0 ± 2.1) kJ·mol^{-1}. In their assessment of the results of Good *et al.* [1964GOO/LAC] for CODATA, Cox *et al.* [1989COX/WAG] used the later measurements of Kilday and Prosen [1973KIL/PRO] on the enthalpy of solution of SiO_2(α-quar) in aqueous HF and obtained:

$$\Delta_f H^\circ_m\,(SiO_2, \alpha\text{-quar}, 298.15\ K) = -\,(910.4 \pm 1.4)\ kJ{\cdot}mol^{-1}.$$

[1964LYA/KHO]

Gibbsite "was obtained by decomposing pure aluminate solutions in ordinary centrifuging conditions". Most experiments were performed at (60 ± 1) °C for 32 to 36 days and although additional data at 95 °C were given in a figure, these were ignored by us because of the high probability that boehmite may have formed. Given $\log_{10}K^{\circ}_{H_2O}$ = − 13.036 at 60 °C, application of the SIT to their data in their Table 1 gave $\log_{10}K^{\circ}_{s,4}$ = – (0.635 ± 0.084), whereas the data in their Table 2 indicated that NaCl was added and yielded $\log_{10}K^{\circ}_{s,4}$ = – (0.688 ± 0.054). The combined datasets yielded $\log_{10}K^{\circ}_{s,4}$ = – (0.658 ± 0.037), which is in surprisingly good agreement with the bulk of results from other studies, and $\Delta\varepsilon$ = – (0.043 ± 0.007) kg·mol^{-1} at 60 °C.

[1964MOR/FOU]

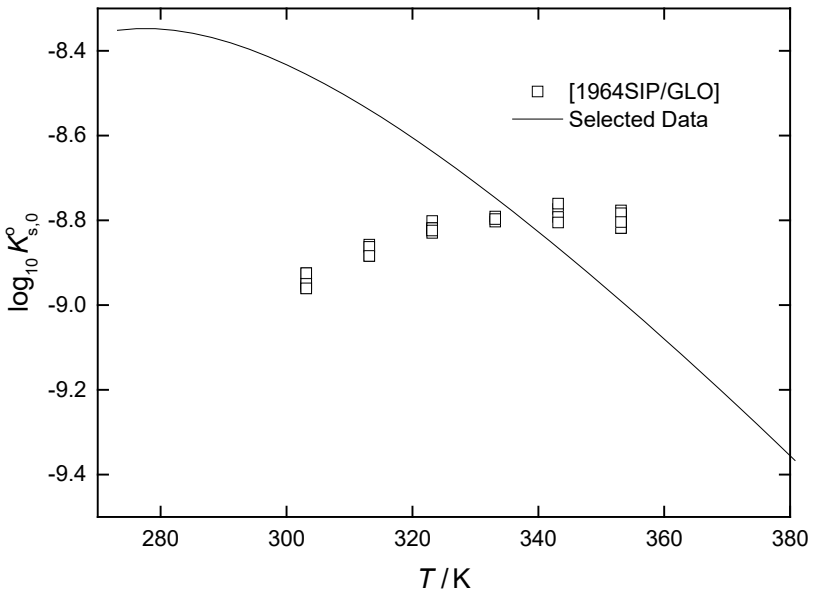
Silica gel was obtained from two sources and silica glass from another. The average solubility of amorphous SiO_2 in water measured from under- and over-saturation was given as 115 ppm (1.927×10^{-3} mol·kg^{-1}) at 25 °C and the differential enthalpy of solution was given as 14.0 kJ·mol^{-1}, which was derived from solubility data to 335 °C in [1935HIT] and [1960KIT]. Morey *et al*. noted that solutions of SiO_2 can become "enormously supersaturated" and that this condition can persist for a long time. They also made the point that attainment of a steady-state concentration during SiO_2 dissolution was not necessarily an indicator of true equilibrium. The solubility limit of 115 ppm was only attained after more than 1000 days from under-saturation and *ca*. 600 days from over-saturation.

[1964SIP/GLO]

The solubility of calcite (Iceland spar) in distilled water and in NaCl solutions was measured between 30 and 80 °C at $p(CO_2)$ = 0.1 and 0.3 bar (equivalent to 0.0033 and 0.01 mol·kg^{-1} CO_2) and at four different total pressures (3.5, 59.3, 118.6 and 178.6 bar) in flow equipment using a flow rate 50 cm^3·h^{-1}. Sippel *et al.* assumed equilibrium was established because no decrease in dissolved calcium was noted at substantially higher flow rates. After filtration, the concentration of calcium in the effluent was determined by photometric titration using EDTA and ammonium purpurate as indicators.

Contrary to the statement of the authors, this review suggests that the dissolution equilibrium was very likely not established during the relatively brief contact time of the solution and calcite crystals. The $\log_{10}K^{\circ}_{s,0}$ ($CaCO_3$, calc) values calculated using the reported solubility data are considerably different from the selected values, being smaller with an unexpected temperature dependence (see Figure A-11). Therefore, all the solubility data reported in this paper were rejected.

Figure A-11: The calculated $\log_{10} K^{\circ}_{s,0}$ ($CaCO_3$, calc) values based on the solubility data reported in [1964SIP/GLO] and the temperature dependence of the selected solubility products.



[1964THO/CUN]

Within the development of a high-temperature calorimeter, Thomasson and Cunningham report enthalpies of decomposition of $NaHCO_3$(cr), $MgCO_3$(natural, cr) and $CaCO_3$(cr) (presumably calcite) to (again presumably) Na_2CO_3(cr) + 0.5 H_2O(g) + CO_2(g), MgO(cr) + CO_2(g) and CaO + CO_2(g), respectively.

The apparatus consisted of two identical platinum crucibles, one containing the sample and the other one 200 B.S. sieve grain-size alumina heated to 1773 K. The heating rate is stated to be 10 K·min^{-1}. From the dimensions of the sketch, and the *modus operandi*, it may be inferred that the sample and standard alumina are gram or sub-gram size. In the case of magnesium and calcium carbonate, fired alumina was used as diluent of the samples. The three compounds studied were of Analar grade furnished by British Drug Houses Ltd. The results were reported as shown in Table A-30. The uncertainty is given by the authors as representing the consistency of twenty determinations for each compound.

Table A-30: Materials studied and measured enthalpies of decomposition.

Compound	Decomposition temperature /K	Enthalpy of decomposition at given T	
		kcal·mol^{-1}	kJ·mol^{-1}
$NaHCO_3$	473	(16.10 ± 0.16)	(67.36 ±0.67)
$MgCO_3$	923	(25.90 ± 0.65)	(108.37 ± 2.72)
$CaCO_3$	1223	(38.80 ± 0.78)	(162.34 ± 3.26)

The authors compare their results with values published in the literature but neither the data used nor the procedure used to bring the literature values to the reaction temperature are unambiguous.

Correction of these values to 298.15 K, using data assessed in this review and TDB auxiliary data, gives:

$$\Delta_f H^\circ_m (MgCO_3, cr, 298.15\ K) = -(1112.14 \pm 3.00)\ kJ\cdot mol^{-1}$$

where the uncertainty has been increased slightly to take account of the well-known slow kinetics of this decomposition and the use of estimated values for $C_p(MgCO_3, cr)$ above 675 K.

$$\Delta_f H^\circ_m (CaCO_3, cr, 298.15\ K) = -(1208.97 \pm 3.50)\ kJ\cdot mol^{-1}$$

For $NaHCO_3$. the correction to 298.15 K is only 58 J·mol$(NaHCO_3)^{-1}$ so

$$\Delta_f H^\circ_m (NaHCO_3, cr, 298.15\ K) = -(949.07 \pm 0.69)\ kJ\cdot mol^{-1}.$$

Thus all three values are in very good agreement with our selected values for the standard enthalpies of formation.

[1965BAR/BAR]

These measurements of the densities of sodium aluminate solutions covered a wide range of NaOH concentrations, but a narrow range of the Na/Al ratios. The precision was $< 2 \times 10^{-4}$ g·cm^{-3} on average. Only a small fraction of the actual experimental data was given graphically. All concentrations were expressed as mass percentages of Na_2O and Al_2O_3. In spite of the absence of original experimental data, the densities are considered to be reliable.

[1965HAN/CHR]

Hansen *et al*. reported the results of calorimetric titrations for simultaneous determinations of the hydrolysis constants of HPO_4^{2-} and HSO_4^- at 25 °C. For the former they determined $\log_{10} K_3^\circ = (1.61 \pm 0.03)$ and $\Delta_r H^\circ_{m,3} = -(37.7 \pm 0.8)$ kJ·mol^{-1}, which correspond to $\log_{10} {}^*K_3^\circ = 12.39$ and $\Delta_r {}^*H^\circ_{m,3} = 18.1$ kJ·mol^{-1}. These values are

accepted for assessment with increased uncertainties of ± 0.20 and ± 5.0 kJ·mol^{-1}, respectively.

[1965HOP/WUL]

The authors measured the enthalpy of solution of $Ca(OH)_2$(cr) both in water and in dilute hydrochloric acid, with the aim of improving the then current knowledge of the thermodynamic functions associated with the reaction:

$$Ca(OH)_2(cr) \rightleftharpoons Ca^{2+} + 2OH^- \quad (A.102)$$

The solution calorimeter used has been described in the literature [1963WU/BIR], and the solvent volume was 0.950 dm^3. Commercial reagent grade calcium hydroxide was used without further purification or analysis. Samples ranging between 0.2 and 1.5 g were used in individual measurements.

For the enthalpy of solution in water 16 determinations (no details given) at (298.15 ± 0.10) K were fitted by the straight line $\Delta_{sln}H_m$/kJ·mol^{-1} = (– 18.974 + 29.957 $m^{0.5}$) ± 0.126, with the molality of the solution m ranging from 0.004 and 0.02 mol·kg^{-1}.

Another series of measurements was performed, in which the samples (0.18-1.25 g) were dissolved in 0.04 molar HCl (0.950 dm^3). The accuracy was lower, as the reaction took longer to complete and the heat effect approached the upper limit of the useful range of the instrument. Ten determinations at (298.15 ± 0.30) K, corrected for the dilution of the unreacted acid, were fitted by the straight line:

$$\Delta_{sln}H_{acid}\,/\text{kJ·mol}^{-1} = (-130.081 + 36.652\ m^{0.5}) \pm 0.753.$$

Three additional divergent determinations were not included in the fit.

[1965PAP/TOR]

The titration calorimetric method, which is purported to allow simultaneous calculation of the acid dissociation constant and the enthalpy of reaction, was examined and applied to the dissociation of HPO_4^{2-}. There is good agreement between the $\log_{10}K_3^\circ$ and $\Delta_r H_{m,3}^\circ$ values found at 25 °C and those of Hansen *et al*. [1965HAN/CHR].

[1965ROB/STO]

The ion interaction coefficient, $\varepsilon(Cs^+, OH^-)$, is unavailable in the NEA TDB and therefore the osmotic coefficients for CsOH(sln) were taken from the tabulated data in this book. Regression of these values, which extended to $I_m = 1.0$, gave:

$\varepsilon(Cs^+, OH^-) = (0.05 \pm 0.01)$ kg·mol^{-1} (2σ) after omitting the scattered data at the three lowest concentrations.

[1965RUP/HOP]

In this study, the enthalpies of solution in water of Na_2CO_3(cr), $NaHCO_3$(cr) and trona, $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$(cr), were determined using a Dewar-type calorimeter. As the results of trona fall outside the scope of this review, they will not be discussed here. Samples were dissolved in 0.95 dm^3 of distilled water at (298.15 ± 0.10) K. The experimental results were extrapolated to standard condition after correction for hydrolysis reactions.

The sodium carbonate employed was Fisher certified reagent grade listed as 99.9 % (minimum). Before use, the sample was dried in an oven at 393 K for at least two days. Fisher sodium bicarbonate, certified to have a purity of 99.98 % or better, was used without further purification.

For the predominant secondary equilibrium in the solutions of sodium carbonate according to Reaction (A.103):

$$CO_3^{2-}(sln) + H_2O(sln) \rightleftharpoons HCO_3^-(sln) + OH^-(sln) \qquad (A.103)$$

the constant K^o((A.103), 298.15 K) = 1.8 × 10^{-4} and the enthalpy change $\Delta_r H_m^o$ ((A.103), 298.15 K) = 40.84 kJ·mol^{-1} taken from the literature were used in the calculations. For any molal concentration, m, of carbonate originally dissolved, the extent of the degree of hydrolysis, $\alpha = m_{HCO_3^-}/m$ was related to K_m((A.103), 298.15 K) and the ionic strength, through the use of the common form of the Debye-Hückel relation for ionic activity coefficients. The activity of water was assumed to be unity in the dilute solutions used. The experimental values for the integral enthalpy of solution of Na_2CO_3 at sixteen molalities, $\Delta_{exp}H_m$, representative values of α and of $\alpha\,\Delta_r H_m^o$ (A.103) are listed in Table A-31, as given in [1965RUP/HOP], with values in kcal·mol^{-1} converted to kJ·mol^{-1}.

Table A-31: Enthalpy of solution of Na_2CO_3(cr) at 298.15 K.

Molality of dissolved carbonate /m [(a)]	Degree of hydrolysis /α	$\alpha\,\Delta_r H_m^o$ (A.103) /kJ·mol^{-1}	Experimental enthalpy of solution, $\Delta_{exp}H_m$ /kJ·mol^{-1}
0.00702	0.126	5.163	– 21.083
0.00848			– 21.426
0.00921			– 21.556
0.01025			– 21.690
0.01138	0.098	4.002	– 22.066
0.01356			– 22.463
0.01664			– 22.410
0.02171			– 22.794
0.02547			– 23.138

(Continued on next page)

Table A-31 (continued)

Molality of dissolved carbonate /m [(a)]	Degree of hydrolysis /α	α $\Delta_r H^\circ_m$ (A.103) /kJ·mol^{-1}	Experimental enthalpy of solution, $\Delta_{exp}H_m$ /kJ·mol^{-1}
0.03219	0.060	2.450	– 23.267
0.03998			– 23.472
0.04885			– 23.962
0.05735			– 23.940
0.06670			– 23.999
0.06833	0.033	1.348	– 24.192
0.07531			– 23.974

(a) The values of m were not corrected for changes in molecular weight which amounted to less than 0.01 %.

The standard enthalpy of solution of Na_2CO_3(cr), $\Delta_{sol}H^\circ$((A.104), 298.15 K)

$$Na_2CO_3(cr) \rightleftharpoons 2Na^+ + CO_3^{2-} \qquad \text{(A.104)}$$

is then written as $\Delta_{sol}H^\circ_m$ ((A.104), 298.15 K) = $\Delta_{exp}H_m - \Phi_L(m) - \alpha\,\Delta_r H^\circ_m$ (A.103), where $\Phi_L(m)$ is the apparent molar heat content of the solution at concentration m. Accepting for $\Phi_L(m)$ values similar to those of the alkali sulfates that are given in the literature, the authors obtain from a least square analysis of their data the equation:

$$\Delta_{exp}H_m - \Phi_L(m) = -26.40 + 37.82\ \alpha \text{ kJ·mol}^{-1}$$

with an average deviation of 0.08 kJ·mol^{-1}, and report from this scheme:

$$\Delta_{sol}H^\circ_m \text{ ((A.104), 298.15 K)} = -(26.40 \pm 0.21) \text{ kJ·mol}^{-1}$$

A least square analysis of $\Delta_{exp}H_m - \Phi_L(m)$ *vs.* $m^{½}$ yields the equation $\Delta_{exp}H_m - \Phi_L(m) = -26.65 + 5.19m^{½}$ kJ·mol^{-1} with an uncertainty of ± 0.13 kJ·mol^{-1} that is the root mean square deviation. From this scheme, the authors thus obtain $\Delta_{sol}H^\circ_m$ ((A.104), 298.15 K) = – 26.65 kJ·mol^{-1}.

Finally, the authors take $\Delta_{sol}H^\circ_m$ ((A.104), 298.15 K) = – (26.61 ± 0.33) kJ·mol^{-1} as the best value.

In solutions of sodium hydrogen carbonate, for the most important secondary reaction:

$$2HCO_3^-(sln) \rightleftharpoons H_2CO_3(sln) + CO_3^{2-}(sln) \qquad \text{(A.105)}$$

the authors use data corresponding to the enthalpy change $\Delta_r H^\circ_m$ ((A.105), 298.15 K) = 7.36 kJ·mol^{-1} and K°((A.105), 298.15 K) = 1.2 × 10^{-4} to obtain $\alpha = m_{HCO_3^-}/m$ They note that the quantity $\alpha\,\Delta_r H^\circ_m$ (A.105) in their solutions (0.011 to 0.09) m amounts only to *ca.* 0.08 kJ·mol^{-1}. The experimental enthalpies of solution of sodium hydrogen carbonate, about which no individual values were given, were fitted as a function of $m^{½}$ to give the linear equation:

$$(\Delta_{exp}H_m - \Phi_L(m))\ /\text{kJ·mol}^{-1} = 18.79 + 1.59\ m^{½}$$

with an uncertainty of ± 0.08 kJ·mol^{-1}.

The standard enthalpy of solution of $NaHCO_3$(cr), $\Delta_{sol}H^{o}_{m}$ ((A.106), 298.15 K)

$$NaHCO_3(cr) \rightleftharpoons Na^+ + HCO_3^- \quad \text{(A.106)}$$

is then reported as $\Delta_{sol}H^{o}_{m}$ ((A.106), 298.15 K) = (18.79 – 0.08) = 18.71 kJ·mol^{-1} with an overall uncertainty of ± 0.21 kJ·mol^{-1} that is not discussed.

[1966CHR/IZA]

An entropy titration-calorimetric method was tested by determining $\Delta_r H_m$ and $\Delta_r S_m$, and hence the ionisation constants of HSO_4^- and HPO_4^{2-}. Christensen *et al.* reported average values of $\log_{10}K^{o}$ ((A.107), 298.15 K) = (1.61 ± 0.03) and $\Delta_r H^{o}_{m}$ ((A.107), 298.15 K) = – (9.14 ± 0.31) kcal·mol^{-1} (– (38.2 ± 1.30) kJ·mol^{-1}).

$$HPO_4^{2-} + OH^- \rightleftharpoons PO_4^{3-} + H_2O(l) \quad \text{(A.107)}$$

[1966FIS/ERT]

The authors measured the enthalpy content of 13 g samples of large single crystals of fused lime at nine temperatures from 693-1989 K in two different drop calorimeters. Four further measurements were made with larger samples between 273.15 K and 336.11 K using two further calorimeters. For measurements up to 1621 K, the samples were contained in nickel capsules, but for the other experiments, the samples were uncovered. No analysis of the CaO crystals is given, either before or after the experiments.

As shown in Figure V-52 cited in Section V.6.2.1.1.2 of the main text, the data above 1200 K, though somewhat scattered, agree reasonably with the consensus of other data, but those below 1200 K seem rather low.

[1966GME2]

The author measured the low-temperature heat capacity of BeO(cr) and CaO(cr) from *ca.* 1.2-65 K; we shall only comment on the data for the latter. The CaO sample was in the form of a crucible weighing 44 g, prepared from 'nuclear grade lime', with an impurity level of *ca.* 300 ppm of which 30 ppm was boron. The detailed measurements are not given, only values at round temperatures (intervals of 2 K from 2 to 20 K and 5 K from 20 to 65 K). The uncertainties in the data were estimated by the author to be 2 % up to 2 K and < 1 % above 4 K.

These data, together with those from [1969GME2] were the basis of the selected data for the low-temperature heat capacity and entropy of CaO(cr) in the CODATA assessment [1989COX/WAG].

[1966IRA/TAU]

The enthalpies of ionisation of $H_2PO_4^-$ and HPO_4^{2-} were measured calorimetrically at 298.15 K with I_c varying from 0.01 to 0.1 and 0.02 to 0.2, respectively. Extrapolation to $I = 0$ yielded: $\Delta_r^* H^o_{m,2} = (2.5 \pm 1.3)$ kJ·mol^{-1} and $\Delta_r^* H^o_{m,3} = (11.3 \pm 1.3)$ kJ·mol^{-1}.

[1966KIT2]

This study of gibbsite solubility according to Reaction (A.108) was one of the first to recognise the need to obtain a well-crystallised solid phase free from boehmite and bayerite impurities. Moreover, Kittrick carried out measurements from over- and under-saturated solutions apparently at room temperature, 23-25 °C.

$$\alpha\text{-Al(OH)}_3\,(\text{cr}) \rightleftharpoons \text{Al}^{3+} + 3\text{OH}^- \qquad \text{(A.108)}$$

ALCOA C-33 and C-730 gibbsite samples were used with surface areas of 0.2 and 10 m^2·g^{-1} corresponding to particle diameters of > 50 and 0.05 μ, respectively. XRD and differential thermal analyses indicated that only one phase was present with the sharper peaks observed for C-33 being attributed to a higher degree of crystallinity. Batch experiments with high loadings of gibbsite were performed with aliquots being centrifuged and analysed for aluminium using a spectrophotometric method. The initial pH of the mixture was adjusted to 3 with HCl and the pH of centrifuged aliquots was measured *versus* standard pH buffers. Solutions were equilibrated for four years with occasional shaking, although the aluminium concentration was constant after one year. A solution containing suspended C-730 gibbsite was allowed to approach equilibrium for 267 days from over-saturation, although the aluminium analyses were constant after 80 days. Table A-32 summarises the data provided in Table 1 of [1966KIT2].

Table A-32: Molar concentrations of aluminium and hydrogen ion for the C-370 and C-33 gibbsite samples at room temperature and the resulting solubility product and Gibbs energy of formation of gibbsite at room temperature.

Quantity	Under-saturation				Over-saturation
	C-730		C-33		C-730
$-\log_{10} c_{Al}$	3.60	3.51	3.57	3.71	3.05
pH	3.92	3.87	3.92	3.96	3.73
$\log_{10} K^o_{sp}$		– 34.02	– 33.99	– 33.97	– 34.14
$\Delta_f G^o_m(\text{Al(OH)}_3, \text{gibb}, 298.15\text{ K})$ [(a)] converted to kJ·mol^{-1}	– 1147.2	– 1147.2	– 1147.0	– 1147.0	– 1147.9

(a) Using $\Delta_f G^o_m(\text{OH}^-, 298.15\text{ K}) = -157.220$ kJ·mol^{-1} and $\Delta_f G^o_m(\text{Al}^{3+}, 298.15\text{ K}) = -481.16$ kJ·mol^{-1}.

The average value of $\log_{10}K^{\circ}_{sp}$ was given as $-(34.03 \pm 0.05)$ and $\Delta_f G^{\circ}_m(Al(OH)_3, gibb, 298.15\ K) = -(1147.3 \pm 0.06)$ kJ·mol^{-1} (1σ). After conversion to the molal scale and application of the SIT (the activity of water was set to unity and $\Delta\varepsilon = 0$ kg·mol^{-1}, because $I_m < 0.0055$; the pH was given on the activity scale) $\log_{10}{}^{*}K^{\circ}_{s,0} = (7.95 \pm 0.12)$ (2σ) which corresponds to an average $\log_{10}K^{\circ}_{sp}$ of $-(34.05 \pm 0.12)$. Bloom and Weaver [1982BLO/WEA] subsequently subjected a sample of gibbsite used by Kittrick to an SEM analysis and observed a coating of finely grained, globular material which suggested the presence of poorly crystalline $Al(OH)_3$. Nevertheless, the value of $\log_{10}K^{\circ}_{s,0} = (7.95 \pm 0.12)$ is consistent with reliable results obtained by other workers for crystalline gibbsite.

[1966MAR/SLU]

The solubility of purified gypsum, $CaSO_4{\cdot}2H_2O$(s), was measured at 0.5, 5, 10, 15, 20, 25, 30, 40, 50, 70, 80, 95 and 110 °C in NaCl solutions ranging from 0 to 6.4 mol·kg^{-1} although some solutions were supersaturated. As in a later study from the same laboratory [1969YEA/MAR] in mixed $NaNO_3$-Na_2SO_4 solutions, a batch technique was employed with equilibration times of 16 to 96 hours required to achieve equilibrium. At 70 to 95 °C, another solid phase was believed to have formed in NaCl saturated solutions, but the two principal equilibria controlling the solubility were still considered to be:

$$CaSO_4{\cdot}2H_2O(s) \rightleftharpoons Ca^{2+} + SO_4^{2-} + 2H_2O(l) \qquad (A.109)$$

$$Ca^{2+} + SO_4^{2-} \rightleftharpoons CaSO_4(aq) \qquad (A.110)$$

Marshall and Slusher incorporated solubility data from other sources and concluded that they could be treated effectively without allowing for hydrolysis. Their results for Reaction (A.109) were limited to $I_m = 2.24$ in our SIT treatment yielding $\log_{10}K^{\circ}_{sp} = -(4.369 \pm 0.013)$ and $\Delta\varepsilon = (0.042 \pm 0.010)$ kg·mol^{-1}, *cf.* $\log_{10}K^{\circ}_{sp} = -4.23$ suggested by the authors from an analysis of their entire dataset and selected literature values. Application of the SIT to data in the subsequent paper [1969YEA/MAR] yielded $\log_{10}K^{\circ}_{sp} = -(4.325 \pm 0.004)$ and $\Delta\varepsilon = (0.040 \pm 0.004)$ kg·mol^{-1}, $I_m < 2.15$ at 25 °C, *cf.* $\log_{10}K^{\circ}_{sp} = -4.52$ suggested by the authors. In this regard, taking $\varepsilon(Ca^{2+}, Cl^-) = (0.14 \pm 0.01)$ kg·mol^{-1} and $\varepsilon(Na^+, SO_4^{2-}) = -(0.12 \pm 0.06)$ kg·mol^{-1} would yield $\Delta\varepsilon = (0.02 \pm 0.06)$ kg·mol^{-1} at 25 °C. By restricting the dataset to $I_m < 1$ and eliminating the second ionic strength extended term in their activity coefficient model, values of $\log_{10}K^{\circ}_1$ (A.110) were determined: (1.8 ± 0.4), (1.9 ± 0.4), (2.5 ± 0.4) and (3.5 ± 0.6) at 0, 25, 100 and 200 °C, respectively. The first two values are considerably larger by *ca.* (0.5 $\log_{10}$) than those from most other studies. It is worth noting that the data of Marshall and Slusher have been thoroughly discussed and re-evaluated using the Pitzer approach in [1984MEI/ROS].

[1966RES]

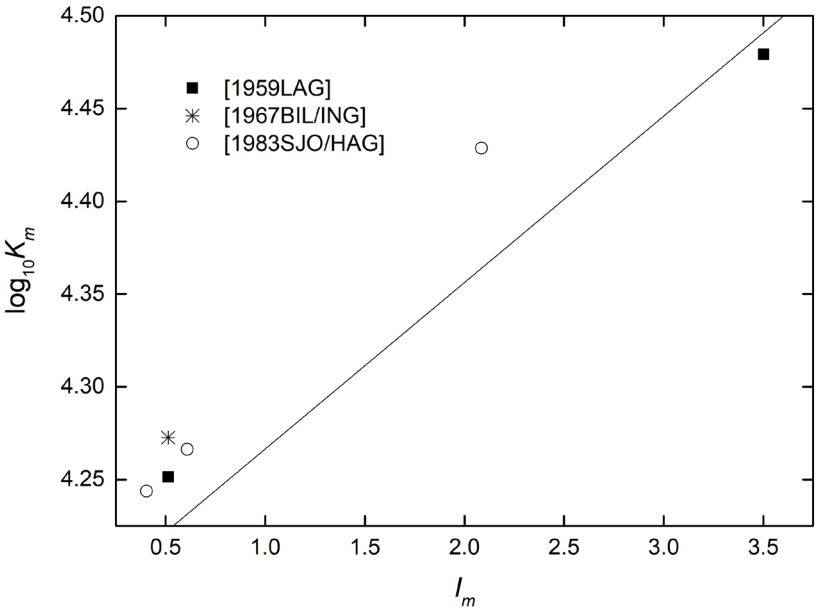
Reshetnikov derived a simple equation of the form $aT^4/(A + BT + T^2)^2$ to fit the 14 randomly selected values of $C_p(Al_2O_3, cr)$ from 40 to 370 K from the data of Furukawa *et al.* [1956FUR/DOU] and compares the derived values with other experimental data up to 2000 K, where, somewhat surprisingly for a three-coefficient equation, the agreement is very good. The author later [1969RES] used the integrated form of this equation to compare with the published enthalpy data for alumina, where again the agreement with the experimental data is excellent.

[1967BIL/ING]

The potentiometric cell and experimental procedure were described elsewhere. Baes and Mesmer [1976BAE/MES] selected a value for the formation constant of $SiO(OH)_3^-$ in 0.5 mol·dm^{-3} $NaClO_4$ at 25 °C, *viz*., $\log_{10}{}^*K_c$ (A.111) = – (9.46 ± 0.02) (3σ), from this study (*i.e.*, $\log_{10}{}^*K_m$ = – (9.47 ± 0.02)) or $\log_{10}K_m$ = 4.28. Lagerström [1959LAG] obtained a similar value and both are retained by this review. The results of SIT treatments of the three values available for this equilibrium in $NaClO_4$ solutions are shown in Figure A-12 together with the corresponding data determined for NaCl solutions, from which it is concluded that there is essentially no difference between the two datasets so that the results for the much larger dataset obtained in NaCl are preferred.

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \qquad \text{(A.111)}$$

Figure A-12: An SIT plot of $\log_{10}K_m$ (A.111) *versus* I_m at 298.15 K [1967BIL/ING] and [1959LAG] in $NaClO_4$ solutions, and [1983SJO/HAG] in NaCl solutions with the line representing the fit of all the values in NaCl solutions.



[1967HOL/KLE]

This paper reports heat capacity measurements on stishovite and coesite in the temperature range 5 to 350 K. The enthalpy of coesite to 1070 K and the transformation for stishovite-silica glass were obtained by a novel drop-calorimetry technique. High-temperature (970 K) melt solution calorimetry was used to determine the enthalpies of transition, quartz to coesite, quartz to cristobalite, quartz to silica glass and cristobalite to silica glass.

Although the present review does not deal with stishovite and coesite, we shall still include here, mainly for comparison purposes, a brief description of the results obtained by [1967HOL/KLE] for these species.

The origin and characteristics of the various materials were rather well described by the authors. All the crystalline samples were identified by X-ray powder diffraction and showed no lines other than those of the crystal in question.

Two samples of quartz were used: natural Brazilian quartz (I) (200-325 mesh) that contained less than 0.003 % B and less than 0.001 % of all other detected impurities (Al, Ca, Ba, Cu) and very finely powdered synthetic quartz (II), with particles ranging

in size from about 1 to 10 μm, prepared by heating silicic acid (Mallinckrodt AR) for 2.5 hours at 973 K and 15 kbar.

Cristobalite was prepared from quartz (I) by heating the sample for 24 hours at (1743 ± 20) K. Three samples of coesite were used: a natural sample isolated from the sandstone of Meteor Crater (AZ, USA) as a very fine powder unsuitable for solution calorimetry, but that was used for the high-temperature heat content measurements; a synthetic sample (I) prepared at (1573-1673) K and 40 kbar from purified crystalline quartz powder, and a second synthetic sample (II) prepared by heating silicic acid (Mallinckrodt AR) for two hours at 973 K and 40 kbar then freed from a graphite impurity by heating the sample in air at 973 K for some hours.

The stishovite was isolated from the sandstone of Meteor Crater (AZ, USA). It contained traces (0.001 to 0.1 %) of Al, Fe and alkaline earths and a residue of 0.6 % TiO_2.

A stishovite glass sample was obtained by heating the stishovite powder in the calorimeter for one hour at 1073 K. Silica glass was prepared by pulverising clean, thin-walled fused silica tubing in an agate mortar.

Heat capacity measurements (5-350 K) of natural stishovite and coesite were made in a previously described cryostat [1962WES]. Two specially built gold-plated copper calorimeters were used: one of 102 mL (mass 60.0 g) for coesite (102.131 g sample) and one of 22.7 mL (mass 13.1 g) for stishovite (27.923 g sample). Adiabatic shield control, circuitry and mode of measurements specific to these experiments were also described.

The reported heat capacities (C_p), entropies ($S°$), and Gibbs energy functions [$-(G° - H°)/T$] at 298.15 K are 45.40, 40.38, 17.25 $J \cdot K^{-1} \cdot mol^{-1}$ at 298.15 K for coesite (about 75 data points) and 42.97, 27.78, 9.883 $J \cdot K^{-1} \cdot mol^{-1}$ for stishovite (about 60 data points). Probable uncertainties in the determinations were reported as 5 % at 5 K, 1 % at 10 K and less than 0.2% above 20 K.

For solution calorimetry, the calorimeter used and the experimental procedures adopted were similar to those described in [1964YOK/KLE]. All experiments were performed at (970 ± 2) K. Calibration was by the gold-drop method, based on the enthalpy equation for pure gold from [1960KEL]. The solvent, prepared from reagent grade lead(II) oxide, cadmium(II) oxide and boric acid, had the approximate composition $9PbO \cdot 3CdO \cdot 4B_2O_3$. In all experiments (repeated 3-5 times), samples of about (0.7-1.0 mmol) were dissolved in 40 g of melt (about 250 mmol of oxides). The following mean values for the enthalpies of solution were obtained:

Table A-33: Enthalpies of solution into a $9PbO{\cdot}3CdO{\cdot}4B_2O_3$ melt at (970 ± 2) K.

Sample	$\Delta_{sol}H_m$ /kJ·mol^{-1}	Sample	$\Delta_{sol}H_m$ /kJ·mol^{-1}
Quartz (I)	– (12.76 ± 0.42)	Coesite (I)	– (15.69 ± 0.42)
Quartz (II)	– (11.72 ± 0.42)	Coesite (II)	– (14.77 ± 0.42)
Cristobalite (I)	– (14.64 ± 0.42)	Stishovite glass	– (18.62 ± 0.63)
		Silica glass	– (18.83 ± 0.42)

The authors note that their stated limits of errors are larger than the random errors due to a potential zero drift in the equipment.

The authors also note, in passing, that Holm and Kleppa [1966HOL/KLE] report a value of 15.23 kJ·mol^{-1} for the dissolution of quartz in the same ternary melt and ascribe the difference from the present results to small differences of composition of the melts.

They potentially ascribe the small systematic (and nearly identical) differences in the results above between the values for quartz(I) and (II) and for coesite (I) and (II), respectively, to differences in particle size. In fact, for the evaluation of the enthalpies of transformation of quartz to the other forms, they appear to take only quartz (I) in consideration.

$$SiO_2(\beta\text{-quartz, 970 K}) \rightleftharpoons SiO_2(\text{coesite, 970 K}) \qquad \text{(A.112)}$$

$$\Delta_r H_m \text{ ((A.112), 970 K)} = (2.9 \pm 0.6) \text{ kJ·mol}^{-1}.$$

Using their own data on the enthalpy of coesite (see below) the authors derive:

$$SiO_2(\alpha\text{-quartz, 298.15 K}) \rightleftharpoons SiO_2(\text{coesite, 298.15 K}) \qquad \text{(A.113)}$$

$$\Delta_r H_m \text{ ((A.113), 298.15 K)} = (5.1 \pm 0.6) \text{ kJ·mol}^{-1}.$$

As their values for stishovite glass and silica glass do not differ significantly, the authors conclude that the two materials are thermodynamically indistinguishable.

They deduce for Reaction (A.114):

$$SiO_2(\beta\text{-quartz, 970 K}) \rightleftharpoons SiO_2(\text{stishovite glass, 970 K}) \qquad \text{(A.114)}$$

the value $\Delta_r H_m$ ((A.114), 970 K) = (5.9 ± 0.6) kJ·mol^{-1}.

Using Reaction (A.115):

$$SiO_2(\alpha\text{-quartz, 298.15 K}) \rightleftharpoons SiO_2(\text{silica glass, 298.15 K}) \qquad \text{(A.115)}$$

$$\Delta_r H_m \text{ ((A.115), 298.15 K)} = (9.0 \pm 0.6) \text{ kJ·mol}^{-1}.$$

For Reaction (A.116):

$$SiO_2(\text{quartz, 970 K}) \rightleftharpoons SiO_2(\text{cristobalite, 970 K}) \qquad \text{(A.116)}$$

the authors take, from their results $\Delta_r H_m$ ((A.116), 970 K) = (1.9 ± 0.6) kJ·mol^{-1} and, on the basis of the enthalpy data from [1960KEL], for Reaction (A.117):

$$SiO_2(\alpha\text{-quartz, } 298.15\text{ K}) \rightleftharpoons SiO_2(\text{cristobalite, } 298.15\text{ K}) \qquad (A.117)$$

$$\Delta_r H_m \text{ ((A.117), } 298.15\text{ K}) = (2.7 \pm 0.6)\text{ kJ·mol}^{-1}.$$

From their data and literature values [1967HOL/KLE] also calculated the enthalpy of melting of cristobalite at 1996 K, to be (10.5 ± 0.8) kJ·mol^{-1}.

Using a calorimetry technique they call "transposed drop calorimetry", Holm *et al.* [1967HOL/KLE] measured the heat content of coesite to 1072 K and the enthalpy effect corresponding to the transformation of stishovite to the glassy phase. In this technique, the material of interest is dropped from a known temperature near 300 K into a high-temperature calorimetric receiver.

In the case of coesite, the heat effect corresponding to the dropping of a 250 mg sample sealed in a platinum capsule, dried to constant weight at 973 K, was compared to the heat effect of an analogous capsule containing the same weight of α-quartz (I). Measurements were made at five calorimeter temperatures, but since the values obtained for quartz differed from those in the JANAF [1965STU/CHA] tables by 0.8, 0.5 and 0.1 % at the three lowest temperatures, the results for coesite were also referred (under the heading "Adjusted") to the JANAF selected values for α-quartz, as shown below in Table A-34 in which the quoted uncertainties are average deviations of the mean.

Table A-34: Enthalpy differences for SiO_2(coesite).

T/K	Number of expts.	$H_m^\circ(T) - H_m^\circ(298\text{ K})$ /kJ·mol^{-1}	
		As found	Adjusted
298.15	0	0	0
577	3	15.02 ± 0.08	15.23
731	3	24.94 ± 0.02	25.06
972	4	41.59 ± 0.12	41.63
1072	2	48.74 ± 0.02	48.74

For the transformation of the stishovite to stishovite glass, a 150 mg sample was sealed in platinum as in the case of coesite. As it was not possible to remove thermally from the stishovite sample the 2.2 % water it contained without initiating the transformation of the stishovite to a glassy phase, measurements were corrected for the vaporisation of water at 1072 K. Corrections were also made for the thermal effect of bringing the platinum capsule and the stishovite glass from 298 to 1072 K.

For Reaction (A.118):

$$SiO_2(\text{stishovite, } 298.15\text{ K}) \rightleftharpoons SiO_2(\text{stishovite glass, } 298.15\text{ K}) \qquad (A.118)$$

[1967HOL/KLE] thus report:

$$\Delta_r H_m \text{ ((A.118), 298.15 K)} = (40.5 \pm 0.5) \text{ kJ·mol}^{-1},$$

and, finally, for Reaction (A.119):

$$SiO_2\,(\alpha\text{-quartz, 298.15 K}) \rightleftharpoons SiO_2\,(\text{stishovite, 298.15 K}) \qquad \text{(A.119)}$$

$$\Delta_r H_m \text{ ((A.119), 298.15 K)} = (49.5 \pm 1.3) \text{ kJ·mol}^{-1}.$$

The authors also discuss their various data in comparison with related literature results and calculate pressure-temperature diagrams for equilibria among the silica polymorphs.

[1967KIN/BAR]

This paper concerns the determination of the enthalpy of formation of forsterite (Mg_2SiO_4) and chrysotile ($Mg_3Si_2O_5(OH)_4$) at 298.15 K, using solution calorimetry. The heat capacities of chrysotile and the compositionally and structurally related antigorite were also measured between 53 and 297 K, and that of antigorite from 405.5 to 847.9 K. The results concerning chrysotile are the principal interest for this review.

Natural samples were used. The reported analytical results correspond to compositions significantly different from stoichiometry, as shown in Table A-35.

Table A-35: Composition of the chrysotile samples, in weight percent.

Constituent	Stoichiometry	Sample 1	Sample 2
SiO_2	43.36	41.16	43.51
Al_2O_3		0.70	0.23
Fe_2O_3		1.31	0.40
FeO		0.95	0.06
MgO	43.64	40.86	41.75
CaO		0.13	0.25
Na_2O		0.04	0.04
H_2O	13.00	13.97	13.55
TiO_2		< 0.01	0
CO_2		–	0.24
MnO		0.07	0.02
Cr_2O_3		0.23	0.02
NiO		0.22	0.02
Total	100.00	99.65	100.09

Sample 1 was used in low-temperature heat capacity measurements and sample 2 in solution calorimetry. The magnesium oxide used in solution calorimetry was 99.93 % pure macrocrystalline material with 0.04 wt % Al_2O_3 and 0.01 wt % SiO_2 as major impurities. The calorimetric equipment was described in earlier publications by the group. The solution medium (sln) was 948.7 g of a solution composed of

10.05 wt % hydrofluoric acid and 10.05 wt % hydrochloric acid. The quantities of the reacting substances were stoichiometric with 0.742 g of silica. The handling of the substances was carefully described. The measured calorimetric reactions corresponded to the conversion of the substances at 298.15 K to reaction products at 346.85 K. It can be seen from the reaction scheme that combination of the measurements results in the evaluation of an enthalpy of reaction at 298.15 K involving compounds in their standard state. 1963 atomic weights were used, but the differences with 1981 atomic weights are negligible.

The thermodynamic cycle used by the authors involves Reactions (A.120)-(A.124) below:

$$2SiO_2(\alpha\text{-quar, } 298.15\text{ K}) + 12HF(\text{sln}, 346.85\text{ K}) \rightleftharpoons 2H_2SiF_6(\text{sln, } 346.85\text{ K}) + 4H_2O(\text{sln, } 346.85\text{ K}) \quad \text{(A.120)}$$

$$3MgO(\text{cr, } 298.15\text{ K}) + 6HF(\text{sln, } 346.85\text{ K}) \rightleftharpoons 3MgF_2(\text{s, } 346.85\text{ K}) + 3H_2O(\text{sln, } 346.85\text{ K}) \quad \text{(A.121)}$$

$$2H_2O(\text{l, } 298.15\text{ K}) \rightleftharpoons 2H_2O(\text{sln, } 346.85\text{ K}) \quad \text{(A.122)}$$

$$3MgF_2(\text{s, } 346.85\text{ K}) + 9H_2O(\text{sln, } 346.85\text{K}) + 2H_2SiF_6(\text{sln, } 346.85\text{ K}) \rightleftharpoons Mg_3Si_2O_5(OH)_4(\text{cr, } 298.15\text{K}) + 18HF(\text{sln, } 346.85\text{ K}) \quad \text{(A.123)}$$

The sum of these four reactions gives:

$$3MgO(\text{cr, } 298.15\text{ K}) + 2SiO_2(\alpha\text{-quar, } 298.15\text{ K}) + 2H_2O(\text{l}, 298.15\text{ K}) \rightleftharpoons Mg_3Si_2O_5(OH)_4(\text{cr, } 298.15\text{K}) \quad \text{(A.124)}$$

All the experimental values are converted from the originally reported values in kcal.

For $\Delta_r H_m$ (A.120) the authors used the value of – (287.441 ± 0.586) kJ·mol^{-1} determined by Barany [1963BAR] who used the same solution medium and experimental conditions in the determination of the enthalpy of formation of talc. As for the other reactions of the scheme, the uncertainties are reported for the 2σ confidence interval.

Based on seven determinations, King *et al.* reported $\Delta_r H_m$ (A.121) = – (444.927 ± 0.628) kJ·mol^{-1} and, based on six measurements, $\Delta_r H_m$ (A.122) = (5.648 ± 0.167) kJ·mol^{-1}. Reaction (A.122) was carried in the final solution resulting from the consecutive Reactions (A.120) and (A.121).

The dissolution of chrysotile (sample 2, (A.123)), was conducted in a fresh batch of acid and the average value of six determinations was $\Delta_r H_m^\circ$ ((A.123), 298.15 K) = (559.945 ± 1.004) kJ·mol^{-1}, with corrections by the authors for impurities in the chrysotile sample.

The enthalpy of formation of chrysotile from its constituent oxides according to Reaction (A.124) is then immediately given as the sum of Reactions (A.120) to (A.123).

$$\Delta_r H^o_m \text{ ((A.124), 298.15 K)} = -(166.774 \pm 2.510) \text{ kJ·mol}^{-1}.$$

The authors indicate that their reported uncertainty, which we accept, contains allowance for uncertainties in the solution measurements, the calibrations and the corrections for impurities.

Using the CODATA values [1989COX/WAG], this review calculates from Reaction (A.124) from this extremely careful work:

$$\Delta_f H^o_m (Mg_3Si_2O_5(OH)_4, \text{ cr, 298.15 K}) = -(4364.0 \pm 3.400) \text{ kJ·mol}^{-1}.$$

The heat capacities of both chrysotile and its structurally-related polymorph antigorite were measured from *ca.* 53 to 296 K (*ca.* 30 measurements for each), with very similar results. The sample 1 of chrysotile used was taken to contain 94 % chrysotile and 6 % impurities, including 1.7 % excess water (ice), while the antigorite sample was taken to be 91.8 % antigorite and 8.2 % impurities; in both cases corrections for the impurities were made. The heat capacities were extrapolated to 0 K using the sum of one Debye and three Einstein functions. The standard entropies at 298.15 K were calculated to be S^o_m (chrysotile, cr, 298.15 K) = (221.3 ± 1.5) J·K^{-1}·mol^{-1} and S^o_m (antigorite, cr, 298.15 K) = (222.6 ± 1.5) J·K^{-1}·mol^{-1}, where the uncertainties have been increased from those given by the authors (± 0.8 J·K^{-1}·mol^{-1}) because of the rather impure materials used. The values of $C^o_{p,m}$ (298.15 K) were not given, but extrapolation of the data just below 298.15 K indicates values of (273.7 ± 1.5) J·K^{-1}·mol^{-1} for chrysotile and (273.9 ± 1.5) J·K^{-1}·mol^{-1} for antigorite.

King *et al.* did not measure the heat capacity of chrysotile above 298.15 K because their samples were not dense enough, but did measure that of the structurally related polymorph antigorite by drop calorimetry from 405.5 to 847.9 K, and fitted their data by an equation corresponding to:

$$C^o_{p,m}/\text{J·K}^{-1}\text{·mol}^{-1} = 317.23 + 0.1323\,(T/\text{K}) - 7.3555 \times 10^6\,(T/\text{K})^{-2}$$

and since the heat capacities of chrysotile and antigorite below 298.15 K differ by less than 1.2 %, this expression, after changing the first coefficient to 317.00 to agree with $C^o_{p,m}$ (chrystotile, 298.15 K) = 273.7 J·K^{-1}·mol^{-1}, can be used for chrysotile without undue error.

[1967NAK/RAS]

This was perhaps the first application of a calcium membrane electrode to investigate the solubility of gypsum and the ion pairing of its component ions in solution at (25 ± 1) °C. An Hg|$HgCl_2$ reference electrode was used. The cell was calibrated with

$CaCl_2$-NaCl solutions which simulated approximately the test solutions of $CaSO_4$, $CaSO_4$-NaCl and $CaSO_4$-Na_2SO_4 thereby minimising the liquid-junction potential which was not quantified. Nakayama and Rasnick incorporated an ion-pair formation constant for $NaSO_4^-$ of 0.19 $dm^3 \cdot mol^{-1}$ into their data treatment, but acknowledged that the activity coefficients could not be adequately accounted for in their electrolyte mixture and that an ionic strength dependence was noted for both the $CaSO_4$(aq) ion pair and $CaSO_4 \cdot 2H_2O$(cr) solubility constants, which they designated as "apparent" constants, K_1°' and K_s°', respectively. The extrapolated values were $K_1^\circ = (188.0 \pm 13.1)$ and $K_s^\circ = (2.50 \pm 0.10) \times 10^{-5}$. Re-calculation with the SIT (the ion interaction contribution was ignored—see Appendix A entry for [1953BEL/GEO]) involved assuming that the "missing" sulfate in solution existed as $CaSO_4$(aq) thereby ignoring the $NaSO_4^-$ ion pair. This led to convergence with I_c varying from 0.0451 to 0.1786 (from the densities in [1985SOH/NOV], $I_m/I_c = (1.0046 \pm 0.0012)$ which is a negligible correction). The apparent values of K_1° and K_s° are ionic-strength dependent. Regarding the former, the linear dependence was generated from the regression, $(\log_{10}K_1 + 4D) = (2.2737 \pm 0.0534) - (2.420 \pm 0.497)\, I_m$. Assuming that this ionic-strength dependence results from the inadequacy of the activity coefficient model as suggested by the authors, $K_1^\circ = (188 \pm 45)$ (2σ), noting the fortuitous agreement with the value obtained by the authors, *i.e.*, (190 ± 20).

Application of the same approach to the solubility data led to the equation $(\log_{10}K_s - 8D) = -(4.5848 \pm 0.0076) - (0.7040 \pm 0.0710)\, I_m$, resulting in $K_s^\circ = (2.60 \pm 0.09) \times 10^{-5}$ (2σ uncertainty). This review proposes a value of $K_s^\circ = (2.6 \pm 0.5) \times 10^{-5}$ as a best estimate in good agreement with the value of $(2.50 \pm 0.10) \times 10^{-5}$ reported by Nakayama and Rasnick. This is another study of the interactions in calcium sulfate solutions where the ionic strength was varied with the addition of inert electrolyte without any attempt to maintain a clear excess. The internal consistency of these data when the SIT Debye-Hückel equation was applied with the empirical correction prompted this review to include these extrapolated values in the overall data treatment.

[1967TAR]

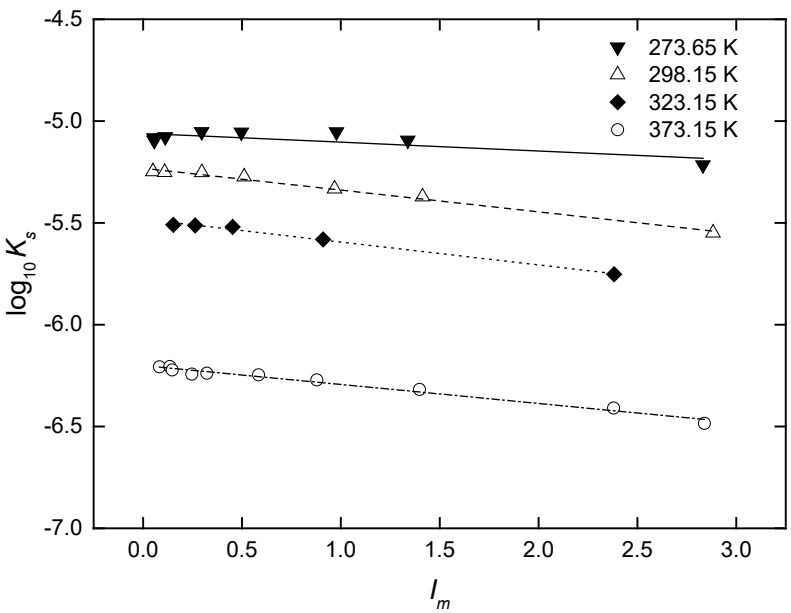
This paper is the only published source of the measurements of the low-temperature heat capacity of vitreous silica measured by Turdakin at the Chemico-Technological Institute in Moscow. No experimental details are given, but the smoothed values of the heat capacity are tabulated at round temperatures from 55 to 300 K. They agree to within 1 % with those measured by [1956WES]—see Section V.3.1.5 on vitreous silica. No values are given for 298.15 K, but the tabulated value of C_p at 300 K is 43.97 $J \cdot K^{-1} \cdot mol^{-1}$.

[1967YEA/MAR]

Extensive pre-treatment of commercial $Ca(OH)_2(s)$ was carried out to remove carbonate and soluble impurities, and to re-crystallise the material. The temperature range of these $Ca(OH)_2(cr)$ solubility measurements was 0.5 to 350 °C, although only data in the range 0.5 to 100 °C (four datasets) were examined in this review in accordance with our imposed limit of 200 °C. Ionic strength was generally controlled with $NaNO_3$ although two experiments were also conducted in pure water. Ample equilibration time of the rocked solutions was allowed. Yeatts and Marshall used a two-term extended Debye-Hückel expression to estimate activity coefficients. The SIT was applied to Reaction (A.125) by this review to the selected datasets which were truncated at $I_m < 3$ and 100 °C. The results are shown in Figure A-13.

$$Ca(OH)_2(s) \rightleftharpoons Ca^{2+} + 2OH^- \qquad \text{(A.125)}$$

Figure A-13: SIT plot for the data in [1967YEA/MAR] pertaining to Reaction (A.125).



The value of $\log_{10} K_s^\circ$ ((A.125), 298.15 K) = – (5.23 ± 0.01) is in excellent agreement with that obtained from [1931JOH/GRO] for a variety of ionic media including $NaNO_3$, *viz.*, – (5.24 ± 0.04), but is noticeably different from the smoothed value of – 5.03 given by the authors. The value of $\Delta\varepsilon$ = (0.107 ± 0.008) kg·mol^{-1} at 25 °C is also in agreement with that calculated from [1931JOH/GRO], (0.113 ± 0.008) kg·mol^{-1}, and with the value below obtained from the NEA TDB recommended ion

interaction coefficients. Yeatts and Marshall were unable to correct their results for $CaOH^+$ formation while acknowledging that hydrolysis would be expected to have an increasing influence at higher temperatures. From the NEA TDB recommended values for the NaCl solution:

$$\Delta\varepsilon = \varepsilon(Ca^{2+}, NO_3^-) + 2\,\varepsilon(Na^+, OH^-) =$$
$$= (0.02 \pm 0.01) + 2\,(0.04 \pm 0.01) = (0.10 \pm 0.02)\ \text{kg·mol}^{-1}.$$

A weighted fit of the $\log_{10} K_s^\circ$ values resulted in $\Delta_r H_m^\circ$((A.125), 298.15 K) = – (26.2 ± 0.1) kJ·mol^{-1}, *cf.* the authors reported $\Delta_r H_m^\circ$((A.125), 298.15 K) = – 12.7 kJ·mol^{-1} based on an empirical fit of all their data. The NEA TDB recommended enthalpies of formation of the ions and our recommended value of $\Delta_f H_m^\circ(Ca(OH)_2, s, 298.15\ K) = -(985.16 \pm 1.00)$ kJ·mol^{-1} led to $\Delta_r H_m^\circ$((A.125), 298.15 K) = – (17.87 ± 1.42) kJ·mol^{-1}.

[1968HOS/CHR]

The solubilities of two samples of chrysotile, one synthetic and one natural from New Idria, California, in distilled water were measured at (90.0 ± 0.7) °C and 1.01 bar. The synthetic material was made from colloidal (SiO_2 + $Mg(NO_3)_2$)(sln) heated to 110 °C and then 800 °C to drive off NO_2 and reheated with H_2O at 350 °C for 14 days. Electron microscopic analysis showed small quantities of impurities, probably $MgCO_3$ and/or SiO_2. Chemical analysis of the solid was reported only for the natural material (apparently no carbonate analysis was made, see below), although the synthetic solid was believed to contain amorphous silica.

The congruent dissolution of chrysotile was represented by the equation:

$$Mg_3Si_2O_5(OH)_4(s) + 5H_2O(l) \rightleftharpoons 3Mg^{2+} + 6OH^- + 2Si(OH)_4(aq) \quad \text{(their Equation 1)}.$$

Uncertainties in analyses of the solution were quite large: Mg^{2+}, ± 4 %; SiO_2(sln), ± (10-50) %; and pH, ± 0.04. The latter was measured at 363 K with a commercial combination glass electrode (the calibration procedure was not given), but for such dilute solutions (I_c estimated to be > 5 × 10^{-4}) with the acknowledged carbonate contamination, the measurements must be considered to be problematic. Furthermore, the Mg/Si ratios in solution, even in the final solutions, were 3.45 and 3.68, much larger than the theoretical 1.5. The final solution was analysed for Ca, Fe, Cl, F, NO_3, CO_3 and HCO_3 and the dominant species balancing the excess Mg^{2+} was HCO_3^- attributed to leakage due to warping of the (polythene) lids of the reaction vessel. Hydrolysis of Mg^{2+} and complexation with HCO_3^- and CO_3^{2-} were considered to be negligible.

Hostetler and Christ used a value for pK_w° of 12.42 at 90 °C (*cf.* 12.43 derived by this review), but they possibly mixed activities with molalities by writing: p_mOH = – 12.42 – pH. The activity coefficients of the total magnesium in solution were estimated using the Debye-Hückel equation and the derived value was (0.91 ± 0.01).

However, the authors chose to simply derive a molal equilibrium constant based on the total molalities of the three product species, as confirmed by the reviewer. The resulting $\log_{10} K_m((l), 363\ \mathrm{K})$ values were plotted against the reciprocal of the time over which samples were taken (1.5-1615 hours) to obtain a final value at infinite time of – (49.2 ± 0.5), which was considered to be in reasonable agreement with an average of the last three samples taken for both solids of – (48.5 ± 0.4).

We note, though, that in view of the high Mg/Si ratios in the solution in these experiments, Bricker *et al.* [1973BRI/NES] suggested that chrysotile actually dissolved incongruently to form talc ($Mg_3Si_4O_{10}(OH)_2$(cr)) and that the solution composition was actually controlled by equilibrium with talc. Indeed, on this basis, they used their estimated $\Delta_f G_m$ to re-interpret the data in [1968HOS/CHR] to calculate $\Delta_f G_m$ ($Mg_3SiO_2O_5(OH)_4$, cr, 298.15 K) = – 4025 kJ·mol^{-1}, in tolerable agreement with our selected value of – (4036.371 ± 3.431) kJ·mol^{-1} in view of the uncertainties in the calculation in [1973BRI/NES].

In view of the assumptions made as to the composition and speciation of the final solutions, including to the presence of significant impurities (principally carbonate and amorphous silica) in these systems, the above value of the solubility of chrysotile, claimed to be equivalent to $\log_{10} K^\circ((l), 363\ \mathrm{K}) = -(49.2 \pm 0.5)$ must be taken with considerable caution and has not been considered further in this review.

[1968JOH/MET]

This paper deals predominantly with equilibria in the ternary MgO-CO_2-H_2O system, but includes two measurements of the phases found at 500 and 1000 bar at temperatures above and below the equilibrium temperatures for these pressures.

The experiments were carried out in pressure bombs; there are no details of their composition, but from the reference given this was probably a Co-Cr-W-C alloy. The temperature inside the bombs was obtained by comparing (unspecified) thermocouple readings inside and outside the bomb, although the authors estimated uncertainty of ± 5 K in the temperature inside the bomb may be too small. The uncertainty in the pressures, measured on a manometer, is given as ± 2 %.

The forward and backward reactions were studied, using *pro analysi* periclase and a natural magnesite sample from Kraubath Sommergraben (Austria), which contained 47.4 wt % MgO (theoretical 47.81), 0.27 wt % CaO and 0.14 wt % FeO. In all the experiments, samples of the periclase and magnesite, contained in perforated gold tubes, were placed side-by-side in the bomb chamber so that the experimental conditions were identical. At the end of each run, the bomb was quenched with water as rapidly as possible to minimise any back-reaction, and the phases present determined by XRD and optical examination.

The phases observed in nine runs for each pressure, from 943 to 993 K at 500 bar and from 1013 to 1053 K at 1000 bar are tabulated, some of which are plotted on a *T-P* diagram showing also the curve predicted from the thermodynamic data then available, and also including the data of [1955HAR/TUT], which are quite similar. From these results, the relevant equilibrium temperatures for the decomposition of magnesite are estimated by this review to be (980 ± 20) K and (1040 ± 15) K at pressures of 500 and 1000 bar respectively. These data have been processed using the thermal functions selected in this review and the corrections for the non-ideality of $CO_2(g)$ and the $p\Delta V$ correction, as described in Appendix A of [1955HAR/TUT]. The derived value of $\Delta_r H_m(298.15\ \text{K})$ for the decomposition of magnesite, Equation (A.126) is $-(116.97 \pm 0.23)$ kJ·mol^{-1}, where the statistical 2σ uncertainty:

$$MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g) \qquad (A.126)$$

does not include any experimental or data analysis uncertainties, which are much larger. The overall uncertainty is estimated to be at least 1.5 kJ·mol^{-1}, thus giving:

$$\Delta_f H^o_m(MgCO_3, \text{cr}, 298.15\ \text{K}) = -(1112.08 \pm 1.54)\ \text{kJ·mol}^{-1}.$$

[1968JOR]

Jørgensen conducted solubility experiments with amorphous silica in 1 mol·dm^{-3} $NaClO_4$ solutions at 25 °C. A solubility of (77.7 ± 1.1) mg(SiO_2)·dm^{-3} or $c_{Si} = (1.293 \pm 0.012) \times 10^{-3}$ (2σ), was measured from under- and supersaturated solutions. This is equivalent to $(1.360 \pm 0.012) \times 10^{-3}$ mol·kg^{-1} in 1.0515 mol·kg^{-1} $(NaClO_4)$. An XRD scan found no evidence of a crystalline phase after equilibration with agitation lasting 13 to 15 months.

[1968KES/PYT]

A cell with a magnesium-sensitive electrode *versus* a calomel reference electrode and a saturated KCl salt bridge was used to determine the association constant of $MgSO_4(aq)$. The test solution consisted of: $c_{Mg^{2+}(tot)} = 0.06567$; $c_{SO_4^{2-}(tot)} = 0.02928$; $c_{Na^+(tot)} = 0.4928$; $c_{Cl^-} = 0.5660$, at a stoichiometric molar ionic strength of 0.670. Titrations in the absence of sulfate allowed Kester and Pytkowicz to compensate for the liquid-junction potential and changes in activity coefficients with addition of titrant. They assumed the presence of $NaSO_4^-$. Parallel titrations in the absence of sulfate yielded an average $K_{c,1}$ value of (5.15 ± 0.5) dm^3·mol^{-1} or $K_{m,1} = (5.2 \pm 0.5)$ kg·mol^{-1} at $I_m = 0.680$ at 25.0 °C.

[1968SWI]

The main purpose of this paper was to measure the solubility of carbon in CaO-SiO_2-Al_2O_3 slags at temperatures from 1823 to 1998 K, by equilibrating these slags in a graphite crucible under a fixed pressure (0.1 bar at 1823 K, 1 bar at higher temperatures) for 2-6 hours. Above a temperature estimated to be *ca.* 1983 K, it was suggested that CaC_2(cr) was formed on the crucible walls, as evidenced by the formation of C_2H_2(g) on rinsing with dilute HCl. However, this does not prove the formation of CaC_2 at unit activity, rather than being dissolved in the slag.

Since this temperature is appreciably lower than that predicted from existing thermodynamic data, the author also reacted CaO(cr) with CO(g) in graphite crucibles for two hours in the absence of any slag, and indicated that CaC_2 was formed above 1923 K. If, as seems to be the case, the indicator for formation of CaC_2 was again the evolution of C_2H_2(g) on reaction with acid, this does not, of course exclude the formation of CaO-CaC_2 solid solutions—as noted elsewhere, these solids have the same NaCl structure at these temperatures.

The author, more cautiously than subsequent authors, suggested that his results indicated a possible error in the data for either CaO(cr) or CaC_2(cr). However, given the uncertainty in the experiments noted above and the appreciable uncertainties in the data for CaC_2(cr), we do not think these data cast any serious doubt on $\Delta_f H$(CaO, cr).

[1968WAT/LIN]

Waterfield *et al*. measured the low-temperature heat capacity of anhydrous Na_2CO_3(cr) from *ca.* 10 to 300 K, as well as the mono- and decahydrates. We shall only note the values for Na_2CO_3(cr) here. The sample used was prepared from analytical reagent material by melting in a platinum boat in a CO_2(g) atmosphere, followed by an anneal at 583 K for six hours to remove any high-temperature phases which might have been present.

The heat capacities were extrapolated to 0 K using a Debye function, but no detailed results of the heat capacity measurements are given, only smoothed values of the thermal functions from 10 to 300 K. The heat capacities agree with the earlier measurements of Anderson [1933AND] to within 1 %. The calculated standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}\,(Na_2CO_3,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = (112.30 \pm 0.50)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}\,(Na_2CO_3,\ \mathrm{cr},\ 298.15\ \mathrm{K}) = (135.00 \pm 0.40)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

where the uncertainties are estimated by this review.

Because this study covered a more extensive temperature range than that in [1933AND] these are the selected data. The same values for both entropies and the heat capacity of Na_2CO_3(cr) were selected in [1982GLU/GUR].

The integral enthalpy of solution of Na_2CO_3(cr) in water (1 mol Na_2CO_3 in 2400 mol H_2O) at 298.15 K, using a Dewar-type calorimeter was also reported to be $-(22.681 \pm 0.084)$ kJ·mol^{-1}, based on three concordant measurements that yielded – 22.656, 22.715 and 22.673 kJ·mol^{-1}.

[1969FUG/SWE]

The authors measured the low-temperature heat capacity of two samples of synthetic sapphire from 2 to 25 K (about 40 measurements) in an adiabatic calorimeter. One sample was relatively small (21 g), and one much larger (220 g). The sapphire was from the same manufacturer as that used in the numerous studies at the U.S. National Bureau of Standards, though from a different batch. The total impurities, as given by the manufacturer, were < 50 ppm, principally *ca.* 10 ppm of Si and Ca, and < 3 ppm each of Ag, Cu, Fe, Mg and Ni. In calculating the heat capacities, the contribution of the container was eliminated by subtracting the values for the two samples.

The individual results are only presented graphically as a plot of C_p/T^3 *vs.* T^2, but the coefficients of two fitted equations are also given. The results below 14 K were anomalous, showing a small (1.4 %) Schottky-like deviation from the expected curve. The source of this anomaly is not clear, given the high purity of the sample, and the indicated absence of ferromagnetic impurities, but may result from the presence of very small quantities of transition metal impurities such as vanadium.

[1969GME2]

Gmelin measured the low-temperature heat capacity of the four alkaline-earth oxides from *ca.* 1.2 to 300 K—the detailed measurements are not given, being presented only in figures and round values of T at 4 K to 300 or 320 K. The preparation and purity of the samples of MgO(cr) and CaO(cr) are given below—we shall not deal with SrO and BaO in this review.

Table A-36: Preparation of MgO(cr) and CaO(cr) crystals.

Material	Preparation	Impurities
MgO(cr)	Sintered powder, heated successively to 1300, 1500 and 1850 °C for 2 h	< 1000ppm, mainly 500 ppm SiO_2, Cr_2O_3 and Na_2O
CaO(cr)	Sintered powder	< 300 ppm

As shown in Figure V-41 in Section V.6.1.1.1.2, the results of this study for MgO(cr) agree closely with those of [1959BAR/BER], and were included in the CODATA assessment [1989COX/WAG] for this oxide. The author's values for CaO(cr) (this study and the earlier study over the temperature range 1 to 65 K [1966GME2] form the basis of the CODATA selection for the heat capacity of CaO(cr), which we have accepted.

[1969GRE/SNE]

The entropies of transfer between heavy and normal water for the dichlorides of magnesium, calcium, strontium and barium have been obtained from the combination of Gibbs energies of transfer using potential difference measurements and the enthalpies of transfer using calorimetric measurements and the evidence for the order of structure-breaking of the four concerned cations is discussed.

We shall, however, restrict our considerations to the enthalpies of solution in normal water of $MgCl_2$(cr) and $CaCl_2$(cr) that are useful for the evaluation of the enthalpies of formation of these salts. $SrCl_2$(cr) and $BaCl_2$(cr) were also studied, but are not relevant for the present review.

The anhydrous chlorides were prepared by the thermal decomposition of the double salt ammonium magnesium chloride hexahydrate under a flow of high-purity (99.99 %) HCl(g) at increasing temperatures up to 873 K. High-purity (99.9 %) N_2(g) replaced HCl(g) in the last step to fully drive off HCl. The chlorides obtained were cooled under N_2(g) and kept in a dry box. No analytical details were given for the individual chlorides. It was only reported that they contained less than 0.013 mol % NH_4Cl, as shown by micro-Kjeldahl analyses and that gravimetric analysis for chloride showed them to be 99.9 % pure. A commercial (LKB) non-isothermal, isoperibol calorimeter was used.

The final concentrations of the chlorides in the calorimetric solutions ranged between 0.008 and 0.02 $mol \cdot dm^{-3}$. The measured enthalpies of solution (five measurements for each chloride) were corrected to a common concentration of 0.015 $mol \cdot dm^{-3}$ by reference to the data tabulated in [1952ROS/WAG]. For this concentration, the values reported at 298.15 K were $-(154.93 \pm 0.17)$ and $-(80.46 \pm 0.17)$ $kJ \cdot mol^{-1}$ (2σ) for the chlorides of magnesium and calcium, respectively.

[1969HOR]

This multifaceted study included solubility measurements of synthetic brucite at 25 °C in 3 $mol \cdot dm^{-3}$ $NaClO_4$. Equilibration times of three to seven days were allowed depending on the acid concentration. The solid phase was examined by XRD before and after three of the solubility experiments to verify that no observable change in its

structure had occurred. A BET analysis yielded a surface area of (14.7 ± 0.5) $m^2 \cdot g^{-1}$ with the particle size estimated at 1 μm, but the presence of smaller particles could not be ruled out. Fortunately, brucite tends to undergo rapid Ostwald ripening even at ambient conditions. The pH_c was measured with a cell containing glass and Ag|AgCl electrodes with a salt bridge of matching ionic strength (2.99 $mol \cdot dm^{-3}$ $NaClO_4$, 0.01 $mol \cdot dm^{-3}$ NaCl). The cell was standardised against stoichiometric HCl + $NaClO_4$ solutions. The results from Tables 3 and 4 in the paper were treated by this review with the SIT after conversion to molal units [1985SOH/NOV]. The mean value of $\log_{10}{}^{*}K_{s,0}$ was calculated to be (16.51 ± 0.04) (2σ), *cf.* a reported molar value of (16.58 ± 0.006), presumably at the 1σ level of uncertainty at I_m = 3.48 ($NaClO_4$). The measured total magnesium concentration and calculated free Mg^{2+} concentration led to a value $\log_{10}{}^{*}K_1 = -(9.64 \pm 0.20)$ for the first hydrolysis equilibrium, *cf.* a reported value of – (9.56 ± 0.07), presumably at the 1σ level of uncertainty. This value was derived from the dependence of $c_{\mathrm{Mg\,total}}$ on pH_c, but is about two orders of magnitude larger than values obtained from numerous other studies. Note that if corrections are made in accordance with the recommended hydrolysis constant values, then an unreasonable value of $\log_{10}{}^{*}K_{s,0}$ is calculated: (17.01 ± 0.62). Despite the internal consistency of these results, they were rejected. Baes and Mesmer [1976BAE/MES] also questioned the pH dependence of the solubility data because it led to an unreasonably high value of $\log_{10}K_1 = -9.74$.

[1969IZA/EAT]

The calorimetric results for the formation of $NiSO_4$(aq) were discussed in [2005GAM/BUG] in which Gamsjäger *et al.* raised concerns about the value of $\Delta_r H_m^o$ and did not attempt to re-calculate these quantities. Izatt *et al*. also reported the following quantities (converted to joules) for the formation of $MgSO_4$(aq), K_1^o = (169 ± 29) $mol \cdot dm^{-3}$, $\Delta_r H_m^o$ = (2.1 ± 0.1) $kJ \cdot mol^{-1}$ and $\Delta_r S_m^o$ = (50 ± 1) $J \cdot K^{-1} \cdot mol^{-1}$; and $CaSO_4$(aq), K_1^o = (269 ± 19) $mol \cdot dm^{-3}$, $\Delta_r H_m^o$ = (3.39 ± 0.04) $kJ \cdot mol^{-1}$ and $\Delta_r S_m^o$ = (57.3 ± 0.04) $J \cdot K^{-1} \cdot mol^{-1}$ (1σ) at 25 °C.

[1969NAZ/NEV]

Nazarenko and Nevskaya used a novel spectrophotometric method to determine the first three hydrolysis constants of Al^{3+} in 0.1 $mol \cdot dm^{-3}$ $NaClO_4$ at 25 °C. They employed arsenazo and pyrocatechol violet as competing ligands while maintaining low aluminium concentrations (*ca.* 2×10^{-5} $mol \cdot dm^{-3}$). They reported $\log_{10}{}^{*}K_{c,n}$ values at I_c = 0.1 of – 4.89 (n = 1), – 5.43 (n = 2) and – 5.86 (n = 3), which are equivalent to $\log_{10}{}^{*}K_{m,n}$ = – 4.90 (n = 1), – 5.44 (n = 2) and – 5.86 (n = 3), respectively, in I_m = 0.1008. No experimental data were provided, but these constants appear to be reliable as suggested by Baes and Mesmer [1976BAE/MES] and are retained by this review.

[1969STA/LIN]

Staveley *et al*. measured the low-temperature heat capacity from (11-303) K of a naturally-occurring (from Bohemia) sample of $CaCO_3$(calc) and from (22-291) K of one sample of aragonite (also from Bohemia). The calcite sample was assumed to be pure, but the aragonite sample was shown by X-ray fluorescence analysis to contain 0.13 % $SrCO_3$.

Detailed tables of the heat capacity measurements (58 for calcite, 48 for aragonite) are given as well as smoothed values of the thermal functions from 10 to 300 K.

The calculated standard values at 298.15 K are:

$$C^{\circ}_{p,\mathrm{m}}(CaCO_3, \text{calc}, 298.15\ \mathrm{K}) = (83.472 \pm 0.200)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(CaCO_3, \text{calc}, 298.15\ \mathrm{K}) = (91.710 \pm 0.210)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$C^{\circ}_{p,\mathrm{m}}(CaCO_3, \text{arag}, 298.15\ \mathrm{K}) = (82.320 \pm 0.200)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(CaCO_3, \text{arag}, 298.15\ \mathrm{K}) = (87.990 \pm 0.210)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

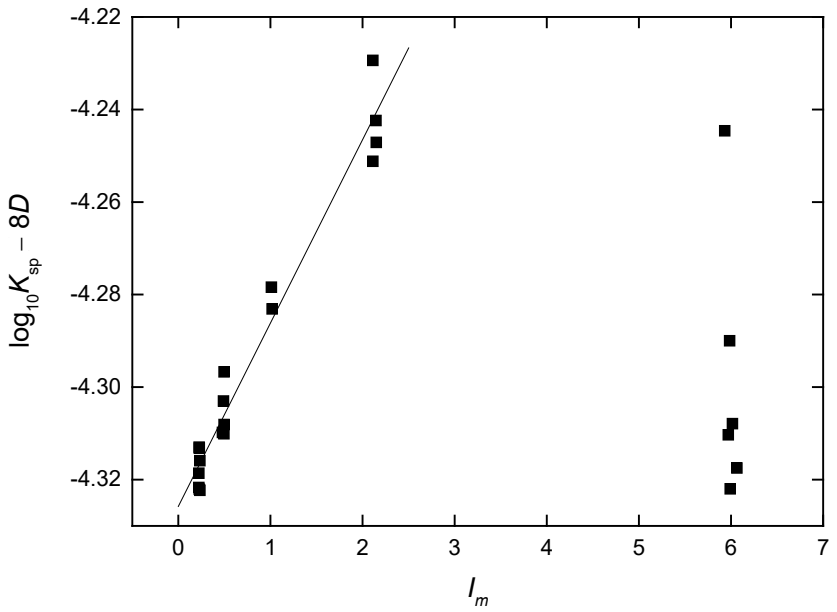
Because this study covered the most extensive temperature range, these are the selected data. The same values for both the entropies and the heat capacities of both calcite and aragonite were selected in [1987GAR/PAR].

The paper also notes that the measured difference in the standard entropy of calcite and aragonite, $(3.72 \pm 0.26)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$, is consistent within the uncertainties with the value calculated by [1925BAC] from the solubility and enthalpy measurements available at that time $(3.1 \pm 0.8)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$.

[1969YEA/MAR]

The solubility of $CaSO_4{\cdot}2H_2O$(s) was measured at 0.5 and 25 °C, and that of anhydrous $CaSO_4$(cr) at 150, 250 and 350 °C in mixed Na_2SO_4, $NaNO_3$ solutions at five ionic strengths from 0.2 to 6 $\mathrm{mol{\cdot}kg^{-1}}$. No characterisation of the pre-treated solid phases was provided. The solubility of the dihydrate was given as (0.0132 ± 0.0002) and $(0.0154 \pm 0.0002)\ \mathrm{mol{\cdot}kg^{-1}}$ at 0.5 and 25 °C, respectively. Neither the possible presence of the ion pairs $CaNO_3^+$ and $NaSO_4^-$, nor hydrolysis of Ca^{2+} and SO_4^{2-} was taken into account. Only in those experiments conducted at the highest ionic strengths (*viz*., 2 and 6 $\mathrm{mol{\cdot}kg^{-1}}$) did nitrate ion make up 90+ % of the ionic strength (five data points). The data in their Table I at 25 °C were examined with the SIT at ratios of $c_{NO_3^-} / I_c < 0.3$ and as shown in Figure A-14, a linear fit to $I_m \approx 2.15\ \mathrm{mol{\cdot}kg^{-1}}$ yielded $\log_{10}K^{\circ}_{\mathrm{sp}} = -(4.325 \pm 0.004)$ and $\Delta\varepsilon = (0.040 \pm 0.004)\ \mathrm{kg{\cdot}mol^{-1}}$, where the uncertainties are statistical (2σ), *cf*. $\log_{10}K^{\circ}_{\mathrm{sp}} = -4.52$ suggested by Yeatts and Marshall. These results do not support a calculation of the ion pairing constant for $CaSO_4$(aq).

Figure A-14: Plot of $\log_{10}K_{sp} - 8D$ *versus* I_m for selected results for the solubility of $CaSO_4{\cdot}2H_2O$(s) at 25 °C, where the solid line is a linear fit to $I_m \approx 2.15$ mol·kg^{-1}.



[1970CHR/HOS]

The solubilities of natural magnesite (from Red Mountain, California and Snarum, Norway) were determined. Measurements were made at 363 K to increase the slow kinetics of dissolution. The solid was suspended in distilled and deionised water in a thermostatted polypropylene vessel and stirred with a PTFE-coated magnetic stirrer. Pure CO_2(g) or a CO_2(g)-N_2 mixture containing 9.7 mol % CO_2, pre-saturated with water at the run temperature, was bubbled through the suspension. The $p(CO_2)$ and pH were determined by a Matheson gauge and by a combined pH electrode. Samples were filtered through a 0.45 mm Millipore filter and analysed for Mg by AAS. An XRD of solid sample before and after experiments did not differ significantly. Trace amounts of Fe, K, Cl, Ca were also found in the filtrates.

The dissolution experiments lasted around 150 days. The solubility of magnesite from Red Mountain at 388.5 K and $p(CO_2) = 0.312$ bar was 0.00198 mol·dm^{-3}, and at 388.5 K and $p(CO_2) = 0.0274$ bar, 0.00095 mol·dm^{-3}. For the magnesite from Norway, equilibrium was apparently not attained after 163 days, therefore this result was discarded from the accepted points (although its solubility (0.00174 mol·dm^{-3}) fits other data very well).

[1970GAR/GLU]

Gardner and Glueckauf reworked the solubility data in [1966MAR/SLU] and the freezing-point depression data in [1955BRO/PRU]. They claimed that the osmotic coefficients derived from the latter study could not be explained in terms of ion pairs alone, although a SIT re-analysis of this work suggests that this is not the case within the accuracy and limited concentration range of these data. Nevertheless, their re-treatment of these two sets of results yielded, $K_1^\circ = 258$ at 25 °C for the formation of $CaSO_4$(aq) ion pairs from Equation (A.127) from 0 to 110 °C.

$$\log_{10} K_1^\circ = -42.927 + 1721.74\,(T/\mathrm{K})^{-1} + 6.9439\,\ln(T/\mathrm{K}) \qquad \text{(A.127)}$$

Similarly, from the K_{sp}° values for the dissolution of $CaSO_4{\cdot}2H_2O$(cr) the following regression gave $\log_{10} K_{sp}^\circ = -4.64$ at 25 °C, *cf.* $\log_{10} K_{sp}^\circ = -4.52$ [1969YEA/MAR].

$$\log_{10} K_{sp}^\circ = 106.826 - 4889.96\,(T/\mathrm{K})^{-1} - 16.6850\,\ln(\mathrm{T/K}) \qquad \text{(A.128)}$$

[1970GRO/CHR]

Hydrofluoric acid solution calorimetry was used to determine the enthalpy of formation of a series of double oxides from their binary constituents and of gibbsite from alumina and liquid water. Only the data concerning gibbsite will be discussed here.

The calorimeter used and its mode of operation were described in detail. The main vessel was made of platinum containing 550 g of 41.4 wt % hydrofluoric acid. The operating temperature was 348 K. The samples encapsulated in Pyrex ampoules (with weight ranging from 120 to 180 g) were dropped from an aluminium block kept at *ca.* 298 K. The temperature of the block was exactly measured at the time of each experiment and small corrections to 298.15 K were made. The calorimetric reactions, including the dissolution of the Pyrex glass containers were completed within two to five minutes.

The gibbsite sample (S6035) from Aluminum Company of America was reported to have an XRD pattern identical to that published for gibbsite, apart from two additional very weak lines. The following impurities (basis Al_2O_3) were reported in wt % as: SiO_2, 0.028; Fe_2O_3, 0.010; TiO_2, 0.001; Na_2O, 0.11; CaO, 0.017; Ga_2O_3, 0.007; B_2O_3, 0.005. The extent of hydration was determined by prolonged heating at 1473 K, as (2.979 ± 0.002) mole of H_2O per mole of Al_2O_3. Anhydrous aluminium chloride used in the cycle was prepared by burning extremely pure aluminium in purified chlorine under pressure in a sealed glass reaction vessel. Excess chlorine was removed by condensation at liquid nitrogen temperature, the final traces being removed by refluxing the product over aluminium wire in the closed reaction vessel. Subsequent operations involving the aluminium chloride were carried out in a dry box under circulating purified argon. Analytical grade hydrofluoric acid (41.4 wt % HF) and hydrochloric

acid (33.88 wt % HCl, as determined by titration and density measurements) were used. Because of the difficulty in introducing a pre-determined amount of the very hygroscopic aluminium chloride into the ampoules, this compound was weighed first and the weights of gibbsite, water and hydrochloric solution adjusted accordingly.

The enthalpy of formation of the gibbsite was thus obtained by comparing the enthalpy of solution of the pair aluminium chloride and water (dropped in the calorimeter from separate sample holders) with that of gibbsite and hydrochloric acid. The mass of gibbsite used in the experiments ranged between 157 and 176 mg.

For the dissolution (Process 1) of [$2AlCl_3$(cr, 298.15 K) + 21.702 H_2O(l, 298.15 K)] in [41.4 wt % HF(348 K)], an average value of $-(619.65 \pm 1.67)$ kJ·mol^{-1} was obtained (2σ uncertainty), on the basis of six itemised experiments. The reported value included corrections for the enthalpy of solution of the glass ampoules (determined in separate experiments as $-(2.590 \pm 1.673)$ kJ·g^{-1})), and for small variations in the initial temperature (*ca.* 298 K) of the samples. For the dissolution (Process 2) of [$2Al(OH)_3$(gibb, 298.15 K) + [$6HCl \cdot 3.617H_2O$ (298.15 K)] in [41.4 wt % HF(348 K)], an average value of $-(250.21 \pm 1.67)$ kJ·mol^{-1} was obtained (2σ uncertainty), on the basis of eleven itemised experiments. This value included the corrections mentioned above and also a correction of 1.67 kJ to take into account the slight substoichiometry in water of the gibbsite sample and the presence of small amounts of impurities. Combination of these two dissolution processes leads to:

$$2AlCl_3(\text{cr, } 298.15\text{ K}) + 6H_2O(\text{l, } 298.15\text{ K}) \rightleftharpoons$$
$$2Al(OH)_3(\text{gibb, } 298.15\text{ K}) + 6HCl(\text{in } HCl \cdot 3.617H_2O, \text{ } 298.15\text{ K}) \qquad \text{(A.129)}$$

with $\Delta_r H^\circ_m$ ((A.129), 298.15 K) = $-(369.44 \pm 2.36)$ kJ·mol^{-1}.

Noting that the correction to 1981 atomic weights is negligible, and making use of $\Delta_f H^\circ_m(AlCl_3, 298.15\text{ K}) = -(706.26 \pm 0.70)$ kJ·mol^{-1} previously determined by the same group [1970GRO/HAY2], $\Delta_f H^\circ_m(H_2O, \text{l}, 298.15\text{ K}) = -(285.830 \pm 0.040)$ kJ·mol^{-1} [1989COX/WAG] and $\Delta_f H^\circ_m(HCl \text{ in } HCl \cdot 3.617H_2O, 298.15\text{ K}) = -(151.441 \pm 0.150)$ kJ·mol^{-1}, (interpolated from the enthalpies of dilution given in [1965PAR], this review obtains $\Delta_f H^\circ_m(Al(OH)_3, \text{gibb}, 298.15\text{ K}) = -(1294.15 \pm 1.45)$ kJ·mol^{-1} from this very careful work. The authors' calculated value was similar, $-(1294.24 \pm 1.41)$ kJ·mol^{-1} (our estimated uncertainty).

[1970KAL]

A synopsis of this paper appeared in Appendix A of [2013LEM/BER].

[1970LAR]

Enthalpy of dilution measurements made by Lange *et al*. [1930LAN/MON], [1931LAN/STR], [1931LAN/STR2], [1933LAN/MON], [1956LAN/MIE], [1959LAN] of the metal(II) ions, Mg^{2+}, Zn^{2+}, Cd^{2+}, Ca^{2+}, Cu^{2+} and Ni^{2+}, were treated by Larson taking ion association into account. The Debye-Hückel equation used contained an empirical parameter for which optimal values were derived by trial and error to give consistent dissociation constants over the entire concentration range. The enthalpies and entropies of ion association are also consistent with the results of most other studies.

[1970RAI/GRE]

Rai and Gregory measured the steady-state pressures generated by heating CaO(cr) and graphite in a graphite effusion cell from 1429 to 1525 K. These pressures were found to decrease with time at a rate dependent on the effusion cell parameters. Rai and Gregory indicated that the pressures calculated by back extrapolation to zero time would correspond to the equilibrium pressures.

They found no evidence for the formation of CaC_2(cr), since XRD examination of the reaction product showed only the lines of CaO and graphite, and no odour of acetylene was detected on moistening these products; there was no mention of the possible formation of a CaC_2-CaO solid solution (both have fcc structures at these temperatures). They therefore processed their results assuming that the cell reaction was the simple reaction:

$$\mathrm{CaO(cr) + C(graphite) \rightleftharpoons Ca(g) + CO(g)} \qquad \text{(A.130)}$$

and thus calculated $\Delta_r H^\circ_m$ ((A.130), 298.15 K) = (169.9 ± 3.0) kcal·mol^{-1} = (710.9 ± 12.6) kJ·mol^{-1} from a Third-Law calculation, using the Gibbs energy functions from the then current assessments by Kelley and King [1961KEL/KIN] and by JANAF [1965STU/CHA]. The use of CODATA values from Cox *et al.* [1989COX/WAG] gives the marginally different value of (710.5 ± 12.6) kJ·mol^{-1}. This is consistent within the combined uncertainty limits with the CODATA value for this reaction $\Delta_r H^\circ_m$ ((A.130), 298.15 K) = (702.19 ± 1.22) kJ·mol^{-1}.

[1970SHA/GLA]

The authors studied the inversion temperatures and heat capacities of nine samples of tridymite, one a natural mineral and eight prepared by heating silicic acid or natural quartz with the tungstates of Li, Na and K, at temperatures from 1100 to 1440 °C; the extent of conversion to tridymite was judged by unspecified petrographic examination and XRD. Final washings in succession with aqueous solutions of 5 % NH_4Cl, 2 %

NaOH and 1 % HCl were said to give very pure preparations, but no chemical analyses are given.

Most samples gave two first-order inversions at *ca.*117 °C (with considerable hysteresis) and 163 °C, but some exhibited an additional second-order inversion in the range 230-250 °C. Calorimetric studies showed that the enthalpies of inversion varied considerably, depending on the particular ordering (and possibly impurity content—added by this review). The heat capacities of two of the samples are also plotted from 50 to 250 °C, but these differ by up to 20 % and are not used by this review.

[1970SPE]

Potentiometric titrations were performed to determine the formation constants of polyborates in solutions of 3 and 5 mol·dm^{-3} NaCl, and in 3 and 4 mol·dm^{-3} KCl from 25 to 90 °C. Other titrations were reported exclusively at 25 °C in 1 mol·dm^{-3} KCl, 1.5 mol·dm^{-3} Na_2SO_4, 3 mol·dm^{-3} NaBr and 2 mol·dm^{-3} CsI, as well as two mixed electrolytes. Spessard titrated 0.4 mol·dm^{-3} $H_3BO_3(aq)$ with 0.4 mol·dm^{-3} $B(OH)_4^-$ at constant ionic strengths. Commercial glass and calomel electrodes were standardised against buffer solutions made up according to NBS standards; the liquid-junction corrections were "small". Dissociation constants (incorrectly tabulated as formation constants) for the following five equilibria were reported on a mixed scale involving hydrogen ion activity and boron molarity (the "dissociation constant" expressions at foot of Table 1 in [1970SPE] are incorrect).

$$B(OH)_3(aq) + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+$$

$$3B(OH)_3(aq) + H_2O(l) \rightleftharpoons B_3O_3(OH)_4^- + H^+$$

$$3B(OH)_3(aq) \rightleftharpoons B_3O_3(OH)_5^{2-} + 2H^+ + H_2O(l)$$

$$4B(OH)_3(aq) \rightleftharpoons B_4O_5(OH)_4^{2-} + 2H^+ + 3H_2O(l)$$

$$5B(OH)_3(aq) \rightleftharpoons B_5O_6(OH)_4^- + H^+ + 5H_2O(l)$$

Edwards and Ross [1960EDW/ROS] discussed criteria for the formation of polyborates based on NMR and XRD evidence and the following species were mentioned: $B_3O_3(OH)_4^-$, $B_3O_3(OH)_5^{2-}$, $B_5O_6(OH)_5^{2-}$ and $B_5O_6(OH)_6^{3-}$. Spessard stated that the pH readings for the same solutions reported in the potentiometric work of Ingri [1963ING2] would have to be corrected by 0.221 to conform to his results. The fundamental problem with this paper is that the boron concentration was held constant, which brings into question the speciation derived by the author.

[1970STU/HIL]

This paper reports the low-temperature heat capacities of 15 inorganic compounds, including K_2CO_3(cr), measured over the 15 years prior to publication, but few details are given. The measurements were made using an automatic adiabatic calorimeter, which the authors note gives values of somewhat lower precision than that achieved with a more conventional calorimeter. The K_2CO_3(cr) sample was a reagent-grade material, and was dried at 573 K before use. The only analysis given is that of the manufacturer, which stated a purity of 99.9 %.

Measurements were carried out over a temperature range specified only as from (10-15) K to (300-310) K, and because the measurements were not of the highest precision, only the smoothed values at round temperatures from 16 to 298.15 K were reported. The heat capacities were extrapolated to 0 K using a simple T^3 law based on the data from 15 to 30 K. Their reported following standard values of C_p and S, estimated from the scant information given to be accurate to within 1 %, are the adopted values:

$$C^{\circ}_{p,\mathrm{m}}(K_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = (114.43 \pm 1.20)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}},$$

$$S^{\circ}_{\mathrm{m}}(K_2CO_3, \mathrm{cr}, 298.15\ \mathrm{K}) = (155.50 \pm 1.60)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}.$$

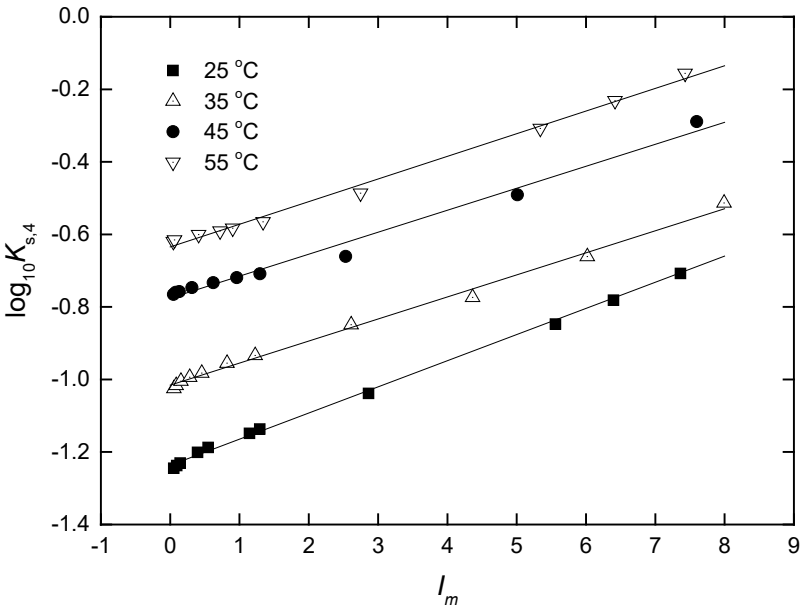
[1970SZI/BER]

This potentiometric investigation of gibbsite solubility at 25, 35, 50 and 60 (± 0.02) °C utilised the cell:

$$\mathrm{Hg,HgO|NaOH(sln),NaAl(OH)_4(sln)|Al(OH)_3(cr)|HgO,Hg}$$

Szita and Berecz used acid-base titrations with phenolphthalein as an indicator to titrate the "free" OH^- molality. The aluminium molality was measured by titration in a two-step process where first the total OH^- was titrated using gluconate to release one OH^- from aluminate ion and then F^- to release the remaining hydroxide ions. They derived $\log_{10}K^{\circ}_{s,4}$ values of – 1.27, – 1.05, – 0.78 and – 0.63 at 25, 35, 50 and 60 °C, respectively. The results of an SIT treatment is indicated by solid lines in Figure A-15. These solutions, at least at 25 °C, appear to be under-saturated, possibly exacerbated by the apparent lack of agitation in the unique cell design employed in this study.

Figure A-15: Plot of $\log_{10}K_{s,4}$ *versus* I_m for the dissolution of gibbsite in the system: $Al(OH)_3$-NaOH-H_2O.



[1970TAK/YOS]

This paper is mainly concerned with the determination of the enthalpies of solution at 298.15 K of crystals and glasses in the systems alkali metal (Li, Na, K) oxides-SiO_2 in 5 % aqueous HF and in distilled water. Most of the reported experimental data fall outside the scope of this review and will not be discussed here. However, results are also reported for the dissolution of crystalline and vitreous quartz. Although no details are given, the reported finely particulate vitreous quartz was of commercial origin, probably obtained from drying aqueous solutions of sparingly soluble silicic acids followed by thermal treatment. The crystalline form of silica was obtained by thermal treatment of the commercial vitreous glass; however, no further information was given as to these crystalline samples. Information on the purity of the raw materials was also missing. Enthalpies of solution of crystalline and vitreous samples, respectively, were reported, as (– 519.4 and – 557.4) cal·(g SiO_2)$^{-1}$, corresponding to (– 130.57 and – 140.13) kJ·mol^{-1}.

These results yield the value of 9.6 kJ·mol^{-1} for the enthalpy of transition of crystalline to vitreous quartz at 298.15 K. In view of the above comments, this review finds it difficult to assign an uncertainty to this result.

[1971CRE/AND]

The solubility of quartz was measured in pure water from under- and over-saturation from 178 to 329 °C (*i.e.*, $SiO_2(cr) + H_2O(l) \rightleftharpoons Si(OH)_4(aq)$) at saturation-vapour pressure. Further measurements were made in 0.01 mol·kg^{-1} NaOH solutions from 125 to 225 °C whereby the $Si(OH)_5^-$ ion was included in the ion-pair model used by the authors. Their measured solubility of quartz decreased with decreasing temperature which raises some concern that precipitation could have occurred during sampling, especially if the sample line was not flushed out between sampling events. The average of four solubility measurements in pure water at 180 °C is (2.80 ± 0.16) × 10^{-3} mol·kg^{-1}.

[1971DIT/DOU]

The relative enthalpy of NBS Standard Reference Material No. 720 (99.98 % pure single-crystal α-Al_2O_3 calorimetric heat capacity standard) was measured from 273 to 1173 K in a very precise Bunsen ice-drop calorimeter. The sample was made by cutting single-crystal rods into small (2-6 mm) pieces. Analyses indicated a purity of 99.98 % by weight, with the major impurities being magnesium, calcium, chromium, iron and silicon. No corrections for these impurities were applied in processing the data. Measurements were also made on a Calorimetry Conference sample of alumina, whose preparation and analysis were the same as those described in the earlier paper by Furukawa *et al.* [1956FUR/DOU]. The precautions observed in order to minimise measurement errors are described in considerable detail.

The results of *ca.* 75 measurements from 323 to 1173 K are tabulated in full, together with the six-term equation used to fit them. The data showed no significant discontinuity with the earlier NBS data on the same substance from 13 to 380 K and 1173 to 2257 K. The average deviation from the mean for all enthalpy measurements on the SRM 720 sample was 0.017 %, and the smoothed enthalpy values derived from the data were estimated to be accurate to 0.1 %. The data were compared with many sets of the most reliable published data and then recommended values for the thermodynamic functions of Al_2O_3 are tabulated at closely spaced intervals from 0 to 1200 K. The values at 298.15 K are C_p = 79.01 J·K^{-1}·mol^{-1} and S = 50.94 J·K^{-1}·mol^{-1}.

[1971GIN/GUB]

This short paper gives the results of the authors' measurements of the enthalpy content of the title compounds using a conventional drop calorimeter. For K_2CO_3, the measurements extended from 298 to 1273 K; the results are presented only as C_p equations over four temperature ranges, and as small graphs. The complete data were deposited in the VINITI document database.

[1971ISO]

A synopsis of this paper appeared in [2005GAM/BUG] in reference to the formation of $NiSO_4$(aq) ion pairs. An SIT treatment of the $MgSO_4$(sln) data was restricted to the concentration range, 0.0058 to 0.0206 mol·kg^{-1}. Note that there is a sign error in the equation at the bottom of page 97 of [1971ISO]: the third term in the expression for σ(x) should read "1/(1 + x)". A series of estimates of the degree of association was required at each concentration to obtain the desired osmotic coefficients, which were fitted by this review as a function of molality using an equation of similar form to that proposed by Isono. The resulting six molal association constants, K_1^o, gave an average of (159 ± 5) (2σ statistical uncertainty), *cf.* (153.8 ± 20.0) estimated from the SIT treatment of the $NiSO_4$(sln) data at 0°C. The value for $MgSO_4$(aq) is in good agreement with that from an earlier freezing-point depression study, [1955BRO/PRU] and is retained here in the overall analysis with an increased uncertainty.

[1971MAC/GEE]

Mackenzie and Gees reported the formation of α-quartz from a suspension of 1 g of finely ground quartz (0.62 μm) in 200 cm^3 sea water with an initial silica concentration of 0.03 ppm at 20 °C. The suspension was shaken for 12 months at room temperature with aliquots being withdrawn every month for analysis by spectrophotometry. The quartz grains before and during their experiments were inspected by SEM and XRD techniques to establish the crystallinity and size of the quartz particles formed. They reported a final silica concentration of (4.4 ± 0.3) ppm (*i.e.*, 7.3×10^{-5} mol·kg^{-1}). A preliminary regression of the solubility data for α-quartz in the literature gave a value of 1.6×10^{-4} mol·kg^{-1}, which indicated that the solution was significantly under-saturated with respect to quartz.

[1971MES/BAE]

The hydrolysis of Al^{3+} was studied from 62.5 to 149.8 °C in 1 mol·kg^{-1} KCl in a hydrogen-electrode concentration cell. The liquid-junction potential was approximated by the Henderson equation [1976BAE/MES] and was generally less than 1 mV. These measurements were designed to elucidate the stoichiometries of the higher-order hydrolysed species and their formation constants as a function of temperature by employing relatively high concentrations of Al, 0.0045-0.045 mol·kg^{-1}. The temperature range was limited at the low end by the slow kinetics of formation of some of these species. Three schemes were presented that included different combinations of species, with the preferred one involving $Al_2(OH)_2^{4+}$, $Al_3(OH)_4^{5+}$ and $Al_{14}(OH)_{34}^{8+}$, where in this case the latter species would have been the dominant aqueous Al species in these experiments. Baes and Mesmer later re-evaluated their data and replaced $Al_{14}(OH)_{34}^{8+}$ in

their scheme 2 by $Al_{13}O_4(OH)_{24}^{7+}$ [1976BAE/MES]. They also re-calculated formation constants from data in [1956KUB], [1965AVE] and [1972SMI/HEM].

[1971MON/PAS]

This paper is mainly concerned with the evaluation of the occurrence of the $MgCl^+$ ion pair in aqueous hydrochloric acid solution and the determination of its enthalpy of formation.

To this aim, the authors measured the enthalpy of reaction of magnesium metal with two hydrochloric acid solutions (HCl, 63.11 H_2O and HCl, 91.16 H_2O) and one perchloric acid solution ($HClO_4$, 85.71 H_2O), taking advantage of the argument that, according to activity coefficient measurements, Mg^{2+} does not form complexes with the perchlorate ion.

The magnesium metal used was of purity superior or equal to 99.9 %, as determined by spectroscopy, and of (99.96 ± 0.05) % as measured by the amount of hydrogen liberated upon its dissolution in hydrochloric acid.

The hermetically sealed silver calorimeter (30 cm^3) used was described previously by the same group. The acid volume was 20 cm^3 and the metal sample weights ranged between 11 and 17 mg. The absence of reduction of the perchloric acid to chloride during dissolution was ascertained. Corrections to the heats measured were made for the heat of solution of hydrogen into the solution and for the vaporisation of water into the gas freed by the breaking of the ampoule containing the metal. Corrections were also made for differences in concentrations between individual measurements. Results were extrapolated to infinite dilution. In the case of $HClO_4$ and $Mg(ClO_4)_2$, dilution data were taken from [1965GLU/MED] and [1965JON/WOO], respectively.

The average values reported for the dissolution of magnesium in water in standard conditions were: – (468.86 ± 0.75) kJ·mol^{-1} from the data (five determinations) in ($HClO_4$, 85.71 H_2O), – (466.01 ± 0.75) kJ·mol^{-1} from the data (nine determinations) in (HCl, 63.11 H_2O) and – (464.72 ± 0.75) kJ·mol^{-1} from the data (five determinations) in (HCl, 91.16 H_2O). The differences between the extrapolated data from the $HClO_4$ solution and the two HCl solutions, – (2.85 ± 1.06) kJ·mol^{-1} and – (4.14 ± 1.06) kJ·mol^{-1} were attributed to the formation of the species $MgCl^+$ in HCl media. To evaluate the enthalpy of complexation at infinite dilution according to the reaction:

$$Mg^{2+} + Cl^- \rightleftharpoons MgCl^+ \qquad \text{(A.131)}$$

the authors used $\log_{10} K°(\text{A.131}) = 0.91$ from [1959KEN2]. From this value, they deduced a degree of association of 0.411 and 0.479 in the two HCl media and $\Delta_r H_m^\circ$ (A.131) = 7.9 kJ·mol^{-1}.

The value of – (468.86 ± 0.75) kJ·mol^{-1} for the dissolution of Mg(cr) in standard conditions obtained in [1971MON/PAS] from the extrapolation of their data in $HClO_4$ solutions (see above) was used by CODATA [1989COX/WAG] in their assessment of $\Delta_f H^o_m$ (Mg^{2+}, aq, 298.15 K).

The authors also refer to an earlier determination [1960VOR/PRI] of their group of the enthalpy of solution of MgO(cr) in ($HClO_4$, 450 H_2O), as – (151.08 ± 0.50) kJ·mol^{-1} that they use to derive another value for $\Delta_f H^o_m$ (Mg^{2+}, aq, 298.15 K).

[1971VOL/PAV]

Volokhov *et al.* used a glass electrode to measure the pH of 1×10^{-5} to 1×10^{-3} mol·dm^{-3} $Al(ClO_4)_3$ solutions at I_c = 0.0025, 0.0050, 0.0075, 0.010, 0.015, 0.020, 0.025, 0.030, 0.040, 0.050, 0.060 and 0.070 (NaCl, NaBr, $NaNO_3$, $NaClO_4$ and Na_2SO_4). The temperature was varied from 10 to 40 °C. A simple Debye-Hückel equation was used to estimate activity coefficients in these relatively dilute solutions (truncated here to $I_c \leq 0.03$). The aluminium concentration may have contributed significantly to the ionic strength. The authors only tabulated the average values of $\log_{10}{}^*K_{c,1}$ obtained in each medium as a function of ionic strength and temperature. Application of the SIT to the values of $\log_{10}{}^*K_{m,1}$ following the trivial correction to molal values gave $\log_{10}{}^*K^o_1 = -(5.07 \pm 0.03)$ compared to the more reasonable value reported by the authors of – (4.99 ± 0.01). By fixing the value of $\Delta\varepsilon$ = 0.30 kg·mol^{-1} [1993PAL/WES], an average $\log_{10}{}^*K^o_1$ value of – (5.00 ± 0.01) was obtained. However, the enthalpy of reaction, $\Delta_r H^o_{m,1}$ = (11.4 ± 2.1) kJ·mol^{-1} was then a factor of five too small.

[1972EDM/TAY]

The authors studied the title reaction by equilibrating lime-saturated CaO-CaF_2 slags with graphite at 1700 to 1773 K, using a Sievert-type apparatus. The effect of the formation of Ca(g) was minimised by enclosing the charge in a thin-walled graphite crucible with a tightly fitting lid, which the authors claimed 'almost suppressed the diffusion of Ca(g) from the reaction site'. Nevertheless, the small random fluctuations due to the presence of Ca(g) did make it very difficult to decide when equilibrium had been reached. It was for this reason, that lime-saturated CaO-CaF_2 mixtures were used rather than pure lime.

However, as Jacob and Varghese [1996JAC/VAR] note in their analysis of these data, CaC_2 is known to dissolve in CaF_2(l) at these temperatures, and it is therefore not certain that the activity of CaC_2 was unity in these experiments, and their derived values for $\Delta_f G$(CaO, cr) cannot be regarded as reliable.

[1972INA/SHI]

This conductivity study of the formation of $CaSO_4(aq)$ ion pairs to high pressures (up to 120 MPa) used a robust glass cell which therefore provided less accurate data than conventional glass conductance cells. Concentrations were in the range 10^{-4} to 10^{-3} mol·dm^{-3}. Their average value of $\Delta_r V_1^\circ$ was given as – 10.2 cm^3·mol^{-1} at 25 °C, *cf.* – 11.6 cm^3·mol^{-1} [1979FIS/FOX]. The ion-association constants at ambient pressure re-evaluated with the Lee-Wheaton equation are retained by this review with increased uncertainties (± 0.20 $\log_{10}K$).

[1972MAR/VAS]

The dissociation constants of $CaOH^+$ in 0.001 mol·dm^{-3} $Ca(NO_3)_2$ solutions were determined by measuring the pH and total calcium concentration; the latter being determined by titration with a Trilon® chelating agent. The total calcium concentration was assumed to be equal to "its ionic concentration", most likely meaning that the total calcium concentration was taken to equal to the concentration of Ca^{2+}. Only the dissociation constants at infinite dilution were reported at temperatures from 25 to 98 °C with the value at 25 °C being taken from [1951DAV/HOY]. Activity coefficients were calculated with the modified Debye-Hückel model in [1966MAR/SLU]. The temperature dependence of the dissociation constant of $CaOH^+$ was presented in the form $\log_{10}K_1^\circ = -13.6 + 3610(T/\mathrm{K})^{-1}$. Re-fitting, excluding the data point at 25 °C, gave the following equation, $\log_{10}K_1^\circ = -13.325 + 3470.49(T/\mathrm{K})^{-1}$. The resulting $\log_{10}K_1^\circ$ value extrapolated to 25 °C is – 1.68.

[1972MES/BAE]

The hydrolysis of boric acid (Reaction (A.132)) was studied potentiometrically with a hydrogen-electrode concentration cell over the temperature range 50-290 °C as a function of ionic strength up to 1.02 mol·kg^{-1} (KCl) while the molality of boric acid was kept $\leq$ 0.02 mol·kg^{-1}. Titrations at much higher boric acid molalities (0.02 to 0.6 mol·kg^{-1}) with I_m = 1.0 were carried out to investigate the formation of polyborates at 50.2, 100.0 and 199.8 °C. Mesmer *et al.* obtained smoothed hydrolysis constants from an empirical equation with seven adjustable parameters (four of which were ionic-strength independent). In applying the SIT, the upper temperature limit was set at 200 °C, yielding the results given in Table A-37.

$$H_3BO_3(aq) + OH^- \rightleftharpoons H_4BO_4^- \qquad \text{(A.132)}$$

Table A-37: Summary of data for the hydrolysis of boric acid according to Reaction (A.132) from the authors' fitting equation and those from application of the SIT where the uncertainties are 2σ.

t /°C	$\log_{10}K_1^\circ$ /reported	$\Delta_r H_m^\circ$ /kJ·mol^{-1} /reported	$\log_{10}K_1^\circ$ /SIT	$\Delta\varepsilon$ /kg·mol^{-1}
50	4.188	– (41.5 ± 0.2)	(4.206 ± 0.015)	– (0.098 ± 0.030)
100	3.314	– (39.0 ± 0.2)	(3.335 ± 0.008)	– (0.091 ± 0.014)
150	2.697	– (35.2 ± 0.3)	(2.712 ± 0.008)	– (0.078 ± 0.012)
200	2.267	– (30.4 ± 0.3)	(2.282 ± 0.010)	– (0.063 ± 0.016)

The corresponding thermodynamic quantities reported in this paper extrapolated to 25 °C were: $\log_{10}K_1^\circ = 4.757$, $\Delta_r H_m^\circ = -(42.3 \pm 0.3)$ kJ·mol^{-1}, $\Delta_r S_m^\circ = -(51 \pm 9)$ J·K^{-1}·mol^{-1} and $\Delta_r C_{p,m}^\circ = (28.0 \pm 1.1)$ J·K^{-1}·mol^{-1}. The good agreement between the reported and SIT $\log_{10}K_1^\circ$ values implies that the former may have been over-fitted. This point is exemplified by the linearity of the SIT values *versus* $(T/\mathrm{K})^{-1}$ including those from [1943OWE/KIN] covering the range 5 to 200 °C. In other words, the value of $\Delta_r C_{p,m}^\circ$ reported may be largely an artefact of over-fitting. From the linear plot of $\Delta\varepsilon$ *versus* $(T/\mathrm{K})^{-1}$, $\Delta\varepsilon = -(0.12 \pm 0.03)$ kg·mol^{-1} at 25 °C. Therefore:

$$\varepsilon(\mathrm{K^+}, \mathrm{H_4BO_4^-}) = \Delta\varepsilon - \varepsilon(\mathrm{K^+}, \mathrm{OH^-}) =$$
$$= -(0.12 \pm 0.03) + (0.09 \pm 0.01) = -(0.03 \pm 0.03)\ \mathrm{kg \cdot mol^{-1}}\ .$$

Mesmer *et al*. concluded that polynuclear anionic borate species exist at least between 50 and 200 °C, at boron concentrations of 0.03 mol·kg^{-1} and higher. They considered fourteen possible boron species in their data analysis algorithms, noting that their terminology used *x*:*y* for the species $\mathrm{B}_x(\mathrm{OH})_{3x+y}^{y-}$. They observed that the (3,1) species, $\mathrm{B_3(OH)_{10}^-}$, was common to all of the best schemes whereas at higher temperatures, inclusion of the (2,1) species, $\mathrm{B_2(OH)_7^-}$, gave a much better fit compared to (4,2), (3,2) or (5,3). At 50 °C any of these species gave a significantly better fit than the (2,1) species. Consequently, they presented two sets of formation constants at 25 °C and $I_m = 1$:

Scheme 1.
$\log_{10}K_1 = (5.278 \pm 0.004)$, $\log_{10}\beta_{2,1} = (5.95 \pm 0.06)$, $\log_{10}\beta_{3,1} = (7.319 \pm 0.002)$ and $\log_{10}\beta_{4,2} = (13.29 \pm 0.07)$.

Scheme 2.
$\log_{10}K_1 = (5.279 \pm 0.003)$, $\log_{10}\beta_{2,1} = (6.02 \pm 0.04)$, $\log_{10}\beta_{3,1} = (7.378 \pm 0.003)$ and $\log_{10}\beta_{5,3} = (19.20 \pm 0.08)$.

Note the second polynuclear anion was referred to in this paper both as the 3,1 and 3,2 species, but the former notation appears to be correct. The results for Reaction (A.132) are consistent with those proposed in lower temperature studies [1935OWE], [1943OWE/KIN]. The authors gave a good critical discussion of previous results for the formation of the polyborates [1949KON/LEB], [1957ING/LAG], [1962ING],

[1963ING2], [1963ING3], [1970SPE]. They cited specifically the results published in [1957ING/LAG] which proposed a different scheme based on measurements at 25 °C, I_m = 3.50 ($NaClO_4$). Mesmer *et al*. were critical of the potentiometric work of Spessard [1970SPE], in particular, because only one borate concentration was employed.

[1972PIT]

Pitzer analysed an array of experimental data from 0 to 25 °C such as heats of dilution, osmotic coefficients and cell potentials for MSO_4 solutions, where M = Mg, Ca, Be, Cu, Zn, Cd, Ni, Co and Mn. This analysis led to an estimation of the ion-pair formation constants given in Table A-38, where K_2^o refers to both $M_2SO_4^{2+}$ and $M(SO_4)_2^{2-}$, which were indistinguishable. Pitzer found that use of a non-unity value for the activity coefficient of the 1:1 ion pair was not justifiable, *cf*. [1976REA/LAN]. The formation of higher-order ion pairs is still controversial and the small values for K_2^o were possibly an artefact of the model. Other papers [1974PIT/MAY], [1981RAR/MIL] used the Pitzer ion interaction model to treat these systems. However, the values of K_1^o in Table A-38 are within the general range of most other experimental values.

Table A-38: Summary of association constants for divalent metal sulfates.

System	K_1^o /0 °C	K_1^o /25 °C	K_2^o /0-25 °C	System	K_1^o /0 °C	K_1^o /25 °C	K_2^o /0-25 °C
$MgSO_4$	213	238	1.5	$CdSO_4$	222	250	5
$CaSO_4$	278	313	1.5	$NiSO_4$	200	222	4
$BeSO_4$		454	0.5	$CoSO_4$	182	204	4.5
$CuSO_4$	222	250	5	$MnSO_4$		222	3
$ZnSO_4$	185	200	4	$CdSO_4$	222	250	5

[1973AIN]

A concentration cell fitted with AgCl|Ag electrodes was used to determine the dissociation constant of the $CaSO_4$(aq) ion pair in aqueous solutions at 25, 40 and 50 (± 0.05) °C. The solutions were buffered near neutral pH using minor additions of Tris. The cell was calibrated on the molal scale against three standard $CaCl_2$ solutions. Unfortunately, the sulfate concentration was substantially higher than that of chloride. Re-treatment with the SIT was made with no $\Delta\varepsilon$ term due to the low ionic strengths employed < 0.013 mol·kg^{-1}. The $CaCl^+$ ion pairs were not considered. The average SIT value of $10^3 K_{dis}^o$ are (3.81 ± 0.19), (3.71 ± 0.20) and (3.85 ± 0.21) at 25, 40, and 50 °C, respectively, from four measurements at each temperature. Ainsworth reported, $\Delta_r H_m^o$ = (10.3 ± 2) kJ·mol^{-1}, $\Delta_r S_m^o$ = (80 ± 7) J·K^{-1}·mol^{-1} and $\Delta_r C_{p,m}^o$ = (260 ± 230) J·K^{-1}·mol^{-1}, which are compromised by over-fitting. The equilibrium constants derived from application of the SIT are retained in the overall assessment.

[1973BRI/NES]

This paper is mainly concerned with properties of talc ($Mg_3Si_4O_{10}(OH)_2$), which are not of relevance for this review, but it also includes some comments and re-analysis of the data by Hostetler and Christ [1968HOS/CHR] on the aqueous solubility of chrysotile. They point out that the fact that the ratio of the Mg/Si compositions of the solution in equilibrium with chrysotile in these experiments was much greater than 1.5 corresponding to congruent dissolution suggests that chrysotile actually dissolved incongruently to form talc ($Mg_3Si_4O_{10}(OH)_2$(cr)) and that the solution composition was in fact controlled by equilibrium with talc. Indeed, on this basis, they used their estimated $\Delta_f G^o_m$(talc) to re-interpret the data in [1968HOS/CHR] to calculate $\Delta_f G^o_m$ ($Mg_3SiO_2O_5(OH)_4$, cr, 298.15 K) = – 4025 kJ·mol^{-1}, in reasonable agreement with that calculated from our selected values, – (4036.37 ± 3.43) kJ·mol^{-1}, in view of the uncertainties in the calculation in [1973BRI/NES].

[1973CHI/DOW]

Isopiestic measurements were made of NaCl, KCl, NaH_2PO_4 and KH_2PO_4 solutions at 25 °C. The molalities of the phosphate salts ranged from 1.0 to 6.5 mol·kg^{-1} and 0.85 to 2.2 mol·kg^{-1}, respectively. The resulting osmotic and activity coefficients are in good agreement with those from other studies.

[1973CHR/HOS]

Two natural samples of sepiolite were used in this solubility study by Christ *et al*., but only the data for the well-crystallised SP1 sample (Balmat, NY, USA) are noted below, since the poorly crystallised sample had almost certainly not reached equilibrium in their experiments. Three runs were made with the SP1 sample, at 324 K for 88 days, 343 K for 112 days, plus a further 182 days, 363 K for 220 days, all at 1 bar. Temperatures were held to within ± 0.3 °C. The pH was monitored with a modified glass electrode throughout the runs and at intervals during the runs aliquots of solution were analysed for Mg and SiO_2 spectrophotometrically with accuracies similar to those given in [1968HOS/CHR]. Christ *et al*. noted that all previous results showed that the Si:Mg ratio for sepiolite is 3:2, but that it has variable H_2O content. The only analyses of their material were by X-ray diffraction and electron-microscopy, so the water content was not accurately defined, but later in the paper the composition was assumed to be both $4H_2O$ (in their Equation 1) and $3.5H_2O$, (in footnote to Table 8 of this paper, and in the estimation of *S*(sepiolite)). Consequently, they gave two equations (designated as Equations 1 and 6 in the paper) for the congruent dissolution of sepiolite:

$$2MgO{\cdot}3SiO_2{\cdot}4H_2O(cr) + 4H_2O(l) \rightleftharpoons 2Mg^{2+} + 3Si(OH)_4(aq) + 4OH^- \qquad (A.133)$$

$$2MgO{\cdot}3SiO_2{\cdot}3.5H_2O(cr) + 4.5H_2O(l) \rightleftharpoons 2Mg^{2+} + 3Si(OH)_4(aq) + 4OH^- \qquad (A.134)$$

where we have corrected the total number of water molecules on the left-hand side of their Equation 1 (A.133). Solution samples were taken after as long as 5300 hours at 70 °C, but within the scatter of K_{sp} values it would appear that equilibrium was attained after only *ca.* 100 hours, and the ratio of the Si/Mg concentrations in the solution was close to 1.5. The authors correctly suggested that essentially all the Mg in solution was present as Mg^{2+} (*i.e.*, < 1 % hydrolysed to $MgOH^+$). The activity coefficients of Mg^{2+} were calculated using a Debye-Hückel equation but apparently not those for OH^-. The reported $\log_{10}K_{sp}$ values for the SP1 sample, based on Equation (A.134), are:

$$\log_{10}K_{sp} \text{ at } 324 \text{ K} = -(38.8 \pm 0.3),$$

$$\log_{10}K_{sp} \text{ at } 343 \text{ K} = -(37.5 \pm 0.3),$$

$$\log_{10}K_{sp} \text{ at } 363 \text{ K} = -(37.2 \pm 0.2).$$

As in the authors' similar paper on the solubility of chrysotile [1968HOS/CHR] there are concerns about the application of activity coefficients and the determination of m_{OH-}. Therefore, this review cannot endorse these values, especially as the water content of the relevant sepiolite is not clear.

The authors also calculated a $\log_{10}K_{sp}$ value of – 40.1 for the solubility of ($2MgO{\cdot}3SiO_2{\cdot}3.5H_2O$) at 298.15 K using estimated values of $\Delta_r S$ and $\Delta_r C_p$ for Reaction (A.134) based on the values listed in their Table 8, in which S^o_m (sepiolite, 298.15 K) was estimated to be either 90.1 $cal{\cdot}K^{-1}{\cdot}mol^{-1}$ (377 $J{\cdot}K^{-1}{\cdot}mol^{-1}$) simply from the sum of those of MgO, SiO_2 and H_2O(l), or 75.4 $cal{\cdot}K^{-1}{\cdot}mol^{-1}$ (315.5 $J{\cdot}K^{-1}{\cdot}mol^{-1}$) if the 1.5 mole of combined, non-zeolitic water is treated as 'ice'. In fact, the entropy of the latter, taken from Latimer [1957LAT], actually refers to the entropy contribution of bound water in hydrates, a more realistic description. However, as noted in the text, a similar calculation for chrysotile (using the entropies of the constituent oxides) gives a value which is 9 % larger than the experimental value, even with the smaller entropy contribution for all the water, so we have not accepted either of these values for the entropy of sepiolite.

[1973DYR/HAN]

The boric acid dissociation equilibrium was studied using a cell fitted with H_2 and Ag|AgCl electrodes without a liquid junction at 5, 15 and 25 °C, and an ionic strength of 0.7 $mol{\cdot}(kg\ NaCl\ solution)^{-1}$ with two additional sets of measurements at ionic strengths of 0.4 and 0.6 $mol{\cdot}(kg\ NaCl\ solution)^{-1}$ at 25 °C. The dissociation constant was calculated with the program LETAGROP in which it is assumed that the units are also $mol{\cdot}(kg\ NaCl\ solution)^{-1}$. These results were re-calculated as molal values and application of the SIT using the $\Delta\varepsilon$ calculated from the ion interaction coefficients in the NEA TDB gave values of $\log_{10}K^o_1 = -9.43$, -9.32 and -9.24 at 5, 15 and 25 °C,

respectively. The disparities between these K_1^o values and those selected in this review amounted to a factor of *ca*. 2. These data are therefore not considered further in this review.

[1973HAV/HOG]

Potentiometric and vapour-pressure measurements of Mg^{2+} in 3 mol·dm^{-3} $NaClO_4$ also containing NaCl, NaBr and NaSCN at 25 °C were undertaken to quantify association between Mg^{2+} and these three anions. The logarithms of the association constants were reported as $-(0.98 \pm 0.19)$, $-(1.45 \pm 0.26)$, and $-(0.91 \pm 0.19)$, respectively. For the former technique, Ag|AgX reference electrodes were used with a number of different working electrodes. The pH was measured with a glass electrode presumably calibrated against commercial buffers. The vapour-pressure measurements were made with a commercial osmometer. So-called "positive" ion measurements were carried out at constant total concentrations of H^+ and ($c_{Na^+} + 2c_{Mg^{2+}}$) so the only change in solution composition during the titrations was the replacement of Na^+ by Mg^{2+}, thereby minimising the liquid-junction potential, which was estimated using the Henderson equation. However, the Mg^{2+} concentration reached almost 1.5 mol·dm^{-3} in some cases. "Negative ion" potentiometric titrations involved replacing perchlorate with the anion of interest up to *ca*. 1 mol·dm^{-3} X^-. Havel and Högfeldt "were tempted to try and interpret differences in the slope" of the linear cell potential *versus* magnesium concentration plots for the three Ag|AgX electrode systems as being due to ion-pair formation. These slopes differed by < 2 mV·dm^3·mol^{-1} despite the fact that the magnesium concentration far exceeded the sodium concentration by the end of the titrations. These results were not considered further.

[1973HOL/GRO]

The authors measured the enthalpy content and enthalpy of fusion of $MgCl_2$ by drop calorimetry. $MgCl_2$(cr) was prepared by heating the hexahydrate in a current of HCl(g) firstly overnight at 648 K, then to 948 K for several hours and finally above its melting point (*ca.* 987 K). Unspecified analysis showed an impurity level of 0.5 %. The results of the 20 measurements at closely spaced temperatures from 906.1 to 1072.0 K (just above and below the melting point) are tabulated and plotted. Their enthalpy content of the solid agreed well with the earlier measurements of [1943MOO] and the assessed data in [1994GUR/VEY], which we have accepted, but the data for the liquid correspond to a lower enthalpy of fusion than those data.

[1973KAT]

Conductance data for $NiSO_4$ solutions were analysed in Section V.5.1.2.1 of [2005GAM/BUG], using a modified version of the Lee-Wheaton equation. Results of the same approach applied to $MgSO_4$ solutions are shown in Table A-39 from 0.0002 to 0.003 mol·dm^{-3}. These re-calculated constants are accepted by this review.

Table A-39: Summary of the original results and those re-calculated according to the Lee-Wheaton equation.

t /°C	K_1° /reported	Λ_0 /reported	K_1° /SIT (2σ)	Λ_0 /SIT	Λ_{obs}-Λ_{calc}
0	(103 ± 1)	68.15	(174 ± 16)	68.74	0.12
5	(108 ± 1)	79.73	(178 ± 16)	80.45	0.16
10	(116 ± 2)	92.15	(183 ± 15)	92.92	0.17
15	(121 ± 2)	105.15	(190 ± 16)	106.06	0.20
20	(129 ± 2)	118.90	(198 ± 16)	119.94	0.25
25	(135 ± 2)	133.30	(205 ± 18)	134.51	0.30
30	(147 ± 3)	148.70	(216 ± 17)	150.03	0.34
35	(151 ± 3)	164.15	(226 ± 22)	165.80	0.41
40	(167 ± 4)	181.20	(244 ± 18)	182.99	0.37
45	(175 ± 3)	198.05	(257 ± 20)	200.18	0.45

[1973KIL/PRO]

The main aim of this paper was to report in a detailed manner the enthalpy of solution of the NBS Standard Reference Material N°1654 (α-quartz or low quartz) for use as reference standard for HF solution calorimetry. The adiabatic platinum calorimeter had been described previously [1973PRO/KIL]. The enthalpy of solution of this material at a concentration of 5 g in 1000 cm^3 of 24.4 wt % HF(aq) was reported as $\Delta_{soln}H$ (353.15 K) = – (2362.2 ± 1.1) J·g^{-1}. This value also applied to the enthalpy of solution of pure α-quartz within the uncertainty limits given. The enthalpy of solution of quartz as a function of temperature in 24.4 wt % HF in the range 298 to 358 K was given as:

$$\Delta_{soln}H_m(T/\mathrm{K}) = -(2275.0 + 1.586\,(T/\mathrm{K} - 298.15))\ \mathrm{J\cdot g^{-1}}.$$

A relation was also given for the enthalpy of solution of α-quartz in aqueous HF in the range 18 to 30 wt %. These results were essential (together with those of [1963WIS/MAR] and [1964GOO/LAC]) for the adoption by CODATA [1989COX/WAG] of $\Delta_f H_m^\circ$ (α-quartz, 298.15 K). However, we shall not discuss them further here, having no reason to question the CODATA assessment. However, in the present paper, [1973KIL/PRO] also reported prior results (obtained in 1966) concerning the transformation α-quartz to quartz glass, from solution calorimetry in 22-23 wt % aqueous HF (*ca.* 400 mL) at about 350 K, using samples of *ca.* 1.5 g. Differences from the calorimeter used in the 1973 study were outlined.

Two batches of crystalline α-quartz (one of Norwegian origin from Hammill and Gillespie, New York and the other of Canadian origin from International Minerals and Chemical Corp, Buckingham, Quebec) were used. They showed the same impurities: 10^{-2} to 10^{-3} % Al and Ti; less than 10^{-2} % Ag, B, Ca, Cu, Fe, Mg and Mn). XRD showed that indeed these samples were α-quartz. The sample of optical quality (Suprasil 2) fused vitreous silica was from Englehard Industries, Hillside, NJ and it contained 0.01 to 0.1 % Ca, Co, Cu and Cr; less than 10^{-3} % Mn. No diffraction lines could be detected in the glass sample by XRD. After hand crushing in a mortar of hardened steel and sieving in a bronze mesh, the particles smaller than 74 μm (#200 sieve) were washed in a 1:1 aqueous HCl solution and decanted for 30 min. A portion of each of these samples was decanted a second time. All these final portions were washed with HCl, rinsed with water until chloride free, with continued washing to remove the finest particles. The portions were then dried for 3-5 hours at 448 K. The following mean values (corrected to 24.4 % HF) of the enthalpies of solution at 353.15 K are given in Table A-40.

In fine, from the analysis of their excellently concordant values, the authors report for the transformation:

$$SiO_2(\text{α-quar, 353.15 K}) \rightleftharpoons SiO_2(\text{vit, 353.15 K}) \qquad \text{(A.135)}$$

$\Delta_r H_m((\text{A.135}), 353.15\ \text{K}) = -(2.330 \pm 0.070)\ \text{kcal·mol}^{-1}$ or $-(9.749 \pm 0.293)\ \text{kJ·mol}^{-1}$.

The authors do not apply any correction to obtain the value at 298.15 K, but as noted in Table V-21 in Section V.3.1.5.2 on vitreous silica, with the heat capacities selected in this review, this only amounts to 0.029 kJ·mol^{-1}.

Table A-40: Enthalpies of solution of silica.

Sample	Number of experiments	$\Delta_{soln}H_m$(353.15 K) in 24.4 wt. % HF (mean value)	
		kcal·mol^{-1}	kJ·mol^{-1}
Norwegian quartz 1 decantation	3	– 33.795	– 141.398
Norwegian quartz 2 decantations	1	– 33.911	– 141.844
Canadian quartz 1 decantation	2	– 33.855	– 141.649
Canadian quartz 2 decantations	5	– 33.909	– 141.875
Vitreous silica 1 decantation	2	– 36.168	– 151.327
Vitreous silica 2 decantations	4	– 36.228	– 151.578

[1973LAR/SOL]

The dissociation constants of the calcium and magnesium complexes with hydroxide, carbonate, and bicarbonate were studied at temperatures of 5, 15, and 25 °C by pH-metric titrations in the ionic strength range of 0.002-0.02 mol·dm^{-3}. The dissociation constants have been determined in terms of activities. The activity coefficients were calculated using the extended Debye-Hückel equation.

In order to determine the dissociation constants of $(Mg/Ca)OH^+$ and $(Mg/Ca)CO_3$ ion pairs the solutions containing $KClO_4$-KOH or $KClO_4$-$KHCO_3$-K_2CO_3 were titrated with $(Mg/Ca)(ClO_4)_2$ solution and changes of pH were followed. Similar titrations of a $KClO_4$-CO_2(aq)-$KHCO_3$ solution with $Mg(ClO_4)_2$ yielded inconclusive results concerning the formation of the $MgHCO_3^+$ species. To avoid this difficulty, an Mg^{2+}-selective electrode was used, instead of a pH-electrode, to follow the concentration of free (uncomplexed) Mg^{2+} ion. From these measurements the authors derived $pK_{d(MgHCO_3)} \sim 1.6$-2.0, which were considered unreliable by the authors themselves.

In general, this is careful work, but the thermodynamic stability constants of $MgOH^+$ determined by the authors ($\log_{10}K^\circ = 2.08$) is considerably smaller than the value selected in this review. Therefore, the uncertainty of the reported constants for the carbonato complexes was increased to ± 0.15.

[1973LIU/NAN]

This determination of the first hydrolysis constant of Mg^{2+} used potentiometric and conductivity methods at 25 °C. Titrations were made under a nitrogen atmosphere and continued until precipitation was detected. The pH measurements were made in mixed $MgCl_2$-NaOH solutions with a glass electrode *versus* a calomel electrode calibrated with buffer solutions prepared according to Bates [1964BAT]. It was assumed that the conductivities of the component ions were additive and a molar conductance of $MgOH^+$ was calculated to give the most consistent hydrolysis constant. No mention was made of the activity coefficients used. The potentiometric and conductivity results are given in Table A-41 and Table A-42 after conversion to molal units.

Table A-41: Summary of potentiometric data for the hydrolysis of Mg^{2+} ions at 25 °C.

$m_{Mg\,total}$	$10^4\, m_{NaOH\,total}$	pH	I_m	$10^4\, m_{MgOH^+}$	pK_{H_2O}	$\log_{10}{}^*K_1^\circ$
0.02311	2.808	9.685	0.0695	1.8694	13.809	11.49
0.02372	2.170	9.664	0.0713	1.2702	13.807	11.64
0.02456	1.498	9.516	0.0738	0.8528	13.805	11.68
0.02519	0.922	9.285	0.0757	0.5406	13.803	11.66

Table A-42: Summary of conductivity data for the association of Mg^{2+} and OH^- ions at 25 °C.

$10^4 m_{Mg\,total}$	$10^4 m_{NaOH\,total}$	I_m	$\log_{10} K_1^\circ$
3.235	1.613	0.00110	2.56
3.209	4.812	0.00137	2.51
3.200	6.399	0.00150	2.54
3.185	3.185	0.00122	2.54
3.176	4.763	0.00135	2.59

From the inadequately described conductance data, the mean value of $\log_{10} K_1^\circ$ was (2.55 ± 0.05) at 25 °C for the association reaction of Mg^{2+} with OH^-. For the potentiometric results it was assumed that the change in the original stoichiometric hydroxide concentration could be attributed entirely to the formation of $MgOH^+$. Activity coefficients are re-calculated according to the SIT without consideration of the ion interaction component due to the low ionic strengths and mixed ionic medium, as well as the scatter in the data. The average of $\log_{10} {}^*K_1^\circ = -(11.61 \pm 0.19)$ compares reasonably well with the value accepted by this review. The reported mean value of $\log_{10} {}^*K_1^\circ$ is – 11.45.

Regarding their brucite solubility data, Liu and Nancollas simply stated: "The solubility product of magnesium hydroxide was obtained by allowing crystal growth and dissolution experiments to proceed to equilibrium. The mean value of pK_{sp} obtained at 25 °C is (10.86 ± 0.03) ($K_{sp} = 1.38 \times 10^{-11}$ mole^3·L^{-3}) corresponding to a total magnesium concentration of 1.67×10^{-4} mole·L^{-1} in pure solution." However, the total magnesium concentration is in error and should have read 1.51×10^{-4} mol·dm^{-3}. The kinetics of crystal growth were also reported showing two distinct stages, both of which could be modelled by rate expressions and the degree of supersaturation. Their solubility product for their uncharacterised brucite was not considered further.

[1973LYA/KRY]

The results of six bayerite solubility experiments at 40, 50 and 60 °C given in terms of g·dm^{-3} Al_2O_3 and Na_2O, were converted by this review to molalities using a modified density function based on that in [2006REY/BER]. Application of the SIT treatment to these results at high ionic strength (I_m = 4.3 to 5.5) yielded acceptable $\log_{10} K_{s,4}$ values with revised uncertainties.

[1973PAP]

This paper summarises and illuminates the prior work on the polymorphism and structures of Na_2CO_3(cr) and K_2CO_3(cr). The author studied these solids by DTA, dilatometry and high-temperature XRD. The materials were 'of high purity' (notably,

free from chlorides) and the studies were carried out in an atmosphere of CO_2(g) to prevent decomposition of the carbonates.

For Na_2CO_3(cr), DTA indicated two phenomena, at 622 and 752 K, and XRD showed that the first transition was a disordering of the monoclinic structure, and the second a transformation to a hexagonal structure. For K_2CO_3(cr), the results indicated two similar transitions at 613 and 694 K; however, the first transition is much less marked than that in Na_2CO_3(cr), there being no noticeable effect in the DTA study, but changes similar to those for Na_2CO_3(cr) in the XRD patterns and the dilatometry were observed.

[1973PIC]

The solubility product of calcite, and the dissociation constants of $CaHCO_3^+$ and $CaCO_3$(aq) ion pairs were determined at 283, 289, 293, 298 and 313 K. The experimental method and original experimental data were published in an earlier paper by the same author [1972PIC]; 10 g calcite (details not specified) were loaded into a 2 dm^3 flask which was almost completely filled with deionised water. CO_2(g) was bubbled for different lengths of time. The flask was then stoppered, sealed and revolved horizontally in a water bath. After five days, the stopper was replaced by another carrying a glass and a saturated calomel electrode in order to determine the pH of the solution (pH values were corrected for liquid-junction potentials). Finally, samples of the solution were analysed for calcium by EDTA titration using Eriochrome Black T.

The experiments were done carefully, and the re-evaluated $\log_{10}K^{\circ}_{s,0}$ (calcite) values are close (± 0.06) to the selected values. The determined formation constants of the $CaCO_3$(aq) ion pair, however, show an opposite temperature dependence to other reports ($\log_{10}K^{\circ}$ ($CaCO_3$(aq)) = 4.00 (283 K), 3.52 (289 K), 3.52 (293 K), 3.40 (298 K), 3.22 (313 K)). Therefore, the latter data are not considered in this review.

[1973PIT/MAY]

Experimental data at 25 °C for 227 pure aqueous electrolytes with one or both ions being univalent are represented within experimental error by a model for dilute solutions up to an ionic strength of 6 mol·dm^{-3}. These include NaH_2PO_4, KH_2PO_4, Na_2HPO_4, K_2HPO_4, Na_3PO_4 and K_3PO_4, and the arsenate analogues.

[1973POW]

A synopsis of this article appeared in [2005GAM/BUG] in which Gamsjäger *et al*. noted that Powell had outlined the limitations of the calorimetric titration method employed by Izatt and coworkers [1969IZA/EAT] with the major concern being the

choice of the activity coefficient model used to derive the formation constant and enthalpy change. Imposing a fixed value of K_1° for $MgSO_4$(aq) association of 178 $dm^3 \cdot mol^{-1}$ from [1958NAI/NAN] led to $\Delta_r H_m^\circ = (5.7 \pm 0.2)$ $kJ \cdot mol^{-1}$ and a K_1° value for $CaSO_4$(aq) of 204 $dm^3 \cdot mol^{-1}$ [1958NAI/NAN] gave $\Delta_r H_m^\circ = (7.8 \pm 0.2)$ $kJ \cdot mol^{-1}$ with different activity-coefficient equations. These are somewhat larger than $\Delta_r H_m^\circ = (2.1 \pm 0.1)$ and (3.39 ± 0.04) $kJ \cdot mol^{-1}$, and $K_1^\circ = 169$ and 269, respectively, reported by Izatt *et al*. For the analogous Ni^{2+} ion-pair equilibrium a value of $\Delta_r H_m^\circ = (5.8 \pm 2.0)$ $kJ \cdot mol^{-1}$ is retained in the NEA with this increased uncertainty. The shortcomings of the treatment of calorimetric titration data have also been pointed out in [1997RAR/CLE].

[1973REE/MEN]

The authors suggested that a synthetic sample of hydroxylapatite has a monoclinic structure (space group $P2_1/b$) at room temperature and from the disappearance of birefringence on a hot-stage microscope inferred that it undergoes a transition to the closely related hexagonal cell (space group $P6_3/m$) at 484.4 K. This was postulated to be an order-disorder transition. Although the authors gave no chemical analysis of the sample, so it is not unambiguously clear that these data refer to pure stoichiometric hydroxylapatite, this structure has been confirmed by the later study of Suda *et al.* [1995SUD/YAS] and other authors have also found a thermal effect on heating hydroxylapatite at very similar temperatures—see Section V.8.2.3.1 in the main text.

[1973SCH/LEV]

The authors studied the transitions in K_2CO_3(cr) by DTA in air and CO_2(g), by high-temperature XRD in air and by DSC in N_2(g). The only transitions observed by DTA in samples heated in CO_2(g) were at 373 and 689 K (with a very small enthalpy change). The peak at 373 K was attributed to the presence of a small amount of the hydrate in the sample. These data were confirmed by the DSC and XRD data, both of which indicated a transition at (693 ± 5) K. A table of the parameters of the monoclinic cell as a function of temperature is given; as the temperature increases, there is a continuous decrease of the β-angle of the low-temperature monoclinic structure to the 90 ° of a hexagonal cell at this temperature. The parameters of the hexagonal cell at 698 K are given as a = 5.70 Å, c = 7.17Å.

[1974CHI/DOW]

Five aqueous binary salt solutions made up of NaCl, KCl, NaH_2PO_4 and KH_2PO_4 were studied by the isopiestic method at 25 °C. All of the data reported are in good agreement with those from other studies and are included in the SIT modelling by this review.

[1974CRA/ROO]

This paper concerns the enthalpies of solution in dilute HCl of hydroxylapatite, fluorapatite, chlorapatite, human enamel, and dicalcium phosphates (both anhydrous and dihydrated $CaHPO_4$). Only the results concerning hydroxylapatite and fluorapatite are discussed in the frame of the present review.

Measurements were made in a commercial calorimeter (LKB 8721-1) (0.1 dm^3 volume) that was calibrated electrically and checked with a suitable NBS standard (Tris(hydroxymethyl)aminomethane, (TRIS), SRM 724a). As to the amount of samples used in the various determinations, the only information available is that they were contained in 1 cm^3 ampoules.

The hydroxylapatite samples were characterised by their $Ca:PO_4$ molar ratio (in all cases well within less than 1 % of the theoretical 1.666), their bulk density, their mean particle size, and their surface area that varied by an order of magnitude from sample to sample.

Dissolution was carried out in 0.1 molar HCl. Results of the enthalpies of solution of four samples differing by their history and characteristics were given as individual numbers with (presumably) one standard deviation, based on an undefined number of measurements. The commercial sample (Victor Div., Stauffer Chemicals, New York) (70.0 $m^2 \cdot g^{-1}$) yielded an enthalpy of solution of – (370.66 ± 0.54) $kJ \cdot mol^{-1}$ for the implied formula $Ca_{10}(PO_4)_6(OH)_2$ (cr), while a sintering of the sample at 1200 °C with a surface area of 1.34 $m^2 \cdot g^{-1}$, resulted in a value of – (335.47 ± 0.04) $kJ \cdot mol^{-1}$. However, no definitive correlation could be established between surface area (or bulk density) and enthalpy of solution, when the samples of various origin are considered. The sample prepared at NBS (22.5 $m^2 \cdot g^{-1}$) according to a described procedure [1968AVN] yielded an enthalpy of solution of – (336.64 ± 2.43) $kJ \cdot mol^{-1}$, while the sample prepared at the Tennessee Valley Authority Laboratories, [TVA Report N°678, Nov. 05, 1956] with a surface area of (3.02 $m^2 \cdot g^{-1}$) gave – (306.69 ± 0.12) $kJ \cdot mol^{-1}$.

In view of the large scatter of the data between different samples, and in absence of any argument to prefer one set to the other, the above results are given for information only.

The fluorapatite sample (0.64 $m^2 \cdot g^{-1}$) was obtained from General Electric Chemical Plant, Cleveland, OH. The authors of the paper indicated that no loss of fluorine was detected during firing of the starting binary component and also mentioned a content of "4.15 % fluorine in fluorapatite", without indicating if this comment was incidental or concerned a value resulting from an analysis; stoichiometric fluorapatite should contain 3.77 % fluorine. The reported $Ca:PO_4$ ratio was 1.70, distinctly different from the stoichiometric 1.666.

The enthalpy of solution of the sample in 1 $mol \cdot dm^{-3}$ HCl was tabulated without details, as – (153.76 ± 0.04) $kJ \cdot mol^{-1}$. Dissolution of the sample in 0.1 and 0.2

mol·dm^{-3} HCl yielded – 155.73 and – 150.08 kJ·mol^{-1} respectively, but, as indicated by the authors, the slow dissolution in these more dilute media affected the reproducibility of the results.

In view of the general context of the reporting of the data in this publication, and also because of the criticism expressed above about the stoichiometry of the fluorapatite sample, this review will also consider the results reported for fluorapatite for information only.

[1974FER/GAL]

This potentiometric study of the ionisation of phosphoric acid in 3 mol·dm^{-3} KCl follows earlier work on the second ionisation equilibrium [1973FER/ANT]. The concentration of phosphate (0.05 to 2.0 mol·dm^{-3}) makes up a significant fraction of the ionic strength, particularly as the pH_c is increased. Nevertheless, Ferroni *et al*. treated the results in terms of four phosphate species, three of which were binuclear anions. If the dissociation constant of $H_3PO_4(aq)$ were unaffected by the mixed ionic medium, then the reported $10^2K_{c,1}$ values of (1.098 ± 0.015), (1.046 ± 0.014), (0.958 ± 0.015) and (0.830 ± 0.011) at 5, 25, 45 and 65 °C, respectively, can be converted to $\log_{10}K_{m,1}$ values of – 1.918, – 1.938, – 1.975 and – 2.037 at I_m = 3.31 using the densities of pure KCl solutions [1985SOH/NOV]. An extrapolation of the SIT treatment well beyond the limit I_m = 1.0 [1974MES/BAE] gives $\log_{10}K_{m,1} = -2.01$ at 25 °C. Despite the reasonable agreement, the use of a mixed electrolyte precludes acceptance of these values, as was the case for the $\log_{10}K_{m,2}$ values in their previous paper [1973FER/ANT].

[1974HAN/ISR]

A cell with a calcium electrode (containing an internal AgC1|Ag reference in $CaCl_2$(sln), reversible to Ca^{2+} across a liquid non-aqueous phase), and a saturated calomel reference electrode with a saturated KC1 bridge, was used at 25 °C to investigate $CaSO_4$ ion pairing. The main focus of this work was on calcium(II) hexacyanidoiron(III) ion pairing with the formation of a $CaSO_4$ ion pair being used as a test case for this cell design. The concentrations of $CaCl_2$ were varied from 1×10^{-4} to 1×10^{-3} mol·dm^{-3} with the assumption that this electrolyte was completely dissociated. Apparently these measurements were repeated with the same concentrations of $CaSO_4$. A value of $K_1^\circ = (234 \pm 8)$ was derived from seven measurements with the SIT treatment and is retained by this review.

[1974JAC/LAN]

The solubility of calcite was measured at 5, 15, 35 and 50 °C in a constant temperature bath in which CO_2 (saturated with water vapour) was bubbled through a solution containing pure water and calcite. The gas-flow rate was constant for any one determination, but was varied from about 1 to 10 $cm^3 \cdot min^{-1}$ for different determinations. Usually, saturation was attained in 1-4 hours as evidenced by constant values of pH and conductivity. Several experiments were run for three months, however, with no further changes observed in pH or conductivity. Equilibrium with calcite was approached from both under- and supersaturation by changing the temperature and/or the CO_2 flow rate. Identical results were obtained when saturation was approached from either direction. Once solutions were saturated with respect to calcite, the final pH and conductivity values were measured and samples were taken for duplicate analysis of the total HCO_3^- concentration (with 0.02 $mol \cdot dm^{-3}$ $KH(IO_3)_2$ standard acid) and total Ca^{2+} concentration (using 0·01 $mol \cdot dm^{-3}$ EDTA).

In addition, Jacobson and Langmuir critically reviewed the available reports on $CaHCO_3^+$ and $CaCO_3$(aq) ion pairs and determined the stability of the former by conductivity measurements in Ca^{2+} containing $KHCO_3$ solutions at 15, 25, 35 and 45 °C.

It was observed that the change in the gas flow from 1 to 10 $cm^3 \cdot min^{-1}$ changed the effective CO_2 gas pressure from approximately 0·8 to 1.2 bar as determined by measuring the pH of dilute KCl and dilute $KHCO_3$ solutions. The authors noted that this could be a potential source of experimental errors in some earlier solubility determinations, where $p(CO_2)$ was only assumed and not measured. Although the authors affirm the existence of $CaHCO_3^+$ and $CaCO_3$(aq) ion pairs (see also [1974REA/LAN]), they also note that, if these ion pairs are taken into account during the evaluation of some earlier reports, the solubility constant of calcite increases with increasing ionic strength (although if only the $CaHCO_3^+$ ion pair is ignored, the above mentioned trend is less important). Due to these difficulties, the authors published two analytical expressions for the temperature dependence of $\log_{10} K^{\circ}_{s,0}(CaCO_3, \text{calc})$ depending on whether the ion pairs are included (or excluded) in the aqueous model.

[1974MES/BAE]

Potentiometric titrations employing a hydrogen-electrode concentration cell were performed from 323.15 to 566.15 K to determine the first two acid dissociation constants of phosphoric acid. Ionic strength was varied from 0.1 to 1 $mol \cdot kg^{-1}$ (KCl). Liquid-junction potentials were estimated using the Henderson equation. Mesmer and Baes fitted their potentiometric data and those of Bates [1951BAT] (273.15 to 333.15 K) for the base hydrolysis reaction. In order to compare their results with the

recommended values in [1992GRE/FUG], the hydrolysis constants were converted to those for Reaction (A.136) using the $\log_{10}K_{H_2O}$ values derived in [1974SWE/MES].

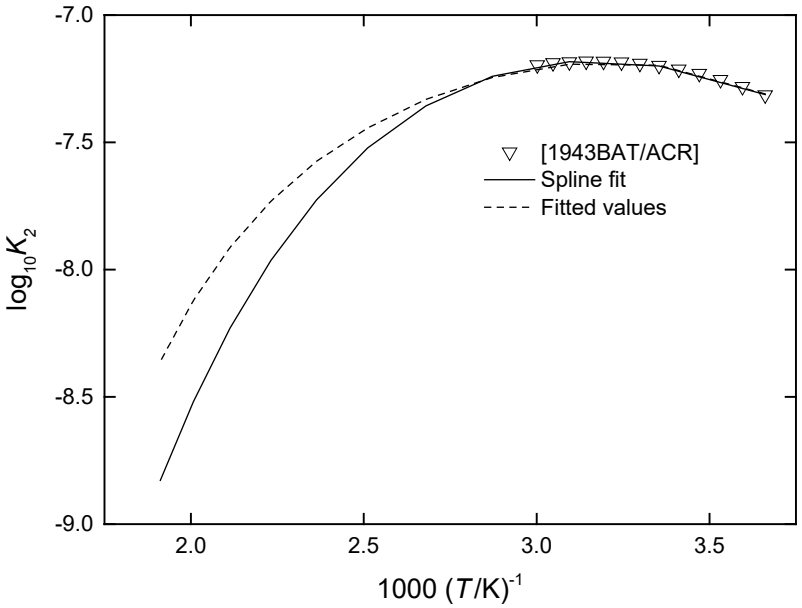
$$H_3PO_4(aq) \rightleftharpoons H_2PO_4^- + H^+ \quad \text{(A.136)}$$

A weighted SIT treatment of the data in [1951BAT], [1974MES/BAE] to 473.15 K yielded: $\log_{10}K_1^\circ = -(2.154 \pm 0.005)$; *cf.* $\log_{10}K_1^\circ = -(2.140 \pm 0.030)$ [1992GRE/FUG] at 298.15 K. In this SIT fit, the ion interaction parameters were considered to be temperature dependent: $\Delta\varepsilon_1 = -(0.05 \pm 0.01)$ and $\Delta\varepsilon_2 = -(0.24 \pm 0.09)$ kg·mol^{-1}. Mesmer and Baes also analysed their experimental data for the second stepwise acid dissociation constant in the basic form of the equilibrium. These results are converted to the acidic form in our SIT analysis:

$$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-} \quad \text{(A.137)}$$

All attempts to apply the SIT treatment to these results yielded ion interaction parameters which were in stark disagreement with those obtained from other sources, [2005PAR/COV] and [2005PAR/COV2] in particular. The high ionic strength, high-temperature data showed the greatest deviations. The extrapolated SIT fit at 298.15 K is compared in the Appendix A entry for [2005PAR/COV2]. The temperature dependence of $\log_{10}K_2$ is illustrated in Figure A-16 and compared to that based only on the data from [1943BAT/ACR], noting that Mesmer and Baes also relied on these low-temperature constants in treating their data.

Figure A-16: Plot of the temperature dependence of $\log_{10}K_2$. The solid curve was generated from a spline fit of the data in Table VI [1974MES/BAE] after conversion from the basic form [1974SWE/MES] while the dashed curve was derived from a fit of the corresponding data in [1943BAT/ACR].



[1974PIT/MAY]

Pitzer and Mayorga made the following statement which is fundamental to the approach taken in this review: "While we have noted that the thermodynamic properties of 2-2 electrolytes do not require the assumption of a discrete chemical species of ion pairs, we recognise that spectroscopic and ultrasonic measurements do indicate a discrete species of inner-shell ion pairs in some cases. Thus, for some purposes it is clear that ion association must be recognised. In many cases, however, the proportion of inner-shell ion pairs is never large, and it may be a satisfactory approximation for thermodynamic purposes to ignore the distinction between inner-shell and solvent-separated ion pairs." They were able to apply their activity coefficient model successfully to electrolytes such as $MgSO_4$ up to a concentration of 3.0 mol·kg^{-1} at 25 °C without the need to include ion association. The corresponding experimental results for $CaSO_4$(sln) were limited so that the treatment was confined to very low concentrations, 0.004 to 0.011 mol·kg^{-1} at 25 °C. Rard and Miller [1981RAR/MIL] reinvestigated the Pitzer model for $CaSO_4$ and $MgSO_4$ solutions incorporating their own isopiestic data. The interaction parameters proposed by Rard and Miller differed only slightly from those reported by Pitzer and

Mayorga, but because of the larger database available to them, the more recent values are preferred here.

[1974PLA]

The solubility of stercorite ($NaNH_4HPO_4 \cdot 4H_2O$) in water and mixtures of sodium and ammonium phosphates was measured at 25 °C. The osmotic coefficients of Na_2HPO_4 and $(NH_4)_2HPO_4$ were also measured at 25 °C relative to an NaCl standard solution. The Na_2HPO_4 dataset proved to be in good agreement with those of an earlier study [1954SCA/BRE] and are included in the SIT treatment by this review with the exception of the lowest concentration, 0.2349 $mol \cdot kg^{-1}$. Interestingly, the osmotic coefficient at the highest concentration used, 2.121 $mol \cdot kg^{-1}$, is in good agreement with the remaining data despite the fact that this solution is supersaturated.

[1974REA/LAN]

Extensive potentiometric measurements of $MgCl_2$-K_2CO_3 and $CaCl_2$-K_2CO_3 solutions at 10, 25, 40 and 50 °C were performed to determine the thermodynamic stability constants of $MgCO_3(aq)$ and $CaCO_3(aq)$ ion pairs.

The experimental procedure is identical to that outlined in [1961GAR/THO]: an aliquot of concentrated HCl was added to a solution of K_2CO_3 to bring the pH to around 10.0, so that the CO_3^{2-}/HCO_3^- mole ratio would be near unity. Various quantities of solid $MgCl_2 \cdot 6H_2O$ or $CaCl_2 \cdot 2H_2O$ were added to these solutions, and the pH change recorded. Activity coefficients for Mg^{2+}, Ca^{2+}, and OH^- were determined using the Debye-Hückel equation at ionic strengths < 0.05, and the mean salt method based on $\gamma(K^+) = \gamma(Cl^-) = \gamma_{\pm}(KCl)$ at higher ionic strength, using mean activity coefficients of $MgCl_2$, $CaCl_2$ and KOH, respectively. Activity coefficients for neutral species were assumed equal to $\gamma(H_2CO_3)$ calculated from the Setchenow expression $\gamma(H_2CO_3) = 10^{0.06 I_m}$.

The plot of pK($MgCO_3(aq)$) *versus* I_m (their Figure 2) shows a significant upward trend at $I_m > 0.05$, which was explained by the probably erroneous expression of $\gamma(MgCO_3(aq)) = 10^{0.06 I_m}$. Therefore, only the data up to $I_m = 0.05$ were used to calculate pK($MgCO_3(aq)$) and pK($CaCO_3(aq)$). This is a careful work, resulting highly reliable pK values.

[1974SIN]

Gibbsite was synthesised and characterised as described later [1976SIN]. The solubility product of gibbsite was measured at 15, 25 and 35 °C (± 0.05 °C) after a two-year equilibration period at each temperature. The pH, which was < 4, of the suspensions

was measured periodically and Al, Na and Cl analyses were performed after centrifugation and filtration. The Al concentration was corrected for hydrolysis according to [1954SCH/TAY]. The ionic strengths converted to molal quantities could only be reconciled in the SIT analysis by assuming the presence of NaCl(sln) (Singh did not state explicitly the electrolyte medium used). The activity coefficient of Al^{3+} was estimated by the Debye-Hückel expression. The $\log_{10}{}^{*}K^{\circ}_{s,0}$ values estimated from the SIT treatment were (8.50 ± 0.12), (7.92 ± 0.12) and (7.41 ± 0.12) at 15, 25 and 35 °C, respectively, *cf.* the average values reported by Singh of (8.59 ± 0.04), (8.04 ± 0.04), and (7.52 ± 0.04). The uncertainties were imposed by this review to be consistent with those in [1966KIT] where the same technique was employed. The SIT values are incorporated into the final data treatment by this review.

[1974TSA/GVE]

The enthalpy content $H(T) - H(298.15\ \mathrm{K})$ of MgO(cr) was measured at thirteen temperatures from 370 to 1591 K. The sample used was a briquette of "optically pure (99.8 %) MgO baked at 1600 K for 5h". There is no mention of any correction for the impurities. The authors fitted the experimental results to the equation:

$$H(T) - H(298.15\ \mathrm{K})\ /\mathrm{J\cdot mol^{-1}} = 46.777(T/\mathrm{K}) + 2.803\times10^{-3}(T/\mathrm{K})^2 + 9.498\times10^{5}(T/\mathrm{K}) - 17308.3 \qquad (298.15\text{-}1600)\ \mathrm{K}$$

These results are in good agreement with those of [1963VIC/DOU], which together form the basis of the selected values in [1994GUR/VEY], which are identical to the CODATA selected values that we have adopted in this review.

[1974VOR/UMY]

This paper concerns the determination of the enthalpies of solution of anhydrous magnesium and lithium chlorides and of their equimolar mixture to determine the enthalpy of interaction of the two chlorides in solutions at the concentration used.

For the present review, we shall only consider the data pertaining to magnesium chloride. This salt was prepared by thermal decomposition of the double salt ammonium magnesium chloride hexahydrate, which was progressively heated under a flow of dry HCl up to 673 K until all ammonium chloride has been driven off. The product was then melted and cooled. Gravimetric analysis by precipitation of AgCl gave 74.12 mass % Cl (calculated 74.47 %). EDTA titration yielded (24.43 ± 0.05) mass % Mg (calculated 25.53 %). A Karl-Fischer determination showed the absence of water within 0.01mass %.

The isothermal jacket calorimeter has been described by the same group. The vessel was a thin-walled platinum beaker with a volume of 0.25 dm^3. The enthalpy of

solution of $MgCl_2$(cr) in 1700 mol H_2O was reported as – (156.12 ± 0.24) kJ·mol^{-1} (2σ), from an average of six determinations. Correction to infinite dilution with the use of the data of Lange and Streeck [1931LAN/STR] leads to the value – (157.49 ± 0.21) kJ·mol^{-1}. These results (with increased uncertainty limits) were used in the CODATA assessment [1989COX/WAG] of $\Delta_f H^\circ_m(Mg^{2+}, aq, 298.15\ K)$.

[1975BUL/IRI]

Bulmer and co-workers used Raman spectroscopy to investigate the hydration structure of a 3.06 mol·kg^{-1} $Mg(NO_3)_2$ solution from – 32 to 99 °C. Three explanations of the observed line-broadening phenomenon with increasing temperature were explored, the favoured one being that a small degree of thermal distortion of the octahedral geometry occurred in the first-hydration shell of the Mg^{2+} cation, rather than being due to ion association.

[1975ELG/WED]

This is the first of three papers on the solubility of $MgSO_4$(s) and $CaSO_4{\cdot}2H_2O$(s) [1978ELG/WED] and [1979ELG/WED] in synthetic sea water whereby the speciation in solution was based on a complete ion-pairing model. Batch experiments with gypsum were described at (25.0 ± 0.5) °C involving low loadings of solid (1 g in 90 cm^3 solution) but for relatively long durations of at least two months under constant shaking. These measurements were made with a variety of solutions containing either one or two electrolytes. The ionic strength was held constant at 0.7 mol·dm^{-3}. Selected solubilities of $CaSO_4{\cdot}2H_2O$(s) at I_c = 0.7 were as follows noting that allowances were made for the formation of $NaSO_4^-$, KSO_4^-, $CsSO_4^-$, $MgCl^+$ ($K_{c,1}$ = 0.48) and $CaCl^+$ ($K_{c,1}$ = 1.2) ion pairs: 0.03992 (NaCl), 0.03194 ($NaClO_4$), 0.04301 (KCl) and 0.04538 (CsCl) mol·dm^{-3}. Their derived $\log_{10} K_{sp}$ value is – 3.542 which resulted in an estimated ion-pair formation constant for $CaSO_4$(aq) of 30.6 kg·mol^{-1}.

[1975LEU/MIL]

The enthalpies of dilution of LiCl, Li_2SO_4, NaCl, Na_2SO_4, KCl, K_2SO_4, $MgCl_2$ and $MgSO_4$ solutions were measured in a batch calorimeter over the range 0.1 to 1.0 mol·kg^{-1} at 30 °C. The relative apparent molal enthalpies, ϕ_L, were estimated from an extended limiting law equation. For the $MgSO_4$ solution, the observed apparent molar enthalpy, ϕ_H, was considered to originate from two contributions, $\alpha\phi_H(Mg^{2+}, SO_4^{2-}) + (1-\alpha)\phi_H(MgSO_4(aq))$, where α is the usual fraction of free ions.

Thus,

$$\phi_L = \phi_H - \phi_H^o \approx \alpha\phi_L(Mg^{2+}, SO_4^{2-}) + (1-\alpha)(\phi_L(MgSO_4(aq)) + \phi_H^o(MgSO_4(aq)) - \phi_H^o(Mg^{2+}, SO_4^{2-}))$$

$$\phi = \alpha\phi_L(Mg^{2+}, SO_4^{2-}) + (1-\alpha)(\phi_L(MgSO_4(aq)) + \Delta_r H_m^o)$$

where $\Delta_r H_m^o$ is the enthalpy of ion-pair formation. The preferred method for estimating $\phi_L(MgSO_4(aq))$ was to use the additivity principle for M = Li, Na and K:

$$\phi_L(Mg^{2+}, SO_4^{2-}) = \phi_L(MgCl_2) + \phi_L(M_2SO_4) - 2\phi_L(MCl) .$$

The fraction of free ions was estimated with an ion-pair formation constant of 223.9 kg·mol^{-1} at 30 °C [1958NAI/NAN]. The resulting value of $\Delta_r H_m^o$ for $MgSO_4(aq)$ formation obtained from this additivity approach was (4.90 ± 0.08) kJ·mol^{-1}, although the maximum expected uncertainty was 0.2 kJ·mol^{-1}; from both approaches for estimating $\phi_L(Mg^{2+}, SO_4^{2-})$. Leung and Millero cited a range of values of $\Delta_r H_m^o$ = (4.81-5.69) kJ·mol^{-1}.

[1975KIN/FER]

Enthalpy increments of $Mg(OH)_2$(cr) down to 298.15 K were determined by drop calorimetry. The sample used was a natural brucite crystal, and after analysis was assumed to contain 98.77 mol % $Mg(OH)_2$, with small amounts (< 0.35 mol %) of each of MgO, $MgCO_3$, SiO_2, Fe_2O_3 and Mn_2O_3. Twelve measurements were made from 350.2 to 699.2 K. The enthalpy values were fitted by an equation, which on differentiation gives:

$$C_{p,m}(Mg(OH)_2, cr, T)/\text{J·K}^{-1}\text{·mol}^{-1} = 101.031 + 1.67360\times10^{-2}(T/\text{K}) - 2.5564\times10^{6}(T/\text{K})^{-2}$$

(298.15-700) K

The enthalpy and entropy increments obtained were combined with the literature data to calculate absolute entropies and Gibbs energy functions and thus to calculate the Gibbs energy for the reaction,

$$MgO(\text{periclase}) + H_2O \rightleftharpoons Mg(OH)_2(\text{brucite})$$

as a function of temperature and pressure. We have re-fitted these data with a constraint of $C_{p,m}(300\text{ K}) = 77.252$ J·K^{-1}·mol^{-1} from the low-temperature measurements of [1937GIA/ARC] to give a $C_{p,m}$ equation with slightly different coefficients, as given in the main text.

[1975MCG/HOS]

The association constant of Mg^{2+} with OH^- was measured potentiometrically as a function of temperature (10, 25, 40, 55, 70 and 90 °C) in $BaCl_2$ solutions where the

barium content was varied by the addition of a $Ba(OH)_2$ titrant. Different glass and reference electrodes were used and were apparently calibrated against commercial buffer solutions. Results were reported for only one titration at each temperature out of the 20 conducted. Some titration curves were discrepant, consistent with CO_2 contamination despite the precautions taken. McGee and Hostetler treated their results with the following assumptions and procedures: formation of $BaOH^+$ was ignored because it occurs *ca*. 2 pH values higher than for $MgOH^+$ formation; high concentrations of Mg(II) were employed to increase the detectability of hydrolysis such that the dominant cations in solution were Mg^{2+} and Ba^{2+}; liquid-junction potentials were ignored although they would be large in these mixed electrolyte solutions. McGee and Hostetler "normalised" their titration curves by assuming that Mg^{2+} hydrolysis is insignificant at neutral pH which is a very reasonable assumption. This normalisation method offsets the presence of protolytic impurities in the $MgCl_2$ starting solution and is almost an essential precaution when dealing with hydrolysis behaviour of metal ions when only a small percentage of the metal is hydrolysed before precipitation occurs (*e.g.*, *ca*. 2 % at 25 °C, run #2); a simple Debye-Hückel expression was used to estimate activity coefficients; corrections were made for the presence of CO_2 in some of the experiments. Their association constants and those re-calculated with the SIT are given in Table A-43.

Table A-43: Summary of the $\log_{10} K_1^\circ$ values for individual titrations (Table 2, 1975MCG/HOS]).

Sam. #	t /°C	$\log_{10} K_1^\circ$ reported	$\log_{10} K_1^\circ$ [(a)] SIT	$-\log_{10} {}^*K_1^\circ$ [(b)] SIT
1	25	2.22		
2	25	2.19	(2.19 ± 0.06)	(11.81 ± 0.06)
3	25	2.20		
4	25	2.16		
5	26	2.14		
6	10	2.15	(2.17 ± 0.02)	(12.35 ± 0.10)
8	10	2.22		
9	40	2.28	(2.33 ± 0.02)	(11.21 ± 0.06)
10	40	2.30		
11	40	2.29		
12	55	2.37		
13	55	2.36	(2.40 ± 0.02)	(10.74 ± 0.06)
14	55	2.38		
15	70	2.46	(2.52 ± 0.01)	(10.28 ± 0.10)
16	70	2.42		
18	90	2.53		
19	90	2.53		
20	90	2.57	(2.61 ± 0.01)	(9.82 ± 0.12)
21	90	2.45		
22	90	2.64		

(a) Uncertainties are statistically based (2σ) and do not reflect the true experimental uncertainties.
(b) The uncertainties were estimated by the reviewer.

[1975SNI/MAN]

The enthalpies of dilution of KCl, $MgCl_2$, Na_2SO_4 and $MgSO_4$ solutions were measured at 40, 60 and 80 °C with additional measurements of $MgCl_2$ solutions at 50 and 70 °C. No mention was made of ion association in the $MgSO_4$ solutions. Mayrath and Wood [1983MAY/WOO] recognised the strong influence of ion pairing in this system and therefore made no attempt to extrapolate the data of Snipes *et al.* to infinite dilution. Results for 0.1 to 2.0 $mol \cdot kg^{-1}$ $MgCl_2$ solutions were also interpreted without consideration of ion pairing.

[1975SUC/CAD]

Potentiometric measurements of the MCl^+ ion pair formation constants, where M = Mg, Ca, Sr and Ba, were reported from 288.15 to 333.15 K using either a chloride ion-selective electrode (Cl-SE) or an Ag|AgCl electrode in conjunction with a calomel reference electrode. Values obtained with both cells were similar at ≥ 303.15 K, but different at ≤ 293.15 K, except for calcium which had similar values at all temperatures. The chloride concentrations were constant at 0.001 $mol \cdot kg^{-1}$ whereas the total metal concentration was either 0.1 or 0.2 $mol \cdot kg^{-1}$, constituting a significant fraction of the total ionic strength, $I_m = 1.0$ ($NaNO_3$). Šucha *et al*. reported standard enthalpies and entropies of reaction at the prevailing ionic strength (activity coefficient corrections were not mentioned). The formation constants were small and the uncertainties large, putting their significance in question. Majer and Štulík [1982MAJ/STU] used similar experimental procedures for the same systems with similar results.

[1975YOK/YAM]

The osmotic coefficients of $MgSO_4$ and $CaSO_4$ solutions were measured by vapour-pressure osmometry at 25 °C. By varying the distance of closest approach parameter in their activity coefficient expression from 0.4 to 1.43 nm, a range of K_1^o values was presented for $MgSO_4$ of 72-194 and 118-219 $kg \cdot mol^{-1}$ for $CaSO_4$ with best estimates given as 110 and 134 $kg \cdot mol^{-1}$, respectively. Rard and Miller [1981RAR/MIL] later noted that these results were not reliable because the experimental method was poorly established. Nevertheless, these data were re-treated here in an iterative procedure. For the $MgSO_4$ data, there is a monotonic increase in the values of K_1^o with concentration, but by limiting the concentration to ≤ 0.03464 $mol \cdot kg^{-1}$, which is similar to the limit imposed on the freezing-point depression data in [1955BRO/PRU] and [1971ISO], $K_1^o = (106 \pm 18)$ $kg \cdot mol^{-1}$ (2σ). For the $CaSO_4$ data, the re-calculated mean value is $K_1^o = (133 \pm 48)$ $kg \cdot mol^{-1}$. Due to concerns about the reliability of the experimental method employed and the variation in the values of the ion-association constant for $MgSO_4$(aq), the results of this study were not considered further.

[1975YOK/YAM2]

This unique spectrophotometric study carried out at 25 °C relied on the molar absorption coefficients of $CuSO_4$ solutions at 250 nm to determine its ion-pair association constant in dilute binary solutions containing H_2SO_4. Ternary solutions containing other metal sulfates were then investigated and the effect of each on the $K_{c,1}$ value of $CuSO_4$(sln) was used to derive their ion-pair association constants in this competitive manner. Ionic strengths of 0.4 mol·dm^{-3} were attained and a simple Debye-Hückel expression with no extended terms was used to determine the activity coefficients of each ionic species. Table A-44 contains results from Tables 2 and 4 in this paper where the uncertainties seem to be significantly underestimated. The estimated K_1° values are within the range of values obtained by other methods, especially for $d = 1.43$ nm, which is the Bjerrum distance used in the NEA TDB analyses. These results were not considered further in this review.

Table A-44: A summary of selected ion-pair association constants at 25 °C and molar absorption coefficients for $CuSO_4$(sln), $MgSO_4$(sln) and $CaSO_4$(sln) as a function of distance of separation of the charges.

d /nm	ε (λ = 250 nm) /cm^2·mol^{-1}	K_1° /$CuSO_4$	K_1° /$MgSO_4$	K_1° /$CaSO_4$
0.40	287	(130 ± 1)	(44 ± 15)	(60 ± 17)
0.50	239	(161 ± 1)	(77 ± 10)	(93 ± 14)
0.60	213	(185 ± 1)	(101 ± 7)	(117 ± 14)
0.70	197	(203 ± 1)	(120 ± 6)	(136 ± 14)
0.80	185	(219 ± 1)	(135 ± 5)	(151 ± 14)
1.00	170	(241 ± 1)	(157 ± 5)	(173 ± 14)
1.20	161	(256 ± 1)	(172 ± 5)	(188 ± 14)
1.43	153	(269 ± 1)	(184 ± 7)	(200 ± 14)

[1976BAE/MES]

Baes and Mesmer based their selected value for the solubility product of $Mg(OH)_2$(cr) on the work of Hostetler [1963HOS], *viz.*, $\log_{10} K_s^\circ = -(11.15 \pm 0.03)$ at 25 °C. They questioned the value for the corresponding molar solubility constant reported by Horn [1969HOR] of $\log_{10} {}^*K_{s,0} = 16.58$ in 3 mol·dm^{-3} $NaClO_4$ because the pH dependence of the solubility led to an unreasonably high value of K_s, *i.e.* $10^{-9.74}$. Their choice of the value for $\log_{10} {}^*K_1^\circ\,(Mg^{2+} + H_2O(l) \rightleftharpoons MgOH^+ + H^+) = -(11.44 \pm 0.1)$ at 25 °C was based on results from [1948STO/DAV] and [1963HOS]. They reviewed the potentiometric study of $MgCl_2$ solutions by Lewis [1963LEW] who maintained a total chloride molarity of 3 mol·dm^{-3} while varying the magnesium concentration from 1 to 1.5 mol·dm^{-3}. Lewis concluded that $Mg_4(OH)_4^{4+}$ was the dominant species in solution with minor amounts of Mg_2OH^{3+}. Baes and Mesmer disputed this speciation scheme, replacing Mg_2OH^{3+} by $MgOH^+$ because the latter hydrolysis constant thus obtained was compatible with the constant derived elsewhere in dilute $MgCl_2$ solutions. They reported the following relationships where m_X is the molality of chloride ions:

$$\log_{10}{}^{*}K_{m,1} = -11.44 - 1.022\sqrt{I_m}\,/(1+\sqrt{I_m}) - 0.034 m_{\mathrm{X}},$$

$$\log_{10}{}^{*}\beta_{4,4} = -39.71 + 2.044\sqrt{I_m}\,/(1+\sqrt{I_m}) - 0.45 m_{\mathrm{X}} \quad \text{(in 1.6 mol·kg}^{-1}\text{ MgCl}_2\text{)},$$

$$\log_{10}{}^{*}K_{s,0} = 16.84 + 1.022\sqrt{I_m}\,/(1+\sqrt{I_m}),$$

$$\log_{10}K_{s} = -11.5 + 3.066\sqrt{I_m}\,/(1+\sqrt{I_m}) - 0.78 m_{\mathrm{X}}$$

For the calcium system, Baes and Mesmer based their derivation of the thermodynamics of formation of the $CaOH^+$ ion pair on the potentiometric work of Harned *et al*. [1937HAR/GEA], [1953HAR/PAX]. Their selected value after consideration of the Cl^-:OH^- activity coefficient ratio was $\log_{10}{}^{*}K_1^{\circ}\,(Ca^{2+} + H_2O(l) \rightleftharpoons CaOH^+ + H^+) = -(12.85 \pm 0.02)$ at 25 °C. Their analysis of $Ca(OH)_2$(s) solubility data was based solely on [1931JOH/GRO] which was also considered to be reliable by this review. They arrived at a value for $\log_{10}{}^{*}K_{s,0}^{\circ}$ of 22.8 at 25 °C.

[1976BER]

Berner's aim was to determine calcite solubility in seawater. However, calcite does not easily equilibrate with seawater (*e.g*., due to the adsorption of dissolution-inhibiting trace species such as orthophosphate, present in seawater). To overcome this problem, the author determined the solubility of aragonite (which behaves more reversibly than calcite) both in pure and seawater (open systems), then the solubility of calcite in pure water. Considering the solubility difference between calcite and aragonite in water (where calcite behaves reversibly), it is possible to calculate the calcite solubility in seawater. The solubility was determined at 5 and 25 °C, from both under- and supersaturated solutions. Gases bubbled through the solution were N_2-CO_2 mixtures containing 100, 10 or 0.32 % CO_2 by volume (the CO_2 content was checked by GC). After equilibration (5-10 hours for calcite, 10-30 hours for aragonite) the pH of the solution was recorded, and determinations of dissolved calcium (with EDTA), dissolved chloride (argentometric titration), and total alkalinity (acid-base titration) were made. The solids used were Fisher reagent calcite and hand-ground aragonite stalactite.

In one run synthetic aragonite was studied. The solubility of the latter was neglected in our re-evaluation, since only a supersaturated solution was examined. Replicate analyses of standards showed a total range of accuracy of ± 2 %. The author noted, that the calculated p_{CO2} values should be slightly lower (not higher) than nominal percent values, because the N_2-CO_2 gas pressure includes the equilibrium vapour pressure of water (0.03 atm at 25 °C; 0.01 atm at 5 °C). Since the total pressure of the laboratory was 1.00 atm, the value of p_{CO2} should have been about 1 to 3 % lower than the nominal 0.32, 10 or 100 %. This was considered in our re-evaluation.

[1976GUR/SOK]

The enthalpies of solution in 1 mol·dm^{-3} HCl of the hydrated calcium borates colemanite and inyoite were measured and the enthalpies of formation for compounds of theoretical stoichiometric composition were reported to be $\Delta_f H^o_m$ ($CaB_3O_4(OH)_3 \cdot H_2O$, cr, 298.15 K) = – (3469.8 ± 2.5) kJ·mol^{-1}, for colemanite and $\Delta_f H^o_m$ ($CaB_3O_3(OH)_5 \cdot 4H_2O$, cr, 298.15 K) = – (4647.2 ± 3.6) kJ·mol^{-1}, for inyoite.

The authors ignored unspecified amounts of impurities associated with the aluminium and iron oxides in these minerals, and expressed the results of the chemical analysis of the samples used as: colemanite: $Ca_{0.97}B_{3.02}O_4(OH)_3 \cdot 0.87H_2O$ and inyoite $Ca_{1.02}B_{2.99}O_3(OH)_5 \cdot 3.94H_2O$. For the composition of the latter, the reviewers note that a minor correction imposed by electroneutrality requirements leads to either $Ca_{1.015}B_{2.99}O_3(OH)_5 \cdot 3.94H_2O$ or $Ca_{1.02}B_{2.9866}O_3(OH)_5 \cdot 3.94H_2O$.

This publication can be criticised on several counts.

1. It is impossible, with the information furnished, to decide whether theoretical or actual compositions of the samples were used in the calculation of the enthalpies of solution or if and how a correction to theoretical composition was made at some stage.

2. The discussion of the statistical approach used to obtain the best value for the enthalpies of solution is not clear.

3. A number of equations used in the calculations and the use of the literature data are ambiguous.

The isoperibol calorimeter used in this study has been described in [1972GUR/SOK]. It has a volume of 92 cm^3. The mass of the samples ranged between 0.17495 and 0.18410 g for colemanite and 0.23100 and 0.24598 g for inyoite.

Given the size (1.5 cm^3) of the calorimetric bulbs used, this review estimates the volume of 1 mol·dm^{-3} HCl used in the experiments to be *ca.* 90 cm^3. Thus, for colemanite, samples averaging 0.18 g correspond to 8.757×10^{-4} mole (using the molecular weight of 205.55 for the theoretical composition). For reasons that will be apparent below, we note here that such a concentration corresponds to the dissolution of 0.0491 g CaO in 0.092 dm^3 or 1 g CaO in 1851 g solvent (assuming a density of *ca* 1.01 g·cm^{-3}).

In view of the above, the reviewers employed the following approach:

1. Use as best value for the enthalpy of solution of the compounds in 1 mol·dm^{-3} HCl the simple averages of the data reported (Table 2 in [1976GUR/SOK]) This gives, for colemanite (eight measurements): $\Delta_{sol}H_m$ (colemanite, 1 mol·dm^{-3} HCl, 298.15 K) = – (4.027 ± 0.060) kJ·mol^{-1} (2σ) and for inyoite (nine measurements): $\Delta_{sol}H_m$(inyoite, 1 mol·dm^{-3} HCl, 298.15 K) = (30.076 ± 0.173) kJ·mol^{-1} (2σ).

2. Use, following [1976GUR/SOK], the enthalpies of solution of CaO(cr) and B_2O_3(cr), in 1 mol·dm^{-3} HCl by Torgeson and Shomate [1947TOR/SHO]. These authors report that the quantities of CaO(cr) used in their experiments correspond to 1 g in 1.8455 kg of acid. Disregarding small differences in dilution effects that we shall neglect in view of other larger uncertainties affecting the cycles to be used, the results in [1947TOR/SHO] can indeed be used as such in the calculations that follow. The authors report, based on six measurements, $\Delta_{sol}H^{o}_{m}$ (CaO, cr, 1 mol·dm^{-3} HCl, 298.15 K) = – (194.054 ± 0.065) kJ·mol^{-1} (the 2σ uncertainty being re-calculated by this review). These authors also report 11 determinations of the enthalpy of solution of B_2O_3(cr), using samples ranging from 0.4 to 2.47 g, thus adequately covering the range of concentrations used in [1976GUR/SOK]. As it is clear that the enthalpies of solution do not vary significantly over the concentration range investigated, we have taken the average of these 11 values, namely, $\Delta_{sol}H^{o}_{m}$ (B_2O_3, cr, 1 mol·dm^{-3} HCl, 298.15 K) = – (13.799 ± 0.038) kJ·mol^{-1} (2σ).

3. Use of $\Delta_{f}H^{o}_{m}$ (CaO, cr, 298.15 K) = – (634.92 ± 0.90) kJ·mol^{-1} and $\Delta_{f}H^{o}_{m}$ (B_2O_3, cr, 298.15 K) = – (1273.5 ± 1.4) kJ·mol^{-1} from [1989COX/WAG].

4. Use of the CODATA compatible partial molar enthalpy of formation of H_2O in 1 mol·dm^{-3} HCl, – (285.848 ± 0.041) kJ·mol^{-1} interpolated from [1982WAG/EVA].

The enthalpy of solution of stoichiometric colemanite in 1 mol·dm^{-3} HCl (solution) may thus be written as:

$$CaO{\cdot}1.5B_2O_3{\cdot}2.5H_2O(cr) \rightleftharpoons \{CaO + 1.5\ B_2O_3 + 2.5\ H_2O\}(sln),$$

leading to $\Delta_{f}H^{o}_{m}$ ($CaB_3O_4(OH)_3{\cdot}H_2O$, 298.15 K) = – (3470.52 ± 2.29) kJ·mol^{-1} through the use of $\Delta_{sol}H^{o}_{m}$ (colemanite, 1 mol·dm^{-3} HCl, 298.15 K) = – (4.0269 ± 0.0602) kJ·mol^{-1} (2σ) calculated above.

The same calculation (using, of course, the only available experimental enthalpy of solution of the compound) for the non-stoichiometric composition yields:

$$\Delta_{f}H^{o}_{m}\ (CaO_{0.97}B_{3.02}O_4(OH)_3{\cdot}0.87H_2O,\ 298.15\ K) = -\ (3421.36 \pm 2.29)\ kJ{\cdot}mol^{-1}.$$

Similar calculations for inyoite give:

$$\Delta_{f}H^{o}_{m}\ (CaB_3O_3(OH)_5{\cdot}4H_2O,\ cr,\ 298.15\ K) = -\ (4648.01 \pm 2.31)\ kJ{\cdot}mol^{-1}$$

assuming a stoichiometric composition or

$$\Delta_{f}H^{o}_{m}\ (Ca_{1.02}B_{2.9866}O_3(OH)_5{\cdot}3.94H_2O,\ cr,\ 298.15\ K) = -\ (4638.81 \pm 2.31)\ kJ{\cdot}mol^{-1}$$

or

$$\Delta_{f}H^{o}_{m}\ (Ca_{1.015}B_{2.99}O_3(OH)_5{\cdot}3.94H_2O,\ cr,\ 298.15\ K) = -\ (4636.86 \pm 2.31)\ kJ{\cdot}mol^{-1}$$

for the possible electroneutral non-stoichiometric compositions.

In view of the lack of definitive information on the way the authors handled their experimental data, as noted above, we suggest their re-calculated results for the stoichiometric compounds be accepted, with increased uncertainty limits, namely:

$$\Delta_f H^o_m (CaB_3O_4(OH)_3 \cdot H_2O, \text{cr}, 298.15\ \text{K}) = -(3470 \pm 20)\ \text{kJ}\cdot\text{mol}^{-1} \text{ and}$$

$$\Delta_f H^o_m (CaB_3O_3(OH)_5 \cdot 4H_2O, \text{cr}, 298.15\ \text{K}) = -(4648 \pm 12)\ \text{kJ}\cdot\text{mol}^{-1}.$$

[1976KAT]

Conductivity measurements were reported of the ion-association constants of calcium, cobalt, zinc and cadmium sulfates in aqueous solutions at temperatures from 0 to 45 °C at 5 ° intervals. Katayama used the revised Fuoss-Onsager equation to treat the data, which incorporated 10 concentrations from (2×10^{-4}) to (3×10^{-3}) mol·dm^{-3}. The SIT treatment was applied by this review based on the modified Lee-Wheaton equation. The results, which are given in Table A-45, are internally consistent, more so than those from a more recent conductance study [2008BES].

Table A-45: Reported and re-calculated limiting molar conductances (Λ_o)/S·cm^2·mol^{-1} and $CaSO_4$(aq) formation constants (uncertainties in the Lee-Wheaton values are 2σ).

t /°C	1/2 Λ_o /S·cm^2·mol^{-1} (reported)	$K°$ (reported)	1/2 Λ_o /S·cm^2·mol^{-1} (Lee-Wheaton)	$K°$ (Lee-Wheaton)
0	72.25	(127 ± 5)	73.03	(211.9 ± 22.3)
5	84.35	(130 ± 4)	85.29	(217.3 ± 23.1)
10	97.25	(134 ± 3)	98.38	(224.0 ± 24.1)
15	109.75	(139 ± 3)	112.07	(232.2 ± 24.0)
20	124.95	(146 ± 2)	126.47	(241.0 ± 23.9)
25	139.85	(150 ± 3)	141.66	(251.5 ± 24.9)
30	155.45	(157 ± 5)	157.65	(265.2 ± 30.3)
35	171.25	(165 ± 2)	173.73	(277.0 ± 27.9)
40	188.00	(179 ± 3)	190.74	(291.3 ± 28.3)
45	204.55	(184 ± 5)	207.81	(306.0 ± 32.7)

[1976LEA/SMI]

The specific heat of cristobalite was measured by adiabatic calorimetry from 313 to 653 K, (thus covering the α-β transition) using a 60 g sample, obtained by heating silica for four hours at 1690 °C. The temperature increments varied from 0.4 K to the surprisingly large value of 12 K. The results are presented only as smoothed values at 10 temperatures from 273 to 673 K. The precision of the results was about ± 0.1 % and the absolute accuracy was estimated to be ± 0.5 %. More detailed measurements were made close to the transition temperature, but these are presented only graphically; these results were less precise (*ca.* 1 %) owing to the smaller temperature steps used and the rapidly changing heat capacity. The results indicated that the transformation region of a bulk sample is spread over a range of temperature and shows marked hysteresis,

suggesting the that individual crystallites have their own transformation temperature, presumably due to different degrees of ordering.

Richet *et al.* [1982RIC/BOT] obtained the detailed results of this study from the authors and included them in their assessment of the data for cristobalite, which we have accepted; the values agree excellently (within 0-0.2 $J \cdot K^{-1} \cdot mol^{-1}$ = < 0.4 %) at the 10 temperatures tabulated by [1982RIC/BOT].

[1976LIC/PIC]

The structure of $CaCl_2$(sln) was established from X-ray diffraction measurements in 1, 2 and 4 $mol \cdot dm^{-3}$ solutions at (18.0 ± 0.5) °C. A coordination number of 6 and octahedral geometry were proposed for Cl^- and Ca^{2+} ions in solution. The Ca-OH_2 distance was given as 0.240-0.245 nm and no ion-ion interactions were detected.

[1976PLA]

The osmotic coefficients of the system H_3PO_4-NaH_2PO_4-H_2O were measured at 0 and 25 °C relative to an NaCl standard solution. The osmotic coefficients of pure NaH_2PO_4 solutions at 25 °C are in very good agreement with those of other earlier studies and are accepted for inclusion in the SIT treatment by this review.

[1976REA/LAN]

Existing solubility data at 25 °C for $CaSO_4 \cdot 2H_2O$ in pure water and in various salt solutions were used to estimate the activity coefficients of the $CaSO_4(aq)$ ion pair. Reardon and Langmuir contended that from the existing experimental solubility data for gypsum [1903CAM/SEI], [1903CAM/BRE], [1905SUL], [1935HIL/YAN], [1938HIL/WIL] and [1966MAR/SLU] and knowledge of the association constant for the $CaSO_4(aq)$ ion pair [1932MON/DAV], [1953BEL/GEO] and [1967NAK/RAS], a consensus existed that $\log_{10} K^\circ_{sp} = -(4.59 \pm 0.03)$ and $\log_{10} K^\circ_1(CaSO_4, aq) =$ (2.31 ± 0.04). Based on these data, they concluded that assuming the activity coefficient of the $CaSO_4(aq)$ ion pair was unity was incorrect. Their treatment included values for $M^+ SO_4^{2-}$ ion pair constants and the assumption that the activity coefficient of the anion so formed was equivalent to that of chloride ion. This in turn relied on the extra-thermodynamic assumption that $\gamma_{K^+} = \gamma_{Cl^-} = \gamma_{\pm KCl}$. They proposed the equation:

$$\log_{10} \gamma_{CaSO_4(aq)} = -(0.45 \pm 0.15) I_m \qquad \text{(A.138)}$$

Therefore, for example, at I_m = 1 inclusion of a non-unity activity coefficient for $CaSO_4$(aq) is equivalent to lowering the ion-pair formation constant by a factor of almost 3. However, more recent gypsum solubility and ion pairing experiments, where

the latter involved a wide variety of techniques, showed that the thermodynamics of these equilibria are significantly less well-established than was proposed by Reardon and Langmuir. The value of $\gamma_{NiSO_4(aq)}$ was assumed to be unity in [2005GAM/BUG] and it is assumed in this review that $\gamma_{CaSO_4(aq)} = 1$.

[1977BUS/MES]

A hydrogen-electrode concentration cell was used to monitor the pH during titrations of silicic acid in NaCl solutions to 300 °C. The initial molality of Si was varied from 0.003 to 0.05 mol·kg^{-1}. The Henderson equation was used to estimate liquid-junction potentials. Application of the SIT by this review for $m_{Si} \approx 0.003$ gave the results in Table A-46 where the dominant reaction is:

$$Si(OH)_4(aq) + OH^- \rightleftharpoons SiO(OH)_3^- + H_2O(l) \quad \text{(A.139)}$$

Table A-46: Results of an isothermal SIT treatment of the potentiometric data given in Table II of this paper, limited to ≤ 200 °C. The uncertainties are (2σ) statistically based.

t /°C	$\log_{10}K^\circ$ (A.139)	$\Delta\varepsilon$ /kg·mol^{-1}	r^2 (fit factor)
60	(3.692 ± 0.024)	– (0.081 ± 0.012)	0.914
100	(3.234 ± 0.016)	– (0.071 ± 0.006)	0.961
150	(2.820 ± 0.014)	– (0.057 ± 0.002)	0.967
200	(2.505 ± 0.014)	– (0.047 ± 0.002)	0.958

Equations (A.140) and (A.141) were used to fit the data in Table A-46:

$$\log_{10}K^\circ(\text{A.139}) = -(0.295 \pm 0.065) + (1321.06 \pm 26.44)(T/\text{K})^{-1} \quad \text{(A.140)}$$

$$\Delta\varepsilon\ (\text{A.139}) = -(0.163 \pm 0.005) + (2.465 \pm 0.117) \times 10^{-4}(T/\text{K}) \quad \text{(A.141)}$$

Therefore, at 25 °C: $\log_{10}K^\circ(\text{A.139}) = (4.14 \pm 0.05)$ (or $\log_{10}{}^*K^\circ(\text{A.142}) = -(9.86 \pm 0.05)$) and $\Delta\varepsilon\ (\text{A.139}) = -(0.089 \pm 0.003)$ kg·mol^{-1}. From Equation (A.140), $\Delta_r H_m^\circ(\text{A.139}) = -(25.3 \pm 1.0)$ kJ·mol^{-1} at 25.0 °C. Grenthe *et al*. [1992GRE/FUG] applied the SIT to data collected in $NaClO_4$ and NaCl media from [1976BAE/MES], [1977BUS/MES], [1981SJO/NOR] and [1983SJO/HAG] to derive values of $\log_{10}{}^*K^\circ(\text{A.142}) = -(9.81 \pm 0.02)$. Busey and Mesmer [1977BUS/MES] proposed $\log_{10}K^\circ(\text{A.139}) = (4.168 \pm 0.022)$, $\Delta_r H_m^\circ(\text{A.139}) = -(30.21 \pm 0.075)$ kJ·mol^{-1} and $\Delta_r C_{p,m}^\circ(\text{A.139}) = (49.4 \pm 7.3)$ J·K^{-1}·mol^{-1} at 25.0 °C (all corrected to 2σ uncertainties) based on an empirical six-parameter fitting equation.

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \quad \text{(A.142)}$$

Since the later potentiometric study by Sjöberg *et al*. [1981SJO/NOR] at 25 °C and I_c = 0.6, gave $\log_{10}{}^*K_1\ (\text{A.142}) = -(9.472 \pm 0.002)$, an alternative fit was conducted that included all the data utilised above and the value from Sjöberg *et al*., resulting in the equation:

$$\log_{10}{}^{*}K^{\circ}\,(\text{A.142}) = -(0.403 \pm 0.032) + (1365.4 \pm 11.7)(T/\text{K})^{-1} - ((0.019 \pm 0.014) - (32.4 \pm 5.5)(T/\text{K})^{-1})I_m \quad \text{(A.143)}$$

From Equation (A.143) at 25 °C: $\log_{10}{}^{*}K^{\circ}\,(\text{A.142}) = -(9.82 \pm 0.05)$, $\Delta_r H^{\circ}_m\,(\text{A.142}) = -(26.1 \pm 0.5)$ kJ·mol^{-1} and $\Delta_r S^{\circ}_m\,(\text{A.142}) = -(7.7 \pm 1.2)$ J·K^{-1}·mol^{-1}.

Similar potentiometric experiments were reported to determine the second dissociation constant to form $SiO_2(OH)_2^{2-}$ (Reaction (A.144)) in 1 mol·kg^{-1} NaCl using higher Si concentrations (*viz.*, 0.005 to 0.05 mol·kg^{-1}) at the same temperatures. Treatment of these data required the inclusion of an additional Si(IV) species, *viz.*, $Si_4O_6(OH)_6^{2-}$, which gave a marginally better fit than either $Si_3O_6(OH)_3^{3-}$, $Si_5O_9(OH)_4^{2-}$ or $Si_6O_{10}(OH)_7^{3-}$. Values of $\log_{10}{}^{*}K_m\,(\text{A.144})$ were tabulated together with those for the formation of $Si_4O_6(OH)_6^{2-}$. Busey and Mesmer did not recommend the latter. In calculations of the 60 °C data, small variations in the values of $\log_{10}{}^{*}K_m\,(\text{A.144})$ were noted depending on the choice of polysilicate species.

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \quad \text{(A.144)}$$

A weighted regression yielded a value of $\log_{10}{}^{*}K_m\,(\text{A.144})$ at 25 °C of (4.98 ± 0.23) and $\Delta_r H_m\,(\text{A.144}) = -(40.4 \pm 12.4)$ kJ·mol^{-1} (2σ), whereas the authors reported $\Delta_r H_m\,(\text{A.144}) = -(36.4 \pm 6.3)$ kJ·mol^{-1}. A value of $\log_{10}{}^{*}K_m\,(\text{A.144})$ at 25 °C of (4.98 ± 0.60) was included in the final SIT assessment with the increased uncertainty to reflect the experimental uncertainty obtained directly at 25 °C.

[1977CHA]

Chang measured the heat capacity of a sample of synthetic sapphire, α-Al_2O_3 (NBS Standard Reference material 720) from 8.61 to 371.3 K in a fully automated vacuum adiabatic calorimeter. The heat capacities were extrapolated below 10 K by assuming a T^3 dependence in this region. The results are in excellent agreement with earlier measurements at the National Bureau of Standards, and all these were combined to provide a table of $C_{p,m}$, $(S - S(0\text{ K}))$, $(H - H(0\text{ K}))$ and $(G - H(0\text{ K}))$ at round values from 0-2250 K, based solely on measurements at this laboratory. The values at 298.15 K are $C_{p,m}$ = 79.06 J·K^{-1}·mol^{-1}, S_m = 50.95 J·K^{-1}·mol^{-1}, essentially the same as the current CODATA values of (79.033 ± 0.200) and (50.92 ± 0.10) J·K^{-1}·mol^{-1}. These heat capacities are also compared to those predicted from the simple three-term equation proposed by Reshetnikov [1966RES] to represent the heat capacity from 0 to 2180 K, which is surprisingly accurate for a relatively simple expression—see the respective Appendix A entry.

[1977FIS/FOX]

Conductance measurements were reported at 25 °C of Na_2SO_4, K_2SO_4 and $MgCl_2$ solutions as a function of pressure (0.10133-202.66 MPa). The stoichiometric concentration range for these measurements was *ca*. 10^{-4} to 0.01 mol·dm^{-3}. It must be recognised that for cells operating at elevated pressures, the cell constants cannot be derived with the same accuracy as standard glass cells. The following results for $MgCl_2$(sln) are taken from Table III in this paper: for the dissociation of $MgCl^+$ at 1 atm $K_1^o = 0.22$ (A) or 0.42 (B) kg·mol^{-1}, $\Delta_r V^o = -5.9$ (A) or -3.1 (B) cm^3·mol^{-1}. Here (A) and (B) refer to the results of two models involving two distances-of-closest-approach differing by a factor of two in the two conductance equations employed for 1:1 electrolytes (*i.e.*, the ion pair) and 2:1 electrolytes (*i.e.*, the free ions). The standard Debye-Hückel equation was utilised to estimate activity coefficients. The Achilles' heel in their treatment appears in the simple statement: "We have set Λ^o for the ion pair to be 0.5 that of the divalent ion". A value for $\Lambda^o(MgCl^+)$ of 101.8 S·cm^2·equiv^{-1} is given.

[1977HEM/ROB]

The solution calorimeter used in this investigation and the methods have been described earlier by the same authors [1972HEM/ROB]. The present comments will be limited to gibbsite as the other compounds studied, low albite and $NaAlO_2$, are outside the scope of this review.

Measurements on gibbsite were conducted at 303 K. The actual measurements corresponded to a stoichiometry of 0.005 mol aluminium metal in 920.0 g of 20.1 wt % HF (Fisher reagent A-147).

The aluminium metal used was in the form of powder filings from certified material (Fisher A-557, 99.85 % Al) that contained 0.07 % Si, 0.006 % Cu, 0.03 % Fe, 0.01 % Ti and 0.03 % insoluble in dilute HCl. Gibbsite was a sample prepared from chemically pure aluminium hydroxide from Baker, using an unpublished method; the sample gave an XRD pattern identical to gibbsite. The material was heated for six hours at 423 K prior to use, which perhaps risked some decomposition to boehmite. Data were reported using 1971 atomic weights, but the corrections to 1981 atomic weights would be insignificant.

The reactions used were, at 303.3 K:

$$Al(cr) + [1849HF + 8163\ H_2O](sln) \rightleftharpoons [AlF_3 + 1846HF + 8163H_2O](sln) + 3/2\ H_2(g) \quad (A.145)$$

$$3H_2O(l) + [AlF_3 + 1846\ HF + 8163\ H_2O](sln) \rightleftharpoons [AlF_3 + 1846\ HF + 8166H_2O](sln) \quad (A.146)$$

and

$$Al(OH)_3(cr) + [1849\ HF + 8163H_2O](sln) \rightleftharpoons [AlF_3 + 1846HF + 8166H_2O](sln) \quad (A.147)$$

For Reaction (A.145) the average value of the enthalpy change from eight measurements was reported as – (592.952 ± 1.192) kJ·mol^{-1}. However, examination of the individual results shows that the reported uncertainty of the above value corresponds to 1σ only.

After correction (as calculated by the authors) for the vaporisation of the solvent by the evolved hydrogen, and taking the 2σ uncertainty limits, the value of the enthalpy change for Reaction (A.145) becomes – (595.195 ± 2.384) kJ·mol^{-1}.

Similarly, the average value of the enthalpy change for Reaction (A.146), based on three measurements becomes – (0.622 ± 0.018) kJ·mol^{-1} and that for Reaction (A.147), based on four measurements becomes – (159.572 ± 0.367) kJ·mol^{-1}.

The enthalpy change for Reaction (A.148) at 303.3 K:

$$\mathrm{Al(cr) + 3H_2O(l) \rightleftharpoons Al(OH)_3(cr) + 3/2\ H_2(g)} \qquad \text{(A.148)}$$

is obtained from Reactions (A.145), (A.146) and (A.147) as – (436.245 ± 2.412) kJ·mol^{-1}.

The authors corrected the enthalpy change for Reaction (A.148) by 0.615 kJ to bring its value to 298.15 K. The correction was based on their own heat capacity data (unpublished when the paper was written, but subsequently published in [1977HEM/ROB3]) and the other required heat capacities from [1976WAG/EVA]. Given the uncertainty affecting Reaction (A.148) we opted not to refine this correction any further.

Use of the standard enthalpy of formation of water from CODATA, leads to

$$\Delta_f H^\circ_m(\mathrm{Al(OH)_3,\ gibb,\ 298.15\ K}) = -\ (1293.120 \pm 2.416)\ \mathrm{kJ\cdot mol^{-1}}.$$

[1977HEM/ROB3]

Hemingway *et al.* used an adiabatic calorimeter to measure the specific heat of fine crystals of $MgCO_3$(cr) from 12.7 to 384 K and of gibbsite $Al(OH)_3$(cr) from 13.4 to 379 K. Additional DSC measurements on gibbsite were made from 340 to 480 K.

The sample of magnesite was a very pure synthetic material prepared by reacting MgO(cr) with CO_2(g) at a pressure of 10 MPa at 548 K for three weeks. Analysis showed it contained 48.01 wt % MgO (theoretical 47.80 %) and 51.97 wt % CO_2 (theoretical 52.19 %) with 0.02 wt % water. The full experimental data (*ca.* 90 measurements) are tabulated, and also smoothed values from 5 to 380 K. The calculated values of the standard heat capacity and entropy are $C^\circ_{p,m}(298.15\ \mathrm{K}) = 76.09\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$ and $S^\circ_m(298.15\ \mathrm{K}) = (65.09 \pm 0.13)\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$. As noted in Section V.6.1.1.5.2 of the main text, in view of the high purity of the sample and the careful investigation, these data, with minor changes in C_p, are the preferred values, and the

latter value was used in the optimisation of the data for $MgCO_3$(cr)—see Section V.6.1.1.6 of the main text.

The sample of gibbsite was prepared from reagent-grade $Al(OH)_3$, whose major cationic impurity was Na^+, by repeated refluxing with distilled water until the aqueous Na^+ level was reduced to that of a control blank. The average particle size varied from 3 to 125 μm, with an average of 30 μm. Only lines of gibbsite were found by XRD. Trivial corrections were made to the heat capacities for the slight deviation of the composition of the gibbsite from the ideal stoichiometry and for minute amounts of impurities (assumed to be Al_2O_3, Fe_2O_3, $FeCO_3$ and NaOH). The full experimental data of the measurements from both the adiabatic calorimeter (72 points) and DSC (five points) are tabulated, together with smoothed values from 5 to 480 K. The heat capacities agreed satisfactorily in the small overlap range.

These heat capacity values agreed with the earlier measurements from 52 to 296.5 K by Shomate and Cook [1946SHO/COO], and as noted in Section V.5.1.2.1.2 of the main text, in view of the high purity of the sample and the careful investigation, the data of [1977HEM/ROB3], with minor changes in C_p, are the selected values. The calculated values of the standard heat capacity and entropy are $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K}) = (91.70 \pm 0.30)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ and $S^{\circ}_{\mathrm{m}}(298.15\ \mathrm{K}) = (68.44 \pm 0.14)\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$, where the uncertainty in C°_p is estimated by this review. The tabulated heat capacities above 298.15 K have been fitted by the equation:

$$C_{p,m}(\mathrm{Al(OH)_3, gibb}, T)\ /\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}} = 77.39936 + 9.87237\times10^{-2}(T/\mathrm{K}) + 5.99677\times10^{-5}(T/\mathrm{K})^2 - 1.819167\times10^{6}(T/\mathrm{K})^{-2} \qquad (298.15\text{-}500\ \mathrm{K})$$

This paper also reports measurements on benzoic acid that are not of direct interest to this review.

[1977MCG/HOS]

Forty two brucite solubility experiments were conducted at 10, 25, 40, 55, 70 and 90 °C with 18 involving dilute HCl solutions (initially 0.005-0.08 $\mathrm{mol{\cdot}dm^{-3}}$) and the remainder involving pure water. Brucite, obtained as reagent grade material, was treated with HCl for several hours, extensively water washed and finally aged as a slurry for three to four weeks at 90 °C. The pH of the suspension was first measured with a combination glass electrode calibrated against commercial buffers and a further sample was taken *via* a 0.45 μm filter for magnesium analysis. McGee and Hostetler provided detailed experimental results for only six of their 42 experiments, one at each temperature, and these results indicated that total equilibration times of up to several hundred hours were allowed. Analysis of the results was done either using their earlier results for K°_1 [1975MCG/HOS] to correct for the amount of $MgOH^+$ present or by extracting K°_1 from their data for each experiment. The former method proved to be more reliable. Activity coefficients were estimated using the Debye-Hückel expression

which was adequate for these dilute solutions. They reported $\log_{10}K_s^o$ = – (10.88 ± 0.10) at 25 °C which was essentially the same as an average of the last four data points, – (10.92 ± 0.06). Application of the SIT to treat the available data was simplified by the low ionic strengths which ensured that contributions of a_{H_2O} and $\Delta\varepsilon$ were negligible. The reported $\log_{10}K_s^o$ values and the corresponding values calculated from the SIT treatment are listed in Table A-47, together with the corresponding $\log_{10}{}^*K_{s,0}^o$ values. It is apparent that equilibrium was established before the first sample was withdrawn so that the values listed in column 5 of Table A-47 are retained. These $\log_{10}{}^*K_{s,0}^o$ values are in accord with those of later reliable studies.

Table A-47: Summary of the reported results for the runs conducted with no added HCl (uncertainties are quoted as 2σ and are statistically based).

Run #	t /°C	$-\log_{10}K_s^o$ /reported	$-\log_{10}K_s^o$ /overall average	$-\log_{10}K_s^o$ /SIT [(a)]	$-\log_{10}K_s^o$ /SIT [(b)]	$\log_{10}{}^*K_{s,0}^o$ SIT
1Br-2	10.0	(10.91 ± 0.18)	(10.89 ± 0.16)	(10.87 ± 0.22)	(10.80 ± 0.19)	(18.24 ± 0.22)
2Br-3	25.0	(10.92 ± 0.06)	(10.88 ± 0.10)	(10.94 ± 0.05)	(10.92 ± 0.03)	(17.07 ± 0.05)
3Br-1	40.0	(10.90 ± 0.12)	(10.90 ± 0.10)	(10.91 ± 0.13)	(10.87 ± 0.12)	(16.15 ± 0.13)
4Br-3	55.0	(10.87 ± 0.15)	(10.90 ± 0.10)	(10.92 ± 0.14)	(10.87 ± 0.18)	(15.36 ± 0.14)
5Br-3	70.0	(10.96 ± 0.16)	(10.99 ± 0.14)	(11.03 ± 0.18)	(10.87 ± 0.21)	(14.58 ± 0.18)
6Br-1	90.0	(11.06 ± 0.16)	(11.10 ± 0.14)	(11.13 ± 0.17)	(10.87 ± 0.18)	(13.71 ± 0.17)

(a) Obtained by fitting all the data in each run.
(b) Obtained by fitting only the data from the last four samples taken.

[1977PET/TAB]

Conductance measurements were made with $MgSO_4$ solutions mainly at 25 °C (0.0001369-0.11907 mol·dm^{-3}) with some additional measurements at 50 °C. Two $MgSO_4$ solid phases were used, an anhydrous form and the more air-stable heptahydrate. No differences in the results obtained were discerned. Three conductance equations and two activity coefficient models were tested, but a combination of the Fuoss [1975FUO], [1975FUO2] conductance equation and a Debye-Hückel expression provided the best fit, albeit over a limited concentration range ≤ 0.003921 mol·dm^{-3}. This combination yielded K_1^o values of 167.7 (d = 1.913 nm) and 207.8 (d = 1.970 nm) at 25 and 50 °C, respectively, with concomitant Λ_0 values of 132.91 and 214.10 S·cm^2·mol^{-1} where the distance of closest approach d was varied stepwise as the third variable. We applied the Lee-Wheaton equation modified to comply with the SIT to the same limited dataset. Bianchi *et al*. [2000BIA/DUJ] obtained K_1^o = 133.01, Λ_0 = 173.5 S·cm^2·mol^{-1} and d = 0.815 nm also from application of the Lee-Wheaton expression to the data at 25 °C for the heptahydrate only, but to a higher concentration limit of ≤ 0.005336 mol·dm^{-3}. Using d as a fit parameter was a common practice and increased the fitting efficiency dramatically, 0.029 S·cm^2·mol^{-1} for (Λ_{obs} – Λ_{calc}) compared with 0.325 S·cm^2·mol^{-1} for a fixed Bjerrum distance of 1.430 nm to force compatibility with

the SIT Debye-Hückel expression. This restriction also caused a monotonic increase in the values of K_1° and Λ_0 with concentration above *ca*. 0.00055 mol·dm^{-3} where the values are (165.8 ± 1.2) and 133.02 S·cm^2·mol^{-1}, respectively, which are much more compatible with those from the analogous treatment in [2000BIA/DUJ]. These results were retained.

[1977SIE/HOS2]

The stability of the magnesium carbonate ion pair was determined by pH-potentiometric titration of a $MgCl_2$-$KHCO_3$ solution with a K_2CO_3 solution (I_m = 0.04-0.20). Experimental runs were performed at 10, 25, 40, 55, 70 and 90 °C. Single ion activity coefficients were estimated from the Debye-Hückel equation.

The experiments were performed at relatively high concentrations of reactants and at high pH values in order to generate large concentrations of $MgCO_3$(aq) by virtue of the large degree of supersaturation with respect to magnesite (3 to 4 orders of magnitude). Siebert and Hostetler mentioned that the supersaturation is tolerated by most of solutions (due to the slow kinetics of magnesite formation). No attempt was made to seal completely the run vessel from contact with atmospheric CO_2, because it was believed that no significant quantity of CO_2 could be absorbed into the solution during the course of runs that lasted less than one hour. The formation of $MgOH^+$ and $MgHCO_3^+$ was taken into account (using the data reported in [1973MCG/HOS] and [1977SIE/HOS], respectively). The method used removed the uncertainties implicit in experimental $p(CO_2)$ values. The reported values are highly reliable, and the selected values above 298.15 K are essentially based on these data.

[1977WOO/HEP]

In order to calculate the heat capacity of completely dissociated $MgSO_4$ in aqueous solution, Woolley and Hepler made use of results for other electrolytes involving M^{n+} and X^{n-} ions whose heat capacities could be extrapolated to zero concentration without the need to consider ion-association effects. Thus, combination of apparent molal heat capacity values for aqueous $MgCl_2$ [1976DES/VIS], NaCl [1976DES/VIS], [[1976SIN/WOO], and Na_2SO_4 [1976DES/VIS], [1977SIN/MCC] and similar combinations of the values for the latter two potassium salts yielded $C_{p,\mathrm{m}}^\circ$(MgSO$_4$, aq, 298.15 K) = – 293 J·K^{-1}·mol^{-1}. Using additional thermodynamic data available in the literature (*viz*., [1964SIL/MAR], [1972PIT] for K_1° = 200 kg·mol^{-1} and $\Delta_r H_m^\circ$ = 5.0 kJ·mol^{-1} from [1964SIL/MAR], [1970LAR], [1975LEU/MIL]), the authors derived a $\Delta_r C_{p,\mathrm{m}}^\circ$ for the ion-association reaction of 125 J·K^{-1}·mol^{-1} at 298.15 K.

[1978BER/VAN]

With the aim of improving the consistency of existing thermochemical data on CO_2(g), $NaHCO_3$(cr), Na_2CO_3(cr), the authors measured the enthalpy of solution of these compounds in excess NaOH solutions. These results were combined with results reported by the same authors [1978BER/VAN2] in a parent paper on the enthalpies of ionisation of carbonic acid and enthalpies of dilution of $NaHCO_3$(cr) and Na_2CO_3(cr). The authors also review measurements involving CO_2(g) in equilibrium with an aqueous phase, as well as results from dissociation pressure measurements on $NaHCO_3$(cr), $Na_2CO_3 \cdot H_2O$(cr), and $Na_2CO_3 \cdot 10H_2O$(cr), the latter two compounds being beyond the scope of the present review.

The carbon dioxide used was Matheson "Coleman Instrument Grade" with a purity greater than 99.99 mass %. Helium (used in the gas transfer) was supplied by Linde , with a purity of at least 99.995 mass %.

Sodium hydroxide solutions were prepared by decanting clear supernatant 50 mass % sodium hydroxide solution and diluting it to the desired composition with CO_2-free distilled water. Molalities of these NaOH solutions were established against a standard perchloric acid solution to better than 0.05 %. Reagent grade sodium hydrogen carbonate (Merck, lot 72400) was used. It assayed 99.99 mass % by duplicate titrations with standard acid solution. Reagent grade anhydrous sodium carbonate (Mallinckrodt, lot AXM) was used. After proper thermal treatment, it assayed, by titration, 99.93 mass % Na_2CO_3.

The calorimeter used has been described by Berg and Vanderzee [1978BER/VAN2] and in earlier papers of the same group [1963VAN/SWA], [1974VAN/ROD] and [1975BER/VAN]. Operations were carried out at known temperatures near 298.15 K and results were corrected to 298.15 K. All calorimetric results were reported with the 2σ confidence interval.

The gas handling procedure for the introduction of the CO_2 and the associated corrections were described in great detail. The volume of the CO_2 gas delivery bulb was (0.588143 ± 0.000020) dm^3 and the amount of CO_2 delivered was obtained by weighing. From the analysis of estimated systematic errors in pressures, temperatures and masses entering the calculations, the authors estimated the contribution to the uncertainty interval for $\Delta_{sol}H^{o}_{m}$ (CO_2, g) to be ± 50 J·mol^{-1} associated with uncertainty in the amount of substance based on weighings.

Thus, for Reaction (A.149):

$$CO_2(g) + (r+2)NaOH(sln) \rightleftharpoons (rNaOH + Na_2CO_3 + H_2O)(sln) \quad \text{(A.149)}$$

16 experiments were carried out using masses of solvent ranging between 1099.97 and 1113.58 g (molalities ranging between 0.20416 and 0.21758). From these, after

accounting for all corrections, the authors report, for Reaction (A.150) in standard conditions:

$$CO_2(g) + 2NaOH(aq) \rightleftharpoons Na_2CO_3(aq) + H_2O(l) \quad \text{(A.150)}$$

a value of $\Delta_r H_m$(A.150) = – (107.508 ± 0.167) kJ·mol^{-1}.

For the dissolution of $NaHCO_3$(cr) in excess NaOH solution, according to Reaction (A.151):

$$NaHCO_3(cr) + (r+1)NaOH(sln) \rightleftharpoons (rNaOH + Na_2CO_3 + H_2O)(sln) \quad \text{(A.151)}$$

five experiments were carried out using masses of NaOH solution ranging between 1060.63 and 1084.90 g and the ratio r of number of moles of NaOH to that of $NaHCO_3$ between 7.737 and 8.219.

From these results, the authors obtain for Reaction (A.152) in standard conditions:

$$NaHCO_3(cr) + NaOH(sln) \rightleftharpoons Na_2CO_3(sln) + H_2O(l) \quad \text{(A.152)}$$

a value of $\Delta_r H_m$(A.152) = (22.389 ± 0.126) kJ·mol^{-1}.

For the dissolution of Na_2CO_3(cr) in excess NaOH solution, according to Reaction (A.153):

$$Na_2CO_3(cr) + rNaOH(sln) \rightleftharpoons \{rNaOH + Na_2CO_3\}(sln) \quad \text{(A.153)}$$

four experiments were carried out using masses of NaOH solution ranging between 1056.61 and 1071.76 g and the ratio r of the number of moles of NaOH to that of Na_2CO_3 between 8.084 and 8.655.

From these results, the authors obtain for Reaction (A.154) in standard conditions:

$$Na_2CO_3(cr) \rightleftharpoons Na_2CO_3(sln) \quad \text{(A.154)}$$

a value of $\Delta_r H_m$(A.154) = – (26.656 ± 0.126) kJ·mol^{-1}.

In a note added in proof, Berg and Vanderzee [1978BER/VAN2] announced the publication of a critical examination of equilibria related to Reactions (A.155) and (A.156):

$$2NaHCO_3(cr) \rightleftharpoons Na_2CO_3(cr) + H_2O(g) + CO_2(g) \quad \text{(A.155)}$$

$$5NaHCO_3(cr) \rightleftharpoons 3NaHCO_3 \cdot Na_2CO_3(cr) + H_2O(g) + CO_2(g) \quad \text{(A.156)}$$

leading to slightly revised values for $NaHCO_3$(cr). These were published subsequently by Vanderzee [1982VAN] as $\Delta_f H^\circ_m$ ($NaHCO_3$, cr, 298.15 K) = – (948.952 ± 0.209) kJ·mol^{-1}, $\Delta_f G^\circ_m$ ($NaHCO_3$, cr, 298.15 K) = – (851.30 ± 0.19) kJ·mol^{-1} and S°_m ($NaHCO_3$, cr, 298.15K) = (102.36 ± 0.80) J·K^{-1}·mol^{-1}.

[1978BER/VAN2]

The authors present strategies and experimental results leading to the standard enthalpies of ionisation of H_2CO_3(aq) and HCO_3^-(aq) and to conventional relative apparent molar enthalpies L_ϕ of $NaHCO_3$ and Na_2CO_3 as "fixed" electrolytes, whose anions are not hydrolysed or ionised. The above enthalpies L_ϕ are referred to the conventional reference state at infinite dilution. The authors also present values of L_ϕ^e, the relative apparent molar enthalpies of the equilibrium states of the solutions referred to the same reference state as L_ϕ. These values of L_ϕ^e are related to the standard enthalpies of solution $\Delta_{sol}H_m^o$ and to experimental enthalpies of solution $\Delta_{sol}H^e$ by the relation $\Delta_{sol}H_m^o = \Delta_{sol}H^e - L_\phi^e$. The results obtained in this paper by the authors are used in their subsequent paper [1978BER/VAN] to extract standard enthalpies of reaction from measurements of the enthalpy of solution of CO_2(g), $NaHCO_3$(cr) and Na_2CO_3(cr). See the relevant Appendix A entry.

In outlining the calorimetric processes, the authors identified and discussed with utmost care the required correction terms and the necessary auxiliary data to allow the extraction of L_ϕ values and of the standard enthalpies of reaction. For sodium carbonate dilutions, the authors stress the need to deal with unavoidable traces of CO_2, especially in the diluent, and with the resulting ionisation process to HCO_3^- and CO_3^{2-}. In the case of sodium hydrogen carbonate dilutions, the perturbations are caused by almost equal opposing influences of hydrolysis and ionisation of HCO_3^- that are of smaller influence because of the internal buffering in these solutions.

Doubly-distilled water was kept under nitrogen. It was shown by titrimetry that its transfer process for the preparation of solutions caused an absorption process that amounted to 1×10^{-5} to 3×10^{-5} mol(CO_2)·dm^{-3}. Standard solutions of hydrochloric and perchloric acid, prepared from reagent grade materials, were standardised with tris(hydroxymethyl)aminomethane, according to recommended procedures: the precision reported was 0.005 % for the HCl solution and 0.02 % for the $HClO_4$ solution, with a normal accuracy of 0.02 % or better.

Sodium hydroxide solutions were prepared by decanting clear supernatant 50 mass % sodium hydroxide solution and diluting it to the desired composition with CO_2-free distilled water.

Reagent grade sodium hydrogen carbonate (Merck, lot 72400) was used. It assayed 99.99 mass % by duplicate titrations with standard acid solution. Two samples of reagent grade sodium carbonate samples (Adamson, lot Y123 and Mallinckrodt, lot AXM) were used. After proper thermal treatment, they assayed, by titration, greater than 99.9 mass % and 99.93 mass % Na_2CO_3, respectively.

The calorimetric equipment has been described previously by the same team and many of the procedures follow directly from a work reported by the same authors [1975BER/VAN2] on relative apparent molar enthalpies of solutions of zinc

perchlorate. Most of the results in the present dilution studies were obtained by the pulse-titration procedures except that batch operations using glass ampoules (0.010 dm^3 and, in a few cases, 0.039 dm^3) were used for solutions above 0.2 $mol \cdot kg^{-1}$ of Na_2CO_3 and 0.3 $mol \cdot kg^{-1}$ of $NaHCO_3$. All the results refer to the isothermal reactions or processes at 298.15 K

The enthalpies of dilution studies for $NaHCO_3$ solutions were obtained from three pulse-titration runs yielding 24 points below 0.2 $mol \cdot kg^{-1}$ and 15 batch-type measurements above 0.2 $mol \cdot kg^{-1}$. Replicate runs at a particular molality yielded L_ϕ^e values agreeing usually within 21 $J \cdot mol^{-1}$. Values of L_ϕ^e and L_ϕ taken from smooth curves based on experimental results were presented as Tables.

The enthalpies of dilution studies for Na_2CO_3 are presented in a similar way. They are based on a total of seven pulse-titration runs, two of which for dilutions into 0.02226 $mol \cdot kg^{-1}$ of NaOH, partially to suppress hydrolysis of CO_3^{2-} . Results at higher ionic strengths are based on 20 batch-type measurements with dilute Na_2CO_3 as diluent.

The validity of the reported data is strongly dependent on the correctness of equilibrium compositions calculated by the authors, *i.e.*, on the thermodynamic deprotonation constants (pK_1^o and pK_2^o) of carbonic acid and their ionic strength dependencies used by the authors. Therefore, for the purpose of this review, we re-evaluated the extrapolation of the relative apparent molar enthalpies of Na_2CO_3 solutions at the equilibrium states of the solutions (L_ϕ^e) using the SIT method and the currently determined ionic strength dependencies of carbonic acid p*K*s (we assumed the same ionic strength dependence of pK_1 and pK_2 in Na_2CO_3 solutions as in NaCl solutions). We calculated L_ϕ^e by the equation:

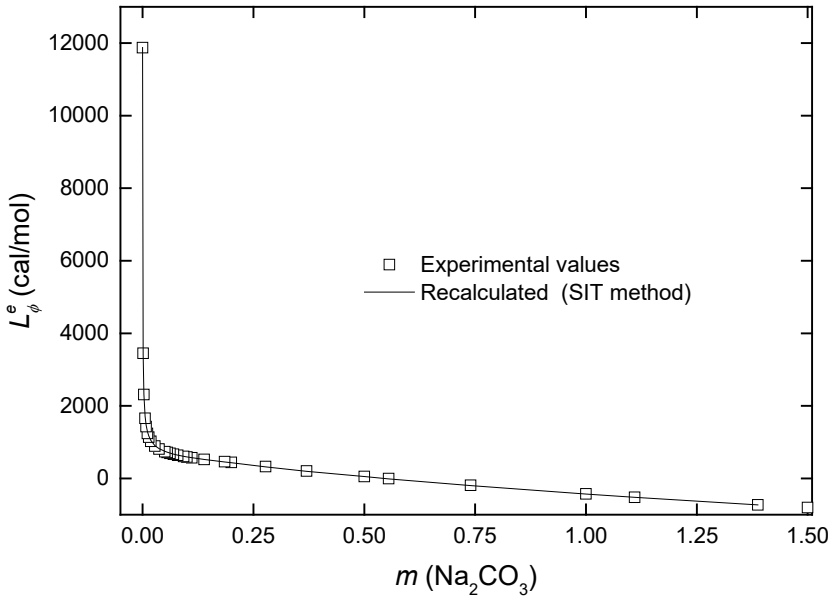
$$L_\phi^e = L_\phi(Na_2CO_3) + \alpha \Delta_r H_m^o((V.7), 298.15\ K) + \beta \Delta_r H_m^o((V.8), 298.15\ K) \quad (A.157)$$

(where α is the extent of hydrolysis of CO_3^{2-}, β is the extent of hydrolysis for HCO_3^-, $\Delta_r H_m^o$((V.7), 298.15 K) and $\Delta_r H_m^o$((V.8), 298.15 K) are the standard enthalpies of first and second dissociation of carbonic acid—see Section V.2.3) using the $L_\phi(Na_2CO_3)$ values listed in Table 4 of the paper, and assuming that the ionic strength dependence of $\Delta_r H_m^o$((V.7), 298.15 K) and $\Delta_r H_m^o$((V.8), 298.15 K) follows the Equation (A.158) given in [1997GRE/PLY2]:

$$\Delta_r H_m = \Delta_r H_m^o + 3/4 A_L \Delta(Z_i)^2 D - RT 2 \Delta\varepsilon_L m \quad (A.158)$$

where A_L is the Debye-Hückel parameter for the enthalpy (at 298.15 K and 1 bar A_L = 1.986 $kJ \cdot kg^{1/2} \cdot mol^{-3/2}$), D is the particular form of Debye-Hückel term used in the SIT model ($D = \sqrt{I_m} / (1 + 1.5\sqrt{I_m})$) and m stands for molality. The experimental L_ϕ^e values listed in Table 3 in the paper can be very well reproduced (solid line in Figure A-17) using $\Delta\varepsilon_L = 1.05 \times 10^{-3}$ $kg \cdot mol^{-1} \cdot K^{-1}$ and -4.61×10^{-3} $kg \cdot mol^{-1} \cdot K^{-1}$ for the first and second dissociation of carbonic acid, respectively.

Figure A-17: Experimental and re-calculated L_ϕ^e values using the SIT method.



The excellent agreement between the experimental and re-calculated L_ϕ^e values in this Figure supports the reliability of the reported data.

[1978ELG/WED]

The $MgSO_4$(aq) ion-pair formation constant was measured potentiometrically at 25 °C using an indirect, competitive method with fluoride as the indicator anion which was monitored with a fluoride sensitive electrode referenced to a glass electrode. An average formation constant of $K_{c,1}$ = (6.1 ± 0.1) at I_c = 0.7 was reported with 0.001 mol·dm^{-3} NaF present initially. Corrections were applied for the ion pairs, $NaSO_4^-$, $MgCl^+$ and MgF^+, where their formation constants were determined empirically at the same conditions. The mixed and variable nature of these media and the lack of any experimental data did not allow re-treatment of these results. The same method with its intrinsic problems was used to determine $K_{c,1}(MgCl^+)$ = (0.34 ± 0.02) at I_c = 0.7. This uncertainty in no way reflects the experimental uncertainty of this value.

[1978EMA/FAR]

This study of magnesium and calcium sulfate ion pairing at 15, 25 and 35 (± 0.05) °C using a cation-sensitive electrode was continued two years later [1980EMA/LIN]. The reference solutions used to calibrate the electrode system contained either $MgCl_2$ or $CaCl_2$ and NaCl at similar "real" ionic strengths to the test solution. The assumption was made that no ion pairing exists between either Mg^{2+} or Ca^{2+} and Cl^- ions. Sodium chloride contributed only 25 to 70 % to the ionic strength of the test solution. Only results for magnesium were tabulated at: I_c = 0.1084 (25 °C) and I_c = 0.0272 (35 °C), and for calcium at I_c = 0.0403 (15 °C), I_c = 0.0207 (25 °C) and I_c = 0.0201 (35 °C). Emara *et al*. used the Davies equation to estimate the mean stoichiometric activity coefficients in their solutions and reported K_1° values for $MgSO_4$ of (160.83 ± 6.00) at 15 °C, (170.46 ± 10.16) at 25 °C, and (182.99 ± 19.67) at 35 °C; and for $CaSO_4$ (172.74 ± 14.18) at 15 °C, (196.51 ± 7.53) at 25 °C, and (221.21 ± 18.91) at 35 °C. These authors also gave mean values for $\Delta_r H_{c,1}^\circ$ at 25 °C for $MgSO_4$ of 1.15 kcal·mol^{-1} (4.81 kJ·mol^{-1}) and $\Delta_r S_{c,1}^\circ$ of 14.1 cal·K^{-1}·mol^{-1} (59.0 J·K^{-1}·mol^{-1}), and for $CaSO_4$ $\Delta_r H_{c,1}^\circ$ of 2.20 kcal·mol^{-1} (9.20 kJ·mol^{-1}) and $\Delta_r S_{c,1}^\circ$ of 17.9 cal·K^{-1}·mol^{-1} (74.9 J·K^{-1}·mol^{-1}). From independent experiments conducted in the absence of NaCl for $MgSO_4$ only, K_1° = (204.40 ± 20.00) at 25 °C. The activity coefficients estimated by the SIT with a small correction to the molality scale [1985SOH/NOV] give: $K_1^\circ(MgSO_4)$ = (228 ± 10) from the I_m = 0.111 data at 25 °C and $K_1^\circ(MgSO_4)$ = (239 ± 11) from the I_m = 0.0274 data at 35 °C, where the uncertainties are statistical only (2σ) and as such underestimate the real experimental uncertainties. The corresponding SIT treatment of the $CaSO_4$ data yields $K_1^\circ(CaSO_4)$ = (211 ± 16) from the I_m = 0.0403 data at 15 °C, $K_1^\circ(CaSO_4)$ = (210 ± 14) from the I_m = 0.0207 data at 25 °C and $K_1^\circ(CaSO_4)$ = (265 ± 20) from the I_m = 0.0201 data at 35 °C.

[1978JOH/PYT]

Potentiometric measurements with a Harned cell were reported of $MgCl_2$ and $CaCl_2$ solutions in the presence of HCl at 298.15 K. The details of this extensive dataset were referenced as being in a manuscript in preparation by Johnson and Pytkowicz, but there is no record of this paper ever being published. The authors took into account HCl(aq) ion pairs in their data treatment. From their Table 6 an average value of $K_1^\circ(Mg^{2+} + Cl^- \rightleftharpoons MgCl^+)$ was given as (1.8 ± 0.3) from the ionic strength dependence (I_c was varied from 0.37 to 0.75 mol·dm^{-3}). De Robertis *et al*. [1987ROB/RIG], Rard and Clegg [1997RAR/CLE], and Méndez de Leo and Wood [2005MEN/WOO] discussed and/or re-treated the calcium data from this study. From Table 7 in this paper an average value of $K_1^\circ(Ca^{2+} + Cl^- \rightleftharpoons CaCl^+)$ was derived (2.3 ± 0.1). The aim of this experimental effort was thwarted by the presence of only the complexing anion, *i.e.*, no supporting electrolyte.

[1978KAL/KRZ]

The $MgSO_4$(aq) ion-pair formation constant was measured by the unusual method of determining the critical coagulation concentration (ccc) of the solution. By studying the electrolyte mixture, $Mg(NO_3)_2$-K_2SO_4, in an AgI(s)|I^- (c_{I^-} = 0.002 mol·dm^{-3}) colloid system at 25 °C, the ccc was determined and the free concentration of Mg^{2+} was found to be 0.0015 mol·dm^{-3} in the absence of added K_2SO_4. After converting the concentrations to molalities [1985SOH/NOV], the ionic strength was determined and used in the SIT to derive the value of $K_1^\circ = (526 \pm 309)$ (2σ), which is considerably larger than obtained by other methods. These results were not considered further, because of the large scatter and uncertainty as to the reliability of the ccc method.

[1979ELG/WED]

This is the second paper by Elgquist and Wedborg using simulated seawater at 25 °C and I_c = 0.7 as the aqueous medium. Here they reported potentiometric measurements with a calcium-sensitive electrode and a calomel double-junction reference electrode aimed at determining the $CaSO_4$(aq) ion-pair association constant. A single standard solution at the same ionic strength was chosen to calibrate their cell so that an ideal Nernstian response was assumed, despite having conducted thirteen titrations with $CaCl_2$(sln) that produced an average slope of 28.6 mV·log$_{10}$(mol·dm^{-3})$^{-1}$ compared to the theoretical slope of 29.6 mV·log$_{10}$(mol·dm^{-3})$^{-1}$. The authors assumed the existence of the ion pairs $NaSO_4^-$, NaCl(aq), KCl(aq), $MgCl_2$(aq), $NaClO_4$(aq), $Mg(ClO_4)_2$(aq) and Na_2SO_4(aq) and recommended a value of $K_{c,1}$ = 25.4 for the formation of $CaSO_4$(aq). Values of $K_{c,1}$ for $MgCl^+$ and $CaCl^+$ from this fully speciated model were taken to be 0.3 and 0.5, respectively, based on results in the literature. The solubility product of gypsum in the simulated seawater was given as 2.87 × 10^{-4} mol^2·dm^{-6}.

[1979FIS/FOX]

A conductance study was made of the ion-pair interactions of some 2:2 electrolytes, including $MgSO_4$ and $CaSO_4$, from 1 to 2000 atm with a total of nine measurements being made at each temperature (15, 20 and 25 °C) using a Pyrex cell with a sliding Teflon piston. The concentration range was quite broad, from 5 × 10^{-4} to 0.1 mol·dm^{-3}. The data were treated with the Fernández-Prini modification of the Fuoss-Hsia equation. Their results at 0.1 MPa were treated with the SIT modified Lee-Wheaton equation by the reviewer. The reported and re-calculated association constants show discrepancies which appear to originate mainly from the concentration range fitted in the case of $MgSO_4$(sln). This trend was not visible for the $CaSO_4$(sln) results. Fisher and Fox fitted their datasets at each temperature with quadratic equations as a function of pressure. As expected, the quality of these $MgSO_4$(sln) data is significantly poorer than the conductivity data reported by Katayama [1973KAT]. However, all the re-treated

results at 0.1 MPa are incorporated into the current analysis with reasonable estimates of the actual experimental uncertainties.

[1979GHI/CAR]

Fifty-two samples of quartz derived from volcanic rocks were investigated by Guinier XRD and DSC. The purity of the samples was determined by neutron activation, and by colorimetry for aluminium, the major impurity. Measurements were made of a number of physical properties, but only the investigation of the variability of the transition temperature and the DSC C_p measurements are of immediate interest to this review. The specimens of volcanic quartz showed a very small variation in their lattice parameters, unrelated to their Al content or transition temperature; the mean values at 295 K were a = (4.1934 ± 0.0004) Å, c = (5.4046 ± 0.0006) Å

The α-β transition temperature varied from 847.2 K (on cooling) to 848.6 K (on heating), and was inversely correlated to the aluminium content of the samples, (which ranged from 15-385 ppm), extrapolating to a value of 848.3 K for pure quartz.

The heat capacity measurements were made on two of the highest purity samples, which were selected as 'standard materials'. These were optically clear single crystals weighing (20-30) mg, containing 65 ppm Al, with an α-β transition temperature of 848.0 K. Below the transition, measurements were made from 400 to 842 K and above the transition from 851 to 900 K.

The detailed data are not included in the paper (but were available from the authors at the time of publication), but are presented graphically and as C_p equations. These latter included terms of the form (T /K – 848) raised to negative powers to force C_p to large values close to the transition point, both below and above 848 K.

The heat capacity values up to 650 K agree excellently with the values calculated from Kelley's assessment of the appreciably earlier C_p and enthalpy data [1960KEL], but are increasingly higher near the transition temperature.

These heat capacity data were included in the assessment in [1989GRO/STO], whose assessed values for α- and β- quartz we are accepting for this review, and will not be considered further here. They have been included in the plot of the heat capacities of quartz in Figure V-19 in Section V.3.1.2.2.1 of the main text.

[1979GUR/KHL]

A calorimeter is described for determination of the low-temperature heat capacities of small samples of minerals and of α-quartz was measured from (7-301) K. The quartz was prepared hydrothermally, and a sample ground to a particle size of 0.25 to 0.5 mm for the measurements. Spectrochemical analyses showed the major impurities to be

Al(0.014 %), and Na(0.009 %), presumably mass %. The detailed results of 237 measurements for the container plus sample and 197 measurements for the container alone are tabulated, together with smoothed values of heat capacity of α-quartz at closely spaced intervals.

At 298.15 K, the values for the heat capacity and entropy are (44.43 ± 0.03) and (41.43 ± 0.02) $J \cdot K^{-1} \cdot mol^{-1}$, respectively. This study was one of the sources on which the CODATA assessment for quartz was based.

[1979MAY/HEL]

Solubility constants were reported for commercial gibbsite at (25.0 ± 0.2) °C in 0.01 $mol \cdot dm^{-3}$ $NaNO_3$ solutions buffered over the pH range 4-9: $\log_{10}{}^{*}K^{\circ}_{s,0} = 8.11$, $\log_{10}{}^{*}K^{\circ}_{s,1} = 3.12$, $\log_{10}{}^{*}K^{\circ}_{s,2} = -2.02$ and $\log_{10}{}^{*}K^{\circ}_{s,4} = -14.05$ (and $\log_{10}{}^{*}K^{\circ}_{s,4} = -14.62$ for an undetermined secondary phase). Gibbsite particles were pre-washed with water and characterised by XRD and SEM resulting in estimated sizes of 2 to 50 μm. The pH buffers were as follows: acetate, 0.001 $mol \cdot dm^{-3}$, pH = 4-6.5; Bis-Tris, 0.002 $mol \cdot dm^{-3}$, pH = 6-7; and Tris, 0.0025 $mol \cdot dm^{-3}$, pH = 7-9. These buffers were assumed not to form ion pairs with Al^{3+} at such low concentrations. It was later shown [1990WES/PAL] that Bis-Tris forms strong complexes with $Al(OH)_4^-$ and that the discontinuity in the plot of $\log_{10} c_{Al}$ *versus* pH at pH ≈ 7 was due to complexation rather than to a solid-state phase change as suggested by May *et al*. Most of their experiments were initiated from supersaturation and reached equilibrium with surprising rapidity under conditions of constant agitation, *viz*., within 31 days. The pH of the centrifuged sample solutions was measured with a glass micro-combination electrode calibrated on the activity scale with commercial standards. The SIT approach led to values of: $\log_{10}{}^{*}K^{\circ}_{s,0} = (8.06 \pm 0.06)$, $\log_{10}{}^{*}K^{\circ}_{s,1} = 3.11$, $\log_{10}{}^{*}K^{\circ}_{s,2} = -2.02$ and $\log_{10}{}^{*}K^{\circ}_{s,4} = -14.62$. The value of $\log_{10}{}^{*}K^{\circ}_{s,2}$ may have been compromised by the effects of complexation of aluminate by Bis-Tris.

[1979SHI/CRI]

The enthalpies of solution of $MgCl_2$(cr) were measured in *ca.* 1 $mol \cdot dm^{-3}$ HCl. The medium and the concentrations used (77.2 HCl, 4190 H_2O) were exactly the same as those used in [1943SHO/HUF] for the determination of the enthalpy of formation of $MgCl_2$(cr), MgO(cr) and several magnesium chloride hydrates. The authors also reported the enthalpy of solution of $MgCl_2$(cr) in pure water.

The anhydrous magnesium chloride was prepared by thermal decomposition of the double salt ammonium magnesium chloride hydrate. The detailed procedure is described and is essentially the same as that given in [1969GRE/SNE]. Gravimetric analysis by AgCl precipitation gave 74.41 mass % Cl (calculated 74.47 %). An aqueous

solution of the salt was clear and the absence of hydrolysis was deduced using phenolphthalein as indicator.

The enthalpy of solution of $MgCl_2$(cr) according to:

$$MgCl_2(cr) + (sln) \rightleftharpoons [Mg^{2+} + 2Cl^-](sln) \quad \text{(A.159)}$$

was reported as $\Delta_{sol}H_m$(A.159) = – (149.98 ± 0.18) kJ·mol^{-1} (2σ), as an average of five measurements. Individual results were corrected for small differences in concentrations. This result is 3 kJ·mol^{-1} less negative than that obtained for the same reaction in [1943SHO/HUF] and it was inferred by the authors that this difference is due to the presence of an MgO impurity in the $MgCl_2$(cr) sample used in [1943SHO/HUF].

Based on four determinations of the enthalpy of solution of $MgCl_2$(cr) in water and after extrapolation to infinite dilution using the equations and parameters given in [1978SIL/PIT], the authors give for Reaction (A.160):

$$MgCl_2(cr) \rightleftharpoons Mg^{2+} + 2Cl^- \quad \text{(A.160)}$$

$$\Delta_{sol}H^\circ_m\text{(A.160)} = – (155.84 ± 0.20) \text{ kJ·mol}^{-1} (2\sigma)$$

The above results were used by CODATA [1989COX/WAG] in their assessment of $\Delta_f H^\circ_m(Mg^{2+}, aq, 298.15\ K)$ and by Gurvich *et al.* [1994GUR/VEY] in their assessment of $\Delta_f H^\circ_m(MgCl_2, cr, 298.15\ K)$.

[1979THO/WEN]

Samples of synthetic tridymite, cristobalite and tridymite-cristobalite mixed phases, were prepared from pure SiO_2 gel using 1 % of K_2CO_3 or Na_2CO_3 mineralisers. Depending on the temperatures and times of heating, single-phase cristobalite and tridymite and mixtures of the polymorphs were obtained (as indicated by Guinier XRD). The samples were not washed after preparation and atomic adsorption spectroscopy indicated they contained up 1.4 % alkaline oxides.

These samples were examined by (DSC) from (360 to 770-840) K, and showed distinct peaks in the heat capacity. Cristobalite showed a very sharp peak at 535 K, due to the second-order transition. For "pure" tridymite, the lambda transition at 390 K and two peaks near 436 and 470 K, which may also be lambda transitions, appear to correlate respectively with the structural changes described by other authors; this may also be true for the weak C_p effect near 653 K. The DSC measurements agree with the heat content measurements by Mosesman and Pitzer [1941MOS/PIT] and the dynamic calorimeter data of Shahid and Glasser [1970SHA/GLA] for tridymite, although the feature observed in both these studies at *ca.* 498 K was absent in the tridymite sample studied. The authors suggest (as did [1956SIN]) that this discontinuity probably results from the inversion of unchanged cristobalite in the "tridymite" samples used in the

previous studies, as suggested by the DSC examination of mixed tridymite-cristobalite phases in the present work.

[1980GUR/SEM]

The low-temperature heat capacities of samples (1.5-8 g) of colemanite ($CaB_3O_4(OH)_3 \cdot H_2O$) and inyoite ($CaB_3O_3(OH)_5 \cdot 4H_2O$) were measured from 12.53 to 313.83 K and 11.54 to 306.16 K, respectively, with full tables of all the experimental values (*ca*. 100 for each mineral) being given. The composition of the colemanite was essentially stoichiometric, *viz.* $Ca_{1.01}B_{2.99}O_4(OH)_3 \cdot 0.99H_2O$, but the inyoite sample, with an analysed composition of $Ca_{1.01}B_{2.99}O_3(OH)_5 \cdot 3.46H_2O$, was distinctly deficient in water with respect to the ideal stoichiometric composition. The measured heat capacity and entropy of inyoite were $C^{o}_{p,m}$ (298.15 K) = 368.61 $J \cdot K^{-1} \cdot mol^{-1}$, and S^{o}_{m} (298.15 K) = 332.34 $J \cdot K^{-1} \cdot mol^{-1}$, and those for colemanite $C^{o}_{p,m}$ (298.15 K) = 217.07 $J \cdot K^{-1} \cdot mol^{-1}$, S^{o}_{m} (298.15 K) = 191.84 $J \cdot K^{-1} \cdot mol^{-1}$. Between 238 and 282 K colemanite showed a heat capacity anomaly representing the transition, with $\Delta_{trs}H^{o}_{m}$ = 120.9 $J \cdot mol^{-1}$ from the less symmetric low-temperature phase to the centrosymmetric phase stable at 298.15 K.

[1980KO/DAU]

The detailed solution calorimetry measurements on magnesium sulfates in this paper are the basis for the selection of the enthalpies of formation of the title compounds. The results of the monohydrate fall outside the scope of the present review and will not be discussed here. The calorimeter used had been described in a prior publication of the same group, but recent modifications were given. The instrument was operated at 298.15 K. The solution medium was 2.260 kg of 4.360 $mol \cdot kg^{-1}$ hydrochloric acid. A quantity of 0.020 mole of each α-$MgSO_4$ and β-$MgSO_4$ was used in each determination and the quantities of other substances involved in the reaction schemes were such as to adequately close the thermodynamic cycles.

α-$MgSO_4$ was obtained by repeatedly heating reagent grade magnesium sulfate in batches of 250 g with 5 cm^3 concentrated sulfuric acid until the latter was completely eliminated. This sulfuric process was analogous to that reported by Rentzeperis and Soldatos [1958REN/SOL], (who used MgO(cr) as starting material) in their study of the structural characterisation of α-$MgSO_4$. The X-ray diffraction pattern of the compound obtained by Ko and Daut was reported to correspond to α-$MgSO_4$. They converted their material to MgO and, in duplicated analyses reported 33.51 and 33.49 wt % MgO (theor., 33.49 %). Sulfur determination yielded 26.65 and 26.67% (theor., 26.64%).

β-$MgSO_4$ was obtained by desiccation to 673 K of reagent grade $MgSO_4 \cdot 7H_2O$(cr), followed by a treatment to 723 K, in a process similar to that reported by Coing-Boyat [1962COI] (who started from a neutral solution of the sulfate) in his study of the structural characterisation of β-$MgSO_4$. The X-ray diffraction pattern identified the material as β-$MgSO_4$ with no trace of the α phase. No appreciable impurities were detected by spectrographic analysis. Conversion of the material to MgO gave 33.48 and 33.47wt % MgO. Sulfur determination gave values of 26.65, 26.65 and 26.61 wt % .

MgO(cr) "specpure" grade was used after drying to 673 K and the acid solutions were prepared from reagent grade materials. A sulfuric acid solution of the composition (H_2SO_4, $6H_2O$) designated hereunder as "sln1" was used as one of the solutes in the heat of solution measurements. As mentioned above, hydrochloric acid, 4.360 molal or (HCl, 12.731 H_2O), was used as solvent (and designated as such in reactions) in all reactions. The media after reactions or dissolution were designated indiscriminately as "sln2" to avoid unnecessary complicated representation.

Noting that, to adequately close the cycle, Reactions (A.161) and (A.162) below were measured consecutively in one batch of solvent and Reaction (A.163) and (A.164) in fresh solvent, the experimental reaction scheme can be represented in a slightly simplified manner as:

$$(H_2SO_4, 6H_2O)(\text{sln1}) + \text{solvent} \rightleftharpoons (2H^+ + SO_4^{2-} + 6H_2O)(\text{sln2}) \quad \text{(A.161)}$$

$$MgO(\text{cr}) + 2H^+(\text{sln2}) \rightleftharpoons (Mg^{2+} + H_2O)(\text{sln2}) \quad \text{(A.162)}$$

$$7H_2O(\text{l}) + \text{solvent} \rightleftharpoons 7H_2O(\text{sln2}) \quad \text{(A.163)}$$

$$MgSO_4(\text{cr}, \alpha \text{ or } \beta) + \text{solvent} \rightleftharpoons (Mg^{2+} + SO_4^{2-})(\text{sln2}) \quad \text{(A.164)}$$

Reactions (A.161) and (A.163) were measured four times, Reaction (A.162) six times, and Reaction (A.164) was measured six times for α-$MgSO_4$ and eight times for β-$MgSO_4$.

These reactions were combined to obtain Reaction (A.165):

$$MgO(\text{cr}) + (H_2SO_4 \text{ in } 6H_2O)(\text{sln1}) \rightleftharpoons MgSO_4(\text{cr}, \alpha \text{ or } \beta) + 7H_2O(\text{l}) \quad \text{(A.165)}$$

giving the values $\Delta_r H_m$(A.165) = – (98.575 ± 0.155) kJ·mol^{-1} and – (94.742 ± 0.180) kJ·mol^{-1} for the α- and β-phase, respectively. These results are given with 2σ uncertainty limits.

From these results, the authors report:

$$\Delta_f H^\circ_m (MgSO_4, \alpha, 298.15 \text{ K}) = – (1288.46 \pm 0.54) \text{ kJ·mol}^{-1} \text{ and}$$

$$\Delta_f H^\circ_m (MgSO_4, \beta, 298.15 \text{ K}) = – (1284.61 \pm 0.54) \text{ kJ·mol}^{-1}$$

using, as auxiliary data, $\Delta_f H^o_m$(MgO, cr, 298.15 K) = – (601.50 ± 0.30) kJ·mol^{-1} from CODATA [1976COD], $\Delta_f H^o_m$(H_2SO_4, 6H_2O, 298.15 K) = – (874.222 ± 0.418) kJ·mol^{-1} and $\Delta_f H^o_m$(H_2O, l, 298.15 K) = – (285.830 ± 0.040) kJ·mol^{-1} from Wagman *et al.* [1968WAG/EVA].

Correcting the data from [1980KO/DAU] to the CODATA values adopted in this review [1989COX/WAG] for the sulfate ion $\Delta_f H^o_m$(SO_4^{2-}, 298.15 K), – (909.34 ± 0.40) and $\Delta_f H^o_m$(MgO, cr, 298.15 K) = – (601.60 ± 0.30) kJ·mol^{-1}, and taking $\Delta_f H^o_m$(H_2SO_4, 6H_2O, 298.15 K) = – (874.292 ± 0.418) kJ·mol^{-1} (this value being 0.07 kJ·mol^{-1} more negative than that used by Ko and Daut, to account of the difference between $\Delta_f H^o_m$(SO_4^{2-}, 298.15 K) in [1989COX/WAG] and [1968WAG/EVA]) gives $\Delta_f H^o_m$($MgSO_4$, α, 298.15 K) = – (1288.64 ± 0.61) and $\Delta_f H^o_m$($MgSO_4$, β, 298.15 K) = – (1284.80 ± 0.61) kJ·mol^{-1}.

[1980EMA/LIN]

This combined spectrophotometric, potentiometric study of $MgSO_4$ ion pairing at 25 °C involved no excess supporting electrolyte (*i.e.*, self-medium with $MgCl_2$ and $(CH_3)_4NCl$ contributing in some cases to the overall ionic strength up to 1 mol·dm^{-3}). Low concentrations of terpyridine, (0.5-2) × 10^{-4} mol·dm^{-3} were employed as a spectrophotometric indicator. Emara *et al*. provided copious data for both media at ionic strengths of 0.38 to 1.00 and 0.076 to 0.800 mol·dm^{-3}. Evidence was provided that terpyridine forms only a 1:1 complex with Mg^{2+}. There is concern in the potentiometric study about the unknown magnitude of the liquid-junction potentials in these highly concentrated mixed electrolyte solutions. The authors stated that these data were not conducive to extrapolation to zero ionic strength. However, they did use the Davies equation to estimate the mean stoichiometric activity coefficients in the 0.107 mol·dm^{-3} ionic strength solutions and offered a value of K_1^o of (282 ± 18). These results were not considered further.

[1980GHO/GHO]

Potentiometric titrations of Na_2HPO_4 solutions were carried out in a Harned cell from 283.15 to 323.15 K. The ionic strength was varied from 0.034 to *ca*. 0.1 mol·kg^{-1} made up principally of NaCl and Na_2HPO_4 where the latter generally varied between 2.4 × 10^{-3} and 7.2 × 10^{-3} mol·kg^{-1}. Activity coefficients were derived from the Davies equation. At 298.15 K, Ghosh *et al*. reported: $\log_{10} K_3^o$ ($HPO_4^{2-} + OH^- \rightleftharpoons PO_4^{3-} + H_2O$) = (2.040 ± 0.023). The acid-dissociation constant and associated thermodynamic quantities at 298.15 K were derived from a four-term fitting equation: $\log_{10} K_3^o$ = (11.962 ± 0.008), $\Delta_r G^o_{m,3}$ = 68.264 kJ·mol^{-1}; $\Delta_r H^o_{m,3}$ = 5.175 kJ·mol^{-1}; $\Delta_r S^o_{m,3}$ = – 212 J·K^{-1}·mol^{-1}; and $\Delta_r C^o_{m,p,3}$ = – 559 J·K^{-1}·mol^{-1}. This is a clear case of over-fitting. Moreover, the $\log_{10} K_3^o$ values (their Table 11) could

not be reproduced by this review and the uncertainties are misleading. Isothermal SIT fits were carried out with the value of $\Delta\varepsilon$ fixed at 0.02 kg·mol^{-1}. The hydrolysis constants obtained with the SIT are compared in Table A-48. Due to the marked disparity between the constants reported by the authors and those derived from the SIT, these data were not included in the final SIT regression.

Table A-48: Reported and re-calculated hydrolysis constants for HPO_4^{2-} ($\pm 2\sigma$).

T /K	$\log_{10}K_3^\circ$ /reported	$\log_{10}K_3^\circ$ /SIT
283.15	(2.485 ± 0.019)	(2.31 ± 0.06)
288.15	(2.345 ± 0.021)	(2.05 ± 0.08)
293.15	(2.195 ± 0.010)	(1.87 ± 0.09)
298.15	(2.040 ± 0.023)	(1.91 ± 0.04)
303.15	(1.885 ± 0.009)	(1.76 ± 0.52)
308.15	(1.725 ± 0.014)	(1.40 ± 0.08)
313.15	(1.580 ± 0.024)	(1.10 ± 0.12)
318.15	(1.420 ± 0.021)	(1.86 ± 0.24)
323.15	(1.285 ± 0.015)	(1.22 ± 0.29)

[1980HIT/MES]

A single solubility measurement with gibbsite (Alcoa Corporation C-33 1) was made in acidic solution and a $\log_{10}{}^*K_{s,0}$ value was reported of (7.20 ± 0.15) (60 °C at a pH of 3.74 in 1 mol·kg^{-1} NaCl) which compares unfavourably with the interpolated value of 5.70 obtained from $\log_{10}{}^*K_{s,0}^\circ = 5.82$ ($\Delta\varepsilon = 0.12$ kg·mol^{-1}) at 60 °C [1992PAL/WES]. Hitch *et al*. conceded that attainment of equilibrium could not be clearly established. Subsequent attempted equilibrations in the near neutral pH region involved removing the solid charge from the column and mixing the recovered solid with large volumes of feed solution before returning it to the column and resuming the slow flow at 0.5 to 2 cm^3·h^{-1}. This attempt to circumvent the problem of slow dissolution kinetics was apparently unsuccessful.

[1980JOH/PER]

Johansson *et al*. used titration calorimetry to determine the enthalpy of formation of $MgSO_4$(aq) and $NaSO_4^-$ ion pairs at 25 °C with $I_c = 0.75$ maintained with combinations of LiCl, NaCl, $NaClO_4$, $MgCl_2$ and $MgSO_4$. Enthalpies of dilution of each reactant were determined independently. The enthalpy of reaction to form $NaSO_4^-$ was given as (0 ± 2) kJ·mol^{-1}. Covariance prevented the authors from determining simultaneously the values of $K_{c,1}$ and $\Delta_r H_c$ for the formation of $MgSO_4$(aq). The range of $\Delta_r H_m^\circ$ values was given as 0.7 to 3.3 kJ·mol^{-1} for the LiCl, NaCl, $NaClO_4$ media depending on the value of $K_{c,1}$ chosen (3-12) with $\Delta_r S_m^\circ$ values varying from 16 to 27 J·K^{-1}·mol^{-1}. An examination of all the thermodynamic data available currently for the formation of $MgSO_4$(aq) indicates that even the maximum value for the enthalpy of formation is too

low.

[1980KIT]

An "immiscible displacement method" was used to measure the solubility of gibbsite (C-370 hydrated Alcoa alumina) at (25.0 ± 0.5) °C. This method allowed samples to be removed from a solid/solution mixture with almost a 1:1 weight to volume ratio, which allowed very rapid equilibration within days compared to years in conventional experiments. Kittrick reported an average value of "$\{pH - (1/3)p(Al^{3+})\}$" = $\log_{10}K_{s,0}$ = (2.66 ± 0.02) from 28 measurements. This corresponds to $\log_{10}{}^{*}K^{\circ}_{s,0}$ = (7.86 ± 0.06), which is in excellent agreement (the actual experimental uncertainty is estimated at ± 0.2) with the results of other more conventional studies involving much longer equilibration periods.

[1980MAR]

Marshall measured the solubility of amorphous silica in aqueous sodium nitrate solutions to 6 mol·kg^{-1} from 25 to 300 °C. These experiments were carried out in small titanium alloy pressure vessels which were slowly rocked to minimise thermal gradients and accelerate the attainment of equilibrium. The results of application of the SIT treatment are summarised in Table A-49, noting that the results at 250 and 300 °C were ignored, and the maximum ionic strength considered was restricted to < 3.5 mol·kg^{-1}.

Table A-49: Results of application of the SIT to the solubility data for amorphous silica in water where the pressure was 0.101 MPa at 25 and 100 °C, and saturation vapour pressure at 150 and 200 °C.

t /°C	m_{NaNO_3} max.	$\log_{10}K^{\circ}_{s,4}$	Δε /kg·mol^{-1}	r^2
25	3.36	– (2.678 ± 0.014)	(0.053 ± 0.008)	0.977
100	2.12	– (2.153 ± 0.047)	(0.046 ± 0.039)	0.722
150	3.31	– (1.975 ± 0.026)	(0.029 ± 0.014)	0.844
200	3.31	– (1.819 ± 0.024)	(0.016 ± 0.012)	0.680

The regression of the $\log_{10}K^{\circ}_{s,4}$ values as a function of (T /K) resulted in the equation:

$$\log_{10}K^{\circ}_{s,4} = -(0.188 \pm 0.157) - (733.33 \pm 58.40)(T/\mathrm{K})^{-1}$$

which yielded $\log_{10}K^{\circ}_{s,4} = -(2.65 \pm 0.08)$ and $\Delta_r H^{\circ}_m = (14.0 \pm 2.2)$ kJ·mol^{-1} (*cf.* $\Delta_r H^{\circ}_m$ = 12.3 kJ·mol^{-1} given in this paper) at 25 °C. These results are in good agreement with other high-temperature values [1935HIT], [1958ELM/NOR]. The $\Delta\varepsilon$ value at 25 °C implies that $\Delta\varepsilon = \varepsilon(\mathrm{Si(OH)_4(aq)}, \mathrm{Na^+ + NO_3^-}) = (0.05 \pm 0.01)$ kg·mol^{-1}. All of these values are retained.

[1980MAR/WAR]

The solubility of amorphous silica was measured at 25 °C in aqueous solutions of KCl, KNO_3, NaCl, LiCl, $LiNO_3$, $MgCl_2$, $CaCl_2$, $MgSO_4$, $NaHCO_3$ and Na_2SO_4, at concentrations varying from zero to saturation. The average silica particle size was 37 nm. Equilibration times varied from 18 to 168 hours with constant rotation. Application of the SIT by this review to the 1:1 salt solutions resulted in $\log_{10} K^{\circ}_{s,4}(SiO_2, \text{am}, 298.15\ \text{K})$ values which are in excellent agreement with those from many other reliable studies and are therefore retained. The limit for the range of I_m treated was due either to the solubility limit of the salt or to the availability of water activity data in the Appendix B, Table B-1. Also the values in pure water were omitted from the SIT treatments because they were outliers from the linear regression, perhaps indicating slower dissolution at low ionic strengths. Dandurand and Schott [1987DAN/SCH] measured the solubility of SiO_2(am) in $LiNO_3$ and other salt solutions at 25 °C that are in good agreement with those of Marshall and Warakomski to high concentrations of salt. They reported a solubility in water of 1.997×10^{-3} mol·kg^{-1} almost identical to 1.98×10^{-3} mol·kg^{-1} obtained from the SIT.

[1980NAV/HON]

The enthalpies of solution in $2PbO \cdot B_2O_3$ melt at 985 K have been measured for glasses in the systems albite-anorthite-diopside, $NaAlO_2$-SiO_2, $Ca_{0.5}AlO_2$-SiO_2, and albite-anorthite-quartz. Most of the reported data are outside the scope of this review. Nevertheless, measurements on the solution of quartz, cristobalite and vitreous silica were also reported and only these will be considered here.

Natural Brazilian quartz in a purified powder form was used; no further details of this material were given. The cristobalite samples were obtained by heating silica glass (prepared from reagent grade SiO_2) for 14 days at 1803 K at 1 atm. The Calvet-type calorimeter used and the calorimetric procedure have been described in an earlier publication [1977NAV]. No further indication is given in the present paper as to the purity of the melt used.

Runs were typically carried out with less than 50 mg samples. The quantity of melt was estimated by this review to be of the order of 30-50 g.

For a stated formula weight of Si_4O_8, enthalpies of solution values of $-(13.355 \pm 0.460)$, $-(16.991 \pm 0.531)$ and $-(41.359 \pm 0.845)$ kJ·mol^{-1} were reported as averages of 17, 8 and 16 experiments respectively, for the dissolution of β-quartz, cristobalite and vitreous silica. The above uncertainties were reported as one standard deviation of the mean.

From these results, the enthalpy of Reaction (A.166):

$$SiO_2(\text{quar}) \rightleftharpoons SiO_2(\text{cris}) \qquad \text{(A.166)}$$

is obtained as $\Delta_r H_m$((A.166), 985 K) = (0.91 ± 0.18) kJ·mol^{-1} and the enthalpy of Reaction (A.167):

$$SiO_2(quar) \rightleftharpoons SiO_2(vit) \qquad \text{(A.167)}$$

is obtained as $\Delta_r H_m$((A.167), 985 K) = (7.00 ± 0.24) kJ·mol^{-1}.

With the selected thermal functions, (including the enthalpies of transitions), these give $\Delta_r H^\circ_m$ ((A.166), 298.15 K) = (1.73 ± 0.18) kJ·mol^{-1} and $\Delta_r H^\circ_m$ ((A.167), 298.15 K) = (9.34 ± 0.24) kJ·mol^{-1}.

[1980PET/TAB]

Pethybridge and Taba estimated the $MgSO_4$(aq) ion-association constant from conductance data using the Fuoss [1978FUO], [1978FUO2] equation and the Lee-Wheaton equation [1978LEE/WHE], [1978LEE/WHE2]. For concentrations < 0.0040 mol·dm^{-3}, both methods yielded an average K_1° value of 174 at 25 °C. No raw experimental data were provided to allow re-calculation.

[1980RIM/BAR]

The rates of dissolution and precipitation of quartz and amorphous silica were studied from 18 to 300 °C using a flow-through packed column at low temperatures and rocking autoclaves at high temperatures. Pressure was shown to have very little effect on the rates of reaction. A reaction mechanism was proposed which depended directly on the stoichiometry of the solid phase. As expected, the reaction rate was directly proportional to the surface area of the silica phase exposed to the solution and inversely proportional to the mass of solution. Rate equations were applied to both over- and under-saturated solutions. No new thermodynamic data were forthcoming.

[1981JAC/KER]

The heat capacity of a natural sample of calcite was measured at 61 temperatures from 345 to 775 K by DSC. The results for a powdered sample and a single-crystal disc agreed closely within the total uncertainty of ± 1 %.

The samples were both prepared from a transparent single crystal of calcite, with lattice parameters of 4.99902 and 1.7065 Å, and which wet analysis showed contained only traces of MgO, Fe_2O_3 and MnO. The samples were encapsulated in gold foil for the measurements using a commercial DSC calorimeter with a heating rate of 10 K·min^{-1}, and the values of C_p measured at approximately 10 K intervals between 345 and 775 K (where calcite starts to decompose) are tabulated.

These data are noticeably higher than those from the equation given in [1960KEL] based on early measurements, but agree very well with those measured by Nan *et al.* [2007NAN/SHI] (up to 390 K), and reasonably well with those of Kobayashi [1951KOB] (up to 700 K), and as noted in Section V.6.2.1.9.1 of the main text, these three studies form the basis for the selected heat capacities of calcite.

[1981OHM/FOR]

Potentiometric titrations were performed in a 0.6 mol·dm^{-3} NaCl solution at 25 °C using a glass electrode to monitor pH defined on the hydrogen ion molarity scale. The pH_c varied from 1.5 to 4.3 during the titrations whereas the initial molarity of aluminium varied from 0.0025 to 0.02 mol·dm^{-3}. These titrations were primarily designed to extract formation constants of carbonatoaluminium(III) complexes. The partial pressure of CO_2 was varied from 9.5 to 101 kPa. Speciation in solution was elucidated as being, Al^{3+}, $AlOH^{2+}$, $Al_3(OH)_4^{5+}$ and $Al_{13}O_4(OH)_{24}^{7+}$, as well as the ternary species $Al_2(OH)_2CO_3^{2+}$ and $Al_3(OH)_4HCO_3^{4+}$ (given as a tentative formula) using the program LETAGROPVRID. Various combinations of species were tried from which Öhman and Forsling proposed average hydrolysis constants of $\log_{10}{}^*K_{c,1,1} = -(5.52 \pm 0.04)$, $\log_{10}{}^*K_{c,4,3} = -(13.57 \pm 0.02)$ and $\log_{10}{}^*K_{c,32,13} = -(109.2 \pm 0.12)$. After conversion to molal units [1985SOH/NOV] and reconciliation of the experimental uncertainties, the following values are retained by this review: $\log_{10}{}^*K_{m,1,1} = -(5.51 \pm 0.12)$, $\log_{10}{}^*K_{m,4,3} = -(13.56 \pm 0.12)$ and $\log_{10}{}^*K_{m,32,13} = -(109.1 \pm 0.4)$ at $I_m = 0.608$.

[1981RAR/MIL]

The isopiestic apparatus [1976SPE/WEB] has been used for decades to produce high-quality osmotic coefficient data at (25.000 ± 0.005) °C. The osmotic coefficients of the standard KCl solutions were taken from [1976SPE/WEB]. Equilibration times were tested to ensure equilibrium was reached. From duplicate measurements the solubility of $MgSO_4{\cdot}7H_2O$ was determined to be (3.0211 ± 0.0022) mol·kg^{-1} after 6-15 days. This value is in good agreement with values of 3.026 and 3.015 mol kg^{-1} reported previously [1965LIN], [1978FIL/ANT]. Some isopiestic measurements extended into the supersaturation range. Rard and Miller combined their results with those from other isopiestic studies [1936ROB/JON], [1968WU/RUS], [1969WU/RUS]. Unit weights were also applied to the freezing-point depression data [1916HAL/HAR], [1955BRO/PRU], [1971ISO]. The isopiestic results of Platford [1968PLA] were given zero weight because the concentration values were systematically low. Similarly, zero weights were applied to the very low results reported from vapour-pressure measurements [1962KAN/GRO].

[1981SJO/NOR]

Potentiometric titrations of silicic acid solutions were performed at 25 °C in 0.6 mol·dm^{-3} NaCl solutions. The silicic acid concentration was varied from 0.0075 to 0.0803 mol·dm^{-3}. The results were interpreted in terms of two acid-dissociation equilibria:

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \quad \text{(A.168)}$$

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \quad \text{(A.169)}$$

Sjöberg *et al*. reported the following values, converted here to molal units: $\log_{10}K_m$ (A.168) = – (9.472 ± 0.002) and $\log_{10}\beta_m$ (A.169) = – (22.07 ± 0.01) at 25 °C and I_c = 0.6. These data were later combined with those from more extensive studies [1977BUS/MES] and [1983SJO/HAG], as well as the treatment by Baes and Mesmer [1976BAE/MES] of Lagerström's results [1959LAG], in an SIT treatment by this review. The results are retained, but the uncertainties are increased to those assigned in [1992GRE/FUG], *viz*., ± 0.08 and ± 0.15, respectively.

[1981VAC/SAB]

Vacca *et al*. used continuous-titration calorimetry to measure the enthalpies of three stepwise protonation reactions of PO_4^{3-} in 0.15 mol·dm^{-3} NaCl solutions at 298 K. These measurements were supported by batch potentiometric titrations under the same conditions to allow determination of the degree of protonation in the calorimetric experiments. The results in Table A-50 were taken from Table 3 in this paper and converted to dissociation equilibria. The enthalpies of reaction appear to be reasonable when compared to results reported at I = 0.

Table A-50: Thermodynamics of the dissociation of phosphoric acid and water at 298 K in 0.15 mol·dm^{-3} NaCl.

Reaction	$\log_{10}K_{c1}$	$\Delta_r H_c$ /kJ·mol^{-1}	$\Delta_r S_c$ /J·K^{-1}·mol^{-1}
$H_3PO_4(aq) \rightleftharpoons H^+ + H_2PO_4^-$	– (1.843 ± 0.003)	– (8.6 ± 0.7)	– (64 ± 2)
$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$	– (6.638 ± 0.003)	(4.70 ± 0.05)	– (111.3 ± 0.2)
$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$	– (11.545 ± 0.005)	(18.6 ± 0.2)	– (158.7 ± 0.2)
$H_2O(l) \rightleftharpoons H^+ + OH^-$	– (13.748 ± 0.001)	(56.49 ± 0.05)	(73.69 ± 0.02)

[1981WAC/RIE]

The enthalpies of dilution (from 1 × 10^{-2} to 5 × 10^{-4} mol·kg^{-1}) of a number of bivalent metal sulfates ($MgSO_4$, $CaSO_4$, $ZnSO_4$, $CdSO_4$, $MnSO_4$, $CoSO_4$ and $NiSO_4$) in water and in 1:1 aqueous propanol mixtures were measured at 25 °C by flow calorimetry. No experimental results were given although a small plot of the relative apparent molal heat content *versus* the square root of molality was presented for the case of $MgSO_4$(sln).

Wachter and Riederer reported the following thermodynamic quantities for the ion-association equilibria: $MgSO_4(aq)$, $K_1^\circ = (161.2 \pm 0.5)$ and $\Delta_r H_m^\circ = (5.776 \pm 0.002)$ kJ·mol^{-1}; and $CaSO_4(aq)$, $K_1^\circ = (192.0 \pm 0.5)$ and $\Delta_r H_m^\circ = (6.665 \pm 0.004)$ kJ·mol^{-1}. The third adjustable parameter was the distance of closest approach, *R*, which was determined to be (9.30 ± 0.05) Å and (9.45 ± 0.05) Å, respectively, and temperature independent. Their corresponding values for the $NiSO_4$ system were $K_1^\circ = (209.6 \pm 0.5)$ and $\Delta_r H_m^\circ = (5.440 \pm 0.003)$ kJ·mol^{-1} with $R = (9.10 \pm 0.05)$ Å whereas Gamsjäger *et al*. [2005GAM/BUG] recommended: $K_1^\circ = (226.1 \pm 15.2)$ and $\Delta_r H_m^\circ = (5.66 \pm 0.81)$ kJ·mol^{-1} at 25 °C. Apart from the unrealistically low uncertainties, this study appears to be quite reliable.

[1981WES/NAV]

The enthalpies of formation of fluorapatite ($Ca_5(PO_4)_3F$), fluorpargasite ($NaCa_2Mg_4AlAl_2Si_6O_{22}F_2$) and fluorphlogopite ($KMg_3AlSi_3O_{10}F_2$) from their binary oxide and fluoride constituents, and tricalcium phosphate for fluorapatite, were determined at 985 K by oxide melt solution calorimetry using a technique previously described involving dissolution in lead borate flux ($2PbO{\cdot}B_2O_3$). For the present experiments, it was necessary to enhance the rate of solution of the some of the fluoride compounds involved (the title compounds, as well as MgF_2, CaF_2) by mixing them, in a 1:1 or a 2:1 molar ratio, with compounds, such as AlF_3, NaF, or SiO_2 that dissolved readily in the flux and of which the enthalpy of solution in the same medium had been determined. Of course, this *modus operandi* implies that there is no reaction between the compound to be studied and the diluent, so that the enthalpy of solution of the compound of interest can be obtained by subtracting the enthalpy effect due to the diluent.

The authors showed that when SiO_2 was added as diluent for the dissolution of the compounds, the enthalpies of solution were very significantly less negative than when $AlCl_3$, Al_2O_3 or NaF were used for the same purpose.

From their experimental observation that no MgO(cr) was found in the X-ray diffraction pattern of a mixture of MgF_2(cr) and SiO_2(cr) that had been heated at 1073 K for 24 hours, the authors concluded that the discrepancy in the heat of solution data could not be ascribed to the loss of SiF_4(g). They also could not detect any significant difference between the enthalpy of solution of NaF(cr) in the lead borate flux to which SiO_2 had been added and a "clean" flux.

Of course, these conclusions can only be drawn while keeping in mind the reported uncertainties of the enthalpies of solution.

The discussion that follows is limited to fluorapatite, $Ca_5(PO_4)_3F$, as the other two compounds studied fall outside the scope of the present review. The preparative technique for fluorapatite, involving the combination of CaF_2 and $Ca_3(PO_4)_2$ was

described. The analytical results by X-ray fluorescence for CaO (theoretical, 55.60 %; found, 54.7 %) and for P_2O_5 (theoretical 42.22 %; found 40.3 %), and for fluorine (3.77 % theoretical, 3.84 % found) by steam distillation/ion selective electrode, indicated a distinct departure from stoichiometry. The authors indicated also that optical studies on their sample of fluorapatite revealed the presence of other oxide phases amounting to 1 to 2 %. The cell parameters were reported as a = (9.330 ± 0.004), c = (6.849 ± 0.004) Å (1σ), β = 120°.

The reported values (for 1σ uncertainty limit) for the enthalpy of solution of CaF_2(cr), $Ca_3(PO_4)_2$(cr) and $Ca_5(PO_4)_3F$(cr) are (92.69 ± 5.23), (108.37 ± 5.27) and (282.96 ± 3.47) kJ·mol^{-1}, respectively, from the dissolution of these compounds in 1:2 molar mixtures with AlF_3(cr). When 1:2 molar mixtures with SiO_2(cr) were used, the corresponding values were (82.13 ± 3.09), (96.23 ± 2.80) and (257.19 ± 2.13) kJ·mol^{-1}, respectively. It was noted by the authors that although $Ca_3(PO_4)_2$(cr) dissolves easily in the flux, its dissolution from mixtures with diluents was made in order to facilitate comparison with the data obtained for fluorapatite.

Using the enthalpies of solution tabulated in the paper our calculated values (with 2σ uncertainties) of the enthalpy of Reaction (A.170) at 985 K are:

$$1/2CaF_2(cr) + 3/2Ca_3(PO_4)_2(cr) \rightleftharpoons Ca_5(PO_4)_3F(cr) \qquad \text{(A.170)}$$

$\Delta_r H_m$((A.170), 985 K) = – (74.06 ± 18.04) kJ·mol^{-1}, from the data with the AlF_3(cr) diluent and

$\Delta_r H_m$((A.170), 985 K) = – (71.78 ± 9.91) kJ·mol^{-1}, from the data with the SiO_2(cr) diluent.

The authors, however, reported somewhat different values (also for 2σ uncertainty):

$\Delta_r H_m$((A.170), 985 K) = – (74.27 ± 17.90) kJ·mol^{-1}, from the data with the AlF_3(cr) diluent and

$\Delta_r H_m$((A.170), 985 K) = – (73.09 ± 6.86) kJ·mol^{-1}, from the data with the SiO_2(cr) diluent.

The reason for these small differences could not be identified by this review, as it is not entirely clear what values Westrich and Navrotsky used in their calculation for the enthalpies of solution of CaF_2(cr) and $Ca_3(PO_4)_2$(cr). In any case, the agreement between the two sets is excellent, *in fine*. Because of its smaller uncertainty limits, the result obtained from the data involving SiO_2 was preferred by the authors. This review is of the opinion that there is no objective reason to discard one set of data and prefers to use the average value of all four values, $\Delta_r H_m$((A.170), 985 K) = – (73.3 ± 10.0) kJ·mol^{-1}, with conservative uncertainty limits covering adequately the two sets of results.

The authors corrected their results to 298.15 K using the heat contents tabulated by Robie *et al.* [1979ROB/HEM] and reported $\Delta_f H_m^\circ(Ca_5(PO_4)_3F, cr, 298.15\ K) = -(6859.39 \pm 8.06)$ kJ·mol^{-1}. Using the heat capacity functions accepted in this review, we re-calculate $\Delta_r H_m(298.15\ K) = -60.721$ kJ·mol^{-1} and thus with $\Delta_f H_m^\circ(CaF_2, cr, 298.15\ K) = -(1228.004 \pm 2.000)$ kJ·mol^{-1} [1987GAR/PAR] and $\Delta_f H_m^\circ(Ca_3(PO_4)_2, cr, \beta, 298.15\ K) = -(4123.6 \pm 12.0)$ kJ·mol^{-1} from this review, we find $\Delta_f H_m^\circ(Ca_5(PO_4)_3F, cr, 298.15\ K) = -(6860.1 \pm 20.6)$ kJ·mol^{-1} from this study.

[1982BLO/WEA]

From 16 under- and over-saturation experiments, Bloom and Weaver reported values of solubility constants of gibbsite samples (C-33 and C-730 Alcoa Corporation, and Fisher ACS grade referred to as FF, FC and L) pre-treated with 0.1 mol·dm^{-3} HCl for 14 days to remove any surface coatings or finely divided material. These gibbsite samples were carefully characterised (XRD, SEM, DTA, BET surface area analyses). The solubility measurements were performed in aqueous NaCl solutions (but the concentrations of NaCl were not provided) at (295 ± 1) K in the pH range 3.82 to 4.16 with samples being withdrawn after one, seven and 28 months, although later in the text it is apparent that samples were also taken after three and 17 days. The pH was measured with a glass electrode *versus* a calomel reference, standardised with three commercial pH buffers. Activity coefficients were estimated with an extended Debye-Hückel expression after correction for formation of $AlOH^{2+}$ [1963FRI/PEE]. The average value of $\log_{10}{}^{*}K_{s,0}^\circ$ was (7.71 ± 0.06) for the acid-treated C-33, C-730 and FF solids with higher solubilities being reported for the Fisher L and untreated FC solids. This value is somewhat lower than generally found in other studies, but this may be an artefact of the activity coefficient model chosen, especially if the NaCl concentrations were high. These results were not considered further by this review.

[1982CHE/MAR]

The solubility of amorphous silica was measured from 100 to 350 °C in water although only those measurements made at 100, 150 and 200 °C were considered in this review. Two experiments from supersaturation gave results consistent with those obtained from undersaturation and these results are therefore retained.

[1982DEV/YEF2]

The enthalpies of solution of $B(OH)_3(cr)$ were measured between 288 and 330 K in water and NaOH solutions at various ionic strengths. The final concentration of boric acid was *ca.* 5.8×10^{-2} mol·dm^{-3}. Four to eight replicate measurements were carried out.

Densities of $B(OH)_3$(sln) were used to calculate the partial molar volume change during the ionisation of $B(OH)_3$(aq) in water, from which the pressure dependence of the ionisation constant was determined up to 473 K. The reported enthalpies of solution of $B(OH)_3$(cr) in water are in excellent agreement with earlier literature results. Devina *et al*. extrapolated to zero ionic strength using the values in [1973WAR/MIL] for the enthalpies of dilution. Taking into account all but one prior literature result, they recommended a value of $\Delta_{sol}H^{o}_{m}(B(OH)_3(cr) \rightleftharpoons B(OH)_3(aq)) = -(22.008 \pm 0.021)$ kJ·mol^{-1} with an implied 2σ uncertainty, which appears to be underestimated. The enthalpies of solution in NaOH solutions were measured at ionic strengths from 0.23 to about 2.5 mol·dm^{-3} by addition of NaCl with the NaOH concentration held constant at 0.23 mol·dm^{-3}. These measurements led to $\Delta_{r}H^{o}_{m}(B(OH)_3(aq) + OH^{-} \rightleftharpoons B(OH)_4^{-}) = -(20.460 \pm 0.042)$ kJ·mol^{-1}. The experiments in NaOH media containing NaCl are in general in agreement with previous results in the literature. Combination of the results yielded, $\Delta_{r}H^{o}_{m}(B(OH)_3(cr) + OH^{-} \rightleftharpoons B(OH)_4^{-}, 298.15\ K) = -(42.468 \pm 0.084)$ kJ·mol^{-1} (2σ), although the way they obtained their uncertainty limits is not clear. This result is in agreement with the value of $-(42.56 \pm 0.13)$ kJ·mol^{-1} (2σ) recommended in [2000PAL/BEN].

[1982DIT/ISH]

This study is the final part of the U.S. National Bureau of Standards (NBS) studies on the heat capacity and enthalpy content of synthetic sapphire (Calorimetry Conference reference material). It reports hitherto unpublished data on the enthalpy content of this synthetic sapphire obtained using a new adiabatic receiving calorimeter which has been described earlier. The samples of about 0.03 mole (3.0 g) were contained in a molybdenum capsule. The enthalpy content of the empty and full container was measured at 13 evenly spaced temperatures between 1173 and 2257 K, each measurement being replicated. The results of all the data are tabulated in full.

However, the main contribution of this paper is an assessment of all the NBS heat capacity and enthalpy data of this standard reference material from 10 to 2250 K. All the data from 0 to 273.15 K were fitted by sixth degree splines; above this temperature the enthalpy data were represented by the nine-term function of T given by Ditmars *et al.* [1971DIT/DOU] (which included the hitherto unpublished experimental data presented in the first part of this paper). The coefficients of all the fits are given and values of C_p and $(H(T) - H(0\ K))$ are tabulated at closely-spaced round temperatures from 10 to 1650 K, and also 273.15 K and 298.15 K. The assessed value of $C_{p,m}(298.15\ K)$ is (79.01 ± 0.08) J·K^{-1}·mol^{-1} where the uncertainty is that estimated by Chang, as reported by Ditmars *et al.*

As noted in the main text, these assessments have been overtaken by the more recent re-assessment by Archer [1993ARC], which includes a number of more recent studies from other laboratories, although the differences are quite small.

[1982FRA/MAR]

This is a conductance study of $MgCl_2$ and $CaCl_2$ solutions (0.001, 0.0025 and 0.005 $mol \cdot kg^{-1}$) from 25 to 600 °C and to pressures up to 400 MPa. The all-metal, static cell design used is inherently less accurate than conventional glass conductance cells so that it is less sensitive to ion pairing until more extreme conditions are encountered. Apparently, hydrolysis and precipitation contributed to hysteresis as the pressure was cycled isothermally. Limiting molar conductances were obtained from an extension of the equation in [1958HAR/OWE]. The first assumption was made that M^{2+}, MCl^+, $MCl_2(aq)$ and Cl^- were the predominant species in solution with the further qualification that only M^{2+}, MCl^+ and Cl^- were present at low temperatures and high solution densities, whereas MCl^+, $MCl_2(aq)$ and Cl^- were dominant at high temperatures and low densities. The second assumption was that the limiting molar conductances of MCl^+, Cl^- are equal to those of Na^+, Cl^-. A simplified Debye-Hückel expression was used to estimate activity coefficients. The dissociation constants of the MCl^+ and $MCl_2(aq)$ were fitted by an equation of the form: $\log_{10} K_n^\circ = a + b(T/\mathrm{K})^{-1} + c \log_{10} \rho_{H_2O}$, where ρ_{H_2O} is the density of water in $g \cdot cm^{-3}$. The uncertainties given by Frantz and Marshall were statistical and were masked by graphical smoothing of the measured conductances. Extrapolation of this equation well beyond the limits intended by Frantz and Marshall yielded $\log_{10} K_n^\circ$ values at 25 °C and 0.1 MPa: $MgCl^+$, (4.86 ± 0.65), $MgCl_2(aq)$, (6.65 ± 0.99); $CaCl^+$, (4.86 ± 0.65), $CaCl_2(aq)$, (5.4 ± 1.2) (1σ uncertainty provided by the authors). These extremely high values suggest that high-temperature measurements do not necessarily provide an alternative route to quantify weak interactions at low temperature.

[1982LAR/ZEE]

Flow calorimetric and flow densitometric measurements were reported of aqueous solutions of H_3PO_4, NaH_2PO_4, Na_2HPO_4, Na_3PO_4, $NaHCO_3$, Na_2CO_3, H_2SO_4 and various buffers at 25 °C. The required dissociation constants for $H_3PO_4(aq)$ were taken from [1979HEP/HOP]: $\log_{10} K_1^\circ = -2.147$, $\log_{10} K_2^\circ = -7.200$ and $\log_{10} K_3^\circ = -12.375$, which are in good agreement with the values selected in this review. Table A-51 gives the results for the phosphate system from which Larson *et al*. calculated the following quantities for the stepwise acid dissociation reactions of $H_3PO_4(aq)$ at 25 ºC: $\Delta_r C_{p,m}^\circ = -128, -220$ and -242 $J \cdot K^{-1} \cdot mol^{-1}$, respectively.

Table A-51: Summary of the reported heat capacity and partial molar volumes data at 25 °C.

Salt	m_{max}	Species	$C^{o}_{p,m}$ /J·K^{-1}·mol^{-1}	V^{o}_{m} /cm^{3}·mol^{-1}
H_3PO_4	1.2763	H_3PO_4(aq)	94	48.1
NaH_2PO_4	0.8286 (a)	$H_2PO_4^-$	– 34	31.3
Na_2HPO_4	0.8286 (a)	HPO_4^{2-}	– 254	5.2
Na_3PO_4	0.1746	PO_4^{3-}	– 496	– 18.8
		Na^+	43 (b)	– 1.2 (c)

(a) Three binary compositions of these two salts were measured to this maximum combined molality
(b) [1978ROU/MUS]
(c) [1978OLO/SPI]

[1982MAJ/STU]

Majer and Štulík used chloride and fluoride ion sensitive electrodes to measure the formation constants of the ion pairs formed with Mg^{2+}, Ca^{2+}, Sr^{2+} and Ba^{2+} from 15 to 85 °C. Five values of $K_{m,1}$ for each metal ion were tabulated ranging from (0.72 ± 0.08) to (1.27 ± 0.14) for $MgCl^+$ and (0.54 ± 0.09) to (0.98 ± 0.09) for $CaCl^+$ (where the uncertainties represent the precision of ten replicates) at an ionic strength of 1.0 mol·kg^{-1} partially made up with $NaNO_3$. Although the chloride ion molality was maintained at a low value of 10^{-3} mol·dm^{-3}, the metal cation concentrations were 0.05, 0.1, 0.2 and 0.3 mol·dm^{-3}. The authors applied a four-parameter fit to represent the temperature dependencies of these data. Furthermore, they reported the changes in the standard molal enthalpies and entropies obtained from these fits, whereas the ionic strength was clearly given as 1.0 mol·kg^{-1}. The primary data have been reprocessed in [2005MEN/WOO] and it is assumed that these authors obtained these data *via* private communication.

[1982PAT/SLO]

The first ionisation constant of aqueous carbon dioxide, which is referred to as carbonic acid in this paper, were determined in NaCl media up to 5 mol·kg^{-1} and from 50 to 300 °C by potentiometric measurements, using the cell H_2, Pt|NaCl(m_1), $NaHCO_3$(m_2), CO_2(aq)(m_3)||NaCl(m_1), HCl(m_2), CO_2(aq)(m_3)|Pt, H_2. The results were combined with selected literature data and used to derive thermodynamic parameters ($\Delta_r G^{o}_{m}$, $\Delta_r H^{o}_{m}$, $\Delta_r S^{o}_{m}$, $\Delta_r C^{o}_{p,m}$, $\Delta_r V^{o}_{m}$) for the first ionisation of aqueous CO_2 at 25 °C intervals.

The $NaHCO_3$ solution was obtained by saturating a NaCl-NaOH solution with a CO_2-H_2 mixture. The ratio of m_1/m_2 was generally greater than 50, while m_2/m_3 was generally between 0.5-2.0. Since the two compartments of the cell were saturated with the same gases, the calculation did not require knowledge of hydrogen fugacity.

[1982PET/TAB]

This paper reported the results of conductance measurements of 23 2:2 electrolytes including $CaSO_4$ in aqueous solutions at 25 °C. For the purposes of the current review, which uses the Lee-Wheaton equation to treat the "equivalent" conductance data ($\Lambda°/2$), the results were truncated at $\leq 2.7392 \times 10^{-3}$ mol·dm^{-3}. This yielded: $K_1^o = (228.5 \pm 19.3)$ and $\Lambda° = 279.64$ S·cm^2·mol^{-1}, *cf.* $K_1^o = (207.5 \pm 0.6)$ and $\Lambda° = (277.36 \pm 0.02)$ S·cm^2·mol^{-1} reported by Pethybridge and Taba.

[1982PLU/BUS]

The solubility of calcite, aragonite and vaterite in CO_2-H_2O solutions between 0 and 90 °C ($p_{CO2} = 0.0002$-1.0 atm) have been determined (*ca*. 350 data points). Based on extensive measurements of the pH of Ca^{2+}-HCO_4^- solutions at 25 °C and $p_{CO2} = 0.956$ atm, the association constant for $CaHCO_3^+$ species was derived. Its temperature dependence between 0 and 90 °C was determined from the p_{CO2} dependence of aragonite solubility. In addition, the CO_2-H_2O equilibria were critically evaluated (based on literature data) and empirical expressions for the temperature dependence of K_H, K_1 and K_2 have been derived.

Materials: calcite (Baker or Fisher Scientific). Both materials were re-crystallised at 65 °C with bubbling of 100 % CO_2(g) for at least 48 hours. Aragonite: precipitated from artificial sea water at 75 °C according to Kinsman and Holland [1969KIN/HOL], aged for at least 24 hours at 75 °C. X-ray analysis showed this aragonite to be free of any other phase.

Water-saturated CO_2(g) and N_2(g) were bubbled through a 1 L reaction vessel, filtered through 0.45 µm filters and titrated for total calcium with EDTA using the fluorescence endpoint of calcein indicator. Times to reach equilibrium varied with temperature and $p(CO_2)$, ranged from one hour to several days. The CO_2(g) composition of the mixtures was determined gravimetrically or by gas chromatography. Several measurements were approached from supersaturation.

This is a very careful and highly appreciated work on $CaCO_3$ solubility, and certainly the most comprehensive concerning aragonite. The results agree well with the $\log_{10} K_{s,0}^o(T)$ calculated from the standard thermodynamic quantities selected by *e.g.*, CODATA. However, as noted in [1999KON/KON], from the work of Plummer and Busenberg a negative value for the phase transition entropy ($-(1.3 \pm 8.4)$ J·K^{-1}·mol^{-1}) is derived, but the generally agreed value is positive. This may be explained by the small differences between measured and calculated $K_{s,0}^o(T)$ values at higher temperature [1999KON/KON].

[1982RIC/BOT]

Although the term "amorphous" is used by Richet *et al*. [1982RIC/BOT] in their title, the abstract and the discussion of their paper, the material the authors describe as AS (amorphous) is, in fact (see p.2641 of [1982RIC/BOT], left column, 2 paragraphs up) termed as "SiO_2 453, a silica glass with a very low…". The authors of this Review used therefore the word "vitreous" rather than "amorphous".

The authors made drop-calorimetric measurements on quartz, cristobalite and vitreous silica from *ca.* 1000 to 1800 K, using an ice calorimeter operated in an argon atmosphere. Measurements on α-alumina reproduced the standard NBS values within 0.2 % (for the enthalpies) and a probably pessimistic 0.5 % (for the derived heat capacities.) Although they emphasise the importance of the knowing the impurity levels and structure of the samples used in studies on silica, the authors give rather few details of the samples they used. The quartz was described merely as "Merck *pro analysi*"; the cristobalite was prepared from a "SiO_2 453" glass of which the main impurity (according to the manufacturer, "*Quartz et Silice*, France") was < 25 ppm Al, by heating in a Pt crucible in air at 1500 °C for *ca.* 13 days. The silica glass used in the measurements was obtained by melting the same material in a graphite furnace and cooling over a period of four hours.

For quartz 13 measurements were made from 1002 to 1676 K, for cristobalite 14 measurements from 1008 to 1834 K and for vitreous silica 63 measurements from 1007 to 1838 K; the detailed results are tabulated.

The rest of the paper is a detailed and thorough assessment of all the heat capacity and enthalpy data for these three polymorphs, including discussions of the transitions in quartz and cristobalite. For the latter, they give a table of numerous experimental inversion temperatures varying from 498 K measured on cooling and 544 K on heating and select a value of 525 K.

We have accepted their assessment of the heat capacities of α- and β-cristobalite, but have preferred the value of (540 ± 15) K for the transition temperature from the thorough investigation of [1958HIL/ROY] which, curiously, is not included in the aforementioned table of [1982RIC/BOT]. [1958HIL/ROY] showed that the inversion temperature depends on the detailed ordering in the material, but found that the most ordered and most stable cristobalite had an inversion temperature of (540 ± 2) K.

[1982SAT/SAS]

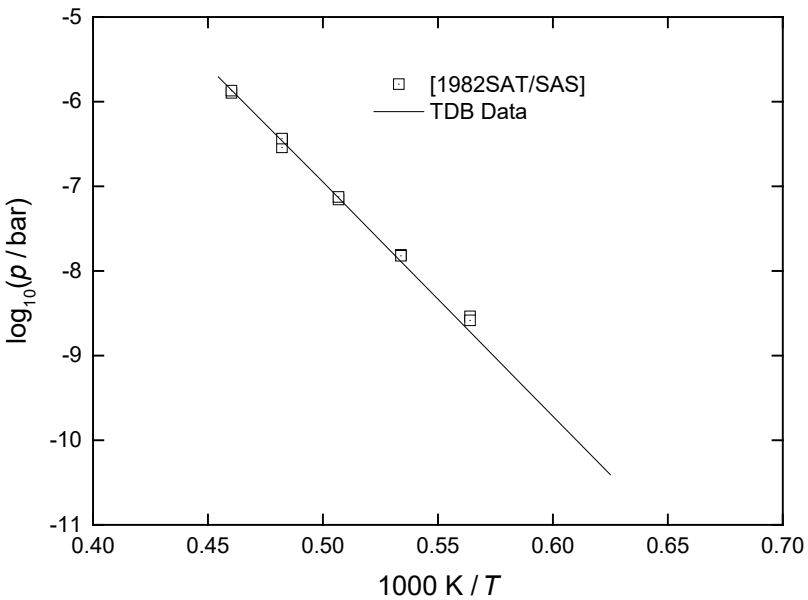
Two sets of measurements on the vaporisation of CaO(cr) were carried out, free evaporation in vacuo (from 1773 to 2074 K) and transportation in mixtures of O_2, H_2O and Ar (from 1678 to 2116 K). The samples of CaO(cr) used were prepared from a

compressed pellet of $CaCO_3$ heated initially to 1573 K and then to *ca.* 2673 K in a hydrogen flame.

For the free evaporation of CaO(cr), the lime sample was resting on rhenium ribbons inside a graphite crucible in a tungsten heater, and the pressures of the main species Ca(g) were obtained from the weight loss of the sample; the vaporisation coefficient was assumed to be unity at these temperatures.

Figure A-18 shows that the measured pressure agree excellently with the selected data for CaO(cr), although the use of a graphite crucible was not ideal experimentally.

Figure A-18: Decomposition pressure of CaO(cr).



In the transpiration experiments, the main vaporisation reaction was $CaO(cr) + H_2O(g) \rightleftharpoons Ca(OH)_2(g)$, enabling the Gibbs energies of formation and standard enthalpies of formation of $Ca(OH)_2(g)$ to be derived. The latter value agreed well with values obtained from flame spectrometry.

[1982THU/MIL]

The stoichiometric pK_1 and pK_2 values for the ionisation of carbonic acid were determined from potential-difference measurements in NaCl solutions (I_m = 0.5-6.0) at

25 °C. Thurmond and Millero used Pitzer's equations to derive the ionic strength dependence. It was necessary to use a triplet ion interaction parameter (ψ) (and higher-order electrostatic terms ($^{E}\Theta$) for pK_2) to obtain reliable calculated values (± 0.01 (pK_1) and ± 0.03 (pK_2) units) over the entire concentration range.

The experimental work in this paper was carried out with high precision. However, as Grenthe and Plyasunov [1997GRE/PLY2] noted, the authors used an erroneous equation for the NaCl concentration dependence of $\ln\gamma(CO_2(aq))$ for the interpretation of their potentiometric data ($\ln\gamma(CO_2(aq)) = 0.2421m - 0.0106m^2$, where m is the molality of NaCl), which gives a poor fit to the experimental $\ln\gamma(CO_2, aq)$ data, especially above 3 mol·kg^{-1} NaCl.

[1983MAY/WOO]

Enthalpy of dilution measurements were made with a flow calorimeter of Na_2SO_4, K_2SO_4 and $MgSO_4$ solutions, in the latter case at initial concentrations of 0.00191 to 1.988 mol·kg^{-1} at 373.15 K and 0.002832 to 2.712 mol·kg^{-1} at 423.65 K. In view of the degree of ion association in these solutions, particularly at elevated temperatures, these results were not treated further.

[1983MIL/THU]

The stoichiometric ionisation constants (pK_1^*, pK_2^*) of carbonic acid have been determined from potential-difference measurements in NaCl solutions (I_m = 0.5 to 6.0) containing small amounts of $MgCl_2$ at 25 °C, in order to study the interaction between Mg^{2+} and HCO_3^-/CO_3^{2-} anions. The results have been analysed using both Pitzer's ion interaction model and an ion-pairing model.

The results at low ionic strengths are in agreement with those reported in [1973DYR/HAN] and [1974PYT/HAW]. Millero and Thurmond described the Mg^{2+}-HCO_3^-/CO_3^{2-} interactions by the corresponding ion interaction coefficients, using Pitzer's formalism. The values of pK_1^* can be reproduced by two Pitzer parameters ($\beta^{(0)}(Mg(HCO_3)_2) = 0.0193$ and $\beta^{(1)}(Mg(HCO_3)_2) = 0.584$. However, four parameters are needed for the pK_2^* values ($\beta^{(0)}(MgCO_3) = 2.833$, $\beta^{(1)}(MgCO_3) = -15.069$, $\beta^{(2)}(MgCO_3) = -204.90$ and $C^{\phi}(MgCO_3) = -0.970$). On the other hand, the Mg^{2+}-CO_3^{2-} interaction could be described to the same precision with fewer parameters using the ion pairing model ($\log_{10}K^{\circ} = (3.000 \pm 0.024)$ and $\log_{10}\gamma(MgCO_3(aq)) = 0.056m_{NaCl}$). The latter values are very similar to the parameters derived in this review by the SIT (which are based on the data reported in this paper as well as in [1973DYR/HAN] and [1974PYT/HAW]).

It should be mentioned that in [1982THU/MIL], in which the authors determined the reference pK values (in NaCl solutions), they used an erroneous

equation for the NaCl concentration dependence of $\ln\gamma(CO_2(aq)) = 0.2421m - 0.0106m^2$) for the interpretation of their potentiometric data (see also in [1997GRE/PLY2]). This error is likely to be propagated to the constant K_{MgCO_3}, too. This may be the reason why the stoichiometric K_{MgCO_3} values calculated in [1983MIL/THU] are somewhat smaller than those re-calculated in this review (see Table V-63 in Section V.6.1.2.7 of the main text), but the extrapolated values to zero ionic strength are identical.

[1983PRA/GHO]

Potentiometric measurements were performed at 5, 15, 25 and 35 °C in a cell fitted with quinhydrone and $Hg|Hg_2SO_4$ electrodes separated by a salt bridge containing H_2SO_4 with a constant ratio $c_{MgSO_4} : c_{H_2SO_4} = 5$ and $I_c < 0.05$ mol·dm^{-3}. Activity coefficients were approximated by an abbreviated Debye-Hückel equation. Despite having a direct measure of the free sulfate concentration, this technique required determination of the amount of sulfate bound as $Hg_2SO_4(aq)$ and knowledge of the dissociation constant of the hydrogen sulfate anion in each medium. It is not clear what values were used for the dissociation constant of hydrogen sulfate and the solubility product of $Hg_2SO_4(s)$. Moreover, from the molarities of various ionic species listed in their Tables 1-4, this reviewer could not establish charge balance in solution. Finally, the dissociation constants of the $MgSO_4(aq)$ ion pair are much lower than those reported elsewhere (by a factor of *ca*. 3) and the enthalpy of reaction is larger than expected. These results were therefore not considered further.

[1983SAS/MOR]

The $K^\circ_{s,0}$ values for calcite and aragonite were determined at 25 °C in water and $CaCl_2$ solutions (up to $I_m = 0.3$). The measured solubility products were extrapolated to infinite dilution using various estimates for the activity coefficients. To obtain values of $K^\circ_{s,0}$ that did not vary with ionic strength, it was necessary to use an ion-pairing model. An apparent increase in $K^\circ_{s,0}$ at very low ionic strengths was found to be the result of solubility control by small particles or surface roughness, which could only be removed by acid leaching. After leaching, the values of $K^\circ_{s,0}$ agreed well with other measurements. In the open system a gas of known CO_2 content ($p(CO_2) = 0.00032$, 0.0032 and 0.96 bar) was bubbled through a stirred suspension of $CaCO_3(s)$. The pH of the solutions was recorded. The equilibration lasted to 1-50 hours and it was found that the minimum time required to reach steady state was three hours for calcite and six hours for aragonite. In some experiments, equilibrium was established from both under- and supersaturation. The closed system experiments were conducted by reacting $CaCO_3$ with HCl of known concentration in sealed vials. Carbonate alkalinity (Alk) for open systems was obtained by acid-base titration and the calcium concentration was

either calculated from the initial calcium present and the alkalinity or by EGTA (ethylene glycol bis(2-aminoethylether)-N,N'-tetraacetic acid) titration.

In all experiments, the Ca^{2+} concentration was calculated as $c(Ca^{2+}) = 0.5(c(Alk) + c(Cl^-))$, which was confirmed, to within experimental error, in several cases by complexometric titrations. Only the solubility data determined in pure water are considered in this review. Although it is careful work, among all accepted data, this paper reports the highest solubility of aragonite at 25 °C. Therefore, the uncertainty of the calculated $\log_{10} K^\circ_{s,0}$ was increased to ± 0.08.

[1983SJO/HAG]

Potentiometric titrations of silicic acid solutions were performed at 25 °C with NaCl as the supporting electrolyte at concentrations of 0.05, 0.1, 0.2, 0.4 and 2 mol·dm^{-3}. The pH sensor was a hydrogen electrode *versus* an Ag|AgCl reference electrode. Hydroxide ions were generated *in situ* by means of innovative coulometry. The silicic acid concentration was varied from 0.007 to 0.080 mol·dm^{-3}. An extended Debye-Hückel expression was used to estimate activity coefficients. There was some discussion on ion pairing of the silicate anions by sodium ions, but Sjöberg *et al*. conceded that the data could be explained in terms of straightforward activity coefficient variations, an approach that is adopted in this review. An SIT treatment (see Figure A-19) by this review using the dissociation constants in their Table II resulted in: $\log_{10}{}^* K^\circ$ (A.171) $= -(9.81 \pm 0.01)$ {*cf.* $-(9.84 \pm 0.02)$ derived by Sjöberg *et al.*} and $\Delta\varepsilon = (0.046 \pm 0.010)$ kg·mol^{-1} (2σ). The former value corresponds to $\log_{10} K^\circ = (4.19 \pm 0.01)$ which compares very favourably to (4.14 ± 0.05) derived from an SIT treatment of the potentiometric results in [1977BUS/MES].

$$Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+ \qquad \text{(A.171)}$$

$$Si(OH)_4(aq) \rightleftharpoons SiO_2(OH)_2^{2-} + 2H^+ \qquad \text{(A.172)}$$

A similar SIT treatment of the tabulated $\log_{10}{}^* \beta_c$ (A.172) values is shown in Figure A-20 with the unweighted fit yielding: $\log_{10}{}^* \beta^\circ_m$ (A.172) $= -(23.18 \pm 0.06)$ (*cf.* $-(23.27 \pm 0.04)$ reported) and $\Delta\varepsilon = (0.12 \pm 0.07)$ kg·mol^{-1} (2σ). The former value corresponds to $\log_{10} \beta^\circ_m = (4.82 \pm 0.07)$. The results of this careful investigation are retained by this review with expanded uncertainties (± 0.05) to match those derived by Grenthe *et al.* [1992GRE/FUG].

Figure A-19: SIT plot for Reaction (A.171).

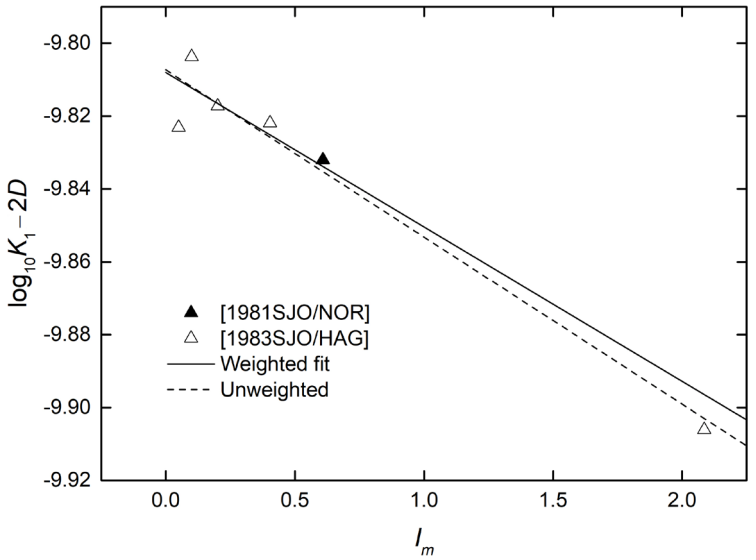
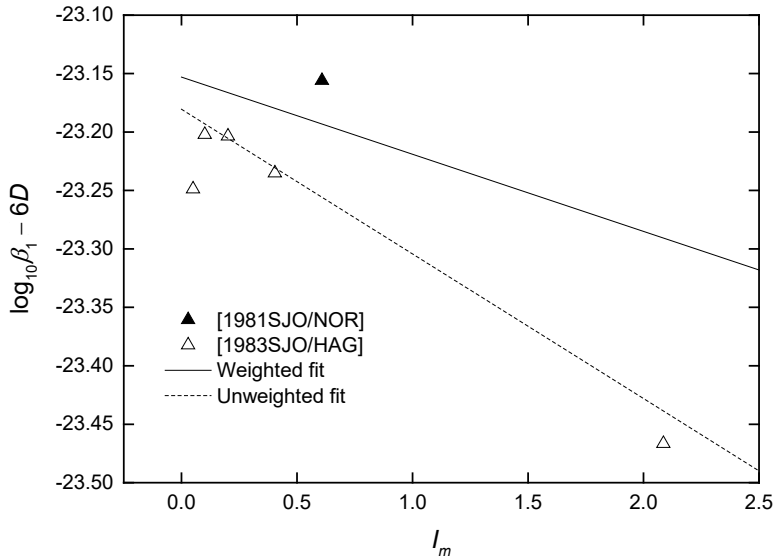


Figure A-20: SIT plot for Reaction (A.172).



[1984COU/MIC]

The stability constants of species $AlOH^{2+}$ and $Al(OH)_4^-$ were reported from 20 to 70 °C at 10 ° intervals in 0.1 mol·kg^{-1} KNO_3 solutions. Solvent extraction (8-hydroxidoquinoline) was used to first quantify the formation of $Al(C_2O_4)_3^{3-}$ (defined in this paper as β_3) and then to determine ${}^*\beta_4$ for Reaction (A.173) by competitive complexation with oxalate.

$$Al^{3+} + 4H_2O(l) \rightleftharpoons Al(OH)_4^- + 4H^+ \qquad (A.173)$$

The tabulated values of $\log_{10}\beta_{m,3}$ varied systematically from 15.91 at 20 °C to 17.01 at 70.0 °C. Couturier *et al*. interpolated a value of $\log_{10}{}^*\beta_4^\circ = -(22.20 \pm 0.15)$ at 25 °C based on activity coefficients for Al^{3+} and H^+ cited by Garrels and Christ [1965GAR/CHR] and the assumption that the activity coefficient of $Al(OH)_4^-$ was equal to that of HCO_3^-. The temperature dependence of $\log_{10}{}^*\beta_4^\circ$ yielded $\Delta_r{}^*H_{m,4}^\circ = (177.4 \pm 2.5)$ kJ·mol^{-1}. These latter data were not included in the final fitting process.

[1984HAR/MOL]

Harvie, Møller and Weare extended earlier calculations [1980HAR/WEA] based on the Pitzer ion interaction model to predict mineral solubilities in the Na-K-Mg-Ca-H-Cl-SO_4-OH-HCO_3-CO_3-CO_2-H_2O system to high ionic strengths at 25 °C. Subsets of this system have been treated subsequently with this model, *viz*. [1995DUC/REA] and [1998FEL/DIX]. A comparison of the values of the ion interaction coefficients for the K-Ca-Cl-OH-H_2O system was given in the Appendix A entry for the former reference. It appears that Harvie *et al*. took the brucite solubility data from [1977MCG/HOS] that were also considered to be very reliable by this review.

[1984IRG/TAR]

The enthalpy content of CaO(cr) was measured at temperatures from 1224 to 2542 K. The sample was prepared from fused high-purity lime; semi-quantitative analysis of the final product indicated that it contained 99.2-99.8 % CaO.

For the measurements, the lime was contained in thin unsealed Mo capsules with a narrow cylindrical hole which served as a black-body for the temperature measurements. These capsules were suspended from tungsten wire in the furnace in an argon atmosphere and dropped into a copper block calorimeter maintained at (298.150 ± 0.001) K. Analysis of the specimens after heating showed small amounts of Mo contamination on their peripheries—0.04 to 0.25 wt % for those heated to 2000 K, and 0.3 to 1.3 wt % for those heated to 2600 K, which had also discoloured on their outside.

The authors made 24 measurements on six different samples of CaO(cr); the detailed results are reported as the average C_p over the range from the temperature drop to 298.15 K and are compared graphically with earlier results. As shown in Figure V-52 cited in Section V.6.2.1.1.2 of the main text, their measurements agree well with the study of [1991RIC/FIQ] over the same temperature range, and were given a high weighting in the C_p assessment of [1994GUR/VEY], whose data we have selected.

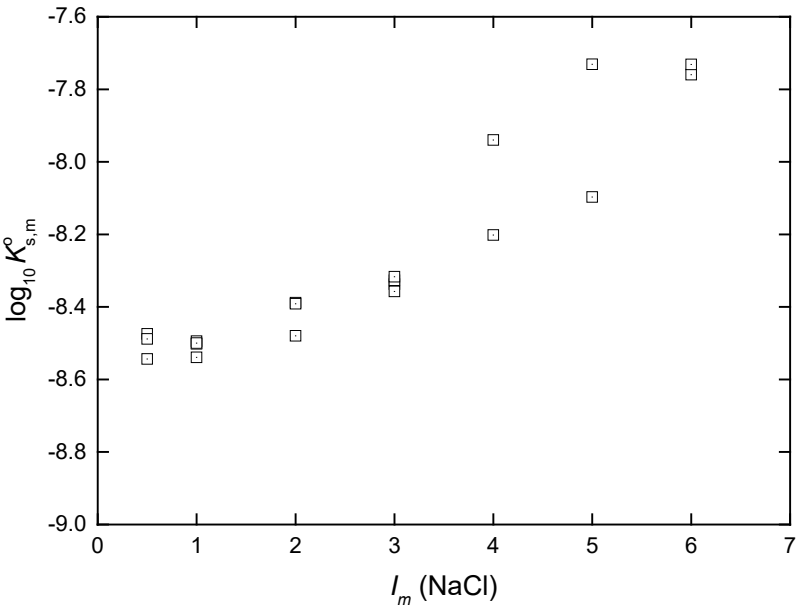
[1984MIL/MIL]

The solubilities of Fisher reagent grade $CaCO_3$(calcite), $SrCO_3$(strontianite), and $BaCO_3$(witherite) were determined in NaCl solutions (0.1 to 6 mol·kg^{-1}) at 298.15 K. One gram of solid was added to the NaCl solutions in 300 cm^3 plastic bottles filled to the top with freshly boiled deionised water (Milli Q). The bottles were rotated in a constant temperature bath (± 0.01 K). Samples were removed after a period of at least two months and were saturated with AgCl then the initial potential difference was measured with a glass electrode. The samples were then filtered through a 0.45 μm Millipore filter and titrated with 0.25 mol·kg^{-1} HCl. The dissolved Ca^{2+} molality was calculated as half of the total alkalinity. The activity coefficients of Ca^{2+} and CO_3^{2-} were estimated from Pitzer's equations with higher-order interaction terms to extrapolate the results to infinite dilution.

Millero *et al.* mentioned that the use of Pitzer parameters for the interaction of Ca^{2+}- CO_3^{2-} is not necessary to reproduce the solubility data (within the experimental error) at low $p(CO_2)$. Indeed, our re-evaluation indicated that the fraction of the total dissolved Ca^{2+} present as the $CaCO_3$(aq) ion pair is only 0.9-5.5 %.

A re-evaluation of the experimental data indicated that reliable solubility products can be obtained only up to $I_m < 3$ (see Figure A-21), which is probably related to the validity range of the SIT treatment used in this review (see Section V.6.2.2.3 of the main text). Consequently, the experimental data obtained in this paper at $I_c \geq 3.0$ NaCl are rejected.

Figure A-21: The re-evaluated $\log_{10} K^{\circ}_{s,0}$ ($CaCO_3$, calc) values at 298.15 K based on the data listed in [1984MIL/MIL] as a function of NaCl molality.



[1984MON/ALE]

The purpose of this paper was to improve our knowledge of the enthalpy of solution in water of magnesium and calcium chlorides, with special emphasis on the preparation and analysis of the samples.

Following [1974VOR/UMY], the double salt ammonium magnesium chloride hexahydrate was the starting material for the preparation of $MgCl_2$(cr). "Chemically pure" calcium chloride hexahydrate (previously dried at (368-373) K was the starting material for the preparation of $CaCl_2$(cr). The process was carried out under a flow of HCl (dried over recently heated silica gel (KSM) and 4 Å molecular sieves). At the end of the treatment the HCl(g) was driven off with a flow of argon and the samples stored in a dry box over P_2O_5.

Table A-52: Analyses for chlorine (gravimetry) and cations (titrimetry, using NTA (Trilon))

Salt	M, wt%		Cl, wt %	
	Found	Calculated	Found	Calculated
$MgCl_2$	(25.48 ± 0.05)	25.53	(74.45 ± 0.05)	74.47
$CaCl_2$	(36.09 ± 0.05)	36.11	(63.90 ± 0.05)	63.89

Atomic absorption spectrometry yielded the following impurities:

For $MgCl_2$: K: 0.016; Na: 0.097; Ca: 0.018 wt %.

For $CaCl_2$: K: 0.007; Na: 0.019; Mg: 0.008 wt %.

The sealed rocking calorimeter (vessel volume: *ca*. 0.055 dm^3) that was used has been described earlier by the same group. For the $MgCl_2$ dissolutions, samples ranging between 33 and 102 mg were used, leading to (0.006 to 0.02) $mol \cdot dm^{-3}$ solutions; for $CaCl_2$, samples between 88 and 128 mg were used leading to (0.014 and 0.021) $mol \cdot dm^{-3}$ solutions. Small corrections were applied to the measured enthalpies of solution (six for $MgCl_2$ and seven for $CaCl_2$) to bring them to a temperature of exactly 298.15 K and a concentration of 0.018 $mol \cdot dm^{-3}$. Corrections were also applied to the mean values for the impurities detected. These resulted in values of – (157.45 ± 0.28) and – (80.95 ± 0.12) $kJ \cdot mol^{-1}$ (2σ), respectively, for the enthalpies of solution of $MgCl_2$ and $CaCl_2$ in water to a concentration of 0.018 $mol \cdot dm^{-3}$.

Correction with the use of the dilution data of [1931LAN/STR] led to the values – (158.59 ± 0.28) and – (82.07 ± 0.12) $kJ \cdot mol^{-1}$ for the enthalpies of solution of the two salts in water at infinite dilution.

[1984PAT/BUS]

The second ionisation constant of carbonic acid was determined in NaCl media up to 5 $mol \cdot kg^{-1}$ and from 50 to 250 °C by potentiometric measurements, using the cell H_2,Pt|NaCl(m_1), Na_2CO_3(m_2), $NaHCO_3$(m_3)||NaCl(m_1), HCl(m_4)|Pt,H_2. The HCO_3^-, CO_3^{2-} solution was prepared by carefully mixing a dilute HC1 solution with a Na_2CO_3 solution containing NaCl to produce values of m_2/m_3 of about unity. The ratios of the molalities of NaCl to the sum of the hydrogen carbonate and carbonate ion molalities $m_1/(m_2 + m_3)$ and NaCl to HCl (m_1/m_4) were usually greater than 50.

Patterson *et al*. noted that at the highest temperatures the free hydroxide molality in the test compartment became very high, which limits the accuracy of the determination of CO_3^{2-} molality (an inherent problem when studying strong bases). Nevertheless, the reported set of stoichiometric constants for the second dissociation of carbonic acid is considered in the literature (and by the reviewers) as the most reliable above 50 °C. Although the expression used for the ionic-strength dependence of activity coefficients is based on the Brønsted-Guggenheim formulation, it is considerably different from the NEA TDB recommended SIT approach. Therefore, the stoichiometric constants were re-evaluated for the purposes of this review. The thermodynamic pK_2° values thus derived are 0.02-0.08 $\log_{10}$ units less than those calculated by the authors.

[1984VAS/KOC]

This paper reports the enthalpy of formation of disodium tetraborate decahydrate (borax) from the enthalpy of solution at 298.15 K of the salt in HCl, NaOH and water, using an isothermally jacketed calorimeter. The authors also report enthalpies of dilution of borax solutions from *ca.* 500 H_2O to *ca.* 45000 H_2O.

Amounts ranging from 0.05 to 0.4 g borax and of 29.94 cm^3 of the dissolving medium were used in the various solution experiments. All experimental dissolution results were extrapolated to zero ionic strength using a Debye-Hückel type relation. This extrapolation also eliminates any requirement to consider polyborate species, since these are of negligible importance at concentrations below 0.025 $mol \cdot dm^{-3}$ boron.

For the dissolution in HCl, seven acid concentrations (ranging from 0.008684 to 0.07014 $mol \cdot dm^{-3}$) were used. As for the other media, measurements were carried out in 3-5 replicates. This scheme yielded $\Delta_{sol}H^{o}_{m}$(A.174) = (86.06 ± 1.10) $kJ \cdot mol^{-1}$ for the dissolution reaction at infinite dilution:

$$Na_2B_4O_7 \cdot 10H_2O(cr) + 2H^+ \rightleftharpoons 4B(OH)_3(aq) + 2Na^+ + 5H_2O(l) \quad (A.174)$$

Accepting the above extrapolated value for the enthalpy of solution and using $\Delta_f H^{o}_{m}(B(OH)_3, aq, 298.15\ K) = -(1072.8 \pm 0.8)$ $kJ \cdot mol^{-1}$, $\Delta_f H^{o}_{m}(Na^+, aq, 298.15\ K) = -(240.34 \pm 0.06)$ $kJ \cdot mol^{-1}$ and $\Delta_f H^{o}_{m}(H_2O, l, 298.15\ K) = -(285.830 \pm 0.040)$ $kJ \cdot mol^{-1}$ from CODATA [1989COX/WAG], we obtain:

$$\begin{aligned}\Delta_f H^{o}_{m}(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) &= -4(1072.8 \pm 0.8) - 2(240.34 \pm 0.06) \\ &\quad - 5(285.830 \pm 0.040) - 2(86.06 \pm 1.10) \\ &= -(6287.09 \pm 3.39)\ kJ \cdot mol^{-1}\ .\end{aligned}$$

For the dissolution in NaOH, four dissolving medium concentrations (ranging from 0.02645 to 0.43594 molar) were used. This scheme yielded $\Delta_{sol}H^{o}_{m}$(A.175) = (31.05 ± 1.60) $kJ \cdot mol^{-1}$ for the dissolution at infinite dilution according to reaction:

$$Na_2B_4O_7 \cdot 10H_2O(cr) + 2OH^- \rightleftharpoons 4B(OH)_4^- + 2Na^+ + 3H_2O(l) \quad (A.175)$$

with $\Delta_f H^{o}_{m}(B(OH)_4^-, aq, 298.15\ K) = -(1345.283 \pm 0.805)$ $kJ \cdot mol^{-1}$ obtained by this review, $\Delta_f H^{o}_{m}(OH^-, aq, 298.15\ K) = -(230.015 \pm 0.040)$ $kJ \cdot mol^{-1}$ from CODATA [1989COX/WAG] we calculate from Reaction (A.175):

$$\begin{aligned}\Delta_f H^{o}_{m}(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) &= -4(1345.283 \pm 0.805) - 2(240.34 \pm 0.06) \\ &\quad - 3(285.830 \pm 0.040) - (31.05 \pm 1.60) + 2(230.015 \pm 0.040)\ kJ \cdot mol^{-1} \\ &= -(6287.09 \pm 3.39)\ kJ \cdot mol^{-1}\ .\end{aligned}$$

For the dissolution in water, the experimental results (seven sets of experiments) were similarly extrapolated and yielded $\Delta_{sol}H^{o}_{m}$(A.176) = (117.1 ± 1.1) $kJ \cdot mol^{-1}$ at infinite dilution:

$$Na_2B_4O_7 \cdot 10H_2O(cr) \rightleftharpoons 2B(OH)_4^- + 2B(OH)_3(aq) + 2Na^+ + 3H_2O(l) \quad (A.176)$$

Use of $\Delta_f H^o_m(BOH_4^-, aq, 298.15\ K) = -(1345.283 \pm 0.805)\ kJ \cdot mol^{-1}$ (see above), and otherwise of CODATA values leads to:

$$\Delta_f H^o_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -2(1345.283 \pm 0.805) - 2(1072.8 \pm 0.8) - 2(240.34 \pm 0.06) - 3(285.830 \pm 0.040) - (117.1 \pm 1.1)\ kJ \cdot mol^{-1} = -(6291.44 \pm 2.53)\ kJ \cdot mol^{-1}.$$

As this review finds no reason to prefer one of the above three excellently agreeing sets of data, their straight average:

$$\Delta_f H^o_m(Na_2B_4O_7 \cdot 10H_2O, cr, 298.15\ K) = -(6289.6 \pm 3.4)\ kJ \cdot mol^{-1},$$

is selected as the best value, keeping the uncertainty limits sufficiently large to take into account that the main sources of error are common to the three schemes.

[1985AKI/ELD]

Aluminium-27 NMR showed that the linewidth attributed to $Al(H_2O)_6^{3+}$ increased as the Al concentration decreased. The spectrum broadened below 0.05 $mol \cdot dm^{-3}$ due to the formation of $AlOH(H_2O)_5^{2+}$. Previous measurements of the hydrogen ion exchange rate between $Al(H_2O)_6^{3+}$ and $AlOH(H_2O)_5^{2+}$ by proton NMR yielded a rate constant of *ca*. $10^9\ s^{-1}$ [1969FON/GRU]. Over the measured range, $0.001 \leq c_{Al} \leq 0.2$, the linewidth was a linear function of pH with an intercept corresponding to $\log_{10}{}^*K_1^o = -(4.93 \pm 0.08)$ which is within the uncertainty limits for the hydrolysis reaction from independent potentiometric and solubility studies. Presumably the temperature of the NMR probe was *ca*. 20-25 °C.

[1985ARC/WOO]

Archer and Wood offered a chemical equilibrium model which took into account the thermodynamic data available for the $MgSO_4$(sln) system from 0 to 150 °C, and discussed an activity expansion model which was less valid at high concentrations. Their model used the Guggenheim activity coefficient equation to calculate the osmotic coefficients of the solutions. Data sources for their model came from freezing-point depression and isopiestic data, vapour-pressure osmometry and enthalpy of dilution measurements. They gave the quantities shown in Table A-53 for three polynuclear magnesium(II) sulfate species proposed to exist at 25 °C over the concentration range 0 to 3 $mol \cdot kg^{-1}$. The existence of a 2:1 species was also proposed based on Raman [2003RUD/IRM] and dielectric-relaxation spectroscopy [2004BUC/CHE] studies. Buchner *et al*. suggested the presence of a higher-order species at high concentrations, *i.e.*, $\leq 2.24\ mol \cdot dm^{-3}$, but could not identify its stoichiometry. Archer and Wood stated that their chemical equilibrium model gave the same results as those of Holmes and Mesmer [1983HOL/MES] at 25 °C.

Table A-53: Thermodynamic quantities for the ion-association equilibria to form $Mg_i(SO_4)_j^{i-j}$ at 25 °C interpolated from a global fit from 0 to 150 °C.

i,j	$\beta^\circ_{i,j}$	$\Delta_r H^\circ_m$ /kJ·mol^{-1}	$\Delta_r C^\circ_{p,m}$ /J·K^{-1}·mol^{-1}
1:1	126.4	5.972	203
2:1	557.3	5.740	350
3:3	3.813×10^6	26.87	797

[1985BRO/SYL]

Brown *et al*. performed automated potentiometric titrations of Al^{3+} solutions at 25 °C and I_c = 0.1 ($NaNO_3$). The titrant was $KHCO_3$(sln) to minimise localised high pH conditions and more time was allowed between additions (5 min) for the formation of polynuclear species. The program MINIQUAD was used to regress the titration data. No unambiguous choice of the stoichiometry of a high molecular weight polymer could be made. On the other hand, the species $AlOH^{2+}$, $Al(OH)_2^+$ and $Al_3(OH)_4^{5+}$ met their acceptance criteria, but $Al_5(OH)_{12}^{3+}$ did not. They concluded that no single technique such as potentiometry can distinguish between all possible high molecular weight polynuclear species. The following hydrolysis constants were tabulated: $\log_{10}{}^*K_{c,1,1} = -(5.41 \pm 0.01)$, $\log_{10}{}^*K_{c,2,1} = -(10.67 \pm 0.02)$ and $\log_{10}{}^*K_{c,4,3} = -(13.130 \pm 0.006)$ at I_c = 0.1 ($NaNO_3$).

[1985GUY/DOR]

The solubility of calcite was reported at 25 °C in four solutions containing NaCl, $MgCl_2$, Na_2SO_4 and $CaSO_4$ with a total ionic strength of 0.015 mol·dm^{-3}. The pH of these solutions was varied from 6.4 to 8.2. An extended Debye-Hückel equation was used to approximate the activity coefficients. A model was adopted that took into account the ion pairs, $CaSO_4$(aq) and $CaHCO_3^+$, as well as that of $NaSO_4^-$, the formation constant for which was taken from the literature. A value of $\log_{10}K^\circ$ ($CaSO_4$, aq, 298.15 K) = (2.35 ± 0.05) was extracted. While this ion-pair association constant is quite reasonable, the complexity of the solutions and the adoption of a comprehensive ion-pairing model prevented their result from being included in the overall data assessment.

[1985HOS/HOW]

Hosieni *et al*. re-analysed the published data on the alpha-beta transition in quartz using Pippard's theory on the volume changes relating to a lambda transition and used the pressure dependence of the transition temperature to evaluate the entropy and enthalpy changes in the transition region, which are in excellent agreement with our selected values based on the later and more comprehensive assessment by Grønvold *et al.*

[1989GRO/STO]. This paper also includes data for cristobalite and coesite, and a calculated *P-T* diagram for silica (which does not include tridymite at ambient pressure). Their data for cristobalite are very close to those of Richet *et al.* [1982RIC/BOT] and thus to our selected values.

[1985KRU/HEM]

The authors measured the heat capacity of seven minerals from 350 to 1000 K, using adiabatic DSC. Only the data for dolomite, from 350 to 620 K will be discussed here.

Two samples of dolomite, $CaMg(CO_3)_2$(cr), were used. Both of these were portions of the materials used in previous investigations, one by Slaughter *et al.* [1975SLA/KER] for studies in higher-order systems, the other by Stout and Robie [1963STO/ROB] for low-temperature heat capacity measurements.

The full analysis of the sample used previously by Slaughter *et al.* [1975SLA/KER] was given (in wt %) by these authors as CaO, 30.99, MgO, 21.19, CO_2, 46.90; Fe_2O_3, 0.79 and MnO, 0.08, yielding a total of 99.95 %. Upper limits were given for SiO_2 0.1 %; Na_2O and K_2O 0.05 %, SrO 0.025 % and TiO_2 0.02 %. The approximate formula was given as $Ca_{1.03}(Mg_{0.98}Fe_{0.02})(CO_3)_{1.98}$. The source of this calcium-rich sample was not stated. As noted in the Appendix A entry of [1963STO/ROB], the second sample was a purified crystal from Binnental, Switzerland which was also slightly CaO-rich, with small quantities (< 0.10 wt %) of FeO, MnO, SrO and H_2O impurities. XRD by Krupka *et al.* [1985KRU/HEM] showed that the lattice parameters of both samples were in good agreement with the values of Goldsmith and Graf [1958GOL/GRA] and compare favourably with the values of Goldsmith *et al.* [1961GOL/GRA] for ordered dolomite.

The results of 54 measurements of C_p from 350.1 to 619.8 K are tabulated; they merge smoothly with the low-temperature data of Stout and Robie [1963STO/ROB], as shown in Figure V-80 in Section V.6.3.2 of the main text. The authors fitted their results by a five-term polynomial, but we have preferred the usual four-term equation with a constraint of $C^{\circ}_{p,\mathrm{m}}(298.15\ \mathrm{K}) = 157.51\ \mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ (see the Appendix A entry of 1963STO/ROB]), shown in the same figure. The maximum deviation of the experimental values from the smoothed curve is consistent with the authors claimed accuracy of ± 1%.

[1985SJO/OEH]

This combined potentiometric (hydrogen electrode in a cell with liquid junction) and ^{29}Si NMR study formed the basis for the selection of formation constants of polysilicate species by Grenthe *et al.* [1992GRE/FUG]. The experimental conditions were: I_c = 0.6 (NaCl); $c_{\mathrm{Si}} = (3\text{-}48) \times 10^{-3}$; the potentiometric titrations covered the range

$2.5 \le pH_c \le 12.2$, whereas the NMR measurements were made at $10.35 \le pH_c \le 12.02$; all at (25.00 ± 0.05) °C. The NMR measurements gave a unique perspective on the equilibrium:

$$Si_3O_6(OH)_3^{3-} \text{ (cyclic)} + H_2O(l) \rightleftharpoons Si_3O_5(OH)_5^{3-} \text{ (linear)} \qquad \text{(A.177)}$$

that could not be detected from potentiometric data. The generalised hydrolysis equilibria postulated to be operative in their experiments are:

$$mH^+ + nSi(OH)_4(aq) \rightleftharpoons (H)_m(Si(OH)_4)_n^{m+} \qquad \text{(A.178)}$$

Two parallel paths were taken by this review to extrapolate to $I = 0$, but both followed the procedure used by Grenthe *et al*. [1992GRE/FUG] who applied the SIT treatment and in the absence of appropriate ion interaction coefficients used the corresponding values for the sodium phosphate system (NEA TDB):

$$\varepsilon(Na^+, Si_2O_2(OH)_5^-) \approx \varepsilon(Na^+, H_2PO_4^-) = -(0.08 \pm 0.04) \text{ kg·mol}^{-1},$$

$$\varepsilon(Na^+, Si_2O_3(OH)_4^{2-}) \approx \varepsilon(Na^+, HPO_4^{2-}) = -(0.15 \pm 0.06) \text{ kg·mol}^{-1},$$

$$\varepsilon(Na^+, Si_3O_6(OH)_3^{3-}) = \varepsilon(Na^+, Si_3O_5(OH)_5^{3-}) = \varepsilon(Na^+, Si_4O_7(OH)_5^{3-}) \approx \varepsilon(Na^+, PO_4^{3-}) = -(0.25 \pm 0.03) \text{ kg·mol}^{-1},$$

and $\varepsilon(Na^+, Si_4O_8(OH)_4^{4-}) \approx \varepsilon(Na^+, P_2O_7^{4-}) = -(0.26 \pm 0.05)$ kg·mol^{-1},

as well as $\varepsilon(H^+, Cl^-) = (0.12 \pm 0.01)$ kg·mol^{-1}.

The second path involved all anionic species (Reactions (A.179)-(A.184)) thereby reducing the magnitude of the activity coefficient approximations with $\varepsilon(Na^+, OH^-) = (0.04 \pm 0.01)$ kg·mol^{-1}. Grenthe *et al.* [1992GRE/FUG] chose to accept the speciation scheme and formation constants in [1985SJO/OEH]. The proposed species and their reported molar formation constants are shown in Table V-29 of Section V.3.2.5 in the main text. Minor corrections were made to convert to molal hydrolysis constants represented by Reactions ((A.179)-(A.184)).

$$2Si(OH)_4(aq) + OH^- \rightleftharpoons Si_2O_2(OH)_5^- + 2H_2O(l) \qquad \text{(A.179)}$$

$$2Si(OH)_4(aq) + 2OH^- \rightleftharpoons Si_2O_3(OH)_4^{2-} + 3H_2O(l) \qquad \text{(A.180)}$$

$$3Si(OH)_4(aq) + 3OH^- \rightleftharpoons Si_3O_6(OH)_3^{3-} + 6H_2O(aq) \qquad \text{(A.181)}$$

$$3Si(OH)_4(aq) + 3OH^- \rightleftharpoons Si_3O_5(OH)_5^{3-} + 5H_2O(l) \qquad \text{(A.182)}$$

$$4Si(OH)_4(aq) + 3OH^- \rightleftharpoons Si_4O_7(OH)_5^{3-} + 7H_2O(l) \qquad \text{(A.183)}$$

$$4Si(OH)_4(aq) + 4OH^- \rightleftharpoons Si_4O_8(OH)_4^{4-} + 8H_2O(l) \qquad \text{(A.184)}$$

As expected the values of $\log_{10}\beta^o$ (Reactions (A.179)-(A.184)) in Table V-29 of Section V.3.2.5 in the main text exhibit only relatively minor differences from those assigned by Grenthe *et al*., perhaps with the exception of those for Reaction (A.184). It should be noted that the ion interaction parameters for the Na^+-P(V) system have been

re-evaluated and are definitely ionic strength dependent. However, for an ionic strength of only 0.6 mol·dm^{-3} this complicating factor can be overlooked in view of the large uncertainties and possible ambiguities as to the real stoichiometries of the polysilicates.

[1986CHE/IRG]

This work extends to 3086 K the earlier measurements of Irgashov *et al.* [1984IRG/TAR] on the enthalpy content of CaO(cr). The preparation of the CaO(cr) (from a fused sample) and the experimental technique were the same as in the earlier work, except that both molybdenum and tungsten capsules (sometimes sealed) were used. The total impurities in CaO samples amounted to 0.3-0.5 wt %. With the tungsten capsules, about 0.1 wt % of tungsten was found in the CaO(cr) sample after a test run, and cross-sections of the capsule wall after an experimental run showed the presence of a CaO layer on the internal surface of a capsule a few tens of μm thick. Trace quantities of Ca_3WO_{6-x} were also found in this layer. 18 measurements (from 2589 to 3086 K) were made at higher temperatures than those in [1984IRG/TAR], and seven (from 1463 to 2377 K) in the same temperature range; the agreement was very good in the overlap range. The uncertainty in the measured enthalpies was estimated to be 0.5-0.6 %.

The authors compare their results to those of earlier studies by plotting the mean $C_p(T - 273\ \mathrm{K})$, similar to the plot in Figure V-54 in Section V.6.2.1.9.1 of the main text. As noted in the main text, the results of this and the earlier study [1984IRG/TAR] were given a high weighting in the C_p assessment of [1994GUR/VEY], whose data we have selected for this review.

[1986KIT/PER]

The solubilities of three natural magnesites were determined in aqueous solutions at 298.15 K (p_{CO_2} = 34.1 Pa) both from under- and supersaturated conditions. Equilibrium was attained within 40-56 days. The CO_2 content of the reaction-chamber atmosphere was determined by IR and the hydrogen ion concentration in solution by a glass electrode. The total soluble Mg and K were determined by AAS.

The solubilities of the three different natural magnesites (from Stevens County, Washington; Nye County, Nevada; Snarum, Norway) were very similar (the averages from 50 solubility determinations starting from both under- and supersaturated conditions were 0.000724, 0.000766 and 0.000764 mol·dm^{-3}). From these experimental data the authors calculated $\Delta_f G_m^\circ$ ($MgCO_3$, cr, 298.15 K) = – (1026.0 ± 2.1) kJ·mol^{-1}.

The attainment of equilibrium in initially undersaturated systems containing either magnesite only or a mixture of magnesite, kaolinite, and montmorillonite was not impeded by the presence of soluble silicon. This observation indirectly discounts the hypothesis that dissolved Si may form magnesium silicate coatings on magnesite

surfaces, thus limiting the rate of dissolution and preventing the establishment of chemical equilibrium.

[1986MUL/PRA]

This study concerns the determination, by high-temperature drop-calorimetry, of the enthalpy of dehydration of $Mg(OH)_2$(cr), $Ca(OH)_2$(cr) and $CaCO_3$(cr) to yield the corresponding oxides.

The instrument used was a Calvet-type twin calorimeter allowing quasi-isothermal measurements according to the heat-flux method in the 600 to 1500 K range. More details are given in a prior publication by the same group. Samples of (50-60) mg were used and the *modus operandi* is described in detail.

For the preparation of their $Mg(OH)_2$(cr) samples the authors refer to the method used in [1937GIA/ARC] (see Appendix A) that involves the heating of the chloride hexahydrate in a mixture of water and potassium hydroxide. For the preparation of $Ca(OH)_2$(cr), the authors used a similar method involving $CaCl_2$, water and sodium hydroxide and refer to [1931JOH/GRO]. Commercial substances (*pro analysi*) were used for the preparation of the hydroxides, which were identified by X-ray diffraction, microscopy and tested for purity, but no analytical results were given, except that the absence of carbonate was verified by acid dissolution. Commercial calcium carbonate (*pro analysi*) was annealed to constant weight in a pure dry CO_2 atmosphere in a platinum crucible at 950 K to yield the required calcite form. No analytical results are given.

At the end of the calorimetric determinations, it was verified that the thermal decomposition of the compounds to oxides was quantitative. For the decomposition of the two hydroxides, the calorimetric temperature was 973 K; after correction to 295 K using auxiliary data from [1973BAR/KNA], [1977BAR/KNA], which were in turn taken from the JANAF assessment, the enthalpies of the reactions:

$$M(OH)_2\,(cr) \rightleftharpoons MO(cr) + H_2O(g)\ \ M = Mg \text{ or } Ca \qquad (A.185)$$

were given without further details as:

$\Delta_r H_m$ ((A.185), Mg, 295 K) = (82.33 ± 2.50) kJ·mol^{-1} (1σ, 22 measurements) and

$\Delta_r H_m$ ((A.185), Ca, 295 K) = (108.84 ± 2.50) kJ·mol^{-1}(1σ, 23 measurements)

The calorimetric decomposition of $CaCO_3$(cr), carried out at 1073 K according to:

$$CaCO_3\,(cr) \rightleftharpoons CaO(cr) + CO_2\,(g) \qquad (A.186)$$

gave:

$\Delta_r H_m$ ((A.186), 1073 K) = (170.5 ± 2.0) kJ·mol^{-1}(1σ, 6 measurements).

There was no report of a correction of the results to room temperature. The values for $\Delta_r H_m$ (A.185) are consistent with those derived from the assessed data (81.02 ± 0.43) kJ·mol^{-1} for Mg and (108.41 ± 1.35) kJ·mol^{-1} for Ca.

[1986SIN/DAH]

Molar conductance data for a number of 2:2 electrolytes, and zinc(II), copper(II) and magnesium(II) sulfates were obtained at 25 °C over the concentration range 0.001 to 0.05 mol·dm^{-3}. The concentration limit of the various conductivity theories, including the Lee-Wheaton equation, examined in [2000BIA/DUJ] for the specific case of $MgSO_4$(sln), was *ca*. 0.006 mol·dm^{-3} for all but the mean spherical approximation equations. Only two measurements were made under this limit so that a re-analysis of these data with the SIT modified Lee-Wheaton equation was not undertaken here. For the formation of $MgSO_4$(aq) Singh *et al*. recorded values of K_1^o = 156.0 and Λ_0 = 130.50 S·cm^2·mol^{-1}.

[1987HEM]

New heat capacity data for quartz have been measured over the temperature interval of 340 to 990 K by differential scanning calorimetry. The quartz sample was a 27.918 mg portion of NBS Standard Reference Material 1654 α-quartz (originating from Brazil), with a size range from 100- to 400+ mesh; no further analysis is given. For the DSC measurements, it was compacted and enclosed in a loosely crimped gold pan. The results of 17 series of measurements, at a total of 109 temperatures at approximately (overlapping) 10 K intervals are tabulated.

The data were combined with recent heat content and heat capacity data to provide a significantly revised set of thermodynamic properties for α-quartz and suggest that the C_p values that are listed in the recent tabulations of the thermodynamic properties of α-quartz (including those selected by CODATA in [1989COX/WAG]) are significantly low between about 600 and 800 K.

These heat capacity data were included in the assessment in [1989GRO/STO], the assessed values from which for α- and β-quartz we accept for this review, and will not be considered further here. They have been included in the plot of the heat capacities of quartz in Figure V-19 in Section V.3.1.2.2.1.

[1987NAV/CAP]

This is a solution calorimetry study of the enthalpy of formation of natural ordered dolomite, synthetically disordered dolomite and four synthetic magnesian calcites from component binary oxides. The results on the magnesian calcites will not be discussed here.

The calorimeter used was of the Tian-Calvet type. A few operating conditions were given and further details on the equipment and the procedure may be found in [1986CAP]. The operating temperature was 358 K and the reacting medium was a solution saturated in NaCl and containing 1.00 molar HCl, 0.75 molar $CaCl_2$, 0.25 molar $MgCl_2$, at room temperature. In all cases, the volume of the solvent (10 cm^3) and the size of the sample (40.0 mg) were kept strictly constant, so as to ensure identical final solutions and CO_2 evolution. Samples for calorimetry were powders of grain size smaller than 150 µm. The enthalpies of solution of the dolomite samples were compared with those of stoichiometric mechanical mixtures of calcium and magnesium carbonates.

The natural dolomite used was an optically transparent, virtually stoichiometric material from Eugui, Spain. Its composition (CaO, 30.65 wt %; MgO, 21.70 wt %; MnO, 0.08 wt %; FeO, 0.41 wt %) determined by electron-microprobe analysis corresponded to published data that showed also that the sample was compositionally homogeneous [1983REE/WEN]. The lattice parameters (hexagonal structure) were given as a = (4.8066 ± 0.0011) Å and c = (16.0016 ± 0.0010) Å.

The disordered sample was obtained by annealing 200 mg amounts of the coarsely crystalline powder from the same Eugui origin at 1523 K in closed platinum capsules under 1.5 GPa pressure. Its lattice parameters (hexagonal structure) were given as a = (4.8037 ± 0.0015) and c = (16.0721 ± 0.0012) Å. X-ray powder diffractometry or single crystal precession techniques did not detect ordering reflections in the quenched sample, indicating that no long-range order persisted. However, the authors noted that, in more detailed structural work, Reeder and Wenk [1983REE/WEN] had observed significant re-ordering upon quenching.

The calcium and magnesium carbonates used for the mechanical mixtures were obtained from reagent grade materials treated at 973 K under 100 MPa CO_2 pressure in open platinum capsules. The authors reported their results as averages of at least six identical measurements; we have doubled their reported 1σ uncertainty. Given the complexity of the medium (sln), no attempt was made to identify solutes.

Thus, for the dissolution of mechanical mixture of binary carbonates:

$$(CaCO_3(calc) + MgCO_3(cr)) + (sln) \rightleftharpoons (CaO + MgO)(sln) + 2CO_2(g) \qquad (A.187)$$

the value is $\Delta_r H_m$ ((A.187), 358 K) = – (0.12 ± 0.96) kJ·(2 mol of cations)$^{-1}$.

For the dissolution of the dolomite:

$$CaMg(CO_3)_2(cr) + (sln) \rightleftharpoons (CaO + MgO)(sln) + 2CO_2(g) \qquad (A.188)$$

the values are $\Delta_r H_m$ ((A.188), ordered dolomite, 358 K) = (11.36 ± 0.32) kJ·mol^{-1} and $\Delta_r H_m$ ((A.188), disordered dolomite, 358 K) = – (2.58 ± 0.84) kJ·mol^{-1}.

From these results, the enthalpy of formation of dolomite from binary carbonates:

$$CaCO_3(calc) + MgCO_3(cr) \rightleftharpoons CaMg(CO_3)_2(cr) \quad (A.189)$$

is obtained as:

$\Delta_r H_m$ ((A.189), ordered dolomite, 358 K) = – (11.36 ± 0.32) – (0.12 ± 0.96) = – (11.48 ± 1.02) kJ·mol^{-1} and $\Delta_r H_m$ ((A.189), disordered dolomite, 358 K) = (2.58 ± 0.84) – (0.12 ± 0.96) = (2.46 ± 1.28) kJ·mol^{-1}.

With the heat capacities selected in this review the correction of $\Delta_r H_m$ (A.189) to 298.15 K is $\Delta_r H_m$ ((A.189), 358 K) – $\Delta_r H_m$ ((A.189), 298.15 K) = – 0.12 kJ·mol^{-1} and thus

$\Delta_r H_m^\circ$ ((A.189), ordered dolomite, 298.15 K) = – (11.60 ± 1.02) kJ·mol^{-1},

$\Delta_r H_m^\circ$ ((A.189), disordered dolomite, 298.15 K) = (2.34 ± 1.28) kJ·mol^{-1},

where the corrections for the ordered and disordered compounds are assumed to be the same.

Thus the value for the enthalpy of disordering of dolomite at 298.15 K is (13.94 ± 1.64) kJ·mol^{-1}, but this should be considered as a lower limit, since, as noted by the authors, the quenching process may have resulted in some short-range ordering.

[1987ROB/RIG]

De Robertis *et al*. calculated Mg^{2+}- and Ca^2-Cl^- ion-pair formation constants using available experimental data. They allowed for the formation of NaCl(aq) and KCl(aq) ion pairs and used a polynomial equation for $K_1^\circ(MCl^+)$ with a Debye-Hückel term that resulted in $\log_{10} K_1^\circ(MCl^+)$ values of (0.57 ± 0.08) and (0.40 ± 0.11) for Mg and Ca, respectively. Finally, there was no discussion of the effect of the high chloride ion concentrations used in most, if not all, the studies treated, *e.g.*, particularly those from the large dataset in [1978JOH/PYT].

[1987SAN/MIC]

Gibbsite solubility results reported at 50 °C were severely criticised in [1990SCH/STR] and [1991HEM], mainly because equilibrium was only approached from supersaturated solutions with a very small initial loading of gibbsite crystals such that either amorphous material, pseudoboehmite or higher energy phases such as bayerite, may have formed. Sanjuan and Michard observed a discontinuity in their solubility curve at a pH of *ca*. 7, which was consistent with earlier findings by May *et al*. [1979MAY/HEL], but this was found subsequently to be an experimental artefact [1990WES/PAL]. Sanjuan and Michard used dilute ammonium acetate to buffer their solutions in the pH range 4 to 10 so that complexation of Al^{3+} by acetate is consistent with the enhanced

solubilities observed below pH 7. Regardless of the cause of their disparate data, they did not tabulate their experimental results and gave no indication of ionic strength. Wesolowski [1992WES] reported a $\log_{10} K_s^\circ$ ($Al(OH)_3(cr) + OH^- \rightleftharpoons Al(OH)_4^-$) = – (0.814 ± 0.005) at 50 °C compared to – (0.1 ± 0.1) given by Sanjuan and Michard.

[1987VER/GOU]

Verdes and Gout found that grinding aluminium oxides under water produced bayerite through a dissolution-reprecipitation process. For bayerite, they reported $\Delta_f G_m^\circ(\beta\text{-}Al(OH)_3, cr, 298.15\ K) = -1158.1\ kJ{\cdot}mol^{-1}$ and a transformation temperature from bayerite to boehmite of *ca*. 102 °C, which appears to be too high. Apps *et al.* [1998APP/NEI] recommended a value of $\Delta_f G_m^\circ(\beta\text{-}Al(OH)_3, cr, 298.15\ K) = -1152.985\ kJ{\cdot}mol^{-1}$.

[1987WEA2]

The $MgSO_4(aq)$ ion-pair formation constant, K_1°, was measured. Weare gave an elegant discussion of the application of the Pitzer ion interaction treatment using a specific interaction approach when possible and one when ion-association equilibria must be included, citing the $CaSO_4(sln)$ system specifically. Comparison of both approaches using only one additional interaction parameter showed that the specific interaction model was "somewhat better". At higher temperatures the latter approach was more strongly favoured. This paper served to show that the existence of ion pairs or complexes in solution cannot be inferred directly from a model unless they are sufficiently stable. Weare accepted previous guidelines that suggested for equilibria, $M^{2+} + L^- \rightleftharpoons ML^+$, with $K_1^\circ > 100$, an ion-association approach should be preferred. For $CaSO_4(sln)$, $K_1^\circ > 200$ at ambient conditions.

[1988HED]

Hedlund conducted potentiometric titrations of Al^{3+} in NaCl and $NaClO_4$ solutions (0.6 to 3.0 $mol{\cdot}dm^{-3}$) at 25 °C to determine the stability constants of the $AlOH^{2+}$ and $Al_3(OH)_4^{5+}$ species:

$$Al^{3+} + H_2O(l) \rightleftharpoons AlOH^{2+} + H^+ \quad \text{(A.190)}$$

$$3Al^{3+} + 4H_2O(l) \rightleftharpoons Al_3(OH)_4^{5+} + 4H^+ \quad \text{(A.191)}$$

The program LETAGROP and the SIT[1] were used to treat the data (four ionic strengths). An unweighted SIT treatment applied to the data for NaCl solutions by the

[1] Hedlund used a version of the SIT with an extended Debye-Hückel term of opposite sign to that used in the TDB NEA series.

reviewer yielded: $\log_{10}{}^{*}K_1^{\circ} = -(4.810 \pm 0.053)$ and $\Delta\varepsilon = -(0.118 \pm 0.026)$ kg·mol^{-1} at 25 °C, *cf.* reported values of $\log_{10}{}^{*}K_1^{\circ} = -(4.86 \pm 0.03)$ and $\Delta\varepsilon = -(0.11 \pm 0.02)$ kg·mol^{-1}. A similar treatment of the $NaClO_4$ solution data gave: $\log_{10}{}^{*}K_1^{\circ} = -(4.801 \pm 0.024)$ and $\Delta\varepsilon = -(0.201 \pm 0.010)$ kg·mol^{-1}, *cf.* reported values of $\log_{10}{}^{*}K_1^{\circ} = -(4.86 \pm 0.04)$ and $\Delta\varepsilon = -(0.19 \pm 0.02)$ kg·mol^{-1}. All uncertainties are 1σ statistically based. Table A-54 shows a comparison of the $\Delta\varepsilon$ values in particular with those for the analogous Fe(III) system. It appears Hedlund made an error in the calculation of $\log_{10}{}^{*}K_1^{\circ}$ because his values could not be reproduced.

Table A-54: Comparison of the results of SIT treatments of the Al(III) $\log_{10}{}^{*}K_{m,1}$ data for $NaClO_4$ solutions, which include those from [1988HED] where appropriate, with those for Fe(III) [2013LEM/BER] at 25 °C; all uncertainties are 1σ.

Metal ion	Condition	$\log_{10}{}^{*}K_1^{\circ}$	$\Delta\varepsilon$ /kg·mol^{-1}	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}
Al(III)	Fit to $I_m \leq 1.1$	$-(4.95 \pm 0.02)$	$-(0.37 \pm 0.05)$		
	Fit of $I_m \leq 3.2$	$-(4.90 \pm 0.03)$	$-(0.17 \pm 0.02)$		
	Fit of $I_m \leq 3.2$	$-(4.92 \pm 0.04)$		$-(0.27 \pm 0.05)$	(0.24 ± 0.08)
Fe(III)	Fit to $I_m \leq 3.2$	$-(2.08 \pm 0.02)$	$-(0.10 \pm 0.02)$		
	Fit of $I_m \leq 3.2$	$-(2.15 \pm 0.02)$		$-(0.22 \pm 0.03)$	(0.21 ± 0.05)

[1988HOV]

Heat capacities and molar volumes of boric acid and sodium borate were measured relative to water with a Picker flow microcalorimeter and densitometer, calibrated against a 1 mol·kg^{-1} NaCl solution at 10, 25, 40 and 55 °C. The values of $C_{p,\mathrm{m}}^{\circ}(298.15\,\mathrm{K})$ for $B(OH)_3(aq)$ and $NaB(OH)_4$ (sln) were tabulated as: (107.4 ± 0.2) and (31.8 ± 0.4) J·K^{-1}·mol^{-1}, but are not published elsewhere apart from a small graph in [1990HOV/HEP]. The $\log_{10}K_1^{\circ}$ values are in excellent agreement with those determined by Mesmer *et al.* [1972MES/BAE] up to 290 °C when coupled with their value of $\Delta_r S_{\mathrm{m}}^{\circ}(298.15\,\mathrm{K})$, *i.e.*, to within $\pm 0.01 \times \log_{10}K_1^{\circ}$. Values of $C_{p,\mathrm{m}}^{\circ}(298.15\,\mathrm{K})$ for NaOH(sln) were taken from Allred and Woolley [1981ALL/WOO], $-(100.5 \pm 0.5)$ J·K^{-1}·mol^{-1}, from Roux *et al.* [1984ROU/PER], $-(97.4 \pm 0.3)$ J·K^{-1}·mol^{-1}, and from Magalhães *et al.* [2002MAG/KON], $-(101.4 \pm 0.7)$ J·K^{-1}·mol^{-1}. Manyá and Antal [2009MAN/ANT] reviewed the available heat capacity measurements for NaOH(sln) and applied the Pitzer ion interaction model to these and other data available over the range 26 to 100 °C and derived a value of $C_{p,\mathrm{m}}^{\circ}(298.15\,\mathrm{K}) = -99.54$ J·K^{-1}·mol^{-1}, without an estimate of the uncertainty. This value is accepted with a statistical uncertainty, *i.e.*, $-(99.5 \pm 3.4)$ J·K^{-1}·mol^{-1}, which gives $\Delta_r C_{p,\mathrm{m}}^{\circ}(H_3BO_3(aq) + OH^- \rightleftharpoons H_4BO_4^-, 298.15\,\mathrm{K}) = (23.9 \pm 3.4)$ J·K^{-1}·mol^{-1} (2σ uncertainty).

[1988PER/KIT]

The solubilities of corundum (T61 Alcoa), gibbsite (C730 Alcoa), boehmite (synthetic) and diaspore (Wards) were reported at 298 K. These solids were all pre-washed for 14 days with 1 mol·dm^{-3} HCl following earlier recommendations for removing unwanted surface phases [1982BLO/WEA] and were characterised from their XRD patterns. Particle sizes were established by SEM: < 5.7, < 1.0, < 0.9 and < 2.5 μm, respectively. The pH was measured with a combination glass electrode presumably calibrated using commercial pH buffers. Experiments were made from under- and oversaturation for up to seven weeks at: pH = 3.29 - 4.16, m_{Na^+} = 0 - 3.61 × 10^{-3} mol·dm^{-3}. Activity coefficients were calculated with the Davies equation. The re-calculated results are summarised in Table A-55 (noting that pAl^{3+} in Tables 2 and 3 of [1988PER/KIT] refers to $-\log_{10} c^{o}_{Al^{3+}}$ corrected for hydrolysis). According to Peryea and Kittrick, hydrolysis accounts for *ca*. 80 % of the total Al in solution, but the authors stated "The relative molar distribution of dissolved Al species predicted from the present model ranged from 89 to 99.9 % Al^{3+}, 0 to 7.5 % $AlOH^{2+}$, 0 to 3.6 % $Al(OH)^{+}_{2}$, and 0.001 % $Al(OH)^{-}_{4}$"; these were derived from: $\log_{10} {}^{*}K^{o}_{1} = -5.02$; $\log_{10} {}^{*}\beta^{o}_{2,1} = -9.30$, $\log_{10} {}^{*}\beta^{o}_{4} = -23.33$.

Table A-55: Average $\log_{10} {}^{*}K^{o}_{s,0}$ values reported in [1988PER/KIT] and re-calculated using the SIT (uncertainties are 2σ and are based on estimates of the experimental uncertainties by this review).

Solid phase	$\log_{10} {}^{*}K^{o}_{s,0}$ [1988PER/KIT]	$\log_{10} {}^{*}K^{o}_{s,0}$ (a) [1988PER/KIT]	$\log_{10} {}^{*}K^{o}_{s,0}$ SIT
corundum	(9.01 ± 0.05)	(9.01 ± 0.05) (9.02 ± 0.05)	(9.05 ± 0.12)
gibbsite	(7.76 ± 0.14)	(7.75 ± 0.14) (7.78 ± 0.14)	(7.83 ± 0.13)
boehmite	(7.49 ± 0.09)	(7.49 ± 0.09) (7.50 ± 0.09)	(7.53 ± 0.20)
diaspore	(6.75 ± 0.24)	(6.77 ±0.25) (7.0 ± 0.5)	(6.9 ± 0.4)

(a) The two entries for each mineral in this column were derived by averaging the $\log_{10} {}^{*}K^{o}_{s,0}$ and ${}^{*}K^{o}_{s,0}$ values given in [1988PER/KIT], respectively, indicating that these authors erroneously used the former.

[1988REA]

Read investigated the effects of pressure and temperature on the first dissociation constant of phosphoric acid using an AC conductivity method. Conductivities of NaH_2PO_4(sln) were measured from 1.662 to 3.192 mol·kg^{-1} at 25, 100, 125, 150, 175 and 200 °C and pressures from up to 200 MPa at 20 MPa intervals. The cell constant was obtained using conductivity values from [1949MAS/CUL] and it was argued that the effects of thermal expansion of the cell and compression under pressure could be ignored. Obviously the unstated accuracy of these data depends strongly on those incurred in the much older work [1949MAS/CUL]. No explanation was given as to why

traditional cell constant determinations with KCl solutions were not attempted. Molar ionic conductivities were extrapolated to zero ionic strength using the Onsager equation. Read listed values of $\log_{10}K_1^\circ$ at 0.1 MPa as follows: 2.15, 2.57, 2.70, 2.84, 3.02 and 3.20 at 25, 100, 125, 150, 175 and 200 °C, respectively; at saturation steam pressure they differ by < 0.1 × $\log_{10}$ at 200 °C.

[1989BOU/GRI]

This compilation of literature data led Bourrié *et al*. to propose a mechanism for the fate of aluminium in low temperature natural waters constrained by the solubility of gibbsite. Three $\log_{10}{}^*K_{s,0}^\circ$ values cited for gibbsite at 0, 10 and 25 °C, *viz*., 9.62, 8.95 and 8.05, respectively, were attributed to [1981BOU]. In the absence of the primary data and due to the low temperatures (0 to 10 °C) at which equilibrium was claimed to have been reached, these results are not included in the final data base, although they were in remarkably good agreement with the extrapolated values from the current assessment.

[1989CAI/CON]

The apparent molar heat capacities of $Al(OH)_4^-$ in aqueous NaOH solutions were measured between 50 and 250 °C in the range 0.3 to 1.7 mol·kg^{-1} (NaOH). These measurements were made with a custom-built differential flow calorimeter. Density measurements were performed with an Anton Paar densitometer with a precision of ± 10^{-5} g·cm^{-3}. Caiani *et al.* determined a value of $C_{p,\mathrm{m}}^\circ(\mathrm{NaAl(OH)_4}, \mathrm{aq}, 298.15\ \mathrm{K}) = 74\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$ by extrapolation, which led to the value of $\Delta_r C_{p,\mathrm{m}}^\circ(\mathrm{Al^{3+} + 4H_2O(l) \rightleftharpoons Al(OH)_4^- + 4H^+}, 298.15\ \mathrm{K}) = -139\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$.

[1989KON/BUG]

This investigation of the aragonite-calcite transition was carried out using two galvanic cells without liquid junction:

Pt |H_2(g, p_{H2});CO_2(g, p_{CO2})|$CaCl_2$(aq, *m*)|$CaCO_3$(calc)|AgCl|Ag(I)

Pt|H_2(g, p_{H2});CO_2(g, p_{CO2})|$CaCl_2$(aq, *m*)|$CaCO_3$(arag)|AgCl|Ag(II)

operating with the same molalities and the same partial pressures of H_2(g) and CO_2(g), thus differing only by the presence of calcite or aragonite. Under such conditions the measured potential differences, *E*, of each of the cells are related to Reaction (A.192) for the selected medium:

$$\mathrm{H_2(g) + 2AgCl(cr) \rightleftharpoons 2H^+ + 2Cl^- + 2Ag(cr)} \qquad \text{(A.192)}$$

the measured pH being given by Reaction (A.193):

$$CaCO_3(\text{calcite or aragonite}) + 2H^+ \rightleftharpoons Ca^{2+} + H_2O(aq) + CO_2(g) \quad \text{(A.193)}$$

that is also an expression of the solubility constants for calcite (K_{calc}) and aragonite (K_{arag}).

Under these conditions, the Gibbs energy change corresponding to Reaction (A.194):

$$CaCO_3(calc) \rightleftharpoons CaCO_3(arag) \quad \text{(A.194)}$$

is obtained as $\Delta_r G(\text{A.194}) = RT\ln(K_{aragonite}/K_{calcite})$ with the proviso that the reaction media are identical.

The reaction vessel, electrode treatment and operating modalities are described in detail. Calcium carbonate (Merck, p.a.), identified as calcite by X-ray diffraction, was used without further purification, either as such or re-crystallised as suggested by Plummer and Busenberg [1982PLU/BUS], with no difference in the observed potential differences. The aragonite sample was prepared as described by Lippmann [1960LIP]. The major impurities were sodium (370-410 ppm) and magnesium (320-430 ppm). X-ray diffraction, SEM and optical microscopy examinations were made and revealed no traces of other phases before and after equilibration with the aqueous solutions.

The potential difference was measured at 21 temperatures between 278.15 and 318.15 K, at intervals of 10 K with 0.10, 0.20, 0.30, and 0.40 mol·kg^{-1} $CaCl_2$. Occasionally, solutions contained also 0.02 mol·kg^{-1} HCl. No more than 50 % of the solid phases dissolved during the experiments. Constant potential difference values (within 0.01 mV) were obtained after 12 to 48 hours equilibration, depending on the temperature. The detailed data were presented in the form of a Table.

The linear regression analysis of the results led the authors to $\Delta_r G_m^o$ ((A.194), 298.15 K) = (0.830 ± 0.020) kJ·mol^{-1}, $\Delta_r H_m^o$ ((A.194), 298.15 K) = – (0.540 ± 0.060) kJ·mol^{-1} and $\Delta_r S_m^o$ ((A.194), 298.15 K) = – (4.60 ± 0.21) J·K^{-1}·mol^{-1}.

However, the well-established calorimetric data on the entropies of aragonite and calcite [1969STA/LIN] indicate $\Delta_r S_m^o$ ((A.194), 298.15 K) = – (3.72 ± 0.32) J·K^{-1}·mol^{-1}, so we have re-fitted their experimental data with this constraint, which gives:

$$\Delta_r G_m^o \text{ ((A.194), 298.15 K)} = (0.834 \pm 0.097) \text{ kJ·mol}^{-1},$$

$$\Delta_r H_m^o \text{ ((A.194), 298.15 K)} = -(0.275 \pm 0.035) \text{ kJ·mol}^{-1}.$$

Their $\Delta_r G_m^o$ ((A.194), 298.15 K) is in good agreement with the solubility data of Plummer and Busenberg [1982PLU/BUS], who reported (0.820 ± 0.130) kJ·mol^{-1}, and more generally, with the value derived from solubilities studied in this review, 0.859 kJ·mol^{-1}.

[1989GRO/STO]

This paper describes a very detailed experimental study of the thermal properties of α and β quartz, using adiabatic shield calorimetry in the temperature range (298.15-1000) K, and a very thorough assessment of all the earlier thermal data on quartz.

Two very pure samples of quartz from Norwegian minerals were purified by unspecified mechanical, magnetic and flotation processes. The maximum impurity levels are given as Al 40, Ca 115, Fe 70, K 20 and Na 70 ppm for the first sample, and Al 20, Ca 0.2, Fe 0.2, K 0.2, Li 4.5, Na 0.4 and Ti 1.5 ppm for the second.

The detailed results of *ca.* 100 points for α-quartz from 319 to 848 K, and 12 for β-quartz, from 862 to 1000 K, plus *ca.* 50 measurements of the enthalpy difference for very small increments in the transition region are tabulated. From the latter, two values of $H(865\ \text{K}) - H(820\ \text{K})$ of (540 ± 40) and $(515.4 \pm 1.5)\ \text{J·mol}^{-1}$ were obtained by summation of the small increments. The α-β transition occurs continuously in a γ-like fashion with maximum heat capacity of about $1700\ \text{J·K}^{-1}\text{·mol}^{-1}$ over a 0.02 K interval near 847.30 K, in complete agreement with the value obtained by [1953SIN], 847.3 K. The temperature of the maximum C_p was (847.25 ± 0.02) K.

This very thorough experimental section is followed by an equally thorough assessment of all the existing data on heat capacity and enthalpy of α and β quartz (in all 83 references are given in the paper) to give finally a table of recommended thermal functions for α and β quartz up to 1700 K; the enthalpy and entropy values are referred to 298.15 K.

We cannot improve on this assessment, and have accepted these values, shown in the comparison Figures V-19 and V-20 in Sections V.3.1.2.2.1 and V.3.1.2.3.1 in the main text for the heat capacity and enthalpy.

[1989GRO/STO] present a more illuminating graph of the deviations of enthalpy from all the earlier relevant studies, which shows that the majority of the studies differ from the assessed line by less than $400\ \text{J·mol}^{-1}$ and all lie within $1200\ \text{J·mol}^{-1}$. This is also true of the subsequent heat capacity measurements of [1995AKA/YUS].

[1989HEM/KIT]

Hemingway, Kittrick and Peryea presented this addendum to an earlier paper by Peryea and Kittrick [1988PER/KIT], which reported $\Delta_f G^\circ_m(\text{Al}^{3+}, \text{aq}, 298.15\ \text{K}) = -487.5\ \text{kJ·mol}^{-1}$. They noted that Hemingway *et al.* [1978HEM/ROB] calculated the Gibbs energy of formation of A1^{3+} as $-489.4\ \text{kJ·mol}^{-1}$, from the corresponding Gibbs energy of gibbsite calculated from calorimetric measurements and solubility data for gibbsite [1966KIT] and [1974SIN]. The principal reason for this difference was postulated to be the presence of Cl^- in the solubility experiments of Kittrick and Peryea, rather than to

different pre-treatments of the solids as proposed previously [1982BLO/WEA], but more recent solubility data favour strongly the argument offered by Bloom and Weaver.

[1989HER/FER]

Automated potentiometric titrations of aqueous phosphate solutions in a cell made up of a glass electrode and an Ag|AgCl reference electrode were performed in (0.5 to 6.0 $mol \cdot kg^{-1}$) NaCl and Na-Mg-Cl solutions at (25.00 ± 0.02) °C. The total phosphate concentration was maintained between 0.003 and 0.005 $mol \cdot dm^{-3}$. The electrodes were calibrated on a molar scale by fitting the apparent standard potential of the cell and the formation constant in each titration using a non-linear regression. Conversions from the measured molar to the molal scale relied on independent density measurements [1982LOS/ALZ]. Hershey *et al*. gave fitting equations for the three stepwise ionisation constants where the infinite dilution values were fixed: $\log_{10} K_1^\circ = -2.146$ [1976PIT/SIL]; $\log_{10} K_2^\circ = -7.198$ [1943BAT/ACR]; and $\log_{10} K_3^\circ = -12.35$ [1964SIL/MAR]. The SIT ion interaction coefficients are applied by this review to the three datasets and these results are given in Table A-56. Note that the value of $\log_{10} K_3^\circ$ is fixed at – 12.35 which is the preferred value derived in this review.

Table A-56: Results of application of the SIT to phosphate solutions at 298.15 K (uncertainties 2σ).

i	$-\log_{10} K_i^\circ$ /fixed	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	(I_m)max	$\Delta\varepsilon$ /kg·mol^{-1}
1	2.153	(0.024 ± 0.020)	(0.013 ± 0.031)	6.0	(0.033 ± 0.005)
2	7.199	– (0.048 ± 0.008)	(0.069 ± 0.029)	2.0	
2	7.199	– (0.056 ± 0.008)	(0.102 ± 0.012)	6.0	
3	12.35	– (0.071 ± 0.017)	(0.124 ± 0.025)	6.0	

The authors went on to apply the Pitzer ion interaction model to these data and also derived an ion-pair formation constant for the $MgHPO_4$ of $10^{2.7}$ $kg \cdot mol^{-1}$. The use of empirical equations to describe the ionic-strength dependencies of the stepwise acid dissociation constants of phosphoric acid without invoking a Debye-Hückel relationship at low ionic strength is difficult to understand. Certainly the second dissociation constants are inconsistent with those from other studies that are very reliable. Therefore, neither these dissociation constants nor the corresponding ion interaction coefficients are accepted by this review (see Appendix A entry for [2005PAR/COV] for comparison with their model for the second acid dissociation constant).

[1989ROB/RUS]

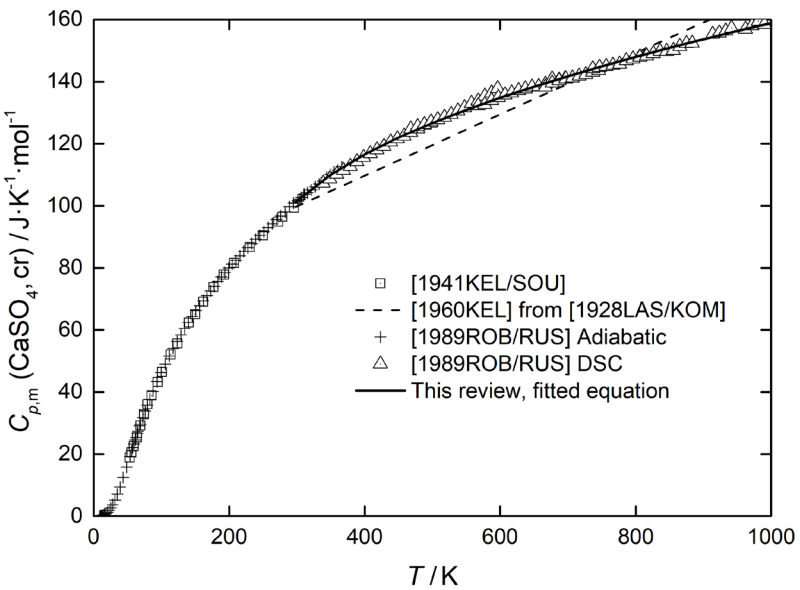
The heat capacities of $CaSO_4$(cr) (anhydrite) were measured by adiabatic shield calorimetry from 8.65 to 366 K (84 measurements) and by DSC from 339 to 900 K (80

measurements). The sample was prepared by heating anhydrous $CaCl_2$ and K_2SO_4 first to 1150 K for three hours and then to 950 K for 16 hours, and, after cooling, washing with hot water to remove any residual $CaCl_2$ and KCl. Spectrographic analysis indicated the presence of a potassium impurity equivalent to 0.09 % K_2O.

Tables of all the individual measurements are given, together with calculated thermal functions at round temperatures from 5 to 360 K; the low-temperature data were systematically greater than the earlier values of [1941KEL/SOU] by 0.4 to 1.4 %. The calculated standard values were $C^{o}_{p,m}(CaSO_4, cr, 298.15\ K) = (101.2 \pm 0.2)$ and $S^{o}_{m}(CaSO_4, cr, 298.15\ K) = (107.4 \pm 0.2)\ J \cdot K^{-1} \cdot mol^{-1}$. The DSC measurements were more scattered than those from the more precise adiabatic calorimeter, but agreed well in the overlap region.

As noted in Section V.6.2.1.4.2, these data form the basis of the selected values for the heat capacity of anhydrite; a comparison of all the experimental and assessed data is shown in Figure A-22.

Figure A-22: Heat capacity of anhydrite.



The heat capacities of natural crystals of gypsum ($CaSO_4 \cdot 2H_2O$) were also studied between 8.1 and 323 K. (*ca.* 72 measurements) Their results are significantly lower than the values of Latimer *et al.* [1933LAT/HIC] up to 50 K, above which the values are consistently larger by 0.2-0.6 % to 300 K. In comparison with the study of

Kelley *et al.* [1941KEL/SOU], the two sets of values are very similar up to *ca.* 200 K, but those of Kelley *et al.* are slightly smaller above this temperature. However, as noted in the main text, the derived standard entropies S^{o}_{m} ($CaSO_4 \cdot 2H_2O$, cr, 298.15 K) from the three studies are essentially the same.

The reported measurements on langbeinite ($K_2Mg_3(SO_4)_3$) from 8.2 to 990 K are of no immediate interest to the present review and are not considered further.

[1989WAK/SAN]

The authors measured the concentration of Ca in Ag(l), equilibrated with a lime-saturated $CaO\text{-}CaF_2$ or $CaO\text{-}CaCl_2$ slags held in a graphite crucible, at eight temperatures from 1570 to 1831 K, under a pressure of CO(g) (mixed with Ar(g)) of 0.1 bar. These results are tabulated; we note that two measurements at temperatures differing by only 1 K differed by 23 %, so the accuracy of these data is difficult to judge. The activity of Ca in Ag(l) was measured in supplementary experiments from 1417 to 1832 K (six measurements) by equilibrating Ag(l) with CaC_2-saturated $CaC_2\text{-}CaF_2$ or $CaC_2\text{-}CaCl_2$ slags. However, since the Ag concentrations are not, of course, the same in the two sets of experiments, it was necessary to assume that Henry's Law was obeyed within the concentration range (x(Ca) = 5.4×10^{-4} to 0.049) and the temperature ranges of the equilibrations with lime. There is some support for this from the more complete measurements by [1985FIS] of a(Ca) in Ag(l) over the whole composition range at lower temperatures (1073-1273 K).

As noted in [1996JAC/VAR], the uncertainties in $\Delta_f H$(CaC_2, cr) and in the activities of calcium and silver mean that these data cannot be used as a strong argument for a considerable error in $\Delta_f H$(CaO, cr).

[1989WIL/SEW]

In the past, Seward and co-workers had taken advantage of the effect of metal cations on the solubility of AgCl(cr) [1976SEW], [1986RUA/SEW] to quantify the stability constants of metal chloride ion pairs and complexes, particularly at high temperatures. In the present paper, solutions of $CaCl_2$+HCl with total chloride molalities ranging from 0.3 to 3.0 mol·kg^{-1} containing 0.1 mol·kg^{-1} Ca(II) were studied from 100 to 360 °C. The only significant anion present was chloride ion which presents the usual problem for delineating activity from mass-action effects. The auxiliary thermodynamic data required to extract the association constants for $CaCl^+$ and $CaCl_2$(aq) were: the solubility of AgCl(cr); the formation constants of the relevant $AgCl_n^{1-n}$ complexes; and the acid-dissociation constants of HCl(aq). They chose the extended Debye-Hückel equation taken from Helgeson *et al.* [1981HEL/KIR] to estimate the activity coefficients at these high ionic strengths. No calcium chloride association constants

could be extracted from the data at 100 °C and only one $\log_{10}K_1^{\circ}(CaCl^+)$ value was determined at 150 °C, *viz*., – (0.07 ± 0.01). Values of $\log_{10}\beta_2^{\circ}(CaCl_2(aq))$ ranged from 0.24 at 150 °C to 4.70 at 360 °C. The $\log_{10}\beta_2^{\circ}(CaCl_2(aq))$ data given in their Table 3 were regressed to give following equation.

$$\log_{10}\beta_2^{\circ}(CaCl_2(aq)) = -12.31 + 821.6\,(T/K)^{-1} + 0.0249(T/K) \quad (A.195)$$

The experimental conditions extend well beyond the range of direct interest to this review, however, they clearly show the difficulty in extrapolation of high-temperature data, where ion association is more readily measured, to low temperatures, *e.g.*, extrapolated values of $\log_{10}\beta_2^{\circ}(CaCl_2(aq))$ at 25 °C were – 2.1 [1989WIL/SEW] and – 5.4 [1982FRA/MAR].

[1989WOL/BRE]

The solubilities of calcite and gypsum (both Merck reagent grade) at 283, 298 and 333 K and at CO_2 partial pressures of ~ 1 kPa were investigated in the systems NaCl-$CaCO_3$-CO_2-H_2O, KCl-$CaCO_3$-CO_2-H_2O, $CaCl_2$-$CaCO_3$-CO_2-H_2O, NaCl-$CaSO_4$-$CaCO_3$-CO_2-H_2O and KCl-$CaSO_4$-$CaCO_3$-CO_2-H_2O up to 6.3 mol·dm^{-3} ionic strengths. The suspensions were stirred magnetically and a N_2-CO_2 stream was bubbled into the solutions. After at least eight days, the solutions were filtered through a ceramic filter under constant CO_2 partial pressure and analysed for total alkalinity and calcium content, but no details are given for the titration procedures. Due to the lack of appropriate Pitzer parameters, only data obtained in the NaCl-$CaCO_3$-CO_2-H_2O system at 298 K were evaluated. The solubility products calculated from these data show good agreement with other reports.

[1990EIK]

Eikenberg provided a critical review of the literature on the solubility of amorphous silica and quartz in aqueous solution with particular emphasis on the high pH region. Eikenberg stated that between pH 10.5 and 11.3, dimers, trimers and tetramers are formed, but that at pH ≤ 10 and $c_{Si} < 0.01$, only monomers are found. Two models were proposed for the speciation of silica in aqueous solution at 25 °C (Models 1A and 1B in his Table 6) that relied on the equilibrium constants for the hydrolysis reactions proposed in a 1989 draft of [1992GRE/FUG] with one exception. These preliminary NEA recommended values were retained in the 1992 paper. The exception, which was recognised by Grenthe *et al*. (see [1992GRE/FUG], p. 399), referred to the equilibrium constant for the reaction:

$$4Si(OH)_4(aq) \rightleftharpoons Si_4O_8(OH)_4^{4-} + 4H^+ + 4H_2O(l) \quad (A.196)$$

A value of $\log_{10}K^{\circ}$((A.196), 298.15 K) of – 34.9 was adopted in Model 1A rather than – (36.3 ± 0.5) [1992GRE/FUG]. Grenthe *et al*. pointed out that if the latter value were accepted, satisfactory agreement between the experimental and calculated solubilities was obtained up to pH ≈ 11. Eikenberg did not recommend values for the first and second hydrolysis constants of $Si(OH)_4$(aq), but did tabulate values from many independent investigations from which it can be concluded that the value of $\log_{10}K^{\circ}(Si(OH)_4(aq) \rightleftharpoons SiO(OH)_3^- + H^+, 298.15\,K) = -(9.81 \pm 0.02)$ [1992GRE/FUG] is reasonable.

[1990HOV/HEP]

Previous papers from these authors reported the standard partial molar heat capacity and volume functions for Al^{3+} and $Al(OH)_4^-$ from 10 to 55 °C. This paper also included values for Fe^{2+}, Fe^{3+}, $B(OH)_4^-$ and $B(OH)_3(aq)$. An objective of this research was to provide data which could extend the use of the Helgeson-Kirkham-Flowers model. The large difference in $C_{p,m}^{\circ}$ for the tetrahedral aluminate and borate anions shown only in their Figure 2 could not be explained and the differences in the non-Born contributions to $C_{p,m}^{\circ}$ could only be removed by assuming unrealistic effective radii according to the HFK model. From their Figure 2, approximate $C_{p,m}^{\circ}(298.15\,K)$ values for $B(OH)_3(aq)$ and $NaB(OH)_4$ were obtained graphically: 108 and 33 $J \cdot K^{-1} \cdot mol^{-1}$. These values were tabulated in Hovey's thesis [1988HOV].

[1990NOR/PLU]

A summary of equilibrium constants and reaction enthalpies for aqueous ion association and mineral solubility reactions was compiled for geochemical modelling purposes pertaining to natural waters between 0 to 100 °C for species containing: Na, K, Li, Ca, Mg, Ba, Sr, Ra, Fe(II/III), Al, Mn, Si, C, Cl, S(VI) and F. For example, Nordstrom *et al*. tabulated $\log_{10}K_1^{\circ}$ and $\Delta_r H_1^{\circ}$ values at 25 °C for the formation of $MgSO_4$(aq) and $CaSO_4$(aq) taken directly from [1958NAI/NAN] and [1953BEL/GEO], respectively. They also tabulated a $\log_{10}{}^{*}K_1^{\circ}$ value of – 12.78 for the hydrolysis of Ca^{2+} at 25 °C stating that it is compatible with the CODATA values in [1987GAR/PAR] and that published in [1960YAT/VAS]. Similarly, a $\log_{10}{}^{*}K_1^{\circ}$ value of – 11.44 for the hydrolysis of Mg^{2+} at 25 °C was given [1976BAE/MES].

[1990WES/PAL]

This investigation was undertaken to explore the interaction of the pH buffer, Bis-Tris, 2,2-bis(hydroxymethyl)-2',2",2"'-nitrilotriethanol, with the aluminate anion in response to an article [1979MAY/HEL] which reported the use of this so-called "non-complexing" buffer in gibbsite solubility experiments at 25 °C. Wesolowski *et al*. used

Raman spectroscopy to verify the existence of a 1:1 complex between $Al(OH)_4^-$ and Bis-Tris at room temperature. Gibbsite solubility experiments were also performed at 25 and 50 °C in 0.1 mol·kg^{-1} NaCl solutions that established $\log_{10}{}^{*}K_{m,1}$ values of (2.63 ± 0.03) and (2.6 ± 0.1), respectively, for the formation of this complex. The interaction of the parent molecule, triethanolamine, with aluminate is well documented. The stoichiometry of the (Bis-Tris) aluminate complex proposed by Wesolowski *et al*. was probably incorrect. Nevertheless, the complex is strong and its formation readily accounts for the discontinuity found in the experiments of May *et al*. [1979MAY/HEL] at pH ≈ 7 that they attributed to the formation of an unknown aluminium oxide/hydroxide solid phase.

[1990WRI]

This paper is a comprehensive review of the data on the Si-O phase diagram, with *ca*. 250 references and includes useful tables of the structure and lattice parameters of the many polymorphs of silica (including coesite and stishovite). There is also a clear summary of the papers of Flörke, Holmquist and Roy regarding the stability or instability of tridymite at ambient pressures. As indicated in Section V.3.1.4.1 (on tridymite) the author also noted that the reported range of stability of tridymite in numerous publications is always close to (870-1470) °C, independent of the mineraliser used. He suggests that if impurities are required to stabilise tridymite, one would expect a more or less continuous range of temperatures at which it is formed. Since this is not the case, he makes the choice (his terminology) that tridymite is indeed stable from (870-1470) °C at ambient pressures, but gives no explanation of the results of the electrolysis experiments of Flörke [1956FLO], [1961FLO] and Holmquist [1958HOL2].

However, this review finds that the evidence of these two electrolysis studies of Flörke and Holmquist, and those of Holmquist [1961HOL] and [1962CYP], that tridymite is only stable in the presence of impurities, more convincing than the argument of Wriedt, and thus accepts that tridymite is not a stable phase in the pure silica system at ambient pressures.

[1991BEN/SOM]

This paper is concerned with the determination enthalpies of formation of calcium-hydroxylapatite, -fluorapatite and -chlorapatite based on the dissolution of these compounds in nitric acid (HNO_3, 35.35H_2O), using an isoperibol calorimeter. However, the authors do not give information on the individual values of the enthalpies of solution nor on uncertainties of their final results. As the thermodynamic cycles used and the reported value for the enthalpies of formation of calcium-hydroxylapatite and fluorapatite are essentially the same as those reported by Jemal *et al.* [1995JEM/BEN] (where more detailed experimental data are given), this review assumes that these do

not pertain to independent studies. In fact, the value for the enthalpy of formation of the calcium hydroxylapatite in [1991BEN/SOM] is identical to that in [1995JEM/BEN]; in the case of the enthalpy of formation of calcium fluorapatite, the values reported in the two papers differ by less than four kJ·mol^{-1} (for the formula with 10 Ca). The reader is thus directed to the discussion in [1995JEM/BEN].

The present paper reports also, without much detail, experimental results on the lattice parameters of $Ca_{10}(PO_4)_6F_2$(cr) as $a = (9.370 \pm 0.005)$ and $c = (6.881 \pm 0.005)$ Å, values reproduced without comment in [1995JEM/BEN]. These values are in agreement with those reported by Hughes *et al.* [1989HUG/CAM] listed in Section V.8.2.4.1 in the main text.

The paper gives a reference [1988BEN/KHA] concerning the analytical results on their sample of $Ca_{10}(PO_4)_6(OH)_2$ (cr).

[1991CHE/XU]

Chen, Xu and Hepler made calorimetric measurements of the enthalpies of solution of gibbsite in aqueous NaOH solutions from 100 to 150 °C. These results were used to obtain the standard enthalpy of formation of $Al(OH)_4^-$ as well as the values of $\Delta_r C^o_{p,m}$ (Table A-57) for the reaction:

$$\alpha\text{-}Al(OH)_3(cr) + OH^- \rightleftharpoons Al(OH)_4^- \qquad (A.197)$$

Table A-57: Values of $\Delta_r H^o_m$ and $\Delta_r C^o_{p,m}$ taken from Table 3 in this paper.

t /°C	$\Delta_r H^o_m$ /kJ·mol^{-1}	$\Delta_r C^o_{p,m}$ /J·K^{-1}·mol^{-1}
100.1	30.07	116.2
112.9	31.56	116.9
125.7	33.06	117.6
138.6	34.59	118.3
150.3	35.97	118.9

This paper also lists the thermodynamic properties of gibbsite from 0 to 150.3 °C with values at 25 °C of $\Delta_f H^o_m = -1293.13$ kJ·mol^{-1} and $C^o_{p,m} = 91.69$ J·K^{-1}·mol^{-1}. The corresponding values for $Al(OH)_4^-$ at 25 °C were given as: $\Delta_f G^o_m = -1305.00$ kJ·mol^{-1}, $\Delta_f H^o_m = -1502.77$ kJ·mol^{-1}, $S^o_m = 101.49$ J·K^{-1}·mol^{-1} and $C^o_{p,m} = 89.6$ J·K^{-1}·mol^{-1}.

[1991CHR/GRI]

The authors studied the reaction:

$$CaO(cr) + 3C(cr) \rightleftharpoons CaC_2(cr) + CO(g) \qquad (A.198)$$

by measuring the CO(g) pressure at which the reaction started at temperatures between 1853 and 1973 K, using a Cartesian manostat. Samples of lime of 99.9 % purity (manufacturer's specification), heat-treated at 1100 °C for 48 hours, mixed with spectroscopic graphite were dropped into a graphite crucible in a heated graphite susceptor and the system allowed to equilibrate. The results are only presented in a small graph, and are reported to correspond to $\Delta_r G_m$ ((A.198), T) /J·mol^{-1} = 424927 –216.145 (T/K). There are unfortunate discrepancies in the reported value for $\Delta_f S_m$ (CaO, cr) between the abstract, main text and conclusions.

The values of $\Delta_r G_m$ (A.198) are appreciably smaller than the values derived from the currently accepted values, and the authors blithely state that the difference "is likely to lie in the ΔH term for CaO". In fact, the uncertainties in many of the data for CaC_2(cr) are, of course, considerably greater than those for CaO(cr).

Moreover, as Jacob and Varghese [1996JAC/VAR] have pointed out in their analysis of this and similar papers, no consideration was given to the possible dissolution of CaC_2 in CaO(cr) (both have fcc structures at these temperatures). Chrysanthou and Grieveson did establish that CaC_2 was indeed present in the product of their reaction, but not unambiguously as a pure phase.

Given the consistency of the enthalpy data for CaO(cr) with that for a wide swathe of other calcium compounds, the uncertainties in the data for CaC_2(cr), and the experimental and reporting uncertainties, we cannot accept this study as proof that $\Delta_f H_m$(CaO, cr) is seriously in error.

[1991DAM/BIS]

The solubility of quartz was measured in seawater at 200, 300, 363, 400 and 450 °C from saturation vapour pressure to 1 kbar. The solutions were equilibrated in a gold bag suspended in a rocking autoclave. Von Damm *et al.* incorporated 394 data points from 17 previous studies to give:

$$\log_{10} K^{\circ}_{s,4}(\text{quartz}) = -1.011 - 983.078\,(T/\text{K})^{-1} + 0.00176910\,(T/\text{K}) + 0.77976 \ln \rho_s + 0.017320\,(p/p_o)\,(T/\text{K})^{-1} \qquad \text{(A.199)}$$

where ρ_s is the solution density (g·cm^{-3}), and p is the pressure in bar and p_o = 1 bar. This empirical equation relies on the correlation of the activity coefficient of $Si(OH)_4$(aq) with the solution density. Equation (A.199) was strictly derived to cover the range 318 to 1173 K and 1 to 9860 bar, but interestingly it produces a value of the solubility of quartz at 25 °C and 1 bar of 1.7 × 10^{-4} mol·kg^{-1} which is quite close to the accepted values from Rimstidt [1997RIM] and others, including this review, with the exception of van Lier *et al*. [1960LIE/BRU].

[1991DAN/DER]

A potentiometric study of the dissociation constants of orthophosphoric acid was carried out using glass electrodes *versus* saturated calomel reference electrodes in NaCl and KCl media (results also obtained in aqueous $(C_2H_5)_4NI$ medium were not re-treated) to I_c = 1 from 10 to 50 (± 0.1) °C. The electrodes were calibrated by titration in the same medium. Liquid-junction potentials were calculated for $pH_c < 3$. Values of $\log_{10}K_{c,n}$ for n = 1-3 corresponding to the stepwise protonation equilibria were tabulated at ionic strengths of 0.04, 0.16, 0.36, 0.64 and 1.00 mol·dm^{-3} at 10, 20, 25, 30, 40 and 50 °C. After conversion to molal values [1985SOH/NOV], the SIT treatment was applied first exclusively to the resulting molal dissociation constants at 25 °C leading to the ionic strength dependencies summarised in Table A-58. There appeared to be a significant problem with the results for $K_{m,2}$. A general SIT equation was then applied by this review to all the data in the following form:

$$\log_{10}K_{m,n} - 2nD = a + b(T/\mathrm{K})^{-1} + c(T/\mathrm{K}) - (d + e\log_{10}(I_m))I_m$$

Deviation plots for this general equation as functions of ionic strength and temperature generated by this review showed no systematic trends. The resulting fit parameters are listed in Table A-59. This paper presents a vast array of data.

Table A-58: SIT values of $\log_{10}K_n^\circ$ and $\Delta\varepsilon$ from the data at 25 °C only (uncertainties are 2σ).

n	Ionic medium	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$-\log_{10}K_n^\circ$ /SIT	r
1	KCl	(0.060 ± 0.091)	(0.107 ± 0.171)	(2.147 ± 0.004)	0.9659
	NaCl	(0.037 ± 0.051)	(0.299 ± 0.095)		0.9689
2	KCl	– (0.080 ± 0.019)	– (0.162 ± 0.100)	(7.200 ± 0.002)	0.9679
	NaCl	– (0.104 ± 0.009)	– (0.173 ± 0.045)		0.9922
3	KCl	(0.234 ± 0.014)	(0.272 ± 0.082)	(12.33 ± 0.08)	0.9994
	NaCl	– (0.094 ± 0.015)	(0.396 ± 0.090)		0.9935

Table A-59: Values of $\log_{10}K_n^\circ$ and the corresponding $\Delta\varepsilon$ at 25 °C from SIT fits of all the data (uncertainties are 2σ).

n	Ionic medium	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$-\log_{10}K_n^\circ$	r
1	KCl	– (0.049 ± 0.080)	(0.13 ± 0.25)	2.148	0.9925
	NaCl	(0.001 ± 0.025)	– (0.089 ± 0.075)	2.141	0.9994
2	KCl	– (0.061 ± 0.007)	– (0.278 ± 0.043)	7.145	0.9909
	NaCl	– (0.083 ± 0.017)	(0.122 ± 0.102)	7.184	0.9096
3	KCl	(0.132 ± 0.085)	(0.39 ± 0.50)	12.38	0.9696
	NaCl	– (0.116 ± 0.035)	(0.41 ± 0.20)	12.33	0.9921

Despite the limited range of ionic strength covered for the reported $\log_{10}K_{c.1}$ values and the fact that the ion interaction parameters for the second dissociation equilibrium appeared to deviate significantly from those correlated in

[2005PAR/COV2], in which the same trends were at least noted, the results were internally consistent. Moreover, the corresponding results for the third dissociation step were the most comprehensive found to date and were included in the final SIT assessment carried out in this review. Goldberg *et al*. [2002GOL/KIS] adopted a value for $\log_{10} K_3^\circ$ of – 12.35 at 25 °C, but referred to the "discordant values" reported in [1991DAN/DER] although it is not clear if this concern refers to all the data from this study or to only that for $\Delta_r H_{m,3}^\circ$.

[1991PAL/KUZ]

Palkin *et al.* measured the heat capacity of a synthetic sample of hydroxylapatite from 58.5 to 305.18 K. The sample was precipitated from $CaCl_2$ and $(NH_4)_2HPO_4$ solutions with hydrated ammonia to increase the pH to 10.7, dried, washed and heated to 1373 K for two hours. Analyses for Ca^{2+}, PO_4^{3-} and OH^- gave values within 0.01 % of the theoretical values, and XRD showed only lines attributable to a structure with a = 9.376 Å, c = 6.862 Å. As noted in the main text, the structure of hydroxylapatite at ambient temperatures was until recently indeed thought to be hexagonal with similar lattice parameters (a = 9.417 Å, c = 6.875 Å), but this structure is now attributed to the closely related high-temperature polymorph (with a transition temperature of 480 K). It thus seems possible that the sample used by Palkin *et al.* was, or contained, this high-temperature form. However, the monoclinic distortion is very small with a corresponding small enthalpy of transition (see main text), so the heat capacities of the two phases should not differ appreciably.

The detailed results of *ca.* 50 measurements are tabulated, and the heat capacities were extrapolated to 0 K using the sum of five Debye and one Einstein function. Curiously, the authors make no mention of the much earlier study of the low-temperature heat capacities of hydroxylapatite by Egan *et al.* [1951EGA/WAK], whose heat capacity values are lower by 1.8 to 2.5 % up to 100 K, but increasingly higher by 1.3 to 2.0 % from 150 to 300 K.

The values of C_p and S found by Palkin *et al.* at 298.15 K are 376.9 and 388.35 $J \cdot K^{-1} \cdot mol^{-1}$ respectively, with maximum errors indicated to be 0.4 %. These values are both slightly smaller than those of Egan *et al.* [1951EGA/WAK], whose measurements extended down to 11 K.

[1991RIC/FIQ]

The authors measured the enthalpy content of six mantle materials, including MgO(cr) and CaO(cr) at temperatures up to *ca.* 1800 K by drop calorimetry into an ice-calorimeter. Their sample of MgO(cr) was a coarse powder from clear single crystals with a reported impurity content of 2000 ppm, mainly Ca (800 ppm), C (490 ppm), Si

(300 ppm), and Al (200 ppm). Prior to the runs, it was heated in air at 1600 K for two hours. CaO(cr) was prepared from $CaCO_3$(cr) by progressive heating to 1900 K. The manufacturer's specifications gave Na (< 0.2 %), Sr (< 0.1 %), Mg (< 0.05 %) as the main impurities. The only other characterisation of MgO(cr) or CaO(cr) was the measurement of lattice parameters (4.212 and 4.909 Å respectively). The latter value is much larger than the accepted parameter, including that measured later by the same authors [1999FIQ/RIC] (4.813 ± 0.002 Å), and since the authors indicate that their values need no further comment, we suspect that 4.909 is a misprint for 4.809.

For the experiments, the samples were contained in Pt-Rh crucibles, and the temperatures measured with Pt-Rh thermocouples. For MgO(cr) 11 runs were made, up to 1755 K, and for CaO(cr) 10 runs up to 1840 K; the detailed results are tabulated. As shown in Figure V-52 in Section V.6.2.1.1.2 of the main text, the experimental values above 1000 K agree very well with the assessed data (from 1994GUR/VEY). The authors' data below 1000 K are lower than the assessed (and other experimental) data.

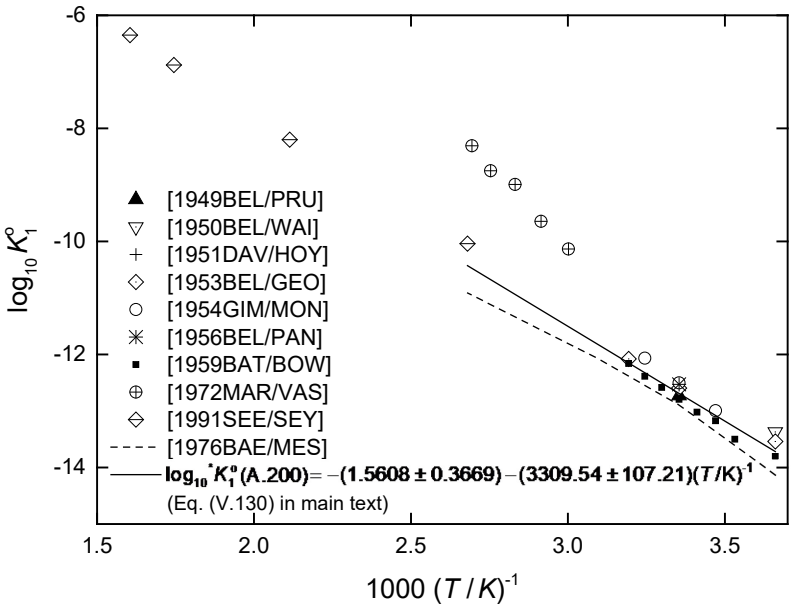
[1991SEE/SEY]

The solubility of $Ca(OH)_2$(s) was studied at 100, 200, 300 and 350 °C with the pressure fixed at 500 bar using the flexible gold-bag batch method. The initial solutions were either pure water or dilute acetic acid, 0.001 to 0.01 mol·kg^{-1}. Activity coefficients were estimated from a simple Debye-Hückel equation [1981HEL/KIR]. The tabulated values for $\log_{10}{}^{*}K_1^{\circ}$ (A.200) were – 10.04, – 8.20, – 6.88 and – 6.35 at 100, 200, 300 and 350 °C, respectively. Acetic acid and water dissociation constants were taken from [1990SCH/HEL].

$$Ca^{2+} + H_2O(l) \rightleftharpoons CaOH^{+} + H^{+} \qquad \text{(A.200)}$$

The respective $\log_{10}{}^{*}K_{s,0}^{\circ}$ values were given as 18.15, 14.40, 11.89 and 10.86. A simple linear regression of these data resulted in a $\log_{10}{}^{*}K_{s,0}^{\circ}$ value at 25 °C (500 bar) of 22.73 and $\Delta_r H_m^{\circ} = -129.4$ kJ·mol^{-1}. As shown in Figure A-23, these results support the choice of the enthalpy of reaction for the hydrolysis of calcium noting that pressure should not have a large effect on this equilibrium at least to 100 °C.

Figure A-23: Temperature dependence of the logarithm of the hydrolysis constant of Ca^{2+} where the solid line was generated by Equation (V.130) in the main text and the dashed curve was taken from [1976BAE/MES].



[1992GIL/OSC]

Flow calorimetry was used to determine the thermodynamic properties (K_1°, $\Delta_r H_{m,1}^\circ$, $\Delta_r S_{m,1}^\circ$ and $\Delta_r C_{p,m,1}^\circ$) for H^+, Mg^{2+} and Ca^{2+} ion pairing with Cl^- ions at 250, 275, 300 and 325 °C, at 10.3, 11.0, 11.0, 13.2 MPa, respectively. For the $MgCl_2$+HCl enthalpies of reaction measurements, the respective concentrations were varied from 1.039 to 0.00969 mol·kg^{-1} and 0.5018 to 0.00582 mol·kg^{-1}. Despite the relatively high ionic strengths employed, Gillespie *et al.* used the Meissner model to estimate activity coefficients. They accepted the presence of $MgCl^+$ and HCl(aq) ion pairs, while noting that inclusion of $MgCl_2$(aq) improved their fit slightly, whereas the inclusion of the $CaCl_2$(aq) species did not. Presumably the authors fitted only their four values of $\log_{10} K_1^\circ$ to which they optimistically assigned an uncertainty of ± 0.04. Rard and Clegg [1997RAR/CLE] wrote "Values of the enthalpy of reaction $\Delta_r H_m^\circ$ and the equilibrium constant K for the reaction are evaluated simultaneously from titration calorimetry data by a non-linear least-squares analysis. However, unless the values of the equilibrium constant are relatively large, of the order of 10^2 or larger (which is not the case for $MgCl_2$ and $CaCl_2$ solutions), the derived reaction enthalpy and equilibrium constant are

highly covariant and usually quite sensitive to the errors in the calorimetric measurements. The resulting least-squares values $\Delta_r H^{\circ}_{m,1}$ and K then depend strongly on the initial estimates, and different estimates may yield significantly different final results".

[1992HAL]

This paper reports an assessment of the Si-O system but with the main emphasis on the fitting and calculation of the phase diagrams. No new data are given; the thermodynamic data used are those from an unpublished assessment by Barry [1987BAR2]. The question of the stability of tridymite at ambient pressure is not discussed, but the data used make it stable from 1143 to 1687 K. These data were not used in our assessment.

[1992JOH/TAS]

The standard molar enthalpy of formation of gibbsite was re-determined in this very careful study because it is a key auxiliary quantity in the evaluation of the enthalpy of formation of mordenite, dehydrated mordenite, and related zeolites studied by the same group of authors. Our comments will be limited to the data concerning gibbsite, the only compound of concern to the present review.

A commercial (LKB-8700) solution calorimeter was used for the enthalpy of solution measurements at 298.15 K in 24.4 wt % hydrofluoric acid (Fisher reagent). The 0.100 dm^3 calorimeter chamber was made of 18-carat gold and the sample holders made of Teflon. Other modifications of the commercial instrument and calorimetry procedures were described. The Fisher reagent gibbsite ($Al(OH)_3$) used was, prior to use, heated for six hours at 423 K which perhaps risked some decomposition to boehmite. The aluminium was part of a single crystal bar (Leytess Chemical Co.). Chemical and spectrochemical analyses revealed the following impurities (in mass fraction × 10^6): O, 663; C, 40; N, 231; H, 33; Mg, 5; and Mn, 5.

The relative amounts of gibbsite and aluminium used were fixed so as to close the reaction cycle below. Seven reactions were carried out with an average weight of 0.01608 g aluminium samples and six reactions with an average weight of 0.04648 g gibbsite. For the aluminium dissolutions, it was assumed that the hydrogen gas that escaped from the calorimeter was saturated with HF and H_2O vapour. Calorimetric results were reported for the 2σ confidence interval.

The thermochemical reactions used were the following:

$$\mathrm{Al(cr) + 2218.39HF\cdot 7632.87H_2O \rightleftharpoons 3/2H_2(g) + solution} \qquad \text{(A.201)}$$

$$\Delta_r H_m\,(\text{A.201}) = -\,(602.93 \pm 1.13)\ \mathrm{kJ\cdot mol^{-1}}$$

$$Al(OH)_3(cr) + 2218.39HF\cdot 7629.87H_2O \rightleftharpoons \text{solution} \quad (A.202)$$
$$\Delta_r H_m\,(A.202) = -\,(166.76 \pm 0.19)\ kJ\cdot mol^{-1}$$

$$2218.39HF\cdot 7629.87H_2O + 3H_2O(l) \rightleftharpoons 2218.39HF\cdot 7632.87H_2O \quad (A.203)$$
$$\Delta_r H_m\,(A.203) = -\,(1.26 \pm 0.20)\ kJ\cdot mol^{-1} \text{ from Johnson } \textit{et al.} \text{ [1973JOH/SMI]}$$

$$3\,H_2(g) + 3/2O_2(g) \rightleftharpoons 3\,H_2O(l) \quad (A.204)$$
$$\Delta_r H_m\,(A.204) = -\,(857.49 \pm 0.13)\ kJ\cdot mol^{-1} \text{ from [1989COX/WAG]}$$

leading to

$$Al(cr) + 3/2\,H_2(g) + 3/2\,O_2(g) \rightleftharpoons Al(OH)_3(cr) \quad (A.205)$$

$\Delta_f H_m^\circ(Al(OH)_3, gibb, 298.15\ K) = \Delta_r H_m\,(A.201) - \Delta_r H_m\,(A.202) + \Delta_r H_m\,(A.203) + \Delta_r H_m\,(A.204) = -\,(1294.92 \pm 1.17)\ kJ\cdot mol^{-1}$.

[1992JUM/ABD]

Potentiometric measurements were made of the ionisation of boric acid in methanol-water and 1,4-dioxane-water mixtures from 5 to 35 °C at 5° intervals. Of direct relevance to this review are the ionisation constants reported in 0.05 $mol\cdot dm^{-3}$ solutions free of organic solvent. The boric acid concentration was maintained at 0.05 $mol\cdot dm^{-3}$ so that the formation of polyborates was ignored. The pH was monitored with a glass combination electrode which was apparently calibrated against boric acid-sodium hydroxide buffers prepared according to a standard procedure. The use of this buffer requires knowledge of the boric acid dissociation constant [1973BAT]. No hysteresis was observed during back-titrations with perchloric acid. The Debye-Hückel expression was used to estimate activity coefficients. The $\log_{10}{}^{*}K_1^\circ$ value of 9.21 at 25 °C compares favourably with the value accepted by this review, but the circular reasoning of the calibration method compromised this result.

[1992NAG/LAS]

Nagy and Lasaga reported a $\log_{10}{}^{*}\beta_{s,0}^\circ$ value of (5.00 ± 0.08) at (80.0 ± 0.1) °C for reaction $\alpha\text{-}Al(OH)_3(cr) + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l)$ obtained from a detailed kinetic study of gibbsite (Alcoa C-31) precipitation and dissolution in dilute chloride and perchlorate media using a stirred-flow reactor. Steady-state concentrations were reached after 0.5 to 10 days. The gibbsite was characterised by XRD, SEM, XPS and BET surface-area analyses $(0.39 \pm 0.06)\ m^2\cdot g^{-1}$ equivalent to particle sizes of (2.2 ± 0.5) µm. The pH of the filtered outlet solutions was measured with a Ross combination glass electrode calibrated against commercial buffer solutions. Hydrolysis was taken into account using the high-temperature results from Bourcier *et al*. [1993BOU/KNA] and the 25 °C results of May *et al*. [1979MAY/HEL] (the $AlOH^{2+}$ species accounted for 24 to 37 % of

the total aluminium). Activity coefficients were estimated with an extended Debye-Hückel equation. The SIT results in both media gave a mean value of $\log_{10}{}^{*}K^{\circ}_{s,0}(\mathrm{Al(OH)_3, gibb}) = (4.95 \pm 0.12)$ (2σ); the uncertainty reflects mainly the experimental uncertainty in the pH and aluminium concentration measurements. The authors supplied evidence for the importance of dislocation defects in controlling the dissolution rate and mechanism.

[1993ARC]

Synthetic sapphire, Al_2O_3(cr) is a standard reference material for the calibration of some types of calorimeters both at low and high temperatures, so it is important that the heat capacity and related values are well-established. Earlier values were based entirely on heat capacity and enthalpy drop measurements made at the U.S. National Bureau of Standards up to *ca.* 1982. In this paper, Archer has very thoroughly re-analysed all the data for corundum, including those from the more recent studies listed in Table A-60 below, taking into account various sources of experimental error in the differing experimental techniques, and has made a more sophisticated fit to all the data from 0 to 2400 K than those used earlier. Archer's fit used 14 cubic splines and he has provided a detailed table of $\{C_p(T),\ H(T) - H(0\ \mathrm{K})\}$ and $S(T)$ at closely spaced intervals over this temperature range. Unfortunately, this table does not include the values at 298.15 K but these have been calculated by fitting the assessed heat capacities from 260 to 320 K, the relevant cubic spline range, by a cubic equation from which these values have been calculated. These $C^{\circ}_{p,\mathrm{m}}$ and S°_{m} values (79.050 and 50.940 $\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$) are both *ca.* 0.02 $\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ greater than the existing CODATA/TDB values. But since the uncertainties are appreciably greater than this, the latter have been retained for this review to avoid tiny meaningless conflicts with other existing assessments.

Table A-60: More recent studies on corundum included in Archer's assessment [1993ARC]

Reference	Temperature range (K)	Technique
[1966MAR/SNO]	300-475	Adiabatic calorimetry
[1967GRO]	298-1048	Adiabatic calorimetry
[1968LEA]	298-773	Adiabatic calorimetry
[1971DIT/DOU]	273-1173	Bunsen ice calorimetry
[1977CHA]	8-375	Adiabatic calorimetry
[1978AND/NOR]	300-550	Adiabatic calorimetry
[1983INA]	70-700	Adiabatic calorimetry
[1992SOR/KAJ]	60-505	Adiabatic calorimetry

Selected points from all these studies are shown in Figure V-29 in Section V.5.1.1.2 of the main text, together with those of three earlier low-temperature studies by Parks and Kelley [1926PAR/KEL] (12 measurements from 91-291 K), Simon and

Swain [1935SIM/SWA] (17 measurements from 30-280 K) and Kerr and Johnson [1950KER/JOH] (50 measurements from 20 to 295K). This Figure illustrates the remarkable consistency of all the data—the less precise data from [1935SIM/SWA] are the only deviating points.

It should be noted that Archer's revised heat capacities above 1000 K deviate noticeably from both the CODATA values and those given earlier by Ditmars *et al.*, [1982DIT/ISH].

[1993CHA/NAV]

The enthalpies of formation of dolomite ($CaMg(CO_3)_2$(cr)), wollastonite ($CaSiO_3$(cr)), and diopside ($CaMgSi_2O_6$(cr)), from their binary oxides, were determined at 977 K by oxide melt drop solution calorimetry using a Calvet-type twin micro-calorimeter previously described [1968NAV/KLE], [1977NAV], involving about 30 g of lead borate ($2PbO{\cdot}B_2O_3$) solvent. Procedures specific to the present drop technique were outlined. Samples for calorimetry were pressed into pellets weighing *ca.* 10 to 30 mg. In the present study, experiments were carried under a flow of CO_2 or argon that was shown experimentally to remove quantitatively (99.5 ± 0.5 %) the CO_2 produced in the reactions. The present discussion will be restricted to data pertaining to the binary carbonates (magnesite and calcite) and dolomite.

The enthalpies of solution of the dolomite samples were compared with those of stoichiometric mechanical mixtures of calcium and magnesium carbonates. Two dolomite samples were used, one natural and one synthetic. The crystalline natural dolomite sample was of the same origin (Euguy, Spain) as that used in [1987NAV/CAP] for their solution calorimetry determination; analytical details are given in the Appendix A for that paper. The synthetic sample (shown to have a Mg/Ca ratio of (1.014 ± 0.001) and to be stoichiometric in carbonate) was obtained by reaction (using cold seal vessels under 0.1-0.2 GPa CO_2 at 973 K) of a mechanical mixture of precipitated calcium carbonate and basic magnesium carbonate. X-ray powder diffraction patterns of the dolomite samples of the two origins showed no differences. The calcite samples were from a commercial product (99.995 %), dried at 373 K prior to calorimetry. Magnesite was obtained by heat treatment of a commercial basic magnesium carbonate (99.999 %) in cold-seal vessels under 0.1-0.2 GPa CO_2 at 673 K for 24 to 72 hours, and also by heating commercial MgO (99.95 %), following the same procedure, at 973 K. The authors noted that the samples were stoichiometric in carbonate and were identical in terms of characterisation. The results were reported as averages (with 2σ uncertainty) of a large number of identical measurements. Given the complexity of the medium (sln), no attempt was made to identify the solute species.

Thus, for the dissolution of mechanical mixture of binary carbonates:

$$\{CaCO_3(calc) + MgCO_3(cr), 298.15\ K\} + (sln, 977\ K) \rightleftharpoons$$
$$\{CaO + MgO\}(sln, 977\ K) + 2CO_2(g, 977\ K) \quad (A.206)$$

the reported value from seven measurements is:

$$\Delta_r H_m\,(A.206) = (373.7 \pm 1.9)\ kJ{\cdot}(2\ mol\ of\ cations)^{-1}$$

For the dissolution of dolomite:

$$CaMg(CO_3)_2(cr, 298.15\ K) + (sln, 977\ K) \rightleftharpoons$$
$$(CaO + MgO)(sln, 977\ K) + 2CO_2(g, 977\ K) \quad (A.207)$$

the values are:

$\Delta_r H_m$ ((A.207), natural sample) = (383.7 ± 1.1) kJ·mol^{-1} from six measurements and

$\Delta_r H_m$ ((A.207), synthetic sample) = (385.8 ± 1.1) kJ·mol^{-1} from 16 measurements.

Noting the limited agreement between these two values, the authors logically indicated that it could not be due to a potential disordering in the synthetic sample, as the corresponding value would have been, in that case, smaller than that for the natural sample. On the other hand, corrections due to differences in stoichiometries and impurities would be distinctly smaller than the uncertainties in each of the two values for Reaction (A.207). Therefore, the authors adopted, as the best value:

$$\Delta_r H_m\,(A.207) = (384.8 \pm 1.6)\ kJ{\cdot}mol^{-1}.$$

From this value and that obtained for Reaction (A.206), the enthalpy of formation of dolomite from the binary carbonates:

$$CaCO_3(calc) + MgCO_3(cr) \rightleftharpoons CaMg(CO_3)_2(cr) \quad (A.208)$$

is obtained as:

$$\Delta_r H^\circ_m\,((A.208), 298.15\ K) = -(384.8 \pm 1.6) + (373.7 \pm 1.9) = -(11.1 \pm 2.5)\ kJ{\cdot}mol^{-1}.$$

This value is in good agreement with (albeit less precise than) the value of – (11.48 ± 1.02) kJ·mol^{-1} at 358 K obtained in the same laboratory for the enthalpy of formation of ordered dolomite from the binary carbonates, by calorimetric dissolution into a concentrated NaCl brine [1987NAV/CAP].

The authors also report the enthalpies of solution of magnesite according to Reaction (A.209):

$$MgCO_3(cr, 298.15\ K) + (sln, 977\ K) \rightleftharpoons MgO(sln, 977\ K) + CO_2(g, 977\ K) \quad (A.209)$$

as $\Delta_r H^\circ_m\,(A.209)$ = (185.6 ± 0.8) kJ·mol^{-1} from 19 measurements and similarly of calcite:

$$CaCO_3(calc, 298.15\ K) + (sln, 977\ K) \rightleftharpoons$$
$$CaO(sln, 977\ K) + CO_2(g, 977\ K) \quad (A.210)$$

as $\Delta_r H^\circ_m$ (A.210) = (189.9 ± 0.7) kJ·mol^{-1} from 15 measurements.

Within the uncertainty limits the sum of these values, (375.5 ± 1.1) kJ·mol^{-1} is consistent with the value obtained for the enthalpy of Reaction (A.206) above.

Making use, in conjunction with the above results and with the enthalpy contents from Robie and Hemingway [1979ROB/HEM], of the enthalpies of solution of MgO(cr) and CaO(cr) in the same medium at 986 K, according to reactions:

$$\mathrm{MgO(cr, 986\ K) + (sln, 986\ K) \rightleftharpoons MgO(sln, 986\ K)} \qquad \text{(A.211)}$$

and

$$\mathrm{CaO(cr, 986\ K) + (sln, 986\ K) \rightleftharpoons CaO(sln, 986\ K)} \qquad \text{(A.212)}$$

reported by Navrotsky and Coons [1976NAV/COO] as $\Delta_r H_m$ (A.211) = (4.81 ± 0.59) kJ·mol^{-1} (five measurements, 1σ) and $\Delta_r H_m$ (A.212) = – (54.52 ± 0.92) kJ·mol^{-1} (six measurements, 1σ), the authors (using 2σ uncertainties) give the enthalpies of formation at 298.15 K of magnesite, calcite and dolomite from the corresponding binary oxides and CO_2(g), as – (116.8 ± 1.0), – (178.3 ± 1.2) and – (304.3 ± 1.9) kJ·mol^{-1}, respectively. These three values yield, for the formation of dolomite from magnesite and calcite at 298.15 K, a value of – (9.2 ± 2.5) kJ·mol^{-1}, in agreement, within the uncertainties, with the value of – (11.1 ± 2.5) kJ·mol^{-1} obtained above from the combination of Reactions (A.206) and (A.207).

We have re-calculated these values using the heat capacity functions adopted in this review, using, as uncertainty, twice the standard deviation for the functions used in [1979ROB/HEM], [1995ROB/HEM], [1994GUR/VEY], [1989COX/WAG] and this review. Thus, to obtain the enthalpy of formation of $MgCO_3$(cr) from MgO(cr) and CO_2(g) at 298.15 K, Reaction (A.213):

$$\mathrm{MgO(cr, 298.15\ K) + CO_2(g, 298.15\ K) \rightleftharpoons MgCO_3(cr, 298.15\ K)} \qquad \text{(A.213)}$$

we need, in addition to the enthalpies for Reactions (A.209) and (A.211) above, the enthalpies for Reactions (A.214)-(A.217):

$$\mathrm{MgO(cr, 298.15\ K) \rightleftharpoons MgO(cr, 986\ K)} \qquad \text{(A.214)}$$

$$\mathrm{CO_2(g, 298.15\ K) \rightleftharpoons CO_2(g, 977\ K)} \qquad \text{(A.215)}$$

$$\mathrm{MgO(sln, 986\ K) \rightleftharpoons MgO(sln, 977\ K)} \qquad \text{(A.216)}$$

$$\mathrm{(sln, 977\ K) \rightleftharpoons (sln, 986\ K)} \qquad \text{(A.217)}$$

We make the very small approximation that the sum of Reactions (A.216) and (A.217) corresponds to Reaction (A.218):

$$\mathrm{MgO(cr, 986\ K) \rightleftharpoons MgO(cr, 977\ K)} \qquad \text{(A.218)}$$

Thus, the sum [(A.214) + (A.216) + (A.217)] corresponds to:

$$\mathrm{MgO(cr, 298.15\ K)} \rightleftharpoons \mathrm{MgO(cr, 977\ K)} \qquad \text{(A.219)}$$

We then obtain:

$$\Delta_r H_m\,(\text{A.213}) = -\ \Delta_r H_m\,(\text{A.209}) + \Delta_r H_m\,(\text{A.211}) + \Delta_r H_m\,(\text{A.215}) + \Delta_r H_m\,(\text{A.219})$$

with $[\Delta_r H_m\,(\text{A.215}) + \Delta_r H_m\,(\text{A.219})] = (63.916 \pm 0.100)\ \mathrm{kJ\cdot mol^{-1}}$ as adopted in this review, this yields:

$$\Delta_r H_m\,(\text{A.213}) = [-(185.6 \pm 0.8) + (4.81 \pm 0.59) + (63.916 \pm 0.100)]\ \mathrm{kJ\cdot mol^{-1}} = -(116.874 \pm 1.500)\ \mathrm{kJ\cdot mol^{-1}},$$

where we have increased by 50 % the uncertainty limits arising from the experimental measurements to account for small variations in the amounts of solute and solvent in the combined two studies of Navrotsky and co-workers, although the authors indicated that they did not observe any dependence of the heat effects due to differences in solute concentrations.

Use of $\Delta_f H_m^o\,(\mathrm{MgO, cr, 298.15\ K}) = -(601.60 \pm 0.30)\ \mathrm{kJ\cdot mol^{-1}}$ and $\Delta_f H_m^o\,(\mathrm{CO_2, g, 298.15\ K}) = -(393.51 \pm 0.13)\ \mathrm{kJ\cdot mol^{-1}}$ from [1989COX/WAG], finally yields:

$$\Delta_f H_m^o\,(\mathrm{MgCO_3, cr, 298.15\ K}) = -(1112.0 \pm 1.5)\ \mathrm{kJ\cdot mol^{-1}}.$$

Similarly, to obtain the enthalpy of formation of calcite from CaO(cr) and CO_2(g) at 298.15 K, according to Reaction (A.220):

$$\mathrm{CaO(cr, 298.15\ K) + CO_2(g, 298.15\ K)} \rightleftharpoons \mathrm{CaCO_3(cr, 298.15\ K)} \qquad \text{(A.220)}$$

we need, besides the enthalpies of Reactions (A.210) and (A.212) above, the enthalpy change corresponding to Reaction (A.221):

$$\mathrm{CaO(cr, 298.15\ K)} \rightleftharpoons \mathrm{CaO(cr, 977\ K)} \qquad \text{(A.221)}$$

Thus:

$$\Delta_r H_m\,(\text{A.220}) = -\ \Delta_r H_m\,(\text{A.210}) + \Delta_r H_m\,(\text{A.212}) + \Delta_r H_m\,(\text{A.215}) + \Delta_r H_m\,(\text{A.221})$$
$$= [-(189.9 \pm 0.7) - (54.52 \pm 0.92) + 65.626 \pm 0.700]\ \mathrm{kJ\cdot mol^{-1}}$$
$$= -(178.794 \pm 2.000)\ \mathrm{kJ\cdot mol^{-1}}$$

(with uncertainty limits increased as above).
With:

$\Delta_f H_m^o\,(\mathrm{CaO, cr, 298.15\ K}) = -(634.92 \pm 0.90)\ \mathrm{kJ\cdot mol^{-1}}$ and $\Delta_f H_m^o\,(\mathrm{CO_2, g, 298.15\ K}) = -(393.51 \pm 0.13)\ \mathrm{kJ\cdot mol^{-1}}$ from [1989COX/WAG], we obtain:

$$\Delta_f H_m^o\,(\mathrm{CaCO_3, calc, 298.15\ K}) = -(1207.2 \pm 2.2)\ \mathrm{kJ\cdot mol^{-1}}.$$

Using the best value given by the authors for Reaction (A.207), our re-calculation of the enthalpy of formation of dolomite from the binary oxides and CO_2(g) at 298.15 K, according to Reaction (A.222):

$$MgO(cr, 298.15\ K) + CaO(cr, 298.15\ K) + 2CO_2(g, 298.15\ K) \rightleftharpoons CaMg(CO_3)_2(cr, 298.15\ K) \quad (A.222)$$

we obtain:

$\Delta_r H_m$ (A.222) = – $\Delta_r H_m$ (A.207) + $\Delta_r H_m$ (A.211) + $\Delta_r H_m$ (A.212) + 2 × $\Delta_r H_m$ (A.215) + $\Delta_r H_m$ (A.219) + $\Delta_r H_m$ (A.221) = [– (384.8 ± 1.6) + (4.81 ± 0.59) – (54.92 ± 0.92) + (63.916 ± 0.100) + (65.626 ± 0.700)] kJ·mol^{-1} = – (304.968 ± 3.100) kJ·mol^{-1} (with uncertainty limits increased as above).

With the values obtained in Reactions (A.213) and (A.220), for the enthalpies of formation of magnesite and calcite from their respective binary oxides and CO_2(g) at 298.15 K, this gives a value of – (9.3 ± 3.0) kJ·mol^{-1} for the enthalpy of formation of dolomite from the binary carbonates, according to Reaction (A.208). The uncertainty limits take into account that some terms are repeated in the equations involved. This value is in agreement, within the uncertainties, with the value of – (11.1 ± 2.5) kJ·mol^{-1} obtained above for Reaction (A.208).

[1993GOU/RAN]

This paper reports a study of the dissociation of MgO(cr) and CaO(cr) into M(g) + {O(g) and O_2(g)} from *ca.* 1850 to 2050 K, from which the authors derived standard enthalpies of formation of *ca.* – 635 kJ·mol^{-1} and – 602 kJ·mol^{-1} respectively. These, it will be noted, would make MgO(cr) noticeably more stable than CaO(cr). This reversal of stabilities has been discredited by the potential-difference study [1996JAC/VAR], who showed that the difference ($\Delta_f H$ (MgO) – $\Delta_f H$ (CaO)) is consistent with the CODATA selected values—see the Appendix A entry for [1996JAC/VAR]. Jacob and Varghese [1996JAC/VAR] also note that Gourishankar *et al.* [1993GOU/RAN] used a graphite susceptor to heat their samples, so their environment was strongly reducing, rather than neutral, leading to the possible formation of CO(g) in their experiments and further that earlier similar free evaporation experiments by Sata *et al.* [1982SAT/SAS] using a non-reducing tungsten heater gave results consistent with the accepted enthalpies of formation.

This accumulated evidence indicates that this study does not give any firm reasons for discarding the currently accepted values for the enthalpies of formation of CaO(cr) and MgO(cr).

[1993HE/MOR]

The CO_2 solubility, as well as the first and second stoichiometric dissociation constants of carbonic acid, was determined in aqueous solutions containing NaCl, KCl, $MgCl_2$, $CaCl_2$ and Na_2SO_4 over a wide range of ionic strength from 0 to 90 °C at 0.10132 MPa

pressure. Activity coefficients of CO_2(aq) were calculated from CO_2 solubility using the Henry's law constants reported in [1982PLU/BUS]. Similarly, the activity coefficients of HCO_3^- and CO_3^{2-} were calculated from the stoichiometric dissociation constants by using the thermodynamic constants (pK_1^o and pK_2^o) reported in [1982PLU/BUS], as well as a_{H_2O} and a_{H+} calculated by a Pitzer ion interaction model. These activity coefficients were then used to derive Pitzer parameters for the interaction between HCO_3^- and CO_3^{2-} with other ions present in the solution between 0 and 90 °C. The resulting Pitzer model for the carbonic acid system was verified by its ability to predict accurately experimentally determined calcite solubility in brines from 0 to 90 °C.

The pK values determined by He and Morse in pure NaCl media notably deviate from those determined by others (see Figures V-9, V-10 and V-13 in Chapter V.2 of the main text). In addition, probably due to the high uncertainties of the dissociation constants obtained in mixed (Na, Mg)Cl and (Na, Ca)Cl background electrolytes, the calculation of the stability constants of (Mg/Ca)CO_3(aq) complexes and ε((Mg/Ca)CO_3(aq), NaCl(aq)) values (in a similar way to that described in [1983MIL/THU]) was not possible. Therefore, the experimental data in this paper were not used in our evaluation.

[1993ONO/NAK]

The authors measured the oxygen content of Nb(cr), mixed with Mg(l) and held in an MgO crucible, from 923 to 1323 K. From their assumed values of the partial $\Delta_f G(O_2)$ in Nb(cr), they derived $\Delta_f G_m$ (MgO(cr) = – 65700 + 141.0 T/K J·mol^{-1}. Using thermal functions from [1985CHA/DAV], they derive second- and third-law values of $\Delta_f H_m^o$ (MgO, cr, 298.15 K) = – (626 ± 6) and – (654 ± 1) kJ·mol^{-1} and a second-law entropy of $\Delta_f S_m^o$ (MgO, cr, 298.15 K) = – (135 ± 4) J·K^{-1}·mol^{-1}. However, the latter value corresponds to S_m^o (MgO, cr, 298.15 K) = (0 ± 4) J·K^{-1}·mol^{-1}, which, together with the strong disagreement between the second- and third-law enthalpies, indicates that their experimental data and/or the assumptions in deriving the Gibbs energies are seriously in error. We note that one of their assumptions is that the oxygen potentials in (Nb, O) are unaffected by the presence of Mg in the system, which seems rather doubtful, and that their data for these oxygen potentials depend, in fact on $\Delta_f G(CaC_2)$(cr), whose value has been thrown into doubt by Jacob and Varghese [1996JAC/VAR]—see the Appendix A entry for this study. Considering the large number of consistent more direct enthalpy measurements of $\Delta_f H$ (MgO, cr), we do not place any credence on these high-temperature results on MgO. We do note, however, that similar measurements on the third-law $\Delta_f H$ (SrO) and $\Delta_f H$ (BaO) reported in this paper, using equilibration with Ag(l) do give values which agree very well with calorimetric data, suggesting the discrepancy in the MgO data may lie in the use of Nb for the Mg/MgO equilibrations.

[1993PAL/WES]

The first hydrolysis constant of Al^{3+} was determined potentiometrically in NaCl solutions using a hydrogen-electrode concentration cell from 25 to 125 °C beyond which drifting potential readings were always observed, probably due to the formation of higher-order multinuclear hydrolysed species. Liquid-junction potentials were accounted for by the Henderson equation. Application of the SIT treatment to the three lowest ionic strengths at all temperatures gave: $\log_{10}{}^{*}K_1^{\circ} = -(4.88 \pm 0.07)$ and $\Delta\varepsilon = -(0.30 \pm 0.11)$ kg·mol^{-1} at 25.0 °C, whereas applying the SIT treatment only to the data at 24.9 °C yielded: $\log_{10}{}^{*}K_1^{\circ} = -(4.89 \pm 0.09)$ and $\Delta\varepsilon = -(0.29 \pm 0.14)$ kg·mol^{-1}. Palmer and Wesolowski reported $\log_{10}{}^{*}K_1^{\circ} = -(4.95 \pm 0.02)$ based on a global fit using five adjustable parameters and a Debye-Hückel term. All uncertainties are at the 2σ level. The authors also used the Pitzer ion interaction model whereby the single electrolyte parameters of NaCl, HCl and $AlCl_3$ were taken from [1984PIT/PEI], [1987HOL/BUS] Model 1 and [1992PAL/WES], respectively. The resulting value of $\log_{10}{}^{*}K_1^{\circ}$ at 25 °C was found to be – 4.91. A plot of $\log_{10}{}^{*}K_1^{\circ}$ *versus* $(T/\mathrm{K})^{-1}$ yielded $\Delta_r H_m^{\circ} = (55.9 \pm 1.2)$ kJ·mol^{-1}, *cf.* $\Delta_r H_m^{\circ} = (55.4 \pm 0.7)$ kJ·mol^{-1} at 25 °C reported by the authors. These results were retained by this review.

[1993VAS/KOC]

The heats of protonation of $H_2PO_4^-$ by aqueous HNO_3 were measured calorimetrically from 298.15 to 318.15 K at ionic strengths 0.5-1.5 mol·kg^{-1} (KNO_3) and 0.25-0.75 mol·kg^{-1} ($(C_2H_5)_4NCl$). The standard thermodynamic functions for dissociation of H_3PO_4(aq) were derived at 298.15 K: $\Delta_r H_{m,1}^{\circ} = -(7.33 \pm 0.17)$ kJ·mol^{-1} (mistakenly reported as J·mol^{-1} in Table 3 by Vasil'ev *et al.*); $\Delta_r S_{m,1}^{\circ} = -(65.7 \pm 0.6)$ J·K^{-1}·mol^{-1}; $\Delta_r C_{p,m,1}^{\circ} = -(124 \pm 0.20)$ J·K^{-1}·mol^{-1}; and $\log_{10} K_1^{\circ} = -(2.148 \pm 0.001)$ taken presumably from [1951BAT].

[1993VAS/KOC2]

The heats of protonation of HPO_4^{2-} at (298.15-318.15) K and ionic strengths 0.5-1.5 mol·dm^{-3} (KNO_3) and 0.25-0.75 mol·dm^{-3} ($(C_2H_5)_4NCl$) were measured calorimetrically. The concentration of K_2HPO_4 was held constant at 0.015 mol·dm^{-3}. The standard thermodynamic functions for dissociation of $H_2PO_4^-$ were virtually identical in both media at 298.15 K: $\log_{10} K_2^{\circ} = -(7.199 \pm 0.004)$; $\Delta_r G_{m,2}^{\circ} = (41.09 \pm 0.11)$ kJ·mol^{-1}; $\Delta_r H_{m,2}^{\circ} = (4.07 \pm 0.27)$ kJ·mol^{-1}; $\Delta_r S_{m,2}^{\circ} = -(124.2 \pm 1.0)$ J·K^{-1}·mol^{-1}; and $\Delta_r C_{p,m,2}^{\circ} = -(160 \pm 40)$ J·K^{-1}·mol^{-1} (adjusted to 2σ).

[1994HEM/ROB]

This is a solution calorimetry study of the enthalpy of formation of natural dolomite $CaMg(CO_3)_2$(cr) through a cycle involving this compound, CaO(cr) and MgO(cr). The experimental study was carried out in 1973, some 20 years before the present publication. The vacuum-jacketed calorimeter used in the study has been described [1972ROB/HEM2]. It was operated in sealed conditions (constant volume) at 300.15 K, using 950 g of 4.00 molar HCl as the solvent and the results corrected to 298.15 K. For the reaction involving CaO(cr), a stoichiometric amount of MgO(cr) (with respect to the dolomite dissolution reaction) was dissolved in the acid before the experimental reaction. These reactions involved 0.020 mol of each cation. The reported results are based on the average of four identical measurements and the uncertainties are for the 2σ confidence limits. The dolomite was taken from the same single crystal (from Binnental, Switzerland) that was used in [1963STO/ROB] for low-temperature heat capacity measurements. The analytical results, in wt %, were (ideal values in parentheses): CaO: 30.77 (30.41); MgO: 21.54 (21.86); CO_2: 47.38 (47.73); MnO: 0.10 (0); FeO:0.08 (0); SrO:0.017 (0). The sample for calorimetry was ground and sieved (100-300 mesh/inch), the powder heated to 388 K and stored in a desiccator.

The CaO(cr) sample was obtained from primary standard $CaCO_3$ heated in a platinum crucible at 1575 K for three hours. The spectroscopically pure reagent MgO used was treated in the same way in order to promote grain growth. These oxide samples were portions of the materials used by [1973HEM/ROB]. In fact, the calorimetry results reported for MgO in the cycle below were taken from this paper.

For the reaction:

$$\begin{aligned} &CaMg(CO_3)_2(cr) + (178HCl + 2276H_2O)(aq) \rightleftharpoons \\ &(CaCl_2 + MgCl_2 + 174HCl + 2278H_2O)(sln) + 2CO_2(g) \end{aligned} \qquad \text{(A.223)}$$

the value $\Delta_r H_m$((A.223), 300.15 K) = – (47.08 ± 0.35) kJ·mol^{-1} was obtained after correction for the CO_2 that remained dissolved (93 %) in the solution in the actual experiment. The authors refer to [1973HEM/ROB] for the discussion of this correction.

Also used in the cycle are the reactions:

$$MgO(cr) + (178HCl + 2276H_2O)(aq) \rightleftharpoons (MgCl_2 + 176HCl + 2277H_2O)(sln) \qquad \text{(A.224)}$$

$\Delta_r H_m$((A.224), 300.15 K) = – (150.30 ± 0.08) kJ·mol^{-1}, from [1973HEM/ROB] for the dissolution under very similar conditions, after correction for a small difference in temperature (from 303.15 to 300.15 K) and:

$$\begin{aligned} &CaO(cr) + (MgCl_2 + 176HCl + 2277H_2O)(aq) \rightleftharpoons \\ &(CaCl_2 + MgCl_2 + 174HCl + 2278H_2O)(sln) \end{aligned} \qquad \text{(A.225)}$$

$$\Delta_r H_m((A.225), 300.15\ K) = – (197.72 \pm 0.34)\ kJ{\cdot}mol^{-1}.$$

These give for the reaction:

$$CaO(cr) + MgO(cr) + 2CO_2(g) \rightleftharpoons CaMg(CO_3)_2(cr) \quad \text{(A.226)}$$

$$\Delta_r H_m((A.226), 300.15\ K) = -(300.94 \pm 0.49)\ kJ{\cdot}mol^{-1}.$$

Using the data from [1978ROB/HEM], the authors corrected the latter result to 298.15 K and calculated the standard enthalpy and Gibbs energy of formation, and solubility constant of dolomite. With the C_p values accepted in this review, we calculate $\Delta_r H_m^o((A.226), 298.15\ K) = -(300.95 \pm 0.49)\ kJ{\cdot}mol^{-1}$ and with CODATA values $\Delta_f H_m^o(CaMg(CO_3)_2, cr, 298.15\ K) = -(2324.49 \pm 1.10)\ kJ{\cdot}mol^{-1}$. With $S_m^o(CaMg(CO_3)_2, cr, 298.15\ K) = (155.17 \pm 0.80)\ J{\cdot}K^{-1}{\cdot}mol^{-1}$ selected in this review, this gives $\Delta_f G_m^o(CaMg(CO_3)_2, cr, 298.15\ K) = -(2161.692 \pm 1.133)\ kJ{\cdot}mol^{-1}$, which corresponds to a solubility of $\log_{10} K_s = -(17.04 \pm 0.38)$.

[1994HOL/BUS]

Following a previous publication by this group [1978HOL/BAE], additional isopiestic molalites and enthalpies of dilution of $CaCl_2$ solutions were measured from 444 to 524 K, giving rise to a comprehensive set of enthalpies of dilution to $m_{Ca} \sim 7.26$ at 40 MPa; the calorimetric measurements actually extended to 674 K but the results above 524 K are not treated here. All of these experimental conditions are beyond the scope of this review, however, it is significant that they found it was only at temperatures greater than 524 K that explicit consideration of ion association was necessary to treat the thermodynamic properties of $CaCl_2$ solutions.

[1994KOC/DUS]

The thermodynamics of formation of the ion pair KPO_4^{2-} were studied calorimetrically at 298.15 K. Effects of ionic strength and supporting electrolyte (KNO_3, $(C_2H_5)_4NCl$) on the reaction enthalpy were elucidated and the standard thermodynamic parameters for the formation reaction were calculated: $\Delta_r G_m^o = -(8.56 \pm 0.23)\ kJ{\cdot}mol^{-1}$, $\Delta_r H_m^o = (16.10 \pm 1.11)\ kJ{\cdot}mol^{-1}$ and $\Delta_r S_m^o = -(25.3 \pm 3.80)\ J{\cdot}K^{-1}{\cdot}mol^{-1}$.

[1994ROB/HEM]

The authors studied the heat capacity of large glassy clear magnesite crystals from Brazil which microprobe analysis indicated to contain 0.5 % $FeCO_3$, 0.3 % $CaCO_3$ and 0.1 % $MnCO_3$ (mole percentages). Measurements were made by adiabatic calorimetry at 84 temperatures from 8 to 343 K and at 49 temperatures from 339 to 676 K by DSC, with good agreement between the two techniques in the overlap region. Full details of these data are tabulated, together with smoothed values from 25 to 340 K. Their C_p values are slightly lower than those measured by Hemingway *et al.* [1977HEM/ROB3] on a purer sample with a smaller crystallite size.

The value of the standard entropy calculated by the authors from this study was 64.98 $J \cdot K^{-1} \cdot mol^{-1}$, no uncertainty limits being given.

As shown in Figure V-42, Section V.6.1.1.5.2 of the main text, the higher temperature DSC measurements are much less self-consistent than the low-temperature measurements, and we also note that the authors do not discuss these results further in the text. Moreover, in their subsequent assessment of the thermodynamic properties of minerals [1995ROB/HEM], the same authors do not quote these data, preferring to rely on the earlier [1977HEM/ROB3] data. From this, we suspect the authors were not entirely confident about the high-temperature data in [1994ROB/HEM] and they accordingly were not taken into account for this later assessment.

This paper also reports similar measurements on synthetic grossular ($Ca_3Al_2Si_3O_{12}$), $YBa_2Cu_3O_{6+x}$ and $BiCaSrCu_2O_{6+x}$, and type-321 stainless steel, which are of no immediate interest to the review.

[1994SWA/SAX]

This paper reports an assessment of the Si-O system valid at pressures up to 15 GPa but with the main emphasis on the fitting and calculation of the phase diagrams, particularly for the polymorphs coesite and stishovite which are stable at high pressures. As with the similar assessment by Hallstedt [1992HAL], the question of the stability of tridymite at ambient pressure is not discussed, but the derived data make it stable from *ca.* 1150 to 1660 K, as read from a small figure of the phase diagram. Since the high-pressure phases are not being included in this review, the data in this paper were not considered in our assessment.

[1994WEN/QIY]

The enthalpy of reaction of Al(s) with aqueous NaOH solutions was measured by calorimetry. When combined with the CODATA values [1989COX/WAG] for $\Delta_f H^\circ_m(OH^-, aq, 298.15\ K)$ and $\Delta_f H^\circ_m(H_2O, l, 298.15\ K)$, Wenming *et al*. derived, $\Delta_f H^\circ_m(Al(OH)_4^-, aq, 298.15\ K) = -(1504.46 \pm 1.00)\ kJ \cdot mol^{-1}$. However, they recommended a value of $\Delta_f H^\circ_m(Al(OH)_4^-, aq, 298.15\ K) = -(1503.61 \pm 1.00)\ kJ \cdot mol^{-1}$.

[1994WES/PAL]

The solubility of acid-washed gibbsite was studied at 50 °C in 0.1 $mol \cdot kg^{-1}$ NaCl using various buffers to control the solution pH which was measured with a Ross combination glass electrode calibrated with 0.1 $mol \cdot kg^{-1}$ ionic strength NaCl solutions. These were all batch-type experiments with periodic sampling (35-144 days). In all but one series of experiments, the total aluminium molality was $\leq 10^{-5}$ $mol \cdot kg^{-1}$. In our SIT re-treatment

of these results a value of $\log_{10} {}^{*}K^{o}_{s,2} = -(3.04 \pm 0.03)$ (2σ) was derived where only aluminium molalities $\leq 10^{-3.614}$ mol·kg^{-1} were considered and values of $\log_{10} {}^{*}K^{o}_{s,0}$ [1992PAL/WES], $\log_{10} {}^{*}K^{o}_{s,1}$ [1993PAL/WES] and $\log_{10} {}^{*}K^{o}_{s,4}$ [1992WES] were imposed, and the $Al(OH)_3(aq)$ species was considered to be insignificant.

[1995AKA/YUS]

This paper deals mainly with data for coesite and stishovite, which are only of marginal interest to this review, but it also contains data for the heat capacity of α-quartz, measured by DSC from *ca.* (250-675) K, which are reported only graphically. The quartz sample is described only as 'reagent-grade quartz powder', and no analysis is given. The heat capacities of coesite and stishovite (prepared from the same quartz powder) were measured down to lower temperatures (*ca.* 183 K) and up to 703 K; those for coesite are reported to differ significantly from previous data. The heat capacities and entropies were also calculated using Kieffer's model. The transition enthalpy for the quartz-coesite at 979 K was measured to be (1.27 ± 0.39) kJ·mol^{-1} by high-temperature solution calorimetry, and the enthalpies for the α-quartz-coesite and coesite-stishovite transitions at 298.15 K were measured to be (3.40 ± 0.56) and (33.62 ± 1.01) kJ·mol^{-1}, respectively, by differential drop-solution calorimetry. The enthalpy of the coesite-stishovite transition obtained in this study is about (10-15) kJ·mol^{-1} smaller than those in previous studies. The phase relations in SiO_2 at high pressures and high temperatures were calculated using these new thermodynamic data and compared to previous calculations.

[1995CHA/NAV]

This paper is concerned with the influence of the excess calcium (up to 56.2 %) on the enthalpy of formation of sedimentary dolomites, using oxide melt ($2PbO \cdot B_2O_3$ at 977 K) drop calorimetry, using a technique previously described by Navrotsky and coworkers (see the AppA for [1993CHA/NAV] and reference therein).

For its most part, the present paper falls beyond the scope of the present review. However, the enthalpies of drop solution of calcite, magnesite and of two samples of nearly stoichiometric dolomite were reported and will be noted here. The enthalpies of drop solution of magnesite and calcite were taken, respectively, as (185.62 ± 0.69) and (190.50 ± 0.73) kJ·mol^{-1}, based on the values obtained by the same group [1993CHA/NAV], (185.6 ± 0.8) and (189.9 ± 0.7) kJ·mol^{-1} and their current study. No additional details were given.

The sample of synthetic dolomite (49.7 mol % $CaCO_3$) was that used also in [1993CHA/NAV]. An average enthalpy of solution (with 2σ uncertainty) of

(385.60 ± 1.22) kJ·mol^{-1} was obtained from 12 measurements. The crystalline natural dolomite sample was of the same origin (Euguy, Spain) as that used in [1987NAV/CAP] for their identical drop solution calorimetry determination and was reported to be almost exactly stoichiometric (50.06 mol % $CaCO_3$) An average enthalpy of solution (with 2σ uncertainty) of (384.37 ± 0.65) kJ·mol^{-1} was obtained from 14 measurements. These values are essentially identical to those obtained in [1993CHA/NAV], (383.7 ± 1.1) kJ·mol^{-1} and (385.8 ± 1.1) kJ·mol^{-1}, respectively, for the same natural and synthetic dolomite samples.

[1995CHE/ZEN2]

Using a high-temperature drop-calorimeter, the authors measured the enthalpy differences of corundum and γ-Al_2O_3, both from 321 to 993 K and also the enthalpy of transformation of the reaction:

$$\gamma\text{-}Al_2O_3 \rightleftharpoons \alpha\text{-}Al_2O_3 \qquad \text{(A.227)}$$

at 1470 K.

The corundum was only described as 'a standard sample', but fuller details are given of the sample of γ-Al_2O_3. It was prepared by combustion of pure diaspore (AlOOH) at 550 °C for 10 hours. TG-DTA and XRD analyses indicated that it was pure Al_2O_3(γ) and spectroscopic analysis showed the only appreciable impurity was 0.012 wt % Ca. The results of eight duplicated runs at identical temperatures for the two samples are tabulated. The results for corundum were in good agreement with the standard reference data given by Ditmars *et al.* [1982DIT/ISH]; these are trivially different from our currently selected values (based on the later assessment by Archer [1993ARC]) in the temperature range studied. For Al_2O_3(γ) the authors fitted the enthalpy data by an expression which corresponds to the equation:

$$C_{p,\mathrm{m}}(Al_2O_3, \gamma, T)\ /\mathrm{J\cdot mol^{-1}\cdot K^{-1}} = 115.25 + 1.953\times10^{-2}(T/\mathrm{K}) - 3.381\times10^{6}(T/\mathrm{K})^{-2} \qquad \text{(A.228)}$$

The enthalpy of the transformation (A.227) was directly measured at 1470 K using a different high-temperature calorimeter. Three measurements were made, with a mean $\Delta_r H_m$ ((A.227), 1470 K) = (26.78 ± 0.41) kJ·mol^{-1}, where the uncertainty is that given in the paper. The authors combined these results with values of C_p(Al_2O_3, cr) from their own measurements (noted above) to give $\Delta_r H^\circ_m$ ((A.227), 298.15 K) = – 18.64 kJ·mol^{-1}, which with the CODATA value of $\Delta_f H^\circ_m$ (Al_2O_3, cr), 298.15 K) gave $\Delta_f H^\circ_m$ (Al_2O_3, γ, 298.15 K) = – (1657.2 ± 1.5) kJ·mol^{-1}. Noting that the values of C_p(Al_2O_3, γ) have to be extrapolated above 1000 K, using uncertainty levels corresponding to a 95 % confidence and our preferred data for C_p(Al_2O_3, cr) (based on Archer's assessment [1993ARC]), the enthalpies of (A.227) at 1470 K and 298.15 K become – (26.78 ± 1.01) and – (18.84 ± 1.65) kJ·mol^{-1}, respectively, and thus $\Delta_f H^\circ_m$ (Al_2O_3, γ, 298.15 K) = – (1656.86 ± 2.10) kJ·mol^{-1}. This is consistent with the

value measured at 978 K by Yokokawa and Kleppa [1964YOK/KLE2] using drop calorimetry, which we calculate gives $\Delta_f H^\circ_m(Al_2O_3, \gamma, 298.15\ K) = -(1657.2 \pm 1.6)$ kJ·mol^{-1}.

[1995DUC/REA]

The solubility of $Ca(OH)_2$(s) was studied at 25 °C in KOH and NaOH solutions to 3.860 and 3.627 mol·kg^{-1}, respectively. Equilibrium was approached from under- and over-saturation to yield a value of $m_{sat} = 0.0222$. Duchesne and Reardon tabulated fifteen solubilities reported in the literature with values ranging from 0.012 mol·kg^{-1} for large well-defined crystals to 0.0230 mol·kg^{-1} for fine or reagent grade material. Excluding the former result, a mean solubility (2σ) of (0.0210 ± 0.0018) mol·kg^{-1} is calculated here. A Pitzer ion interaction routine developed by Harvey *et al*. [1984HAR/MOL] was utilised to treat the solubility data (for the K^+ solutions only) with 33 ion interaction parameters, 21 of which were known (see Table A-61), eight were found to be negligible leaving only four to be evaluated, *viz*., $\beta^{(0)}(Ca^{2+}, OH^-)$, $\beta^{(1)}(Ca^{2+}, OH^-)$, $\beta^{(2)}(Ca^{2+}, OH^-)$ and $C^\phi(Ca^{2+}, OH^-)$. They did not include specifically the hydrolysis species $CaOH^+$ although an equilibrium constant was derived from $\beta^{(2)}(Ca^{2+}, OH^-) = -K_a/2$ (*i.e*., $\log_{10}K^\circ = -1.34$, *cf*. -1.17 [1998FEL/DIX]). Included in the Pitzer treatment were data from seven other sources, *viz*., solubility values from [1911SCH], [1931JOH/GRO], [1956BAT/BOW], [1967YEA/MAR], [1988KIM/FRE], [1995DUC/REA] and potential-difference measurements in [1959BAT/BOW]. Three values in [1931JOH/GRO] were rejected.

Table A-61: Ion interaction parameters at 25 °C for the system Ca-K-OH-Cl-H_2O.

Parameter	[1991PIT]	[1984HAR/MOL]	[1995DUC/REA]
$\beta^{(0)}(Ca^{2+},OH^-)$		– 0.1747	– 0.1421 [(a)]
$\beta^{(1)}(Ca^{2+},OH^-)$		– 0.2303	– 0.3727 [(a)]
$\beta^{(2)}(Ca^{2+},OH^-)$		– 5.72	– 11.052 [(a)]
$C^\phi(Ca^{2+},OH^-)$			– 0.0092 [(a)]
$\beta^{(0)}(Ca^{2+},Cl^-)$	0.3053	0.3159	0.3053
$\beta^{(1)}(Ca^{2+},Cl^-)$	1.7085	1.614	1.7085
$\beta^{(2)}(Ca^{2+},Cl^-)$			
$C^\phi(Ca^{2+},Cl^-)$	0.00215	– 0.00034	0.00215
$\beta^{(0)}(K^+,OH^-)$	0.1298	0.1298	0.1298
$\beta^{(1)}(K^+,OH^-)$	0.320	0.320	0.320
$\beta^{(2)}(K^+,OH^-)$			

(Continued on the next page)

Table A-61 (continued)

Parameter	[1991PIT]	[1984HAR/MOL]	[1995DUC/REA]
$C^{\phi}(K^+,OH^-)$	0.041	0.041	0.041
$\beta^{(0)}(K^+,Cl^-)$	0.4835	0.4835	0.4835
$\beta^{(1)}(K^+,Cl^-)$	0.2122	0.2122	0.2122
$\beta^{(2)}(K^+,Cl^-)$			
$C^{\phi}(K^+,Cl^-)$	– 0.00084	– 0.00084	– 0.00084
$\beta^{(0)}(H^+,Cl^-)$	0.1775	0.1775	0.1775
$\beta^{(1)}(H^+,Cl^-)$	0.2945	0.2945	0.2945
$\beta^{(2)}(H^+,Cl^-)$			
$C^{\phi}(H^+,Cl^-)$	0.00080	0.00080	0.00080
$\theta(Ca^{2+},K^+)$	0.032	0.032	0.032
$\theta(Ca^{2+},H^+)$	0.092	0.092	0.092
$\theta(H^+,K^+)$	0.005	0.005	0.005
$\theta(Cl^-,OH^-)$	– 0.050	– 0.050	– 0.050
$\psi(Ca^{2+},H^+,OH^-)$			
$\psi(Ca^{2+},H^+,Cl^-)$	– 0.015	– 0.015	– 0.015
$\psi(Ca^{2+},K^+,OH^-)$			
$\psi(Ca^{2+},K^+,Cl^-)$	– 0.025	– 0.025	– 0.025
$\psi(H^+,K^+,OH^-)$			
$\psi(H^+,K^+,Cl^-)$	– 0.011	– 0.011	– 0.011
$\psi(Ca^{2+},OH^-,Cl^-)$	– 0.025	– 0.025	– 0.025
$\psi(H^+,OH^-,Cl^-)$			
$\psi(K^+,OH^-,Cl^-)$	– 0.006	– 0.006	– 0.006

(a) These four values were designated in Table 3 of this article with a superscript "b" which was not defined but presumably these were derived by the authors from their fit.

[1995JEM/BEN]

This study concerns the synthesis, characterisation and the determination of the enthalpy of formation of calcium-, strontium-, cadmium- and lead-hydroxlyapatites and fluorapatites. Results are also reported for mixed (calcium-cadmium) and (calcium-lead)-hydroxylapatites. Lattice parameters are also given. Only the results concerning the calcium-hydroxylapatite and fluorapatite will be discussed here.

In fact, the calorimetric results concerning these two pure phases were already reported in [1991BEN/SOM], but with far less detail (no report of the individual measurements). The lattice parameters appearing in [1995JEM/BEN] may also be traced back to [1991BEN/SOM]. Information on the analysis of the samples of calcium hydroxylapatite used by this group was given in [1988BEN/KHA].

An isoperibol type calorimeter was used at 298.15 K. The dissolution medium (350 cm^3) was (HNO_3 in 35.35H_2O), hereunder designed by (sln1). Samples ranging between *ca.* 40 and 200 mg of the apatites were used, leading to rather dilute solutions. It was not possible to reproduce average values given by the authors for the enthalpies of solution of the two apatites, as they mention using a weighted average, without indicating the weights given to individual results. We have thus resorted to a straight average of the individual measurements.

For the calcium hydroxylapatite, eight measurements of the enthalpies of solution were reported while seven measurements were made for calcium fluorapatite. We calculate:

$$\Delta_{sol}H_m(Ca_{10}(PO_4)_6(OH)_2, \text{cr, in sln1}) = -(364.157 \pm 2.660)\ \text{kJ·mol}^{-1}\ (2\sigma),$$

with (sln1) ≡ (HNO_3, in 35.35H_2O). The corresponding value reported in Table 1 of [1995JEM/BEN], – (326.87 ± 1.70) J·g^{-1} is presumably a misprint.

We also obtain $\Delta_{sol}H_m(Ca_{10}(PO_4)_6F_2, \text{cr, in sln1}) = -(196.121 \pm 1.645)$ kJ·mol^{-1} (2σ).

The cycle that follows to obtain $\Delta_f H^o_m(Ca_{10}(PO_4)_6(OH)_2, \text{cr})$ is that used by the authors, with a few changes, principally in the symbols. Thus:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 20HNO_3(sln1) \rightleftharpoons (10Ca(NO_3)_2 + 6H_3PO_4 + 2H_2O)(sln1) \quad \text{(A.229)}$$

with ΔH_m (A.229) = $\Delta_{sol}H_m(Ca_{10}(PO_4)_6(OH)_2, \text{cr, in sln1}) = -(364.157 \pm 2.660)$ kJ·mol^{-1},

$$10Ca(cr) + 10N_2(g) + 50O_2(g) + 40H_2(g) \rightleftharpoons 10(Ca(NO_3)_2 \cdot 4H_2O)(cr) \quad \text{(A.230)}$$

with ΔH_m (A.230) = 10 $\Delta_f H^o_m(Ca(NO_3)_2 \cdot 4H_2O, \text{cr}) = 10\ (-(2132.03 \pm 1.30))$ kJ·mol^{-1} from [1987GAR/PAR],

$$(Ca(NO_3)_2 \cdot 4H_2O(cr) + (sln1)) \rightleftharpoons 10(Ca(NO_3)_2 + 4H_2O(sln1)) \quad \text{(A.231)}$$

with ΔH_m (A.231) = 10 $\Delta_{sol}H_m(Ca(NO_3)_2 \cdot 4H_2O, \text{cr, in sln1}) = 10\ (33.3 \pm 0.2)$ kJ·mol^{-1} that the review takes as that given by the authors and given, with no details, as an experimental result in [1991BEN/SOM]:

$$10H_2(g) + 10N_2(g) + 30O_2(g) + (sln1) \rightleftharpoons 20(\Delta_f H_m(HNO_3, \text{in sln1})) \quad \text{(A.232)}$$

with ΔH_m (A.232) = 20 (– (206.30 ± 0.40)) kJ·mol^{-1}, derived from $\Delta_f H^o_m(HNO_3, \text{aq}) = -(206.85 \pm 0.40)$ kJ·mol^{-1} [1989COX/WAG] and 0.55 kJ·mol^{-1} for the transfer of one mol of HNO_3 (aq, ∞ dilution) to HNO_3(sln1), using the interpolation from the values in [1982WAG/EVA].

$$9H_2(g) + 6P(cr, white) + 12O_2(g) + sln2 \rightleftharpoons 6(H_3PO_4, in\ sln2) \quad \text{(A.233)}$$

taking (sln2) ≡ (H_3PO_4, in 0.756 H_2O) with ΔH_m (A.233) = 6($\Delta_f H_m(H_3PO_4, \text{in soln2})$) = 6 (– (1280.22 ± 1.616)) kJ·mol^{-1} (where $\Delta_f H_m(H_3PO_4, \text{in soln2})$ was obtained from – 1273.43 [1982WAG/EVA] – 6.792 [conversion to CODATA]) kJ·mol^{-1}),

$$6(H_3PO_4 \cdot 0.756H_2O, \text{partial, in sln 2}) + (sln1) \rightleftharpoons (6\ H_3PO_4 + 4.536\ H_2O, \text{partial, in sln 1}) \quad \text{(A.234)}$$

that the review takes as that given by the authors and found as experimental result, but without details in [1991BEN/SOM], ΔH_m (A.234) = 6(– (7.87 ± 0.04)) kJ·mol^{-1},

$$(42.536\ H_2O, \text{partial, in sln1}) \rightleftharpoons 42.536\ H_2O(l) + (\text{sln1}) \qquad \text{(A.235)}$$

with ΔH_m (A.235) = 42.536 $\Delta_{tr} H_m$ (H_2O, from (sln1) to standard state) = (42.536 × 0.001269) = 0.0540 kJ·mol^{-1}, an estimate that we prefer to include rather than the authors' neglect of this correction, and:

$$38H_2(g) + 19O_2(g) \rightleftharpoons 38\ H_2O(l) \qquad \text{(A.236)}$$

with ΔH_m (A.236) = 38(– (285.830 ± 0.040)) kJ·mol^{-1} from CODATA.

Thus, $\Delta_f H^o_m$ ($Ca_{10}(PO_4)_6(OH)_2$, cr) = – ΔH_m (A.229) + ΔH_m (A.230) + ΔH_m (A.231) – ΔH_m (A.232) + ΔH_m (A.233) + ΔH_m (A.234) + ΔH_m (A.235) – ΔH_m (A.236) = – (13364.089 ± 18.451) kJ·mol^{-1}.

This value is in reasonable agreement with that reported by the authors, – 13305 kJ·mol^{-1}, considering the difference in auxiliary data used, in particular for phosphoric acid.

We thus obtain, for the representation of hydroxylapatite used in this review, $\Delta_f H^o_m$ ($Ca_5(PO_4)_3OH$, cr) = – (6682.044 ± 9.226) kJ·mol^{-1}.

In the case of fluorapatite the following sequence is used:

$$Ca_{10}(PO_4)_6F_2(cr) + 20HNO_3(\text{sln1}) \rightleftharpoons (10Ca(NO_3)_2 + 6H_3PO_4 + 2HF)(\text{sln1}) \qquad \text{(A.237)}$$

with ΔH_m (A.237) = $\Delta_{sol} H_m$ ($Ca_{10}(PO_4)_6F_2$, cr, in sln1) = – (196.121 ± 1.645) kJ·mol^{-1}, Reactions ((A.230) to (A.234)) remain the same as for hydroxylapatite;

$$(47.952\ H_2O, \text{partial, in sln1}) \rightleftharpoons 47.952\ H_2O(l) + (\text{sln1}) \qquad \text{(A.238)}$$

with ΔH_m (A.238) = 47.952 $\Delta_{tr} H_m$ (H_2O, from (sln1) to standard state) = (47.952 × 0.001269) = 0.0609 kJ·mol^{-1},

$$40H_2(g) + 20O_2(g) \rightleftharpoons 40H_2O(l) \qquad \text{(A.239)}$$

with ΔH_m (A.239) = 40 (– (285.830 ± 0.040)) kJ·mol^{-1} from CODATA,

$$2(HF \cdot 1.708\ H_2O, \text{partial, in sln3}) + \text{sln1} \rightleftharpoons (2HF + 3.416\ H_2O, \text{partial in sln1}) \qquad \text{(A.240)}$$

taking (sln3) ≡ (HF, in 1.708H_2O) and ΔH(A.240) = 2 (– (4.52 ± 0.04)) kJ·mol^{-1} from the authors, and:

$$H_2(g) + F_2(g) + 3.416H_2O(l) \rightleftharpoons 2(HF, \text{in } 1.708H_2O) \qquad \text{(A.241)}$$

We have corrected the authors' value for ΔH_m (A.241) = 2 (– 316.85) kJ·mol^{-1} by the well-established difference of – 2.72 kJ·mol(F)$^{-1}$ between the values for fluorides in [1982WAG/EVA] and CODATA. Thus, ΔH_m (A.241) = 2 (– 319.57) = – (639.14 ± 2.00) kJ·mol^{-1}, with an estimated uncertainty.

Thus $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6F_2, cr) = -\Delta H_m(A.237) + \Delta H_m(A.230) + \Delta H_m(A.231) - \Delta H_m(A.232) + \Delta H_m(A.233) + \Delta H_m(A.234) + \Delta H_m(A.238) - \Delta H_m(A.239) + \Delta H_m(A.240) + \Delta H_m(A.241) = -(13608.638 \pm 18.448)$ kJ·mol^{-1}.

This value is in reasonable agreement with that reported by the authors, – 13545 kJ·mol^{-1}, considering the differences in the auxiliary data used, in particular for phosphoric acid. We thus obtain, for the representation of fluorapatite used in this review, $\Delta_f H^\circ_m(Ca_5(PO_4)_3F, cr, 298.15\ K) = -(6804.319 \pm 9.224)$ kJ·mol^{-1}.

[1995KAB/ZAF]

Solution densities and water activities of KH_2PO_4 and K_2HPO_4 solutions were measured at 25 °C.

[1995KOZ/NEW]

This paper reports the equilibrium *P-T* curve of the $MgCO_3 + SiO_2 \rightleftharpoons MgSiO_3 + CO_2(g)$ reaction and the determination of the reversal pressure at 988 K of the reaction $MgCO_3(cr) \rightleftharpoons MgO(cr) + CO_2(g)$. This review concerns the latter reaction only; the only details given of the apparatus used was that it was carried out using an internally heated gas-pressure vessel with argon pressure medium.

The original periclase was in the form of large limpid crystals made by arc-fusion with a maximum impurity of 0.1 % CaO. Coarsely crystalline periclase was used to minimise back reaction to magnesite during quenching. Magnesite was synthesised from commercial hydrated magnesium carbonate by re-crystallisation under $CO_2(g)$ at 400 °C and *ca.* 1 kbar pressure for two days, which gave well-crystallised magnesite with an average grain size of about l5 μm and unit-cell constants of a = (4.636 ± 0.001) Å and c = (15.016 ± 0.001) Å.

The starting material for the measurements consisted of a homogeneous mixture of nearly equal masses of ground synthetic magnesite and periclase. The sample powders were heated at 623 K immediately before sealing with silver oxalate (to produce more $CO_2(g)$) in a small platinum tube. Double capsules of (Fe_2O_3 + Fe_3O_4) buffers (possibly to prevent the formation of CO(g)) were positioned in the gas-pressure vessel. The uncertainties in T and p were given as ± 2 K and ± 4 bar.

The extent of the reaction in the quenched samples was determined both by mass loss upon puncturing the sample capsules and by comparing the XRD peak heights before and after the run. The results of five experiments at (989 ± 1) K from 560-621 K are tabulated and indicate that the reversal pressure at 988 K was (590 ± 30) bar.

With the thermal functions selected in this review, the derived value of the enthalpy of formation of magnesite from the oxides at 298.15 K is – 116.46 kJ·mol^{-1} and thus $\Delta_f H^\circ_m$ ($MgCO_3$, cr, 298.15 K) = – 1111.57 kJ·mol^{-1} from the elements. The uncertainty in the pressure gives an uncertainty of ± 0.46 kJ·mol^{-1}, but the total uncertainty is appreciably bigger, owing to the use of extrapolated heat capacities of magnesite and the approximated method of calculating the contribution of the $p\Delta V$ term to the Gibbs energy—see the Appendix A for [1955HAR/TUT] for details.

[1995SHA/FED]

Using a Setaram DSC apparatus the authors made *ca.* 170 measurements of the heat capacity of hydroxylapatite from 299.2 to 1068.6 K at intervals of 5 K or less.

Several samples of hydroxylapatite were prepared from $CaCl_2$ and $(NH_4)_2HPO_4$ solutions with added aqueous hydrated ammonia to increase the pH to 10.7, the same method as used by Palkin *et al.* [1991PAL/KUZ] for their measurements of the low-temperature heat capacity. No analytical data were given for the samples other than their structure, a hexagonal lattice with a = (9.376 ± 0.004) Å and c = (6.862 ± 0.005) Å.

Sharpataya *et al.* first measured the C_p of single crystal Al_2O_3 and compared their values with the NBS standard reference data given in [1956FUR/DOU], and their Figure 1 shows the (smoothed) differences (between + 1.5 % below 750 K and – 1.5 % from 750 to 1050 K). These differences were used to correct their experimental data for hydroxylapatite.

A transition with a very small $\Delta_{trs}H$ of (187.5 ± 7.5) J·mol$(Ca_5(PO_4)_3OH)^{-1}$ was observed at (474.5 ± 5.0) K, the mean value of seven measurements between 471.4 and 476.7 K found for the exo- and endothermic transitions. This was assumed to be related to the reversible phase transition.

A table of the (somewhat scattered) raw data for hydroxylapatite is given, together with two tables of the smoothed uncorrected and corrected data from 298.15 to 1100 K and the corresponding Figures (including data down to 200 K). These show that their high-temperature measurements merge smoothly with the low-temperature C_p measurements of Palkin *et al.* [1991PAL/KUZ] from the same laboratory.

[1995SUD/YAS]

This paper mainly reports an XRD study of hydroxylapatite, but also gives details of DSC experiments around the temperature of the monoclinic to hexagonal transition. The hydroxylapatite was prepared by reacting a mixture of commercial samples of $CaHPO_4·2H_2O$(cr) and $CaCO_3$(cr), which were pelletised with methanol, heated to

1273 K *in vacuo* for 10 hours, followed by a further 10 hours at 1173 K in pure argon saturated with water vapour, the latter heating being repeated three times. The calcium and phosphate analyses of the product were very close to the theoretical values for hydroxylapatite and X-ray diffraction, presumably at room temperature, showed only those lines for hydroxylapatite with a structure with $a = 9.4187(5)$ and $c = 6.8805(2)$ Å. High-temperature diffractometer examination indicated a transition to a hexagonal phase at (478 ± 5) K from the disappearance of small additional peaks due to the monoclinic distortion below this temperature.

This was confirmed by DSC studies, which indicated a transition at (480.0 ± 0.3) K from exo- and endothermic peaks, with a transition enthalpy of $\Delta_{trs}H = (315.0 \pm 12.5)$ J·mol^{-1}. This is noticeably larger than the value of (188 ± 8) J·mol^{-1} at (476.7 ± 0.1) K found by Sharpataya *et al.* in [1995SHA/FED], also from DSC measurements.

Suda *et al.* also *g*ive a small plot of their measured heat capacities from 443 to 505 K, (from DSC measurements) showing a sharp peak at 483 K (slightly higher than the value of 480 K obtained using a lower heating rate). A plot of the heat capacities of hydroxylapatite, including these data, is included in the main text, which shows that these C_p values agree well with the data in [1950EGA/WAK], [1951EGA/WAK], [1991PAL/KUZ] and [1995SHA/FED].

[1995SWA/DOV]

Detailed neutron powder diffraction data are presented which show that the phase transition at 754 K in sodium carbonate, Na_2CO_3(cr), is a second-order ferroelastic transition from the low-temperature $P6_3/mmc$ phase to a hexagonal $C2/m$ phase.

[1996BRO/DRU]

A hydrogen-electrode concentration cell was used to detect the advent of precipitation from Mg^{2+} solutions that occurred after *ca*. 4 % of the total magnesium concentration had been titrated with a NaOH solution of matching ionic strength. Precipitation was clearly observed when the rapid change in potential upon addition of titrant was followed by a slower return to almost the original value (*i.e.*, a pH buffering effect). Back extrapolation of the titration curve to the inflection point allowed the pH at which precipitation first occurred to be defined accurately. This technique can only be utilised for the rare cases when formation of a stable solid phase is very rapid. The recovered solid was shown to be crystalline brucite by XRD, but no particle size examination could be made. The titrations were reversed quantitatively corresponding to approaching equilibrium from under-saturation. The titrations were performed in $I_m = 0.1$ and 1.0 NaCl solutions at 60, 100, 150 and 200 °C. Initially $m_{Mg} = 0.005$

mol·kg^{-1} when I_m = 0.1, but was varied from 0.001 to 0.010 mol·kg^{-1} when I_m = 1.0. Application of the SIT treatment according to Equation (A.242) gave the results listed in Table A-62.

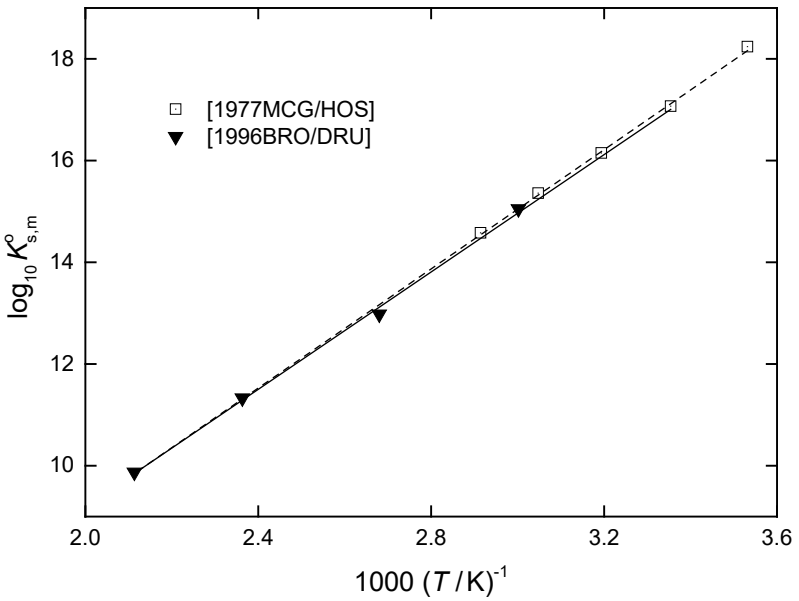
$$\log_{10}{}^{*}K_{s,0} - 2D + \log_{10} a_{H_2O} = \log_{10}{}^{*}K^{o}_{s,0} - \Delta\varepsilon I_m \qquad \text{(A.242)}$$

Table A-62: Summary of the reported experimental conditions and solubility constants, as well as the water activity and the results of application of the SIT according to Equation (A.242).

t /°C	I_m NaCl	$\log_{10}{}^{*}K_{s,0}$ /reported	$\log_{10}{}^{*}K^{o}_{s,0}$ /reported	a_{H_2O}	$\log_{10}{}^{*}K^{o}_{s,0}$ /SIT	$\Delta\varepsilon$ /kg·mol^{-1}
60	0.1	(15.29 ± 0.2)	15.07	0.9967	(15.05 ± 0.21)	– (0.02 ± 0.25)
60	1.0	(15.52 ± 0.1)		0.9666		
100	0.1	(13.25 ± 0.2)	13.18	0.9967	(12.98 ± 0.21)	– (0.18 ± 0.25)
100	1.0	(13.64 ± 0.1)		0.9669		
150	0.1	(11.62 ± 0.1)	11.33	0.9967	(11.33 ± 0.11)	– (0.01 ± 0.15)
150	1.0	(11.88 ± 0.1)		0.9678		
200	0.1	(10.21 ± 0.1)	9.87	0.9968	(9.88 ± 0.11)	(0.06 ± 0.15)
200	1.0	(10.44 ± 0.1)		0.9693		

The $\log_{10}{}^{*}K^{o}_{s,0}$ value obtained from the SIT treatment extrapolated to 25 °C is 17.01 which is consistent with those obtained in other studies, *e.g.*, [1977MCG/HOS], [1999KON/KON] and [2003ALT/MET]. An error was apparently made in estimating the activity coefficient of Mg^{2+} at 100 °C. This error may have contributed to criticism [1999KON/KON] of these results. When combined with the results of McGee and Hostetler (dashed line in Figure A-24) $\log_{10}{}^{*}K^{o}_{s,0}$ = 17.02 at 25 °C and the standard enthalpy of reaction is – (112 ± 3) kJ·mol^{-1}, in good agreement with – 111.2 kJ·mol^{-1} derived subsequently by Königsberger *et al*. [1999KON/KON] from solubility measurements in 3.5 and 3.0 mol·kg^{-1} ($NaClO_4$) at 25 and 50 °C, respectively.

Figure A-24: Solubility constants for $Mg(OH)_2$(cr) obtained from the SIT treatment of the data in this paper and those taken from [1977MCG/HOS], where the solid and dashed lines were derived from linear regressions of the former and combined data sets, respectively.



[1996CHE/ZEN2]

The standard enthalpies of formation of gibbsite and boehmite were reported, based on the enthalpy changes upon decomposition of these compounds to γ-alumina; only the data for gibbsite will be discussed here. For gibbsite, a Tian-Calvet heat flow calorimeter operating from room temperature to 1273 K was used. The melting enthalpies of various metals and compounds were measured and shown to be in agreement within 0.5 % with literature values. The samples of gibbsite were prepared by the authors as described in prior publications. The total amount of impurities detected by spectral analyses was less than 0.03 %. X-ray diffraction and thermogravimetric analyses were used to ascertain the purity of the compound.

For gibbsite, three measurements of the enthalpy change of the decomposition reaction:

$$Al(OH)_3(\text{gibb, } 298.15\text{ K}) \rightleftharpoons (1/2Al_2O_3(\text{cr, } \gamma) + 3/2\,H_2O(\text{g}),\ 958\text{ K}) \qquad \text{(A.243)}$$

were made, each using *ca.*170 g of sample and the results listed, with a mean value of (176.72 ± 0.90) kJ·mol^{-1} (the uncertainty limits being increased by this review to cover the 95 % confidence interval). XRD of the products showed only the lines of γ-Al_2O_3. To obtain the enthalpy change for Reaction (A.243) at 298.15 K, the authors subtracted the endothermic effect (given as 104.99 kJ·mol^{-1} from the C_p measurements of Hemingway *et al.* [1977HEM/ROB3]) due to the heating of gibbsite in the calorimeter from 298.15 K to 958 K, and then subtracted the enthalpy effect of Reaction (A.243) from 958 to 298.15 K, using the same heat capacity of gibbsite, that for H_2O(g) from Barin *et al.* [1977BAR/KNA] and for γ-Al_2O_3 that measured by the same group of authors [1995CHE/ZEN2]. However, this is equivalent to just subtracting from $\Delta_r H$ (A.243) the enthalpy difference for (1/2 γ-Al_2O_3 + 3/2 H_2O(g)) from 298.15 to 958 K, which we calculate to be (74.664 ± 1.000) kJ·mol^{-1}, using the heat capacity of γ-Al_2O_3 from [1995CHE/ZEN2] and that for H_2O(g) from CODATA [1989COX/WAG]. This simpler method also avoids the need to estimate the C_p of gibbsite above 480 K, the highest temperature of the measurements in [1977HEM/ROB3]. Thus, the enthalpy of the decomposition Reaction (A.243) at 298.15 K is (176.72 ± 0.90) – (74.664 ± 1.000) = (102.06 ± 1.35) kJ·mol^{-1}, compared to the authors' value of (101.89 ± 1.20) kJ·mol^{-1}.

Making use of our re-calculated value $\Delta_f H^o_m$(Al_2O_3, cr, γ, 298.15 K) = – (1656.86 ± 2.10) kJ·mol^{-1} from the same team [1995CHE/ZEN2] (see the relevant Appendix A review), and of $\Delta_f H^o_m$(H_2O, g, 298.15 K) = – (241.826 ± 0.040) kJ·mol^{-1} from [1989COX/WAG], $\Delta_f H^o_m$($Al(OH)_3$, gibb) becomes – (1293.23 ± 1.71) kJ·mol^{-1} from our calculation.

[1996HOL/MES]

New isopiestic measurements of $MgCl_2$ solutions were made from 383 to 473 K to check on experiments made previously with the same apparatus [1978HOL/BAE] after modifications had been made. Sodium chloride reference solutions were used. The Pitzer ion interaction treatment was utilised in modelling the thermodynamic properties of $MgCl_2$ solutions as well as those of $SrCl_2$ and $BaCl_2$. The value of $\alpha_1 = 2.0$ commonly used in the Pitzer model did not produce an acceptable fit for the isopiestic data for all the alkaline metal chlorides. Instead they used a general temperature-dependent equation: $\alpha_1 = 2.0 - 0.00181((T/\text{K}) - 298.15)$. Their earlier results had been criticised (*e.g.*, [1984EMO/VOI]) as being in error by several percent. The new results agree well with those published earlier [1978HOL/BAE], [1981RAR/MIL2], [1990FAN/GRJ]. However, the vapour-pressure measurements of $MgCl_2$ solutions [1972LIU/LIN] could not be fitted with their model. There was also substantial disagreement with the high-temperature osmotic coefficients for $MgCl_2$ solutions measured by Valyashko *et al*. [1988VAL/URU] that could not be explained. Although the criticism raised by Emons *et al*. [1984EMO/VOI] could not be rationalised, the mere fact that the comprehensive

experimental datasets measured by Holmes *et al*. are both internally consistent and agree with the results of other major studies gave reasonable confidence in their osmotic coefficient data.

[1996JAC/VAR]

The authors have analysed in some detail the various papers, particularly [1991CHR/GRI] and [1993GOU/RAN], which suggest that the CODATA values of the enthalpies of formation of CaO(cr) are in error by (35-40) $kJ \cdot mol^{-1}$, and have made new potential-difference measurements to throw more light on this situation.

Their measurements involved the potential difference of the solid electrolyte cells:

$$Pt, O_2(g), CaO + CaF_2|CaF_2|\ MgO + MgF_2, O_2(g), Pt \text{ and}$$

$$Pt, O_2(g), SrO + SrF_2|SrF_2\ |\ CaO + CaF_2, O_2(g), Pt$$

in the range (950-1280) K, the choice of the solid electrolyte being governed by compatibility considerations at the electrodes.

From the potential difference of these cells, the Gibbs energies of the reactions:

$$CaO(cr) + MgF_2(cr) \rightleftharpoons CaF_2(cr) + MgO(cr) \qquad (A.244)$$

$$SrO(cr) + CaF_2(cr) \rightleftharpoons SrF_2(cr) + CaO(cr) \qquad (A.245)$$

were found to be:

$$\Delta_r G_m\,(A.244) = -\,68050 - 2.47T\ J \cdot mol^{-1}\ (ca.\ 950\text{-}1280)\ K \text{ and}$$

$$\Delta_r G_m\,(A.245) = -\,35010 + 6.39T\ J \cdot mol^{-1}\ (ca.\ 950\text{-}1280)\ K.$$

We have compared these to the values from [1994GUR/VEY] (which are essentially the CODATA values where these exist) in Figures A-25 and A-26 below. The uncertainties in the values in [1994GUR/VEY] may be slightly over-estimated, since they are just a combination of the uncertainties in the Gibbs energies of formation of the four solids, which are not entirely independent.

Figure A-25: Data for $\Delta_r G_m$(A.244).

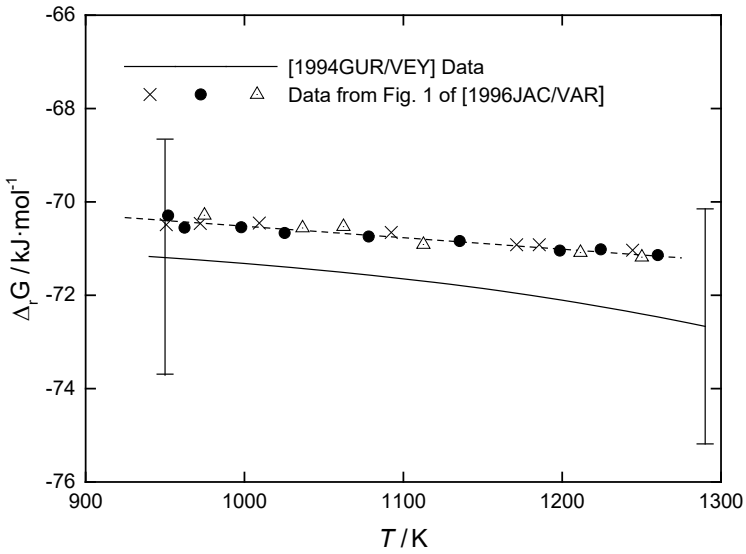
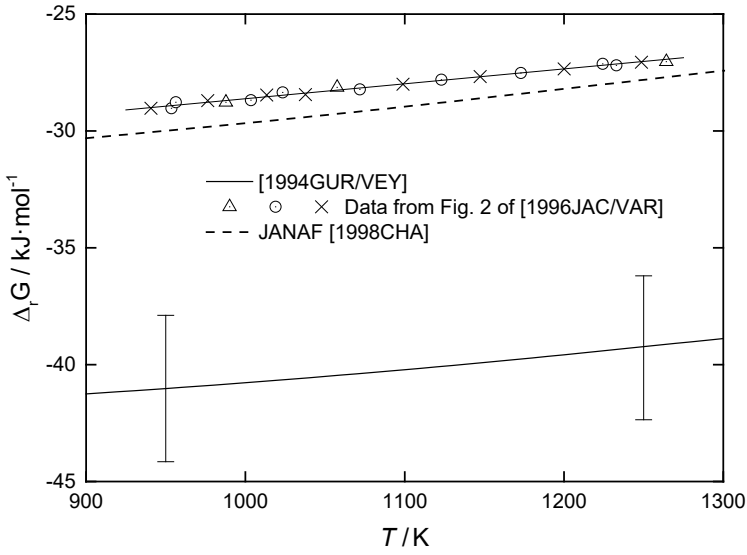


Figure A-26: Data for $\Delta_r G_m$(A.245).



For Reaction (A.244), it is clear that the authors' values are totally consistent with the values in [1989COX/WAG] and [1994GUR/VEY], unlike those from the

second reaction. However, we note that the value of $\Delta_f H^o_m$ (SrF_2, cr, 298.15 K) selected in [1994GUR/VEY] is appreciably more negative than those selected in other assessments. As an additional comparison, the values for $\Delta_r G_m$ (A.245) from the JANAF compilation [1998CHA] are also included in the relevant Figure.

These results suggest that the difference, at least, between the CODATA values of $\Delta_r G^o_m$ (MgO, cr, 298.15 K) and $\Delta_f H^o_m$ (CaO, cr, 298.15 K), is not seriously in error. This throws particular doubt on the study by [1993GOU/RAN] which would suggest that this difference is approximately – 33 kJ·mol^{-1} rather than + 33 kJ·mol^{-1}, and the value of $\Delta_f H^o_m$ (SrF_2, cr, 298.15 K) selected in [1994GUR/VEY] may be too negative. Jacob and Varghese also critically examined the various papers which suggest that the CODATA value of $\Delta_f H^o_m$ (CaO, cr, 298.15 K), in particular, is seriously in error. They highlight a number of experimental uncertainties in all of this (particularly those involving CaC_2(cr)), and conclude that the CODATA values are not seriously in error.

[1996KHA/ISS]

Potentiometric measurements were performed on the ionisation of boric acid and potassium hydrogen phthalate in acetonitrile-water mixtures from 5 to 40 °C at 5 ° intervals similar to those reported in [1992JUM/ABD]. A $\log_{10}{}^* K^o_1$ value of – 8.91 at 25 °C was reported, compared to – 9.24, which is the average of those obtained from other papers. The difference could be due to carbonate contamination which is always a major concern with these titrations. These results were not considered further in the current review.

[1996MOG/GAN]

A study was made of the dissolution kinetics of a natural gibbsite at 25 °C using a flow-through column. The solid particles were monitored by the BET surface-area technique, as well as XRD and SEM methods with no change being observed after the experiments. The concentrations of $HClO_4$ in the feed solution and of aluminium in the outlet solution were < 10^{-3} mol·dm^{-3}. Activity coefficient and hydrolysis corrections were made. A value of $\log_{10}{}^* K_{s,0} = (7.83 \pm 0.03)$ was reported and is accepted by this review.

[1996ORL/ION2]

The enthalpy of formation of hydroxylapatite $Ca_{10}(PO_4)_6(OH)_2$(cr) was calculated based on the X-ray structure analysis of the compound using bond-energies from a structure-thermochemical model for phosphorus compounds proposed earlier by the same authors We shall not go in detail in this purely calculational paper, and the reported value,

$\Delta_f H_m$ ($Ca_{10}(PO_4)_6(OH)_2$, cr, 0 K) = – 13488 kJ·mol^{-1}, is mentioned here for information only. This result was compared by the authors with the experimental value $\Delta_f H_m^o$ ($Ca_{10}(PO_4)_3(OH)_2$, cr, 298.15 K) = – (13510.1 ± 20.9) kJ·mol^{-1} in [1963JAC] and converted to $\Delta_f H_m$ ($Ca_{10}(PO_4)_3(OH)_2$, cr, 0 K) = – 13361 kJ·mol^{-1} using the heat capacity data of [1991PAL/KUZ]. The authors do not mention any of the other experimental results on the enthalpy of formation of this compound.

The authors' value $\Delta_f H_m^o$ ($Ca_5(PO_4)_3OH$, cr, 0 K) = – 6744 kJ·mol^{-1} for the simpler formula is, in fact, in good agreement with our selected value of $\Delta_f H_m^o$ ($Ca_5(PO_4)_3OH$, cr, 298.15 K) = – (6740.4 ± 15.0) kJ·mol^{-1}.

[1996TIK/FAW]

Tikanen and Fawcett took dielectric-relaxation results from [1992BAR/HET] for a number of electrolytes, including those for $MgCl_2$ and $MgSO_4$ solutions, and applied the restricted version of the mean spherical approximation, MSA, to estimate the activity coefficients in these solutions with only two adjustable parameters: a single effective diameter and the association constant, K_1^o. The dielectric-relaxation results for $MgSO_4$(sln) have been superseded by higher resolution data [2004BUC/CHE], but the resulting ion-pair formation constant was only slightly affected, *i.e.*, K_1^o = 167 compared to 164 in [1996TIK/FAW]; the later study differentiated between contact and solvent-separated ion pairs. The results of Barthel *et al*. [1992BAR/HET] extended from 0.2 to 1.2 mol·kg^{-1}. The restricted MSA treatment involved an iterative procedure with a fixed effective diameter for the ion pair whereby the difference between the calculated activity coefficients of the electrolyte was minimised compared to those in [1955ROB/STO]. The activity coefficient of $MgSO_4$(aq) was allowed to vary iteratively, unlike most other treatments where a value of unity was maintained. The value of K_1^o = 164 is in good agreement with those from other experimental and computational investigations, indicating the activity coefficient of the neutral ion pair had little effect on the formation constant. Tikanen and Fawcett reported a value of K_1^o = 0.22 for the formation of $MgCl^+$ that is insignificant for a 2:1 electrolyte considering the uncertainties in any activity coefficient model used.

[1996WOL/LER]

This paper concerns a determination of the enthalpies of the aragonite-to-calcite transition by various dissolution techniques in aqueous media, differential-scanning calorimetry and heat capacity measurements.

Reagent grade commercial calcite was used. An aragonite sample prepared according to literature [1960LIP] and a natural sample (from Bilin, Czechia) were used; the different aragonite samples were indistinguishable with respect to the

thermochemical experiments. The composition and structure of all samples were ascertained by chemical analyses and X-ray methods, but no details were given except to indicate that they were pure phases.

Two Dewar type twin calorimeters (25 and 50 cm^3), using samples of 25 and 100 mg, and a commercial (LKB 8700) calorimeter (with similar sample/solution ratio) were used. Only typical results were listed with no indication of the number of measurements.

Dissolution in acidic media (0.7 $mol \cdot kg^{-1}$ HCl saturated with NaCl) led to a very limited accuracy because of the difficulty in defining precisely how much CO_2 remained in solution at the end of the reactions.

The values of $\Delta_r H_m^o$ ((A.246), 298.15 K) were calculated to be $-(0.21 \pm 0.46)$ $kJ \cdot mol^{-1}$ if the CO_2 is completely evolved and $-(0.10 \pm 0.54)$ $kJ \cdot mol^{-1}$ if the CO_2 is completely retained in solution.

$$CaCO_3(\text{calc, 298.15 K}) \rightleftharpoons CaCO_3(\text{arag, 298.15 K}) \qquad \text{(A.246)}$$

Dissolution (at 313 K) in aqueous alkaline EDTA-media (0.5 $mol \cdot kg^{-1}$, pH = 10.3) led to more reproducible values, typically, $\Delta_r H_m^o$ ((A.246), 313.15 K) = $-(0.05 \pm 0.18)$ $kJ \cdot mol^{-1}$.

For the determination of the high-temperature enthalpy change upon transition, a DSC (SETARAM, model 111) under a CO_2 atmosphere was used. Depending on the heating rates (2 to 25 $K \cdot min^{-1}$) the temperature of the onset of the transition increased from 721 K to 742 K. The enthalpy effect, $\Delta_r H_m$ ((A.246), T) = $-(0.403 \pm 0.008)$ $kJ \cdot mol^{-1}$ was reported.

Heat capacities and heat capacity differences between calcite and aragonite were measured using a heat-flux calorimeter. Only typical values were given that resulted in the values of $\Delta_r C_{p,m}$ (A.246) = (0.0 ± 0.2), (0.0 ± 0.2), and $-(0.6 \pm 0.2)$ $J \cdot K^{-1} \cdot mol^{-1}$ at 303, 353 and 403 K respectively. For comparison, those from the equations of Kobayashi [1951KOB], [1951KOB2], are – 1.45, – 0.70 and – 0.45 $J \cdot K^{-1} \cdot mol^{-1}$ at these temperatures, and Staveley *et al.* [1969STA/LIN] found $\Delta_r C_{p,m}$ (A.246) = $-(1.15 \pm 0.28)$ $J \cdot K^{-1} \cdot mol^{-1}$ at 298.15 K.

Using the minimisation routine in ChemSage [1990ERI/HAC], [1995KON/ERI], the authors then optimised all their own data for this reaction, *viz.* those from this work and the earlier studies of this group, in particular [1989KON/BUG], obtaining, finally their recommended values:

$$\Delta_r G_m^o \text{ ((A.246), 298.15 K)} = (0.840 \pm 0.020) \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_r H_m^o \text{ ((A.246), 298.15 K)} = -(0.440 \pm 0.050) \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_r S_m^o \text{ ((A.246), 298.15 K)} = -(4.3 \pm 0.2) \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \text{ and}$$

$$\Delta_r C^{\circ}_{p,m}((A.246), 298.15\ K) = (0.0 \pm 0.2)\ J \cdot K^{-1} \cdot mol^{-1}.$$

[1997KRI/ORL]

The standard enthalpy of formation of hydroxylapatite, $Ca_{10}(PO_4)_6(OH)_2$(cr) at 298.15 K was obtained from the enthalpy of solution of this compound, of $CaCl_2$(cr) and of β-$Ca_3(PO_4)_2$ in 14.04 % HCl, using a silver-coated double Dewar vessel containing 0.300 dm^3 of HCl solvent. The calorimeter assembly was properly calibrated electrically and its performances verified using the enthalpy of solution of KCl(cr). The preparation of the compounds involved was described and analyses were reported. For β-$Ca_3(PO_4)_2$ the calculated composition (%) is Ca, 38.76; PO_4, 61.23, while the experimental results were Ca, 38.9; PO_4, 61.1. Infrared spectroscopy and X-ray diffraction analyses of β-$Ca_3(PO_4)_2$ revealed no detectable calcium hydroxylapatite, nor any other calcium phosphates.

For $Ca_5(PO_4)_3(OH)$(cr) the calculated composition (%) is Ca, 39.8; PO_4, 56.74; OH, 3.38, while the experimental results were Ca, 39.89; PO_4, 56.73; OH, 3.38. Pelletised reagent grade $CaCl_2$(cr) was used, after drying at 600-620 °C.

In the final calorimetric solutions (xHCl, $y$$H_2O$), the ratio y/x was equal to 12.4 and given the sample size (gram amounts), x ranged between 360 and 660.

The authors based their experimental cycle on the enthalpy change corresponding to the process:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 2HCl(g) \rightleftharpoons CaCl_2(cr) + 3\ \beta\text{-}Ca_3(PO_4)_2 + 2H_2O(l) \qquad (A.247)$$

using the following reactions:

$$CaCl_2(cr) + (xHCl, yH_2O)(sln) \rightleftharpoons (CaCl_2, xHCl, yH_2O)(sln) \qquad (A.248)$$

for which this review calculates $\Delta_r H_m$ (A.248) = – (68.70 ± 0.22) kJ·mol^{-1} (95 % confidence) from the straight average of seven reported measurements (while the authors give – (68.74 ± 0.23) kJ·mol^{-1} as the average value).

$$3\ \beta\text{-}Ca_3(PO_4)_2 + (CaCl_2, xHCl, yH_2O)(sln) \rightleftharpoons$$
$$(3\ Ca_3(PO_4)_2 + CaCl_2 + xHCl + yH_2O)(sln) \qquad (A.249)$$

for which this review calculates $\Delta_r H_m$ (A.249) = – (306.35 ± 2.64) kJ·mol^{-1} (95 % confidence) from the straight average of three reported measurements (while the authors give – (306.0 ± 1.2) kJ·mol^{-1} as the average value).

$$Ca_{10}(PO_4)_6(OH)_2(cr) + ((x+2)HCl, (y-2)H_2O)(sln) \rightleftharpoons$$
$$(3Ca_3(PO_4)_2 + CaCl_2 + xHCl + yH_2O)(sln) \qquad (A.250)$$

for which this review calculates $\Delta_r H_m$ (A.250) = – (279.70 ± 1.00) kJ·mol^{-1} (95 % confidence) from the straight average of seven reported measurements (while the authors give – (279.5 ± 1.4) kJ·mol^{-1} as the average value).

It should be noted that the number of measurements for these three reactions are seven, three, and seven, respectively. Thus 1σ does not represent the same degree of confidence. To obtain 95 % confidence, we have multiplied the standard deviation on the average by the appropriate Student-t factor that is 2.45 for seven measurements and 4.30 for three measurements. In order to complete the cycle leading to Reaction (A.247), the authors used:

$$2HCl(g) + (xHCl, yH_2O)(sln) \rightleftharpoons ((x+2)HCl, yH_2O)(sln) \qquad (A.251)$$

with $\Delta_r H_m$ (A.251) = – (133.47 ± 0.08) kJ·mol^{-1} and

$$2H_2O(l) + ((x+2)HCl, (y-2)H_2O)(sln) \rightleftharpoons ((x+2)HCl, yH_2O)(sln) \qquad (A.252)$$

with $\Delta_r H_m$ (A.252) = – (0.44 ± 0.08) kJ·mol^{-1}, both from [1979GLU/MED].

This review re-calculates the above $\Delta_r H_m$ (A.251) and $\Delta_r H_m$ (A.252) partial values for the medium (HCl, 12.4H_2O) as $\Delta_r H_m$ (A.251) = – (131.648 ± 0.28) kJ·mol^{-1} and $\Delta_r H_m$ (A.252) = – (0.66 ± 0.20) kJ·mol^{-1} from Parker [1965PAR].

Thus, $\Delta_r H_m$ (A.247) = – $\Delta_r H_m$ (A.248) – $\Delta_r H_m$ (A.249) + $\Delta_r H_m$ (A.250) + $\Delta_r H_m$ (A.251) – $\Delta_r H_m$ (A.252) = – (35.638 ± 2.852) kJ·mol^{-1} from our re-calculated values.

Use of $\Delta_f H_m^o$ (H_2O, l, 298.15 K) = – (285.830 ± 0.040) and $\Delta_f H_m^o$ (HCl, g, 298.15 K) = – (92.310 ± 0.100) kJ·mol^{-1} from CODATA [1989COX/WAG], $\Delta_f H_m^o$ ($CaCl_2$, cr, 298.15 K) = – (795.39 ± 1.00) kJ·mol^{-1} from this review, and $\Delta_f H_m^o$ ($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – (4123.6 ± 12.0) kJ·mol^{-1} selected by this review, leads to $\Delta_f H_m^o$ ($Ca_{10}(PO_4)_6(OH)_2$, cr, 298.15 K) = – (13517.592 ± 36.127) kJ·mol^{-1} or, adopting, as in the text of this review, the use of the formula $Ca_5(PO_4)_3OH$(cr) and rounding:

$$\Delta_f H_m^o (Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6758.8 \pm 18.1)\ kJ \cdot mol^{-1}.$$

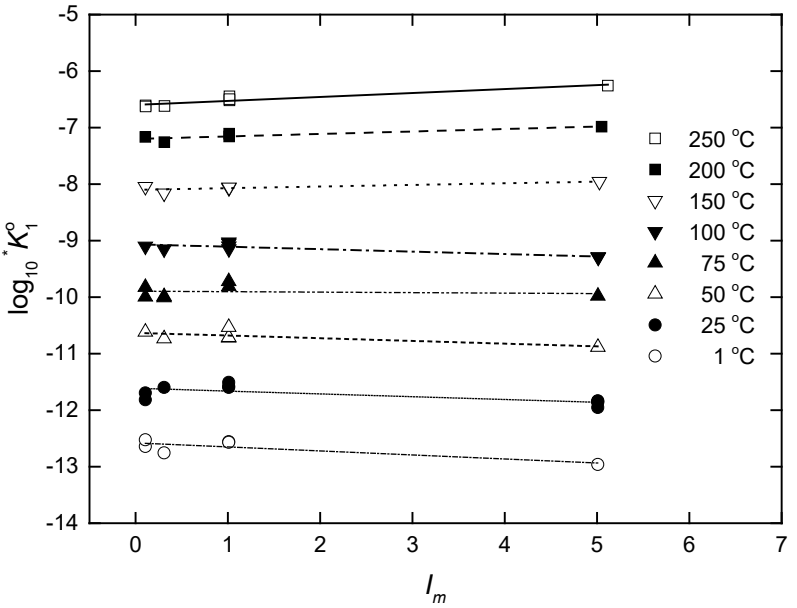
[1997MAL/ZAM]

Potential-difference measurements were reported with a “liquid-membrane” concentration cell using four different test compositions. For $MgSO_4$ and $CaSO_4$ solutions the molality was increased from 4.4 × 10^{-5} to 2.5 mol·kg^{-1} and 1.285 × 10^{-4} to 1.273 × 10^{-2} mol·kg^{-1}, respectively. Malatesta and Zamboni used the Pitzer ion interaction model to treat their results within a precision that did not require consideration of ion association for any of the solutes studied.

[1997PAL/WES]

Potentiometric titrations were performed to determine the first hydrolysis constant of Mg^{2+} using a hydrogen-electrode concentration cell from 1 to 250 °C at ionic strengths of 0.11, 0.31, 1.01, and 5.0 mol·kg^{-1} (NaCl). Magnesium concentrations were *ca.* (4-5) × 10^{-3} mol·kg^{-1} in *ca.* 2 × 10^{-3} mol·kg^{-1} HCl solutions. The initial small potential reading (< 1 mV) attributed to protolytic contamination was adjusted to zero by varying m_{HCl}; a similar procedure was adopted by McGee and Hostetler [1975MCG/HOS]. Liquid-junction potentials were small (< 0.05 mV) and were approximated by the Henderson equation [1976BAE/MES]. The pH of the test solution, defined as $-\log_{10} m_{H^+}$, was calculated from the Nernst expression. The reported thermodynamic quantities at 25 °C for hydrolysis of Mg^{2+} were: $\log_{10}{}^{*}K_1^{\circ} = -(11.68 \pm 0.03)$; $\Delta_r H_m^{\circ} = (70.1 \pm 0.8)$ kJ·mol^{-1} and $\Delta_r S_m^{\circ} = (11 \pm 0.3)$ J·K^{-1}·mol^{-1} (2σ). These values of $\log_{10}{}^{*}K_1^{\circ}$ are in good agreement with those reported with larger uncertainties resulting from fewer adjustable parameters allowed by the SIT treatment. Figure A-27 shows that the plots are linear to ionic strengths of 5 mol·kg^{-1}, which is higher than normally expected for equilibria involving 2:1 electrolytes.

Figure A-27: Plot illustrating application of the SIT to the data summarised in Table A-63:



The ion interaction parameters, which trend monotonically with temperature, were treated to yield the value of $\varepsilon(MgOH^+,Cl^-)$ at 25 °C as follows:

$\varepsilon(MgOH^+,Cl^-) = \Delta\varepsilon - \varepsilon(H^+,Cl^-) + \varepsilon(Mg^{2+},Cl^-) = (0.050 \pm 0.025) - (0.12 \pm 0.01) + (0.19 \pm 0.02) = (0.12 \pm 0.03)$ kg·mol^{-1}.

Equation (A.253) results from an unweighted regression of the $\log_{10}{}^*K_1^o$ values in Table A-63:

$$\log_{10}{}^*K_1^o = (0.1374 \pm 0.1235) - (3481.93 \pm 43.90)(T/\mathrm{K})^{-1} \qquad (A.253)$$

The resulting thermodynamic parameters at 25 °C are: $\log_{10}{}^*K_1^o = -(11.54 \pm 0.19)$; $\Delta_r G_m^o = (65.9 \pm 1.1)$ kJ·mol^{-1}, $\Delta_r H_m^o = (66.7 \pm 0.8)$ kJ·mol^{-1} and $\Delta_r S_m^o = (2.6 \pm 0.5)$ J·K^{-1}·mol^{-1}, where the uncertainties are 2σ. The $\log_{10}{}^*K_1^o$ value at 25 °C is in good agreement with potentiometric values of Stock and Davies [1948STO/DAV], and McGee and Hostetler [1975MCG/HOS], but significantly lower than were obtained previously.

Table A-63: Summary of the fit parameters obtained from application of Equation (A.253)

t /°C	p_{sat} /MPa	$-\log_{10}{}^*K_1^o$	$\Delta\varepsilon$ /kg·mol^{-1}
1.0	0.1	(12.58 ± 0.11)	(0.071 ± 0.052)
24.5	0.1	(11.61 ± 0.09)	(0.050 ± 0.025)
49.9	0.1	(10.63 ± 0.10)	(0.048 ± 0.045)
74.9	0.1	(9.89 ± 0.12)	(0.009 ± 0.058)
99.8	0.1	(9.06 ± 0.05)	(0.043 ± 0.021)
149.8	0.474	(8.10 ± 0.06)	– (0.029 ± 0.026)
199.8	1.548	(7.20 ± 0.06)	– (0.044 ± 0.026)
249.9	3.970	(6.60 ± 0.04)	– (0.070 ± 0.022)

[1997RAR/CLE]

Rard and Clegg estimated that the minimum threshold concentration of chloride for detecting the formation of contact $CaCl^+$ ion pairs was 3 mol·kg^{-1}. Smoothed osmotic coefficients at 298.15 K were given in Table 11 in this paper based on critical analyses of a wide range of experimental results taken from the literature with additional results measured by the authors. The extended Pitzer equation was used in the smoothing process. An SIT treatment of these smoothed values limited to $I_m \leq 1.0$ gave $\varepsilon(Ca^{2+},Cl^-) = (0.140 \pm 0.003)$ kg·mol^{-1}, whereas expanding the range to $I_m = 3.0$ required an ionic strength-dependent ion interaction term which by the NEA TDB convention would led to: $\varepsilon_1(Ca^{2+},Cl^-) = (0.102 \pm 0.04)$ and $\varepsilon_2(Ca^{2+},Cl^-) = (0.099 \pm 0.010)$ kg·mol^{-1} (2σ). The former matches exactly the recommended value in the NEA data base and in [1980CIA]. The ion-pair association constants derived from enthalpies of dilution and mixing of $MgCl_2$ solutions by Gillespie *et al*. [1992GIL/OSC] from 523 to 598 K were considered by Rard and Clegg to be compromised by

covariance in their fitting procedures (see the Appendix A entry for [1992GIL/OSC] for more details).

[1997RIM]

The solubility of quartz was measured in pure water from undersaturation at 21, 50, 74 and 96 °C ($i.e., \mathrm{SiO_2(cr) + 2H_2O(l) \rightleftharpoons Si(OH)_4(aq)}$). Rimstidt used a high solid to solution mass ratio to enhance the rate of dissolution of large pre-treated particulate Brazilian quartz (50-400 μm) and equilibration times of up to 13.5 years. The results are shown in Table A-64. When combined with selected quartz solubility data from the literature, a new solubility expression was proposed by Rimstidt from 0 to 300 °C:

$$\log_{10} m_{\mathrm{Si}} = -(0.0254 \pm 0.0247) - (1107.12 \pm 10.77)(T/\mathrm{K})^{-1}$$

where m_{Si} is the molality of silica. This function predicts the solubility of quartz at 25 °C to be $(1.83 \pm 0.30) \times 10^{-4}$ mol·kg^{-1} (all uncertainties are adjusted to 2σ). Rimstidt proposed revised values of $\Delta_f G^\circ_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = -(1309.231 \pm 1.196)$ kJ·mol^{-1}, $\Delta_f H^\circ_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = -(1460.913 \pm 1.062)$ kJ·mol^{-1} and $S^\circ_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = (180.87 \pm 0.96)$ J·K^{-1}·mol^{-1}.

Table A-64: Solubility measurements of α-quartz (same material as was used in [1980RIM/BAR] where the reported ppm refer to SiO_2(aq)).

t /°C	Equilibration time /days	Reported solubility /ppm	Solubility /mol·kg^{-1}
(21 ± 4)	4917	(9.59 ± 0.25)	1.60×10^{-4}
(21 ± 4)	4917	(9.87 ± 0.29)	1.64×10^{-4}
(50 ± 1)	210	19	3.16×10^{-4}
(50 ± 1)	173	22	3.66×10^{-4}
(75 ± 1)	54	40	6.66×10^{-4}
(74 ± 1)	54	35	5.83×10^{-4}
(96 ± 2)	108	54	8.99×10^{-4}
(96 ± 2)	108	53	8.82×10^{-4}
(96 ± 2)	108	56	9.32×10^{-4}
(96 ± 2)	108	54	8.99×10^{-4}

[1998FEL/DIX]

The hydrolysis and carbonate complexation of calcium and strontium cations were investigated using a combined experimental and modelling approach. The latter used both macroscopic thermodynamic models and molecular modelling based on the density functional theory from which Felmy *et al.* calculated that calcium ions are surrounded by six water molecules in keeping with the later findings from XAFS results [2003FUL/HEA]. Batch solubility measurements were made of $Ca(OH)_2$(cr), $Sr(OH)_2 \cdot 8H_2O$(cr), $CaCO_3$(cr) and $SrCO_3$(cr) in 0.01 to 5 mol·kg^{-1} (NaOH and Na_2CO_3) solutions at 298.15 K. The solid phases were characterised by XRD before and after the

equilibrations which in the case of portlandite apparently occurred within 18 days. Their seven results for the solubility of $Ca(OH)_2$(cr) in NaOH solutions were in good agreement with those of Duchesne and Reardon [1995DUC/REA], but disagreed with previously published data at high hydroxide concentrations. Introduction of $\beta^{(2)}_{CaOH} = -K^\circ_1/2$ according to Pitzer [1991PIT] with values taken from [1984HAR/MOL] and [1995DUC/REA], viz., – 11.05 and – 7.52, respectively, gave $\log_{10}K^\circ_1(CaOH^+ \rightleftharpoons Ca^{2+} + OH^-)$ values of – 1.34 and – 1.06, respectively. Felmy *et al.* provided an interesting linear correlation in their Figure 7 of the logarithm of the formation constants of $CaOH^+$, $CaCO_3(aq)$ and $Ca(CO_3)_2^{2-}$ with the gas phase binding energies.

[1998GAL]

Gallup conducted solubility experiments with amorphous silica in water from 25 to 250 °C in connection with scale formation in geothermal wells. He used a stirred pressure vessel for these studies and withdrew aliquots for analysis after five days. Solubilities were reported in the units of $mg(SiO_2)\cdot kg^{-1}$ (molar mass of SiO_2 = 60.0843). The values of $\log_{10}K^\circ_{s,4} = \log_{10}m_{Si}$ listed in Table A-65 are in excellent agreement with those from other reliable studies and are therefore retained by this review with the uncertainties assigned by this review.

Table A-65: Reported SiO_2(am) solubility values in water.

t /°C	mg $SiO_2\cdot kg^{-1}$	m_{Si}	$\log_{10}m_{Si}$
25.0	(115 ± 6)	0.001914	– (2.72 ± 0.04)
100.0	(375 ± 6)	0.006243	– (2.20 ± 0.03)
150.0	(620 ± 6)	0.010325	– (1.99 ± 0.03)
200.0	(950 ± 6)	0.015827	– (1.80 ± 0.03)

[1998GUI/LIT]

Guiomar *et al*. developed a global minimisation program to calculate pH, pOH, m_{OH^-}, activity coefficients and speciation in saturated $Ca(OH)_2$ solutions from 10 to 40 °C. Special attention was given to calculations of the dissociation constants of $Ca(OH)_2$(aq) and $CaOH^+$, K_{b1} and K_{b2} , respectively, which are equivalent to K°_2 and K°_1, respectively. Analysis of the K_{b2} data was based solely on the potentiometric data in [1959BAT/BOW]. Guiomar *et al*. used the Debye-Hückel model following the Bates-Guggenheim convention to estimate the activity coefficients. Assuming that $Ca(OH)_2$(aq) is completely dissociated led to K_{b2} = 0.0393, *cf*. 0.071-0.074 reported by Bates and Bower for their model, which assumes the equality of the activity coefficients of chloride and hydroxide ions, and 0.062 obtained by this review using an identical approach to these authors. Based on the solubility product of $Ca(OH)_2$(s) [1934KIL] and Harned-cell data for HCl(sln) [1998LID], Guiomar *et al*. calculated K_{b1} = 0.0342, which

resulted in a corrected K_{b2} value of 0.071. Applying their optimisation routine to the data in [1959BAT/BOW] yielded a preferred value of K_{b1} of 0.0102. This modelling work was difficult to accept, especially considering that basically they quantified deviations in the mean ionic activity coefficient of the ions in solution derived from the simple Debye-Hückel expression in terms of a solution species existing only at very high pH. This concern was magnified by their significantly lower value of K_{b2} = 0.0393 derived even when $Ca(OH)_2(aq)$ was ignored. Perhaps most telling is the lack of any sensitivity analysis (uncertainty) values.

[1998NOR/SUL]

Nordin *et al*. reported on the kinetics of water exchange between hydrated aluminium ions and the bulk aqueous solution from an ^{17}O-NMR study over the temperature range 298-348 K in 2 mol·dm^{-3} NaCl. The observed pseudo-first-order rate constant of exchange at any given temperature was described by $k^t_{ex,obs} = k^t_1 + k^t_2 / c_{H^+}$, where k^t_1 is the rate constant for the exchange of inner-sphere water molecules in $Al(H_2O)_6^{3+}$ with bulk solution and k^t_2 / c_{H^+} is the pH-dependent contribution from $AlOH(H_2O)_5^{2+}$. At 298.15 K, k^t_1 = 1.3 s^{-1} and k^t_2 = 0.072 mol·dm^{-3}·s^{-1}. Hydrolysis constants for $Al(H_2O)_6^{3+}$ were taken from [1993PAL/WES] and [1994WES/PAL]. Deprotonation of $Al(H_2O)_6^{3+}$ enhanced the reactivity of the other inner-sphere oxygen atoms by a factor of ~ 10^4. Presumably this lability promotes the kinetics of formation of higher-order hydrolysed Al(III) species and ensures that equilibrium is reached within the time frame of a potentiometric titration in the determination of $^*K_1$ for example.

[1998PYE/RUD]

Raman spectroscopic measurements of aqueous magnesium perchlorate, chloride and sulfate solutions (3.0, 3.3 and 3.05 mol·kg^{-1}, respectively) revealed a weak polarised band which was assigned to the υ_1-MgO_6 mode of the $Mg(H_2O)_6^{2+}$ cation. Over the range, 25 to 125 °C, this 356 cm^{-1} symmetrical stretching frequency decreased by about 3 cm^{-1} but broadened by 13 cm^{-1} above 100 °C. A depolarised band at 235 cm^{-1} was assigned to the υ_2 mode. These data were interpreted to suggest that the $Mg(H_2O)_6^{2+}$ cation was thermodynamically stable in perchlorate and chloride solutions, *i.e.*, there was no observable indication of the formation of a contact ion pair, although it appears that a solvent-separated ion pair existed in sulfate solutions [1998ROB/STE].

[1998ROB/STE]

This potentiometric study utilised a standard autotitration technique, a cell equipped with a chloride ion-sensitive electrode (ISE-Cl) and a calomel reference electrode. The temperature was controlled at (25.0 ± 0.1) °C. De Robertis and de Stefano relied on a

computer software package BSTAC to process their data, stating that the activity coefficients were estimated using an extended Debye-Hückel expression. The resulting formation constants for $MgCl^+$ and $CaCl^+$ are: $\log_{10} K_1^\circ = (0.49 \pm 0.05)$ and (0.42 ± 0.15), respectively. This investigation of weak ion pairing employed mixed electrolytes with chloride ion concentrations reaching 0.23 $mol \cdot dm^{-3}$ and metal ion concentrations reaching 0.12 $mol \cdot dm^{-3}$ with sufficient $NaNO_3$ to produce an I_c(max) of 0.94. This ubiquitous problem was encountered in virtually all of the studies of Mg^{2+}-Cl^- interactions. There was also a question as to whether the ISE-Cl responds uniquely to the free chloride ion concentration [1997RAR/CLE].

[1998WAN/PIT]

A Pitzer ion interaction model of the thermodynamic properties of $MgCl_2$ solutions (isopiestic and vapour-pressure, freezing-point depressions, enthalpies of dilution, heat capacity and density measurements) embraced a wide range of temperature (250-600 K) to 10 MPa. The global nature of this treatment meant that no emphasis was placed on obtaining accurate infinite dilution thermodynamic data at 298.15 K and indeed such values were not reported. Wang *et al.* provided an excellent critique of papers available at that time on the formation of $MgCl^+$ ion pairs. In brief, mention was made of the ion-pair association constants derived from enthalpies of dilution and mixing of $MgCl_2$ solutions by Gillespie *et al*. [1992GIL/OSC] from 523 to 598 K that were considered by Rard and Clegg [1997RAR/CLE] to be compromised by covariance in their fitting procedures. Wang *et al*. pointed out that interpretation of these data was based on the assumption that the activity coefficient of $MgCl^+$ could be set equal to that of NaCl(sln) whereas the mean activity coefficient of $MgCl_2$(sln) was taken from the isopiestic data of Holmes *et al*. [1978HOL/BAE], which were in turn criticised [1986EMO/VOI], then superseded in [1996HOL/MES]. Solubility measurements [1990SAC/SEY] of talc-quartz mixtures from 573 to 673 K were interpreted by taking the formation constants of $MgCl^+$ ion pairs into account in this mixed-electrolyte medium. Here again for such weak interactions the values derived were highly sensitive to the choice of activity coefficient models. The electrical conductivity data for $MgCl_2$ and $CaCl_2$ solutions obtained from 298 to 873 K [1982FRA/MAR] at pressures up to 400 MPa yielded 2:1 formation constants which are considerably lower than those from other studies with the difference becoming larger with increasing temperature. The model developed in this study could not assign specific ion-pair formation constants in this system largely because of interference of the ion association of HCl(aq), although the comment was made that "The dilution enthalpy measurements of Simonson *et al*.[12] (where reference 12 refers to unpublished data) in 0.02 $mol \cdot kg^{-1}$ HCl solution are essentially consistent with the assumption of complete dissociation of $MgCl_2$, extending to low molalities".

[1999BEN/BEN]

This work concerns the preparation, the characterisation and the determination of the enthalpy of formation of $Mg_3(PO_4)_2$ and β-$Ca_3(PO_4)_2$. Only the results concerning β-$Ca_3(PO_4)_2$ are of relevance to the present review and will be discussed**.** The method of preparation was described and the XRD pattern was that of the β-phase (space group *R*3*c*) with lattice parameters in agreement with literature data. Analyses by atomic absorption (for Ca) and colorimetry (for P) gave a Ca/P mol ratio of 1.504. Infrared spectroscopy revealed parasitic bands due to nitrates and water that the authors attributed without further say to impurities in the KBr used to pelletise the samples.

The solution calorimeter used has been described previously by authors from the same group [1984BEN/OME]. The instrument was operated at (298.15 ± 0.20) K, with 0.350 dm^3 of 9 mass % HNO_3 solution (HNO_3, 35.36H_2O) as the solvent. The results of eight experiments, with sample weights of 10 to 25 mg, were listed, and an average value of – (130.3 ± 1.2) kJ·mol^{-1} for the enthalpy of solution of $Ca_3(PO_4)_2$(cr) was reported. As the cycle used to reach $\Delta_f H^\circ_m$($Ca_3(PO_4)_2$, cr, β, 298.15 K) involved $Ca(NO_3)_2$·4H_2O(cr), and (H_3PO_4, 0.756H_2O), the enthalpies of solution of these species in (HNO_3, 35.36H_2O) were required. These values were taken by the authors from an earlier publication by Ben Cherifa [1988BEN] that was not available to this review. The value for $\Delta_f H^\circ_m$((H_3PO_4, 0.756H_2O), sln, 298.15 K) was taken from Wagman *et al.* [1968WAG/EVA] that reports values identical to those in [1982WAG/EVA] and that for $\Delta_f H^\circ_m$($Ca(NO_3)_2$·4H_2O, cr, 298.15 K) from Green *et al.* [1984GRE/MAL]. The latter value, – (2132.3) kJ·mol^{-1} was essentially identical to that, – (2132.03 ± 1.30) kJ·mol^{-1}, selected in [1987GAR/PAR] and used in this review.

The authors reported $\Delta_f H^\circ_m$($Ca_3(PO_4)_2$, cr, β, 298.15 K) = – 4056 kJ·mol^{-1} without any indication of uncertainty limits. They noted, without further comments, that this result was some 60 kJ·mol^{-1} less negative than literature values. They also observed that a very large discrepancy in the same direction was affecting their result on $Mg_3(PO_4)_2$(cr). The use of the CODATA-compatible [1989COX/WAG] values for the phosphoric acid species (which are 6.792 kJ·mol^{-1} more negative than those given in [1982WAG/EVA]) would result in a significant change in the calculated values, leading to $\Delta_f H^\circ_m$($Ca_3(PO_4)_2$, cr, β, 298.15 K) ≈ – 4070 kJ·mol^{-1}, but this shift is far from sufficient to bring these data close to the consensus of other values. These data are thus given for information only.

[1999BUC/HEF]

This dielectric-relaxation study of NaOH, $NaAl(OH)_4$ and $NaB(OH)_4$ solutions addressed the question of ion pairing, leading Buchner *et al*. to state that because of the larger radii of the $Al(OH)_4^-$ and $B(OH)_4^-$ anions compared to that of OH^-, contact ion pairs should have been readily detected. However, the lack of a detectable solute

relaxation process indicated that the ion-pair lifetimes are similar or even shorter than the water-relaxation time. They concluded that rapid hydrogen exchange controlled the dynamics in these solutions and cited two publications which gave ion-pair formation constants for the Na^+- $B(OH)_4^-$ of 1.66 [1976REA/LAN] and 1.91 [1995POK/SCH]. Other experiments using sodium amalgam electrodes were claimed to have identified weak ion association for both NaOH and $NaAl(OH)_4$ with apparent formation constants values of 0.6 and 2.0 $dm^3 \cdot mol^{-1}$ in 5 $mol \cdot dm^{-3}$ (Me_4NCl).

[1999KON/KON]

A thermodynamic model for the Na_2CO_3-$MgCO_3$-$CaCO_3$-H_2O system was created based on calorimetric data for $\Delta_f H_m^o$ (298.15 K), S_m^o (298.15 K) and $C_{p,m}^o(T)$ functions, and μ^o (298.15 K) values of the solids derived from their solubility measurements and from those in the literature. These thermodynamic quantities were combined with temperature-dependent Pitzer ion interaction parameters. Phase equilibria were obtained from the Gibbs energy minimiser, ChemSage [1990ERI/HAC], whereas the ChemSage optimiser was used to optimise the thermodynamic quantities with respect to experimental data [1995KON/ERI]. Königsberger *et al*. tabulated all the relevant $\Delta_f H_m^o$ (298.15 K) and S_m^o (298.15 K) values, the Pitzer parameters $\beta^{(0)}$, $\beta^{(1)}$ and C^φ, the corresponding first and second temperature derivatives, and the higher-order terms, ${}^s\theta$, Ψ and λ. The $\Delta_f H_m^o$ (298.15 K) and S_m^o (298.15 K) values for 19 magnesium and calcium solid phases were included from CODATA [1987GAR/PAR]. Königsberger *et al*. combined solubility data [1977MCG/HOS], [1996BRO/DRU] with CODATA values for Mg^{2+} and H_2O(l) [1989COX/WAG] to give $\Delta_f G_m^o$ ($Mg(OH)_2$, cr, 298.15 K) = – 831.9 $kJ \cdot mol^{-1}$, $\Delta_f H_m^o$ ($Mg(OH)_2$, cr, 298.15 K) = – 926.7 $kJ \cdot mol^{-1}$, and S_m^o ($Mg(OH)_2$, cr, 298.15 K) = 50.6 $J \cdot K^{-1} \cdot mol^{-1}$. They proposed that the precipitation experiments of Brown *et al*. [1996BRO/DRU] could have generated smaller, less-crystalline brucite particles with higher surface energies. However, Altmaier *et al*. [2003ALT/MET] reported a similar solubility constant to that measured by Brown *et al.* corresponding to $\Delta_f G_m^o$ ($Mg(OH)_2$, cr, 298.15 K) = – 832.1 $kJ \cdot mol^{-1}$ for well-crystallised brucite (0.3-1.0 μm) for which the Gibbs surface energy contribution was known to be negligible [1963HOS].

[1999PAR/KIM]

This paper reports spectrophotometric study in order to determine the second dissociation constant of carbonic acid between 25-175 °C under vapour-saturated pressure. The authors first determined the temperature dependence of the acid dissociation constant for 2-naphthol in borate buffer solutions (based on the data in [1972MES/BAE]). Thereafter, the spectral changes of 2-naphthol, as pH indicator, were used to determine the second dissociation constant of carbonic acid.

From the thermodynamic dissociation constants determined at seven temperatures, the following temperature dependence was deduced:

$$\mathrm{p}K_2^\circ = 4201.48 - 229200\,(T/\mathrm{K})^{-1} - 661.21\ln(T/\mathrm{K}) + 0.64753\,(T/\mathrm{K}) + 13498200\,(T/\mathrm{K})^{-2}$$

To calculate the equilibrium constant from the measured spectrum, it is assumed that the activity coefficient

(i) for neutral chemical species is unity;

(ii) for singly charged ionic species it is equal with the mean activity coefficient, $\gamma_\pm$, which was approximated by the simplified Pitzer's formula; for doubly charged ionic species, the mean activity coefficient, γ_z, was calculated from $\ln \gamma_z = 4 \ln \gamma_\pm$.

$$\ln \gamma_\pm = -A_\phi \left[\frac{\sqrt{I}}{1+1.2\sqrt{I}} + \frac{2}{1.2} \ln\left(1+1.2\sqrt{I}\right) \right]$$

Since background (strong) electrolyte was not used during the measurements, the activity coefficient calculated on the basis of the above is only a rough approximation. In addition, no numerical values of the determined $\mathrm{p}K_2^\circ$ are given, the constants can only be read from Figure 8 of [1999PAR/KIM], which obviously increases the uncertainties.

Several further problems are likely associated with the experimental results of this work as noted in [2013STE/BEN]:

(i) the pH range of the experimental solutions was limited (pH 10.00-10.17 at 25 °C), thus increasing the uncertainties on the derived equilibrium constants;

(ii) the pH values measured by Park *et al.* are on the margin of the optimal pH range of the pH indicator, particularly at high temperatures.

As a conclusion, large uncertainties can be assigned to the $\mathrm{p}K_2^\circ$ values determined in [1999PAR/KIM] above 373.15 K. In addition, the numerical values considerably deviate from those reported in [1984PAT/BUS] or [2013STE/BEN], considered as the two most reliable source. Therefore, the constants reported in [1999PAR/KIM] were not considered in our evaluation.

[1999PAV/NIN]

The osmotic coefficients of mixed electrolyte solutions containing $\{y\mathrm{NaH_2PO_4} + (1-y)\mathrm{Na_2SO_4}\}$ were measured at 298.15 K using the isopiestic method. Two datasets were tabulated using NaCl and $\mathrm{Na_2SO_4}$ reference solutions. The activity coefficients of $\mathrm{NaH_2PO_4}$ and $\mathrm{Na_2SO_4}$ solutions were estimated by Scatchard's neutral-electrolyte

method [1961SCA], and by Pitzer and Kim's treatment of mixed-electrolyte solutions [1974PIT/KIM]. A subset of the second series of measurements was first chosen for treatment with the SIT that limited the ionic strength to ≤ 1.2524 mol·kg^{-1} and yielded: $\varepsilon(Na^+, H_2PO_4^-) = -(0.145 \pm 0.006)$ kg·mol^{-1} (seven measurements). Increasing the range of ionic strength treated to 2.891 mol·kg^{-1} (thirteen measurements including some with a NaCl reference solution) led to: $\varepsilon_1(Na^+, H_2PO_4^-) = (0.129 \pm 0.011)$ kg·mol^{-1} and $\varepsilon_2(Na^+, H_2PO_4^-) = (0.129 \pm 0.023)$ kg·mol^{-1}. A comparison with the bulk of earlier osmotic coefficients of NaH_2PO_4 solutions clearly shows that the results above $I_m \approx 1$ are too low and therefore the results in this paper were not considered further in this review.

[2000BIA/DUJ]

Bianchi *et al*. used the conductance data in [1977PET/TAB] for $MgSO_4$ solutions to test the performance of various conductivity equations (FHFP [1967FUO/HSI], [1969FER]; LW [1987BIA/COR]; MSA [1995TUR/BLU], Leist [1955LEI]; and MSAp) for symmetrical electrolytes. Table A-66 is taken from their Table III. The utility of the Lee-Wheaton equation, which was adopted in the NEA reviews, was established at least in terms of a low standard deviation, σ_Λ, albeit over a relatively narrow concentration range: (0.1-20) × 10^{-3} mol·dm^{-3}. This is a valuable article which provided guidance for the evaluation of conductance results in the current review, although the values of these ion-pair formation constants were not used in the overall data assessment.

Table A-66: Analysis of the conductivity of multi-charged symmetrical electrolytes over the same concentration range (0.13-5.20) × 10^{-3} mol·dm^{-3} at 25 °C.

Equation abbreviations	Equation names	d /nm	Λ_0 /S·cm^2·mol^{-1}	K_1°	σ_Λ /S·cm^2·mol^{-1}
FHFP	Fuoss-Hsia-Fernández-Prini	1.095	132.77	157.2	0.037
LW	Lee-Wheaton	0.815	133.01	173.5	0.029
MSA	mean spherical approximation	0.385	132.44	175	0.68
Leist	Leist	0.585	131.74	148	0.241
MSAp	an abridged MSA	0.195	132.37	162.2	0.078

[2000GUN/ARN]

Gunnarsson and Arnórsson measured the solubility of amorphous silica (commercial grade silica gel pre-washed with water twice) in pure water from 8 to 310 °C (*i.e.*, $SiO_2(s) + 2H_2O(l) \rightleftharpoons Si(OH)_4(aq)$) where the pressure above 100 °C was p_{sat}. A maximum temperature limit of 200 °C was imposed in this review. The reported expression to fit these data was:

$$\log_{10} K_{s,4} = -8.476 - 485.24(T/\mathrm{K})^{-1} + 3.068\log_{10}(T/\mathrm{K}) - (2.268\times10^{-6})(T/\mathrm{K})^2 \quad \text{(A.254)}$$

There was good agreement between the solubility data for amorphous silica from near room temperature to 473 K obtained in this study and those obtained from the literature as illustrated in Figure 2 of this paper.

Table A-67: Solubility measurements of amorphous SiO_2 where the reported ppm refer to SiO_2(aq).

t /°C	Equilibration time /days	Reported solubility /ppm	Approach to equilibrium	Solubility /mol·kg^{-1}
8	399	76	↑	0.00127
8	556	85	↓	0.00141
20	123	103	↑	0.00171
20	122	132	↓	0.00220
50	40	165	↑	0.00275
51	38	179	↑	0.00298
51	38	176	↓ (a)	0.00293
89	4.2	318	↑	0.00529
90	17	329	↑	0.00548
130	15	497	↑	0.00828
150	14	604	↑	0.0101
150	7	607	↑	0.0101
170	7	793	↑	0.0132
200	1	931	↑	0.0155

(a) Equilibrium approached from oversaturation (↓) with solid silica gel present.

Solubility data for crystalline quartz were taken from Rimstidt [1997RIM] and the literature values cited therein and were re-fitted to give the equation:

$$\log_{10} K_{s,4} = -34.188 + 197.47(T/\mathrm{K})^{-1} + 12.245\log_{10}(T/\mathrm{K}) - (5.851\times10^{-6})(T/\mathrm{K})^{2} \quad \text{(A.255)}$$

The thermodynamic properties for $Si(OH)_4$(aq) are consistent with the solubility data for amorphous silica and quartz [1997RIM]: $\Delta_f G^{\circ}_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = -(1309.181)$ kJ·mol^{-1}, $\Delta_f H^{\circ}_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = -(1461.722)$ kJ·mol^{-1} and $S^{\circ}_m(\mathrm{Si(OH)_4, aq, 298.15\ K}) = 178.85$ J·K^{-1}·mol^{-1}.

[2000MEI/GAL]

In the context of nuclear waste disposal, the authors investigated the possibility of incorporating selectively plutonium and caesium into mono-silicate neodymium-fluorapatite $Ca_9Nd(SiO_4)(PO_4)_5F_2$(britholite). For this purpose, a force field for the modelling of fluorapatite was established, based upon the experimental data of crystallographic parameters and elastic constants, and in order to check the validity of the potentials, the heat capacity of fluorapatite was measured from 293 to 1073 K and compared to the calculated values.

The calculated cell parameters of fluorapatite a = 9.3688 Å, c = 6.8694 Å were close to the experimental values, 9.397 and 6.878 Å. The C_p measurements were carried out on a SETARAM DSC 111 differential calorimeter calibrated with a sapphire NBS standard. A step size of 20 K in flowing argon was used for the fluorapatite powders (100 mg) in alumina crucibles. The only details of the three fluorapatite samples given are that they were 'obtained by the sintering route at 1100 °C at atmospheric pressure'. Some water was absorbed on cooling, whose evolution was observed in the DSC measurements. The results of the C_p measurements (reported to have an uncertainty of *ca.* ± 3%) were only presented in two fairly small graphs, which are not completely consistent. For a comparison with data for fluorapatite from other studies we have taken both the experimental and calculated values read from the curves in their Figure 3. As shown in Figure V-86 in Section V.8.2.4.2 of the main text, the calculated values are in quite good agreement with our selected values, whereas the experimental values are consistently about 4 % greater than the calculated values.

As, in general, this review does not utilise data from *ab initio* calculations, and because of the lack of detail in the samples used, and the disagreement of the experimental heat capacities with other consistent measurements, no data from this study were included in our fitting for the selected values.

[2000PAL/BEN]

The available data (equilibrium constants, heat capacity and volume changes) for the base hydrolysis of boric acid (Reaction (A.256)) over the temperature range 273-623 K were critically reviewed as a function of ionic strength up to 1 mol·kg^{-1}(KCl) and re-assessed based on potentiometric results from 323 to 563 K in [1972MES/BAE] and subsequent experimental results. Consideration was also given to the formation of polyborates.

$$B(OH)_3(aq) + OH^- \rightleftharpoons B(OH)_4^- \qquad \text{(A.256)}$$

The following values for Reaction (A.256) at 298.15 K with 2σ uncertainties were recommended:

$$\log_{10}K^\circ\,((\text{A.256}),\,298.15\ \text{K}) = (4.757 \pm 0.003),$$

$$\Delta_r H_m^\circ\,((\text{A.256}),\,298.15\ \text{K}) = -(42.7 \pm 0.1)\ \text{kJ·mol}^{-1},$$

$$\Delta_r S_m^\circ\,((\text{A.256}),\,298.15\ \text{K}) = -(51.1 \pm 0.5)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r C_{p,m}^\circ\,((\text{A.256}),\,298.15\ \text{K}) = (28 \pm 5)\ \text{J·K}^{-1}\text{·mol}^{-1},$$

$$\Delta_r V_m^\circ\,((\text{A.256}),\,298.15\ \text{K}) = -(10.5 \pm 2.5)\ \text{cm}^3\text{·mol}^{-1}.$$

Hydrolysis constants for the polyborates were reported in [1972MES/BAE] at I_m = 1.0 (KCl) for temperatures of 323.15, 373.15 and 473.15 K. The notation used by these authors is defined in the Appendix A entry for [1972MES/BAE]. Four hydrolysis equilibria were identified whereby the last two are alternatives such that three hydrolysis equilibria were quantified at each temperature, *viz*.,

$$2B(OH)_3(aq) + OH^- \rightleftharpoons H_7B_2O_7^- \quad \text{(A.257)}$$

$$3B(OH)_3(aq) + OH^- \rightleftharpoons H_{10}B_3O_{10}^- \quad \text{(A.258)}$$

$$4B(OH)_3(aq) + 2OH^- \rightleftharpoons H_{14}B_4O_{14}^{2-} \quad \text{(A.259)}$$

or

$$5B(OH)_3(aq) + 3OH^- \rightleftharpoons H_{18}B_5O_{18}^{3-} \quad \text{(A.260)}$$

Palmer *et al*. used the ionic-strength dependencies of the hydrolysis constants for phosphoric acid as models to approximate the activity changes incurred in Reactions (A.257) through (A.260). The estimated values of these constants at 298.15 K are given as: $\log_{10}K^\circ_{7,2}$ (A.257) = 4.39, $\log_{10}K^\circ_{10,3}$ (A.258) = 6.66, $\log_{10}K^\circ_{14,4}$ (A.259) = 11.74 and $\log_{10}K^\circ_{18,5}$ (A.260) = 16.85. No new experimental data were provided in this paper and the results of this data re-assessment were mainly intended for water chemistry applications in nuclear power plants.

[2000SHE/BAR]

In this study, the solubility of a high-purity dolomite was measured directly by monitoring dissolution in solutions with varying concentrations of Ca^{2+}, Mg^{2+} and HCO_3^-/CO_3^{2-} designed to determine the approximate solubility product of dolomite at 0.094 MPa at saturated water vapour pressure and 298.15 K. The equilibration lasted 672 days. The dolomite dissolved congruently in Ca^{2+}/Mg^{2+}-HCO_3^-/CO_3^{2-} solutions with initial ion activity products $pIAP_{dolomite}$ greater than 17.5, where $pIAP_{dolomite} = -\log_{10}(Ca^{2+})(Mg^{2+})(CO_3^{2-})^2$, and the values in parentheses are the measured activities. Samples with initial $pIAP_{dolomite} < 17.0$ showed no evidence of either dolomite dissolution or precipitation, and were therefore either at equilibrium or supersaturated. Based on statistical inference by comparison of alkalinity in unseeded (control) and dolomite-seeded solutions, $\log_{10}K^\circ_{s,0}$ (dolomite) = (17.2 ± 0.2) was suggested. A previously proposed kinetic model of successive reactions for dolomite dissolution near equilibrium—a fairly rapid dissolution of the $CaCO_3$ component in equilibrium and a rate-limiting protonation reaction dependent on the activity of the $MgCO_3$ component—appears to fit the experimental data.

The experimental work was done very carefully. The use of gas-permeable and water-impermeable membranes over the dissolution vessels allowed for maintenance of an open system for CO_2, with minimal water loss during the course of the long

equilibration period. Dolomite powder was pre-washed with HCl to eliminate fine particles. The method used in this paper allowed the equilibrium concentrations for dolomite dissolution to be estimated in a reasonable time. The three most concentrated solutions in which dolomite dissolved, showed considerable undersaturation with respect to calcite, magnesian calcite and magnesite. Therefore, it may be concluded that these solutions were not in equilibrium with other carbonate solids and were not determining the dolomite solubility.

[2000TSI/MOL]

Electrolytic conductivities of dilute MSO_4 solutions (where M = Ni^{2+}, Cd^{2+}, Mg^{2+} and Cu^{2+}) were measured in binary mixtures of methanol and water at 293.15 K. Tsierkezos and Molinou employed a platinised dipping electrode type cell, but only made measurements at one frequency, *viz*., 1 kHz. The use of one frequency and a single calibration solution limited the accuracy of these measurements, and the quoted uncertainty in K_1^o is substantially underestimated. The limiting molar conductivities and the ion-pair association constants were re-analysed using the Lee-Wheaton equation of Pethybridge and Taba [1980PET/TAB]. For $MgSO_4$ solutions, Tsierkezos and Molinou reported $K_1^o = (164 \pm 1)$ and $\Lambda_0 = (225.2 \pm 0.9)$ S·cm^2·mol^{-1} at 293.15 K, whereas re-treatment with the SIT-modified Lee-Wheaton equation gave $K_1^o = (253 \pm 32)$ and $\Lambda_0 = 240.2$ S·cm^2·mol^{-1} (2σ). The reason for the large difference between these K_1^o values is not clear. Ironically, treatment of their data for $NiSO_4$ using the same algorithm led to a smaller value of K_1^o [2005GAM/BUG] than reported by these authors.

[2001FEL/CHO]

Silicon-29 NMR and potentiometric experiments were carried out with basic silica solutions (≤ 0.15 mol·dm^{-3}) at 15 ºC. Felmy *et al*. suggested that the stability of the cyclic tetramer $Si_4O_8(OH)_4^{4-}$ proposed by Sjöberg *et al*. [1985SJO/OEH] may have been overestimated because of the lower resolution of the NMR instrument available to Sjöberg *et al*. However, their re-analysis indicated that the measurements [1985SJO/OEH] of the cyclic tetramer were probably correct and the unidentified species in that study were probably substituted trimers, linear tetramers and prismatic hexamers. The Pitzer ion interaction model was used to treat the experimental data to $I_c \leq 3$ and molecular simulations were made with a molecular dynamics model which included interatomic potentials for NaOH + H_2O + SiO_2 solutions. Formation constants for all of these Si(IV) species were tabulated and are probably the most reliable values currently available.

[2001KRA/HEF]

Titrations of $MgCl_2$ solutions with a Cs_2SO_4(sln) titrant were made in a cell with liquid junction and Mg-ISE and Ag|AgCl reference electrodes at (25.000 ± 0.005) °C under a nitrogen atmosphere. The non-Nernstian behaviour of the ISE at low magnesium concentrations (< 0.002 mol·dm^{-3}) was compensated for by assuming a requisite amount of an Mg^{2+} impurity was present although surprisingly it was not detected by ICP-AES analyses. The initial total magnesium concentration was 0.005 to 0.010 mol·dm^{-3}, but the final sulfate concentration was not given and may have exceeded 10 % of the ionic strength by the end of some of these titrations. Table A-68 was taken from Table II in this paper and the results of an SIT treatment are included.

Table A-68: Magnesium sulfate association constants at 25 °C (from an average of 3 to 5 titrations) are given after conversion to molality units.

salt	I_c	$\log_{10}K_{c,1}$	ρ /g·cm^{-3}	I_m (a)	$\log_{10}K_{m,1}$/SIT	$\log_{10}K_{m,1}+8D$
CsCl	0.25	(1.286 ± 0.003)	1.0292	0.253	1.282	2.448
	0.5	(1.017 ± 0.003)	1.0610	0.512	1.007	2.412
	1.0	(0.750 ± 0.001)	1.1242	1.046	0.730	2.374
	2.0	(0.558 ± 0.002)	1.2496	2.191	0.519	2.390
	4.0	(0.416 ± 0.002)	1.4967	4.859	0.332	2.416
	6.0	(0.343 ± 0.003)				
Me_4NCl	1.0	(0.977 ± 0.002)				

(a) Using molecular masses for CsCl and Me_4NCl of 168.358 and 101.535, respectively.

Application of an SIT treatment to the molal association constants in Table A-68 showed that if the lower two values were compromised by high relative sulfate concentrations, a linear regression of the remaining three points resulted in K_1° = (231 ± 7) (2σ) and $\Delta\varepsilon$ = – (0.011 ± 0.002) kg·mol^{-1}. Conversely, if all five points are considered and $\Delta\varepsilon$ = 0 kg·mol^{-1}, then an average value of K_1° = (256 ± 33) (2σ) was calculated, *cf.* (239.9 ± 3.1) proposed by Kratsis *et al*. The possibilities of formation of $MgCl^+$ and $CsSO_4^-$ ion pairs were ignored. From $\Delta\varepsilon$ = – (0.011 ± 0.002) kg·mol^{-1} and $\varepsilon(Mg^{2+}, Cl^-)$ = (0.19 ± 0.02) kg·mol^{-1} (NEA TDB), $\varepsilon(Cs^+, SO_4^{2-})$ = – (0.20 ± 0.02) kg·mol^{-1} which is reasonable but should be verified independently. The value of K_1° = (231 ± 40) was included in the overall data treatment.

[2001ROC/MAN]

This paper reported new data for the Gibbs energy of dolomite at 298.15 K and 0.1 MPa, based on electrochemical cell measurements and on lattice energy calculations.

Details of the electrochemical measurement method are given in earlier publications of the group, [1996CAS/ROC], [1998MCB/ROC]. The cell used was:

$$\mathrm{Cd(Hg)(two\ phase)\ |\ CdCO_3(cr),\ Ca_{0.50}Mg_{0.50}CO_3(cr)\ |\ CaCl_2\ (aq,\ }m_1\mathrm{),\ MgCl_2(aq,\ }m_2\mathrm{)\ |\ Hg_2Cl_2(cr)\ |\ Hg(l)} \quad \text{(A.261)}$$

(although in the paper, $HgCl_2$ was written instead of Hg_2Cl_2 here and elsewhere), where Cd(Hg) represents a saturated cadmium-mercury amalgam, and m_1, m_2, respectively, are the molalities of calcium and magnesium in the aqueous medium. The Mg/Ca ratio ranged between 5.88 and 111.85. Two samples of dolomite of different origin (Selasvan, Norway and Euguy, Spain) were used with no difference in the results. No other details were given on the samples.

The net cell reaction was written as:

$$\mathrm{Cd(Hg) + Ca_{0.50}Mg_{0.50}CO_3(s)} \rightleftharpoons \mathrm{2Hg(l) + CdCO_3(s) + 0.50Ca^{2+}(aq, }m_1\mathrm{) + 0.50Mg^{2+}(aq, }m_2\mathrm{) + 2Cl^-(aq,\ }2m_1 + 2m_2\mathrm{)} \quad \text{(A.262)}$$

Application of the Nernst equation and introduction of the mean activity coefficients for $CaCl_2$(sln) and $MgCl_2$(sln) as mixed electrolytes, calculated with the speciation programme of Plummer *et al.* [1988PLU/PAR], (using the Pitzer model), resulted in twelve standard potential values E°(A.262) that we have averaged to give (0.74246 ± 0.00309) V (2σ, statistical error only), and thus $\Delta_r G^\circ_m$ (A.262) = $-(143.273 \pm 0.596)$ kJ·mol^{-1}.

For their calculation of $\Delta_f G^\circ_m$ ($CaMg(CO_3)_2$, cr) the authors did not give any details of the auxiliary data used, but reported $\Delta_f G^\circ_m$ ($CaMg(CO_3)_2$, cr, 298.15 K) = $-(2147.82 \pm 2.20)$ kJ·mol^{-1}.

Using the equation:

$$\Delta_f G^\circ(\mathrm{CaMg(CO_3)_2,\ cr}) = 2\,[-\Delta_r G^\circ(\text{A.262}) - \Delta_f G^\circ(\mathrm{Cd(Hg),\ two\ phase}) - \Delta_f G^\circ(\mathrm{Hg_2Cl_2,\ cr}) + \Delta_f G^\circ(\mathrm{CdCO_3,\ cr}) + 2\,\Delta_f G^\circ(\mathrm{Cl^-,\ aq}) + 0.5\,\Delta_f G^\circ(\mathrm{Ca^{2+},\ aq}) + 0.5\,\Delta_f G^\circ(\mathrm{Mg^{2+},\ aq})\,] \quad \text{(A.263)}$$

we have re-calculated their results with TDB data (including our revised value for $\Delta_f G^\circ_m(Mg^{2+}, 298.15\ K)$), or where these are lacking (*viz.* for Cd(Hg), two phase) and $CdCO_3$(cr)), values from Wagman *et al.* [1982WAG/EVA], to give $\Delta_f G^\circ_m$ ($CaMg(CO_3)_2$, cr) = $-(2144.82 \pm 2.90)$ kJ·mol^{-1}.

For an unknown reason, this value is *ca.* 20 kJ·mol^{-1} more positive than that derived from the three consistent calorimetric studies discussed in Chapter V.6 of the main text and is not considered in our final selection of $\Delta_f H^\circ_m$ ($CaMg(CO_3)_2$, 298.15 K) and $\Delta_f G^\circ_m$ ($CaMg(CO_3)_2$, 298.15 K). We note the authors also refer to their unpublished value for $\Delta_f G^\circ_m$ ($MgCO_3$, cr) by a similar electrochemical study, which is also *ca.* 20 kJ·mol^{-1} more positive than the well-established value assessed in this review.

The lattice energy calculations reported by the authors were by an extension of a calculational technique for pure carbonate minerals developed in [1998MAN/ROC]

and led to the value $\Delta_f G^o_m(CaMg(CO_3)_2, cr, 298.15\ K) = -2148.90\ kJ\cdot mol^{-1}$, which the authors noted was in "extraordinary" agreement with their experimental value. This theoretical value is quoted here for information only, without further discussion.

[2001STE]

The solubilities of a range of minerals and Fe-Ti oxide solid solutions of fixed composition were derived from 0 to 350 °C at saturated water vapour pressure using a thermodynamic database for the end-member minerals [1995ROB/HEM]. The solubility of amorphous silica is reported to be $1.936 \times 10^{-3}\ mol\cdot kg^{-1}$ at 25 °C noting that consistent values are generally reported for this metastable phase. The reported thermodynamic properties for $Si(OH)_4(aq)$ are compatible with those derived by Rimstidt: $\Delta_f G^o_m(Si(OH)_4, aq, 298.15\ K) = -1309.257\ kJ\cdot mol^{-1}$, $\Delta_f H^o_m(Si(OH)_4, aq, 298.15\ K) = -1458.861\ kJ\cdot mol^{-1}$, $S^o_m(Si(OH)_4, aq, 298.15\ K) = 188.70\ J\cdot K^{-1}\cdot mol^{-1}$, $C^o_{p,m}(Si(OH)_4, aq, 298.15\ K) = 62.8\ J\cdot K^{-1}\cdot mol^{-1}$ and $V^o_m(Si(OH)_4, aq, 298.15\ K) = 52.3\ cm^3\cdot mol^{-1}$.

[2002HUM/BER]

From a critical review of the available experimental data, Hummel *et al*. selected the following thermodynamic quantities for the dissolution of gibbsite and consequently those for the formation of gibbsite and Al^{3+}:

$$\alpha\text{-}Al(OH)_3(cr) + 3H^+ \rightleftharpoons Al^{3+} + 3H_2O(l) \qquad (A.264)$$

$$\log_{10} {}^*K^o_{s,0}((A.264), 298.15\ K) = 7.756,$$

$$\Delta_r G^o_m((A.264), 298.15\ K) = -44.271\ kJ\cdot mol^{-1},$$

$$\Delta_r H^o_m((A.264), 298.15\ K) = -102.784\ kJ\cdot mol^{-1},$$

$$S^o_m((A.264), 298.15\ K) = -196.648\ J\cdot K^{-1}\cdot mol^{-1},$$

$$\Delta_r C^o_{p,m}((A.264), 298.15\ K) = -1.255\ J\cdot K^{-1}\cdot mol^{-1},$$

and

$$\Delta_f G^o_m(Al^{3+}, aq, 298.15\ K) = -487.740\ kJ\cdot mol^{-1},$$

$$\Delta_f H^o_m(Al^{3+}, aq, 298.15\ K) = -538.424\ kJ\cdot mol^{-1},$$

$$S^o_m(Al^{3+}, aq, 298.15\ K) = -337.71\ J\cdot K^{-1}\cdot mol^{-1},$$

$$C^o_{p,m}(Al^{3+}, aq, 298.15\ K) = -133.07\ J\cdot K^{-1}\cdot mol^{-1},$$

and

$$\Delta_f G^o_m(Al(OH)_3, cr, 298.15\ K) = -487.740\ kJ\cdot mol^{-1},$$

$$\Delta_f H^\circ_m(\mathrm{Al(OH)_3}, \mathrm{cr}, 298.15\ \mathrm{K}) = -538.424\ \mathrm{kJ\cdot mol^{-1}},$$

$$S^\circ_m(\mathrm{Al(OH)_3}, \mathrm{cr}, 298.15\ \mathrm{K}) = -337.71\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}},$$

$$C^\circ_{p,m}(\mathrm{Al(OH)_3}, \mathrm{cr}, 298.15\ \mathrm{K}) = 92.72\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

The values for gibbsite with the exception of $C^\circ_{p,m}(\mathrm{Al(OH)_3}, \mathrm{cr}, 298.15\ \mathrm{K})$ are obviously in error as they correspond to those given for Al^{3+}.

[2002GOL/KIS]

In this review of ionisation equilibria involving buffers, Goldberg *et al*. selected values at 298.15 K for the phosphate system:

$$\mathrm{H_3PO_4(aq)} \rightleftharpoons \mathrm{H^+ + H_2PO_4^-}$$

$\log_{10}K^\circ_1 = -2.148$, $\Delta_r G^\circ_m = 12.261\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r H^\circ_m = -8.0\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r C^\circ_{p,m} = -141\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$,

$$\mathrm{H_2PO_4^-} \rightleftharpoons \mathrm{H^+ + HPO_4^{2-}}$$

$\log_{10}K^\circ_2 = -7.198$, $\Delta_r G^\circ_m = 41.087\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r H^\circ_m = 3.6\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r C^\circ_{p,m} = -230\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$,

$$\mathrm{HPO_4^{2-}} \rightleftharpoons \mathrm{H^+ + PO_4^{3-}}$$

$\log_{10}K^\circ_3 = -12.35$, $\Delta_r G^\circ_m = 70.49\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r H^\circ_m = 16.0\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_r C^\circ_{p,m} = -242\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$.

The selected $\log_{10}K^\circ_1$ and $\log_{10}K^\circ_2$ values were taken largely from [1951BAT] and therefore are in close agreement with those selected by the current review. The selected $\log_{10}K^\circ_3$ value was based on [1909ABB/BRA], [1929BJE/UNM], [1954BEU/RIE], [1958ELL/SHA], [1961VAN/QUI], [1965HAN/CHR], [1965PAP/TOR], [1966CHR/IZA], [1991DAN/DER], and was accepted by Grenthe *et al*. [1992GRE/FUG]. The enthalpies of reaction selected in the current review are significantly different from those proposed by Goldberg *et al*.

[2002KON/NAN]

The solubility of $Ca(OH)_2$, aragonite and calcite in pure water and in NaOH solution was determined, and the crystallisation behaviour of aragonite was investigated at 25, 50 and 75 °C. The solubility of $CaCO_3$ in a NaOH solution reached a maximum value around 0.5-1.0 $\mathrm{mol\cdot dm^{-3}}$ NaOH. The subsequent decrease in the solubility was due to the conversion of $CaCO_3$ to $Ca(OH)_2$. Pure NaOH solution (0-3 $\mathrm{mol\cdot dm^{-3}}$) containing various amounts of Na_2CO_3 (0.00-0.04 $\mathrm{mol\cdot dm^{-3}}$) was poured into a flask. An excess of solid $CaCO_3$ was added to the solution which was stirred slowly. Samples were taken with a syringe equipped with a 0.45 mm membrane filter over a period of 1-6 hours, and

the filtrate was neutralised with HCl. The Ca^{2+} concentration in the neutralised filtrate was measured by ICP-AES.

Only graphical representation of the solubility data is provided, which strongly limits the accuracy of data. At 75 °C the measured solubility of calcite exceeded the solubility of aragonite under identical conditions, indicating a relatively high experimental error. Only the solubility data reported for 25 and 50 °C were accepted in this review, with relatively high uncertainties.

[2002TOM/BES]

This conductance study of the formation of $MgSO_4$(aq) ion pairs was carried out at 10 concentrations from 0.00025 to about 0.2 mol·dm^{-3} at 5 ° intervals from 5 to 35 °C. Multiple KCl solutions were used to determine the cell constant following two approaches: a "low concentration chemical model" designated as "lcCM" (limited to concentrations < 0.0032 mol·dm^{-3}), and a mean spherical approximation "MSA" for the moderate to high concentration data. This review only considers those data in the former range because accurate values of K_1° are desired. Concentrations were tabulated in terms of the measured quantity, $\tilde{m}$ with the units, mol·kg(solution)$^{-1}$, and used in the following equation, where the solution density ρ and water density ρ_0 have the units g·cm^{-3}.

$$\rho = \rho_0 + 0.1169\tilde{m} + 0.00863\tilde{m}^2$$

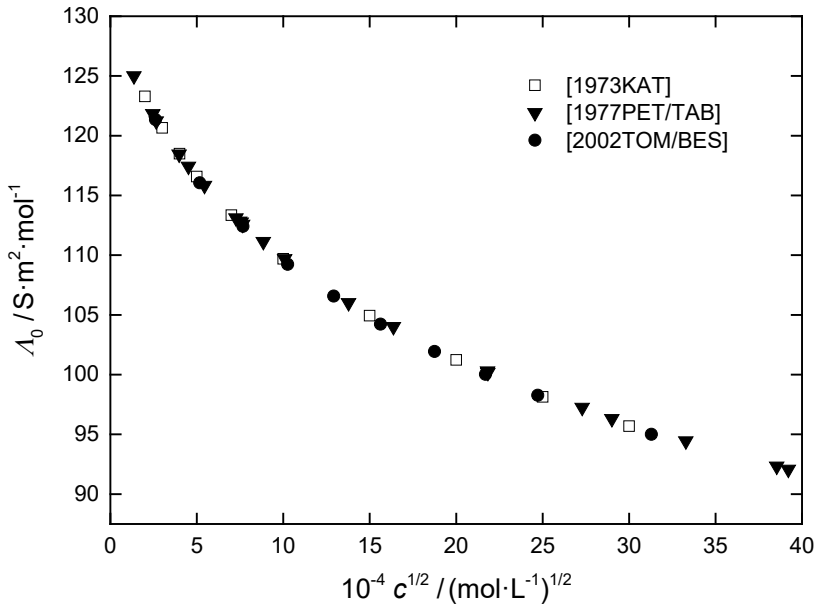
The density of water was taken from [1998HAR]. The ion-pair formation constants and the molar conductivities reported by Tomšič *et al*. are given in Table A-69 as well as those re-calculated with the Lee-Wheaton equation modified to conform to the SIT.

Table A-69: Summary of the original results for the formation of $MgSO_4$(aq) ion pairs taken from Table II (model n = 2, which represents the sum of all postulated 2:2 ion pairs) and those re-calculated according to the SIT modified Lee-Wheaton equation.

t /°C	K_1° /reported	Molar Λ_0 reported /S·m^2·mol^{-1}	K_1° /SIT (2σ)	Molar Λ_0 SIT /S·m^2·mol^{-1}
5	144.6	79.68	(180.8 ± 15.7)	80.60
10	146.9	91.96	(185.5 ± 16.5)	93.06
15	148.7	104.89	(190.9 ± 17.7)	106.23
20	152.0	118.46	(196.6 ± 17.7)	120.02
25	157.2	132.70	(204.7 ± 19.0)	134.53
30	165.5	147.66	(214.3 ± 19.5)	149.71
35	177.6	163.37	(225.6 ± 16.5)	165.67

The SIT results are in good agreement with those from the conductance study carried out almost thirty years earlier [1973KAT] and are accepted by this review. A value of $\Delta_r H_m^\circ = 6.63$ kJ·mol^{-1} was given at 25 °C. A plot of the molar conductance values *versus* concentration at 25 °C is shown in Figure A-28.

Figure A-28: The temperature dependence of the molar conductances of $MgSO_4$(sln) at 25 °C from this study (solid symbols) and those from [1973KAT] (open symbols).



[2003ALT/MET]

The solubility equilibria involving brucite and magnesium oxychloride were investigated in reference particularly to the values in the HMW database [1984HAR/MOL]. Altmaier *et al*. identified a discrepancy in the data for $Mg(OH)_2$(cr) in two databases commonly used in geochemical modelling [1984HAR/MOL] and [1992JOH/OEL]. Their experiments used commercial brucite with particle sizes from 0.1 to 1.0 μm based on SEM measurements. Solubility measurements were made in an argon-filled glove box at (22 ± 2) °C after several days to fourteen months. The solids were characterised by XRD before and after the equilibration period. pH measurements were made with a combination Ross glass electrode calibrated against commercial buffers at $I_m < 0.1$. They employed the Pitzer ion interaction model, to estimate activity coefficients. At higher ionic strengths, the electrode was calibrated empirically using NaCl and $MgCl_2$ solutions [1990KNA/WOL], [1991FEL/RAI], [1996FAN/NEC]. The

solubility decrease observed at high $MgCl_2$ molalities from 2.67 to 5.15 mol·kg^{-1} was postulated to result from transformation of brucite to an anhydrous magnesium oxychloride as confirmed by XRD analyses. The authors discussed reasons for finding that $m_{Mg^{2+}} > 2m_{OH^-}$ in NaCl solutions and could only assume that magnesium impurities were responsible. The mean values of $\log_{10}{}^{*}K^{\circ}_{s,0}$ from experiments in which no NaCl was added and the ten cases where NaCl was present, are given as (17.16 ± 0.10) and (17.01 ± 0.10) (2σ), respectively, giving an average value of (17.1 ± 0.2) at 22 °C. Justification was given for the selection of thermodynamic data yielding $\log_{10}{}^{*}K^{\circ}_{s,0}$ = 16.30 at 25 °C calculated from $\Delta_f G^{\circ}_m$, $\Delta_f H^{\circ}_m$ and S°_m at 298.15 K selected for brucite by Helgeson *et al*. [1978HEL/DEL] who used high-pressure, high-temperature data for the equilibrium between $Mg(OH)_2$(cr) and MgO(cr) in combination with calorimetric data for MgO(cr) and auxiliary data for the standard molal volume, heat capacity and entropy of brucite. The discrepant solubility constant was thought to be primarily due to the choice of the standard entropy of brucite (S°_m = 63.14 J·K^{-1}·mol^{-1}) at 298.15 K [1968ROB/WAL] which apparently stemmed from a collection of entropy data [1961KEL/KIN] which were included with an uncertainty of ± 0.21 J·K^{-1}·mol^{-1} calculated by Giauque and Archibald [1937GIA/ARC] from experimental heat capacities for $Mg(OH)_2$(cr) from 20 to 300 K. Königsberger *et al*. [1999KON/KON] combined earlier solubility data [1977MCG/HOS], [1996BRO/DRU] with CODATA values for Mg^{2+} and H_2O(l) [1989COX/WAG] to give $\Delta_f G^{\circ}_m$ ($Mg(OH)_2$, cr, 298.15 K) = – 831.9 kJ·mol^{-1}, $\Delta_f H^{\circ}_m$ ($Mg(OH)_2$, cr, 298.15 K) = – 926.7 kJ·mol^{-1}, and S°_m ($Mg(OH)_2$, cr, 298.15 K) = 50.6 J·K^{-1}·mol^{-1}. Königsberger *et al*. also proposed that the precipitation experiments of Brown *et al*. [1996BRO/DRU] could have generated less-crystalline, small brucite particles with higher Gibbs surface energy and solubility. However, Altmaier *et al*. refuted this suggestion because they found a similar solubility constant to that reported by Brown *et al*. corresponding to $\Delta_f G^{\circ}_m$ ($Mg(OH)_2$, cr, 298.15 K) = – 832.1 kJ·mol^{-1} for well-crystallised brucite (0.3-1.0 μm) for which the Gibbs energy contribution from the surface energy is known to be negligible [1963HOS]. Application of the SIT treatment in this review resulted in the following values for the NaCl-free and NaCl-containing solutions: $\log_{10}{}^{*}K^{\circ}_{s,0}$ = (17.12 ± 0.04) and (17.06 ± 0.05); $\Delta\varepsilon$ = – (0.092 ± 0.036) and – (0.081 ± 0.010) kg·mol^{-1}, respectively.

The second system studied involved the equilibrium:

$$Mg_2(OH)_3Cl\cdot 4H_2O(cr) + 3H^+ \rightleftharpoons 2Mg^{2+} + Cl^- + 7H_2O(l) \qquad (A.265)$$

According to the authors, this hydrated oxychlorido phase is formed at room temperature only at magnesium molalities > 1.7 mol·kg^{-1} (see Figure 5 in this paper). The activity coefficients at these high ionic strengths were estimated from the Pitzer ion interaction model. The mean $\log_{10}{}^{*}K^{\circ}_{s,0}$ (A.265) value from six independent experiments was given as (26.0 ± 0.2) at 22 °C.

[2003DUS/CHA]

This is a very detailed study of the structures of the monoclinic phases of Na_2CO_3(cr) stable at 298.15 K and the very closely related structure of the phase to which it transforms at 170 K, using single crystal XRD. Only the basic structures are of interest to this review, and Dusek *et al.* give the following parameters:

- Phase stable at 298.15 K: an incommensurate monoclinic structure, space group $C2/m$, with parameters $a = (8.920 \pm 0.007)$ Å, $b = (5.245 \pm 0.005)$ Å, $c = (6.050 \pm 0.005)$ Å, $\gamma = (101.35 \pm 0.08)$ ° at 295 K.
- Phase formed at 170 K: a commensurate monoclinic structure, space group $C2/m$, with parameters: $a = (8.898 \pm 0.007)$ Å, $b = (5.237 \pm 0.005)$ Å, $c = (5.996 \pm 0.005)$ Å and $\gamma = (101.87 \pm 0.08)$ ° at 110 K.

[2003FUL/HEA]

X-ray absorption fine structure (XAFS) spectroscopy was used to probe the effects of concentration on the first-shell structure of Ca^{2+} in aqueous solution under ambient conditions using a bending magnet beamline. The XAFS spectrum for 6 mol·kg^{-1} $CaCl_2$ yielded no evidence for the formation of significant numbers of Ca^{2+}-Cl^- contact ion pairs, confirmed by comparison with the data for a dilute (0.2 mol·kg^{-1}) reference solution of the perchlorate salt. A mean coordination number of (7.2 ± 1.2) was determined for 6 mol·kg^{-1} $CaCl_2$ with an average Ca-O distance of (0.2437 ± 0.0010) nm, consistent with earlier XAFS measurements on dilute Ca^{2+} solutions. Comparison of the pre-edge and near-edge (XANES) spectra with those from various references, including those for crystalline hydrates, provided further confirmation of the lack of change in the Ca^{2+} first-shell structure and symmetry. The combined results suggested the formation of Ca^{2+}-H_2O-Cl^- solvent-shared ion pairs. An XAFS spectrum of a 9.2 mol·kg^{-1} $CaCl_2 \cdot 6H_2O$ melt did show the presence of contact ion pairs.

[2003PAL/BEN]

Bayerite was synthesised by dissolving aluminium powder in an NaOH solution (6 mol·dm^{-3}) to produce a 4:1 mole ratio of OH:Al [2000PHA/HUM], [2002PHA/HUM]. The precipitate was thoroughly washed with deionised water. Characterisation was made by XRD, SEM, Raman spectroscopy and BET before and after the experiments. Solubility measurements were made at 30, 63 and 89.8 °C in $NaOH/NaNO_3$ solutions with ionic strengths ranging from 0.1 to 5 mol·kg^{-1}. A parallel set of experiments was also conducted with gibbsite. Application of the SIT treatment to the results for bayerite led to the constants given in Table A-70 and for gibbsite the corresponding results are listed in Table A-71. The average of both $\varepsilon(Na^+, Al(OH)_4^-)$ values at 30 °C where the datasets are largest is – (0.017 ± 0.006) kg·mol^{-1}. Alternative

values of $\log_{10}K^{\circ}_{s,4}$ are given in these tables based on this average value of $\varepsilon(Na^+, Al(OH)_4^-)$. These revised values are given in the fifth column of Table A-70 and Table A-71.

Table A-70: Application of the SIT treatment to the bayerite solubility results (2σ uncertainties; actual experimental uncertainties estimated at ± 0.08 in $\log_{10}K^{\circ}_{s,4}$).

t /°C	$(I_m)_{max}$	$-\log_{10}K^{\circ}_{s,4}$ /SIT	$\varepsilon(Na^+, Al(OH)_4^-)$ /kg·mol^{-1}	$-\log_{10}K^{\circ}_{s,4}$ [(a)]
30.0	5.000	(0.848 ± 0.013)	– (0.020 ± 0.004)	(0.839 ± 0.008)
63.0	4.999	(0.515 ± 0.023)	– (0.021 ± 0.007)	(0.503 ± 0.012)
89.8	5.007	(0.158 ± 0.020)	(0.003 ± 0.006)	(0.217 ± 0.045)

(a) Assuming a mean value: $\varepsilon(Na^+, Al(OH)_4^-) = -(0.017 \pm 0.006)$ kg·mol^{-1}.

A weighted linear fit of $\log_{10}K^{\circ}_{s,4}$ (Table A-70, column 5) *versus* $(T/K)^{-1}$ yields:

$$\log_{10}K^{\circ}_{s,4}(Al(OH)_3, baye) = (2.651 \pm 0.153) - (1058.4 \pm 47.8)(T/K)^{-1}\ (1\sigma) \qquad (A.266)$$

Equation (A.266) for the dissolution of bayerite gives values of $\log_{10}K^{\circ}_{s,4} = -(0.90 \pm 0.44)$ and $\Delta_r H^{\circ}_m = (20.3 \pm 1.8)$ kJ·mol^{-1} (2σ) for bayerite at 25 °C.

Table A-71: Application of the SIT treatment to the gibbsite solubility results (2σ uncertainties; actual experimental uncertainties estimated at ± 0.08 in $\log_{10}K^{\circ}_{s,4}$)

t /°C	$(I_m)_{max}$	$-\log_{10}K^{\circ}_{s,4}$ /SIT	$\varepsilon(Na^+, Al(OH)_4^-)$ /kg·mol^{-1}	$-\log_{10}K^{\circ}_{s,4}$ [(a)]
30.0	4.904	(1.061 ± 0.019)	– (0.014 ± 0.006)	(1.071 ± 0.009)
63.0	4.900	(0.657 ± 0.043)	– 0.014 (fixed)	(0.663 ± 0.038)
89.8	5.007	(0.302 ± 0.014)	– (0.015 ± 0.004)	(0.308 ± 0.006)

(a) As in Table A-70.

A weighted linear fit of $\log_{10}K^{\circ}_{s,4}$ (Table A-71, column 5) *versus* $(T/K)^{-1}$ yields:

$$\log_{10}K^{\circ}_{s,4}(Al(OH)_3, gibb) = (3.561 \pm 0.071) - (1404.5 \pm 24.2)(T/K)^{-1}\ (1\sigma) \qquad (A.267)$$

Equation (A.267) gives values of $\log_{10}K^{\circ}_{s,4} = -(1.15 \pm 0.22)$ and $\Delta_r H^{\circ}_m = (26.90 \pm 0.93)$ kJ·mol^{-1} (2σ) for gibbsite at 25 °C.

[2003PAR/MOR]

Partanen *et al.* found that the osmotic coefficient data for KH_2PO_4 solutions in [1945STO2] could be fitted adequately with an extended Debye-Hückel equation to a molality of 1.00 mol·kg^{-1} and then applied the Pitzer ion interaction treatment to fit the remaining data at 298.15 K to 1.83 mol·kg^{-1}, which is the saturation limit of this salt at this temperature. They also measured the solubility of the salt at 293.15, 298.15, 303.15 and 308.15 K in pure water: 1.645, 1.830, 2.022 and 2.231 mol·kg^{-1}, respectively. The

activity coefficients derived by these authors differed only marginally from those derived by Stokes [1945STO2], and Hamer and Wu [1972HAM/WU].

[2003RAP/ZUL]

Stoichiometric formation constants for H_3PO_4(aq) at 25 °C in NaCl, KCl and $NaClO_4$ solutions were taken from the literature and treated by a modified Bromley equation. This approach yielded values of the corresponding acid dissociation constants: $\log_{10}K_1^\circ = -(2.15 \pm 0.03)$ [1956BAE], [1957THA], [1963GAL/STU], [1971PET], [1974CIA], [1981VAC/SAB], [1989HER/FER], [1991DAN/DER], [1991MAT], [1993GLA/MAJ]; $\log_{10}K_2^\circ = -(7.17 \pm 0.04)$ [1958MAD], [1961BRU/NEL], [1963GAL/STU], [1971PET], [1981VAC/SAB], [1989BIN/BYR], [1989HER/FER], [1991DAN/DER], [1993GLA/MAJ]; and $\log_{10}K_3^\circ = -(12.31 \pm 0.03)$ [1971PET], [1981VAC/SAB], [1989HER/FER], [1991DAN/DER], [1993GLA/MAJ] that are in good agreement with those obtained using the SIT.

[2003RUD/IRM]

Raman spectra were recorded of aqueous $Mg(ClO_4)_2$, $MgCl_2$, $(NH_4)_2SO_4$ and $MgSO_4$ solutions to low concentrations (6×10^{-5} mol·kg^{-1}) at temperatures up to 200 °C. In the case of $MgSO_4$(sln), asymmetry in the $\nu_1 - SO_4^{2-}$ stretching frequency at *ca*. 980 cm^{-1} increased with concentration and was considered to be due to formation of an $MgOSO_3$ contact ion pair. Rudolph *et al*. detected another $\nu_1 - SO_4^{2-}$ mode at 982.2 cm^{-1} which they assigned to a solvent-shared ion pair. Moreover, at high Mg^{2+}: SO_4^{2-} concentration ratios, yet another $\nu_1 - SO_4^{2-}$ mode was observed at 1005 cm^{-1}, which was tentatively assigned to an $Mg_2SO_4^{2+}$ ion triplet [1985ARC/WOO]. Reference was made to a dielectric spectroscopic relaxation study [2004BUC/CHE], which demonstrated that three types of ion-pair interactions coexisted in solution: contact, solvent-separated and double solvent-separated ion pairs. Rudolph *et al.* confirmed that the former is dominant at high temperatures. Moreover, if any association occurs in $MgCl_2$ solutions then only weak outer-sphere ion pairs are formed so that as established in other studies, the observed spectral features represent those of $Mg(H_2O)_6^{2+}$ only in chloride solutions. In this work, the usual assumption was made that the $\nu_1 - SO_4^{2-}$ band represented only contributions from the "free" SO_4^{2-} anion at *ca*. 980 cm^{-1} and the contact ion pair at *ca*. 993 cm^{-1}. The authors derived equations for the formation constant of an $MgSO_4$(aq) contact ion pair and presented speciation diagrams for the various associated species in solution as a function of concentration; the actual ion-pair association constant was not given at any conditions.

[2004BRU/MOR]

Brunet *et al.* describe a new adiabatic calorimeter for measuring low-temperature heat capacities, and give results of measurements for lazulite. They also measured values for α-alumina as a check on their apparatus. The data for lazulite are of no immediate interest to the review, but the authors used this value to estimate the entropies of hydroxylapatite and β-$Ca_3(PO_4)_2$.

However, the sample of α-alumina used was not artificial sapphire as normally used for heat capacity calibrations, but a sample of Merck high-purity alumina which was annealed at 1473 K overnight before use; no analyses of the material are given. The detailed results of *ca.* 200 measurements from 34.25-299.2 K are tabulated. The heat capacities are very close to those recommended for the standard reference material, but the derived entropy of $S^{o}_{m}(298.15\ \mathrm{K})$ of 51.2 $J{\cdot}K^{-1}{\cdot}mol^{-1}$ (in the text) or 51.37 $J{\cdot}K^{-1}{\cdot}mol^{-1}$ (in their Figure 3a), is slightly higher than the recommended value (50.92 ± 0.10) $J{\cdot}K^{-1}{\cdot}mol^{-1}$, presumably due to the smaller particle size of the material used or a slight difference in the much longer extrapolation to 0 K required in this work. Brunet *et al.* used a simple cubic extrapolation from 34.25 to 0 K.

[2004BUC/CHE]

Dielectric-relaxation spectroscopy was used to infer the coexistence of contact (CIP), solvent-shared (SIP) and double-solvent-separated (2SIP) $MgSO_4$(aq) ion pairs over a wide range of concentration (0.017 to 2.24 $mol{\cdot}dm^{-3}$) and frequency (0.2-89 GHz) at 25 °C. Their data were interpreted in terms of the classic Eigen mechanism:

$$\underset{\text{free ions}}{Mg^{2+} + SO_4^{2-}} \rightleftharpoons \underset{\text{2SIP}}{Mg(H_2O)(H_2O)SO_4(aq)} \rightleftharpoons \underset{\text{SIP}}{Mg(H_2O)SO_4(aq)} \rightleftharpoons \underset{\text{CIP}}{MgSO_4(aq)}$$

The extent of ion pairing increased to a maximum of ~ 40 % (~ 14 % 2SIP, ~ 18 % SIP, ~ 8 % CIP) at $c \approx 0.1$ $mol{\cdot}dm^{-3}$ then decreased monotonically to ~ 8 % (~ 1 % SIP, ~ 7 % CIP) at the approach to saturation, reflecting the effect of increasing ionic strength. Buchner *et al*. provided a plot of the stepwise and overall association constants as a function of concentration, but only reported a value of (167 ± 11) for the latter at infinite dilution. This value resulted from an extended Debye-Hückel fit, although the value of the extended term was not quantified. They stressed that Raman measurements usually led to a smaller estimate of the association constant, the exact mathematical reason for this trend having been described in [2003RUD/IRM]. Rard and Miller [1981RAR/MIL], among others [1997MAL/ZAM] were sceptical about the existence of ion triplets.

[2004SCH/GRO]

Schnurre *et al*. present a comprehensive thermodynamic assessment of SiO_2(l), SiO(am) and of the solubility of oxygen in Si(cr) and Si(l), together with experimental data on the decomposition and condensation of SiO(am), mainly from XRD. Equations for the thermodynamic data for the Gibbs energies of the crystalline SiO_2 phases are given, taken from the otherwise unpublished assessment by Seifert [1999SEI]; these data imply that tridymite is not a stable polymorph at ambient pressure. From these, and their assessed data for SiO(am) and the solubility of oxygen in silica, a consistent thermodynamic description of the entire Si–O system is constructed, covering the complete composition range. The authors also present a useful table of the crystallographic data for most of the known crystalline forms of silica. A total of 248 references are quoted.

[2005ARA/CHA]

This paper gives a good summary of the phase transitions in Na_2CO_3(cr). From 1000 K down, four distinct phases were identified:

- T = 754-1000 K: space group $P6_3/mmc$; 10 refinements were reported by Swainson *et al.* [1995SWA/DOV] for different temperatures; unit-cell parameters: a = (5.209 ± 0.002) Å and c varies from 6.454 to 6.676 Å.
- T = 605-754 K: space group $C2/m$; about 20 refinements were performed by Swainson *et al.* [1995SWA/DOV] for different temperatures; unit-cell parameters: a = (9.00 ± 0.02), b = (5.23 ± 0.02) Å, the c parameter varies from 6.21 to 6.44 Å and the monoclinic angle β varies from 90 to 99.33°.
- T = 170-605 K (incommensurate monoclinic): space group $C2/m$; the study was performed at 295 K [2003DUS/CHA]; unit-cell parameters a = (8.920 ± 0.007), b = (5.245 ± 0.005), c = (6.050 ± 0.005) Å, β = (101.35 ± 0.08) °.
- T < 170 K (lock-in monoclinic): space group $P2_1/n$ [equal to superspace group $C2/m$] the structural data were obtained at 110 K by Dusek *et al.* [2003DUS/CHA]; unit-cell parameters a = (8.898 ± 0.007), b = (5.237 ± 0.005), c = (5.996 ± 0.005) Å, β = (101.87 ± 0.08).

[2005CRU/CAN]

The authors made an *ab initio* study of the hexagonal form of hydroxyl- and fluorapatite and report values for the lattice parameters, lattice enthalpies and $C^{\circ}_{p,\mathrm{m}}$(298.15 K). The only values they report for the lattice energy are those from the reactions from the gaseous ions Ca^{2+}, PO_4^{3-} and X^- where X = OH or F. The substantial uncertainties in the enthalpies of formation of these ions, combined with the considerable uncertainties

in the *ab initio* calculations themselves, mean that these are not of any immediate interest to this review. Their calculated $C^{o}_{p,m}$ (298.15 K) for the OH^- and F^- apatites are (347 ± 34) and (323 ± 13) $J \cdot K^{-1} \cdot mol^{-1}$ respectively, compared to the our selected values of (381.15 ± 3.00) and (373.0 ± 3.5) $J \cdot K^{-1} \cdot mol^{-1}$.

We cannot, at the moment, understand the lattice parameters reported by the authors, since their computed values for the *a* and *b* parameters are not equal, implying that their calculated structure was monoclinic rather than hexagonal. But in general, their calculated lattice parameters differ appreciably from the well-established crystallographic data.

This paper does not provide any useful additional information for our purposes and is documented here only for completeness.

[2005CRU/MIN]

The standard enthalpies of formation of the apatites of general formula $Ca_{10}(PO_4)_6(X)_2$(cr) with X = OH, Cl, and Br were determined from reaction-solution calorimetry in (HCl, 553.41H_2O) and (HNO_3, 35.35 H_2O) at a stated temperature of 298.15 K. As the other compounds are not in the charter of this review, only the data pertaining to $Ca_{10}(PO_4)_6(OH)_2$ (cr) will be discussed here. The sample preparation was described in detail and the analyses by several techniques did not reveal presence of extraneous phases, nor that of carbon-containing species. The Dewar type calorimeter assembly used was described earlier by the same group, but a number of details were given in the present publication. Reagents were of analytical grade. Unfortunately, the calorimetric results were not reported individually and the uncertainties quoted for the experimental data was said to "correspond to twice the standard deviation of the mean of a minimum of five determinations".

From the dissolutions in HCl, the authors derive the enthalpy of formation of $Ca_{10}(PO_4)_6(OH)_2$ (cr) from Reaction (A.268) (their reaction 8):

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 20(HCl,\ 553.41H_2O)(sln) + 18H_2O(l) \rightleftharpoons 10CaCl_2 \cdot 2H_2O(cr) + 6(H_3PO_4,\ 0.77H_2O) \quad \text{(A.268)}$$

for which they report $\Delta_r H_m$ (A.268) = (237.1 ± 10.6) $kJ \cdot mol^{-1}$.

However, this overall reaction is derived from other reactions whose enthalpies are reported by the authors, namely Reactions (A.269), (A.270) and (A.271) below (the authors' reactions 4, 5, and 6, respectively):

Cruz *et al.* write for these reactions:

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 135.63(HCl,\ 553.41H_2O)(sln) \rightleftharpoons \text{Solution A} \quad \text{(A.269)}$$

$$10CaCl_2 \cdot 2H_2O(cr) + 115.63(HCl,\ 648.94\ H_2O)(sln) \rightleftharpoons \text{Solution B} \quad \text{(A.270)}$$

and

$$6(H_3PO_4, 0.77\ H_2O) + \text{Solution B} \rightleftharpoons \text{Solution A} \qquad \text{(A.271)}$$

However, as written, Reaction (A.271) is unbalanced since our calculation indicates that the composition of the H_3PO_4 solution should in fact be (H_3PO_4, 0.677 H_2O). As, unfortunately, this review was not able to ascertain the origin of this discrepancy, we have resorted to evaluate $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K)$ from Reaction (A.268) as written by the authors, and increasing the uncertainty of the resultant value.

Using the CODATA compatible values in Table A-72 this review re-calculates $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(13482.0 \pm 19.0)\ kJ\cdot mol^{-1}$ and thus for the smaller formula used in this review $\Delta_f H^\circ_m(Ca_5(PO_4)_3OH, cr, 298.15\ K) = -(6741.0 \pm 15.0)\ kJ\cdot mol^{-1}$, where the uncertainty has been increased substantially to allow for the unbalanced reaction scheme noted above. From this scheme involving dissolutions in HCl solution, the authors reported $\Delta_f H^\circ_m(Ca_{10}(PO_4)_6(OH)_2, cr, 298.15\ K) = -(13430 \pm 81)\ kJ\cdot mol^{-1}$.

Table A-72: Auxiliary data used by this review in the re-calculation of the enthalpy of formation of ($Ca_5(PO_4)_3OH$, cr, 298.15 K).

Species	$\Delta_f H^\circ_m$ (298.15 K) /kJ·mol^{-1}	Reference
HCl (aq)	– (167.080 ± 0.100)	[1989COX/WAG]
HCl, 553.41 H_2O	– (166.598 ± 0.100)	Interpolated from [1982WAG/EVA]
H_2O(l)	– (285.830 ± 0.040)	[1989COX/WAG]
H_2O in {HCl, 553.41H_2O}	– (285.8346 ± 0.040)	Interpolated from [1982WAG/EVA]
$CaCl_2\cdot 2H_2O$(cr)	– (1404.0 ± 1.8)	[1987GAR/PAR]
{H_3PO_4, 0.77 H_2O }	– (1280.302 ± 1.000)	(a)

(a) From the interpolated value of – 1273.51 kJ·mol^{-1} from [1982WAG/EVA], corrected to be consistent with the CODATA value of $\Delta_f H^\circ_m(H_3PO_4, 100\ H_2O, 298.15\ K) = -(1296.2 \pm 1.0)\ kJ\cdot mol^{-1}$.

For their dissolutions in HNO_3, the authors derived their enthalpy of formation from a reaction scheme analogous to that used for HCl, making use of Reaction (A.272) (their reaction (15)):

$$Ca_{10}(PO_4)_6(OH)_2(cr) + 20(HNO_3, 35.35H_2O)(sln) + 38H_2O(l) \rightleftharpoons$$
$$10Ca(NO_3)_2\cdot 4H_2O(cr) + 6(H_3PO_4, 0.77\ H_2O) \qquad \text{(A.272)}$$

for which they reported $\Delta_r H_m$ (A.272) = – (568.6 ± 6.0) kJ·mol^{-1}.

However, in this case, we could be even less certain than in the dissolutions in HCl about the stoichiometries in the cycle involved and, therefore we do not consider this cycle further, merely noting that the authors reported a value of

$\Delta_f H_m^o$ ($Ca_{10}(PO_4)_6(OH)_2$, cr, 298.15 K) = – (13398 ± 11) kJ·mol^{-1} from it, in limited agreement with their own value of – (13430 ± 81) kJ·mol^{-1} from the dissolutions in HCl.

[2005KUM/PAR]

Kumzerov *et al.* measured the heat capacity of two samples of chrysotile of unstated origin from 3 to 65 K by a pulsed quasi-adiabatic method. However, the results are only presented on a small graph with non-linear axes from which it is impossible to derive any precise data. These results were therefore not included further in this review.

[2005MEN/WOO]

A platinum-rhodium flow-through conductance apparatus was used to determine the first association constants of Ca^{2+} with Cl^- and CH_3COO^-. The $CaCl_2$ molality was varied from 10^{-4} to 0.090 mol·kg^{-1}. Corrections for Ca^{2+} hydrolysis were applied based on solubility results taken from [1991SEE/SEY]. Conductivities were treated with the TBBK model [1995TUR/BLU] and activity coefficients were estimated using the mean spherical approximation. Implicit in their data treatment was the extra-thermodynamic assumption that $\lambda_{Cl^-} = \lambda_{CaCl^+} = \lambda_{CaOH^+}$ which is an unusual assumption. Presented in Table A-73 are also the re-calculated results from [1982MAJ/STU] using Pitzer's model with interaction parameters taken from [1991PIT], whereas the dataset from [1978JOH/PYT] was re-calculated using an extended Debye-Hückel equation.

Table A-73: Selected ion-pair association constants for $CaCl^+$ calculated in this article.

T /K	p /MPa	$\log_{10} K_1^o$	T /K	p /MPa	$\log_{10} K_1^o$
[2005MEN/WOO]			[1982MAJ/STU]		
348.0	8.45	0.85	288.15	0.1	0.72
398.7	9.37	1.08	298.15	0.1	0.82
473.7	9.38	1.85	318.15	0.1	0.81
522.5	9.40	2.06	338.15	0.1	0.94
			358.15	0.1	1.05
			[1978JOH/PYT]		
			298.15	0.1	0.60

Méndez de Leo and Wood argued that their association constants were in good agreement with those from [1982MAJ/STU] and [1978JOH/PYT]. However, whereas [1978JOH/PYT] contained detailed experimental data, such as concentrations and conductivities required to account for the activity coefficients and extrapolation to infinite dilution, [1982MAJ/STU] contained only values for the association constants. Thus, it is assumed that the required data were obtained *via* private communication. Values for the association constants in [1982MAJ/STU] differ significantly from the

corrected values given in Table A-73. There is a typographical error in their Table 5: reference 41 is actually reference 36.

[2005PAR/COV]

Partanen and Covington applied their Hückel activity coefficient model [2005PAR/COV2] to Reaction (A.273) based on the Harned cell data from 273.15 to 333.15 K in [1943BAT/ACR] and on data from [1933NIM], [1945BAT/ACR], [1958GRZ], [1974MES/BAE], [1989HER/FER].

$$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-} \quad \text{(A.273)}$$

The general SIT-based fitting Equation (A.274) was applied by this review to the tabulated $\log_{10}K_{m,2}$ values, restricted to the temperature range 273.15 to 333.15 K in both media. The parameters obtained from various fitting strategies are given in Table A-74, while Table A-75 gives the thermodynamic values.

$$\log_{10}K_2 = a + b\,(T/\mathrm{K})^{-1} + c(T/\mathrm{K}) - (d + e\log_{10}I_m)I_m \quad \text{(A.274)}$$

Table A-74: Parameters obtained from weighted fits according to Equation (A.274) for the two supporting electrolytes.

a	b	c	d	e	$(I_m)_{max}$
		NaCl			
4.218	– 1794.5	– 0.018105	0.0151	0.1008	1.0
3.721	– 1713.2	– 0.017351	0.0148	0.0958	2.0
– 7.198			0.0180	0.0920	2.0 [(a)]
		KCl			
5.097	– 1953.3	– 0.019366	0.2304	0.0806	2.0
5.094	– 1945.7	– 0.019344	0.2313	0.1016	1.0
– 7.199			0.2313	0.0858	2.0 [(a)]
– 7.197			0.2334	0.081	1.0 [(a)]

(a) Only data at 298.15 K were treated.

Table A-75: Results of application of the SIT and comparison with values reported in the literature at 298.15 K (uncertainties 2σ) for the two supporting electrolytes.

$-\log_{10}K_2^\circ$ SIT	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$(I_m)_{max}$	$-\log_{10}K_2^\circ$	Reference
NaCl					
7.199	(0.015 ± 0.004)	(0.101 ± 0.018)	1.0	7.199	This study [(a)]
7.199	(0.015 ± 0.005)	(0.096 ± 0.015)	2.0		This study [(a)]
(7.198 ± 0.002)	(0.018 ± 0.003)	(0.092 ± 0.007)	2.0		This study [(a), (b)]
(7.212 ± 0.013)	(0.039 ± 0.014)	(0.02 ± 0.03)			Current NEA
				(7.198 ± 0.002)	[1943BAT/ACR]
	(0.030 ± 0.013)	– (0.001 ± 0.025)	1.0		[1965ROB/STO] [(c)]
7.198	–(0.056 ± 0.008)	(0.102 ± 0.012)	6	7.198	[1989HER/FER]

(Continued on next page)

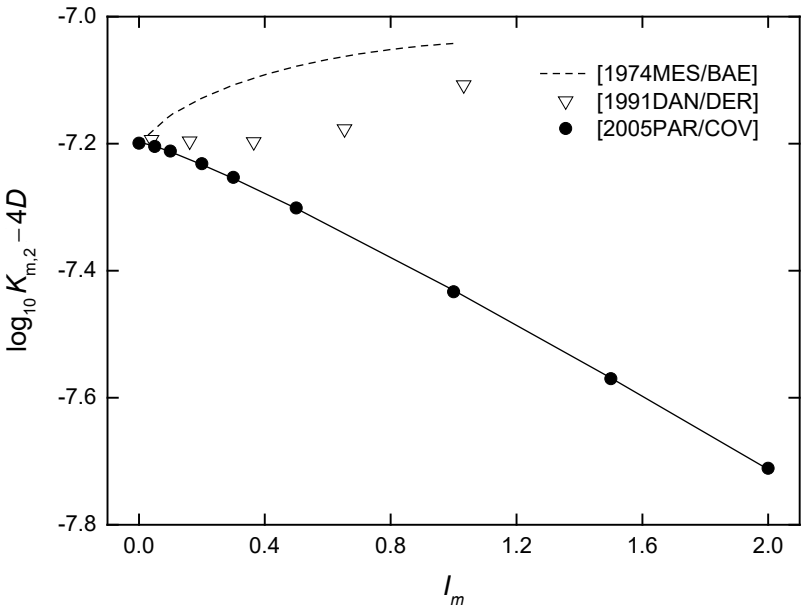
Table A-75 (continued)

$-\log_{10}K_2^\circ$ SIT	$\Delta\varepsilon_1$ /kg·mol^{-1}	$\Delta\varepsilon_2$ /kg·mol^{-1}	$(I_m)_{max}$	$-\log_{10}K_2^\circ$	Reference
KCl					
7.198	(0.230 ± 0.004)	(0.081 ± 0.012)	2.0	7.199	This study [(a)]
7.199	(0.231 ± 0.002)	(0.102 ± 0.012)	1.0		This study [(a)]
(7.199 ± 0.001)	(0.234 ± 0.001)	(0.103 ± 0.006)	1.0		This study [(a), (b)]
(7.197 ±0.002)	(0.231 ± 0.004)	(0.081 ± 0.011)	2.0		This study [(a), (b)]
	(0.111 ± 0.012)	– (0.001 ± 0.017)	1.0		[1965ROB/STO] [(c)]
(7.212 ± 0.013)	(0.179 ± 0.012)	(0.002 ± 0.016)			Current NEA

(a) Only the data in Table 3 [2005PAR/COV] from 273.15 to 333.15 K were treated.
(b) Only data at 298.15 K were treated.
(c) Only from isopiestic data.

Figure A-29 gives a comparison of the fits obtained from the SIT and from equations reported in this paper and by Mesmer and Baes [1974MES/BAE] for KCl solutions.

Figure A-29: A comparison of fits obtained from application of the SIT and from equations reported in [2005PAR/COV] and [1974MES/BAE] for KCl solutions at 298.15 K.



[2005PAR/COV2]

This is a companion paper to [2005PAR/COV]. The second molal dissociation constant ($K_{m,2}$) of phosphoric acid was calculated in buffer solutions containing NaH_2PO_4, Na_2HPO_4 and NaCl, and in solutions containing mixed potassium and sodium salts. Previous data [1943BAT/ACR] and [1945BAT/ACR] were treated based on single-ion activity coefficient Hückel equations covering the range from 0 to 60 °C up to ionic strengths of *ca*. 0.5 mol·kg^{-1}. Partanen and Covington tabulated the experimental $\log_{10}K_{m,2}$ *versus* (T /K) and I_m data from [1943BAT/ACR] and [1945BAT/ACR] which were unavailable in these original publications. This allowed the SIT treatment to be applied to the original data (see the two relevant Appendix A entries).

[2005WAH/MAH]

Wahab *et al.* summarised the results of studies involving a variety of experimental approaches (Raman and IR spectroscopies, NMR, XRD, conductivity, vapour-pressure, hygroscopicity or isopiestic, and ultrasonic-absorption measurements) to identify ion pairing in $Mg(NO_3)_2$ and $Mg(CH_3COO)_2$ solutions. This summary emphasised the disparity between the findings of various researchers in not only observing whether weak ion association even exists in these solutions, but when it was identified, also in claiming different modes of association (*e.g.*, contact, solvent-separated ion pairs), and mono- *versus* bi-dentate modes of interaction. These discussions apply to other 2:1 Mg^{2+}:X^- interactions, such as Cl^- ion pairing with Mg^{2+}. Raman results at 28 °C conducted in pure water and in concentrated ($\leq$ 5.410 mol·kg^{-1} $Mg(CH_3COO)_2$) solutions, were interpreted as demonstrating the presence of contact ion pairs including mono- and bi-dentate complexes. In similar experiments with $Mg(NO_3)_2$ ($\leq$ 4.883 mol·kg^{-1}) solutions, only solvent-separated ion pairs were postulated to have formed. Results of molecular dynamics simulations supported the Raman evidence for much stronger association of CH_3COO^- compared to NO_3^-.

[2005WOL]

The purpose of this paper was to point out discrepancies in the thermodynamic properties of key species in the U.S. Yucca Mountain Project (YMP) thermodynamic database. These differences are reflected in the standard Gibbs energies of formation at 298.15 K, while acknowledging that baseline differences of 0.05 to 0.20 kJ·mol^{-1} are comparable to the minimal uncertainties and are therefore usually not a problem. However, differences of $\approx$ 1 kJ·mol^{-1} exist for some species important to the current review, such as Mg^{2+} and Ca^{2+} with significantly higher differences for Al^{3+}, $H_2PO_4^-$ and HPO_4^{2-} (see Table A-76). Similar discrepancies exist between the $\Delta_f G_m^o$ values for all but Ca^{2+} in CODATA89 [1989COX/WAG] and SUPCRT92 [1992JOH/OEL] which was the code underpinning the YMP database.

Table A-76: Comparison of the standard Gibbs energies of formation of selected aqueous species at 298.15 K, CODATA89 [1989COX/WAG] *versus* NIST90 [1982WAG/EVA]; [1989WAG/EVA] taken from Table 3 in this paper.

	$\Delta_f G_m^\circ$ /kJ·mol^{-1}		
Species	CODATA89	NIST90	Discrepancy
Al^{3+}	– 491.5070	– 485.0000	– 6.5070
Ca^{2+}	– 552.8062	– 553.5800	0.7738
Mg^{2+}	– 455.3751	– 454.8000	– 0.5751
$H_2PO_4^-$	– 1137.1524	– 1130.2800	– 6.8724
HPO_4^{2-}	– 1095.9855	– 1089.1500	– 6.8355

Wolery stated that SUPCRT92-CODATA89 discrepancies in the data for $H_2PO_4^-$ and HPO_4^{2-} could be traced to a standard-state problem, and likely led to some inconsistencies in the $\log_{10}K$ values for aqueous phosphate complexes and solids in both the NEA TDB and YMP thermodynamic databases. He stated that consequently, "the NEA should at least be made cognisant of the problem so that they can issue appropriate corrections and keep the problem from propagating into future work". Moreover, the recommendation is made that for some common ions such as Mg^{2+}, the uncertainties could possibly be reduced by incorporating solubility results in the data assessment. Altmaier *et al*. [2003ALT/MET] also addressed the discrepancy in the Mg^{2+} data as discussed in the Appendix A entry. Subsequently, Rard and Wolery [2007RAR/WOL] argued that discrepancies in the data for $H_2PO_4^-$ and HPO_4^{2-} are more likely due to the use by the NBS of an incorrect $\Delta_f H_m^\circ$ value for P_2O_5(cr).

[2006AKI/ROH]

This is an extension of an earlier study of $MgSO_4$ solutions [2004BUC/CHE] involving the principal authors over similar prevailing conditions but extending the temperature from 25 °C to the range 5 to 65 °C. Activity coefficients were treated as previously described [2004BUC/CHE] with a second virial coefficient in $I^{3/2}$ being invoked for the overall formation constant at 65 °C. At 25 °C, a value of $K_1^\circ = (162 \pm 12)$ was tabulated whereas the temperature dependence was described by the quantities at 25 °C: $K_1^\circ = (166 \pm 30)$, $\Delta_r H_m^\circ = (13.3 \pm 0.2)$ kJ·mol^{-1} and $\Delta_r C_{p,m}^\circ = (614 \pm 20)$ J·K^{-1}·mol^{-1}. However, these values appear to be affected adversely by over-fitting. Their fit included conductance results from [2005BES/BAB], but other precise calorimetric data available in the literature were not taken into account. A simple weighted regression of the values at the three lowest temperatures (the value at 65 °C is simply too large) gave a similar enthalpy change but with a larger statistical uncertainty, $\Delta_r H_m^\circ = (13.0 \pm 4.0)$ kJ·mol^{-1} (2σ). Akilan *et al*. commented that earlier dielectric-relaxation spectroscopic studies, *e.g.*, [1992BAR/HET] and [1992BAR/HET2], were limited to one concentration and therefore did not observe the formation of three 2:2 ion pair types [2004BUC/CHE]. Further information is provided, namely a summary of the thermodynamic quantities for

ion association with the stepwise formation of each ion pair type. The values for $\Delta_r G_m^\circ$ and K_1° in Table A-77 were calculated from the enthalpy and entropy change values provided in their Table 3.

Table A-77: Thermodynamic quantities for ion association of $MgSO_4$(sln) with the formation of double-solvent-separated ion pairs, 2SIP, solvent-shared ion pairs, SIP, and contact ion pairs, CIP, as well as an ion triplet, Tl ≡ $Mg_2SO_4^{2+}$.

Quantity	2SIP	SIP	CIP	Tl
$\Delta_r H_m^\circ$ /kJ·mol^{-1}	– 42	55	19	11
$\Delta_r S_m^\circ$ /J·K^{-1}·mol^{-1}	– 170	210	70	30
$\Delta_r C_{p,m}^\circ$ /J·K^{-1}·mol^{-1}	– 3000	3000	0 (a)	0 (a)
$\Delta_r G_m^\circ$ /kJ·mol^{-1}	– 8.7	– 7.6	– 1.9	– 2.06
K_1°	0.030	21.6	2.15	0.44

(a) For reasons not given, in these cases a two-parameter fit was applied excluding the 5 °C data.

This study provided definitive proof for the existence of distinct ion-pair types in $MgSO_4$ solutions and quantified their stabilities. It opens the door to the interpretation of traditional physical chemical methods which rely on fitting criteria to identify and quantify the existence of species in solution, albeit usually in terms of an overall interaction or as one specific species whose stoichiometry is a matter of conjecture.

[2006CHO/FEL]

Silicon-29 NMR data were collected at 99.304 MHz and (15.0 ± 0.2) °C. NMR spectral simulations were computed by numerical integration of the equations of motion of the ^{29}Si spin-density matrix and then Fourier transformed to obtain the spectra. Evidence was presented for the existence of nine new aqueous silicate oligomers based on these ^{29}Si NMR homonuclear correlation experiments of ^{29}Si-enriched samples. No thermodynamic data were given.

[2006HUA/PAP]

The conductivity measurements given in this article extended from 15 to 250 °C for $MgSO_4$-H_2SO_4 aqueous solutions, but also included independent data for each solute. However, in the case of $MgSO_4$(sln), the lowest concentration investigated was 0.01 mol·kg^{-1} (the other concentrations being 0.20, 0.30 and 0.40 mol·kg^{-1}), which is already too high for this 2:2 electrolyte to extract an accurate ion-pair formation constant by the Lee-Wheaton equation. Huang and Papangelakis noted the effect of ion pairing on the conductivity of $MgSO_4$ solutions in the absence of acid, but did not quantify the extent

of ion pairing. In the mixed $MgSO_4$-H_2SO_4 solutions, the concentrations of $MgSO_4$(aq) were considered to be insignificant and there was no discernible ion pairing between Mg^{2+} and HSO_4^-.

[2006OGO/KIS]

The enthalpy of formation of chrysotile (theoretical formula: $Mg_3Si_2O_5(OH)_4$(cr)) was obtained from high-temperature melt solution calorimetry, using a natural sample of chrysotile-asbestos (Eastern Sayan Mountains, Russia) analysed as $(Mg_{2.96}Ca_{0.04})(Al_{0.05}Si_{1.94}Fe^{3+}_{0.02})O_{5.03}(OH)_{3.91}$ and a synthetic sample (prepared hydrothermally according to a process in the literature) analysed as $Mg_{2.96}Si_{2.02}O_5(OH)_4 \cdot 0.66H_2O$. These tubular crystals were also characterised by optical and electron microscopic techniques, as well as by differential thermal and thermogravimetric analyses.

The thermochemical experiments (Calvet microcalorimeter, Setaram, France), were performed by the "drop" method into a melt of $2PbO \cdot B_2O_3$ at 973 K, yielding the global heat effect $\Delta_r H_m$ as per Equation (A.275) (namely, $\Delta_r H_m$ (A.275)) that includes the heat content and the enthalpy of solution:

$$\Delta_r H_m = \Delta_{sol} H^o_m (973\ K) + (H^o_m (973\ K) - H^o_m (298.15\ K)) \qquad (A.275)$$

Samples of 6 to17 (± 0.002) mg of chrysotile were used.

For the natural (six measurements) and synthetic (five measurements), respectively, average values (95 % confidence) for $\Delta_r H_m$ (A.275) of (423.7 ± 16.4) and (389.0 ± 5.8) kJ·mol^{-1} were reported without details (possibly due to a difference in rounding, the latter value was given as (389.2 ± 5.8) kJ·mol^{-1}).

The enthalpies of formation at 298.15 K of the chrysotile samples from the constituent oxides $\Delta_r H_m$ as per Equation (A.276) (namely, $\Delta_r H_m$ (A.276)) were obtained by the authors from the heat contents and enthalpies of solution of the constituent oxides, taken from earlier data either from their group (some of them estimated) [1976KIS], [1984KIS/OGO], [2003OGO/MEL] or from others [1976NAV/COO] and [1995ROB/HEM]. However, this review cannot reproduce some of their listed values. Then:

$$\Delta_r H_m = \sum_i \nu_i \left(\Delta_{sol} H_m (973\ K) + (H^o_m (973\ K) - H^o_m (298.15\ K)) \right) - \Delta_r H_m \text{ (A.275)} \quad (A.276)$$

where ν_i are the stoichiometric coefficients for the constituting binary oxides.

Using these values, the standard enthalpies of formation of the chrysotile samples $\Delta_f H^o_m$ (chrysotile, 298.15 K), were then obtained by the authors as:

$$\Delta_f H^o_m \text{ (chrysotile, 298.15 K)} = \Delta_r H_m \text{ (A.276)} + \sum \nu_i (\Delta_f H^o_m \text{ (binary oxides, 298.15 K)});$$

$\Delta_f H^o_m$ ((Mg$_{2.96}$Ca$_{0.04}$)(Al$_{0.05}$Si$_{1.94}$Fe$^{3+}_{0.02}$)O$_{5.03}$(OH)$_{3.91}$, cr, 298.15 K) = – (4338 ± 17) kJ·mol^{-1}

and

$\Delta_f H^o_m$ (Mg$_{2.96}$ Si$_{2.02}$ O$_5$ (OH)$_4$·0.66H$_2$O, cr, 298.15 K)) = – (4473.7 ± 9.3) kJ·mol^{-1}.

However, in a later publication by the same authors [2007OGO/KIS] concerning the system (Mg,Ni)$_3$Si$_2$O$_5$(OH)$_4$(cr), that is beyond the scope of the present review, they list $\Delta_f H^o_m$ (Mg$_{2.96}$Si$_{2.02}$O$_5$(OH)$_4$·0.66H$_2$O, cr, 298.15 K) = – (4479.3 ± 9.9) kJ·mol^{-1}, based on the same experimental results as those in [2006OGO/KIS], but these are only in marginal agreement.

Using a procedure that is not entirely clear to this review, the authors reported for the stoichiometric composition $\Delta_f H^o_m$ (Mg$_3$Si$_2$O$_5$(OH)$_4$, cr, 298.15 K) = – (4353 ± 17) kJ·mol^{-1} from the data on the natural sample and $\Delta_f H^o_m$ (Mg$_3$Si$_2$O$_5$(OH)$_4$, cr, 298.15 K) = – (4291 ± 26) kJ·mol^{-1} from the data on the synthetic sample.

The value of the reported standard enthalpy of formation of the natural sample is, according to the authors, some 60 kJ·mol^{-1} more negative. This greater stability, attested in the DTA and TGA measurements by a 100 K higher temperature of decomposition, was ascribed by the authors to structural differences. The enthalpy of formation for their natural sample is in reasonable agreement with our selected value $\Delta_f H^o_m$ (Mg$_3$Si$_2$O$_5$(OH)$_4$, cr, 298.15 K) = – (4364.0 ± 3.4) kJ·mol^{-1} obtained by this review from the solution calorimetry results of King *et al.* [1967KIN/BAR] and of Bennington [1956BEN], also on natural samples. However, in view of the numerous difficulties we have encountered in trying to analyse the results of Ogorodova *et al.* (which as noted above also include some estimated data), the results of this paper were not included in our assessment.

[2006REY/BER]

The density data for the system NaOH-NaAl(OH)$_4$-H$_2$O from six studies [1955RUS/EDW], [1965BAR/BAR], [1988HOV/HEP], [1988SAN/MIC], [2001SIP/STA] and [2005KON/BEV], were analysed. Other data were rejected generally because of a lack of reported accuracy. Moreover, of the six datasets analysed, only [1965BAR/BAR], [1988SAN/MIC] and [2005KON/BEV] gave acceptable correlation coefficients. These three studies embraced the temperature range 25 to 90 °C which covered the range of stability of gibbsite and metastable bayerite. Reynolds and Bernards proposed the following equation (their Equation 2):

$$\rho_{calc} = at + b(MF_{NaOH}) + c(MF_{NaAl(OH)_4}) + d(MF_{H_2O}) \quad \text{(A.277)}$$

where ρ is the density in g·cm^{-3}, t is the temperature in °C, (MF_x) is the mass fraction of species X in solution. The parameters were as follows: $a = -6.5 \times 10^{-4}$, $b = 2.1068$ (NaOH), $c = 1.7769$ ($NaAl(OH)_4$), $d = 1.0134$ (H_2O). A re-analysis was made by this review with the exclusion of the data in [1965BAR/BAR], because only a portion of their data were given, using the same form as Equation (A.277). The combined data, 168 values, gave deviations of < 1 % in $100(\rho_{expr.} - \rho_{calc})/\rho_{expr.}$, with a slightly better fit: $a = -(5.837 \pm 0.072) \times 10^{-4}$, $b = (2.1169 \pm 0.0037)$ (NaOH), $c = (1.7406 \pm 0.0023)$ ($NaAl(OH)_4$), $d = (1.0112 \pm 0.0004)$ (H_2O) (1σ).

[2007BEN/PAL]

Bénézeth *et al*. measured the solubility of synthetic dawsonite ($NaAlCO_3(OH)_2$) according to Reaction (A.278) from 323.15 to 473.15 K in 1.0 mol·kg^{-1} NaCl solutions from under- and over-saturation. Relevant to the current review are the results of eleven additional bayerite (β-$Al(OH)_3$) solubility experiments conducted at 100 °C in basic solutions in the absence and presence of 0.1 mol·kg^{-1} HCO_3^- ($I_m = 1.1$(NaCl)). Their mean value of $\log_{10}{}^{*}K_{s,4}(Al(OH)_3, \text{baye}, 373.15\ K) = -(11.8 \pm 0.30)$ at 373.15 K is compatible with values at lower temperatures accepted in this review.

$$NaAlCO_3(OH)_2(cr) + 2H_2O(l) \rightleftharpoons Al(OH)_4^- + HCO_3^- + Na^+ + H^+ \quad (A.278)$$

[2007CHR/DIC]

Christov *et al*. developed a Pitzer ion interaction model of the system, H-Al-Na-K-Cl-H_2O, to high molalities from 0 to 100 °C. They incorporated osmotic coefficients, equilibrium constant and solubility data. These were supplemented by new potential-difference measurements from 0 to 45 °C with ionic strengths ranging from 0.1 to 3 mol·kg^{-1}. Gibbsite and boehmite solubility data were used to test their model. The calculated $\log_{10}K^{\circ}_{sp}$ value at 25 °C for $AlCl_3 \cdot 6H_2O$(cr) was 7.196, *cf*. 6.970 [2001CHR]. They also predicted a value for $\log_{10}K^{\circ}_{s,0}(Al(OH)_3, \text{gibb}, 298.15\ K) = 7.612$ in agreement with 7.646 [1992PAL/WES] which also used higher-order electrostatic mixing terms in the Pitzer model.

[2007MIL/HUA]

Stoichiometric constants for the dissociation of carbonic acid (K_1^* and K_2^*) in NaCl solutions were determined by potentiometric measurements as a function of molality (0-6 mol·kg^{-1}) and temperature (0-50 °C). The results were fitted by the equation:

$$pK_i^* - pK_i = A_i + B_i(T/K)^{-1} + C_i \ln(T/K)$$

The values of thermodynamic dissociation constants (pK_i , i = 1 or 2) in pure water are taken from the literature and the empirical parameters A_i, B_i and C_i are functions of molality:

$A_1 = 35.2911m^{0.5} + 0.8491m - 0.32m^{1.5} + 0.055m^2$, $B_1 = -1583.09m^{0.5}$, $C_1 = -5.4366m^{0.5}$

$A_2 = 38.2746m^{0.5} + 1.6057m - 0.647m^{1.5} + 0.113m^2$, $B_2 = -1738.16m^{0.5}$, $C_2 = -6.0346m^{0.5}$

The results were combined with previous measurements to derive equations that are valid from 0 to 250 °C and 0 to 5 $mol \cdot kg^{-1}$. This large dataset was used to determine the Pitzer parameters ($\beta^{(0)}$, $\beta^{(1)}$ and C^{ϕ}) for the interactions of Na^+ with HCO_3^- and CO_3^{2-} from 0 to 250 °C.

[2007NAN/SHI]

This paper reports the formation and characterisation (by XRD, FTIR and SEM) of crystalline metastable vaterite and stable calcite, formed by reacting 0.1 molar aqueous solutions of $CaCl_2$ and Na_2CO_3 at 303 K, with rapid stirring. The samples taken after 10 s and 10 min contained respectively 11.4 and 7.3 % vaterite, but those taken after 3.3 hours contained only calcite. However, no chemical analyses of the samples used are given.

The heat capacity of a small sample (0.75 g) of the calcite thus formed was studied in an automatic adiabatic calorimeter. Previous measurements on standard reference Al_2O_3(cr) agreed with the recommended values to within 0.2 %. A considerable number of measurements were made on calcite between 80 and 390 K, but the individual results are not quoted, just presented in a rather small graph and a fitted seven-term polynomial as a function of T, from which the calculated thermal functions were tabulated at round temperatures in the same temperature range. The authors do not seem to be aware of the earlier low-temperature measurements on calcite in [1934AND2] and [1969STA/LIN], and their entropy values are calculated from a value of S^{o}_{m} (298.15 K) = 92.902 $J \cdot K^{-1} \cdot mol^{-1}$, for which the reference is a Handbook which we have not been able to obtain. This entropy value is intermediate between the values measured by Anderson [1934AND2], (93.1 ± 1.7) $J \cdot K^{-1} \cdot mol^{-1}$ and by Staveley and Linford [1969STA/LIN], (91.71 ± 0.21) $J \cdot K^{-1} \cdot mol^{-1}$. The reported C_p values from this study are consistently higher than those reported in [1969STA/LIN] over most of the temperature range, by about 3.85 % at 100 K. 1.03 % at 200 K and 0.29 % at 300 K. But the C_p values above 300 K agree reasonably well with those given in [1981JAC/KER], and were included in our fitting of the high-temperature C_p data.

[2007RAR/WOL]

Rard and Wolery pointed out that the NBS (NIST) Tables (1982 with a 1989 erratum) and the 1989 CODATA evaluation reported enthalpies of formation and Gibbs energies of formation for key phosphate compounds and aqueous species, especially $H_2PO_4^-$ and HPO_4^{2-}, that showed a systematic and nearly constant difference of 6.3 to 6.9 kJ·mol^{-1} per phosphorus atom. They concluded from their analysis that the CODATA evaluation was more reliable and a slightly different value of $S_m^o(H_2PO_4^-, 298.15\ K) = (90.6 \pm 1.5)$ J·K^{-1}·mol^{-1} was suggested, based on additional literature which appeared after publication of the CODATA review which recommended $S_m^o(H_2PO_4^-, 298.15\ K) = (92.5 \pm 1.5)$ J·K^{-1}·mol^{-1}. Rard and Wolery also noted that polymeric phosphate ions existed, including the pyrophosphate ions, $P_2O_7^{4-}$ and $P_3O_{10}^{5-}$, as well as their protonated forms, and higher-order polymers. They concluded that the systematic differences in $\Delta_f G_m^o(298.15\ K)$ resulted from differences in $\Delta_f H_m^o(298.15\ K)$ obtained from different enthalpies of combustion of P(cr) to yield H_3PO_4(aq) that were reflected in the values of $\Delta_f H_m^o(P_4O_{10}, cr, 298.15\ K)$. Dissolution of P_4O_{10}(cr) is represented by:

$$P_4O_{10}(cr) + 6H_2O(l) \rightleftharpoons 4H_3PO_4(sln) \qquad (A.279)$$

Their examination of various data sources established that the NBS compilation used exclusively $\Delta_f H_m^o(P_4O_{10}, cr, 298.15\ K) = -2984.0$ kJ·mol^{-1} from Holmes [1962HOL] and that this value had been disputed. Rard and Wolery derived the following preferred value of $\Delta_f H_m^o(P_4O_{10}, cr, 298.15\ K) = -3008.2$ kJ·mol^{-1} from the CODATA Key values, but were only able to derive one new value, *viz.*, $S_m^o(H_2PO_4^-, 208.15\ K) = (90.6 \pm 1.5)$ J·K^{-1}·mol^{-1}. They gave a comparative table, which includes the values in Table A-78.

Table A-78: Standard molar thermodynamic quantities of formation for the aqueous phosphate species at 298.15 K.

Species	$\Delta_f G_m^o$ /kJ·mol^{-1}	$\Delta_f H_m^o$ /kJ·mol^{-1}	S_m^o /J·K^{-1}·mol^{-1}	Reference
H_3PO_4(aq)	– (1149.367 ± 1.576)	– (1294.120 ± 1.616)	(161.912 ± 2.575)	TDB/CODATA
$H_2PO_4^-$	– (1137.152 ± 1.567)	– (1302.600 ± 1.500)	(92.500 ± 1.500) (90.600 ± 1.500)	TDB [2007RAR/WOL]
HPO_4^{2-}	– (1095.985 ± 1.567)	– (1299.000 ± 1.500)	– (33.500 ± 1.500)	TDB
PO_4^{3-}	– (1025.491 ± 1.576)	– (1284.400 ± 4.085)	– (220.970 ± 12.846)	TDB

[2008BES]

Electric conductivity measurements were made of dilute aqueous calcium and manganese sulfate solutions from 5 to 35 °C at intervals of 5 ° in the concentration range $6 \times 10^{-4} < c/\text{mol·dm}^{-3} < 5.4 \times 10^{-2}$. The cell was calibrated against standard KCl solutions. Bešter-Rogač actually gave the concentration *versus* molar conductance using the term "molonity" (moles of solute per 1 kg solution) and used the "low concentration chemical model (lcCM)" to treat the data. This model gave values for the formation

constant, $K°(CaSO_4, aq)$, which were consistent with values obtained by other authors using different techniques. However, the Lee-Wheaton approach adopted by the TDB NEA gave higher values with the disparity increasing to a factor of 1.86 at 35 °C. At 25 °C the latter equation gave $K° = (273.2 \pm 7.4)$.

[2008RAT/KUZ]

Rat'ko *et al*. used SEM and TEM to monitor the growth of bayerite on aluminium powder exposed to water at 373 K. The growth rate is governed by diffusion of hydroxidoaluminium species through the bayerite layer, *i.e.*, the growth rate of the bayerite layer is diffusion controlled. No thermodynamic data are provided in this paper.

[2008XIO]

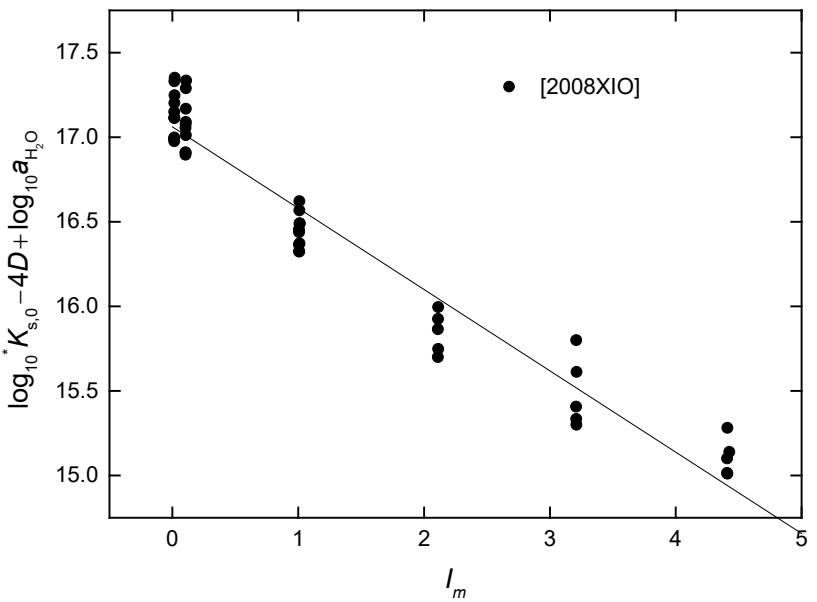
Experiments on the solubility of synthetic brucite (size range 0.57 to 15 μm) were reported in solutions containing up to 4.4 mol·kg^{-1} NaCl at (22.5 ± 1.5) °C. Additional experiments from over-saturation produced crystals in the size range from 0.63 to 23 μm. The pH of the reaction mixture was measured as a function of time with a Ross combination glass electrode calibrated with three commercial buffer solutions and converted to molality units using the empirical relationship, $pH_c = pH + A$, where $A = 0.159\, c_{NaCl}$ [1995RAI/FEL], which was presumably generated for 25 °C and is claimed to be valid to 6 mol·kg^{-1} (NaCl). Magnesium and sodium concentrations were measured by ICP-AES. The solid was characterised by XRD and SEM. One experiment in deionised water and the single reversal experiment, in which no NaCl was added, appeared to show trends with time, up to 10000 hours in the latter case, and therefore these two results were not treated in this review. For the bulk of the experiments in NaCl solutions, the analytical and stoichiometric concentrations given in Xiong's Table 3 could only be reconciled if significant HCl was initially present in the starting solutions because the molality of magnesium was greater by more than a factor of ten than that of hydroxide ion, such that $I_m \approx 3m_{Mg^{2+}} + m_{NaCl}$. The last magnesium molality shown in Table 1 of this paper was very high and may be a typographical error. The brucite solubility equilibrium is:

$$Mg(OH)_2(cr) + 2H^+ \rightleftharpoons Mg^{2+} + 2H_2O(l) \qquad \text{(A.280)}$$

A weighted SIT treatment of the data for Reaction (A.280) with equilibration times > 2000 hours is illustrated in Figure A-30 and results in the following parameters at 22.5 °C: $\log_{10}{}^*K^\circ_{s,0} = (17.06 \pm 0.07)$ and $\Delta\varepsilon = (0.481 \pm 0.035)$ kg·mol^{-1} (2σ), *cf.* $\log_{10}{}^*K^\circ_{s,0} = (17.05 \pm 0.2)$ given in this paper. The author used NONLIN, which is based on the Pitzer ion interaction model, to derive the activity coefficients. The $\Delta\varepsilon$ value is significantly larger than those determined in [1996BRO/DRU]. The author tried to shed

more light on the question raised by Altmaier *et al*. [2003ALT/MET] regarding the validity of the standard entropy of brucite $S^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = 63.14\ J{\cdot}K^{-1}{\cdot}mol^{-1}$ which was reported in [1937GIA/ARC]. Altmaier *et al*. used $\Delta_f H^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = -926.7\ kJ{\cdot}mol^{-1}$ calculated from the $\Delta_r H^{o}_{m}$((A.280), 298.15 K) determined in [1977MCG/HOS] and [1996BRO/DRU] that when combined with a $\Delta_f G^{o}_{m}(Mg(OH)_2, cr, 298.15\,K)$ value of $-831.9\ kJ{\cdot}mol^{-1}$, produced a value of $S^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = 50.6\ J{\cdot}K^{-1}{\cdot}mol^{-1}$. Apparently, they ignored the calorimetric studies tabulated by Gurvich *et al*. [1994GUR/VEY] who recommended $\Delta_f H^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = -(924.35 \pm 0.3)\ kJ{\cdot}mol^{-1}$. Xiong noted that by using $\Delta_f G^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = -832.2\ kJ{\cdot}mol^{-1}$ (his value) and $S^{o}_{m}(Mg(OH)_2, cr, 298.15\,K)$ from calorimetry [1937GIA/ARC] and the auxiliary data from CODATA [1989COX/WAG], $\Delta_f H^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = -(923.3 \pm 1.1)$ $kJ{\cdot}mol^{-1}$, which is consistent with the value adopted by Gurvich *et al*. Therefore, Xiong concluded that the concern raised by Altmaier *et al*. as to the value of $S^{o}_{m}(Mg(OH)_2, cr, 298.15\,K) = 63.14\ J{\cdot}K^{-1}{\cdot}mol^{-1}$ was largely unfounded and this opinion is upheld by this review.

Figure A-30: Plot of $\log_{10}{}^{*}K'_{s,0} - 4D + \log_{10}a_{H_2O}$ *versus* ionic strength at 22.5 °C, where the solid line results from application of the SIT model.



[2009SIP]

Studies of the structure of aluminium in strongly basic solutions were reviewed in relation to synthetic homogeneous Bayer liquors. This review supported the existence of the hydrated ions: Na^+, OH^-, $Al(OH)_4^-$, a dimer with the probable formula $Al_2(O)(OH)_6^{2-}$, and ion pairs thereof up to 100 °C. The existence of higher-order mononuclear Al-OH species was unconfirmed.

[2010ARA/BIN]

The crystallographic parameters of two samples of natural natrite, Na_2CO_3(cr) from Russia were determined at 293 and 120 K using synchrotron X-ray data. At 293 K, the monoclinic cell parameters were (8.851 ± 0.002), (5.24 ± 0.02) and (6.021 ± 0.002) Å, β = 101.08 ° for one sample and (8.882 ± 0.002), (5.235 ± 0.010) and (6.046 ± 0.011) Å, β = 101.464 ° for the other. In the sample measured at low temperatures, no transition was observed above 120 K, lower than the values of 130 K reported in [1979PAT/HEL] and 170 K in [2003DUS/CHA].

[2010DAC/HAR]

This paper mainly concerns the thermal properties of the solid solution between fluorapatite and chlorapatite, but includes the properties of the essentially pure two end members. In particular, the heat capacities of fluorapatite have been measured from 5 to 764 K using relaxation calorimetry and differential-scanning calorimetry (DSC) The fluorapatite was prepared by heating $Ca_3(PO_4)_2$ in an excess of CaF_2 at 1375 °C for 15 hours, followed by slow cooling to 1220 °C to allow crystals to form. After cooling, the flux and crystals were boiled in an aqueous 20 % solution of $Al(NO_3)_3 \cdot 9H_2O$ to remove the excess CaF_2. No chemical analyses of the final products are given. The relaxation calorimetric measurements were made between 5 and 300 K with a 10 mg powdered sample, and the DSC measurements from 282 to 764 K on a 25 mg powdered sample.

All the experimental data measurements are presented in a small graph, with a table of smoothed values of C_p and S at round temperatures from 5 to 298.15 K, and the coefficients of a fitted equation for C_p from 298.15 K to 768 K.

They give the following values at 298.15 K:

$$C_{p,m}(Ca_5(PO_4)_3F, \text{cr}, T) = (369.6 \pm 1.1) \text{ J K}^{-1} \cdot \text{mol}^{-1}$$

$$S_m(Ca_5(PO_4)_3F, \text{cr}, T) = (383.2 \pm 1.5) \text{ J K}^{-1} \cdot \text{mol}^{-1}.$$

Their values of C_p agree excellently with the adiabatic calorimetric measurements of [1951EGA/WAK2] up to about *ca.* 150 K, but are increasingly smaller above this, being 1.7% lower at 298.15 K.

A fuller comparison of the two studies is given in Section V.8.2.4.2 of the main text.

[2011BEN/SAL]

In this study, the solubility of synthetic magnesite was determined from 50 to 200 °C in 0.1 mol·kg^{-1} NaCl solutions, and in some cases under constant CO_2 partial pressure (4-30 bar) both by means of a hydrogen-electrode concentration cell (HECC) and a traditional batch reactor. Solution samples were analysed for Mg, alkalinity and/or total dissolved inorganic carbon (TDIC). Magnesium was analysed by flame atomic absorption spectroscopy. TDIC was determined by using a non-dispersive infrared (NDIR) CO_2 gas analyser.

The thermodynamic solubility products ($K^{\circ}_{s,0}$) determined at different temperatures were fitted by $\log_{10}K^{\circ}_{s,0} = a + b\,(T/\mathrm{K})^{-1} + c\,(T/\mathrm{K})$ with $a = 7.267$, $b = -1476.604$ and $c = -0.033918$. Based on this equation and its first and second derivatives, the following thermodynamic parameters were determined:

$$\Delta_r G^{\circ}_m\,(298.15\ \mathrm{K}) = 44.511\ \mathrm{kJ\cdot mol^{-1}},$$
$$\Delta_r H^{\circ}_m\,(298.15\ \mathrm{K}) = -29.453\ \mathrm{kJ\cdot mol^{-1}},$$
$$\Delta_r S^{\circ}_m\,(298.15\ \mathrm{K}) = -248.075\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$$
and
$$\Delta_r C^{\circ}_{p,m}\,(298.15\ \mathrm{K}) = -387.197\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}.$$

Combining these values with the thermodynamic parameters for Mg^{2+} and CO_3^{2-} from Shock *et al.* [1997SHO/SAS], the authors calculated the following thermodynamic functions for $MgCO_3$(cr) at 298.15 K and 1 bar: $\Delta_f G^{\circ}_m = -(1026.48 \pm 2.00)$ kJ·mol^{-1}, $\Delta_f H^{\circ}_m = -(1111.75 \pm 2.00)$ kJ·mol^{-1}, $S^{\circ}_m = (60 \pm 2)$ J·K^{-1}·mol^{-1} and $\Delta_f C^{\circ}_{p,m} = (75.91 \pm 2.00)$ J·K^{-1}·mol^{-1}.

In their Table 5, the last two items of $\log_{10}[Mg^{2+}]_{meas}$ are probably erroneous (– 3.094 and – 3.068 instead of – 2.094 and – 2.068). The derived apparent solubility products were extrapolated to infinite dilution using activity coefficients calculated by the relation $\gamma_{|z|} = \gamma^{z^2}_{\pm(NaCl)}$. Although the experimental difficulties related to systems closed to CO_2 are well known, the data reported in this paper for closed systems are accepted, since the authors determined accurately and in parallel the pH, $[Mg^{2+}]_{tot}$ and the total dissolved inorganic carbon.

[2011FOR/RAD]

To determine the surface energy of calcite, the enthalpies of acid dissolution and water adsorption were measured, using one bulk and four commercial nanophase calcite samples. We shall, however, restrict the present comments to acid dissolution of the bulk sample.

The sample was prepared by precipitation from a chloride solution using potassium carbonate. After filtering and thorough washing (deionised water, alcohol), the sample was heated in an alumina crucible for 16 hours at 773 K. Infrared spectra and X-ray diffraction patterns did not reveal any extraneous phase or extraneous material. An impurity of 1.61 % of potassium was detected in the sample by ICP-MS.

Five to 10 mg samples of calcite were dissolved in 25 g of 5 molar HCl at 299.15 K in a commercial (CSC 4400) isothermal calorimeter. The enthalpy of solution was reported to be $-(25.56 \pm 0.41)$ kJ·mol^{-1} as the average of five to nine determinations, with no further details.

[2011KUM/KAR]

Kumzerov *et al.* measured the heat capacity of a natural chrysotile sample from Uzbekistan from 80 to 300 K using an automated adiabatic calorimeter. No further details of the sample or apparatus were given and the results are only presented in two small graphs from which it is impossible to derive any precise data. These results were therefore not used in this review.

[2011POP/MIL2]

The osmotic coefficients of K_2HPO_4 solutions were measured at 298.15 K by the isopiestic technique from 1.3846 mol·kg^{-1} to supersaturated concentrations of 13.939 mol·kg^{-1} with a $CaCl_2$ reference solution. These results appear to be very reliable and complemented existing lower concentration results [1954SCA/BRE]. Therefore, some were included in the final SIT treatment to extract the ion interaction coefficient, $\varepsilon(K^+, HPO_4^{2-})$.

[2012VIS/VAN]

This reference compiles and critically evaluates solubility data (published up to 2009) of three magnesium carbonates (magnesite ($MgCO_3$), nesquehonite ($MgCO_3 \cdot 3H_2O$), and lansfordite ($MgCO_3 \cdot 5H_2O$)) in water and in simple aqueous electrolyte solutions. This volume also outlines the complete thermodynamic models applied for the re-evaluation of primary experimental data, using the Pitzer ion interaction formalism.

Thermodynamic data of magnesium carbonates consistent with two thermodynamic model variants are proposed. Model 2, which describes the $Mg^{2+}-HCO_3^-$ interaction with (Pitzer) ion interaction parameters, is more consistent with the solubility data and with other thermodynamic data than the Model 1 that describes the interaction with a stability constant. Useful side products of the compilation and evaluation of the data are new relationships for the Henry constant of CO_2 with Setchenow parameters, for various equilibria in the aqueous phase including the dissociation constants of CO_2(aq) and the stability constant of the ion pairs $MHCO_3^+$ / MCO_3(aq) (M = Mg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}).

The equation derived for the Henry constant of CO_2 valid between 0-360 °C,

$$\ln(k_H/p_{sat}) = A/T_r + B\tau^{0.355}/T_r + C T_r^{-0.41}\exp(\tau) + D$$

where k_H is the Henry constant defined as f(CO_2(g)) = x(CO_2(aq)) (bar) with f(CO_2(g)) the fugacity of CO_2(g), x(CO_2(aq)) the mole fraction of CO_2(aq), p_{sat} is the saturated vapour pressure of water (bar), T_r is the reduced temperature, T/T_c with T_c the critical temperature of water, $\tau = 1 - T_r$, and A = – 9.14122, B = 2.81920, C = 11.28516 and D = – 0.80660. This equation is also used in the present re-evaluation.

The temperature dependence of stability constants for Mg^{2+}/Ca^{2+} bicarbonate/carbonate ion pairs were described with the general equation

$$\log_{10} K = A + B/(T/K) + C \log_{10}(T/K) + D\,(T/K) + E/(T/K)^2,$$

where for $MgHCO_3^+$

A = 437.909531, B = – 27415.3023;

for $CaHCO_3^+$

A = 439.872327, B = – 27988.839;

for $MgCO_3$(aq)

A = 2403.544158, B = – 133162.9686, C = – 869.0072054, D = 0.36380638, E = 7808760.2 ;

and for $CaCO_3$(aq)

A = 3423.002821, B = – 198599.9198, C = – 1226.6370290, D = 0.48383483, E = 12202744.3 .

Based on the seven accepted data for magnesite solubility (t = 18-91 °C) the following thermodynamic parameters are derived (due to the relatively narrow temperature range $\Delta_{sol}H_m^o$(298.15 K) was set to – 30.54 kJ·mol^{-1} (based on the CODATA and NIST-JANAF values [1998CHA], $\Delta_{sol}C_p^o$ was set to – 387.197 J·K^{-1}·mol^{-1} [2011BEN/SAL]):

Model 1:
$\log_{10}K^{\circ}_{s,0} = 122.5203 - 4446.81/(T/\mathrm{K}) - 44.66201\log_{10}(T/\mathrm{K})$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -(7.8565 \pm 0.2)$, $\Delta_{sol}G^{\circ} = 44.84\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}S^{\circ} = -252.84\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$.

Model 2:
$\log_{10}K^{\circ}_{s,0} = 122.5940 - 4446.81/(T/\mathrm{K}) - 44.66201\log_{10}(T/\mathrm{K})$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -(7.7828 \pm 0.2)$, $\Delta_{sol}G^{\circ} = 44.42\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}S^{\circ} = -251.43\ \mathrm{J\cdot K^{-1}\cdot mol^{-1}}$.

[2012VIS/VAN2]

This review comprehensively compiles and critically evaluates the solubility data of all forms of calcium carbonate (calcite, aragonite, vaterite, monohydrocalcite and ikaite) in water and in single electrolytes in the presence and the absence of external CO_2. The data were described using two model variants: Model 1 (which describes the $CaHCO_3$ ion-pair with a stability constant) and Model 2 (which describes the interactions between the Ca^{2+} and HCO_3^- ions with Pitzer parameters). The complete thermodynamic model used has been described in detail in [2012VIS/VAN]. While there is no compelling evidence in favour one of the models, it is clear that Model 2 performs better overall, similarly as was found in [2012VIS/VAN] for magnesium carbonates.

For calcite the following thermodynamic parameters have been derived:

Model 1: $\log_{10}K^{\circ}_{s,0} = -1080.01714 + 61063.868/(T/\mathrm{K}) + 391.579367\log_{10}(T/\mathrm{K}) - 0.191678473(T/\mathrm{K}) - 4007199.35/(T/\mathrm{K})^2$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -8.501$, $\Delta_{sol}G^{\circ}(298.15\ \mathrm{K}) = 48.506\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}H^{\circ}(298.15\ \mathrm{K}) = -9.932\ \mathrm{kJ\cdot mol^{-1}}$.

Model 2: $\log_{10}K^{\circ}_{s,0} = -1626.44427 + 90452.543/(T/\mathrm{K}) + 452.534\log_{10}(T/\mathrm{K}) - 0.263581445(T/\mathrm{K}) - 5533912.2/(T/\mathrm{K})^2$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -8.438$, $\Delta_{sol}G^{\circ}(298.15\ \mathrm{K}) = 48.144\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}H^{\circ}(298.15\ \mathrm{K}) = -11.450\ \mathrm{kJ\cdot mol^{-1}}$.

For aragonite the following thermodynamic properties have been obtained:

Model 1: $\log_{10}K^{\circ}_{s,0} = -1086.13322 + 61462.647/(T/\mathrm{K}) + 393.747358\log_{10}(T/\mathrm{K}) - 0.192163253(T/\mathrm{K}) - 4033482.02/(T/\mathrm{K})^2$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -8.352$, $\Delta_{sol}G^{\circ}(298.15\ \mathrm{K}) = 47.674\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}H^{\circ}(298.15\ \mathrm{K}) = -9.642\ \mathrm{kJ\cdot mol^{-1}}$.

Model 2: $\log_{10}K^{\circ}_{s,0} = -1632.56035 + 90851.313/(T/\mathrm{K}) + 590.371259\log_{10}(T/\mathrm{K}) - 0.264066225(T/\mathrm{K}) - 5560194.87/(T/\mathrm{K})^2$; $\log_{10}K^{\circ}_{s,0}(298.15\ \mathrm{K}) = -8.289$, $\Delta_{sol}G^{\circ}(298.15\ \mathrm{K}) = 47.312\ \mathrm{kJ\cdot mol^{-1}}$, $\Delta_{sol}H^{\circ}(298.15\ \mathrm{K}) = -11.160\ \mathrm{kJ\cdot mol^{-1}}$.

[2013SCH/MUN]

Isopiestic measurements were carried out at 25 ºC using NaCl and KCl reference solutions based on the water activities given in [1985CLA/GLE] and [1999ARC2],

respectively. Equilibration was only attained after two weeks or more. The solutes studied were NaH_2PO_4, Na_2HPO_4, KH_2PO_4 and K_2HPO_4. However, water activities were also tabulated from many other studies of these electrolytes and good agreement was found, except for K_2HPO_4, which may have been contaminated with KH_2PO_4 [2014POP/MIL]. Water activities of K_3PO_4(sln) were taken from [1954SCA/BRE] and [1974REZ/VIT], but the quality of the latter data could not be confirmed by this review. The measured osmotic coefficients for solutions of NaH_2PO_4, Na_2HPO_4 and KH_2PO_4 were included in the SIT treatments conducted by this review. Average solubilities for all of these salts were also calculated by Scharge *et al*.

[2013STE/BEN]

Dissociation constants of carbonic acid and formation constants of $NaHCO_3$(aq) and $NaCO_3^-$ ion pairs have been determined in dilute hydrothermal solutions to 200 °C. Two experimental approaches were applied, potentiometric titrations at 10-60 °C and spectrophotometric pH measurements using pH indicators (2-napthol and 4-nitrophenol) at 25-200 °C. At a given temperature, the first and second ionisation constants of carbonic acid and the ion-pair formation constants were simultaneously fitted to the data. The formation constants of $NaHCO_3$(aq) and $NaCO_3^-$ ion pairs increase as expected with the temperature (*e.g.*, at 200 °C $\log_{10}K$ = 0.58 and $\log_{10}K$ = 2.21, respectively).

The ionic-strength corrections of equilibrium constants were made using an SIT-related method used by Robinson and Stokes [1968ROB/STO]. The results of this study compare well with previously determined values of pK_1 and pK_2 (and even with the pK_2 values reported in [1984PAT/BUS] for higher temperatures). Significant errors were associated with the equilibrium constants derived from the spectrophotometric pH measurements above 100 °C, mostly related to the errors assigned to the ionisation constants of the two pH indicators used, and to the relatively narrow available pH-range. The increasing importance of ion-pair formation above 100 °C is in line with the conclusions drawn in this review.

[2014OGO/KIS]

In the frame of a broad study of various physicochemical properties of a natural sepiolite from the Akkermanovskoe field (Southern Ural, Russia), Ogorodova *et al.* reported the enthalpies of formation, dehydration and dehydroxylation of this mineral that, from a full analysis, possessed the chemical formula (calculated on 64 charges) $Mg_{7.93}Ca_{0.03}Fe^{2+}_{0.03}Ni_{0.03}Al_{0.03}Fe^{3+}_{0.04}Si_{11.94}O_{30}(OH)_4(H_2O)_4{\cdot}9.9H_2O$, close to the ideal $Mg_8Si_{12}O_{30}(OH)_4(H_2O)_4{\cdot}nH_2O$. To remain within the charter of this review, we shall limit the discussion to the enthalpy of formation of two materials: $Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O$(cr) formed after removal of the adsorbed water ($5.9H_2O$) by

heating for 30 min at 328 K, and $Mg_4Si_6O_{15}(OH)_2$ formed after removal of further water at 873 K, where we use, as in related sections, the reduced formulae with only 4 Mg atoms.

The thermochemical experiments (Calvet microcalorimeter, Setaram, France), were performed by the "drop" method into a melt of $2PbO{\cdot}B_2O_3$ at 973 K, using samples of (6-15) (± 0.002) mg. The experiments yielded the global heat effect given by $\Delta_r H_m$ (A.281) that includes the heat content and the enthalpy of solution, namely:

$$\Delta_r H_m \text{ (A.281)} = \Delta_{sol} H^o_m \text{ (973 K)} + (H^o_m \text{ (973 K)} - H^o_m \text{ (298.15 K)}) \qquad \text{(A.281)}$$

Literature data [1976KIS], [1979KIS/OGO], [1994CAR/YAZ], [2003OGO/MEL] were used to evaluate the corrections for the differences in the composition between the compounds studied and sepiolite of "ideal" formula, but since the total correction was less than the uncertainty in their experimental results, no corrections were applied.

Moreover, the authors did not discuss any experimental problems relating to the probable loss of water from the sepiolites during the dissolution process. The authors thus reported, $\Delta_r H_m$ (A.281) = (659.0 ± 9.5) and (477.0 ± 8.0) $kJ{\cdot}mol^{-1}$ for the dissolution of the two species $Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O$ and $Mg_4Si_6O_{15}(OH)_2$, respectively. These values were given without further details as averages (five and six experiments, respectively) at a declared 95 % confidence level. Using literature data for the enthalpies of solution under the same conditions from [1976NAV/COO] [1979KIS/OGO] [1984KIS/OGO] and the heat capacities and standard enthalpies of formation of the constituent oxides from [1995ROB/HEM], the authors finally reported, without further explanation,

$$\Delta_f H^o_m (Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O, \text{ cr, } 298.15 \text{ K}) = -(9386.5 \pm 14.0) \text{ kJ}{\cdot}\text{mol}^{-1}$$

and $\Delta_f H^o_m (Mg_4Si_6O_{15}(OH)_2, \text{ cr, } 298.15 \text{ K}) = -(8213.0 \pm 10.5) \text{ kJ}{\cdot}\text{mol}^{-1}$.

However, because of the complete absence of the actual experimental results, and the experimental uncertainties, this review cannot select these values, which are given for information only.

In order to evaluate the Gibbs energies of formation of their sepiolite samples, Ogorodova *et al.* [2014OGO/KIS] estimated the entropies of these on the basis of those of talc ($Mg_3Si_4O_{10}(OH)_2$), enstatite ($MgSiO_3$), quartz and MgO from Robie and Hemingway [1995ROB/HEM]. For this, they used entropy contributions of the bound and zeolitic water of 39.5 and 50.2 $J{\cdot}K^{-1}{\cdot}mol^{-1}$, respectively, from the values listed by Fyfe *et al.* [1958FYF/TUR] (which in turn seem to be taken from Latimer [1952LAT]). With no further details, Ogorodova *et al.* reported:

$$S^o_m (Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O, \text{ cr, } 298.15 \text{ K}) = 547.5 \text{ J}{\cdot}\text{K}^{-1}{\cdot}\text{mol}^{-1}$$

and

$$S^{o}_{m}(Mg_4Si_6O_{15}(OH)_2, cr, 298.15\ K) = 368.5\ J{\cdot}K^{-1}{\cdot}mol^{-1},$$

but as noted in the text, similar calculations for chrysotile and talc (using the entropies of the constituent oxides) give values which are 5-10 % larger than the experimental values, so we have not accepted these entropy values for the sepiolites.

However, we also note, for information only, the Gibbs energies of formation that Ogorodova *et al.* [2014OGO/KIS] reported from their measured enthalpies of formation and estimated entropies: $\Delta_f G^{o}_{m}(Mg_4Si_6O_{15}(OH)_2{\cdot}4H_2O, cr, 298.15\ K) = -8640.0\ kJ{\cdot}mol^{-1}$ and $\Delta_f G^{o}_{m}(Mg_4Si_6O_{15}(OH)_2, cr, 298.15\ K) = -7691.5\ kJ{\cdot}mol^{-1}$.

[2014STE/BEN]

The formation constants of magnesium hydrogencarbonate and carbonate ion pairs have been experimentally determined in dilute hydrothermal solutions between 10 and 150 °C. Two experimental approaches were applied. Potentiometric titrations at 10-60 °C under an Ar atmosphere were carried out on four systems: (1) acid titrations with 0.1 $mol{\cdot}kg^{-1}$ HCl of aqueous Na_2CO_3 and NaCl solutions, (2) acid titrations with 0.1 $mol{\cdot}kg^{-1}$ HCl of aqueous Na_2CO_3, $MgCl_2$ and NaCl solutions, (3) base titrations with 0.1 $mol{\cdot}kg^{-1}$ NaOH of aqueous $NaHCO_3$, $MgCl_2$ and NaCl solutions and (4) Mg titrations with 0.2 $mol{\cdot}kg^{-1}$ $MgCl_2$ solutions of aqueous $NaHCO_3$ and NaCl solutions. At 25 and 80-150 °C, spectrophotometric pH measurements of solutions with different compositions were also performed using two pH indicators, 2-naphthol and 4-nitrophenol. At a given temperature, the first- and second-ionisation constants of carbonic acid (K_1, K_2) and the ion-pair formation constants for $MgHCO_3^+$ ($K_{MgHCO_3^+}$) and $MgCO_3(aq)$ (K_{MgCO_3}) were fitted simultaneously to the data.

The equilibrium constants for both $MgHCO_3^+$ and $MgCO_3$(aq) ion pairs were accurately determined between 25-100 °C. At $t > 100$ °C, only $K_{MgHCO_3^+}$ could be determined from measurements in neutral to mildly alkaline solutions, whereas the determination of K_{MgCO_3} is prevented by the precipitation of brucite and/or magnesite in alkaline solutions where $MgCO_3$(aq) contributes significantly to the aqueous carbonate speciation. Based on these findings, the optimum conditions for CO_2 sequestration by Mg-carbonate mineralisation are at pH < 6.5 where Mg^{2+} is available for carbonate formation.

Individual ion-activity coefficients were estimated using a modified Debye–Hückel equation, which is similar but not identical to the SIT version. Therefore, the uncertainties of the reported thermodynamic formation constants are slightly increased (see Table V-59). Nevertheless, the experiments were carefully designed and performed. This is also evidenced by the fact that the obtained K_{MgCO_3} values compare well with those reported earlier (see Figure V-51). In addition, the determined ionisation constants of carbonic acid (K_1 and K_2) are close to those selected in this review (ΔpK is < 0.02 up to 80 °C, and < 0.04 between 80-150 °C).

[2016ARC/FER]

Molar conductivity measurements were made of aqueous $KB(OH)_4$ and $NaB(OH)_4$ solutions from 298.15 to 623.16 K at a constant pressure of 20 MPa, using a flow-through instrument. These data were treated with the Turq-Blum-Bernard-Kunz model [1995TUR/BLU]. The limiting conductivities for the borate anion its ion-pair formation constants with K^+ and Na^+ were derived over the entire temperature range. However, for the purposes of this review, the results show that these ion pairs can be ignored at ambient conditions up to moderate concentrations.

[2016ARC/FER2]

Molar conductivities of aqueous $LiB(OH)_4$, $NaB(OH)_4$ and $KB(OH)_4$ solutions were reported from 298.15 to 623.16 K at 20 MPa. This paper is an extension of work presented in a companion papers [2016ARC/FER], [2017ARC/FER]. As in the other studies, these data were treated with the Turq-Blum-Bernard-Kunz ("TBBK") model [1995TUR/BLU]. The formation constants of the ion pairs, $LiB(OH)_4(aq)$, $NaB(OH)_4(aq)$ and $KB(OH)_4(aq)$, were also derived. In addition, Raman spectra were measured using cell configurations with an internal perchlorate standard to derive quantitatively the boron speciation in concentrated solutions. The principal species in the concentrated boric acid solutions (an example was given of spectra at 353 K for m_B = 1 and with varying ratios of added NaOH) from 298 to 353 K were found to be $B(OH)_3(aq)$ and $B(OH)_4^-$, as well as the polyborate species, $B_2O(OH)_5^-$, $B_3O_3(OH)_4^-$, $B_4O_5(OH)_4^{2-}$, and $B_5O_6(OH)_4^-$. The trimer was observed also at 523 and 573 K. No thermodynamic data were presented.

[2017ARC/FER]

The ionisation constants of boric acid were measured from 298 to 623 K at a constant pressure ~ 20 MPa by conductance. In addition, quantitative reduced isotropic Raman spectra for a ~ 1 mol·kg^{-1} boric acid solution were reported over the same temperature range at a constant pressure of 25 MPa. For the former technique, molar conductivities are reported at concentrations from 10^{-4} to 10^{-1} mol·kg^{-1}, which when analysed with the Turq-Blum-Bernard-Kunz model [1995TUR/BLU], yield the base-hydrolysis constants of boric acid, $K^\circ_{b,1}$, up to 623 K. At 298.15 K, Arcis *et al*. reported a $\log_{10}K^\circ_{b,1}$ value at 19.767 MPa of (4.92 ± 0.02). Using a $\log_{10}K^\circ_{H_2O}$ value of – 13.926 at this condition derived by the authors from Bancroft and Lvov [2006BAN/LVO], the corresponding acid dissociation equilibrium is represented by $\log_{10}K^\circ_{a,1} = -9.005$. This review selected $\log_{10}K^\circ_{b,1} = (4.777 \pm 0.121)$ at 298.15 K and 0.1 MPa which is good agreement with previous assessments, *viz*., 4.76 [1972MES/BAE] [2000PAL/BEN] and that derived from Equation (6) in this paper, *viz*., 4.76. From the conductance data at 598 and 623 K, the authors were able to extract the formation constant for the reaction:

$$2B(OH)_3(aq) + OH^- \rightleftharpoons B_2(OH)_7^- \text{ (or } B_2O(OH)_5^-)$$

However, these conditions are beyond the scope of the present review.

[2018BEN/BER]

The solubility of natural dolomite ($CaMg(CO_3)_2$) has been measured from 50 to 253 °C in 0.1 mol·kg^{-1} NaCl solutions using a hydrogen electrode concentration cell (HECC), which avoids any contact and equilibration of the experimental fluids with atmospheric CO_2. For all experiments performed in this study, the same initial reference and test solution was used: an aqueous solution containing 2.500×10^{-3} mol·kg^{-1} HCl and 0.0975 mol·kg^{-1} NaCl. Each measurement was initiated by placing ~ 3 g of the initial dolomite seeds into the cell. Steady-state fluid compositions were attained within a few weeks at temperatures ≥ 100 °C; but samples were taken over longer time intervals to verify attainment of equilibrium. Equilibrium was approached both from under and supersaturation demonstrating its reversibility, at least at temperatures above 100 °C.

Fluid samples were analyzed for pH, total magnesium, calcium, and total dissolved inorganic carbon (TDIC) concentrations. Mg and Ca were analyzed using a Perkin Elmer Analyst 400 Atomic Absorption Spectrometer (AAS) with an uncertainty of ± 2 % and a detection limit of 0.007 ppm. Note that the aqueous calcium concentration is always higher than that of aqueous magnesium, which may stem from the initial preferential release of Ca^{2+} due to its faster aquo-ion exchange rate. TDIC was analyzed using a LI-820 (LICOR Inc.) non-dispersive infrared (NDIR) CO_2 gas analyser, after acidification of the samples. The uncertainty of the TDIC analyses in this study is generally lower than 5 % with a detection limit of 0.44 ppm. Combining this TDIC with *in situ* pH measurements allowed the calculation of aqueous carbonate speciation at the temperature of the experiment, using the dissociation constants of carbonic acid taken from Patterson *et al.* ([1982PAT/SLO], [1984PAT/BUS]).

The apparent solubility products for the reaction, $CaMg(CO_3)_2 = Ca^{2+} + Mg^{2+} + 2\,CO_3^{2-}$, generated from under or supersaturation, were generally in close agreement with each other. The obtained apparent solubility products were extrapolated to infinite dilution using the equation $\log_{10}K^{\circ}_{s,0} = \log_{10}K_{s,0} + 16\log_{10}(\gamma_{\pm(NaCl)})$. The derived solubility products were accurately described by $\log_{10}K^{\circ}_{s,0} = a + b/T\,(K) + cT\,(K)$ where $a = 17.502$, $b = -4220.119$ and $c = -0.0689$. These parameters were used to determine the following thermodynamic quantities for dolomite at 298.15 K: $\Delta_r G^{\circ}_m = -(2160.9 \pm 2.0)$ kJ·mol^{-1} ($\log_{10}K^{\circ}_{s,0} = -(17.19 \pm 0.3)$), $\Delta_r H^{\circ}_m = -(2323.1 \pm 2.0)$ kJ·mol^{-1} and $S^{\circ}_m = (156.9 \pm 2.0)$ J·K^{-1}·mol^{-1} (uncertainties are 3σ).

The study of dolomite solids recovered after experiments indicated the atomic Ca:Mg ratio of 0.97 of the precipitated dolomitic over-growth. On the other hand, XPS results demonstrated an enrichment in Mg^{2+} in the uppermost growth layer and/or a

depletion of Mg^{2+}. One potential explanation of these observations is the initial precipitation of a Ca^{2+}-enriched layer followed by the precipitation of a slightly less reactive and possibly less soluble Mg^{2+}-enriched layer.

The experiments were carefully performed, the authors accurately determined the pH as well as the $[Mg^{2+}]_{tot}$, $[Ca^{2+}]_{tot}$ and total dissolved inorganic carbon (TDIC) concentrations. The calculation of apparent solubility products did not consider the formation of carbonato complexes. The extrapolation to infinite dilution is different from that recommended by NEA TDB (although it gives nearly the same results). However, the re-evaluation of reported data is not possible due to some lacking experimental details. As can be seen from Table V-76, the $([Mg^{2+}]_{tot} + [Ca^{2+}]_{tot})/[TDIC]$ ratios are, in many cases, considerably different from unity, which were not mentioned by the authors. In Table 5, the last four items of $\log_{10} K_{app\text{-}sp\text{-}dol}$ are erroneous (the corresponding $\log_{10} K_{sp^{\circ}\text{-}dol}$ values are given).

Appendix B

Ionic strength corrections[1]

Thermodynamic data always refer to a selected standard state. The definition given by IUPAC [1993MIL/CVI] is adopted in this review as outlined in Section II.3.1. According to this definition, the standard state for a solute B in a solution is the (hypothetical) state of solute at the standard molality $m°$, standard pressure $p°$ or standard concentration $c°$ and exhibiting infinitely dilute solution behaviour. However, for many reactions, measurements cannot be made accurately (or at all) in dilute solutions from which the necessary extrapolation to the standard state would be simple. This is invariably the case for reactions involving ions of high charge, which have usually strong hydrolysis and a strong dependence of activity coefficients on the ionic strength. Precise thermodynamic information for these systems can only be obtained based on the ionic medium principle introduced by Bjerrum, *i.e.,* by carrying out the measurements in the presence of an inert electrolyte of sufficiently high concentration that ensures activity factors are reasonably constant throughout the measurements. This appendix describes and illustrates the method used in this review for the extrapolation of experimental equilibrium data to zero ionic strength.

The activity factors of all the species participating in reactions in high ionic strength media must be estimated in order to reduce the thermodynamic data obtained from the experiment to the state $I = 0$. Two alternative methods can be used to describe the ionic medium dependence of equilibrium constants:

- One method takes into account the individual characteristics of the ionic media by using a medium dependent expression for the activity coefficients of the species involved in the equilibrium reactions. The medium dependence is described by virial equations (in analogy with real gas equations), where the first term is based on Debye-Hückel theory and other terms involving ion interaction

[1] This appendix contains much of the text of the TDB-2 Guideline written by Grenthe and Wanner [2013GRE/MOM], earlier versions of which have been printed in the previous NEA TDB reviews. The equations presented here are an essential part of the review procedure and are required to use the selected thermodynamic values. Parts of Section B.1.1 and the contents of Tables B-4, B-5, B-6 and B-7 have been revised by Rand *et al.* [2008RAN/FUG] and in the present review.

coefficients as used in the Pitzer equations [1973PIT] and in the specific ion interaction treatment.

- The other method accounts for the medium specific properties by introducing ion pairing between the medium ions and the species involved in the equilibrium reactions and calculating an effective ionic strength. Given the large number of species considered, the activity coefficients are expressed through an extended Debye-Hückel expression in which the activity coefficients of reactants and products depend only on the ionic charge and the effective ionic strength. Earlier, this approach has been used extensively in marine chemistry, *cf.* Refs. [1979JOH/PYT], [1979MIL], [1979PYT], [1979WHI2].

The activity factor estimates in the ion association approaches are thus based on the use of Debye-Hückel type equations, the simplest of which is the Davies equation [1962DAV]. However, the Davies equation should in general not be used at ionic strengths larger than 0.1 mol·kg^{-1}. The method preferred in the NEA Thermochemical Database review is a medium-dependent expression for the activity coefficients, which is the specific ion interaction treatment in the form of the Brønsted-Guggenheim-Scatchard approach. Other forms of specific ion interaction methods (the Pitzer and Brewer "B-method" [1961LEW/RAN] and the Pitzer virial coefficient method [1979PIT]) are described in the NEA Guidelines for the extrapolation to zero ionic strength [2013GRE/MOM].

The specific ion interaction methods are reliable for intercomparison of experimental data in a given concentration range. In many cases this includes data at rather low ionic strengths, I_c = 0.01 to 0.1 M, while in other cases, notably for cations of high charge ($\geq +4$ and ≤ -4), the lowest available ionic strength is often 0.2 M or higher, see for example Figures V.12 and V.13 in [1992GRE/FUG]. It is reasonable to assume that the extrapolated equilibrium constants at $I = 0$ are more precise in the former than in the latter cases. The extrapolation error is composed of two parts, one due to experimental errors, and the other due to model errors. The model errors seem to be rather small for many systems, less than 0.1 in $\log_{10} K^\circ$. For reactions involving ions of high charge, which may be extensively hydrolysed, one cannot perform experiments at low ionic strengths. Hence, it is impossible to estimate the extrapolation error. This is true for all methods used to estimate activity corrections. Systematic model errors of this type are not included in the uncertainties assigned to the selected data in this review.

B.1 The specific ion interaction equations

B.1.1 Background

The Debye-Hückel term, which is the dominant term in the expression for the activity coefficients in dilute solution, accounts for electrostatic, non-specific long-range interactions. At higher concentrations, short range, non-electrostatic interactions have to be taken into account. This is usually done by adding molality dependent terms to the Debye-Hückel expression. This method was first outlined by Brønsted [1922BRO], [1922BRO2] and elaborated by Scatchard [1936SCA] and Guggenheim [1966GUG]. Biedermann [1975BIE] highlighted its practical value, especially for the estimation of ionic medium effects on equilibrium constants. The two basic assumptions in the specific ion interaction treatment are described below.

- **Assumption 1:** The activity coefficient γ_j of an ion j of charge z_j in the solution of ionic strength I_m may be described by Eq. (B.1):

$$\log_{10} \gamma_j = -z_j^2 D + \sum_k \varepsilon(j,k,I_m) m_k \tag{B.1}$$

D is the Debye-Hückel term:

$$D = \frac{A\sqrt{I_m}}{1+1.5\sqrt{I_m}} \tag{B.2}$$

A is a constant which is temperature and pressure dependent, and I_m is the molal ionic strength:

$$I_m = \frac{1}{2}\sum_i m_i z_i^2$$

The Debye-Hückel limiting slope, A, has a value of (0.509 ± 0.001) $kg^{1/2} \cdot mol^{-1/2}$ at 298.15 K and 1 bar, (*cf*. Section B.1.2). The value 1.5 was proposed by Scatchard [1976SCA] to minimise the ionic strength dependence of $\varepsilon(j,k)$ for a number of electrolytes, and it was found to be particularly appropriate between I_m = 0.5 and 3.5 m. A constant value of 1.5 for all species simplifies modelling of both binary and multicomponent aqueous electrolyte systems, and makes it easier to give a consistent description of mean activity coefficient both in binary and multicomponent solutions ([1959ROB/STO], pp.435-441). Thus,

$$D = \frac{A I_m^{1/2}}{1+1.5 I_m^{1/2}} \tag{B.2a}$$

It should be mentioned that some authors have proposed different values for the coefficient in the denominator, ranging from 1.0 [1935GUG] to 1.6 [1962VAS]. However, this parameter is empirical and as such is correlated to the value of

$\varepsilon(j,k,I_m)$. Hence, this variety of values for the coefficient does not represent an uncertainty range, but rather indicates that several different sets of coefficients and $\varepsilon(j,k,I_m)$ may describe equally well the experimental mean activity coefficients of a given electrolyte. The ion interaction coefficients at 298.15 K listed in Table B-4, Table B-5, Table B-6, Table B-7 and Table B-8 have thus to be used with 1.5 $kg^{1/2} \cdot mol^{-1/2}$.

The summation in Eq. (B.1) extends over all ions *k* present in solution. Their molality is denoted by m_k, and the specific ion interaction parameters, $\varepsilon(j,k,I_m)$, in general depend only slightly on the ionic strength. The concentrations of the ions of the ionic medium are often very much larger than those of the reacting species. Hence, the ionic medium ions will make the main contribution to the value of $\log_{10}\gamma_j$ for the reacting ions. This fact often makes it possible to simplify the summation $\sum_k \varepsilon(j,k,I_m)m_k$, so that only ion interaction coefficients between the participating ionic species and the ionic medium ions are included, as shown in Eqs. (B.5) to (B.9).

- **Assumption 2:** The ion interaction coefficients, $\varepsilon(j,k,I_m)$ are zero for ions of the same charge sign and for uncharged species. The rationale behind this is that ε, which describes specific short-range interactions, must be small for ions of the same charge since they are usually far from one another due to electrostatic repulsion. This holds to a lesser extent also for uncharged species, but in some cases the experimental data allow determination of interaction coefficients for uncharged species with electroneutral combinations of ions, *cf.* Table B-7.

Equation (B.1) will allow fairly accurate estimates of the activity coefficients in mixtures of electrolytes if the ion interaction coefficients are known. Ion interaction coefficients for simple ions can be obtained from tabulated data of mean activity coefficients of strong electrolytes or from the corresponding osmotic coefficients. Ion interaction coefficients for complexes can either be estimated from the charge and size of the ion or determined experimentally from the variation of the equilibrium constant with the ionic strength.

Ion interaction coefficients are not strictly constant but may vary slightly with the ionic strength. The extent of this variation depends on the charge type and is small for 1:1, 1:2 and 2:1 electrolytes for molalities less than 3.5 m. The concentration dependence of the ion interaction coefficients can thus often be neglected. This point was emphasised by Guggenheim [1966GUG], who has presented a considerable amount of experimental material supporting this approach. The concentration dependence is larger for electrolytes of higher charge. In order to reproduce accurately their activity coefficient data, concentration dependent ion interaction coefficients have to be used, *cf.* Lewis *et al.* [1961LEW/RAN], Baes and Mesmer [1976BAE/MES], or Ciavatta [1980CIA].

For cases where the uncertainties in the epsilon values collected in Table B-4 and Table B-5 are ± 0.03 $kg \cdot mol^{-1}$ or greater, Ciavatta [1980CIA] proposed the use of Eq. (B.3)

$$\varepsilon = \varepsilon_1 + \varepsilon_2 \log_{10} I_m \tag{B.3}$$

However, even if the value of ε calculated in this way describes the variation with ionic strength slightly better than a constant value, this equation has no theoretical basis; ε_2 is a fitting parameter and the term $\varepsilon_2 \log_{10} I_m$ goes to minus infinity at the limiting value $I_m = 0$, even though the term $\varepsilon \cdot m = (\varepsilon_1 + \varepsilon_2 \log_{10} I_m) \cdot m$ (in the calculation of activity coefficients) is zero at $I_m = 0$. There may be cases where reviewers will still want to use Eq. (B.3) to describe the ionic-strength variation of the interaction parameters, but the rationale behind this should then be described. However, as discussed by Rand *et al.* [2008RAN/FUG] (*cf.* Section IX.1.3.3) the one- and two-epsilon models usually will provide similar calculated results for values of $\log_{10} K^\circ$. As the two-epsilon model has been used in the current and previous volumes, the relevant parameters have been retained and augmented in Table B-6.

By using a more elaborate virial expansion, which includes terms depending on the second power of molality, Pitzer and co-workers [1973PIT], [1973PIT/MAY], [1974PIT/KIM], [1974PIT/MAY], [1975PIT], [1976PIT/SIL], [1978PIT/PET], [1979PIT] have managed to describe measured activity coefficients of a large number of electrolytes with high precision over a large concentration range. Pitzer's model generally contains three parameters as compared to one in the specific ion interaction treatment. The use of the treatment requires the knowledge of all these parameters. The derivation of Pitzer coefficients for many complexes, such as those of the actinides would require a very large amount of additional experimental work, since few data of this type are currently available.

The way in which the activity-coefficient corrections are performed in this review according to the specific ion interaction treatment is illustrated below for a general case of a complex formation reaction. Charges on all species except the hydrogen ions are omitted for brevity.

$$m\,\mathrm{M} + q\,\mathrm{L} + n\,\mathrm{H_2O(l)} \rightleftharpoons \mathrm{M}_m\mathrm{L}_q(\mathrm{OH})_n + n\,\mathrm{H}^+$$

The formation constant of $\mathrm{M}_m\mathrm{L}_q(\mathrm{OH})_n$, ${}^*\beta_{q,n,m}$, determined in an ionic medium (1:1 salt NX) of the ionic strength I_m, is related to the corresponding value at zero ionic strength, ${}^*\beta^\circ_{q,n,m}$ by Eq. (B.4).

$$\begin{aligned}\log_{10} {}^*\beta_{q,n,m} = \log_{10} {}^*\beta^\circ_{q,n,m} + m \log_{10} \gamma_{\mathrm{M}} + q \log_{10} \gamma_{\mathrm{L}} \\ + n \log_{10} a_{\mathrm{H_2O}} - \log_{10} \gamma_{q,n,m} - n \log_{10} \gamma_{\mathrm{H}^+}\end{aligned} \tag{B.4}$$

The subscript (q,n,m) denotes the complex ion, $\mathrm{M}_m\mathrm{L}_q(\mathrm{OH})_n$. If the concentrations of N and X are much greater than the concentrations of M, L, $\mathrm{M}_m\mathrm{L}_q(\mathrm{OH})_n$ and H^+, only the molalities m_{N} and m_{X} have to be taken into account for the calculation of the term, $\sum_k \varepsilon(j,k,I_m) m_k$ in Eq. (B.1). For example, for the activity coefficient of the metal cation M, γ_{M}, Eq. (B.5) is obtained at 298.15 K and 1 bar.

$$\log_{10}\gamma_{\mathrm{M}} = \frac{-z_{\mathrm{M}}^2\, 0.509\sqrt{I_m}}{1 + 1.5\sqrt{I_m}} + \varepsilon(\mathrm{M},\mathrm{X},I_m)m_{\mathrm{X}} \qquad \text{(B.5)}$$

Under these conditions, $I_m \approx m_{\mathrm{X}} = m_{\mathrm{N}}$ Substituting the $\log_{10}\gamma_j$ values in Eq. (B.4) with the corresponding forms of Eq. (B.5) and rearranging leads to:

$$\log_{10}{}^{*}\beta_{q,n,m} - \Delta z^2 D - n\log_{10} a_{\mathrm{H_2O}} = \log_{10}{}^{*}\beta^{\circ}_{q,n,m} - \Delta\varepsilon I_m \qquad \text{(B.6)}$$

where, at 298.15 K and 0.1 MPa:

$$\Delta z^2 = (m z_{\mathrm{M}} - q z_{\mathrm{L}} - n)^2 + n - m z_{\mathrm{M}}^2 - q z_{\mathrm{L}}^2 \qquad \text{(B.7)}$$

$$D = \frac{0.509\sqrt{I_m}}{1 + 1.5\sqrt{I_m}} \qquad \text{(B.8)}$$

$$\Delta\varepsilon = \varepsilon(q,n,m,\mathrm{N\ or\ X}) + n\,\varepsilon(\mathrm{H},\mathrm{X}) - q\,\varepsilon(\mathrm{N},\mathrm{L}) - m\,\varepsilon(\mathrm{M},\mathrm{X}) \qquad \text{(B.9)}$$

Here $(m z_{\mathrm{M}} - q z_{\mathrm{L}} - n)$, z_{M} and z_{L} are the charges of the complex, $\mathrm{M}_m\mathrm{L}_q(\mathrm{OH})_n$, the metal ion M and ligand L, respectively.

Equilibria involving H_2O(l) as a reactant or product require a correction for the activity of water, $a_{\mathrm{H_2O}}$. The activity of water in an electrolyte mixture can be calculated as:

$$\log_{10} a_{\mathrm{H_2O}} = \frac{-\phi_m \sum_k m_k}{\ln(10)\, M_{\mathrm{H_2O}}^{-1}} \qquad \text{(B.10)}$$

where ϕ_m is the osmotic coefficient of the mixture, $M_{\mathrm{H_2O}}$ is the molar mass of H_2O, and the summation extends over all solute species *k* with molality m_k present in the solution. In the presence of an ionic medium NX as the dominant species, Eq. (B.10) can be simplified by neglecting the contributions of all minor species, *i.e.*, the reacting ions. Hence, for a 1:1 electrolyte of ionic strength $I_m \approx m_{\mathrm{NX}}$, Eq. (B.10) becomes:

$$\log_{10} a_{\mathrm{H_2O}} = \frac{-2\, m_{\mathrm{NX}}\, \phi_m}{\ln(10)\, M_{\mathrm{H_2O}}^{-1}} \qquad \text{(B.11)}$$

Alternatively, water activities can be taken from Table B-1. These have been calculated for the most common ionic media at various concentrations applying Pitzer's ion interaction model and the interaction parameters given in [1991PIT]. Data in *italics* have been calculated for concentrations beyond the validity of the parameter set applied. These data are therefore extrapolations and should be used with care.

Values of osmotic coefficients for single electrolytes have been compiled by various authors, *e.g.*, Robinson and Stokes [1959ROB/STO]. The activity of water can also be calculated from the known activity coefficients of the dissolved species. In the presence of an ionic medium, $\mathrm{N}_{\nu_+}\mathrm{X}_{\nu_-}$, of a concentration much larger than those of the

reacting ions, the osmotic coefficient can be calculated according to Eq. (B.12) (*cf.* Eqs. (23-39), (23-40) and (A4-2) in [1961LEW/RAN]).

$$1-\phi_m = \frac{A\ln(10)\left|z_+z_-\right|}{I_m(1.5)^3}\left[1+1.5\sqrt{I_m}-2\ln(1+1.5\sqrt{I_m})-\frac{1}{1+1.5\sqrt{I_m}}\right] - \ln(10)\ \varepsilon(\mathrm{N,X})\,m_{\mathrm{NX}}\left(\frac{\nu_+\nu_-}{\nu_++\nu_-}\right) \tag{B.12}$$

Here ν_+ and ν_- are the number of cations and anions in the salt formula ($\nu_+z_+ = \left|\nu_-z_-\right|$) and in this case:

$$I_m = \frac{1}{2}\left|z_+z_-\right| m_{\mathrm{NX}}(\nu_+ + \nu_-)\,.$$

Table B-1: Water activities $a_{\mathrm{H_2O}}$ for the most common ionic media at various concentrations applying Pitzer's ion interaction approach and the interaction parameters given in [1991PIT] and using density conversions from Table II-5. Data in *italics* have been calculated for concentrations beyond the validity of the parameter set applied. These data are therefore extrapolations and should be used with care.

	Water activities $a_{\mathrm{H_2O}}$ at 298.15 K							
c (M)	$HClO_4$	$NaClO_4$	$LiClO_4$	NH_4ClO_4	$Ba(ClO_4)_2$	HCl	NaCl	LiCl
0.100	0.9966	0.9966	0.9966	0.9967	0.9953	0.9966	0.9966	0.9966
0.250	0.9914	0.9917	0.9912	0.9920	0.9879	0.9914	0.9917	0.9915
0.500	0.9821	0.9833	0.9817	0.9844	0.9740	0.9823	0.9833	0.9826
0.750	0.9720	0.9747	0.9713	0.9769	0.9576	0.9726	0.9748	0.9731
1.000	0.9609	0.9660	0.9602	0.9694	0.9387	0.9620	0.9661	0.9631
1.500	0.9357	0.9476	0.9341	0.9542	0.8929	0.9386	0.9479	0.9412
2.000	0.9056	0.9279	0.9037		*0.8383*	0.9115	0.9284	0.9167
3.000	0.8285	0.8840	0.8280		*0.7226*	0.8459	0.8850	0.8589
4.000	0.7260	0.8331	*0.7309*			0.7643	0.8352	0.7991
5.000	*0.5982*	*0.7744*				0.6677	0.7782	0.7079
6.000	*0.4513*	*0.7075*				*0.5592*		*0.6169*
c (M)	KCl	NH_4Cl	$MgCl_2$	$CaCl_2$	NaBr	HNO_3	$NaNO_3$	$LiNO_3$
0.100	0.9966	0.9966	0.9953	0.9954	0.9966	0.9966	0.9967	0.9966
0.250	0.9918	0.9918	0.9880	0.9882	0.9916	0.9915	0.9919	0.9915
0.500	0.9836	0.9836	0.9744	0.9753	0.9830	0.9827	0.9841	0.9827
0.750	0.9754	0.9753	0.9585	0.9605	0.9742	0.9736	0.9764	0.9733
1.000	0.9671	0.9669	0.9399	0.9436	0.9650	0.9641	0.9688	0.9635
1.500	0.9500	0.9494	0.8939	0.9024	0.9455	0.9439	0.9536	0.9422
2.000	0.9320	0.9311	0.8358	0.8507	0.9241	0.9221	0.9385	0.9188

(Continued on next page)

Table B-1 (continued)

Water activities a_{H_2O} at 298.15 K								
c (M)	KCl	NH_4Cl	$MgCl_2$	$CaCl_2$	NaBr	HNO_3	$NaNO_3$	$LiNO_3$
3.000	0.8933	0.8918	0.6866	0.7168	0.8753	0.8737	0.9079	0.8657
4.000	0.8503	0.8491	0.5083	0.5511	*0.8174*	0.8196	0.8766	0.8052
5.000		*0.8037*		*0.3738*	*0.7499*	0.7612	0.8446	0.7390
6.000					*0.6728*	*0.7006*	*0.8120*	*0.6696*
c (M)	NH_4NO_3	Na_2SO_4	$(NH_4)_2SO_4$	Na_2CO_3	K_2CO_3	NaSCN		
0.500	0.9843	0.9813	0.9814	0.9805	0.9789	0.9828		
0.750	0.9768	0.9732	0.9728	0.9720	0.9683	0.9736		
1.000	0.9694	0.9653	0.9640	0.9637	0.9570	0.9641		
1.500	0.9548	0.9491	0.9455	0.9467	0.9316	0.9438		
2.000	0.9403		0.9247	0.9283	0.9014	0.9215		
3.000	0.9115		0.8735		0.8235	0.8708		
4.000	0.8829		0.8050		0.7195	*0.8115*		
5.000	*0.8545*				*0.5887*	*0.7436*		
6.000	*0.8266*					*0.6685*		

The activity of water is obtained by inserting Eq. (B.12) into Eq. (B.11). It should be mentioned that in mixed electrolytes with several components at high concentrations, it might be necessary to use Pitzer's equation to calculate the activity of water. On the other hand, a_{H_2O} is nearly constant in most experimental studies of equilibria in dilute aqueous solutions, where an ionic medium is used in large excess with respect to the reactants. The medium electrolyte thus determines the osmotic coefficient of the solvent.

In natural waters the situation is similar; the ionic strength of most surface waters is so low that the activity of $H_2O(l)$ can be set equal to unity. A correction may be necessary in the case of seawater, where a sufficiently good approximation for the osmotic coefficient may be obtained by considering NaCl as the dominant electrolyte.

In more complex solutions of high ionic strengths with more than one electrolyte at significant concentrations, *e.g.*, (Na^+, Mg^{2+}, Ca^{2+}) (Cl^-, SO_4^{2-}), Pitzer's equation (*cf.* [2013GRE/MOM]) may be used to estimate the osmotic coefficient; the necessary interaction coefficients are known for most systems of geochemical interest.

Note that in all ion interaction approaches, the equation for the mean activity coefficients can be split up to give equations for conventional single ion activity coefficients in mixtures, *e.g.*, Eq. (B.1). The latter are strictly valid only when used in combinations which yield electroneutrality. Thus, while estimating medium effects on standard potentials, a combination of redox equilibria with, $H^+ + e^- \rightleftharpoons \frac{1}{2}H_2(g)$, is necessary (*cf.* Example B.3).

B.1.2 Ionic-strength corrections at temperatures other than 25 °C (298.15 K)

Values of the Debye-Hückel parameter A in Eqs. (B.2), (B.2a) and (B.12) are listed in Table B-2 for several temperatures at a pressure of 1 bar below 100 °C and at the steam saturated pressure for $t \geq 100$ °C. The values in Table B-2 can be calculated from the static dielectric constant and the density of water as a function of temperature and pressure, as discussed in various references [1974HEL/KIR], [1979BRA/PIT], [1981HEL/KIR], [1984ANA/ATK], [1990ARC/WAN], [2009PAT/HRU]. The values in Table B-2 are those from Helgeson *et al*. [1981HEL/KIR]. More recent expressions for the density and dielectric constant (*e.g.*, those of Archer and Wang [1990ARC/WAN] or Pátek *et al*. [2009PAT/HRU]) would generate very slightly different values, but the Table B-2 values should be used to provide consistent TDB activity coefficient expressions.

Table B- 2: Values of the Debye-Hückel constant A as a function of temperature at a pressure of 1 bar below 100 °C and at the steam saturated pressure for $t \geq 100$ °C. The uncertainty in the A parameter is estimated by this review to be ± 0.001 $kg^{1/2} \cdot mol^{-1/2}$ at 25 °C, and ± 0.006 $kg^{1/2} \cdot mol^{-1/2}$ at 300 °C.

t/°C	p/bar	A/ $kg^{1/2} \cdot mol^{-1/2}$
0	1.00	0.491
5	1.00	0.494
10	1.00	0.498
15	1.00	0.501
20	1.00	0.505
25	1.00	0.509
30	1.00	0.514
35	1.00	0.518
40	1.00	0.523
50	1.00	0.534
75	1.00	0.564
100	1.013	0.600
125	2.32	0.642
150	4.76	0.690
175	8.92	0.745
200	15.5	0.810
250	39.7	0.979
300	85.8	1.256

In this review the term in the denominator of the Debye-Hückel term, D of Eq. (B.2), has been assigned a value of 1.5 $kg^{1/2} \cdot mol^{-1/2}$ (Eq. (B.2a)), and is kept constant at all temperatures. Assuming the ion size is in reality constant, it becomes obvious that this simplification introduces an error in D, which increases with temperature and ionic strength. However, this error is less than ± 0.01 $kg^{1/2} \cdot mol^{-1/2}$ at $t \leq 100$ °C and $I_m < 6$, and less than ± 0.006 $kg^{1/2} \cdot mol^{-1/2}$ at $t \leq 50$ °C and $I_m \leq 4$.

The values of $\varepsilon(j,k,I_m)$, obtained with the methods described in Section B.1.3 at temperatures other than 25 °C, will depend on the value adopted for Ba_{j^-}. Nevertheless, as long as a consistent approach is followed, values of $\varepsilon(j,k,I_m)$ absorb the choice of Ba_j, and for moderate temperature intervals (between 0 and 200 °C) the choice Ba_j = 1.5 $kg^{1/2}\cdot mol^{-1/2}$ is simple and is recommended by this review.

The variation of $\varepsilon(j,k,I_m)$ with temperature is discussed by Lewis *et al.* [1961LEW/RAN], Millero [1979MIL], Helgeson *et al.* [1981HEL/KIR], [1990OEL/HEL], Giffaut *et al.* [1993GIF/VIT2] and Grenthe and Plyasunov [1997GRE/PLY]. The absolute values for the reported ion interaction parameters differ in these studies due to the fact that the Debye-Hückel term used by these authors is not exactly the same. Nevertheless, common to all these studies is the fact that values of $(\partial\varepsilon/\partial T)_p$ are usually ≤ 0.005 $kg\cdot K^{-1}\cdot mol^{-1}$ for temperatures below 200 °C. Therefore, if values of $\varepsilon(j,k,I_m)$ obtained at 25 °C are used in the temperature range 0 to 50 °C to perform ionic strength corrections, the error in $\log_{10}\gamma_j/I_m$ will be ≤ 0.13 $kg\cdot mol^{-1}$. It is clear that in order to reduce the uncertainties in solubility calculations at $t \neq 25$ °C, studies on the variation of $\varepsilon(j,k,I_m)$ values with temperature should be undertaken.

B.1.3 Estimation of ion interaction coefficients

B.1.3.1 Estimation from mean activity coefficient data

Example B.1:

The ion interaction coefficient $\varepsilon(H^+,Cl^-)$ can be obtained from published values of $\gamma_{\pm,\,HCl}$ *versus* m_{HCl}:

$$\begin{aligned} 2\log_{10}\gamma_{\pm,\,HCl} &= \log_{10}\gamma_{H^+} + \log_{10}\gamma_{Cl^-} \\ &= -D + \varepsilon(H^+,Cl^-)\,m_{Cl^-} \; -D \; + \; \varepsilon(H^+,Cl^-)\,m_{H^+} \\ \log_{10}\gamma_{\pm,\,HCl} &= -D + \varepsilon(H^+,Cl^-)\,m_{HCl} \end{aligned}$$

By plotting $(\log_{10}\gamma_{\pm,HCl} + D)$ *versus* m_{HCl} a straight line with the slope $\varepsilon(H^+,Cl^-)$ is obtained. The degree of linearity should in itself indicate the range of validity of the specific ion interaction approach. Osmotic coefficient data can be treated in an analogous way.

B.1.3.2 Estimations based on experimental values of equilibrium constants at different ionic strengths

Example B.2:

Equilibrium constants are given in Table B-3 for the reaction

$$UO_2^{2+} + 2CO_3^{2-} \rightleftharpoons UO_2(CO_3)_2^{2-} \qquad \text{(B.13)}$$

The following formula is deduced from Eq. (B.6) for the extrapolation to $I = 0$:

$$\log_{10} \beta_2 + 8D = \log_{10} \beta_2^{\circ} - \Delta\varepsilon I_m \,. \tag{B.14}$$

A linear regression gives

$$\log_{10} \beta_2^{\circ} = (16.94 \pm 0.12)$$

$$\Delta\varepsilon \text{ (B.13)} = -\,(0.32 \pm 0.06) \text{ kg·mol}^{-1}$$

where the uncertainties are calculated as described in the NEA Guidelines for the Assignment of Uncertainties [1999NEA].

The experimental data are depicted in Figure B-1.

Table B- 3: The experimental equilibrium constants for the extrapolation to $I = 0$ with the specific ion interaction method at 25 °C and 1 bar according to Reaction (B.13). The linear regression of this set of data is shown in Figure B-1.

Method	Ionic medium	$\log_{10} \beta_2$ [(a)]	$\log_{10} \beta_2^{\circ}$ [(a)]	Reference
sol	0		14.6	[1956MCC/BUL]
sol	0.2 M NH_aNO_3	15.6		[1960BAB/KOD]
emf[(b)], gl	0.1 M $NaClO_4$	16.2 ± 0.3 [(c,e)]		[1969TSY]
sol	0.03 M	16.7 ± 0.3 [(e)]		[1972SER/NIK]
sol	var.	4.0		[1973ALM/GAR]
dis	0.1 M $NaNO_3$	16.22 ± 0.30 [(e)]		[1977SCA]
emf, gl	0.1 M $NaClO_4$	16.15 ± 0.30 [(e)]		[1982MAY]
sol	0.5 M $NaClO_4$	15.56 ± 0.15 [(c,d)]		[1984GRE/FER]
	3.0 M $NaClO_4$	16.20 ± 0.15 [(c,d)]		
emf, gl	0.5 M $NaClO_4$	14.93 ± 0.30		[1991GRE/LAG]
rev	0		17.0	[1978LAN]
rev	0		17.1 ± 0.4	[1980LEM/TRE]

(a) Refers to the reaction indicated, $\log_{10} K$ in the ionic medium given and $\log_{10} K^{\circ}$ (molal units) at $I = 0$ and 25 °C.

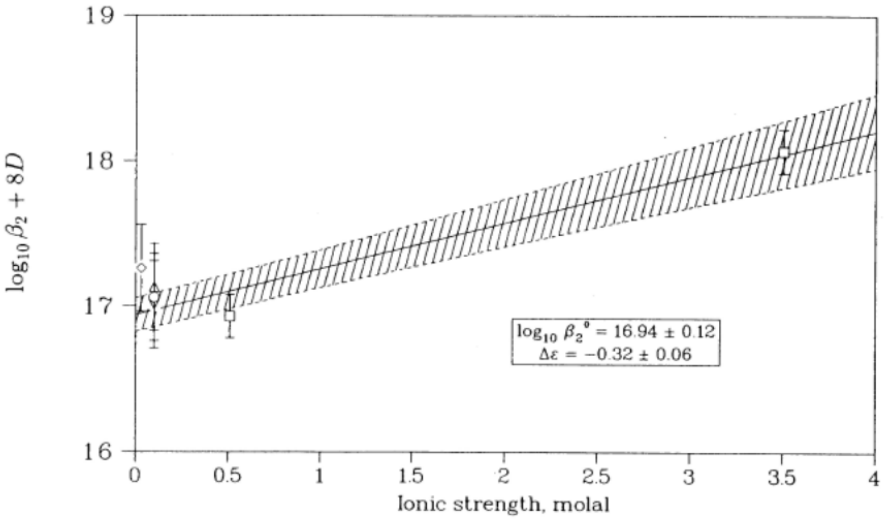
(b) emf = electromotive force

(c) Re-evaluated in [1992GRE/FUG].

(d) The reported constant is corrected for the different protonation constants of carbonate used in [1992GRE/FUG].

(e) Uncertainties estimated in [1992GRE/FUG].

Figure B- 1: Extrapolation to $I = 0$ of experimental data for the formation of $UO_2(CO_3)_2^{2-}$ using the specific ion interaction theory. The shaded area represents the uncertainty range obtained by propagating the resulting uncertainties at $I = 0$ back to $I_m = 4$.



Example B.3:

When using the specific ion interaction treatment, the relationship between the redox potential of the couple, UO_2^{2+}/U^{4+}, in a perchlorate medium of ionic strength, I_m, and the corresponding quantity at $I = 0$ should be calculated in the following way. The reaction in the galvanic cell:

$$Pt \mid H_2(g) \mid H^+ \vdots\vdots UO_2^{2+}, U^{4+} \mid Pt$$

is:

$$UO_2^{2+} + H_2(g) + 2H^+ \rightleftharpoons U^{4+} + 2H_2O(l)\,. \qquad (B.15)$$

For this reaction

$$\log_{10} K^\circ = \log_{10}\left(\frac{a_{U^{4+}} \cdot a_{H_2O}^2}{a_{UO_2^{2+}} \cdot a_{H^+}^2 \cdot f_{H_2}}\right).$$

$$\log_{10} K^\circ = \log_{10} K + \log_{10}\gamma_{U^{4+}} - \log_{10}\gamma_{UO_2^{2+}} - 2\log_{10}\gamma_{H^+} - \log_{10}\gamma_{f,H_2} + 2\log_{10} a_{H_2O},$$

$f_{H_2} \approx p_{H_2}$ at reasonably low partial pressure of $H_2(g)$, $a_{H_2O} \approx 1$ (which is a reasonable approximation for $I_m \leq 3$), and

$$\log_{10}\gamma_{U^{4+}} = -16D + \varepsilon(U^{4+}, ClO_4^-)\, m_{ClO_4^-}$$

$$\log_{10}\gamma_{UO_2^{2+}} = -4D + \varepsilon(UO_2^{2+}, ClO_4^-)\, m_{ClO_4^-}$$

$$\log_{10}\gamma_{H^+} = -D + \varepsilon(H^+, ClO_4^-)\, m_{ClO_4^-}$$

Hence,

$$\log_{10} K^\circ = \log_{10} K - 10D + (\varepsilon(U^{4+}, ClO_4^-) - \varepsilon(UO_2^{2+}, ClO_4^-) - 2\varepsilon(H^+, ClO_4^-))\, m_{ClO_4^-} \quad \text{(B.16)}$$

The relationship between the equilibrium constant and the redox potential is:

$$\ln K = \frac{nF}{RT} E \quad \text{(B.17)}$$

$$\ln K^\circ = \frac{nF}{RT} E^\circ. \quad \text{(B.18)}$$

E is the redox potential in a medium of ionic strength I, E° is the corresponding standard potential at $I = 0$, and n is the number of transferred electrons in the reaction considered. Combining Eqs. (B.16), (B.17) and (B.18) and rearranging them leads to Eq. (B.19):

$$E - 10D\left(\frac{RT\ln(10)}{nF}\right) = E^\circ - \Delta\varepsilon\, m_{ClO_4^-}\left(\frac{RT\ln(10)}{nF}\right) \quad \text{(B.19)}$$

For $n = 2$ in the present example and $T = 298.15$ K, Eq. (B.19) becomes:

$$E/\text{mV} - 295.8\ D = E^\circ/\text{mV} - 29.58\,\Delta\varepsilon m_{ClO_4^-}$$

where

$$\Delta\varepsilon = \varepsilon(U^{4+}, ClO_4^-) - \varepsilon(UO_2^{2+}, ClO_4^-) - 2\varepsilon(H^+, ClO_4^-).$$

B.1.4 On the magnitude of ion interaction coefficients

Ciavatta [1980CIA] made a compilation of ion interaction coefficients for a large number of electrolytes. Similar data for complex formation reactions of various kinds were reported by Spahiu [1983SPA] and Ferri *et al.* [1983FER/GRE]. These and some other data for 25 °C and 0.1 MPa have been collected and are listed in Section B.3.

It is obvious from the data in these tables that the charge of an ion is of great importance for determining the magnitude of the ion interaction coefficient. Ions of the same charge type have similar ion interaction coefficients with a given counter-ion. Based on the tabulated data, Grenthe *et al*. [1992GRE/FUG] proposed that it is possible to estimate, with an error of at most ± 0.1 kg·mol^{-1} in ε, ion interaction coefficients for cases where there are insufficient experimental data for an extrapolation to $I = 0$. The error that is made by this approximation is estimated to be ± 0.1 kg·mol^{-1} in $\Delta\varepsilon$ in most cases, based on comparison with $\Delta\varepsilon$ values of various reactions of the same charge type.

B.2 Weak complexes within an ion interaction approach: Ion interaction coefficients versus equilibrium constants for ion pairs

A problem often encountered in practice is that of adequately modelling systems where weak complexes are formed. In these cases the distinction between complex formation and activity coefficient variations is unclear or even arbitrary. As long as the fraction associated is rather small, the mathematical effect of an ion association is the same as that of a negative contribution to a virial coefficient [1981PIT]. This means that chloride salts have lower interaction coefficients than perchlorates for the same cation, since they are partially associated.

Usually, large changes of the ionic medium are necessary to study weak complex formation by substituting in most cases completely the medium anion by the ligand. This causes changes of the activity coefficients of the species involved, even though the stoichiometric ionic strength is usually maintained constant. In this case, it is recommended to use the general formulation of the SIT equation (see Eq. B.1), and not the simplified form (Eq. B.5). Then, the reviewer has to decide what terms, *e.g.* $\varepsilon(i,j)\,[j]$, can be neglected among all the SIT terms. If only L^- has a non negligible concentration compared with that of ClO_4^-, then the procedure of extrapolation should be as follows.

The reviewer is advised to estimate first these changes of activity coefficients during the experimental determination of the constants at each ionic strength. For practical reasons [1985SPA] it is advantageous to carry out this part of the calculation in molar scale. The problem involves estimation of the activity coefficient changes for the equilibrium:

$$M^{m+} + n\,L^- \rightleftharpoons ML_n^{(m-n)},\ \beta_n^{\circ} = \frac{[ML_n^{(m-n)}]}{[M^{m+}]\cdot[L^-]^n}\cdot\frac{\gamma_{c,ML_n^{(m-n)}}}{\gamma_{c,M^{m+}}\cdot\gamma_{c,L}^n} \qquad \text{(B.20)}$$

in mixtures with a common cation N and of total ionic strength I_c during the substitution of perchlorate by the anion L of the weak complex. The reference state is I_c $NClO_4$, *i.e.* activity coefficients tend to unity when [L] goes to zero, and the composition of the medium becomes that of the reference state. For both the SIT and Pitzer approaches, at the concentrations relevant for most studies of weak complexes, the ion interaction part is dominated by linear terms in $[ClO_4^-]$ or $[L^-]$ (ε in SIT and $\beta^{(0)} + \theta_{i,j}$ in the Pitzer approach) as all the other species are at trace concentrations. Since $[ClO_4^-] = I_c - [L^-]$ and because I and also $D_H(I)$ are constant, the value of $\log_{10}\beta_n$ at constant ionic strength must vary linearly with $[L^-]$. In some cases, as in spectrophotometric studies [1949HIN], this change in $\log_{10}\beta_n$ during the titration has been interpreted as an experimental error, since the spectra did not show evidence for new complexes. In many cases, however, the variation of activity coefficients at constant ionic strength has been misinterpreted as the formation of one or several (up to 3) complexes.

By denoting the ratio of molar activity coefficients $G = \gamma_{M^{m+}} \cdot \gamma_L^n \Big/ \gamma_{ML_n}$, Eq.(B.20) can be written also:

$$\beta_n = \frac{[ML_n]}{[M^{m+}][L^-]^n} = \beta_n^o \frac{\gamma_{M^{m+}} \cdot \gamma_L^n}{\gamma ML_n} = \beta_n^o \cdot G$$

By taking the logarithm of the ratio of $\beta_{n,\text{mix}} = \beta_n^o \cdot G_{\text{mix}}$ to $\beta_{n,\text{perchl}} = \beta_n^o \cdot G_{\text{perchl}}$, we obtain the expression:

$$\log_{10} \beta_{n,\text{ mix}} = \log_{10} \beta_{n,\text{ perchl}} + \log_{10} G_{\text{mix}} - \log_{10} G_{\text{perchl}}$$

By using as a reference state the pure background electrolyte (*e.g.*, $NaClO_4$) at an ionic strength, I, it is possible to obtain an expression for activity coefficient of species i in ($NaClO_4$ – NaL) mixtures at the same ionic strength:

$$\begin{aligned}\log_{10} \gamma_{c\,(i,\text{ mix})} - \log_{10} \gamma_{c\,(i,\text{ perchl})} &= -z_i^2 \{D_H(I) - D_H(I)\} + \varepsilon(i, L^-)\,[L^-] \\ &\quad + \varepsilon(i, ClO_4^-)\,(I - [L^-]) - \varepsilon(i, ClO_4^-)\,I \\ &= -\{\varepsilon(i, ClO_4^-) - \varepsilon(i, L^-)\}\,[L^-].\end{aligned}$$

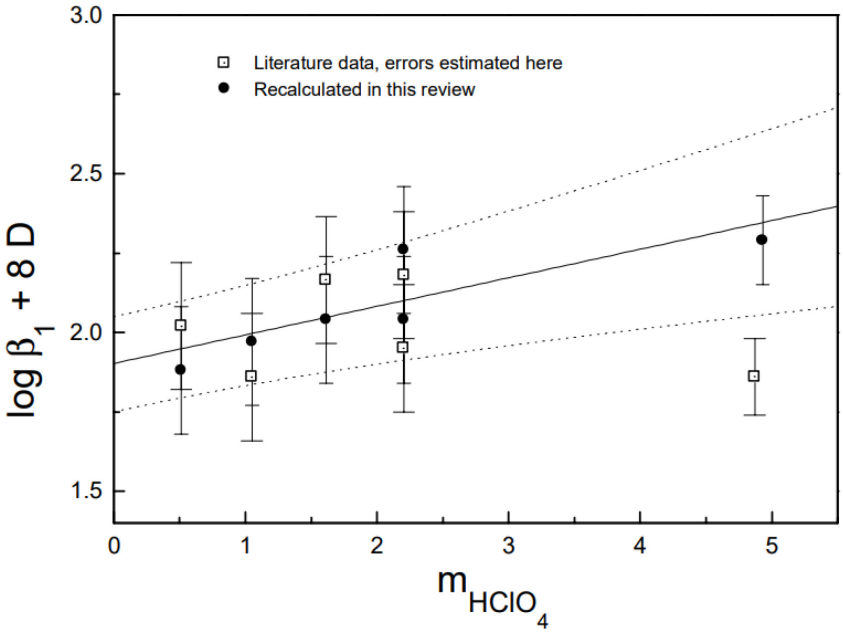
Substitution of these expressions for all species participating in the complex formation equilibrium (B.20) gives relatively simple linear expressions relating $\log_{10} \beta_n$ to $\log_{10} \beta_n^o$ from which a value of the constant β_n^o at the given ionic medium level and trace concentrations of the ligand is obtained.

The values of the constants thus obtained may then be converted to molal scale and used in the standard SIT extrapolation at zero ionic strength, an example of which for the formation of $NpNO_3^{3+}$ is shown in Figure B-2.

As can be seen from Figure B-2 which contains reported and corrected formation constants for $Np^{4+} + NO_3^- \rightleftharpoons NpNO_3^{3+}$, such corrections are not very large (compare filled and empty symbols at each ionic strength level) and no real loss of accuracy in the value of the constant at zero ionic strength which is finally selected in the review is expected to originate from not carrying out this kind of corrections.

However, what the reviewer gains is: firstly, a more consistent set of data; secondly, in the process several very weak higher-order complexes which have no independent chemical validation are excluded and thirdly, the values of $\Delta\varepsilon$ obtained do not result in the inconsistencies commented in Chapter V of the Organics review [2005HUM/AND].

Figure B- 2: Extrapolation to infinite dilution using the SIT (open squares) and data re-calculated in [2001LEM/FUG] at trace nitrate concentrations (filled circles).



In conclusion, for the analysis of systems where weak complexes are suspected, the following procedure should be followed:

- The review of the original work should preferably include an estimation of the activity coefficient changes occurring during the addition of the weak ligand at each constant ionic strength. If during the course of an experiment more than 10 % of the background electrolyte is replaced by the ligand, calculation of an independent association constant value cannot usually be justified because it violates the ionic medium principle and large changes in activity coefficients during the measurements.
- A standard SIT treatment on a molar scale at each ionic strength level should allow the value of the equilibrium constant at trace ligand concentration to be estimated. If this calculation shows that the system can equally well be described without recourse to complexation, this should be stated in the Review. Otherwise the data thus obtained for each ionic strength should be used to obtain by extrapolation the value at zero ionic strength with the usual SIT procedure (and statistical tests).

B.3 Tables of ion interaction coefficients

Table B-4, Table B-5, Table B-6 and Table B-7 contain the selected specific ion interaction coefficients used in this review, according to the specific ion interaction treatment described. Table B-4 contains cation interaction coefficients with Cl^-, ClO_4^- and NO_3^-, Table B-5 anion interaction coefficients with Li^+, Na^+ (or NH_4^+) and K^+, and Table B-7 neutral species – electroneutral combination of ions. The coefficients have the units of kg·mol^{-1} and are valid for 298.15 K and 0.1 MPa. The species are ordered by charge and appear, within each charge class, in the standard order of arrangement, *cf.* Section II.1.8.

Table B- 4: Ion interaction coefficients $\varepsilon(j,k)$ (kg·mol^{-1}) for cations j with k = Cl^-, ClO_4^- and NO_3^- at 298.15 K. The uncertainties represent the 95% confidence level. The ion interaction coefficients marked with † can be described more accurately with an ionic-strength dependent function, listed in Table B-6. The coefficients $\varepsilon(M^{n+}, Cl^-)$ and $\varepsilon(M^{n+}, NO_3^-)$ reported by Ciavatta [1980CIA] were evaluated without taking chloride and nitrate complexation into account, as discussed in Section B.2.

j	k	$\varepsilon(j,k)$	Comments
H^+	Cl^-	0.12 ± 0.01	
	ClO_4^-	0.14 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	0.07 ± 0.01	
NH_4^+	Cl^-	-0.01 ± 0.01	Reported by Ciavatta [1980CIA].
	ClO_4^-	$-0.08 \pm 0.04^\dagger$	
	NO_3^-	$-0.06 \pm 0.03^\dagger$	
H_2gly^+	Cl^-	-0.06 ± 0.02	Reported by Ciavatta [1988CIA].
	ClO_4^-	—	
	NO_3^-	—	
H_5edta^+	Cl^-	-0.23 ± 0.15	
	ClO_4^-	-0.23 ± 0.15	See Section VIII.3.7 of [2005HUM/AND].
	NO_3^-	-0.23 ± 0.15	
$SnOH^+$	Cl^-	—	
	ClO_4^-	-0.07 ± 0.13	See Section VII.1.1 of [2012GAM/GAJ].
	NO_3^-	—	
SnF^+	Cl^-	—	
	ClO_4^-	0.14 ± 0.10	See Section VIII.3.1.1 of [2012GAM/GAJ].
	NO_3^-	—	
$SnCl^+$	Cl^-	—	
	ClO_4^-	0.08 ± 0.07	See Section VIII.3.2.1 of [2012GAM/GAJ].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$SnBr^+$	Cl^-	—	
	ClO_4^-	0.15 ± 0.07	See Section VIII.3.3.1 of [2012GAM/GAJ].
	NO_3^-	—	
$SnNO_3^+$	Cl^-	—	
	ClO_4^-	0.17 ± 0.09	See Section X.1.1 of [2012GAM/GAJ].
	NO_3^-	—	
$SnSCN^+$	Cl^-	—	
	ClO_4^-	0.17 ± 0.29	See Section XI.1.1 of [2012GAM/GAJ].
	NO_3^-	—	
Tl^+	Cl^-	—	
	ClO_4^-	−0.21 ± 0.06†	
	NO_3^-	—	
$ZnHCO_3^+$	Cl^-	0.2	Taken from Ferri *et al.* [1985FER/GRE].
	ClO_4^-	—	
	NO_3^-	—	
$CdCl^+$	Cl^-	—	
	ClO_4^-	0.25 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	—	
CdI^+	Cl^-	—	
	ClO_4^-	0.27 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	—	
$CdSCN^+$	Cl^-	—	
	ClO_4^-	0.31 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	—	
$HgCl^+$	Cl^-	—	
	ClO_4^-	0.19 ± 0.02	Reported by Ciavatta [1988CIA].
	NO_3^-	—	
Cu^+	Cl^-	—	
	ClO_4^-	0.11 ± 0.01	Reported by Ciavatta [1980CIA].
	NO_3^-	—	
Ag^+	Cl^-	—	
	ClO_4^-	0.00 ± 0.01	Reported by Ciavatta [1980CIA].
	NO_3^-	−0.12 ± 0.05†	
$NiOH^+$	Cl^-	−0.01 ± 0.07	Evaluated in [2005GAM/BUG] (Section V.3.1.1) for the reaction $Ni^{2+} + H_2O \rightleftharpoons NiOH^+ + H^+$ from $\Delta\varepsilon$ in chloride media/perchlorate media.
	ClO_4^-	0.14 ± 0.07	
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
NiF^+	Cl^-	—	
	ClO_4^-	0.34 ± 0.08	Derived from $\Delta\varepsilon = \varepsilon(NiF^+,ClO_4^-) - \varepsilon(Ni^{2+},ClO_4^-) - \varepsilon(Na^+,F^-) = -(0.049 \pm 0.060)$ kg·mol^{-1} (see Section V.4.2.3 of [2005GAM/BUG]).
	NO_3^-	—	
$NiCl^+$	Cl^-	—	
	ClO_4^-	0.47 ± 0.06	See details in Section V.4.2.4 of [2005GAM/BUG].
	NO_3^-	—	
$NiNO_3^+$	Cl^-	—	
	ClO_4^-	0.44 ± 0.14	See details in Section V.6.1.2 of [2005GAM/BUG], specially sub-section V.6.1.2.1 for an alternative treatment of this system.
	NO_3^-	—	
$Ni(H_2cit)^+$	Cl^-	—	
	ClO_4^-	0.12 ± 0.50	See Section VII.7 in [2005HUM/AND].
	NO_3^-	—	
$NiBr^+$	Cl^-	—	
	ClO_4^-	0.59 ± 0.10	See details in [2005GAM/BUG], *cf*. Section V.4.2.5, specially sub-section V.4.2.5.1 for an alternative treatment of this system.
	NO_3^-	—	
$NiHS^+$	Cl^-	—	
	ClO_4^-	−0.85 ± 0.39	See details in [2005GAM/BUG], Section V.5.1.1.2.
	NO_3^-	—	
$NiSCN^+$	Cl^-	—	
	ClO_4^-	0.31 ± 0.04	Derived from $\Delta\varepsilon = \varepsilon(NiSCN^+,ClO_4^-) - \varepsilon(Na^+,SCN^-) - \varepsilon(Ni^{2+},ClO_4^-) = -(0.109 \pm 0.025)$ kg·mol^{-1} (see [2005GAM/BUG], Section V.7.1.3.1).
	NO_3^-	—	
$Fe(OH)_2^+$	Cl^-	—	
	ClO_4^-	0.37 ± 0.18	Determined in Section VII.1.3.2 of [2013LEM/BER].
	NO_3^-	—	
FeF^+	Cl^-		
	ClO_4^-	0.34 ± 0.07	Determined in Section VIII.2.1.3 of [2013LEM/BER].
	NO_3^-		
$FeCl_2^+$	Cl^-	—	
	ClO_4^-	0.52 ± 0.05	Determined in Section VIII.2.3.2.2.1 of [2013LEM/BER].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$FeSO_3^+$	Cl^-	—	
	ClO_4^-	0.4	Determined in Section IX.2.2.3 of [2020LEM/PAL].
	NO_3^-	—	
$FeSO_4^+$	Cl^-	—	
	ClO_4^-	0.4 ± 0.1	Determined in Section IX.1.2.1.4.1 of [2013LEM/BER].
	NO_3^-	—	
$Zr_4(OH)_{15}^+$	Cl^-	—	
	ClO_4^-	0.09 ± 0.92	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.
	NO_3^-	– 0.02 ± 1.46	
ZrF_3^+	Cl^-	—	
	ClO_4^-	0.20 ± 0.06	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
YCO_3^+	Cl^-	—	
	ClO_4^-	0.17 ± 0.04	Taken from Spahiu [1983SPA].
	NO_3^-	—	
$Am(OH)_2^+$	Cl^-	– 0.27 ± 0.20	Evaluated in [2003GUI/FAN] (*cf.* Section 12.3.1.1) from $\Delta\varepsilon$ (in NaCl solution) for the reactions $An^{3+} + nH_2O(l) \rightleftharpoons An(OH)_n^{(3-n)} + nH^+$ (An = Am, Cm).
	ClO_4^-	0.17 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
AmF_2^+	Cl^-	—	
	ClO_4^-	0.17 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmSO_4^+$	Cl^-	—	
	ClO_4^-	0.22 ± 0.08	Evaluated in [1995SIL/BID].
	NO_3^-	—	
$AmCO_3^+$	Cl^-	0.01 ± 0.05	Evaluated in [2003GUI/FAN] (Section 12.6.1.1.1) from $\Delta\varepsilon$ (in NaCl solution) for the reactions $An^{3+} + n\,CO_3^{2-} \rightleftharpoons An(CO_3)_n^{(3-2n)}$ (based on $\varepsilon(Am^{3+}, Cl^-) = (0.23 \pm 0.02)$ kg·mol^{-1} and $\varepsilon(Na^+, CO_3^{2-}) = -(0.08 \pm 0.03)$ kg·mol^{-1}.
	ClO_4^-	0.17 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$Am(ox)^+$	Cl^-	—	
	ClO_4^-	0.08 ± 0.10	See Section VI.13 of [2005HUM/AND].
	NO_3^-	—	
PuO_2^+	Cl^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
	ClO_4^-	0.24 ± 0.05	Derived from $\Delta\varepsilon = \varepsilon(PuO_2^{2+}, ClO_4^-) - \varepsilon(PuO_2^+, ClO_4^-)$ = (0.22 ± 0.03) kg·mol^{-1} [1995CAP/VIT]. In [1992GRE/FUG], $\varepsilon(PuO_2^+, ClO_4^-)$ = (0.17 ± 0.05) kg·mol^{-1} was tabulated based on [1989ROB], [1989RIG/ROB] and [1990RIG]. Capdevila and Vitorge's data [1992CAP], [1994CAP/VIT] and [1995CAP/VIT] were unavailable at that time.
	NO_3^-	—	
PuO_2F^+	Cl^-	—	
	ClO_4^-	0.29 ± 0.11	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding Np(IV) reaction.
	NO_3^-	—	
PuO_2Cl^+	Cl^-	—	
	ClO_4^-	0.50 ± 0.09	From $\Delta\varepsilon$ evaluated by Giffaut [1994GIF].
	NO_3^-	—	
NpO_2^+	Cl^-	0.09 ± 0.05	See Section 12.1 of [2001LEM/FUG].
	ClO_4^-	0.25 ± 0.05	Derived from $\Delta\varepsilon = \varepsilon(NpO_2^{2+}, ClO_4^-) - \varepsilon(NpO_2^+, ClO_4^-)$ = (0.21 ± 0.03) kg·mol^{-1} [1987RIG/VIT], [1989RIG/ROB] and [1990RIG].
	NO_3^-	—	
NpO_2OH^+	Cl^-	—	
	ClO_4^-	– 0.06 ± 0.40	Estimated in [2001LEM/FUG].
	NO_3^-	—	
$(NpO_2)_3(OH)_5^+$	Cl^-	—	
	ClO_4^-	0.45 ± 0.20	See Section 8.1.2 of [2001LEM/FUG].
	NO_3^-	—	
NpO_2F^+	Cl^-	—	
	ClO_4^-	0.29 ± 0.12	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction.
	NO_3^-	—	
NpO_2Cl^+	Cl^-	—	
	ClO_4^-	0.50 ± 0.14	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding Pu(VI) reaction.
	NO_3^-	—	
$NpO_2IO_3^+$	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Estimated in [2001LEM/FUG] by assuming $\varepsilon(NpO_2IO_3^+, ClO_4^-) \approx \varepsilon(UO_2IO_3^+, ClO_4^-)$.
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Np(SCN)_3^+$	Cl^-	—	
	ClO_4^-	0.17 ± 0.04	Estimated in [2001LEM/FUG] by assuming $\varepsilon(Np(SCN)_3^+, ClO_4^-) \approx \varepsilon(AmF_2^+, ClO_4^-)$.
	NO_3^-	—	
$Ca[NpO_2(OH)_2]^+$	Cl^-	– 0.07 ± 0.08	Estimated in [2020GRE/GAO].
	ClO_4^-	—	
	NO_3^-	—	
UO_2^+	Cl^-	—	
	ClO_4^-	0.26 ± 0.03	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+}, ClO_4^-)$ = (0.46 ± 0.03) kg·mol^{-1}.
	NO_3^-	—	
UO_2OH^+	Cl^-	—	
	ClO_4^-	– 0.06 ± 0.40	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+}, X)$ = (0.46 ± 0.03) kg·mol^{-1}, where X = Cl^-, ClO_4^- and NO_3^-
	NO_3^-	0.51 ± 1.4	
$(UO_2)_3(OH)_5^+$	Cl^-	0.81 ± 0.17	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+}, X)$ = (0.46 ± 0.03) kg·mol^{-1}, where X = Cl^-, ClO_4^- and NO_3^-
	ClO_4^-	0.45 ± 0.15	
	NO_3^-	0.41 ± 0.22	
UF_3^+	Cl^-	0.1 ± 0.1	Estimated in [1992GRE/FUG].
	ClO_4^-	0.1 ± 0.1	
	NO_3^-	—	
UO_2F^+	Cl^-	0.04 ± 0.07	Taken from Riglet *et al.* [1989RIG/ROB], where the following assumptions were made: $\varepsilon(Np^{3+}, ClO_4^-) \approx \varepsilon(Pu^{3+}, ClO_4^-)$ = 0.49 kg·mol^{-1} as for other (M^{3+}, ClO_4^-) interactions, and $\varepsilon(NpO_2^{2+}, ClO_4^-) \approx \varepsilon(PuO_2^{2+}, ClO_4^-) \approx \varepsilon(UO_2^{2+}, ClO_4^-)$ = 0.46 kg·mol^{-1}.
	ClO_4^-	0.28 ± 0.04	See Section 9.4.2.2.1 of [2003GUI/FAN].
	NO_3^-	—	
UO_2Cl^+	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+}, X)$ = (0.46 ± 0.03) kg·mol^{-1}, where X = Cl^-, ClO_4^- and NO_3^-.
	NO_3^-	—	
$UO_2ClO_3^+$	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
UO_2Br^+	Cl^-	—	
	ClO_4^-	0.24 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$UO_2BrO_3^+$	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$UO_2IO_3^+$	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$UO_2N_3^+$	Cl^-	—	
	ClO_4^-	0.3 ± 0.1	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$UO_2NO_3^+$	Cl^-	—	
	ClO_4^-	0.33 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	
UO_2SCN^+	Cl^-	—	
	ClO_4^-	0.22 ± 0.04	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$Th(OH)_3^+$	Cl^-	0.06 ± 0.05	See Table VII-16 in Section VII.3.6.1 of [2008RAN/FUG].
	ClO_4^-	0.15 ± 0.10	
	NO_3^-	0.05 ± 0.15	
ThF_3^+	Cl^-	—	
	ClO_4^-	0.1 ± 0.1	See Table VIII-8 in Section VIII.1.2.1 of [2008RAN/FUG].
	NO_3^-	0.0 ± 0.2	
$Th(NO_3)_3^+$	Cl^-	—	
	ClO_4^-	0.25 ± 0.15	Evaluated in Section X.1.3.3 of [2008RAN/FUG], using $\varepsilon(Th^{4+},X) = (0.70 \pm 0.10)$ kg·mol^{-1} where X = ClO_4^- and NO_3^-.
	NO_3^-	0.25 ± 0.15	
$MgOH^+$	Cl^-	0.12 ± 0.03	This review. $I_m \leq 5.0$. See Section V.6.1.2.
	ClO_4^-	—	
	NO_3^-	—	
H_6edta^{2+}	Cl^-	−0.20 ± 0.16	Evaluated in [2005HUM/AND] (Section VIII.3.7).
	ClO_4^-	−0.20 ± 0.16	
	NO_3^-	−0.20 ± 0.16	
Sn^{2+}	Cl^-	0.19 ± 0.04	See Section VI.2.1 of [2012GAM/GAJ].
	ClO_4^-	0.19 ± 0.04	See Section VI.2.1 of [2012GAM/GAJ].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Sn_3(OH)_4^{2+}$	Cl^-	—	
	ClO_4^-	-0.02 ± 0.16	See Section VII.1.1 of [2012GAM/GAJ].
	NO_3^-	—	
Pb^{2+}	Cl^-	—	
	ClO_4^-	0.15 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	$-0.20 \pm 0.12^\dagger$	
$AlOH^{2+}$	Cl^-	$0.040 \pm 0.051^\dagger$	This review. $I_m \leq 1.0$. See Section V.5.2.5.
	ClO_4^-	0.31	Taken from Hedlund [1988HED].
	NO_3^-	—	
$Al_2CO_3(OH)_2^{2+}$	Cl^-	0.26	Taken from Hedlund [1988HED].
	ClO_4^-	—	
	NO_3^-	—	
Zn^{2+}	Cl^-	—	
	ClO_4^-	0.33 ± 0.03	Reported by Ciavatta [1980CIA].
	NO_3^-	0.16 ± 0.02	
$ZnCO_3^{2+}$	Cl^-	0.35 ± 0.05	Taken from Ferri *et al.* [1985FER/GRE].
	ClO_4^-	—	
	NO_3^-	—	
Cd^{2+}	Cl^-	—	
	ClO_4^-	—	
	NO_3^-	0.09 ± 0.02	Reported by Ciavatta [1980CIA].
Hg^{2+}	Cl^-	—	
	ClO_4^-	0.34 ± 0.03	Reported by Ciavatta [1980CIA].
	NO_3^-	$-0.1 \pm 0.1^\dagger$	
Hg_2^{2+}	Cl^-	—	
	ClO_4^-	0.09 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	$-0.2 \pm 0.1^\dagger$	
Cu^{2+}	Cl^-	0.08 ± 0.01	Reported by Ciavatta [1980CIA].
	ClO_4^-	0.32 ± 0.02	
	NO_3^-	0.11 ± 0.01	
Ni^{2+}	Cl^-	0.17 ± 0.02	Reported by Ciavatta [1980CIA].
	ClO_4^-	0.370 ± 0.032	Derived from the ionic strength dependence of the osmotic and mean activity coefficient of $Ni(ClO_4)_2$ solution ([2005GAM/BUG], Section V.4.3).
	NO_3^-	0.182 ± 0.010	Derived from the ionic strength dependence of the osmotic and mean activity coefficient of $Ni(NO_3)_2$ solution ([2005GAM/BUG], Section V.6.1.2.1).

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
Co^{2+}	Cl^-	0.16 ± 0.02	
	ClO_4^-	0.34 ± 0.03	Reported by Ciavatta [1980CIA].
	NO_3^-	0.14 ± 0.01	
Fe^{2+}	Cl^-	0.17 ± 0.01	Determined in Section VI.4.3 of [2013LEM/BER].
	ClO_4^-	0.37 ± 0.04	Determined in Section VI.4.4 of [2013LEM/BER].
	NO_3^-	—	
$FeOH^{2+}$	Cl^-	—	
	ClO_4^-	0.46 ± 0.05	Section VII.1.3.1 of [2013LEM/BER]
	NO_3^-	—	
$FeCl^{2+}$	Cl^-	0.64 ± 0.06	Determined in Section VIII.2.3.1.3 of [2013LEM/BER].
	ClO_4^-	0.63 ± 0.05	Determined in Section VIII.2.3.2.2.1 of [2020LEM/PAL].
	NO_3^-	—	
$FeSCN^{2+}$	Cl^-	—	
	ClO_4^-	0.49 ± 0.05	Determined in Section X.1.2.1 of [2020LEM/PAL].
	NO_3^-	0.13 ± 0.04	Determined in Section X.1.2.1 of [2020LEM/PAL].
Mn^{2+}	Cl^-	0.13 ± 0.01	Reported by Ciavatta [1980CIA].
	ClO_4^-	—	
	NO_3^-	—	
$Zr(OH)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.62 ± 0.39	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.
	NO_3^-	—	
ZrF_2^{2+}	Cl^-	—	
	ClO_4^-	0.47 ± 0.08	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
$ZrCl_2^{2+}$	Cl^-	—	
	ClO_4^-	0.84 ± 0.11	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
$Zr(NO_3)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.84 ± 0.11	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
$YHCO_3^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Taken from Spahiu [1983SPA].
	NO_3^-	—	
$AmOH^{2+}$	Cl^-	−0.04 ± 0.07	Evaluated in [2003GUI/FAN] (*cf.* Section 12.3.1.1) from $\Delta\varepsilon$ (in NaCl solution) for the reactions $An^{3+} + n\,H_2O(l) \rightleftharpoons An(OH)_n^{(3-n)} + n\,H^+$.

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
AmF^{2+}	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmCl^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
AmN_3^{2+}	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmNO_2^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmNO_3^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmH_2PO_4^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
$AmSCN^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [1995SIL/BID].
	NO_3^-	—	
PuO_2^{2+}	Cl^-	—	
	ClO_4^-	0.46 ± 0.05	By analogy with $\varepsilon(UO_2^{2+},ClO_4^-)$ as derived from isopiestic measurements in [1992GRE/FUG].The uncertainty is increased because the value is estimated by analogy.
	NO_3^-	—	
PuF_2^{2+}	Cl^-		
	ClO_4^-	0.36 ± 0.17	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction.
	NO_3^-	—	
$PuCl^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.16	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding Am(III) reaction.
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
PuI^{2+}	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [2001LEM/FUG] by assuming $\varepsilon(PuI^{2+},ClO_4^-) \approx \varepsilon(AmSCN^{2+},ClO_4^-)$ and $\varepsilon(NH_4^+,I^-) \approx \varepsilon(Na^+,SCN^-)$.
	NO_3^-	—	
$PuSCN^{2+}$	Cl^-	—	
	ClO_4^-	0.39 ± 0.04	Estimated in [2001LEM/FUG] by assuming $\varepsilon(PuSCN^{2+},ClO_4^-) \approx \varepsilon(AmSCN^{2+},ClO_4^-)$.
	NO_3^-	—	
NpO_2^{2+}	Cl^-	—	
	ClO_4^-	0.46 ± 0.05	By analogy with $\varepsilon(UO_2^{2+},ClO_4^-)$ as derived from isopiestic measurements noted in [1992GRE/FUG]. The uncertainty is increased because the value is estimated by analogy.
	NO_3^-	—	
$(NpO_2)_2(OH)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.57 ± 0.10	See Section 8.1.2 in [2001LEM/FUG].
	NO_3^-	—	
NpF_2^{2+}	Cl^-	—	
	ClO_4^-	0.38 ± 0.17	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction.
	NO_3^-	—	
$NpSO_4^{2+}$	Cl^-	—	
	ClO_4^-	0.48 ± 0.11	Estimated on Section 10.1.2.1 of [2001LEM/FUG].
	NO_3^-	—	
$Np(SCN)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.38 ± 0.20	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction.
	NO_3^-	—	
$Ca_3[NpO_2(OH)_5]^{2+}$	Cl^-	– 0.20 ± 0.11	Estimated in [2020GRE/GAO].
	ClO_4^-	—	
	NO_3^-	—	
UO_2^{2+}	Cl^-	0.21 ± 0.02	Reported by Ciavatta [1980CIA].
	ClO_4^-	0.46 ± 0.03	Reported by Ciavatta [1980CIA].

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
	NO_3^-	0.24 ± 0.03	Reported by Ciavatta [1980CIA]. These coefficients were not used in [1992GRE/FUG] because they were evaluated by Ciavatta [1980CIA] without taking chloride and nitrate complexation into account. Instead, Grenthe *et al.* used $\varepsilon(UO_2^{2+},X)$ = (0.46 ± 0.03) kg·mol^{-1}, for X = Cl^-, ClO_4^- and NO_3^-.
$(UO_2)_2(OH)_2^{2+}$	Cl^-	0.69 ± 0.07	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+},X)$ = (0.46 ± 0.03) kg·mol^{-1}, where X = Cl^-, ClO_4^- and NO_3^-.
	ClO_4^-	0.57 ± 0.07	
	NO_3^-	0.49 ± 0.09	
$(UO_2)_3(OH)_4^{2+}$	Cl^-	0.50 ± 0.18	Evaluated in [1992GRE/FUG], using $\varepsilon(UO_2^{2+},X)$ = (0.46 ± 0.03) kg·mol^{-1}, where X = Cl^-, ClO_4^- and NO_3^-.
	ClO_4^-	0.89 ± 0.23	
	NO_3^-	0.72 ± 1.0	
UF_2^{2+}	Cl^-	—	
	ClO_4^-	0.3 ± 0.1	Estimated in [1992GRE/FUG].
	NO_3^-	—	
USO_4^{2+}	Cl^-	—	
	ClO_4^-	0.3 ± 0.1	Estimated in [1992GRE/FUG].
	NO_3^-	—	
$U(NO_3)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.49 ± 0.14	Evaluated in [1992GRE/FUG] using $\varepsilon(U^{4+},ClO_4^-)$ = (0.76 ± 0.06) kg·mol^{-1}.
	NO_3^-	—	
$Th(OH)_2^{2+}$	Cl^-	0.13 ± 0.05	Calculated in Section VII.3.6.1 of [2008RAN/FUG].
	ClO_4^-	0.33 ± 0.10	
	NO_3^-	0.10 ± 0.15	
ThF_2^{2+}	Cl^-	—	
	ClO_4^-	0.3 ± 0.1	See Table VIII-8 in Section VIII.1.2.1 of [2008RAN/FUG].
	NO_3^-	0.15 ± 0.20	
$ThSO_4^{2+}$	Cl^-	0.14 ± 0.15	See Section IX.1.3.2 of [2008RAN/FUG].
	ClO_4^-	0.3 ± 0.1	
	NO_3^-	—	
$Th(N_3)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.40 ± 0.15	Estimated in Section X.1.2 of [2008RAN/FUG].
	NO_3^-	—	
$Th(NO_3)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.43 ± 0.18	Estimated in Section X.1.3.3 of [2008RAN/FUG], using $\varepsilon(Th^{4+}, X)$ = (0.70 ± 0.10) kg·mol^{-1} for X = ClO_4^- and NO_3^-.
	NO_3^-	0.43 ± 0.18	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Th(H_2PO_4)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.4 ± 0.1	Estimated in Section X.2.3.2 of [2008RAN/FUG].
	NO_3^-	—	
$Th(SCN)_2^{2+}$	Cl^-	—	
	ClO_4^-	0.38 ± 0.20	See Section XI.1.3.6.1 of [2008RAN/FUG].
	NO_3^-	—	
Be^{2+}	Cl^-	—	
	ClO_4^-	0.30 ± 0.04	Taken from [1986BRU], where it appears to have been based on the average of the values for $\varepsilon(Mg^{2+}, ClO_4^-)$ and $\varepsilon(Ca^{2+}, ClO_4^-)$ [1980CIA].
	NO_3^-	—	
Mg^{2+}	Cl^-	0.19 ± 0.02	
	ClO_4^-	0.33 ± 0.03	Reported by Ciavatta [1980CIA].
	NO_3^-	0.17 ± 0.01	
Ca^{2+}	Cl^-	0.140 ± 0.003†	This review. $I_m \leq 3.0$.
	ClO_4^-	0.27 ± 0.03†	Reported by Ciavatta [1980CIA]. Confirmed in this review. $I_m \leq 3.0$.
	NO_3^-	0.02 ± 0.01	Reported by Ciavatta [1980CIA].
Ba^{2+}	Cl^-	0.07 ± 0.01	
	ClO_4^-	0.15 ± 0.02	Reported by Ciavatta [1980CIA].
	NO_3^-	−0.28 ± 0.03	
Al^{3+}	Cl^-	0.34 ± 0.02†	This review. $I_m \leq 5.4$. See Section V.5.2.5.
	ClO_4^-	0.52 ± 0.01	This review. $I_m \leq 1.0$. See Section V.5.2.5.
	NO_3^-	0.25 ± 0.02	This review. $I_m \leq 6.0$. See Section V.5.2.5.
Ni_2OH^{3+}	Cl^-	—	
	ClO_4^-	0.59 ± 0.15	By assuming $\varepsilon(Ni_2OH^{3+}, ClO_4^-) \approx \varepsilon(Be_2OH^{3+}, ClO_4^-)$, see Section V.3.1.1 in [2005GAM/BUG].
	NO_3^-	—	
Fe^{3+}	Cl^-	0.76 ± 0.03	Determined in Section VI.4.2.2 of [2013LEM/BER].
	ClO_4^-	0.73 ± 0.04	Determined in Sections VI.1.2.1.1.1 and VI.4.1 of [2013LEM/BER].
	NO_3^-	0.26 ± 0.08	Determined in Section IX.4.1.3 of [2020LEM/PAL].
Cr^{3+}	Cl^-	0.30 ± 0.03	Reported by Ciavatta [1980CIA].
	ClO_4^-	—	
	NO_3^-	0.27 ± 0.02	Reported by Ciavatta [1980CIA].
$ZrOH^{3+}$	Cl^-	—	
	ClO_4^-	0.57 ± 0.13	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
	NO_3^-	—	
$Zr_3(OH)_9^{3+}$	Cl^-	—	
	ClO_4^-	0.93 ± 0.35	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.
	NO_3^-	—	
ZrF^{3+}	Cl^-	—	
	ClO_4^-	0.63 ± 0.10	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
$ZrCl^{3+}$	Cl^-	—	
	ClO_4^-	0.87 ± 0.10	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
$Zr(NO_3)^{3+}$	Cl^-	—	
	ClO_4^-	0.88 ± 0.11	Determined from SIT plot in [2005BRO/CUR].
	NO_3^-	—	
La^{3+}	Cl^-	0.22 ± 0.02	Reported by Ciavatta [1980CIA].
	ClO_4^-	0.47 ± 0.03	
	NO_3^-	—	
$La^{3+} \rightarrow Lu^{3+}$	Cl^-	—	
	ClO_4^-	0.47 → 0.52	Taken from Spahiu [1983SPA].
	NO_3^-	—	
Am^{3+}	Cl^-	0.23 ± 0.02	The $\varepsilon(An^{3+},Cl^-)$ for An=Am and Cm is assumed to be equal to $\varepsilon(Nd^{3+},Cl^-)$ which is calculated from trace activity coefficients of Nd^{3+} ion in 0–4 m NaCl. These trace activity coefficients are based on the Pitzer ion interaction parameters evaluated in [1997KON/FAN] from osmotic coefficients in aqueous $NdCl_3$ – NaCl and $NdCl_3$ – $CaCl_2$.
	ClO_4^-	0.49 ± 0.03	Estimated in [1995SIL/BID].
	NO_3^-	—	
Pu^{3+}	Cl^-	—	
	ClO_4^-	0.49 ± 0.05	Estimated by analogy with $\varepsilon(Ho^{3+},ClO_4^-)$ [1983SPA] as in [1992GRE/FUG], [1995SIL/BID]. The uncertainty is increased because the value is estimated by analogy.
	NO_3^-	—	
$PuOH^{3+}$	Cl^-	—	
	ClO_4^-	0.50 ± 0.05	Estimated in [2001LEM/FUG].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
PuF^{3+}	Cl^-	—	
	ClO_4^-	0.56 ± 0.11	Estimated in [2001LEM/FUG].
	NO_3^-	—	
$PuCl^{3+}$	Cl^-	—	
	ClO_4^-	0.85 ± 0.09	Derived from the $\Delta\varepsilon$ evaluated in [2001LEM/FUG].
	NO_3^-	—	
$PuBr^{3+}$	Cl^-	—	
	ClO_4^-	0.58 ± 0.16	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction, and by assuming $\varepsilon(H^+,Br^-)\approx\varepsilon(Na^+,Br^-)$.
	NO_3^-	—	
Np^{3+}	Cl^-	—	
	ClO_4^-	0.49 ± 0.05	Estimated by analogy with $\varepsilon(Ho^{3+},ClO_4^-)$ [1983SPA] as in previous books in this series [1992GRE/FUG], [1995SIL/BID]. The uncertainty is increased because the value is estimated by analogy.
	NO_3^-	—	
$NpOH^{3+}$	Cl^-	—	
	ClO_4^-	0.50 ± 0.05	Estimated in [2001LEM/FUG].
	NO_3^-	—	
NpF^{3+}	Cl^-	—	
	ClO_4^-	0.58 ± 0.07	Evaluated in [2001LEM/FUG].
	NO_3^-	—	
$NpCl^{3+}$	Cl^-	—	
	ClO_4^-	0.81 ± 0.09	Derived from the $\Delta\varepsilon$ selected in [2001LEM/FUG].
	NO_3^-	—	
NpI^{3+}	Cl^-	—	
	ClO_4^-	0.77 ± 0.26	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding Np(IV) chloride reaction, and by assuming $\varepsilon(H^+,I^-)\approx\varepsilon(Na^+,I^-)$.
	NO_3^-	—	
$NpSCN^{3+}$	Cl^-	0.76 ± 0.12	Estimated in [2001LEM/FUG] by analogy with $\Delta\varepsilon$ of the corresponding U(IV) reaction.
	ClO_4^-	—	
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
U^{3+}	Cl^-	—	
	ClO_4^-	0.49 ± 0.05	Evaluated in [2001LEM/FUG] by analogy with $\varepsilon(Am^{3+}, ClO_4^-)$.
	NO_3^-	—	
UOH^{3+}	Cl^-	—	
	ClO_4^-	0.48 ± 0.08	Evaluated in [1992GRE/FUG].
	NO_3^-	—	
UF^{3+}	Cl^-	—	
	ClO_4^-	0.48 ± 0.08	Estimated in [1992GRE/FUG].
	NO_3^-	—	
UCl^{3+}	Cl^-	—	
	ClO_4^-	0.50 ± 0.10	Estimated in [2003GUI/FAN].
	NO_3^-	—	
UBr^{3+}	Cl^-	—	
	ClO_4^-	0.52 ± 0.10	Estimated in [1992GRE/FUG] using $\varepsilon(U^{4+},X)$ = (0.76 ± 0.06) kg·mol^{-1}, for X = Br^- and ClO_4^-.
	NO_3^-	—	
UI^{3+}	Cl^-	—	
	ClO_4^-	0.55 ± 0.10	Estimated in [1992GRE/FUG] using $\varepsilon(U^{4+},X)$ = (0.76 ± 0.06) kg·mol^{-1}, for X = I^- and ClO_4^-.
	NO_3^-	—	
UNO_3^{3+}	Cl^-	—	
	ClO_4^-	0.62 ± 0.08	Evaluated in [1992GRE/FUG] using $\varepsilon(U^{4+},X)$ = (0.76 ± 0.06) kg·mol^{-1} for X = NO_3^- and ClO_4^-.
	NO_3^-	—	
$ThOH^{3+}$	Cl^-	0.19 ± 0.05	See Table VII-18 in Section VII.3.6.1 of [2008RAN/FUG].
	ClO_4^-	0.48 ± 0.08	
	NO_3^-	0.20 ± 0.15	
ThF^{3+}	Cl^-	—	
	ClO_4^-	0.48 ± 0.08	Estimated in Section VIII.1.2.1 of [2008RAN/FUG] (Table VIII-8).
	NO_3^-	0.25 ± 0.20	
$ThCl^{3+}$	Cl^-	0.62 ± 0.11	Calculated in Section VIII.2.2.1.2 of [2008RAN/FUG] using $\varepsilon(Th^{4+},X)$ = (0.70 ± 0.10) kg·mol^{-1}, for X = Cl^- and ClO_4^-
	ClO_4^-	0.62 ± 0.11	
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$ThClO_3^{3+}$	Cl^-	—	
	ClO_4^-	0.62 ± 0.11	Calculated in Section VIII.2.2.2 of [2008RAN/FUG] using $\varepsilon(Th^{4+},X) = (0.70 \pm 0.10)$ kg·mol^{-1}, for X = ClO_3^- and ClO_4^-
	NO_3^-	—	
$ThBr^{3+}$	Cl^-	—	
	ClO_4^-	0.62 ± 0.11	Calculated in Section VIII.3.2.1 of [2008RAN/FUG] using $\varepsilon(Th^{4+},X) = (0.70 \pm 0.10)$ kg·mol^{-1}, for X = Br^- and ClO_4^-
	NO_3^-	—	
$ThBrO_3^{3+}$	Cl^-	—	
	ClO_4^-	0.62 ± 0.08	Calculated in Section VIII.3.2.2 of [2008RAN/FUG] using $\varepsilon(Th^{4+},X) = (0.70 \pm 0.10)$ kg·mol^{-1}, for X = BrO_3^- and ClO_4^-
	NO_3^-	—	
ThN_3^{3+}	Cl^-	—	
	ClO_4^-	0.55 ± 0.15	See Section X.1.2 of [2008RAN/FUG].
	NO_3^-	—	
$ThNO_3^{3+}$	Cl^-	—	
	ClO_4^-	0.56 ± 0.14	Calculated in Section X.1.3.3 of [2008RAN/FUG] using
	NO_3^-	0.56 ± 0.14	$\varepsilon(Th^{4+},X) = (0.70 \pm 0.10)$ kg·mol^{-1}, for X = ClO_4^- and NO_3^-.
$Th(H_2PO_4)^{3+}$	Cl^-	—	
	ClO_4^-	0.5 ± 0.1	Estimated in Section X.2.3.2 of [2008RAN/FUG].
	NO_3^-	—	
$Th(H_2PO_4)(H_3PO_4)^{3+}$	Cl^-	—	
	ClO_4^-	0.5 ± 0.1	Estimated in Section X.2.3.2 of [2008RAN/FUG].
	NO_3^-	—	
$ThSCN^{3+}$	Cl^-	—	
	ClO_4^-	0.50 ± 0.10	See Section XI.1.3.6.1 of [2008RAN/FUG].
	NO_3^-	—	
Be_2OH^{3+}	Cl^-	—	
	ClO_4^-	0.50 ± 0.05	Taken from [1986BRU], where the assumption was made that $\varepsilon(Be^{2+},ClO_4^-) = 0.30$ kg·mol^{-1}, apparently based on the average of the values for $\varepsilon(Mg^{2+}, ClO_4^-)$ and $\varepsilon(Ca^{2+}, ClO_4^-)$ [1980CIA].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Be_3(OH)_3^{3+}$	Cl^-	—	
	ClO_4^-	0.51 ± 0.05	Taken from [1986BRU], where the assumption was made that $\varepsilon(Be^{2+},ClO_4^-) = 0.30$ kg·mol^{-1}, apparently based on the average of the values for $\varepsilon(Mg^{2+}, ClO_4^-)$ and $\varepsilon(Ca^{2+}, ClO_4^-)$ [1980CIA].
	NO_3^-	—	
Sn^{4+}	Cl^-	—	
	ClO_4^-	0.7 ± 0.2	See Section VI.3.1 of [2012GAM/GAJ].
	NO_3^-	—	
$Al_3HCO_3(OH)_4^{4+}$	Cl^-	0.41	Taken from Hedlund [1988HED].
	ClO_4^-	—	
	NO_3^-	—	
$Ni_4(OH)_4^{4+}$	Cl^-	—	
	ClO_4^-	1.08 ± 0.08	Derived from $\Delta\varepsilon = 4\varepsilon(H^+,ClO_4^-) - \varepsilon(Ni_4OH_4^{4+},ClO_4^-) - 4\varepsilon(Ni^{2+},ClO_4^-) = (0.16 \pm 0.05)$ kg·mol^{-1} (see [2005GAM/BUG], Section V.3.1.1.1).
	NO_3^-	—	
$Fe_2(OH)_2^{4+}$	Cl^-	—	
	ClO_4^-	1.04 ± 0.10	Determined in Section VII.1.3.3 of [2013LEM/BER].
	NO_3^-	—	
Zr^{4+}	Cl^-	0.33 ± 0.09	
	ClO_4^-	0.89 ± 0.10	Determined in [2005BRO/CUR]
	NO_3^-	0.33 ± 0.35	
$Y_2CO_3^{4+}$	Cl^-	—	
	ClO_4^-	0.80 ± 0.04	Taken from Spahiu [1983SPA].
	NO_3^-	—	
Pu^{4+}	Cl^-	0.37 ± 0.10	Estimated in [2020GRE/GAO].
	ClO_4^-	—	
	NO_3^-	—	
Np^{4+}	Cl^-	—	
	ClO_4^-	0.84 ± 0.06	Derived from $\Delta\varepsilon = \varepsilon(Np^{4+},ClO_4^-) - \varepsilon(Np^{3+},ClO_4^-) = (0.35 \pm 0.03)$ kg·mol^{-1} [1989ROB], [1989RIG/ROB], [1990RIG].
	NO_3^-	—	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
U^{4+}	Cl^-	—	
	ClO_4^-	0.76 ± 0.06	Estimated in [1992GRE/FUG]. Using the measured value of $\Delta\varepsilon = \varepsilon(U^{4+},ClO_4^-) - \varepsilon(U^{3+},ClO_4^-) = (0.35 \pm 0.06)$ kg·mol^{-1} p.89 [1990RIG], where the uncertainty is re-calculated in [2001LEM/FUG] from the data given in this thesis, and $\varepsilon(U^{3+},ClO_4^-) = (0.49 \pm 0.05)$ kg·mol^{-1}, a value for $\varepsilon(U^{4+},ClO_4^-)$ can be calculated in the same way as is done for $\varepsilon(Np^{4+},ClO_4^-)$ and $\varepsilon(Pu^{4+},ClO_4^-)$. This value, $\varepsilon(U^{4+},ClO_4^-) = (0.84 \pm 0.06)$ kg·mol^{-1} is consistent with that tabulated $\varepsilon(U^{4+},ClO_4^-) = (0.76 \pm 0.06)$ kg·mol^{-1}, since the uncertainties overlap. The authors of [2001LEM/FUG] do not believe that a change in the previously selected value for $\varepsilon(U^{4+},ClO_4^-)$ is justified at present.
	NO_3^-	—	
Th^{4+}	Cl^-	0.25 ± 0.03	Reported by Ciavatta [1980CIA].
	ClO_4^-	0.70 ± 0.10	Evaluated in Section VI.3.1 of [2008RAN/FUG].
	NO_3^-	0.31 ± 0.12	Evaluated in Section VI.3.1 of [2008RAN/FUG].
$Th_4(OH)_{12}^{4+}$	Cl^-	0.25 ± 0.20	See Section VII.3.4.1.3 of [2008RAN/FUG].
	ClO_4^-	0.56 ± 0.42	
	NO_3^-	0.42 ± 0.50	
$Th(H_3PO_4)^{4+}$	Cl^-	—	
	ClO_4^-	0.7 ± 0.1	Estimated in Section X.2.3.2 of [2008RAN/FUG].
	NO_3^-	—	
$Al_3(OH)_4^{5+}$	Cl^-	0.66	Taken from Hedlund [1988HED]
	ClO_4^-	1.30	Taken from Hedlund [1988HED]
	NO_3^-	—	
$Th_2(OH)_3^{5+}$	Cl^-	0.29 ± 0.09	Calculated in Section VII.3.4.1.1 of [2008RAN/FUG].
	ClO_4^-	0.91 ± 0.21	
	NO_3^-	0.69 ± 0.25	
$Th_2(OH)_2^{6+}$	Cl^-	0.40 ± 0.16	Evaluated in Section VII.3.4.1.1 of [2008RAN/FUG].
	ClO_4^-	1.22 ± 0.24	
	NO_3^-	0.69 ± 0.26	
$Zr_3(OH)_4^{8+}$	Cl^-	0.33 ± 0.28	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.
	ClO_4^-	1.89 ± 0.31	
	NO_3^-	2.28 ± 0.35	

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Table B-4 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Zr_4(OH)_8^{8+}$	Cl^-	—	
	ClO_4^-	3.61 ± 1.02	Determined in [2005BRO/CUR] from the overall fit of the hydrolysis data as described in Appendix D.
	NO_3^-	—	
$Th_4(OH)_8^{8+}$	Cl^-	0.70 ± 0.20	
	ClO_4^-	1.69 ± 0.42	Evaluated in Section VII.3.4.1.3 of [2008RAN/FUG].
	NO_3^-	1.59 ± 0.51	
$Th_6(OH)_{15}^{9+}$	Cl^-	0.72 ± 0.30	
	ClO_4^-	1.85 ± 0.74	See details in Section VII.3.4.1.4 of [2008RAN/FUG].
	NO_3^-	2.20 ± 0.77	
$Th_6(OH)_{14}^{10+}$	Cl^-	0.83 ± 0.30	
	ClO_4^-	2.2 ± 0.3	Estimated in Section VII.3.4.1.4 of [2008RAN/FUG].
	NO_3^-	2.9 ± 0.5	

Table B- 5: Ion interaction coefficients, $\varepsilon(j,k)$ (kg·mol^{-1}), for anions j with k = Li^+, Na^+ and K^+ at 298.15 K. The uncertainties represent the 95 % confidence level. The ion interaction coefficients marked with † can be described more accurately with an ionic strength dependent function, listed in Table B-6.

j	k	$\varepsilon(j,k)$	Comments
OH^-	Li^+	$-0.02 \pm 0.03^\dagger$	
	Na^+	0.04 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	0.09 ± 0.01	
F^-	Li^+	—	
	Na^+	0.02 ± 0.02	Evaluated in [1992GRE/FUG].
	K^+	0.03 ± 0.02	[1988CIA]
HF_2^-	Li^+	—	
	Na^+	-0.11 ± 0.06	Evaluated in [1992GRE/FUG].
	K^+	—	
Cl^-	Li^+	0.10 ± 0.01	
	Na^+	0.03 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	0.00 ± 0.01	
ClO_3^-	Li^+	—	
	Na^+	-0.01 ± 0.02	Reported by Ciavatta [1980CIA].
	K^+	—	
ClO_4^-	Li^+	0.15 ± 0.01	
	Na^+	0.01 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	—	
Br^-	Li^+	0.13 ± 0.02	
	Na^+	0.05 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	0.01 ± 0.02	
BrO_3^-	Li^+	—	
	Na^+	-0.06 ± 0.02	Reported by Ciavatta [1980CIA].
	K^+	—	
I^-	Li^+	0.16 ± 0.01	
	Na^+	0.08 ± 0.02	Reported by Ciavatta [1980CIA].
	K^+	0.02 ± 0.01	
IO_3^-	Li^+	—	
	Na^+	-0.06 ± 0.02	Estimated in [1992GRE/FUG].
	K^+	—	
HSO_4^-	Li^+	—	
	Na^+	-0.01 ± 0.02	Reported by Ciavatta [1980CIA].
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
N_3^-	Li^+	—	
	Na^+	0.015 ± 0.020	See Section X.1.2 of [2008RAN/FUG].
	K^+	—	
NO_2^-	Li^+	$0.06 \pm 0.04^\dagger$	
	Na^+	0.00 ± 0.02	Reported by Ciavatta [1980CIA].
	K^+	-0.04 ± 0.02	Reported by Ciavatta [1988CIA].
NO_3^-	Li^+	0.08 ± 0.01	Reported by Ciavatta [1980CIA].
	Na^+	$-0.04 \pm 0.03^\dagger$	
	K^+	$-0.11 \pm 0.04^\dagger$	
$H_2PO_4^-$	Li^+	—	
	Na^+	$-0.063 \pm 0.005^\dagger$	This review. $I_m \leq 2.0$. See Section V.1.2.6.
	K^+	$-0.141 \pm 0.005^\dagger$	This review. $I_m \leq 1.8$. See Section V.1.2.6.
$H_2AsO_4^-$	Li^+	—	
	Na^+	$-0.08 \pm 0.00^\dagger$	This review. $I_m \leq 1.3$. See Section V.1.2.6.
	K^+	$0.13 \pm 0.02^\dagger$	
HCO_3^-	Li^+	—	
	Na^+	$0.00 \pm 0.02^\dagger$	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	-0.06 ± 0.05	Calculated in [2001LEM/FUG] from Pitzer coefficients [1998RAI/FEL].
Hox^-	Li^+	-0.28 ± 0.09	Evaluated in Section VI.3.5 of [2005HUM/AND].
	Na^+	-0.07 ± 0.01	
	K^+	-0.01 ± 0.08	
H_2cit^-	Li^+	-0.11 ± 0.03	Evaluated in Section VII.3.6 of [2005HUM/AND].
	Na^+	-0.05 ± 0.01	
	K^+	-0.04 ± 0.01	
CN^-	Li^+	—	
	Na^+	0.07 ± 0.03	As reported in [1992BAN/BLI].
	K^+	—	
SCN^-	Li^+	—	Reported by Ciavatta [1980CIA].
	Na^+	0.05 ± 0.01	
	K^+	-0.01 ± 0.01	
$HCOO^-$	Li^+	—	
	Na^+	0.03 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
CH_3COO^-	Li^+	0.05 ± 0.01	
	Na^+	0.08 ± 0.01	Reported by Ciavatta [1980CIA].
	K^+	0.09 ± 0.01	
H_3edta^-	Li^+	—	
	Na^+	-0.33 ± 0.14	Evaluated in Section VIII.3.7 of [2005HUM/AND].
	K^+	-0.14 ± 0.17	
$SiO(OH)_3^-$	Li^+	—	
	Na^+	0.00 ± 0.05	This review. $I_m \leq 5.0$. See Section V.3.2.4.
	K^+	—	
$Si_2O_2(OH)_5^-$	Li^+	—	
	Na^+	-0.08 ± 0.04	Estimated in [1992GRE/FUG].
	K^+	—	
$SiO_2(OH)_2^{2-}$	Li^+	—	
	Na^+	-0.07 ± 0.07	This review. $I_m \leq 3.0$. See Section V.3.2.4.
	K^+	—	
$Sn(OH)_3^-$	Li^+	—	
	Na^+	0.22 ± 0.03	See Section VII.1.1 of [2012GAM/GAJ].
	K^+	—	
$SnCl_3^-$	Li^+	—	
	Na^+	0.04 ± 0.07	See Section VIII.3.2.1 of [2012GAM/GAJ].
	K^+	—	
$SnBr_3^-$	Li^+	—	
	Na^+	0.16 ± 0.08	See Section VIII.3.3.1 of [2012GAM/GAJ].
	K^+	—	
$B(OH)_4^-$	Li^+	—	
	Na^+	$-0.09 \pm 0.01^\dagger$	This review. $I_m \leq 1.0$. See Section V.4.1.1.
	K^+	-0.03 ± 0.03	
$Al(OH)_4^-$	Li^+	—	
	Na^+	0.00 ± 0.02	This review. $I_m \leq 5.0$. See Section V.5.2.3.
	K^+	0.06 ± 0.01	This review. $I_m \leq 2.5$. See Section V.5.2.3.
$Ni(SCN)_3^-$	Li^+	—	
	Na^+	0.66 ± 0.13	Evaluated in [2005GAM/BUG] (see Section V.7.1.3.1).
	K^+	—	
$Ni(cit)^-$	Li^+	—	
	Na^+	0.22 ± 0.50	Evaluated in Section VII.7 of [2005HUM/AND].
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Fe(SO_4)_2^-$	Li^+	—	
	Na^+	0.24 ± 0.14	Estimated in Section IX.1.2.1.4.3 of [2013LEM/BER].
	K^+	—	
$TcO(OH)_3^-$	Li^+	—	
	Na^+	– 0.09 ± 0.09	Estimated in [2020GRE/GAO].
	K^+	—	
ZrF_5^-	Li^+	—	
	Na^+	– 0.14 ± 0.03	Determined from SIT plots in [2005BRO/CUR].
	K^+	—	
$Am(SO_4)_2^-$	Li^+	—	
	Na^+	– 0.05 ± 0.05	Estimated in [1995SIL/BID].
	K^+	—	
$Am(CO_3)_2^-$	Li^+	—	
	Na^+	– 0.14 ± 0.06	Evaluated Section 12.6.1.1.1 [2003GUI/FAN], from $\Delta\varepsilon_n$ in NaCl solution for the reactions $An^{3+} + n\,CO_3^{2-} \rightleftharpoons An(CO_3)_n^{(3-2n)}$ (An = Am, Cm) based on $\varepsilon(Am^{3+},Cl^-)$ = (0.23 ± 0.02) kg·mol^{-1} and $\varepsilon(Na^+,CO_3^{2-})$ = – (0.08 ± 0.03) kg·mol^{-1}.
	K^+	—	
$Am(ox)_2^-$	Li^+	—	
	Na^+	– 0.21 ± 0.08	Evaluated in Section VI.13 of [2005HUM/AND].
	K^+	—	
$Am(edta)^-$	Li^+	—	
	Na^+	0.01 ± 0.16	Evaluated in Section VIII.13.2.1 of [2005HUM/AND].
	K^+	0.01 ± 0.16	Estimated in [2005HUM/AND] Section VIII.13.2.1 by assuming $\varepsilon(K^+,Am(edta)^-) \approx \varepsilon(Na^+,Am(edta)^-)$.
$PuO_2CO_3^-$	Li^+	—	
	Na^+	– 0.18 ± 0.18	Estimated in [2001LEM/FUG] by analogy with $\varepsilon(Na^+,NpO_2CO_3^-)$.
	K^+	—	
$Pu(edta)^-$	Li^+	—	
	Na^+	—	
	K^+	0.01 ± 0.16	Estimated in [2005HUM/AND], Section VIII.12.2.1 by assuming $\varepsilon(K^+,Pu(edta)^-) \approx \varepsilon(Na^+,Am(edta)^-)$.
$NpO_2(OH)_2^-$	Li^+	—	
	Na^+	– 0.01 ± 0.07	Estimated in [2001LEM/FUG] (Section 8.1.3).
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$NpO_2(OH)_3^-$	Li^+	—	
	Na^+	-0.20 ± 0.02	Estimated in [2020GRE/GAO].
	K^+	—	
$NpO_2SO_4^-$	Li^+	—	
	Na^+	0.07 ± 0.13	Estimated in [2020GRE/GAO].
	K^+	—	
$NpO_2HPO_4^-$	Li^+	—	
	Na^+	-0.05 ± 0.11	Estimated in [2020GRE/GAO].
	K^+	—	
$NpO_2CO_3^-$	Li^+	—	
	Na^+	-0.18 ± 0.15	Calculated in [2001LEM/FUG] (Section 12.1.2.1.3).
	K^+	—	
$NpO_2(ox)^-$	Li^+	—	
	Na^+	-0.4 ± 0.1	Evaluated in Section VI.11.2.3 of [2005HUM/AND].
	K^+	—	
$NpO_2(H_2edta)^-$	Li^+	—	
	Na^+	-0.18 ± 0.16	Evaluated in Section VIII.11.2.3 of [2005HUM/AND].
	K^+	—	
$(NpO_2)_2CO_3(OH)_3^-$	Li^+	—	
	Na^+	0.00 ± 0.05	Estimated by analogy in [2001LEM/FUG] (Section 12.1.2.1.2).
	K^+	—	
$UO_2(OH)_3^-$	Li^+	—	
	Na^+	-0.24 ± 0.09	Estimated in [2020GRE/GAO].
	K^+	—	
$UO_2F_3^-$	Li^+	—	
	Na^+	-0.14 ± 0.05	Evaluated in [2003GUI/FAN], Section 9.4.2.2.1.1.
	K^+	—	
$UO_2(N_3)_3^-$	Li^+	—	
	Na^+	0.0 ± 0.1	Estimated in [1992GRE/FUG].
	K^+	—	
$(UO_2)_2CO_3(OH)_3^-$	Li^+	—	
	Na^+	0.00 ± 0.05	Estimated in [1992GRE/FUG].
	K^+	—	
UO_2cit^-	Li^+	—	
	Na^+	-0.11 ± 0.09	Evaluated in [2005HUM/AND].
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Th(OH)_3(CO_3)^-$	Li^+	—	
	Na^+	-0.05 ± 0.20	See Section XI.1.3.2 of [2008RAN/FUG].
	K^+	—	
$Mg(cit)^-$	Li^+	—	
	Na^+	0.03 ± 0.03	Evaluated in [2005HUM/AND].
	K^+	—	
$UO_2(Hedta)^-$	Li^+	—	
	Na^+	-0.18 ± 0.16	Evaluated in Section VIII.10.2.4 of [2005HUM/AND].
	K^+	—	
$Mg(Hedta)^-$	Li^+	—	
	Na^+	0.11 ± 0.20	Estimated in Section VIII.5.1 of [2005HUM/AND]
	K^+	—	
SO_3^{2-}	Li^+	—	
	Na^+	$-0.08 \pm 0.05^\dagger$	
	K^+	—	
SO_4^{2-}	Li^+	$-0.03 \pm 0.04^\dagger$	
	Na^+	$-0.12 \pm 0.06^\dagger$	
	K^+	-0.06 ± 0.02	Reported by Ciavatta [1988CIA].
$S_2O_3^{2-}$	Li^+	—	
	Na^+	$-0.08 \pm 0.05^\dagger$	
	K^+	—	
HPO_4^{2-}	Li^+	—	
	Na^+	$-0.168 \pm 0.008^\dagger$	This review. $I_m \leq 3$. See Section V.1.2.6.
	K^+	$-0.122 \pm 0.008^\dagger$	This review. $I_m \leq 2.62$. See Section V.1.2.6.
$HAsO_4^{2-}$	Li^+	—	
	Na^+	$-0.11 \pm 0.006\dagger$	This review. $I_m \leq 3$. See Section V.1.2.6.
	K^+	$-0.04 \pm 0.00\dagger$	
CO_3^{2-}	Li^+	—	
	Na^+	$-0.08 \pm 0.03^\dagger$	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	0.02 ± 0.01	Reported by Ciavatta [1980CIA].
ox^{2-}	Li^+	-0.51 ± 0.09	
	Na^+	-0.08 ± 0.01	Evaluated in Section VI.3.5 of [2005HUM/AND].
	K^+	0.07 ± 0.08	
$Hcit^{2-}$	Li^+	-0.17 ± 0.04	
	Na^+	-0.04 ± 0.02	Evaluated in Section VII.3.6 of [2005HUM/AND].
	K^+	-0.01 ± 0.02	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
H_2edta^{2-}	Li^+	—	
	Na^+	-0.37 ± 0.14	Evaluated in Section VII.3.7 of [2005HUM/AND].
	K^+	-0.17 ± 0.18	
$SiO_2(OH)_2^{2-}$	Li^+	—	
	Na^+	-0.10 ± 0.07	Evaluated in [1992GRE/FUG].
	K^+	—	
$Si_2O_3(OH)_4^{2-}$	Li^+	—	
	Na^+	-0.15 ± 0.06	Estimated in [1992GRE/FUG].
	K^+	—	
$Ni(ox)_2^{2-}$	Li^+	—	
	Na^+	-0.26 ± 0.03	Evaluated in Section VI.7.2 of [2005HUM/AND].
	K^+	—	
$Ni(CN)_4^{2-}$	Li^+	—	
	Na^+	0.185 ± 0.081	Evaluated in [2005GAM/BUG] (see Section V.7.1.2.1.1).
	K^+	—	
$Fe(CO_3)_2^{2-}$	Li^+		
	Na^+	-0.05 ± 0.05	By analogy. See Appendix A entry for [1992BRU/WER].
	K^+		
CrO_4^{2-}	Li^+	—	
	Na^+	$-0.06 \pm 0.04^\dagger$	
	K^+	$-0.08 \pm 0.04^\dagger$	
$Zr(OH)_6^{2-}$	Li^+	—	
	Na^+	-0.10 ± 0.10	Determined in [2005BRO/CUR] by analogy to known interaction coefficients of doubly charged anionic species with alkali ions (source: [2001LEM/FUG]).
	K^+	—	
ZrF_6^{2-}	Li^+	—	
	Na^+	-0.15 ± 0.06	Determined from SIT plots in [2005BRO/CUR].
	K^+	—	
$NpO_2(OH)_4^{2-}$	Li^+	—	
	Na^+	-0.12 ± 0.04	Estimated in [2020GRE/GAO].
	K^+	—	
$NpO_2(HPO_4)_2^{2-}$	Li^+	—	
	Na^+	-0.1 ± 0.10	Estimated in [2001LEM/FUG].
	K^+	—	
$NpO_2(CO_3)_2^{2-}$	Li^+	—	
	Na^+	-0.02 ± 0.14	Estimated by analogy in [2001LEM/FUG] Section 12.1.2.1.2
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
NpO_2cit^{2-}	Li^+	—	
	Na^+	-0.06 ± 0.03	Evaluated in Section VII.11 of [2005HUM/AND].
	K^+	—	
$NpO_2(Hedta)^{2-}$	Li^+	—	
	Na^+	0.07 ± 0.16	Estimated in Section VIII.11.2.3 of [2005HUM/AND].
	K^+	—	
$UO_2(OH)_4^{2-}$	Li^+	—	
	Na^+	0.01 ± 0.04	Estimated in [2020GRE/GAO].
	K^+	—	
$UO_2F_4^{2-}$	Li^+	—	
	Na^+	-0.30 ± 0.06	Evaluated in [2003GUI/FAN], Section 9.4.2.2.1.1.
	K^+	—	
$UO_2(SO_4)_2^{2-}$	Li^+	—	
	Na^+	-0.12 ± 0.06	Estimated in [1992GRE/FUG].
	K^+	—	
$UO_2(N_3)_4^{2-}$	Li^+	—	
	Na^+	-0.1 ± 0.1	Estimated in [1992GRE/FUG].
	K^+	—	
$UO_2(ox)_2^{2-}$	Li^+	—	
	Na^+	-0.18 ± 0.07	Estimated in Section VI.1.2.4.1 of [2005HUM/AND].
	K^+	—	
UO_2edta^{2-}	Li^+	—	
	Na^+	-0.22 ± 0.18	Estimated in Section VIII.10.2.4 of [2005HUM/AND].
	K^+	—	
$UO_2(CO_3)_2^{2-}$	Li^+	—	
	Na^+	-0.02 ± 0.09	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	—	
$(UO_2)_2(OH)_2(SO_4)_2^{2-}$	Li^+	—	
	Na^+	-0.14 ± 0.22	Evaluated in Section 9.5.1.1.2 of [2003GUI/FAN].
	K^+	—	
ThF_6^{2-}	Li^+	—	
	Na^+	-0.30 ± 0.06	See Table VIII-8 in Section VIII.1.2.1 of [2008RAN/FUG].
	K^+	—	
$Th(SO_4)_3^{2-}$	Li^+	-0.068 ± 0.003	In combination with $\varepsilon_2 = (0.093 \pm 0.007)$.
	Na^+	-0.091 ± 0.038	See Section IX.1.3.2 of [2008RAN/FUG].
	K^+	-0.091 ± 0.038	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Th(OH)_2(CO_3)_2^{2-}$	Li^+	—	
	Na^+	-0.1 ± 0.2	See Section XI.1.3.2 of [2008RAN/FUG].
	K^+	—	
$Th(OH)_4(CO_3)^{2-}$	Li^+	—	
	Na^+	-0.1 ± 0.2	See Section XI.1.3.2 of [2008RAN/FUG].
	K^+	—	
$Mg(ox)_2^{2-}$	Li^+	—	
	Na^+	-0.15 ± 0.03	Estimated in Section VI.5.1 of [2005HUM/AND].
	K^+	-0.15 ± 0.10	Estimated in [2005HUM/AND], Section VI.5.1 by assuming $\varepsilon(K^+, Mg(ox)_2^{2-}) \approx \varepsilon(Na^+, Mg(ox)_2^{2-})$.
$Mg(edta)^{2-}$	Li^+	—	
	Na^+	-0.01 ± 0.15	Evaluated in Section VIII.5.2 of [2005HUM/AND].
	K^+	—	
$Ca(ox)_2^{2-}$	Li^+	—	
	Na^+	-0.15 ± 0.10	Estimated in [2005HUM/AND], Section VI.5.2 by assuming $\varepsilon(Na^+, Ca(ox)_2^{2-}) \approx \varepsilon(Na^+, Mg(ox)_2^{2-})$.
	K^+	-0.15 ± 0.10	Estimated in [2005HUM/AND], Section VI.5.2 by assuming $\varepsilon(K^+, Ca(ox)_2^{2-}) \approx \varepsilon(Na^+, Mg(ox)_2^{2-})$.
cit^{3-}	Li^+	$-0.44 \pm 0.15^\dagger$	
	Na^+	$-0.076 \pm 0.030^\dagger$	
	K^+	0.02 ± 0.02	Evaluated in Section VI.3.6 of [2005HUM/AND].
$Hedta^{3-}$	Li^+	—	
	Na^+	-0.10 ± 0.14	Evaluated in Section VIII.3.7 of [2005HUM/AND].
	K^+	0.31 ± 0.18	
PO_4^{3-}	Li^+	—	
	Na^+	$-0.23 \pm 0.02^\dagger$	This review. $I_m \leq 4.2$. See Section V.1.2.6.
	K^+	$-0.09 \pm 0.01^\dagger$	
AsO_4^{3-}	Li^+	—	
	Na^+	$-0.17 \pm 0.02\dagger$	This review. $I_m \leq 4.2$. See Section V.1.2.6.
	K^+	$-0.01 \pm 0.01\dagger$	
$Si_3O_6(OH)_3^{3-}$	Li^+	—	
	Na^+	-0.25 ± 0.03	Estimated in [1992GRE/FUG].
	K^+	—	
$Si_3O_5(OH)_5^{3-}$	Li^+	—	
	Na^+	-0.25 ± 0.03	Estimated in [1992GRE/FUG].
	K^+	—	
$Si_4O_7(OH)_5^{3-}$	Li^+	—	
	Na^+	-0.25 ± 0.03	Estimated in [1992GRE/FUG].
	K^+	—	

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Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$Ni(CN)_5^{3-}$	Li^+	—	
	Na^+	0.25 ± 0.14	Evaluated in [2005GAM/BUG] (see Section V.7.1.2.1.1).
	K^+	—	
$Fe(CO_3)_3^{3-}$	Li^+	—	
	Na^+	-0.23 ± 0.07	By analogy. See Appendix A entry for [2005GRI].
	K^+	—	
$Am(CO_3)_3^{3-}$	Li^+	—	
	Na^+	-0.23 ± 0.07	Evaluated Section 12.6.1.1.1 [2003GUI/FAN], from $\Delta\varepsilon_n$ in NaCl solution for the reactions $An^{3+} + n\,CO_3^{2-} \rightleftharpoons An(CO_3)_n^{(3-2n)}$ (An = Am, Cm) based on $\varepsilon(Am^{3+},Cl^-) = (0.23 \pm 0.02)\,kg\cdot mol^{-1}$ and $\varepsilon(Na^+, CO_3^{2-}) = -(0.08 \pm 0.03)\,kg\cdot mol^{-1}$.
	K^+	—	
$Am(ox)_3^{3-}$	Li^+	—	
	Na^+	-0.23 ± 0.10	Estimated in [2005HUM/AND], Section VI.13.2.1 by assuming $\varepsilon(Na^+, Am(ox)_3^{3-}) \approx \varepsilon(Na^+, Am(CO_3)_3^{3-})$.
	K^+	—	
$Np(CO_3)_3^{3-}$	Li^+	—	
	Na^+	—	
	K^+	-0.15 ± 0.07	Estimated by analogy in [2001LEM/FUG] (Section 12.1.2.1.5).
$NpO_2(CO_3)_2^{3-}$	Li^+	—	
	Na^+	-0.33 ± 0.17	Calculated in [2001LEM/FUG] (Section 12.1.2.1.3).
	K^+	—	
$NpO_2(CO_3)_2OH^{3-}$	Li^+	—	
	Na^+	-0.40 ± 0.19	Estimated in [2001LEM/FUG] by analogy with $NpO_2(CO_3)_3^{4-}$.
	K^+	—	
$NpO_2(ox)_2^{3-}$	Li^+	—	
	Na^+	-0.3 ± 0.2	Evaluated in Section VI.11.2.3 of [2005HUM/AND].
	K^+	—	
NpO_2edta^{3-}	Li^+	—	
	Na^+	0.20 ± 0.16	Estimated in Section VIII.11.2.3 of [2005HUM/AND].
	K^+	—	
$edta^{4-}$	Li^+	—	
	Na^+	0.32 ± 0.14	Evaluated in Section VIII.3.7 of [2005HUM/AND].
	K^+	1.07 ± 0.19	

(Continued on next page)

Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$P_2O_7^{4-}$	Li^+	—	
	Na^+	-0.26 ± 0.05	Reported by Ciavatta [1988CIA].
	K^+	-0.15 ± 0.05	Reported by Ciavatta [1988CIA].
$Fe(CN)_6^{4-}$	Li^+	—	
	Na^+	—	
	K^+	-0.17 ± 0.03	
$Tc_2OCl_{10}^{4-}$	Li^+	—	
	Na^+	0.78 ± 0.45	Estimated in [2020GRE/GAO].
	K^+	—	
$Zr(CO_3)_4^{4-}$	Li^+	—	
	Na^+	-0.09 ± 0.20	This value, referring to $\varepsilon(NH_4^+, Zr(CO_3)_4^{4-})$ has been estimated in [2005BRO/CUR] by analogy with the homovalent ion pair $(Na^+, U(CO_3)_4^{4-})$, see [1997GRE/PLY].
	K^+	—	
$NpO_2(CO_3)_3^{4-}$	Li^+	—	
	Na^+	-0.40 ± 0.19	Calculated in [2001LEM/FUG] (Section 12.1.2.1.2).
	K^+	-0.62 ± 0.42	$\varepsilon(NH_4^+, NpO_2(CO_3)_3^{4-}) = -(0.78 \pm 0.25)$ kg·mol^{-1} is calculated in [2001LEM/FUG] (Section 12.1.2.2.1).
$U(CO_3)_4^{4-}$	Li^+	—	
	Na^+	-0.09 ± 0.10	These values differ from those estimated in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	—	
$UO_2(CO_3)_3^{4-}$	Li^+	—	
	Na^+	-0.01 ± 0.11	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	—	
$UO_2(ox)_3^{4-}$	Li^+	—	
	Na^+	-0.01 ± 0.11	Estimated in [2005HUM/AND], Section VI.10.2.4.1 by assuming $\varepsilon(Na^+, UO_2(ox)_3^{4-}) \approx \varepsilon(Na^+, UO_2(CO_3)_3^{4-})$.
	K^+	—	
$(UO_2)_3(OH)_4(SO_4)_3^{4-}$	Li^+	—	
	Na^+	0.6 ± 0.6	Estimated in Section 9.5.1.1.2 of [2003GUI/FAN].
	K^+	—	
$NpO_2(CO_3)_3^{5-}$	Li^+	—	
	Na^+	-0.53 ± 0.19	Calculated in [2001LEM/FUG] (Section 12.1.2.1.3).

(Continued on next page)

Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
	K^+	-0.22 ± 0.03	Evaluated in [2003GUI/FAN] (discussion of [1998ALM/NOV] in Appendix A) from $\Delta\varepsilon$ for the reactions $KNpO_2CO_3(s) + 2\,CO_3^{2-} \rightleftharpoons NpO_2(CO_3)_3^{5-} + K^+$ (in K_2CO_3–KCl solution) and $K_3NpO_2(CO_3)_2(s) + CO_3^{2-} \rightleftharpoons NpO_2(CO_3)_3^{5-} + 3K^+$ (in K_2CO_3 solution) (based on $\varepsilon(K^+, CO_3^{2-}) = (0.02 \pm 0.01)$ kg·mol^{-1}).
$UO_2(CO_3)_3^{5-}$	Li^+	—	
	Na^+	-0.62 ± 0.15	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	—	
$Th(OH)(CO_3)_4^{5-}$	Li^+	—	
	Na^+	-0.22 ± 0.13	Evaluated in Section XI.1.3.2.1 of [2008RAN/FUG].
	K^+	—	
$Np(CO_3)_5^{6-}$	Li^+	—	
	Na^+	—	
	K^+	-0.73 ± 0.68	Calculated in [2001LEM/FUG] (Section 12.1.2.1.4).
$(NpO_2)_3(CO_3)_6^{6-}$	Li^+	—	
	Na^+	-0.46 ± 0.73	Calculated in [2001LEM/FUG] (Section 12.1.2.1.2).
	K^+	—	
$U(CO_3)_5^{6-}$	Li^+	—	
	Na^+	-0.30 ± 0.15	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	-0.70 ± 0.31	Calculated in [2001LEM/FUG] from Pitzer coefficients [1998RAI/FEL].
$(UO_2)_3(CO_3)_6^{6-}$	Li^+	—	
	Na^+	0.37 ± 0.11	These values differ from those reported in [1992GRE/FUG]. See the discussion in [1995GRE/PUI]. Values for CO_3^{2-} and HCO_3^- are based on [1980CIA].
	K^+	—	
$(UO_2)_2NpO_2(CO_3)_6^{6-}$	Li^+	—	
	Na^+	0.09 ± 0.71	Estimated by analogy in [2001LEM/FUG] (Section 12.1.2.2.1).
	K^+	—	
$(UO_2)_5(OH)_8(SO_4)_4^{6-}$	Li^+	—	
	Na^+	1.10 ± 0.5	Estimated in Section 9.5.1.1.2 of [2003GUI/FAN].
	K^+	—	
$Th(CO_3)_5^{6-}$	Li^+	—	
	Na^+	-0.30 ± 0.15	Estimated in Section XI.1.3.2.1 of [2008RAN/FUG].
	K^+	—	

(Continued on next page)

Table B-5 (continued)

j	k	$\varepsilon(j,k)$	Comments
$(UO_2)_4(OH)_7(SO_4)_4^{7-}$	Li^+	—	
	Na^+	2.80 ± 0.7	Estimated in Section 9.5.1.1.2 of [2003GUI/FAN].
	K^+	—	

Table B- 6: Ion interaction coefficients, $\varepsilon_1(j,k)$ and $\varepsilon_2(j,k)$, both in kg·mol^{-1}, for cations j with $k = Cl^-$, ClO_4^- and NO_3^- (first part), and for anions j with $k = Li^+$, Na^+ and K^+ (second part), according to the relationship $\varepsilon = \varepsilon_1 + \varepsilon_2 \log_{10} I_m$ at 298.15 K. The data are taken from Ciavatta [1980CIA], [1988CIA] unless indicated otherwise. The uncertainties represent the 95% confidence level.

j k →	Cl^-		ClO_4^-		NO_3^-	
↓	ε_1	ε_2	ε_1	ε_2	ε_1	ε_2
NH_4^+			−0.088 ± 0.002	0.095 ± 0.012	−0.075 ± 0.001	0.057 ± 0.004
Tl^+			−0.18 ± 0.02	0.09 ± 0.02		
Ag^+					−0.1432 ± 0.0002	0.0971 ± 0.000
Pb^{2+}					−0.329 ± 0.007	0.288 ± 0.018
Hg^{2+}					−0.145 ± 0.001	0.194 ± 0.002
Hg_2^{2+}					−0.2300 ± 0.0004	0.194 ± 0.002
Al^{3+}	0.34 ± 0.04 [a]	0.40 ± 0.04 [a]				
$AlOH^{2+}$	−0.05 ± 0.10 [b]	0.64 ± 0.16 [b]				
Ca^{2+}	0.117 ± 0.010 [c]	0.072 ± 0.010 [c]	0.253 ± 0.020 [c]	0.031 ± 0.020 [c]		

j k →	Li^+		Na^+		K^+	
↓	ε_1	ε_2	ε_1	ε_2	ε_1	ε_2
OH^-	−0.039 ± 0.002	0.072 ± 0.006				
NO_2^-	0.02 ± 0.01	0.11 ± 0.01				
NO_3^-			−0.049 ± 0.001	0.044 ± 0.002	−0.131 ± 0.002	0.082 ± 0.006
$H_2PO_4^-$			−0.113 ± 0.001 [d]	0.107 ± 0.002 [d]	−0.147 ± 0.001 [d]	0.122 ± 0.003 [d]
$H_2AsO_4^-$			−0.08 ± 0.00 [d]	0.02 ± 0.00 [d]	−0.13 ± 0.02 [d]	0.06 ± 0.01 [d]
$B(OH)_4^-$			−0.156 ± 0.001 [e]	0.053 ± 0.001 [e]		
SO_3^{2-}			−0.125 ± 0.008	0.106 ± 0.009		
SO_4^{2-}	−0.068 ± 0.003	0.093 ± 0.007	−0.184 ± 0.002	0.139 ± 0.006		
$S_2O_3^{2-}$			−0.125 ± 0.008	0.106 ± 0.009		
HPO_4^{2-}			−0.196 ± 0.004 [d]	0.095 ± 0.010 [d]	−0.159 ± 0.002 [d]	0.129 ± 0.008 [d]
$HAsO_4^{2-}$			−0.12 ± 0.01 [d]	0.03 ± 0.02 [d]	−0.04 ± 0.01 [d]	−0.01 ± 0.02 [d]
CrO_4^{2-}			−0.090 ± 0.005	0.07 ± 0.01	−0.123 ± 0.003	0.106 ± 0.007
cit^{3-}	−0.55 ± 0.11 [6]	0.3 ± 0.2 [6]	−0.15 ± 0.03 [f]	0.13 ± 0.03 [f]		
PO_4^{3-}			−0.32 ± 0.02 [d]	0.16 ± 0.04 [d]	−0.13 ± 0.02 [d]	0.08 ± 0.03 [d]
AsO_4^{3-}			−0.25 ± 0.02 [d]	0.15 ± 0.04 [d]	0.01 ± 0.02 [d]	−0.04 ± 0.03 [d]
CO_3^{2-}			−0.209 ± 0.010 [g]	0.243 ± 0.010 [g]		
HCO_3^-			−0.054 ± 0.010 [g]	0.098 ± 0.010 [g]		

(a) This review. See Section V.5.2.1;
(b) This review. See Section V.5.2.2;
(c) This review. See Section V.6.2.2.5;
(d) This review. See Section V.1.2.6;
(e) This review. See Section V.4.1.1;
(f) See Section VII.3.6 of [2005HUM/AND];
(g) This review. See Section V.2.2.

Table B- 7: SIT interaction coefficients $\varepsilon(j,k)$ (kg·mol^{-1}) for neutral species, *j,* with *k,* electroneutral combination of ions at 298.15 K.

j $k\rightarrow$ ↓	$Na^+ + ClO_4^-$	$Na^+ + Cl^-$	$K^+ + NO_3^-$
H_2ox(aq)	0.00 ± 0.01 [b]	0.00 ± 0.01 [b]	0.00 ± 0.01 [b]
H_3cit(aq)	0.00 ± 0.01 [b]	0.00 ± 0.01 [b]	0.00 ± 0.01 [b]
H_4edta(aq)	– 0.29 ± 0.14	– 0.29 ± 0.14	– 0.29 ± 0.14
$Si(OH)_4$(aq)	0.10 ± 0.08 [h]		
$SnBr_2$(aq)	0.14 ± 0.07		
$SnSO_4$(aq)	0.19 ± 0.35		
$Sn(NO_3)_2$(aq)	0.130 ± 0.111		
Ni(ox)(aq)	– 0.07 ± 0.03	– 0.07 ± 0.03	
Ni(Hcit)(aq)	– 0.07 ± 0.5		
Ni(SCN)$_2$(aq)	0.38 ± 0.06 [a]		
Am(cit)(aq)		0.00 ± 0.05	
Np(edta)(aq)	– 0.19 ± 0.19 [g]		
UO_2ox(aq)	– 0.05 ± 0.06	– 0.05 ± 0.06	
Uedta(aq)	– 0.19 ± 0.19		
Mg(ox)(aq)		0.00 ± 0.03	0.0 ± 0.1[c]
Mg(Hcit)(aq)	0.02 ± 0.05	0.02 ± 0.05	
Ca(ox)(aq)	0.0 ± 0.1 [d]	0.0 ± 0.1 [e]	0.0 ± 0.1 [f]
$Si(OH)_4$(aq)	0.10 ± 0.08 [h]		– 0.05 ± 0.01 [i]

(a) See Section V.7.1.3.1 in [2005GAM/BUG].
(b) Basic assumption made in [2005HUM/AND]. See Sections VI.3.5 and VII.3.6 of that review for discussions.
(c) Estimated in Section VI.5.2 of [2005HUM/AND] assuming ε(Mg(ox)(aq), KNO_3) ≈ ε(Mg(ox)(aq), NaCl).
(d) Estimated in Section VI.5.2 of [2005HUM/AND] assuming ε(Ca(ox)(aq), $NaClO_4$) ≈ ε(Mg(ox)(aq), NaCl).
(e) Estimated in Section VI.5.2 of [2005HUM/AND] assuming ε(Ca(ox)(aq), NaCl) ≈ ε(Mg(ox)(aq), NaCl).
(f) Estimated in Section VI.5.2 of [2005HUM/AND] assuming ε(Ca(ox)(aq), KNO_3) ≈ ε(Mg(ox)(aq), NaCl).
(g) Estimated in Section VIII.11.2.2 of [2005HUM/AND] by assuming ε(Np(edta)(aq),$NaClO_4$) ≈ ε(Uedta(aq), $NaClO_4$).
(h) Evaluated in this review, $I_m \leq 1.05$ (based on only one value of Δε).
(i) Evaluated in this review, $I_m \leq 3.5$.

Table B-8: Other ion interaction coefficients $\varepsilon(j, k)$ (kg·mol^{-1}).

j	k	$\varepsilon(j,k)$	Comments
H^+	HSO_4^-	0.14 ± 0.05	Determined in [2013LEM/BER], p. 476, footnote.
NH_4^+	F^-	– 0.04 ± 0.03	Determined in Section VIII.1.1.1.1 of [2020LEM/PAL] (preferred to a value derived earlier in [2008RAN/FUG]).
Mg^{2+}	HCO_3^-	– 0.06 ± 0.05	Determined in this review. $I_m \leq 6.0$. See Section V.6.1.2.7.
Ca^{2+}	HCO_3^-	0.113 ± 0.030	Determined in this review. $I_m \leq 3.0$. See Section V.6.2.2.4.
Cs^+	SO_4^{2-}	– 0.20 ± 0.02	Determined in this review. $I_m \leq 4.86$. See Section V.6.1.2.5.

Appendix C

Assigned uncertainties[1]

This appendix describes the origin of the uncertainty estimates that are given in the TDB tables of selected data. The original text in [1992GRE/FUG] has been retained in [1995SIL/BID], [1999RAR/RAN] and [2001LEM/FUG], except for some minor changes. Because of the importance of the uncertainty estimates, the present review offers a more comprehensive description of the procedures used. The primary experimental data on which the thermodynamic data and their uncertainties are based consist of equilibrium constants, reported as $\log_{10}K$ values, with an accuracy that in general is in the range 0.01 to 0.1 $\log_{10}$-units (corresponding to 0.06 and 0.57 kJ·mol^{-1}, ΔG_r^o) respectively. In calorimetric data of various types the accuracy is at best 0.1 kJ·mol^{-1}, but the composite uncertainty in the enthalpy of reaction in most thermodynamic cycles is often much larger, the uncertainty in molar entropies is typically around 1-3 %. The uncertainty in the reported Gibbs energies and enthalpies of formation is larger because they also include the uncertainties in the auxiliary data used. The thermodynamic data based on these experimental values are, however, reported with three decimals, in the present and previous reviews, where only one and in exceptional cases two decimals are justified. Nevertheless the reported values are given with three decimals because the current reviews are partly based on CODATA values, which for unknown reasons, are listed with three decimals. We have kept the current three decimals to avoid re-calculation of the selected values. Users of the thermodynamic data should be aware that the three decimals do not correspond to the experimental uncertainty.

C.1 The general problem

The focus of this section is on the uncertainty estimates of equilibria in solution, where the key problem is analytical, *i.e.*, the determination of the stoichiometric composition

[1] This appendix essentially contains the text of the TDB-3 Guideline, [1999WAN/OST], earlier versions of which have been printed in all the previous NEA TDB reviews. Because of its importance in the selection of data and to guide the users of the values in Chapters 3, 4, 5, 6, 7 and 8, the text is reproduced here with minor revisions.

and equilibrium constants of complexes that are in rapid equilibrium with one another. We can formulate analyses of the experimental data in the following way: From N measurements, y_i, of the variable y we would like to determine a set of n equilibrium constants k_r, $r = 1, 2,\ldots, n$, assuming that we know the functional relationship:

$$y = f(k_1, k_2, \ldots k_r \ldots k_n; a_1, a_2, \ldots.) \tag{C.1}$$

where a_1, a_2 , *etc.* are quantities that can be varied but whose values (a_{1i}; a_{2i}; *etc.*) are assumed to be known accurately in each experiment from the data sets (y_i, a_{1i}, a_{2i},…), $i = 1, 2, \ldots N$. The functional relationship (C.1) is obtained from the chemical model proposed and in general several different models have to be tested before the "best" one is selected. Details of the procedures are given in Rossotti and Rossotti [1961ROS/ROS].

When selecting the functional relationship (C.1) and determining the set of equilibrium constants that best describes the experiments one often uses a least-squares method. Within this method, the “best” description is the one that will minimise the residual sum of squares, U:

$$U = \sum_i w_i \left[y_i - f(k_1 \ldots k_n;\ a_{1i},\ a_{2i} \ldots) \right]^2 \tag{C.2}$$

where w_i is the weight of each experimental measurement y_i.

The minimum of the function (C.2) is obtained by solving a set of normal equations:

$$\frac{\partial U}{\partial k_r} = 0,\ r = 1, \ldots.n \tag{C.3}$$

A “true” minimum is only obtained if:

- the functional relationship (C.1) is correct, *i.e*., if the chemical model is correct;
- all errors are random errors in the variable y, in particular there are no systematic errors;
- the random errors in y follow a Gaussian (normal) distribution;
- the weight $w_i(y_i,\ a_{1i},\ a_{2i}, \ldots.)$ of an experimental determination is an exact measure of its inherent accuracy.

To ascertain that the first condition is fulfilled requires chemical insight, such as information of the coordination geometry, relative affinity between metal ions and various donor atoms, *etc*. It is particularly important to test if the chemical equilibrium constants of complexes that occur in small amounts are chemically reasonable. Too many experimentalists seem to look upon the least-squares refinement of experimental data more as an exercise in applied mathematics than as a chemical venture. One of the

tasks in the review of the literature is to check this point. An erroneous chemical model is one of the more serious types of systematic error.

The experimentalist usually selects the variable that he/she finds most appropriate to fulfill the second condition. If the estimated errors in a_{1i}, a_{2i} … are smaller than the error in y_i, the second condition is reasonably well fulfilled. The choice of the error-carrying variable is a matter of choice based on experience, but one must be aware that it has implications, especially in the estimated uncertainty.

The presence of systematic errors is, potentially, the most important source of uncertainty. There is no possibility to handle systematic errors using statistics; statistical methods may indicate their presence, no more. Systematic errors in the chemical model have been mentioned. In addition there may be systematic errors in the methods used. By comparing experimental data obtained with different experimental methods one can obtain an indication of the presence and magnitude of such errors. The systematic errors of this type are accounted for both in the review of the literature and when taking the average of data obtained with different experimental methods. This type of systematic error does not seem to affect the selected data very much, as judged by the usually very good agreement between the equilibrium data obtained using spectroscopic, potentiometric and solubility methods.

The electrode calibration, especially the conversion between measured pH and $-\log_{10}[H^+]$ is an important source of systematic error. The reviewers have when possible corrected this error, as seen in many instances in Appendix A.

The assumption of a normal distribution of the random errors is a choice made in the absence of better alternatives.

Finally, a comment on the weights used in least-squares refinements; this is important because it influences the uncertainty estimate of the equilibrium constants. The weights of individual experimental points can be obtained by repeating the experiment several times and then calculating the average and standard deviation of these data. This procedure is rarely used, instead most experimentalists seem to use unit weight when making a least-squares analysis of their data. However, also in this case there is a weighting of the data by the number of experimental determinations in the parameter range where the different complexes are formed. In order to have comparable uncertainty estimates for the different complexes, one should try to have the same number of experimental data points in the concentration ranges where each of these complexes is predominant; a procedure very rarely used.

As indicated above, the assignment of uncertainties to equilibrium constants is not a straightforward procedure and it is complicated further when there is lack of primary experimental data. The uncertainty estimates given for the individual equilibrium constants reported by the authors and for some cases re-estimated by this review are given in the tables of this and previous reviews. The procedure used to obtain

these estimates is given in the original publications and in the Appendix A discussions. However, this uncertainty is still a subjective estimate and to a large extent based on "expert judgment".

C.2 Uncertainty estimates in the selected thermodynamic data

The uncertainty estimate in the selected thermodynamic data is based on the uncertainty of the individual equilibrium constants or other thermodynamic data, calculated as described in the following sections. A weighted average of the individual $\log_{10}K$ values is calculated using the estimated uncertainty of the individual experimental values to assign its weight. The uncertainty in this average is then calculated using the formulae given in the following text. This uncertainty depends on the number of experimental data points – for N data points with the same estimated uncertainty, σ, the uncertainty in the average is $\sigma / \sqrt{N}$. The average and the associated uncertainty reported in the tables of selected data are reported with many more digits than justified only in order to allow the users to back-track the calculations. The reported uncertainty is much smaller than the estimated experimental uncertainty and the users of the tables should look at the discussion of the selected constants in order to get a better estimate of the uncertainty in an experimental determination using a specific method.

One of the objectives of the NEA Thermochemical Database (TDB) project is to provide an idea of the uncertainties associated with the data selected in this review. As a rule, the uncertainties define the range within which the corresponding data can be reproduced with a probability of 95 % at any place and by any appropriate method. In many cases, the statistical treatment is limited or impossible due to the availability of only one or few data points. A particular problem has to be solved when significant discrepancies occur between different source data. This appendix outlines the statistical procedures, which were used for fundamentally different problems, and explains the philosophy used in this review when statistics were inapplicable. These rules are followed consistently throughout the series of reviews within the TDB Project. Four fundamentally different cases are considered:

1. One source datum available
2. Two or more independent source data available
3. Several data available at different ionic strengths
4. Data at non-standard conditions: Procedures for data correction and re-calculation.

C.3 One source datum

The assignment of an uncertainty to a selected value that is based on only one experimental source is a highly subjective procedure. In some cases, the number of data points, on which the selected value is based, allows the use of the "root mean square" [1982TAY] deviation of the data points, X_i, to describe the standard deviation, s_X, associated with the average, $\overline{X}$:

$$s_X = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(X_i - \overline{X})^2} \ . \tag{C.4}$$

The standard deviation, s_X, is thus calculated from the dispersion of the equally weighted data points, X_i, around the average $\overline{X}$, and the probability is 95 % that an X_i is within $\overline{X} \pm 1.96\ s_X$, see Taylor [1982TAY] (pp. 244-245). The standard deviation, s_X, is a measure of the precision of the experiment and does not include any systematic errors.

Many authors report standard deviations, s_X, calculated with Eq. (C.4) (but often not multiplied by 1.96), but these do not represent the quality of the reported values in absolute terms. Therefore, it is thus important not to confuse the standard deviation, s_X, with the uncertainty, σ. The latter reflects the reliability and reproducibility of an experimental value and also includes all kinds of systematic errors, s_j, that may be involved. The uncertainty, σ, can be calculated with Eq. (C.5), assuming that the systematic errors are independent.

$$\sigma_X = \sqrt{s_X^2 + \sum_j (s_j^2)} \tag{C.5}$$

The estimation of the systematic errors s_j (which, of course, have to relate to $\overline{X}$ and be expressed in the same units) can only be made by a person who is familiar with the experimental method. The uncertainty, σ, has to correspond to the 95 % confidence level preferred in this review. It should be noted that for all the corrections and re-calculations made (*e.g.*, temperature or ionic strength corrections) the rules of the propagation of errors have to be followed, as outlined in Section C.6.2.

More often, the determination of s_X is impossible because either only one or two data points are available, or the authors did not report the individual values. The uncertainty σ in the resulting value can still be estimated using Eq. (C.5) assuming that s_X^2 is much smaller than $\sum_j (s_j^2)$, which is usually the case anyway.

C.4 Two or more independent source data

Frequently, two or more experimental data sources are available, reporting experimental determinations of the desired thermodynamic data. In general, the quality of these determinations varies widely, and the data have to be weighted accordingly for the calculation of the mean. Instead of assigning weight factors, the individual source data,

X_i, are provided with an uncertainty, σ_i, that also includes all systematic errors and represents the 95 % confidence level, as described in Section C.3. The weighted mean $\overline{X}$ and its uncertainty, $\sigma_{\overline{X}}$, are then calculated according to Eqs. (C.6) and (C.7).

$$\overline{X} \equiv \frac{\sum_{i=1}^{N}\left(\frac{X_i}{\sigma_i^2}\right)}{\sum_{i=1}^{N}\left(\frac{1}{\sigma_i^2}\right)}, \tag{C.6}$$

$$\sigma_{\overline{X}} = \sqrt{\frac{1}{\sum_{i=1}^{N}\left(\frac{1}{\sigma_i^2}\right)}} \tag{C.7}$$

Equations (C.6) and (C.7) may only be used if all the X_i belong to the same parent distribution. If there are serious discrepancies among the X_i, one proceeds as described below under Section C.4.1 . It can be seen from Eq. (C.7) that $\sigma_{\overline{X}}$ is directly dependent on the absolute magnitude of the σ_i values, and not on the dispersion of the data points around the mean. This is reasonable because there are no discrepancies among the X_i, and because the σ_i values already represent the 95 % confidence level. The selected uncertainty, $\sigma_{\overline{X}}$, will therefore also represent the 95 % confidence level.

In cases where all the uncertainties are equal, $\sigma_i = \sigma$, Eqs. (C.6) and (C.7) reduce to Eqs. (C.8) and (C.9).

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i \tag{C.8}$$

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{N}} \tag{C.9}$$

Example C.1:

Five data sources report values for the thermodynamic quantity, *X*. The reviewer has assigned uncertainties that represent the 95 % confidence level as described in Section C.3.

i	X_i	σ_i
1	25.3	0.5
2	26.1	0.4
3	26.0	0.5
4	24.85	0.25
5	25.0	0.6

According to Eqs. (C.6) and (C.7), the following result is obtained:

$$\overline{X} = (25.3 \pm 0.2).$$

The calculated uncertainty, $\sigma_{\bar{X}} = 0.2$, appears relatively small, but is statistically correct, as the values are assumed to follow a Gaussian distribution. As a consequence of Eq. (C.7), $\sigma_{\bar{X}}$ will always come out smaller than the smallest σ_i. Assuming $\sigma_4 = 0.10$ instead of 0.25 would yield $\bar{X} = (25.0 \pm 0.1)$ and $\sigma_4 = 0.60$ would result in $\bar{X} = (25.6 \pm 0.2)$. In fact, the values $(X_i \pm \sigma_i)$ in this example are at the limit of consistency, *i.e.*, the range $(X_4 \pm \sigma_4)$ does not overlap with the ranges $(X_2 \pm \sigma_2)$ and $(X_3 \pm \sigma_3)$. There might be a better way to solve this problem. Three possible choices seem more reasonable:

i. The uncertainties, σ_i, are re-assigned because they appear too optimistic after further consideration. Some assessments may have to be reconsidered and the uncertainties re-assigned. For example, multiplying all the σ_i by 2 would yield $\bar{X} = (25.3 \pm 0.3)$.

ii. If reconsideration of the previous assessments gives no evidence for re-assigning the X_i and σ_i (95 % confidence level) values listed above, the statistical conclusion will be that all the X_i do not belong to the same parent distribution and cannot therefore be treated in the same group (*cf.* item iii below for a non-statistical
explanation). The values for i =1, 4 and 5 might be considered as belonging to Group A and the values for i = 2 and 3 to Group B. The weighted average of the values in Group A is X_A (i = 1, 4, 5) = (24.95 ± 0.21) and of those in Group B, X_B (i = 2, 3) = (26.06 ± 0.31), the second digit after the decimal point being carried over to avoid loss of information. The selected value is now determined as described below under "Discrepancies" (Section C.4.1, Case I). X_A and X_B are averaged (straight average, there is no reason for giving X_A a larger weight than X_B), and $\sigma_{\bar{X}}$ is chosen in such a way that it covers the complete ranges of expectancy of X_A and X_B. The selected value is then $\bar{X} = (25.5 \pm 0.9)$.

iii. Another explanation could be that unidentified systematic errors are associated with some values. If this seems likely to be the case, there is no reason for splitting the values up into two groups. The correct way of proceeding would be to calculate the unweighted average of all the five points and assign an uncertainty that covers the whole range of expectancy of the five values. The resulting value is then $\bar{X} = (25.4 \pm 1.0)$, which is rounded according to the rules in Section C.6.3 to $\bar{X} = (25.4 \pm 1.1)$.

C.4.1 Discrepancies

Two data are called discrepant if they differ significantly, *i.e.*, their uncertainty ranges do not overlap. In this context, two cases of discrepancies are considered. Case I: Two significantly different source data are available. Case II: Several, mostly consistent source data are available, one of them being significantly different, *i.e.*, an "outlier".

Case I. Two discrepant data: This is a particularly difficult case because the number of data points is obviously insufficient to allow the preference of one of the two values. If there is absolutely no way of discarding one of the two values and selecting the other, the only solution is to average the two source data in order to obtain the selected value, because the underlying reason for the discrepancy must be unrecognised systematic errors. There is no point in calculating a weighted average, even if the two source data have been given different uncertainties, because there is obviously too little information to give even only limited preference to one of the values. The uncertainty, $\sigma_{\bar{X}}$, assigned to the selected mean, $\bar{X}$, has to cover the range of expectation of both source data, X_1, X_2, as shown in Eq. (C.10),

$$\sigma_{\bar{X}} = \left| X_i - \bar{X} \right| + \sigma_{\max} \tag{C.10}$$

where i =1, 2, and $\sigma_{\max}$ is the larger of the two uncertainties σ_i, see Example C.1.ii and Example C.2.

Example C.2:

The following credible source data are given:

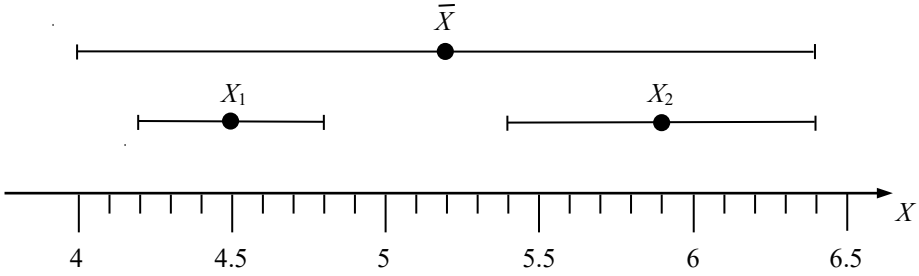
$$X_1 = (4.5 \pm 0.3)$$

$$X_2 = (5.9 \pm 0.5).$$

The uncertainties have been assigned by the reviewer. Both experimental methods are satisfactory and there is no justification to discard one of the data. The selected value is then:

$$\bar{X} = (5.2 \pm 1.2).$$

Figure C-1: Illustration for Example C.2.



Case II. Outliers: This problem can often be solved by either discarding the outlying data point, or by providing it with a large uncertainty to lower its weight. If, however, the outlying value is considered to be of high quality and there is no reason to

discard all the other data, this case is treated in a way similar to Case I. Example C.3 illustrates the procedure.

Example C.3:

The following data points are available. The reviewer has assigned the uncertainties and sees no justification for any change.

i	X_i	σ_i
1	4.45	0.35
2	5.9	0.5
3	5.7	0.4
4	6.0	0.6
5	5.2	0.4

There are two data sets that, statistically, belong to different parent distributions, A and B. According to Eqs. (C.6) and (C.7), the following average values are found for the two groups: $X_A(i = 1) = (4.45 \pm 0.35)$ and $X_B(i = 2, 3, 4, 5) = (5.62 \pm 0.23)$. The selected value will be the straight average of X_A and X_B, analogous to Example C.1:

$$\overline{X} = (5.0 \pm 0.9).$$

C.5 Several data at different ionic strengths

The extrapolation procedure for aqueous equilibria used in this review is the specific ion interaction model outlined in Appendix B. The objective of this review is to provide selected data sets at standard conditions, *i.e.,* among others, at infinite dilution for aqueous species. Equilibrium constants determined at different ionic strengths can, according to the specific ion interaction equations, be extrapolated to $I = 0$ with a linear regression model, yielding as the intercept the desired equilibrium constant at $I = 0$, and as the slope the stoichiometric sum of the ion interaction coefficients, $\Delta\varepsilon$. The ion interaction coefficient of the target species can usually be extracted from $\Delta\varepsilon$ and would be listed in the corresponding table of Appendix B.

The available source data may sometimes be sparse or may not cover a sufficient range of ionic strengths to allow a proper linear regression. In this case, the correction to $I = 0$ should be carried out according to the procedure described in Section C.6.1.

If sufficient data are available at different ionic strengths and in the same inert salt medium, a weighted linear regression will be the appropriate way to obtain both the constant at $I = 0$, $\bar{X}^{\circ}$, and $\Delta\varepsilon$. The first step is the conversion of the ionic strength from the frequently used molar (mol·dm^{-3}, M) to the molal (mol·kg^{-1}, m) scale, as described in Section 2.2. The second step is the assignment of an uncertainty, σ_i, to each data point X_i at the molality, $m_{k,i}$, according to the rules described in Section C.3. A large number of commercial and public domain computer programs and routines exist for weighted linear regressions. The subroutine published by Bevington [1969BEV] (pp.104 – 105) has been used for the calculations in the examples of this appendix. Eqs. (C.11) through (C.15) present the equations that are used for the calculation of the intercept $\bar{X}^{\circ}$ and the slope $-\Delta\varepsilon$:

$$\bar{X}^{\circ} = \frac{1}{\Delta}\left(\sum_{i=1}^{N}\frac{m_{k,i}^2}{\sigma_i^2}\sum_{i=1}^{N}\frac{X_i}{\sigma_i^2} - \sum_{i=1}^{N}\frac{m_{k,i}}{\sigma_i^2}\sum_{i=1}^{N}\frac{m_{k,i}X_i}{\sigma_i^2}\right) \tag{C.11}$$

$$-\Delta\varepsilon = \frac{1}{\Delta}\left(\sum_{i=1}^{N}\frac{1}{\sigma_i^2}\sum_{i=1}^{N}\frac{m_{k,i}X_i}{\sigma_i^2} - \sum_{i=1}^{N}\frac{m_{k,i}}{\sigma_i^2}\sum_{i=1}^{N}\frac{X_i}{\sigma_i^2}\right) \tag{C.12}$$

$$\sigma_{\bar{X}^{\circ}} = \sqrt{\frac{1}{\Delta}\sum_{i=1}^{N}\frac{m_{k,i}^2}{\sigma_i^2}} \tag{C.13}$$

$$\sigma_{\Delta\varepsilon} = \sqrt{\frac{1}{\Delta}\sum_{i=1}^{N}\frac{1}{\sigma_i^2}} \tag{C.14}$$

where

$$\Delta = \sum_{i=1}^{N}\frac{1}{\sigma_i^2}\sum_{i=1}^{N}\frac{m_{k,i}^2}{\sigma_i^2} - \left(\sum_{i=1}^{N}\frac{m_{k,i}}{\sigma_i^2}\right)^2. \tag{C.15}$$

In this way, the uncertainties, σ_i, are not only used for the weighting of the data in Eqs. (C.11) and (C.12), but also for the calculation of the uncertainties, $\sigma_{\bar{X}^{\circ}}$ and $\sigma_{\Delta\varepsilon}$, in Eqs. (C.13) and (C.14). If the σ_i represents the 95 % confidence level, $\sigma_{\bar{X}^{\circ}}$ and $\sigma_{\Delta\varepsilon}$ will also do so. In other words, the uncertainties of the intercept and the slope do not depend on the dispersion of the data points around the straight line, but rather directly on their absolute uncertainties, σ_i.

Example C.4:

Ten independent determinations of the equilibrium constant, $\log_{10}{}^{*}\beta$, for the reaction:

$$UO_2^{2+} + HF(aq) \rightleftharpoons UO_2F^{+} + H^{+} \tag{C.16}$$

are available in $HClO_4/NaClO_4$ media at different ionic strengths. Uncertainties that represent the 95 % confidence level have been assigned by the reviewer. A weighted linear regression, ($\log_{10}{}^{*}\beta + 2D$) *vs.* m_k, according to the formula, $\log_{10}{}^{*}\beta$ (C.16) + $2D$ = $\log_{10}{}^{*}\beta^{\circ}$ (C.16) $- \Delta\varepsilon\, m_k$, will yield the correct values for the intercept, $\log_{10}{}^{*}\beta^{\circ}$ (C.16), and the slope, $\Delta\varepsilon$. In this case, m_k corresponds to the molality of ClO_4^-. D is the Debye-Hückel term, *cf.* Appendix B.

i	$m_{ClO_4^-,i}$	$\log_{10} {}^*\beta + 2D$	σ_i
1	0.05	1.88	0.10
2	0.25	1.86	0.10
3	0.51	1.73	0.10
4	1.05	1.84	0.10
5	2.21	1.88	0.10
6	0.52	1.89	0.11
7	1.09	1.93	0.11
8	2.32	1.78	0.11
9	2.21	2.03	0.10
10	4.95	2.00	0.32

The results of the linear regression are:

$$\text{intercept} = (1.837 \pm 0.054) = \log_{10} {}^*\beta^\circ \text{ (C.16)}$$

$$\text{slope} = (0.029 \pm 0.036) = -\Delta\varepsilon \, .$$

Calculation of the ion interaction coefficient $\varepsilon(UO_2F^+, ClO_4^-) = \Delta\varepsilon + \varepsilon(UO_2^{2+}, ClO_4^-) - \varepsilon(H^+, ClO_4^-)$: from $\varepsilon(UO_2^{2+}, ClO_4^-) = (0.46 \pm 0.03)$ kg·mol^{-1}, $\varepsilon(H^+, ClO_4^-) = (0.14 \pm 0.02)$ kg·mol^{-1} (see Appendix B) and the slope of the linear regression, $\Delta\varepsilon = -(0.03 \pm 0.04)$ kg·mol^{-1}, it follows that $\varepsilon(UO_2F^+, ClO_4^-) = (0.29 \pm 0.05)$ kg·mol^{-1}. Note that the uncertainty (± 0.05) kg·mol^{-1} is obtained based on the rules of error propagation as described in Section C.6.2:

$$\sigma = \sqrt{(0.04)^2 + (0.03)^2 + (0.02)^2} \, .$$

The resulting selected values are thus:

$$\log_{10} {}^*\beta^\circ \text{ (C.16)} = (1.84 \pm 0.05)$$

$$\varepsilon(UO_2F^+, ClO_4^-) = (0.29 \pm 0.05) \text{ kg·mol}^{-1}.$$

C.5.1 Discrepancies or insufficient number of data points

Discrepancies are principally treated as described in Section C.4. Again, two cases can be defined. Case I: Only two data points are available. Case II: An "outlier" cannot be discarded. If only one data point is available, the procedure for correction to zero ionic strength outlined in Section C.6 should be followed.

Case I. Too few molalities: If only two source data are available, there will be no straightforward way to decide whether or not these two data points belong to the same parent distribution unless either the slope of the straight line is known or the two data refer to the same ionic strength. Drawing a straight line right through the two data points is an inappropriate procedure because all the errors associated with the two

source data would accumulate and may lead to highly erroneous values of $\log_{10} K^{\circ}$ and $\Delta\varepsilon$. In this case, an ion interaction coefficient for the key species in the reaction in question may be selected by analogy (charge is the most important parameter), and a straight line with the slope $\Delta\varepsilon$ as calculated may then be drawn through each data point. If there is no reason to discard one of the two data points based on the quality of the underlying experiment, the selected value will be the unweighted average of the two standard state data points obtained by this procedure, and its uncertainty must cover the entire range of expectancy of the two values, analogous to Case I in Section C.4. It should be mentioned that the ranges of expectancy of the corrected values at $I = 0$ are given by their uncertainties, which are based on the uncertainties of the source data at $I \neq 0$ and the uncertainty in the slope of the straight line. The latter uncertainty is not an estimate, but is calculated from the uncertainties in the ion interaction coefficients involved, according to the rules of error propagation outlined in Section C.6.2. The ion interaction coefficients estimated by analogy are listed in the table of selected ion interaction coefficients (Appendix B), but they are flagged as estimates.

Case II. Outliers and inconsistent data sets: This case includes situations where it is difficult to decide whether or not a large number of points belong to the same parent distribution. There is no general rule on how to solve this problem, and decisions are left to the judgment of the reviewer. For example, if eight data points follow a straight line reasonably well and two lie way out, it may be justified to discard the "outliers". If, however, the eight points are scattered considerably and two points are just a bit further out, one can probably not consider them as "outliers". It depends on the particular case and on the judgment of the reviewer whether it is reasonable to increase the uncertainties of the data to reach consistency, or whether the slope, $\Delta\varepsilon$, of the straight line should be estimated by analogy.

Example C.5:

Six reliable determinations of the equilibrium constant, $\log_{10} \beta$, of the reaction:

$$UO_2^{2+} + SCN^- \rightleftharpoons UO_2SCN^+ \qquad \text{(C.17)}$$

are available in different electrolyte media:

I_c = 0.1 M (KNO_3) $\quad \log_{10} \beta$ (C.17) = (1.19 ± 0.03)

I_c = 0.33 M (KNO_3) $\quad \log_{10} \beta$ (C.17) = (0.90 ± 0.10)

I_c = 1.0 M ($NaClO_4$) $\quad \log_{10} \beta$ (C.17) = (0.75 ± 0.03)

I_c = 1.0 M ($NaClO_4$) $\quad \log_{10} \beta$ (C.17) = (0.76 ± 0.03)

I_c = 1.0 M ($NaClO_4$) $\quad \log_{10} \beta$ (C.17) = (0.93 ± 0.03)

I_c = 2.5 M ($NaNO_3$) $\quad \log_{10} \beta$ (C.17) = (0.72 ± 0.03)

The uncertainties are assumed to represent the 95 % confidence level. From the values at I_c = 1 M, it can be seen that there is a lack of consistency in the data, and that a linear regression similar to that shown in Example C.4 would be inappropriate. Instead, the use of $\Delta\varepsilon$ values from reactions of the same charge type is encouraged. Analogies with $\Delta\varepsilon$ are more reliable than analogies with single ε values due to canceling effects. For the same reason, the dependency of $\Delta\varepsilon$ on the type of electrolyte is often smaller than for single ε values.

A reaction of the same charge type as Reaction (C.17), and for which $\Delta\varepsilon$ is well known, is:

$$UO_2^{2+} + Cl^- \rightleftharpoons UO_2Cl^+ . \qquad (C.18)$$

The value of $\Delta\varepsilon$ (C.18) = – (0.25 ± 0.02) kg·mol^{-1} was obtained from a linear regression using 16 experimental values between I_c = 0.1 M and I_c = 3 M Na(Cl,ClO_4) [1992GRE/FUG]. It is thus assumed that:

$$\Delta\varepsilon \text{ (C.17)} = \Delta\varepsilon \text{ (C.18)} = -(0.25 \pm 0.02) \text{ kg·mol}^{-1}.$$

The correction of $\log_{10}\beta$ (C.17) to I = 0 is done using the specific ion interaction equation, *cf.* TDB-2, which uses molal units:

$$\log_{10}\beta + 4D = \log_{10}\beta^\circ - \Delta\varepsilon\, I_m. \qquad (C.19)$$

D is the Debye-Hückel term and I_m the ionic strength converted to molal units by using the conversion factors listed in Table 2-5. The following list gives the details of this calculation. The resulting uncertainties in $\log_{10}\beta$ are obtained based on the rules of error propagation as described in Section C.6.2.

Table C-1: Details of the calculation of the equilibrium constant for Reaction (C.17) corrected to I = 0, using (C.19).

I_m	electrolyte	$\log_{10}\beta$	$4D$	$\Delta\varepsilon\, I_m$.	$\log_{10}\beta^\circ$
0.101	KNO_3	(1.19 ± 0.03)	0.438	– 0.025	(1.68 ± 0.03)[a]
0.335	KNO_3	(0.90 ± 0.10)	0.617	– 0.084	(1.65 ± 0.10)[a]
1.050	$NaClO_4$	(0.75 ± 0.03)	0.822	– 0.263	(1.31 ± 0.04)
1.050	$NaClO_4$	(0.76 ± 0.03)	0.822	– 0.263	(1.32 ± 0.04)
1.050	$NaClO_4$	(0.93 ± 0.03)	0.822	– 0.263	(1.49 ± 0.04)
2.714	$NaNO_3$	(0.72 ± 0.03)	0.968	– 0.679	(1.82 ± 0.13)[a]

a: These values were corrected for the formation of the nitrate complex, $UO_2NO_3^+$, by using $\log_{10}\beta^\circ(UO_2NO_3^+)$ = (0.30 ± 0.15) [1992GRE/FUG].

As was expected, the resulting values, $\log_{10}\beta^\circ$, are inconsistent and have therefore to be treated as described in Case I of Section C.4. That is, the selected value will be the unweighted average of $\log_{10}\beta^\circ$, and its uncertainty will cover the entire

range of expectancy of the six values. A weighted average would only be justified if the six values of $\log_{10} \beta^{\circ}$ were consistent. The result is:

$$\log_{10} \beta^{\circ} = (1.56 \pm 0.39).$$

C.6 Procedures for data handling

C.6.1 Correction to zero ionic strength

The correction of experimental data to zero ionic strength is necessary in all cases where a linear regression is impossible or appears inappropriate. The method used throughout the review involves application of the specific ion interaction equations described in detail in Appendix B. Two variables are needed for this correction, and both have to be provided with an uncertainty at the 95 % confidence level: the experimental source value, $\log_{10} K$ or $\log_{10} \beta$, and the stoichiometric sum of the ion interaction coefficients, $\Delta\varepsilon$. The ion interaction coefficients (see Tables B-4, B-5, B-6 and B-7 of Appendix B) required to calculate $\Delta\varepsilon$ may not all be known. Missing values therefore need to be estimated. It is recalled that the electric charge has the most significant influence on the magnitude of the ion interaction coefficients, and that it is in general more reliable to estimate $\Delta\varepsilon$ from known reactions of the same charge type, rather than to estimate single ε values. The uncertainty of the corrected value at $I = 0$ is calculated by taking into account the propagation of errors, as described below. It should be noted that the ionic strength is frequently given in moles per dm^3 of solution (molar, M) and has to be converted to moles per kg H_2O (molal, *m*), as the model requires. Conversion factors for the most common inert salts are given in Table 2.5.

Example C.6:

For the equilibrium constant of the reaction:

$$M^{3+} + 2H_2O(l) \rightleftharpoons M(OH)_2^+ + 2\,H^+, \qquad \text{(C.20)}$$

only one credible determination in 3 M $NaClO_4$ solution is known to be, $\log_{10}{}^{*}\beta\,(\text{C.20}) = -6.31$, to which an uncertainty of ± 0.12 has been assigned. The ion interaction coefficients are as follows:

$$\varepsilon(M^{3+}, ClO_4^-) = (0.56 \pm 0.03)\ \text{kg·mol}^{-1},$$

$$\varepsilon(M(OH)_2^+, ClO_4^-) = (0.26 \pm 0.11)\ \text{kg·mol}^{-1},$$

$$\varepsilon(H^+, ClO_4^-) = (0.14 \pm 0.02)\ \text{kg·mol}^{-1}.$$

The values of $\Delta\varepsilon$ and $\sigma_{\Delta\varepsilon}$ can be obtained readily (*cf.* Eq. (C.22)):

$$\Delta\varepsilon = \varepsilon(M(OH)_2^+, ClO_4^-) + 2\varepsilon(H^+, ClO_4^-) - \varepsilon(M^{3+}, ClO_4^-) = -0.22\ \text{kg·mol}^{-1},$$

$$\sigma_{\Delta\varepsilon} = \sqrt{(0.11)^2 + (2\times 0.02)^2 + (0.03)^2} = 0.12 \text{ kg·mol}^{-1}.$$

The two variables are thus:

$$\log_{10}{}^{*}\beta\,(\text{C.20}) = -(6.31 \pm 0.12),$$

$$\Delta\varepsilon = -(0.02 \pm 0.12) \text{ kg·mol}^{-1}.$$

According to the specific ion interaction model the following equation is used to correct for ionic strength for the reaction considered here:

$$\log_{10}{}^{*}\beta\,(\text{C.20}) + 6D = \log_{10}{}^{*}\beta^{\circ}\,(\text{C.20}) - \Delta\varepsilon\, m_{\text{ClO}_4^-}$$

D is the Debye-Hückel term:

$$D = \frac{0.509\sqrt{I_m}}{(1+1.5\sqrt{I_m})}.$$

The ionic strength, I_m, and the molality, $m_{\text{ClO}_4^-}$ ($I_m \approx m_{\text{ClO}_4^-}$), have to be expressed in molal units, 3 M $NaClO_4$ corresponding to 3.5 m $NaClO_4$ (see Section 2.2), giving $D = 0.25$. This results in:

$$\log_{10}{}^{*}\beta^{\circ}\,(\text{C.20}) = -4.88.$$

The uncertainty in $\log_{10}{}^{*}\beta^{\circ}$ is calculated from the uncertainties in $\log_{10}{}^{*}\beta$ and $\Delta\varepsilon$ (*cf*. Eq. (C.22)):

$$\sigma_{\log_{10}{}^{*}\beta^{\circ}} = \sqrt{\sigma^2_{\log_{10}{}^{*}\beta} + (m_{\text{ClO}_4^-}\sigma_{\Delta\varepsilon})^2} = \sqrt{(0.12)^2 + (3.5\times 0.12)^2} = 0.44$$

The selected, rounded value is:

$$\log_{10}{}^{*}\beta^{\circ}\,(\text{C.20}) = -(4.9 \pm 0.4).$$

C.6.2 Propagation of errors

Whenever data are converted or re-calculated, or other algebraic manipulations are performed that involve uncertainties, the propagation of these uncertainties has to be taken into account in a correct way. A clear outline of the propagation of errors is given by Bevington [1969BEV]. A simplified form of the general formula for error propagation is given by Eq. (C.21), supposing that X is a function of $Y_1, Y_2, \ldots, Y_N$.

$$\sigma_X^2 = \sum_{i=1}^{N}\left(\frac{\partial X}{\partial Y_i}\sigma_{Y_i}\right)^2 \qquad \text{(C.21)}$$

Equation (C.21) can be used only if the variables, Y_1, Y_2,…,Y_N , are independent or if their uncertainties are small, *i.e.*, the covariances can be disregarded. One of these two assumptions can almost always be made in chemical thermodynamics, and Eq. (C.21) can thus almost universally be used in this review. Eqs. (C.22) through (C.26) present explicit formulae for a number of frequently encountered algebraic expressions, where c, c_1, c_2 are constants.

$$X = c_1 Y_1 \pm c_2 Y_2 : \qquad \sigma_X^2 = (c_1 \sigma_{Y_1})^2 + (c_2 \sigma_{Y_2})^2 \qquad \text{(C.22)}$$

$$X = \pm c Y_1 Y_2 \text{ and } X = \pm \frac{c Y_1}{Y_2} : \qquad \left(\frac{\sigma_X}{X}\right)^2 = \left(\frac{\sigma_{Y_1}}{Y_1}\right)^2 + \left(\frac{\sigma_{Y_2}}{Y_2}\right)^2 \qquad \text{(C.23)}$$

$$X = c_1 Y^{\pm c_2} : \qquad \frac{\sigma_X}{X} = c_2 \frac{\sigma_Y}{Y} \qquad \text{(C.24)}$$

$$X = c_1 e^{\pm c_2 Y} : \qquad \frac{\sigma_X}{X} = c_2 \sigma_Y \qquad \text{(C.25)}$$

$$X = c_1 \ln\left(\left|c_2 Y\right|\right) : \qquad \sigma_X = c_1 \frac{\sigma_Y}{Y} \qquad \text{(C.26)}$$

Example C.7:

A few simple calculations illustrate how these formulae are used. The values have not been rounded.

Eq. (C.22) : $\Delta_r G_m = 2[-(277.4 \pm 4.9)]$ kJ·mol^{-1} – [– (467.3 ± 6.2)] kJ·mol^{-1}

$= -(87.5 \pm 11.6)$ kJ·mol^{-1}.

Eq. (C.23) : $K = \dfrac{(0.038 \pm 0.002)}{(0.0047 \pm 0.0005)} = (8.09 \pm 0.96)$

Eq. (C.24) : $K = 4(3.75 \pm 0.12)^3 = (210.9 \pm 20.3)$

Eq. (C.25) : $K^\circ = e^{\frac{-\Delta_r G_m^\circ}{RT}}$; $\Delta_r G_m^\circ = -(2.7 \pm 0.3)$ kJ·mol^{-1}

$R = 8.3145$ J·K^{-1}·mol^{-1}

$T = 298.15$ K

$K^\circ = (2.97 \pm 0.36)$.

Note that powers of 10 have to be reduced to powers of *e*, *i.e.*, the variable has to be multiplied by ln(10), *e.g.*,

$$\log_{10} K = (2.45 \pm 0.10);\ K = 10^{\log_{10} K} = e^{(\ln(10)\cdot\log_{10} K)} = (282 \pm 65).$$

Eq. (C.26): $\Delta_r G_m^o = -RT \ln K^o$; $K^o = (8.2 \pm 1.2) \times 10^6$

$R = 8.3145\ \text{J·K}^{-1}\text{·mol}^{-1}$

$T = 298.15\ \text{K}$

$$\Delta_r G_m^o = -(39.46 \pm 0.36)\ \text{kJ·mol}^{-1}$$

$$\ln K^o = (15.92 \pm 0.15)$$

$$\log_{10} K^o = \ln K^o / \ln(10) = (6.91 \pm 0.06).$$

Again, it can be seen that the uncertainty in $\log_{10} K^o$ cannot be the same as in $\ln K^o$. The constant conversion factor of ln(10) = 2.303 is also to be applied to the uncertainty.

C.6.3 Rounding

The standard rules to be used for rounding are:

1. When the digit following the last digit to be retained is less than 5, the last digit retained is kept unchanged.
2. When the digit following the last digit to be retained is greater than 5, the last digit retained is increased by 1.
3. When the digit following the last digit to be retained is 5 and
    a) there are no digits (or only zeroes) beyond the 5, an odd digit in the last place to be retained is increased by 1 while an even digit is kept unchanged;
    b) other non-zero digits follow, the last digit to be retained is increased by 1, whether odd or even.

This procedure avoids introducing a systematic error from always dropping or not dropping a 5 after the last digit retained.

When adding or subtracting, the result is rounded to the number of decimal places (not significant digits) in the term with the least number of places. In multiplication and division, the results are rounded to the number of significant digits in the term with the least number of significant digits.

In general, all operations are carried out in full, and only the final results are rounded, in order to avoid the loss of information from repeated rounding. For this reason, several additional digits are carried in all calculations until the final selected data set is developed, and only then are data rounded.

C.6.4 Significant digits

The uncertainty of a value basically defines the number of significant digits a value should be given.

Example: (3.478 ± 0.008)

(3.48 ± 0.01)

(2.8 ± 0.4)

(10 ± 1)

(105 ± 20).

In the case of auxiliary data or values that are used for later calculations, it is often inconvenient to round to the last significant digit. In the value (4.85 ± 0.26), for example, the "5" is close to being significant and should be carried along a re-calculation path in order to avoid loss of information. In particular cases, where the rounding to significant digits could lead to slight internal inconsistencies, digits with no significant meaning in absolute terms are nevertheless retained. The uncertainty of a selected value always contains the same number of digits after the decimal point as the value itself.

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Bogacheva, V. V.	[1995SHA/FED]
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Bol, W.	[1970BOL/GER]
Bolden, L.	[2000SIP/BOL]
Bologni, L.	[1981VAC/SAB]
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Booth, H. S.	[1950BOO/BID]
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Bosbach, D.	[2007BRU/BOS]
Bossant, M.	[2019MAR/SAN]
Bottinga, Y.	[1982RIC/BOT]

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Bourgeois, L.	[1886BOU]
Bourrié, G.	[1981BOU2], [1989BOU/GRI]
Bovet, N.	[2018BEN/BER]
Bowden, M. E.	[2018GRA/DEM]
Bowen, J. S.	[1971DIC/BOW]
Bower, V. E.	[1954BAT/BOW], [1956BAT/BOW], [1959BAT/BOW]
Boyer, L.	[2000MEI/GAL]
Bradley, D. J.	[1979BRA/PIT]
Bradley, S. M.	[1994BRA/HAN]
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Brand, H. E. A.	[2007FOR/WOO]
Brassinnes, S.	[2015RAG/BRA]
Bray, W. C.	[1909ABB/BRA]
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Breitkopf, O.	[1989WOL/BRE]
Brett, C. M. A.	[2002BUC/RON]
Bricke, E. V.	[1937BRI/VES]
Bricker, O. P.	[1973BRI/NES]
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Brito, F.	[2015MAR/ARA]
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Brown, A. S.	[1934SHE/BRO]
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Bruckenstein, S.	[1961BRU/NEL]
Brunet, F.	[2004BRU/MOR]
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Brönsted, J. N.	[1922BRO], [1922BRO2]
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Bychkov, A. Yu.	[2020BYC/BEN]
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Calvado, J. C. G.	[2005CRU/MIN]
Calvarin, G.	[1993DES/GRE]
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Cvitaš, T.	[1993MIL/CVI], [2007COH/CVI]
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Cyprès, R.	[1962CYP]
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Davis, R., Jr	[1943HAR/DAV]
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Delegard, C. H.	[2018GRA/DEM]
Dembowski, M.	[2018GRA/DEM]
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Dinsdale, A. T.	[1991DIN]
Ditmars, D. A.	[1971DIT/DOU], [1982DIT/ISH]
Dixon, D. A.	[1998FEL/DIX], [2006CHO/FEL]
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Dobrogowska, C.	[1997WU/DOB]
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Dorange, G.	[1983GUY/DOR], [1985GUY/DOR], [1990DOR/MAR]
Dorrepaal, J.	[1984DOR]
Douglas, T. B.	[1956FUR/DOU], [1963VIC/DOU], [1971DIT/DOU]
Dove, M. T.	[1995SWA/DOV]
Downes, C. J.	[1973CHI/DOW], [1974CHI/DOW]
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Drever, J. I.	[1982DRE]
Drummond, S. J., Jr.	[1996BRO/DRU]
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Du Plessis, L. A.	[1965STU/CHA]
Du, H.	[2000DU]

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Duchesne, J.	[1995DUC/REA]
Dujovne, I.	[2000BIA/DUJ]
Dunkle, S. A.	[1988PLU/PAR]
Durst, R. A.	[1985COV/BAT]
Dusek, M.	[2003DUS/CHA]
Dushina, S. V.	[1993VAS/KOC], [1993VAS/KOC2], [1994KOC/DUS], [1994VAS/KOC]
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Döll, W.	[1939DOL/KLE]
Eatough, D.	[1969IZA/EAT]
Edmunds, D. M.	[1972EDM/TAY]
Edwards, J. D.	[1955RUS/EDW]
Edwards, J. O.	[1960EDW/ROS]
Egan, E. P., Jr.	[1950EGA/WAK], [1951EGA/WAK], [1951EGA/WAK2], [1956EGA/WAK], [1961EGA/LUF]
Ehlers, R. W.	[1932HAR/EHL]
Ehlert, H.	[1912EHL/HEM]
Ehrlich, P.	[1963EHR/PEI]
Eick, H.	[1955ANS/BRE]
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Eitel, W.	[1957EIT]
Elders, J. M.	[1985AKI/ELD], [1989AKI/ELD2]
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Elmore, K. L.	[1946ELM/MAS], [1950EGA/WAK], [1951EGA/WAK], [1951EGA/WAK2], [1956EGA/WAK]
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Elyard, C. A.	[1968WAT/LIN]
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Ertmer, W.	[1966FIS/ERT]
Evans, J. C.	[1970MOO/EVA]
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Ezhova, Zh. A.	[1991PAL/KUZ], [1997KRI/ORL]
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Fang, C. H.	[2016MIA/FAN]
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Farid, N. A.	[1978EMA/FAR]
Farr, T. D.	[1950FAR]
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Favre, P. A.	[1873FAV/VAL]
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Felmy, A. R.	[1991FEL/RAI], [1995RAI/FEL], [1998FEL/DIX], [1998RAI/FEL], [2001FEL/CHO], [2006CHO/FEL], [2018GRA/DEM]
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Fernández-Prini, R.	[1969FER], [1987BIA/COR], [2000BIA/DUJ], [2003FER/ALV]
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Ferri, D.	[1983FER/GRE], [1984GRE/FER], [1985FER/GRE]

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Fil, V. D.	[2005KUM/PAR]
Filippov, V. K.	[1978FIL/ANT]
Filomena, M.	[1998GUI/LIT]
Finch, B.	[2007WOL/JOV]
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Frinck, C. R.	[1962FRI/PEE], [1963FRI/PEE]
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Fuentes-Montero, L.	[2011SAN/GAR]
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Gajda, T.	[2005GAM/BUG], [2005POW/BRO], [2012GAM/GAJ], [2022RAN/PAL]
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Ganor, J.	[1996MOG/GAN]
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Homann, K.	[1993MIL/CVI]

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Hynek, V.	[1997OBS/MAJ]

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Ingri, N.	[1957ING/LAG], [1961DYR/ING], [1962ING], [1962ING/SIL], [1963ING2], [1963ING3], [1967BIL/ING], [1981SJO/NOR], [1983SJO/HAG], [1985SJO/OEH]
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Manyà, J. J.	[2009MAN/ANT]
Mao, S.	[2019SHI/MAO]
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Marchal, G.	[1925MAR]
Marchand, A.	[1983GUY/DOR], [1985GUY/DOR], [1990DOR/MAR]
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Marinenko, J. W.	[1980HEM/MON]
Marion, G. M.	[2001MAR]
Markgraf, S. A.	[1985MAR/REE]
Markham, G. D.	[1996MAR/GLU]
Marquardt, R.	[2007COH/CVI]
Marshall, W. L.	[1964MAR/SLU], [1966MAR/SLU], [1967YEA/MAR], [1968MAR/SLU], [1969YEA/MAR], [1973KAL/YEA], [1973KAL/YEA2], [1980MAR], [1980MAR/WAR], [1982CHE/MAR], [1982FRA/MAR], [1984MAR/MES]

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Mason, C. M.	[1941MAS], [1946ELM/MAS], [1949MAS/CUL]
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Mathur, J. N.	[1991MAT]
Matignon, C.	[1932MAT/SEO]
Matraszek, A.	[2014MAT/RAD]
Matsui, M.	[1928MAT/NAK], [1932MAT/KAM]
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Matteoli, E.	[1989CAI/CON]
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Mayorga, G.	[1973PIT/MAY], [1974PIT/MAY]
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Seyl, R. G.	[1933THO/SEY]

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Taylor, P.	[2013LEM/BER], [2020LEM/PAL]
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Thonstad, J.	[1968THO]
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Tomson, M. B.	[2017DAI/KAN]

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Varghese, V.	[1996JAC/VAR]
Varzina, A. G.	[1973LYA/KRY]
Vasil'ev, V. P.	[1960YAT/VAS], [1962VAS], [1981VAS/VAS2], [1984VAS/KOC], [1993VAS/KOC], [1993VAS/KOC2], [1994VAS/KOC]
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Veits, I. V.	[1979GLU/GUR], [1981GLU/GUR], [1982GLU/GUR]
Verdes, G.	[1987VER/GOU], [1990VER], [1992VER/GOU]
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Vesala, S.	[1969SAL/HAK]
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Wahyudi, O.	[2011GIU/LEV]
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